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SOME BASIC CONCEPT OF CHEMISTRY

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NEET SYLLABUS

SOME BASIC CONCEPTS OF CHEMISTRY : General Introduction: Importance and scope of chemistry. concept of empirical and molecular formula; chemical reactions, stoichiometry and calculations based on stoichiometry. Laws of chemical combination, Dalton's atomic theory: Precision and accuracy, Significant figures

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OBJECTIVES

After studying this unit, we will be able to :

- understand and appreciate the role of chemistry in different spheres of life;
- explain the characteristics of three states of matter;
- classify different substances into elements, compounds and mixtures;
- explain various laws of chemical combination;
- appreciate significance of atomic mass, average atomic mass and molecular mass.
- describe the terms – mole and molar mass;
- calculate the mass percentage of different elements constituting a compound;
- determine empirical formula and molecular formula for a compound from the given experimental data;
- perform the stoichiometric calculations.

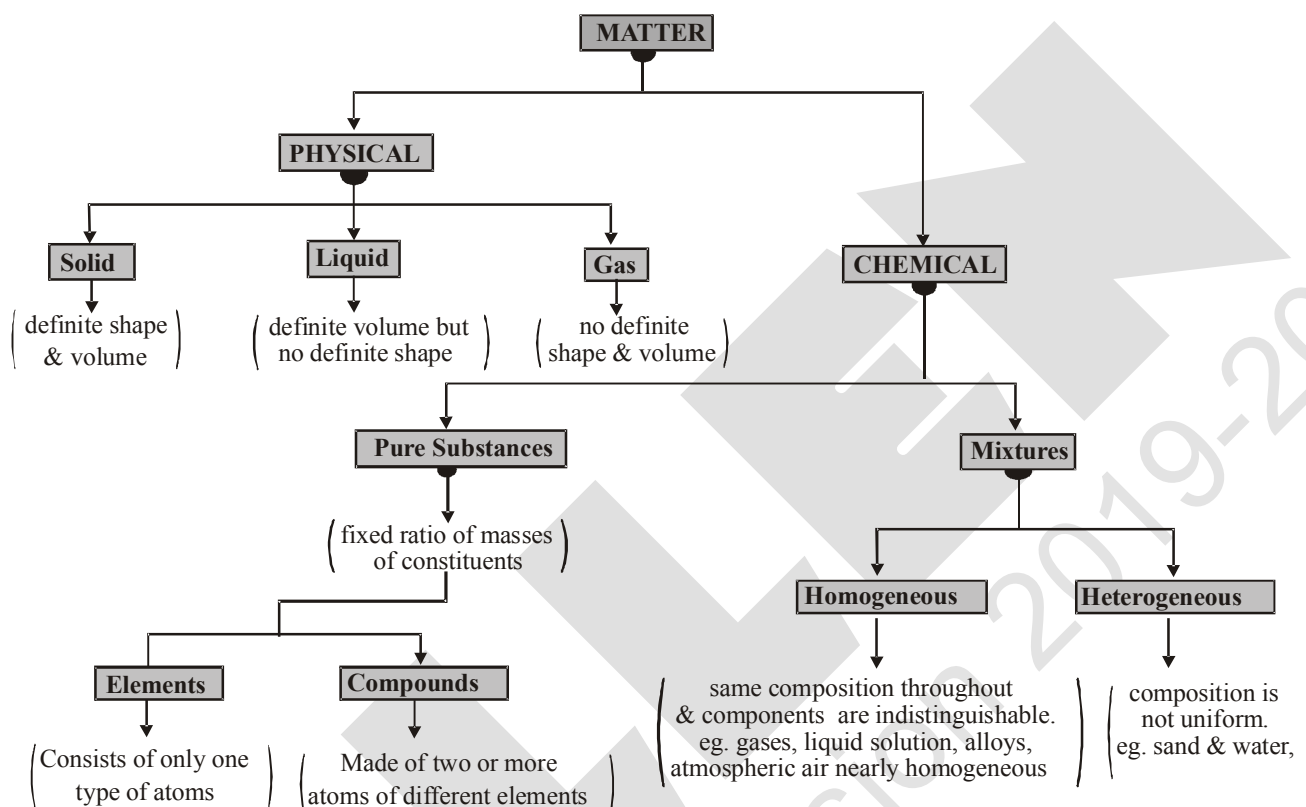
"Chemistry is the melodies you can play on vibrating strings"

Michio Kaku

SOME BASIC CONCEPTS OF CHEMISTRY

1.0 INTRODUCTION

Chemistry deals with the composition, structure and properties of matter. These aspects can be best described and understood in terms of basic constituents of matter: **atoms** and **molecules**. That is why chemistry is called the science of atoms and molecules. Can we see, weight and perceive these entities? Is it possible to count the number of atoms and molecules in a given mass of matter and have a quantitative relationship between the mass and number of these particles (atoms and molecules)? We will like to answer some of these questions in this Unit. We would further describe how physical properties of matter can be quantitatively described using numerical values with suitable units.

**Classification of universe**

Universe is classified into two types i.e. matter and energy.

- (A) **MATTER** : The thing which occupy space and having mass which can be felt by our five senses is called matter.

Matter is further classified into two categories :

(I) Physical classification

(II) Chemical classification

PHYSICAL CLASSIFICATION

It is based on physical state under ordinary conditions of temperature and pressure, **so on the basis of two nature of forces matter** can be classified into the following three ways :

(a) Solid

(b) Liquid

(c) Gas

- (a) **Solid** : A substance is said to be solid if it possesses a definite volume and a definite shape.

e.g. Sugar, Iron, Gold, Wood etc.

- (b) **Liquid** : A substance is said to be liquid if it possesses a definite volume but not definite shape. They take the shape of the vessel in which they are placed.

e.g. Water, Milk, Oil, Mercury, Alcohol etc.

- (c) **Gas** : A substance is said to be gas if it neither possesses a definite volume nor a definite shape. This is because they completely occupy the whole vessel in which they are placed.

e.g. Hydrogen(H_2), Oxygen(O_2), Carbon dioxide(CO_2) etc.

Chemical Classification

It may be classified into two types :

- (a) Pure Substances (b) Mixtures

(a) Pure Substance : A material containing only one type of substance. Pure Substance can not be separated into simpler substance by physical method.

e.g. : Elements = Na, Mg, Ca etc.

Compounds = HCl, H₂O, CO₂, HNO₃ etc.

Pure substances are classified into two types :

- (a) Elements (b) Compounds

(i) Elements : The pure substances containing only one kind of atoms.

It is classified into 3 types (depend on physical and chemical property)

(i) Metal → Zn, Cu, Hg, Ag, Sn, Pb etc.

(ii) Non-metal → N₂, O₂, Cl₂, Br₂, F₂, P₄, S₈ etc.

(iii) Metalloids → B, Si, As, Te etc.

(ii) Compounds : It is defined as pure substances containing more than one kind of elements or atoms which are combined together in a fixed proportion by weight and which can be decomposed into simpler substances by the suitable chemical methods. The properties of a compound are completely different from those of its constituent elements.

e.g. HCl, H₂O, H₂SO₄, HClO₄, HNO₃ etc.

(b) Mixtures : A material which contains more than one type of substance and which are mixed in any ratio by weight are known as mixtures. The properties of a mixture are same as the property individual components. The components of a mixture can be separated by simple physical methods.

Mixtures are classified into two types :

(i) Homogeneous mixtures : The mixtures in which all the components are present **uniformly** are called as homogeneous mixtures. Components of a mixture are present in single phase.

e.g. Water + Salt, Water + Sugar, Water + alcohol,

(ii) Heterogeneous mixtures : The mixtures in which all the components are present **non-uniformly** are called as Heterogeneous mixture.

e.g. Water + Sand, Water + Oil, blood, petrol etc.

Illustrations

Illustration 1. Which is an example of matter according to physical state at room temperature and pressure.
(1) solid (2) liquid (3) gas (4) all of these

Solution **Ans. (4)** According to the physical state at room temperature and pressure, the matter is present in 3 state solid, liquid & gas

Illustration 2. Which of the following are the types of the compound.

- (1) Organic compound (2) Inorganic compound
(3) Both (1) and (2) (4) None of these

Solution **Ans. (3)** Compound is divided into 2 types. Inorganic compound & Organic compound

Illustration 3. Which of the following is an example of a homogeneous mixture.

- (1) Water + Alcohol (2) Water + Sand (3) Water + Oil (4) None of these

Solution **Ans. (1)** Water and alcohol are completely mixed and form uniform solution.

Illustration 4. Which of the following is a solution.

- (1) Heterogeneous mixture (2) Homogeneous mixture
(3) Both (1) and (2) (4) None of these

Solution **Ans. (2)** Homogeneous mixture is a solution.

Illustration 5. Which of the following is a compound

- (1) graphite (2) producer gas (3) cement (4) marble

Solution **Ans. (4)** Marble = CaCO₃ = compound.

Illustration 6. Which of the following statements is/are true :

- (1) An element of a substance contains only one kind of atoms.
- (2) A compound can be decomposed into its components.
- (3) All homogeneous mixtures are solutions.
- (4) All of these

Solution **Ans. (4)**

Illustration 7. A pure substance can only be :-

- (1) A compound
- (2) An element
- (3) An element or a compound
- (4) A heterogenous mixture

Solution **Ans. (3)**

Illustration 8. Which one of the following is not a mixture :

- (1) Tap water
- (2) Distilled water
- (3) Salt in water
- (4) Oil in water

Solution **Ans. (2)**

1.1 S.I. UNITS (INTERNATIONAL SYSTEM OF UNITS)

Different types of units of measurements have been in use in different parts of the world e.g. kilograms, pounds etc. for mass ; miles, furlongs, yards etc. for distance.

To have a common system of units throughout the world. French Academy of Science, in 1791, introduced a new system of measurements called metric system in which the different units of a physical quantity are related to each other as multiples of powers of 10, e.g. 1 km = 10^3 m, 1cm = 10^{-2} m etc. This system of units was found to be so convenient that scientists all over the world adopted this system for scientific data.

(A) Seven Basic Units

The seven basic physical quantities in the International System of Units, their symbols, the names of their units (called the base units) and the symbols of these units are given in Table.

TABLE : SEVEN BASIC PHYSICAL QUANTITIES AND THEIR S.I. UNITS

Physical Quantity	Symbol	S.I. Unit	Symbol
Length	l	metre	m
Mass	m	kilogram	kg
Time	t	second	s
Electric current	I	ampere	A
Temperature	T	kelvin	K
Luminous intensity	I_u	candela	cd
Amount of the substance	n	mole	mol

(B) Prefixes Used With Units

The S.I. system recommends the multiples such as 10^3 , 10^6 , 10^9 etc. and fraction such as 10^{-3} , 10^{-6} , 10^{-9} etc., i.e. the powers are the multiples of 3. These are indicated by special prefixes. These along with some other fractions or multiples in common use, along with their prefixes are given below in Table and illustrated for length (m)

TABLE : SOME COMMONLY USED PREFIXES WITH THE BASE UNITS.

Prefix	Symbol	Multiplication Factor	Example
deci	d	10^{-1}	1 decimetre (dm) = 10^{-1} m
centi	c	10^{-2}	1 centimetre (cm) = 10^{-2} m
milli	m	10^{-3}	1 millimetre (mm) = 10^{-3} m
micro	μ	10^{-6}	1 micrometre (μ m) = 10^{-6} m
nano	n	10^{-9}	1 nanometre (nm) = 10^{-9} m
pico	p	10^{-12}	1 picometre (pm) = 10^{-12} m
femto	f	10^{-15}	1 femtometre (fm) = 10^{-15} m
atto	a	10^{-18}	1 attometre (am) = 10^{-18} m
deca	da	10^1	1 dekametre (dam) = 10^1 m
hecto	h	10^2	1 hectometre (hm) = 10^2 m
kilo	k	10^3	1 kilometre (km) = 10^3 m
mega	M	10^6	1 megametre (Mm) = 10^6 m
giga	G	10^9	1 gigametre (Gm) = 10^9 m
tera	T	10^{12}	1 terametre (Tm) = 10^{12} m
peta	P	10^{15}	1 petametre (Pm) = 10^{15} m
exa	E	10^{18}	1 exametre (Em) = 10^{18} m

As volume is very often expressed in litres, it is important to note that the equivalence in S.I. units for volume is as under: **1 litre (1L) = 1 dm³ = 1000 cm³**

and **1 millilitre (1mL) = 1 cm³ = 1cc**

(C) SOME IMPORTANT UNIT CONVERSIONS

- Length :**
 - 1 mile = 1760 yards
 - 1 yard = 3 feet
 - 1 foot = 12 inches
 - 1 inch = 2.54 cm
 - 1 Å = 10^{-10} m or 10^{-8} cm
- Mass :**
 - 1 Ton = 1000 kg
 - 1 Quintal = 100 kg
 - 1 kg = 2.205 Pounds (lb)
 - 1 kg = 1000 g
 - 1 gram = 1000 milli gram
 - 1 amu = 1.67×10^{-24} g
- Volume :**
 - 1 L = $1 \text{ dm}^3 = 10^{-3} \text{ m}^3 = 10^3 \text{ cm}^3 = 10^3 \text{ mL} = 10^3 \text{ cc}$
 - 1 mL = $1 \text{ cm}^3 = 10^{-6} \text{ m}^3$
 - = 1 cc
- Energy :**
 - 1 calorie = 4.184 joules $\approx$ 4.2 joules
 - 1 joule = 10^7 ergs
 - 1 litre atmosphere (L-atm) = 101.3 joule
 - 1 electron volt (eV) = 1.602×10^{-19} joule
- Pressure :**
 - 1 atmosphere (atm) = 760 torr
 - = 760 mm of Hg
 - = 76 cm of Hg
 - = 1.01325×10^5 pascal (Pa)
 - = $1.01325 \times 10^5 \text{ N/m}^2$
- Temperature :** $^{\circ}\text{C} + 273.15 = \text{K}$; $\frac{5}{9}(^{\circ}\text{F} - 32) = ^{\circ}\text{C}$

Some More Prefixes :

Semi	=	$\frac{1}{2}$	Mono	=	1
Sesqui	=	$\frac{3}{2} = 1.5$	Di or Bi	=	2
Tri	=	3	Tetra	=	4
Penta	=	5	Hexa	=	6
Hepta	=	7	Octa	=	8
Nona	=	9	Deca	=	10
Undeca	=	11	Do deca	=	12
Trideca	=	13	Tetra deca	=	14
Pentadeca	=	15	Hexa deca	=	16
Hepta deca	=	17	Octa deca	=	18
Nonadeca	=	19	Eicosa/Icoso	=	20

GOLDEN KEY POINTS

- The unit named after a scientist is started with a small letter and not with a capital letter e.g. unit of force is written as newton and not as Newton.
Likewise unit of heat and work is written as joule and not as Joule.
- Symbols of the units do not have a plural ending like 's'. For example we have 10 cm and not 10 cms.
- Words and symbols should not be mixed e.g. we should write either joules per mole or J mol^{-1} and not joules mol^{-1}
- Prefixes are used with the basic units e.g. kilometer means 1000 m (because meter is the basic unit).
Exception. Though kilogram is the basic unit of mass, yet prefixes are used with gram because in kilogram, kilo is already a prefix.
- A unit written with a prefix and a power is a power for the complete unit e.g. cm^3 means (centimeter)³ and not centi (meter)³.

Illustrations

Illustration 9. Which one of the following forms part of seven basic SI units :

- (1) Joule (2) Candela (3) Newton (4) Pascal

Solution **Ans. (2)**

Illustration 10 Convert 2 litre atmosphere into erg.

Solution 2 litre atmosphere = $2 \times 101.3 \text{ joule} = 2 \times 101.3 \times 10^7 \text{ erg.} = 202.6 \times 10^7 \text{ erg.}$
{1 litre atmosphere = 101.3J}

Illustration 11 Convert 2 atm into cm of Hg.

Solution 2 atm = $2 \times 76 \text{ cm of Hg} = 152 \text{ cm of Hg}$
{1 atmosphere = 76 cm of Hg}

Illustration 12 Convert 20 dm^3 into mL.

Solution 20 $\text{dm}^3 = 20 \text{ L} = 20 \times 1000 \text{ mL} = 2 \times 10^4 \text{ mL}$
1 $\text{dm}^3 = 1 \text{ L} = 1000 \text{ mL}$

Illustration 13 Convert 59 F into $^{\circ}\text{C}$.

Solution $^{\circ}\text{C} = \frac{5}{9} (\text{F} - 32) = \frac{5}{9} (59 - 32) = \frac{5}{9} \times 27 = 15^{\circ}\text{C}$

1.2 MOLE CONCEPT

In SI Units we represent mole by the symbol 'mol'. It is defined as follows :

- (i) **A mole is the amount of a substance that contains as many entities (atoms, molecules or other particles) as there are atoms in exactly 12g of the carbon - 12 isotope.**

It may be emphasised that the mole of a substance always contains the same number of entities, no matter what the substance may be. In order to determine this number precisely, the mass of a carbon-12 atom was determined by a mass spectrometer and found to be equal to 1.992648×10^{-23} g. Knowing that 1 mole of carbon weighs 12g, the number of atoms in it is equal to :

$$\frac{12\text{g/mol C}^{12}}{1.992648 \times 10^{-23}\text{g/C}^{12}\text{ atom}} = 6.0221367 \times 10^{23} \text{ atoms/mol}$$

- (ii) **In a simple way, we can say that mole has 6.0221367×10^{23} entities (atoms, molecules or ions etc.)**

The number of entities in 1 mole is so important that it is given a separate name and symbol, known as '**Avogadro constant**' denoted by N_A .

Here entities may represent atoms, ions, molecules or other subatomic entities. Chemists count the number of atoms and molecules by weighing. In a reaction we require these particles (atoms, molecules and ions) in a definite ratio. We make use of this relationship between numbers and masses of the particles for determining the stoichiometry of reactions.

Formula to get moles are following :

(i)
$$\text{Number of moles (n)} = \frac{\text{weight (g)}}{\text{molar mass}}$$

Where molar mass = gram atomic mass or gram molecular mass or gram ionic mass

(ii)
$$\text{Number of moles (n)} = \frac{V_{(L)}}{22.4} \quad (\text{Where } V = \text{Volume of gas in L at NTP or STP})$$

(iv)
$$\text{Number of moles (n)} = \frac{N}{N_A} \quad (\text{Where } N = \text{Number of particles})$$

$$\text{No. of moles of atoms} = \frac{\text{number of atoms}}{N_A} \quad \text{and} \quad \text{No. of moles of molecules} = \frac{\text{number of molecules}}{N_A}$$

SOME RELATED DEFINITIONS :**Atomic Mass (Relative Atomic Mass)**

It is defined as the number which indicates how many times the mass of one atom of an element is heavier in comparison to $1/12^{\text{th}}$ part of the mass of one atom of C^{12} .

Atomic mass unit (amu) : The quantity $1/12^{\text{th}}$ mass of an atom of C^{12} is known as atomic mass unit.

Since mass of 1 atom of $\text{C}^{12} = 1.9924 \times 10^{-23}$ g

$$\therefore 1/12^{\text{th}} \text{ part of the mass of 1 atom} = \frac{1.9924 \times 10^{-23} \text{ g}}{12} = 1.67 \times 10^{-24} \text{ g} = 1 \text{ a.m.u.} = \frac{1}{6.023 \times 10^{23}}$$

It may be noted that the atomic masses as obtained above are the relative atomic masses and not the actual masses of the atoms. These masses on the atomic mass scale are expressed in terms of atomic mass units (abbreviated as amu). Today, 'amu' has been replaced by 'u' which is known as unified mass.

One atomic mass unit (amu) is equal to $1/12^{\text{th}}$ of the mass of an atom of C^{12} isotope.

Thus the atomic mass of hydrogen is 1.008 amu while that of oxygen is 15.9994 amu (or taken as 16 amu).

Gram Atomic Mass (or Mass of 1 Gram Atom)

When numerical value of atomic mass of an element is expressed in grams then the value becomes gram atomic mass.

$$\text{gram atomic mass} = \text{mass of 1 gram atom} = \text{mass of 1 mole atom}$$

$$= \text{mass of } N_A \text{ atoms} = \text{mass of } 6.023 \times 10^{23} \text{ atoms.}$$

Ex. gram atomic mass of oxygen = mass of 1 **g atom** of oxygen = mass of 1 **mol atom** of oxygen.

$$= \text{mass of } N_A \text{ atoms of oxygen.} = \left(\frac{16}{N_A} \text{ g} \right) \times N_A = 16 \text{ g}$$

Molecular Mass (Relative Molecular Mass)

The number which indicates how many times the mass of one molecule of a substance is heavier in comparison to $1/12^{\text{th}}$ part of the mass of an atom of C^{12} .

Gram Molecular Mass (Mass of 1 Gram Molecule)

When numerical value of molecular mass of the substance is expressed in grams then the value becomes gram molecular mass.

$$\text{gram molecular mass} = \text{mass of 1 gram molecule} = \text{mass of 1 mole molecule}$$

$$= \text{mass of } N_A \text{ molecules} = \text{mass of } 6.023 \times 10^{23} \text{ molecules}$$

Ex. gram molecular mass of H_2SO_4 = mass of 1 **gram molecule** of H_2SO_4

= mass of 1 **mole molecule** of H_2SO_4

= mass of N_A molecules of H_2SO_4

$$= \left(\frac{98}{N_A} \text{ g} \right) \times N_A = 98 \text{ g}$$

Actual Mass

The mass of one atom or one molecule of a substance is called as actual mass.

Ex. (i) Actual mass of O_2 = 32 amu = $32 \times 1.67 \times 10^{-24} \text{ g}$ → Actual mass

(ii) Actual mass of H_2O = (2 + 16) amu = $18 \times 1.67 \times 10^{-24} \text{ g} = 2.99 \times 10^{-23} \text{ g}$

Atomicity – Total number of atoms in a **molecule** of elementary substance is called as atomicity.

Ex.

Molecule	Atomicity
H_2	2
O_2	2
O_3	3
NH_3	4

Illustrations

Illustration 14. Find out the volume and mole in 56 g nitrogen at STP**Solution**Molecular weight of N_2 is 28 g

$$(a) \quad \text{Calculation of volume : } \because 28 \text{ g of } N_2 \text{ occupies} = 22.4 \text{ L at STP}$$

$$\therefore 56 \text{ g of } N_2 \text{ occupies} = \frac{22.4}{28} \times 56 \text{ L} = 44.8 \text{ L at STP}$$

$$(b) \quad \text{Calculation of mole : } \because 28 \text{ g of } N_2 = 1 \text{ mol of } N_2$$

$$\therefore 56 \text{ g of } N_2 = \frac{1}{28} \times 56 = 2 \text{ mol of } N_2$$

Illustration 15. Calculate the volume and mass of 0.2 mol of O_3 at STP.**Solution**

$$(a) \quad \text{Calculation of volume : } \because \text{volume of 1 mole of } O_3 \text{ at STP} = 22.4 \text{ L}$$

$$\therefore \text{volume of 0.2 mole of } O_3 \text{ at STP} = 22.4 \times 0.2 = 4.48 \text{ L}$$

$$(b) \quad \text{Calculation of mass : } \because \text{mass of 1 mol of } O_3 = 48 \text{ g}$$

$$\therefore \text{mass of 0.2 mol of } O_3 = 48 \times 0.2 \text{ g} = 9.6 \text{ g}$$

Illustration 16. Find out the moles & mass in 1.12 L O_3 at STP.**Solution**

$$(a) \quad \text{Calculation of mole: } \because \text{at STP } 22.4 \text{ L of } O_3 \text{ contain} = 1 \text{ mol of } O_3$$

$$\therefore \text{at STP } 1.12 \text{ L of } O_3 \text{ contain} = \frac{1}{22.4} \times 1.12 = 0.05 \text{ mol of } O_3$$

$$(b) \quad \text{Calculation of mass : Molecular weight of } O_3 = 48 \text{ g}$$

$$\therefore \text{weight of } 22.4 \text{ L of } O_3 \text{ at STP is} = 48 \text{ g}$$

$$\therefore \text{weight of } 1.12 \text{ L of } O_3 \text{ at STP is} = \frac{48}{22.4} \times 1.12 = 2.4 \text{ g}$$

Illustration 17. Find out the mass of 10^{21} molecules of Cu.**Solution**

For Cu (i.e. mono atomic substance) number of atoms = number of molecules

$$\text{Number of moles of Cu} = \frac{N}{N_A} = \frac{10^{21}}{6.023 \times 10^{23}} = \frac{\text{weight}}{\text{Atomic weight}} = \frac{\text{weight}}{63.5}$$

$$\text{weight of Cu} = \frac{10^{21}}{6.023 \times 10^{23}} \times 63.5 = 0.106 \text{ g}$$

Illustration 18. Calculate the number of molecules and number of atoms present in 1 g of nitrogen ?**Solution**

$$\text{Number of moles (n)} = \frac{\text{weight}}{M_w} = \frac{1}{28} \Rightarrow \text{Number of molecules (N)} = \frac{N_A}{28}$$

$$\therefore 1 \text{ molecule of } N_2 \text{ gas contain} = 2 \text{ atoms}$$

$$\therefore \frac{N_A}{28} \text{ molecules of } N_2 \text{ gas contain} = 2 \times \frac{N_A}{28} = \frac{N_A}{14} \text{ atoms}$$

Illustration 19. Calculate the number of moles in 11.2 L at STP of oxygen.**Solution**

$$\text{Number of moles of } O_2 (n) = \frac{V}{22.4} = \frac{11.2}{22.4} = 0.5 \text{ mol}$$

Illustration 20. $\frac{1}{2}$ g molecule of oxygen. Find (i) mass, (ii) number of molecules, (iii) volume at STP. (iv) No. of oxygen atoms.

Solution

(i) $n = \frac{1}{2} \text{ mol} = \frac{\text{weight}}{M_w} = \frac{\text{weight}}{32} \Rightarrow \text{weight of oxygen} = 16 \text{ g}$

(ii) $n = \frac{1}{2} \text{ mol} = \frac{N}{N_A} \Rightarrow \text{Number of molecules of oxygen (N)} = \frac{N_A}{2}$

(iii) $n = \frac{1}{2} \text{ mol} = \frac{V}{22.4} \Rightarrow V = 11.2 \text{ L}$

(iv) 1 molecule of O_2 contain = 2 oxygen atoms.

$\frac{N_A}{2}$ molecules of O_2 contain = $\frac{N_A}{2} \times 2 = N_A$ oxygen atoms.

BEGINNER'S BOX-1

- The modern atomic weight scale is based on.
(1) C^{12} (2) O^{16} (3) H^1 (4) C^{13}
- Gram atomic weight of oxygen is
(1) 16 amu (2) 16 g (3) 32 amu (4) 32 g
- Molecular weight of SO_2 is :
(1) 64 g (2) 64 amu (3) 32 g (4) 32 amu
- 1 amu is equal to :-
(1) $\frac{1}{12}$ of C^{12} (2) $\frac{1}{14}$ of O^{16} (3) 1 g of H_2 (4) $1.66 \times 10^{-24} \text{ kg}$
- The actual molecular mass of chlorine is :
(1) $58.93 \times 10^{-24} \text{ g}$ (2) $117.86 \times 10^{-24} \text{ g}$ (3) $58.93 \times 10^{-24} \text{ kg}$ (4) $117.86 \times 10^{-24} \text{ kg}$

RELATION BETWEEN MOLECULAR WEIGHT AND VAPOUR DENSITY :

Vapour density (V.D) : Vapour density of a gas is the ratio of densities of gas & hydrogen at the same temperature & pressure.

$$\text{Vapour Density (V.D)} = \frac{\text{Density of gas}}{\text{Density of hydrogen}} = \frac{d_{\text{gas}}}{d_{H_2}} \quad \left\{ d = \frac{m(\text{mass})(g)}{V(\text{Volume})(mL)} \right.$$

$$\text{V.D} = \frac{(m_{\text{gas}}) \text{ for certain } V \text{ litre volume}}{(m_{H_2}) \text{ for certain } V \text{ litre volume}}$$

If N molecules are present in the given volume of a gas and hydrogen under similar condition of temperature and pressure.

$$\text{V.D.} = \frac{(m_{\text{gas}}) \text{ of } N \text{ molecules}}{(m_{H_2}) \text{ of } N \text{ molecules}} = \frac{(m_{\text{gas}}) \text{ of } 1 \text{ molecule}}{(m_{H_2}) \text{ of } 1 \text{ molecule}} = \frac{\text{Molecular mass of gas}}{2}$$

$$\therefore \boxed{\text{Molecular mass of gas (M}_w) = 2 \times \text{V.D}}$$

RELATION BETWEEN MOLAR MASS (M_w) & VOLUME :

$$\text{At STP, } M_w = 2 \times V.D = 2 \times \frac{d_{\text{gas}}}{d_{H_2}} = 2 \times \frac{(m_{\text{gas}}) \text{ for certain } V \text{ litre volume}}{(m_{H_2}) \text{ for certain } V \text{ litre volume}}$$

$$\text{or } M_w = 2 \times \frac{\text{mass of 1 litre gas}}{\text{mass of 1 litre } H_2}$$

$$\text{or } M_w = 2 \times \frac{\text{Mass of 1 litre gas}}{0.089g}$$

$$M_w(g) = 22.4 \times \text{mass of 1 litre gas}$$

$$d_{H_2} = 0.000089 \frac{g}{mL} = \frac{m}{V} = \frac{m}{1000mL}$$

$$V = 1 L = 1000 mL$$

$$\text{then } m_{H_2} = 0.089g$$

$$M_w(g) = \text{Mass of 22.4 litre gas}$$

$$\text{or } M_w(g) \equiv 22.4 \text{ litre (at STP)}$$

GRAM MOLECULAR VOLUME (GMV)

At NTP, the volume of 1 mole of gaseous substance is 22.4 litre is called as gram molecular volume.

At NTP, $d_{H_2} = 0.000089 \text{ g/mL} = \text{mass/volume} = \text{mass}/1000 \text{ mL}$

If volume = 1 L = 1000 mL then mass = 0.089 g

$\therefore 0.089g \text{ } H_2 \text{ occupies} = 1 \text{ L at STP}$

$\therefore 1 \text{ g } H_2 \text{ occupies} = \frac{1 \text{ litre}}{0.089} \text{ at STP}$

$\therefore 2 \text{ g or } 1 \text{ mol } H_2 \text{ occupies} = \frac{1 \text{ litre}}{0.089} \times 2 = 22.4 \text{ L at STP}$

1 mole of any gaseous substance occupy 22.4 litre of volume at NTP or STP

$$1 \text{ mol} \equiv 22.4 \text{ L (at STP)}$$

Illustrations

Illustration 21. Calculate the number of atoms of chlorine in 2.08 g of $BaCl_2$. (Atomic weight of Ba = 137, Cl = 35.5)

Solution Number of moles of $BaCl_2$ (n) = $\frac{\text{weight}}{M_w} = \frac{2.08}{208} = 0.01 \text{ mol} = \frac{N}{N_A}$
 Number of molecules of $BaCl_2$ (N) = $0.01 N_A$
 1 molecule of $BaCl_2$ contain = 2 chlorine atoms.
 $0.01 N_A$ molecules $BaCl_2$ contain = $2 \times 0.01 N_A$ Chlorine atoms. = $2 \times 10^{-2} N_A$ Chlorine atoms.

Illustration 22. Calculate the number of molecules and number of atoms present in 1.2 g of ozone.

Solution Number of moles of O_3 (n) = $\frac{\text{weight}}{M_w} = \frac{1.2}{48} = \frac{1}{40} \text{ mol}$

$\Rightarrow$ number of molecules of O_3 (N) = $\frac{N_A}{40}$

$\therefore$ 1 molecule of O_3 contain = 3 atoms, $\therefore \frac{N_A}{40}$ molecules O_3 contain = $\frac{3N_A}{40}$ atoms.

Illustration 23. Calculate the number of atoms present in one drop of water having mass 1.8 g.

Solution Number of moles of H_2O (n) = $\frac{\text{weight}}{M_w} = \frac{1.8}{18} = 0.1 \text{ mol}$

Number of molecules of H_2O (N) = $0.1 N_A$

$\therefore$ 1 molecule of H_2O contain = 3 atoms

$\therefore$ $0.1 N_A$ molecules of H_2O contain = $3 \times (0.1 N_A) = 0.3 N_A$ atoms.

Illustration 24. Calculate the number of atoms present in one litre of water (density of water is 1 g/mL).

Solution 1 litre = 1000 mL = 1000 g

Moles of H_2O (n) = $\frac{\text{weight}}{M_w} = \frac{1000}{18} = 55.5 \text{ mol} = \frac{N}{N_A}$

$\Rightarrow$ number of molecules of H_2O (N) = $55.5 N_A$

$\therefore$ 1 molecule of H_2O contain = 3 atoms

$\therefore$ $55.5 N_A$ molecules H_2O contain = $3 \times (55.5 N_A) \text{ atoms} = 166.5 N_A \text{ atoms}$

Illustration 25. At NTP the density of a gas is 0.00445 g/mL then find out its V.D. and molecular mass.

Solution $\text{V.D.} = \frac{\text{Density of gas}}{\text{Density of } \text{H}_2} = \frac{0.004450}{0.000089} = 50$

Molecular mass = $2 \times \text{V.D.} = 2 \times 50 = 100$

Illustration 26. Weight of 1 L gas is 2 g then find out its V.D. and molecular mass

Solution Density of gas = $\frac{\text{Mass}}{\text{Volume}} = \frac{2}{1000} = 0.002 \text{ g/mL}$

$\text{V.D.} = \frac{\text{Density of gas}}{\text{Density of } \text{H}_2} = \frac{0.002000}{0.000089} = 22.4$

Molecular mass = $2 \times \text{V.D.} = 44.8$

GOLDEN KEY POINTS

- Term molar mass means mass of 1 mol particles.
- Vapour density is calculated with respect to H_2 gas under similar conditions of temperature and pressure.
- Relative density = $\frac{\text{Density of gas A}}{\text{Density of gas B}}$
- Specific gravity : It is density of material with respect to water.
- Vapour density, relative density and specific gravity are ratios so they are unitless.
- The term STP means 273.15 K (0°C) and 1 bar pressure. The term NTP means 273.15 K (0°C) and 1 atm.

BEGINNER'S BOX-2

- Calculate the number of atoms in 11.2 L of SO_2 gas at STP :
 (1) $\frac{N_A}{2}$ (2) $\frac{3N_A}{2}$ (3) $3N_A$ (4) N_A
- Which of the following has maximum mass :
 (1) 0.1 gram atom of carbon (2) 0.1 mol of ammonia
 (3) 6.02×10^{22} molecules of hydrogen (4) 1120 cc of carbon dioxide at STP
- The total number of electrons present in 18 mL of water :-
 (1) 6.02×10^{22} (2) 6.02×10^{23} (3) 6.02×10^{24} (4) 6.02×10^{25}
- The volume of 1.0 g of hydrogen at NTP is :
 (1) 2.24 L (2) 22.4 L (3) 1.12 L (4) 11.2 L
- 11 grams of a gas occupy 5.6 litres of volume at STP. The gas is :-
 (1) NO (2) N_2O_4 (3) CO (4) CO_2
- At NTP, 5.6 L of a gas weight 8 grams. The vapour density of gas is :-
 (1) 32 (2) 40 (3) 16 (4) 8
- The vapour densities of two gases are in the ratio of 1 : 3. Their molecular masses are in the ratio of :-
 (1) 1 : 3 (2) 1 : 2 (3) 2 : 3 (4) 3 : 1

1.3 PERCENTAGE COMPOSITION, EMPIRICAL FORMULA & MOLECULAR FORMULA

Percentage formula (% by mass)

(In a molecule or compound) Mass % of an element = $\frac{\text{Number of atom (Atomicity)} \times \text{atomic mass}}{\text{molecular mass}} \times 100$

If number of atom = 1 : Molecular mass = **minimum molecular mass**

Empirical Formula

The empirical formula of a compound express the *simplest whole number ratio of atoms* of various elements present in 1 molecule of the compound.

Ex.	Molecular Formula	→	H_2O_2	CH_4	C_2H_6	$\text{C}_2\text{H}_4\text{O}_2$
			2:2	1:4	2:6	2:4:2
			1:1	1:4	1:3	1:2:1
	Empirical Formula	→	$\boxed{\text{HO}}$	$\boxed{\text{CH}_4}$	$\boxed{\text{CH}_3}$	$\boxed{\text{CH}_2\text{O}}$

Molecular Formula

The molecular formula of a compound represents the **actual number of atoms** present in 1 molecule of the compound i.e. it shows the real formula of its 1 molecule.

Relationship between Empirical & Molecular Formula

Molecular Formula = $n \times$ Empirical Formula

[Where n = natural no. (1, 2, 3,.....)]

$$\text{or } n = \frac{\text{Molecular Formula}}{\text{Empirical Formula}} \quad \text{or } n = \frac{\text{Molecular formula mass}}{\text{Empirical formula mass}}$$

Determination of Empirical Formula

Following steps are involved to determine the empirical formula of the compounds –

- (1) First of all find the % by weight of each element present in 1 molecule of the compound
- (2) The % by weight of each element is divided by its atomic weight. It gives atomic ratio of elements present in the compounds.
- (3) Atomic ratio of each element is divided by the minimum value of atomic ratio as to get simplest ratio of atoms.
- (4) If the value of simplest atomic ratio is fractional then raise the value to the nearest whole number.
or Multiply with suitable coefficient to convert it into nearest whole number.
- (5) Write the Empirical formula as we get the simplest ratio of atoms.

Illustrations

Illustration 28. Find out percentage composition of each element present in glucose ?

Solution

$$\% \text{ of C} = \frac{12 \times 6}{180} \times 100 = 40\%$$

$$\% \text{ of H} = \frac{12 \times 1}{180} \times 100 = 6.67\%$$

$$\% \text{ of O} = \frac{16 \times 6}{180} \times 100 = 53.33\%$$

Illustration 29. In a compound x is 75.8% and y is 24.2% by weight present. If atomic weight of x and y are 24 and 16 respectively. Then calculate the empirical formula of the compound.

Solution

Elements	%	Atomic weight	$\frac{\%}{\text{Atomic weight}}$	Simplest ratio	Ratio
x	75.8%	24	$\frac{75.8}{24} = 3.1$	$\frac{3.1}{1.5} = 2$	2
y	24.2%	16	$\frac{24.2}{16} = 1.5$	$\frac{1.5}{1.5} = 1$	1

Empirical formula = x_2y

Illustration 30. In a compound Carbon is 52.2%, Hydrogen is 13%, Oxygen is 34.8% are present and molecular mass of the compound is 92. Calculate molecular formula of the compound ?

Solution

Elements	%	Atomic weight	$\frac{\%}{\text{Atomic weight}}$	Simplest ratio	Ratio
C	52.2	12	$\frac{52.2}{12} = 4.35 = 4.4$	$\frac{4.4}{2.2} = 2$	2
H	13	1	$\frac{13}{1} = 13$	$\frac{13}{22} = 5.9$	6
O	34.8	16	$\frac{34.8}{16} = 2.2$	$\frac{2.2}{2.2} = 1$	1

Empirical formula = C_2H_6O

Empirical formula mass = $12 \times 2 + 16 + 6 = 46$

$$n = \frac{\text{Molecular formula mass}}{\text{Empirical formula mass}} = \frac{92}{46} = 2$$

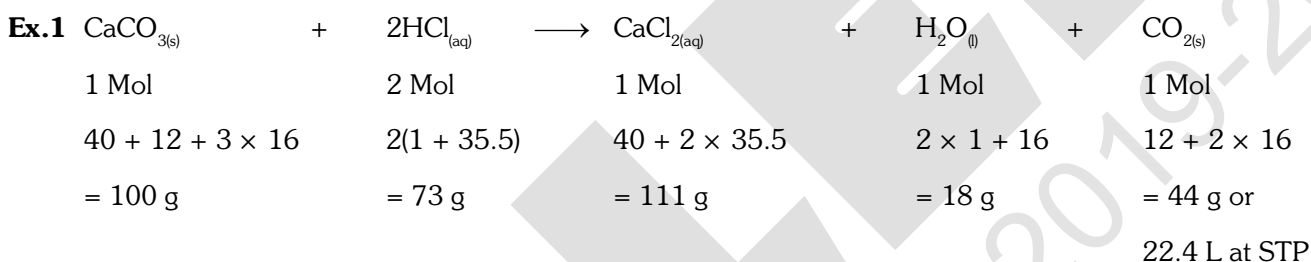
molecular formula = $2 \times (C_2H_6O) = C_4H_{12}O_2$

BEGINNER'S BOX-3

- A hydrocarbon contain 80% C. The vapour density of compound is 30. Empirical formula of compound is :-
 (1) CH_3 (2) C_2H_6 (3) C_4H_{12} (4) C_4H_8
- Two elements X (Atomic weight = 75) and Y (Atomic weight = 16) combine to give a compound having 75.8% of X. The empirical formula of compound is :
 (1) XY (2) X_2Y (3) X_2Y_2 (4) X_2Y_3
- In a compound element A (Atomic weight = 12.5) is 25% and element B (Atomic weight = 37.5) is 75% by weight. The Empirical formula of the compound is :
 (1) AB (2) A_2B (3) A_2B_2 (4) A_2B_3

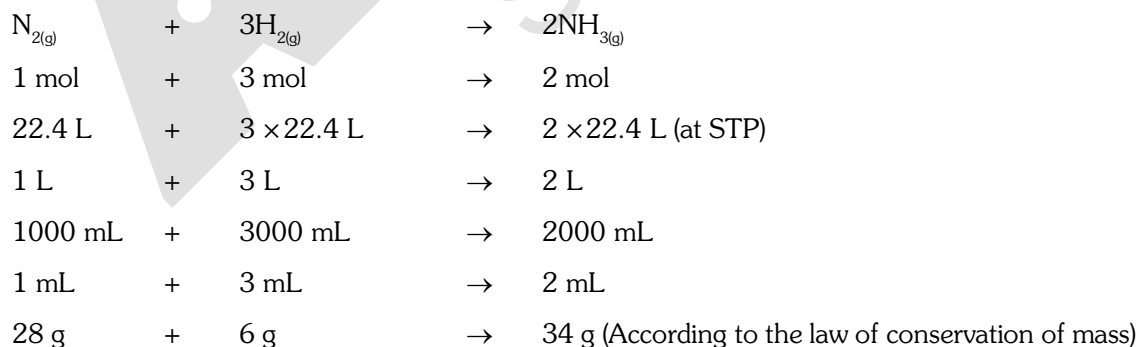
1.4 STOICHIOMETRY BASED CONCEPT (PROBLEMS BASED ON CHEMICAL REACTION)

One of the most important aspects of a chemical equation is that when it is written in the balanced form, it gives quantitative relationships between the various reactants and products in terms of moles, masses, molecules and volumes. This is called stoichiometry (Greek word, meaning 'to measure an element'). For example, a balanced chemical equation along with the quantitative information conveyed by it is given below:



Thus,

- 1 mole of calcium carbonate reacts with 2 moles of hydrochloric acid to give 1 mole of calcium chloride, 1 mole of water and 1 mole of carbon dioxide.
- 100 g of calcium carbonate react with 73 g hydrochloric acid to give 111 g of calcium chloride, 18 g of water and 44 g (or 22.4 litres at STP) of carbon dioxide.



- Gram can not be represented according to stoichiometry.

The quantitative information conveyed by a chemical equation helps in a number of calculations. The problems involving these calculations may be classified into the following two different types :

(a) Single reactant based

(b) More than one reactant based

(A) SINGLE REACTANT BASED :

- (1) Mass - Mass Relationships i.e. mass of one of the reactants or products is given and the mass of some other reactant or product is to be calculated.
- (2) Mass - Volume Relationships i.e. mass/volume of one of the reactants or products is given and the volume/mass of the other is to be calculated.
- (3) Volume - Volume Relationships i.e. volume of one of the reactants or the products is given and the volume of the other is to be calculated.

General method : Calculations for all the problems of the above types consists of the following steps :-

- (i) Write down the balanced chemical equation.
- (ii) Write the relative number of moles or the relative masses (gram atomic or molecular masses) of the reactants and the products below their formula.
- (iii) In case of a gaseous substance, write down 22.4 litres at STP below the formula in place of 1 mole
- (iv) Apply unitary method to make the required calculations.

Quite often one of the reactants is present in larger amount than the other as required according to the balanced equation. The amount of the product formed then depends upon the reactant which has reacted completely. This reactant is called the limiting reactant. The excess of the other is left unreacted.

Combustion reaction : (Problem based on combustion reactions) :

For balancing the combustion reaction: First of all balance C atoms, Then balance H atom, Finally balance Oxygen atom.

For Example : Combustion reaction of C_2H_6 : $C_2H_6 + O_2 \longrightarrow CO_2 + H_2O$ (skeleton equation)

First balance C atoms $C_2H_6 + O_2 \longrightarrow 2CO_2 + H_2O$

Now balance H atoms $C_2H_6 + O_2 \longrightarrow 2CO_2 + 3H_2O$

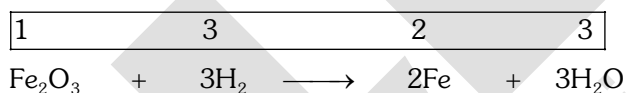
Now balance Oxygen atoms $C_2H_6 + \frac{7}{2}O_2 \longrightarrow 2CO_2 + 3H_2O$

Illustrations

TYPE-I (INVOLVING MASS-MASS RELATIONSHIP)

Illustration 31. How much iron can be theoretically obtained in the reduction of 1 kg of Fe_2O_3

Solution



$$n = \frac{\text{weight}}{M_w} = \frac{1000}{160} \text{ mol}$$

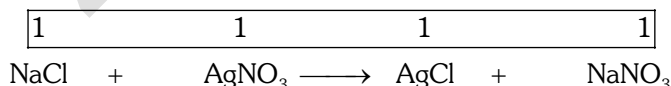
The equation shows that 2 mol of iron are obtained from 1 mol of ferric oxide.

$$\text{Hence, the obtained no. of moles of Fe} = \frac{2 \times 1000}{160} = 12.5 \text{ mol} = \frac{\text{weight}}{\text{Atomic weight}} = \frac{\text{weight}}{56}$$

$$\text{Weight of iron obtained} = 12.5 \times 56 \text{ g} = 700 \text{ g}$$

Illustration 32. What amount of silver chloride is formed by the action of 5.850 g of sodium chloride on an excess of silver nitrate?

Solution



$$n = \frac{\text{weight}}{M_w} = \frac{5.85}{58.5} = 0.1 \text{ mol}$$

1 mol of $AgCl$ is obtained from 1 mol of $NaCl$

Hence, the number of moles of $AgCl$ obtained with 0.1 mol of $NaCl$ = 0.1 mol

$$\therefore n = \frac{\text{weight}}{M_w} \Rightarrow 0.1 \text{ mol} = \frac{\text{weight}}{M_w} = \frac{\text{weight}}{143.5} \Rightarrow \text{weight} = 0.1 \times 143.5 \text{ g} = 14.35 \text{ g.}$$

TYPE-II (MASS - VOLUME RELATIONSHIP)

Illustration 33. For complete combustion of 3g ethane the required volume of O_2 & produced volume of CO_2 at STP will be.

Solution



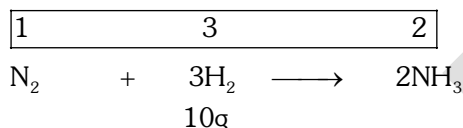
$$n = \frac{\text{weight}}{M_w} = \frac{3}{30} = \frac{1}{10} = 0.1 \text{ mol}$$

(a) Required moles of $O_2 = \frac{7}{2} \times 0.1 = 0.35 \text{ mol}$
 volume of O_2 at STP = $0.35 \times 22.4 = 7.84 \text{ L}$

(b) Produced moles of $CO_2 = \frac{4}{2} \times 0.1 = 0.2 \text{ mol}$
 volume of CO_2 at STP = $0.2 \times 22.4 = 4.48 \text{ L}$

Illustration 34. In the following reaction, if 10 g of H_2 reacts with N_2 . What will be the volume of NH_3 at STP.
 $N_2 + 3H_2 \longrightarrow 2NH_3$

Solution



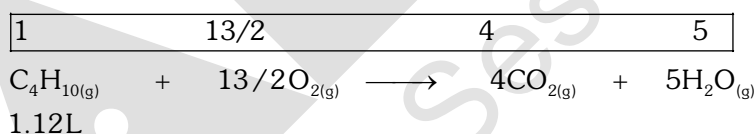
$$n = \frac{\text{weight}}{M_w} = \frac{10}{2} = 5 \text{ mol.}$$

Produced moles of $NH_3 = \frac{2}{3} \times 5 = \frac{10}{3}$, Volume of NH_3 at STP = $\frac{10}{3} \times 22.4 = 74.67 \text{ litre}$

TYPE-III (VOLUME-VOLUME RELATIONSHIP)

Illustration 35. For complete combustion of 1.12 L of butane (C_4H_{10}), the produced volume of $H_2O_{(g)}$ & $CO_{2(g)}$ at STP will be.

Solution



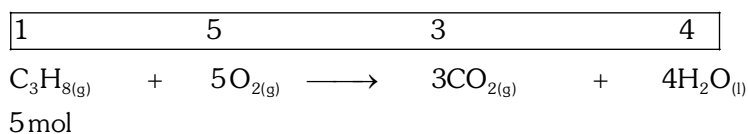
Volume of $H_2O_{(g)}$ at STP = $5 \times 1.12 = 5.6 \text{ L}$

Volume of $CO_{2(g)}$ at STP = $4 \times 1.12 = 4.48 \text{ L}$

Illustration 36. For complete combustion of 5 mol propane (C_3H_8). The required volume of O_2 at STP will be.

Solution

For C_3H_8 , the combustion reaction is



Required moles of $O_2 = 5 \times 5 = 25 \text{ mol} = \frac{V}{22.4}$

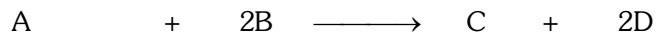
volume of O_2 gas at STP = $25 \times 22.4 = 560 \text{ L}$

(B) MORE THAN ONE REACTANT BASED :

Limiting reagent (L.R.) concept

Limiting Reagent (L.R.) : The reactant which is completely consumed in a reaction is called as limiting reagent.

Ex. 1 2 1 2 ← Stoichiometry



given 3 mol

9mol

$$3 - 3 = 0 \text{ mol}$$

$$9 - 6 = 3 \text{ mol}$$

3 mol

6 mol

L.R. = A

$$X = \frac{\text{given value (moles, volume, or molecules)}}{\text{Stoichiometry Co-efficient}}$$

Reactants having least value of x are limiting reagents.

Ex. A + 2B → P

$$\frac{3}{1} = 3$$

$$\frac{9}{2} = 4.5$$

3 < 4.5 So A is L.R.

Illustrations

Illustration 37. A + 5B → C + 3D In this reaction which is a L.R.

Solution A + 5B → C + 3D Given 10 mol of A and 10 mol of B.

10 mol

10 mol

$$x = \frac{10}{1} = 10$$

$$x = \frac{10}{5} = 2$$

2 < 10 So B is L.R.

Illustration 38. $H_{2(g)} + \frac{1}{2} O_{2(g)} \longrightarrow H_2O_{(g)}$; In the above reaction what is the volume of water vapour at STP.

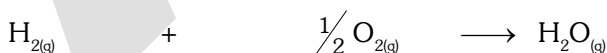
Given 4 g of H_2 and 32 g of O_2

Solution

1

$\frac{1}{2}$

1



4 g

32 g

For H_2

For O_2

$$n = \frac{4}{2} = 2 \text{ mol}$$

$$n = \frac{32}{32} = 1 \text{ mol}$$

$$\frac{2}{1} = 2$$

$$\frac{1}{\frac{1}{2}} = 2 \text{ mol}$$

$$\text{Moles of } H_2O_{(g)} = 2 \text{ mol} = \frac{V}{22.4}$$

2 = 2 So Both H_2 & O_2 are L.R.

$$\text{Volume of } H_2O_{(g)} \text{ at STP} = 22.4 \times 2 = 44.8 \text{ litre}$$

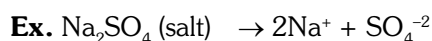
Solution	Balanced equation will be	N_2	+	3H_2	$\longrightarrow$	2NH_3
	Given	100mL		100mL		

$$\frac{100}{1} = 100 \qquad \frac{100}{3} = 33.3 \text{ (Limiting reagent)}$$

That is from 100 mL of H_2 produced volume of $\text{NH}_3 = \frac{2}{3} \times 100 = 66.6 \text{ mL}$

- 1.5 mol of O_2 combine with Mg to form oxide MgO . The mass of Mg (At. mass 24) that has combined is :
(1) 72 g (2) 36 g (3) 24 g (4) 94 g
- What quantity of lime stone on heating will give 56 kg of CaO :-
(1) 1000 kg (2) 56 kg (3) 44 kg (4) 100 kg
- For reaction $\text{A} + 2\text{B} \rightarrow \text{C}$. The amount of product formed by starting the reaction with 5 mol of A and 8 mol of B is :
(1) 5 mol (2) 8 mol (3) 16 mol (4) 4 mol

(v) Equivalent weight of salt = $\frac{\text{Molecular weight}}{\text{Total charge on cation or anion}}$



Total charge on cation or anion is 2

molecular weight of Na_2SO_4 is $= (2 \times 23 + 32 + 16 \times 4) = 142$

$$\text{Equivalent weight of Na}_2\text{SO}_4 = \frac{142}{2} = 71$$

(vi) Equivalent weight of an oxidizing or reducing agent

$$= \frac{\text{Molecular weight of the substance}}{\text{Number of electrons gain/lost by one molecule}}$$

(b) Concept of gram equivalent and law of chemical equivalence :

$$\begin{aligned}\text{Number of gram equivalent} &= \frac{W_{(\text{gram})}}{E} \\ &= \frac{W_{(\text{gram})} \times \text{Valence factor}}{M} \\ &= n \times \text{valence factor}\end{aligned}$$

According to it, in a reaction equal number of gram equivalents of reactants react to give equal number of gram equivalents of products.

For a reaction



Number of gram equivalents of A = Number of gram equivalents of B = Number of gram equivalents of C = Number of gram equivalents of D

(c) METHODS FOR DETERMINATION OF EQUIVALENT WEIGHT

(i) Hydrogen displacement method : This method is used for those elements which can evolve hydrogen from acids i.e. active metals.

$$\text{equivalent weight of metal} = \frac{\text{weight of metal}}{\text{weight of H}_2 \text{ gas (displaced)}} \times 1.008$$

(ii) **Oxide formation method :** A known mass of the element is changed into oxide directly or indirectly. The mass of oxide is noted.

$$\text{Mass of oxygen} = (\text{Mass of oxide} - \text{Mass of element})$$

$$\text{equivalent weight of element} = \frac{\text{weight of element}}{\text{weight of oxygen}} \times 8$$

(iii) **Chloride formation method :** A known mass of the element is changed into chloride directly or indirectly. The mass of the chloride is determined.

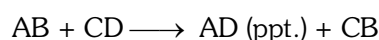
$$\text{equivalent weight of element} = \frac{\text{weight of element}}{\text{weight of chlorine}} \times 35.5$$

- (iv) **Metal to metal displacement method** : More active metal can displace less active metal from its salt's solution. The mass of the displaced metal bear the same ratio as their equivalent weights.

$$\frac{m_1}{m_2} = \frac{E_1}{E_2}$$

- (v) **Double decomposition method** : This method is based on the following points -

- The mass of the compound reacted and the mass of product formed are in the ratio of their equivalent masses.
- The equivalent mass of the compound (electrovalent) is the sum of equivalent masses of its radicals.
- The equivalent mass of a radical is equal to the formula mass of the radical divided by its charge.



$$\frac{\text{Mass of AB}}{\text{Mass of AD}} = \frac{\text{Equivalent mass of AB}}{\text{Equivalent mass of AD}} = \frac{\text{Equivalent mass of A} + \text{Equivalent mass of B}}{\text{Equivalent mass of A} + \text{Equivalent mass of D}}$$

- (vi) **Silver salt method** : This method is used for finding the equivalent weight of carbonic (organic) acids. A known mass of the RCOOAg is changed into Ag through combustion. The mass of Ag is determined.

$$\frac{\text{Equivalent weight of RCOOAg}}{\text{Equivalent weight of Ag}} = \frac{\text{weight of RCOOAg}}{\text{weight of Ag}}$$

$$\text{equivalent weight of RCOOAg} = \frac{\text{weight of RCOOAg}}{\text{weight of Ag}} \times 108$$

- (vii) **By electrolysis** : $\frac{w_1}{w_2} = \frac{E_1}{E_2}$

Where w_1 & w_2 are deposited weight of metals at electrodes and E_1 and E_2 are equivalent weight respectively.

1.6 METHODS FOR CALCULATION OF ATOMIC WEIGHT AND MOLECULAR WEIGHT

- (a) **Methods for Determination of Atomic Weight**

- (i) **Atomic weight = equivalent weight $\times$ valency**

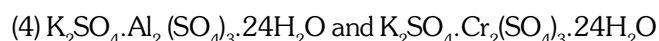
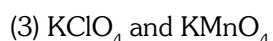
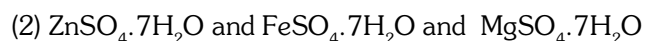
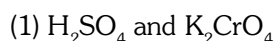
- (ii) **Dulong and Petit's law** - This law is applicable only for solids (except Be, B, Si, C)

$$\text{Atomic mass} \times \text{specific heat (Cal g}^{-1} \text{ } ^\circ\text{C)} \approx 6.4$$

$$\text{or atomic mass (approximate)} = \frac{6.4}{\text{specific heat}}$$

- (iii) **Law of isomorphism** : Isomorphous substances form crystals which have same shape and size and can grow in the saturated solution of each other.

Examples of isomorphous compounds -



Conclusions -

- Masses of two elements that combine with same mass of other elements in their respective compounds are in the ratio of their atomic masses.

$$\frac{\text{Mass of one elements (A) that combines with a certain mass of other element}}{\text{Mass of other element (B) that combines with the same mass of other element}} = \frac{\text{Atomic mass of A}}{\text{Atomic mass of B}}$$

- The valencies of the elements forming isomorphous compounds are the same.

(iv) Volatile chloride method

Required condition – chloride of element should be vapour.

Required data - (i) Vapour density of chloride. (ii) Equivalent weight of element.

Let the valency of the element be x . The formula of its chloride will be MCl_x .

$$\text{Molecular weight} = \text{Atomic weight of M} + 35.5 \times$$

$$\therefore \text{Atomic weight} = \text{Equivalent weight} \times \text{valency} \text{ or } A = E \times x$$

$$\therefore \text{Molecular weight} = E \times + 35.5 \times \text{or } 2 \times \text{V.D.} = x(E + 35.5) \text{ or } x = \frac{2 \times \text{V.D.}}{E + 35.5}$$

(v) **Specific heat method :** If $\frac{C_P}{C_V} = \gamma$ is given, then

Case I. If $\gamma = 5/3 = 1.66$

Atomicity will be one

Case II. If $\gamma = 7/5 = 1.4$

Atomicity will be two

Case III. If $\gamma = 4/3 = 1.33$

Atomicity will be three

$$\text{Atomic weight} = \frac{\text{Molecular weight}}{\text{Atomicity}}$$

(b) Method for Determination of Molecular Weight :

(i) Molecular weight = $2 \times \text{V.D.}$

(ii) Victor Mayer's method is used to determine molecular weight of volatile compound.

Illustrations

Illustration 40. Specific heat of a metal is $0.031 \frac{^{\circ}\text{C} \times \text{Cal}}{\text{g}}$, and its equivalent weight is 103.6. Calculate the exact atomic weight of the metal.

Solution According to Dulong and Petit's law - approximate atomic weight = $\frac{6.4}{0.031} = 206.45$

$$\text{Valency of metal} = \frac{\text{Approximate atomic weight}}{\text{Equivalent weight}} = \frac{206.45}{103.6} = 1.99 \approx 2$$

So, the exact atomic weight of the element = Equivalent weight $\times$ valency
 $= 103.6 \times 2 = 207.2$

Illustration 41. A chloride of an element contains 49.5% chlorine. The specific heat of the element is $0.064\text{ }^{\circ}\text{C cal g}^{-1}$. Calculate the equivalent mass, valency and atomic mass of the element.

Solution Mass of chlorine in the metal chloride = 49.5
 Mass of metal = $(100 - 49.5) = 50.5$

$$\text{Equivalent weight of metal} = \frac{\text{weight of metal}}{\text{weight of chlorine}} \times 35.5 = \frac{50.5}{49.50} \times 35.5 = 36.21$$

Now according to Dulong and Petit's law,

$$\text{Approximate at. wt. of the metal} = \frac{6.4}{\text{specific heat}} = \frac{6.4}{0.064} = 100$$

$$\text{Valency} = \frac{\text{Approximate atomic weight}}{\text{Equivalent weight}} = \frac{100}{36.21} = 2.7 \approx 3$$

Hence, exact atomic weight = $\frac{\text{Equivalent weight}}{3} = \frac{36.21}{3} = 108.63$

Illustration 42. The oxide of an element contains 67.67% of oxygen and the vapour density of its volatile chloride is 79. Calculate the atomic weight of the element.

Solution

Calculation of equivalent weight : weight of oxygen = 67.67 g

weight of element = $100 - 67.67 = 32.33$ g

$\therefore$ 67.67 g of oxygen combines with 32.33 g of element

$\therefore$ 8 g of oxygen combines with = $\frac{32.33 \times 8}{67.67} = 3.82$ g of element

$\therefore$ Equivalent weight of the element = 3.82

Suppose M represents one atom of the element and x is its valency. The molecular formula of the volatile chloride would be MCl_x .

Formula weight of chloride = $3.82 \times x + 35.5 \times x = 39.32 \times x$

But molecular weight of Chloride = $2 \times \text{V.D.} \Rightarrow 39.32 \times x = 2 \times 79 \Rightarrow x = \frac{2 \times 79}{39.32} = 4$

Now atomic weight = Equivalent weight $\times$ valency of element = $3.82 \times 4 = 15.28$

Illustration 43. Vapour density of a gas is 16. If the ratio of specific heat at constant pressure and specific heat at constant volume is 1.4. Then find out its atomic weight.

Solution

Given : $\frac{C_p}{C_v} = 1.4 = \gamma$ and vapour density = 16

We know that Molecular weight = $2 \times$ vapour density

$\therefore$ Molecular weight = $2 \times 16 = 32$

Here $\gamma = 1.4$ so atomicity will be 2.

Atomic weight = $\frac{\text{Molecular weight}}{\text{Atomicity}} = \frac{32}{2} = 16$

GOLDEN KEY POINTS

- Equivalent weight of a species changes with reaction in which it gets involved.
- Amount of substance which loses or gains 1 mole electrons or 96500 coulomb electricity will always be its equivalent weight.

BEGINNER'S BOX-5

- Molecular weight of dibasic acid is W. Its equivalent weight will be :
 (1) $\frac{W}{2}$ (2) $\frac{W}{3}$ (3) W (4) 3W
- 0.126 g of an acid requires 20 ml of 0.1 N NaOH for complete neutralization. Eq. wt. of the acid is:
 (1) 45 (2) 53 (3) 40 (4) 63
- In a metal oxide 32% oxygen is present what will be equivalent mass of metal ?
 (1) 17 (2) 34 (3) 32 (4) 52
- 1 mol O_2 will be equal to :
 (1) 4 g equivalent oxygen (2) 2 g equivalent oxygen
 (3) 32 g equivalent oxygen (4) 8 g equivalent oxygen
- Volume of one gram equivalent of H_2 at NTP is :
 (1) 5.6 L (2) 11.2 L (3) 22.4 L (4) 44.8 L

1.7 LAWS OF CHEMICAL COMBINATION

(a) Law of Mass Conservation (Law of Indestructibility of Matter)

"It was given by **Lavoisier** and tested by **Landolt**"

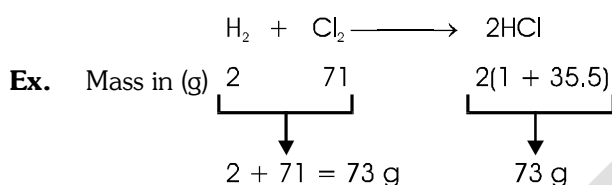
According to this law, the mass can neither be created nor be destroyed in a balanced chemical reaction or physical reaction. But one form is changed into another form is called as law of mass conservation.

If the reactants are completely converted into products, then the sum of the mass of reactants is equal to the sum of the mass of products.

Total mass of reactants = Total mass of products.

If reactants are not completely consumed then the relationship will be :

Total mass of reactants = Total mass of products + Mass of unreacted reactants



Illustrations

Illustration 44. What weight of BaCl_2 would react with 24.4 g of sodium sulphate to produce 46.6 g of barium sulphate and 23.4 g of sodium chloride ?

Solution Barium chloride and sodium sulphate react to produce barium sulphate and sodium chloride according to the equation :



Let the weight of BaCl_2 be x g. According to law of conservation of mass :

Total mass of reactants = Total mass of products

Total mass of reactants = (x + 24.4) g

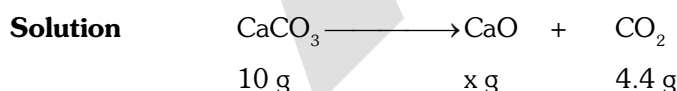
Total mass of products = (46.6 + 23.4) g

Equating the two masses $\Rightarrow x + 24.4 = 46.6 + 23.4$

$x = 46.6 + 23.4 - 24.4$ or $x = 45.6 \text{ g}$

Hence, the weight of BaCl_2 is 45.6 g

Illustration 45. 10g of CaCO_3 on heating gives 4.4 g of CO_2 then determine weight of produced CaO in quintal.



According to law of conservation of mass

10 = 4.4 + x

10 - 4.4 = x

x = 5.6 g

$\begin{cases} 1 \text{ quintal} = 100 \text{ kg} \\ 1 \text{ kg} = 1000 \text{ g} \end{cases}$

weight of $\text{CaO}(x) = 5.6 \times \frac{\text{kg}}{1000} = 5.6 \times 10^{-3} \text{ kg} = 5.6 \times 10^{-3} \times \frac{1}{100} \text{ quintal} = 5.6 \times 10^{-5} \text{ quintal}$

(c) **Law of Multiple Proportion**

"It was given by **John Dalton**"

According to law of Multiple proportion if two elements combine to form more than one compound then the different mass of one element which combine with a fixed mass of other element bear a simple ratio to one another.

The following examples illustrate this law.

- (i) **Nitrogen and oxygen combine to form five oxides, which are :** Nitrous oxide (N_2O), nitric oxide (NO), nitrogen trioxide (N_2O_3), nitrogen tetraoxide (N_2O_4) and nitrogen pentaoxide (N_2O_5).

Weight of oxygen which combine with the fixed weight of nitrogen in these oxides are calculated as under:

Oxide Ratio of weight of nitrogen and oxygen in each compound

N_2O 28 : 16 NO 14 : 16 N_2O_3 28 : 48

N_2O_4 28 : 64 N_2O_5 28 : 80

Number of parts by weight of oxygen which combine with 14 parts by weight of nitrogen from the above are 8, 16, 24, 32 and 40 respectively. Their ratio is 1 : 2 : 3 : 4 : 5, which is a simple ratio. Hence, the law is illustrated.

- (ii) Sulphur combines with oxygen to form two oxides SO_2 and SO_3 , the weights of oxygen which combine with a fixed weight of sulphur, i.e. 32 parts by weight of sulphur in two oxides are in the ratio of 32 : 48 or 2 : 3 which is a simple ratio. Hence the law of multiple proportions is illustrated.

Illustrations

Illustration 48. Hydrogen peroxide and water contain 5.93% and 11.2% of hydrogen respectively. Show that the data illustrate the law of multiple proportions.

Solution

Compound H_2O_2

H : O
5.93 : 94.07

$\frac{5.93}{5.93} : \frac{94.07}{5.93}$

① : 15.86

Compound H_2O

H : O
11.2 : 88.8

$\frac{11.2}{11.2} : \frac{88.8}{11.2}$

① : 7.92

Thus the ratio of weights of oxygen which combine with the fixed weight (1.0 gram) of hydrogen in H_2O_2 and H_2O is 15.86 : 7.92 = 2 : 1 (Which is simple ratio). So the law of multiple proportion is illustrated.

Illustration 49. Carbon combines with hydrogen in P, Q and R. The % of hydrogen in P, Q and R are 25, 14.3, and 7.7 respectively. Which law of chemical combination is illustrated?

Solution

P

H : C
25 : 75

1 : $\frac{75}{25}$

① : 3

Q

H : C
14.3 : 85.7

1 : $\frac{85.7}{14.3}$

① : 6

R

H : C
7.7 : 92.3

1 : $\frac{92.3}{7.7}$

① : 12

Ratio of C in compounds P, Q and R is = 3 : 6 : 12 = 1 : 2 : 4

Which is a simple ratio so the data illustrate the law of multiple proportion.

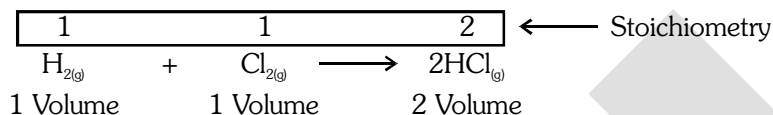
(d) Law of Gaseous Volume

"It was given by **Gay Lussac**"

According to this law, in the gaseous reaction, the reactants are always combined in a simple ratio by volume and form products, which is **simple ratio by volume** at same temperature and pressure.

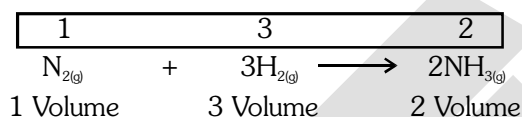
Ex.1 One volume of hydrogen combines with one volume of chlorine to produce 2 volumes of hydrogen chloride.

Simple ratio = 1 : 1 : 2.



Ex.2 One volume of nitrogen combines with 3 volumes of hydrogen to form 2 volumes of ammonia.

Simple ratio 1 : 3 : 2



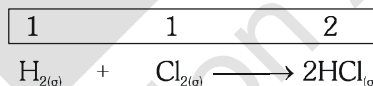
Special Note : This law is used only for gaseous reaction. It relate volume to mole or molecules. But not relate with mass.

Illustrations

Illustration 50. For the gaseous reaction : $H_{2(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)}$. If 40 mL of hydrogen completely reacts with chlorine then find out the required volume of Chlorine & volume of produced $HCl_{(g)}$?

Solution

According to Gay Lussac's Law :



$\therefore$ 1 mL of $H_{2(g)}$ will react with 1 mL of $Cl_{2(g)}$ and 2 mL of $HCl_{(g)}$ will produce

$\therefore$ 40 mL of $H_{2(g)}$ will react with 40 mL of $Cl_{2(g)}$ and 80 mL of $HCl_{(g)}$ will produce

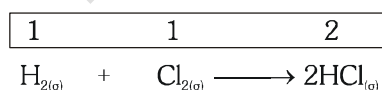
required volume of $Cl_{2(g)}$ = 40 mL

produced volume of $HCl_{(g)}$ = 80 mL

Illustration 51. For the gaseous reaction : $H_{2(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)}$. If initially 20 mL of $H_{2(g)}$ and 30 mL of $Cl_{2(g)}$ are present then find out the volume of $HCl_{(g)}$ and unreacted part of $Cl_{2(g)}$.

Solution

According to Gay-Lussac's Law



$\therefore$ 1 mL of $H_{2(g)}$ will react with 1 mL of $Cl_{2(g)}$ and 2 mL of $HCl_{(g)}$ will produce

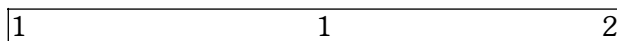
$\therefore$ 20 mL of $H_{2(g)}$ will react with 20 mL of $Cl_{2(g)}$ and 40 mL of $HCl_{(g)}$ will produce

Given volume of $Cl_{2(g)}$ is 30 mL but its 20 mL reacts with $H_{2(g)}$. So 10 mL of $Cl_{2(g)}$ remains unreacted.

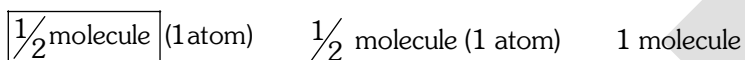
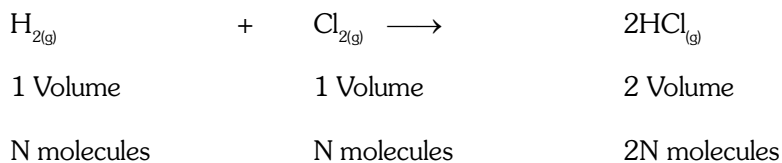
(e) Avogadro's law

"Equal volume of all gases contain equal number of molecules at same temperature and pressure."

Ex.



← Stoichiometry



It is correct due to molecule is divisible.

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5					
	Ans.	1	2	2	1	2					
BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7			
	Ans.	2	4	3	4	4	3	1			
BEGINNER'S BOX-3	Que.	1	2	3							
	Ans.	1	4	1							
BEGINNER'S BOX-4	Que.	1	2	3							
	Ans.	1	4	4							
BEGINNER'S BOX-5	Que.	1	2	3	4	5					
	Ans.	1	4	1	1	2					

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

QUESTIONS BASED ON MOLES

- The number of atoms present in 16 g of oxygen is
 (1) $6.02 \times 10^{11.5}$ (2) 3.01×10^{23}
 (3) $3.01 \times 10^{11.5}$ (4) 6.02×10^{23}
- The number of atoms in 4.25 g of NH_3 is approx:-
 (1) 1×10^{23} (2) 1.5×10^{23}
 (3) 2×10^{23} (4) 6×10^{23}
- Which of the following contains maximum number of oxygen atoms ?
 (1) 1 g of O
 (2) 1 g of O_2
 (3) 1 g of O_3
 (4) all have the same number of atoms
- The number of atoms present in 0.5 g atom of nitrogen is same as the atoms in -
 (1) 12 g of C (2) 32 g of S
 (3) 8 g of oxygen (4) 24 g of Mg
- Which of the following contains maximum number of atoms ?
 (1) 4 g of H_2 (2) 16 g of O_2
 (3) 28 g of N_2 (4) 18 g of H_2O
- Number of neutrons present in 1.7 g of ammonia is -
 (1) N_A (2) $N_A/10 \times 4$
 (3) $(N_A/10) \times 7$ (4) $N_A \times 10 \times 7$
- 5.6 L of oxygen at STP contains -
 (1) 6.02×10^{23} atoms (2) 3.01×10^{23} atoms
 (3) 1.505×10^{23} atoms (4) 0.7525×10^{23} atoms
- Number of oxygen atoms in 8 g of ozone is -
 (1) 6.02×10^{23} (2) $\frac{6.02 \times 10^{23}}{2}$
 (3) $\frac{6.02 \times 10^{23}}{3}$ (4) $\frac{6.02 \times 10^{23}}{6}$
- The number of atoms in "n" mole of gas can be given by :-
 (1) $n \times \text{Av. No.} \times \text{atomicity}$ (2) $\frac{n \times \text{Av. No.}}{\text{Atomicity}}$
 (3) $\frac{\text{Av. No.} \times \text{Atomicity}}{n}$ (4) None
- Sum of number of protons, electrons and neutrons in 12g of $^{12}_6\text{C}$ is :-
 (1) 1.8 (2) 12.044×10^{23}
 (3) 1.084×10^{25} (4) 10.84×10^{23}
- The weight of one atom of Uranium is 238 amu. Its actual weight is g.
 (1) 1.43×10^{26} (2) 3.94×10^{-22}
 (3) 6.99×10^{-23} (4) 1.53×10^{-22}
- The actual weight of a molecule of water is -
 (1) 18 g
 (2) 2.99×10^{-23} g
 (3) both (1) & (2) are correct
 (4) 1.66×10^{-24} g
- What is the mass of a molecule of CH_4 :-
 (1) 16 g (2) 26.6×10^{22} g
 (3) 2.66×10^{-23} g (4) $16 N_A$ g
- Which of the following has the highest mass ?
 (1) 1 g atom of C
 (2) 1/2 mole of CH_4
 (3) 10 mL of water
 (4) 3.011×10^{23} atoms of oxygen
- Which of the following contains the least number of molecules ?
 (1) 4.4 g CO_2 (2) 3.4 g NH_3
 (3) 1.6 g CH_4 (4) 3.2 g SO_2
- The number of molecule in 4.25 g of NH_3 is -
 (1) 1.505×10^{23} (2) 3.01×10^{23}
 (3) 6.02×10^{23} (4) None of these
- Elements A and B form two compounds B_2A_3 and B_2A . 0.05 moles of B_2A_3 weight 9.0 g and 0.10 mole of B_2A weight 10 g. Calculate the atomic weight of A and B :-
 (1) 20 and 30 (2) 30 and 40
 (3) 40 and 30 (4) 30 and 20
- 5.6 L of oxygen at NTP is equivalent to -
 (1) 1 mol (2) 1/2 mol
 (3) 1/4 mol (4) 1/8 mol
- 4.4 g of an unknown gas occupies 2.24 L of volume at STP. The gas may be :-
 (1) N_2O (2) CO
 (3) CO_2 (4) 1 & 3 both

20. Which contains least number of molecules :-

- (1) 1 g CO₂ (2) 1 g N₂
(3) 1 g O₂ (4) 1 g H₂

21. If V mL of the vapours of substance at NTP weight W g. Then molecular weight of substance is:-

- (1) $(W/V) \times 22400$ (2) $\frac{V}{W} \times 22.4$

- (3) $(W - V) \times 22400$ (4) $\frac{W \times 1}{V \times 22400}$

22. If 3.01×10^{20} molecules are removed from 98 mg of H₂SO₄, then the number of moles of H₂SO₄ left are :-

- (1) 0.1×10^{-3} (2) 0.5×10^{-3}
(3) 1.66×10^{-3} (4) 9.95×10^{-2}

23. A gas is found to have the formula (CO)_x. It's VD is 70. The value of x must be:-

- (1) 7 (2) 4 (3) 5 (4) 6

24. Vapour density of gas is 11.2. Volume occupied by 2.4 g of this at STP will be -

- (1) 11.2 L (2) 2.24 L
(3) 22.4 L (4) 2.4 L

25. The volume of a gas in discharge tube is 1.12×10^{-7} mL at STP. Then the number of molecule of gas in the tube is -

- (1) 3.01×10^4 (2) 3.01×10^{15}
(3) 3.01×10^{12} (4) 3.01×10^{16}

26. A person adds 1.71 gram of sugar (C₁₂H₂₂O₁₁) in order to sweeten his tea. The number of carbon atoms added are (mol. mass of sugar = 342)

- (1) 3.6×10^{22} (2) 7.2×10^{21}
(3) 0.05 (4) 6.6×10^{22}

27. The total number of ions present in 1 mL of 0.1 M barium nitrate Ba(NO₃)₂ solution is -

- (1) 6.02×10^{18} (2) 6.02×10^{19}
(3) $3.0 \times 6.02 \times 10^{19}$ (4) $3.0 \times 6.02 \times 10^{18}$

28. The weight of 1 mole of a gas of density 0.1784 g L⁻¹ at NTP is -

- (1) 0.1784 g (2) 1 g
(3) 4 g (4) 4 amu

29. Given that one mole of N₂ at NTP occupies 22.4 L the density of N₂ is -

- (1) 1.25 g L⁻¹ (2) 0.80 g L⁻¹
(3) 2.5 g L⁻¹ (4) 1.60 g L⁻¹

30. The number of gram molecules of oxygen in 6.02×10^{24} CO molecules is -

- (1) 10 g molecules (2) 5 g molecules
(3) 1 g molecules (4) 0.5 g molecules

QUESTIONS BASED ON PERCENTAGE, EMPIRICAL FORMULA & MOLECULAR FORMULA

31. A compound of X and Y has equal mass of them. If their atomic weights are 30 and 20 respectively. Molecular formula of the compound is :-

- (1) X₂Y₂ (2) X₃Y₃
(3) X₂Y₃ (4) X₃Y₂

32. An oxide of sulphur contains 50% of sulphur in it. Its empirical formula is -

- (1) SO₂ (2) SO₃
(3) SO (4) S₂O

33. A hydrocarbon contains 80% of carbon, then the hydrocarbon is -

- (1) CH₄ (2) C₂H₄
(3) C₂H₆ (4) C₂H₂

34. Empirical formula of glucose is -

- (1) C₆H₁₂O₆ (2) C₃H₆O₃
(3) C₂H₄O₂ (4) CH₂O

35. An oxide of metal M has 40% by mass of oxygen. Metal M has atomic mass of 24. The empirical formula of the oxide is :-

- (1) M₂O (2) M₂O₃ (3) MO (4) M₃O₄

36. A compound contains 38.8% C, 16.0% H and 45.2% N. The formula of the compound would be

- (1) CH₃NH₂ (2) CH₃CN
(3) C₂H₅CN (4) CH₂(NH)₂

37. The simplest formula of a compound containing 50% of element X (at wt. = 10) and 50% of element Y (at wt. = 20) is:-

- (1) XY (2) X₂Y (3) XY₂ (4) X₃Y

38. Which of the following compounds has same empirical formula as that of glucose:-

- (1) CH₃CHO (2) CH₃COOH
(3) CH₃OH (4) C₂H₆

39. A gas is found to contain 2.34 g of Nitrogen and 5.34 g of oxygen. Simplest formula of the compound is -

- (1) N₂O (2) NO (3) N₂O₃ (4) NO₂

40. 2.2 g of a compound of phosphorous and sulphur has 1.24 g of 'P' in it. Its empirical formula is -

- (1) P₂S₃ (2) P₃S₂
(3) P₃S₄ (4) P₄S₃

41. On analysis, a certain compound was found to contain iodine and oxygen in the ratio of 254:80. The formula of the compound is :
(At mass I = 127, O = 16)
(1) IO (2) I₂O (3) I₅O₂ (4) I₂O₅
42. The number of atoms of Cr and O are 4.8×10^{10} and 9.6×10^{10} respectively. Its empirical formula is –
(1) Cr₂O₃ (2) CrO₂
(3) Cr₂O₄ (4) CrO₅
43. Insulin contains 3.4% sulphur ; the minimum molecular weight of insulin is :
(1) 941.176 (2) 944
(3) 945.27 (4) None
44. A giant molecule contains 0.25% of a metal whose atomic weight is 59. Its molecule contains one atom of that metal. Its minimum molecular weight is -
(1) 5900 (2) 23600
(3) 11800 (4) $\frac{100 \times 59}{0.4}$
45. Caffeine has a molecular weight of 194. It contains 28.9% by mass of nitrogen. Number of atoms of nitrogen in one molecule of it is :-
(1) 2 (2) 3
(3) 4 (4) 5

QUESTIONS BASED ON STOICHIOMETRY

46. In a gaseous reaction of the type
 $aA + bB \longrightarrow cC + dD$,
which statement is wrong ?
(1) a litre of A combines with b litre of B to give C and D
(2) a mole of A combines with b moles of B to give C and D
(3) a g of A combines with b g of B to give C and D
(4) a molecules of A combines with b molecules of B to give C and D
47. Assuming that petrol is octane (C₈H₁₈) and has density 0.8 g mL⁻¹. 1.425 L of petrol on complete combustion will consume.
(1) 50 mole of O₂ (2) 100 mole of O₂
(3) 125 mole of O₂ (4) 200 mole of O₂
48. In a given reaction, 9 g of Al will react with
 $2Al + \frac{3}{2}O_2 \rightarrow Al_2O_3$
(1) 6 g O₂ (2) 8 g O₂
(3) 9 g O₂ (4) 4 g O₂
49. The equation :
 $2Al_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow Al_2O_{3(s)}$ shows that :-
(1) 2 mol of Al reacts with $\frac{3}{2}$ mol of O₂ to produce $\frac{7}{2}$ mol of Al₂O₃
(2) 2 g of Al reacts with $\frac{3}{2}$ g of O₂ to produce one mol of Al₂O₃
(3) 2 g of Al reacts with $\frac{3}{2}$ L of O₂ to produce 1 mol of Al₂O₃
(4) 2 mol of Al reacts with $\frac{3}{2}$ mol of O₂ to produce 1 mol of Al₂O₃
50. 1 L of CO₂ is passed over hot coke. When the volume of reaction mixture becomes 1.4 L, the composition of reaction mixture is–
(1) 0.6 L CO
(2) 0.8 L CO₂
(3) 0.6 L CO₂ and 0.8 L CO
(4) None
51. 26 cc of CO₂ are passed over red hot coke. The volume of CO evolved is :-
(1) 15 cc (2) 10 cc
(3) 32 cc (4) 52 cc
52. If $\frac{1}{2}$ mol of oxygen combine with Aluminium to form Al₂O₃ then weight of Aluminium metal used in the reaction is (Al = 27) –
(1) 27 g (2) 18 g
(3) 54 g (4) 40.5 g
53. The number of litres of air required to burn 8 litres of C₂H₂ is approximately–
(1) 40 (2) 60 (3) 80 (4) 100
54. If 0.5 mol of BaCl₂ is mixed with 0.2 mol of Na₃PO₄, the maximum number of moles of Ba₃(PO₄)₂ that can be formed is -
 $3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$
(1) 0.7 (2) 0.5
(3) 0.3 (4) 0.1

55. If 8 mL of uncombined O_2 remain after exploding O_2 with 4 mL of hydrogen, the number of mL of O_2 originally were -
 (1) 12 (2) 2
 (3) 10 (4) 4
56. 4 g of hydrogen are ignited with 4 g of oxygen. The weight of water formed is -
 (1) 0.5 g (2) 3.5 g
 (3) 4.5 g (4) 2.5 g
57. For the reaction $A + 2B \longrightarrow C$,
 5 mol of A and 8 mol of B will produce
 (1) 5 mole of C
 (2) 4 mole of C
 (3) 8 mole of C
 (4) 13 mole of C
58. If 1.6 g of SO_2 and 1.5×10^{22} molecules of H_2S are mixed and allowed to remain in contact in a closed vessel until the reaction
 $2H_2S + SO_2 \longrightarrow 3S + 2H_2O$,
 proceeds to completion. Which of the following statement is true ?
 (1) Only 'S' and ' H_2O ' remain in the reaction vessel.
 (2) ' H_2S ' will remain in excess
 (3) ' SO_2 ' will remain in excess
 (4) None
59. 12 L of H_2 and 11.2 L of Cl_2 are mixed and exploded. The composition by volume of mixture is—
 (1) 24 L of HCl (g)
 (2) 0.8 L Cl_2 and 20.8 L HCl (g)
 (3) 0.8 L H_2 and 22.4 L HCl (g)
 (4) 22.4 L HCl (g)
60. 10 mL of gaseous hydrocarbon on combustion give 40 mL of CO_2 (g) and 50 mL of H_2O (vap.). The hydrocarbon is -
 (1) C_4H_5 (2) C_8H_{10}
 (3) C_4H_8 (4) C_4H_{10}
61. 500 mL of a gaseous hydrocarbon when burnt in excess of O_2 gave 2.5 L of CO_2 and 3.0 L of water vapours under same conditions. Molecular formula of the hydrocarbon is -
 (1) C_4H_8 (2) C_4H_{10}
 (3) C_5H_{10} (4) C_5H_{12}

QUESTIONS BASED ON EQUIVALENT WEIGHTS

62. Molecular weight of tribasic acid is W. Its equivalent weight will be :
 (1) $\frac{W}{2}$ (2) $\frac{W}{3}$
 (3) W (4) 3W
63. A, E, M and n are the atomic weight, equivalent weight, molecular weight and valency of an element. The correct relation is :
 (1) $A = E \times n$ (2) $A = \frac{M}{E}$
 (3) $A = \frac{M}{n}$ (4) $M = A \times n$
64. Sulphur forms two chlorides S_2Cl_2 and SCl_2 . The equivalent mass of sulphur in SCl_2 is 16. The equivalent weight of sulphur in S_2Cl_2 is -
 (1) 8 (2) 16
 (3) 32 (4) 64
65. If equivalent weight of S in SO_2 is 8 then equivalent weight of S in SO_3 is -
 (1) $\frac{8 \times 2}{3}$ (2) $\frac{8 \times 3}{2}$
 (3) $8 \times 2 \times 3$ (4) $\frac{2 \times 3}{8}$
66. Which property of an element is not variable :
 (1) Valency (2) Atomic weight
 (3) Equivalent weight (4) None
67. One g equivalent of a substance is present in -
 (1) 0.25 mol of O_2 (2) 0.5 mol of O_2
 (3) 1.00 mol of O_2 (4) 8.00 mol of O_2
68. In a compound $AxBy$,
 (1) Mole of A = mole of B = mole of $AxBy$
 (2) eq. of A = eq. of B = eq. of $AxBy$
 (3) yx mole of A = yx mole of B = $(x + y) \times$ mole of $AxBy$
 (4) $y \times$ mole of A = $y \times$ mole of B
69. 0.45 g of acid (molecular wt. = 90) was exactly neutralised by 20 mL of 0.5 N NaOH. Basicity of the acid is -
 (1) 1 (2) 2
 (3) 3 (4) 4

- 70.** 0.5 g of a base was completely neutralised by 100 mL of 0.2 N acid. Equivalent weight of the base is
(1) 50 (2) 100
(3) 25 (4) 125
- 71.** 0.126 g of an acid requires 20 mL of 0.1 N NaOH for complete neutralisation. Equivalent weight of the acid is –
(1) 45 (2) 53 (3) 40 (4) 63
- 72.** 2g of a base whose equivalent weight is 40 reacts with 3 g of an acid. The equivalent weight of the acid is :
(1) 40 (2) 60
(3) 10 (4) 80
- 73.** Equivalent weight of a divalent metal is 24. The volume of hydrogen liberated at STP by 12 g of the same metal when added to excess of an acid solution is –
(1) 2.8 litres (2) 5.6 litres
(3) 11.2 litres (4) 22.4 litres
- 74.** 0.84 g of a metal carbonate reacts exactly with 40 mL of N/2 H_2SO_4 . The equivalent weight of the metal carbonate is –
(1) 84 (2) 64
(3) 42 (4) 32
- 75.** 1.0 g of a metal combines with 8.89 g of Bromine. Equivalent weight of the metal is nearly :
(at.wt. of Br = 80)
(1) 8 (2) 9 (3) 10 (4) 7
- 76.** H_3PO_4 is a tribasic acid and one of its salt is NaH_2PO_4 . What volume of 1M NaOH solution should be added to 12 g NaH_2PO_4 to convert it into Na_3PO_4 ? (at.wt of P=31)
(1) 100 mL (2) 200 mL
(3) 80 mL (4) 300 mL
- 77.** 0.84 g of metal hydride contains 0.04 g of hydrogen. The equivalent wt. of the metal is
(1) 80 (2) 40 (3) 20 (4) 60
- 78.** A_1 g of an element give A_2 g of its oxide. The equivalent mass of the element is –
(1) $\frac{A_2 - A_1}{A_1} \times 8$ (2) $\frac{A_2 - A_1}{A_2} \times 8$
(3) $\frac{A_1}{A_2 - A_1} \times 8$ (4) $(A_2 - A_1) \times 8$
- 79.** When an element forms an oxide in which oxygen is 20% of the oxide by mass, the equivalent mass of the element will be –
(1) 32 (2) 40 (3) 60 (4) 128
- 80.** If 1.2 g of a metal displaces 1.12 L of hydrogen at NTP, equivalent mass of the metal would be –
(1) 1.2×11.2 (2) 12
(3) 24 (4) $1.2 + 11.2$
- 81.** 1 g of hydrogen is found to combine with 80 g of bromine. 1 g of calcium (valency = 2) combines with 4 g of bromine. The equivalent weight of calcium is –
(1) 10 (2) 20
(3) 40 (4) 80
- 82.** 2.8 g of iron displaces 3.2 g of copper from a solution of copper sulphate. If the equivalent mass of iron is 28, then equivalent mass of copper will be –
(1) 16 (2) 32
(3) 48 (4) 64
- 83.** A metal oxide is reduced by heating it in a stream of hydrogen. It is found that after complete reduction 3.15 g of the oxide have yielded 1.05 g of the metal. We may conclude that.
(1) Atomic weight of the metal is 4
(2) Equivalent weight of the metal is 8
(3) Equivalent weight of the metal is 4
(4) Atomic weight of the metal is 8
- 84.** If m_1 g of a metal A displaces m_2 g of another metal B from its salt solution and if their equivalent weight are E_2 and E_1 respectively then the equivalent weight of A can be expressed by:-
(1) $\frac{m_1}{m_2} \times E_2$ (2) $\frac{m_2}{m_1} \times E_2$
(3) $\frac{m_1}{m_2} \times E_1$ (4) $\frac{m_2}{m_1} \times E_1$
- 85.** 14 g of element X combines with 16 g of oxygen. On the basis of this information, which of the following is a correct statement:-
(1) The element X could have an atomic weight of 7 and its oxide is XO
(2) The element X could have an atomic weight of 14 and its oxide is X_2O
(3) The element X could have an atomic weight of 7 and its oxide is X_2O
(4) The element X could have an atomic weight of 14 and its oxide is XO_2

86. If 2.4 g of a metal displaces 1.12 L hydrogen at normal temperature and pressure equivalent weight of metal would be:-
 (1) 12 (2) 24
 (3) 1.2×11.2 (4) $1.2 \div 11.2$
87. 45 g of acid of molecular weight 90 neutralised by 200 mL of 5 N caustic potash. The basicity of the acid is :-
 (1) 1 (2) 2
 (3) 3 (4) None
88. The weights of two elements which combine with one another are in the ratio of their :-
 (1) Atomic weight (2) Molecular weight
 (3) Equivalent weight (4) None
89. The oxide of a metal has 32% oxygen. It's equivalent weight would be:-
 (1) 34 (2) 32
 (3) 17 (4) 16
90. 1.6 g of Ca and 2.60 g of Zn when treated with an acid in excess separately, produced the same amount of hydrogen. If the equivalent weight of Zn is 32.6, what is the equivalent weight of Ca:-
 (1) 10 (2) 20 (3) 40 (4) 5
91. 74.5 g of a metallic chloride contains 35.5 g of chlorine. The equivalent mass of the metal is -
 (1) 19.5 (2) 35.5
 (3) 39.0 (4) 78.0

QUESTIONS BASED ON CALCULATION OF ATOMIC WEIGHTS AND MOLECULAR WEIGHTS

92. The equivalent weight of an element is 4. It's chloride has a V.D. 59.25. Then the valency of the element is -
 (1) 4 (2) 3 (3) 2 (4) 1
93. Vapour density of metal chloride is 77. Equivalent weight of metal is 3, then its atomic weight will be-
 (1) 3 (2) 6 (3) 9 (4) 12
94. Specific heat of a solid element is $0.1 \text{ Cal g}^{-1} \text{ }^{\circ}\text{C}$ and its equivalent weight is 31.8. Its exact atomic weight is -
 (1) 31.8 (2) 63.6 (3) 318 (4) 95.4
95. The specific heat of an element is $0.214 \text{ Cal g}^{-1} \text{ }^{\circ}\text{C}$. The approximate atomic weight is -
 (1) 0.6 (2) 12 (3) 30 (4) 65

96. A metal M forms a sulphate which is isomorphous with $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. If 0.6538 g of metals M displaced 2.16 g of silver from silver nitrate solution, then the atomic weight of the metal M is
 (1) 32.61 (2) 56.82 (3) 65.38 (4) 74.58
97. The carbonate of a metal is isomorphous with MgCO_3 and contains 6.091% of carbon. Atomic weight of the metal is nearly -
 (1) 48 (2) 68.5
 (3) 137 (4) 120
98. 71 g of chlorine combines with a metal giving 111 g of its chloride. The chloride is isomorphous with $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$. The atomic mass of the metal is:-
 (1) 20 (2) 30 (3) 40 (4) 69
99. The atomic weight of a metal (M) is 27 and its equivalent weight is 9, the formula of its chloride will be:-
 (1) MCl (2) MCl_2
 (3) M_3Cl (4) MCl_3
100. The chloride of a metal contains 71% chlorine by weight and the vapour density of it is 50, the atomic weight of the metal will be:-
 (1) 29 (2) 58 (3) 35.5 (4) 71
101. The specific heat of a metal M is 0.25. Its equivalent weight is 12. What is it's correct atomic weight :-
 (1) 25.6 (2) 36 (3) 24 (4) 12
102. The density of air is $0.001293 \text{ g mL}^{-1}$. It's vapour density is -
 (1) 143 (2) 14.3 (3) 1.43 (4) 0.143
103. Relative density of a volatile substance with respect to CH_4 is 4. Its molecular weight would be -
 (1) 8 (2) 32 (3) 64 (4) 128
104. Vapour density of a gas is 16. The ratio of specific heat at constant pressure to specific heat at constant volume is 1.4, then its atomic weight is -
 (1) 8 (2) 16 (3) 24 (4) 32
105. The weight of substance that displaces 22.4 L air at NTP is :
 (1) Mol. wt. (2) At. wt.
 (3) Eq. wt. (4) All
106. 0.39 g of a liquid on vapourisation gave 112 mL of vapour at STP. Its molecular weight is -
 (1) 39 (2) 18.5 (3) 78 (4) 112
107. In victor Mayer's method 0.2 g of a volatile compound on volatilisation gave 56 mL of vapour at STP. Its molecular weight is -
 (1) 40 (2) 60 (3) 80 (4) 120

- 108.** 510 mg of a liquid on vapourisation in Victor Mayer's apparatus displaces 67.2 cc of dry air (at NTP). The molecular weight of liquid is -
 (1) 130 (2) 17 (3) 1700 (4) 170
- 109.** 5 L of gas at STP weighs 6.25 g. What is its gram molecular weight ?
 (1) 1.25 (2) 14 (3) 28 (4) 56
- 110.** 0.44 g of a colourless oxide of nitrogen occupies 224 mL at STP. The compound is -
 (1) N_2O (2) NO (3) N_2O_4 (4) NO_2
- 111.** One litre of a certain gas weighs 1.16 g at STP. The gas may possibly be -
 (1) C_2H_2 (2) CO (3) O_2 (4) NH_3
- 112.** Equivalent weight of bivalent metal is 32.7. Molecular weight of its chloride is :-
 (1) 68.2 (2) 103.7 (3) 136.4 (4) 166.3
- 113.** The oxide of an element possess the molecular formula M_2O_3 . If the equivalent mass of the metal is 9, the molecular mass of the oxide will be -
 (1) 27 (2) 75 (3) 102 (4) 18

QUESTIONS BASED ON LAWS OF CHEMICAL COMBINATION

- 114.** The law of multiple proportion was proposed by :
 (1) Lavoisier (2) Dalton
 (3) Proust (4) Gaylussac
- 115.** Which one of the following pairs of compound illustrate the law of multiple proportions ?
 (1) H_2O , Na_2O (2) MgO , Na_2O
 (3) Na_2O , BaO (4) SnCl_2 , SnCl_4
- 116.** In the reaction $\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$, ratio by volume of N_2 , H_2 and NH_3 is 1 : 3 : 2. This illustrates law of -
 (1) Definite proportion
 (2) Multiple proportion
 (3) Law of conservation of mass
 (4) Gaseous volumes
- 117.** Different proportions of oxygen in the various oxides of nitrogen prove the law of -
 (1) Equivalent proportion
 (2) Multiple proportion
 (3) Constant proportion
 (4) Conservation of matter
- 118.** Oxygen combines with two isotopes of carbon ^{12}C and ^{14}C to form two sample of carbon dioxide. The data illustrates -
 (1) Law of conservation of mass
 (2) Law of multiple proportions
 (3) Law of gaseous volume
 (4) None of these
- 119.** The law of conservation of mass holds good for all of the following except -
 (1) All chemical reactions
 (2) Nuclear reactions
 (3) Endothermic reactions
 (4) Exothermic reactions
- 120.** Number of molecules in 100 mL of each of O_2 , NH_3 and CO_2 at STP are -
 (1) in the order $\text{CO}_2 < \text{O}_2 < \text{NH}_3$
 (2) in the order $\text{NH}_3 < \text{O}_2 < \text{CO}_2$
 (3) the same
 (4) $\text{NH}_3 = \text{CO}_2 < \text{O}_2$
- 121.** The empirical formula of an organic compound containing carbon and hydrogen is CH_2 . The mass of one litre of this organic gas is exactly equal to that of one litre of N_2 at same temperature and pressure. Therefore, the molecular formula of the organic gas is -
 (1) C_2H_4 (2) C_3H_6
 (3) C_6H_{12} (4) C_4H_8
- 122.** Four one litre flasks are separately filled with the gases hydrogen, helium, oxygen and ozone at same room temperature and pressure. The ratio of total number of atoms of these gases present in the different flasks would be -
 (1) 1 : 1 : 1 : 1 (2) 1 : 2 : 2 : 3
 (3) 2 : 1 : 2 : 3 (4) 2 : 1 : 3 : 2
- 123.** A container of volume V, contains 0.28 g of N_2 gas. If same volume of an unknown gas under similar condition of temperature and pressure weighs 0.44 g, the molecular mass of the gas is
 (1) 22 (2) 44
 (3) 66 (4) 88
- 124.** A and B are two identical vessels. A contains 15 g ethane at 1 atm and 298 K. The vessel B contains 75 g of a gas X_2 at same temperature and pressure. The vapour density of X_2 is -
 (1) 75 (2) 150
 (3) 37.5 (4) 45

125. When 100 g of ethylene polymerizes to polyethylene according to equation

$n\text{CH}_2 = \text{CH}_2 \rightarrow -(\text{CH}_2 - \text{CH}_2)_n-$. The weight of polyethylene produced will be:-

- (1) $\frac{n}{2}$ g (2) 100 g (3) $\frac{100}{n}$ g (4) 100n g

126. If law of conservation of mass was to hold true, then 20.8 g of BaCl_2 on reaction with 9.8 g of H_2SO_4 will produce 7.3 g of HCl. Determine the weight of BaSO_4 produced ?

- (1) 11.65 g (2) 23.3 g
(3) 25.5 g (4) 30.6 g

127. A chemical equation is balanced according to the law of -

- (1) Multiple proportion
(2) Constant composition
(3) Gaseous volume
(4) Conservation of mass

128. Two flasks A & B of equal capacity of volume contain NH_3 and SO_2 gas respectively under similar conditions. Which flask has more number of moles:-

- (1) A
(2) B
(3) Both have same moles
(4) None

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	4	4	3	1	3	2	2	1	3	2	2	3	1	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	3	3	4	1	1	2	3	4	3	1	3	3	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	1	3	4	3	1	2	2	4	4	4	2	1	2	3
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	3	2	4	3	4	2	4	4	3	3	2	3	3	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	4	2	1	3	1	2	1	2	2	3	4	2	2	3	2
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	3	3	1	2	2	2	3	3	3	2	2	3	3	2
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	3	2	4	2	3	3	3	3	4	1	3	2	3	2	1
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	3	3	4	3	1	1	3	3	2	4	4	2	4	2	3
Que.	121	122	123	124	125	126	127	128							
Ans.	1	3	2	1	2	2	4	3							

EXERCISE-II (Previous Year Questions)**AIPMT/NEET & AIIMS (2006-2018)****AIPMT 2008**

- What volume of oxygen gas (O_2) measured at $0^\circ C$ and 1 atm, is needed to burn completely 1L of propane gas (C_3H_8) measured under the same conditions:-
 (1) 5 L (2) 10 L
 (3) 7 L (4) 6 L
- Volume occupied by one molecule of water (density = 1 g cm^{-3}) is :-
 (1) $3.0 \times 10^{-23} \text{ cm}^3$ (2) $5.5 \times 10^{-23} \text{ cm}^3$
 (3) $9.0 \times 10^{-23} \text{ cm}^3$ (4) $6.023 \times 10^{-23} \text{ cm}^3$
- How many moles of lead (II) chloride will be formed from a reaction between 6.5 g of PbO and 3.2 g of HCl ? (Atomic wt. of Pb=207)
 (1) 0.011 (2) 0.029
 (3) 0.044 (4) 0.333
- An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gives 38.71% of C and 9.67% of H. The empirical formula of the compound would be :-
 (1) CHO (2) CH_4O
 (3) CH_3O (4) CH_2O

AIPMT 2009

- 10 g of hydrogen and 64 g of oxygen were filled in a steel vessel and exploded. Amount of water produced in this reaction will be :-
 (1) 1 mol (2) 2 mol
 (3) 3 mol (4) 4 mol

AIPMT 2010

- The number of atoms in 0.1 mol of a triatomic gas is :- ($N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)
 (1) 1.800×10^{22} (2) 6.026×10^{22}
 (3) 1.806×10^{23} (4) 3.600×10^{23}

AIPMT Mains 2011

- Which has the maximum number of molecules among the following ?
 (1) 64 g SO_2 (2) 44 g CO_2
 (3) 48 g O_3 (4) 8 g H_2

NEET UG 2013

- An excess of $AgNO_3$ is added to 100 mL of a 0.01 M solution of dichlorotetraaquachromium(III) chloride. The number of moles of AgCl precipitated would be :-
 (1) 0.01 (2) 0.001 (3) 0.002 (4) 0.003

AIPMT 2014

- Equal masses of H_2 , O_2 and methane have been taken in a container of volume V at temperature $27^\circ C$ at identical conditions. The ratio of the volumes of gases $H_2 : O_2 : CH_4$ would be :
 (1) 8 : 16 : 1 (2) 16 : 8 : 1
 (3) 16 : 1 : 2 (4) 8 : 1 : 2
- When 22.4 L of $H_2(g)$ is mixed with 11.2 L of $Cl_2(g)$ at S.T.P., the moles of HCl (g) formed is equal to:-
 (1) 1 mol of HCl (g)
 (2) 2 mol of HCl (g)
 (3) 0.5 mol of HCl (g)
 (4) 1.5 mol of HCl (g)
- 1.0 g of magnesium is burnt with 0.56 g O_2 in a closed vessel. Which reactant is left in excess and by how much ?
 (At. wt. Mg = 24 ; O = 16)
 (1) Mg, 0.16 g (2) O_2 , 0.16 g
 (3) Mg, 0.44 g (4) O_2 , 0.28 g

AIPMT 2015

- A mixture of gases contains H_2 and O_2 gases in the ratio of 1 : 4 (w/w). What is the molar ratio of the two gases in the mixture ?
 (1) 4 : 1 (2) 16 : 1 (3) 2 : 1 (4) 1 : 4

Re-AIPMT 2015

- The number of water molecules is maximum in :-
 (1) 18 g of water
 (2) 18 mol of water
 (3) 18 molecules of water
 (4) 1.8 g of water
- If avogadro number N_A , is changed from $6.022 \times 10^{23} \text{ mol}^{-1}$ to $6.022 \times 10^{20} \text{ mol}^{-1}$, this would change :
 (1) the ratio of chemical species to each other in a balanced equation
 (2) the ratio of elements to each other in a compound
 (3) the definition of mass in units of grams
 (4) the mass of one mole of carbon

15. 20.0 g of a magnesium carbonate sample decomposes on heating to give carbon dioxide and 8.0 g magnesium oxide. What will be the percentage purity of magnesium carbonate in the sample ?

(Atomic weight of Mg = 24)

- (1) 60 (2) 84 (3) 75 (4) 96

NEET-II 2016

16. Suppose the elements X and Y combine to form two compounds XY_2 and X_3Y_2 . When 0.1 mole of XY_2 weighs 10 g and 0.05 mole of X_3Y_2 weighs 9 g, the atomic weights of X and Y are

- (1) 20, 30 (2) 30, 20
(3) 40, 30 (4) 60, 40

AIIMS 2016

17. An organic compound, on analysis, was found to contain 71.7% of chlorine, 4.04% of hydrogen and rest is carbon. If its molecular weight is 99. Then calculate molecular formula :-

- (1) $CHCl_3$ (2) $C_2H_4Cl_2$
(3) $C_2H_2Cl_2$ (4) CH_3Cl

NEET(UG) 2018

18. A mixture of 2.3 g formic acid and 4.5 g oxalic acid is treated with conc. H_2SO_4 . The evolved gaseous mixture is passed through KOH pellets. Weight (in g) of the remaining product at STP will be
(1) 1.4 (2) 3.0
(3) 2.8 (4) 4.4
19. In which case is the number of molecules of water maximum?
(1) 18 mL of water
(2) 0.18 g of water
(3) 0.00224 L of water vapours at 1 atm and 273 K
(4) 10^{-3} mol of water

AIIMS 2018

20. Initially in a container 1 g of gas A has 4 atm pressure at constant temperature. If 2 g of gas B is added in same container at same temperature then pressure becomes 6 atm what will be the ratio of molecular weight of A and B :-
(1) $M_A = 4M_B$ (2) $M_A = 2M_B$
(3) $M_B = 2M_A$ (4) $M_B = 4M_A$
21. 2 g mixture of two divalent metals A (at wt = 30) and B (at wt = 15) on reacting with dilute HCl solution gives 2.24 L H_2 gas at NTP then composition of A (in g) :-
(1) 1 (2) 0.5 (3) 1.5 (4) 1.2

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	2	3	4	3	4	2	3	1	1	1	2	4	2
Que.	16	17	18	19	20	21									
Ans.	3	2	3	1	4	1									

EXERCISE-III (Analytical Questions)

Check Your Understanding

- Number of HCl molecules present in 10 mL of 0.1 M solution is :
 (1) 6.022×10^{23} (2) 6.023×10^{22}
 (3) 6.022×10^{21} (4) 6.022×10^{20}
- The volume of a gas at 0°C and 700 mm pressure is 760 cc. The no. of molecules present in this volume is
 (1) 1.88×10^{22} (2) 6.022×10^{23}
 (3) 18.8×10^{23} (4) 18.8×10^{22}
- Rearrange the following (I to IV) in the order of increasing masses and choose the correct answer. (Atomic masses : N = 14, O = 16, Cu = 63)
 I 1 molecule of oxygen
 II 1 atom of Nitrogen
 III $1 \times 10^{-10} \times (\text{g molecular weight of oxygen})$
 IV $1 \times 10^{-10} \times (\text{g atomic weight of copper})$
 (1) II < I < III < IV (2) IV < III < II < I
 (3) II < III < I < IV (4) III < IV < I < II
- The number of moles of carbon dioxide which contain 8 g of oxygen is –
 (1) 0.5 mole (2) 0.20 mole
 (3) 0.40 mole (4) 0.25 mole
- If 224 mL of a triatomic gas has a mass of 1g at 273 K and 1 atm pressure, then the mass of one atom is –
 (1) 8.30×10^{-23} g (2) 2.08×10^{-23} g
 (3) 5.53×10^{-23} g (4) 6.24×10^{-23} g
- The maximum number of molecules are present in
 (1) 5 L of N_2 gas at STP
 (2) 0.5 g of H_2 gas
 (3) 10 g of O_2 gas
 (4) 15 L of H_2 gas at STP
- How many moles of magnesium phosphate, $\text{Mg}_3(\text{PO}_4)_2$ will contain 0.25 mol of oxygen atoms?
 (1) 2.5×10^{-2} (2) 0.02
 (3) 3.125×10^{-2} (4) 1.25×10^{-2}
- The mass of carbon anode consumed (giving only carbondioxide) in the production of 270 Kg of aluminium metal from bauxite by the Hall's process is : $2\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow 4\text{Al} + 3\text{CO}_2$
 (1) 180 kg (2) 270 kg
 (3) 240 kg (4) 90 kg
- 22.4 L of water vapour at NTP, When condensed to water occupies an approximate volume of –
 (1) 18 L (2) 1 L
 (3) 1 mL (4) 18 mL
- 0.01 mol of iodoform (CHI_3) reacts with Ag to produce a gas whose volume at NTP is
 $2\text{CHI}_3 + 6\text{Ag} \rightarrow \text{C}_2\text{H}_2 + 6\text{AgI(s)}$
 (1) 224 mL (2) 112 mL
 (3) 336 mL (4) None of these
- The minimum quantity in grams of H_2S needed to precipitate 63.5 g of Cu^{2+} will be nearly :
 $\text{Cu}^{+2} + \text{H}_2\text{S} \rightarrow \text{CuS} + \text{H}_2$
 (1) 63.5 g (2) 31.75 g
 (3) 34 g (4) 20 g
- 2.76 g of silver carbonate on being strongly heated yields a residue weighing –
 $\text{Ag}_2\text{CO}_3 \rightarrow 2\text{Ag} + \text{CO}_2 + \frac{1}{2}\text{O}_2$
 (1) 2.16 g (2) 2.48 g
 (3) 2.32 g (4) 2.64 g
- The volume of gas at NTP produced by 100 g of CaC_2 with water is :–
 $\text{CaC}_2 + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{C}_2\text{H}_2$
 (1) 70 L (2) 35 L
 (3) 17.5 L (4) 22.4 L
- 90 mL of pure dry O_2 is subjected to silent electric discharge. If only 10% of it is converted to O_3 , volume of the mixture of gases (O_2 and O_3) after the reaction will be ----- and after passing through turpentine oil will be
 (1) 84 mL and 78 mL (2) 81 mL and 87 mL
 (3) 78 mL and 84 mL (4) 87 mL and 81 mL
- Element 'A' reacts with oxygen to form a compound A_2O_3 . If 0.359 g of 'A' reacts to give 0.559 g of the compound, then atomic weight of 'A' will be :–
 (1) 51 (2) 43.08
 (3) 49.7 (4) 47.9
- 1.12 mL of a gas is produced at STP by the action of 4.12 mg of alcohol ROH with methyl magnesium iodide. The molecular mass of alcohol is –
 $\text{R-OH} + \text{CH}_3\text{MgI} \rightarrow \text{CH}_4 + \text{Mg(OR)I}$
 (1) 16 (2) 41.2
 (3) 82.4 (4) 156.0
- CaCO_3 is 90% pure. Volume of CO_2 collected at STP when 10 g of CaCO_3 is decomposed is –
 (1) 2.016 L (2) 1.008 L
 (3) 10.08 L (4) 20.16 L
- 50 g CaCO_3 will react with g of 20% HCl by weight .
 (1) 36.5 g (2) 73 g
 (3) 109.5 g (4) 182.5 g

19. Two oxides of a metal contains 50% and 40% of the metal respectively. The formula of the first oxide is MO . Then the formula of the second oxide is
 (1) MO_2 (2) M_2O_3
 (3) M_2O (4) M_2O_5
20. A gas mixture of 3 L of propane and butane on complete combustion at $25^\circ C$ produces 10 L of CO_2 . Initial composition of the propane & butane in the gas mixture is –
 (1) 66.67%, 33.33% (2) 33.33%, 66.67%
 (3) 50%, 50% (4) 60%, 40%
21. The atomic mass of an element is 27. If valency is 3, the vapour density of the volatile chloride will be:-
 (1) 66.75 (2) 6.675 (3) 667.5 (4) 81
22. 1 L of a hydrocarbon weighs as much as 1 L of CO_2 under similar conditions. Then the molecular formula of the hydrocarbon is –
 (1) C_3H_8 (2) C_2H_6
 (3) C_2H_4 (4) C_3H_6
23. There are two oxides of sulphur. They contain 50% and 60% of oxygen respectively by weight. The weight of sulphur which combine with 1 g of oxygen is in the ratio of –
 (1) 1 : 1 (2) 2 : 1
 (3) 2 : 3 (4) 3 : 2
24. A litre of air containing 1% Ar is repeatedly passed over hot Cu and hot Mg till no reduction of volume takes place. The final volume of Ar shall be :
 (1) 0 mL (2) 230 mL
 (3) 770 mL (4) 10 mL
25. An organic compound having molecular mass 60 is found to contain 20% of C, 6.67% of H and 46.67% of O while rest is oxygen. On heating it gives NH_3 along with a solid residue. The solid residue gives violet colour with alkaline copper sulphate solution. The compound is –
 (1) $(NH_2)_2CO$ (2) $CH_3CH_2CONH_2$
 (3) CH_3NCO (4) CH_3CONH_2
26. Percentage composition of an organic compound is as follows :
 C=10.06, H=0.84, Cl=89.10
 Which of the following corresponds to its molecular formula if the vapour density is 60.0
 (1) CH_2Cl_2 (2) $CHCl_3$
 (3) CH_3Cl (4) None
27. The ratio of masses of oxygen and nitrogen in a particular gaseous mixture is 1 : 4. The ratio of number of molecules is :
 (1) 1 : 8 (2) 3 : 16
 (3) 1 : 4 (4) 7 : 32
28. A gaseous hydrocarbon on combustion gives 0.72 g of water and 3.08 g of CO_2 . The empirical formula of the hydrocarbon is
 (1) C_2H_4 (2) C_3H_4
 (3) C_6H_6 (4) C_7H_8

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	1	1	4	3	4	3	4	4	2	3	1	2	4	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28		
Ans.	3	1	4	2	1	1	1	4	4	1	2	4	4		

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** 16 g each of O_2 and O_3 contains $\frac{N_A}{2}$ and $\frac{N_A}{3}$ atoms respectively
Reason :- 16 g O_2 and O_3 contains same no. of molecules.
 (1) A (2) B (3) C (4) D
- Assertion :-** Volume occupied by 1 mol $H_2O_{(l)}$ is equal to 22400 cc at NTP.
Reason :- 1 mol of any substance occupies 22.4 L volume at N.T.P.
 (1) A (2) B (3) C (4) D
- Assertion :-** 44 g of CO_2 28 g of CO have same volume at STP.
Reason :- Both CO_2 and CO are formed by C and oxygen.
 (1) A (2) B (3) C (4) D
- Assertion :-** Equivalent wt. of Cu in both CuO and Cu_2O is different.
Reason :- Equivalent wt. of an element is constant.
 (1) A (2) B (3) C (4) D
- Assertion :-** On compressing a gas to half the volume, the number of moles is halved.
Reason :- Number of moles decreases with decrease in volume.
 (1) A (2) B (3) C (4) D
- Assertion :-** The amount of the products formed in a reaction depends upon the limiting reactant.
Reason :- Limiting reactant reacts completely in the reaction.
 (1) A (2) B (3) C (4) D
- Assertion :-** Carbon and oxygen combined together only in one fixed ratio.
Reason :- In a chemical compound the elements are combined together in a fixed ratio.
 (1) A (2) B (3) C (4) D
- Assertion :-** At same temp & pressure 1 L O_2 and 1 L SO_2 contains equal no. of molecules.
Reason :- According to Avogadro's hypothesis equal volume of all gases under similar condition of temperature and pressure contains equal number of molecules.
 (1) A (2) B (3) C (4) D
- Assertion :-** Law of conservation of mass holds good for nuclear reaction.
Reason :- Law states that mass can be neither created nor destroyed in a chemical reaction.
 (1) A (2) B (3) C (4) D
- Assertion :-** The balancing of chemical equations is based on law of conservation of mass.
Reason :- Total mass of reactants is equal to total mass of products in a chemical reaction.
 (1) A (2) B (3) C (4) D
- Assertion :-** Pure water obtained from different sources such as river, well, spring, sea etc. always contains hydrogen and oxygen combined in the ratio of 1 : 8 by mass
Reason :- A chemical compound always contains elements combined together in same proportion by mass.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	
Ans.	4	4	2	3	4	1	4	1	4	1	1	

ATOMIC STRUCTURE

E

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NEET SYLLABUS

STRUCTURE OF ATOM : Atomic number, isotopes and isobars. Concept of shells and subshells, dual nature of matter and light, de Broglie's relationship, Heisenberg uncertainty principle, concept of orbital, quantum numbers, shapes of s, p and d orbitals, rules for filling electrons in orbitals- Aufbau principle, Pauli's exclusion principle and Hund's rule, electronic configuration of atoms, stability of half filled and completely filled orbitals.

@neetfocusquizz

OBJECTIVES

After studying this unit, you will be able to :

- describe Thomson, Rutherford and Bohr atomic models;
- understand the important features of the quantum mechanical model of atom;
- understand nature of electromagnetic radiation and Planck's quantum theory;
- state the de Broglie relation and Heisenberg uncertainty principle;
- define an atomic orbital in terms of quantum numbers;
- state aufbau principle, Pauli's exclusion principle and Hund's rule of maximum multiplicity;
- write the electronic configurations of atoms.

"Imagination is more important than knowledge."

Albert Einstein

ATOMIC STRUCTURE

2.0 INTRODUCTION

Atom is a Greek word and its meaning is indivisible i.e. an ultimate particle which cannot be further subdivided. John Dalton considered that "all matter are composed of smallest indivisible particle called atom.

Dalton's Atomic Theory :

This theory is based on law of mass conservation and law of definite proportions. The salient features of this theory are :-

- (1) Each element is composed of extremely small particles called atoms.
- (2) Atoms of a particular element are like but differ from atoms of other element.
- (3) Atom of each element is an ultimate particle and it has a characteristic mass but is structureless
- (4) Atoms are indestructible i.e. they can neither be created nor be destroyed.
- (5) Atoms of different elements take part in chemical reaction to form molecule.

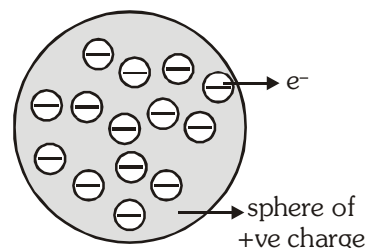
GOLDEN KEY POINTS

- Particles carrying negative charge were called negatrons by Thomson. The name negatron was changed to 'electron' by Stoney.
In cathode ray experiment, particles (electron) forming the rays have same specific charge (e/m) which is independent of the nature of gas and electrode used. It points out that electrons are present in all atoms.
- Mass of electron is $\frac{1}{1837}$ times that of proton.
- Mass of moving electron = $\frac{\text{rest mass of electron}}{\sqrt{1-(v/c)^2}}$
(Where v is the velocity of the electron and c is the velocity of light.)
When $v = c \Rightarrow$ mass of electron = ∞ and if $v > c \Rightarrow$ mass of electron = imaginary
- In anode ray experiment, the particles forming rays have e/m value that is dependent on the nature of the gas taken in the discharge tube, i.e. +ve particles are different in different gases. Therefore, the mass of the proton can be calculated.

2.1 ATOMIC MODELS

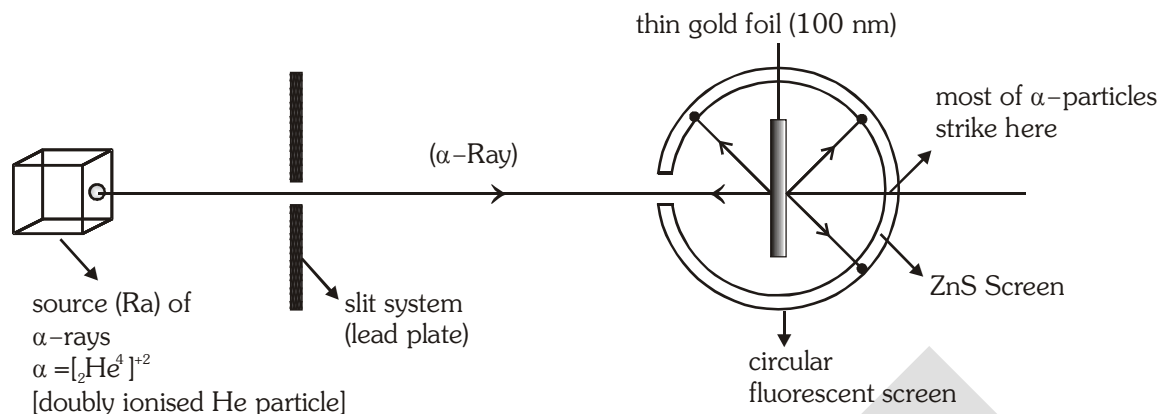
(A) Thomson's Model of Atom [1904]

- Thomson was the first to propose a detailed model of the atom.
- Thomson proposed that an atom consists of a uniform sphere of positive charge in which the electrons are distributed more or less uniformly.
- This model of atom is known as "Plum-Pudding model" or "Raisin Pudding Model" or "Water Melon Model".



Drawbacks :

- An important drawback of this model is that the mass of the atoms is considered to be evenly spread over that atom.
- It is a static model. It does not reflect the movement of electron.
- It couldn't explain the stability of an atom.

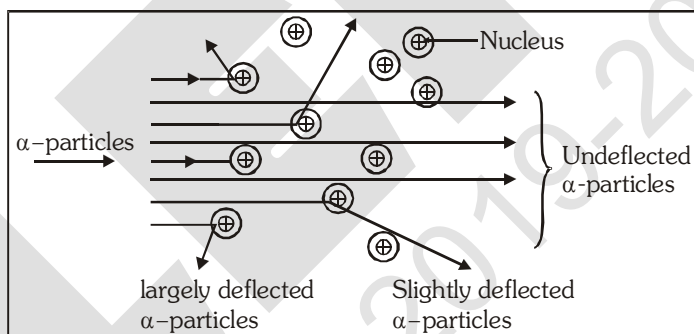
(B) Rutherford's Scattering Experiment

Rutherford observed that -

- Most of the α -particles passed through the gold foil undeflected.
- A small fraction of the α -particles were deflected by small angles.
- A very few α -particles (~ 1 in 20,000) bounced back, that is, were deflected by nearly 180° .

Following conclusions were drawn from the above observations -

- Since most of the α -particles went straight through the metal foil undeflected, it means that there must be very large empty space within the atom.
 - Since few of the α -particles were deflected from their original paths through moderate angles; it was concluded that whole of the +ve charge is concentrated and the space occupied by this positive charge is very small in the atom.
- When α -particles come closer to this point, they suffer a force of repulsion and deviate from their paths.
 - The positively charged heavy mass which occupies only a small volume in an atom is called **nucleus**. It is supposed to be present at the centre of the atom.
- A very few of the α -particles suffered strong deflections or even returned on their path indicating that the nucleus is rigid and α -particles recoil due to direct collision with the heavy positively charged mass.

**(C) RUTHERFORD'S ATOMIC MODEL**

On the basis of scattering experiments, Rutherford proposed model of the atom, which is known as nuclear atomic model. According to this model -

- An atom consists of a heavy positively charged nucleus where all the protons and neutrons are present. Protons & neutrons are collectively referred to as nucleons. Almost whole of the mass of the atom is contributed by these nucleons. The magnitude of the +ve charge on the nucleus is different for different atoms.
- The volume of the nucleus is very small and is only a minute fraction of the total volume of the atom. Nucleus has a diameter of the order of 10^{-12} to 10^{-13} cm and the atom has a diameter of the order of 10^{-8} cm.

$$\frac{D_A}{D_N} = \frac{\text{Diameter of the atom}}{\text{Diameter of the nucleus}} = \frac{10^{-8}}{10^{-13}} = 10^5, \quad D_A = 10^5 D_N$$

Thus diameter (size) of the atom is 10^5 times the diameter of the nucleus.

- The radius of a nucleus is proportional to the cube root of the number of nucleons within it.

$$R \propto A^{1/3} \Rightarrow R = R_0 A^{1/3}$$

Where $R_0 = 1.33 \times 10^{-13}$ cm (a constant) and A = mass number ($p + n$) and R = radius of the nucleus.

$$R = 1.33 \times 10^{-13} \times A^{1/3} \text{ cm}$$

- (iii) There is an empty space around the nucleus called extra nuclear part. In this part electrons are present. The number of electrons in an atom is always equal to number of protons present in the nucleus. As the nuclear part of atom is responsible for the mass of the atom, the extra nuclear part is responsible for its volume.

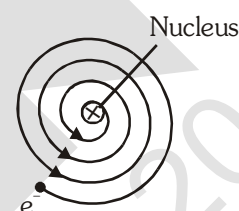
The volume of the atom is about 10^{15} times the volume of the nucleus.

$$\frac{\text{Volume of the atom}}{\text{Volume of the nucleus}} = \frac{(10^{-8})^3}{(10^{-13})^3} = 10^{15}$$

- (iv) Electrons revolve around the nucleus in closed orbits with high speeds. The centrifugal force acting on the revolving electron is being counter balanced by the force of attraction between the electrons and the nucleus.
- This model was similar to the solar system, the nucleus representing the sun and revolving electrons as planets.

Drawbacks of rutherford model -

- (i) This theory could not explain the stability of atom. According to Maxwell, electron loose its energy continuously in the form of electromagnetic radiations. As a result of this, the e^- should loose energy at every turn and move closer and closer to the nucleus following a spiral path. The ultimate result will be that it will fall into the nucleus, thereby making the atom unstable.
- (ii) If the electrons loose energy continuously, the observed spectrum should be continuous but the actual observed spectrum consists of well defined lines of definite frequencies. Hence, the loss of energy by electron is not continuous in an atom.



2.2 ATOMIC NUMBER AND MASS NUMBER

(a) Atomic Number

It is represented by Z. The number of protons present in the nucleus is called atomic number of an element.

For neutral atom : Number of electrons = Number of protons

For an ion : Number of electrons = $Z - (\text{charge on ion})$

$Z =$ number of protons only

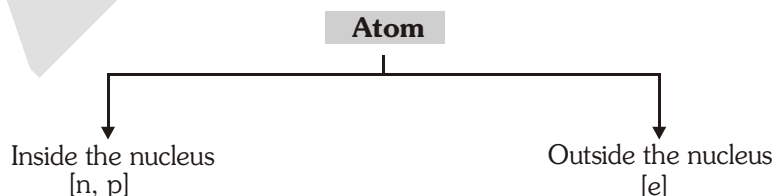
(b) Mass Number

It is represented by capital A. The sum of number of neutrons and protons is called the mass number of the element. It is also known as number of nucleons because neutrons & protons are present in nucleus.

Formula : $A = \text{number of protons} + \text{number of neutrons}$

Number of neutrons = $A - Z$

Note : A is always a whole number.

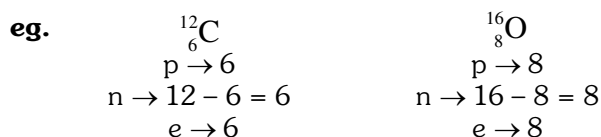
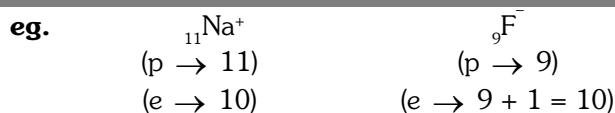


An atom of the element is represented by ${}_Z^AX$

Where, X = Symbol of element

Z = Atomic number = no. of protons = no. of electrons (If atom is neutral)

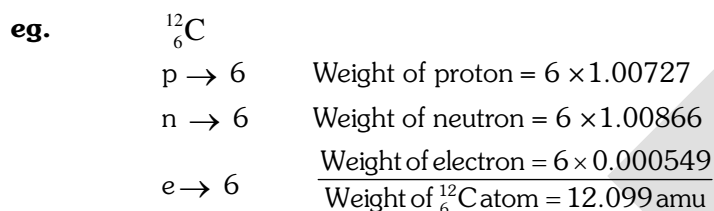
A = Mass number = no. of neutrons + Atomic no.



Mass no. [A] and atomic weight (amu = atomic mass unit)

Mass of Proton (m_p)	Mass of Neutron (m_n)	Mass of Electron (m_e)
1.672×10^{-27} kg	1.675×10^{-27} kg	9.1×10^{-31} kg
1.672×10^{-24} g	1.675×10^{-24} g	9.1×10^{-28} g
1.00727 amu	1.00866 amu	0.000549 amu

Method for Analysis of atomic weight $\rightarrow$

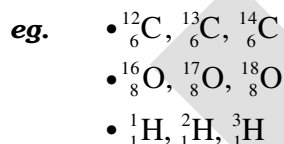


Mass no. of ${}_{6}^{12}\text{C}$ atom = 12 [p and n]

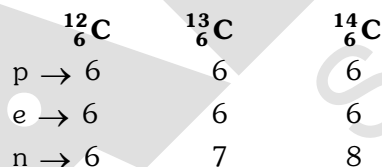
Note : Mass no. of atom is always a whole no. but atomic weight may be in decimal.

2.3 SOME IMPORTANT DEFINITIONS

(a) **Isotopes :** They are atoms of a given element which have the same atomic number but differ in their mass number.

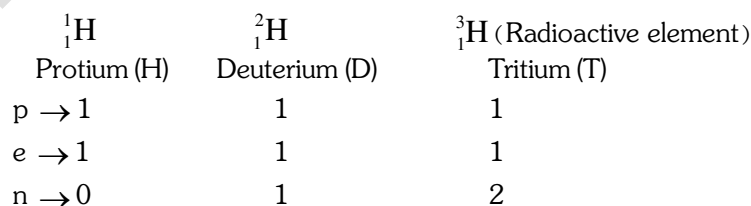


Explanation 1:



[**Note :** Isotopes have the same number of protons but differ in the number of neutrons in the nucleus]

Explanation 2:



- Neutron is not available in Protium
- No. of Nucleons = No. of Neutrons + No. of Protons
 $= n + p$

Atomic Weight : The atomic weight of an element is the average of mass of all the isotopes of that element.

- If an element have three isotopes y_1 , y_2 and y_3 and their isotopic weights are w_1 , w_2 , w_3 and their percentage/possibility/probability/ratio of occurrence in nature are x_1 , x_2 , x_3 respectively, then the average atomic weight of element is

$$\text{Average atomic weight} = \frac{w_1x_1 + w_2x_2 + w_3x_3}{x_1 + x_2 + x_3}$$

eg.	^{35}Cl	^{37}Cl
Probability	75%	25%
ratio	3	1

$$\text{Average atomic weight} = \frac{35 \times 3 + 37 \times 1}{3 + 1} = \frac{142}{4} = 35.5$$

(b) Isobars

Isobars are the atoms of different element which have the same mass number but different atomic number i.e they have different number of electrons, protons & neutrons but sum of number of neutrons & protons remains same.

Ex.1 ^3_1H	^3_2He	Ex.2 $^{40}_{19}\text{K}$	$^{40}_{20}\text{Ca}$
p = 1	p = 2	p = 19	p = 20
e = 1	e = 2	e = 19	e = 20
n = 2	n = 1	n = 21	n = 20
p + n = 3	p + n = 3	n + p = 40	n + p = 40

(c) Isodiaphers

They are the atoms of different element which have the same difference of the number of Neutrons & protons.

Ex.1 $^{11}_5\text{B}$	$^{13}_6\text{C}$	Ex.2 $^{15}_7\text{N}$	$^{19}_9\text{F}$
p = 5	p = 6	p = 7	p = 9
e = 5	e = 6	e = 7	e = 9
n = 6	n = 7	n = 8	n = 10
n - p = 1	n - p = 1	n - p = 1	n - p = 1

(d) Isotones/Isoneutronic Species/Isotonic

They are the atoms of different element which have the same number of neutrons.

Ex.1 ^3_1H	^4_2He	Ex.2 $^{39}_{19}\text{K}$	$^{40}_{20}\text{Ca}$
p = 1	p = 2	p = 19	p = 20
e = 1	e = 2	e = 19	e = 20
n = 2	n = 2	n = 20	n = 20

(e) Isosters

They are the molecules which have the same number of atoms & electrons.

Ex.1	CO_2	N_2O	Ex.2	CaO	KF
Atoms	= 1 + 2 = 3	Atoms = 2 + 1 = 3	Atoms	= 2	Atoms = 2
Electrons	= 6 + 8 × 2 = 22 e	Electrons = 7 × 2 + 8 = 22e	Electrons	= 20 + 8 = 28 e	Electrons = 19 + 9 = 28 e

(f) Isoelectronic Species

They are the atoms, molecules or ions which have the same number of electrons.

Ex.1	Cl^-	Ar
	18 e	18 e
Ex.2	H_2O	NH_3
	(2 + 8) = 10 e	(7 + 3) = 10 e
Ex.3	BF_3	SO_2
	(5 + 9 × 3) = 32 e	(16 + 8 × 2) = 32 e

GOLDEN KEY POINTS

- Isotopes have same chemical property but different physical property.
- Isotopes do not have the same value of e/m .
- Isobars do not have the same chemical & physical property.
- Isobars do not have the same value of e/m
- For isotones, $A_1 - Z_1 = A_2 - Z_2$
- For isodiaphers, $A_1 - 2Z_1 = A_2 - 2Z_2$

Illustrations

Illustration 1. If the mass of neutron is doubled & mass of electron is halved then find out the new atomic mass of ${}^{12}_6\text{C}$ and the percent by which it is increased.

Solution **Step-1** ${}^{12}_6\text{C} \rightarrow$ $e = 6$
 $p = 6 = 6 \text{ amu}$
 $n = 6 = 6 \text{ amu}$ $\left. \vphantom{\begin{matrix} p = 6 \\ n = 6 \end{matrix}} \right\} = 12 \text{ amu}$

If the mass of neutron is doubled and mass of electron is halved then,

$$\left. \begin{matrix} n = 12 \text{ amu} \\ p = 6 \text{ amu} \end{matrix} \right\} = 18 \text{ amu}$$

Note : mass of electron is negligible, so it is not considered in atomic mass.

Step-2 $\% \text{ Increment} = \frac{\text{Final mass} - \text{Initial mass}}{\text{Initial mass}} \times 100 = \frac{18 - 12}{12} \times 100 = 50\%$

Illustration 2. If mass of neutron is doubled, mass of proton is halved and mass of electron is doubled then find out the new atomic weight of ${}^{12}_6\text{C}$.

Solution **Step-1** ${}^{12}_6\text{C} \rightarrow$ $p = 6$
 $n = 6$ $\left. \vphantom{\begin{matrix} p = 6 \\ n = 6 \end{matrix}} \right\} = 12 \text{ amu}$

If mass of neutron is doubled, mass of proton is halved and mass of electron is doubled, then new atomic mass will be :

$$\left. \begin{matrix} n = 12 \text{ amu} \\ p = 3 \text{ amu} \end{matrix} \right\} = 15 \text{ amu}$$

Step-2 $\% \text{ Increment} = \frac{\text{Final mass} - \text{Initial mass}}{\text{Initial mass}} \times 100 = \frac{15 - 12}{12} \times 100 = 25\%$

Illustration 3. If no. of protons in X^{-2} is 16. then no. of electrons in X^{+2} will be—
 (1) 14 (2) 16 (3) 18 (4) None

Solution $\therefore$ No. of protons in X^{-2} is = 16
 $\therefore$ No. of electrons in X^{+2} is = 14

Illustration 4. Assuming that atomic weight of ${}^{12}\text{C}$ is 150 unit from atomic table, then according to this assumption, the weight of ${}^{16}\text{O}$ will be :-

Solution $\therefore 12 \text{ amu} = 150$
 $\therefore 1 \text{ amu} = \frac{150}{12}$
 $\therefore 16 \text{ amu} = \frac{150}{12} \times 16 = 200 \text{ Unit}$

Illustration 5. An element have three isotopes and their isotopic weights are 11, 12, 13 unit and their percentage of occurrence in nature is 85, 10, 5 respectively, then calculate the average atomic weight of element.

Solution Average Atomic weight = $\frac{11 \times 85 + 12 \times 10 + 13 \times 5}{85 + 10 + 5} = \frac{935 + 120 + 65}{100}$

Average weight = $\frac{1120}{100} = 11.2$

Illustration 6. Average atomic weight of an element M is 51.7. If two isotopes of M are ^{50}M and ^{52}M , then calculate the percentage of occurrence of ^{50}M in nature.

Solution

$$\begin{array}{rcl} ^{50}\text{M} & + & ^{52}\text{M} \\ x_1 & + & x_2 = 100 \\ & & x_2 = (100 - x_1) \end{array}$$

$$\begin{aligned} \text{average atomic weight} &= \frac{w_1 x_1 + w_2 x_2}{x_1 + x_2} = 51.7 = \frac{50 \times x_1 + 52 \times x_2}{x_1 + x_2} \\ 51.7 &= \frac{50x_1 + 52(100 - x_1)}{x_1 + (100 - x_1)} \end{aligned}$$

$$5170 = 50x_1 + 5200 - 52x_1$$

$$5170 = -2x_1 + 5200$$

$$2x_1 = 30$$

$$x_1 = 15$$

$$^{50}\text{M} = 15\% \quad ^{52}\text{M} = 85\%$$

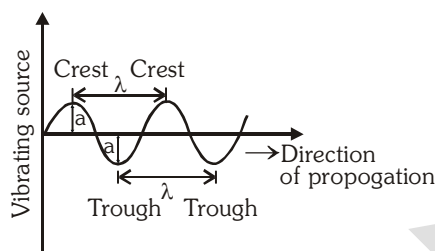
BEGINNER'S BOX-1

- Which of the following statements is incorrect for anode rays?
 - (1) They are deflected by electric and magnetic fields.
 - (2) Their e/m ratio depends on the gas in the discharge tube used to produce the anode rays.
 - (3) The e/m ratio of anode rays is constant.
 - (4) They are produced by the ionisation of the gas in the discharge tube.
- Which of the following pairs have identical value of e/m ?
 - (1) A proton and a neutron
 - (2) A proton and deuteron
 - (3) Deuteron and an α - particle
 - (4) An electron and γ - rays
- Rutherford's α - particle scattering experiments led to the conclusion that
 - (1) mass and energy are related together
 - (2) the mass and the positive charge of an atom are concentrated in the nucleus
 - (3) neutrons are present in the nucleus
 - (4) atoms are electrically neutral
- The radius of nucleus $^{27}_{13}\text{Al}$ will be
 - (1) $1.2 \times 10^{-15} \text{ m}$
 - (2) $2.7 \times 10^{-15} \text{ m}$
 - (3) $10.8 \times 10^{-15} \text{ m}$
 - (4) $4 \times 10^{-15} \text{ m}$
- Which of the following elements has maximum density of nucleus.
 - (1) $^{30}_{14}\text{SO}$
 - (2) $^{31}_{15}\text{P}$
 - (3) $^{16}_8\text{O}$
 - (4) All have same density
- Select iso electronic set :-
 - (a) Na^+ , H_3O^+ , NH_4^+
 - (b) CO_3^{2-} , NO_3^- , H_2CO_3
 - (c) P^{3-} , HCl , C_2H_5^- , PH_3
 - (d) F^- , Ne , Na^+
 - (1) a, b, d
 - (2) b, c, d
 - (3) a, b, c
 - (4) a, b, c, d
- If the table of atomic masses were established with the oxygen atom and assigned value of 100, then the mass of carbon atom would be, approximately:-
 - (1) 24
 - (2) 75
 - (3) 50
 - (4) 112

2.4 ELECTROMAGNETIC WAVES (EM WAVES) OR RADIANT ENERGY

According to this theory, the energy is transmitted from one body to another in the form of waves and these waves travel in the space with the same speed as light (3×10^8 m/s). These waves are known as Electro magnetic waves or radiant energy. **Ex :** Radio waves, micro waves, Infra red rays, visible rays, ultraviolet rays, X-rays, gamma rays.

- The radiant energy do not need any medium for propagation.
- The radiant energy have electric and magnetic fields and travel at right angle to these fields.
- The upper most point of the wave is called crest and the lower most point is called trough.



Some of the terms employed in dealing with the waves are described below.

- (1) **Wavelength (λ) (Lambda) :** It is defined as the distance between two nearest crest or trough.

It is measured in terms of Å (Angstrom), pm (picometre), nm (nanometer), cm (centimetre), m (metre)

$$1\text{Å} = 10^{-10}\text{ m}, \quad 1\text{ pm} = 10^{-12}\text{ m}, \quad 1\text{ nm} = 10^{-9}\text{ m}, \quad 1\text{ cm} = 10^{-2}\text{ m}$$

- (2) **Wave number ($\bar{\nu}$) (nu bar) :** It is the reciprocal of the wavelength, that is number of waves present in

unit length

$$\bar{\nu} = \frac{1}{\lambda}$$

It is measured in terms of cm^{-1} , m^{-1} etc.

- (3) **Frequency (ν) (nu) :** Frequency of a wave is defined as the number of waves which pass through a point in 1 s. It is measured in terms of Hertz (Hz), s^{-1} or cycle/s(cps) (1 Hertz = 1 s^{-1})

- (4) **Time period (T) :** Time taken by a wave to pass through one point. $T = \frac{1}{\nu} \text{ second}$

- (5) **Velocity (c) :** Velocity of a wave is defined as distance covered by a wave in 1 second

$$c = \lambda / T = \lambda \nu \text{ or } \nu = c / \lambda \text{ or } c = \nu (\text{s}^{-1}) \times \lambda (\text{m}) \text{ or } c = \nu \lambda (\text{m s}^{-1})$$

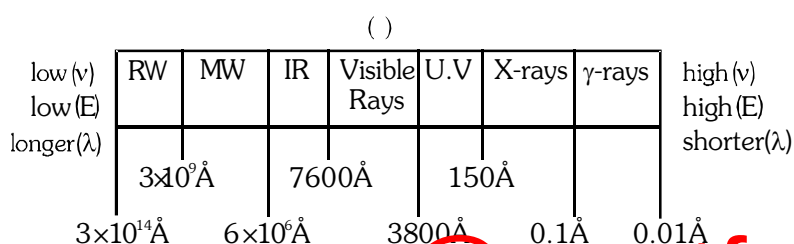
Since c is constants i.e. frequency is inversely proportional to λ

- (6) **Amplitude (a) :** The amplitude of a wave is defined as the height of crust or depth of trough.

Important note : $\nu = \frac{c}{\lambda} = c\bar{\nu} \quad \left(\bar{\nu} = \frac{1}{\lambda} \right)$

Electromagnetic spectrum or EM spectrum :

The arrangement obtained by arranging various types of EM waves in order of their increasing frequency or decreasing wave length is called as EM SPECTRUM



Illustrations

Illustration 7. The Vividh Bharti station of All India Radio broadcasts on a frequency of 1368 kilo hertz. Calculate the wavelength of the electromagnetic waves emitted by the transmitter.

Solution As we know velocity of light (c) = 3×10^8 m/s

Given ν (frequency) = 1368 kHz = 1368×10^3 Hz = 1368×10^3 s⁻¹

$$\therefore \lambda = \frac{c}{\nu} \quad \therefore \lambda = \frac{3 \times 10^8 \text{ ms}^{-1}}{1368 \times 10^3 \text{ s}^{-1}} = 219.3 \text{ m}$$

Illustration 8. Calculate $\bar{\nu}$ in cm⁻¹ and ν of yellow radiation having a wavelength of 5800 Å

Solution As we know $\bar{\nu} = \frac{1}{\lambda} = \frac{1}{5800 \text{ Å}} = \frac{1}{5800 \times 10^{-8} \text{ cm}} = \frac{10^8}{5800} \text{ cm}^{-1} = 17241.37 \text{ cm}^{-1}$

$$\nu = c \bar{\nu} = 3 \times 10^{10} \text{ cm s}^{-1} \times 1.7 \times 10^4 \text{ cm}^{-1} = 3 \times 1.7 \times 10^{14} = 5.1 \times 10^{14} \text{ s}^{-1}$$

Illustration 9. A particular radiostation broadcast at a frequency of 1120 kilo hertz. Another radio station broadcast at a frequency of 98.7 mega hertz. What are the wavelength of radiations from each station.

Solution Station 1st $\lambda = \frac{c}{\nu} = \frac{3 \times 10^8 \text{ ms}^{-1}}{1120 \times 10^3 \text{ s}^{-1}} = 267.86 \text{ m}$

Station 2nd $\lambda = \frac{c}{\nu} = \frac{3 \times 10^8 \text{ ms}^{-1}}{98.7 \times 10^6 \text{ s}^{-1}} = 3.0395 \text{ m}$

Illustration 10. How long would it take a radio wave of frequency 6×10^3 s⁻¹ to travel from mars to earth, that is a distance of 8×10^7 km ?

Solution Distance to be travelled from mars to earth = 8×10^7 km = 8×10^{10} m

$\therefore$ Velocity of EM waves = 3×10^8 m/s

$$\therefore \text{Time} = \frac{\text{Distance}}{\text{Velocity}} = \frac{8 \times 10^{10} \text{ m}}{3 \times 10^8 \text{ m/s}} = 2.66 \times 10^2 \text{ s} = 4 \text{ min } 26 \text{ s}$$

2.5 PLANCK'S QUANTUM THEORY

According to planck's quantum theory :

- (1) The radiant energy emitted or absorbed by a body not continuously but discontinuously in the form of small discrete packets of energy and these packets are called quantum.
- (2) In case of light, the smallest packet of energy is called as 'photon' but in general case the smallest packet of energy is called as quantum.
- (3) The energy of each quantum is directly proportional to frequency of the radiation i.e.

$$E \propto \nu \quad \Rightarrow \quad E = h\nu \quad \text{or} \quad E = \frac{hc}{\lambda} \left\{ \because \nu = \frac{c}{\lambda} \right\}$$

h is proportionality constant or Planck's constant

$$h = 6.626 \times 10^{-37} \text{ kJ s} \quad \text{or} \quad 6.626 \times 10^{-34} \text{ J s} \quad \text{or} \quad 6.626 \times 10^{-27} \text{ erg s}$$

- (4) Total amount of energy transmitted from one body to another will be some integral multiple of energy of

a quantum.

$$E = nh\nu = \frac{nhc}{\lambda} = nhc\bar{\nu}$$

where n = Positive integer

= Number of quanta

Illustrations

Illustration 11. Calculate the energy of a photon of sodium light of wave length 5.862×10^{-16} m in joule.

Solution $\lambda = 5.862 \times 10^{-16}$ m, $c = 3 \times 10^8$ m s⁻¹

$$E = nh\nu \quad \text{or} \quad \frac{nhc}{\lambda} \quad \{ \because n = 1 \}$$

$$\therefore E = \frac{hc}{\lambda} = \frac{1 \times 6.6 \times 10^{-34} \text{ J s} \times 3 \times 10^8 \text{ ms}^{-1}}{5.862 \times 10^{-16} \text{ m}} = \frac{6.6 \times 3}{5.862} \times 10^{-10} \text{ J} = 3.38 \times 10^{-10} \text{ J}$$

Illustration 12. Calculate the frequency & energy of a photon of wave length 4000 Å.

Solution (a) Calculation of frequency : $\lambda = 4000 \text{ Å} = 4000 \times 10^{-10} \text{ m}$

$$\therefore \nu = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m/s}}{4 \times 10^{-7} \text{ m}} = 0.75 \times 10^{15} \text{ s}^{-1} = 7.5 \times 10^{14} \text{ s}^{-1}$$

(b) Calculation of energy :

$$E = h\nu = 6.626 \times 10^{-34} \text{ joule second} \times 7.5 \times 10^{14} \text{ s}^{-1} = 4.96 \times 10^{-19} \text{ joule}$$

Illustration 13. Calculate the λ and frequency of a photon having an energy of 2 electron volt

Solution $\therefore 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J} \therefore 2 \text{ eV} = 3.204 \times 10^{-19} \text{ J} = E$

$$\begin{aligned} \text{(a) Calculation of wavelength } (\lambda) : E &= \frac{hc}{\lambda} \text{ or } \lambda = \frac{hc}{E} = \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{3.204 \times 10^{-19} \text{ J}} \\ &= 6.204 \times 10^{-7} \text{ m} \end{aligned}$$

$$\text{(b) Calculation of frequency } (\nu) : \nu = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ ms}^{-1}}{6.204 \times 10^{-7} \text{ m}} = 0.49 \times 10^{15} \text{ s}^{-1} = 4.9 \times 10^{14} \text{ s}^{-1}$$

Illustration 14. Which has a higher energy ?

(a) A photon of violet light with wave length 4000 Å

(b) A photon of red light with wave length 7000 Å

Solution (a) Violet light : $E_{\text{violet}} = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{4000 \times 10^{-10} \text{ m}} = 4.97 \times 10^{-19} \text{ joule}$

$$\text{(b) Red light : } E_{\text{red}} = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{7000 \times 10^{-10} \text{ m}} = 2.8 \times 10^{-19} \text{ joule}$$

$$\text{So, } E_{\text{violet}} > E_{\text{red}}$$

2.6 BOHR'S ATOMIC MODEL

Some Important formulae :

- This model was based on quantum theory of radiation and classical laws of physics.
- Bohr model is applicable only for single electron species like H, He⁺, Li²⁺ etc.
- Bohr model is based on particle nature of electron.

$$\text{Coulombic force} = \frac{kq_1q_2}{r^2}$$

$$\text{Centrifugal force} = \frac{mv^2}{r}$$

$$\text{Angular momentum} = mvr$$

Important postulates :

1st Postulate :

- Atom has a nucleus where all protons and neutrons are present.
- The size of nucleus is very small and it is present at the centre of the atom.

2nd Postulate :

- Negatively charged electron revolve around the nucleus in the same way as the planets revolve around the sun.
- The path of electron is circular.
- The attraction force (Coulombic or electrostatic force) between nucleus and electron is equal to the centrifugal force on electron.

i.e. Attraction force towards nucleus = centrifugal force away from nucleus.

3rd Postulate :

- Electrons can revolve only in those orbits in which angular momentum (mvr) of electron is integral multiple

of $\frac{h}{2\pi}$ i.e.

$$mvr = \frac{nh}{2\pi} = n\hbar$$

$$\hbar = \frac{h}{2\pi}$$

where : $n = +ve$ integer number ($n = 1, 2, 3, 4, \dots$) or ($n \in \mathbb{I}^+$)

h = Planck's constant

π = Constant

- Angular momentum can have values such as $\frac{h}{2\pi}, 2\frac{h}{2\pi}, 3\frac{h}{2\pi}, 4\frac{h}{2\pi}, 5\frac{h}{2\pi}, \dots$ but cannot have fractional values such as $1.5\frac{h}{2\pi}, 1.2\frac{h}{2\pi}, 0.5\frac{h}{2\pi}, \dots$

4th Postulate :

- The orbits in which electron can revolve are known as **stationary orbits** because in these orbits energy of electron is always constant.

5th Postulate :

- Each stationary orbit is associated with definite amount of energy therefore these orbits are also called as energy levels and are numbered as 1, 2, 3, 4, 5, or K, L, M, N, O, from the nucleus outwards.

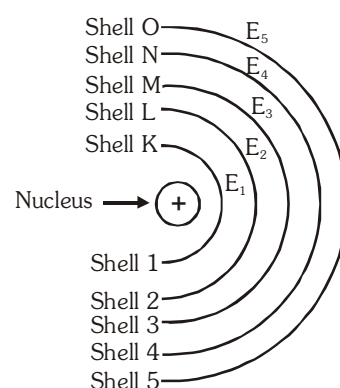
6th Postulate

- The emission or absorption of energy in the form of photon can only occur when electron jumps from one stationary state to another & it is

$$\Delta E = E_{\text{higher}} - E_{\text{lower}} = E_{n_2} - E_{n_1} = \text{Energy of a quantum}$$

$$= h\nu = \text{Bohr's frequency condition}$$

- Energy is absorbed when electron jumps from inner to outer orbit and is emitted when electron moves from outer to inner orbit.
- $n_2 > n_1$ whether emission or absorption of energy will occur.



BEGINNER'S BOX-2

- Electromagnetic radiation travels through vacuum at a speed of
 (1) 186000 m/s (2) 125 m/s (3) 3.00×10^8 m/s (4) It depends upon wavelength
- What is the wavelength (Å) of a photon that has an energy of 4.38×10^{-18} J
 (1) 454 Å (2) 2.3×10^7 Å (3) 6.89×10^{15} Å (4) 1.45×10^{-15} Å
- A 1kw radio transmitter operates at a frequency of 800 Hz. How many photons per second does it emit.
 (1) 1.71×10^{21} (2) 1.88×10^{33} (3) 6.02×10^{23} (4) 2.85×10^{20}
- Bohr's theory is not applicable to
 (1) He (2) Li^{+2} (3) He^+ (4) H atom

2.7 APPLICATION OF BOHR'S MODEL

(A) Radius of Various Orbits (Shell)

$$\begin{aligned} \text{Coulombic force} &= \frac{Kq_1q_2}{r^2} \\ &= \frac{KZe \cdot e}{r^2} = \frac{KZe^2}{r^2} \end{aligned}$$

Where $K = 9 \times 10^9 \text{ Nm}^2/\text{coulomb}^2$

As we know – Coulombic force = Centrifugal force

$$\frac{KZe^2}{r^2} = \frac{mv^2}{r} \quad \text{or} \quad v^2 = \frac{KZe^2}{mr} \quad \dots\dots\dots(1)$$

$$\text{As we know – } mvr = \frac{nh}{2\pi} \quad \text{or} \quad v = \frac{nh}{2\pi mr} \quad \dots\dots\dots(2)$$

Putting the value of v from eqⁿ. (2) to eqⁿ. (1)

$$\left(\frac{nh}{2\pi mr} \right)^2 = \frac{KZe^2}{mr} \quad \text{or} \quad \frac{n^2 h^2}{4\pi^2 m^2 r^2} = \frac{KZe^2}{mr}$$

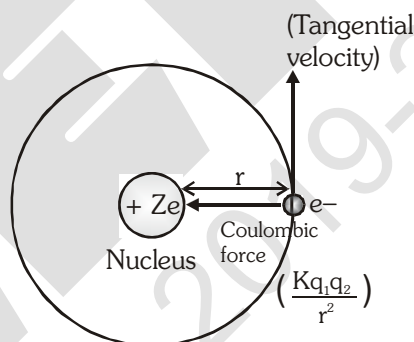
$$\boxed{r = \frac{n^2 h^2}{4\pi^2 m KZe^2}} \quad \dots\dots\dots(3)$$

Putting the value of π , h , m , K , & e (Constants) in the above eqⁿ. (3)

$$r = 0.529 \times 10^{-8} \times \frac{n^2}{Z} \text{ cm} \quad \{1\text{Å} = 10^{-10}\text{m} = 10^{-8}\text{cm}\}$$

$$\boxed{r_n = 0.529 \times \frac{n^2}{Z} \text{ Å}}$$

This formula is only applicable for hydrogen and hydrogen like species i.e. species containing single electron.



(B) Velocity of an electron

Since coulombic force = Centrifugal force

$$\frac{KZe^2}{r^2} = \frac{mv^2}{r} \quad \text{or} \quad v^2 = \frac{KZe^2}{mr} \quad \dots\dots\dots(1)$$

Putting the value of Angular momentum

$$mvr = \frac{nh}{2\pi} \quad \text{or,} \quad KZe^2 = \frac{nh}{2\pi}(v)$$

$$v = \frac{2\pi KZe^2}{nh}$$

Putting the value of π , k , e & h

$$v = 2.188 \times 10^6 \frac{Z}{n} \text{m/s}$$

Illustrations

Illustration 16. Calculate the radius of 1st, 2nd, 3rd, 4th Bohr's orbit of hydrogen.

Solution Radius of Bohr's orbit $r = 0.529 \times \frac{n^2}{Z} \text{Å}$

(a) Radius of Ist orbit : $r = 0.529 \times \frac{1^2}{1} \text{Å} = 0.529 \text{Å}$

(b) Radius of IInd orbit : $r = 0.529 \times \frac{2^2}{1} = 0.529 \times 4 = 2.116 \text{Å}$

(c) Radius of IIIrd orbit : $r = 0.529 \times \frac{3^2}{1} = 0.529 \times 9 = 4.761 \text{Å}$

(d) Radius of 4th orbit : $r = 0.529 \times \frac{4^2}{1} = 0.529 \times 16 = 8.464 \text{Å}$

Illustration 17. Calculate the radius ratio of 3rd & 5th orbit of He⁺.

Solution $\therefore r = 0.529 \times \frac{n^2}{Z} \text{Å}$ and Atomic Number of He = 2

$\therefore r_3 = 0.529 \times \frac{(3)^2}{2} = 0.529 \times \frac{9}{2}$ and $r_5 = 0.529 \times \frac{(5)^2}{2} = 0.529 \times \frac{25}{2}$

Therefore $\frac{r_3}{r_5} = \frac{0.529 \times \frac{(3)^2}{2}}{0.529 \times \frac{(5)^2}{2}} = \frac{9}{25}$ or $r_3 : r_5 = 9 : 25$

Illustration 18. Calculate the radius ratio of 2nd orbit of hydrogen and 3rd orbit of Li⁺².

Solution Atomic number of H = 1, Atomic number of Li = 3,

2nd orbit radius of Hydrogen $(r_2)_H = 0.529 \times \frac{2^2}{1}$

3rd orbit radius of Li⁺² $(r_3)_{Li^{+2}} = 0.529 \times \frac{3^2}{3}$

$\therefore \frac{(r_2)_H}{(r_3)_{Li^{+2}}} = \frac{0.529 \times \frac{2^2}{1}}{0.529 \times \frac{3^2}{3}} = \frac{4}{3} \quad \therefore (r_2)_H : (r_3)_{Li^{+2}} = 4 : 3$

Illustration 18. Calculate the radius ratio of 2nd excited state of H & 1st excited state of Li⁺

Solution 2nd excited state, means e⁻ is present in 3rd shell of hydrogen $r_3 = 0.529 \times \frac{(3)^2}{1} = 0.529 \times 9$

1st excited state, means e⁻ exist in 2nd shell of Li⁺ $r_2 = 0.529 \times \frac{(2)^2}{3} = 0.529 \times \frac{4}{3}$

$$\frac{\text{radius of 2}^{\text{nd}} \text{ excited state of hydrogen}}{\text{radius of 1}^{\text{st}} \text{ excited state of Li}^{+2}} = \frac{(r_3)_H}{(r_2)_{\text{Li}^{+2}}} = \frac{0.529 \times \frac{9}{1}}{0.529 \times \frac{4}{3}} = \frac{27}{4}$$

Illustration 19. Calculate velocity of an electron placed in the third orbit of the hydrogen atom. Also calculate the number of revolutions per second that this electron makes around the nucleus.

Solution Velocity of electron in 3rd orbit :

$$\therefore V_n = 2.182 \times 10^6 \times \frac{Z}{n} \text{ ms}^{-1}$$

$$\therefore V_3 = 2.182 \times 10^6 \times \frac{1}{3} \text{ ms}^{-1} = 7.27 \times 10^5 \text{ ms}^{-1}$$

No. of revolution per second

$$= \frac{v_n}{2\pi r_3} = \frac{v_n}{2\pi \left(\frac{n^2 a_0}{Z} \right)} = \frac{7.27 \times 10^5}{2 \times 3.14 \times 9 \times 0.529 \times 10^{-10}} = 2.43 \times 10^{14} \text{ r.p.s.}$$

Illustration 20. How much time an e⁻ will take for one complete revolution in 2nd orbit of He⁺?

Solution time taken = $\frac{\text{distance}}{\text{velocity}} = \frac{2\pi r}{v} = \frac{2 \times 3.14 \times 0.529 \times \frac{4}{2} \times 10^{-10} \text{ m}}{2.18 \times 10^6 \times \frac{2}{2} \text{ ms}^{-1}} = 3.05 \times 10^{-16} \text{ s}$

(C) Energy of an electron

Let the total energy of an electron be E. It is the sum of kinetic and potential energy.

$$\text{i.e. } E = \text{K.E.} + \text{P.E.}$$

$$E = \left(\frac{1}{2} mv^2 \right) + \left(\frac{Kq_1 q_2}{r} \right) \quad \left[\text{P.E.} = -\frac{KZe^2}{r} \right]$$

$$E = \frac{1}{2} mv^2 + \frac{KZe(-e)}{r} = \frac{1}{2} mv^2 - \frac{KZe^2}{r} \quad \left[\text{KE} = \frac{1}{2} mv^2 = \frac{KZe^2}{2r} \right]$$

$$E = \frac{KZe^2}{2r} - \frac{KZe^2}{r} = -\frac{KZe^2}{2r}$$

Putting the value of r from eq. (3)

$$E_n = -\frac{KZe^2 \times 4\pi^2 m KZe^2}{2n^2 h^2} \quad \text{or} \quad E_n = -\frac{2\pi^2 m K^2 Z^2 e^4}{n^2 h^2}$$

Putting the value of π , K, e, m, h, we get :

$$E_n = -2.18 \times 10^{-18} \times \frac{Z^2}{n^2} \text{ J/atom} \quad \text{or} \quad E_n = -13.6 \times \frac{Z^2}{n^2} \text{ eV/atom}$$

This formula is applicable for hydrogen atom & hydrogen like species i.e. single electron species.

Since n can have only integral values, it follows that total energy of the e⁻ is quantised.

The -ve sign indicates that the electron is bonded towards nucleus.

Some extra points :

- (i) $K.E = \frac{KZe^2}{2r}$ i.e. $K.E. \propto \frac{1}{r}$ On increasing radius, K.E. decreases.
- (ii) $P.E. = -\frac{KZe^2}{r}$ i.e. $P.E. \propto -\frac{1}{r}$ On increasing radius, P.E. increases.
- (iii) $E = -\frac{KZe^2}{2r}$ i.e. $E \propto -\frac{1}{r}$ On increasing radius, total energy increases.

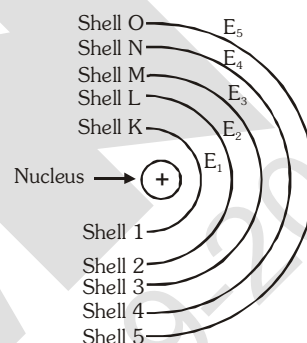
Conclusion : $P.E = (-)2KE$ $KE = (-)E$ $P.E = 2E$

Energy difference between two energy levels :

$$E_{n_2} - E_{n_1} = -13.6 \times Z^2 \left[\frac{1}{n_2^2} - \frac{1}{n_1^2} \right]$$

Energy level for H atom can be represented as follows :

$n = 6$ or P	$E_6 = -0.38 \text{ eV}$	
$n = 5$ or O	$E_5 = -0.54 \text{ eV}$	
$n = 4$ or N	$E_4 = -0.85 \text{ eV}$	$E_5 - E_4 = 0.31 \text{ eV}$
$n = 3$ or M	$E_3 = -1.51 \text{ eV}$	$E_4 - E_3 = 0.66 \text{ eV}$
$n = 2$ or L	$E_2 = -3.4 \text{ eV}$	$E_3 - E_2 = 1.89 \text{ eV}$
$n = 1$ or K	$E_1 = -13.6 \text{ eV}$	$E_2 - E_1 = 10.2 \text{ eV}$
i.e. $(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3) > (E_5 - E_4) \dots$		



Important Definitions :-

- (i) **Ionization energy :** Minimum amount of energy required to liberate an electron from the ground state of an isolated atom is called as ionization energy.
 $n_1 = 1; n_2 = \infty$
- (ii) **Separation energy :** Minimum energy required to remove an electron from its excited state is called as separation energy.
 $n_1 = 2, 3, 4, 5, \dots; n_2 = \infty$
- (iii) **Excitation energy :** Amount of energy required to shift an electron from ground state to any excited state is called as excitation energy.
 $n_1 = 1; n_2 = 2, 3, 4, 5, \dots$

Note : All these kinds of energy are always positive.

Illustrations

Illustration 21. If the total energy of an electron is -1.51 eV in hydrogen atom then find out K.E, P.E, orbit, radius and velocity of the electron in that orbit.

Solution

- (i) $K.E = -E = 1.51 \text{ eV}$
- (ii) $PE = 2 \times E = -2 \times 1.51 = -3.02 \text{ eV}$
- (iii) $\therefore E = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$ or $-1.51 = -13.6 \times \frac{1^2}{n^2}$
 $\Rightarrow n^2 = \frac{-13.6}{-1.51} = 9 \quad \therefore n = 3 \quad \text{i.e. } 3^{\text{rd}} \text{ orbit}$
- (iv) $r = 0.529 \times \frac{n^2}{Z} = 0.529 \times \frac{3 \times 3}{1} = 0.529 \times 9 = 4.761 \text{ \AA}$
- (v) $v = 2.188 \times 10^8 \times \frac{Z}{n} = 2.188 \times 10^8 \times \frac{1}{3} \text{ cm/s} = 0.729 \times 10^8 \text{ cm/s}$

Illustration 22. Calculate the energy of Li^{+2} ion for 2nd excited state

Solution $E = -13.6 \times \frac{Z^2}{n^2}$ $\therefore Z = 3$ and electron exist in 2nd excited state, means electron present in 3rd shell
i.e. $n = 3$

$$\therefore E = -13.6 \times \frac{(3)^2}{(3)^2} = -13.6 \text{ eV/atom}$$

Illustration 23. Calculate the ratio of energies of He^+ for 1st & 2nd excited state .

Solution $\frac{\text{Energy of } (\text{He}^+) \text{ 1}^{\text{st}} \text{ Excited state}}{\text{Energy of } (\text{He}^+) \text{ 2}^{\text{nd}} \text{ Excited state}} = \frac{\text{Energy of } (\text{He}^+) \text{ 2}^{\text{nd}} \text{ shell}}{\text{Energy of } (\text{He}^+) \text{ 3}^{\text{rd}} \text{ shell}} = \frac{-13.6 \times \frac{(2)^2}{(2)^2}}{-13.6 \times \frac{(2)^2}{(3)^2}} = \frac{9}{4}$

Illustration 24. The ionization energy for the hydrogen atom is 13.6 eV then the required energy in eV to excite it from the ground state to 1st excited state

Solution Ionization energy = 13.6 eV i.e. Energy in ground state = -13.6 eV
Energy of 1st excited state i.e. 2nd orbit = -3.4 eV
so, $E_2 - E_1 = -3.4 + 13.6 = 10.2 \text{ eV}$

GOLDEN KEY POINTS

- Bohr's atomic model is applicable only for monoelectronic species like H, He^+ , Li^{+2} , Na^{10+} , U^{91+} etc.

$$E_{z,n} = E_H \times \frac{Z^2}{n^2}$$

if z is same

 $E_n = E_H \times \frac{1}{n^2}$

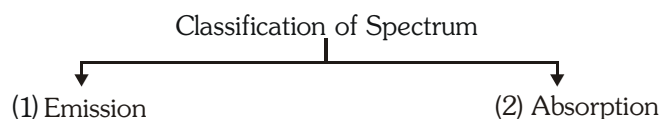
if n is same

 $E_z = E_H \times Z^2$

BEGINNER'S BOX-3

- In which of the following is the radius of the first orbit minimum ?
(1) A Hydrogen atom (2) A tritium atom
(3) Triply ionized beryllium (4) Double ionized helium
- The energy needed to excite a hydrogen atom from its ground to its third excited state is
(1) 12.1 eV (2) 10.2 eV (3) 0.85 eV (4) 12.75 eV
- The ionisation energy of a hydrogen atom is 13.6 eV. The energy of the ground level in doubly ionised lithium is
(1) -28.7 eV (2) -54.4 eV (3) -122.4 eV (4) -13.6 eV
- What would be the radius of 2nd excited state in Li^{+2} ion ?
(1) $0.529 \text{ \AA}^\circ$ (2) $1.51 \text{ \AA}^\circ$ (3) $0.2645 \text{ \AA}^\circ$ (4) $0.5299 \text{ \AA}^\circ$
- 2nd separation energy of an electron in H atom
(1) 27.2 eV (2) 1.57 eV (3) 3.4 eV (4) 13.6 eV
- How much energy would be required by an electron while moving from ground state to 3rd excited state of He^+ ion.
(1) 40.8 eV (2) 10.2 eV (3) 51 eV (4) 48.35 eV

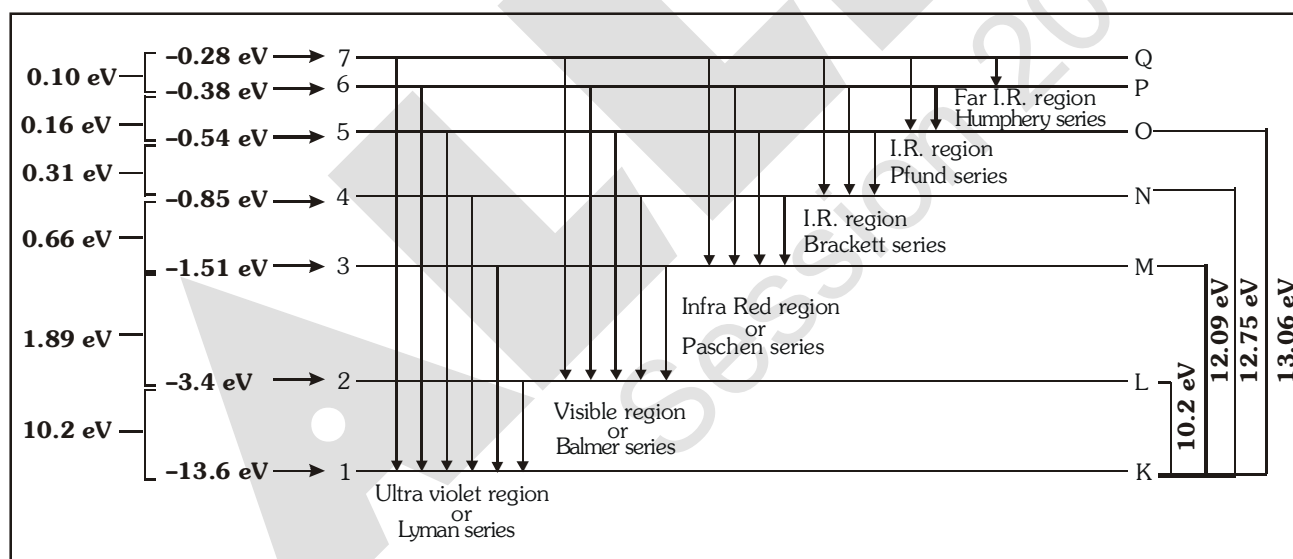
When a radiation is passed through a spectroscope (prism) for the dispersion of the radiation, the pattern (photograph) obtained on the screen (photographic plate) is called as spectrum of the given radiation



HYDROGEN SPECTRUM

When an electric excitation is applied on hydrogen atomic gas at low pressure, a bluish light is emitted. When a ray of this light is passed through a prism, a spectrum of several isolated sharp lines is obtained. The wavelength of various lines show that spectrum lines lie in Visible, Ultraviolet and Infra red region. These lines are grouped into different series.

Series	Discovered by	regions	$n_2 \rightarrow n_1$	No. of lines
Lyman	Lyman	U.V. region	$n_2 = 2, 3, 4 \dots / n_1 = 1$	$n_2 - 1$
Balmer	Balmer	Visible region	$n_2 = 3, 4, 5 \dots / n_1 = 2$	$n_2 - 2$
Paschen	Paschen	Infra red (I.R.)	$n_2 = 4, 5, 6 \dots / n_1 = 3$	$n_2 - 3$
Brackett	Brackett	I.R. region	$n_2 = 5, 6, 7 \dots / n_1 = 4$	$n_2 - 4$
Pfund	Pfund	I.R. region	$n_2 = 6, 7, 8 \dots / n_1 = 5$	$n_2 - 5$
Humphery	Humphery	Far I.R. region	$n_2 = 7, 8, 9 \dots / n_1 = 6$	$n_2 - 6$



Similar words

- First line / Starting line / Initial line ($\lambda_{\max}$ and $\nu_{\min}$)
- Last line / limiting line / marginal line ($\lambda_{\min}$ and $\nu_{\max}$)
- First line of any series = α line
- Second line of any series = β line
- Third line of any series = γ line

Calculation of number of spectral lines

$$(a) \text{ Total number of spectral lines} = 1 + 2 + \dots (n_2 - n_1) = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

Where : n_2 = higher energy level; n_1 = lower energy level

If $n_1 = 1$ (ground state)

$$\text{Total number of spectral lines} = \frac{(n_2 - 1)n_2}{2} = \frac{n(n-1)}{2}$$

$$(b) \text{ Number of spectral lines which falls in a particular series} = (n_2 - n_1)$$

where n_2 = higher energy level, n_1 = Fixed lower energy level of each series.

RYDBERG FORMULA

In 1890, Rydberg gave a very simple theoretical equation for the calculation of the wavelength of various lines of hydrogen like spectrum

$$\bar{\nu} = \frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

where R = Rydberg constant = $109678 \text{ cm}^{-1} = 109700 \text{ cm}^{-1} = 10970000 \text{ m}^{-1} = 1.1 \times 10^7 \text{ m}^{-1}$

$$\frac{1}{R} = 9.12 \times 10^{-6} \text{ cm} = 912 \text{ \AA}$$

Derivation of Rydberg formula :

$$\Delta E = E_{n_2} - E_{n_1}$$

$$\Delta E = \frac{-2\pi^2 m K^2 Z^2 e^4}{n_2^2 h^2} - \left[\frac{-2\pi^2 m K^2 Z^2 e^4}{n_1^2 h^2} \right]$$

$$= \frac{2\pi^2 m K^2 Z^2 e^4}{n_1^2 h^2} - \frac{2\pi^2 m K^2 Z^2 e^4}{n_2^2 h^2} \quad \left(\because \Delta E = h\nu = \frac{hc}{\lambda} \right)$$

$$\frac{hc}{\lambda} = \frac{2\pi^2 m K^2 Z^2 e^4}{h^2} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \quad \text{or} \quad \frac{1}{\lambda} = \frac{2\pi^2 m K^2 e^4 Z^2}{ch^3} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

where $\frac{2\pi^2 m K^2 e^4}{ch^3}$ is a constant which is equal to Rydberg constant (R).

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Illustrations

Illustration 25. Calculate the wavelength of 1st line of Balmer series in Hydrogen spectrum.

Solution For first line of Balmer series $n_1 = 2, n_2 = 3$

$$\frac{1}{\lambda} = R(1)^2 \left[\frac{1}{4} - \frac{1}{9} \right] = R \left[\frac{9-4}{36} \right] = R \left[\frac{5}{36} \right]$$

$$\Rightarrow \lambda = \frac{36}{5R} = \frac{36}{5} \times \frac{1}{R} = \frac{36}{5} \times 9.12 \times 10^{-6} \text{ cm} = 65.66 \times 10^{-6} \text{ cm} = 6566 \text{ \AA}$$

Illustration 26. Calculate the frequency of the last line of the Lyman series in hydrogen spectrum.

Solution For last line of Lyman series $n_1 = 1$, $n_2 = \infty$

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = R \left(\frac{1}{1} - 0 \right) = R$$

$$\Rightarrow \nu = \frac{c}{\lambda} = c \times \frac{1}{\lambda} = c \times R = 3 \times 10^{10} \text{ cm s}^{-1} \times 109678 \text{ cm}^{-1} = 3.29 \times 10^{15} \text{ s}^{-1}$$

Illustration 27. Calculate the wavelength of 3rd line of Brackett series in hydrogen spectrum.

Solution For 3rd line of Brackett series $n_1 = 4$, $n_2 = 7$

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{(4)^2} - \frac{1}{(7)^2} \right] = R \left[\frac{1}{16} - \frac{1}{49} \right] = R \left[\frac{49-16}{16 \times 49} \right] = R \frac{33}{784}$$

$$\text{Therefore, } \lambda = \frac{784}{33R} = \frac{784}{33} \times 912 \text{ \AA} = 21667 \text{ \AA}$$

Illustration 28. The wave number of 1st line of Balmer series of hydrogen spectrum is 15200 cm^{-1} . The wave number of 1st line of Balmer series of Li^{+2} spectrum will be ?

Solution Wave number of 1st line of Balmer series of hydrogen spectrum. $\bar{\nu} = \frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

$$(\text{for H, } Z = 1) \quad \bar{\nu} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = 15200 \text{ cm}^{-1}$$

Wave number of 1st line of Balmer series of Li^{+2} ion is.

$$\bar{\nu} = Z^2 \times R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad \{ \because Z = 3 \text{ for } \text{Li}^{+2} \}$$

$$\therefore \bar{\nu} = 3^2 \times 15200 = 9 \times 15200 = 136800 \text{ cm}^{-1}$$

Illustration 29. Calculate the ratio of maximum λ of Lyman & Balmer series ?

Solution $E \propto \nu \propto \frac{1}{\lambda}$

$$\frac{\text{Maximum } \lambda \text{ of Lyman series}}{\text{Maximum } \lambda \text{ of Balmer series}} = \frac{1^{\text{st}} \text{ line of Lyman series}}{1^{\text{st}} \text{ line of Balmer series}}$$

$$\frac{\lambda_{\text{Lyman}}}{\lambda_{\text{Balmer}}} = \frac{\lambda_B}{\lambda_L} = \frac{R \left[\frac{1}{2^2} - \frac{1}{3^2} \right]}{R \left[\frac{1}{1^2} - \frac{1}{2^2} \right]} = \frac{\frac{1}{4} - \frac{1}{9}}{\frac{1}{1} - \frac{1}{4}} = \frac{\frac{5}{36}}{\frac{3}{4}} = \frac{5}{36} \times \frac{4}{3} = \frac{5}{27}$$

Illustration 30. In a hydrogen spectrum if electron moves from 7 to 1 orbit by transition in multi steps then find out the total number of lines in the spectrum.

Solution	Lyman	=	$(n_2 - 1) = 7 - 1 = 6$
	Balmer	=	$(n_2 - 2) = 7 - 2 = 5$
	Paschen	=	$(n_2 - 3) = 7 - 3 = 4$
	Bracket	=	$(n_2 - 4) = 7 - 4 = 3$
	Pfund	=	$(n_2 - 5) = 7 - 5 = 2$
	Humphrey	=	$(n_2 - 6) = 7 - 6 = 1$
	Total	=	21

Total number of lines can be calculated as follows :

$$\text{Total number of lines} = \frac{(n_2 - n_1)[(n_2 - n_1) + 1]}{2} = \frac{(7 - 1)(6 + 1)}{2} = \frac{42}{2} = 21$$

Illustration 31. In a hydrogen spectrum if electron moves from 6th to 2nd orbit by transition in multi steps then find out the number of lines in spectrum

Solution Total number of line = $4 + 3 + 2 + 1 = 10$

$$\text{or Total number of lines} = \frac{(n_2 - n_1)[(n_2 - n_1) + 1]}{2} = \frac{(6 - 2)(4 + 1)}{2} = \frac{4 \times 5}{2} = 10$$

Illustration 32. A certain electronic transition from an excited state to Ground state of the Hydrogen atom in one or more steps gives rise to 5 lines in the ultra violet region of the spectrum. How many lines does this transition produce in the Infra red region of the spectrum?

Solution **(Lyman Series) ultra violet region :** 5 Lines i.e. e^- is coming from 6th to 1st Orbit

$$n_2 - 1 = 5 \quad \text{or} \quad n_2 = 6$$

Infrared region line

$$(i) \text{ Paschen series} = (6 - 3) = 3 \quad (ii) \text{ Bracket} = (6 - 4) = 2 \quad (iii) \text{ Pfund} = (6 - 5) = 1$$

Total Number of lines are = 6

Illustration 33. In H atom if the electron moves from n^{th} orbit to 1st orbit by transition in multi steps, then the total number of lines observed in the spectrum are 10, then find out the value of n.

Solution
$$\text{Total number of lines} = \frac{(n_2 - n_1)[(n_2 - n_1) + 1]}{2}$$

$$\text{So,} \quad 10 = \frac{(n - 1)(n - 1 + 1)}{2}$$

$$\text{or} \quad 20 = (n - 1)(n)$$

$$n^2 - n - 20 = 0$$

$$n^2 - 5n + 4n - 20 = 0$$

$$n(n - 5) + 4(n - 5) = 0$$

$$(n + 4)(n - 5) = 0 \quad \Rightarrow \quad n = 5$$

Limitation of the Bohr's model :

- (1) Bohr's theory does not explain the spectrum of multi electron atom.
- (2) Why the Angular momentum of the revolving electron is equal to $\frac{nh}{2\pi}$, has not been explained by Bohr's theory.
- (3) Bohr inter related quantum theory of radiation and classical laws of physics without any theoretical explanation.
- (4) Bohr's theory does not explain the fine structure of the spectral lines. Fine structure of the spectral line is obtained when spectrum is viewed by spectroscopy of more resolution power.
- (5) Bohr theory does not explain the splitting of spectral lines in the presence of magnetic field (Zeeman's effect) or electric field (Stark's effect)

BEGINNER'S BOX-4

- The line spectra of two elements are not identical because
 - (1) The elements don't have the same number of neutrons.
 - (2) They have different mass numbers
 - (3) Their outermost electrons are at different energy levels.
 - (4) They have different valencies.
- In which of the following transition will the wavelength be minimum.
 - (1) $n = 6$ to $n = 4$
 - (2) $n = 4$ to $n = 2$
 - (3) $n = 3$ to $n = 1$
 - (4) $n = 2$ to $n = 1$
- The wavelength of third line of the Balmer series for a H atom is
 - (1) $\frac{21}{100R}$
 - (2) $\frac{100}{21R}$
 - (3) $\frac{21R}{100}$
 - (4) $\frac{100R}{21}$
- When the electron of a hydrogen atom jumps from $n = 4$ to $n = 1$ state, the number of spectral lines emitted is
 - (1) 15
 - (2) 6
 - (3) 3
 - (4) 4

2.9 WAVE MECHANICAL MODEL OF AN ATOM

This model consists of following

- de-Broglie concept (Dual nature of Matter)
- Heisenberg's Uncertainty principle.

(A) THE DUAL NATURE OF MATTER (THE WAVE NATURE OF ELECTRON)

In 1924, a French physicist, **Louis de-Broglie** suggested that if the nature of light is both that of a particle and of a wave, then this dual behavior should be true also for the matter.

- The wave nature of light rays and X-rays is proved on the basis of their interference and diffraction and many facts related to radiations can only be explained when the beam of light rays is regarded as composed of energy corpuscles or photons whose velocity is 3×10^{10} cm/s.
- According to de-Broglie, the wavelength λ of an electron is inversely proportional to its momentum p .

$$\lambda \propto \frac{1}{p} \quad \text{or} \quad \lambda = \frac{h}{p} \quad (\text{Here } h = \text{Planck's constant, } p = \text{momentum of electron})$$

$$\therefore \text{Momentum } (p) = \text{Mass } (m) \times \text{Velocity } (v) \quad \therefore \lambda = \frac{h}{mv}$$

- The above relation can be proved as follows by using Einstein's equation, Planck's quantum theory and wave theory of light.

Einstein's equation, $E = mc^2$ where E is energy, m is mass of a body and c is its velocity.

$$\therefore E = hv = h \times \frac{c}{\lambda} \quad (\text{According to Planck's quantum theory}) \quad \dots(i)$$

$$\text{and } c = v\lambda \quad (\text{According to wave theory of light}) \quad \therefore v = \frac{c}{\lambda}$$

But according to Einstein's equation $E = mc^2 \dots(ii)$

$$\text{From equation (i) \& (ii) : } mc^2 = h \times \frac{c}{\lambda} \quad \text{or} \quad mc = \frac{h}{\lambda} \quad \text{or} \quad p = \frac{h}{\lambda} \quad \text{or} \quad \boxed{\lambda = \frac{h}{p}}$$

- It is clear from the above equation that the value of λ decreases on increasing either m or v or both. The wavelength of many fast-moving objects like an aeroplane or a cricket ball, is very low because of their high mass.

Bohr's theory and de-broglie concept :

- (1) According to de-Broglie, the nature of an electron moving around the nucleus is like a wave that flows in circular orbits around the nucleus.
- (2) If an electron is regarded as a wave, the quantum condition as given by Bohr in his theory is readily fulfilled.
- (3) If the radius of a circular orbit is r , then its circumference will be $2\pi r$.
- (4) We know that according to Bohr theory, $mvr = \frac{nh}{2\pi}$

$$\text{or } 2\pi r = \frac{nh}{mv} \quad (\because mv = p \text{ momentum})$$

$$\text{or } 2\pi r = \frac{nh}{p} \quad \left(\because \frac{h}{p} = \lambda \text{ de-Broglie equation} \right)$$

$$\therefore 2\pi r = n\lambda \quad (\text{where } n = \text{total number of waves } 1, 2, 3, 4, 5, \dots, \infty \text{ and } \lambda = \text{Wavelength})$$

$$(5) \quad \therefore 2\pi r = \frac{nh}{mv} \quad \text{or} \quad mvr = \frac{nh}{2\pi} \quad \therefore mvr = \text{Angular momentum}$$

Thus $mvr = \text{Angular momentum}$, which is a integral multiple of $\frac{h}{2\pi}$.

- (6) It is clear from the above description that according to de-Broglie there is similarity between wave theory and Bohr theory.



figure : Similarity between de-Broglie waves and Bohr's orbit

(B) HEISENBERG UNCERTAINTY PRINCIPLE

Bohr's theory considers an electron as a material particle. Its position and momentum can be determined with accuracy. But, when an electron is considered in the form of wave as suggested by de-Broglie, it is not possible to ascertain simultaneously the exact position and velocity of the electron more precisely at a given instant since the wave extends throughout a region of space.

In 1927, Werner Heisenberg presented a principle known as Heisenberg uncertainty principle which states that : "It is impossible to measure simultaneously the exact position and exact momentum of a body as small as an electron."

The uncertainty in measurement of position, (Δx) , and the uncertainty in momentum (Δp) are related by Heisenberg's relationship as

$$\Delta E \times \Delta t \geq \frac{h}{4\pi} \quad \text{or} \quad \Delta E \times \Delta t \geq \frac{h}{4\pi}$$

where h is Planck's constant.

- (i) When $\Delta x = 0$, $\Delta v = \infty$
- (ii) When $\Delta v = 0$, $\Delta x = \infty$ So, if the position is known quite accurately, i.e., Δx is very small, Δv becomes large and vice-versa.

GOLDEN KEY POINTS

- **de-Broglie wavelength in terms of kinetic energy.**

$$\text{Kinetic Energy (K.E.)} = \frac{1}{2}mv^2 \text{ or } m \times \text{K.E.} = \frac{1}{2}m^2v^2 \text{ or } m^2v^2 = 2m \text{ K.E. or } mv = \sqrt{2m \text{ K.E.}}$$

$$\text{But } \lambda = \frac{h}{mv} \quad \therefore \quad \lambda = \frac{h}{\sqrt{2m \text{ K.E.}}} \quad (\because mv = \sqrt{2m \text{ K.E.}})$$

- **When a charged particle carrying Q coulomb charge is accelerated by applying potential difference of V volts, then :-**

$$\text{K.E.} = Q \times V \text{ Joule}$$

$$\text{But } \lambda = \frac{h}{\sqrt{2m \text{ K.E.}}} \quad \therefore \quad \lambda = \frac{h}{\sqrt{2m QV}}$$

$$\text{For electron } \left(\lambda = \sqrt{\frac{150}{V}} \text{ \AA} \right) = \frac{12.25}{\sqrt{V}} \text{ \AA}$$

- The wave nature of electron was verified experimentally by Davisson and Germer.
- de-Broglie hypothesis is applicable to macroscopic as well as microscopic objects but it has no physical significance for macroscopic objects.

- Remember $\frac{h}{4\pi} = 0.527 \times 10^{-34} \text{ J sec}$

Illustrations

Illustration 34. The mass of a particle is 1 mg and its velocity is $4.5 \times 10^5 \text{ cm per second}$. What should be the wavelength of this particle if $h = 6.652 \times 10^{-27} \text{ erg second}$.

(1) $1.4722 \times 10^{-24} \text{ cm}$ (2) $1.4722 \times 10^{-29} \text{ cm}$ (3) $1.4722 \times 10^{-32} \text{ cm}$ (4) $1.4722 \times 10^{-34} \text{ cm}$

Solution

Given that $m = 1 \text{ mg} = 1 \times 10^{-3} \text{ g}$, $v = 4.5 \times 10^5 \text{ cm s}^{-1}$, $h = 6.652 \times 10^{-27} \text{ erg s}$.

$$\therefore \lambda = \frac{h}{mv} = \frac{6.625 \times 10^{-27} \text{ erg s}}{1 \times 10^{-3} \text{ g} \times 4.5 \times 10^5 \text{ cm s}^{-1}} = 1.4722 \times 10^{-29} \text{ cm}$$

Illustration 35. Which of the following should be the wavelength of an electron if its mass is $9.1 \times 10^{-31} \text{ kg}$ and its velocity is $1/10$ of that of light and the value of h is $6.6252 \times 10^{-34} \text{ joule second}$?

(1) $2.446 \times 10^{-7} \text{ metre}$ (2) $2.246 \times 10^{-9} \text{ metre}$ (3) $2.246 \times 10^{-11} \text{ metre}$ (4) $2.246 \times 10^{-13} \text{ metre}$

Solution

Given that $m = 9.1 \times 10^{-31} \text{ kg}$, $v = \frac{1}{10}$ of velocity of light

or $v = \frac{1}{10} \times 3 \times 10^8 \text{ metre second}^{-1} \text{ i.e. } 3 \times 10^7 \text{ metre second}^{-1}$

$$h = 6.6252 \times 10^{-34} \text{ joule second}$$

$$\lambda = \frac{h}{mc} = \frac{6.6252 \times 10^{-34} \text{ J.s}}{9.1 \times 10^{-31} \text{ kg} \times 3 \times 10^7 \text{ ms}^{-1}} = \frac{6.6252 \times 10^{-34}}{27.3 \times 10^{-24}}$$

$$= 0.2426 \times 10^{-10} \text{ metre} = 2.426 \times 10^{-11} \text{ metre}$$

Illustration 36. A ball weighing 25 g moves with a velocity of $6.6 \times 10^4 \text{ cm s}^{-1}$ then find out the de-Broglie λ associated with it.

Solution

$$\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \times 10^7 \text{ erg s}}{25 \times 6.6 \times 10^4 \text{ cm s}^{-1}} = 0.04 \times 10^{-31} \text{ cm} = 4 \times 10^{-33} \text{ cm}$$

- Greater the value of n , greater is the energy of shell

$$E = -13.6 \times \frac{Z^2}{n^2} \text{ eV/atom}$$

$$E_1 < E_2 < E_3 < E_4 \text{ -----}$$
 - Velocity of electron $v = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}$

$$v_1 > v_2 > v_3 \text{}$$
 - The angular momentum of a revolving electron is $mvr = \frac{nh}{2\pi}$
 Where n = Principal quantum number.
 - The number of electrons in a particular shell is equal to $2n^2$
- (b) **Azimuthal quantum number / Angular quantum number / Secondary quantum number / Subsidiary quantum number (ℓ)**
 Given by – Sommerfeld
- It represents the name of the subshell, shape of orbital and orbital angular momentum
 - Possible values of ' ℓ ' are :-
 i.e. $\ell = 0, 1, 2 \text{ ----- } (n-1)$
 $\ell = 0$ (s Subshell)
 $\ell = 1$ (p Subshell)
 $\ell = 2$ (d Subshell)
 $\ell = 3$ (f Subshell)
 - Value of ℓ lies between 0 to $(n-1)$ in a particular n^{th} shell :-
Ex. If $n = 1$ then $\ell = 0 \Rightarrow 1s$ i.e. in $n = 1$ shell, only one subshell 's' is present.
 If $n = 2$ then $\ell = 0, 1 \Rightarrow 2s, 2p$ i.e. in $n = 2$ shell, two subshell 's' & 'p' are present.
 If $n = 3$ then $\ell = 0, 1, 2 \Rightarrow 3s, 3p, 3d$ i.e. in $n = 3$ shell, three subshell 's', 'p' & 'd' are present.
 If $n = 4$ then $\ell = 0, 1, 2, 3 \Rightarrow 4s, 4p, 4d, 4f$ i.e. in $n = 4$ shell, four subshell 's', 'p', 'd' & 'f' are present.
 - If the value of n is same then the order of energy of the various subshell will be $s < p < d < f$ [valid only for multi-electron species]
Ex. $4s < 4p < 4d < 4f$, $3s < 3p < 3d$, $2s < 2p$
 - If Value of ℓ is same but value of n is different then the order of energy will be.
Ex. $1s < 2s < 3s < 4s < 5s < 6s$
 $3d < 4d < 5d < 6d$
 $4p < 5p < 6p$
 - The orbital angular momentum = $\sqrt{\ell(\ell+1)} \frac{h}{2\pi}$ or $\sqrt{\ell(\ell+1)} \hbar$ $\left\{ \because \hbar = \frac{h}{2\pi} \right\}$ { $\hbar$ is called as 'hash' }
 Orbital angular momentum : For s subshell = 0
 For p subshell = $\sqrt{2} \frac{h}{2\pi}$ or $\sqrt{2} \hbar$
 - The number of electrons in a particular subshell is equal to $2(2\ell + 1)$
 for s subshell number of electrons = $2 e^-$
 for p subshell number of electrons = $6 e^-$
 for d subshell number of electrons = $10 e^-$
 for f subshell number of electrons = $14 e^-$
 - Shape of the orbital :

s	→	spherical
p	→	dumb bell shape
d	→	double dumb bell shape
f	→	complex shape

BEGINNER'S BOX-5

1. de-Broglie wavelength is related to applied voltage as :-

(1) $\lambda = \frac{12.3}{\sqrt{h}} \text{ \AA}$ (2) $\lambda = \frac{12.3}{\sqrt{V}} \text{ \AA}$ (3) $\lambda = \frac{12.3}{\sqrt{r}} \text{ \AA}$ (4) $\lambda = \frac{12.3}{\sqrt{m}} \text{ \AA}$

2. Select the incorrect statements among the following.

(1) $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$ (2) $\Delta x \cdot \Delta p \geq \frac{h}{4\pi m}$ (3) $\Delta x \cdot \Delta V \geq \frac{h}{4\pi m}$ (4) $\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$

3. If the kinetic energy of an electron is increased 4 times, the wavelength of the de-Broglie wave associated with it would become :-

- (1) four times (2) two times (3) half times (4) one fourth times

4. Velocity of de-Broglie wave is given by :-

(1) $\frac{c^2}{v}$ (2) $\frac{h\nu}{mc}$ (3) $\frac{mc^2}{h}$ (4) $v\lambda$

5. The representation of an orbital with $n = 4$ and $\ell = 1$:-

- (1) 4d (2) 4s (3) 4f (4) 4p

6. Maximum number of electrons present in M shell is :

- (1) 8 (2) 18 (3) 32 (4) 10

(c) **Magnetic Quantum Number / Orientation Quantum Number (m) :**

Given by Linde

- It represents the orientation of electron cloud (orbital)
- Under the influence of magnetic field each subshell is further subdivided into orbitals (The electron cloud is known as orbital)
Magnetic quantum number describes these different distributions of electron cloud.
- Value of m = all integral value from $-\ell$ to $+\ell$ including zero.
i.e. Value of $m = -\ell$ to $+\ell$

Orbital : 3D space around the nucleus where the probability of finding electrons is maximum is called an orbital. An orbital can be represented by 3 set of quantum numbers

$= \Psi_{n,\ell,m}$

Ex. 1 : $2p_x$; $n=2$, $\ell=1$, $m=-1$ or $m=+1$

Ex. 2 : $3d_{z^2}$; $n=3$, $\ell=2$, $m=0$

Ex. 3 : $\Psi_{(3,2,0)}$; $n=3$, $\ell=2$, $m=0$; $3d_{z^2}$

Node : It is point / line / plane / surface in which probability of finding electron is zero.

Total numbers of nodes = $n-1$

They are of 2 types.

(i) Radial nodes / Spherical nodes / Nodal surface number of radial nodes = $n-\ell-1$

(ii) Angular nodes / Nodal planes number of angular nodes / nodal planes = ℓ

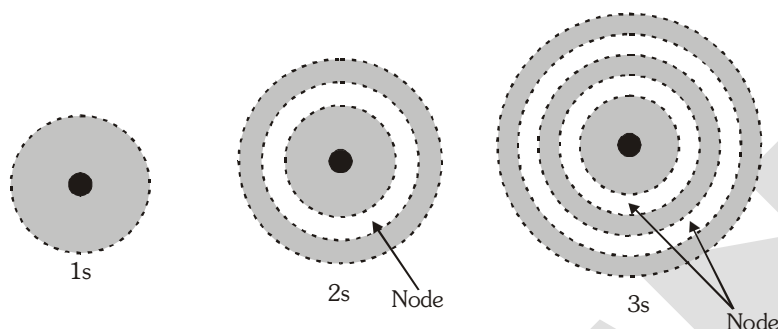
* Nucleus and ∞ (infinite) are not considered as node.

Types of orbitals :

Case-I : If $\ell = 0$ then $m = 0$, it implies that s subshell has only one orbital called as s orbital.

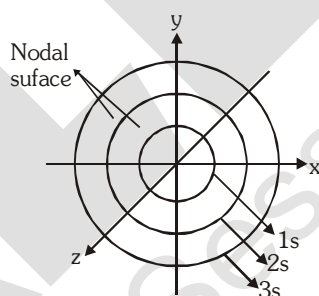
Shapes of s-orbitals :

The s-orbitals are spherically symmetrical about the nucleus, i.e., the probability of finding electron is same in all directions from the nucleus. The size of the orbital depends on the value of principal quantum number. The 1s orbital is smaller than 2s-orbital and 2s-orbital is smaller than 3s, but all are spherical in shape as shown in figure.



Although the s-orbitals belonging to different shells are spherically symmetrical, yet they differ in certain respects as explained below :

- The probability of finding 1s electron is found to be maximum near the nucleus and decreases as the distance from the nucleus increases. In case of 2s electrons, the probability is again maximum near the nucleus and then decreases to zero as the distance from the nucleus increases. The intermediate region (a spherical shell) where the probability is zero is called a nodal surface or simply node. Thus, 2s-orbital differs from 1s-orbital in having one node within it. Similarly, 3s has two nodes. In general, any ns orbital has $(n - 1)$ nodes.
- The size and energy of the s-orbital increases as the principal quantum number increases, i.e., the size and energy of s-orbital increases in the order $1s < 2s < 3s \dots$



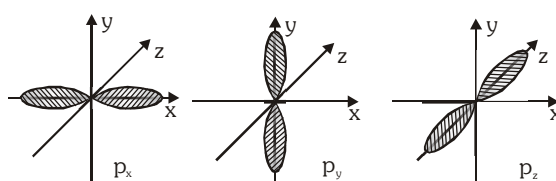
The s orbital of higher energy levels are also symmetrically spherical and can be represented as above

Case-II If $\ell = 1$ (p - subshell) then $m = \begin{matrix} -1 & 0 & +1 \\ p_x & p_z & p_y \end{matrix}$

It implies that, p subshell have three orbitals called as p_x , p_y and p_z .

Shape of p-orbitals :

There are three p-orbitals, commonly referred to as p_x , p_y and p_z . These three p-orbitals, possess equivalent energy and therefore, have same relation with the nucleus. They, however, differ in their direction & distribution of the charge.



These three p-orbitals are situated at right angle to one another and are directed along x, y and z axis (figure)

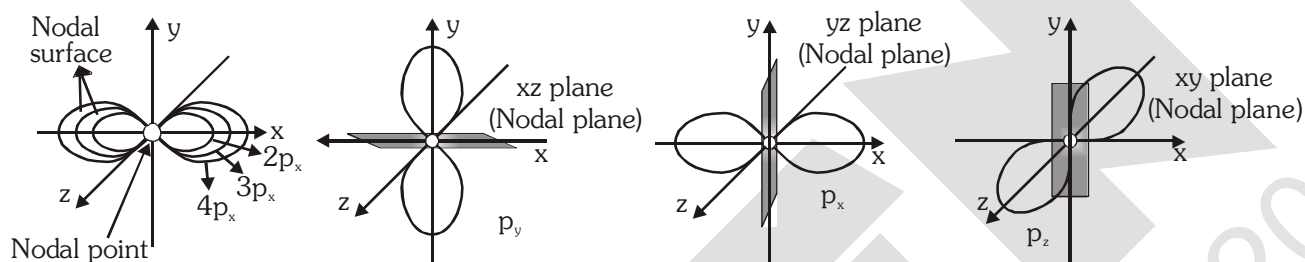
- Each p orbital has dumb bell shape (2 lobes which are separated from each other by a point of zero probability called nodal point or node or nucleus).
- The two lobes of each orbital are separated by a plane of zero electron density called nodal plane.
- Each p orbital of higher energy level are also dumb bell shape but they have nodal surface.

Nodal surface :

Orbital	Nodal surface
3 p _x	1
4 p _x	2
np _x	(n - 2)

Nodal Plane :

Orbital	Nodal plane
p_x	yz plane
p_y	xz plane
p_z	xy plane

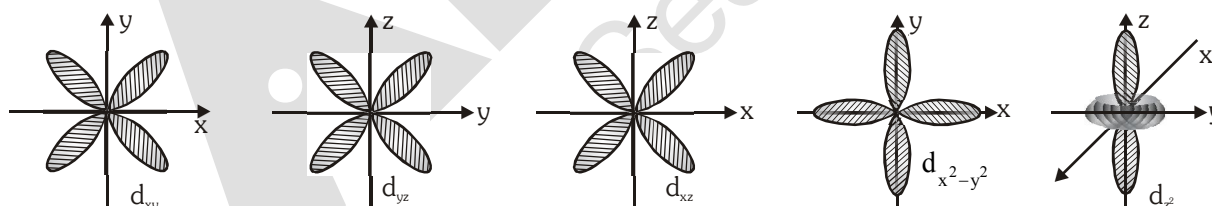


Case III When $\ell = 2$, m' has five values $-2, -1, 0, +1, +2$. It implies that d subshell of any energy shell has five orbitals. All the five orbitals are not identical in shape. Four of the d orbitals $d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}$ contain four lobes while fifth orbital d_{z^2} consists of only two lobes. The lobes of d_{xy} orbital lie between x and y axes. Similar is the case for d_{yz} and d_{xz} . Four lobes of $d_{x^2-y^2}$ orbital are lying along x and y axes while the two lobes of d_{z^2} orbital are lying along z axis and contain a ring of negative charge surrounding the nucleus in xy plane. Geometry of d orbital is double dumbbell

$$m = \begin{array}{ccccc} \begin{array}{|c|} \hline -2 \\ \hline \end{array} & \begin{array}{|c|} \hline -1 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline +1 \\ \hline \end{array} & \begin{array}{|c|} \hline +2 \\ \hline \end{array} \\ \begin{array}{|c|} \hline d_{xy} \\ \hline \end{array} & \begin{array}{|c|} \hline d_{yz} \\ \hline \end{array} & \begin{array}{|c|} \hline d_{z^2} \\ \hline \end{array} & \begin{array}{|c|} \hline d_{zx} \\ \hline \end{array} & \begin{array}{|c|} \hline d_{x^2-y^2} \\ \hline \end{array} \end{array}$$

Shape of d-orbitals :

It implies that d subshell has 5 orbitals i.e. five electron cloud and can be represented as follows

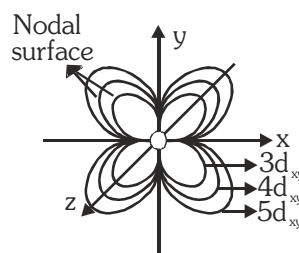


Each d-orbital of higher energy level also have double dumbbell shape but they have nodal surface.

In d-orbitals :

- (i) Nodal Point $\rightarrow 1$
- (ii) Nodal Surface $\rightarrow$
- | | |
|------------|-------------------------------|
| $3 d_{xy}$ | $\rightarrow 0$ Nodal surface |
| $4 d_{xy}$ | $\rightarrow 1$ Nodal surface |
| $5 d_{xy}$ | $\rightarrow 2$ Nodal surface |
| $n d_{xy}$ | $\rightarrow (n - 3)$ |

Number of nodal surface = $n - \ell - 1$



- (iii) Nodal plane : $d_{xy} \rightarrow xz$ & yz nodal plane :
 $d_{xz} \rightarrow xy$ & zy nodal plane :
 $d_{zy} \rightarrow zx$ & yx nodal plane :
 $d_{x^2-y^2} \rightarrow 2$, nodal plane :
 $d_{z^2} \rightarrow 0$, nodal plane :

Note: Orbitals of d subshell are equivalent in energy.

(d) Spin Quantum number (s) :

Given by **Goudsmit** and **Uhlenbeck**

- It represents the direction of electron spin around its own axis
- For clockwise spin/spin up($\uparrow$) electron $\rightarrow \pm \frac{1}{2}$
- For anti-clockwise spin/spin down($\downarrow$) electron $\rightarrow \mp \frac{1}{2}$

Spin angular momentum of an electron = $\sqrt{s(s+1)} \cdot \frac{h}{2\pi}$ or $\sqrt{s(s+1)} \hbar$

- Each orbital can accommodate 2 electrons with opposite spin or spin paired.

Correct $\boxed{\uparrow\downarrow}$ Spin paired e^-

Wrong $\boxed{\uparrow\uparrow}$ Spin parallel e^-

Illustrations

Illustration 41. Calculate the value of n , ℓ and m for $7p_y$ orbital ?

Solution $n = 7, \ell = 1, m = +1$ or -1

Illustration 42. Calculate the value of n , ℓ and m for $3s$ orbital ?

Solution $n = 3, \ell = 0, m = 0$

Illustration 43. Calculate the value of n , ℓ and m for $5d_{z^2}$ orbital ?

Solution $n = 5, \ell = 2, m = 0$

Illustration 44. Which of the following set of quantum numbers is not possible ?

(a) $n = 2, \ell = 0, m = -1, s = -\frac{1}{2}$

(b) $n = 3, \ell = 2, m = 0, s = \pm \frac{1}{2}$

(c) $n = 2, \ell = 3, m = -2, s = \pm \frac{1}{2}$

Solution (a) not possible (b) possible (c) not possible

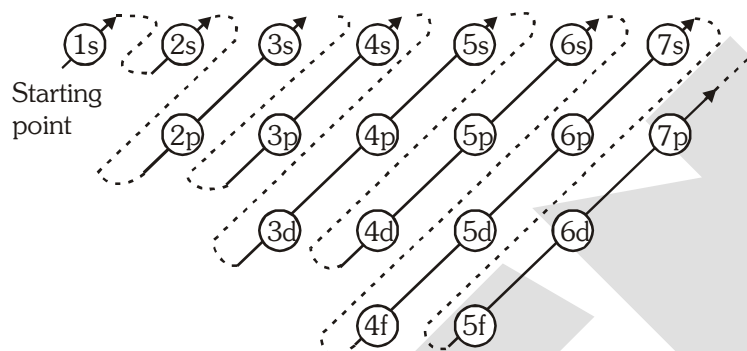
2.11 RULES FOR FILLING OF ELECTRONS

- (a) Aufbau Principle (b) $(n + \ell)$ rule
 (c) Hund's maximum multiplicity principle (d) Pauli's exclusion principle

(a) Aufbau Principle

Aufbau is a German word and its meaning is 'Building up'

- Aufbau principle gives a sequence in which various subshell are filled up depending on the relative order of the energies of various subshell.
- Principle : The subshell with minimum energy is filled up first when this subshell obtained maximum quota of electrons then the next subshell of higher energy starts filling.
- The sequence in which various subshell are filled are as follows.



$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^2, 4d^{10}, 5p^6, 6s^2, 4f^{14}, 5d^{10}, 6p^6, 7s^2, 5f^{14}, 6d^{10}, \dots$

For Example

${}^1\text{H}$	$\rightarrow$	$1s^1$
${}^2\text{He}$	$\rightarrow$	$1s^2$
${}^3\text{Li}$	$\rightarrow$	$1s^2, 2s^1$
${}^4\text{Be}$	$\rightarrow$	$1s^2, 2s^2$
${}^5\text{B}$	$\rightarrow$	$1s^2, 2s^2, 2p^1$
${}^6\text{C}$	$\rightarrow$	$1s^2, 2s^2, 2p^2$
${}^7\text{N}$	$\rightarrow$	$1s^2, 2s^2, 2p^3$
${}^8\text{O}$	$\rightarrow$	$1s^2, 2s^2, 2p^4$
${}^9\text{F}$	$\rightarrow$	$1s^2, 2s^2, 2p^5$
${}^{10}\text{Ne}$	$\rightarrow$	$1s^2, 2s^2, 2p^6$
${}^{11}\text{Na}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^1$
${}^{12}\text{Mg}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2$
${}^{13}\text{Al}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^1$
${}^{14}\text{Si}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^2$
${}^{15}\text{P}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^3$
${}^{16}\text{S}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^4$
${}^{17}\text{Cl}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^5$
${}^{18}\text{Ar}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6$
${}^{19}\text{K}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$
${}^{20}\text{Ca}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2$
${}^{21}\text{Sc}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^1$
${}^{22}\text{Ti}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^2$
${}^{23}\text{V}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^3$
${}^{24}\text{Cr}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^5$ [Exception]
${}^{25}\text{Mn}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^5$
${}^{26}\text{Fe}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^6$
${}^{27}\text{Co}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^7$
${}^{28}\text{Ni}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^8$
${}^{29}\text{Cu}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^{10}$ [Exception]
${}^{30}\text{Zn}$	$\rightarrow$	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}$

Electronic configuration can be written by following different methods :

- $_{26}\text{Fe} \rightarrow$
 - $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^6$
 - $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^6, 4s^2$
 - $1s^2, 2s^2p^6, 3s^2p^6d^6, 4s^2$
 - $$\begin{array}{cccc} & 2 & 8 & 14 & 2 \\ (4) & [\text{Ar}] & 4s^2 & 3d^6 & \end{array}$$

- $_{26}\text{Fe} \rightarrow$

$$1s^2 \underbrace{2s^2 2p^6}_{(n-2)} \underbrace{3s^2 3p^6 3d^6}_{(n-1)} \underbrace{4s^2}_n$$

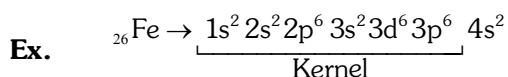
$n \rightarrow$ Outer most Shell or Ultimate Shell or Valence Shell

In this Shell electrons are called as Valence electrons or this is called core charge

$(n-1) \rightarrow$ Penultimate Shell or core or pre valence Shell

$(n-2) \rightarrow$ Pre Penultimate Shell

- If we remove the last 'n' Shell (ultimate Shell) then the remaining shells are collectively called as Kernel.



(b) $(n + \ell)$ Rule (For multi electron species)

According to it the sequence in which various subshell are filled up can also be determined with the help of $(n + \ell)$ value for a given subshell.

Principle of $(n + \ell)$ rule :

The subshell with lowest $(n + \ell)$ value is filled up first. When two or more subshell have same $(n + \ell)$ value then the subshell with lowest value of n is filled up first.

In case of H-atom :

Energy only depends on principal quantum number

$$1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f < \dots\dots\dots$$

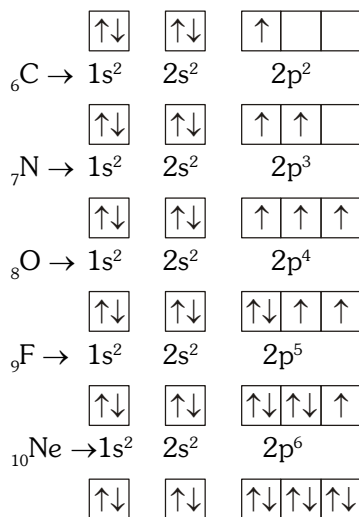
Sub Shell	n	ℓ	$n + \ell$	
1s	1	0	1	
2s	2	0	2	
2p	2	1	3	(1)
3s	3	0	3	(2)
3p	3	1	4	(1)
4s	4	0	4	(2)
3d	3	2	5	(1)
4p	4	1	5	(2)
5s	5	0	5	(3)
4d	4	2	6	(1)
5p	5	1	6	(2)
6s	6	0	6	(3)

Order : $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^2, 4d^{10}, 5p^6, 6s^2, 4f^{14}, 5d^{10}, 6p^6, 7s^2, 5f^{14}, 6d^{10}, \dots$

(c) Hund's Maximum Multiplicity Rule (Multiplicity : Many of the same kind)

- This rule deals with the filling of electrons into the orbitals belonging to the same subshell (that is, orbitals of equal energy, called degenerate orbitals).
- It states : pairing of electrons in the orbitals belonging to the same subshell (p, d or f) does not take place until each orbital belonging to that subshell has got one electron each i.e., it is singly occupied.
- Since there are three p, five d and seven f orbitals, therefore, the pairing of electrons will start in the p, d and f orbitals with the entry of 4th, 6th and 8th electron, respectively.

Ex. $B \rightarrow 1s^2 \quad 2s^2 \quad 2p^1$



(d) Pauli's Exclusion Principle

In 1925 Pauli stated that no two electron in an atom can have same values of all four quantum numbers i.e., an orbital can accomodate maximum 2 electrons with opposite spin.

Ex.1. $^{12}_6\text{C} \rightarrow$

	$1s^2$	$2s^2$	$2p^2$
	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow}\boxed{\uparrow}\boxed{}$
n	= 1	2	2
ℓ	= 0	0	1
m	= 0	0	+1, 0, -1
s	= $+\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, +\frac{1}{2}$

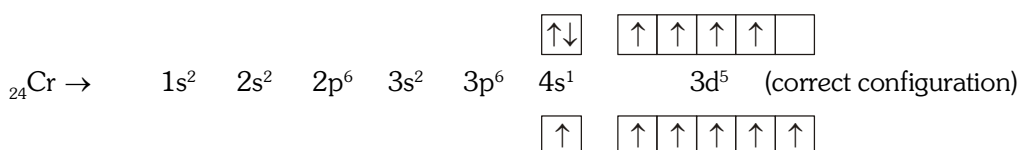
Ex.2 ${}_{17}\text{Cl} \rightarrow$

	$1s^2$	$2s^2$	$2p^6$	$3s^2$	$3p^5$
	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow\downarrow}\boxed{\uparrow\downarrow}\boxed{\uparrow}$
n =	1	2	2	3	3
ℓ =	0	0	1	0	1
m =	0	0	+1, -1, 0	0	+1, -1, 0
s =	$+\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}$

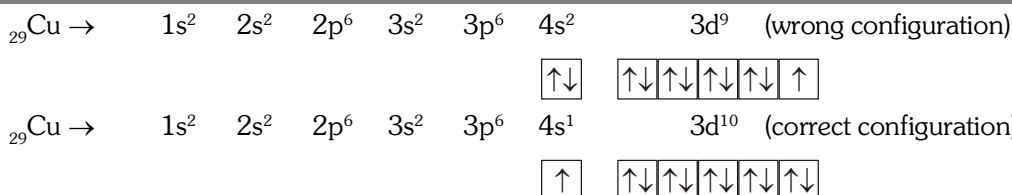
Exception of Aufbau principle :

In some cases it is seen that the electronic configuration is slightly different from the arrangement given by Aufbau principle. A simple reason behind this is that half filled & full filled subshell have got extra stability.

Ex.1 $_{24}\text{Cr} \rightarrow 1s^2 \ 2s^2 \ 2p^6 \ 3s^2 \ 3p^6 \ 4s^2 \ 3d^4$ (wrong configuration)



Ex.2



Illustrations

Illustration 45. Calculate the number of unpaired electrons in Cr

Solution $_{24}\text{Cr} \rightarrow 1s^2 \ 2s^2 \ 2p^6 \ 3s^2 \ 3p^6 \ 4s^1 \ 3d^5$
 in $_{24}\text{Cr}$, 6 electrons are unpaired.

Illustration 46. The number of unpaired electrons in Cr^{+3}

Solution $\text{Cr}^{+3} \rightarrow 1s^2 \ 2s^2 \ 2p^6 \ 3s^2 \ 3p^6 \ 4s^0 \ 3d^3$
 in Cr^{+3} , 3 electrons are unpaired.

Illustration 47. The number of unpaired electrons in 3d subshell of Cr^{+3}

Solution 3

Illustration 48. The number of unpaired electrons in Fe^{+2} & Fe^{+3}

Solution $\text{Fe}^{+2} \rightarrow 1s^2 \ 2s^2 \ 2p^6 \ 3s^2 \ 3p^6 \ 4s^0 \ 3d^6 = 4$ unpaired electrons
 $\text{Fe}^{+3} \rightarrow 1s^2 \ 2s^2 \ 2p^6 \ 3s^2 \ 3p^6 \ 4s^0 \ 3d^5 = 5$ unpaired electrons

BEGINNER'S BOX-6

- A neutral atom of an element has 2K, 8L, 11M and 2N electrons. The number of p-electrons in the atom are :-
 (1) 2 (2) 12 (3) 10 (4) 6
- An atom has 2 electrons in K-shell, 8 electrons in L-shell & 8 electrons in M-shell. The number of p-electrons presents in the element is :-
 (1) 10 (2) 7 (3) 12 (4) 4
- The maximum number of such electrons in an atom with quantum number $n = 3$, $\ell = 2$ is :-
 (1) 2 (2) 6 (3) 10 (4) 14
- The number of orbitals in $n=3$ are :-
 (1) 1 (2) 4 (3) 9 (4) 16
- In potassium the probable order of energy level for 19th electron is :-
 (1) $3s > 3d$ (2) $4s < 3d$ (3) $4s > 4p$ (4) $4s = 3d$

ANSWER KEY

BEGINNER'S BOX-1		Que.	1	2	3	4	5	6	7			
		Ans.	3	3	2	4	4	4	2			
BEGINNER'S BOX-2		Que.	1	2	3	4						
		Ans.	3	1	2	1						
BEGINNER'S BOX-3		Que.	1	2	3	4	5	6				
		Ans.	3	4	3	2	2	3				
BEGINNER'S BOX-4		Que.	1	2	3	4						
		Ans.	3	3	2	2						
BEGINNER'S BOX-5		Que.	1	2	3	4	5	6				
		Ans.	2	2	3	2	4	2				
BEGINNER'S BOX-6		Que.	1	2	3	4	5					
		Ans.	2	3	3	3	2					

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

INTRODUCTION

- Rutherford's α -particle scattering experiment proved that atom has :-
(1) Electrons (2) Neutrons
(3) Nucleus (4) Orbitals
- A and B are two elements which have same atomic weight and are having atomic number 27 and 30 respectively. If the atomic weight of A is 57 then number of neutron in B is :-
(1) 27 (2) 33 (3) 30 (4) 40
- Find out the atoms which are isoneutronic :-
(1) $^{14}_6\text{C}$, $^{15}_7\text{N}$, $^{17}_9\text{F}$ (2) $^{12}_6\text{C}$, $^{14}_7\text{N}$, $^{19}_9\text{F}$
(3) $^{14}_6\text{C}$, $^{14}_7\text{N}$, $^{17}_9\text{F}$ (4) $^{14}_6\text{C}$, $^{14}_7\text{N}$, $^{19}_9\text{F}$
- Species which are isoelectronic to one another are
(a) CN^- (b) OH^-
(c) CH_3^+ (d) N_2
(e) CO
Correct answer is :-
(1) a, b, c (2) a, c, d
(3) a, d, e (4) b, c, d
- For any anion X^{-3} , the mass number is 14. If anion has 10 electrons, then number of neutrons in X_2 nucleus :-
(1) 10 (2) 14
(3) 7 (4) 5
- Which of the following pairs is correctly matched :
(1) Isotopes $^{40}_{20}\text{Ca}$, $^{40}_{19}\text{K}$
(2) Isotones $^{30}_{14}\text{Si}$, $^{31}_{15}\text{P}$, $^{32}_{16}\text{S}$
(3) Isobars $^{16}_8\text{O}$, $^{17}_8\text{O}$, $^{18}_8\text{O}$
(4) Isoelectronic N^{-3} , O^{-2} , Cr^{+3}
- The atom A, B, C have the configuration
 $\text{A} \rightarrow [\text{Z}(90) + n(146)]$, $\text{B} \rightarrow [\text{Z}(92) + n(146)]$,
 $\text{C} \rightarrow [\text{Z}(90) + n(148)]$ So that :-
(a) A and C - Isotones (b) A and C - Isotopes
(c) A and B - Isobars (d) B and C - Isobars
(e) B and C - Isotopes
The wrong statements are:-
(1) a, b only (2) c, d, e only
(3) a, c, d only (4) a, c, e only

- (i) $^{54}_{26}\text{Fe}$, $^{56}_{26}\text{Fe}$, $^{57}_{26}\text{Fe}$, $^{58}_{26}\text{Fe}$ (a) Isotopes
(ii) ^3_1H , ^3_2He (b) Isotones
(iii) $^{76}_{32}\text{Ge}$, $^{77}_{33}\text{As}$ (c) Isodiaphers
(iv) $^{235}_{92}\text{U}$, $^{231}_{90}\text{Th}$ (d) Isobars
(v) ^1_1H , ^2_1D , ^3_1T

Match the above correct terms:-

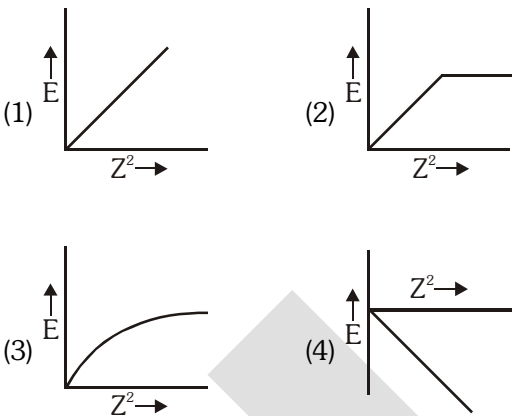
- (1) [(i), - a], [(ii) - d], [(iii) - b], [(iv) - c], [(v) - a]
(2) [(i) - a] [(ii) - d], [(iii) - d] [(iv) - c] [v - a]
(3) [v - a] [(iv) - c]. [(iii) - d] [(ii) - b] [(i) - a]
(4) None of them
- Choose the false statement about deuterium :-
(1) It is an isotope of hydrogen
(2) It contains [(1 e) + (1 p) + (1 n)]
(3) It contains only [(1 p) + (1 n)]
(4) D_2O is called as heavy water
- If the table of atomic masses is established with the oxygen atom and assigned value of 200, then the mass of carbon atom would be, approximately:-
(1) 24 (2) 150 (3) 50 (4) 112
- The relative abundance of two rubidium isotopes of atomic weights 85 and 87 are 75% and 25% respectively. The average atomic weight of rubidium is:-
(1) 75.5 (2) 85.5 (3) 86.5 (4) 87.5
- The ratio of specific charge of a proton and an α -particle is :-
(1) 2 : 1 (2) 1 : 2 (3) 1 : 4 (4) 1 : 1
- In an atom $^{27}_{13}\text{Al}$, number of proton is (a), electron is (b) and neutron is (c). Hence ratio will be [in order c : b : a] :-
(1) 13 : 14 : 13 (2) 13 : 13 : 14
(3) 14 : 13 : 13 (4) 14 : 13 : 14
- Atomic weight of Ne is 20.2. Ne is mixture of ^{20}Ne and ^{22}Ne , relative abundance of heavier isotope is :-
(1) 90 (2) 20 (3) 40 (4) 10
- Number of protons, neutrons & electrons in the element $^{231}_{89}\text{Y}$ is :-
(1) 89, 231, 89 (2) 89, 89, 242
(3) 89, 142, 89 (4) 89, 71, 89

16. Atoms $^{13}_6\text{C}$ and $^{17}_8\text{O}$ are related to each other as:-
 (1) Isotones (2) Isoelectronic
 (3) Isodiaphers (4) Isosters
17. The e/m ratio is maximum for :-
 (1) D^+ (2) He^+ (3) H^+ (4) He^{2+}
18. Let mass of electron is half, mass of proton is two times and mass of neutron is three fourth of original masses, then new atomic weight of ^{16}O atom:-
 (1) increases by 37.5 %
 (2) remain constant
 (3) increases by 12.5 %
 (4) decreases by 25 %
19. An isotone of $^{76}_{32}\text{Ge}$ is :-
 (i) $^{77}_{32}\text{Ge}$ (ii) $^{77}_{33}\text{As}$
 (iii) $^{77}_{34}\text{Se}$ (iv) $^{78}_{34}\text{Se}$
 (1) (ii) & (iii) (2) (i) & (ii)
 (3) (ii) & (iv) (4) (ii) & (iii) & (iv)
20. In $^{14}_7\text{N}$ if mass attributed to electrons were doubled & the mass attributed to protons were halved, the atomic mass would become approximately :-
 (1) Halved
 (2) Doubled
 (3) Reduced by 25%
 (4) Remain same
21. The value of planck's constant is $6.63 \times 10^{-34}\text{Js}$. The velocity of light is $3.0 \times 10^8\text{ms}^{-1}$. Which value is closest to the wavelength in metres of a quantum of light with frequency of $8 \times 10^{15}\text{s}^{-1}$?
 (1) 3×10^{-7} (2) 2×10^{-25}
 (3) 5×10^{-18} (4) 3.75×10^{-8}
22. If change in energy (ΔE) = $3 \times 10^{-8}\text{J}$, $h = 6.64 \times 10^{-34}\text{J-s}$ and $c = 3 \times 10^8\text{m/s}$, then wavelength of the light is :-
 (1) $6.64 \times 10^3\text{\AA}$ (2) $6.64 \times 10^5\text{\AA}$
 (3) $6.64 \times 10^{-8}\text{\AA}$ (4) $6.64 \times 10^{18}\text{\AA}$

BOHR'S ATOMIC MODEL

23. Angular momentum in second Bohr orbit of H-atom is x. Then find out angular momentum in 1st excited state of Li^{+2} ion :
 (1) $3x$ (2) $9x$ (3) $\frac{x}{2}$ (4) x
24. Angular momentum for P-shell electron is :-
 (1) $\frac{3h}{\pi}$ (2) Zero (3) $\frac{\sqrt{2}h}{2\pi}$ (4) None

25. Multiplication of electron velocity and radius for a orbit in an atom is :-
 (1) Proportional to mass of electron
 (2) Proportional to square of mass of electron
 (3) Inversely proportional to mass of electron
 (4) Does not depend upon mass of electron
26. The radius of a shell for H-atom is $4.761\text{\AA}$. The value of n is :-
 (1) 3 (2) 9 (3) 5 (4) 4
27. In Bohr's atomic model radius of 1st orbit of Hydrogen is 0.053 nm then radius of 3rd orbit of Li^{+2} is :
 (1) 0.159 (2) 0.053
 (3) 0.023 (4) 0.026
28. The first three radius ratio of Bohr orbits :-
 (1) 1 : 0.5 : 0.5 (2) 1 : 2 : 3
 (3) 1 : 4 : 9 (4) 1 : 8 : 27
29. For Li^{+2} ion, $r_2 : r_5$ will be :-
 (1) 9 : 25 (2) 4 : 25 (3) 25 : 4 (4) 25 : 9
30. The ratio of the radii of two Bohr orbits of H-atom is 4 : 1, what would be their nomenclature :-
 (1) K & L (2) L & K
 (3) N & L (4) 2 & 3 both
31. The velocity of electron in third excited state of Be^{3+} ion will be :-
 (1) $\frac{3}{4}(2.188 \times 10^8)\text{ms}^{-1}$ (2) $\frac{3}{4}(2.188 \times 10^6)\text{ms}^{-1}$
 (3) $(2.188 \times 10^6)\text{Kms}^{-1}$ (4) $(2.188 \times 10^3)\text{Kms}^{-1}$
32. The Bohr orbit radius for the hydrogen atom ($n = 1$) is approximately $0.530\text{\AA}$. The radius for the first excited state ($n = 2$) will be :-
 (1) $0.13\text{\AA}$ (2) $1.0\text{\AA}$ (3) $4.77\text{\AA}$ (4) $2.12\text{\AA}$
33. According to Bohr theory, the radius (r) and velocity (v) of an electron vary with the increasing principal quantum number 'n' as :-
 (1) r increases, v decreases
 (2) r and v both increases
 (3) r & v both decreases
 (4) r decreases, v increases
34. The ratio of radius of first orbit in hydrogen to the radius of first orbit in deuterium will be :-
 (1) 1 : 1 (2) 1 : 2 (3) 2 : 1 (4) 4 : 1

35. For any H like system, the ratio of velocities of I, II & III orbit i.e., $V_1 : V_2 : V_3$ will be :-
 (1) 1 : 2 : 3 (2) 1 : 1/2 : 1/3
 (3) 3 : 2 : 1 (4) 1 : 1 : 1
36. The energy of H-atom in n^{th} orbit is E_n , then energy in n^{th} orbit of singly ionised helium atom will be :-
 (1) $4E_n$ (2) $E_n/4$
 (3) $2E_n$ (4) $E_n/2$
37. The energy of second Bohr orbit of the hydrogen atom is -328 kJ/mol . Hence the energy of fourth Bohr orbit should be :
 (1) -41 kJ/mol
 (2) -1312 kJ/mol
 (3) -164 kJ/mol
 (4) -82 kJ/mol
38. In a hydrogen atom, if energy of an electron in ground state is -13.6 eV , then energy in the 2^{nd} excited state is :-
 (1) -1.51 eV (2) -3.4 eV
 (3) -6.04 eV (4) -13.6 eV
39. The ratio between kinetic energy and the total energy of the electron of hydrogen atom according to Bohr's model is :-
 (1) 2 : 1 (2) 1 : 1
 (3) 1 : -1 (4) 1 : 2
40. Potential energy is -27.2 eV in second orbit of He^+ , then calculate double of total energy in first excited state of hydrogen atom :-
 (1) -13.6 eV
 (2) -54.4 eV
 (3) -6.8 eV
 (4) -27.2 eV
41. The energy levels for ${}_Z\text{A}^{(+Z-1)}$ can be given by :-
 (1) E_n for $\text{A}^{(+Z-1)} = Z^2 \times E_n$ for H
 (2) E_n for $\text{A}^{(+Z-1)} = Z \times E_n$ for H
 (3) E_n for $\text{A}^{(+Z-1)} = \frac{1}{Z^2} \times E_n$ for H
 (4) E_n for $\text{A}^{(+Z-1)} = \frac{1}{Z} \times E_n$ for H
42. The graphical representation of energy of electron and atomic number is :-

43. Going from K-shell to N-shell in case of H-atom :-
 (1) Kinetic energy decreases
 (2) Total energy decreases
 (3) Potential energy decreases
 (4) None of these
44. Maximum frequency of emission is obtained for the transition :-
 (1) $n = 2$ to $n = 1$ (2) $n = 6$ to $n = 2$
 (3) $n = 1$ to $n = 2$ (4) $n = 2$ to $n = 6$
45. If the ionization energy of hydrogen is $313.8 \text{ kcal per mole}$, then the energy of the electron in 2^{nd} excited state will be :-
 (1) -113.2 kcal/mole (2) -78.45 kcal/mole
 (3) -313.8 kcal/mole (4) -35 kcal/mole
46. Which of the following electron transition will require the largest amount of energy in a hydrogen atom :-
 (1) From $n = 1$ to $n = 2$ (2) From $n = 2$ to $n = 3$
 (3) From $n = \infty$ to $n = 1$ (4) From $n = 3$ to $n = 5$
47. If the potential energy (PE) of hydrogen electron is -3.02 eV then in which of the following excited level is electron present :-
 (1) 1^{st} (2) 2^{nd} (3) 3^{rd} (4) 4^{th}
48. The radiation of low frequency will be emitted in which transition of hydrogen atom :-
 (1) $n = 1$ to $n = 4$ (2) $n = 2$ to $n = 5$
 (3) $n = 3$ to $n = 1$ (4) $n = 5$ to $n = 2$
49. A single electron orbits a stationary nucleus ($z = 5$). The energy required to excite the electron from third to fourth Bohr orbit will be :-
 (1) 4.5 eV (2) 8.53 eV
 (3) 25 eV (4) 16.53 eV

50. The ratio of energies of hydrogen atom for first and second excited state is :-
 (1) $4/1$ (2) $1/4$ (3) $4/9$ (4) $9/4$
51. $E_n = -313.6/n^2$. If the value of $E_n = -34.84$ then to which of the following values does 'n' correspond :-
 (1) 1 (2) 2 (3) 3 (4) 4
52. The ratio of potential energy and total energy of an electron in a Bohr orbit of hydrogen like species is :-
 (1) 2 (2) -2 (3) 1 (4) -1
53. Which is not a correct order of energy for 1, 2nd & 3rd orbit :-
 (1) $E_1 > E_2 > E_3$
 (2) $(PE)_1 < (PE)_2 < (PE)_3$
 (3) $(KE)_1 > (KE)_2 > (KE)_3$
 (4) '1' & '3' both
54. Which of the following is a correct relationship :-
 (1) E_1 of H = $1/2 E_2$ of He^+ = $1/3 E_3$ of Li^{+2} = $1/4 E_4$ of Be^{+3}
 (2) E_1 (H) = E_2 (He^+) = E_3 (Li^{+2}) = E_4 (Be^{+3})
 (3) E_1 (H) = $2 E_2$ (He^+) = $3 E_3$ (Li^{+2}) = $4 E_4$ (Be^{+3})
 (4) No relation
55. Which is correct for any H like species :-
 (1) $(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3)$
 (2) $(E_2 - E_1) < (E_3 - E_2) < (E_4 - E_3)$
 (3) $(E_2 - E_1) = (E_3 - E_2) = (E_4 - E_3)$
 (4) $(E_2 - E_1) = 1/4 (E_3 - E_2) = 1/9 (E_4 - E_3)$
56. Which of the following is a correct graph :-
- (1)

(2)
- (3)

(4)
57. First excitation potential of H atom is :
 (1) 10.2 eV (2) 3.4 eV
 (3) 0 (4) - 3.4 eV
58. Energy required to remove an e^- from M shell of H-atom is 1.51 eV, then energy of first excited state will be :-
 (1) -1.51 eV (2) +1.51 eV
 (3) -3.4 eV (4) -13.6 eV

59. The ionisation potential of the hydrogen atom is 13.6 eV. The energy needed to ionise a hydrogen atom which is in its second excited state is about:-
 (1) 13.6 eV (2) 10.2 eV
 (3) 3.4 eV (4) 1.5 eV
60. The ionisation energy for excited hydrogen atom in eV will be :-
 (1) 13.6 (2) Less than 13.6
 (3) Greater than 13.6 (4) 3.4 or less
61. The energy required to excite an electron of H-atom from first orbit to second orbit is :-
 (1) $\frac{3}{4}$ of its ionisation energy
 (2) $\frac{1}{2}$ of its ionisation energy
 (3) $\frac{1}{4}$ of its ionisation energy
 (4) None
62. The ionisation potential of a singly ionised helium ion is equivalent to :-
 (1) Kinetic energy of first orbit
 (2) Energy of last orbit
 (3) Average energy in orbits
 (4) Maximum energy in orbits
63. The ionisation energy for the H-atom is 13.6 eV, then the required energy to excite it from the ground state to next higher state will be :- (in eV)
 (1) 3.4 (2) 10.2 (3) 12.1 (4) 1.5

SPECTRUM AND SPECTRAL LINES

64. The spectrum of He is expected to be similar to that of :-
 (1) H (2) Na
 (3) He^+ (4) Li^+
65. Third line of Balmer series is produced by which transition in spectrum of H-atom
 (1) 5 to 2 (2) 5 to 1
 (3) 4 to 2 (4) 4 to 1
66. Which one of the following electron transitions between energy levels produces the line of shortest wavelength in hydrogen spectrum ?
 (1) $n_2 \rightarrow n_1$ (2) $n_3 \rightarrow n_1$
 (3) $n_4 \rightarrow n_1$ (4) $n_4 \rightarrow n_3$

67. Which series have highest energy in hydrogen spectrum :-
 (1) Balmer (2) Brackett
 (3) Pfund (4) Lyman
68. The ratio of minimum frequency of Lyman & Balmer series will be :-
 (1) 1.25 (2) 0.25 (3) 5.4 (4) 10
69. Which transition emits photon of maximum frequency :-
 (1) second spectral line of Balmer series
 (2) second spectral line of Paschen series
 (3) fifth spectral line of Humphery series
 (4) first spectral line of Lyman series
70. Which one of the following species will give a series of spectral lines similar to that of Mg^{2+} :-
 (1) Al^{3+} (2) Na (3) Mg^+ (4) F
71. The ratio of minimum wavelengths of Lyman & Balmer series will be :-
 (1) 1.25 (2) 0.25 (3) 5 (4) 10
72. The wavelength of photon obtained by electron transition between two levels in H-atom and singly ionised He are λ_1 and λ_2 respectively, then :-
 (1) $\lambda_2 = \lambda_1$ (2) $\lambda_2 = 2\lambda_1$
 (3) $\lambda_2 = \lambda_1/2$ (4) $\lambda_2 = \lambda_1/4$
73. Find out ratio of following for photon
 $(\nu_{\max})_{\text{Lyman}} : (\nu_{\max})_{\text{Brackett}}$
 (1) 1 : 16 (2) 16 : 1 (3) 4 : 1 (4) 1 : 4
74. The ratio of wavelengths of first line of Lyman series in Li^{+2} and first line of Lyman series in deuterium (2_1H) is :-
 (1) 1 : 9 (2) 9 : 1 (3) 1 : 4 (4) 4 : 1
75. In an electronic transition atom cannot emit :-
 (1) Visible light (2) γ - rays
 (3) Infra red light (4) Ultra violet light
76. The first Lyman transition in the hydrogen spectrum has $\Delta E = 10.2$ eV. The same energy change is observed in the second Balmer transition of :-
 (1) Li^{2+} (2) Li^+ (3) He^+ (4) Be^{3+}
77. The limiting line in Balmer series will have a frequency of :-
 (1) $3.65 \times 10^{14} \text{sec}^{-1}$ (2) $3.29 \times 10^{15} \text{sec}^{-1}$
 (3) $8.22 \times 10^{14} \text{sec}^{-1}$ (4) $-8.22 \times 10^{14} \text{sec}^{-1}$
78. If the shortest wavelength of Lyman series of H atom is x, then the wave length of first line of Balmer series of H atom will be :-
 (1) $\frac{9x}{5}$ (2) $\frac{36x}{5}$ (3) $\frac{5x}{9}$ (4) $\frac{5x}{36}$
79. The first emission line in the H-atom spectrum in the Balmer series will have wave number :-
 (1) $\frac{5R}{36} \text{cm}^{-1}$ (2) $\frac{3R}{4} \text{cm}^{-1}$
 (3) $\frac{7R}{144} \text{cm}^{-1}$ (4) $\frac{9R}{400} \text{cm}^{-1}$
80. What transition in He^+ will have the same λ as the I line in Lyman series of H - atom :-
 (1) $5 \rightarrow 3$ (2) $3 \rightarrow 2$
 (3) $6 \rightarrow 4$ (4) $4 \rightarrow 2$
81. In H-atom, electron transits from 6th orbit to 2nd orbit in multi step. Then total spectral lines (without Balmer series) will be :-
 (1) 6 (2) 10 (3) 4 (4) 0
82. An atom has x energy level, then total number of lines in its spectrum are:-
 (1) $1 + 2 + 3 + \dots + (x-1)$
 (2) $1 + 2 + 3 + \dots + (x^2)$
 (3) $1 + 2 + 3 + \dots + (x-1)$
 (4) $(x+1)(x+2)(x+4)$
83. The figure indicates the energy level diagram for the origin of six spectral lines in emission spectrum (e.g. line no. 5 arises from the transition from level B to X) which of the following spectral lines will not occur in the absorption spectrum :-
 (1) 1, 2, 3 (2) 3, 2
 (3) 4, 5, 6 (4) 3, 2, 1
-
84. A certain electronic transition from an excited state to ground state of the H atom in one or more step gives rise to three lines in the ultra violet region of the spectrum. How many lines does this transition produce in the infrared region of the spectrum :-
 (1) 1 (2) 2 (3) 3 (4) 4

85. Four lowest energy levels of H-atom are shown in the figure. The number of emission lines could be :-
- | | |
|-------|-----------|
| (1) 3 | 4 _____ |
| (2) 4 | 3 _____ |
| (3) 5 | 2 _____ |
| (4) 6 | n=1 _____ |
86. In the above problem, the number of absorption lines could be :-
- (1) 3 (2) 4 (3) 5 (4) 6
87. If 9.9 eV energy is supplied to H atom, the no. of spectral lines emitted is equal to :-
- (1) 0 (2) 1 (3) 2 (4) 3

DE-BROGLIE CONCEPT AND HEISENBERG PRINCIPLE

88. An electron has a kinetic energy of 2.8×10^{-23} J. de-Broglie wavelength will be nearly :-
($m_e = 9.1 \times 10^{-31}$ kg)
- (1) 9.28×10^{-24} m (2) 9.28×10^{-7} m
(3) 9.28×10^{-8} m (4) 9.28×10^{-10} m
89. What is the de-Broglie wavelength associated with the hydrogen electron in its third orbit :-
- (1) 9.96×10^{-10} cm (2) 9.96×10^{-8} cm
(3) 9.96×10^4 cm (4) 9.96×10^8 cm
90. If the de-Broglie wavelength of the fourth Bohr orbit of hydrogen atom is $4\text{\AA}$, the circumference of the orbit will be :-
- (1) $4\text{\AA}$ (2) 4 nm (3) $16\text{\AA}$ (4) 16 nm
91. Number of waves in fourth orbit :-
- (1) 4 (2) 5 (3) 0 (4) 1
92. What is the ratio of the de-Broglie wavelengths for electrons accelerated through 200 volts and 50 volts :-
- (1) 1 : 2 (2) 2 : 1
(3) 3 : 10 (4) 10 : 3
93. For a valid Bohr orbit, its circumference should be
- (1) $= n\lambda$ (2) $= (n-1)\lambda$
(3) $> n\lambda$ (4) $< n\lambda$
94. A particle X moving with a certain velocity has a debroglie wavelength of $1\text{\AA}$. If particle Y has a mass of 25% that of X and velocity 75% that of X, de-Broglie wavelength of Y will be :-
- (1) $3\text{\AA}$ (2) $5.33\text{\AA}$
(3) $6.88\text{\AA}$ (4) $48\text{\AA}$

95. The number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number +2:-
- (1) 3 (2) 4 (3) 2 (4) 1
96. The uncertainty in position of an electron & helium atom are same. If the uncertainty in momentum for the electron is 32×10^5 , then the uncertainty in momentum of helium atom will be
- (1) 32×10^5 (2) 16×10^5
(3) 8×10^5 (4) None
97. Calculate the uncertainty in the position of an electron (mass 9.1×10^{-28} g) moving with a velocity of 3×10^4 cm sec⁻¹, if the uncertainty in velocity is 0.011% ?
- (1) 1.92 cm (2) 7.68 cm
(3) 0.175 cm (4) 3.84 cm
98. Heisenberg Uncertainty principle is not valid for
- (1) Moving electron (2) Motor car
(3) Stationary particles (4) 2 & 3 both
99. What should be the momentum (in gram centimetre per second) of a particle if its de-Broglie wavelength is $1\text{\AA}$ and the value of h is 6.6252×10^{-27} erg second ?
- (1) 6.6252×10^{-19} gcm/s
(2) 6.6252×10^{-21} gcm/s
(3) 6.6252×10^{-24} gcm/s
(4) 6.6252×10^{-27} gcm/s
100. What should be the mass of the photon of sodium if its wavelength is $5894\text{\AA}$, the velocity of light is 3×10^8 metre/second and the value of h is 6.6252×10^{-34} kg m²/s ?
- (1) 3.746×10^{-26} kg (2) 3.746×10^{-30} kg
(3) 3.746×10^{-34} kg (4) 3.746×10^{-36} kg
101. Which of the following has least de-Broglie λ ?
- (1) e⁻ (2) p (3) CO₂ (4) SO₂

QUANTUM NUMBERS

102. The following quantum no. are possible for how many orbitals $n = 3$, $\ell = 2$, $m = +2$?
- (1) 1 (2) 2 (3) 3 (4) 4
103. Number of possible orbitals (all types) in $n = 3$ energy level is :-
- (1) 1 (2) 3 (3) 4 (4) 9
104. Which sub-shell is not permissible :-
- (1) 2d (2) 4f (3) 6p (4) 3s

- 105.** Nodal plane is found in which orbital :-
 (1) $n = 2, \ell = 0$ (2) $n = 3, \ell = 0$
 (3) $n = 2, \ell = 1$ (4) $n = 1, \ell = 0$
- 106.** No. of nodal surface in 2s orbital :-
 (1) 0 (2) 1 (3) 2 (4) 3
- 107.** Number of orbitals in h sub-shell is
 (1) 11 (2) 15 (3) 17 (4) 19
- 108.** How many quantum numbers are required to specify the position of electron :-
 (1) 1 (2) 2 (3) 3 (4) 4
- 109.** Which of the following is correct for a 4d-electron
 (1) $n = 4, \ell = 2, s = + \frac{1}{2}$
 (2) $n = 4, \ell = 2, s = 0$
 (3) $n = 4, \ell = 3, s = 0$
 (4) $n = 4, \ell = 3, s = + \frac{1}{2}$
- 110.** If $n = 3$, then which value of ' ℓ ' is correct :-
 (1) 0 (2) 1
 (3) 2 (4) All of them
- 111.** Energy of atomic orbitals in a particular shell is in order:-
 (1) $s < p < d < f$ (2) $s > p > d > f$
 (3) $p < d < f < s$ (4) $f > d > s > p$
- 112.** Which statement is not correct for $n = 5, m = 2$:-
 (1) $\ell = 4$
 (2) $\ell = 0, 1, 2, 3; s = + 1/2$
 (3) $\ell = 3$
 (4) $\ell = 2, 3, 4$
- 113.** Spin angular momentum for electron :-
 (1) $\sqrt{s(s+1)} \frac{h}{2\pi}$ (2) $\sqrt{2s(s+1)} \frac{h}{2\pi}$
 (3) $\sqrt{s(s+2)} \frac{h}{2\pi}$ (4) None
- 114.** The maximum number of electrons in a p-orbital with $n = 6$ and $m = 0$ can be :-
 (1) 14 (2) 6 (3) 2 (4) 10
- 115.** The total number of value of m for the electrons in $n = 4$ is -
 (1) 4 (2) 8 (3) 16 (4) 32
- 116.** In an atom, for how many electrons, the quantum numbers will be $n = 3, \ell = 2, m = + 2, s = + \frac{1}{2}$:-
 (1) 18 (2) 6 (3) 24 (4) 1
- 117.** Which orbital is represented by the complete wave function ψ_{420} :-
 (1) 4d (2) 3d (3) 4p (4) 4s
- 118.** An electron is in one of 4d orbital. Which of the following orbital quantum number value is not possible :-
 (1) $n = 4$ (2) $\ell = 1$ (3) $m = 1$ (4) $m = 2$
- 119.** A neutral atom of an element has 2K, 8L, 11M and 2N electrons. The number of s-electron in the atom are
 (1) 2 (2) 8 (3) 10 (4) 6
- 120.** If $\ell = 3$ then type and number of orbital is :-
 (1) 3p, 3 (2) 4f, 14 (3) 5f, 7 (4) 3d, 5
- 121.** Any nf-orbital can accomodate upto :-
 (1) 14 electron
 (2) Six electrons
 (3) Two electrons with parallel spin
 (4) Two electrons with opposite spin
- 122.** n, ℓ and m values of an electron in $3p_y$ orbital are :-
 (1) $n = 3; \ell = 1$ and $m = 1$
 (2) $n = 3; \ell = 1$ and $m = -1$
 (3) Both 1 and 2 are correct
 (4) None of these
- 123.** $^{36}_{36}\text{Kr}$ has the electronic configuration $(_{18}\text{Ar}) 4s^2 3d^{10} 4p^6$. The 39th electron will go into which one of the following sub-levels :-
 (1) 4f (2) 4d (3) 3p (4) 5s
- 124.** The maximum probability of finding an electron in the d_{xy} orbital is :-
 (1) Along the x-axis
 (2) Along the y-axis
 (3) At an angle of 45° from the x and y axis
 (4) At an angle of 90° from the x and y axis
- 125.** Which orbital has two angular nodal planes :-
 (1) s (2) p (3) d (4) f
- 126.** An orbital with $\ell = 0$ is symmetrical about the :-
 (1) x-axis only (2) y-axis only
 (3) z-axis only (4) The nucleus

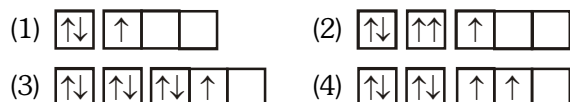
- 127.** If n & ℓ are principal and azimuthal quantum no. respectively then the expression for calculating the total no. of electron in any energy level is :-

$$(1) \sum_{\ell=0}^{\ell=n} 2(2\ell+1) \quad (2) \sum_{\ell=1}^{\ell=n-1} 2(2\ell+1)$$

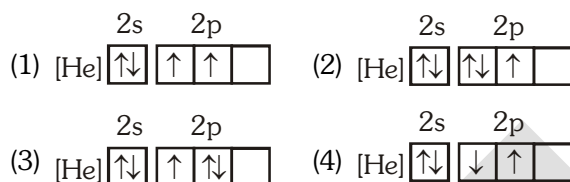
$$(3) \sum_{\ell=0}^{\ell=n+1} 2(2\ell+1) \quad (4) \sum_{\ell=0}^{\ell=n-1} 2(2\ell+1)$$

RULES FOR FILLING OF ORBITALS

- 128.** Which configuration does not obey pauli's exclusion principle :-



- 129.** Which of the following configuration follows the Hund's rule :-



- 130.** The basis of three unpaired electrons present in the configuration of nitrogen is :-

- (1) Aufbau principle (2) Pauli's principle
 (3) Hund's principle (4) Uncertainty principle

- 131.** The orbital with maximum energy is :-

- (1) 3d (2) 5p (3) 4s (4) 6d

- 132.** n and ℓ values of an orbital 'A' are 3 and 2 and for another orbital 'B' are 5 and 0. The energy of :-

- (1) B is more than A
 (2) A is more than B
 (3) A and B are of same energy
 (4) None

- 133.** No. of all subshells of $n + \ell = 7$ is:-

- (1) 4 (2) 5 (3) 6 (4) 7

- 134.** Electronic configuration $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \square$ has violated :-

- (1) Hund's rule (2) Pauli's principle
 (3) Aufbau principle (4) $(n + \ell)$ rule

- 135.** The total spin resulting from a d^9 configuration is:-

- (1) $\frac{1}{2}$ (2) 2 (3) 1 (4) $\frac{3}{2}$

- 136.** Which of the following transition neither shows absorption nor emission of energy in case of hydrogen atom :-

- (1) $3p_x \rightarrow 3s$ (2) $3d_{xy} \rightarrow 3d_{yz}$
 (3) $3s \rightarrow 3d_{xy}$ (4) All the above

- 137.** In ground state of ${}_{24}\text{Cr}$, number of orbitals with paired and unpaired electron :-

- (1) 10 (2) 12 (3) 15 (4) 18

- 138.** For Na ($Z = 11$) set of quantum numbers for last electron is:-

- (1) $n = 3, \ell = 1, m = 1, s = +\frac{1}{2}$
 (2) $n = 3, \ell = 0, m = 0, s = +\frac{1}{2}$
 (3) $n = 3, \ell = 0, m = 1, s = +\frac{1}{2}$
 (4) $n = 3, \ell = 1, m = 1, s = -\frac{1}{2}$

- 139.** Which of the following set of quantum numbers is correct for the 19th electron of Chromium :-

n	ℓ	m	s
(1) 3	0	0	$1/2$
(2) 3	2	-2	$1/2$
(3) 4	0	0	$1/2$
(4) 4	1	-1	$1/2$

- 140.** Which set of quantum number is correct for an electron in 3p orbital :-

- (1) $n = 3, \ell = 2, m = 0, s = +\frac{1}{2}$
 (2) $n = 3, \ell = 0, m = +1, S = +\frac{1}{2}$
 (3) $n = 3, \ell = -2, m = -1, s = +\frac{1}{2}$
 (4) $n = 3, \ell = 1, m = 0, s = +\frac{1}{2}$

- 141.** An atom of Cr [$Z = 24$] loses 2 electrons. How many unpaired electrons shall be there in Cr^{+2} :

- (1) 4 (2) 3 (3) 2 (4) 1

- 142.** The atomic weight of an element is double its atomic number. If there are three electrons in 2p sub-shell, the element is :-

- (1) C (2) N (3) O (4) Ca

- 143.** The atomic number of an element is 17, the number of orbitals containing electron pairs in the valence shell is:-

- (1) 8 (2) 2 (3) 3 (4) 6

144. A transition metal 'X' has a configuration $[\text{Ar}] 3d^5$ in its +3 oxidation state. Its atomic number is:-
 (1) 22 (2) 26 (3) 28 (4) 19

145. $4s^2$ is the configuration of the outermost orbit of an element. Its atomic number would be :-
 (1) 29 (2) 24 (3) 30 (4) 19

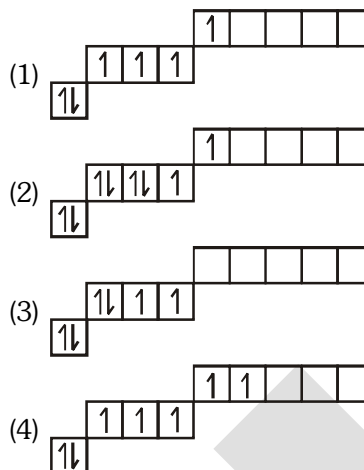
146. Sum of the paired electrons present in the orbital with $\ell = 2$ in all the species Fe^{2+} , Co^{2+} and Ni^{+2} are:-
 (1) 9 (2) 12 (3) 6 (4) 15

147. What is the electronic configuration of an element in its first excited state which is isoelectronic with O_2
 (1) $[\text{Ne}] 3s^2 3p^3 3d^1$ (2) $[\text{Ne}] 3s^2 3p^4$
 (3) $[\text{Ne}] 3s^1 3p^3 3d^2$ (4) $[\text{Ne}] 3s^1 3p^5$

148. The quantum number of 20th electron of $\text{Fe}(Z = 26)$ would be :-
 (1) 3, 2, -2, $-\frac{1}{2}$ (2) 3, 2, 0, $\frac{1}{2}$
 (3) 4, 0, 0, $+\frac{1}{2}$ (4) 4, 1, -1, $+\frac{1}{2}$

149. The atomic number of the element having maximum number of unpaired 3p electrons is (in ground state):-
 (1) 15 (2) 10 (3) 12 (4) 8

150. Which one represent ground state configuration :-



151. The electronic configuration of a dipositive metal ion M^{2+} is 2, 8, 14 and its ionic weight is 58 a.m.u. The number of neutrons in its nucleus would be :-
 (1) 30 (2) 32 (3) 34 (4) 42

152. In an atom having 2K, 8L, 8M and 2N electrons, the number of electrons with $m = 0$; $s = +\frac{1}{2}$ are :-
 (1) 6 (2) 2 (3) 8 (4) 16

153. The number of electrons in the M-shell of the element with atomic number 24 is :-
 (1) 24 (2) 12 (3) 8 (4) 13

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	1	1	3	2	2	4	1	3	2	2	1	3	4	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	1	3	3	4	3	4	1	3	1	1	3	2	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	4	1	1	2	1	4	1	3	3	1	4	1	1	4
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	2	4	4	4	3	1	1	2	1	3	1	3	4	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	1	1	2	4	1	3	4	3	4	1	2	4	2	1	2
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	3	3	2	1	4	1	3	3	1	4	1	1	3	2	3
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	1	1	1	2	1	1	3	4	1	4	4	1	4	1	3
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	2	1	3	1	4	1	2	1	3	3	4	1	2	2	3
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	4	3	2	3	3	4	4	2	1	3	4	1	1	1	1
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	4	3	2	3	4	1	2	3	2	3	2	1	3	1	3
Que.	151	152	153												
Ans.	2	1	4												

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2008

- If uncertainty in position and momentum are equal, then uncertainty in velocity is ?
 (1) $\sqrt{\frac{h}{\pi}}$ (2) $\frac{1}{2m} \sqrt{\frac{h}{\pi}}$
 (3) $\sqrt{\frac{h}{2\pi}}$ (4) $\frac{1}{m} \sqrt{\frac{h}{\pi}}$
- The measurement of the electron position is associated with an uncertainty in momentum, which is equal to $1 \times 10^{-18} \text{ g cm s}^{-1}$. the uncertainty in electron velocity is : (mass of electron = $9 \times 10^{-28} \text{ g}$)
 (1) $1 \times 10^{11} \text{ cm s}^{-1}$ (2) $1 \times 10^9 \text{ cm s}^{-1}$
 (3) $1 \times 10^6 \text{ cm s}^{-1}$ (4) $1 \times 10^5 \text{ cm s}^{-1}$

AIPMT 2009

- Maximum number of electrons in a subshell of an atom is determined by the following :-
 (1) $2n^2$ (2) $4\ell + 2$ (3) $2\ell + 1$ (4) $4\ell - 2$
- Which of the following is not permissible arrangement of electrons in an atom ?
 (1) $n = 3, \ell = 2, m = -2, s = -1/2$
 (2) $n = 4, \ell = 0, m = 0, s = -1/2$
 (3) $n = 5, \ell = 3, m = 0, s = +1/2$
 (4) $n = 3, \ell = 2, m = -3, s = -1/2$

AIPMT 2010

- A 0.66 kg ball is moving with a speed of 100 m/s. The associated wavelength will be ($h = 6.6 \times 10^{-34} \text{ Js}$) :-
 (1) $6.6 \times 10^{-34} \text{ m}$ (2) $1.0 \times 10^{-35} \text{ m}$
 (3) $1.0 \times 10^{-32} \text{ m}$ (4) $6.6 \times 10^{-32} \text{ m}$

AIPMT Pre-2011

- The total number of atomic orbitals in fourth energy level of an atom is :-
 (1) 8 (2) 16 (3) 32 (4) 4
- The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relation between their wavelengths i.e. λ_1 and λ_2 will be :
 (1) $\lambda_1 = \lambda_2$ (2) $\lambda_1 = 2\lambda_2$
 (3) $\lambda_1 = 4\lambda_2$ (4) $\lambda_1 = \frac{1}{2}\lambda_2$

- If $n = 6$, the correct sequence for filling of electrons will be :
 (1) $ns \rightarrow (n-2)f \rightarrow (n-1)d \rightarrow np$
 (2) $ns \rightarrow (n-1)d \rightarrow (n-2)f \rightarrow np$
 (3) $ns \rightarrow (n-2)f \rightarrow np \rightarrow (n-1)d$
 (4) $ns \rightarrow np \rightarrow (n-1)d \rightarrow (n-2)f$

AIPMT Mains 2011

- According to the Bohr Theory, which of the following transitions in the hydrogen atom will give rise to the least energetic photon ?
 (1) $n = 5$ to $n = 3$ (2) $n = 6$ to $n = 1$
 (3) $n = 5$ to $n = 4$ (4) $n = 6$ to $n = 5$

AIIMS 2011

- Smallest wavelength occurs for :-
 (1) Lyman series (2) Balmer series
 (3) Paschen series (4) Brackett series
- Which of the following is wrong for Bohr model:-
 (1) It establishes stability of atom
 (2) It is contradicted with Heisenberg uncertainty principle
 (3) It explain the concept of spectral lines
 (4) e^- behaves as particle & wave

AIPMT PRE 2012

- Maximum number of electrons in a subshell with $\ell = 3$ and $n = 4$ is :-
 (1) 10 (2) 12
 (3) 14 (4) 16
- The correct set of four quantum numbers for the valence electron of rubidium atom ($Z = 37$) is:-
 (1) 5, 0, 0, $+\frac{1}{2}$ (2) 5, 1, 0, $+\frac{1}{2}$
 (3) 5, 1, 1, $+\frac{1}{2}$ (4) 6, 0, 0, $+\frac{1}{2}$

AIPMT MAINS 2012

- The orbital angular momentum of a p-electron is given as :-
 (1) $\sqrt{\frac{3}{2}} \frac{h}{\pi}$ (2) $\sqrt{6} \cdot \frac{h}{2\pi}$
 (3) $\frac{h}{\sqrt{2}\pi}$ (4) $\sqrt{3} \frac{h}{2\pi}$

AIIMS 2012

15. Threshold frequency of a metal is $5 \times 10^{13} \text{ sec}^{-1}$ upon which $1 \times 10^{14} \text{ sec}^{-1}$ frequency light is focused then maximum kinetic energy of emitted electron :-
 (1) 3.3×10^{-21} (2) 3.3×10^{-20}
 (3) 6.6×10^{-21} (4) 6.6×10^{-20}
16. In Bohr's orbit $\frac{nh}{2\pi}$ indicates :-
 (1) Momentum
 (2) Kinetic energy
 (3) Potential energy
 (4) Angular momentum

NEET UG 2013

17. The value of Planck's constant is $6.63 \times 10^{-34} \text{ Js}$. The speed of light is $3 \times 10^{17} \text{ nm s}^{-1}$. Which value is closest to the wavelength in nanometer of a quantum of light with frequency of $6 \times 10^{15} \text{ s}^{-1}$?
 (1) 75 (2) 10 (3) 25 (4) 50
18. Based on equation $E = -2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$ certain conclusions are written. Which of them is **not** correct ?
 (1) For $n = 1$, the electron has a more negative energy than it does for $n = 6$ which means that the electron is more loosely bound in the smallest allowed orbit.
 (2) The negative sign in equation simply means that the energy of electron bound to the nucleus is lower than it would be if the electrons were at the infinite distance from the nucleus
 (3) Larger the value of n , the larger is the orbit radius
 (4) Equation can be used to calculate the change in energy when the electron change orbit
19. What is the maximum numbers of electrons that can be associated with the following set of quantum numbers $\Rightarrow n = 3; l = 1$ and $m = -1$?
 (1) 2 (2) 10
 (3) 6 (4) 4

AIIMS 2013

20. A particle is moving 3 times faster than speed of e^- . Ratio of wavelength of particle and electron is 1.8×10^{-4} , then particle is :-
 (1) Neutron (2) α -particle
 (3) Deuteron (4) Tritium

AIPMT 2014

21. What is the maximum number of orbitals that can be identified with the following quantum numbers. $n = 3, l = 1, m_l = 0$?
 (1) 1 (2) 2 (3) 3 (4) 4
22. Calculate the energy in joule corresponding to light of wavelength 45 nm : (Planck's constant $h = 6.63 \times 10^{-34} \text{ Js}$; speed of light $c = 3 \times 10^8 \text{ ms}^{-1}$)
 (1) 6.67×10^{15} (2) 6.67×10^{11}
 (3) 4.42×10^{-15} (4) 4.42×10^{-18}
23. Magnetic moment 2.83 BM is given by which of the following ions ?
 (At. no. Ti = 22, Cr = 24, Mn = 25, Ni = 28):-
 (1) Ti^{3+} (2) Ni^{2+} (3) Cr^{3+} (4) Mn^{2+}

AIIMS 2014

24. The energy of an electron of $2p_y$ orbital is
 (1) greater than $2p_x$ orbital
 (2) Less than $2p_z$ orbital
 (3) same as that of $2p_x$ and $2p_z$ orbital
 (4) Equal to 2s orbital

AIPMT 2015

25. Which of the following pairs of ions are isoelectronic and isostructural ?
 (1) $\text{ClO}_3^-, \text{CO}_3^{2-}$ (2) $\text{SO}_3^{2-}, \text{NO}_3^-$
 (3) $\text{ClO}_3^-, \text{SO}_3^{2-}$ (4) $\text{CO}_3^{2-}, \text{SO}_3^{2-}$
26. The number of d-electrons in Fe^{2+} ($Z = 26$) is not equal to the number of electrons in which one of the following?
 (1) p-electrons in Cl ($Z = 17$)
 (2) d-electrons in Fe ($Z = 26$)
 (3) p-electrons in Ne ($Z = 10$)
 (4) s-electrons in Mg ($Z = 12$)

27. Magnetic moment 2.84 B.M. is given by :-
(At. no.), Ni = 28, Ti = 22, Cr = 24, Co = 27)
(1) Ti^{3+} (2) Cr^{2+}
(3) Co^{2+} (4) Ni^{2+}

28. The angular momentum of electron in 'd' orbital is equal to :-
(1) $\sqrt{2} \hbar$ (2) $2\sqrt{3} \hbar$
(3) $0 \hbar$ (4) $\sqrt{6} \hbar$

RE-AIPMT 2015

29. Which is the correct order of increasing energy of the listed orbitals in the atom of titanium ?
(At. no. Z = 22)
(1) 3s 3p 3d 4s (2) 3s 3p 4s 3d
(3) 3s 4s 3p 3d (4) 4s 3s 3p 3d

AIIMS 2015

30. In which transition of hydrogen will atom have same wavelength as in Balmer series transition of He^+ ion ($n = 4$ to $n = 2$) :-
(1) 4 to 2 (2) 3 to 2
(3) 2 to 1 (4) 4 to 1

NEET-I 2016

31. Two electrons occupying the same orbital are distinguished by :-
(1) Principal quantum number
(2) Magnetic quantum number
(3) Azimuthal quantum number
(4) Spin quantum number

NEET-II 2016

32. Which of the following pairs of d-orbitals will have electron density along the axis ?
(1) $d_{z^2}, d_{x^2-y^2}$ (2) $d_{xy}, d_{x^2-y^2}$
(3) d_{z^2}, d_{xz} (4) d_{xz}, d_{yz}
33. How many electrons can fit in the orbital for which $n = 3$ and $l = 1$?
(1) 10 (2) 14 (3) 2 (4) 6

AIIMS 2016

34. Bohr's model is applicable to explain :-
(1) Zeeman effect
(2) Stark effect
(3) Bond formation of H_2
(4) Hydrogen spectrum

NEET(UG) 2017

35. Which one is the wrong statement ?
(1) The uncertainty principle is $\Delta E \times \Delta t \geq h/4\pi$
(2) Half filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement.
(3) The energy of 2s orbital is less than the energy of 2p orbital in case of Hydrogen like atoms
(4) de-Broglie's wavelength is given by $\lambda = \frac{h}{mv}$, where m = mass of the particle, v = velocity of the particle

AIIMS 2017

36. For H-spectrum, electron transition takes place from $n = 5$ to $n = 2$ then emitted wavelength of photon is 434 nm. The wave length of photon in electron transition from $n = 4$ to $n = 2$ will be :-
(1) 586.16 nm
(2) 48.608 nm
(3) 486 nm
(4) 400.16 nm
37. The statement, an atom possesses a spherical shape in which the positive charge is uniformly distributed and the electrons are embedded into it in such a manner so as to give stable arrangement, was given by :-
(1) Thomson's atomic model
(2) Rutherford's atomic model
(3) Bohr's atomic model
(4) Chadwick's atomic model

NEET(UG) 2018

38. Which one is a **wrong** statement ?
(1) Total orbital angular momentum of electron in 's' orbital is equal to zero
(2) An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers.
(3) The electronic configuration of N atom is

$$\begin{array}{ccc} 1s^2 & 2s^2 & 2p_x^1 \ 2p_y^1 \ 2p_z^1 \\ \boxed{\uparrow\downarrow} & \boxed{\uparrow\downarrow} & \boxed{\uparrow} \ \boxed{\uparrow} \ \boxed{\downarrow} \end{array}$$

(4) The value of m for d_{z^2} is zero

AIIMS 2018

39. Maximum wave length of balmer series for H-atom ($R_H = 10^7 m^{-1}$) :-
(1) 720 nm (2) 520 nm
(3) 700 nm (4) 400 nm

40. An electron is moving in 2nd orbit of He⁺ then what will be velocity of electron :-

- (1) $\frac{2\pi K e^2}{h}$ (2) $\frac{16\pi K e^2}{h}$
 (3) $\frac{8\pi K e^2}{h}$ (4) $\frac{\pi K e^2}{4h}$

41. What will be velocity of an electron revolving in second orbit of H-atom :-

- (1) $5.45 \times 10^5 \text{ m/sec}^{-1}$ (2) $21.8 \times 10^5 \text{ m/sec}^{-1}$
 (3) $10.9 \times 10^5 \text{ m/sec}^{-1}$ (4) $43.6 \times 10^5 \text{ m/sec}^{-1}$

42. An incident UV light having wave length λ_1 is incident on metal surface and the ejected electron has the velocity of V_1 . Another UV light having wave length λ_2 is incident on the same metal surface ejects electron having velocity V_2 then $V_2^2 - V_1^2$ is :-

- (1) $\frac{2hc}{m_e} \left[\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right]$ (2) $\frac{2hc}{m_e} \left[\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right]^2$
 (3) $\frac{2hc}{m_e} [\lambda_2 - \lambda_1]$ (4) $\frac{2hc}{m_e} [\lambda_2 - \lambda_1]^2$

43. In Bohr's model of H-atom what will be shortest wave length in paschen series :-

- (1) 820.8 nm (2) 100.8 nm
 (3) 520.8 nm (4) 700.8 nm

EXERCISE-II (Previous Year Questions)

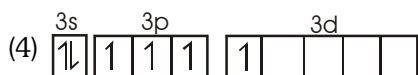
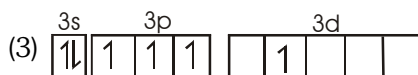
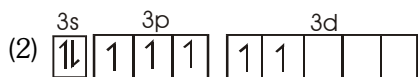
ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	2	2	4	2	2	2	1	4	1	4	3	1	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	1	1	1	1	4	2	3	3	1	4	4	2	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43		
Ans.	4	1	3	4	3	3	1	3	1	1	3	1	1		

EXERCISE-III (Analytical Questions)

Check Your Understanding

1. Which of the following has maximum energy :-



2. The frequency of radiation emitted when the electron falls from $n = 4$ to $n = 1$ in a hydrogen atom will be (Given ionization energy of $H = 2.18 \times 10^{-18} \text{ J atom}^{-1}$) :-

- (1) $1.03 \times 10^{15} \text{ s}^{-1}$ (2) $3.08 \times 10^{15} \text{ s}^{-1}$
(3) $2.00 \times 10^{15} \text{ s}^{-1}$ (4) $1.54 \times 10^{15} \text{ s}^{-1}$

3. Which orbital diagram does not obey Aufbau principle :-



4. Quantum number which cannot be calculated by schrodinger wave equation is:-

- (1) n (2) ℓ (3) m (4) s

5. $\frac{h}{2\pi}$ is angular momentum in.....orbit of He^+

- (1) First (2) Second
(3) Third (4) Infinite

6. Which element contain non-spherical electron density :-

- (1) He (2) B
(3) Be (4) Li

7. First shell energy of He^+ ion is -54.4 eV . Then energy of its second shell is :-

- (1) -54.4 eV (2) -13.6 eV
(3) -27.2 eV (4) $+27.2 \text{ eV}$

8. If $\ell = 2$ and $n = 3$ for any atom then maximum number of electron is :-

- (1) 2 (2) 6
(3) 12 (4) 10

9. A metal in its dipositive state has the electronic configuration 2, 8, 14 and has the atomic weight equal to 56. Number of neutrons in its nucleus would be :-

- (1) 30 (2) 32 (3) 34 (4) 28

10. What is the maximum number of electrons which can be accommodated in an atom in which the highest principal quantum number value is 4 :-

- (1) 10 (2) 18 (3) 36 (4) 54

11. The quantum numbers $+1/2$ and $-1/2$ for the electron spin represent.

- (1) Rotation of the electron in clockwise and anticlockwise direction respectively.
(2) Rotation of the electron in anticlockwise and clockwise direction respectively.
(3) Magnetic moment of the electron pointing up and down respectively.
(4) Two quantum mechanical spin states which have no classical analogue.

12. Uncertainty in position of a 0.25 g particle is 10^{-5} m . Then uncertainty in its velocity will be:- ($h = 6.6 \times 10^{-34} \text{ Js}$) :-

- (1) 1.2×10^{34} (2) 2.1×10^{-26}
(3) 1.6×10^{-20} (4) 1.7×10^{-9}

13. The wavelength of radiation emitted when an electron in a hydrogen atom makes a transition from an energy level with $n = 3$ to a level with

$n = 2$ is : [Given that $E_n = \frac{-1312}{n^2} \text{ kJ mol}^{-1}$]

- (1) $6.56 \times 10^{-7} \text{ m}$ (2) 65.6 nm
(3) $65.6 \times 10^{-7} \text{ m}$ (4) any of the above

14. Electronic energy of hydrogen atom depends on the quantum number :

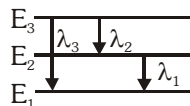
- (1) n , ℓ and m (2) n and ℓ only
(3) n and m only (4) n only

15. The energy required to escape the electron from ground state of H is 13.6 eV then the same for 1^{st} excited state of H atom :-

- (1) 3.4 (2) 13.6
(3) 27.2 (4) can't say anything

- 16.** A gas absorbs a photon of 355 nm and emits at two wavelengths. If one of the emissions is at 680 nm, the other is at :-
 (1) 743 nm (2) 518 nm
 (3) 1035 nm (4) 325 nm
- 17.** The frequency of light emitted for the transition $n = 4$ to $n = 2$ of He^+ is equal to the transition in H atom corresponding to which of the following :-
 (1) $n = 3$ to $n = 1$ (2) $n = 2$ to $n = 1$
 (3) $n = 3$ to $n = 2$ (4) $n = 4$ to $n = 3$
- 18.** The electrons identified by quantum numbers n and ℓ :-
 (a) $n = 4, \ell = 1$ (b) $n = 4, \ell = 0$
 (c) $n = 3, \ell = 2$ (d) $n = 3, \ell = 1$
 Can be placed in order of increasing energy as:-
 (1) (a) < (c) < (b) < (d) (2) (c) < (d) < (b) < (a)
 (3) (d) < (b) < (c) < (a) (4) (b) < (d) < (a) < (c)
- 19.** Energy of an electron is given by $E = -2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$. Wavelength of light required to excite an electron in an hydrogen atom from level $n = 1$ to $n = 2$ will be :-
 ($h = 6.62 \times 10^{-34} \text{ Js}$ and $c = 3.0 \times 10^8 \text{ ms}^{-1}$)
 (1) $1.214 \times 10^{-7} \text{ m}$ (2) $2.816 \times 10^{-7} \text{ m}$
 (3) $6.500 \times 10^{-7} \text{ m}$ (4) $8.500 \times 10^{-7} \text{ m}$
- 20.** A certain negative ion X^{2-} has in its nucleus 18 neutrons and 18 electrons in its extra nuclear structure. What is the mass number of the most abundant isotope of 'X' :-
 (1) 35.46 (2) 32 (3) 36 (4) 39
- 21.** Supposing the I.P. of hydrogen atom is 960 eV. Find out the value of principal quantum number having the energy equal to -60 eV :-
 (1) $n = 2$ (2) $n = 3$ (3) $n = 4$ (4) $n = 5$
- 22.** If the ionisation potential of an atom is 20V, its first excitation potential will be :-
 (1) 5 V (2) 10 V (3) 15 V (4) 20 V
- 23.** A single electron orbits a stationary nucleus of charge $+Ze$, where Z is a constant. It requires 47.2 eV to excite electron from second Bohr orbit to third Bohr orbit, find the value of Z :-
 (1) 1 (2) 3 (3) 5 (4) 4
- 24.** A photon of energy 12.75 eV is completely absorbed by a hydrogen atom initially in ground state. The principal quantum number of the excited state is :-
 (1) 1 (2) 3 (3) 4 (4) ∞
- 25.** An hydrogen atom (ionisation energy 13.6 eV) jumps from third excited state to first excited state. The energy of photon emitted in the process is :-
 (1) 1.89 eV (2) 2.55 eV
 (3) 12.09 eV (4) 12.75 eV
- 26.** If a photon of energy 14 eV is incident on an H-atom, what is true :-
 (1) Atom will be ionised and electron will have a kinetic energy of 14 eV
 (2) Atom will be ionised and electron will have a kinetic energy of 0.4 eV
 (3) Photon passes through atom without interacting with it
 (4) More than one electrons will make transitions
- 27.** An electron of energy 10.8 eV is incident on an H-atom then :-
 (1) The electron will come out with 10.8 eV energy.
 (2) The electron will be completely absorbed
 (3) 10.2 eV. of the electron would be absorbed by H atom and it would come out with 0.6 eV energy.
 (4) Can't be predicted
- 28.** The ratio of the difference in energy between the first and second Bohr orbit to that between second and third Bohr orbit in H-atom is :-
 (1) 4/9 (2) 1/3 (3) 27/5 (4) 1/2
- 29.** Match the following :-
 (A) Energy of ground state of He^+ (i) +6.04 eV
 (B) Potential energy of I orbit of H-atom (ii) -27.2 eV
 (C) Kinetic energy of II excited state of He^+ (iii) $8.72 \times 10^{-18} \text{ J}$
 (D) Ionisation potential of He^+ (iv) -54.4 eV
 (1) A - (i), B - (ii), C - (iii), D - (iv)
 (2) A - (iv), B - (iii), C - (ii), D - (i)
 (3) A - (iv), B - (ii), C - (i), D - (iii)
 (4) A - (ii), B - (iii), C - (i), D - (iv)

30. In the following transition which statement is correct
- (1) $E_{3-1} = E_{3-2} - E_{2-1}$ (2) $\lambda_3 = \lambda_1 + \lambda_2$ (3) $\nu_3 = \nu_2 + \nu_1$ (4) All of these
31. In which transition, one quantum of energy is emitted :-
- (1) $n = 4 \rightarrow n = 2$ (2) $n = 3 \rightarrow n = 1$ (3) $n = 4 \rightarrow n = 1$ (4) All of them
32. When a hydrogen sample in ground state is bombarded then what potential is required to accelerate electron so that first Paschen line is emitted :-
- (1) 2.55 V (2) 0.65 V (3) 12.09 V (4) 12.75 V
33. The binding energy of electron in ground state of hydrogen atom is 13.6 eV. The energies required to eject out an electron from three lowest states of He^+ will be - (in eV)
- (1) 13.6, 10.2, 3.4 (2) 13.6, 3.4, 1.5 (3) 13.6, 27.2, 40.8 (4) 54.4, 13.6, 6
34. The transition of electron in H-atom that will emit maximum energy is :-
- (1) $n_3 \rightarrow n_2$ (2) $n_4 \rightarrow n_3$ (3) $n_5 \rightarrow n_4$ (4) All have same energy
35. Given that in the H-atom the transition energy for $n = 1$ to $n = 2$ Rydberg states is 10.2 eV. The energy for the same transition in Be^{3+} is :-
- (1) 20.4 eV (2) 163.2 eV (3) 30.6 eV (4) 40.8 eV
36. When a electron in H-atom jumps from $n = 4$ to $n = 1$, ultra violet light is emitted. If the transition corresponds to $n = 4$ to $n = 2$ then which of the following colours will be emitted :-
- (1) Ultra violet (2) Green (3) Infra red (4) No colour
37. The wavelength of first line of the Lyman series for hydrogen is 1216 Å. The wavelength for the first line of this series for a 10 time ionised sodium atom ($z = 11$) will be :-
- (1) 1000 Å (2) 100 Å (3) 10 Å (4) 1 Å

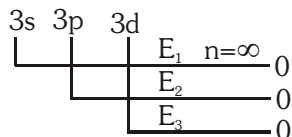


38. A certain electronic transition from an excited state to the ground state of the H atom in one or more steps gives rise to four lines in the ultra violet region of the spectrum, how many lines does this transition produce in the infrared region of the spectrum :-
- (1) 1 (2) 2 (3) 3 (4) 4
39. For the azimuthal quantum number (ℓ), the total number of magnetic quantum number is given by:-
- (1) $\ell = \frac{(m+1)}{2}$ (2) $\ell = \frac{(m-1)}{2}$ (3) $\ell = \frac{(2m+1)}{2}$ (4) $\ell = \frac{(2m-1)}{2}$
40. In P-atom find out the no. of paired electrons for $\ell = 1$ and $m = 0$:-
- (1) 3 (2) 1 (3) 2 (4) 0
41. When the value of principal quantum number n is 3, the permitted value of azimuthal quantum numbers ℓ and magnetic quantum numbers 'm' are:-
- (1) ℓ m
 0 0
 1 +1, 0, -1
 2 +2, +1, 0, -1, -2
- (2) ℓ m
 0 1
 2 +2, 1, -2
 3 +3, +2, +1, -2, -3
- (3) ℓ m
 0 0
 1 1 2 3 0, 1, 2
 2 +3, +2, 1, -2, -3
- (4) ℓ m
 1 0, 1
 2 0, 1, 2
 3 0, 1, 2, 3
42. A filled or half-filled set of p or d orbitals is spherically symmetric. Point out the species which has spherical symmetry :-
- (1) Na (2) C (3) Cl^- (4) Fe

43. Remaining part of atom except outer orbit is called:-

- (1) Kernel (2) Core
(3) Empty space (4) None of these

44. For H atom, the energy required for the removal of electron from various sub-shells is given as under:-



The order of the energies would be :-

- (1) $E_1 > E_2 > E_3$ (2) $E_3 > E_2 > E_1$
(3) $E_1 = E_2 = E_3$ (4) None of these

45. Radius of H-atom in its ground state is 5.3×10^{-11} m. After collision with an electron it is found to have a radius of 21.2×10^{-11} m. What is the principal quantum no. 'n' of the final state of the atom:-

- (1) $n = 2$ (2) $n = 3$
(3) $n = 4$ (4) $n = 16$

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	2	2	4	1	2	2	4	1	3	4	2	1	4	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	3	1	2	3	3	3	3	2	2	3	3	3	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	4	4	1	2	2	3	3	2	3	1	3	1	3	1

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** In Rutherford's gold foil experiment, very few α - particles are deflected back.
Reason :- Nucleus present inside the atom is heavy.
 (1) A (2) B (3) C (4) D
- Assertion :-** Mass numbers of most of the elements are fractional.
Reason :- Mass numbers are obtained by comparing with the mass number of carbon taken as 12.
 (1) A (2) B (3) C (4) D
- Assertion :-** In an atom, the velocity of electron in the higher orbits keeps on decreasing.
Reason :- Velocity of electron is inversely proportional to radius of the orbit.
 (1) A (2) B (3) C (4) D
- Assertion :-** Total energy of electron in hydrogen atom is negative.
Reason :- It is in bound state.
 (1) A (2) B (3) C (4) D
- Assertion :-** Limiting line in the Balmer series has a wavelength of 364.7 nm.
Reason :- Limiting line is obtained for a jump of electron from $n=\infty$ to $n=2$ for Balmer series.
 (1) A (2) B (3) C (4) D
- Assertion :-** A spectral line will be seen for a $2p_x - 2p_y$ transition.
Reason :- Only Balmer lines are observed in the visible region.
 (1) A (2) B (3) C (4) D
- Assertion :-** Bohr model is not suitable in case of multielectron species.
Reason :- It does not tell about electron-electron interaction.
 (1) A (2) B (3) C (4) D
- Assertion :-** p-orbital has dumbbell shape.
Reason :- Electrons present in p-orbital can have one of three values for 'm', i.e. 0, +1, -1
 (1) A (2) B (3) C (4) D
- Assertion :-** Nodal plane of p_x atomic orbital is yz plane.
Reason :- In p_x atomic orbital, electron density is zero in the yz plane.
 (1) A (2) B (3) C (4) D
- Assertion :-** 2p orbitals do not have spherical nodes.
Reason :- The number of spherical nodes in p-orbitals is given by $(n-2)$.
 (1) A (2) B (3) C (4) D
- Assertion :-** There are two spherical nodes in 3s-orbital.
Reason :- There is no angular node in 3s-orbital.
 (1) A (2) B (3) C (4) D
- Assertion :-** Number of radial and angular node for 3p-orbital are 1, 1 respectively.
Reason :- No. of radial and angular node depends only on principal quantum no.
 (1) A (2) B (3) C (4) D
- Assertion :-** For hydrogen 2s & 2p have same energy.
Reason :- For an atom, for same principal quantum number s, p, d & f have same energy.
 (1) A (2) B (3) C (4) D
- Assertion :-** No two electrons in an atom can have the same values of four quantum numbers.
Reason :- No two electrons in an atom can be simultaneously in the same shell, same subshell, same orbitals and have same spin.
 (1) A (2) B (3) C (4) D

15. Assertion :- An orbital cannot have more than two electrons.

Reason :- The two electrons with opposite spin in an orbital create opposite magnetic field.

- (1) A (2) B (3) C (4) D

16. Assertion :- In hydrogen energy of 4s is more than 3d.

Reason :- An orbital with lower value of $(n+l)$ has smaller energy than the orbital with higher value of $(n+l)$.

- (1) A (2) B (3) C (4) D

17. Assertion :- The configuration of B atom cannot be $1s^2 2s^3$.

Reason :- Hund's rule demands that the configuration should display maximum multiplicity.

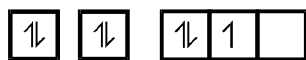
- (1) A (2) B (3) C (4) D

18. Assertion :- The electronic configuration of nitrogen atom is represented as :



1s 2s 2p

not as



1s 2s 2p

Reason :- The configuration of ground state of an atom is the one which has the greatest multiplicity.

- (1) A (2) B (3) C (4) D

19. Assertion :- The ground state configuration of Cr is $3d^5 4s^1$.

Reason :- A set of exactly half filled orbitals containing parallel spin arrangement provide extra stability.

- (1) A (2) B (3) C (4) D

20. Assertion :- When we give electron beam of 10.60 eV energy on hydrogen sample it will excite to first excited state.

Reason :- The difference between n_2 and n_1 is 10.2005 eV only.

- (1) A (2) B (3) C (4) D

21. Assertion :- Bohr's model is not applicable for He atom.

Reason :- Bohr's theory does not consider inter electronic repulsion.

- (1) A (2) B (3) C (4) D

22. Assertion :- Probability of finding an electron is directly proportional to wave function.

Reason :- In Bohr orbit revolving electron makes same number of waves.

- (1) A (2) B (3) C (4) D

23. Assertion :- Heisenberg uncertainty principle is not applicable for a moving tennis ball.

Reason :- Tennis ball in motion does not exhibit wave nature.

- (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	4	3	1	1	4	1	2	1	1	2	3	3	1	2
Que.	16	17	18	19	20	21	22	23							
Ans.	2	2	1	1	2	1	4	3							

CHEMICAL EQUILIBRIUM

E

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NEET SYLLABUS

EQUILIBRIUM : Equilibrium in physical and chemical processes, dynamic nature of equilibrium, law of chemical equilibrium, equilibrium constant, factors affecting equilibrium-Le Chatelier's principle.

@neetfocusquizz

OBJECTIVES

After studying this unit, we will be able to :

- *Identify dynamic nature of equilibrium involved in physical and chemical processes;*
 - *State the law of equilibrium;*
 - *Explain characteristics of equilibria involved in physical and chemical processes;*
 - *Write expressions for equilibrium constants;*
 - *Establish a relationship between K_p and K_c ;*
 - *Explain various factors that affect the equilibrium state of a reaction;*
- Understand physical equilibrium*

"You cannot teach a man anything; you can only help him discover it in himself."

Galileo

CHEMICAL EQUILIBRIUM

3.0 INTRODUCTION

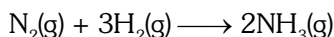
Chemical reaction : Symbolic representation of any chemical change in terms of reactants and products is called chemical reaction.

Types of chemical reaction :

(a) On the basis of physical state

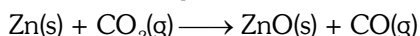
(I) Homogeneous reactions

All reactants and products are in same phase.



(II) Heterogeneous reactions

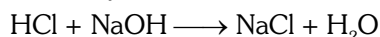
Reactants and products are in two or more phases.



(b) On the basis of speed.

(I) Fast reactions

(i) Generally these reactions are ionic reactions.

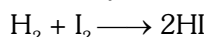


Acid Base Salt Water

(ii) Rate determination is almost impossible.

(II) Slow reactions

(i) Generally these reactions are molecular reactions.



(ii) Rate determination is possible.

(c) On the basis of heat

(I) Exothermic reactions

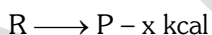
(i) Heat is evolved in these type of chemical reactions.



(ii) Change in enthalpy, $\Delta H = (-)$ ve

(II) Endothermic reactions

Heat is absorbed in these type of chemical reactions.

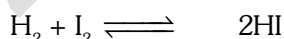
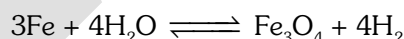
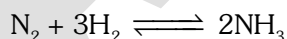


$\Delta H = (+)$ ve

(d) On the basis of direction

(I) Reversible reactions

(i) Chemical reaction in which products can be converted back into reactants.



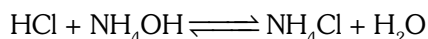
(ii) Proceed in forward as well as in backward direction.

(iii) Possible in closed container.

(iv) These can attain equilibrium.

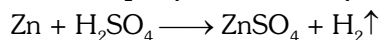
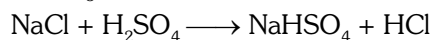
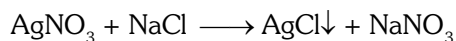
(v) Reactants are never completely converted into products.

(vi) Neutralisation reactions except of strong acid and strong base.

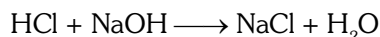


(II) Irreversible reactions

- (i) Chemical reaction in which products cannot be converted back into reactants.



- (ii) Proceed only in one direction (forward direction).
 (iii) Generally possible in open container.
 (iv) These do not attain equilibrium.
 (v) Reactants are nearly completely converted into products.
 (vi) Neutralisation reactions of strong acid and strong base.

**GOLDEN KEY POINTS**

- We always take forward direction if direction is not specified.
- In a reversible reaction if forward reaction is exothermic then the backward reaction will be endothermic and vice-versa.

- **Rate of Reaction**

The change in concentration of reactants or products in unit time is known as rate of the reaction.

$$\text{Rate of reaction} = (\pm) \frac{\text{change in concentration}}{\text{time taken for the change}}$$

3.1 EQUILIBRIUM AND CHEMICAL PROCESS**(A) Chemical Equilibrium**

The most important characteristic property of a reversible reaction is that it always attains a state of chemical equilibrium.

Consider a general reversible reaction in a closed vessel,

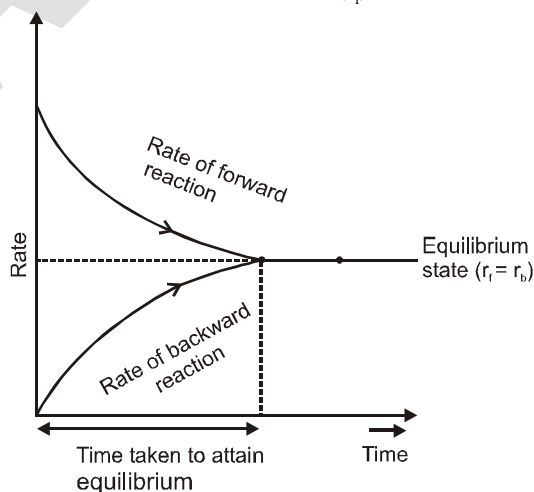


r_b = rate of backward reaction

Initially reaction occurs in forward direction but as the concentration of products increases reaction also starts in backward direction.

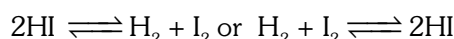
At a certain stage, rate of forward reaction becomes equal to the rate of backward reaction called equilibrium state.

At equilibrium state : Rate of forward reaction (r_f) = Rate of backward reaction (r_b)



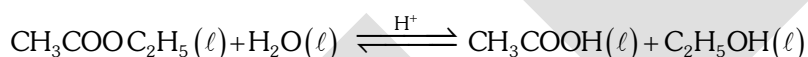
(B) Characteristics of equilibrium :

- The concentrations of the reactants and products do not change with time.
- At this stage, number of moles of substances produced per second in the forward reaction is equal to the number of moles of substances which disappear per second in the backward reaction.
- Chemical equilibrium is dynamic in nature i.e. the reaction although appears to be stopped but actually takes place in both the directions with the same speed.
- Chemical equilibrium can be approached from both sides



At equilibrium, each reactant and product has a constant concentration and this is independent of the fact whether the reaction starts from forward direction or backward direction with the reactant or with the product.

- Equilibrium is not affected by the presence of catalyst. The catalyst only helps in attaining equilibrium rapidly.
- The measurable properties of the system like temperature, concentration, colour, density etc. don't undergo any change with time at the chemical equilibrium conditions.
- Homogeneous equilibrium is the equilibrium in which the reactants and products are in the same phase.



- Heterogeneous equilibrium is the equilibrium in which the reactants and products are in two or more phases.



- **ACTIVE MASS :** The term active mass means the concentration of the reactants expressed in moles per litre (molar concentration) or the pressure of the reacting gas in atmosphere. In case of gases and solutions, the molar concentration means the number of gram molecules present per litre.

Active mass is usually expressed by enclosing the symbol of the reactant in square bracket [].

$$\begin{aligned} \text{Active mass} &= \frac{\text{Number of gram moles of the substance}}{\text{Volume (L)}} \\ &= \frac{\text{Weight of substance (in grams)}}{\text{Molecular weight (M}_w\text{)} \times \text{Volume (L)}} = \frac{w}{M_w \times V(\text{L})} = \frac{w \times 1000}{M_w \times V(\text{mL})} \end{aligned}$$

- The active mass of solids and pure liquids is a constant quantity (unity) because it is an intensive property i.e. number of molecules present per unit volume do not change because density and molecular weight of solids and pure liquids are constant. But it does not apply for gaseous substances because for them number of molecules present per unit volume change with change in volume of vessel.

$$\text{Molar concentration} = \frac{w}{M_w \times V(\text{L})} = \frac{\rho}{M_w} \quad (\text{where } \rho = \text{density (in gL}^{-1}\text{)})$$

$$= \frac{\rho}{M_w} \times 1000 \quad (\text{where } \rho = \text{density (in gmL}^{-1}\text{)})$$

$$\text{Active mass} = \frac{\text{density of the substance}}{\text{molecular mass of the substance}}$$

Following other names of active mass can also be used :

- | | | |
|-----------------------|----------------------|------------------------------|
| (i) mole/litre | (ii) gram mole/litre | (iii) gram molecules/litre |
| (iv) molarity | (v) Concentration | (vi) Effective concentration |
| (vii) active quantity | (viii) n/v | (ix) C |
| (x) M | (xi) $[]$ | |

Illustrations

Illustration 1. In any chemical reaction, equilibrium is supposed to be established when :

- (1) Mutual opposite reaction undergo.
- (2) concentration of reactants and resulting products are equal.
- (3) Velocity of mutual reactions become equal.
- (4) The temperature of mutual opposite reactions becomes equal.

Solution. **Ans. (3)**

Illustration 2. **Assertion :** The active mass of pure solids and pure liquids is taken unity.

Reason: The active mass of pure solids and pure liquids depends on density and molecular mass. The density and molecular mass of pure solids and pure liquids are constant.

Solution. **Ans. (1)**

Illustration 3. 8.5 g ammonia is present in a vessel of 0.5 litre capacity then find out the active mass of ammonia?

Solution. $[\text{NH}_3] = \frac{8.5}{17 \times 0.5} = 1 \text{ mol L}^{-1}$

BEGINNER'S BOX-1

1. Which of the following statement is correct regarding with chemical equilibrium :-
 - (1) Based on extent to which the reactions proceed to reach the equilibrium we may have negligible concentrations of reactants are left
 - (2) Equilibrium is not static
 - (3) Concentration of reactants and products becomes constant at equilibrium
 - (4) All of these
2. Find out the correct statement :-
 - (1) Equilibrium condition is a state of reversible reaction
 - (2) Chemical equilibrium are important in numerous biological process like transport and delivery of O_2
 - (3) Reversible reactions can be homogeneous and heterogeneous both
 - (4) All of these
3. Which of the following reaction is endothermic reaction :-
 - (1) Bond formation by two unstable atoms at certain condition
 - (2) Combustion reactions
 - (3) Conversion of more stable allotrope to less stable allotropic element
 - (4) Condensation of vapour to its liquid state
4. Active mass of 2 mol of NaCl kept in 4 litre vessel at NTP is :-

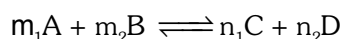
(1) 1	(2) 2	(3) $\frac{1}{2}$	(4) Not defined
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3.2 LAW OF MASS ACTION OR LAW OF CHEMICAL EQUILIBRIUM

- The law of mass action was given by Guldberg and Waage (1864).
- It states that the rate of a chemical reaction is directly proportional to the product of active masses of the reacting substances raised to a power equal to the stoichiometric coefficient in the balanced chemical equation.

(A) Derivation of equilibrium constant :-

Consider a reversible homogeneous chemical reaction which has attained equilibrium state at a particular temperature :



Let the active masses of A, B, C and D be [A] [B] [C] and [D] respectively at equilibrium.

According to law of mass action :-

$$\text{Rate of forward reaction} \quad (r_f) \propto [A]^{m_1} [B]^{m_2}$$

$$\text{Rate of backward reaction} \quad (r_b) \propto [C]^{n_1} [D]^{n_2}$$

$$r_f = k_f[A]^{m_1}[B]^{m_2} \quad \text{and} \quad r_b = k_b[C]^{n_1}[D]^{n_2}$$

Where k_f and k_b are forward and backward rate or velocity constants respectively.

At equilibrium state –

$$r_f = r_b$$

$$k_f[A]^{m_1}[B]^{m_2} = k_b[C]^{n_1}[D]^{n_2}$$

$$\frac{k_f}{k_b} = \frac{[C]^{n_1}[D]^{n_2}}{[A]^{m_1}[B]^{m_2}}$$

$$K = \frac{[C]^{n_1}[D]^{n_2}}{[A]^{m_1}[B]^{m_2}} \quad \therefore K = \frac{k_f}{k_b}$$

K is known as equilibrium constant and has a definite value for every chemical reaction at particular temperature.

- The equilibrium constant at a given temperature is the ratio of the rate constants of forward and backward reactions,

$$K = \frac{k_f}{k_b}$$

Forms of K

Concentration
(K_c)

Partial pressure
(K_p)

For reaction $m_1A + m_2B \rightleftharpoons n_1C + n_2D$

$$K_c = \frac{[C]^{n_1} [D]^{n_2}}{[A]^{m_1} [B]^{m_2}}$$

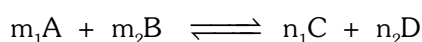
Unit of [] = mol L⁻¹

$$K_p = \frac{(P_C)^{n_1} \times (P_D)^{n_2}}{(P_A)^{m_1} \times (P_B)^{m_2}}$$

Unit of P = atm.

(B) Relation between K_p and K_c :

Consider a reversible homogeneous chemical equilibrium reaction



According to law of mass action (LOMA)

$$K_C = \frac{[C]^{n_1} [D]^{n_2}}{[A]^{m_1} [B]^{m_2}}$$

$$K_P = \frac{(P_C)^{n_1} (P_D)^{n_2}}{(P_A)^{m_1} (P_B)^{m_2}}$$

For an ideal gas **PV = nRT**

Where –

P = Pressure in atm

V = Volume in litres

n = Number of gaseous moles

R = Gas constant = 0.0821 L atm mol⁻¹ K⁻¹

T = Temperature in kelvin

$$P = \frac{n}{V} RT = \text{active mass} \times RT$$

$$\frac{n}{V} = \text{molar concentration or active mass}$$

$$P_A = [A]RT, P_B = [B]RT, P_C = [C]RT \text{ and } P_D = [D]RT$$

Put all these values in K_P expression

$$\text{So } K_P = \frac{[C]^{n_1} (RT)^{n_1} \times [D]^{n_2} (RT)^{n_2}}{[A]^{m_1} (RT)^{m_1} \times [B]^{m_2} (RT)^{m_2}} = \frac{[C]^{n_1} [D]^{n_2}}{[A]^{m_1} [B]^{m_2}} \times \frac{(RT)^{n_1+n_2}}{(RT)^{m_1+m_2}}$$

$$K_P = K_C (RT)^{(n_1+n_2)-(m_1+m_2)}$$

$$\Delta n_g = (n_1 + n_2) - (m_1 + m_2)$$

= Sum of stoichiometric coefficient of gaseous products

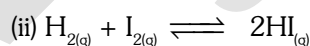
– sum of stoichiometric coefficient of gaseous reactants

$$K_P = K_C (RT)^{\Delta n_g}$$

- The K_C is expressed by the units $(\text{mol L}^{-1})^{\Delta n_g}$ and K_P by $(\text{atm})^{\Delta n_g}$.
- Three cases may arise :-

(a) When $\Delta n_g = 0$

$$K_P = K_C (RT)^0 = K_C$$

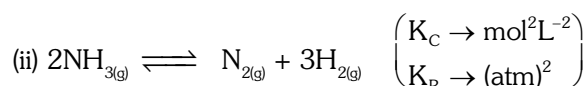
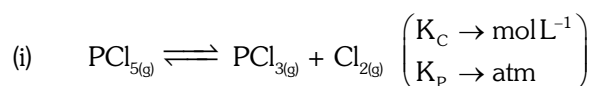


- K_C and K_P are unit less in this case.

(b) When $\Delta n_g = +ve$

$$K_P > K_C$$

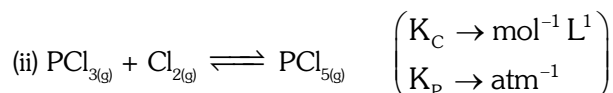
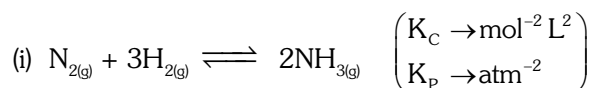
For example :



(c) When $\Delta n_g = -ve$

$$K_p < K_c$$

For example :-



(d) Special point :

$$\text{If } T = \frac{1}{R} \text{ then}$$

$$K_p = K_c \left(R \times \frac{1}{R} \right)^{\Delta n_g}$$

$$K_p = K_c (1)^{\Delta n_g}$$

$$K_p = K_c$$

For any value of Δn_g

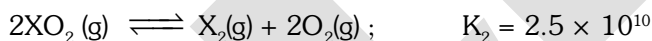
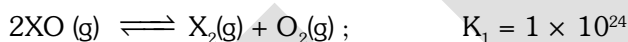
(C) Application of K -

Stability of reactants and products :

Stability of reactants increases when value of K decreases

Stability of products increases when value of K increases

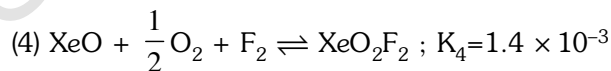
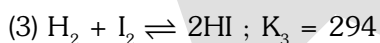
Ex : In the following reactions which one oxide is more stable.



$\therefore K_1 > K_2$ So the stability of $XO_2 > XO$

BEGINNER'S BOX-2

1. In which of the following reaction product is more stable :-



2. Equilibrium constant is :-

$$(1) \frac{k_b}{k_f}$$

$$(2) \frac{k_f}{k_b}$$

$$(3) k_f \times k_b$$

$$(4) \frac{1}{k_f k_b}$$

3. At 527°C, the reaction $NH_3(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{3}{2} H_2(g)$ has $K_c = 4$ then what is the value of K_p for the same reaction :-

$$(1) 16 \times (800 R)^2$$

$$(2) \left(\frac{800R}{4} \right)^{-2}$$

$$(3) \left(\frac{1}{4 \times 800 R} \right)^2$$

(4) None of these

4. For the equilibrium $\text{SO}_2\text{Cl}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$, what is the temperature at which $\frac{K_p(\text{atm})}{K_c(\text{M})} = 3$:-

- (1) 0.027 K (2) 0.36 K (3) 36.54 K (4) 273 K

(D) Factors affecting the equilibrium constant –

- (a) Temperature :** The value of equilibrium constant changes with the change of temperature. If K_1 and K_2 be the equilibrium constants of a reaction at absolute temperatures T_1 and T_2 and ΔH is the change in enthalpy then

$$\log\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^\circ}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] \text{ or } \log K_2 - \log K_1 = \frac{\Delta H^\circ}{2.303R} \left[\frac{T_2 - T_1}{T_1 \cdot T_2} \right] \text{ (According to van't hof equation)}$$

If the temperature T_2 is higher than T_1 then $\left(\frac{T_2 - T_1}{T_1 \cdot T_2} \right) > 0$.

- (i)** When $\Delta H = +ve$ (endothermic reaction)

$$\log K_2 - \log K_1 > 0 \Rightarrow \log K_2 > \log K_1$$

$$\Rightarrow K_2 > K_1$$

$\therefore$ The value of equilibrium constant increases when temperature increases in case of endothermic reactions.

- (ii)** When $\Delta H = -ve$ (exothermic reaction)

$$\log K_2 - \log K_1 < 0$$

$$\Rightarrow \log K_2 < \log K_1$$

$$\Rightarrow K_2 < K_1$$

$\therefore$ The value of equilibrium constant decreases when temperature increases in the case of exothermic reactions.

(b) The mode of representation of the reaction :

Consider the reversible chemical equilibrium reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$

$$\text{Equilibrium constant } K_c = \frac{[\text{C}][\text{D}]}{[\text{A}][\text{B}]}$$

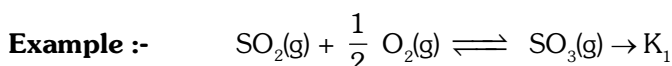
If the reaction is reversed $\text{C} + \text{D} \rightleftharpoons \text{A} + \text{B}$

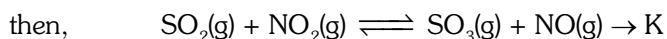
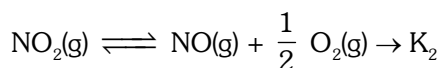
$$\text{Equilibrium constant } K'_c = \frac{[\text{A}][\text{B}]}{[\text{C}][\text{D}]}$$

The equilibrium constant K'_c is actually the reciprocal of K_c

$$\text{Thus, the two equilibrium constants are related as } \rightarrow K'_c = \frac{1}{K_c}$$

- (c) Multi step reaction :** If a reaction can be expressed as the sum of two or more reactions then overall K_c will be equal to the product of the individual equilibrium constants of the reactions.





So, $K = K_1 \times K_2$

(d) Stoichiometry of the reaction :-

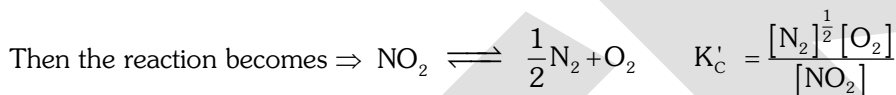
When a reversible reaction is multiplied or divided by a coefficient then the value of equilibrium constant will be numerically different in these cases.

For example the dissociation of NO_2 can be represented as :



$$K_c = \frac{[\text{N}_2][\text{O}_2]^2}{[\text{NO}_2]^2}$$

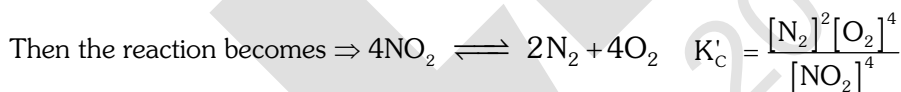
(1) If reaction (i) is divided by 2 -



Thus, the two equilibrium constants are related as $K'_c = \sqrt{K_c}$

So if reaction is divided by n then $K'_c = (K_c)^{\frac{1}{n}}$

(2) If reaction (i) is multiplied by 2



Thus, the two equilibrium constants are related as $K'_c = (K_c)^2$

So if reaction is multiplied by n then $K'_c = (K_c)^n$

GOLDEN KEY POINTS

● **Factors not affecting the equilibrium constant :**

The value of equilibrium constant is independent of the following factors—

- (a) Concentration of reactants and products. (b) Pressure
- (c) Volume (d) The presence of a catalyst.
- (e) Presence of inert materials.

● **The value of equilibrium constant depends only on temperature.**

Illustrations

Illustration 4. $\text{XeF}_6 + \text{H}_2\text{O} \rightleftharpoons \text{XeOF}_4 + 2\text{HF}$ constant = K_1 , $\text{XeO}_4 + \text{XeF}_6 \rightleftharpoons \text{XeOF}_4 + \text{XeO}_3\text{F}_2$ constant = K_2 . Then equilibrium constant for the reaction $\text{XeO}_4 + 2\text{HF} \rightleftharpoons \text{XeO}_3\text{F}_2 + \text{H}_2\text{O}$ will be—

- (1) $\frac{K_1}{K_2}$ (2) $K_1 + K_2$ (3) $\frac{K_2}{K_1}$ (4) $\frac{K_1}{(K_2)^2}$

Solution

Ans. (3)

Illustration 5 **Assertion:-** In the presence of catalyst, the value of equilibrium constant K increases.

Reason :- Catalysts increases the rate of forward and backward reaction to same extent.

- (1) A (2) B (3) C (4) D

Solution **Ans. (4)**

3.3 DEGREE OF DISSOCIATION

It is the fraction of moles of reactant dissociated $\alpha = \frac{x}{a}$

$$\% \alpha = \frac{x}{a} \times 100$$

Where α = Degree of dissociation
 x = Number of dissociated moles
 a = Initial number of moles (given)

Illustrations

Illustration 6 40% of PCl_5 is not dissociated at 300°C . The reaction is carried out in a flask of 1 litre capacity. The value of K_c would be :-

- (1) 3.2 (2) 1.6 (3) $(3.2)^{-1}$ (4) 0.9

Solution **Ans. (4)**

Illustration 7 In the beginning of the reaction, $\text{A} \rightleftharpoons \text{B} + \text{C}$, 2 moles of A are taken, out of which 0.5 moles gets dissociated. What is the amount of dissociation of A ?

- (1) 0.5 (2) 1 (3) 0.25 (4) 4.2

Solution **Ans. (3)**

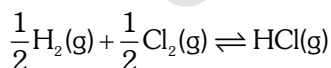
	A	$\rightleftharpoons$	B	+	C	
Initially	2		0		0	$x = 0.5$
Moles at eq.	$2 - x$		x		x	
	$2 - 0.5$		0.5		0.5	

Since, two moles dissociated into 0.5

Therefore, one mole will dissociated into 0.25

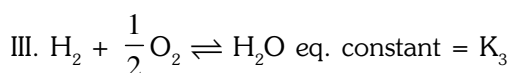
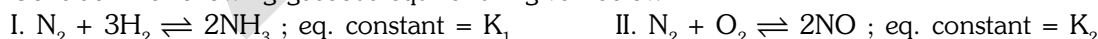
BEGINNER'S BOX-3

1. The equilibrium constant (K_c) for the reaction $2\text{HCl}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{Cl}_2(\text{g})$ is 4×10^{-34} at 25°C . What is the equilibrium constant for the reaction :-



- (1) 2×10^{-17} (2) 2.5×10^{33} (3) 5×10^6 (4) None of these

2. Consider the following gaseous equilibrium given below



The equilibrium constant for the reaction $2\text{NH}_3 + \frac{5}{2}\text{O}_2 \rightleftharpoons 2\text{NO} + 3\text{H}_2\text{O}$ in terms of K_1 , K_2 and K_3 will be :-

- (1) $K_1 K_2 K_3$ (2) $\frac{K_1 K_2}{K_3}$ (3) $\frac{K_1 K_3^2}{K_2}$ (4) $\frac{K_2 K_3^3}{K_1}$

3. Using molar concentrations, what is the unit of K_c for the reaction $\text{CH}_3\text{OH}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 2\text{H}_2(\text{g})$:-
 (1) M^{-2} (2) M^2 (3) M^{-1} (4) M
4. If temperature is increased then equilibrium constant will be :-
 (1) Increased
 (2) Decreased
 (3) Remains constant
 (4) May increased or decreased depends on exothermic or endothermic nature
5. What will be the equilibrium constant at 127°C . If equilibrium constant at 27°C is 4 for reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$; $\Delta H = -46.06 \text{ kJ}$:-
 (1) 4×10^{-2} (2) 2×10^{-3} (3) 10^2 (4) 4×10^2
6. In which of the following equilibrium equation, $K_p > K_c$
 (1) $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$ (2) $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$
 (3) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ (4) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
7. If $\text{CoO}(\text{s}) + \text{H}_2(\text{g}) \rightleftharpoons \text{Co}(\text{s}) + \text{H}_2\text{O}(\text{g})$, $K_1 = 60$; $\text{CoO}(\text{s}) + \text{CO}(\text{g}) \rightleftharpoons \text{Co}(\text{s}) + \text{CO}_2(\text{g})$, $K_2 = 180$ then the equilibrium constant of the reaction $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g})$ will be—
 (1) 0.44 (2) 0.11 (3) 0.22 (4) 0.33

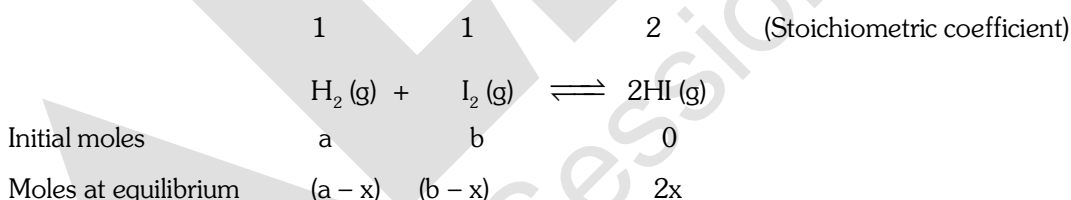
3.4 APPLICATIONS OF LAW OF MASS ACTION

[Relation of dissociation (x) with volume (V) and pressure (P)]

(A) Homogeneous Gaseous Reactions of Type-I ($\Delta n_g = 0$)

Synthesis of HI :-

- (i) **Expression for K_c** : The formation of HI from H_2 and I_2 is represented by following reaction



Let us start with 'a' moles of H_2 and 'b' moles of I_2 in a closed bulb of V volume. If at equilibrium x moles of each of H_2 and I_2 have reacted, then 2x moles of HI will be formed so active masses.

$$[\text{H}_2] = \frac{(a-x)}{V}; [\text{I}_2] = \frac{(b-x)}{V}; [\text{HI}] = \frac{2x}{V}$$

Applying law of mass action

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{a-x}{V}\right)\left(\frac{b-x}{V}\right)}$$

$$\Rightarrow K_c = \frac{4x^2}{(a-x)(b-x)}$$

- When $a = b = 1$, x becomes degree of formation of HI or degree of dissociation of H_2 (or I_2). $K_C = \frac{4x^2}{(1-x)^2}$

Let if $x \ll 1$ then $1-x \approx 1$ So $K_C = 4x^2$

$$x = \sqrt{\frac{K_C}{4}} \quad \text{i.e.} \quad \boxed{x \propto \sqrt{V}}$$

At equilibrium the degree of dissociation is independent to the volume.

- (ii) **Expression for K_p** : The equilibrium constant K_p can also be calculated considering partial pressures of reactants and products at equilibrium.

Total number of moles at equilibrium = $(a-x) + (b-x) + 2x = (a+b)$

If total pressure of the system at equilibrium be P then

$$\text{Partial pressure of } H_2 = \frac{(a-x)}{(a+b)} P ; \text{ Partial pressure of } I_2 = \frac{(b-x)}{(a+b)} P ; \text{ Partial pressure of HI} = \frac{2x}{(a+b)} P$$

$$K_p = \frac{(p_{HI})^2}{(p_{H_2})(p_{I_2})} = \frac{\left(\frac{2x}{a+b}\right)^2 P^2}{\left(\frac{a-x}{a+b}\right)\left(\frac{b-x}{a+b}\right) P^2}$$

$$K_p = \frac{4x^2}{(a-x)(b-x)} \quad \text{Thus } K_p = K_C$$

Let if $x \ll 1$ then $1-x \approx 1$ So $K_p = 4x^2$

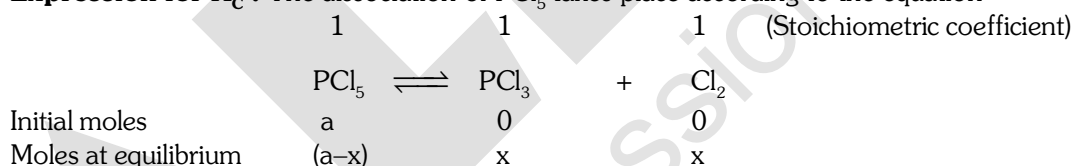
$$x = \sqrt{\frac{K_p}{4}} \quad \text{i.e.} \quad \boxed{x \propto \sqrt{P}}$$

At equilibrium the degree of dissociation is independent to the pressure also.

(B) Homogeneous Gaseous reactions of Type-II ($\Delta n_g > 0$)

Dissociation of PCl_5 :-

- (i) **Expression for K_C** : The dissociation of PCl_5 takes place according to the equation



Let a moles of PCl_5 be taken in a closed vessel of volume V . At equilibrium x moles of PCl_5 are dissociated into x moles of each PCl_3 and Cl_2 .

$$[PCl_5] = \frac{(a-x)}{V} ; \quad [PCl_3] = \frac{x}{V} ; \quad [Cl_2] = \frac{x}{V}$$

$$\text{Apply law of mass action } K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left(\frac{x}{V}\right)\left(\frac{x}{V}\right)}{\left(\frac{a-x}{V}\right)} = \frac{x^2}{(a-x)V}$$

- When $a = 1$, x becomes degree of dissociation (α) $K_C = \frac{\alpha^2}{(1-\alpha)V}$

$$\text{If } \alpha \ll 1 \text{ then } 1-\alpha \approx 1 \quad K_C \approx \frac{\alpha^2}{V} \quad \text{or} \quad \boxed{\alpha \propto \sqrt{V}}$$

The degree of dissociation of PCl_5 at equilibrium is directly proportional to the square root of the volume.

(ii) **Expression for K_p :**

Let the total pressure at equilibrium be P .

Total number of moles at equilibrium = $(a - x) + x + x = a + x$

$$p_{\text{PCl}_5} = \left(\frac{a-x}{a+x} \right) P, \quad p_{\text{PCl}_3} = \left(\frac{x}{a+x} \right) P, \quad p_{\text{Cl}_2} = \left(\frac{x}{a+x} \right) P$$

Apply law of mass action

$$K_p = \frac{p_{\text{PCl}_3} \cdot p_{\text{Cl}_2}}{p_{\text{PCl}_5}} = \frac{x^2 P}{(a+x)(a-x)}$$

When $a = 1$, x becomes degree of dissociation (α)

$$K_p = \frac{\alpha^2 P}{(1+\alpha)(1-\alpha)} = \frac{\alpha^2 P}{1-\alpha^2}$$

If $\alpha \ll 1$ then $1 - \alpha^2 \approx 1$, $K_p \approx \alpha^2 P$

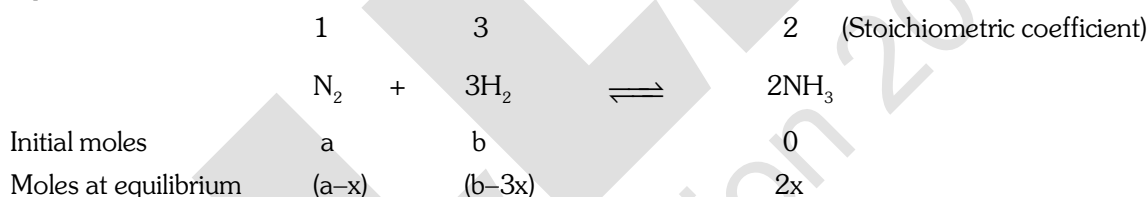
$$\alpha^2 \propto \frac{1}{P} \Rightarrow \boxed{\alpha \propto \frac{1}{\sqrt{P}}}$$

- The degree of dissociation of PCl_5 is inversely proportional to the square root of the total pressure at equilibrium.

(C) **Homogeneous Gaseous reactions of Type-III ($\Delta n_g < 0$)**

Synthesis of Ammonia :-

- (i) **Expression for K_c :** The formation of ammonia from nitrogen and hydrogen is represented by the equation :



Let us start with 'a' moles of N_2 and 'b' moles of H_2 in a closed vessel of Volume V . At equilibrium x moles of N_2 has combined with $3x$ moles of H_2 and produced $2x$ moles of NH_3 .

At equilibrium $[\text{N}_2] = \frac{(a-x)}{V}$; $[\text{H}_2] = \frac{(b-3x)}{V}$; $[\text{NH}_3] = \frac{2x}{V}$

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3} = \frac{\left(\frac{2x}{V} \right)^2}{\left(\frac{a-x}{V} \right) \left(\frac{b-3x}{V} \right)^3} = \frac{4x^2 V^2}{(a-x)(b-3x)^3}$$

If $a = 1$, $b = 3$ then $K_c = \frac{4x^2 V^2}{27(1-x)^4}$

If $x \ll 1$ then $1 - x \approx 1$

$$K_c = \frac{4x^2 V^2}{27} \quad \text{i.e.} \quad \boxed{x \propto \frac{1}{V}}$$

At equilibrium, the degree of dissociation is inversely proportional to the volume of vessel.

(ii) Expression for K_p :-

Total number of moles at equilibrium = $a - x + b - 3x + 2x = a + b - 2x$

If total pressure is P at equilibrium then

$$p_{N_2} = \frac{(a-x)}{(a+b-2x)}P, \quad p_{H_2} = \frac{(b-3x)}{(a+b-2x)}P, \quad p_{NH_3} = \frac{2x}{(a+b-2x)}P$$

According to Law of mass action

$$K_p = \frac{P_{NH_3}^2}{P_{N_2} P_{H_2}^3} = \frac{\left(\frac{2x}{a+b-2x}P\right)^2}{\left(\frac{a-x}{a+b-2x}\right)\left(\frac{b-3x}{a+b-2x}\right)^3}$$

$$K_p = \frac{4x^2(a+b-2x)^2}{(a-x)(b-3x)^3 P^2}$$

If $a = 1, b = 3$ then $K_p = \frac{16x^2(2-x)^2}{27(1-x)^4 P^2}$

If $x \ll 1$ then $2-x \approx 2$ and $1-x \approx 1$

$$K_p = \frac{64x^2}{27P^2} \text{ i.e. } x^2 \propto P^2 \Rightarrow \boxed{x \propto P}$$

At equilibrium, the degree of dissociation is directly proportional to the pressure.

GOLDEN KEY POINTS

- If inert gas mixed at constant temperature and constant volume in an equilibrium chemical reaction then total number of moles of gases are present in a container increases i.e. total pressure of gases increases but concentration in terms of mol L^{-1} and partial pressure of reacting substances are unchanged so dissociation (x) unchanged.

- $\alpha \propto (V)^{\frac{\Delta n_g}{\text{sum of stoichiometric coefficient of gaseous products}}}$ or $\alpha \propto \left(\frac{1}{P}\right)^{\frac{\Delta n_g}{\text{sum of stoichiometric coefficient of gaseous products}}}$

Effect	$\Delta n_g = 0$ $H_2 + I_2 \rightleftharpoons 2HI$ $x \propto (v)^0 \propto (P)^0$	$\Delta n_g > 0$ or +ve $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ $x \propto (v)^{1/2} \propto \left(\frac{1}{P}\right)^{1/2}$	$\Delta n_g < 0$ or -ve $N_2 + 3H_2 \rightleftharpoons 2NH_3$ $x \propto \left(\frac{1}{v}\right) \propto (P)$
(i) Pressure (increases)	x unchanged	x decreases	x increases
(ii) Volume (increases)	x unchanged	x increases	x decreases
(iii) Mixing of inert gas at			
(a) constant pressure	x unchanged	x increases	x decreases
(b) constant volume	x unchanged	x unchanged	x unchanged

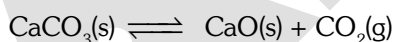
BEGINNER'S BOX-4

1. $A + B \rightleftharpoons C + D$ If initially of A and B are both are taken in equal amount but at equilibrium concentration of D will be twice of that of A then what will be the equilibrium constant of reaction :-
 (1) $\frac{4}{9}$ (2) $\frac{9}{4}$ (3) $\frac{1}{9}$ (4) 4
2. At a certain temperature, only 50% HI is dissociated at equilibrium in the reaction $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ The equilibrium constant for the reaction is :-
 (1) 0.25 (2) 1.0 (3) 3.0 (4) 0.5
3. The equilibrium constant K_p for the reaction $H_2(g) + CO_2(g) \rightleftharpoons H_2O(g) + CO(g)$ is 4.0 at 1660°C. Initially 0.80 mole H_2 and 0.80 mole CO_2 are injected into a 5.0 liter flask. What is the equilibrium concentration of $CO_2(g)$:-
 (1) 0.533 M (2) 0.0534 M
 (3) 5.34 M (4) None of these
4. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ for the reaction initially the mole ratio was 1 : 3 of N_2 : H_2 . At equilibrium 50% of each has reacted. If the equilibrium pressure is p, the partial pressure of NH_3 at equilibrium is :-
 (1) $\frac{p}{3}$ (2) $\frac{p}{4}$ (3) $\frac{p}{6}$ (4) $\frac{p}{8}$
5. For the reaction $H_2(g) + CO_2(g) \rightleftharpoons CO(g) + H_2O(g)$, if the initial concentration of $[H_2] = [CO_2]$ and x moles/litre of hydrogen is consumed at equilibrium, the correct expression of K_p is :-
 (1) $\frac{x^2}{(1-x)^2}$ (2) $\frac{(1+x)^2}{(1-x)^2}$ (3) $\frac{x^2}{(2+x)^2}$ (4) $\frac{x^2}{1-x^2}$

(D) Law of Mass Action as Applied to Heterogeneous Equilibrium :-

In such cases the active mass of pure solids and pure liquids is taken as unity and the value of equilibrium constant is determined by the gaseous substances only.

- (i) The dissociation of $CaCO_3$ in closed vessel.



$$K_p = p_{CO_2}$$

- (ii) $2H_2O(l) \rightleftharpoons 2H_2(g) + O_2(g)$

$$K_p = (p_{H_2})^2 (p_{O_2})$$

- (iii) $3Fe(s) + 4H_2O(g) \rightleftharpoons Fe_3O_4(s) + 4H_2(g)$

$$K_p = \frac{(p_{H_2})^4}{(p_{H_2O})^4}$$

Illustrations

Illustration 8 Two sample of HI each of 5 g were taken separately into vessels of volume 5 and 10 litres respectively at 27°C. The extent of dissociation of HI will be :-

- (1) More in 5 litre vessel (2) More in 10 litre vessel
(3) Equal in both vessel (4) None of these

Solution **Ans. (3)**

Illustration 9 What will be the amount of dissociation, if the volume is increased 16 times of initial volume in the reaction $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$?

- (1) 4 times (2) $\frac{1}{4}$ times
(3) 2 times (4) $\frac{1}{5}$ times

Solution **Ans. (1)**

$x \propto \sqrt{V}$ or $x \propto \sqrt{16}$ Thus, 4 times.

Illustration 10 **Assertion:-** For the reaction, $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$, increase in pressure at equilibrium has no effect on the reaction.

Reason :- The reaction is not accompanied by any change in number of moles of gaseous species.

- (1) A (2) B (3) C (4) D

Solution **Ans. (1)**

3.5 LE-CHATELIER'S PRINCIPLE

PRINCIPLE :- According to this principle, if a system at equilibrium is subjected to a change of concentration, pressure or temperature then the equilibrium is shifted in such a way as to nullify the effect of change.

- Le-Chatelier's principle is applicable for both chemical and physical equilibrium.

(A) CHEMICAL EQUILIBRIUM

(a) Change in concentration:-

In an equilibrium increasing the concentrations of reactants results in shifting the equilibrium in favour of products while increasing concentrations of the products results in shifting the equilibrium in favour of the reactants.

(b) Change in pressure :-

When the pressure on the system is increased, the volume decreases proportionately i.e. the total number of moles present per unit volume increases. According to Le-Chatelier's principle, the equilibrium shifts in that direction in which there is decrease in number of moles.

- If there is no change in number of moles of gases in a reaction then a pressure change does not affect the equilibrium.

(c) Change in temperature :-

If the temperature of the system at equilibrium is increased then reaction will proceed in that direction in which heat can be used. Thus increase in temperature will favour the forward reaction for endothermic reaction.

Similarly, increase in temperature will favour the backward reaction for exothermic reactions.

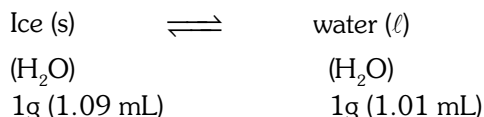
(B) **PHYSICAL EQUILIBRIUM**

Physical reaction :- Those reaction in which change in only and only physical states (solid, liquid and gas) of substance takes place without any chemical change, is called physical reaction.

Example :

(a) **Ice-water system (melting of ice) :**

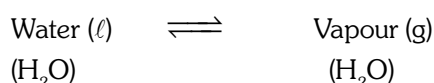
Melting of ice is accompanied by absorption of heat (endothermic) and decrease in volume



Hence both increase of temperature and pressure will favour the melting of ice into water.

(b) **Water-water vapour system (Vapourisation of water) :**

Vapourisation of water is an endothermic and condensation of vapour into water is an exothermic reaction:



- The equilibrium shifts towards right side when the temperature is increased so rise in temperature will increase the vapour.
- The equilibrium shifts towards left side when the pressure is increased (i.e. volume is decreased) so increase in pressure will favour the rate of condensation of vapour into water.
- Thus favourable conditions for conversion of water into vapour are high temperature and low pressure.

(c) **Solubility of gases :**



- **Effect of pressure** → Solubility of such gases increases with increasing pressure which dissolves in a solvent with a decrease in volume.

Illustrations

Illustration 11 On applying pressure to the equilibrium $\text{ice} \rightleftharpoons \text{water}$, which phenomenon will happen :

- (1) More ice will be formed
- (2) More water will be formed
- (3) Equilibrium will not be disturbed
- (4) Water will evaporate

Solution **Ans. (2)**

Illustration 12 Which of the following conditions should be more favourable for increasing the rate of forward reaction in the equilibrium $\text{H}_2 \rightleftharpoons \text{H} + \text{H}$ ($\Delta H = +ve$) ?

- (1) 2000° C temperature and 760 mm of Hg pressure.
- (2) 3500° C temperature and 100 cm of Hg pressure.
- (3) 3500° C temperature and 1 mm of Hg pressure.
- (4) All are wrong.

Solution **Ans. (3)**

In $\text{H}_2 \rightleftharpoons \text{H} + \text{H}$, heat has to be provided to dissociate H_2 into H . Therefore, the reaction is endothermic (ΔH will positive). So, temperature should be high. Since, one mole of H_2 forms two atoms of H , so volume is increasing (Δn is positive) so pressure should be low for increasing the rate of forward reaction.

Illustration 13 Assertion : $\text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g}) + \text{heat}$; Forward reaction is favoured at low temperature and high pressure.

Reason : Reaction is endothermic.

- (1) A (2) B (3) C (4) D

Solution Ans. (3)

Illustration 14. The reaction $\text{CaCO}_{3(\text{s})} \rightleftharpoons \text{CaO}_{(\text{s})} + \text{CO}_{2(\text{g})}$ goes to completion in lime kiln because :

- (1) of the high temperature (2) CaO is more stable than CaCO_3
(3) CaO is not dissociated (4) CO_2 escapes continuously

Solution Ans. (4)

Illustration 15. Assertion: For a reversible exothermic reaction, extent of reaction decreases with increase in temperature.

Reason: In reversible exothermic reaction temperature is favourable for more formation of product.

Solution Ans. (3)

BEGINNER'S BOX-5

- Which of the following equilibrium remains unaffected by a change in pressure (or volume) ?
 (1) $2\text{NOCl}(\text{g}) \rightleftharpoons 2\text{NO}(\text{s}) + \text{Cl}_2(\text{g})$ (2) $\text{H}_2(\text{g}) + \text{CO}_2(\text{g}) \rightleftharpoons \text{H}_2\text{O}(\text{g}) + \text{CO}(\text{g})$
 (3) $3\text{PbS}(\text{s}) + 3\text{O}_2(\text{g}) \rightleftharpoons 2\text{PbO}(\text{s}) + 2\text{SO}_2(\text{g})$ (4) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
- Consider the following equilibrium system; $2\text{SO}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightleftharpoons 2\text{SO}_{3(\text{g})}$; some inert gas is added to the above system at constant volume. Predict which of the following is true?
 (1) More of SO_3 is produced.
 (2) Less SO_2 is produced.
 (3) Addition of inert gas does not affect equilibrium.
 (4) system moves to new equilibrium position which cannot be predicted theoretically.
- Which of the following is not true for the equilibrium reaction; $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$; $\Delta H = 180 \text{ kJ mol}^{-1}$.
 (1) the formation of NO is increased at higher temperature.
 (2) The volume change at constant pressure does not affect the equilibrium.
 (3) The pressure change at constant volume does not affect the equilibrium.
 (4) The formation of NO is decreased at higher temperature.
- Consider the following equilibrium system; $\text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$; set up in a cylinder fitted with a piston. Some inert gas is added and the piston is moved outwards to keep the total gaseous pressure constant. Predict which of the following is true?
 (1) Addition of inert gas does not affect the equilibrium.
 (2) Less $\text{SO}_3(\text{g})$ is produced.
 (3) More $\text{SO}_3(\text{g})$ is produced.
 (4) The system moves to new equilibrium position which cannot be predicted theoretically.
- When a volatile liquid is introduced into an evacuated closed vessel at a particular temperature, both evaporation and condensation take place simultaneously. The system reaches equilibrium state when—
 (1) The liquid is completely transformed into the corresponding vapour
 (2) Equal amounts of liquid and vapour are present in the system
 (3) The rate of evaporation becomes equal to the rate of condensation
 (4) Liquid cannot be converted into vapour and vice versa.
- Which of the following equilibrium is dynamic ?
 (1) $\text{Solid} \rightleftharpoons \text{Liquid}$ (2) $\text{Liquid} \rightleftharpoons \text{Vapor}$ (3) $\text{Solid} \rightleftharpoons \text{Vapor}$ (4) All of these

7. Which of the following is not true for solid-liquid equilibrium ?
 (1) It can be established at any given temperature.
 (2) The mass of solid does not change with time.
 (3) The mass of liquid does not change with time.
 (4) There is no exchange of heat between the system and its surrounding.
8. Which of the following substances can be placed in a closed vessel to establish (solid $\rightleftharpoons$ vapour) equilibrium?
 (1) Ammonium chloride (2) Camphor (3) Iodine (4) All of these
9. Which of the following solutions kept in contact with undissolved solute is an example of solid-solution equilibrium?
 (1) Aqueous solution (2) Saturated solution (3) Unsaturated solution (4) Nonaqueous solution
10. Which of the following is correct regarding the gas-solution equilibrium ?
 (1) The solubility of the dissolved gas increases with the increase of pressure and decreases with the increase of temperature.
 (2) The solubility of the dissolved gas increases with the increase of pressure as well as temperature.
 (3) The solubility of the dissolved gas decreases with the increase of pressure and increases with the increase of temperature.
 (4) The solubility of the dissolved gas decreases with the increase of pressure as well as temperature.

3.6 REACTION QUOTIENT (Q)

Consider a general homogeneous reversible reaction :



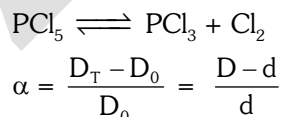
$$\text{Reaction Quotient (Q)} = \frac{[C]^{n_1} [D]^{n_2}}{[A]^{m_1} [B]^{m_2}}, \text{ (Applied at any stage of reaction)}$$

$$\text{Equilibrium constant K} = \frac{[C]^{n_1} [D]^{n_2}}{[A]^{m_1} [B]^{m_2}}, \text{ (Applied only at equilibrium state)}$$

- (i) When $Q = K$ then reaction is in equilibrium state.
 (ii) When $Q < K$ then rate of forward reaction increases.
 (iii) When $Q > K$ then rate of backward reaction increases.

3.7 Calculation of degree of dissociation from vapour density :-

Ex.



Where : D_T or D = Principle or theoretical vapour density or normal vapour density
 D_0 or d = Observed or practical vapour density or experimental vapour density or vapour density at higher temp.
 α = Degree of dissociation

$\text{Vapour density} = \frac{\text{Molecular weight}}{2}$

Reversible reaction	$\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$			Total moles	Volume at NTP	Vapour density
Let initial moles	1	0	0	1	$V_T = 22.4$	$D_T \propto \frac{1}{V_T}$
Moles at equilibrium	$(1-\alpha)$	α	α	$1+\alpha$	$V_0 = 22.4 (1+\alpha)$	$D_0 \propto \frac{1}{V_0}$

If α is the degree of dissociation

$$\frac{D_T}{D_0} = \frac{V_0}{V_T} = \frac{22.4(1+\alpha)}{22.4}$$

$$\frac{D_T}{D_0} = 1 + \alpha \quad \text{or} \quad \alpha = \frac{D_T}{D_0} - 1 \quad \boxed{\alpha = \frac{D_T - D_0}{D_0}}$$

So for a general reversible reaction $n_1 A \rightleftharpoons n_2 B + n_3 C$

$$\alpha = \frac{n_1 \left(\frac{D_T - D_0}{D_0} \right)}{\Delta n} \quad \Delta n = (n_2 + n_3) - (n_1)$$

OR

$$\alpha = \frac{n_1 \left(\frac{M_T - M_0}{M_0} \right)}{\Delta n}$$

M_T = Theoretical molecular weight

M_0 = Observed or experiment molecular weight

Illustrations

Illustration 16 The vapour density of undecomposed N_2O_4 is 46. When heated, vapour density decreases to 24.5 due to its dissociation to NO_2 . The percentage dissociation of N_2O_4 at the final temperature is -
 (1) 87 (2) 60 (3) 40 (4) 70

Solution **Ans. (1)**

Illustration 17 If PCl_5 is 80% dissociated at 250°C then its vapour density at room temperature will be
 (1) 56.5 (2) 104.25 (3) 101.2 (4) 52.7

Solution **Ans. (2)**

$$\alpha = \frac{D_T - D_0}{D_0}; D_T = \frac{\text{Molecular weight}}{2}$$

Vapour density at room temperature (D_T) is 104.25, which is fixed.

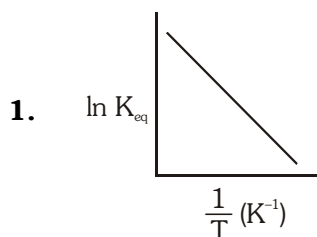
Illustration 18 **Assertion:-** For a reaction, reaction quotient (Q) is equal to K when the reaction is in equilibrium.

Reason:- If a catalyst is added to the reaction at equilibrium, the value of Q remains no longer equal to K.

(1) A (2) B (3) C (4) D

Solution **Ans. (3)**

BEGINNER'S BOX-6



According to this graph reaction will be :-

- (1) Endothermic (2) Exothermic
(3) Spontaneous at room temperature (4) ΔH is negligible

2. For the reaction $\text{SO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$

If $K_p = K_c (\text{RT})^x$, when the symbols have usual meaning the value of x is (assuming ideality) :-

- (1) - 1 (2) $-\frac{1}{2}$ (3) $+\frac{1}{2}$ (4) + 1

3. For the equilibrium $\text{C}(\text{s}) + \text{CO}_2(\text{g}) \rightleftharpoons 2\text{CO}(\text{g})$ $K_p = 63 \text{ atm}$ at 1000 K. If at equilibrium $P_{\text{CO}} = 10 P_{\text{CO}_2}$ then total pressure at equilibrium is :-

- (1) 6.30 atm (2) 0.693 atm (3) 6.93 atm (4) 69.3 atm

4. $\text{A}(\text{g})$ is 90 % converted in to B according to the reaction $\text{A}(\text{g}) \rightleftharpoons 3\text{B}(\text{g})$ value of $\left(\frac{D}{d}\right)$ at this point is :-

- (1) 1.0 (2) 2.0 (3) 2.5 (4) 2.8

5. What will be the direction of reaction if concentration of H_2 , I_2 and HI are 2 mol L^{-1} , 2 mol L^{-1} and 8 mol L^{-1} respectively. K_c for reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ is 4.

- (1) forward direction (2) backward direction
(3) equilibrium condition (4) reaction will be completed

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4						
	Ans.	4	4	3	1						
BEGINNER'S BOX-2	Que.	1	2	3	4						
	Ans.	3	2	4	3						
BEGINNER'S BOX-3	Que.	1	2	3	4	5	6	7			
	Ans.	4	4	2	4	1	1	4			
BEGINNER'S BOX-4	Que.	1	2	3	4	5					
	Ans.	4	1	2	1	1					
BEGINNER'S BOX-5	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	2	3	4	2	3	4	1	4	2	1
BEGINNER'S BOX-6	Que.	1	2	3	4	5					
	Ans.	1	2	3	4	2					

EXERCISE-I (Conceptual Questions)**Build Up Your Understanding****FACTORS AFFECTING RATE OF REACTION**

- In an elementary reaction $A + 2B \rightarrow 2C + D$. If the concentration of A is increased four times and B is decreased to half of its initial concentration then the rate becomes
 - Twice
 - Half
 - Unchanged
 - One fourth of the rate
- The role of catalyst in a chemical reaction is :-
 - To help attain equilibrium in a shorter time.
 - To lower the activation energy.
 - To shift the equilibrium in such a way as to increase the concentration of the product
 - Both 1 & 2

EQUILIBRIUM AND CHEMICAL PROCESS

- $x \rightleftharpoons y$ reaction is said to be in equilibrium, when:-
 - Only 10% conversion x to y takes place.
 - Complete conversion of x to y takes place
 - Conversion of x to y is only 50% complete
 - The rate of change of x to y is just equal to the rate of change of y to x in the system
- In the chemical reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ at equilibrium, state whether :-
 - Equal volumes of N_2 & H_2 are reacting
 - Equal masses of N_2 & H_2 are reacting
 - The reaction has stopped
 - The same amount of ammonia is formed as is decomposed into N_2 and H_2
- Active mass of 5 g CaO :-

(1) 56	(2) 1
(3) 3.5	(4) 2
- Ratio of active masses of 22g CO_2 , 3g H_2 and 7g N_2 in a gaseous mixture :-
 - 22 : 3 : 7
 - 0.5 : 3 : 7
 - 1 : 3 : 1
 - 1 : 3 : 0.5

- Which of the following example shows effect of catalyst on reversible reaction
 - It gives new reaction path with low activation energy.
 - It shifts equilibrium right side.
 - It decrease kinetic energy of activated molecules.
 - It decrease rate of backward reaction.
- Select the correct statement from the following :
 - Equilibrium constant changes with addition of catalyst
 - Catalyst increases the rate of forward reaction.
 - The ratio of mixture at equilibrium does not changed by catalyst
 - Catalyst are active only in solution.
- In reversible chemical reaction equilibrium will be establish when -
 - Reactant completely converted into product
 - Rate of forward and backward reaction is equal
 - Minimum yield of product
 - concentration of reactant and product is equal

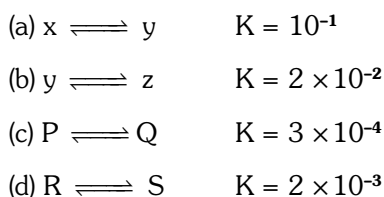
LAW OF MASS ACTION

- In a chemical equilibrium, the rate constant for the backward reaction is 7.5×10^{-4} and the equilibrium constant is 1.5. The rate constant for the forward reaction is:-
 - 2×10^{-3}
 - 5×10^{-4}
 - 1.12×10^{-3}
 - 9.0×10^{-4}
- The equilibrium concentration of $[B]_e$ for the reversible reaction $A \rightleftharpoons B$ can be evaluated by the expression:-

(1) $K_c [A]_e^{-1}$	(2) $\frac{k_f}{k_b} [A]_e^{-1}$
(3) $k_f k_b^{-1} [A]_e$	(4) $k_f k_b [A]^{-1}$
- In this reaction $Ag^+ + 2NH_3 \rightleftharpoons Ag(NH_3)_2^+$ at 298K molar concentration of Ag^+ , $Ag(NH_3)_2^+$ and NH_3 is 10^{-1} , 10^{-1} , and 10^3 . The value of K_c at 298K for this equilibrium :-

(1) 10^{-6}	(2) 10^6
(3) 2×10^{-3}	(4) 2×10^6

13. Equilibrium constant of some reactions are given as under :

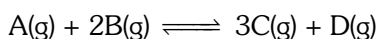


Initial concentration of the reactants for each reaction was taken to be equal :

Review the above reaction and indicate the reactions in which the reactants and products respectively were of highest concentration :-

- (1) d, c (2) c, a
 (3) a, d (4) b, c

14. At 1000 K, the value of K_p for the reaction :

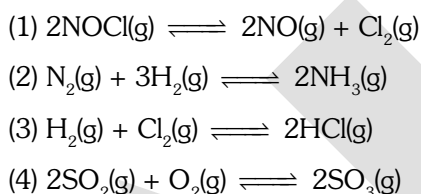


is 0.05 atm. The value of K_c in terms of R would be :-

- (1) 20000 R (2) 0.02 R
 (3) 5×10^{-5} R (4) $5 \times 10^{-5} \times R^{-1}$

15. For the reaction $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$ the partial pressure of CO and CO_2 are 2.0 and 4.0 atm respectively at equilibrium. The K_p for the reaction is
 (1) 0.5 (2) 4.0 (3) 8.0 (4) 1

16. For which reaction is $K_p = K_c$:-



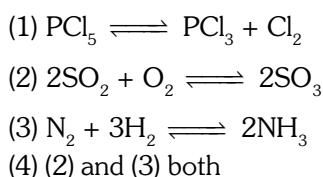
17. For the reaction



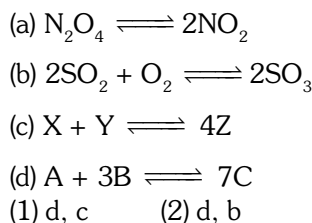
Which one is correct representation :-

- (1) $K_p = p_{H_2O}^2$ (2) $K_c = [H_2O]^2$
 (3) $K_p = K_c(RT)^2$ (4) All

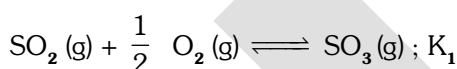
18. $\log \frac{K_p}{K_c} + \log RT = 0$ is true relationship for the following reaction:-



19. For which reaction at 298 K, the value of $\frac{K_p}{K_c}$ is maximum and minimum respectively:-



20. Consider the two gaseous equilibrium involving SO_2 and the corresponding equilibrium constants at 299 K



The value of the equilibrium constant are related by :-

- (1) $K_2 = \frac{1}{(K_1)^4}$ (2) $K_2 = K_1^4$
 (3) $K_2 = \left(\frac{1}{K_1}\right)^{\frac{1}{4}}$ (4) $K_2 = \frac{1}{K_1}$

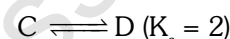
21. For the reactions :- $A \rightleftharpoons B ; K_C = 2$,



K_C for the reaction $A \rightleftharpoons D$:-

- (1) 12 (2) 4/3
 (3) 24 (4) 48

22. If $A \rightleftharpoons B (K_c = 3)$, $B \rightleftharpoons C (K_c = 5)$,

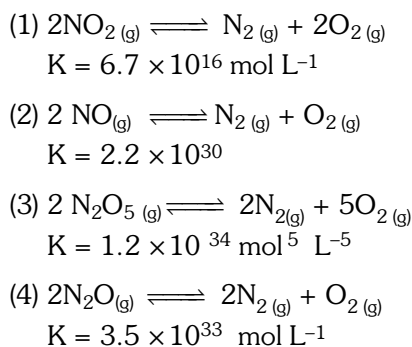


The value of equilibrium constant for the above reaction are given, the value of equilibrium constant

for $D \rightleftharpoons A$ will be:-

- (1) 15 (2) 0.3 (3) 30 (4) 0.03

23. Which Oxide of Nitrogen is most stable :-



- 24.** The equilibrium constant in a reversible reaction at a given temperature:-
 (1) Depends on initial concentration of the reactants.
 (2) Depends on the concentration of the products at equilibrium.
 (3) Does not depend on the initial concentrations.
 (4) It is not characteristic of the reaction.
- 25.** Which one of the following statements is correct about equilibrium constant:-
 (1) Equilibrium constant of a reaction changes with temperature.
 (2) Equilibrium constant of a reaction depends upon the concentration of reactants with which we start.
 (3) Equilibrium constant of a reaction,

$$3\text{Fe(s)} + 4\text{H}_2\text{O(g)} \rightleftharpoons \text{Fe}_3\text{O}_4\text{(s)} + 4\text{H}_2\text{(g)}$$
 is same whether, the reaction is carried out in an open vessel or a closed vessel.
 (4) Equilibrium constant of a reaction becomes double if the reaction is multiplied by 2 throughout.
- 26.** For a reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, the value of K_c does not depends upon :-
 (a) Initial concentration of the reactants
 (b) Pressure
 (c) Temperature
 (d) Catalyst
 (1) Only c (2) a, b, c
 (3) a, b, d (4) a, b, c, d
- 27.** For any reversible reaction if concentration of reactants increases then effect on equilibrium constant :-
 (1) Depends on amount of concentration
 (2) Unchange
 (3) Decrease
 (4) Increase
- 28.** Effect of increasing temperature on equilibrium constant is given by $\log K_2 - \log K_1 = \frac{-\Delta H}{2.303R} \left[\frac{1}{T_2} - \frac{1}{T_1} \right]$. Then for an endothermic reaction the false statement is:-
 (1) $\left[\frac{1}{T_2} - \frac{1}{T_1} \right] = \text{positive}$ (2) $\log K_2 > \log K_1$
 (3) $\Delta H = \text{positive}$ (4) $K_2 > K_1$
- 29.** The equilibrium constant for the reaction
 $\text{Br}_2 \rightleftharpoons 2\text{Br}$ at 500 K and 700 K are 1×10^{-10} and 1×10^{-5} respectively. The reaction is:-
 (1) Endothermic
 (2) Exothermic
 (3) Fast
 (4) Slow
- 30.** In an experiment the equilibrium constant for the reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ is K when the initial concentration of A and B each is 0.1 mol L^{-1} . Under the similar conditions in an another experiment if the initial concentration of A and B are taken 2 and 3 mol L^{-1} respectively then the value of equilibrium constant will be:-
 (1) $\frac{K}{2}$ (2) K
 (3) K^2 (4) $\frac{1}{K}$
- 31.** In system $\text{A(s)} \rightleftharpoons 2\text{B(g)} + 3\text{C(g)}$ at equilibrium if concentration of 'C' is doubled then concentration of B at equilibrium.
 (1) Double its original concentration
 (2) Half its original concentration
 (3) $2\sqrt{2}$ its original concentration
 (4) $\frac{1}{2\sqrt{2}}$ its original concentration
- 32.** For the reaction, $\text{H}_2\text{(g)} + \text{I}_2\text{(g)} \rightleftharpoons 2\text{HI(g)}$ equilibrium constant, K_p changes with :-
 (1) Temperature
 (2) Total pressure
 (3) Catalyst
 (4) Amount of H_2 and I_2 present
- 33.** The equilibrium constant (K_p) for the reaction $\text{PCl}_5\text{(g)} \rightleftharpoons \text{PCl}_3\text{(g)} + \text{Cl}_2\text{(g)}$ is 16. If the volume of the container is reduced to one-half its original volume, the value of K_p for the reaction at the same temperature will be :-
 (1) 32 (2) 64 (3) 16 (4) 4
- 34.** The equilibrium constant for the reaction :
 $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightleftharpoons 2\text{NO(g)}$ at 2000 K is 4×10^4 . In presence of catalyst the equilibrium is established ten times faster at the same temperature. What is the value of equilibrium constant in presence of catalyst :-
 (1) 40×10^{-4} (2) 4×10^{-4}
 (3) 4×10^4 (4) None

35. The equilibrium constant of the reaction $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ is 64. If the volume of the container is reduced to one fourth of its original volume, the value of the equilibrium constant will be
- (1) 16 (2) 32
(3) 64 (4) 128

36. If some He gas is introduced into the equilibrium $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$ at constant pressure and temperature then equilibrium constant of reaction :
- (1) Increase
(2) Decrease
(3) Unchange
(4) Nothing can be said

37. List X

- (A) Active mass
(B) Dynamic nature
(C) $\text{A} + \text{heat} \rightleftharpoons \text{B}$
(D) $\log(K_{p_2}/K_{p_1})$

List Y

- (I) $\Delta n = 0$
(II) Molar concentration
(III) Vant hoff's equation
(IV) adaptation if temperature increases

$$= \frac{\Delta H}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

- (E) $2\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 3\text{C}(\text{g})$ (v) Chemical equilibrium
Correct match list X and Y

- (1) A - (V), B - (III), C - (III), D - (I), E - (IV)
(2) A - (V), B - (IV), C - (III), D - (II), E - (I)
(3) A - (III), B - (V), C - (IV), D - (III), E - (I)
(4) None of these

DEGREE OF DISSOCIATION AND APPLICATION OF LAW OF MASS ACTION

38. For the reaction : $\text{P} \rightleftharpoons \text{Q} + \text{R}$. Initially 2 mol of P was taken. Up to equilibrium 0.5 mol of P was dissociated. What would be the degree of dissociation :-
- (1) 0.5 (2) 1 (3) 0.25 (4) 4.2
39. The dissociation of CO_2 can be expressed as $2\text{CO}_2 \rightleftharpoons 2\text{CO} + \text{O}_2$. If the 2 mol of CO_2 is taken initially and 40% of the CO_2 is dissociated completely. What is the total number of moles at equilibrium:-
- (1) 2.4 (2) 2.0 (3) 1.2 (4) 5

40. In $\text{A}_3(\text{g}) \rightleftharpoons 3\text{A}(\text{g})$ reaction, the initial concentration of A_3 is "a" mol L^{-1} . If x is degree of dissociation of A_3 . The total number of moles at equilibrium will be:-

- (1) $a - \frac{ax}{3}$ (2) $\frac{a}{3} - x$
(3) $\left(\frac{a - ax}{2} \right)$ (4) $a + 2ax$

41. In the reaction $2\text{P}(\text{g}) + \text{Q}(\text{g}) \rightleftharpoons 3\text{R}(\text{g}) + \text{S}(\text{g})$. If 2 mol each of P and Q taken initially in a 1 L flask. At equilibrium which is true:-
- (1) $[\text{P}] < [\text{Q}]$ (2) $[\text{P}] = [\text{Q}]$
(3) $[\text{Q}] = [\text{R}]$ (4) None of these

42. In a 13 L vessel initially following reaction occur $\text{C}(\text{s}) + \text{S}_2(\text{g}) \rightleftharpoons \text{CS}_2(\text{g})$ by 12 g C, 64 g S_2 , 76 g CS_2 at 1027°C temperature then total pressure is.
- (1) 200R (2) 158R (3) 100R (4) 79R

43. The reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ is studied in a one litre Vessel at 250°C . The initial concentration of A was 3n and of B was n. After equilibrium was attained then equilibrium concentration of C was found to be equal to equilibrium concentration of B. What is the concentration of D at equilibrium :-

- (1) $\frac{n}{2}$ (2) $\left(3n - \frac{n}{2} \right)$
(3) $\left(n + \frac{n}{2} \right)$ (4) n

44. $\text{X}_2 + \text{Y}_2 \rightleftharpoons 2\text{XY}$ reaction was studied at a certain temperature. In the beginning 1 mole of X_2 was taken in a one litre flask and 2 moles of Y_2 was taken in another 2 litre flask. What is the equilibrium concentration of X_2 and Y_2 ? (Given equilibrium concentration of $[\text{XY}] = 0.6 \text{ mol L}^{-1}$).

- (1) $\left(\frac{1}{3} - 0.3 \right), \left(\frac{2}{3} - 0.3 \right)$
(2) $\left(\frac{1}{3} - 0.6 \right), \left(\frac{2}{3} - 0.6 \right)$
(3) $(1 - 0.3), (2 - 0.3)$
(4) $(1 - 0.6), (2 - 0.6)$

45. If the initial pressure of N_2 and H_2 mixture in a closed apparatus is 100 atm. At equilibrium 20% of the mixture reacts then total pressure at the same temperature would be -
 (1) 100 (2) 90 (3) 85 (4) 80
46. In a 20 litre vessel initially 1 - 1 mole CO , H_2O , CO_2 is present, then for the equilibrium of $CO + H_2O \rightleftharpoons CO_2 + H_2$ following is true:-
 (1) H_2 , more than 1 mole
 (2) CO , H_2O , H_2 less than 1 mole
 (3) CO_2 & H_2O both more than 1 mole
 (4) All of these
47. 4 mol of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 then calculate total number of moles at equilibrium
 (1) 4.5 (2) 6 (3) 3 (4) 4
48. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
 In above reaction, at equilibrium condition mole fraction of PCl_5 is 0.4 and mole fraction of Cl_2 is 0.3. Then find out mole fraction of PCl_3
 (1) 0.3 (2) 0.7 (3) 0.4 (4) 0.6
49. If 8 mol of PCl_5 heated in a closed vessel of 10 L capacity and 25% of its dissociates into PCl_3 and Cl_2 at the equilibrium then value of K_p will be equal to:-
 (1) $P/30$ (2) $P/15$ (3) $2/3P$ (4) $3/2P$
50. In the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ the partial pressure of PCl_3 , Cl_2 and PCl_5 are 0.3, 0.2 and 0.6 atm respectively at equilibrium. If partial pressure of PCl_3 and Cl_2 was increased twice, what will be the partial pressure of PCl_5 is in atm at new equilibrium condition :-
 (1) 0.3 (2) 1.2 (3) 2.4 (4) 0.15
51. 'a' mol of PCl_5 , undergoes, thermal dissociation as: $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the mole fraction of PCl_3 at equilibrium is 0.25 and the total pressure is 2.0 atm. The partial pressure of Cl_2 at equilibrium is :-
 (1) 2.5 (2) 1.0 (3) 0.5 (4) None
52. In a 0.25 L tube dissociation of 4 mol of NO is take place. If its degree of dissociation is 10%. The value of K_p for reaction $2 NO \rightleftharpoons N_2 + O_2$ is :-
 (1) $\frac{1}{(18)^2}$ (2) $\frac{1}{(8)^2}$ (3) $\frac{1}{16}$ (4) $\frac{1}{32}$
53. In a chemical equilibrium $A + B \rightleftharpoons C + D$ when one mole each of the two reactants are mixed, 0.4 mol each of the products are formed. The equilibrium constant is :-
 (1) 1 (2) 0.36 (3) 2.25 (4) $\frac{4}{9}$
54. K_c for the esterification reaction :
 $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$ is 4. If 4 mol each of acid and alcohol are taken initially, what is the equilibrium concentration of the acid :-
 (1) $\frac{2}{3}$ (2) $\frac{4}{3}$ (3) $\frac{3}{4}$ (4) $\frac{3}{2}$
55. Evaluate K_p for the reaction : $H_2 + I_2 \rightleftharpoons 2HI$. If 2 moles each of H_2 and I_2 are taken initially. At equilibrium moles of HI are 2.
 (1) 2.5 (2) 4 (3) 0.25 (4) 1.0
56. 4 moles of A are mixed with 4 moles of B, when 2 mol of C are formed at equilibrium, according to the reaction, $A + B \rightleftharpoons C + D$. The equilibrium constant is :-
 (1) 4 (2) 1 (3) $\sqrt{2}$ (4) $\sqrt{4}$
57. Two moles of ammonia is introduced in a evacuated 500 mL vessel at high temperature. The decomposition reaction is :
 $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$
 At the equilibrium NH_3 becomes 1 mole then the K would be :-
 (1) 0.42 (2) 6.75 (3) 1.7 (4) 1.5
58. 4.5 mol each of hydrogen and iodine heated in a sealed 10 litre vessel. At equilibrium, 3 mol of HI were found. The equilibrium constant for $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ is:-
 (1) 1 (2) 10
 (3) 5 (4) 0.33
59. 1.50 mol each of hydrogen and iodine were placed in a sealed 10 L container maintained at 717 K. At equilibrium 1.25 mol each of hydrogen and iodine were left behind. The equilibrium constant, K_c for the reaction
 $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ at 717 K is
 (1) 0.4 (2) 0.16
 (3) 25 (4) 50

60. AB dissociates as $2AB(g) \rightleftharpoons 2A(g) + B_2(g)$
When the initial pressure of AB is 500 mm, the total pressure becomes 625 mm when the equilibrium is attained. Calculate K_p for the reaction assuming volume remains constant.
(1) 500 (2) 125 (3) 750 (4) 375

LE-CHATLIER'S PRINCIPLE

61. Cis -2- pentene $\rightleftharpoons$ Trans -2- pentene for the above equilibrium the value of standard free energy change at 400 K is $-3.67 \text{ kJ mol}^{-1}$. If excess of trans -2- pentene is added to the system then :-
(1) Additional trans -2- pentene will form
(2) Excess of cis -2- pentene will form
(3) Equilibrium will proceed in the forward
(4) Equilibrium will remain unaffected
62. When NaNO_3 is heated in a closed vessel, O_2 is liberated and NaNO_2 is left behind. At equilibrium -
(1) Addition of NaNO_3 favours forward reaction
(2) Addition of NaNO_2 favours reverse reaction
(3) Increasing pressure favours reverse reaction.
(4) Decreasing temperature favours forward reaction.
63. The equilibrium $2\text{SO}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{SO}_3(g)$ shifts forward if :-
(1) A catalyst is used.
(2) An adsorbent is used to remove SO_3 as soon as it is formed.
(3) Small amounts of reactants are used.
(4) None
64. In manufacture of NO, the reaction of N_2 and O_2 to form NO is favourable if :-
(1) Pressure is increased
(2) Pressure is decreased
(3) Temperature is increased
(4) Temperature is decreased
65. In which of the following equilibrium reactions, the equilibrium would shift to right side, if total pressure is decreased :-
(1) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (2) $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$
(3) $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ (4) $\text{H}_2 + \text{Cl}_2 \rightleftharpoons 2\text{HCl}$
66. The oxidation of SO_2 by O_2 to SO_3 is exothermic reaction. The yield of SO_2 will be minimum if :-
(1) Temperature is increased and pressure is kept constant
(2) Temperature is reduced and pressure is increased
(3) Both temperature and pressure are increased
(4) Both temperature and pressure are decreased

67. For the manufacture of ammonia by the reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3 + 21.9 \text{ k Cal}$, the favourable conditions are :-
(1) Low temperature, low pressure & catalyst
(2) Low temperature, high pressure & catalyst
(3) High temperature, low pressure & catalyst
(4) High temperature, high pressure & catalyst
68. In the reaction $2\text{A}_{(g)} + \text{B}_{(g)} \rightleftharpoons \text{C}_{(g)} + 362 \text{ kCal}$. Which combination of pressure and temperature gives the highest yield of C at equilibrium:-
(1) 1000 atm and 500°C
(2) 500 atm and 500°C
(3) 1000 atm and 50°C
(4) 500 atm and 100°C
69. Does Le chatelier's principle predict a change of equilibrium concentration for the following reaction if the gas mixture is compressed
 $\text{N}_2\text{O}_{4(g)} \rightleftharpoons 2\text{NO}_{2(g)}$
(1) Yes, backward reaction is favoured
(2) Yes, forward reaction is favoured
(3) No change
(4) No information
70. $a\text{A} \rightleftharpoons b\text{B} + c\text{C}$, $\Delta H = -x \text{ kCal}$.
If high pressure and low temperature are the favourable condition for the formation of the product in above reaction, hence:-
(1) $a > b + c$ (2) $a < b + c$
(3) $a = b + c$ (4) None of them
71. The reaction in which yield of production cannot be increased by the application of high pressure is :-
(1) $\text{PCl}_3(g) + \text{Cl}_2(g) \rightleftharpoons \text{PCl}_5(g)$
(2) $\text{N}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{NO}(g)$
(3) $\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g)$
(4) $2\text{SO}_2(g) + \text{O}_2(g) \rightleftharpoons 2\text{SO}_3(g)$
72. In a vessel containing SO_3 , SO_2 and O_2 at equilibrium, some helium gas is introduced so that the total pressure increases while temperature and volume remain constant. According to Le-Chatelier principle, the dissociation of SO_3 ,
(1) Increases
(2) Decreases
(3) Remains unaltered
(4) None of these

PHYSICAL EQUILIBRIUM

73. For the equilibrium reaction, $\text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_2\text{O}(\text{g})$,
What happens, if pressure is applied:-
(1) More water evaporates
(2) The boiling point of water is increased
(3) No effect on boiling point
(4) None of the above
74. On cooling of following system at equilibrium
 $\text{CO}_{2(\text{s})} \rightleftharpoons \text{CO}_{2(\text{g})}$:-
(1) There is no effect on the equilibrium state
(2) More gas is formed
(3) More gas is solidifies
(4) None of above

CALCULATION OF DEGREE OF DISSOCIATION BY V.D. METHOD

75. Vapour density of PCl_5 is 104.25 at $t^\circ\text{C}$. Then degree of dissociation of PCl_5 is. (Mw = 208.5)
(1) 20% (2) 0% (3) 30% (4) 15%
76. When heating PCl_5 then it decompose PCl_3 and Cl_2 in form of gas, The vapour density of gas mixture is 70.2 and 57.9 at 200°C and 250°C . The degree of dissociation of PCl_5 at 200°C and 250°C is
(1) 48.50% & 80% (2) 60% & 70%
(3) 70% & 80% (4) 80% & 90%

77. Vapour density of PCl_5 is 104.16 but when heated to 230°C its vapour density is reduced to 62. The degree of dissociation of PCl_5 at this temperature will be :
(1) 6.8% (2) 68% (3) 46% (4) 64%

78. The equation $\alpha = \frac{D-d}{(n-1)d}$ is correctly matched for

Where D = Theoretical vapour density
 d = Observed vapour density

(1) $A \rightleftharpoons \frac{nB}{2} + \frac{nC}{3}$

(2) $A \rightleftharpoons \frac{nB}{3} + \left(\frac{2n}{3}\right)C$

(3) $A \rightleftharpoons \left(\frac{n}{2}\right)B + \left(\frac{n}{4}\right)C$

(4) $A \rightleftharpoons \left(\frac{n}{2}\right)B + C$

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	4	4	4	2	4	1	3	2	3	3	1	2	4	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	4	2	2	1	4	4	1	3	1	3	2	1	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	1	3	3	3	3	3	3	1	4	1	1	1	1	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	2	1	2	3	3	1	4	2	2	2	2	1	2	2
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	2	3	2	3	3	2	2	3	1	1	2	3	2	3	2
Que.	76	77	78												
Ans.	1	2	2												

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

1. For the reaction :
- $$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$$
- $$\Delta_r H = -170.8 \text{ kJ mol}^{-1}$$
- Which of the following statements is not true:-
- (1) At equilibrium, the concentrations of $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$ are not equal
- (2) The equilibrium constant for the reaction is given by $K_p = \frac{[\text{CO}_2]}{[\text{CH}_4][\text{O}_2]}$
- (3) Addition of $\text{CH}_4(\text{g})$ or $\text{O}_2(\text{g})$ at equilibrium will cause a shift to the right
- (4) The reaction is exothermic

AIPMT 2007

2. The following equilibrium are given
- $$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3 \quad \text{-----} K_1$$
- $$\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO} \quad \text{-----} K_2$$
- $$\text{H}_2 + \frac{1}{2} \text{O}_2 \rightleftharpoons \text{H}_2\text{O} \quad \text{-----} K_3$$
- The equilibrium constant of the reaction
- $$2\text{NH}_3 + \frac{5}{2} \text{O}_2 \rightleftharpoons 2\text{NO} + 3 \text{H}_2\text{O}, \text{ in terms of } K_1, K_2 \text{ and } K_3 \text{ is :}$$
- (1) $\frac{K_1 K_2}{K_3}$ (2) $\frac{K_1 K_3^2}{K_2}$
- (3) $\frac{K_2 K_3^3}{K_1}$ (4) $K_1 K_2 K_3$

AIPMT 2008

3. The dissociation equilibrium of a gas AB_2 can be represented as : $2\text{AB}_2(\text{g}) \rightleftharpoons 2\text{AB}(\text{g}) + \text{B}_2(\text{g})$
- The degree of dissociation is 'x' and is small compared to 1. The expression relating the degree of dissociation (x) with equilibrium constant K_p and total pressure P is :
- (1) $(2K_p/P)^{1/3}$ (2) $(2K_p/P)^{1/2}$
- (3) (K_p/P) (4) $(2K_p/P)$

4. The value of equilibrium constant of the reaction $\text{HI}(\text{g}) \rightleftharpoons \frac{1}{2} \text{H}_2(\text{g}) + \frac{1}{2} \text{I}_2$ is 8.0
- The equilibrium constant of the reaction $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ will be :
- (1) $\frac{1}{64}$ (2) 16 (3) $\frac{1}{8}$ (4) $\frac{1}{16}$
5. The values of K_{p_1} and K_{p_2} for the reactions
- $$\text{X} \rightleftharpoons \text{Y} + \text{Z} \quad \text{-----} (1) \text{ and}$$
- $$\text{A} \rightleftharpoons 2\text{B} \quad \text{-----} (2)$$
- are in ratio of 9:1. If degree of dissociation of X and A be equal, then total pressure at equilibrium (1) and (2) are in the ratio :
- (1) 1:9 (2) 36:1 (3) 1:1 (4) 3:1
6. If the concentration of OH^- ions in the reaction $\text{Fe}(\text{OH})_3(\text{s}) \rightleftharpoons \text{Fe}^{3+}(\text{aq.}) + 3\text{OH}^-(\text{aq.})$ is decreased upto $\frac{1}{4}$ times, then equilibrium concentration of Fe^{3+} will increase upto:
- (1) 16 times (2) 64 times
- (3) 4 times (4) 8 times

AIPMT-Pre 2010

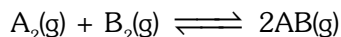
7. The reaction :
- $$2\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 3\text{C}(\text{g}) + \text{D}(\text{g})$$
- is start with the concentrations of A and B both at an initial value of 1.00 M. When equilibrium is reached, the concentration of D is measured and found to be 0.25 M. The value for the equilibrium constant for this reaction is given by the expression.
- (1) $[(0.75)^3 (0.25)] \div [(0.50)^2 (0.75)]$
- (2) $[(0.75)^3 (0.25)] \div [(0.50)^2 (0.25)]$
- (3) $[(0.75)^3 (0.25)] \div [(0.75)^2 (0.25)]$
- (4) $[(0.75)^3 (0.25)] \div [(1.00)^2 (1.00)]$

AIPMT-Mains 2010

8. In which of the following equilibrium K_c and K_p are not equal ?
- (1) $2\text{C}(\text{s}) + \text{O}_{2(\text{g})} \rightleftharpoons 2\text{CO}_{2(\text{g})}$
- (2) $2\text{NO}_{(\text{g})} \rightleftharpoons \text{N}_{2(\text{g})} + \text{O}_{2(\text{g})}$
- (3) $\text{SO}_{2(\text{g})} + \text{NO}_{2(\text{g})} \rightleftharpoons \text{SO}_{3(\text{g})} + \text{NO}_{(\text{g})}$
- (4) $\text{H}_{2(\text{g})} + \text{I}_{2(\text{g})} \rightleftharpoons 2\text{HI}_{(\text{g})}$

AIIMS 2010

9. Two moles of gas A_2 are mixed with two moles of gas B_2 in a flask of volume 1 lit. If at equilibrium 0.5 moles of A_2 are obtained. Then find out K_p for reaction



- (1) 12 (2) 9
(3) 4 (4) 36

AIPMT-Pre 2011

10. For the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, the equilibrium constant is K_1 . The equilibrium constant is K_2 for the reaction $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$.

What is K for the reaction $NO_2(g) \rightleftharpoons \frac{1}{2} N_2(g) + O_2(g)$?

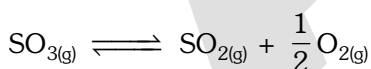
- (1) $1/(2K_1K_2)$ (2) $1/(4K_1K_2)$
(3) $[1/K_1K_2]^{1/2}$ (4) $1/(K_1K_2)$

AIIMS 2011

11. Which of the following is not a characteristics of equilibrium:
- (1) Rate is equal in both direction
 - (2) Measurable quantities are constant at equilibrium
 - (3) Equilibrium occurs in reversible condition
 - (4) Equilibrium occurs only in open vessel at constant temperature

AIPMT-Mains 2012

12. Given that the equilibrium constant for the reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ has a value of 278 at a particular temperature. What is the value of the equilibrium constant for the following reaction at the same temperature?



- (1) 6.0×10^{-2} (2) 1.3×10^{-5}
(3) 1.8×10^{-3} (4) 3.6×10^{-3}

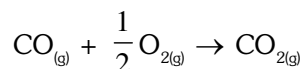
13. Given the reaction between 2 gases represented by A_2 and B_2 to give the compound $AB(g)$.
 $A_{2(g)} + B_{2(g)} = 2AB_{(g)}$
 At equilibrium, the concentration
 of $A_2 = 3.0 \times 10^{-3} M$
 of $B_2 = 4.2 \times 10^{-3} M$
 of $AB = 2.8 \times 10^{-3} M$

If the reaction takes place in a sealed vessel at $527^\circ C$, then the value of K_C will be :-

- (1) 0.62 (2) 4.5
(3) 2.0 (4) 1.9

AIIMS 2012

14. $\frac{K_p}{K_C}$ for following reaction will be :-



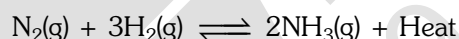
- (1) RT (2) $\frac{1}{RT}$ (3) $\frac{1}{\sqrt{RT}}$ (4) $\frac{RT}{2}$

15. At $60^\circ C$ initial pressure 1 atm of N_2O_4 is 50% dissociated into NO_2 then K_p is :-

- (1) 1.33 atm (2) 2 atm
(3) 2.67 atm (4) 3 atm

AIPMT 2014

16. For the reversible reaction :



The equilibrium shifts in forward direction :

- (1) By increasing the concentration of $NH_3(g)$
- (2) By decreasing the pressure
- (3) By decreasing the concentrations of $N_2(g)$ and $H_2(g)$
- (4) By increasing pressure and decreasing temperature

17. For a given exothermic reaction, K_p and K'_p are the equilibrium constants at temperatures T_1 and T_2 , respectively. Assuming that heat of reaction is constant in temperature range between T_1 and T_2 , it is readily observed that :-

- (1) $K_p > K'_p$ (2) $K_p < K'_p$
(3) $K_p = K'_p$ (4) $K_p = \frac{1}{K'_p}$

AIIMS 2014

18. For a reaction $\frac{1}{2} H_2 + \frac{1}{2} I_2 \rightleftharpoons HI$;

Equilibrium constant is K .

Then what will be equilibrium constant for

$2HI \rightleftharpoons H_2 + I_2$ will be :-

- (1) $\frac{1}{K^2}$ (2) $\frac{1}{\sqrt{K}}$ (3) K (4) $2K$

AIPMT-2015

19. If the value of an equilibrium constant for a particular reaction is 1.6×10^{12} , then at equilibrium the system will contain :-
 (1) mostly reactants
 (2) mostly products
 (3) similar amounts of reactants and products
 (4) all reactants

Re-AIPMT-2015

20. If the equilibrium constant for $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is K, the equilibrium constant for $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g)$ will be:-
 (1) K (2) K^2 (3) $K^{1/2}$ (4) $\frac{1}{2}K$

AIIMS-2016

21. For a reaction $A_{(g)} \rightleftharpoons B_{(g)}$ if initial pressure at 300°C is 1 atm and its K_p is 4, Then calculate P_B at equilibrium state.
 (1) $\frac{1}{5}$ atm (2) $\frac{2}{5}$ atm (3) $\frac{4}{5}$ atm (4) $\frac{3}{5}$ atm

NEET(UG) 2017

22. The equilibrium constant of the following are :
 $N_2 + 3H_2 \rightleftharpoons 2NH_3$; K_1
 $N_2 + O_2 \rightleftharpoons 2NO$; K_2
 $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$; K_3
 The equilibrium constant (K) of the reaction :
 $2NH_3 + \frac{5}{2}O_2 \xrightleftharpoons{K} 2NO + 3H_2O$, will be :
 (1) $K_2 K_3^3 / K_1$ (2) $K_2 K_3 / K_1$
 (3) $K_2^3 K_3 / K_1$ (4) $K_1 K_3^3 / K_2$
23. A 20 litre container at 400 K contains $CO_2(g)$ at pressure 0.4 atm and an excess of SrO (neglect the volume of solid SrO). The volume of the container is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value, will be :-

(Given that : $SrCO_3(s) \rightleftharpoons SrO(s) + CO_2(g)$, $K_p = 1.6 \text{ atm}$)
 (1) 10 litre (2) 4 litre
 (3) 2 litre (4) 5 litre

AIIMS-2017

24. At a certain temperature and total pressure of 1 atm., Equilibrium mixture contains 20% by volume of A atoms $A_2(g) \rightleftharpoons 2A(g)$. The K_p of reaction is-
 (1) 5×10^{-2} (2) 2×10^{-3}
 (3) 5×10^{-3} (4) 4×10^{-4}
25. In a reversible reaction $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)}$. Initially 1 mol of each reactant is taken in 2L container if at equilibrium 20% of A reacts with B. Then what will be equilibrium constant K_c for reaction :-
 (1) 0.225 (2) 0.412
 (3) 0.625 (4) 0.925

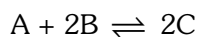
NEET(UG) 2018

26. Which one of the following conditions will favour maximum formation of the product in the reaction, $A_2(g) + B_2(g) \rightleftharpoons X_2(g)$ $\Delta_r H = -X \text{ kJ}$?
 (1) Low temperature and high pressure
 (2) Low temperature and low pressure
 (3) High temperature and high pressure
 (4) High temperature and low pressure

AIIMS-2018

27. $2ICl_{(g)} \rightleftharpoons I_{2(g)} + Cl_{2(g)}$
 Equilibrium constant K_c for given reaction is 0.14
 If initially 0.6M ICl was taken then what will equilibrium concentration of I_2 is :-
 (1) 0.1275 M (2) 0.3250 M
 (3) 0.35 M (4) 0.2550 M
28. At constant temperature on increasing pressure which is correct :-
 (1) Solid $\rightleftharpoons$ Liquid more liquid formed
 (2) Formation of ammonia from N_2 and H_2 decreases
 (3) $CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$
 reaction will shifted in forward direction
 (4) Solubility of gas in liquid increases

29. For reaction

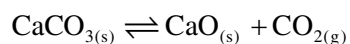


Initially 2 mole of A and B are taken in 10 lit flask at equilibrium 1 mole of C is formed then calculate

K_c for given reaction :-

- (1) 6.66 (2) 3.33 (3) 2.22 (4) 1.11

30. At constant temperature in a closed container CaCO_3 was decomposed as per following reaction



Which of the following will give change in pressure?

- (1) Size of container
(2) Temperature
(3) Addition of $\text{CaO}_{(s)}$
(4) Amount of $\text{CaCO}_{3(s)}$

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	3	1	1	2	2	1	1	4	3	4	1	1	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	1	1	2	3	3	1	4	1	3	1	1	4	1	2

EXERCISE-III (Analytical Questions)

Check Your Understanding

1. For which of the following reaction the degree of dissociation (α) and equilibrium constant (K_p) are

related as $K_p = \frac{4\alpha^2 P}{(1-\alpha^2)}$:-

- (1) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
- (2) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
- (3) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
- (4) $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

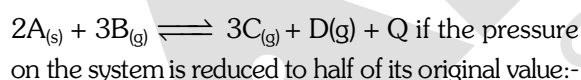
2. Pure ammonia is placed in a vessel at a temperature where its dissociation constant is appreciable. At equilibrium:-

- (1) K_p does not change significantly with pressure
- (2) Degree of dissociation does not change with pressure
- (3) concentration of NH_3 does not change with pressure
- (4) concentration of H_2 is less than that of N_2

3. For the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, the degree of dissociation varies inversely as the square root of pressure of the system. Supposing at constant temperature If the volume is increased 16 times the initial volume, the degree of dissociation for this reaction will becomes

- (1) 4 times
- (2) $\frac{1}{4}$ times
- (3) 2 times
- (4) $\frac{1}{2}$ times

4. A reaction in equilibrium is represented by the following equation -



- (1) The amounts of C and D decreases
- (2) The amounts of C and D increases
- (3) The amount of D decreases
- (4) All the amounts remain constant

5. Two systems $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ and $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$ are simultaneously in equilibrium in a vessel at constant volume. If some $CO(g)$ is introduced in the vessel at constant volume, then at new equilibrium the concentration of :

- (1) PCl_5 is increases
- (2) PCl_3 remains unchanged
- (3) PCl_5 is decreases
- (4) Cl_2 is increases

6. The effect of adding krypton (Kr) gas on position of equilibrium, keeping the volume of the system constant is :-

- (1) If $\Delta n = 0$, backward reaction is favoured.
- (2) If $\Delta n = +ve$, forward reaction is favoured
- (3) If $\Delta n = -ve$, forward reaction is favoured
- (4) No effect whatever be the value of Δn

7. Match list -I (equilibrium) with List -II (conditions for reaction) and select the correct answer using the codes given below the lists :-

List-I (Equilibrium)	List-II (Conditions)
P. $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$ Endothermic	1. High temperature
Q. $2AB_2(g) + B_2(g) \rightleftharpoons 2AB_3(g)$ Exothermic	2. Low temperature
R. $2AB_3(g) \rightleftharpoons A_2(g) + 3B_2(g)$ Endothermic	3. High pressure
	4. Low pressure
	5. Independent of pressure

CODE :

P	Q	R
(1) 1 & 3	2 & 3	2 & 4
(2) 2 & 3	1 & 4	1 & 3
(3) 1 & 5	2 & 3	1 & 4
(4) 2 & 4	1 & 5	1 & 3

8. $aA + bB \rightleftharpoons cC + dD$

in above reaction low pressure and high temperature, conditions shift equilibrium in backward direction so correct set is :-

- (1) $(a + b) > (c + d)$, $\Delta H > 0$
- (2) $(a + b) < (c + d)$, $\Delta H > 0$
- (3) $(a + b) < (c + d)$, $\Delta H < 0$
- (4) $(a + b) > (c + d)$, $\Delta H < 0$

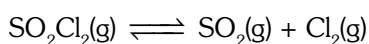
9. For reaction $aA \rightleftharpoons \ell L + mM$. In condition of suddenly volume increase, degree of dissociation decreases it represents that.

- (1) $a < (\ell + m)$
- (2) $a = (\ell + m)$
- (3) $a = (\ell - m)$
- (4) $a > (\ell + m)$

10. For the reaction, $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ the forward reaction at constant temperature is favoured by :-

(a) Introducing an inert gas at constant volume
 (b) Introducing chlorine gas at constant volume
 (c) Introducing an inert gas at constant pressure
 (d) Increasing volume of the container
 (e) Introducing PCl_5 at constant volume
 (1) a, b, c (2) b, c, d
 (3) c, d, e (4) a, c, d, e

11. Following equilibrium is present in a closed container at the temperature of 25°C .



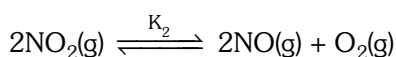
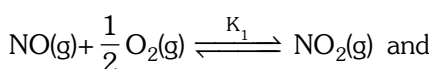
When Cl_2 is added to the equilibrium mixture, the following statements will be correct for the system.

(a) Concentrations of SO_2 , Cl_2 and SO_2Cl_2 change.
 (b) Cl_2 is formed in more amount.
 (c) Concentration of SO_2 decreases and that of SO_2Cl_2 increases.
 (1) a, c (2) a, b
 (3) b, c (4) a, b, c

12. A reaction mixture containing H_2 , N_2 and NH_3 has partial pressures 2 atm, 1 atm and 3 atm respectively at 725K. If the value of K_p for the reaction, $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ is $4.28 \times 10^{-5} \text{ atm}^{-2}$ at 725K, in which direction the net reaction will go
 (1) Forward
 (2) Backward
 (3) No net reaction
 (4) Direction of reaction cannot be predicted

13. For reaction $2\text{NOCl}(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{Cl}_2(\text{g})$, K_c at 427°C is $3 \times 10^{-6} \text{ L}^{-1} \text{ mol}$. The value of K_p is nearly :-
 (1) 7.50×10^{-5} (2) 2.50×10^{-5}
 (3) 2.50×10^{-4} (4) 1.75×10^{-4}

14. Equilibrium constants K_1 and K_2 for the following equilibrium :



are related as :-

$$(1) K_2 = \frac{1}{K_1} \quad (2) K_2 = \frac{K_1}{2}$$

$$(3) K_2 = \frac{1}{K_1^2} \quad (4) K_2 = K_1^2$$

15. For the reversible reaction

$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ at 500°C , the value of K_p is 1.44×10^{-5} when partial pressure is measured in atmospheres. The corresponding value of K_c , with concentration in mol L^{-1} is:-

(1) $1.44 \times 10^{-5} / (0.082 \times 500)^{-2}$
 (2) $1.44 \times 10^{-5} / (8.314 \times 773)^{-2}$
 (3) $1.44 \times 10^{-5} / (0.082 \times 773)^{-2}$
 (4) $1.44 \times 10^{-5} / (0.082 \times 773)^{-2}$

16. For the reaction, $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$, $K_c = 9$. If A and B are taken 1 mole each, then amount of C at equilibrium is :-

(1) 1
 (2) 0.25
 (3) 0.75
 (4) None of these

17. In which reaction equilibrium moves in left hand side when pressure is increased :-

(1) $\text{H}_2 + \text{Cl}_2 \rightleftharpoons 2\text{HCl}$
 (2) $2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{MgO}(\text{s})$
 (3) $2\text{H}_2\text{O} \rightleftharpoons 2\text{H}_2 + \text{O}_2$
 (4) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$

18. $K_p \rightarrow 10^{-2} \quad 10^{-3}$
 Temperature $\rightarrow 400\text{K} \quad 450\text{K}$

What would you consider by above information :-

(1) Equilibrium constant increases with increase in concentration
 (2) more molecules form on left hand side
 (3) Energy is released
 (4) None

19. For the reaction $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$, K_c is 100

then K_c for reaction $2\text{NO} \rightleftharpoons \text{N}_2 + \text{O}_2$ will be :-

(1) 0.01 (2) 0.1
 (3) 10 (4) 100

20. 3.1 mol of FeCl_3 and 3.2 mol of NH_4SCN are added to one litre of water. At equilibrium 3.0 mol of FeSCN^{2+} are formed. The equilibrium constant K_c of the reaction $\text{Fe}^{3+} + \text{SCN}^- \rightleftharpoons \text{FeSCN}^{2+}$ will be
 (1) 6.66×10^{-3} (2) 0.30
 (3) 3.30 (4) 150
21. One mole of N_2O_4 in a 1 L flask decomposes to attain the equilibrium $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$. At the equilibrium the mole fraction of NO_2 is $1/2$. Hence K_c will be :
 (1) $1/3$ (2) $1/2$
 (3) $2/3$ (4) 1
22. In reversible chemical reaction equilibrium will be establish when –
 (1) Reactant completely converted into product
 (2) Rate of forward and backward reaction is equal
 (3) Minimum yield of product
 (4) concentration of reactant and product is equal
23. In the manufacture of NH_3 by Haber's process which condition give maximum yield–
 (1) High temperature, High pressure and high concentration of reactants
 (2) High temperature, low pressure and low concentration of reactants
 (3) Low temperature, high pressure and high concentration of reactants
 (4) Low temperature, low pressure and low concentration of reactants
24. Increase in temperature in a reversible equilibrium reaction favours –
 (1) Forward reaction only
 (2) Backward reaction only
 (3) Either forward or backward reaction
 (4) Neither forward nor backward reaction
25. For which of the following reaction $K_p = K_c$ –
 (1) $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$
 (2) $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$
 (3) $2\text{NH}_3 \rightleftharpoons 3\text{H}_2 + \text{N}_2$
 (4) $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$
26. K for the synthesis of HI is 50. What is K for its dissociation
 (1) 50 (2) 5
 (3) 0.2 (4) 0.02
27. Which of the following is in favour of forward reaction
 (1) $Q = K_c$ (2) $Q > K_c$
 (3) $Q < K_c$ (4) None
28. At equilibrium 500mL vessel contains 1.5 M of each A, B, C, D. If 0.5M of C and D expelled out than what would be the K_c :–
 $\text{A}_{(\text{g})} + \text{B}_{(\text{g})} \rightleftharpoons \text{C}_{(\text{g})} + \text{D}_{(\text{g})}$
 (1) 1 (2) $\frac{1}{9}$
 (3) $\frac{4}{9}$ (4) $\frac{5}{9}$
29. If K_c is 41 for, $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ then for $\text{NH}_3 \rightleftharpoons \frac{1}{2} \text{N}_2 + \frac{3}{2} \text{H}_2$ K_c will be :–
 (1) 41 (2) $\sqrt{41}$
 (3) 20.5 (4) $\sqrt{\frac{1}{41}}$
30. For which of the following reaction value of K_p and K_c is equal :–
 (1) $2\text{NOCl} \rightleftharpoons 2\text{NO} + \text{Cl}_2$
 (2) $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$
 (3) $\text{H}_2 + \text{Cl}_2 \rightleftharpoons 2\text{HCl}$
 (4) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$
31. In a reaction, equilibrium proceeds towards reactants then K will be :–
 (1) $K > 1$ (2) $K < 1$
 (3) $K = 0$ (4) $K = 1$
32. For following reaction $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ value of K_c depends on
 (1) Initial concentration of reactant
 (2) Pressure
 (3) Temperature
 (4) All of these
33. $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO} - \text{Heat}$
 reaction shift in right hand direction on :–
 (1) On decreasing pressure
 (2) On increasing pressure
 (3) On decreasing temperature
 (4) On increasing temperature

34. If K_p for the reaction $N_2O_4 \rightleftharpoons 2NO_2$ is 0.66 then what is the equilibrium pressure of N_2O_4 . (Total pressure at equilibrium is 0.5 atm)
 (1) 0.168 (2) 0.322 (3) 0.1 (4) 0.5
35. For the reaction $2HBr \rightleftharpoons H_2 + Br_2$ which relation is true :-
 (1) $2K_p = K_c$ (2) $\frac{1}{K_p} = K_c$
 (3) $K_p = K_c$ (4) None of these
36. For the process $A(g) \rightleftharpoons 2B(g)$. If pressure is doubled then which is true information about A:-
 (1) Increase in A
 (2) Decrease in A
 (3) No effect
 (4) can't say because data is insufficient
37. For the reaction $2A + 3B \rightleftharpoons 2C$ the expression for K_c is :
 (1) $\frac{[A]^2[B]^3}{[C]^2}$ (2) $\frac{[C]}{[A][B]}$
 (3) $\frac{[C]^2}{[A]^2[B]^3}$ (4) $\frac{[C]^2}{[A]^3[B]^2}$
38. For the equation $2A + B \rightleftharpoons BA_2$, the equilibrium concentration of A, B, BA_2 is 4, 2 and 2 respectively. The value of K_c will be :-
 (1) 0.0625 (2) 0.625 (3) 6.280 (4) 6.250
39. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.8 atm, the value of K is :-
 (1) 0.3 atm (2) 0.18 atm
 (3) 1.8 atm (4) 3 atm
40. The equilibrium constant (K_c) for the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ at temperature T is 4×10^{-4} . The value of K_c for the reaction. $NO(g) \rightleftharpoons \frac{1}{2} N_2(g) + \frac{1}{2} O_2(g)$ at the same temperature is :-
 (1) 50.0 (2) 0.02
 (3) 2.5×10^2 (4) 4×10^{-4}
41. For the reaction $SO_{2(g)} + \frac{1}{2} O_{2(g)} \rightleftharpoons SO_{3(g)}$, if $K_p = K_c(RT)^x$ where the symbols have usual meaning then the value of x is (assuming ideality)
 (1) $\frac{1}{2}$ (2) 1
 (3) -1 (4) $-\frac{1}{2}$
42. If the amount of dissociation is $\sqrt{0.5}$, the value of K_p for the reaction $N_2O_3 \rightleftharpoons NO + NO_2$ will be
 (1) equal to the pressure of the system.
 (2) $\frac{2}{8}$ of the pressure of the system.
 (3) $\frac{8}{3}$ of the pressure of the system.
 (4) 5 times of the pressure of the system.
43. How many moles per litre of PCl_5 has to be taken to obtain 0.1 mol of Cl_2 , if the value of equilibrium constant K_c is 0.04 ?
 (1) 0.15 (2) 0.25
 (3) 0.35 (4) 0.05
44. For the reaction, $N_2O_3 \rightleftharpoons NO + NO_2$, the value of equilibrium constant K_p at fixed temperature is 4. What will be the amount of dissociation at same temperature and 5 atmospheric pressure ?
 (1) $\frac{1}{3}$ (2) $\frac{2}{3}$
 (3) $\frac{7}{9}$ (4) $\frac{2}{4}$
45. One mole of PCl_5 is heated in a closed container of one litre capacity. At equilibrium, 20% PCl_5 is not dissociated. What should be the value of K_c ?
 (1) $(3.2)^{-1}$ (2) 3.2
 (3) 2.4 (4) 42

46. For $\text{N}_2\text{O}_3 \rightleftharpoons \text{NO} + \text{NO}_2$, if total pressure is P atm and amount of dissociation is 50%, the value of K_p will be

- (1) 3 P (2) 2 P
(3) $\frac{P}{3}$ (4) $\frac{P}{2}$

47. If $\frac{2}{9}$ of 1 mol of HI dissociates, the equilibrium constant of disintegration of acid at same temperature will be

- (1) 64 (2) $\frac{1}{64}$
(3) 49 (4) $\frac{1}{49}$

48. 1.1 mol of A mixed with 2.2 mol of B and the mixture is kept in a 1 L flask and the equilibrium,

$\text{A} + 2\text{B} \rightleftharpoons 2\text{C} + \text{D}$ is reached. If at equilibrium 0.2 mol of C is formed then the value of K_c will be.

- (1) 0.1 (2) 0.01
(3) 0.001 (4) 0.0001

49. If 0.5 mol of H_2 is reacted with 0.5 mol of I_2 in a 10 L container at 444°C and at same temperature value of equilibrium constant K_c is 49, the ratio of $[\text{HI}]$ and $[\text{I}_2]$ will be :-

- (1) 7 (2) $\frac{1}{7}$
(3) $\sqrt{\frac{1}{7}}$ (4) 49

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	1	2	3	4	3	4	4	3	1	2	4	3	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	3	1	4	3	2	3	3	1	4	3	1	4	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	3	4	1	3	1	3	1	3	1	4	1	3	2	2
Que.	46	47	48	49											
Ans.	3	4	3	1											

EXERCISE-IV (Assertion & Reason)

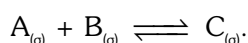
Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

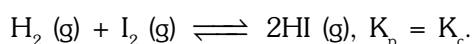
1. **Assertion :-** There is no effect on equilibrium constant if inert gas is added to the reaction



Reason :- Equilibrium constant changes only with temperature.

- (1) A (2) B (3) C (4) D

2. **Assertion :-** For the reaction



Reason :- K_p of all gaseous reactions is equal to K_c .

- (1) A (2) B (3) C (4) D

3. **Assertion :-** $K_p = K_c$ for all reactions.

Reason :- At constant temperature, the pressure of the gas is proportional to the concentration.

- (1) A (2) B (3) C (4) D

4. **Assertion :-** The value of K increases when concentration of the reactants are increased.

Reason :- With increases of concentration of reactants the equilibrium shifts in forward direction.

- (1) A (2) B (3) C (4) D

5. **Assertion :-** The effect of temperature on equilibrium constant is given by vant Hoff's equation.

Reason :- vant Hoff's equation is

$$\log \frac{K_1}{K_2} = \frac{\Delta H}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

- (1) A (2) B (3) C (4) D

6. **Assertion :-** Effect of temperature on K_c or K_p depends on enthalpy change.

Reason :- Increase in temperature shifts the equilibrium in exothermic direction and decrease in temperature shifts the equilibrium position in endothermic direction.

- (1) A (2) B (3) C (4) D

7. **Assertion :-** On opening a sealed soda bottle dissolved carbon dioxide gas escapes.

Reason :- Gas escapes to reach the new equilibrium condition of lower pressure.

- (1) A (2) B (3) C (4) D

8. **Assertion :-** Solubility of a gas in liquids increases with increase in pressure of the gas in equilibrium with solution.

Reason :- The dissolution of a gas in liquids is an exothermic process.

- (1) A (2) B (3) C (4) D

9. **Assertion :-** Solubility of a gas in water decreases with increase in temperature.

Reason :- Dissolution of a gas in water is an exothermic process.

- (1) A (2) B (3) C (4) D

10. **Assertion :-** Catalyst affects the final state of the equilibrium.

Reason :- It enables the system to attain a new equilibrium state by complexing with the reagents.

- (1) A (2) B (3) C (4) D

11. **Assertion :-** For the reaction $A \rightleftharpoons B + C + x \text{ kCal}$ at equilibrium state $[A] = [B] = [C] = 3 \times 10^{-4} \text{ M}$ and equilibrium constant is 3×10^{-4} .

Reason :- Given reaction is exothermic.

- (1) A (2) B (3) C (4) D

12. **Assertion :-** For a reaction $A_{(g)} + B_{(g)} \rightleftharpoons AB_{(g)}$ if inert gas is added in a container at constant volume. The equilibrium shifts to left side.

Reason :- Because partial pressure of A, B and AB decreases.

- (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	
Ans.	1	3	4	4	3	3	1	2	1	4	2	4	

IONIC EQUILIBRIUM

E

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NEET SYLLABUS

IONIC EQUILIBRIUM : ionization of acids and bases, strong and weak electrolytes, degree of ionization, ionization of polybasic acids, acid strength, concept of pH, Hydrolysis of salts (elementary idea), buffer solutions, Henderson equation, solubility product, common ion effect (with illustrative examples).

@neetfocusquizz

OBJECTIVES

After studying this unit, you will be able to :

- Classify acids and bases as weak or strong in terms of their ionization constants;
- Explain the dependence of degree of ionization on concentration of the electrolyte and that of the common ion;
- Describe pH scale for representing hydrogen ion concentration;
- Explain ionisation of water and its dual role as acid and base;
- Describe ionic product (K_w) and pK_w for water;
- Appreciate use of buffer solutions;
- Calculate solubility product constant.
- Classify substances as acids or bases according to Arrhenius, Bronsted-Lowry and Lewis concepts;

"If you're not part of the solution, you're part of the precipitate."

Henry J. Tillman

IONIC EQUILIBRIUM

4.0 INTRODUCTION

Formulae :

$$(i) \ln x = \log_e x = 2.303 \log_{10} x = 2.303 \log x$$

$$(ii) \log (x \times y) = \log x + \log y$$

$$(iii) \log \left(\frac{x}{y} \right) = \log x - \log y$$

$$(iv) \log x^y = y \log x$$

$$\begin{aligned} \text{Ex. (i)} \quad \log 6 &= \log (2 \times 3) \\ &= \log 2 + \log 3 \\ &= 0.3010 + 0.4771 = 0.7781 \end{aligned}$$

$$\begin{aligned} (ii) \quad \log 30 &= \log (3 \times 10) \\ &= \log 3 + \log 10 \\ &= \log 0.4771 + 1 = 1.4771 \end{aligned}$$

$$\begin{aligned} (iii) \quad \log 1000 &= \log 10^3 \\ &= 3 \log 10 = 3 \times 1 = 3 \end{aligned}$$

A. Some values of Log :

$$\log 1 = 0$$

$$\log 2 = 0.3010$$

$$\log 3 = 0.4771$$

$$\log 4 = 0.6020$$

$$\log 5 = 0.699$$

$$\log 6 = 0.7781$$

$$\log 7 = 0.8451$$

$$\log 8 = 0.9030$$

$$\log 9 = 0.9542$$

$$\log 10 = 1$$

$$\log 11 = 1.04$$

$$\log 100 = 2$$

$$\log 1000 = 3$$

Some terms which are used in ionic equilibrium : **Antilog :** $\text{Antilog } (x) = 10^x$

$$\text{Ex. } \text{Antilog } (2) = 10^2 = 100$$

$$\text{Antilog } (0.3010) = 10^{0.3010} = 2$$

$$\text{Antilog } [\log (2)] = \text{Antilog } (0.3010) = 2$$

pH - Scale : Given by – **Sorenson**

pH Scale is called Sorenson scale.

pH scale is a measuring scale used to measure strength of acid and base and its value is equal to $-\log[H^+]$

$$\text{i.e. } \text{pH} = -\log[H^+] = \log \frac{1}{[H^+]}$$

$$\begin{aligned} \text{Ex. } [H^+] &= 10^{-3} \\ \text{pH} &= -\log 10^{-3} = +3 \log 10 = 3 \end{aligned}$$

Conclusion :

$$\text{If } \text{pH} = x \quad \text{then } [H^+] = 10^{-x} \quad \text{or Vice versa}$$

$$\text{i.e. } \text{If } [H^+] = 10^{-x} \quad \text{then } \text{pH} = x$$

$$\text{pOH} \rightarrow \text{It is equal to } -\log [OH^-] \quad \text{i.e. } \text{pOH} = -\log [OH^-] = \log \frac{1}{[OH^-]}$$

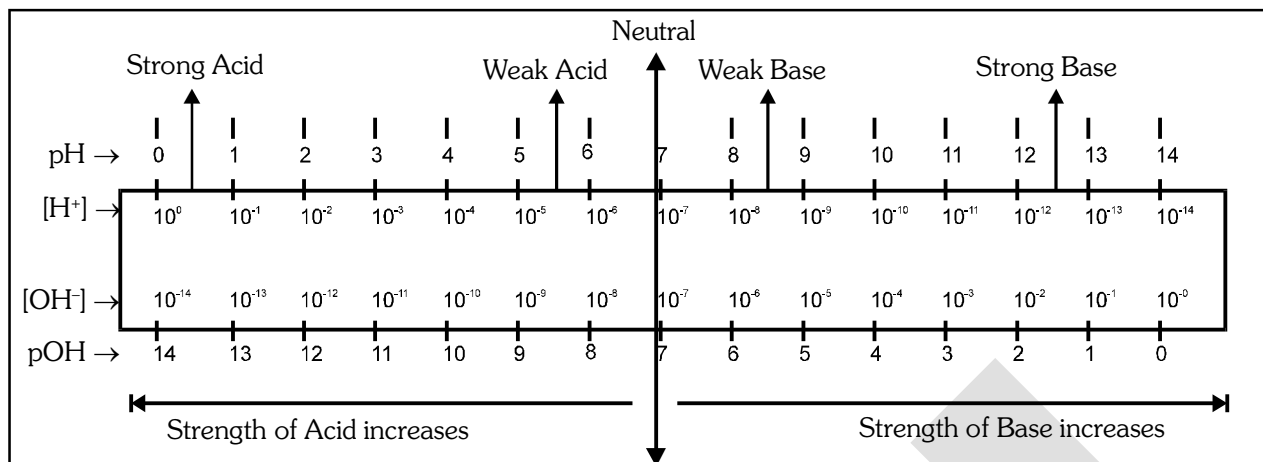
$$\text{Ex. } \text{If } X = \frac{a}{b} \times 10^{-c}, \text{ then find pX ?}$$

$$\text{Sol. } \text{pX} = -\log X = -\log \left(\frac{a}{b} \times 10^{-c} \right)$$

$$\text{pX} = - \left[\log \frac{a}{b} + \log 10^{-c} \right] = - [\log a - \log b - c]$$

$$\boxed{\text{pX} = c + \log b - \log a}$$

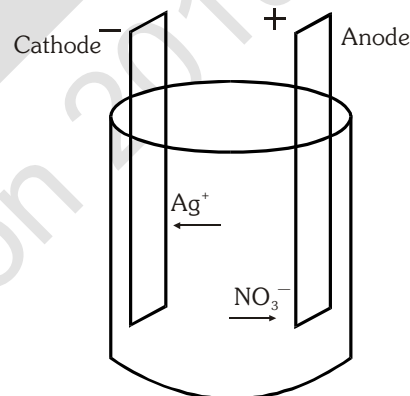
pH scale at 25°C



According to conductivity substances are of two types:

- (1) **Non-Conductor** - Those substances which do not show the flow of current or electricity.
Ex. Non - metals, plastic, rubber, wood etc.
Exception - Graphite is a non-metal but show conductivity due to motion of free electrons.
- (2) **Conductors** - Those substances which show conductivity or flow of current are called conductors and these are of two types :
 - (a) **Metallic conductor** - Those conductor which show conductivity due to motion of free electrons.
Ex. All metals, Graphite
 - (b) **Ionic conductors** -

Those conductor which show conductivity due to movement of free ions. Ions are in free state in the solutions of ionic compounds. On passing electric current through the solution, ions move towards oppositely charged electrodes, i.e., the cation moves towards cathode (negative electrode) and the anion moves towards anode (positive electrode). Due to this reason, they are called cations and anions respectively. The current flows through the solution due to the movement of the ion. Movement of ions through the solution of electrolyte ($AgNO_3$) towards oppositely charged electrodes.



According to strength, ionic conductors are of two types -

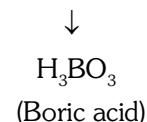
- (1) **Strong electrolytes** - Those ionic conductors which are **completely ionized in aqueous solution** are called as strong electrolytes.

For strong electrolyte the value of degree of ionisation is 100% i.e. $\alpha = 1$

- Ex.**
- (a) Strong acid $\rightarrow H_2SO_4, HCl, HNO_3, HClO_4, HBr, HI$
 - (b) Strong base $\rightarrow KOH, NaOH, Ba(OH)_2, CsOH, RbOH$
 - (c) All Salts $\rightarrow NaCl, KCl, CuSO_4, \dots$

- (2) **Weak electrolytes** - Those electrolytes which are **partially ionized in aqueous solution** are called as weak electrolytes. For weak electrolytes the value of α is less than one.

- Ex.** (a) Weak acids $\rightarrow HCN, CH_3COOH, HCOOH, H_2CO_3, H_3PO_3, H_3PO_2, B(OH)_3$, etc.



- (b) Weak bases $\rightarrow NH_4OH, Cu(OH)_2, Zn(OH)_2, Fe(OH)_3, Al(OH)_3$ etc.

4.1 ARRHENIUS CONCEPT :

- (a) According to Arrhenius when an electrolyte dissolves in water it splits up into two oppositely charged particles i.e. cation and anion.
- (b) In an electrolytic solution (Aqueous solution of electrolyte), total +ve charge is equal to total -ve charge i.e. solution is electrically neutral.
 $\Rightarrow$ In an electrolytic solution number of +ve ions may or may not be equal to the number of negative ions.
- (c) Properties of an electrolytic solution are defined by its ions.
- Ex.** Blue colour of CuSO_4 aqueous solution is due to Cu^{+2} ion (dark blue colour)
- (d) When electric current is pass in aqueous solution of electrolyte then cation shows migration towards cathode where as anion shows migration towards anode.
- (e) When a weak electrolyte is dissolved in water an equilibrium is set up between unionized moles and ionized moles.

This condition of the reversible ionic reaction is known as ionic equilibrium. $\text{AB (Aq.)} \rightleftharpoons \text{A}^+(\text{Aq.}) + \text{B}^-(\text{Aq.})$

According to L.M.A.

$$\text{Ionisation constant } K = \frac{[\text{A}^+][\text{B}^-]}{[\text{AB}]}$$

Illustrations

Illustration 1 If $[\text{H}^+] = \frac{5}{3} \times 10^{-4}$ then find pH ?

Solution

$$\begin{aligned} \text{pH} &= -\log [\text{H}^+] = -\log \left(\frac{5}{3} \times 10^{-4} \right) \\ &= - \left[\log \frac{5}{3} + \log 10^{-4} \right] = - [\log 5 - \log 3 - 4] \\ &= - [0.699 - 0.4771 - 4] = - [-3.778] = 3.778 \end{aligned}$$

Illustration 2 **Assertion :** H_2SO_4 is a strong acid.

Reason : H_2SO_4 undergoes almost completely ionised in aqueous solution.

(1) A

(2) B

(3) C

(4) D

Solution **Ans. (1)**

4.2 OSTWALD'S DILUTION LAW

$\Rightarrow$ Ostwald dilution law is applicable only for weak electrolytes.

Statement : According to Ostwald when solution of weak electrolyte is diluted then degree of ionisation of solution is increased, is called dilution law.

Let $\text{AB (Aq.)} \rightleftharpoons \text{A}^+(\text{Aq.}) + \text{B}^-(\text{Aq.})$

Initial concentration $\frac{C}{C}$ $\frac{0}{0}$ $\frac{0}{0}$

Degree of ionisation (α) $(C - C\alpha)$ $(C\alpha)$ $(C\alpha)$

According to L.M.A.

Ionisation constant $K = \frac{[\text{A}^+][\text{B}^-]}{[\text{AB}]}$; $K = \frac{C\alpha \times C\alpha}{C(1-\alpha)} = \frac{C\alpha^2}{(1-\alpha)}$

If $\alpha \ll \ll \ll 1$ then $1 - \alpha \simeq 1$

$$K = C\alpha^2 \quad \text{or} \quad \alpha = \sqrt{\frac{K}{C}} \quad (K = \text{constant, At constant temperature})$$

$$\alpha \propto \frac{1}{\sqrt{C}} \quad (C \propto \frac{1}{V}) \quad \alpha \propto \sqrt{V}$$

volume $\equiv$ dilution $\alpha \propto \sqrt{\text{dilution}}$ by dilution law

dilution $\uparrow$ $\alpha \uparrow$

At infinite dilution, $\alpha = 100\%$

Thus for weak electrolytes the degree of ionisation is directly proportional to square root of dilution or inversely proportional to square root of concentration. This law is known as Ostwald Dilution Law.

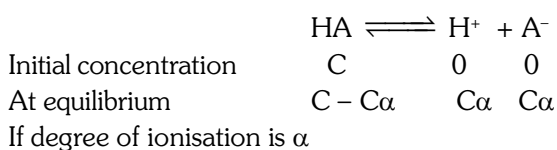
GOLDEN KEY POINTS

- At infinite dilution the value of α becomes equal to one.
- The value of concentration should be in terms of normality for calculation of pH.

Application of Ostwald's Dilution Law : $K = C\alpha^2$

(a) For mono basic weak acid (HA)

(i). Ionisation constant (K_a)



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} = \frac{C\alpha \times C\alpha}{C - C\alpha}$$

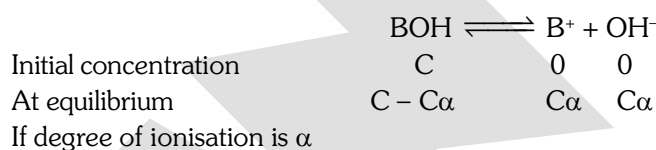
$$K_a = \frac{C^2\alpha^2}{C(1-\alpha)} = \frac{C\alpha^2}{(1-\alpha)}$$

$$\because \alpha \ll \ll \ll 1 \quad \therefore (1 - \alpha) \approx 1$$

$$\therefore K_a = C\alpha^2$$

(b) For mono acidic weak base (BOH)

Ionisation constant (K_b)



$$K_b = \frac{[\text{B}^+][\text{OH}^-]}{[\text{BOH}]} = \frac{C\alpha \times C\alpha}{C - C\alpha}$$

$$K_b = \frac{C^2\alpha^2}{C(1-\alpha)} = \frac{C\alpha^2}{(1-\alpha)}$$

$$\because \alpha \ll \ll \ll 1 \quad \therefore (1 - \alpha) \approx 1$$

$$\therefore K_b = C\alpha^2$$

Where

$K_a \rightarrow$ Ionisation constant of weak acid	}	Both K_a and K_b depends only on temperature
$K_b \rightarrow$ Ionisation constant of weak base		

(ii). $[\text{H}^+]$ (Concentration of H^+)

$$[\text{H}^+] = C\alpha \quad \dots\dots(1)$$

$$K_a = C\alpha^2 \quad \text{or} \quad \alpha = \sqrt{\frac{K_a}{C}} \quad \dots\dots(2)$$

$$\text{from Eq. (1) and (2) } [\text{H}^+] = C \times \frac{\sqrt{K_a}}{\sqrt{C}}$$

$$[\text{H}^+] = \sqrt{K_a \times C}$$

(iii). $\text{pH} = -\log [\text{H}^+]$

put the value of $[\text{H}^+]$

$$\text{pH} = -\log(\sqrt{K_a \times C}) = -\log(K_a \times C)^{1/2}$$

$$\text{pH} = -\frac{1}{2}[\log K_a + \log C]$$

$$\text{pH} = -\frac{1}{2}\log K_a - \frac{1}{2}\log C$$

$$\text{pH} = \frac{1}{2}\text{p}K_a - \frac{1}{2}\log C$$

$[\text{OH}^-]$ (Concentration of OH^-)

$$[\text{OH}^-] = C\alpha \quad \dots\dots(1)$$

$$K_b = C\alpha^2 \quad \text{or} \quad \alpha = \sqrt{\frac{K_b}{C}} \quad \dots\dots(2)$$

$$\text{from Eq. (1) and (2) } [\text{OH}^-] = C \times \frac{\sqrt{K_b}}{\sqrt{C}}$$

$$[\text{OH}^-] = \sqrt{K_b \times C}$$

$\text{pOH} = -\log [\text{OH}^-]$

put the value of $[\text{OH}^-]$

$$\text{pOH} = -\log(\sqrt{K_b \times C}) = -\log(K_b \times C)^{1/2}$$

$$\text{pOH} = -\frac{1}{2}[\log K_b + \log C]$$

$$\text{pOH} = -\frac{1}{2}\log K_b - \frac{1}{2}\log C$$

$$\text{pOH} = \frac{1}{2}\text{p}K_b - \frac{1}{2}\log C$$

In summary :

$$1. \quad K_a = C\alpha^2$$

$$2. \quad [H^+] = C\alpha = \sqrt{K_a \times C}$$

$$3. \quad pH = -\log [H^+]$$

$$\text{or} \quad pH = \frac{1}{2}pK_a - \frac{1}{2}\log C$$

$$1. \quad K_b = C\alpha^2$$

$$2. \quad [OH^-] = C\alpha = \sqrt{K_b \times C}$$

$$3. \quad pOH = -\log [OH^-]$$

$$\text{or} \quad pOH = \frac{1}{2}pK_b - \frac{1}{2}\log C$$

Limitation of Ostwald Dilution Law :

It is not applicable for strong electrolytes.

Factors affecting the Value of Degree of ionisation :

(1) Temperature $\rightarrow$ On increasing temperature, ionization increases so α increases because dissociation is endothermic process.

(2) Dilution $\rightarrow \alpha \propto \sqrt{V}$ so on dilution, α increases.

(3) Nature of electrolytes

(i) Strong electrolytes

$$\alpha = 100\%$$

(ii) Weak electrolytes

$$\alpha < 100\%$$

(4) Nature of solvent

If Dielectric constant μ of solvent increases then the value of α increases.

$$H_2O \rightarrow \mu = 81$$

$$D_2O \rightarrow \mu = 79$$

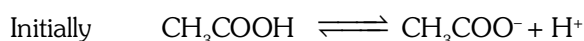
$$C_6H_6 \rightarrow \mu = 2.5$$

$$CCl_4 \rightarrow \mu = 0$$

Mixing of ions :

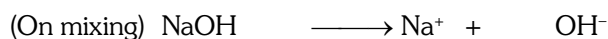
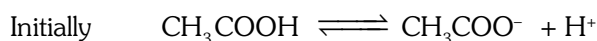
(a) Common ion effect	(b) Odd ion effect
When a strong electrolyte having a common ion, is mixed with weak electrolyte then the degree of ionisation (α) of weak electrolyte is decreased. This effect is called common ion effect.	When a strong electrolyte having a odd ion, is mixed with weak electrolyte then the degree of ionisation (α) of weak electrolyte is increased. This effect is called odd ion effect.

Common ion : On mixing CH_3COONa with CH_3COOH solution



(On mixing) $CH_3COONa \longrightarrow CH_3COO^- + Na^+$; $[CH_3COO^-]$ increases so $[H^+]$ decreases respectively.

Odd ion : On mixing, $NaOH$ with CH_3COOH solution



Due to odd ion effect $H^+ + OH^- \rightarrow \text{Water}$

Illustrations

Illustration 3 For which, dilution law is applicable :

- (1) NaCl (SASB salt) (2) HCl (SA)
 (3) CH₃COONa (WASB salt) (4) None

Solution **Ans. (4)****Illustration 4** Find out the value of α of 10^{-2} M HCN solution if $[H^+] = 10^{-3}$.**Solution** $[H^+] = C\alpha$

$$10^{-3} = 10^{-2}\alpha \text{ or } \alpha = \frac{10^{-3}}{10^{-2}} = 10^{-1} \text{ or } \alpha = 0.1$$

$$\alpha = 10\%$$

Illustration 5 For 10 M CH₃COOH solution if $K_a = 10^{-5}$ then find out :

- (i) α (ii) $[H^+]$ (iii) pH

Solution (i) α (Degree of ionisation) $\therefore K_a = C\alpha^2$

$$10^{-5} = 10\alpha^2 \text{ or } \alpha^2 = \frac{10^{-5}}{10} = 10^{-6} \text{ or } \alpha = 10^{-3}$$

$$(ii) [H^+] = C\alpha = 10 \times 10^{-3} = 10^{-2}$$

$$(iii) \text{pH} = -\log [H^+] = -\log 10^{-2} = 2$$

Illustration 6 For 10^{-3} M H₂CO₃ if $\alpha = 10\%$ then find out the value of pH ?**Solution** Normality = M $\times$ V.F. = $10^{-3} \times 2 = 2 \times 10^{-3}$

$$[H^+] = C\alpha = 2 \times 10^{-3} \times 0.1 = 2 \times 10^{-4}$$

$$\text{pH} = -\log [H^+] = -\log (2 \times 10^{-4}) = -\log 2 - \log 10^{-4} = -0.3010 + 4 = 3.7$$

Illustration 7 **Assertion :** NaCl + HCl does not experience common ion effect.**Reason :** Both NaCl and HCl are strong electrolytes.

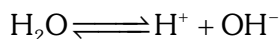
- (1) A (2) B (3) C (4) D

Solution **Ans. (1)**

4.3 EXPLANATION OF WATER

- a.** Nature of water is neutral. $[H^+] = [OH^-] [=10^{-7}\text{M (at } 25^\circ\text{C)}]$
 i.e. at 25°C $\text{pH} = 7$ and $\text{pOH} = 7$
 $\therefore \text{pH} = \text{pOH}$
- b.** No. of H₂O moles in 1 litre water = $\frac{1000}{18} = 55.5$ moles
- c.** Molar concentration of H₂O = 55.5 mol L^{-1}
- d.** Number of H₂O molecules in 1 litre water = $55.5 N_A$ (N_A = AVOGADRO'S NUMBER)
- e.** Concentration of H⁺ and OH⁻ ions in 1 litre water
 $[H^+] = 10^{-7} \text{ mol L}^{-1}$ and $[OH^-] = 10^{-7} \text{ mol L}^{-1}$
- f.** No. of H⁺ and OH⁻ ions in 1 litre water
 No. of H⁺ ions = $10^{-7}N_A$ and No. of OH⁻ ions = $10^{-7}N_A$
- g.** In water (Number of H₂O molecules : Number of H⁺ ions)
 $= 55.5 N_A : 10^{-7}N_A$
 $= 55.5 \times 10^7 : 1$
 i.e. one H⁺ ion is obtained from $55.5 \times 10^7 \text{ H}_2\text{O}$ molecules

h. Degree of ionisation of water (α) :



According to Ostwald's dilution law

$$[\text{H}^+] = c \alpha$$

$$\alpha = \frac{10^{-7}}{55.5} = 1.8 \times 10^{-9}$$

$$\alpha = 1.8 \times 10^{-7}\%$$

Hence, water is a very weak electrolyte.

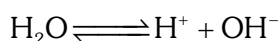
i. K (Ionisation constant of water) :



$$K = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$$

$$K = \frac{10^{-7} \times 10^{-7}}{55.5} \quad \text{or} \quad K = 1.8 \times 10^{-16}$$

j. Ionic product of water K_w :



$$K = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$$

$$K \times [\text{H}_2\text{O}] = [\text{H}^+][\text{OH}^-]$$

Since, dissociation takes place to a very small extent, $[\text{H}_2\text{O}]$ may be regarded as constant.

Thus, the product $K \cdot [\text{H}_2\text{O}]$ gives another constant which designated as K_w . So,

$$K_w = [\text{H}^+][\text{OH}^-]$$

$$\text{At } 25^\circ\text{C} - K_w = 10^{-7} \times 10^{-7} = 10^{-14}$$

$$K [\text{H}_2\text{O}] = K_w \Rightarrow K_w > K \quad (\text{always})$$

Various forms of K_w

(a) $K_w = [\text{H}^+][\text{OH}^-]$ for water $[\text{H}^+] = [\text{OH}^-]$

(b) $K_w = [\text{H}^+]^2$

(c) $K_w = [\text{OH}^-]^2$

(d) $K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$

(e) $K_w = [\text{H}_3\text{O}^+]^2 \quad \{[\text{H}_3\text{O}^+] = [\text{H}^+]\}$
Hydronium ion Proton

• Relation in between pH and pOH :

$$K_w = [\text{H}^+][\text{OH}^-]$$

taking $-\log$ on both sides

$$-\log K_w = -\log [\text{H}^+] - \log [\text{OH}^-]$$

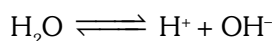
$$\text{p}K_w = \text{pH} + \text{pOH}$$

- Nature of water is neutral so,
 $[pH = pOH]$
 $pK_w = pH + pOH$
 $2pH = pK_w$
 $pH = \frac{pK_w}{2}$
 - $pK_w = pOH + pOH$
 $2pOH = pK_w$
 $pOH = \frac{pK_w}{2}$
- $pH = pOH = \frac{pK_w}{2}$

At 25°C, $K_w = 10^{-14}$ or $pK_w = 14$

$\therefore pH + pOH = 14$ or $pH = pOH = 7$

k. Effect of temperature :-



Ionization of water is endothermic process so, on increasing temperature α increases or $[H^+]$ and $[OH^-]$ increases or $[H^+][OH^-]$ increases i.e. K_w increases means pH decreases or pOH decreases.

$$T \uparrow = \alpha \uparrow = [H^+][OH^-] \uparrow = K_w \uparrow \Rightarrow pK_w \downarrow$$

At 25°C, $K_w = 10^{-14}$

At 90°C, $K_w = 10^{-12}$

Parameters	At 25°C	At 90°C
K_w	10^{-14}	10^{-12}
pK_w	14	12
$pH = \frac{pK_w}{2} = pOH$	7	6
$[H^+] = [OH^-] = \sqrt{K_w}$	10^{-7}	10^{-6}
$pH + pOH = pK_w$	14	12

GOLDEN KEY POINTS

- On increasing temperature, both $[H^+]$ and $[OH^-]$ increases equally so water remains neutral but neutral pH changes from 7 to 6 at 90°C.

Illustrations

Illustration 8 Dissociation constant of water at 25°C is

(1) $10^{-14} \times (55.5)^{-1}$ (2) $10^{-7} \times (18)^{-1}$ (3) $10^{-14} \times (18)^{-1}$ (4) $10^{-7} \times (55.4)^{-1}$

Solution **Ans. (1)**

Illustration 9 What should be the number of H^+ ions in 1 mL of distilled water, if the number of H^+ ions in 1 L is 6.023×10^{16} ?

Solution Number of H^+ ions in 1 mL distilled water = $\frac{6.023 \times 10^{16}}{1000} = 6.023 \times 10^{13}$

BEGINNER'S BOX-1

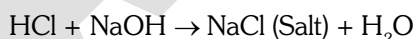
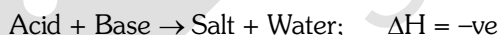
- The pH of a 0.005 M H_2SO_4 solution is—
 (1) 3.3 (2) 5.0 (3) 2.0 (4) 4.0
- If pure water has $pK_w = 13.36$ at 50°C, the pH of pure water will be—
 (1) 6.68 (2) 7.0 (3) 7.13 (4) 6.0

3. How many H^+ ions are present in 1 ml of a solution whose pH is 13 ?
 (1) 10^{-16} (2) 6.022×10^{13} (3) 6.022×10^7 (4) 6.022×10^{23}
4. The pH of solutions A, B, C and D are 9.5, 2.5, 3.5 and 5.5 respectively. The most acidic solution is—
 (1) D (2) C (3) A (4) B
5. Calculate the concentration of the formate ion present in 0.100 M formic acid ($HCOOH$) solution at equilibrium ($K_a = 1.7 \times 10^{-4}$).
 (1) 4.1×10^{-3} M (2) 3.1×10^{-3} M (3) 2.1×10^{-3} M (4) 5.1×10^{-3} M
6. Which of the following is the weakest acid ?
 (1) Phenol ($K_a = 1.3 \times 10^{-10}$) (2) Hydrocyanic acid ($K_a = 4.9 \times 10^{-10}$)
 (3) Acetic acid ($K_a = 1.8 \times 10^{-5}$) (4) Benzoic acid ($K_a = 6.5 \times 10^{-5}$)
7. The pH of 0.1 M monobasic acid is 4.50. The acidity constant (K_a) of the monobasic acid is—
 (1) 1.0×10^{-7} (2) 1.0×10^{-5} (3) 1.0×10^{-4} (4) 1.0×10^{-8}
8. Which of the following is the strongest base ?
 (1) $C_6H_5NH_2$ ($pK_b = 9.42$) (2) $C_6H_5NHCH_3$ ($pK_b = 9.15$)
 (3) $C_6H_5N(CH_3)_2$ ($pK_b = 8.94$) (4) $C_6H_5NHC_2H_5$ ($pK_b = 8.89$)
9. Value of dissociation constant of acetic acid is 10^{-6} , where as dissociation constant of formic acid is 10^{-5} . Which of the following will be the value of pK_a (acetic acid) - pK_a (formic acid)
 (1) 10 (2) +1 (3) 10^{-1} (4) -1
10. A solution has pOH equal to 13 at 298 K. The solution will be
 (1) highly acidic (2) highly basic (3) moderately basic (4) unpredictable
11. What would be $[H^+]$ of 0.006 M benzoic acid ($K = 6 \times 10^{-5}$)
 (1) 0.6×10^{-4} (2) 6×10^{-4} (3) 6×10^{-3} (4) 3.6×10^{-4}

4.4 SALTS, TYPES OF SALT AND CONJUGATE THEORY

(A) **SALT** : Salts are regarded as compounds made up of positive and negative ions. The positive part comes from a base while negative part from an acid. Salts are ionic compounds.

i.e. A compound formed by the combination of acid and base is known as salt.



(B) TYPES OF SALT

(a) **Normal/general salts** :- The salts formed by the loss of all possible protons. (replaceable hydrogen ions as H^+) are called normal salts. Such a salt does not contain either a replaceable hydrogen ion or a hydroxyl (OH^-) group.

Ex.: $NaCl$, $NaNO_3$, K_2SO_4 , $Ca_3(PO_4)_2$, Na_2HPO_3 , NaH_2PO_2 etc.

(b) **Acid salts** :- Salts formed by incomplete neutralisation of poly basic acids are called acid salts. Such salts still contain one or more replaceable hydrogen ions (H^+).

Ex.: $NaHCO_3$, $NaHSO_4$, NaH_2PO_4 , Na_2HPO_4 etc.

(c) **Basic salts** :- Salts are formed by incomplete neutralisation of poly acidic base are called basic salts. Such salts still contain one or more hydroxyl ion group.

Ex.: $Zn(OH)Cl$, $Mg(OH)Cl$, $Fe(OH)_2Cl$, $Bi(OH)_2Cl$ etc.

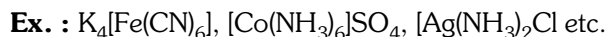
- (d) **Double salts** :- The addition compounds formed by the combination of two simple salts are termed double salts. Such salts are stable in solid state only i.e. when dissolved in water they break into simple ions.



(Ferrous ammonium sulphate)

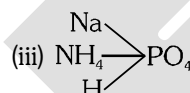
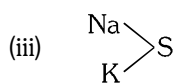
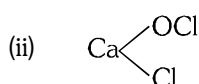
(Potash alum)

- (e) **Complex salts** :- These salts when dissolved in water give complex ions. Complex ions are stable in solid state as well as in solutions.



- (f) **Mixed salts** :- The salts which furnishes more than one type of cations or more than one type of anions when dissolved in water are called mixed salts.

These are formed by the neutralisation of more than one acids and bases.



(a) **Types of general salts :**

(i) SASB

(ii) SAWB

(iii) WASB

(iv) WAWB

(C) **Conjugate acid-base pair :**

- (i) Conjugate acid-base pairs means difference between two species of only one H^+ ion.
- (ii) When an acid loses a proton, the residual part of it has a tendency to regain a proton. Therefore it behaves as a base.

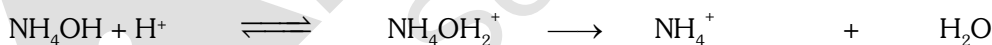


Weak acid

Conjugate strong base

($K_a = 1.85 \times 10^{-5}$)

(acceptor ion)

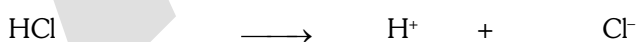


Weak base

Conjugate strong acid

($K_b = 1.85 \times 10^{-5}$)

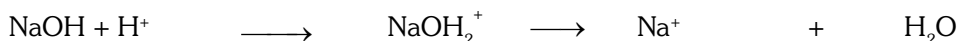
(acceptor ion)



Strong acid

Conjugate weak base (spectator ion)

Accepting tendency of $\text{H}^+ \simeq 0$



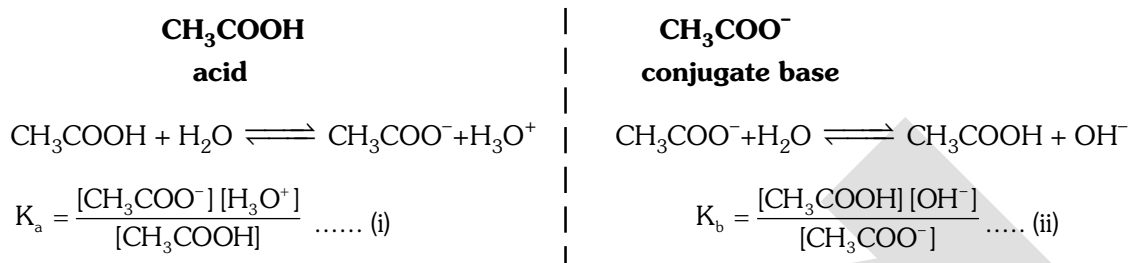
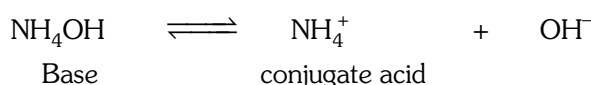
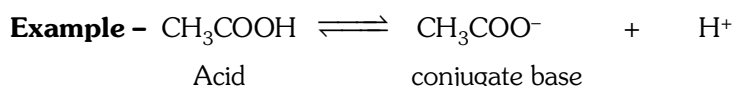
Strong base

Conjugate weak acid

(spectator ion)

Note : Strong acids have weak conjugate bases while weak acids have strong conjugate bases. Similarly, strong bases have weak conjugate acids while weak bases have strong conjugate acids.

(D) Relation between conjugate acid-base pair :-



In both the reactions H_2O in excess quantity so active mass of H_2O is one.

Now multiply the equation (i) and (ii)

$$K_a \times K_b = [\text{H}^+][\text{OH}^-]$$

we know $[\text{H}^+] \times [\text{OH}^-] = K_w$ (Ionic product of water)

$$K_a \times K_b = K_w$$

Taking $-\log$ on both sides

$$\text{p}K_a + \text{p}K_b = \text{p}K_w$$

we know that for water at 25°C ,

$$K_w = 10^{-14} \text{ or } \text{p}K_w = 14$$

So $K_a \times K_b = 10^{-14}$ or $\text{p}K_a + \text{p}K_b = 14$

Above relation is applicable only for conjugate acid-base pairs.

Illustrations

Illustration 10 Which salt is not an example of acidic salt :-

(1) HCOONa

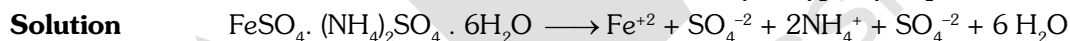
(2) NaH_2PO_2

(3) NaHS

(4) (1) and (2) both

Solution **Ans. (4)**

Illustration 11 Calculate the no. of ions in Mohr's salt $[\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}]$ or Ferrous Ammonium Sulphate



Total number of ions = 5

4.5. HYDROLYSIS OF SALTS

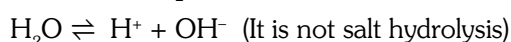
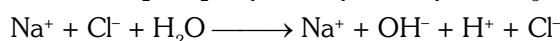
Salt hydrolysis is defined as the process in which water reacts with cation or anion or both of a salt to change the concentration of H^+ and OH^- ions of water.

Salt hydrolysis is reverse process of neutralization.



(A) Hydrolysis of strong acid and strong base [SA - SB] types of salt -

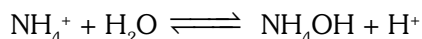
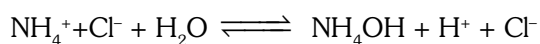
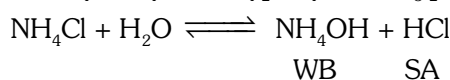
Ex. NaCl , BaCl_2 , Na_2SO_4 , KClO_4 , BaSO_4 , NaNO_3 , KBr , KCl etc.



- (i) Hydrolysis of salt of [SA - SB] is not possible as both cation and anion are not reactive.
- (ii) Aqueous solution of these type of salt is neutral in nature. ($\text{pH} = \text{pOH} = 7$)
- (iii) pH of the solution is 7.
- (iv) No effect on litmus paper.

(B) Hydrolysis of strong acid and weak base [SA - WB] types of salt -

Ex. CaSO_4 , NH_4Cl , $(\text{NH}_4)_2\text{SO}_4$, $\text{Ca}(\text{NO}_3)_2$, ZnCl_2 , CuCl_2 , CaCl_2 , AgCl , AgI , AgNO_3 etc



- (i) In this type of salt hydrolysis, cation reacts with H_2O therefore called as cationic hydrolysis. The cation of the salt which has come from weak base is reactive.
- (ii) Solution is acidic in nature as $[\text{H}^+]$ is increased.
- (iii) pH of the solution is less than 7.
- (iv) Solution turns blue litmus paper red.

K_h = Hydrolysis constant

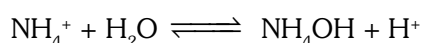
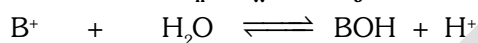
K_w = Ionic product of water

K_a = Ionisation constant of acid

K_b = Ionisation constant of base

h = Degree of hydrolysis

C = Concentration of salt (concentration of ions)

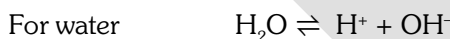
(a) Relation between K_h , K_w and K_b 

Hydrolysis constant $[K_h]$

$$K_h = \frac{[\text{NH}_4\text{OH}][\text{H}^+]}{[\text{NH}_4^+]} \quad \dots\dots(1)$$



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} \quad \dots\dots(2)$$



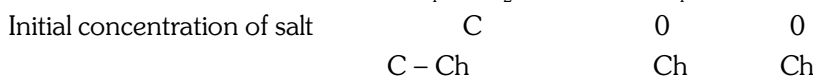
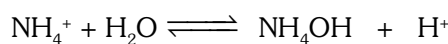
$$K_w = [\text{H}^+][\text{OH}^-] \quad \dots\dots(3)$$

Now multiplying Eq. (1) and (2) = Eq. (3)

$$\frac{[\text{NH}_4\text{OH}][\text{H}^+]}{[\text{NH}_4^+]} \times \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} = [\text{H}^+][\text{OH}^-]$$

i.e. $K_h \times K_b = K_w$

$$K_h = \frac{K_w}{K_b} \quad \dots\dots(4)$$

(b) Degree of hydrolysis - Represented by h 

$$K_h = \frac{[\text{NH}_4\text{OH}][\text{H}^+]}{[\text{NH}_4^+]} = \frac{Ch \times Ch}{C - Ch} = \frac{C^2 h^2}{C(1 - h)} = \frac{Ch^2}{(1 - h)}$$

Since $h \ll \ll \ll 1$ then $(1 - h) \approx 1$

$$\therefore K_h = Ch^2 \quad \dots\dots(5)$$

$$h^2 = \frac{K_h}{C} \quad \Rightarrow \quad h = \sqrt{\frac{K_h}{C}} \quad \dots\dots(6)$$

$$\therefore K_h = \frac{K_w}{K_b} \quad \Rightarrow \quad h = \sqrt{\frac{K_w}{K_b C}}$$

$$h = \sqrt{\frac{K_w}{K_b \times C}} \quad \dots\dots(7)$$

(c) **pH of the solution :** $\text{pH} = -\log [\text{H}^+]$

$$[\text{H}^+] = Ch = C \sqrt{\frac{K_w}{K_b \times C}} \quad \Rightarrow \quad [\text{H}^+] = \sqrt{\frac{K_w \times C}{K_b}} \quad \dots\dots(8)$$

taking - log on both sides

$$-\log [\text{H}^+] = -\log \sqrt{\frac{K_w \times C}{K_b}} \quad \Rightarrow \quad \text{pH} = -\log \left(\frac{K_w \times C}{K_b} \right)^{1/2}$$

$$\text{pH} = -\frac{1}{2} [\log K_w + \log C - \log K_b]$$

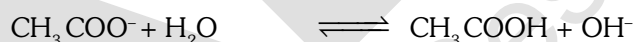
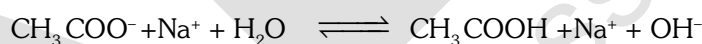
$$\text{pH} = -\frac{1}{2} \log K_w - \frac{1}{2} \log C - \frac{1}{2} (-\log K_b)$$

$$\text{pH} = \frac{1}{2} \text{p}K_w - \frac{1}{2} \log C - \frac{1}{2} \text{p}K_b$$

$$\boxed{\text{pH} = 7 - \frac{1}{2} \text{p}K_b - \frac{1}{2} \log C} \quad \dots\dots(9)$$

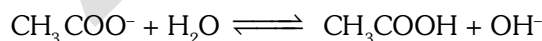
(C) **Hydrolysis of weak acid and strong base [WA - SB] types of salt -**

Ex. CH_3COONa , HCOONa , KCN , NaCN , K_2CO_3 , BaCO_3 , K_3PO_4 etc.



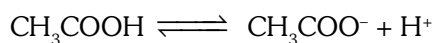
- (i) In this type of salt hydrolysis, anion reacts with water therefore called as anionic hydrolysis. The anion of the salt which has come from weak acid is reactive.
- (ii) Solution is basic in nature as $[\text{OH}^-]$ increases.
- (iii) pH of the solution is greater than 7.
- (iv) Solution turns red litmus paper blue.

(a) **Relation between K_h , K_w and K_a**



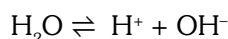
$$K_h = \frac{[\text{CH}_3\text{COOH}][\text{OH}^-]}{[\text{CH}_3\text{COO}^-]} \quad \dots\dots(1)$$

For weak acid



$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]} \quad \dots\dots(2)$$

For water



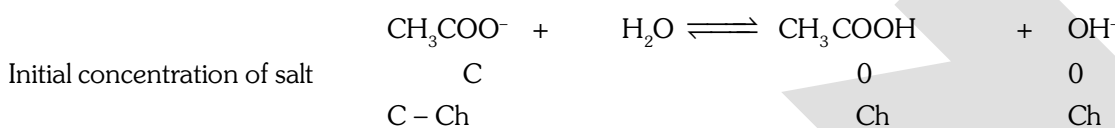
$$K_w = [\text{H}^+][\text{OH}^-] \quad \dots\dots(3)$$

Now multiply eq. (1) $\times$ eq. (2) = eq. (3)

$$\frac{[\text{CH}_3\text{COOH}][\text{OH}^-]}{[\text{CH}_3\text{COO}^-]} \times \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]} = [\text{H}^+][\text{OH}^-]$$

$$K_h \times K_a = K_w$$

$$\boxed{K_h = \frac{K_w}{K_a}} \quad \dots\dots(4)$$

(b) Degree of hydrolysis (h) :

$$K_h = \frac{[\text{CH}_3\text{COOH}][\text{OH}^-]}{[\text{CH}_3\text{COO}^-]} = \frac{Ch \times Ch}{C - Ch} = \frac{C^2 h^2}{C(1-h)}$$

$$K_h = \frac{Ch^2}{(1-h)}$$

Since $h \ll 1$ then $(1-h) \approx 1$

$$\therefore \boxed{K_h = Ch^2} \quad \dots\dots(5)$$

$$h^2 = \frac{K_h}{C} \quad \text{or} \quad h = \sqrt{\frac{K_h}{C}} \quad \dots\dots(6)$$

$$\boxed{h = \sqrt{\frac{K_w}{K_a \times C}}} \quad \dots\dots(7)$$

(c) pH of the solution

$$[\text{OH}^-] = Ch$$

$$[\text{OH}^-] = C \times \sqrt{\frac{K_w}{K_a \times C}} \quad \text{or} \quad [\text{OH}^-] = \sqrt{\frac{K_w \times C}{K_a}} \quad \dots\dots(8)$$

taking -log on both sides

$$-\log [\text{OH}^-] = -\log \left(\frac{K_w \cdot C}{K_a} \right)^{1/2}$$

$$\text{pOH} = -\frac{1}{2} [\log K_w + \log C - \log K_a]$$

$$\text{pOH} = \frac{1}{2} \text{p}K_w - \frac{1}{2} \text{p}K_a - \frac{1}{2} \log C \quad \text{or} \quad \text{pOH} = 7 - \frac{1}{2} \text{p}K_a - \frac{1}{2} \log C$$

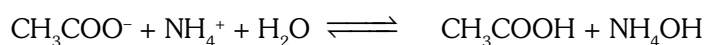
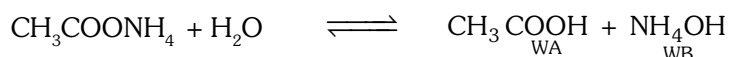
$$\therefore \text{pH} + \text{pOH} = 14$$

$$\text{pH} = 14 - \text{pOH}$$

$$\therefore \text{pH} = 7 + \frac{1}{2} \text{p}K_a + \frac{1}{2} \log C \quad \dots\dots(9)$$

(D) **Hydrolysis of weak acid and weak base (WA - WB) type of salt :**

Ex. $\text{CH}_3\text{COONH}_4$, AgCN , NH_4CN , CaCO_3 , $[\text{NH}_4]_2\text{CO}_3$, ZnHPO_3 etc.

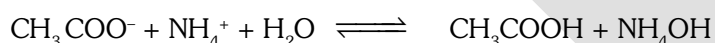


- Maximum hydrolysis occurs of the salt of (WA – WB) as both the cation and anion are reactive.
- Solution is almost neutral but it may be acidic or basic depending upon the nature of acid base
- pH of the solution is near to 7.

For WA - WB types of salt :

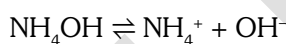
Terms	$K_a > K_b$	$K_b > K_a$	$K_a = K_b$
1. Hydrolysis	Cationic-anionic	Anionic-cationic	Neutral hydrolysis
2. Nature	Acidic	Basic	Neutral
3. pH	$\text{pH} < 7$	$\text{pH} > 7$	$\text{pH} = 7$

(a) **Relation between K_h , K_w , K_a and K_b**



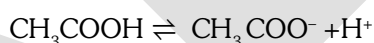
$$K_h = \frac{[\text{CH}_3\text{COOH}][\text{NH}_4\text{OH}]}{[\text{CH}_3\text{COO}^-][\text{NH}_4^+]} \quad \text{.....(1)}$$

For weak base



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} \quad \text{.....(2)}$$

For weak acid



$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]} \quad \text{.....(3)}$$

For water



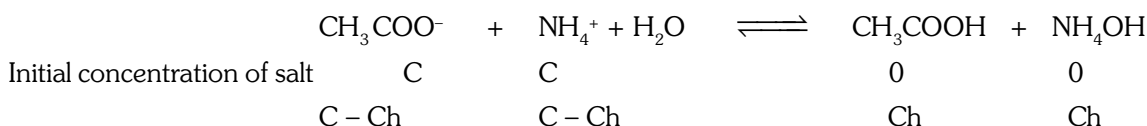
$$K_w = [\text{H}^+][\text{OH}^-] \quad \text{.....(4)}$$

Multiply Eq. (1) $\times$ Eq. (2) $\times$ Eq. (3) = Eq. (4)

$$\frac{[\text{CH}_3\text{COOH}][\text{NH}_4\text{OH}]}{[\text{CH}_3\text{COO}^-][\text{NH}_4^+]} \times \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} \times \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]} = [\text{H}^+][\text{OH}^-]$$

$$K_h \times K_b \times K_a = K_w$$

$$K_h = \frac{K_w}{K_a \times K_b} \quad \text{.....(5)}$$

(b) Degree of hydrolysis (h) –

$$K_h = \frac{[\text{CH}_3\text{COOH}][\text{NH}_4\text{OH}]}{[\text{CH}_3\text{COO}^-][\text{NH}_4^+]} = \frac{\text{Ch} \times \text{Ch}}{(C - \text{Ch})(C - \text{Ch})} = \frac{C^2 h^2}{C(1-h) \times C(1-h)}$$

Since $h \ll \ll \ll 1$ then $(1 - h) \approx 1$

$$\therefore \boxed{K_h = h^2} \quad \text{.....(6)}$$

$$\text{or } h^2 = \frac{K_w}{K_a \times K_b} \quad \text{or } \boxed{h = \sqrt{\frac{K_w}{K_a \times K_b}}} \quad \text{.....(7)}$$

(c) pH of the solution

from equation (3)

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

$$[\text{H}^+] = \frac{K_a \times [\text{CH}_3\text{COOH}]}{[\text{CH}_3\text{COO}^-]} = \frac{K_a \times \text{Ch}}{C - \text{Ch}} = \frac{K_a \times h}{1 - h}$$

Since $h \ll \ll \ll 1$ then $(1 - h) \approx 1$

$$[\text{H}^+] = K_a \times h \quad [\text{Now put the value of } h \text{ from eq. (5)}]$$

$$= K_a \times \sqrt{\frac{K_w}{K_a \times K_b}}$$

$$\boxed{[\text{H}^+] = \sqrt{\frac{K_w \times K_a}{K_b}}} \quad \text{.....(8)}$$

$$\text{taking } -\log \text{ on both sides } -\log [\text{H}^+] = -\log \left(\frac{K_w \times K_a}{K_b} \right)^{\frac{1}{2}}$$

$$\text{pH} = -\frac{1}{2} [\log (K_w \times K_a) - \log K_b]$$

$$\text{pH} = -\frac{1}{2} [\log K_w + \log K_a - \log K_b]$$

$$\text{pH} = -\frac{1}{2} [\log K_w] - \frac{1}{2} [\log K_a] - \frac{1}{2} [-\log K_b]$$

$$\text{pH} = +\frac{1}{2} \text{p}K_w + \frac{1}{2} \text{p}K_a - \frac{1}{2} \text{p}K_b$$

$$\boxed{\text{pH} = 7 + \frac{1}{2} \text{p}K_a - \frac{1}{2} \text{p}K_b} \quad \text{.....(9)}$$

GOLDEN KEY POINTS

- Degree of hydrolysis, pH of weak acid and weak base [WA – WB] type salts do not depend on the concentration of salt.

	SA SB salts	SA WB salts	WA SB salts	WA WB salts
1.	Neutral solution	Acidic solution	Basic solution	Almost neutral solution
2.	No hydrolysis	Cationic hydrolysis	Anionic hydrolysis	Hydrolysis by both cation and anion
3.	–	$K_h = \frac{K_w}{K_b}$	$K_h = \frac{K_w}{K_a}$	$K_h = \frac{K_w}{K_a \times K_b}$
4.	–	$h = \sqrt{\frac{K_w}{K_b \cdot C}}$	$h = \sqrt{\frac{K_w}{K_a \cdot C}}$	$h = \sqrt{\frac{K_w}{K_a \cdot K_b}}$
5.	–	$[H^+] = C \cdot h = \sqrt{\frac{K_w \cdot C}{K_b}}$	$[OH^-] = C \cdot h = \sqrt{\frac{K_w \cdot C}{K_a}}$	$[H^+] = K_a \cdot h = \sqrt{\frac{K_w \cdot K_a}{K_b}}$
6.	–	$pH = 7 - \frac{1}{2} pK_b - \frac{1}{2} \log C$	$pH = 7 + \frac{1}{2} pK_a + \frac{1}{2} \log C$	$pH = 7 + \frac{1}{2} pK_a - \frac{1}{2} pK_b$

Illustrations

Illustration 12 What is the pH of 1 M CH_3COONa solution? K_a of acetic acid = 1.8×10^{-5} , $K_w = 10^{-14} \text{ mol}^2 \text{ L}^{-2}$
 (1) 2.4 (2) 3.6 (3) 4.8 (4) 9.4

Solution **Ans. (4)**

Illustration 13 Calculate the degree of hydrolysis of a mixture containing 0.1N NH_4OH and 0.1N HCN
 If $K_a = 10^{-5}$ and $K_b = 10^{-5}$

Solution Salt is [WA – WB]

$$h = \sqrt{\frac{K_w}{K_a \times K_b}} = \sqrt{\frac{10^{-14}}{10^{-5} \times 10^{-5}}} = \sqrt{10^{-14} \times 10^{+10}} = \sqrt{10^{-4}} = 10^{-2}$$

Illustration 14 **Assertion :** An aqueous solution of NH_4NO_3 is acidic in characters.

Reason : NH_4NO_3 in an aqueous solution undergoes anionic hydrolysis.

(1) A (2) B (3) C (4) D

Solution **Ans. (3)**

BEGINNER'S BOX-2

- When sodium acetate (CH_3COONa) is added to aqueous solution of acetic acid (CH_3COOH), the–
 (1) the pH value becomes zero (2) pH value remains unchanged
 (3) pH value decreases (4) pH value increases
- Which of the following cations is not hydrolyzed in aqueous solution ?
 (i) Mg^{2+} (ii) Ca^{2+} (iii) Na^+ (iv) K^+
 (1) (i), (ii) (2) (iii), (iv) (3) (i), (ii), (iii), (iv) (4) (i), (ii), (iii)

3. Which of the anions is not hydrolyzed in aqueous solution ?
 (i) Cl^- (ii) NO_3^- (iii) Br^- (iv) ClO_4^-
 (1) (i), (ii), (iii), (iv) (2) (ii), (iii), (iv) (3) (i), (ii), (iii) (4) (ii), (iv)
4. Which of the following salts does not undergo hydrolysis ?
 (1) KCN (2) KCl (3) NH_4NO_3 (4) $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$
5. Which of the following salts undergoes anionic hydrolysis ?
 (1) AlCl_3 (2) CuSO_4 (3) Na_2CO_3 (4) NH_4Cl
6. For cationic hydrolysis, pH is given by—
 (1) $\text{pH} = \frac{1}{2}\text{pK}_w + \frac{1}{2}\text{pK}_a + \frac{1}{2}\log C$ (2) $\text{pH} = \frac{1}{2}\text{pK}_w - \frac{1}{2}\text{pK}_b - \frac{1}{2}\log C$
 (3) $\text{pH} = \frac{1}{2}\text{pK}_w + \frac{1}{2}\text{pK}_a - \frac{1}{2}\text{pK}_b$ (4) $\text{pH} = \frac{1}{2}\text{pK}_w + \frac{1}{2}\text{pK}_b + \frac{1}{2}\log C$
7. Which of the following salts is neutral in water ?
 (1) KCl (2) NH_4NO_3 (3) NH_4CN (4) NH_4OH

4.6 SOLUBILITY AND SOLUBILITY PRODUCT (K_{sp})

(A) SOLUBILITY

- (a) **Definition** At constant temperature the maximum number of moles of solute which can be dissolved in a solvent to obtain 1 litre of solution (i.e. saturated solution) is called solubility.

$$S(M) = \frac{\text{Number of moles of solute}}{\text{Volume of solution (L)}}$$

$$S = \frac{x}{M_w \times V_L} \text{ mol L}^{-1}$$

$$S_{g/l} = S_M \times \text{molar mass}$$

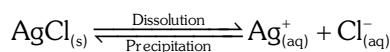
GOLDEN KEY POINTS

- Solubility does not depend on amount of substances and volume of solution where as depends on the following—
 (i) Temperature
 (ii) Presence of common ion
 (iii) Nature of solvent (Molecular wt. of $\text{AgCl} = 143.5$)
 (Molecular wt. of $\text{BaSO}_4 = 233$)
- Solubility can be expressed in terms of molarity.

(B) SOLUBILITY PRODUCT (K_{sp}) :

When a sparingly soluble salt such as AgCl is put into water, a very small amount of AgCl dissolves in water and most of the salt remains undissolved in its saturated solution.

- The salt AgCl is an electrolyte, its dissociation occurs in solution. Hence, the quantity of AgCl that dissolves in water dissociates into Ag^+ and Cl^- ions. Thus, in the saturated solution of AgCl an equilibrium exists between undissolved solid AgCl and its ions, Ag^+ and Cl^- ions.



according to law of mass action
$$K = \frac{[\text{Ag}^+][\text{Cl}^-]}{[\text{AgCl}]}$$

Since, the concentration of undissolved solid AgCl is constant. Thus, the product $K.[AgCl]$ gives another constant which is designated as K_{sp} .

$$\text{So, } K.[AgCl] = [Ag^+].[Cl^-] \quad \therefore K_{sp} = [Ag^+] \cdot [Cl^-]$$

At constant temperature product of concentrations of ions in a saturated solution of substance is called solubility product of that substance. (Saturated solution is that solution in which solid solute and the ions in solution remain in equilibrium with each other.

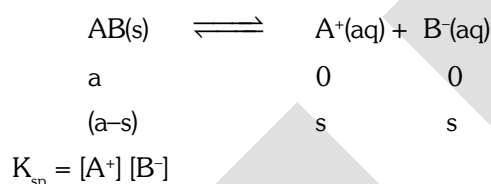
- **K_{sp} for $CaCl_2$** $CaCl_2(s) \rightleftharpoons Ca^{+2}(aq) + 2Cl^-(aq)$
Solubility product in terms of concentration of ions $K_{sp} = [Ca^{+2}] [Cl^-]^2$
- **K_{sp} for $AlCl_3$** $AlCl_3(s) \rightleftharpoons Al^{+3}(aq) + 3Cl^-(aq)$
Solubility product in terms of concentration of ions $K_{sp} = [Al^{+3}] [Cl^-]^3$
- **General form** $A_xB_y(s) \rightleftharpoons xA^{+y}(aq) + yB^{-x}(aq)$
 $K_{sp} = [A^{+y}]^x [B^{-x}]^y$

4.7 APPLICATION OF SOLUBILITY PRODUCT (K_{sp})

(A) To find out the solubility (S) :

(i) **K_{sp} of AB (Mono-mono, di-di, tri-tri valency) type salt -**

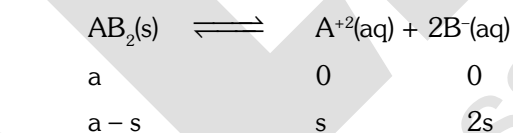
Ex. NaCl, $BaSO_4$, CH_3COONa , $CaCO_3$, NaCN, KCN, NH_4CN , NH_4Cl etc.



$$K_{sp} = s^2 \quad \text{or} \quad s = \sqrt{K_{sp}}$$

(ii) **K_{sp} of AB_2 or A_2B (Mono-di or di-mono valency) type salt -**

Ex. $CaCl_2$, $CaBr_2$, K_2S , $(NH_4)_2SO_4$, K_2SO_4 , K_2CO_3 etc.



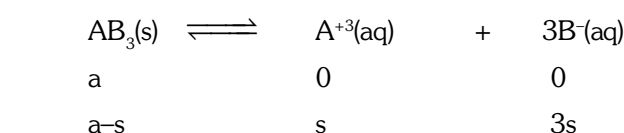
$$K_{sp} = [A^{+2}] [B^-]^2$$

$$K_{sp} = s \times (2s)^2 = s \times 4s^2 = 4s^3$$

$$s = \left(\frac{K_{sp}}{4} \right)^{\frac{1}{3}}$$

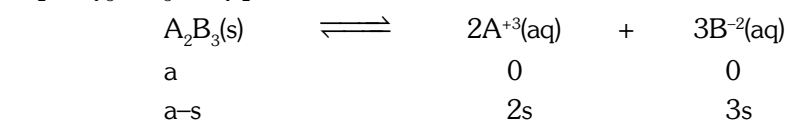
(iii) **K_{sp} of AB_3 or A_3B (Mono-tri or tri-mono valency) type salt -**

Ex. $FeCl_3$, $AlCl_3$, K_3PO_4 etc.



$$K_{sp} = [A^{+3}] [B^-]^3 = s \times (3s)^3 = 27s^4$$

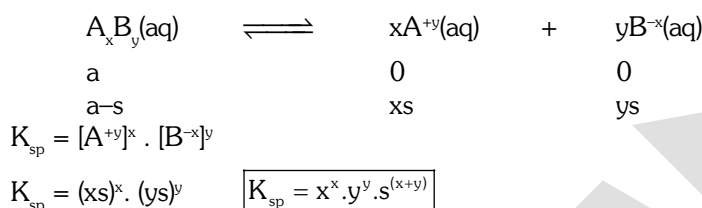
$$s = \left(\frac{K_{sp}}{27} \right)^{\frac{1}{4}}$$

(iv) K_{sp} of A_2B_3 or A_3B_2 (Di-tri or tri-di valency) type saltEx. $Al_2(SO_4)_3$, $Ba_3(PO_4)_2$ etc.

$$K_{sp} = [A^{+3}]^2 [B^{-2}]^3 = 2s \times 2s \times 3s \times 3s \times 3s = 108s^5$$

$$s = \left(\frac{K_{sp}}{108} \right)^{1/5}$$

(v) General form :



Illustrations

Illustration 15 What will be the solubility product of following in terms of solubility (mol L⁻¹)(i) $Al_2(SO_4)_3(s)$ (ii) $Na_2KPO_4(s)$ (iii) $NaKRbPO_4(s)$ **Solution**(i) $Al_2(SO_4)_3(s) \rightleftharpoons 2Al^{+3}(aq) + 3SO_4^{-2}(aq)$

$$K_{sp} = 2^2 \times 3^3 \times (S)^{2+3} = 4 \times 27 \times S^5 = 108 S^5$$

(ii) $Na_2KPO_4(s) \rightleftharpoons 2Na^{+}(aq) + K^{+}(aq) + PO_4^{-3}(aq)$

$$K_{sp} = 2^2 \times 1^1 \times 1^1 (S)^{2+1+1} = 4S^4$$

(iii) $NaKRbPO_4(s) \rightleftharpoons Na^{+}(aq) + K^{+}(aq) + Rb^{+}(aq) + PO_4^{-3}(aq)$

$$K_{sp} = 1^1 \times 1^1 \times 1^1 \times 1^1 \times (S)^{1+1+1+1} = S^4$$

Illustration 16 If solubility product of the base $M(OH)_3$ is 2.7×10^{-11} , the concentration of OH^{-1} will be
(1) 3×10^{-3} (2) 3×10^{-4} (3) 10^{-3} (4) 10^{-11} **Solution****Ans. (1)****Illustration 17 Assertion :** For a sparingly soluble salt, S is the maximum dissolved value of solute in a solution.**Reason :** K_{sp} corresponds to the ionic product of the salt in a saturated solution.

(1) A (2) B (3) C (4) D

Solution**Ans. (2)****Illustration 18** The solubility of $BaSO_4$ in water is 1.07×10^{-5} mol dm⁻³. Estimate its solubility product.**Solution**Solubility equilibrium for $BaSO_4$ is $BaSO_{4(s)} \rightleftharpoons Ba_{(aq)}^{2+} + SO_{(aq)}^{2-}$ The solubility product is, $K_{sp} = [Ba^{2+}] [SO_4^{2-}]$ If S is the molar solubility of $BaSO_4$, then $K_{sp} = S^2$ because $x = 1 = y$.Now, $S = 1.07 \times 10^{-5}$ M.

$$\text{Hence, } K_{sp} = (1.07 \times 10^{-5})^2 = 1.145 \times 10^{-10}$$

Illustration 19 The solubility product of $AgBr$ is 5.2×10^{-13} . Calculate its solubility in mol dm⁻³ and g dm⁻³. (Molar mass of $AgBr$ = 187.8 g mol⁻¹)**Solution**The solubility equilibrium of $AgBr$ is $AgBr_{(s)} \rightleftharpoons Ag_{(aq)}^{+} + Br_{(aq)}^{-}$ and $K_{sp} = [Ag^{+}] [Br^{-}] = S^2$ because $x = 1 = y$.

$$\text{The molar solubility } S \text{ of } AgBr \text{ is given by } S = \sqrt{K_{sp}} = \sqrt{5.2 \times 10^{-13}} = 7.2 \times 10^{-7} \text{ mol dm}^{-3}$$

The solubility in g dm⁻³ = molar solubility (mol dm⁻³) $\times$ molar mass (g mol⁻¹)

$$\text{Hence, solubility} = 7.2 \times 10^{-7} \text{ (mol dm}^{-3}) \times 187.8 = 1.35 \times 10^{-4} \text{ g dm}^{-3}.$$

(B) **Condition of precipitation /Ionic product (IP or Q_{sp}) :**

- Ionic product (IP) of an electrolyte is defined in the same way as K_{sp} . The only difference is that ionic product expression contains the initial concentration of ions or the concentration at any time whereas the expression of K_{sp} contains only equilibrium concentration. Thus, for AgCl.
 $Q_{sp} = IP = [Ag^+]_i [Cl^-]_i$ and $K_{sp} = [Ag^+]_{eq} [Cl^-]_{eq}$
- Ionic product changes with concentration but K_{sp} does not. K_{sp} is applicable for saturated solution of the sparingly soluble electrolyte.
- To decide whether an ionic compound will precipitate, its K_{sp} is compared with the value of ionic product. The following three cases arise :
 - $Q_{sp} < K_{sp}$: The solution is unsaturated and precipitation will not occur.
 - $Q_{sp} = K_{sp}$: The solution is saturated and solubility equilibrium exists.
 - $Q_{sp} > K_{sp}$: The solution is supersaturated and hence precipitation of the compound will occur.

Thus, a salt is precipitated when its ionic product exceeds the solubility product of the salt.

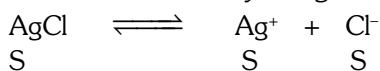
(C) **Common ion effect on solubility :**

Important point :- Solubility of substances always decreases in the presence of common ion. According to Le-Chatelier's principle on increasing common ion concentration equilibrium shifts in backward direction until the equilibrium is re-established so, the solubility of substances decreases but K_{sp} remains same because it is an equilibrium constant which depends only on temperature.

Illustrations

Illustration 20 Find out the solubility of AgCl in the presence of C molar NaCl solution ?

Solution.

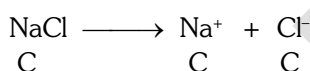


(Let solubility of AgCl is S mol L⁻¹)

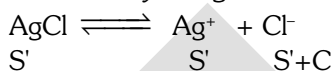
$$K_{sp} = [Ag^+] [Cl^-]$$

$$K_{sp} = S^2$$

In NaCl solution



Let solubility of AgCl in the presence of NaCl solution is S' mol L⁻¹.



According to L.M.A.

$$K_{sp} = [Ag^+] [Cl^-]$$

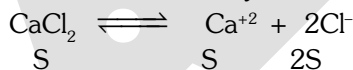
$$K_{sp} = S' (S' + C) = S'^2 + S'C \quad (\text{Neglecting the higher power terms of } S')$$

$$K_{sp} = S' C$$

$$S' = \frac{K_{sp}}{C}$$

Illustration 21 Find out the solubility of CaCl₂ solution in the presence of C NaCl solution ?

Solution



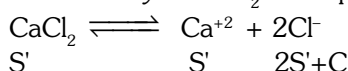
[Let solubility of CaCl₂ is S mol L⁻¹]

$$K_{sp} = [Ca^{+2}] [Cl^-]^2 = 4S^3$$

For NaCl solution



Let solubility of CaCl₂ in the presence of NaCl solution is S' mol L⁻¹.



According to L.M.A.

$$K_{sp} = [Ca^{+2}] [Cl^-]^2$$

$$K_{sp} = S' (2S' + C)^2 = S' (4S'^2 + 4S'C + C^2)$$

$$K_{sp} = 4S'^3 + 4S'^2 C + S' C^2 \quad (\text{Neglecting the higher power terms of } S')$$

$$S' = \frac{K_{sp}}{C^2}$$

$$\begin{array}{ccc} \text{CaCl}_2 & \longrightarrow & \text{Ca}^{+2} + 2\text{Cl}^- \\ \text{C} & & \text{C} \quad 2\text{C} \end{array}$$
$$\text{NaCl} \rightleftharpoons \text{Na}^+ + \text{Cl}^-$$
$$S' \qquad S' \qquad S'+2C$$
$$K_{sp} = [Na^+]^+ [Cl^-]^+$$
$$K_{sp} = S' (S' + 2C) = S'^2 + 2S'C \text{ (Neglecting the higher power terms of } S')$$

$$S' = \frac{K_{sp}}{2C}$$

***● Group precipitation of Salt**

Group	Radicals	Condition for precipitation (Group reagent)	Forms of precipitation
Zero	NH_4^+		
First	$\text{Pb}^{+2}, \text{Hg}^{+1}, (\text{Hg}_2^{+2}), \text{Ag}^+$	By mixing of dilute HCl	Chloride $\text{AgCl}, \text{Hg}_2\text{Cl}_2, \text{PbCl}_2$
Second II(A)	$\text{Pb}^{+2}, \text{Cu}^{+2}, \text{Hg}^{+2}, \text{Cd}^{+2}, \text{Bi}^{+3}$	H_2S gas passed in	Sulphide
Second II(B)	$\text{As}^{+3}, \text{Sb}^{+3}, \text{Sn}^{+2}, \text{Sn}^{+4}$	the presence of acidic medium	$\text{PbS}, \text{HgS}, \text{CuS}, \text{CdS},$ $\text{SnS}, \text{SnS}_2, \text{As}_2\text{S}_3,$ $\text{Sb}_2\text{S}_3, \text{Bi}_2\text{S}_3$
Third	$\text{Al}^{+3}, \text{Cr}^{+3}, \text{Fe}^{+3}$	NH_4OH mixed in the presence of NH_4Cl	Hydroxide $\text{Al}(\text{OH})_3, \text{Fe}(\text{OH})_3, \text{Cr}(\text{OH})_3$
Fourth	$\text{Zn}^{+2}, \text{Ni}^{+2}, \text{Mn}^{+2}, \text{Co}^{+2}$	H_2S gas passed in presence of basic medium	Sulphide $\text{MnS}, \text{CoS}, \text{NiS}, \text{ZnS}$
Fifth	$\text{Ba}^{+2}, \text{Sr}^{+2}, \text{Ca}^{+2}$	$(\text{NH}_4)_2\text{CO}_3$ mixed in the presence of NH_4Cl	Carbonate $\text{BaCO}_3, \text{SrCO}_3, \text{CaCO}_3$
Sixth	$\text{Na}^+, \text{K}^+, \text{Mg}^{+2}$	By mixing of Na_2HPO_4	Hydrogen phosphate (MgHPO_4)

- $\text{Hg}^{+2}(\text{ic}) \rightarrow \text{HgCl}_2 \rightarrow \text{stable in aqueous solution}$

$$\text{HgCl}_2 \rightleftharpoons \text{Hg}^{+2} + 2\text{Cl}^-$$
$$\text{Hg}_2\text{Cl}_2 \rightleftharpoons \text{Hg}_2^{+2} + 2\text{Cl}^-$$
$$S \qquad S \qquad 2S$$
$$S \qquad S \qquad 2S$$
$$\boxed{K_{sp} = 4S^3}$$
$$K_{sp} = [Hg_2^{+2}] [Cl^-]^2$$

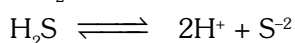
$$K_{sp} = 4S^3$$

$$= (S) (2S)^2$$

$$K_{sp} = 4S^3$$

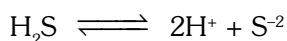
- For precipitation of II group, H_2S gas is passed in acidic medium to decrease S^{2-} concentration by common ion effect. So that ionic product of cations of group II and S^{2-} ions exceed the solubility product of their corresponding metal sulphides. Therefore only the cations of group II gets precipitated. But the sulphides of the cation of the other groups (III, IV & V) are not precipitated under these conditions because their solubility products are quite high.

When H_2S gas is directly passed through solution then IVth group is also precipitated with IInd group.



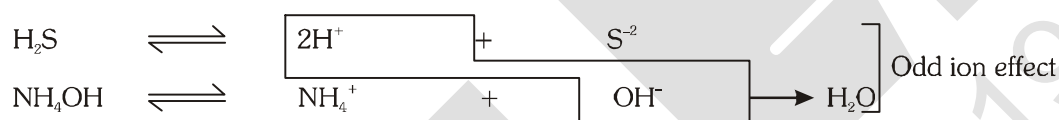
$$K_{sp_{II}} < K_{sp_{IV}} < [\text{radicals of II}^{\text{nd}} \text{ and IV group}] [\text{S}^{2-}]$$

When H_2S gas pass in acidic medium $[\text{H}^+]$ then only IInd group radicals are precipitated.



$$K_{sp_{II}} < [\text{radicals of II}^{\text{nd}} \text{ and IV group}] [\text{S}^{2-}] < K_{sp_{IV}}$$

- For precipitation of group III, NH_4OH is added in the presence of NH_4Cl to decrease the concentration of OH^- by common ion effect so that ionic product of cations of group III and hydroxides ions exceed the solubility product of their corresponding metal hydroxides and only group III cations will precipitated. On the other hand cations of group IV, V and VI, which require large concentration of OH^- due to their high solubility product, will not be precipitated.
- In IV group, H_2S gas is passed in basic medium to increase S^{2-} ion concentration by odd ion effect, so that the ionic product of cations of group IV and S^{2-} ions exceed the solubility product of their corresponding metal sulphide and hence gets precipitated.



So $[\text{S}^{2-}] \uparrow$

$$K_{sp_{IV}} < [\text{Radicals of group IV}] [\text{S}^{2-}]$$

(D) Precipitation of soap and salt (NaCl)

(i) Precipitation of NaCl (common salt) :

For the precipitation of NaCl, HCl gas passed in saturated solution of NaCl to increase the concentration of Cl^- ion by common ion, so ionic product of concentration of Na^+ and Cl^- ion becomes more than K_{sp} of NaCl and NaCl easily precipitated.

Saturated solution of NaCl $\text{NaCl} \rightleftharpoons \text{Na}^+ + \text{Cl}^-$

$$K_{sp} = [\text{Na}^+] [\text{Cl}^-]$$

By passing HCl gas



due to common ion

$$K_{sp} < [\text{Na}^+] [\text{Cl}^-] \uparrow$$

(ii) Precipitation of soap :

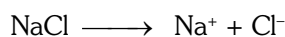
For precipitation of soap, NaCl mixed in saturated solution of soap to increase the concentration of Na^+ ion by common ion. So ionic product of concentration of Na^+ and stearate ions is more than K_{sp} of soap and soap is easily precipitated.

Ex. $\text{C}_{17}\text{H}_{35}\text{COONa}$ (Sodium stearate)

Saturated solution of $\text{C}_{17}\text{H}_{35}\text{COONa} \Rightarrow \text{C}_{17}\text{H}_{35}\text{COONa}^+ \rightleftharpoons \text{C}_{17}\text{H}_{35}\text{COO}^- + \text{Na}^+$

$$K_{sp} = [\text{C}_{17}\text{H}_{35}\text{COO}^-] [\text{Na}^+]$$

By mixing NaCl solution



due to common ion

$$K_{sp} < [\text{C}_{17}\text{H}_{35}\text{COO}^-] [\text{Na}^+] \uparrow$$

Illustrations

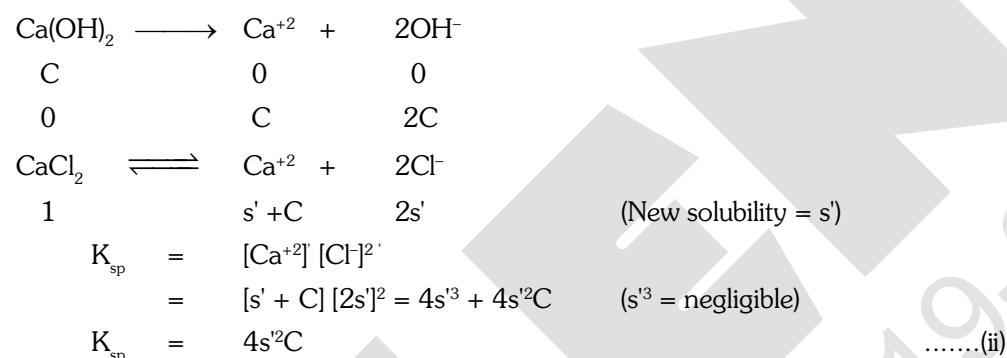
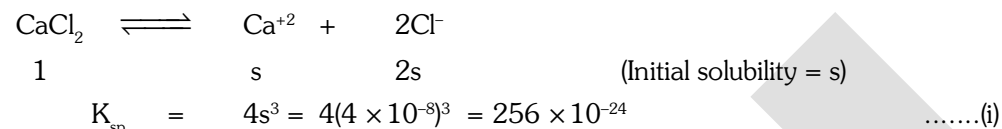
Illustration 23 Solubility products of $M(OH)_3$ and $M(OH)_2$ are 10^{-23} and 10^{-14} respectively. Which will be precipitated first on adding NH_4OH , if M^{+2} and M^{+3} both the ions are in solution ?

(1) M^{+2} (2) M^{+3} (3) Both M^{+2} and M^{+3} together

(4) Precipitation will not take place.

Solution**Ans. (2)**

Illustration 24 Solubility of $CaCl_2$ is 4×10^{-8} , then find out its K_{sp} and its new solubility in the presence of 10^{-2} M $Ca(OH)_2$.

Solution

$$\begin{aligned}
 \text{From equation (i) and (ii)} \quad s'^2 &= \frac{K_{sp}}{4C} = \frac{256 \times 10^{-24}}{4 \times 10^{-2}} = 64 \times 10^{-22} \\
 s' &= 8 \times 10^{-11} \text{ mol L}^{-1}
 \end{aligned}$$

4.8 FEW IMPORTANT POINTS

(A) Isohydric solution – If different-different solution has same pH are called isohydric solution.

Sp. point :- Isohydric conditions for two given weak acids HA_1 and HA_2 which has concentration C_1 and C_2 and ionisation constants are K_{a1} and K_{a2} respectively.

$$\text{Then } [H^+]_1 = \sqrt{K_{a1} C_1} \quad \text{and} \quad [H^+]_2 = \sqrt{K_{a2} C_2}$$

According to given condition for isohydric solution pH of different solution is same.

$$\text{Hence } [H^+]_1 = [H^+]_2$$

$$\sqrt{K_{a1} C_1} = \sqrt{K_{a2} C_2}$$

$$\boxed{K_{a1} C_1 = K_{a2} C_2} \quad \text{or} \quad \text{If } n = 1 \text{ mole then} \quad \boxed{\frac{K_{a1}}{V_1} = \frac{K_{a2}}{V_2}} \quad \therefore \quad \boxed{C \propto \frac{1}{V}}$$

(B) Relative strength of Acids :- It indicates that how many times an acid is stronger than the other acid.

$$\text{We know } \boxed{\text{Strength of acid} \propto [H^+]}$$

If there are two weak acids HA_1 and HA_2 which has concentration C_1 and C_2 , degree of ionisation α_1 and α_2 and ionisation constants K_{a1} and K_{a2} respectively then ratio of their strength of acids.

$$\text{We know } [H^+] = C\alpha = \sqrt{K_a C}$$

$$\text{So } \frac{\text{Strength of weak acid } HA_1}{\text{Strength of weak acid } HA_2} = \frac{[H^+]_1}{[H^+]_2} = \frac{C_1 \alpha_1}{C_2 \alpha_2} = \frac{\sqrt{K_{a1} C_1}}{\sqrt{K_{a2} C_2}}$$

If $C_1 = C_2$

$$\text{R.S.} = \frac{\text{Strength of weak acid HA}_1}{\text{Strength of weak acid HA}_2} = \sqrt{\frac{K_{a1}}{K_{a2}}} = \frac{\alpha_1}{\alpha_2}$$

Thus the relative strength of two acids of equimolar concentration can be compared by taking square root of the ratio of their ionisation constants.

Illustrations

Illustration 25 Which of the following solvents will undergo self-ionization ?

- (1) H_2O (2) NH_3 (3) HF (4) All of these

Solution **Ans. (4)**

Illustration 26 For two acids A and B, $\text{p}K_{a1} = 1.2$, $\text{p}K_{a2} = 2.8$ respectively in value, then which is true:-

- (1) A & B both are equally acidic
(2) A is stronger than B
(3) B is stronger than A
(4) None of these

Solution **Ans. (2)**

Illustration 27 pH values of two acids A and B are 4 and 5. The strengths of these two acids are related as :-

- (1) The strengths of the two acids cannot be compared.
(2) Acid B is 10 times stronger than acid A.
(3) Strength of acid A : Strength of acid B = 4 : 5
(4) Acid A is 10 times stronger than acid B.

Solution **Ans. (4)**

BEGINNER'S BOX-3

- The units of solubility product of silver chromate (Ag_2CrO_4) will be-
(1) $\text{mol}^2\text{L}^{-2}$ (2) $\text{mol}^3\text{L}^{-3}$ (3) mol L^{-1} (4) mol^{-1}L
- At a certain temperature, the solubility of the salt A_xB_y is S moles per liter. The general expression for its solubility product will be-
(1) $K_{sp} = x^y y^x S^{x+y}$ (2) $K_{sp} = (xy)^{x+y} S^{x+y}$ (3) $K_{sp} = (x^x y^y) S^{x+y}$ (4) $K_{sp} = x^x y^y S^{xy}$
- The molar solubility of silver sulphate is $1.5 \times 10^{-2} \text{ mol L}^{-1}$. The solubility product of the salt will be-
(1) 2.25×10^{-4} (2) 1.35×10^{-5} (3) 1.7×10^{-6} (4) 3.0×10^{-3}
- The precipitate of CaF_2 ($K_{sp} = 1.7 \times 10^{-10}$) is obtained when equal volumes of the following are mixed:
(1) $10^{-3} \text{ M Ca}^{2+} + 10^{-5} \text{ M F}^-$ (2) $10^{-5} \text{ M Ca}^{2+} + 10^{-3} \text{ M F}^-$
(3) $10^{-2} \text{ M Ca}^{2+} + 10^{-3} \text{ M F}^-$ (4) $10^{-4} \text{ M Ca}^{2+} + 10^{-4} \text{ M F}^-$
- If S_0 , S_1 , S_2 and S_3 are the solubilities of AgCl in water, 0.01 M CaCl_2 , 0.01 M NaCl and 0.5 M AgNO_3 solutions, respectively, then which of the following is true?
(1) $S_0 > S_2 > S_1 > S_3$ (2) $S_0 = S_2 = S_1 > S_3$
(3) $S_3 > S_1 > S_2 > S_0$ (4) $S_0 > S_2 > S_3 > S_1$
- Given $K_{sp}(\text{AgI}) = 8.5 \times 10^{-17}$. The solubility of AgI in 0.1 M KI solution is-
(1) 0.1 M (2) $8.5 \times 10^{-16} \text{ M}$
(3) $8.5 \times 10^{-17} \text{ M}$ (4) $8.5 \times 10^{-18} \text{ M}$

4.9 pH

1. INTRODUCTION

pH of some important substance :-

- | | | | |
|-------|---------------|---|------------|
| (i) | Blood | → | 7.4 |
| (ii) | Tear | → | 7.4 |
| (iii) | Gastric Juice | → | 1 to 3 |
| (iv) | Soft drink | → | 2 to 4 |
| (v) | Acidic rain | → | 6 |
| (vi) | Milk | → | 6.3 to 6.6 |
| (vii) | Water | → | 7 |

2. TYPES OF pH

- (i) pH of SA and SB. $[H^+] = C$, $[OH^-] = C$
- (ii) pH of WA and WB. $[H^+] = Ca$, $[OH^-] = Ca$
- (iii) pH of very dilute solution. (Consider H^+ & OH^- of H_2O also)
- (iv) pH of salt solution.
- | | | | | | |
|-----|----|----|------|------|------------|
| (a) | SA | SB | Type | Salt | (always 7) |
| (b) | SA | WB | Type | Salt | (< 7) |
| (c) | WA | SB | Type | Salt | (> 7) |
| (d) | WA | WB | Type | Salt | (Almost 7) |

3. pH of mixture of acid and base.

- (a) pH of mixture of strong acids :

$$N_1V_1 + N_2V_2 + N_3V_3 + \dots = NV$$

V = Volume of final solution = $V_1 + V_2 + V_3 + \dots$ N = Normality of final solution = $[H^+]$ in final solution.

- (b) pH of mixture of strong base :

$$N_1V_1 + N_2V_2 + N_3V_3 + \dots = NV$$

V = Volume of final solution = $V_1 + V_2 + V_3 + \dots$ N = Normality of final solution = $[OH^-]$ in final solution.

- (c) pH of mixture of strong acids and strong bases :

For acid :

$$N_1V_1 + N_2V_2 + N_3V_3 + \dots = (NV)_{\text{Acid}}$$

For base :

$$N_1V_1 + N_2V_2 + N_3V_3 + \dots = (NV)_{\text{Base}}$$

- (i) If
- $(NV)_{\text{Acid}} > (NV)_{\text{Base}}$
- then solution is acidic.

$$NV = (NV)_{\text{Acid}} - (NV)_{\text{Base}} \quad \text{and}$$

$$[H^+] = N$$

- (ii) If
- $(NV)_{\text{Base}} > (NV)_{\text{Acid}}$
- then solution is basic.

$$NV = (NV)_{\text{Base}} - (NV)_{\text{Acid}} \quad \text{and}$$

$$[OH^-] = N$$

- (iii) If,
- $(NV)_{\text{acid}} = (NV)_{\text{base}}$
- then solution is neutral.

- (d) pH of mixture of (WA + SB); when equivalent of WA > SB]

pH of mixture of (SA + WB); When equivalent of WB > SA]

↓

gives rise to concept of buffer solution

4.10 BUFFER SOLUTION

- (A) **Definition** : A solution which resist the change in pH and pH does not change significantly on addition of small amount of strong acid or strong base is called buffer solution.
- (B) **Properties of buffer solution** :
- The pH of buffer solution does not change appreciably upon the addition of small amount of either strong acid or strong base.
 - The pH of buffer solution does not depend on the volume of solution. Hence, solution can be diluted without change in pH.
 - The pH of buffer solution remains constant even if it is kept for a long time.
- (C) **Buffer solutions are used in**
- Qualitative analysis of mixture
 - Quantitative analysis of mixture
 - Digestion of food
 - Preservation of food and fruits.
 - Agriculture and dairy product preservation.
 - Industrial process such as manufacture of paper, dyes inks, paints, drugs etc.

TYPES OF BUFFER SOLUTION

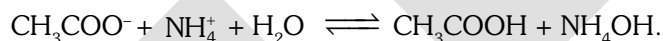
- Simple buffer solution
- Mixed buffer solution :-
 - Acidic buffer solution
 - Basic buffer solution

- (A) **Simple buffer solution :-** Aqueous solution of weak acid-weak base (WA – WB) types of salt.

Ex. $\text{CH}_3\text{COONH}_4$, NH_4CN , AgCN etc.

$$\text{pH} = 7 + \frac{1}{2} \text{pK}_a - \frac{1}{2} \text{pK}_b$$

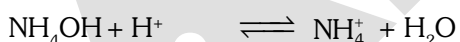
pH does not depend on concentration.



Buffer Action :

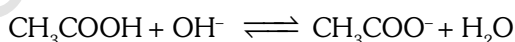
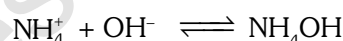
Case 1

When mixing of acid $[\text{H}^+]$



Case 2.

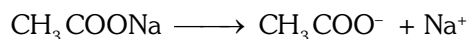
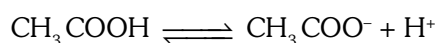
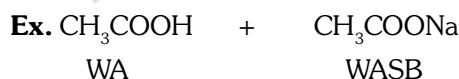
When mixing of base $[\text{OH}^-]$



- (B) **Mixed buffer solution :**

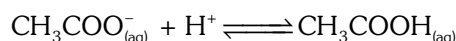
- (i) **Acidic buffer solution :-** The solution in which weak acid and its conjugate base are present.
OR

Aqueous solution of mixture of weak acid and salt of same weak acid with any strong base is called acidic buffer solution.

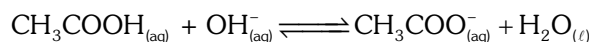


BUFFER ACTION :-

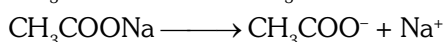
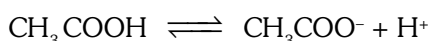
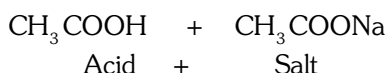
Case 1. If a small amount of acid is added to the buffer solution. The H^+ ions of the acid react with CH_3COO^- ions and produce CH_3COOH and most of the added H^+ ions are consumed so there is no appreciable change in pH.



Case 2. If small amount of base is added to the buffer solution, the OH^- ions are consumed by CH_3COOH .



So, there is no appreciable change in pH.

(a) pH of acidic buffer solution :

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

$$\text{or } [H^+] = \frac{K_a[CH_3COOH]}{[CH_3COO^-]} = \frac{K_a[\text{Acid}]}{[\text{Conjugate base}]}$$

taking $-\log$ on both sides

$$pH = pK_a - \log \frac{[\text{Acid}]}{[\text{Conjugate base}]} \quad \text{or} \quad pH = pK_a + \log \frac{[\text{Conjugate base}]}{[\text{Acid}]}$$

Henderson's equation :

$$pH = pK_a + \log \frac{[\text{Salt}]}{[\text{Acid}]} \quad \text{or} \quad \frac{[\text{Conjugate base}]}{[\text{Acid}]}$$

- $[\text{Conjugate base}]_{eq} \approx [\text{Salt}]$ because CH_3COO^- mainly comes from salt since dissociation of CH_3COOH in presence of CH_3COONa is appreciably decreased.
- $[\text{Acid}]_{aq} \approx$ initial concentration of acid since it is almost unionised in presence of salt due to common ion effect.

(b) pH range of acidic buffer solution : It depends on pK_a of acid and ratio of salt to acid concentrations.

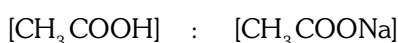
$$[CH_3COOH] : [CH_3COONa] \Rightarrow pH = pK_a + \log \frac{[CH_3COONa]}{[CH_3COOH]}$$

$$(i) \quad \text{If, } 1 : 10 \Rightarrow pH = pK_a + \log \frac{10}{1} = pK_a + 1$$

$$(ii) \quad \text{If, } 10 : 1 \Rightarrow pH = pK_a - 1$$

So pH range

$$pH = pK_a \pm 1$$

(c) Maximum buffer action condition of acidic buffer solution :

$$1 : 1 \Rightarrow pH = pK_a + \log \frac{1}{1}$$

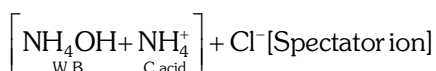
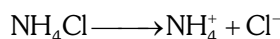
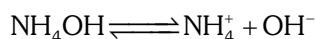
$$pH = pK_a$$

(ii) **Basic buffer solution :**

Definition : The solution in which weak base and its conjugate acid are present. OR

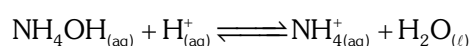
Aqueous solution of mixture of weak base and salt of same weak base with any strong acid is called basic buffer solution.

Ex. $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$



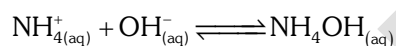
BUFFER ACTION :-

Case 1. If a small amount of the acid is added to the buffer solution, the H^+ ions are consumed by NH_4OH .



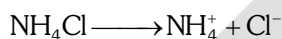
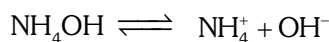
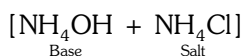
So there is no appreciable change in pH.

Case 2. If small amount of base is added to the buffer solution, the OH^- ions react with NH_4^+ ions to produce NH_4OH .



Hence most of the added OH^- ions are consumed. so there is no appreciable change in pH.

(a) **pOH of basic buffer solution :**



$$K_b = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_4\text{OH}]} \quad \text{or} \quad [\text{OH}^-] = \frac{K_b [\text{NH}_4\text{OH}]}{[\text{NH}_4^+]}$$

taking $-\log$ on both sides $\text{pOH} = \text{p}K_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_4\text{OH}]}$

Henderson's equation :

$$\boxed{\text{pOH} = \text{p}K_b + \log \frac{[\text{Salt}]}{[\text{Base}]} \text{ or } \frac{[\text{Conjugate acid}]}{[\text{Base}]}}$$

(b) **pOH range of basic buffer solution :** It depends on $\text{p}K_b$ of base and ratio of salt to base concentrations.

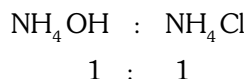
$$[\text{NH}_4\text{OH}] : [\text{NH}_4\text{Cl}] \Rightarrow \text{pOH} = \text{p}K_b + \log \frac{[\text{NH}_4\text{Cl}]}{[\text{NH}_4\text{OH}]}$$

(i) If, 1 : 10 $\Rightarrow \text{pOH} = \text{p}K_b + 1$

(ii) If, 10 : 1 $\Rightarrow \text{pOH} = \text{p}K_b - 1$

So, pOH range : $\boxed{\text{pOH} = \text{p}K_b \pm 1}$

(c) Maximum buffer action condition of basic buffer solution :



$$\boxed{\text{pOH} = \text{pK}_b}$$

3. BUFFER CAPACITY

Definition :-

- (i) It is defined as the number of moles of strong acid (or strong base) added to one litre of a buffer solution to change its pH by one unit.
- (ii) It measures the effectiveness of a buffer.
- (iii) Larger the value of buffer capacity more resistant is the solution to pH change.

$$\text{Buffer capacity} = \frac{\text{Number of moles of acid or base added per litre}}{\text{Change in pH of buffer solution}}$$

Illustrations

Illustration 28 When 2 moles of HCl is added to 1 L. of an acidic buffer solution, its pH changes from 3.9 to 3.4. Find its buffer capacity.

Solution $\text{B.C.} = \frac{2}{0.5} = 4$

BEGINNER'S BOX-4

- A buffer solution is one which has—
 - (1) reserved acid
 - (2) reserved base
 - (3) constant pH
 - (4) pH equal to 7
- Which of the following solutions cannot act as a buffer system ?
 - (1) $\text{KH}_2\text{PO}_4/\text{H}_3\text{PO}_4$
 - (2) $\text{NaClO}_4/\text{HClO}_4$
 - (3) $\text{C}_5\text{H}_5\text{N}/\text{C}_5\text{H}_5\text{N}^+\text{HCl}^-$
 - (4) $\text{Na}_2\text{CO}_3/\text{NaHCO}_3$
- A buffer solution can not be prepared by mixing equimolar amounts of—
 - (1) B(OH)_3 and $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$
 - (2) NH_3 and NH_4Cl
 - (3) HCl and NaCl
 - (4) CH_3COOH and CH_3COONa
- Which of the following salt solution will act as a buffer ?
 - (1) $\text{CH}_3\text{COONH}_4$ (aq.)
 - (2) NH_4Cl (aq.)
 - (3) CH_3COONa (aq.)
 - (4) NaCl (aq.)
- Which of the following combinations will make a buffer solutions ?
 - (i) CH_3COONa (2 mol) + HCl (1 mol)
 - (ii) CH_3COOH (2 mol) + NaOH (1 mol)
 - (iii) CH_3COOH (1 mol) + CH_3COONa (1 mol)
 - (1) (iii)
 - (2) (i), (ii)
 - (3) (ii), (iii)
 - (4) (i), (ii), (iii)
- The pH of blood circulating in a human body is maintained around 7.4 by the action of the buffer system—
 - (1) $\text{CH}_3\text{COOH}/\text{CH}_3\text{COONa}$
 - (2) $\text{NH}_4\text{Cl}/\text{NH}_3$
 - (3) $\text{H}_2\text{PO}_4^{2-}$
 - (4) $\text{H}_2\text{CO}_3/\text{HCO}_3^-$

4.11 INDICATOR

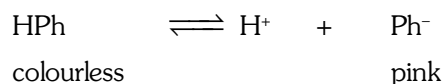
(A) **Definition :-** An indicator is a substance that undergoes change in its colour to show end point of an acid-base titration.

- Titration involves neutralization of an acid and base.

(B) **Types of indicators :**

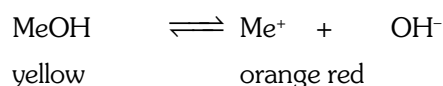
(a) **Acidic indicator (HIn)**

Ex. Phenolphthalein (HPh)



(b) **Basic indicator (InOH)**

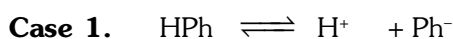
Ex. Methyl orange (MeOH)



OSTWALD THEORY OF INDICATORS

- According to this theory, all the indicators are weak organic acid or base.
- Every indicator has light colour in its unionised form and dark colour in ionised form.
- Every indicator has one colour in acidic medium and an altogether different colour in basic medium due to common ion effect and odd ion effect.

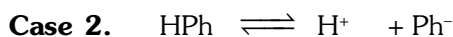
Ex. Acidic indicator [HPh]



In acidic medium $[\text{H}^+]$

$$[\text{HPh}] > [\text{Ph}^-]$$

Colourless due to common ion effect.

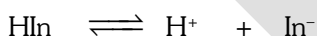


In basic medium $[\text{OH}^-]$

$$[\text{Ph}^-] > [\text{HPh}]$$

Pink colour due to odd ion effect.

(a) **pH of acidic indicator -**



$$K_a \text{ or } K_1 = \frac{[\text{H}^+][\text{In}^-]}{[\text{HIn}]}$$

$$[\text{H}^+] = \frac{K_1[\text{HIn}]}{[\text{In}^-]}$$

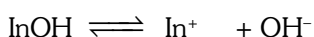
Taking - log on both sides

$$\text{pH} = \text{p}K_1 - \log [\text{HIn}] + \log [\text{In}^-]$$

$$\text{pH} = \text{p}K_1 + \log \frac{[\text{In}^-]}{[\text{HIn}]}$$

$$\boxed{\text{pH} = \text{p}K_1 + \log \frac{[\text{Ionised form}]}{[\text{Unionised form}]}}$$

(b) **pOH of basic indicator -**



$$K_b \text{ or } K_1 = \frac{[\text{In}^+][\text{OH}^-]}{[\text{InOH}]}$$

$$\boxed{\text{pOH} = \text{p}K_1 + \log \frac{[\text{Ionised form}]}{[\text{Unionised form}]}}$$

(c) pH range of indicators :

Indicators show colour change for a certain pH range not at a definite pH which is called pH range of indicator.

(i) pH range of acidic indicator :

$$\begin{array}{ccc} [\text{HIn}] & : & [\text{In}^-] \\ 1 & : & 10 \\ 10 & : & 1 \end{array} \quad \boxed{\text{pH} = \text{pK}_i \pm 1}$$

(ii) pOH range of basic indicator :

$$\begin{array}{ccc} [\text{InOH}] & : & [\text{In}^+] \\ 1 & : & 10 \\ 10 & : & 1 \end{array} \quad \boxed{\text{pOH} = \text{pK}_i \pm 1}$$

(iii) Neutral condition for indicators (No working condition)

For acidic indicator

$$\begin{array}{ccc} [\text{HIn}] & : & [\text{In}^-] \\ 1 & : & 1 \end{array}$$

$$\boxed{\text{pH} = \text{pK}_i}$$

For basic indicators

$$\begin{array}{ccc} [\text{InOH}] & : & [\text{In}^+] \\ 1 & : & 1 \end{array}$$

$$\boxed{\text{pOH} = \text{pK}_i}$$

S.No.	Name of indicator	Colour in acidic medium	Colour in basic medium	Working pH range of indicators
1.	Methyl orange (MeOH)	Orange red	Yellow	3.1 to 4.5
2.	Methyl red	Red	Yellow	4.2 to 6.2
3.	Phenol red	Yellow	Red	6.2 to 8.2
4.	Phenolphthalein (HPh)	Colourless	Pink	8.2 to 10.2

3. ACID-BASE TITRATION

S.No.	Type of titration	pH range of titration	Suitable indicators
1.	SA/SB.	3 – 11	All indicators (MeOH, HPh etc.)
2.	SA/WB	3 – 7	Methyl orange (MeOH) and methyl red
3.	WA/SB	7 – 11	Phenolphthalein (HPh)

Important points of titration :

- Titration is always possible in two opposite solution. (i.e. one is acidic and other is basic).
- For any titration only that indicator is suitable if their working pH range is in pH range of titration.
- At the end point of titration equivalents of acids and bases are always same.
- At the equivalence point, nature of solution depends on the type of titration.

SA/SB	-	Neutral (pH = 7)
SA/WB	-	Acidic
WA/SB	-	Basic
WA/WB	-	Any one possible

4.12 ACID AND BASE

(1) INTRODUCTION

The earliest classification of substances into acids and bases was based upon the characteristic properties possessed by each one of them. For example,

Acid was defined as a substance whose aqueous solution possesses the following characteristic properties:

- (i) conducts electricity,
- (ii) reacts with active metals like zinc, magnesium etc. to give hydrogen,
- (iii) turns blue litmus red.
- (iv) has a sour taste.
- (v) whose acidic properties disappear on reaction with a base.

Base was defined as a substance whose aqueous solution possessed the following characteristic properties:

- (i) conducts electricity,
- (ii) reacts with active metals like zinc, magnesium etc. to give hydrogen,
- (iii) turns red litmus blue,
- (iv) has a bitter taste
- (v) whose basic properties are destroyed on reaction with an acid.

The above definitions of acids and bases are called operational definitions as they are based on certain operations (i.e. tests) to be performed on the substances. However, these have been replaced by conceptual definitions (put forward by Arrhenius, Bronsted-Lowry and Lewis) which go into the causes of the observed behavior, based upon structure and composition of the substances.

(2) Arrhenius Concept (1884)

- (a) **Acid:-** According to this concept, those substances which produce free H^+ ions in aqueous solution are called acid.

Example $HCl, HNO_3, H_2SO_4, H_3PO_4, H_2CO_3, H_2S, CH_3COOH$ etc.



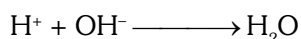
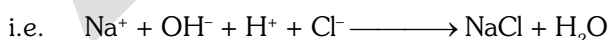
- (b) **Base :-** Those substances which produce free OH^- ions in aqueous solution are called base.

Example $NaOH, KOH, Cs(OH), Rb(OH), NH_4OH, Ba(OH)_2, Ca(OH)_2, Al(OH)_3$ etc.



- (c) **Nature of water :-** According to this concept nature of water is neutral and acts as a solvent.

- (d) **Neutralisation Reaction :-** Those reactions in which acid and base react together to form water molecule are called neutralisation reactions.



- (e) **Strength of acids and bases :-** This concept explains the strength of acids and bases depending upon the basis of degree of ionisation.

Example For strong electrolytes $\alpha \simeq 100\%$

For weak electrolytes $\alpha < 100\%$

- (f) **Advantage :-** This concept explains the acids and bases practically. i.e. To find out the pH, ionisation constant, hydrolysis constants, heat of neutralisation etc.

(g) Disadvantage :-

It explains the behaviour of acids and bases only in aqueous (water) solvents.

Illustrations

Illustration 29 Gaseous hydrogen chloride is a very poor conductor of electricity but a solution of hydrogen chloride in water is a good conductor. This is due to the fact that :-

- (1) Water is a good conductor of electricity
- (2) Hydrogen chloride ionises in water
- (3) A gas cannot conduct electricity but a liquid can
- (4) HCl does not obey Ohm's law where as the solution does

Solution (2)

Illustration 30 Which is acid in the following pairs according to Arrhenius concept ?

- | | |
|-------------------------|--|
| (1) HCl(g) and HCl (aq) | (2) CH ₃ COOH(l) and CH ₃ COOH(aq) |
|-------------------------|--|

Solution (1) HCl(aq.) (2) CH₃COOH(aq.)

(3) Bronsted-Lowry Concept (1923)

It is based upon the exchange of proton.

(a) Acid :- According to this concept those substances which have tendency to donate the proton (H⁺) by any method in any solvent are called acids.

Example (i) Neutral molecules - HCl, HNO₃, H₂SO₄, H₂CO₃, H₂S, CH₃COOH, H₃PO₃ etc.

(ii) Anions - HS⁻, HCO₃⁻, HSO₄⁻, H₂PO₄⁻, HPO₄⁻², H₂O etc.

(iii) Cations - NH₄⁺, H₃O⁺, PH₄⁺, CH₃COOH₂⁺ etc.

[Al(H₂O)₆]⁺³, [Ag(H₂O)₂]⁺¹, [Fe(H₂O)₆]⁺³ etc.

Ex. 1 HCl (Acid) + H₂O (Solvent) $\longrightarrow$ H₃O⁺ + Cl⁻

Ex.2 HS⁻ (Acid) + H₂O (Solvent) $\longrightarrow$ H₃O⁺ + S⁻²

Ex.3 NH₄⁺ (Acid) + H₂O (Solvent) $\longrightarrow$ NH₃ + H₃O⁺

Ex.4 [Al(H₂O)₆]⁺³ (Acid) + H₂O (Solvent) $\longrightarrow$ [Al(H₂O)₅OH]⁺² + H₃O⁺

(b) Base :- Those substances which have tendency to accept the proton by any method in any solvent are called the bases.

(1) HS⁻, HCO₃⁻, HSO₄⁻, H₂PO₄⁻, HPO₄⁻², O⁻², SO₄⁻², CO₃⁻², Cl⁻, Br⁻, I⁻, CN⁻ etc.

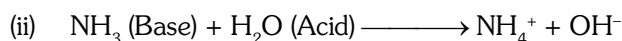
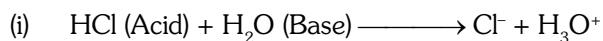
(2) NH₃, RNH₂, R₂NH, R₃N, C₆H₅NH₂, C₅H₅N, H₂N-NH₂ etc.

Example (i) HS⁻ (Base) + H₂O (Solvent) $\longrightarrow$ H₂S + OH⁻

(ii) NH₃ (Base) + H₂O (Solvent) $\longrightarrow$ NH₄⁺ + OH⁻

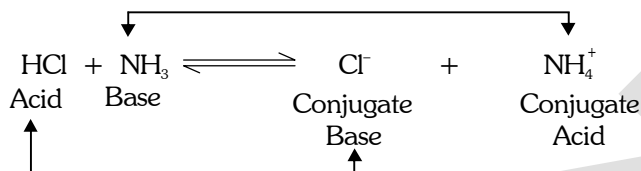
(iii) CO₃⁻² (Base) + H₂O (Solvent) $\longrightarrow$ HCO₃⁻ + OH⁻

(c) **Nature of water :-** According to this concept nature of water is amphoteric or amphiprotic i.e. water can act as both acid and base.

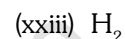
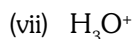
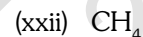
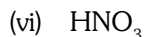
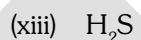
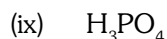
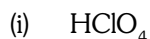


(d) **Neutralisation Reaction :-** According to this concept those reaction in which acid and base react together and convert into their respective conjugate acid and base are called neutralisation reactions.

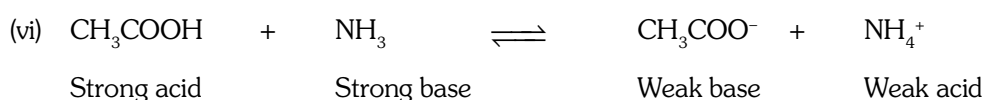
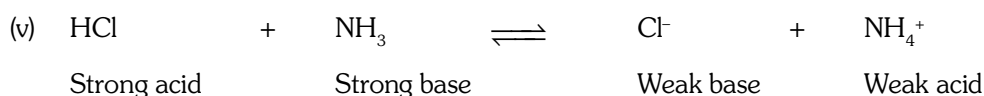
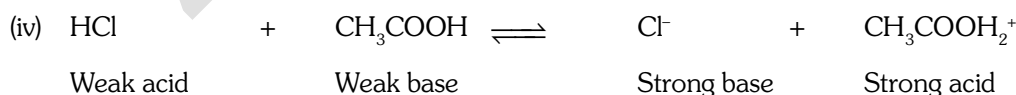
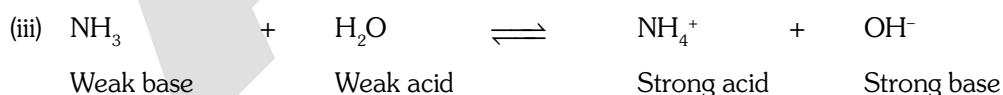
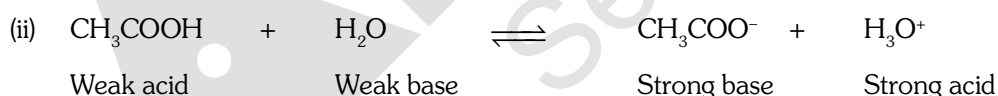
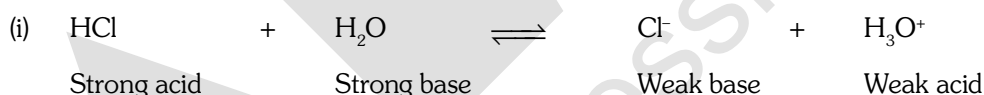
Example



(e) **Strength of acids and bases :-** This concept explain the strength of acid and base depending upon the basis of relative tendency to accept or donate the proton.



Example



GOLDEN KEY POINTS

Types of Solvent

(a) **Protogenic or acidic solvent** :- They have a tendency to generate or donate the protons.

Example CH_3COOH , HCl , HNO_3 , H_2SO_4 etc.

(b) **Protophilic or basic solvent** :- They have a tendency to accept the protons.

Example NH_3 , CH_3OH , RNH_2 etc.

(c) **Amphiprotic or amphoteric solvents** :- They have the tendency to accept or donate the protons or they are either protophilic or protogenic.

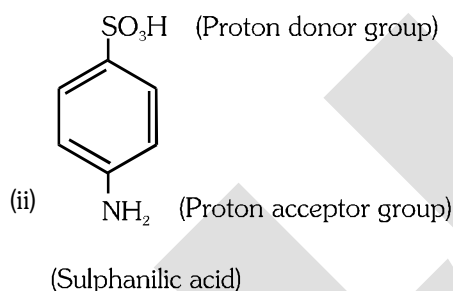
Example H_2O , HS^- , HSO_4^- , HCO_3^- etc.

(d) **Aprotic solvents** :- They neither donate nor accept the protons.

Example C_6H_6 , CCl_4 , CHCl_3 , BrF_3 , NO_2 , COCl_2 etc.

(e) **Ampholyte solvents** - Those solvents in which the same molecule acts as proton donor and proton acceptor.

Example (i) $\text{H}_2\text{N} - \text{CH}_2 - \text{COOH} \longrightarrow \text{H}_3^+\text{N} - \text{CH}_2 - \text{COO}^-$
(Glycine) (Zwitter ion)



(f) **Advantage** :-

- (i) It explain the behaviour of acids and bases in any type of solvent.
- (ii) It explain the stability of proton (H^+).

(g) **Disadvantage** :-

- (i) It does not explain the acid and base practically.
- (ii) It does not explain acidic behaviour of aprotic acid. (SO_2 , SO_3 , CO_2 , AlCl_3 , SiCl_4 etc.)

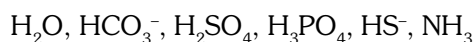
Illustrations

Illustration 31 In the process : $\text{NH}_3 + \text{NH}_3 \rightleftharpoons \text{NH}_2^- + \text{NH}_4^+$, The nature of ammonia is:-

- (1) Acidic (2) Basic (3) Amphoteric (4) None

Solution. **Ans.(3)**

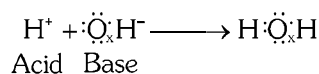
Illustration 32 Which of the following behave both as Bronsted acid as well as Bronsted bases ?



Solution. H_2O , HCO_3^- , HS^- , NH_3

(4) **Lewis Concept (1939)**

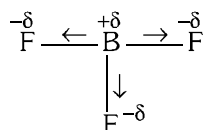
- (a) **Lewis Acid :-** According to this concept those species which have self tendency to accept the lone pair of electrons are called acids. i.e. Lewis acid is an electron pair acceptor (electrophilic).



Classification of Lewis Acids :-

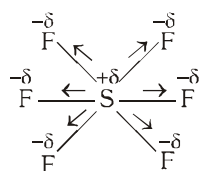
- (i) Compounds whose central atom have an incomplete octet (electron deficient)

Example BF_3 , BBr_3 , BCl_3 , BI_3 , $\text{B}(\text{CH}_3)_3$, $\text{B}(\text{OH})_3$, AlCl_3 , GeCl_3 etc.

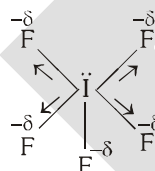


- (ii) Compounds whose central atom have vacant d-orbitals and can accept one or more pair of lone electrons.

Example SF_4 , SF_6 , SnCl_2 , SnCl_4 , PX_3 , PX_5 , GeX_4 , TeX_4 , IF_5 , IF_7 , etc.



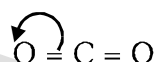
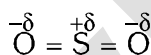
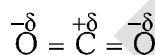
SF_6



IF_5

- (iii) Molecules with a multiple bond between atoms of different electronegativities.

Example CO_2 , SO_2 , SO_3 etc.

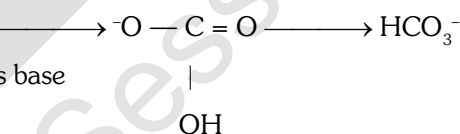


Lewis acid

+



Lewis base



Lewis acid

+



Lewis base

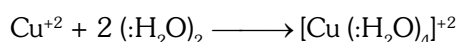


- (iv) **Cations :-**

Example Ag^+ , Na^+ , Li^+ , Al^{+3} , Be^{+2} , Mg^{+2} , I^+ , Cl^+ , H^+ etc.

False cations :- Example NH_4^+ , H_3O^+ , $\text{CH}_3\text{COOH}_2^+$, PH_4^+ etc.

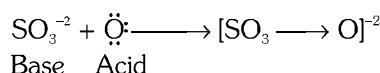
Example $\text{Ag}^+ + 2 : \text{NH}_3 \rightleftharpoons [\text{Ag} (: \text{NH}_3)_2]^+$



- (v) Elements which have six electrons in their outermost shell or valence shell.

O-Family (O, S, Se, Te)

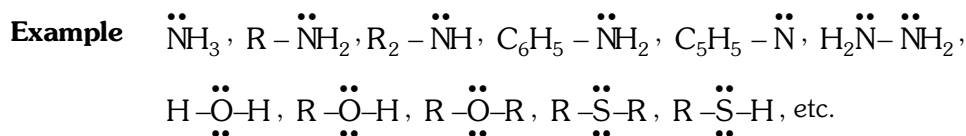
Po – Radioactive element



- (b) Lewis Base :-** Those species which have self tendency to donate the lone pair of electrons are called bases. i.e. a base is an electron pair (lone pair) donor (nucleophile).

Classification of Lewis Bases

- (i) Those species whose central atom have lone pair of electrons and have self tendency to donate them –



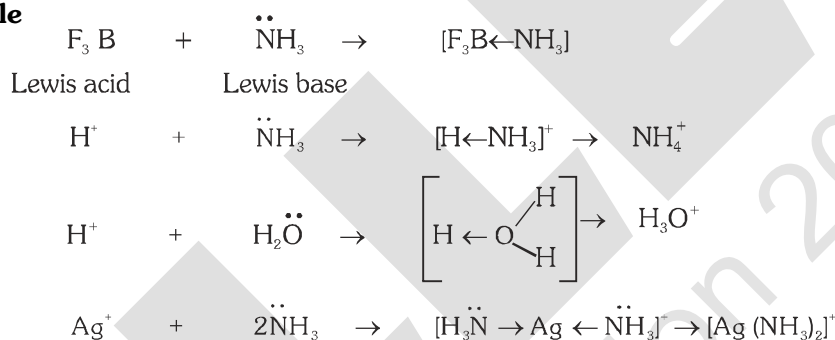
- (ii) **Anions –**



- (c) Nature of water :-** According to this concept nature of water is basic i.e. water is a base.

- (d) Neutralisation Reaction :-** According to this concept those reactions in which acid and base react together to form a co-ordinate bond between them are called neutralisation reactions.

Example



- (e) Strength of acids and bases :-** This concept explain the strength of acids and bases depending upon the basis of self tendency. To accept or donate the lone pair of electrons (e^-).

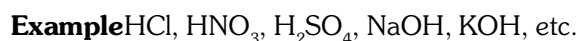


(f) Advantage :-

- This concept explains the acid and base without any solvent.
- This concept explains those acids and bases which are not explained by other concepts.

(g) Disadvantage :-

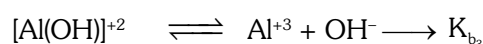
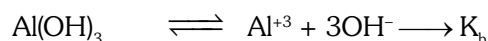
- This concept fails to explain the acid and base practically.
- According to this concept forming of co-ordinate bond in neutralisation reaction of acid and base is a slow process, whereas neutralisation reaction of acids and bases is actually fast process.
- This concept fails to explain the real acids and bases



- All the lewis bases are bronsted bases but all the lewis acids are not bronsted acids.
- All Arrhenius acids are Bronsted acids but it is not so for bases.

(6) POLYACIDIC BASE AND POLYBASIC ACIDS

(a) Polyacidic Base



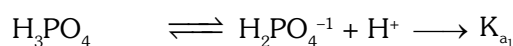
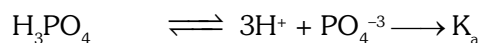
$$K_b = K_{b_1} \times K_{b_2} \times K_{b_3}$$

taking $-\log$ on both sides

$$\text{p}K_b = \text{p}K_{b_1} + \text{p}K_{b_2} + \text{p}K_{b_3}$$

$$\text{i.e. } \boxed{K_{b_1} > K_{b_2} > K_{b_3}} \Rightarrow \boxed{\text{p}K_{b_1} < \text{p}K_{b_2} < \text{p}K_{b_3}}$$

(b) Polybasic Acid



$$K_a = K_{a_1} \times K_{a_2} \times K_{a_3}$$

taking $-\log$ on both sides

$$\text{p}K_a = \text{p}K_{a_1} + \text{p}K_{a_2} + \text{p}K_{a_3}$$

$$\boxed{K_{a_1} > K_{a_2} > K_{a_3}} \Rightarrow \boxed{\text{p}K_{a_1} < \text{p}K_{a_2} < \text{p}K_{a_3}}$$

Illustrations

Illustration 33 In the dissociation, $\text{H}_2\text{A} \xrightleftharpoons{K_1} \text{H}^+ + \text{HA}^-$; $\text{HA}^- \xrightleftharpoons{K_2} \text{H}^+ + \text{A}^{2-}$

(1) K_1 is equal to K_2

(2) K_1 is smaller than K_2

(3) K_1 is greater than K_2

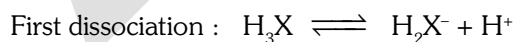
(4) K_1 is negligible

Solution.

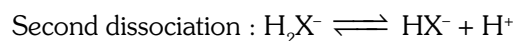
Ans. (3)

Illustration 34 Calculate the pH of 2.0×10^{-4} M H_3X Solution assuming first dissociation to be 100%, second to be 50% and third to be negligible :

Solution.



$$\therefore \text{H}^+ \text{ ion concentration} = 2 \times 10^{-4} \text{ M}$$



$$\text{H}^+ \text{ ion concentration} = C\alpha = 2 \times 10^{-4} \times 0.5$$

$\therefore$ Third dissociation is negligible

$$\therefore [\text{H}^+] = 2 \times 10^{-4} + 10^{-4} = 3 \times 10^{-4}$$

$$\text{pH} = 4 - \log 3 = 3.52$$

BEGINNER'S BOX-5

- Which of the following is a Bronsted acid ?
 (i) HCN (ii) H_2PO_4^- (iii) NH_4^+ (iv) HCl
 (1) (i), (iii) (2) (i), (ii), (iii), (iv) (3) (ii), (iii) (4) (i), (iii), (iv)
- Which of the following is a Bronsted base ?
 (i) NH_3 (ii) CH_3NH_2 (iii) HCO_3^- (iv) SO_4^{2-}
 (1) (i), (ii), (iii), (iv) (2) (i), (ii) (3) (i), (ii), (iii) (4) (i), (iii), (iv)
- The conjugate base of hydroxide ion is—
 (1) H_2O (2) H_3O^+ (3) O^{2-} (4) O_2
- The conjugate acid of amide ion (NH_2^-) is—
 (1) N_2H_4 (2) NH_2OH (3) NH_4^+ (4) NH_3
- Which of the following can act both as a Bronsted acid as well as a Bronsted base ?
 (1) H_2SO_4 (2) HCO_3^- (3) O^{2-} (4) NH_4^+
- Which of the following acid-base reactions cannot be explained by the Bronsted theory ?
 (1) $\text{CO}_2 + \text{CaO} \longrightarrow \text{CaCO}_3$ (2) $\text{BF}_3 + \text{NH}_3 \longrightarrow \text{BF}_3^+ \text{NH}_3^-$
 (3) $\text{Ni} + 4\text{CO} \longrightarrow \text{Ni}(\text{CO})_4$ (4) All of these
- Which of the following Bronsted acid has the weakest conjugate base ?
 (1) H_2O (2) HCN (3) HCOOH (4) HF

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	1	3	4	1	1	4	4	2	1
BEGINNER'S BOX-2	Que.	11									
	Ans.	2									
BEGINNER'S BOX-3	Que.	1	2	3	4	5	6				
	Ans.	2	3	2	3	1	2				
BEGINNER'S BOX-4	Que.	1	2	3	4	5	6				
	Ans.	3	2	3	1	4	4				
BEGINNER'S BOX-5	Que.	1	2	3	4	5	6	7			
	Ans.	2	1	3	4	2	4	4			

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

INTRODUCTION

- The formula weight of H_2SO_4 is 98. The weight of the acid in 400mL of 0.1 M solution is:-
(1) 2.45 g (2) 3.92 g (3) 4.90 g (4) 9.8 g
- Normality of 2M sulphuric acid is:-
(1) 2 N (2) 4N (3) N/2 (4) N/4
- If $\text{pH} = 3.31$, then find out $[\text{H}^+]$ (Approxy)
(1) 3.39×10^{-4} (2) 5×10^{-4}
(3) 3.0×10^{-3} (4) None
- If $[\text{OH}^-] = 5.0 \times 10^{-5} \text{ M}$ then pH will be :-
(1) $5 - \log 5$ (2) $9 + \log 5$
(3) $\log 5 - 5$ (4) $\log 5 - 9$
- Basicity of H_3PO_3 and H_3PO_2 are respectively :-
(1) 1 and 2 (2) 2 and 3
(3) 3 and 2 (4) 2 and 1
- Find out pH of solution having 2×10^{-3} moles of OH^- ion's in 2 litre solution :-
(1) $\text{pH} = 3$ (2) $\text{pH} = 3 + \log 2$
(3) $\text{pH} = 3 - \log 2$ (4) $\text{pH} = 11$
- pH of tomato juice is 4.4. Then concentration of H_3O^+ will be :-
(1) 39×10^{-4} (2) 3.9×10^{-5}
(3) 3.9×10^{-4} (4) 3.9×10^5
- 8 g NaOH is dissolved in one litre of solution, the molarity of the solution is:-
(1) 0.2 M (2) 0.4 M (3) 0.02 M (4) 0.8 M
- The amount of acetic acid present in 100 mL of 0.1M solution is :-
(1) 0.30 g (2) 3.0 g (3) 0.60 g (4) None
- The number of milli equivalents of acid in 100 mL of 0.5N HCl solution is:-
(1) 50 (2) 100 (3) 25 (4) 200
- If the molar concentration of PbI_2 is $1.5 \times 10^{-3} \text{ mol L}^{-1}$, the concentration of iodide ions in g ion L^{-1} is:-
(1) 3.0×10^{-3} (2) 6.0×10^{-3}
(3) 0.3×10^{-3} (4) 0.6×10^{-6}

OSTWALD'S DILUTION LAW

- Order of dissociation of 0.1 N CH_3COOH is :-
(Dissociation constant = 1×10^{-5})
(1) 10^{-5} (2) 10^{-4} (3) 10^{-3} (4) 10^{-2}

- If α is the degree of dissociation of weak dibasic organic acid and y is the hydrogen ion concentration, what is the initial concentration of acid :-
(1) $\frac{\alpha(y)^{-1}}{2} \text{ M}$ (2) $y(\alpha)^{-1} \text{ M}$
(3) $\frac{y(\alpha)^{-1}}{2} \text{ M}$ (4) None of them
- The degree of dissociation of acetic acid is given by the expression $\alpha = 0.1 \times C^{-1}$ (where C = concentration of the acid) What is the pH of the solution :-
(1) 1 (2) 2 (3) 3 (4) 4
- Ostwald's dilution law is not applicable for strong electrolytes because:-
(1) Strong electrolytes are completely ionised
(2) Strong electrolytes are volatile
(3) Strong electrolytes are unstable
(4) Strong electrolytes often contain metal ions
- The degree of ionisation of a compound depends upon :
(1) Size of the solute molecules
(2) Nature of the solute molecules
(3) Nature of the container taken
(4) The amount of current passed
- Find out K_a for 10^{-2} M HCN acid, having pOH is 10 :-
(1) $K_a = 10^{-4}$ (2) $K_a = 10^{-2}$
(3) $K_a = 10^{-5}$ (4) None of them
- Which of the following will occur if a 1.0 M solution of a weak acid is diluted to 0.01 M at constant temperature:-
(1) Percentage ionisation will increase
(2) $[\text{H}^+]$ will decrease to 0.01M
(3) K_a will increase
(4) pH will decrease by 2 units
- The pH of 0.15 M solution of HOCl ($K_a = 9.6 \times 10^{-6}$) is:-
(1) 4.42 (2) 2.92 (3) 3.42 (4) None
- The extent of ionisation increases (weak electrolytes)
(1) With the increase in concentration of solute
(2) On decreasing the temp. of solution
(3) On addition of excess of water to the solution
(4) On stirring the solution vigorously

21. If K_a of $\text{HCN} = 4 \times 10^{-10}$, then the pH of 2.5×10^{-1} molar HCN (aq) is:-
 (1) 4.2 (2) 4.7 (3) 0.47 (4) 5.0
22. The molarity of nitrous acid at which its pH becomes 2. ($K_a = 4.5 \times 10^{-4}$) :-
 (1) 0.3333 (2) 0.4444
 (3) 0.6666 (4) 0.2222
23. Correct statement for HCN weak acid at 25°C temperature:-
 (1) $\alpha = \frac{K_a}{[\text{H}^+]}$ (2) $\alpha = \frac{K_a \times [\text{OH}^-]}{K_w}$
 (3) (1) & (2) both (4) $K_b = C\alpha^2$

EXPLANATION OF WATER

24. Ionic product of water will increase, if :-
 (1) Dissociation the pressure
 (2) Add H^+
 (3) Add OH^-
 (4) Increase the temperature
25. For water at 25°C , 2×10^{-7} moles per litre is the correct answer for which one of the following
 (1) $[\text{H}^+] + [\text{OH}^-]$ (2) $[\text{H}^+]^2$
 (3) $[\text{OH}^-]^2$ (4) $[\text{H}^+] - [\text{OH}^-]$
26. At 25°C , the dissociation constant for pure water is given by :-
 (1) $(55.4 \times 10^{14})^{-1}$ (2) 1×10^{-14}
 (3) $\frac{1 \times 10^{-14}}{18}$ (4) None of these
27. Ionic product of water is equal to :-
 (1) Dissociation constant of water $\times [\text{H}_2\text{O}]$
 (2) Dissociation constant of water $\times [\text{H}^+]$
 (3) Product of $[\text{H}_2\text{O}]$ and $[\text{H}^+]$
 (4) Product of $[\text{OH}^-]^2$ and $[\text{H}^+]$
28. Addition of H^+ and OH^- ion's concentration at 90°C
 (1) 10^{-14} (2) 10^{-12}
 (3) 2×10^{-6} (4) 2×10^{-7}
29. At 90°C , pure water has $[\text{H}_3\text{O}^+] = 10^{-6.7} \text{ mol L}^{-1}$ what is the value of K_w at 90°C :-
 (1) 10^{-6} (2) 10^{-12}
 (3) 10^{-67} (4) $10^{-13.4}$
30. At 373 K, temp. the pH of pure H_2O can be:-
 (1) < 7 (2) > 7 (3) $= 7$ (4) $= 0$

31. The common ion effect is shown by which of the following sets of solutions :-
 (1) $\text{BaCl}_2 + \text{Ba}(\text{NO}_3)_2$ (2) $\text{NaCl} + \text{HCl}$
 (3) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$ (4) None
32. Basic strength of NH_4OH in presence of NH_4Cl
 (1) Increases
 (2) Remains unchanged
 (3) Decreases
 (4) Some times increases or sometimes decreases
33. Which of the following is a true statement :
 (1) The ionisation constant and ionic product of water are same.
 (2) Water is a strong electrolyte.
 (3) The value of ionic product of water is less than that of its ionisation constant.
 (4) At 298K, the number of H^+ ions in a litre of water is 6.023×10^{16} .
34. If it is known that H_2S is a weak acid and it is ionised into 2H^+ and S^{2-} . Then in this solution HCl is added so, pH becomes less, then what will happen :-
 (1) Decrease in S^{2-} ion concentration
 (2) Concentration of S^{2-} is not affected
 (3) Increase in S^{2-} ion concentration
 (4) It is not possible, to add HCl in solution

SALTS, TYPES OF SALT & CONJUGATE THEORY

35. Which of the following is not an acidic salt :-
 (1) NaHSO_4 (2) HCOONa
 (3) NaH_2PO_3 (4) None of them
36. Which is a basic salt :-
 (1) PbS (2) PbCO_3
 (3) PbSO_4 (4) $2\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$
37. The process of neutralisation invariably results in the production of :-
 (1) H^+ ions
 (2) OH^- ions
 (3) Both H^+ and OH^- ions
 (4) Molecules of water
38. Which of the following is an acid salt :-
 (1) Na_2S (2) Na_2SO_3
 (3) NaHSO_3 (4) Na_2SO_4
39. The mixed salt among the following is :-
 (1) $\text{CH}(\text{OH})\text{COONa}$ (2) NaKSO_4
 (3) CaCl_2 (4) All

HYDROLYSIS OF SALTS

40. At 90°C, the pH of 0.1M NaCl aqueous solution is :-
 (1) < 7 (2) > 7 (3) 7 (4) 0.1
41. What will be the pH of 1.0 M ammonium formate solution, If $K_a = 1 \times 10^{-4}$ acid $K_b = 1 \times 10^{-5}$:-
 (1) 6.5 (2) 7.5 (3) 8.0 (4) 9.0
42. Which salt will not undergo hydrolysis :-
 (1) KCl (2) Na_2SO_4 (3) NaCl (4) All
43. Maximum efficiency of cationic hydrolysis will be shown by :-
 (1) Al^{+3} (2) Ga^{+3} (3) Tl^{+1} (4) Tl^{+3}
44. $\text{HCOO}^- + \text{H}_2\text{O} \rightleftharpoons \text{HCOOH} + \text{OH}^-$ is related :-
 (1) $h = \sqrt{K_h}$ (2) $h = \sqrt{\frac{K_h}{C}}$
 (3) $h = \sqrt{\frac{K_h}{V}}$ (4) $K_h = \sqrt{hc}$
45. The pH of aqueous solution of sodium acetate is
 (1) 7 (2) Very low
 (3) > 7 (4) < 7
46. If $\text{p}K_b$ for CN^- at 25°C is 4.7. The pH of 0.5M aqueous NaCN solution is :-
 (1) 12 (2) 10 (3) 11.5 (4) 11
47. The highest pH value is of :-
 (1) 0.1 M NaCl
 (2) 0.1 M NH_4Cl
 (3) 0.1 M CH_3COONa
 (4) 0.1 M $\text{CH}_3\text{COONH}_4$
48. pH of K_2S solution is :-
 (1) 7 (2) Less than 7
 (3) More than 7 (4) 0
49. For anionic hydrolysis, pH is given by :-
 (1) $\text{pH} = \frac{1}{2} \text{p}K_w - \frac{1}{2} \text{p}K_b - \frac{1}{2} \log c$
 (2) $\text{pH} = \frac{1}{2} \text{p}K_w + \frac{1}{2} \text{p}K_a - \frac{1}{2} \text{p}K_b$
 (3) $\text{pH} = \frac{1}{2} \text{p}K_w + \frac{1}{2} \text{p}K_a + \frac{1}{2} \log c$
 (4) None of above
50. A weak acid react with strong base, ionisation constant of weak acid is 10^{-4} . Find out equilibrium constant for this reaction :-
 (1) 10^{-10} (2) 10^{10} (3) 10^{-9} (4) 10^9

51. Hydroxyl ion concentration $[\text{OH}^-]$ in the case of sodium acetate can be expressed as (where K_a is dissociation constant of CH_3COOH and C is the concentration of sodium acetate) :-
 (1) $[\text{OH}^-] = (\text{C}K_w \cdot K_a)^{1/2}$ (2) $[\text{OH}^-] = \text{C} \cdot K_w \sqrt{K_a}$
 (3) $[\text{OH}^-] = \left(\frac{\text{C} \cdot K_w}{K_a} \right)^{1/2}$ (4) $[\text{OH}^-] = \text{C} \cdot K_a \cdot K_w$
52. Consider :-
 (a) FeCl_3 in water - Basic
 (b) NH_4Cl in water - Acidic
 (c) Ammonium acetate in water - Acidic
 (d) Na_2CO_3 in water - Basic
 Which is/are not correctly matched :-
 (1) b and d (2) b only
 (3) a and c (4) d only
53. Which of the following salts undergoes hydrolysis in water :-
 (1) Na_3PO_4 (2) CH_3COONa
 (3) NaNO_3 (4) Both of (1) and (2)
54. A salt 'X' is dissolved in water of pH = 7. The resulting solution becomes alkaline in nature. The salt is made up of :-
 (1) A strong acid and strong base
 (2) A strong acid and weak base
 (3) A weak acid and weak base
 (4) A weak acid and strong base
55. K_a for cyano acetic acid is 3.5×10^{-3} . Then the degree of hydrolysis of 0.05 M. sodium cyano acetate solution will have the following value :-
 (1) 4.559×10^{-6} (2) 5.559×10^{-6}
 (3) 6.559×10^{-6} (4) 7.559×10^{-6}
56. Degree of Hydrolysis of $\frac{N}{100}$ solution of KCN is
 (Given $K_a = 1.4 \times 10^{-9}$)
 (1) 2.7×10^{-3} (2) 2.7×10^{-2}
 (3) 2.7×10^{-4} (4) 2.7×10^{-5}

SOLUBILITY & SOLUBILITY PRODUCT (K_{sp})

57. The solubility product of sparingly soluble univalent salt is defined as the product of ionic concentration in a :-
 (1) 1 M solution
 (2) Concentration solution
 (3) Very dilute solution
 (4) Saturated solution

58. In solubility of salts M_2X , QY_2 and PZ_2 equal, then the relation between their K_{sp} will be :-

- (1) $K_{sp}(M_2X) > K_{sp}(QY_2) > K_{sp}(PZ_2)$
 (2) $K_{sp}(M_2X) = K_{sp}(QY_2) < K_{sp}(PZ_2)$
 (3) $K_{sp}(M_2X) > K_{sp}(QY_2) = K_{sp}(PZ_2)$
 (4) $K_{sp}(M_2X) = K_{sp}(QY_2) = K_{sp}(PZ_2)$

59. The expression of solubility product of mercurous iodide is :-

- (1) $[2 \text{Hg}^{+}]^2 \times 2 [\text{I}^-]^2$ (2) $[\text{Hg}^{2+}]^2 \times [2\text{I}^-]^2$
 (3) $[\text{Hg}_2^{2+}] \times [\text{I}^-]^2$ (4) $[\text{Hg}^{2+}]^2 \times [\text{I}^-]^2$

60. At 25°C , the K_{sp} value of AgCl is 1.8×10^{-10} . If 10^{-5} moles of Ag^+ are added to solution then K_{sp} will be :-

- (1) 1.8×10^{-15} (2) 1.8×10^{-10}
 (3) 1.8×10^{-5} (4) $18 \times 10^{+10}$

61. At 25°C , required volume of water, to dissolve 1g BaSO_4 ($K_{sp} = 1.1 \times 10^{-10}$) will be (Molecular weight of $\text{BaSO}_4 = 233$) :-

- (1) 820 L. (2) 1 L.
 (3) 205 L. (4) 430 L.

62. Concentration of Ag^+ ions in saturated solution of Ag_2CrO_4 at 20°C is $1.5 \times 10^{-4} \text{ mol L}^{-1}$. At 20°C , the solubility product of Ag_2CrO_4 is :-

- (1) 3.3750×10^{-12} (2) 1.6875×10^{-10}
 (3) 1.68×10^{-12} (4) 1.6875×10^{-11}

63. How many grams of CaC_2O_4 will dissolve in distilled water to make one litre saturated solution? solubility product of CaC_2O_4 is $2.5 \times 10^{-9} \text{ mol}^2 \text{ L}^{-2}$ and its molecular weight is 128.

- (1) 0.0064 g (2) 0.0128 g
 (3) 0.0032 g (4) 0.0640 g

64. If the concentration of CrO_4^{2-} ion in a saturated solution of silver chromate will be $2 \times 10^{-4} \text{ M}$, solubility product of silver chromate will be -

- (1) 4×10^{-8} (2) 8×10^{-12}
 (3) 32×10^{-12} (4) 6×10^{-12}

65. If the solubility of AgCl (formula mass=143) in water at 25°C is $1.43 \times 10^{-4} \text{ g/100 mL}$ of solution then the value of K_{sp} will be :-

- (1) 1×10^{-5} (2) 2×10^{-5}
 (3) 1×10^{-10} (4) 2×10^{-10}

66. If the salts M_2X , QY_2 and PZ_3 have the same solubilities, their K_{sp} values are related as - ($S < 1$)

- (1) $K_{sp}(M_2X) = K_{sp}(QY_2) < K_{sp}(PZ_3)$
 (2) $K_{sp}(M_2X) > K_{sp}(QY_2) = K_{sp}(PZ_3)$
 (3) $K_{sp}(M_2X) = K_{sp}(QY_2) > K_{sp}(PZ_3)$
 (4) $K_{sp}(M_2X) > K_{sp}(QY_2) > K_{sp}(PZ_3)$

67. The solubility product of As_2S_3 is given by the expression :-

- (1) $K_{sp} = [\text{As}^{3+}] \times [\text{S}^{2-}]$ (2) $K_{sp} = [\text{As}^{3+}]^1 [\text{S}^{2-}]^1$
 (3) $K_{sp} = [\text{As}^{3+}]^3 [\text{S}^{2-}]^2$ (4) $K_{sp} = [\text{As}^{3+}]^2 [\text{S}^{2-}]^3$

68. If the solubility of PbBr_2 is 'S' g molecules per litre, considering 100% ionisation its solubility product is :-

- (1) 2S^3 (2) 4S^2 (3) 4S^3 (4) 2S^4

69. If the solubility of lithium sodium hexafluoroaluminate $\text{Li}_3\text{Na}_3(\text{AlF}_6)_2$ is 'S' mol L^{-1} . Its solubility product is equal to :-

- (1) S^8 (2) 12S^3
 (3) 18S^3 (4) 2916S^8

70. One litre of saturated solution of CaCO_3 is evaporated to dryness, when 7.0 g of residue is left. The solubility product for CaCO_3 is:-

- (1) 4.9×10^{-3} (2) 4.9×10^{-5}
 (3) 4.9×10^{-9} (4) 4.9×10^{-7}

APPLICATION OF SOLUBILITY PRODUCT (K_{sp})

71. At 30°C , In which of the one litre solution, the solubility of Ag_2CO_3 (solubility product = 8×10^{-12}) will be maximum :-

- (1) 0.05 M Na_2CO_3 (2) Pure water
 (3) 0.05 M AgNO_3 (4) 0.05 M NH_3

72. Solubility of AgBr will be minimum in :-

- (1) Pure water (2) 0.1 M CaBr_2
 (3) 0.1 M NaBr (4) 0.1 M AgNO_3

73. In which of the following, the solution of AgSCN will be unsaturated :-

- (1) $[\text{Ag}^+] [\text{SCN}^-] = K_{sp}$ (2) $[\text{Ag}^+] \times [\text{SCN}^-] < K_{sp}$
 (3) $[\text{Ag}^+] \times [\text{SCN}^-] > K_{sp}$ (4) $[\text{Ag}^+] [\text{SCN}^-]^2 < K_{sp}$

74. If 's' and 'S' are respectively solubility and solubility product of a sparingly soluble binary electrolyte then :-

- (1) $s = S$ (2) $s = S^2$
 (3) $s = S^{1/2}$ (4) $s = \frac{1}{2} S$

75. The solubility product of CuS , Ag_2S and HgS are 10^{-37} , 10^{-44} and 10^{-54} respectively. The solubility of these sulphides will be in the order

- (1) $\text{HgS} > \text{Ag}_2\text{S} > \text{CuS}$ (2) $\text{Ag}_2\text{S} > \text{HgS} > \text{CuS}$
 (3) $\text{CuS} > \text{Ag}_2\text{S} > \text{HgS}$ (4) $\text{Ag}_2\text{S} > \text{CuS} > \text{HgS}$

76. If the maximum concentration of PbCl_2 in water is 0.01 M at 298 K, Its maximum concentration in 0.1 M NaCl will be:-

- (1) $4 \times 10^{-3} \text{ M}$ (2) $0.4 \times 10^{-4} \text{ M}$
 (3) $4 \times 10^{-2} \text{ M}$ (4) $4 \times 10^{-4} \text{ M}$

77. M_2SO_4 (M^+ is a monovalent metal ion) has a K_{sp} of 1.2×10^{-5} at 298 K. The maximum concentration of M^+ ion that could be attained in a saturated solution of this solid at 298 K is :-

- (1) 3.46×10^{-3} M (2) 2.89×10^{-2} M
(3) 2.8×10^{-3} M (4) 7.0×10^{-3} M

78. Which of the following has maximum solubility (K_{sp} value is given in brackets) :-

- (1) HgS (1.6×10^{-54}) (2) $PbSO_4$ (1.3×10^{-8})
(3) ZnS (7.0×10^{-26}) (4) $AgCl$ (1.7×10^{-10})

79. Maximum soluble is :- (K_{sp} is given)

- (1) CuS (8.5×10^{-36}) (2) CdS (3.6×10^{-28})
(3) ZnS (1.2×10^{-28}) (4) MnS (1.4×10^{-10})

80. In which of the following, the solubility of $AgCl$ will be maximum :-

- (1) 0.1 M $AgNO_3$ (2) Water
(3) 0.1 M $NaCl$ (4) 0.1 M KCl

81. The solubility product of three sparingly soluble salts are given below :

No.	Formula	Solubility product
1	PQ	4.0×10^{-20}
2	PQ_2	3.2×10^{-14}
3	PQ_3	2.7×10^{-35}

The correct order of decreasing molar solubility is:-

- (1) 1, 2, 3 (2) 2, 1, 3 (3) 3, 2, 1 (4) 2, 3, 1

82. The K_{sp} value for $Gd(OH)_3$ is 2.8×10^{-23} , the pH of $Gd(OH)_3$ in saturated solution :-

- (1) 6.08 (2) 5.08 (3) 8.47 (4) 4.08

83. If the solubility product of $AgBrO_3$ and Ag_2SO_4 are 5.5×10^{-5} and 2×10^{-5} respectively, the relationship between the solubilities of these can be correctly represented as:-

- (1) $s_{AgBrO_3} > s_{Ag_2SO_4}$
(2) $s_{AgBrO_3} = s_{Ag_2SO_4}$
(3) $s_{AgBrO_3} < s_{Ag_2SO_4}$
(4) Can't predict

84. 0.5 M HCl solution has ions- Hg^{++} , Cd^{++} , Sr^{++} , Fe^{++} , Cu^{++} . To pass the H_2S gas in this solution, which are precipitated out :-

- (1) Cd^{++} , Fe^{++} , Sr^{++} (2) Cd^{++} , Hg^{++} , Cu^{++}
(3) Hg^{++} , Cu^{++} , Fe^{++} (4) Cu^{++} , Sr^{++} , Fe^{++}

85. Solubility product of $Mg(OH)_2$ is 1×10^{-11} . At what pH, precipitation of $Mg(OH)_2$ will begin from 0.1 M Mg^{2+} solution :-

- (1) 9 (2) 5 (3) 3 (4) 7

86. A solution, containing 0.01 M Zn^{+2} and 0.01 M Cu^{2+} is saturated by passing H_2S gas. The S^{-2} concentration is 8.1×10^{-21} M, K_{sp} for ZnS and CuS are 3.0×10^{-22} and 8.0×10^{-36} respectively. Which of the following will occur in the solution:-

- (1) ZnS will precipitate
(2) CuS will precipitate
(3) Both ZnS and CuS will precipitate
(4) Both Zn^{2+} and Cu^{2+} will remain in the solution

87. Consider (i) $Zn(OH)_2$ (ii) $Cr(OH)_3$ (iii) $Mg(OH)_2$ (iv) $Al(OH)_3$ which hydroxide is precipitated by NH_4OH containing NH_4Cl :-

- (1) i, ii (2) ii, iv
(3) Only iv (4) i, ii, iii and iv

88. What will happen if the pH of the solution of 0.001 M $Mg(NO_3)_2$ solution is adjusted to pH = 9 ($K_{sp} Mg(OH)_2 = 8.9 \times 10^{-12}$)

- (1) ppt will take place
(2) ppt will not take place
(3) Solution will be saturated
(4) None of these

89. The solubility product constant K_{sp} of $Mg(OH)_2$ is 9.0×10^{-12} . If a solution is 0.010 M with respect to Mg^{2+} ion, what is the maximum hydroxide ion concentration which could be present without causing the precipitation of $Mg(OH)_2$:-

- (1) 1.5×10^{-7} M (2) 3.0×10^{-7} M
(3) 1.5×10^{-5} M (4) 3.0×10^{-5} M

90. When HCl gas is passed through a saturated solution of common salt, pure $NaCl$ is precipitated because:-

- (1) The impurities dissolve in HCl
(2) HCl is slightly soluble in water
(3) The ionic product $[Na^+] \times [Cl^-]$ exceeds the solubility product of $NaCl$
(4) The solubility product of $NaCl$ is lowered by Cl^- from aq. HCl

91. A solution is a mixture of 0.06 M KCl and 0.06 M KI . $AgNO_3$ solution is being added drop by drop till $AgCl$ starts precipitating ($K_{sp} AgCl = 1 \times 10^{-10}$ and $K_{sp} AgI = 4 \times 10^{-16}$). The concentration of Iodide ion at this stage will be nearly equal to :-

- (1) 4.0×10^{-5} M (2) 2.4×10^{-7} M
(3) 2.0×10^{-8} M (4) 4×10^{-8} M

92. Why only As^{3+} gets precipitated as As_2S_3 and not Zn^{2+} as ZnS when H_2S is passed through an acidic solution containing As^{3+} and Zn^{2+} :-

- (1) Enough As^{3+} are present in acidic medium
(2) Zinc salt does not ionise in acidic medium
(3) Solubility product of As_2S_3 is less than that of ZnS
(4) Solubility product changes in presence of an acid

93. H_2S is passed through a solution of cations in HCl medium to precipitate cation of :-
 (1) II-A group of cation analysis
 (2) II - B group of cation analysis
 (3) IV group of cation analysis
 (4) Both II - A and II-B gps.
94. To have more sulphide ion concentration, H_2S should be passed through :-
 (1) 1 N HCl solution
 (2) 0.1 M HCl solution
 (3) A neutral solution such as water
 (4) An ammoniacal solution
95. The solubility product of hydroxides of Mg^{+2} , Zn^{+2} , and Fe^{+3} decreases as
 $K_{\text{sp}} \text{Mg}(\text{OH})_2 > K_{\text{sp}} \text{Zn}(\text{OH})_2 > K_{\text{sp}} \text{Fe}(\text{OH})_3$ The order of precipitation of hydroxides is:-
 (1) $\text{Fe}(\text{OH})_3$, $\text{Zn}(\text{OH})_2$, $\text{Mg}(\text{OH})_2$
 (2) $\text{Mg}(\text{OH})_2$, $\text{Zn}(\text{OH})_2$, $\text{Fe}(\text{OH})_3$
 (3) $\text{Zn}(\text{OH})_2$, $\text{Fe}(\text{OH})_3$, $\text{Mg}(\text{OH})_2$
 (4) $\text{Zn}(\text{OH})_2$, $\text{Mg}(\text{OH})_2$, $\text{Fe}(\text{OH})_3$

FEW IMPORTANT POINTS

96. Two monobasic weak acids have the same concentration of H^+ ions. What is the relationship between dissociation constant and dilution:-
 (1) $K_{\text{a}1}V_1 = K_{\text{a}2}V_2$ (2) $K_{\text{a}1}V_2 = K_{\text{a}2}V_1$
 (3) $[K_{\text{a}1}V_1]^{\frac{1}{2}} = K_{\text{a}2}V_2$ (4) $K_{\text{a}1}V_1 = [K_{\text{a}2}V_2]^{\frac{1}{2}}$
97. What is the molar concentration of chloride ion in the solution obtained by mixing 300 mL of 3.0M NaCl and 200 mL of 4.0 M solution of BaCl_2 :-
 (1) 5.0 M (2) 1.8 M
 (3) 1.6 M (4) None of these
98. The pH of a 0.1 M formic acid 0.1% dissociated is equal to 4. What will be the pH of another weak monobasic acid (same concentration) which is 1% dissociated
 (1) 2 (2) 3 (3) 1 (4) 4

pH

99. pH of water is 7. When any substance Y is dissolved in water then pH becomes 13. Substance Y is a salt of :-
 (1) Strong acid and strong base
 (2) Weak acid and weak base
 (3) Strong acid and weak base
 (4) Weak acid and strong base

100. Minimum pH is shown by aqueous solution of :-
 (1) 0.1 M BaCl_2 (2) 0.1 M $\text{Ba}(\text{NO}_3)_2$
 (3) 0.1 M BeCl_2 (4) 0.1 M $\text{Ba}(\text{OH})_2$
101. Given :-
 (a) 0.005 M H_2SO_4 (b) 0.1 M Na_2SO_4
 (c) 10^{-2} M NaOH (d) 0.01 M HCl
 Choose the correct code having same pH :-
 (1) a, c, d (2) b, d (3) a, d (4) a, c
102. What is H^+ ion concentration of 5×10^{-3} M H_2CO_3 solution having a 10% dissociation:-
 (1) 10^{-3} (2) 10^{-2}
 (3) 10^{-1} (4) 5×10^{-2}
103. A metal hydroxide of molecular formula $\text{M}(\text{OH})_4$ is 50% ionised. Its 0.0025M solution will have the pH :-
 (1) 12 (2) 2 (3) 4 (4) 11.7
104. In the following solutions, the conc. of different acids are given, which mixture of the acid has highest pH :-
 (1) $\frac{M}{10} \text{H}_2\text{SO}_4$, $\frac{M}{20} \text{HNO}_3$, $\frac{M}{10} \text{HClO}_4$
 (2) $\frac{M}{20} \text{H}_2\text{SO}_4$, $\frac{M}{10} \text{HNO}_3$, $\frac{M}{20} \text{HClO}_4$
 (3) $\frac{M}{20} \text{H}_2\text{SO}_4$, $\frac{M}{10} \text{HNO}_3$, $\frac{M}{40} \text{HClO}_4$
 (4) $\frac{M}{20} \text{H}_2\text{SO}_4$, $\frac{M}{5} \text{HNO}_3$, $\frac{M}{5} \text{HClO}_4$
105. If 100 mL of pH = 3 and 400 mL of pH = 3 is mixed, what will be the pH of the mixture
 (1) 3.2 (2) 3.0 (3) 3.5 (4) 2.8
106. 10^{-6} M HCl is diluted to 100 times. Its pH is :-
 (1) 6.0 (2) 8.0 (3) 6.95 (4) 9.5
107. pH of 0.001M acetic acid would be :-
 (1) 2 (2) > 3 (3) 7 (4) 14
108. At 90°C , the pH of 0.001M KOH solution will be
 (1) 3 (2) 11 (3) 5 (4) 9
109. The pH of solution is increased from 3 to 6. Its H^+ ion concentration will be :-
 (1) Reduced to half
 (2) Doubled
 (3) Reduced by 1000 times
 (4) Increased by 1000 times
110. A solution has pOH equal to 13 at 298 K. The solution will be:-
 (1) Highly acidic (2) Highly basic
 (3) Moderately basic (4) Unpredictable

- 111.** The pH of the solution containing 10 mL of a 0.1M NaOH and 10 mL of 0.05M H_2SO_4 would be
(1) Zero (2) 1 (3) >7 (4) 7
- 112.** Which of the following statements are (is) correct
(a) The pH of 1.0×10^{-8} M solution of HCl is 8.
(b) The conjugate base of H_2PO_4^- is HPO_4^{2-} .
(c) Autoprotolysis constant of water increases with temperature.
(d) When a solution of a weak monoprotic acid is titrated against a strong base, at half neutralization point $\text{pH} = 1/2 \text{ pK}_a$
(1) a (2) a, b (3) a, b, d (4) b, c
- 113.** In a solution of $\text{pH} = 5$, more acid is added in order to reduce the $\text{pH} = 2$. The increase in hydrogen ion concentration is:-
(1) 100 times (2) 1000 times
(3) 3 times (4) 5 times
- 114.** The hydrogen ion concentration in a given solution is 6×10^{-4} M. Its pH will be :-
(1) 6 (2) 3.22 (3) 4 (4) 2.
- 115.** The pOH of beer is 10.0. The hydrogen ion concentration will be :-
(a) 10^{-10} (b) $\frac{K_w}{10^{-10}}$ (c) $\frac{K_w}{10^{-8}}$ (d) 10^{-4}
(1) a, d (2) b, c (3) a, b, c (4) None
- 116.** An aqueous solution whose $\text{pH} = 0$ is :-
(1) Basic (2) Acidic
(3) Neutral (4) Amphoteric
- 117.** The pH of the solution produced when an aqueous solution of strong acid pH 5 is mixed with equal volume of an aqueous solution of strong acid of pH 3 is :-
(1) 3.3 (2) 3.5 (3) 4.5 (4) 4.0
- 118.** Following five solution of KOH were prepare as-
First $\rightarrow$ 0.1 moles in 1 L
Second $\rightarrow$ 0.2 moles in 2 L
Third $\rightarrow$ 0.3 moles in 3 L
Fourth $\rightarrow$ 0.4 moles in 4 L
Fifth $\rightarrow$ 0.5 moles in 5 L
The pH of resultant solution is :-
(1) 2 (2) 1 (3) 13 (4) 7
- 119.** The pH of a 0.02 M ammonia solution which is 5% ionised will be :-
(1) 2 (2) 11 (3) 5 (4) 7
- 120.** For $\frac{N}{10}$ H_2SO_4 , pH value is :-
(1) 1 (2) 0.586 (3) 0.856 (4) None
- 121.** An aqueous solution of HCl is 10^{-9} M HCl. The pH of the solution should be:-
(1) 9 (2) Between 6 and 7
(3) 7 (4) Unpredictable
- 122.** H_2X is a dibasic acid which dissociates completely in water. Which one of the following is the molarity of an aqueous solution of this acid which has a pH of 1 :-
(1) 0.1 (2) 0.05 (3) 0.2 (4) 0.5
- 123.** How many moles of HCl must be removed from 1 litre of aqueous HCl solution to change its pH from 2 to 3 :-
(1) 1 (2) 0.02 (3) 0.009 (4) 0.01
- 124.** 8 g NaOH and 4.9 g H_2SO_4 are present in one litre of the solution. What is its pH
(1) 1 (2) 13 (3) 12 (4) 2
- 125.** Calculate pH of a solution whose 100 mL contains 0.2 g NaOH dissolved in it :-
(1) 10.699 (2) 11.699
(3) 12.699 (4) 13.699
- 126.** One litre solution contains 1M HOCl [$K_a = 10^{-8}$] and 1 M NaOH. What is the pH of the solution :-
(1) 8 (2) 11 (3) 5 (4) 2
- 127.** What is the quantity of NaOH present in 250 cc of the solution, so that it gives a $\text{pH} = 13$:-
(1) 10^{-13} g (2) 10^{-1} g (3) 1.0 g (4) 4.0 g
- 128.** 0.001 mol of the strong electrolyte $\text{M}(\text{OH})_2$ has been dissolved to make a 20 mL of its saturated solution. Its pH will be :- [$K_w = 1 \times 10^{-14}$]
(1) 13 (2) 3.3 (3) 11 (4) 9.8
- 129.** Choose the wrong statement :-
(1) For a neutral solution : $[\text{H}^+] = [\text{OH}^-] = \sqrt{K_w}$
(2) For an acidic solution : $[\text{H}^+] > \sqrt{K_w}$ and $[\text{OH}^-] < \sqrt{K_w}$
(3) For a basic solution : $[\text{H}^+] < \sqrt{K_w}$ and $[\text{OH}^-] > \sqrt{K_w}$
(4) For a neutral solution at all temperatures : $[\text{H}^+] = [\text{OH}^-] = 10^{-7}$ M
- 130.** The pH of 0.1 M solution of the following salts increases in order :-
(1) $\text{NaCl} < \text{NH}_4\text{Cl} < \text{NaCN} < \text{HCl}$
(2) $\text{NaCN} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{HCl}$
(3) $\text{HCl} < \text{NaCl} < \text{NaCN} < \text{NH}_4\text{Cl}$
(4) $\text{HCl} < \text{NH}_4\text{Cl} < \text{NaCl} < \text{NaCN}$

BUFFER SOLUTIONS and INDICATOR

- 131.** In a buffer solution the ratio of concentration of NH_4Cl and NH_4OH is 1 : 1 when it changes in 2 : 1 what will be the value of pH of buffer :-
 (1) Increase (2) Decrease
 (3) No effect (4) N.O.T.
- 132.** To a 50 mL of 0.05M formic acid how much volume of 0.10M sodium formate must be added to get a buffer solution of pH = 4.0 ? (pK_a of the acid is 3.8)
 (1) 50 mL (2) 4 mL (3) 39.6 mL (4) 100 mL
- 133.** In the volumetric estimation of HCl, if we make use of phenolphthalein as an indicator, which base is unsuitable for the titration :-
 (1) NaOH (2) RbOH
 (3) KOH (4) NH_4OH
- 134.** In a mixture of weak acid and its salt, the ratio of concentration of acid to salt is increased ten-fold. The pH of the solution :-
 (1) Decreases by one (2) Increases by one-tenth
 (3) Increases by one (4) Increases ten-fold
- 135.** pK_b for NH_4OH at certain temperature is 4.74. The pH of basic buffer containing equimolar concentration of NH_4OH and NH_4Cl will be:-
 (1) 7.74 (2) 4.74 (3) 2.37 (4) 9.26
- 136.** What is the suitable indicator for titration of NaOH and oxalic acid:-
 (1) Methyl orange (2) Methyl red
 (3) Phenolphthalein (4) Starch solution
- 137.** Phenolphthalein does not act as an indicator for the titration between :-
 (1) KOH and H_2SO_4
 (2) NaOH and CH_3COOH
 (3) Oxalic acid and KMnO_4
 (4) $\text{Ba}(\text{OH})_2$ and HCl
- 138.** Which can act as buffer :-
 (1) $\text{NH}_4\text{OH} + \text{NaOH}$
 (2) $\text{HCOOH} + \text{CH}_3\text{COONa}$
 (3) 40 mL 0.1 M NaCN + 20 mL of 0.1 M HCl
 (4) None of them
- 139.** The buffer solution play an important role in :-
 (1) Increasing the pH value
 (2) Decreasing the pH value
 (3) Keeping the pH constant
 (4) Solution will be neutral
- 140.** K_a for HCN is 5×10^{-10} at 25°C . For maintaining a constant pH of 9, the volume of 5M KCN solution required to be added to 10mL of 2M HCN solution is-
 (1) 4 mL (2) 7.95 mL
 (3) 2 mL (4) 9.3 mL
- 141.** Buffering action of a mixture of CH_3COOH and CH_3COONa is maximum when the ratio of salt to acid is equal to -
 (1) 1.0 (2) 100.0 (3) 10.0 (4) 0.1
- 142.** The pink colour of phenolphthalein in alkaline medium is due to -
 (1) Negative ion (2) Positive ion
 (3) OH^- ions (4) Neutral form
- 143.** Which indicator works in the pH range 8 – 9.8
 (1) Phenolphthalein (2) Methyl orange
 (3) Methyl red (4) Litmus
- 144.** A basic - buffer will obey the equation $\text{pOH} - \text{pK}_b = 1$ only under condition:-
 (1) [Conjugate acid] : [base] = 1 : 10
 (2) [Conjugate acid] = [base]
 (3) [Conjugate acid] : [base] = 10 : 1
 (4) N.O.T
- 145.** For weak acid strong base titration, the indicator used is :-
 (1) Potassium di-chromate
 (2) Methyl orange
 (3) Litmus
 (4) Phenolphthalein
- 146.** From the following in which titration methyl orange is a best indicator :-
 (1) $\text{CH}_3\text{COOH} + \text{NaOH}$ (2) $\text{H}_2\text{C}_2\text{O}_4 + \text{NaOH}$
 (3) $\text{HCl} + \text{NaOH}$ (4) $\text{CH}_3\text{COOH} + \text{NH}_4\text{OH}$
- 147.** The total number of different kind of buffers obtained during the titration of H_3PO_4 with NaOH are :-
 (1) 3 (2) 1 (3) 2 (4) 0
- 148.** The H^+ ion concentration in 0.001 M acetic acid is $1.34 \times 10^{-4} \text{ g ion L}^{-1}$. The H^+ ion concentration of 0.164 g of CH_3COONa is added to a litre of 0.001 M CH_3COOH will be :-
 (1) 9×10^{-6} (2) 18×10^{-6}
 (3) 4.5×10^{-6} (4) 5×10^{-6}
- 149.** A certain acidic buffer solution contains equal concentration of X^- and HX . The K_b for X^- is 10^{-10} . The pH of the buffer is :-
 (1) 4 (2) 7 (3) 10 (4) 14
- 150.** When 1.0 mL of dil. HCl acid is added to 100 mL of a buffer solution of pH 4.0. The pH of the solution
 (1) Becomes 7 (2) Does not change
 (3) Becomes 2 (4) Becomes 10
- 151.** The pH of blood is maintained by CO_2 and H_2CO_3 in the body and chemical constituents of blood. This phenomenon is called :-
 (1) Colloidal (2) Buffer action
 (3) Acidity (4) Salt balance

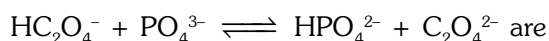
- 152.** Phenolphthalein is not a good indicator for titrating
(1) NaOH against oxalic acid
(2) NaOH against HCl
(3) NaOH against H_2SO_4
(4) Ferrous sulphate against KMnO_4
- 153.** Which of the following solutions does not act as buffer :-
(1) $\text{H}_3\text{PO}_4 + \text{NaH}_2\text{PO}_4$
(2) $\text{NaHCO}_3 + \text{H}_2\text{CO}_3$
(3) $\text{NH}_4\text{Cl} + \text{HCl}$
(4) $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$
- 154.** 50 mL of 2N acetic acid mixed with 10 mL of 1N sodium acetate solution will have an approximate pH of ($K_a = 10^{-5}$) :-
(1) 4 (2) 5 (3) 6 (4) 7
- 155.** On addition of NaOH to CH_3COOH solution, 60% of the acid is neutralised. If $\text{p}K_a$ of CH_3COOH is 4.7 then the pH of the resulting solution is :-
(1) More than 4.7 but less than 5.0
(2) Less than 4.7 but more than 4.0
(3) More than 5.0
(4) Remains unchanged
- 156.** 500 mL of 0.2 M acetic acid are added to 500 mL of 0.30 M sodium acetate solution. If the dissociation constant of acetic acid is 1.5×10^{-5} then pH of the resulting solution is:-
(1) 5.0 (2) 9.0 (3) 3.0 (4) 4.0
- 157.** Half of the formic acid solution is neutralised on addition of a KOH solution to it. If $K_a(\text{HCOOH}) = 2 \times 10^{-4}$ then pH of the solution is : ($\log 2 = 0.3010$)
(1) 3.6990 (2) 10.3010 (3) 3.85 (4) 4.3010
- 158.** A solution contains 0.2M NH_4OH and 0.2M NH_4Cl . If 1.0 mL of 0.001 M HCl is added to it. What will be the $[\text{OH}^-]$ of the resulting solution [$K_b = 2 \times 10^{-5}$] :-
(1) 2×10^{-5} (2) 5×10^{-10}
(3) 2×10^{-3} (4) None of these
- 159.** Henderson equation $\text{pH} - \text{p}K_a = 1$ will be applicable to an acidic buffer when :-
(1) $[\text{Acid}] = [\text{Conjugate base}]$
(2) $[\text{Acid}] \times 10 = [\text{Conjugate base}]$
(3) $[\text{Acid}] = [\text{Conjugate base}] \times 10$
(4) None of these
- 160.** 0.05 M ammonium hydroxide solution is dissolved in 0.001 M ammonium chloride solution. What will be the OH^- ion concentration of this solution: $K_b(\text{NH}_4\text{OH}) = 1.8 \times 10^{-5}$
(1) 3.0×10^{-3} (2) 9.0×10^{-4}
(3) 9.0×10^{-3} (4) 3.0×10^{-4}
- 161.** When 0.02 moles of NaOH are added to a litre of buffer solution, its pH changes from 5.75 to 5.80. What is its buffer capacity :-
(1) 0.4 (2) 0.05 (3) - 0.05 (4) 2.5
- 162.** Calculate the pH of a buffer prepared by mixing 300 cc of 0.3 M NH_3 and 500 cc of 0.5 M NH_4Cl . K_b for $\text{NH}_3 = 1.8 \times 10^{-5}$:-
(1) 8.1187 (2) 9.8117
(3) 8.8117 (4) None of these
- 163.** What amount of sodium propanoate should be added to one litre of an aqueous solution containing 0.02 mole of propanoic acid ($K_a = 1.34 \times 10^{-5}$ at 25°C) to obtain a buffer solution of pH 4.75 :-
(1) 4.52×10^{-2} M (2) 3.52×10^{-2} M
(3) 2.52×10^{-2} M (4) 1.52×10^{-2} M
- 164.** What will be the pH of the solution, if 0.01 moles of HCl is dissolved in a buffer solution containing 0.02 moles of propanoic acid ($K_a = 1.34 \times 10^{-5}$) and 0.0152 moles of salt, at 25°C : [$\log(0.173) = -0.76$]
(1) 3.11 (2) 4.11 (3) 5.11 (4) 6.11
- 165.** Calculate the pH of a buffer solution prepared by dissolving 30g of Na_2CO_3 in 500 mL of an aqueous solution containing 150 mL of 1M HCl. K_a for $\text{HCO}_3^- = 5.63 \times 10^{-11}$ [$\log\left(\frac{133}{150}\right) = -0.05$]
(1) 8.197 (2) 9.197
(3) 10.197 (4) 11.197
- 166.** Calculate the ratio of pH of a solution containing 1 mole of $\text{CH}_3\text{COONa} + 1$ mole of HCl per litre and of other solution containing 1 mole $\text{CH}_3\text{COONa} + 1$ mole of acetic acid per litre :-
(1) 1 : 1 (2) 2 : 1 (3) 1 : 2 (4) 2 : 3
- 167.** When 20 mL of $\frac{M}{20}$ NaOH are added to 10 mL of $\frac{M}{10}$ HCl, the resulting solution will:-
(1) Turn blue litmus red
(2) Turn phenolphthalein solution
(3) Turn methyl orange red
(4) Will have no effect on either red or blue litmus
- 168.** 10 mL of a solution contains 0.1 M $\text{NH}_4\text{Cl} + 0.01$ M NH_4OH . Which addition would not change the pH of solution :-
(1) Adding 1 mL water
(2) Adding 5 mL of 0.1 M NH_4Cl
(3) Adding 5 mL of 0.1 M NH_4OH
(4) Adding 10 mL of 0.1 M NH_4Cl

169. $\frac{N}{10}$ acetic acid was titrated with $\frac{N}{10}$ NaOH. When 25%, 50% and 75% of titration is over then the pH of the solution will be :- [$K_a = 10^{-5}$]
- (1) $5 + \log 1/3$, 5, $5 + \log 3$
 - (2) $5 + \log 3$, 4, $5 + \log 1/3$
 - (3) $5 - \log 1/3$, 5, $5 - \log 3$
 - (4) $5 - \log 1/3$, 4, $5 + \log 1/3$

ACID AND BASE

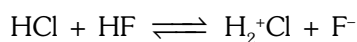
170. The conjugated acid of O^{2-} ion's is :-
 (1) O_2^+ (2) H^+ (3) H_3O^+ (4) OH^-
171. Ionization constant of AOH and BOH base are K_{b1} and K_{b2} . Their relation is $pK_{b1} < pK_{b2}$. Conjugate of following base, does not show maximum pH :
 (1) AOH (2) BOH
 (3) Both of them (4) NOT
172. Select the species which can function as - Lewis base, bronsted acid and bronsted base:-
 (a) H_2O (b) NH_4^+ (c) N^{3-}
 Correct code is :-
 (1) Only a (2) a, b (3) a, c (4) b, c
173. Which ion does not show acid behaviour :-
 (1) $[Al(H_2O)_6]^{+3}$ (2) $[Fe(H_2O)_6]^{+3}$
 (3) HPO_4^{-2} (4) ClO_3^-
174. An example of Lewis acid is:-
 (1) CaO (2) CH_3NH_2
 (3) SO_3 (4) None of these
175. In the reaction $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$ water behaves as :-
 (1) Acid (2) Base
 (3) Neutral (4) Both acid & Base
176. Which acts as Lewis base in the reaction $BCl_3 + :PH_3 \rightarrow Cl_3B \leftarrow PH_3$
 (1) PH_3 (2) BCl_3
 (3) Both 1 & 2 (4) None
177. Which acts as Lewis acid in the reaction $SnCl_2 + 2Cl^- \longrightarrow [SnCl_4]^{2-}$
 (1) Cl^- (2) $SnCl_2$ (3) $SnCl_4$ (4) None
178. The conjugated base of $(CH_3)_2NH_2^+$ is :-
 (1) CH_3NH_2 (2) $(CH_3)_2N^+$
 (3) $(CH_3)_2N$ (4) $(CH_3)_2NH$
179. Which equilibrium can be described as Lewis acid base reaction but not Bronsted acid base reaction:-
 (1) $H_2O + CH_3COOH \rightleftharpoons H_3O^+ + CH_3COO^-$
 (2) $2NH_3 + H_2SO_4 \rightleftharpoons 2NH_4^+ + SO_4^{2-}$
 (3) $NH_3 + CH_3COOH \rightleftharpoons NH_4^+ + CH_3COO^-$
 (4) $Cu^{+2} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+}$
180. Conjugate base of hydrazoic acid is :-
 (1) HN_3^- (2) N_3^- (3) N^{3-} (4) N_2^-
181. NH_3 gas dissolves in water to give NH_4OH , in this reaction, water acts as :-
 (1) An acid (2) A base
 (3) A salt (4) A conjugate base
182. Conjugate acid of $Zn(OH)_2$ is :-
 (1) $Zn(OH)^+$ (2) $Zn(OH)_3^-$ (3) Zn^{2+} (4) None
183. When ammonia is added to water it decreases the concentration of which of the following ion
 (1) OH^- (2) H_3O^+ (3) NH_4^+ (4) None
184. The strongest acid among the following is -
 (1) $ClO_3(OH)$ (2) $ClO_2(OH)$
 (3) $SO(OH)_2$ (4) $SO_2(OH)_2$
185. Which of the following is not a Bronsted acid :-
 (1) $CH_3NH_4^+$ (2) CH_3COO^-
 (3) H_2O (4) HSO_4^-
186. Which of the following example behave as a Lewis acid BF_3 , $SnCl_2$, $SnCl_4$:-
 (1) Stanus chloride, stanic chloride
 (2) BF_3 , Stanus chloride
 (3) Only BF_3
 (4) BF_3 , stanus chloride, stanic chloride
187. In the reaction $HNO_3 + H_2O \rightleftharpoons H_3O^+ + NO_3^-$, the conjugate base of HNO_3 is :-
 (1) H_2O (2) H_3O^+
 (3) NO_3^- (4) H_3O^+ and NO_3^-
188. The conjugate base of the weak acid in the reaction $HBr + H_2O \rightleftharpoons H_3O^+ + Br^-$ is
 (1) HBr (2) H_2O (3) Br^- (4) H_3O^+
189. In the reaction, $AlCl_3 + Cl^- \rightarrow [AlCl_4]^-$, $AlCl_3$ acts as :-
 (1) Salt (2) Lewis base
 (3) Bronsted acid (4) Lewis acid
190. Mg^{2+} is ----- than Al^{3+} :-
 (1) Strong Lewis acid (2) Strong Lewis base
 (3) Weak Lewis acid (4) Weak Lewis base

191. The two Bronsted bases in the reaction



- (1) HC_2O_4^- and PO_4^{3-}
 (2) HPO_4^{2-} and $\text{C}_2\text{O}_4^{2-}$
 (3) PO_4^{3-} and $\text{C}_2\text{O}_4^{2-}$
 (4) HC_2O_4^- and HPO_4^{2-}

192. The compound HCl behaves as ---- in the reaction,



- (1) Strong acid (2) Strong base
 (3) Weak acid (4) Weak base

193. Which of the following is not a Lewis base:-

- (1) NH_3 (2) O^{2-} (3) H_2O (4) I^+

194. Which of the following is Bronsted Lowry acid:-

- (1) SO_4^{2-} (2) H_3O^+ (3) OH^- (4) Cl^-

195. The conjugated base for bicarbonate ion is:-

- (1) CO_3^{2-} (2) HCO_3^-
 (3) CO_2 (4) H_2CO_3

196. Conjugated base of OH^- is :-

- (1) H_2O (2) H_3O^+ (3) H^+ (4) O^{2-}

197. HCl does not behave as acid in :-

- (1) NH_3 (2) $\text{C}_2\text{H}_5\text{OH}$ (3) H_2O (4) C_6H_6

198. Which of the following is a base according to Lowry-Bronsted concept :-

- (1) I^- (2) H_3O^+ (3) HCl (4) NH_4^+

199. In which of the following reactions NH_3 acts as acid

- (1) $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$
 (2) $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$
 (3) $\text{NH}_3 + \text{Na} \rightarrow \text{NaNH}_2 + \frac{1}{2}\text{H}_2$
 (4) NH_3 cannot act as acid

200. According to Bronsted concept, the acids in the

following reaction $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ are :-

- (1) NH_3 and NH_4^+ (2) H_2O and OH^-
 (3) H_2O and NH_4^+ (4) NH_3 and OH^-

201. Consider the following reactions :-

- (i) $\text{CO}_3^{2-} + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{OH}^-$
 (ii) $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$
 (iii) $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4\text{OH}$
 (iv) $\text{HCl} + \text{H}_2\text{O} \rightleftharpoons \text{Cl}^- + \text{H}_3\text{O}^+$

Which of the pairs of reactions proves that water is amphoteric in character :-

- (1) (i) and (ii) (2) (ii) and (iii)
 (3) (iii) and (iv) (4) (i) and (iii)

202. CH_3COO^- ion is a :-

- (1) Weak conjugate base
 (2) Strong conjugate base
 (3) Weak conjugate acid
 (4) Strong conjugate acid

203. Which of the following is strongest conjugate base

- (1) ClO_4^- (2) HCO_3^- (3) F^- (4) HSO_4^-

204. Which of the following species can act as Lewis base :-

- (1) Cu^{2+} (2) AlCl_3 (3) NH_3 (4) BF_3

205. A compound having the formula $\text{NH}_2\text{CH}_2\text{COOH}$ may behave :-

- (1) Only as an acid
 (2) Only as a base
 (3) Both as an acid and base
 (4) Neither acid nor base

206. BF_3 is acid according to :-

- (1) Lewis
 (2) Arrhenius
 (3) Bronsted and Lowry
 (4) Madam Curie

207. Which of the following can act both as Bronsted acid and Bronsted base :-

- (1) Na_2CO_3 (2) O^{2-}
 (3) CO_3^{2-} (4) NH_3

208. The strongest conjugate base is :-

- (1) NO_3^- (2) Cl^-
 (3) SO_4^{2-} (4) CH_3COO^-

209. Aluminium chloride is :-

- (1) Bronsted Lowry acid (2) Arrhenius acid
 (3) Lewis acid (4) Lewis base

210. Water is a :-

- (1) Protogenic solvent (2) Protophilic solvent
 (3) Amphoteric solvent (4) Aprotic solvent

211. Ammonium ion is :-

- (1) A conjugate acid
 (2) A conjugate base
 (3) Neither an acid nor a basic
 (4) Both an acid and a base

212. Species which do not act both as Bronsted acid and base is :-

- (1) $(\text{HSO}_4^-)^{-1}$ (2) Na_2CO_3
 (3) NH_3 (4) OH^-

213. Which one of the following is strong Lewis base & Bronsted acid & Bronsted base:-

- (1) NH_3 (2) PH_3 (3) CH_4 (4) BH_3

214. Which of the following pair is Lewis acid & Lewis base & product of these is also Lewis base

- (1) BF_3 , NH_3 (2) SiCl_4 , 2Cl^-
 (3) $\text{CH}_3^\oplus$, $^\ominus\text{OC}_2\text{H}_5$ (4) None of these

215. Which of the following is not a correct statement

- (1) Arrhenius theory of acids-bases is capable of explaining the acidic or basic nature of the substances in the solvents other than water
 (2) Arrhenius theory does not explain acidic nature of AlCl_3
 (3) The aqueous solution of Na_2CO_3 is alkaline although it does not contain OH^- ions
 (4) Aqueous solution of CO_2 is acidic although it does not contain H^+ ions

216. For the reaction $\text{NH}_4^+ + \text{S}^{2-} \rightleftharpoons \text{NH}_3 + \text{HS}^-$, NH_3 and S^{2-} are a group of :-

- (1) Acids (2) Bases
 (3) Acid-base pair (4) None of these

217. According to Lewis concept acid & base pair is-

- (1) HO^- , H^+ (2) Ag^+ , Cl^-
 (3) BF_3 , NH_3 (4) None of these

EXERCISE-I (Conceptual Questions)**ANSWER KEY**

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	2	2	2	4	4	2	1	3	1	1	4	3	1	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	1	2	3	4	4	3	4	1	1	1	3	4	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	3	4	1	2	4	4	3	2	1	1	4	1	2	3
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	3	3	3	2	3	3	4	4	4	2	4	4	3	2
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	4	3	1	3	3	3	4	3	4	1	4	2	2	3	4
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	4	2	2	4	2	4	3	3	2	1	2	2	2	4	3
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	2	3	4	4	1	2	1	2	4	3	3	1	4	3	2
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	3	2	4	3	1	4	4	2	2	4	2	1	3	2	1
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	2	2	3	2	3	2	3	1	4	4	2	3	4	1	4
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	3	3	3	3	3	1	1	1	3	4	3	1	1	1	2
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	2	4	3	1	1	1	1	1	2	2	1	3	4	2	3
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	3	4	1	1	4	2	1	4	3	1	1	2	4	4	2
Que.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
Ans.	1	1	2	1	2	4	3	2	4	3	3	4	4	2	1
Que.	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
Ans.	4	4	1	3	3	3	2	2	3	3	1	4	4	3	3
Que.	211	212	213	214	215	216	217								
Ans.	1	2	1	3	1	2	3								

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

- Which of the following pairs constitutes a buffer :-
(1) HNO_2 & NaNO_2 (2) NaOH & NaCl
(3) HNO_3 & NH_4NO_3 (4) HCl & KCl
- The hydrogen ion concentration of a 10^{-8} M HCl aqueous solution at 298 K ($K_w = 10^{-14}$) is :-
(1) 1.0×10^{-6} M (2) 1.0525×10^{-7} M
(3) 9.525×10^{-8} M (4) 1.0×10^{-8} M

AIIMS 2006

- 40 mL of 0.1 M ammonia solution is mixed with 20 mL of 0.1 M HCl . What is the pH of the mixture (pK_b of ammonia solution is 4.74) :-
(1) 4.74 (2) 2.26 (3) 9.26 (4) 5.00

AIPMT 2007

- Calculate the pOH of a solution at 25°C that contains 1×10^{-10} M of hydronium ions, i.e., H_3O^+
(1) 7.000 (2) 4.000 (3) 9.000 (4) 1.000
- A weak acid HA has a K_a of 1.00×10^{-5} . If 0.100 moles of this acid is dissolved in one litre of water the percentage of acid dissociated at equilibrium is closed to :-
(1) 99.0% (2) 1.00%
(3) 99.9% (4) 0.100%

AIPMT 2008

- Equimolar solutions of the following were prepared in water separately. Which one of the solutions will record the highest pH?
(1) BaCl_2 (2) MgCl_2 (3) CaCl_2 (4) SrCl_2
- Equal volumes of three acid solutions of pH 3, 4 and 5 are mixed in a vessel. What will be the H^+ ion concentration in the mixture ?
(1) 3.7×10^{-4} M (2) 3.7×10^{-3} M
(3) 1.11×10^{-3} M (4) 1.11×10^{-4} M

AIPMT 2009

- The ionization constant of ammonium hydroxide is 1.77×10^{-5} at 298 K. Hydrolysis constant of ammonium chloride is :-
(1) 5.65×10^{-12} (2) 5.65×10^{-10}
(3) 6.50×10^{-12} (4) 5.65×10^{-13}
- What is the $[\text{OH}^-]$ in the final solution prepared by mixing 20.0 mL of 0.050 M HCl with 30.0 mL of 0.10M $\text{Ba}(\text{OH})_2$?
(1) 0.12 M (2) 0.10 M
(3) 0.40 M (4) 0.0050M

- The dissociation constants for acetic acid and HCN at 25°C are 1.5×10^{-5} and 4.5×10^{-10} , respectively. The equilibrium constant for the equilibrium
 $\text{CN}^- + \text{CH}_3\text{COOH} \rightleftharpoons \text{HCN} + \text{CH}_3\text{COO}^-$
would be :-
(1) 3.0×10^4 (2) 3.0×10^5
(3) 3.0×10^{-5} (4) 3.0×10^{-4}

AIPMT 2010

- If pH of a saturated solution of $\text{Ba}(\text{OH})_2$ is 12, the value of its K_{sp} is :-
(1) 5.00×10^{-7} M³ (2) 4.00×10^{-6} M³
(3) 4.00×10^{-7} M³ (4) 5.00×10^{-6} M³
- In a buffer solution containing equal concentration of B^- and HB the K_b for B^- is 10^{-10} . The pH of buffer solution is :-
(1) 4 (2) 10 (3) 7 (4) 6

AIIMS 2010

- The pH of blood from 7.26 to 7.42 is maintained by buffer :-
(1) $\text{H}_2\text{CO}_3/\text{HCO}_3^-$
(2) $\text{H}_3\text{PO}_4/\text{H}_2\text{PO}_4^-$
(3) $\text{CH}_3\text{COOH}/\text{CH}_3\text{COO}^-$
(4) $\text{NH}_2\text{-CH}_2\text{COOH}/\text{NH}_2\text{CH}_2\text{COO}^-$
- K_{sp} of a salt $\text{Ni}(\text{OH})_2$ is 2×10^{-15} then molar solubility of $\text{Ni}(\text{OH})_2$ in 0.01M NaOH is :-
(1) 2×10^{-15} M (2) $2^{1/3} \times 10^{-5}$ M
(3) 2×10^{-11} M (4) 10^{-7} M

AIPMT Mains 2011

- In qualitative analysis, the metals of Group I can be separated from other ions by precipitating them as chloride salts. A solution initially contains Ag^+ and Pb^{2+} at a concentration of 0.10 M. Aqueous HCl is added to this solution until the Cl^- concentration is 0.10 M. What will the concentrations of Ag^+ and Pb^{2+} be at equilibrium ?
(K_{sp} for $\text{AgCl} = 1.8 \times 10^{-10}$, K_{sp} for $\text{PbCl}_2 = 1.7 \times 10^{-5}$)
(1) $[\text{Ag}^+] = 1.8 \times 10^{-11}$ M ; $[\text{Pb}^{2+}] = 1.7 \times 10^{-4}$ M;
(2) $[\text{Ag}^+] = 1.8 \times 10^{-7}$ M ; $[\text{Pb}^{2+}] = 1.7 \times 10^{-6}$ M;
(3) $[\text{Ag}^+] = 1.8 \times 10^{-11}$ M ; $[\text{Pb}^{2+}] = 8.5 \times 10^{-5}$ M;
(4) $[\text{Ag}^+] = 1.8 \times 10^{-9}$ M ; $[\text{Pb}^{2+}] = 1.7 \times 10^{-3}$ M;

16. A buffer solution is prepared in which the concentration of NH_3 is 0.30 M and the concentration of NH_4^+ is 0.20 M. If the equilibrium constant, K_b for NH_3 equals 1.8×10^{-5} , what is the pH of this solution ? ($\log 2.7 = 0.43$)
- (1) 9.08 (2) 9.43
(3) 11.72 (4) 8.73

AIIMS 2011

17. K_{sp} of a $\text{CaSO}_4 \cdot 5\text{H}_2\text{O}$ is 9×10^{-6} , Find the volume of CaSO_4 for 1gm (Mw = 136)
- (1) 2.45 litre (2) 5.1 litre
(3) 4.52 litre (4) 3.2 litre
18. 25ml, 0.2M Ca(OH)_2 is neutralised by 10ml of 1M HCl. Then pH of resulting solution is—
- (1) 1.37 (2) 9
(3) 12 (4) 7

AIPMT Mains 2012

19. Buffer solutions have constant acidity and alkalinity because:
- (1) they have large excess of H^+ or OH^- ions
(2) they have fixed value of pH
(3) these give unionised acid or base on reaction with added acid or alkali
(4) acids and alkalies in these solutions are shielded from attack by other ions
20. Equimolar solutions of the following substances were prepared separately. Which one of the these will record the highest pH value?
- (1) LiCl (2) BeCl_2
(3) BaCl_2 (4) AlCl_3

AIIMS 2012

21. Which will not be precipitated after addition of $(\text{NH}_4)_2\text{CO}_3$ in presence of NH_4Cl :-
- (1) Mg (2) Ba
(3) Ca (4) Sr

NEET UG 2013

22. Which is the strongest acid in the following ?
- (1) H_2SO_3 (2) H_2SO_4
(3) HClO_3 (4) HClO_4

AIIMS 2013

23. What will be the solubility product of AX_3 :-
- (1) $27 s^4$ (2) $4s^3$
(3) $36 s^4$ (4) $9s^3$

AIPMT 2014

24. Which of the following salts will give highest pH in water ?
- (1) KCl (2) NaCl
(3) Na_2CO_3 (4) CuSO_4

AIPMT 2015

25. The K_{sp} of Ag_2CrO_4 , AgCl , AgBr and AgI are respectively, 1.1×10^{-12} , 1.8×10^{-10} , 5.0×10^{-13} , 8.3×10^{-17} . Which one of the following salts will precipitate last if AgNO_3 solution is added to the solution containing equal moles of NaCl, NaBr, NaI and Na_2CrO_4 ?
- (1) AgCl (2) AgBr
(3) Ag_2CrO_4 (4) AgI

Re-AIPMT 2015

26. Which one of the following pairs of solution is not an acidic buffer ?
- (1) H_2CO_3 and Na_2CO_3
(2) H_3PO_4 and Na_3PO_4
(3) HClO_4 and NaClO_4
(4) CH_3COOH and CH_3COONa
27. What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.01 M HCl are mixed ?
- (1) 7.0 (2) 1.04
(3) 12.65 (4) 2.0

AIIMS 2015

28. What will be solubility of A_2X_3 if its solubility product (K_{sp}) is equal to 1.08×10^{-23}
- (1) $10^{-5} \text{ mol L}^{-1}$
(2) $3.7 \times 10^{-4} \text{ mol L}^{-1}$
(3) $1.2 \times 10^{-3} \text{ mol L}^{-1}$
(4) $7.5 \times 10^{-4} \text{ mol L}^{-1}$
29. pH of salt of weak acid with strong base at 25°C .
- (1) = 7
(2) > 7
(3) < 7
(4) None

NEET-I 2016

30. MY and NY_3 , two nearly insoluble salts, have the same K_{sp} values of 6.2×10^{-13} at room temperature. Which statement would be **true** in regard to MY and NY_3 ?

- (1) The molar solubilities of MY and NY_3 in water are identical.
- (2) The molar solubility of MY in water is less than that of NY_3
- (3) The salts MY and NY_3 are more soluble in 0.5 M KY than in pure water.
- (4) The addition of the salt of KY to solution of MY and NY_3 will have no effect on their solubilities.

NEET-II 2016

31. The percentage of pyridine (C_5H_5N) that forms pyridinium ion ($C_5H_5N^+H$) in a 0.10 M aqueous pyridine solution (K_b for $C_5H_5N = 1.7 \times 10^{-9}$) is

- (1) 0.77% (2) 1.6%
- (3) 0.0060% (4) 0.013%

32. The solubility of $AgCl(s)$ with solubility product 1.6×10^{-10} in 0.1 M NaCl solution would be

- (1) 1.6×10^{-11} M (2) zero
- (3) 1.26×10^{-5} M (4) 1.6×10^{-9} M

AIIMS 2016

33. Calculate the degree of ionization of 0.04 M HOCl solution having ionization constant 1.25×10^{-4} ?

- (1) 0.025 (2) 0.25 (3) 0.5 (4) 0.055

34. Calculate pH of a salt of weak monobasic acid and weak monoacidic base having concentration 0.1M at $25^\circ C$ (Given :- $pK_a = 4.8$, $pK_b = 5.2$)

- (1) 7.2 (2) 6.8 (3) 9.6 (4) 7

35. The K_{sp} of salt $AgCl$ at $25^\circ C$ is 2.56×10^{-10} . Then how much volume of H_2O is required to dissolve 0.01 mole of salt.

- (1) 800 L (2) 400 L
- (3) 625 L (4) 50 L

36. Calculate pOH of $10^{-2}M$ monobasic acid if value of K_w at $90^\circ C$ is 10^{-12} ?

- (1) 2 (2) 10
- (3) 12 (4) 7

37. K_{sp} of $Sr(OH)_2$ is x, then calculate value of $[OH^-]$ in saturated solution?

- (1) $\left(\frac{3x}{4}\right)^{1/3}$ (2) $2\left(\frac{x}{4}\right)^{1/3}$
- (3) $\left(\frac{x}{4}\right)^{1/3}$ (4) $\left(\frac{x}{3}\right)^{1/4}$

NEET(UG) 2017

38. Concentration of the Ag^+ ions in a saturated solution of $Ag_2C_2O_4$ is $2.2 \times 10^{-4} \text{ mol L}^{-1}$ Solubility product of $Ag_2C_2O_4$ is :-

- (1) 2.66×10^{-12} (2) 4.5×10^{-11}
- (3) 5.3×10^{-12} (4) 2.42×10^{-8}

AIIMS 2017

39. Calculate the ratio of solubility of $AgCl$ in 0.1M $AgNO_3$ and in pure water.

Given : K_{sp} of $AgCl = 1.8 \times 10^{-10}$:-

- (1) 1.34×10^{-5} M (2) 1.34×10^{-4} M
- (3) 2×10^{-8} M (4) 3×10^{-6} M

40. 200 ml of 0.005 M $AgNO_3$ reacts with 300 mL of 0.01 M KCl. If K_{sp} of $AgCl$ is 1.8×10^{-10} . Then maximum conc. of Ag^+ in mixture is :-

- (1) 2×10^{-8} (2) 4.5×10^{-8}
- (3) 4.8×10^{-5} (4) 1.34×10^{-5}

NEET(UG) 2018

41. Following solutions were prepared by mixing different volumes of NaOH and HCl of different concentrations :

a. $60\text{mL} \frac{M}{10} \text{HCl} + 40\text{mL} \frac{M}{10} \text{NaOH}$

b. $55\text{mL} \frac{M}{10} \text{HCl} + 45\text{mL} \frac{M}{10} \text{NaOH}$

c. $75\text{mL} \frac{M}{5} \text{HCl} + 25\text{mL} \frac{M}{5} \text{NaOH}$

d. $100\text{mL} \frac{M}{10} \text{HCl} + 100\text{mL} \frac{M}{10} \text{NaOH}$

pH of which one of them will be equal to 1 ?

- (1) b (2) a
- (3) d (4) c

42. The solubility of BaSO_4 in water $2.42 \times 10^{-3} \text{ g L}^{-1}$ at 298 K. The value of solubility product (K_{sp}) will be

(Given molar mass of $\text{BaSO}_4 = 233 \text{ g mol}^{-1}$)

- (1) $1.08 \times 10^{-10} \text{ mol}^2 \text{ L}^{-2}$
- (2) $1.08 \times 10^{-12} \text{ mol}^2 \text{ L}^{-2}$
- (3) $1.08 \times 10^{-14} \text{ mol}^2 \text{ L}^{-2}$
- (4) $1.08 \times 10^{-8} \text{ mol}^2 \text{ L}^{-2}$

AIIMS 2018

43. Mixture of ester and HCl is titrated with NaOH using phenolphthalein as an acidic indicator at end point pink color disappear after some time due to :-

- (1) CH_3COOH is formed
- (2) Due to weak acidic nature of CH_3OH
- (3) Regeneration of HCl
- (4) Due to ionization of phenolphthalein

44. 50 ml of 0.1 M NH_3 solution and 10 mL of 0.1 M HCl mixed together then calculate pH of mixture. If $\text{p}K_b$ of NH_3 is 4.75 :-

- (1) 8.86
- (2) 4.74
- (3) 9.86
- (4) 5.32

45. Which of following will have maximum pH :-

- (1) Human saliva
- (2) Black coffee
- (3) Human blood
- (4) Gastric juice

46. Find concentration of $[\text{H}^+]$ after mixing of 15 mL, 0.1M H_2SO_4 and 15ml 0.1M NaOH :-

- (1) 5×10^{-2}
- (2) 10^{-7}
- (3) 1×10^{-2}
- (4) 2×10^{-2}

47. Find the concentration of $[\text{H}^+]$ after mixing 20 ml, 0.1M CH_3COOH and 10 ml, 0.1 M NaOH. Given K_a for CH_3COOH 1.7×10^{-5} .

- (1) 1.7×10^{-5}
- (2) 3.4×10^{-5}
- (3) 3.4×10^{-6}
- (4) 1.7×10^{-4}

48. Solubility of XB_2 is X mol/lit in water find out solubility of XB_2 in 10^{-3} M YB . Consider YB is completely soluble salt :-

- (1) $4X^3 \times 10^{+6}$
- (2) $4X^3 \times 10^{-6}$
- (3) 10^{-3}
- (4) X^3

49. 0.02M of pyridium hydrochloride having $[\text{H}^+]$ is 5.55×10^{-6} what is ionisation constant of pyridine.

- (1) 1.54×10^{-9}
- (2) 1.54×10^{-7}
- (3) 1.54×10^{-6}
- (4) 1.54×10^{-5}

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	2	3	2	2	1	1	2	2	1	1	1	1	3	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	4	3	3	1	4	1	3	3	3	3	1	2	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	4	4	2	3	2	2	3	2	2	4	1	4	3	3
Que.	46	47	48	49											
Ans.	1	1	1	1											

EXERCISE-III (Analytical Questions)

Check Your Understanding

- The concentration of $[H^+]$ and concentration of $[OH^-]$ of a 0.1 M aqueous solution of 2% ionised weak acid is [ionic product of water = 1×10^{-14}] :-
 (1) 0.02×10^{-3} M and 5×10^{-11} M
 (2) 1×10^{-3} M and 3×10^{-11} M
 (3) 2×10^{-3} M and 5×10^{-12} M
 (4) 3×10^{-2} M and 4×10^{-13} M
- The solubility of $BaSO_4$ in water, is 2.33×10^{-3} gL⁻¹. Its solubility product will be (molecular weight of $BaSO_4$ = 233) :-
 (1) 1×10^{-5} (2) 1×10^{-10}
 (3) 1×10^{-15} (4) 1×10^{-20}
- What will be the H^+ ion concentration, when 4 g NaOH dissolved in 1000 mL of water:
 (1) 10^{-1} (2) 10^{-13} (3) 10^{-4} (4) 10^{-10}
- When 10 mL of 0.1 M acetic acid ($pK_a=5.0$) is titrated against 10 mL of 0.1 M ammonia solution ($pK_b = 5.0$), the equivalence point occurs at pH:
 (1) 5.0 (2) 6.0 (3) 7.0 (4) 9.0
- At 25°C, the dissociation constant of a base, BOH, is 1.0×10^{-12} . The concentration of hydroxyl ions in 0.01M aqueous solution of the base would be:
 (1) 1.0×10^{-6} mol L⁻¹ (2) 1.0×10^{-7} mol L⁻¹
 (3) 2.0×10^{-6} mol L⁻¹ (4) 1.0×10^{-5} mol L⁻¹
- The solubility product of $BaSO_4$ at 25°C is 1.0×10^{-9} . What would be the concentration of H_2SO_4 necessary to precipitate $BaSO_4$ from a solution of 0.01 M Ba^{+2} ions :-
 (1) 10^{-9} (2) 10^{-8} (3) 10^{-7} (4) 10^{-6}
- pH of the solution of $HCOONH_4$ is 6.48 this can be explained by :-
 (1) Hydrolysis of both cation and anion
 (2) Hydrolysis of cation
 (3) Hydrolysis of anion
 (4) Hydrolysis of water
- The correct representation of solubility product of SnS_2 is :-
 (1) $[Sn^{4+}][S^{2-}]^2$ (2) $[Sn^{4+}][S^{2-}]$
 (3) $[Sn^{4+}][2S^{2-}]$ (4) $[Sn^{4+}][2S^{2-}]^2$
- A solution of $FeCl_3$ in water acts as acidic due to:-
 (1) Acidic impurities
 (2) Ionisation
 (3) Hydrolysis of Fe^{3+}
 (4) Dissociation
- The pK_a of HNO_2 is 3.37. The pH of HNO_2 in its 0.01 mol L⁻¹ aqueous solution will be :-
 (1) 5.37 (2) 2.69 (3) 1.69 (4) 0.69
- When 0.01 M HCl is added in aqueous solution of acetic acid then :-
 (1) $[CH_3COO^-]$ decreases
 (2) $[CH_3COOH]$ decreases
 (3) $[CH_3COO^-]$ increases
 (4) NOT
- On passing H_2S gas through a solution of Cu^{+2} and Zn^{+2} ions, CuS is precipitated first because:-
 (1) Solubility product of CuS is equal to the ionic product of ZnS ;
 (2) Solubility product of CuS is equal to the solubility product of ZnS .
 (3) Solubility product of CuS is lower than the solubility product of ZnS .
 (4) Solubility product of CuS is greater than the solubility product of ZnS .
- Solubility of MX_2 - type electrolytes is 0.5×10^{-4} mol L⁻¹ then find out K_{sp} of electrolytes:-
 (1) 5×10^{-12} (2) 25×10^{-10}
 (3) 1×10^{-13} (4) 5×10^{-13}
- A solution of $MgCl_2$ in water has pH :
 (1) < 7 (2) > 7 (3) 7 (4) 14.2
- When H_2S gas is passed through the HCl containing aqueous solutions of $CuCl_2$, $HgCl_2$, $BiCl_3$ and $CoCl_2$, it does not precipitate out :
 (1) CuS (2) HgS (3) $Bi_2 S_3$ (4) CoS
- Mark the correct statement :
 (1) I group basic radicals precipitate as chloride.
 (2) IV group basic radicals precipitate as sulphides
 (3) V group basic radicals precipitate as carbonates
 (4) All the above statements are correct
- The role of NH_4Cl in group III for analysis of cations is :-
 (1) that it acts as a catalyst.
 (2) that it increases the solubility of hydroxides.
 (3) that it lowers the OH^- concentration.
 (4) that it causes the precipitation of cations as chlorides.
- K_{sp} of $Ca_3(PO_4)_2$ is :-
 (1) $[Ca^{+2}][PO_4^{-3}]^2$ (2) $[Ca^{+2}]^3 [PO_4^{-3}]^2$
 (3) $[Ca^{+2}]^2 [PO_4^{-3}]^3$ (4) $[Ca^{+2}] [PO_4^{-3}]$

- 19.** An acid HA has dissociated as following manner
 $HA \rightleftharpoons H^+ + A^-$
 It has concentration 1 M and pH = 5 then find out dissociation constant :-
 (1) 1×10^{-10} (2) 1×10^{-5}
 (3) 5×10^{-5} (4) 5
- 20.** Which of the following group of cations will be precipitated when passing of H_2S gas in the presence of acidic medium:-
 (1) Cu^{+2} , Cr^{+3} (2) Zn^{+2} , Co^{+2}
 (3) Cu^{+2} , Cd^{+2} (4) Al^{+3} , Cd^{+2}
- 21.** Solution of sodium carbonate is :-
 (1) Strongly acidic (2) Weakly basic
 (3) Strongly basic (4) Weakly acidic
- 22.** Aqueous solution of $Al_2(SO_4)_3$ is :-
 (1) Basic & acidic (2) Neutral
 (3) Basic (4) Acidic
- 23.** In a saturated solution of the sparingly soluble strong electrolyte $AgIO_3$ (Molecular mass = 283). The equilibrium which sets in is
 $AgIO_{3(s)} \rightleftharpoons Ag^+_{(aq)} + IO_{3(aq)}^-$ If the solubility product constant K_{sp} of $AgIO_3$ at a given temperature is 1.0×10^{-8} , what is the mass of $AgIO_3$ contained in 100 mL of its saturated solution :-
 (1) 2.83×10^{-3} g (2) 1.0×10^{-7} g
 (3) 1.0×10^{-4} g (4) 28.3×10^{-2} g
- 24.** If k_{sp} of CaF_2 in pure water is 1.70×10^{-10} , then find the solubility of CaF_2 in 0.10M NaF solution :-
 (1) 1.70×10^{-10} (2) 1.70×10^{-9}
 (3) 1.70×10^{-8} (4) 0.10 M
- 25.** The solubility product of $AgCl$ is 1×10^{-10} , then solubility of $AgCl$ is :-
 (1) 1×10^{-10} (2) 1×10^{-7}
 (3) 1×10^{-5} (4) 1×10^{-8}
- 26.** To precipitate Zn in form of ZnS , Why NH_4OH is first added in the solution before H_2S gas is passed through it :-
 (1) To convert Zn into Zn^{+2}
 (2) To reduce Zinc
 (3) To decrease the dissociation of H_2S
 (4) To increase the dissociation of H_2S
- 27.** K_a for CH_3COOH is 1.8×10^{-5} . Find out the percentage dissociation of 0.2M CH_3COOH in 0.1M HCl solution
 (1) 0.018 (2) 0.36 (3) 18 (4) 36
- 28.** The pK_a of a weak acid HA is 4.80. The pK_b of weak base BOH is 4.78. The pH of an aqueous solution of the corresponding salt BA will be :
 (1) 9.58 (2) 4.79 (3) 7.01 (4) 9.22
- 29.** 0.2M solution of $HCOOH$ is 3.2% ionised then find ionisation constant of acid :-
 (1) 4.2×10^{-4} (2) 4.2×10^{-5}
 (3) 2.1×10^{-4} (4) 2.1×10^{-5}
- 30.** Three reactions involving $H_2PO_4^-$ are given below :-
 (i) $H_3PO_4 + H_2O \rightarrow H_3O^+ + H_2PO_4^-$
 (ii) $H_2PO_4^- + H_2O \rightarrow HPO_4^{2-} + H_3O^+$
 (iii) $H_2PO_4^- + OH^- \rightarrow H_3PO_4 + O^{2-}$
 In which of the above does $H_2PO_4^-$ act as an acid ?
 (1) (i) only (2) (ii) only
 (3) (i) and (ii) (4) (iii) only
- 31.** Given that for HA acid, $K_a = 10^{-6}$ and for MOH base $K_b = 10^{-6}$. The pH of 0.1 M MA salt solution will be :-
 (1) 5 (2) 7 (3) 9 (4) 2
- 32.** The K_{sp} for $Cr(OH)_3$ is 1.6×10^{-30} . The molar solubility of this compound in water is :-
 (1) $\sqrt[3]{1.6 \times 10^{-30}}$ (2) $\sqrt[4]{1.6 \times 10^{-30}}$
 (3) $\sqrt[4]{1.6 \times 10^{-30} / 27}$ (4) $1.6 \times 10^{-30} / 27$
- 33.** An acid HA ionises as
 $HA \rightleftharpoons H^+ + A^-$
 The pH of 1.0 M solution is 5. Its dissociation constant would be :-
 (1) 1×10^{-10} (2) 5
 (3) 5×10^{-8} (4) 1×10^{-5}
- 34.** The pH of a 0.1 molar solution of the acid HQ is 3. The value of the ionization constant, K_a of this acid is :-
 (1) 1×10^{-7} (2) 3×10^{-7}
 (3) 1×10^{-3} (4) 1×10^{-5}
- 35.** How many litres of water must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2 ?
 (1) 0.1 L (2) 0.9 L (3) 2.0 L (4) 9.0 L
- 36.** What volume of 0.1M H_2SO_4 is needed to completely neutralize 40 mL of 0.2M NaOH solution-
 (1) 10 mL (2) 40 mL (3) 20 mL (4) 80 mL

37. If pH value of a solution is 3 and by adding water, it becomes 6, then the dilution is increased by :
 (1) 10 times (2) 100 times
 (3) 500 times (4) 1000 times
38. The aqueous solution of which of the following salt will have the lowest pH
 (1) NaClO (2) NaClO₂
 (3) NaClO₃ (4) NaClO₄
39. In the reaction $I_2 + I^- \rightarrow I_3^-$, the Lewis base is :
 (1) I⁻ (2) I₂
 (3) I₃⁻ (4) None of these
40. In which of the following acid-base titration, pH is greater than 8 at equivalence point:-
 (1) Acetic acid versus ammonia
 (2) Acetic acid versus sodium hydroxide
 (3) Hydrochloric acid versus ammonia
 (4) Hydrochloric acid versus sodium hydroxide
41. Which one of the following compounds is not a protonic acid :-
 (1) B(OH)₃ (2) PO(OH)₃
 (3) SO(OH)₂ (4) SO₂(OH)₂
42. Which one of the following is NOT a buffer solution:-
 (1) 0.8M H₂S + 0.8M KHS
 (2) 2M C₆H₅NH₂ + 2M C₆H₅N⁺H₃Br
 (3) 3M H₂CO₃ + 3M KHCO₃
 (4) 0.05M KClO₄ + 0.05M HClO₄
43. The rapid change of pH near the stoichiometric point of an acid-base titration is the basis of indicator detection. pH of the solution is related to ratio of the concentrations of the conjugate acid (HIn) and base (In⁻) forms of the indicator by the expression:-
 (1) $\log \frac{[HIn]}{[In^-]} = pK_{in} - pH$
 (2) $\log \frac{[HIn]}{[In^-]} = pH - pK_{in}$
 (3) $\log \frac{[In^-]}{[HIn]} = pH + pK_{in}$
 (4) $\log \frac{[In^-]}{[HIn]} = pK_{in} - pH$
44. What is the pH of 0.01M glycine solution? For glycine, $K_{a1} = 4.5 \times 10^{-3}$ and $K_{a2} = 1.7 \times 10^{-10}$ at 298 K.
 (1) 3.0 (2) 10.0 (3) 6.1 (4) 7.2
45. The correct order of acid strength is -
 (1) HClO₄ < HClO₃ < HClO₂ < HClO
 (2) HClO₂ < HClO₃ < HClO₄ < HClO
 (3) HClO₄ < HClO < HClO₂ < HClO₃
 (4) HClO < HClO₂ < HClO₃ < HClO₄
46. Concentrations of NH₄Cl and NH₄OH in a buffer solution are in the ratio 1 : 10. If K_b for NH₄OH is 10^{-10} , then pH of the buffer is :-
 (1) 4 (2) 5 (3) 9 (4) 11
47. 10 mL concentrated H₂SO₄ (18 molar) is diluted to 1 litre. Concentration of diluted acid is:-
 (1) 0.18N (2) 0.09N (3) 0.36N (4) 18N
48. When HF is dissolved in formic acid, the equilibrium established is :-
 $HF + HCOOH \rightleftharpoons F^- + HCOOH_2^+$
 the true pair of conjugate acid base is as a [acid, conjugate acid] [Base, conjugate base]:-
 (1) (HF, HCOOH) and (HCOOH₂⁺, F⁻)
 (2) (HF, HCOOH₂⁺) and (HCOOH, F⁻)
 (3) (HCOOH₂⁺, HF) and (F⁻, HCOOH)
 (4) (HF, F⁻) and (HCOOH₂⁺, HCOOH)
49. How many grams of dibasic acid (mol. wt. = 200) should be present in 100 mL the aqueous solution to give strength of (N/10) :-
 (1) 1g. (2) 2g. (3) 5g. (4) 10g.
50. In a mixture of equimolar solutions of NaHCO₃ and NaOH the species present in solution shall be:-
 (1) Na₂CO₃ (2) NaHCO₃ + NaOH
 (3) NaOH (4) NaHCO₃ + Na₂CO₃
51. An aqueous solution contains 10^{-4} [H⁺]. If it is diluted by mixing equal volume of water then the concentration of OH⁻ in mol dm⁻³ will be :-
 (1) 0.5×10^{-10} (2) 2×10^{-10}
 (3) 10^{-6} (4) 10^{-8}
52. Which of the following is right for diprotic acid :
 (1) $K_{a2} > K_{a1}$ (2) $K_{a1} > K_{a2}$
 (3) $K_{a2} > \frac{1}{K_{a1}}$ (4) $K_{a2} = K_{a1}$

53. The first and second dissociation constants of an acid H_2A are 1.0×10^{-5} and 5.0×10^{-10} respectively. The overall dissociation constant of the acid will be:-
 (1) 5.0×10^{15} (2) 5.0×10^{-15}
 (3) 0.2×10^5 (4) 5.0×10^{-5}
54. The pK_a of a weak acid (HA) is 4.5. The pOH of an aqueous buffered solution of HA in which 50% of the acid is ionized is :-
 (1) 2.5 (2) 9.5 (3) 7.0 (4) 4.5
55. 50 mL solution of 0.1M CH_3COOH ($pK_a = 4.73$) is titrated with 0.1M NaOH solution, pH of solution when half of CH_3COOH is neutralized
 (1) 4.53 (2) 4.63 (3) 4.73 (4) 4.83
56. pK_a of Quinoline base is 4.88. What will be the pK_a of 0.01M solution of it
 (1) 4.88 (2) 0.01 (3) 9.12 (4) 14
57. The pH of an aqueous solution of a 1×10^{-7} M solution of HCl will be :-
 (1) 7 (2) slightly less than 7
 (3) slightly greater than 7 (4) 1
58. What will be the concentration of H^+ ions in 0.1M acetic acid and 0.1M sodium acetate solution, if the dissociation constant of acetic acid is 1.8×10^{-5} ?
 (1) 1.8×10^{-7} (2) 1.8×10^{-5}
 (3) 1.8×10^{-2} (4) 1.8×10^{-3}
59. Calculate the pH of 4×10^{-3} M, $Y(OH)_2$ solution assuming the first dissociation to be 100% second dissociation to be 50%:-
 (1) 11.78 (2) 9.9 (3) 2.5 (4) 2.22

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	2	3	2	4	1	1	3	2	1	3	4	1	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	3	2	1	3	2	4	1	3	3	4	1	3	3	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	3	1	4	4	2	4	4	1	2	1	4	1	3	4
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	
Ans.	2	3	2	1	1	2	2	2	2	3	1	2	2	1	

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** When small amount of acid or base is added to pure water, its pH undergoes a change.
Reason :- Addition of an acid or base increases the degree of ionisation of water.
 (1) A (2) B (3) C (4) D
- Assertion :-** Solubility of AgBr decreases in the presence of sodium bromide.
Reason :- Sodium bromide undergoes hydrolysis in water.
 (1) A (2) B (3) C (4) D
- Assertion :-** The pH of an aqueous solution of acetic acid remains unchanged on the addition of sodium acetate.
Reason :- The ionisation of acetic acid is suppressed by the addition of sodium acetate.
 (1) A (2) B (3) C (4) D
- Assertion :-** If HCl gas is passed through saturated NaCl solution, solid NaCl starts to precipitate.
Reason :- HCl decreases the solubility product of NaCl.
 (1) A (2) B (3) C (4) D
- Assertion :-** The aqueous solution of Na_3PO_4 is alkaline in nature.
Reason :- Na_3PO_4 in its aqueous solution undergoes anionic hydrolysis.
 (1) A (2) B (3) C (4) D
- Assertion :-** If $K_{sp} <$ ionic product, precipitate is formed.
Reason :- Solubility product (K_{sp}) is the highest limit of ionic product of the electrolyte in saturated solutions.
 (1) A (2) B (3) C (4) D
- Assertion :-** To precipitate the cations of fourth group in qualitative analysis, medium is made alkaline before passing H_2S gas.
Reason :- This is done to suppress the ionisation of H_2S .
 (1) A (2) B (3) C (4) D
- Assertion :-** When an acid or a base is added to water at constant temperature the pH changes.
Reason :- This is due to change in ionic product of water.
 (1) A (2) B (3) C (4) D
- Assertion :-** Addition of NH_4OH to an aqueous solution of BaCl_2 in the presence of NH_4Cl (excess) precipitates Ba(OH)_2 .
Reason :- Ba(OH)_2 is insoluble in water.
 (1) A (2) B (3) C (4) D
- Assertion :-** Sb(III) is not precipitated as sulphide when in its alkaline solution H_2S is passed.
Reason :- The concentration of S^{2-} ion in alkaline medium is inadequate for precipitation.
 (1) A (2) B (3) C (4) D
- Assertion :-** For H_2SO_4 , K_{a1} value is much greater than K_{a2} .
Reason :- H_2SO_4 is a strong acid.
 (1) A (2) B (3) C (4) D
- Assertion :-** Solubility of salt of SB and WA changes when pH of the solution is changed.
Reason :- Solubility product (K_{sp}) depends on pH.
 (1) A (2) B (3) C (4) D
- Assertion :-** A mixture of a weak acid CH_3COOH and sodium acetate forms a buffer solution.
Reason :- A buffer solution reacts with small quantities of hydrogen or hydroxyl ions and keeps the pH almost same.
 (1) A (2) B (3) C (4) D
- Assertion :-** At 25°C the pH of 10^{-8} M HCl is 8.
Reason :- pH of acidic solution is always below 7 at 25°C .
 (1) A (2) B (3) C (4) D

15. **Assertion :-** In the acid base titration involving a strong base and a weak acid methyl orange can be used as an indicator.

Reason :- Methyl orange changes its colour in pH range 3 to 5.

(1) A (2) B (3) C (4) D

16. **Assertion :-** pH of a buffer changes with temperature.

Reason :- Ionic product of water (K_w) changes with temperature.

(1) A (2) B (3) C (4) D

17. **Assertion :-** H_3PO_3 is a dibasic compound.

Reason :- The two H-atom are directly attached to P.

(1) A (2) B (3) C (4) D

18. **Assertion :** Boric acid behaves as a weak monobasic acid.

Reason : Boric acid contains hydrogen bonds in its structure.

(1) A (2) B (3) C (4) D

19. **Assertion :** H_2O is amphoteric in nature.

Reason : H_2O can accept a proton to form H_3O^+ ion and can donate a proton to form OH^- ion.

(1) A (2) B (3) C (4) D

20. **Assertion :** All Arrhenius acids are also Bronsted acids.

Reason : All Bronsted bases are also Lewis bases.

(1) A (2) B (3) C (4) D

21. **Assertion :-** The buffer solution has a capacity to resist the change in pH value on addition of small amount of acid or base to it.

Reason :- pH value of buffer solution does not change on dilution or on keeping for long.

(1) A (2) B (3) C (4) D

22. **Assertion :-** The species in the buffer must not react with each other.

Reason :- The pH of a buffer depends on the value of K_a of the weak acid and the relative concentration of that acid and its conjugate base.

(1) A (2) B (3) C (4) D

23. **Assertion :-** Only a very small amount of indicator should be used.

Reason :- So that addition of the indicator does not effect the pH of the solution.

(1) A (2) B (3) C (4) D

24. **Assertion :-** Solubility of sparingly soluble salt increases with increase in temperature.

Reason :- Enthalpy of solution is negative for all salts.

(1) A (2) B (3) C (4) D

25. **Assertion :-** HCl is titrated with NaOH using phenolphthalein as indicator, pink colour appears in solution when OH^- ion is present in excess in solution.

Reason :- At end point, both reacting species, neutralises each other.

(1) A (2) B (3) C (4) D

26. **Assertion :-**



IInd is more suitable for titration.

Reason :- It is difficult to add titrant in first.

(1) A (2) B (3) C (4) D

27. **Assertion :** In $NaHCO_3$ solution phenolphthalein is colorless.

Reason : Phenolphthalein is colorless in basic medium and pH of $NaHCO_3$ is less than 10.

(1) A (2) B (3) C (4) D

28. **Assertion :-** Na_2SO_3 is alkaline towards litmus.

Reason :- H_2SO_3 is formed due to hydrolysis.

(1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	4	3	1	1	3	3	4	4	2	3	1	4	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28		
Ans.	2	3	2	1	3	2	2	1	3	2	4	4	2		

CHEMICAL THERMODYNAMICS AND ENERGETICS

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NEET SYLLABUS

CHEMICAL THERMODYNAMICS : First law of thermodynamics-internal energy and enthalpy, heat capacity and specific heat, measurement of ΔU and ΔH , Hess's law of constant heat summation, enthalpy of : bond dissociation, combustion, formation, atomization, sublimation, phase transition, ionization, solution and dilution. Introduction of entropy as state function, Second law of thermodynamics, Gibbs energy change for spontaneous and non-spontaneous process, criteria for equilibrium and spontaneity. Third law of thermodynamics- Brief introduction.

OBJECTIVES

After studying this unit, we will be able to :

- explain the terms : system and surroundings;
- discriminate between close, open and isolated systems;
- explain internal energy, work and heat;
- state first law of thermodynamics and express it mathematically;
- calculate energy changes as work and heat contributions in chemical systems;
- explain state functions: U , H .
- correlate ΔU and ΔH ;
- define standard states for ΔH ;
- calculate enthalpy changes for various types of reactions;
- state and apply Hess's law of constant heat summation;
- differentiate between extensive and intensive properties;
- define spontaneous and nonspontaneous processes;
- explain entropy as a thermodynamic state function and apply it for spontaneity;
- explain Gibbs energy change (ΔG);
- establish relationship between ΔG and spontaneity, ΔG and equilibrium constant.

"Thermodynamics cannot be fathomed in all its profundity in a single pass"

Joseph Kestin

CHEMICAL THERMODYNAMICS

5.0 INTRODUCTION :

Thermo (Heat/energy) + Dynamics (Flow/motion)

Thermodynamics is the branch of science which deals with the energy changes taking place in all physical and chemical processes but **Chemical thermodynamics** is the branch of thermodynamics which deals with the study of energy changes taking place in chemical processes.

Advantages of thermodynamics :

- It gives information about various **thermodynamic laws**.
- It helps us to predict whether a given **chemical reaction will take place or not** under the given set of conditions.
- It gives information about various **energy changes**.

Limitations of thermodynamics :

- Thermodynamics deals with the properties like temperature, pressure, volume, etc of matter in bulk but doesn't tell us anything about the individual properties of atoms or molecules.

or

Thermodynamics deals with **macroscopic system** but not with **microscopic system**.

- It tells us whether a given chemical reaction will take place or not under the given set of conditions but doesn't tell us anything about the **rate of reaction**.

5.1 SOME IMPORTANT TERMS :

- System** : A system is defined as a specific part of universe or specified portion of the matter which is under experimental investigation
- Surrounding** : The rest part of the universe excluding the system is called surrounding.
Universe = System + Surrounding
- Boundary** : Anything which separates system and surrounding is called boundary.

Types of boundary :

- Boundary can be conducting or non-conducting.
- Boundary can be rigid or non-rigid.
- Boundary can be real or imaginary.

For example :

A reaction is carried out in a beaker. The contents of beaker constitute the system, beaker serves as boundary and anything which is outside the beaker is called surroundings.

5.2 TYPES OF SYSTEM :

System are of 3 types :

Open system : This type of system can exchange energy as well as matter with the surrounding. The boundary is neither sealed nor insulated. Total mass will not remain constant.

Eg.1 Coffee in open glass. Hot water in open glass.

Eg.2 All living systems. human being, plants, animals.

Eg.3 Classroom, earth.

Closed system : This type of systems can exchange energy, (in the form of heat, work or radiations) but not matter with its surroundings. The boundary is sealed but not insulated. Amount of the system will remain constant.

Eg.1 Coffee in closed vessel. Hot water in closed vessel.

Eg.2 Glowing bulb, tube light.

Eg.3 A satellite in orbit.

Isolated system : This type of system are perfectly insulated systems and cannot interact in any way with its surrounding i.e. neither matter nor energy can be exchanged with the surrounding. The boundary is sealed and insulated. Universe can be considered as an isolated system.

Eg.1 Coffee in thermosflask.

5.3 STATE OF THE SYSTEM :

- Properties which define state of any system are called its state variables or thermodynamic variables or thermodynamic quantities.
- The state of the system is defined by their measurable properties like temperature, pressure, volume etc.
- If any of these properties change, state of the system is said to be changed.

State Function :

- Those state variables which depend only upon initial and final state of the system but doesn't depend upon the path or mechanism followed by the system to achieve final state are called state function.
- State functions are denoted by capital letters.

Ex. E, H, S, G, T, P, V etc.

Path function :

- Properties of the system which depend upon the initial and final state of the system as well as the path or mechanism followed by the system to achieve final state are called path function.
- Path functions are denoted by small letters.

Eg. Work done (w), heat (q)

Thermodynamic properties :

- Intensive properties :** The properties of the system which are independent of matter (size and mass) present in system are called intensive properties.
- Extensive properties :** The properties of the system which are dependent on matter (size and mass) present in system are called extensive properties

Extensive Properties	Intensive Properties
Volume (V)	Molar volume (V_m)
Number of moles (n)	Density (d)
Mass (m)	Gibb's energy per mole (G_m)
Gibb's Energy (G)	Specific heat
Entropy (S)	Pressure (P)
Enthalpy (H)	Temperature (T)
Internal energy (E or U)	All concentration terms (M, N)
Heat capacity (C)	Boiling point, freezing point (T_b , T_f)
Force (F)	Cell potential (E_{cell})
Surface Area (A)	Specific conductance (κ)
	Refractive index
	Surface tension, Viscosity
	pH value
	Vapour pressure

Special Points :

- (i) The ratio of two extensive properties indicates the intensive property Eg. $d = \frac{\text{mass}}{\text{volume}}$
- (ii) An extensive property can be converted into intensive property when it is defined for unit amount of the substance.

Eg. Mass per unit volume = density; $\left(d = \frac{m}{V}\right)$

- (iii) Intensive properties of a substance are non-additive in nature while extensive properties are additive in nature.

	$\boxed{\text{H}_2\text{O}}$	$\boxed{\text{H}_2\text{O}}$	On adding
Vessel	A	B	
Mass	m_1	m_2	$m_1 + m_2$ (Total mass)
Moles	n_1	n_2	$n_1 + n_2$ (Total moles)
Volume	V_1	V_2	$V_1 + V_2$ (Total volume)
Density	d	d	Remains same (d)
Boiling point	T	T	Remains same (T)

5.4 TYPES OF THERMODYNAMIC PROCESSES :

When a system changes from one state to another, the operation is called a thermodynamic process. Thermodynamic processes may be in form of expansion or compression.

(a) Isothermal Process ($n, T = \text{constant}$)

Isothermal processes are those processes in which

- (a) Temperature of system during entire process remains constant i.e. $\Delta T = 0$
 - (b) Heat is exchanged with surroundings
 - (c) Volume and pressure are variable
- For ideal gas in Isothermal process $\Delta E = 0$ and $\Delta H = 0$
 - All phase transitions are isothermal process but $\Delta E \neq 0$ and $\Delta H \neq 0$

(b) Isobaric Process ($n, P = \text{constant}$) :

Isobaric processes are those processes in which

- (a) Pressure of system during entire process remains const i.e. $\Delta P = 0$
- (b) Volume and temperature are variable.
- (c) Process in open system is isobaric in nature.

(c) Isochoric Process ($n, V = \text{constant}$) :

Isochoric processes are those processes in which

- (a) Volume remains constant i.e. $\Delta V = 0$
- (b) Pressure and temperature are variable
- (c) Work, $w = -P\Delta V$, $\therefore \Delta V = 0$, $\therefore w = 0$ (Zero)
- (d) Process in closed system is isochoric in nature.

(d) Adiabatic Process ($n = \text{constant}, q = 0$)

Adiabatic processes are those processes in which

- (a) No exchange of heat between system and surrounding takes place during entire process i.e. $q = 0$
- (b) The temperature pressure, volume of the system varies.
- (c) The system is thermally insulated by keeping the system in an insulated container.

(e) Cyclic Process :

When a system undergoes a number of different processes and finally returns to its initial state, it is termed as cyclic process.

In cyclic process change in all state function will be zero. i.e. $\Delta E = 0$, $\Delta H = 0$, $\Delta P = 0$, $\Delta T = 0$

(f) Reversible Process (quasi-static) :

- (1) Process in which all changes occurring at any part of the system are exactly reversed when small changes in variables are carried out in opposite direction.
- (2) Driving force should be infinitesimally greater than opposing force.
- (3) Process takes place in infinitesimal small steps or in many steps and takes infinite time to complete the process.
- (4) It is an ideal process.
- (5) Work obtained in expansion is maximum.
- (6) System is in virtual equilibrium at any state.
- (7) $P_{\text{ext}} = P_{\text{int}} \pm dP$; P_{ext} is variable.

Irreversible Process :

- (1) Process in which direction of change cannot be reversed by small changes in variables.
- (2) Driving force is much greater than opposing force.
- (3) It takes finite time and finite/usually single step.
- (4) Process takes place in short time
- (5) All natural processes are irreversible
- (6) System is in equilibrium only at initial and final state
- (7) $P_{\text{ext}} = P_{\text{int}} \pm \Delta P$; P_{ext} is constant.

5.5 WORK AND HEAT :

(A) Work : Product of force and displacement is known as work.

$$\text{work (w)} = \text{force (F)} \times \text{displacement (}\ell\text{)}$$

Consider a gas enclosed in a cylinder fitted with a frictionless piston.

Suppose area of cross section of cylinder = A and pressure on the piston = P

Initial volume of the gas = V_1 and final volume of the gas = V_2

(By expansion) displacement of piston = ℓ

$$\text{work done by the gas (in expansion)} = w = F \cdot \ell$$

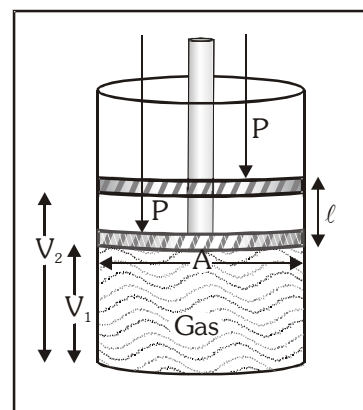
$$\therefore P = \frac{F}{A} \quad \therefore F = P \times A$$

$$w = P \times A \times \ell \quad (\text{change in volume} = A \times \ell = V_2 - V_1)$$

$$w = P \times (V_2 - V_1)$$

$$w = P\Delta V \quad (\text{According to Physics})$$

$$\text{In general } \boxed{w = -P_{\text{external}} DV} \quad (\text{According to Chemistry})$$



SIGN CONVENTIONS :

- According to latest sign conventions
 - (a) Work done is taken negative if it is done by the system since energy of system is decreased.
Eg. Expansion of gas.
 $\therefore V_2 > V_1$; $\Delta V = \text{positive}$; $w = \text{negative}$
 - (b) Work done is taken positive if it is done on the system, since energy of system is increased.
Eg. Compression work.
 $\therefore V_1 > V_2$; $\Delta V = \text{negative}$; $w = \text{positive}$

(B) Heat exchange (q) :

Heat is defined as the energy that flow in or out of a system because of a difference in temperature between the thermodynamic system and its surrounding. It is a path function.

According to IUPAC convention, heat given by the system is expressed with negative sign and heat given to the system is expressed with positive sign.

- Heat always flows from high temperature to low temperature.
- Heat flowing into the system $q = \text{positive}$
- Heat flowing out of the system $q = \text{negative}$

Units of heat & work :

Calorie : It is defined as the quantity of heat required to raise the temperature of 1 g of water by 1°C (14.5 to 15.5°C)

$$1 \text{ cal} = 4.184 \text{ J} \simeq 4.2 \text{ J}$$

$$1 \text{ L-atm} = 101.3 \text{ J} = 24.206 \text{ cal} = 101.3 \times 10^7 \text{ erg}$$

$$1 \text{ L-atm} > 1 \text{ Cal} > 1 \text{ J} > 1 \text{ erg}$$

GOLDEN KEY POINTS

- Molar properties like ΔH_m , ΔG_m , ΔS_m , ΔU_m are intensive properties.
- In cyclic process change in all state function will be equal to zero.
 $\Delta E = 0$; $\Delta H = 0$ $\Delta P = 0$, $\Delta T = 0$ etc.
- For ideal gases $\Delta E = 0$ (For isothermal process)
- All natural process are irreversible in nature.
- Both q and w are (+) **to** system.
- Both q and w are (–) **by** the system.

Illustrations

Illustration 1. Find the work done in each case :

- When one mol of ideal gas in 10 litre container at 1 atm is allowed to enter a **vacuum bulb** of capacity 100 litre.
- When 1 mol of gas expands from 1 litre to 5 litre against constant atmospheric pressure.

Solution

(a) $W = -P\Delta V$ but since gas enters the vacuum bulb and pressure in vacuum is zero. This type of expansion is called **free expansion** and work done is zero.

Note :- Work done in free expansion is always zero.

(b) $W = -P\Delta V = -1(5 - 1) = -4 \text{ L-atm}$.

Illustration 2. A 5 litre cylinder contained 10 mol of oxygen gas at 27°C. Due to sudden leakage through the hole, all the gas escaped into the atmosphere and the cylinder got empty. If the atmosphere pressure is 1.0 atm. Calculate the work done by the gas ? ($R = 0.083 \text{ lit atm mol}^{-1} \text{ K}^{-1}$)

Solution

$$V_{\text{initial}} = 5 \text{ L}$$

$$T = 27^\circ\text{C} = 27 + 273 = 300 \text{ K}$$

$$V_{\text{final}} = \frac{nRT}{P} = \frac{10 \times 0.083 \times 300}{1} = 249 \text{ L}$$

$$\Delta V = V_{\text{final}} - V_{\text{initial}} = 249 - 5 = 244 \text{ L}$$

$$W_{\text{exp}} = -P\Delta V = -1 \times 244 \text{ L-atm.} = -244 \text{ L-atm.}$$

BEGINNER'S BOX-1

- Which of the following open system
(1) Animals and plants (2) A fridge (3) A solar cooker (4) None of these
- One mole of gas occupying 3 litre volume is expanded against a constant external pressure of one atm to a volume of 15 litre. The work done by the system is :-
(1) $-1.215 \times 10^3 \text{ J}$ (2) $+12.15 \times 10^3 \text{ J}$ (3) $+121.5 \times 10^3 \text{ J}$ (4) $+1.215 \times 10^3 \text{ J}$
- The work done during the expansion of a gas from a volume of 4 dm³ to 6 dm³ against a constant external pressure of 3 atm is :-
(1) -608 J (2) +304 J (3) -304 J (4) -6 J
- The work done during the expansion of a gas from a volume of 14 dm³ to 16 dm³ against a constant external pressure of 2 atm is :-
(1) -405.2 J (2) +304 J (3) -304 J (4) -6 J

5.6 INTERNAL ENERGY (E/U) :

Internal energy of a system is defined as the sum of different energies associated with its atoms and molecules like Potential energy, Kinetic energy (due to translatory, rotatory and vibratory motion), electronic energy, nuclear energy etc.

$$E = E_{\text{PE}} + E_{\text{T}} + E_{\text{R}} + E_{\text{V}} + E_{\text{e}} + E_{\text{N}}$$

- Internal energy is an **extensive property**.
- We can **never find out the absolute value** of internal energy (E) of system.

We can only calculate the change in internal energy of the system (ΔE) by using an instrument which is called as Bomb calorimeter. In Bomb calorimeter reactions are carried out at constant volume.

- Internal energy is a **state function**.

$$\Delta E = E_f - E_i$$

$$(\text{In reaction}) \quad \Delta E = E_p - E_R$$

$$\Delta E \text{ is + ve if } E_f > E_i$$

$$\Delta E \text{ is - ve if } E_f < E_i$$

5.7 LAWS OF THERMODYNAMICS :

GENERAL POINT :

Laws of thermodynamics are based on human experiences there is no formal proof for them.

FIRST LAW OF THERMODYNAMICS (FLOT) :

- This law is based on law of conservation of energy and was given by **Robert Mayer and Helmholtz**.
- Energy can neither be created nor destroyed but can be transformed from one form to another.
- The total energy of the universe is always constant. i.e. total energy of an isolated system is always conserved.
- The mathematical form of first law of thermodynamics can be expressed as follows

$$\boxed{\Delta E = q + w} : \text{where } q, w \text{ are path function and } E \text{ is state function.}$$

Here q is the energy given to the system and w is the work done on the system ; ΔE is change in internal energy.

Note :- Put the value of q and w with proper sign.

CONCLUSIONS FROM THE FIRST LAW OF THERMODYNAMICS :

(a) During isothermal process of an ideal gas :

During an isothermal process the temperature of the system remains constant and hence

$$\Delta E = 0$$

Therefore FLOT : $\Delta E = q + w$

$$\therefore \boxed{+q = -w} \text{ or } -q = +w$$

In isothermal process –

(I) Heat absorbed by the system is equal to work done by the system.

OR

(II) Heat evolved by the system is equal to work done on the system.

(b) During isochoric process :

At constant volume $V_1 = V_2$ i.e. $\Delta V = 0$

Hence $W = -P\Delta V = 0$ No work done at constant volume therefore,

$$\text{FLOT: } \Delta E = q + w \quad \boxed{\Delta E = q_v}$$

(i) In isochoric process

- At constant volume, heat absorbed by the system is equal to increase in internal energy of the system.

OR

- At constant volume, heat evolved by the system is equal to decrease in internal energy of the system.

(ii) Heat at constant volume (q_v) = ΔE

(iii) **In isochoric process heat is independent of path.**

(c) During adiabatic process :

During adiabatic process the system acts as an isolated system and hence $q = 0$ in such cases. Therefore

$$\text{FLOT : } \Delta E = q + w \quad \because q = 0 \quad \boxed{\Delta E = w}$$

- (i) Work done on the system is equal to increase in internal energy of the system i.e., when a gas is compressed adiabatically its internal energy increases.

OR

Work done by the system is equal to decrease in internal energy of the system, i.e., when a gas is expanded adiabatically its internal energy decreases.

- (ii) **In adiabatic process work is independent of path.**

(d) During Cyclic Process :

$$\Delta E = 0 \quad \text{FLOT ; } \Delta E = q + w \quad \boxed{+q = -w}$$

In cyclic process.

- (I) Work done by the system is equal to heat absorbed by the system.

OR

- (II) Work done on the system is equal to heat evolved by the system.

(e) During Isobaric process :

$P = \text{constant}$

$$\text{FLOT : } \Delta E = q + W$$

$$\Delta E = q + (-P\Delta V)$$

$$q_p = \Delta E + P\Delta V$$

$$q_p = (E_2 - E_1) + P(V_2 - V_1)$$

$$q_p = (E_2 + PV_2) - (E_1 + PV_1)$$

$$\therefore \text{Enthalpy } H = E + PV$$

$$\therefore q_p = H_2 - H_1 \quad \boxed{q_p = \Delta H}$$

(i) In isobaric process :

- At constant pressure, heat absorbed by the system is equal to increase in enthalpy of the system.

OR

- At constant pressure, heat evolved by the system is equal to decrease in enthalpy of the system.

- (ii) Heat at constant pressure (q_p) = ΔH

- (iii) **In isobaric process heat is independent of path.**

Illustrations

Illustration 3. 1g of water changes from liquid to vapour phase at constant pressure of 1 atmosphere, the volume increases from 1 mL to 1671 mL. The heat of vaporisation at this pressure is 540 Cal/g. Find the increase in internal energy of water. (1 L atm = 101 J)

Solution Work done $w = -P\Delta V = -P(V_2 - V_1)$

$$= -1(1671 - 1) \times \frac{1}{1000} = \frac{-1670}{1000} \text{ L-atm}$$

$$= \frac{-1670}{1000} \times 101 \text{ J} = -168.67 \text{ J}$$

given that $q = 540 \text{ Cal} = 540 \times 4.2 \text{ J} = 2268 \text{ J}$

$\therefore \Delta E = q + w = 2268 - 168.67 = 2099.33 \text{ J}$

Illustration 4. A gas occupies 2 L at STP. It is provided 300 J heat so that its volume becomes 2.5 L at 1 atm. Calculate change in its internal energy.

Solution $w = -P\Delta V = -1 \times (2.5 - 2) = -0.5 \text{ L-atm}$ or $w = -0.5 \times 101.3 = -50.65 \text{ J}$.

$\Delta E = q + w = 300 + (-50.65)$

$\Delta E = 249.35 \text{ J}$

Illustration 5. A sample of gas present in a cylinder fitted with a frictionless piston expands against a constant pressure of 1 atm from a volume of 2L to 12L. During the process, it absorbs 600 J of heat from the surroundings. Calculate the change in internal energy of the system.

Solution During the process,

$q = 600 \text{ J}$, $\Delta V = 12 - 2 = 10 \text{ L}$, $P = 1 \text{ atm}$

$w = -P\Delta V$

$= -1 \times 10 = -10 \text{ L atm}$

Now, $1 \text{ L atm} = 101.3 \text{ J}$

$\therefore w = -10 \times 101.3 = -1013 \text{ J}$

According to first law of thermodynamics,

$\Delta E = q + w = 600 - 1013 = -413 \text{ J}$

Illustration 6. Two moles of an ideal gas at 2 atm and 27°C is compressed isothermally to one half of its volume by a constant external pressure of 4 atm. Calculate q , w & ΔE . ($R = 0.082 \text{ L atm mol}^{-1} \text{ K}^{-1}$)

Solution Work done on the system,

$w = -P_{\text{ext}}\Delta V = -P_{\text{ext}}(V_f - V_i)$

$V_i = \frac{nRT}{P}$

$n = 2 \text{ mol}$ and $R = 0.082 \text{ L atm mol}^{-1} \text{ K}^{-1}$,

$T = 273 + 27 = 300 \text{ K}$, $P = 2 \text{ atm}$

$V_i = \frac{2 \times 0.082 \times 300}{2} = 24.6 \text{ L}$ and $V_f = \frac{V_i}{2} = \frac{24.6}{2} = 12.3 \text{ L}$

$$w = -4 \text{ atm} \times (12.3 - 24.6) \text{ L} = 49.2 \text{ L atm} = 49.2 \times 101.3 \text{ J} = 4984 \text{ J}.$$

Since, it is isothermal compression, $\Delta E = 0$

Now, $\Delta E = q + w$

$$0 = q + 4984 \text{ J or } q = -4984 \text{ J}$$

Illustration 7 A system is provided with 100 J of heat. Work done on the system is 20 J. What is the change in internal energy.

Solution $q = 100 \text{ J}$ and $w = +20 \text{ J}$
 $\Delta E = q + w = 100 + 20 \text{ J} = 120 \text{ J}$

Illustration 8. An insulated container is divided into two equal portions. One portion contains an ideal gas at pressure P and temperature T , while the other portion is a perfect Vacuum. If a hole is opened between the two portions, Calculate the –

(i) Change in internal energy of the gas (ii) Change in temperature of the gas

Solution For insulated system, $q = 0$

The gas is allowed to expand against vacuum, the process of free expansion and thus

$$w = -P\Delta V = 0 \text{ (for vacuum } = P_{\text{ext}} = 0)$$

Thus from I law of thermodynamics, $\Delta E = q + w$ or $\Delta E = 0$

i.e. internal energy change is zero or internal energy of gas remains constant during free expansion.

Also $E \propto T$ and thus temperature of the gas will also remains constant.

BEGINNER'S BOX-2

- A system absorb 300 cal of heat with the result of that, the volume of the system becomes double of its initial volume and temperature changes from 273K to 546K. The work done by the system on the surroundings is 200.0 Cal Calculate ΔE :-
 (1) 273 kCal (2) 500 Cal (3) 100 Cal (4) -500 Cal
- One mol of an ideal gas at 300 K is expanded isothermally from an initial volume of 1 litre to 10 litre. The ΔE for the process is :- ($R = 2 \text{ Cal K}^{-1} \text{ mol}^{-1}$)
 (1) 163.7 Cal (2) 1381.1 Cal (3) 9 L-atm (4) Zero
- In an adiabatic process which of the following is true :-
 (1) $q = +w$ (2) $-\Delta E = -w$ (3) $P\Delta V = 0$ (4) $q = \Delta E$
- In an isochoric process, the increase in internal energy is :-
 (1) Equal to the heat absorbed (2) Equal to the heat evolved
 (3) Equal to the work done (4) Equal to zero

5.8 ENTHALPY (H) :

Mathematically it is heat contained in the system measured at constant pressure.

The sum of internal energy and pressure volume (PV) energy is known as enthalpy.

$$H = E + PV$$

- It is impossible to determine absolute value of enthalpy so we determine change in enthalpy (ΔH).

$$\Delta H = H_{\text{final}} - H_{\text{initial}}$$

- Enthalpy is an extensive property because E and V are extensive properties.
- It is a state function because E, P and V are state functions.

$$\therefore H = E + PV$$

$$\therefore \Delta H = \Delta E + \Delta(PV) \dots(i)$$

(when P, V and T are variables)

$$\text{At constant pressure : } \Delta H = \Delta E + P.\Delta V \dots(ii)$$

$$\text{At constant volume : } \Delta H = \Delta E + V.\Delta P \dots(iii)$$

For chemical reactions at constant temperature and pressure $\therefore P.\Delta V = \Delta n_g RT$

$$\text{So from equation (i) } \Delta H = \Delta E + \Delta n_g RT \dots(iv)$$

where $\Delta H = q_p$; at constant P ; $\Delta E = q_v$; at constant V

So equation (iv) can be also written as $q_p = q_v + \Delta n_g RT \dots(v)$

GOLDEN KEY POINTS

- If, $\Delta n_g = 0 \rightarrow \Delta H = \Delta E$ eg. $H_2(g) + I_2(g) \rightarrow 2HI(g)$
- If, $\Delta n_g > 0 \rightarrow \Delta H > \Delta E$ eg. $PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$
- If, $\Delta n_g < 0 \rightarrow \Delta H < \Delta E$ eg. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

Illustrations

Illustration 9. The heat of reaction for $C_{10}H_8(s) + 12O_2(g) \longrightarrow 10CO_2(g) + 4H_2O(l)$ at constant volume is -1228.2 kCal at 25°C . Calculate the heat of reaction at constant pressure and at 25°C .

Solution

$$\Delta n_g = [10] - [12] = -2$$

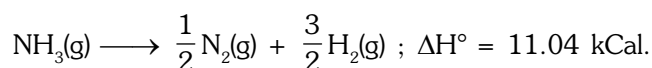
$$\Delta H = \Delta E + \Delta n_g RT$$

$$= -1228.2 \times 10^3 + (-2)(2) \times 298$$

$$= -1229392 \text{ Cal}$$

$$\Delta H = -1229.392 \text{ kCal}$$

Illustration 10. For the reaction at 25°C



Calculate ΔE° of the reaction at the given temperature.

Solution

$$\Delta H^\circ = \Delta E^\circ + \Delta n_g RT$$

$$\Delta n_g = 2 - 1 = 1 \text{ mol}$$

$$\Delta E^\circ = \Delta H^\circ - \Delta n_g RT$$

$$= 11.04 \text{ Kcal} - 1 \text{ mol} \times \frac{2}{1000} \text{ kCal mol}^{-1} \text{K}^{-1} \times 298 \text{K}$$

$$= 11.04 - 0.596 = 10.44 \text{ kCal}$$

Illustration 11 At 27°C the internal energy change of reaction $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$ is 2Cal. What is the enthalpy change of this reaction.

Solution

$$\Delta H = \Delta E + \Delta n_g RT$$

$$\Delta H = \Delta E + 0 \times RT$$

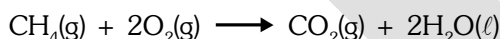
$$\Delta H = \Delta E$$

$$\Delta H = 2\text{Cal}$$

Illustration 12 The heat of combustion of gaseous methane (CH_4) at constant volume is measured in bomb calorimeter at 298K is found to be $-885.4 \text{ kJ mol}^{-1}$. Find the value of enthalpy change at the same temperature.

Solution

Combustion of methane gives $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\ell)$ as



$$\Delta E = -885.4 \text{ kJ mol}^{-1} = -885400 \text{ J mol}^{-1}$$

$$\Delta n_g = 1 - (1 + 2) = -2 \text{ mol}$$

$$T = 298\text{K}, R = 8.314 \text{ J mol}^{-1} \text{K}^{-1}$$

$$\text{Now, } \Delta H = \Delta E + \Delta n_g RT$$

$$= -885400 + (-2 \text{ mol}) \times (8.314 \text{ J mol}^{-1} \text{K}^{-1}) \times (298\text{K})$$

$$= -885400 - 4955$$

$$= -890355 = -890.355 \text{ kJ}$$

Illustration 13. The enthalpy change (ΔH) for the reaction : $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ is -92.38 kJ at 298 K. What is ΔE at 298 K ?

Solution

$$\Delta H \text{ and } \Delta E \text{ are related as } \Delta H = \Delta E + \Delta n_g RT$$

$$\text{for the reaction, } \text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$$

$$\Delta n_g = 2 - (1 + 3) = -2 \text{ mol}, T = 298 \text{ K}$$

$$\Delta H = -92.38 \text{ kJ} = -92380 \text{ J}, R = 8.314 \text{ JK}^{-1} \text{mol}^{-1}$$

$$-92380 = \Delta E + (-2 \text{ mol}) \times (8.314 \text{ J mol}^{-1} \text{K}^{-1}) \times (298 \text{ K})$$

$$-92380 = \Delta E - 4955$$

$$\Delta E = -92380 + 4955$$

$$= -87425 \text{ J} = -87.425 \text{ kJ.}$$

Illustration 14 The enthalpy change for the reaction $\text{CaCO}_3(\text{s}) \longrightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ at 1000K is 176 kJ mol^{-1} . Calculate the change in internal energy.

Solution $\Delta H = \Delta E + \Delta n_g RT$

$$176 = \Delta E + (+1) \times 8.314 \times 10^{-3} \times 1000$$

$$\Delta E = 167.686 \text{ kJ}$$

BEGINNER'S BOX-3

- When a solid melts, there is :-
 (1) No increase in enthalpy (2) Increase in enthalpy
 (3) Decrease in enthalpy (4) Anything can happen
- For the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$ which of the following is valid :-
 (1) $\Delta H = \Delta E$ (2) $\Delta H < \Delta E$
 (3) $\Delta H > \Delta E$ (4) None of these
- Heat exchanged in a chemical reaction at constant pressure is called :-
 (1) Internal energy (2) Enthalpy
 (3) Entropy (4) Free energy
- Latent heat of vaporisation of a liquid at 500 K and 1 atm pressure is $10.0 \text{ kCal mol}^{-1}$. What will be the change in internal energy of 3 mol of liquid at same temperature and pressure
 (1) 13.0 kCal (2) -13.0 kCal
 (3) 27.0 kCal (4) -27.0 kCal
- What is the value of Δn_g if we consider the combustion of 1 mol of liquid ethanol if reactants and products are at 298 K :-
 (1) -1 (2) -2 (3) +1 (4) +2
- If a reaction involves only solids and liquids, which of the following is true
 (1) $\Delta H < \Delta E$ (2) $\Delta H = \Delta E$
 (3) $\Delta H > \Delta E$ (4) $\Delta H = \Delta E + RT\Delta n_g$
- The value of $\Delta H - \Delta E$ for the following reaction at 27°C will be, $2\text{NH}_3(\text{g}) \longrightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$:-
 (1) $8.314 \times 273 \times (-2) \text{ J}$
 (2) $8.314 \times 300 \times (-2) \text{ J}$
 (3) $8.314 \times 27 \times (+2) \text{ J}$
 (4) $8.314 \times 300 \times (+2) \text{ J}$
- At constant temperature for the reaction $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \longrightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\ell)$, $\Delta E - \Delta H$ is :-
 (1) +RT (2) -3RT (3) +3RT (4) -RT

5.9 HEAT CAPACITY / MOLAR HEAT CAPACITY / SPECIFIC HEAT CAPACITY :

- (i) **Heat capacity (C)** : Amount of heat required to raise the temperature of given amount of a substance by 1°C or 1K is called heat capacity.

$$\text{Heat capacity} = \frac{\text{Heat required}}{\text{rise in temp.}}$$

$$C = \frac{dq}{dT}$$

Units :- JK⁻¹, Cal K⁻¹, J °C⁻¹, Cal °C⁻¹

- Heat capacity is extensive property.

- (ii) **Molar heat capacity (C_m)** : Amount of heat required to raise the temperature of 1 mole of substance by 1°C or 1 K is called as molar heat capacity.

$$\text{Molar heat capacity} = \frac{\text{Heat capacity}}{\text{mole of substance}} \Rightarrow C_m = \frac{C}{n}$$

Units :- Jmol⁻¹ K⁻¹, Cal mol⁻¹ K⁻¹, J mol⁻¹ °C⁻¹, Cal mol⁻¹ °C⁻¹

- Molar heat capacity is an intensive property.

- (iii) **Specific heat capacity (c)** : Amount of heat required to raise the temperature of 1 g of substance by 1°C or 1K is called as specific heat capacity.

$$c = \frac{C_m}{\text{molecular weight}}$$

Units :- Jg⁻¹ K⁻¹, Cal g⁻¹ K⁻¹, J g⁻¹ °C⁻¹, Cal g⁻¹ °C⁻¹

- Specific heat capacity is an intensive property.

- If heat is supplied at constant pressure, then $C_p = \left(\frac{dq}{dT} \right)_p = \frac{dH}{dT} \dots (i)$

- If heat is supplied at constant volume, then $C_v = \left(\frac{dq}{dT} \right)_v = \frac{dE}{dT} \dots (ii)$

From equation (i) and (ii) :

From equation (i)	Unit	From equation (ii)
$\Delta H = C_p dT$ [Here C_p is heat capacity at constant P]	JK ⁻¹	$\Delta E = C_v dT$ [Here C_v is heat capacity at constant V]
For n moles $\Delta H = nC_p dT$ [Here C_p is molar heat capacity at constant P]	J mol ⁻¹ K ⁻¹	$\Delta E = nC_v dT$ [Here C_v is molar heat capacity at constant V]
For m gram $\Delta H = mC_p dT$ [Here C_p is gram specific heat (specific heat capacity) at constant P]	J g ⁻¹ K ⁻¹	$\Delta E = mC_v dT$ [Here C_v is gram specific heat (specific heat capacity) at constant V]

Relation between C_p and C_v for 1 mole of an ideal gas :

$$\therefore H = E + PV$$

for ideal gas,

$$PV = nRT$$

$$PV = RT \quad \text{for 1 mole}$$

$$\therefore H = E + RT \text{ differentiate w.r.t. temperature}$$

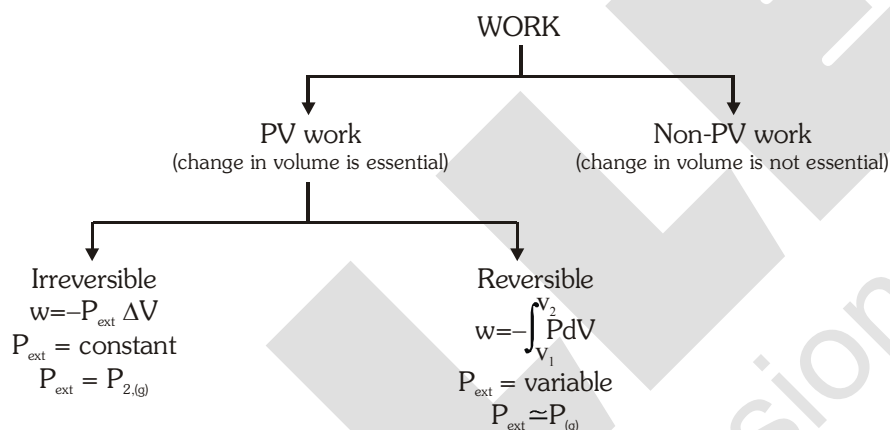
$$\left(\frac{dH}{dT}\right) = \left(\frac{dE}{dT}\right) + R$$

$$C_p = C_v + R$$

(i) $C_P - C_V = R$ Mayer's formula

(ii) $\frac{C_P}{C_V} = \gamma$, γ = Poisson's ratio

5.10 WORK DONE IN DIFFERENT PROCESS :



Here $P_{2(g)}$ = Pressure of gas outside piston.

$P_{(g)}$ = Pressure of gas inside piston.

(a) Work done in reversible isothermal process :

P_{external} = Variable

For reversible process

$$P_{\text{ext}} = P_{\text{gas}} \pm dP$$

$$\therefore w_{\text{rev.}} = - \int_{V_1}^{V_2} P_{\text{ext.}} dV$$

$$W_{\text{rev.}} = - \int_{V_1}^{V_2} (P_{\text{gas}} \pm dP).dV$$

Both dP and dV are very small so, $(dP.dV)$ is negligible.

$$w_{\text{rev.}} = - \int_{V_1}^{V_2} P_{\text{gas}} \cdot dV$$

$$w_{\text{rev.}} = - \int_{V_1}^{V_2} \frac{nRT}{V} . dV$$

$$w_{\text{rev.}} = -nRT \int_{V_1}^{V_2} \frac{dV}{V}$$

$$w_{\text{rev.}} = -nRT \ln \left(\frac{V_2}{V_1} \right)$$

$$\text{or, } \boxed{w_{\text{rev.}} = -2.303nRT \log_{10} \left(\frac{V_2}{V_1} \right)} \quad \dots(i)$$

according to Boyle's law at constant temperature $P \propto \frac{1}{V}$ or $PV = \text{constant}$

$$\therefore P_1 V_1 = P_2 V_2 \Rightarrow \frac{V_2}{V_1} = \frac{P_1}{P_2}$$

$$\boxed{w_{\text{rev.}} = -2.303nRT \log \left(\frac{P_1}{P_2} \right)} \quad \dots(ii)$$

(b) Work done in reversible adiabatic process :

In adiabatic process $q = 0$

FLOT : $\Delta E = q + w$

$$\Rightarrow \boxed{w = \Delta E} \Rightarrow w = nC_V \Delta T$$

$$\boxed{w = nC_V (T_2 - T_1)} \quad \dots(i)$$

$$\therefore C_p - C_v = R$$

$$\frac{C_p}{C_v} - 1 = \frac{R}{C_v}$$

$$\gamma - 1 = \frac{R}{C_v} \left(\because \gamma = \frac{C_p}{C_v} \right) \Rightarrow C_v = \frac{R}{\gamma - 1} \quad \dots(ii)$$

From equation (i) and (ii) $\boxed{w = \frac{nR}{\gamma - 1} (T_2 - T_1)} = \boxed{w = \frac{P_2 V_2 - P_1 V_1}{(\gamma - 1)}}$

State equations of reversible adiabatic processes are :

$$PV^\gamma = \text{constant}$$

$$TV^{\gamma-1} = \text{constant}$$

$$T^\gamma P^{1-\gamma} = \text{constant}$$

GOLDEN KEY POINTS

- Work in closed vessel is zero because volume remains constant.
- Work during free expansion of an ideal gas in vacuum is zero, because P_{ext} is zero.
- Work in a chemical reaction $w = -P\Delta V = -\Delta n_g \cdot RT$

Illustrations

Illustration 15 5 moles of oxygen are heated at constant volume from 10°C to 20°C. What will be the change in the internal energy of gas? The molar heat capacity of oxygen at constant pressure,

$$C_p = 7.03 \frac{\text{Cal}}{\text{molK}} \text{ and } R = 2 \text{ Cal mol}^{-1} \text{ K}^{-1}$$

Solution We know mayer's relation is $C_p - C_v = R$

$$C_v = C_p - R = 7.03 - 2 = 5.03 \text{ Cal mol}^{-1} \text{ K}^{-1}$$

$$\text{As we know } \Delta E = nC_v dT = 5 \times 5.03 \times 10 = 251.5 \text{ Cal}$$

Illustration 16 At 27°C, one mole of an ideal gas compressed isothermally and reversibly from a pressure of 2 atm to 10 atm. Calculate ΔE and q in calorie.

Solution For isothermal process $\Delta E = 0$ and $w = -2.303 nRT \log_{10} \frac{P_1}{P_2}$

$$w = -2.303 \times 1 \times 2 \times 300 \times \log_{10} \frac{2}{10}$$

$$w = + 2.303 \times 600 \times \log 5$$

$$w = + 2.303 \times 600 \times 0.699$$

$$w = + 965.87 \text{ Cal}$$

$$\text{For isothermal process } \therefore w = -q$$

$$\therefore q = -965.87 \text{ Cal}$$

Illustration 17 A gas expands from 3 dm³ to 5 dm³ against a constant pressure of 3 atm. The work done during expansion is used to heat 10 mol of water of temperature 290 K. Calculate final temperature of water (if specific heat of water is 4.184 Jg⁻¹K⁻¹)

Solution Since work is done against constant P and thus, irreversible

$$\Delta V = 5 - 3 = 2 \text{ dm}^3 = 2 \text{ L}, P = 3 \text{ atm}$$

$$w = -P\Delta V = -3 \times 2 \text{ L atm} = -6 \times 101.3 \text{ J} = -607.8 \text{ Joule}$$

Now this work is used up in heating water

$$w = n \times C \times \Delta T$$

$$607.8 = 10 \times (4.184 \times 18) \times \Delta T$$

$$\Delta T = 0.81 = T_2 - T_1$$

$$\therefore \text{Final temperature} = T_1 + \Delta T = 290 + 0.81 = 290.81 \text{ K}$$

Illustration 18 A sample of 3 mol of an ideal gas at 200K and 2 atm is compressed reversibly and adiabatically until the temperature reaches 250K, given that molar heat capacity is $27.5 \text{ J K}^{-1} \text{ mol}^{-1}$ at constant volume, calculate w .

Solution

$$C_v = 27.5 \text{ J K}^{-1} \text{ mol}^{-1}$$

During reversible adiabatic process

$$w = nC_v (T_2 - T_1) = 3 \times 27.5 \times 50 = 4125 \text{ Joule}$$

Illustration 19 10 moles of an ideal gas at 27°C and 10 atm., pressure occupying a volume of 24.6 L undergoes the following changes.

- (i) Isothermal & reversible expansion to 246 L
- (ii) Isothermal and irreversible expansion to 246 L.
- (iii) Isochoric heating to 177°C .

Calculate the work done in each transformation in kJ.

Solution

(i) Work done in isothermal reversible expansion

$$w = -2.303 \times nRT \log \frac{V_2}{V_1} = -2.303 \times 10 \times 8.31 \times 300 \times \log \frac{246}{24.6}$$

$$= -57413.79 \text{ J} = -57.41 \text{ kJ}$$

(ii) Work done in isothermal irreversible expansion

$$w = -P (V_2 - V_1) = -1 (246 - 24.6) = -221.4 \text{ L-atm} = -221.4 \times 101.3 \text{ J} = -22.43 \text{ kJ}$$

(iii) Work done in isochoric change

$$\text{Since } \Delta V = 0 \quad \therefore w = 0$$

Illustration 20 Find the work done when 2 mol of a gas expands isothermally from 5 dm^3 to 40 dm^3 against a constant external pressure of 2 atm at 298K. Also calculate w_{rev} for the change.

Solution

(i) $w = -P\Delta V$

$$w = -2 \times (40 - 5)$$

$$w = -70 \text{ L atm} = -70 \times 101.3 \text{ J}$$

$$w = -7091 \text{ J}$$

(ii) $w = -2.303 nRT \log \frac{V_2}{V_1}$

$$w = -2.303 \times 2 \times 8.314 \times 298 \log \frac{40}{5}$$

$$w = -10.3 \times 10^3 \text{ J}$$

BEGINNER'S BOX-4

- Calculate w for the isothermal reversible expansion of 1mol of an ideal gas from an initial pressure of 1.0 bar to a final pressure of 0.1 bar at a constant temperature of 273 K :-
 (1) -5227.2 J (2) $+5227.2 \text{ J}$ (3) -2257 J (4) $+2257 \text{ J}$
- When 229 J of energy is supplied as heat at constant pressure to 3 mol Ar(g), the temperature of the sample is increased by 2.55K. Calculate the molar heat capacity at constant volume :-
 (1) $30 \text{ kJ K}^{-1} \text{ mol}^{-1}$ (2) $30 \text{ J K}^{-1} \text{ mol}^{-1}$ (3) $21.7 \text{ J K}^{-1} \text{ mol}^{-1}$ (4) $21.7 \text{ kJ K}^{-1} \text{ mol}^{-1}$

5.11 SPONTANEOUS PROCESS AND NON-SPONTANEOUS PROCESS :

(i) Spontaneous process :

- The process which has a natural tendency to occur in a particular direction either of its own or after proper initiation under the given set of conditions.
- All natural processes are ir-reversible and spontaneous processes. The natural processes take place of their own in one direction only.

(ii) Non-spontaneous process :

- The process which does not occur of its own in a particular direction i.e. a process which does not have a natural tendency to occur in a particular direction either of its own or after initiation is called as non-spontaneous process.
- Non-spontaneous process may be made to occur when energy from some external source is supplied continuously throughout the process.

Examples of spontaneous process that need no initiation :

Ex.1 Flow of water from high level to low level.

Flow of heat from hot body to cold body.

Flow of charge from high potential to low potential.

Flow of gas from high pressure to low pressure.

Ex.2 Melting of ice at 25° C

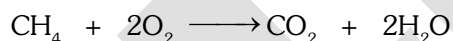
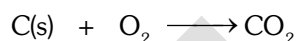
Evapouration of water at 25° C

Dissolution of common salt in water.

Ex.3 Mixing of different non reacting gases

Examples of spontaneous process that need initiation :

Ex.1 Burning of fuel (coal, petrol)



Ex.2 Lighting of candle

Criteria for a process to be spontaneous :

- (i) Tendency to attain minimum energy or maximum stability (Energy Factor).
- (ii) Tendency to attain maximum randomness (Entropy Factor)

5.12 ENTROPY (S) :

- (i) The thermodynamic quantity, which is used to measure **degree of randomness** or **disorderness** of the system is called as entropy.
Entropy (s) $\propto$ Randomness or disorderness
- (ii) More is the disorderness, higher is the entropy.
- (iii) The ratio of heat absorbed by the system in isothermal and reversible manner to the temperature at which heat is absorbed is equals to the change in entropy.

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

Where q_{rev} = heat absorbed by the system in a reversible manner at the temperature T

Unit : J K⁻¹ or Cal K⁻¹

(iv) $\Delta S = S_{\text{final}} - S_{\text{initial}}$

If $S_{\text{final}} > S_{\text{initial}}$: Then $\Delta S = \text{positive}$

If $S_{\text{final}} < S_{\text{initial}}$: Then $\Delta S = \text{negative}$

(v) Entropy is an extensive property and state function.

(vi) Entropy change in a chemical reaction

$$\Delta S = \sum S_{\text{product}} - \sum S_{\text{reactant}}$$

(vii) Entropy change for a process : $\Delta S = \frac{q_{\text{rev}}}{T}$

$$\Delta S = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$$

$$\text{or } \Delta S = nC_p \ln \frac{T_2}{T_1} + nR \ln \frac{P_1}{P_2}$$

Case I : For an ideal gas reversible isothermal process; : $T_2 = T_1$

$$\Delta S = nR \ln \frac{V_2}{V_1} = nR \ln \frac{P_1}{P_2}$$

$$\Delta S = 2.303nR \log \frac{V_2}{V_1} = 2.303nR \log \frac{P_1}{P_2}$$

Case II : For an isochoric process $V_2 = V_1$

$$\Delta S = nC_v \ln \frac{T_2}{T_1} ; \Delta S = 2.303 nC_v \log \frac{T_2}{T_1}$$

(viii) **For reversible adiabatic process :**

Entropy remains constant so process is also known as **isoentropic process**.

Factors affecting entropy of system :

(I) If $\Delta n_g > 0$ then $\Delta S > 0$

If $\Delta n_g < 0$ then $\Delta S < 0$

(II) Physical state : $S_{\text{solid}} < S_{\text{liquid}} < S_{\text{gas}}$

(III) On increasing gaseous moles entropy increases.

(IV) On increasing temperature, S will increase.

Ex. $\text{Fe(s)} \rightarrow \text{Fe(s)}$: $\Delta S = \text{positive}$
 300K 400K

(V) On decreasing pressure, S increases.

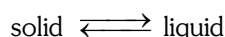
Ex. $\text{N}_2(\text{g}) \rightarrow \text{N}_2(\text{g})$: $\Delta S = \text{positive}$
 5atm 2atm

(VI) Mixture : $\left. \begin{array}{l} \text{Solid} + \text{solid} \\ \text{liquid} + \text{liquid} \\ \text{gas} + \text{gas} \end{array} \right\} S \uparrow$

Entropy change during phase transition :

(I) Entropy of fusion $[(\Delta S)_f]$ mole :

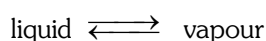
The entropy change, when 1 mol solid changes into liquid at its melting point temperature.



$$(\Delta S)_f = \frac{\Delta H_{\text{fusion}}}{T}$$

(II) Entropy of vapourisation $[(\Delta S)_{\text{vap}}]$ mole :

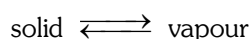
The entropy change, when 1 mol liquid changes into vapour at its boiling point temperature.



$$(\Delta S)_{\text{vap}} = \frac{\Delta H_{\text{vap}}}{T}$$

(III) Entropy of sublimation $[(\Delta S)_{\text{sub}}]$ mole :

The entropy change, when 1 mol solid changes into vapour at a particular temperature.



$$(\Delta S)_{\text{sub}} = \frac{\Delta H_{\text{sub}}}{T}$$

Some famous or extra ordinary examples of entropy change :

- (i) Entropy of graphite > Entropy of diamond.
- (ii) In rusting of iron entropy increases.
- (iii) $\text{NH}_4\text{Cl(s)} + \text{aq} \longrightarrow \text{NH}_4^+(\text{aq}) + \text{Cl}^-(\text{aq})$

In this process NH_4^+ and Cl^- ions are free to move in solution where as they are not free to move in solid NH_4Cl . Hence ΔS is positive for this type of dissolution process.

- (iv) On addition of HCl in the aqueous solution of Ag^+ ions entropy decreases due to precipitation of AgCl.
- (v) **On boiling of egg :** Denaturation of proteins occur. Thus entropy increases.
- (vi) **Stretching of rubber :** During stretching of rubber band its long flexible macromolecules get uncoiled. The uncoiled form has more specific geometry and more ordered arrangement. Thus entropy decreases.

Total entropy change in reversible process :

In reversible process, at every step system and surroundings remain in thermal equilibrium with each other.

Let a system, releases q heat to the surroundings at temperature T .

$$\Delta S_{\text{system}} = \frac{-q}{T}; \quad \Delta S_{\text{surroundings}} = \frac{+q}{T}$$

$$\therefore \Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

$$\Delta S_{\text{total}} = \frac{-q}{T} + \frac{q}{T} \Rightarrow \boxed{\Delta S_{\text{total}} = 0}$$

Total entropy change in irreversible process :

Let a system is at high temperature T_1 and surroundings are at low temperature T_2 .

Let q amount of heat is released by the system.

$$\Delta S_{\text{system}} = \frac{-q}{T_1}, \quad \Delta S_{\text{surroundings}} = \frac{+q}{T_2}$$

$$\therefore \Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} = \frac{-q}{T_1} + \frac{q}{T_2}$$

$$\therefore \boxed{\Delta S_{\text{total}} = +ve} \quad (\because T_1 > T_2)$$

Total entropy change for irreversible spontaneous process is always greater than zero.

Spontaneity of a process in terms of total entropy change :

If, $\Delta S_{\text{total}} = +ve \Rightarrow$ spontaneous process

If, $\Delta S_{\text{total}} = -ve \Rightarrow$ non spontaneous process

If, $\Delta S_{\text{total}} = 0 \Rightarrow$ process is at equilibrium

Illustrations

Illustration 21 The enthalpy change for transition of liquid water to steam is 40.8 kJ mol^{-1} at 373K . Calculate ΔS for the process.

Solution The entropy change for the vapourization of water is given by $\Delta S = \frac{\Delta H_{\text{vap.}}}{T}$

Given $\Delta H_{\text{vap.}} = 40.8 \text{ kJ mol}^{-1} = 40.8 \times 1000 \text{ J mol}^{-1}$ and $T = 373 \text{ K}$

$$\Delta S = \frac{40.8 \times 1000 \text{ J mol}^{-1}}{373\text{K}} = 109.38 \text{ JK}^{-1} \text{ mol}^{-1}$$

Illustration 22 Calculate the change in entropy for the fusion of 1 mol of ice. The melting point of ice is 273K and molar enthalpy of fusion of ice = 6 kJ mol^{-1}

Solution $\Delta S_f = \frac{\Delta H_f}{T} = \frac{6 \times 10^3}{273} = 21.97 \text{ JK}^{-1} \text{ mol}^{-1}$

Illustration 23 The enthalpy of vapourisation of liquid diethyl ether $(\text{C}_2\text{H}_5)_2\text{O}$, is 26.0 kJ mol^{-1} at its boiling point (35.0°C). Calculate ΔS for conversion of :-

(i) Liquid to vapour and (ii) Vapour to liquid at 35°C

Solution (i) $\Delta S_{\text{vap.}} = \frac{\Delta H_{\text{vap.}}}{T} = \frac{26 \times 10^3}{308} = 84.41 \text{ JK}^{-1} \text{ mol}^{-1}$

(ii) $\Delta S_{\text{cond.}} = \frac{\Delta H_{\text{cond.}}}{T} = \frac{-26 \times 10^3}{308} = -84.41 \text{ JK}^{-1} \text{ mol}^{-1}$

Illustration 24 Which of the following processes are accompanied by increase of entropy :

- (i) Dissolution of iodine in a solvent $[I_2(s) \longrightarrow I_2(aq.)]$
- (ii) HCl is added to $AgNO_3$ and a precipitate of AgCl is obtained.
- (iii) A partition is removed to allow two gases to mix.

Solution Increase of entropy : (i) and (iii)

BEGINNER'S BOX-5

- In any natural process, occurring in the universe :-
 - (1) Entropy is conserved
 - (2) Entropy increases
 - (3) Entropy decreases
 - (4) Entropy remains unchanged
- The most random state of H_2O system is :-
 - (1) Ice
 - (2) $H_2O(l)$ at $80^\circ C$; 1 atm
 - (3) Steam
 - (4) $H_2O(l)$ at $25^\circ C$; 1 atm
- Change in entropy is negative for :-
 - (1) $Br_2(l) \longrightarrow Br_2(g)$
 - (2) $C(s) + H_2O(g) \longrightarrow CO(g) + H_2(g)$
 - (3) $M_2(g, 10 \text{ atm}) \longrightarrow M_2(g, 1 \text{ atm})$
 - (4) $Fe \text{ (at } 400K) \longrightarrow Fe \text{ (at } 300K)$
- Entropy change in spontaneous adiabatic process is :-
 - (1) Zero
 - (2) < 0
 - (3) > 0
 - (4) None of these
- 5 mole of an ideal gas expand reversibly from a volume of 8 dm^3 to 80 dm^3 at a temperature of $27^\circ C$. The change in entropy is :-
 - (1) 41.57 JK^{-1}
 - (2) -95.73 JK^{-1}
 - (3) 95.73 JK^{-1}
 - (4) -41.57 JK^{-1}
- The latent heat of vapourisation of water at $100^\circ C$ is 540 Cal g^{-1} . Calculate the entropy increase when one mole of water at $100^\circ C$ is evaporated :-
 - (1) $26 \text{ Cal K}^{-1} \text{ mol}^{-1}$
 - (2) $1.45 \text{ Cal K}^{-1} \text{ mol}^{-1}$
 - (3) $367 \text{ Cal K}^{-1} \text{ mol}^{-1}$
 - (4) $1.82 \text{ Cal K}^{-1} \text{ mol}^{-1}$
- Calculate enthalpy of vapourization per mole of ethanol. Given $\Delta S = 109.8 \text{ J K}^{-1} \text{ mol}^{-1}$ and B.P. of ethanol is $78.5^\circ C$:-
 - (1) Zero
 - (2) $38.594 \text{ kJ mol}^{-1}$
 - (3) 3.85 kJ mol^{-1}
 - (4) None of these

5.13 SECOND LAW OF THERMODYNAMICS (SLOT) :

- (i) It states about the direction of flow of heat
- (ii) All natural processes in universe are ir-reversible process or natural processes are spontaneous process.
- (iii) Due to spontaneous process entropy of universe is increasing continuously i.e. entropy of an isolated system increases.

$$(\Delta S)_T = +ve \quad \text{or} \quad (\Delta S)_T > 0 \quad \text{or} \quad (\Delta S)_{\text{system}} + (\Delta S)_{\text{surr.}} > 0$$

5.14 GIBB'S ENERGY (G or F) :

- Gibb's energy is defined at constant temperature and pressure to predict spontaneity of a process.
- Gibb's energy is a thermodynamic quantity which is used to measure the capacity of system to do useful work or Gibb's energy is that part of the total energy of system which can be converted into useful work.
- The term Gibbs energy was introduced to explain criteria of spontaneity in terms of system.
 - Since ; energy = useful work + randomness energy

$$H = G + TS$$

$$\boxed{G = H - TS} \quad \dots(i)$$

So the function that takes both enthalpy and entropy of system into account is called Gibbs energy.

- Gibbs energy is an extensive property and state function.
- Absolute value of G can't be measured but change can be measured, So we discuss (ΔG)

$$\Delta G = G_{\text{final}} - G_{\text{initial}}$$

From eq. (i) $\boxed{\Delta G = \Delta H - T\Delta S}$

Where ΔG = Change in Gibb's energy

ΔH = Change in enthalpy

ΔS = Change in entropy

Relation between Gibb's energy change and non expansion work or useful work :

$$\text{FLOT : } \Delta E = q + W$$

If, work is done by the system, then, $\Delta E = q - W \quad \dots(i)$

According to Gibb's, system does both expansion and non expansion work.

$$\therefore W = W_{\text{expansion}} + W_{\text{non expansion}}$$

$$W = P\Delta V + W_{\text{non expansion}}$$

Put W in equation (i)

$$\Rightarrow \Delta E = q - (P\Delta V + W_{\text{non expansion}})$$

$$q = \Delta E + P\Delta V + W_{\text{non expansion}}$$

$$q = \Delta H + W_{\text{non expansion}} \quad (\because \Delta H = \Delta E + P\Delta V)$$

$$T\Delta S = \Delta H + W_{\text{non expansion}} \quad (\because \Delta S = q/T)$$

$$W_{\text{non expansion}} = T\Delta S - \Delta H \quad (\because \Delta G = \Delta H - T\Delta S)$$

$$\Rightarrow \boxed{W_{\text{non expansion}} = -\Delta G} \text{ or } \boxed{W_{\text{useful}} = -\Delta G}$$

The decrease in Gibb's energy of system is equal to the non expansion work or useful work.

Relation in between ΔG of system and ΔS_{total} or Gibb's energy change and spontaneity :

$$\therefore \Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

Let system releases heat at constant temperature T and pressure P.

$$\therefore q = q_p = \Delta H$$

$$q_{\text{system}} = -q_{\text{surroundings}}$$

$$\Rightarrow \Delta H_{\text{system}} = -\Delta H_{\text{surroundings}}$$

$$\Delta S_{\text{surr.}} = \frac{\Delta H_{\text{surr.}}}{T}$$

$$\Delta S_{\text{surr.}} = \frac{-\Delta H_{\text{sys.}}}{T}$$

$$\therefore \Delta S_{\text{total}} = \Delta S_{\text{sys.}} + \left(\frac{-\Delta H_{\text{sys.}}}{T} \right)$$

$$T\Delta S_{\text{total}} = T\Delta S_{\text{sys.}} - \Delta H_{\text{sys.}}$$

$$T\Delta S_{\text{total}} = -(\Delta H_{\text{sys.}} - T\Delta S_{\text{sys.}})$$

$$T\Delta S_{\text{total}} = -\Delta G_{\text{sys.}}$$

or $\Delta G_{\text{sys.}} = -T\Delta S_{\text{total}}$

- (i) If, $\Delta S_{\text{total}} = +ve \Rightarrow \Delta G_{\text{system}} = -ve \Rightarrow$ spontaneous process
 (ii) If, $\Delta S_{\text{total}} = -ve \Rightarrow \Delta G_{\text{system}} = +ve \Rightarrow$ non spontaneous process
 (iii) If, $\Delta S_{\text{total}} = 0 \Rightarrow \Delta G_{\text{system}} = 0 \Rightarrow$ process is at equilibrium.

$\Delta_r H^\circ$	$\Delta_r S^\circ$	$\Delta_r G^\circ$	Description
-	+	-	Reaction spontaneous at all temperature
-	-	-	(at low T) Reaction spontaneous at low temperature
-	-	+	(at high T) Reaction nonspontaneous at high temperature
+	+	+	(at low T) Reaction nonspontaneous at low temperature
+	+	-	(at high T) Reaction spontaneous at high temperature
+	-	+	(at all T) Reaction nonspontaneous at all temperatures

Relationship between standard Gibbs's energy change (ΔG°) and Equilibrium constant (K_{eq}) :-

For a reaction $m_1A + m_2B \rightleftharpoons n_1C + n_2D$

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium - $\Delta G = 0$ and $Q = K_{\text{eq}}$

$$0 = \Delta G^\circ + RT \ln K_{\text{eq}}$$

$$\therefore \Delta G^\circ = -RT \ln K_{\text{eq}} \quad \dots(i)$$

$$\text{or } \Delta G^\circ = -2.303RT \log_{10} K_{\text{eq}} \quad \dots(ii)$$

from equation (i)

$$\log K_{\text{eq}} = -\frac{\Delta G^\circ}{RT}$$

$$\therefore K_{\text{eq}} = e^{-\Delta G^\circ/RT} \quad \dots(iii)$$

Illustrations

Illustration 25. For a certain reaction the change in enthalpy and change in entropy are $40.63 \text{ kJ mol}^{-1}$ and 100 JK^{-1} . What is the value of ΔG at 27°C and indicate whether the reaction is possible or not?

Solution We know that :

$$\Delta G = \Delta H - T\Delta S$$

$$T = 27 + 273 = 300\text{K}$$

$$\Delta H = 40.63 \times 10^3 \text{ J mol}^{-1} = 40630 \text{ J mol}^{-1}$$

$$\Delta S = 100 \text{ JK}^{-1}$$

$$\Delta G = 40630 - 300 \times 100 = 40630 - 30000 = + 10630 \text{ J}$$

Positive value of ΔG indicates that the reaction is not possible.

Illustration 26 For a reaction at 25°C enthalpy change (ΔH) and entropy change (ΔS) are $-11.7 \times 10^3 \text{ J mol}^{-1}$ and $-105 \text{ J mol}^{-1} \text{ K}^{-1}$ respectively. Find out whether this reaction is spontaneous or not.

Solution $\Delta G = \Delta H - T\Delta S$

$$= -11700 - 298 \times (-105)$$

$$= +19590 \text{ J}$$

$\Delta G = +ve$, so reaction is non-spontaneous.

Illustration 27 Calculate the equilibrium constant for the reaction given below at 400K .

If $\Delta H^\circ = 77.2 \text{ kJ mol}^{-1}$ and $\Delta S^\circ = 122 \text{ JK}^{-1} \text{ mol}^{-1}$



Solution $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = 77200 - 400 \times 122 = 28400 \text{ J}$

$$\Delta G^\circ = -2.303 RT \log K_c$$

$$\text{or } 28400 = -2.303 \times 8.31 \times 400 \log K_c$$

$$\text{or } K_c = 1.958 \times 10^{-4}$$

Illustration 28 For the reaction, $\text{N}_2(g) + 3\text{H}_2(g) \longrightarrow 2\text{NH}_3(g)$; $\Delta H = -95.4 \text{ kJ}$ and $\Delta S = -198.3 \text{ J K}^{-1}$. Calculate the temperature at which the reaction will proceed in forward direction.

Solution $\Delta G = \Delta H - T\Delta S$

$\therefore$ At equilibrium $\Delta G = 0$

$$\therefore \Delta H = T\Delta S \quad \text{so } T = \frac{\Delta H}{\Delta S} = \frac{-95.4 \times 1000 \text{ J}}{-198.3 \text{ J K}^{-1}} = 481 \text{ K}$$

For this reaction ΔH is $-ve$ and ΔS is $-ve$, so it will be spontaneous at low temperature.

$\therefore$ Below 481K the reaction would be spontaneous.

Illustration 29 Enthalpy and entropy changes of a reaction are $40.63 \text{ kJ mol}^{-1}$ and $108.8 \text{ J K}^{-1} \text{ mol}^{-1}$ respectively. Analyse the feasibility of the reaction at 27°C .

Solution

$$\Delta H = 40.63 \text{ kJ mol}^{-1} = 40630 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta S = 108.8 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$T = 27^\circ\text{C} = 27 + 273 = 300 \text{ K}$$

$$\begin{aligned} \text{Now } \Delta G &= \Delta H - T\Delta S \\ &= 40630 \text{ J mol}^{-1} - (300 \text{ K}) \times (108.8 \text{ J K}^{-1} \text{ mol}^{-1}) \end{aligned}$$

$$\Delta G = 7990 \text{ J mol}^{-1}.$$

Since ΔG is positive, the reaction is not feasible in the forward direction.

Illustration 30 For a certain reaction the change in enthalpy and change in entropy are $40.63 \text{ kJ mol}^{-1}$ and 100 JK^{-1} . Show that the reaction at 27°C is possible or not.

Solution

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = 40.63 - 300 \times 100 \times 10^{-3}$$

$$\Delta G = + 10.630 \text{ kJ}$$

ΔG is positive so reaction is not possible.

Illustration 31 Zinc reacts with dilute hydrochloric acid to give hydrogen at 17°C . The enthalpy of the reaction is $-12.55 \text{ kJ mol}^{-1}$ and entropy change is $5 \text{ JK}^{-1} \text{ mol}^{-1}$ for the reaction. Calculate the free energy change and predict whether the reaction is spontaneous or not.

Solution

$$\text{Given } \Delta H = -12.55 \text{ kJ mol}^{-1} \text{ and } \Delta S = 5 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$T = 17 + 273 = 290 \text{ K}$$

$$\text{Applying } \Delta G = \Delta H - T \Delta S$$

$$= -12.55 \frac{\text{kJ}}{\text{mol}} - 290\text{K} \times \frac{5}{1000} \frac{\text{kJ}}{\text{K mol}}$$

$$= -12.55 - 1.45$$

$$= -14 \frac{\text{kJ}}{\text{mol}}$$

Since ΔG is negative, the reaction will be spontaneous

Illustration 32 For a reaction both ΔH and ΔS are positive under what condition will the reaction occur spontaneously.

Solution

The reaction will occur spontaneously only when $T\Delta S > \Delta H$.

$$\Delta G = \Delta H - T\Delta S = (+) - T(+)$$

For ΔG to be negative, $T\Delta S$ must be $> \Delta H$

Illustration 33 Which of the following are state function ?

- (i) q (ii) Entropy (iii) Specific heat capacity (iv) H (v) w

Solution

Ans. (ii), (iii) and (iv)

BEGINNER'S BOX-6

- If $\Delta G^\circ > 0$ for a reaction then :-
 (1) $K_p > 1$ (2) $K_p < 1$ (3) $K_p = 1$ (4) None
- For an endothermic reaction to be spontaneous :-
 (1) ΔG must be +ve (2) ΔS must be > 0
 (3) $T\Delta S$ must be -ve (4) $T\Delta S$ must be equal to ΔG
- The value of ΔG for the process $\text{H}_2\text{O(s)} \longrightarrow \text{H}_2\text{O(l)}$ at 1 atm and 260 K is :-
 (1) < 0 (2) $= 0$ (3) > 0 (4) Unpredictable
- In a certain chemical reaction $\Delta H = 150 \text{ KJ}$ and $\Delta S = 10 \text{ JK}^{-1}$ at 300 K. The value of ΔG would be :-
 (1) -2850 J (2) Zero
 (3) +2850 J (4) 147 kJ
- The standard Gibb's energy change for a gaseous reaction at 27°C is X kCal. If equilibrium constant for a reaction is 100 and R is $2 \text{ Cal K}^{-1} \text{ mol}^{-1}$. Then X is :-
 (1) -2.7636 (2) +2.7636 (3) + 807 (4) - 807
- The favourable conditions for a spontaneous reaction are :-
 (1) $T\Delta S > \Delta H$, $\Delta H = +ve$, $\Delta S = +ve$ (2) $T\Delta S > \Delta H$, $\Delta H = +ve$, $\Delta S = -ve$
 (3) $T\Delta S = \Delta H$, $\Delta H = -ve$, $\Delta S = -ve$ (4) $T\Delta S = \Delta H$, $\Delta H = +ve$, $\Delta S = +ve$

5.15 THIRD LAW OF THERMODYNAMICS (TLOT)

At zero kelvin (absolute zero temperature), the entropy of pure perfect crystalline solid is taken as zero.

Exceptions :

- (i) NO, N_2O (ii) CO, CO_2 (iii) Mixture of isotopes (iv) Ice

ENERGETICS

5.16 INTRODUCTION :

Thermochemistry is the branch of physical chemistry which deals with the transfer of heat between a chemical system and its surrounding when a change of phase or chemical reaction takes place within the system.

Depending upon the conditions under which the reaction is carried out, the quantity of heat transferred is related to energy or enthalpy change due to changes of states which occur in the system.

In this chapter we will introduced enthalpies of some specific reaction. Like, Enthalpy of formation (ΔH_f), Enthalpy of combustion (ΔH_{comb}), Bond dissociation enthalpy (ΔH_{BDE}) & Enthalpy of Neutralisation ($\Delta H_{\text{neutralization}}$)

THERMOCHEMICAL REACTION :

The balanced chemical reaction which give information about the physical states of reactants & products and heat change is called as thermo chemical reaction.



Thermo chemical reaction are of 2 types.

(i) Endothermic reaction :

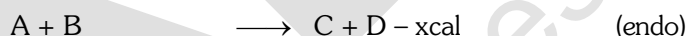
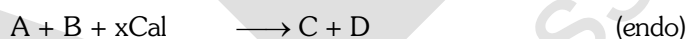
The reaction which absorbs heat is called as endothermic reaction.

$$\Delta H = +ve$$

$$\Delta H = H_p - H_R = +ve \text{ i.e. } H_p > H_R$$

- (a) Stability of reactant > Stability of product because more heat is required to break the bonds of reactant.
- (b) The product formed in the endothermic reaction is called endothermic compound.
- (c) If more heat is absorbed then the product formed in the reaction will be less stable or the reactant is more stable.

Representation of endothermic reaction :



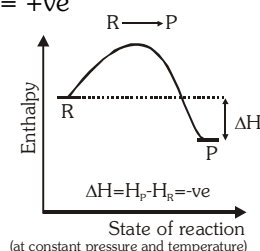
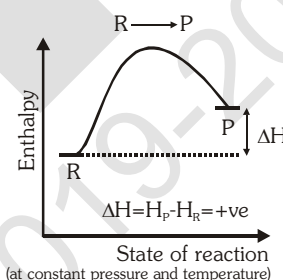
Examples :

- (I) Dissociation reactions (mostly)
- (II) Fusion reactions
- (III) Vaporization reactions
- (IV) Sublimation reactions
- (V) Photosynthesis $6\text{CO}_2 + 6\text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$, $\Delta H = +ve$

(ii) Exothermic reaction :

The reaction which evolves heat is called as exothermic reaction.

$$\Delta H = -ve \quad \Delta H = H_p - H_R = -ve \text{ i.e. } H_p < H_R$$



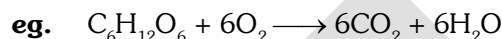
- (a) Stability of reactant < Stability of product because less heat is required to break the bonds of reactant.
 (b) The product formed in the exothermic reaction is called exothermic compound.
 (c) If more heat is released then the product formed in the reaction will be more stable or the reactant is less stable.

Representation of exothermic reaction.**Examples :**

(I) Combustion reactions

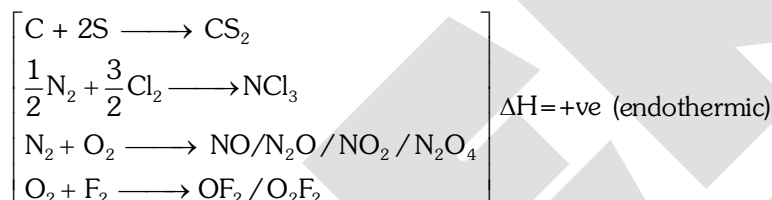
(II) Neutralisation reactions

(III) Respiration reaction



(IV) Formation reaction $\begin{cases} \nearrow \text{endo} \\ \searrow \text{exo (generally)} \end{cases}$

Exceptions of formation reaction :

**GOLDEN KEY POINTS**

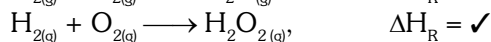
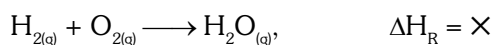
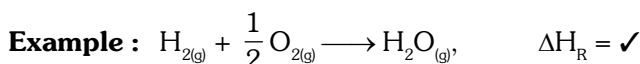
- (i) If conditions are not given then, ΔH is considered to be ΔH° .
 (ii) If thermochemical reaction is multiplied by a coefficient then, ΔH of reaction is also multiplied by that coefficient.
 e.g. $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{H}_2\text{O}(\ell) \quad \Delta H = -285 \text{ kJ mol}^{-1}$
 $2\text{H}_2(\text{g}) + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}(\ell) \quad \Delta H = -2 \times 285 \text{ kJ}$
 (iii) If reaction is reversed then numerical value of ΔH remains same but sign is changed.
 $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{H}_2\text{O}(\ell) \quad \Delta H = -285 \text{ kJ mol}^{-1}$
 $\text{H}_2\text{O}(\ell) \longrightarrow \text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \quad \Delta H = +285 \text{ kJ mol}^{-1}$

BEGINNER'S BOX-7

- An endothermic reaction is one in which :
 (1) heat is converted into electricity
 (2) heat is absorbed
 (3) heat is evolved
 (4) heat is converted into mechanical work
- If heat of reaction $A + 5B \longrightarrow 2C + 3D$, is -50 kJ . What is the heat of the reaction $2A + 10B \longrightarrow 4C + 6D$.
 (1) -50 kJ (2) -25 kJ (3) -100 kJ (4) $+100 \text{ kJ}$
- The process $\text{CH}_3\text{COOH} \longrightarrow \text{CH}_3\text{COO}^- + \text{H}^+$, should be :
 (1) exothermic (2) endothermic
 (3) neither exothermic nor endothermic (4) exothermic or endothermic depending upon temperature
- For the given reaction :
 $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \longrightarrow \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) ; \Delta H = 40 \text{ kJ}$
 The ΔH is specifically called
 (1) Heat of formation of CO (2) Heat of combustion
 (3) Heat of reaction (4) Heat of hydrogenation of C = O bond

5.17 HEAT OF REACTION OR (ENTHALPY OF REACTION) OR (ΔH_R) :

The amount of heat evolved or absorbed when number of moles of the reactant according to the balanced chemical reaction had completely reacted is called as heat of reaction.



Note : Heat of reaction at constant pressure is ΔH and heat of reaction at constant volume is ΔE .

Factors affecting heat of reaction :

(i) Reaction condition :

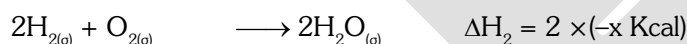
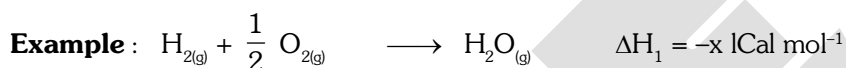
- (a) The chemical reactions are carried out at constant temperature with either pressure or volume constant.

At constant pressure $q_p = \Delta H_{\text{reaction}}$

At constant volume $q_v = \Delta E_{\text{reaction}}$

$$\Delta H = \Delta E + \Delta n_g RT$$

(ii) Quantity of reactant :



If equation is multiplied by coefficient then value of ΔH is also multiplied by that coefficient.

(iii) Physical state of products and reactants :



If the physical state of product is different then the value of ΔH is different.

Note : For H_2O (liq.), ΔH is more negative in comparison to the formation of H_2O (vap.) because when vapours convert into liquid then some heat is released.

(iv) Allotropic form : (Physical nature of reactant)



(v) Temperature :

Effect of temperature on heat of reaction is given by **Kirchoff equation**

(i) at constant pressure : $\frac{\Delta H_{T_2} - \Delta H_{T_1}}{T_2 - T_1} = \Delta C_{Pm}$

$$\Delta C_{Pm} = \Sigma(C_{Pm/P}) - \Sigma(C_{Pm/R})$$

ΔH_{T_1} = Heat of reaction at T_1 temperature

ΔH_{T_2} = Heat of reaction at T_2 temperature

(ii) at constant volume : $\frac{\Delta E_{T_2} - \Delta E_{T_1}}{T_2 - T_1} = \Delta C_{Vm}$

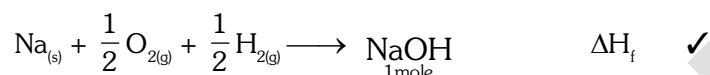
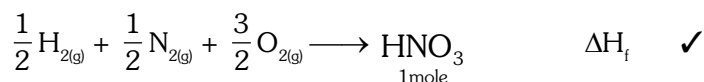
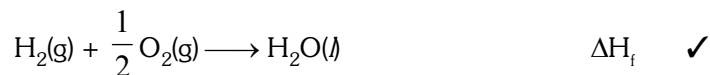
$$\Delta C_{Vm} = \Sigma(C_{Vm/P}) - \Sigma(C_{Vm/R})$$

5.18 TYPES OF HEAT OF REACTION :

(A) Heat of formation, Enthalpy of formation (ΔH_f) or ($\Delta_f H$) :

It is the enthalpy change when one mole of a substance is formed from its elements in their most abundant naturally occurring form or in their standard and stable state form (also called reference states).

The reference state of oxygen, carbon and sulphur are O_2 gas, C_{graphite} and S_{rhombic} , respectively some reactions with standard molar enthalpies of formation are :

APPLICATION OF ΔH_f :Calculation of ΔH of any general reaction.

Let us considered a general reaction $aA + bB \rightarrow cC + dD$

$$\Delta H_{\text{reaction}} = \Sigma \Delta H_{f(\text{products})} - \Sigma \Delta H_{f(\text{reactant})} = [c\Delta H_{f(C)} + d\Delta H_{f(D)}] - [a\Delta H_{f(A)} + b\Delta H_{f(B)}]$$

GOLDEN KEY POINTS

- Standard condition means, $P = 1 \text{ atm}$, $T = 25^\circ\text{C}$ or 298 K
- Standard heat of formation is represent by ΔH_f° .
- If no condition is given then value of ΔH_f is considered as ΔH_f° .
- Standard heat of formation of all the elements in stable standard state is taken to be zero.
- The reference state of commonly used elements are

Elements	Reference state
C	$C_{(\text{graphite})}$
S	$S_{8(\text{Rhombic})}$ (Rhombic sulphur is energy wise more stable as compared to monoclinic sulphur)
P	$P_{4(\text{white})}$
O	$O_{2(g)}$
H	$H_{2(g)}$
Br	$Br_{2(l)}$
Metal	$M_{(s)}$ [except $Hg_{(l)}$]

- The formation reaction may be exothermic or endothermic.

Illustrations

Illustration 34 Since enthalpy of elements in their natural state is taken as zero. The value of ΔH_f of compounds :

(1) is always negative (2) is always positive
(3) may be positive or negative (4) is zero

Solution **Ans. (3)**

Illustration 35 The enthalpy of formation of ammonia at 298K is given as $\Delta H_f^\circ = -46.11 \text{ kJ per mol of } NH_3(g)$. To which of the following equation does this value apply ?



Solution **Ans. (1)**
224

Illustration 36 Which of the following equation represents the standard heat of formation :

- (1) $\text{C(diamond)} + 2\text{H}_2(\text{g}) \longrightarrow \text{CH}_4(\text{g})$ (2) $\text{C(graphite)} + 2\text{H}_2(\text{g}) \longrightarrow \text{CH}_4(\text{g})$
 (3) $\text{C(diamond)} + 4\text{H}(\text{g}) \longrightarrow \text{CH}_4(\text{g})$ (4) $\text{C(graphite)} + 4\text{H}(\text{g}) \longrightarrow \text{CH}_4(\text{g})$

Solution **Ans. (2)**

Illustration 37 Which of the following reaction defines ΔH_f°

- (1) $\text{C}_{(\text{diamond})} + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$ (2) $\frac{1}{2} \text{H}_2(\text{g}) + \frac{1}{2} \text{F}_2(\text{g}) \longrightarrow \text{HF}(\text{g})$
 (3) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$ (4) $\text{CO}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$

Solution **Ans. (2)**

Illustration 38 How much heat will be required at constant pressure to form 1.28 kg of CaC_2 from $\text{CaO}(\text{s})$ & $\text{C}(\text{s})$?

Given : $\Delta_f H^\circ (\text{CaO}, \text{s}) = -152 \text{ kCal mol}^{-1}$

$\Delta_f H^\circ (\text{CaC}_2, \text{s}) = -14 \text{ kCal mol}^{-1}$

$\Delta_f H^\circ (\text{CO}, \text{g}) = -26 \text{ kCal mol}^{-1}$

- (1) + 112 kCal (2) 224 kCal (3) 3840 kCal (4) 2240 kCal

Solution. $\text{CaO}(\text{s}) + 3\text{C}(\text{s}) \longrightarrow \text{CaC}_2(\text{s}) + \text{CO}(\text{g})$
 $\Delta_f H^\circ = (-14 - 26) - (-152) = +112 \text{ kCal mol}^{-1}$

$$\text{Total heat required} = \left(\frac{1280}{64} \right) \times 112 \Rightarrow 2240 \text{ kCal}$$

Illustration 39 The $\Delta_f H^\circ (\text{N}_2\text{O}_5, \text{g})$ in kJ mol^{-1} on the bases of the following data is :

$2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$ $\Delta_r H^\circ = -114 \text{ kJ mol}^{-1}$

$4\text{NO}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{N}_2\text{O}_5(\text{g})$ $\Delta_r H^\circ = -102.6 \text{ kJ mol}^{-1}$

$\Delta_f H^\circ (\text{NO}, \text{g}) = 90.2 \text{ kJ mol}^{-1}$

- (1) 15.1 (2) 30.2 (3) -36.2 (4) none of these

Solution $\frac{1}{2} \text{N}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \longrightarrow \text{NO}(\text{g})$ $\Delta_f H^\circ = 90.2$
 $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}(\text{g})$ $\Delta_r H^\circ = 90.2 \times 2$... (1)
 $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{NO}_2(\text{g})$ $\Delta_r H^\circ = -114$... (2)
 $2\text{NO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \longrightarrow \text{N}_2\text{O}_5(\text{g})$ $\Delta_r H^\circ = \frac{-102.6}{2} = -51.3$... (3)

From Equations (1) + (2) + (3)

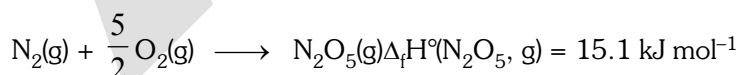
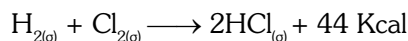


Illustration 40 Calculate ΔH° for $2\text{Al}_{(\text{s})} + \text{Fe}_2\text{O}_3 \longrightarrow 2\text{Fe}_{(\text{s})} + \text{Al}_2\text{O}_3$ given that standard enthalpy of Fe_2O_3 and Al_2O_3 are -196.5 and -399.1 kCal.

Solution $\Delta H_{\text{Reaction}}^\circ = \Sigma \Delta H_{\text{P}}^\circ - \Sigma \Delta H_{\text{R}}^\circ$
 $= [2 \times \Delta H_{\text{Fe}(\text{s})}^\circ + \Delta H_{\text{Al}_2\text{O}_3}^\circ] - [2 \times \Delta H_{\text{Al}(\text{s})}^\circ + \Delta H_{\text{Fe}_2\text{O}_3}^\circ] = 2 \times 0 + (-399.1) - [2 \times 0 + (-196.5)]$
 $\Delta H_{\text{Reaction}}^\circ = -202.6 \text{ kCal}$

Illustration 41 The heat of formation of the compound in the following reaction is :



- (1) $-44 \text{ kCal mol}^{-1}$ (2) $-22 \text{ kCal mol}^{-1}$ (3) $+11 \text{ kCal mol}^{-1}$ (4) $-88 \text{ kCal mol}^{-1}$

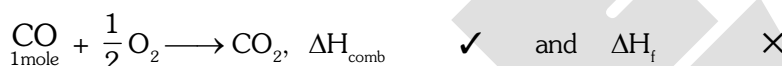
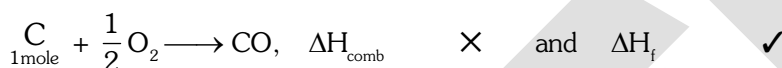
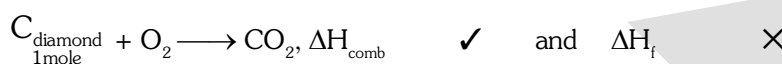
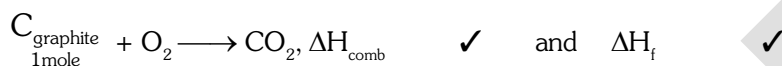
Solution **Ans. (2)**

For the formation of 1 mol of HCl from elements $\Delta H_f^\circ = -\frac{44}{2} = -22 \text{ Kcal}$

(B) Heat of combustion (ΔH_{comb}) :

Amount of heat evolved when 1 mole of substance is completely burnt (or oxidised) in excess of oxygen.

Example :



Note :

- (I) Heat of combustion reaction is always exothermic
- (II) If conditions are not given then ΔH_{comb} considered as $\Delta H_{\text{comb}}^\circ$.
- (III) If in a reaction heats of combustion of reactants and products are given then heat of that reaction can be measured as follows

$$\Delta H = \sum (\Delta H_{\text{comb}})_R - \sum (\Delta H_{\text{comb}})_P$$

APPLICATION OF HEAT OF COMBUSTION :

Calorific value or fuel value (C.V.) :

The amount of heat evolved when 1 g of a substance (food or fuel) is completely burnt (or oxidised)

$$\text{Calorific value} = \frac{\Delta H_{\text{comb}}}{\text{Molecular weight}}$$

Unit :- kJ g^{-1} or kCal g^{-1}

GOLDEN KEY POINTS

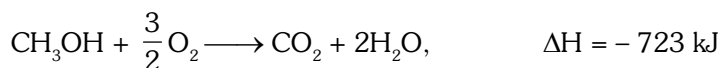
- Heat of combustion reaction is always exothermic
- If conditions are not given then ΔH_{comb} is considered as $\Delta H_{\text{comb}}^\circ$.
- Maximum value of calorific value = Maximum efficiency or best fuel
- H_2 has the highest calorific value (150 kJ/g) but it is not used as domestic or industrial fuel due to some technical problems.

Illustrations

Illustration 42 1 mole of methanol, when burnt in oxygen, gives out -723 kJ mol^{-1} heat. If 1 mole of oxygen is used what will be the amount of heat evolved?

- (1) 723 kJ (2) 964 kJ (3) 482 kJ (4) 241 kJ

Solution **Ans. (3)**



with 1 mole of O_2 , $\Delta H = -\frac{2}{3} \times 723 = -482 \text{ kJ}$

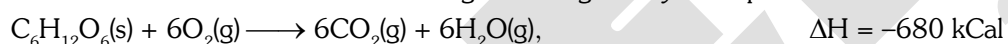
Illustration 43 Combustion of methane :

- (1) is an exothermic reaction (2) is an endothermic reaction
(3) requires a catalyst (4) gives H_2

Solution **Ans. (1)**

Combustion is always exothermic

Illustration 44 The heat evolved in the combustion of glucose is given by the equation



The wt. of $\text{CO}_2(\text{g})$ produced when 170 kCal of heat is evolved in the combustion of glucose is

- (1) 264 g (2) 66 g (3) 11 g (4) 44 g

Solution **Ans. (2)**

Evolution of 680 Kcal is accompanied by $\text{CO}_2 = 6 \times 44 = 264 \text{ g}$

Evolution of 170 Kcal is accompanied by $\text{CO}_2 = \frac{264}{680} \times 170 = 66 \text{ g}$

Illustration 45 Find out the calorific value of Glucose



Solution

$\therefore$ Heat evolved from 1 mol glucose = 2900 kJ

or Heat evolved from 180 gram glucose = 2900 kJ

$\therefore$ Heat evolved from 1 gram glucose = $\frac{2900}{180} = 16.11 \text{ kJ g}$

or another method $\text{C.V.} = \frac{\Delta H_{\text{comb}}}{M_w} = \frac{2900}{180} = 16.11 \text{ kJ g}$

Illustration 46 Enthalpy of combustion of a substance is always :

- (1) > 0 (2) ≥ 0 (3) ≤ 0 (4) < 0

Solution **Ans. (4)**

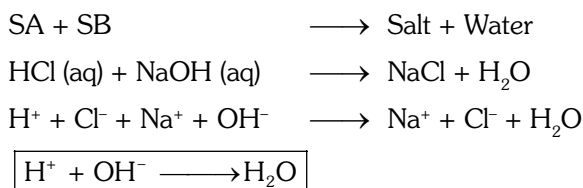
Illustration 47 The heat change for a reaction : $\text{CO}(\text{g}) + \frac{1}{2}\text{O}_2 \longrightarrow \text{CO}_2(\text{g})$ refers to

- (1) enthalpy of formation of carbon dioxide (2) enthalpy of combustion of carbon dioxide
(3) enthalpy of vapourisation (4) enthalpy of combustion of carbon monoxide

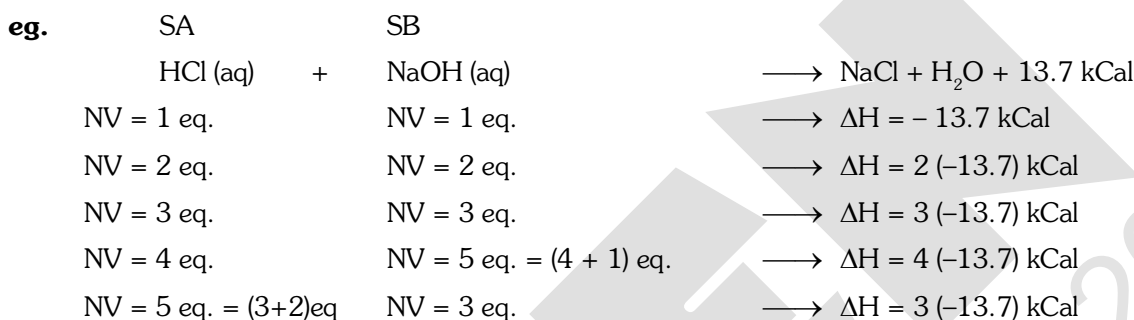
Solution **Ans. (4)**

(C) Heat of neutralisation (ΔH_{neut}) :

The heat evolved when one equivalent of an acid is completely neutralised by one equivalent of a base in dilute solution is called as heat of neutralisation.

**Note :**

- (i) When one equivalent of SA is neutralised by one equivalent of SB then evolve heat remain constant and its value is $-13.7 \text{ kCal/equivalent}$ or $-57.2 \text{ kJ equivalent}^{-1}$.



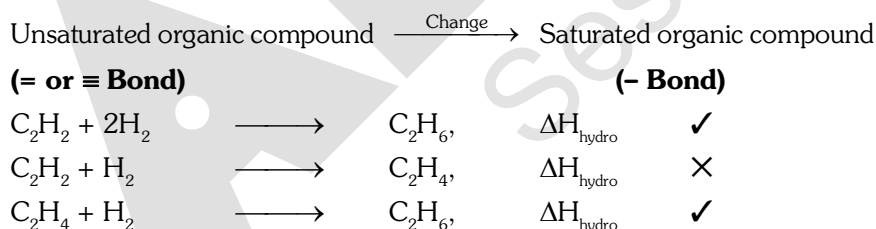
- (ii) If one of the acid or base or both are weak then heat of neutralization is usually less than $-13.7 \text{ KCal eq}^{-1}$ or -57.3 kJ eq^{-1} because some part of the heat released in neutralization is absorbed to dissociate the weak electrolyte completely.

**Exception :**

For a reaction $\text{HF} + \text{NaOH} \rightarrow \text{NaF} + \text{H}_2\text{O}; \Delta H = -16.7 \text{ Kcal}$; this is because of hydration of F^- ion.

(D) Heat of hydrogenation ($\Delta H_{\text{Hydrogenation}}$) :

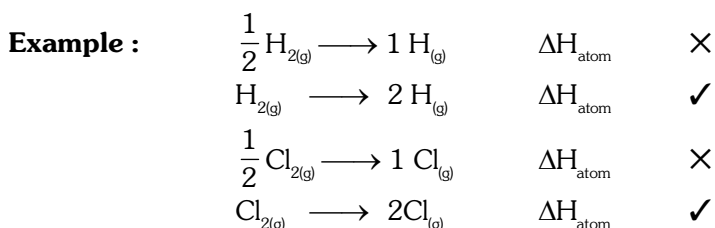
The heat evolved during the complete hydrogenation of one mol unsaturated organic compound into its saturated compound is called as heat of hydrogenation.



Note : Heat of hydrogenation is exothermic process.

(E) Heat of atomization (ΔH_{atom}) :

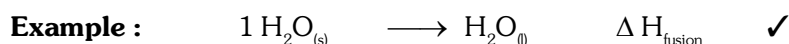
The amount of heat required to dissociate 1 mol substance into gaseous atoms is called as heat of atomization.



Note : It is an endothermic reaction.

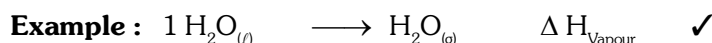
(F) HEAT OF TRANSFORMATION :

- (i) **Heat of fusion (ΔH_{fusion})**: The required amount of heat to convert 1 mole solid into liquid at its melting point is called as heat of fusion.



Note : Heat of fusion is always endothermic reaction i.e. ($\Delta H = +ve$)

- (ii) **Heat of vapourization (ΔH_{vapour})**: The required amount of heat to convert 1 mole liquid into gas at its boiling point is called as heat of vapourization.



Note : Heat of vapourization is always endothermic reaction i.e. ($\Delta H = +ve$)

- (iii) **Heat of sublimation (ΔH_{Sub})**:

The required amount of heat to convert 1 mole solid into gas at a certain temperature is called as heat of sublimation.



Note : Heat of sublimation is always endothermic reaction i.e. ($\Delta H = +ve$)

Illustrations

Illustration 48 Heat of neutralisation of an acid by a base is maximum when :

- (1) Both the acid and base are weak
- (2) Both the acid and base are strong
- (3) The acid is strong and the base is weak
- (4) The acid is weak and the base is strong

Solution

Ans. (2)

Heat of neutralisation is maximum (57.2 kJ eq^{-1} or $13.7 \text{ kCal eq}^{-1}$) when both acid and base are strong.

Illustration 49 The enthalpy change for the process $\text{C}_{(s)} \longrightarrow \text{C}_{(g)}$ corresponds to the enthalpy of

- (1) fusion
- (2) vapourization
- (3) combustion
- (4) sublimation

Solution

Ans. (4)

Solid $\longrightarrow$ gas, is sublimation.

Illustration 50 If $\text{H}^+ + \text{OH}^- \longrightarrow \text{H}_2\text{O} + 13.7 \text{ kCal}$, then heat of complete neutralisation of 1 gm mol of H_2SO_4 with base in excess will be :

- (1) -13.7 kCal
- (2) -27.4 kCal
- (3) -6.85 kCal
- (4) -3.425 kCal

Solution

Ans. (2)

Moles of $\text{H}_2\text{SO}_4 = 1 \text{ mol}$

g eq. of $\text{H}_2\text{SO}_4 = \text{moles} \times \text{V.F.} = 1 \times 2 = 2 \text{ g eq.}$

Heat evolve due to 2 g eq. = $-13.7 \times 2 = -27.4 \text{ kCal}$

Illustration 51 200 cm^3 of $0.1 \text{ M H}_2\text{SO}_4$ is mixed with 150 cm^3 of 0.2 M KOH . Find the value of evolved heat.

Solution

H_2SO_4

KOH

eq. = $NV = (0.1 \times 2) \times 0.2$

$(0.2 \times 1) \times (0.15)$

= 0.04

= 0.03

Heat liberated by 1 eq. = 57.2 kJ

So heat liberated by 0.03 eq. = $57.2 \times 0.03 = 1.7 \text{ kJ}$

BEGINNER'S BOX-8

- Enthalpy of neutralisation of acetic acid with KOH will be numerically :
 (1) = 57.2 kJ (2) > 57.2 kJ (3) < 57.2 kJ (4) unpredictable
- The vapourisation process is always :
 (1) exothermic (2) endothermic
 (3) can be exothermic or endothermic (4) none of these
- One mol of H_2SO_4 is completely neutralised with 2 mole of NaOH in dilute solutions. The amount of heat evolved during the process is :
 (1) 57.2 kJ (2) $\frac{57.2 \text{ kJ}}{2}$ (3) 13.7 kCal (4) 114.4 kJ
- Which of the following data represents the value of heat of neutralisation of strong acid against strong base ?
 (1) – 13.7 kCal (2) – 57.2 kJ (3) – $5.72 \times 10^4 \text{ J}$ (4) All the above
- Fusion of ice is :
 (1) exothermic change (2) endothermic change
 (3) a process that does not involve any heat change (4) unpredictable

(G) Heat of hydration (ΔH_{hydra}) :

Amount of heat evolved when **one mole** of anhydrous salt combines with fixed number of water molecules to convert into its specific hydrated crystal is called as heat of hydration.

Example :

- $1 \text{ CuSO}_4(\text{s}) + 5\text{H}_2\text{O}(\ell) \longrightarrow \text{CuSO}_4 \cdot 5 \text{H}_2\text{O}(\text{s}) \quad \Delta H = -ve$
 anhydrous salt hydrated salt
- $1 \text{ MgSO}_4(\text{s}) + 7\text{H}_2\text{O}(\ell) \longrightarrow \text{MgSO}_4 \cdot 7\text{H}_2\text{O}(\text{s}) \quad \Delta H = -ve$
 anhydrous salt hydrated salt
- $1 \text{ CaCl}_2(\text{s}) + 6\text{H}_2\text{O}(\ell) \longrightarrow \text{CaCl}_2 \cdot 6 \text{H}_2\text{O}(\text{s}) \quad \Delta H = -ve$
 anhydrous salt hydrated salt

Special Note : Heat of hydration is exothermic

(H) Heat of solution ($\Delta H_{\text{sol.}}$) :

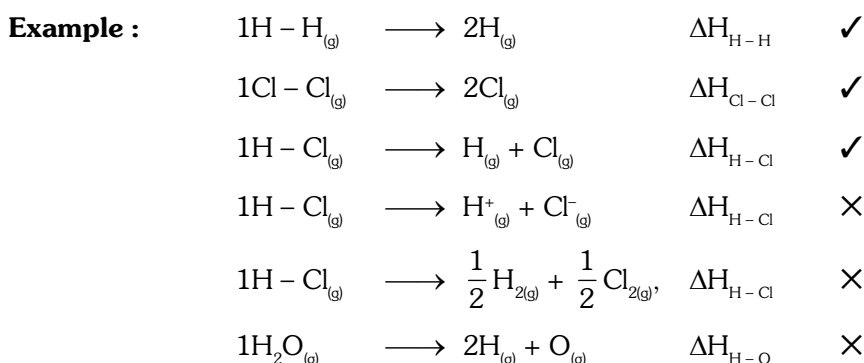
Amount of heat absorbed or evolved when **one mol** of substance is dissolved in such a large volume of solvent that further addition of solvent does not produce any more heat change is called as '**Heat of solution**'.

- Example 1:**
- | | | | | | | |
|--|---|----|-------------------|---|-------------------------------|---|
| $1 \text{ CuSO}_4(\text{s})$ | + | aq | $\longrightarrow$ | $\text{CuSO}_4(\text{aq})$ | $\Delta H_{\text{solution}}$ | ✓ |
| $1 \text{ CuSO}_4(\text{s}) + 5\text{H}_2\text{O}(\ell)$ | | | $\longrightarrow$ | $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$ | $\Delta H_{\text{hydration}}$ | ✓ |
- Example 2:**
- | | | | | | | |
|---|---|----|-------------------|----------------------------|------------------------------|---|
| $\text{MgSO}_4(\text{s}) + 7\text{H}_2\text{O}(\ell)$ | + | aq | $\longrightarrow$ | $\text{MgSO}_4(\text{aq})$ | $\Delta H_{\text{solution}}$ | ✓ |
| $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}(\text{s})$ | + | aq | $\longrightarrow$ | $\text{MgSO}_4(\text{aq})$ | $\Delta H_{\text{solution}}$ | ✓ |

Sp. Note : Heat of solution may be endothermic or exothermic.

(I) **Bond energy / Bond dissociation energy :**

The required amount of energy to dissociate **one mole gaseous bond** into separate **gaseous atoms** is called as **bond dissociation energy**.



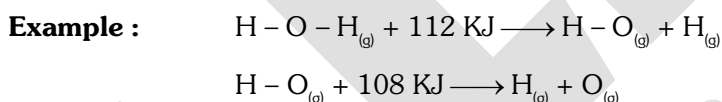
GOLDEN KEY POINTS

- The **bond energy** may be defined as the **average** amount of energy required to dissociate one mole gaseous bond into separate gaseous atoms.
- Bond dissociation process is an endothermic process.
- If bond energy of various bonds present in the reactants and products are given then ΔH of that reaction can be calculate as follows.

$$\Delta H = \Sigma (\text{B.E.})_{\text{R}} - \Sigma (\text{B.E.})_{\text{P}}$$

- In the case of poly atomic molecule** we calculate the average bond energy.

$$(\text{BE})_{\text{av}} = \text{Average bond energy} = \frac{\text{Total energy required with all bonds}}{\text{Number of bond dissociation}}$$



$$(\text{BE})_{\text{av}} = \text{Average bond energy} = \frac{112 + 108}{2} = 110 \text{ kJ mol}^{-1}$$

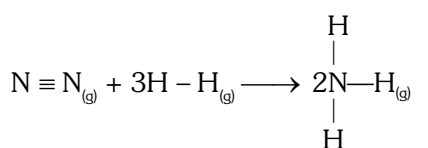
Illustrations

Illustration 52 Given the bond energy of $\text{N} \equiv \text{N}$, $\text{H} - \text{H}$ and $\text{N} - \text{H}$ bonds are 945, 436 and 391 kJ mol^{-1} respectively, the enthalpy of the reaction $\text{N}_{2(\text{g})} + 3\text{H}_{2(\text{g})} \longrightarrow 2\text{NH}_{3(\text{g})}$ is :

(1) - 93 kJ (2) 102 kJ (3) 90 kJ (4) 105 kJ

Solution

Ans. (1)



$$945 + 3 \times 436 \qquad 2 \times (3 \times 391)$$

$$= 2253 \text{ kJ} \qquad = 2346 \text{ kJ}$$

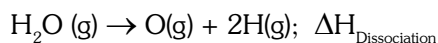
$$\Delta H = \Sigma (\text{B.E.})_{\text{R}} - \Sigma (\text{B.E.})_{\text{P}} = 2253 - 2346 = - 93 \text{ kJ}$$

- Illustration 53** The enthalpy changes at 298 K in successive breaking of O – H bonds of H – O – H are
- $$\text{H}_2\text{O}_{(g)} \longrightarrow \text{H}_{(g)} + \text{OH}_{(g)}, \quad \Delta H = 498 \text{ kJ mol}^{-1}$$
- $$\text{OH}_{(g)} \longrightarrow \text{H}_{(g)} + \text{O}_{(g)}, \quad \Delta H = 428 \text{ kJ mol}^{-1}$$
- The bond enthalpy of the O – H bond is
- (1) 498 kJ mol⁻¹ (2) 463 kJ mol⁻¹ (3) 428 kJ mol⁻¹ (4) 70 kJ mol⁻¹

Solution **Ans. (2)**

$$(\text{B.E.})_{\text{av}} = \frac{498 + 428}{2} = 463 \text{ kJ}$$

- Illustration 54** The required heat for dissociation of 1 mol H₂O into its atoms (H and oxygen) is $\Delta H_{\text{Dis.}}$. Then calculate the bond energy of O – H bond.



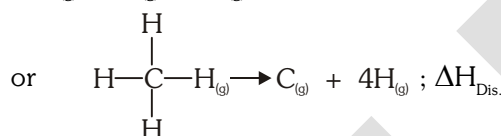
Solution $\text{H} - \text{O} - \text{H}_{(g)} \longrightarrow \text{O}_{(g)} + 2\text{H}_{(g)}; \Delta H_{\text{Dissociation}}$

$\therefore$ required energy for breaking the 2 mol O – H bond = $\Delta H_{\text{Dis.}}$

$$\therefore \text{required energy for 1 mole} = \frac{\Delta H_{\text{Dis.}}}{2}; \Delta H_{\text{O-H}} = \frac{\Delta H_{\text{Dis.}}}{2}$$

- Illustration 55** Calculate the bond energy of C – H Bond in methane.

Solution $\text{CH}_{4(g)} \rightarrow \text{C}_{(g)} + 4\text{H}_{(g)}; \Delta H_{\text{Dis.}}$



$\therefore$ Bond energy of 4 mol C – H = $\Delta H_{\text{Dis.}}$

$$\therefore \text{Bond energy of 1 mol C – H bond} = \frac{\Delta H_{\text{Dis.}}}{4}$$

- Illustration 56** The energy change of reaction $\text{C}_2\text{H}_{6(g)} \longrightarrow 2\text{C}_{(g)} + 6\text{H}_{(g)}$ is X kJ. The bond energy of C – H bond is:

- (1) $\frac{X}{6}$ kJ mol⁻¹ (2) $\frac{X}{3}$ kJ mol⁻¹ (3) X kJ /mol⁻¹ (4) unpredictable from data

Solution **Ans. (4)**

Illustration 57 $\text{CuSO}_4(\ell) + 5\text{H}_2\text{O}(\text{s}) \longrightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}); \Delta H = -x \text{ kJ}$

The value of ΔH represents :

- (1) enthalpy of solution of copper (II) sulphate
 (2) enthalpy of hydration of copper (II) sulphate
 (3) enthalpy of hydrolysis of copper (II) sulphate
 (4) lattice energy of copper (II) sulphate

Solution **Ans. (2)**

Illustration 58 The bond energy of hydrogen is 103 kCal mol⁻¹. This means that :

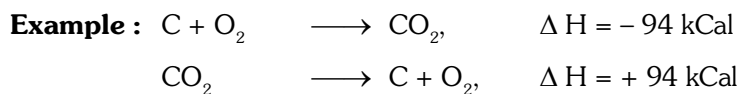
- (1) 103 kCal are required to break 6.023×10^{23} gaseous H₂ molecules into gaseous atoms
 (2) 103 kCal are required to break the bonds in one gram of hydrogen
 (3) 103 kCal are required to break one bond to form two atoms of hydrogen
 (4) 103 kCal are required to break one mole of gaseous hydrogen molecules into ions.

Solution **Ans. (1)**

5.19 LAWS OF THERMOCHEMISTRY :

(I) LAVOISIER AND LAPLACE LAW :

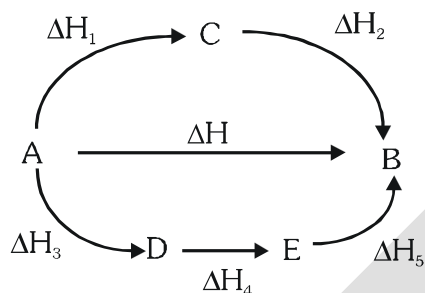
Enthalpy of formation of compound is **numerically equal** to the enthalpy of decomposition of that compound with **opposite sign**.



(II) HESS LAW OF CONSTANT HEAT SUMMATION :

The heat change in a complete chemical reaction always remain same whether reaction completes in one step or more.

Example - 1 :



$$\Delta H = \Delta H_1 + \Delta H_2$$

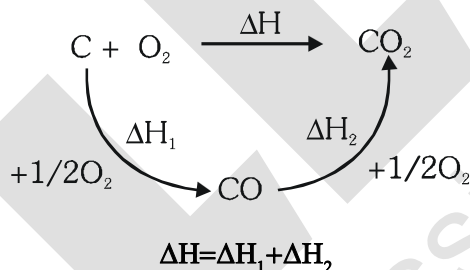
or

$$\Delta H = \Delta H_3 + \Delta H_4 + \Delta H_5$$

or

$$\Delta H = \Delta H_1 + \Delta H_2 = \Delta H_3 + \Delta H_4 + \Delta H_5$$

Example - 2 :



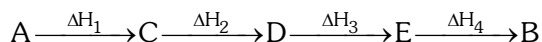
GOLDEN KEY POINTS

- Heat change of a reaction does not depend on the **number of steps** used in the reaction.
- Heat change of a reaction does not depend on **intermediate position**, it depend only on initial and final state.
- Heat change of a chemical reaction does not depend on **time of reaction**.

Illustrations

Illustration 59 Single step reaction $A \rightarrow B$; $\Delta H = ?$

Multi step reaction to produce B from A is given



Solution

According to Hess's law $\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4$

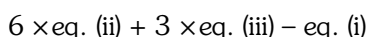
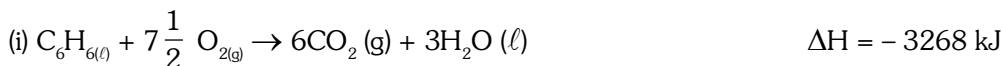
Illustration 60 Calculate the heat of formation of Benzene. The reaction is given below :

$6\text{C(s)} + 3\text{H}_{2(\text{g})} \rightarrow \text{C}_6\text{H}_{6(\text{l})}$ and -3268 , -393.5 and $-285.8 \text{ kJ mol}^{-1}$ are the heats of combustion of benzene, heat of formation of CO_2 and heat of formation of $\text{H}_2\text{O}(\text{l})$ respectively.

Solution

Target reaction $6\text{C}_{(\text{s})} + 3\text{H}_{2(\text{g})} \rightarrow \text{C}_6\text{H}_{6(\text{l})}$

Given

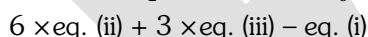
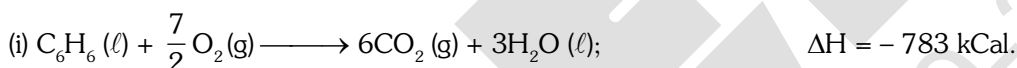


$$\Delta H = 6 \times -393.5 + 3 \times -285.8 - (-3268) = +49.6 \text{ kJ mol}^{-1}$$

Illustration 61 The heats of formation of $\text{CO}_{2(\text{g})}$ and $\text{H}_2\text{O}(\text{l})$ are -97 and $-68 \text{ kcal mol}^{-1}$. The heat of combustion of benzene is $-783 \text{ kcal mol}^{-1}$. What will be the heat of formation of benzene ?

Solution

Given :

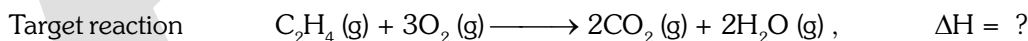
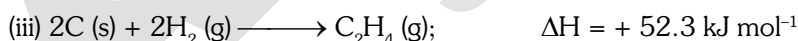
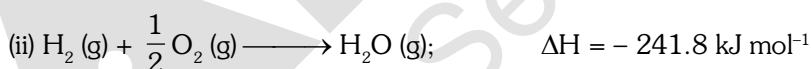


$$\Delta H = 6 \times -97 + 3 \times -68 - (-783) = -3 \text{ kcal mol}^{-1}$$

Illustration 62 Calculate the enthalpy of combustion of ethylene (gas) to form CO_2 (gas) and H_2O (gas) at 298 K and 1 atmospheric pressure. The enthalpies of formation of $\text{CO}_{2(\text{g})}$, $\text{H}_2\text{O}_{(\text{g})}$ and $\text{C}_2\text{H}_{4(\text{g})}$ are -393.7 , -241.8 , $+52.3 \text{ kJ per mol}$ respectively.

Solution

We are given :

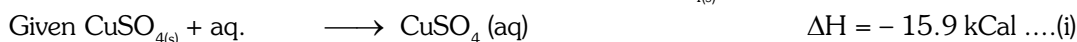


$2 \times \text{Equation (i)} + 2 \times \text{Equation (ii)} - \text{Equation (iii)}$ gives

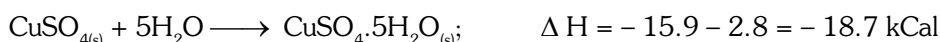
$$\Delta H = 2(-393.7) + 2(-241.8) - (52.3) = -1323.3 \text{ kJ mol}^{-1}$$

Illustration 63 The heat of solution of anhydrous $\text{CuSO}_{4(\text{s})}$ is $-15.9 \text{ kcal mol}^{-1}$ and that of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}_{(\text{s})}$ is $2.8 \text{ kcal mol}^{-1}$. Calculate the heat of hydration of $\text{CuSO}_{4(\text{s})}$.

Solution



Subtracting Eq. (ii) from Eq. (i)



Heat of hydration of $\text{CuSO}_4 = -18.7 \text{ kcal mol}^{-1}$

BEGINNER'S BOX-9

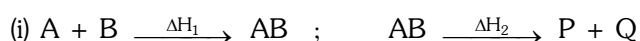
1. A hypothetical reaction, $X \longrightarrow 2Y$ proceeds by the following sequence of steps



The value of ΔH of reaction is :

- (1) $q_1 + q_2 + q_3$ (2) $2q_1 + 2q_2 + 3q_3$ (3) $2(q_1 + q_2 + 2q_3)$ (4) $2(q_1 + q_2 + q_3)$

2. Consider two paths of a certain reaction



(1) $(\Delta H_1 + \Delta H_2) > (\Delta H_3 + \Delta H_4)$

(2) $(\Delta H_1 + \Delta H_2) = (\Delta H_3 + \Delta H_4)$

(3) $(\Delta H_2 + \Delta H_3) = (\Delta H_1 + \Delta H_4)$

(4) $(\Delta H_1 + \Delta H_2) < (\Delta H_3 + \Delta H_4)$

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4						
	Ans.	1	1	1	1						
BEGINNER'S BOX-2	Que.	1	2	3	4						
	Ans.	3	4	2	1						
BEGINNER'S BOX-3	Que.	1	2	3	4	5	6	7	8		
	Ans.	2	2	2	3	1	2	4	3		
BEGINNER'S BOX-4	Que.	1	2								
	Ans.	1	3								
BEGINNER'S BOX-5	Que.	1	2	3	4	5	6	7			
	Ans.	2	3	4	3	3	1	2			
BEGINNER'S BOX-6	Que.	1	2	3	4	5	6				
	Ans.	2	2	3	4	1	1				
BEGINNER'S BOX-7	Que.	1	2	3	4						
	Ans.	2	3	2	3						
BEGINNER'S BOX-8	Que.	1	2	3	4	5					
	Ans.	3	2	4	4	2					
BEGINNER'S BOX-9	Que.	1	2								
	Ans.	3	2								

EXERCISE-I (Conceptual Questions)**Build Up Your Understanding****INTRODUCTION**

- Thermodynamics is concerned with :-
 - (1) Total energy of a system
 - (2) Energy changes in a system
 - (3) Rate of a chemical change
 - (4) Mass changes in nuclear reactions
- A well stoppered thermos flask contains some ice cubes. This is an example of :-
 - (1) Closed system
 - (2) Open system
 - (3) Isolated system
 - (4) Non-thermodynamic system
- Identify the intensive quantity from the following -
 - (1) Enthalpy and temperature
 - (2) Volume and temperature
 - (3) Enthalpy and volume
 - (4) Temperature and refractive index
- Which of the following is an extensive property
 - (1) Mass
 - (2) Enthalpy
 - (3) Energy
 - (4) All of these
- For an adiabatic process which of the following relations is correct
 - (1) $\Delta E = 0$
 - (2) $P\Delta V = 0$
 - (3) $q = 0$
 - (4) $q = +W$
- In which of the following process work is independent of path :
 - (1) Isothermal
 - (2) Isochoric
 - (3) Adiabatic
 - (4) Isobaric
- When a gas is compressed adiabatically and reversibly, the final temperature is-
 - (1) Higher than the initial temperature
 - (2) Lower than the initial temperature
 - (3) The same as initial temperature
 - (4) Dependent upon the rate of compression
- Which one is dependent only on initial and final state ?
 - (1) Heat supplied at constant pressure
 - (2) Heat supplied at constant volume
 - (3) Enthalpy
 - (4) All of the above
- Out of boiling point (I), entropy (II), pH (III) and emf of a cell (IV), intensive properties are :
 - (1) I, III, IV
 - (2) I, II
 - (3) I, II, III
 - (4) All of these

- The work done by a weightless piston in causing an expansion ΔV (at constant temperature), when the opposing pressure P is variable, is given by :
 - (1) $W = - \int PdV$
 - (2) $W = 0$
 - (3) $W = -P\Delta V$
 - (4) None
- The work done by 100 calorie of heat in isothermal expansion of ideal gas is :-
 - (1) 418.4 J
 - (2) 4.184 J
 - (3) 41.84 J
 - (4) None
- Temperature and volume are not :-
 - (1) Extensive properties
 - (2) Intensive properties
 - (3) Intensive and extensive properties respectively
 - (4) Extensive and intensive properties respectively
- $q = -w$ is not true for :-
 - (1) Isothermal process
 - (2) Adiabatic process
 - (3) Cyclic process
 - (4) 1 and 3 both
- The temperature of an ideal gas increase in an -
 - (1) Adiabatic compression
 - (2) Adiabatic expansion
 - (3) Isothermal expansion
 - (4) Isothermal compression
- Enthalpy of 1 mole monoatomic ideal gas is equals to :-
 - (1) $\frac{3}{2}RT$
 - (2) $\frac{5}{2}RT$
 - (3) RT
 - (4) $2RT$
- Which statement is true for reversible process :-
 - (1) It takes place in single step
 - (2) Driving force is much greater than opposing force
 - (3) Work obtain is minimum
 - (4) None

FIRST LAW OF THERMODYNAMICS ($\Delta E = q + W$)

- Both q & w are _ _ _ _ function :-
 - (1) State
 - (2) State, Path
 - (3) Path, State
 - (4) Path
- If work done by the system is 300 joule when 100 cal. heat is supplied to it. The change in internal energy during the process is :-
 - (1) - 200 Joule
 - (2) 400 Joule
 - (3) 720 Joule
 - (4) 120 Joule

19. A system has internal energy equal to E_1 , 450 J of heat is taken out of it and 600 J of work is done on it. The final energy of the system will be -
 (1) $(E_1 + 150)$ (2) $(E_1 + 1050)$
 (3) $(E_1 - 150)$ (4) None of these
20. The work done by a system is 8J when 40J heat is supplied to it. The change in internal energy of the system during the process :
 (1) 32 J (2) 40 J
 (3) 48 J (4) -32 J
21. If a gas absorbs 100 J of heat and expands by 500cm^3 against a constant pressure of $2 \times 10^5 \text{Nm}^{-2}$, the change in internal energy is:-
 (1) - 300 J (2) - 100 J
 (3) + 100 J (4) None of these

ENTHALPY $[\Delta H = \Delta E + P\Delta V/\Delta H = \Delta E + \Delta n_g RT]$

22. Internal energy change during a reversible isothermal expansion of an ideal gas is :-
 (1) Always negative
 (2) Always positive
 (3) Zero
 (4) May be positive or negative
23. Under which of the following conditions is the relation, $\Delta H = \Delta E + P\Delta V$ valid for a system :-
 (1) Constant pressure
 (2) Constant temperature
 (3) Constant temperature and pressure
 (4) Constant temperature, pressure and composition
24. The difference between heats of reaction at constant pressure and constant volume for the reaction $2\text{C}_6\text{H}_6(\text{l}) + 15\text{O}_2(\text{g}) \longrightarrow 12\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$ at 25°C in KJ is
 (1) + 7.43 (2) +3.72
 (3) - 7.43 (4) - 3.72
25. For a gaseous reaction,
 $\text{A}(\text{g}) + 3\text{B}(\text{g}) \longrightarrow 3\text{C}(\text{g}) + 3\text{D}(\text{g})$
 ΔE is 17 kCal at 27°C assuming $R = 2 \text{Cal K}^{-1} \text{mol}^{-1}$, the value of ΔH for the above reaction is:
 (1) 15.8 Kcal (2) 18.2 Kcal
 (3) 20.0 Kcal (4) 16.4 Kcal
26. Which of the following statements is correct for the reaction $\text{CO}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$ at constant temperature and pressure
 (1) $\Delta H = \Delta E$ (2) $\Delta H < \Delta E$
 (3) $\Delta H > \Delta E$ (4) None of the above

27. For the reaction $\text{Ag}_2\text{O}(\text{s}) \longrightarrow 2\text{Ag}(\text{s}) + \frac{1}{2}\text{O}_2(\text{g})$, which one of the following is true :
 (1) $\Delta H = \Delta E$ (2) $\Delta H = \frac{1}{2}\Delta E$
 (3) $\Delta H < \Delta E$ (4) $\Delta H > \Delta E$
28. A mixture of 2 moles of carbon monoxide and one mole of oxygen in a closed vessel is ignited to get carbon dioxide. If ΔH is the enthalpy change and ΔE is the change in internal energy, then :-
 (1) $\Delta H > \Delta E$ (2) $\Delta H < \Delta E$
 (3) $\Delta H = \Delta E$ (4) Not definite
29. For the gaseous reaction involving the complete combustion of isobutane -
 (1) $\Delta H = \Delta E$ (2) $\Delta H > \Delta E$
 (3) $\Delta H = \Delta E = 0$ (4) $\Delta H < \Delta E$
30. For the reversible isothermal expansion of one mole of an ideal gas at 300 K, from a volume of 10dm^3 to 20dm^3 , ΔH is -
 (1) 1.73 kJ (2) -1.73 kJ
 (3) 3.46 kJ (4) Zero
31. For $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ at 977°C , $\Delta H = 174 \text{KJ/mol}$; then ΔE is :-
 (1) 160 kJ (2) 163.6 kJ
 (3) 186.4 kJ (4) 180 kJ
32. Heat of reaction for, $\text{CO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$ at constant V is -67.71Kcal at 17°C . The heat of reaction at constant P at 17°C is :-
 (1) -68.0kCal (2) $+ 68.0 \text{kCal}$
 (3) $- 67.42 \text{kCal}$ (4) None
33. The reaction :-

$$\text{NH}_2\text{CN}(\text{s}) + \frac{3}{2}\text{O}_{2(\text{g})} \rightarrow \text{N}_{2(\text{g})} + \text{CO}_{2(\text{g})} + \text{H}_2\text{O}_{(\text{l})}$$

 was carried out in a bomb calorimeter. The heat released was 743kJ mol^{-1} . The value of $\Delta H_{300\text{K}}$ for this reaction would be :-
 (1) $- 740.5 \text{kJ mol}^{-1}$ (2) $- 741.75 \text{kJ mol}^{-1}$
 (3) $- 743.0 \text{kJ mol}^{-1}$ (4) $- 744.25 \text{kJ mol}^{-1}$
34. The enthalpy of vaporisation of water at 100°C is 40.63kJ mol^{-1} . The value ΔE for this process would be:-
 (1) 37.53kJ mol^{-1} (2) 39.08kJ mol^{-1}
 (3) 42.19kJ mol^{-1} (4) 43.73kJ mol^{-1}
35. For the system $\text{S}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{SO}_2(\text{g})$:-
 (1) $\Delta H = \Delta E$ (2) $\Delta H > \Delta E$
 (3) $\Delta E > \Delta H$ (4) $\Delta H = 0$

36. For the reaction $\text{CO (g)} + \frac{1}{2} \text{O}_2 \text{ (g)} \rightarrow \text{CO}_2 \text{ (g)}$
Which one of the statement is correct at constant T and P ?
(1) $\Delta H = \Delta E$
(2) $\Delta H < \Delta E$
(3) $\Delta H > \Delta E$
(4) ΔH is Independent of physical state of reactants
37. Which is true for the combustion of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) at 25°C :-
(1) $\Delta H > \Delta E$ (2) $\Delta H < \Delta E$
(3) $\Delta H = \Delta E$ (4) None
38. For which change $\Delta H \neq \Delta E$:-
(1) $\text{H}_2\text{(g)} + \text{I}_2\text{(g)} \rightleftharpoons 2\text{HI(g)}$
(2) $\text{HCl (l)} + \text{NaOH (l)} \rightarrow \text{NaCl (s)} + \text{H}_2\text{O (l)}$
(3) $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$
(4) $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightarrow 2\text{NH}_3\text{(g)}$
39. The heat of combustion of ethanol determined in a bomb calorimeter is $-670.48 \text{ kCal mole}^{-1}$ at 27°C . What is ΔH at 27°C for the reaction :-
(1) -335.24 kCal (2) -671.08 kCal
(3) -670.48 kCal (4) $+670.48 \text{ kCal}$
40. The difference in ΔH and ΔE for the combustion of methane at 25°C would be :-
(1) Zero (2) $2 \times 298 \times -2 \text{ Cals.}$
(3) $2 \times 298 \times -3 \text{ Cals.}$ (4) $2 \times 25 \times -3 \text{ Cals.}$
41. For which of the following reactions ΔH is less than ΔE :-
(1) $\text{C}_{12}\text{H}_{22}\text{O}_{11}\text{(s)} + 6\text{O}_2\text{(g)} \rightarrow 6\text{CO}_2\text{(g)} + 6\text{H}_2\text{O(l)}$
(2) $2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{SO}_3\text{(g)}$
(3) $\text{N}_2\text{O}_4\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
(4) $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO(g)}$
42. For a reaction $2\text{X(s)} + 2\text{Y(s)} \rightarrow 2\text{C(l)} + \text{D(g)}$
The q_p at 27°C is $-28 \text{ kCal mol}^{-1}$.
The q_v is ----- kCal mol^{-1} :-
(1) -27.4 (2) $+27.4$
(3) -28.6 (4) 28.6

WORK DONE IN DIFFERENT PROCESS

43. The work done in ergs for a reversible expansion of one mole of an ideal gas from a volume of 10 litres to 20 litres at 25°C is :
(1) $-2.303 \times 8.31 \times 10^7 \times 298 \log 2$
(2) $-2.303 \times 0.0821 \times 298 \log 2$
(3) $-2.303 \times 0.0821 \times 298 \log 0.5$
(4) $-2.303 \times 2 \times 298 \log 2$

44. Two litre of N_2 at 0°C and 5 atm are expanded isothermally against a constant external pressure of 1 atm until the pressure of gas reaches 1 atm. Assuming the gas to be ideal calculate work of expansion ?
(1) -504.2 Joule (2) -405.2 Joule
(3) $+810.4 \text{ Joule}$ (4) -810.4 Joule
45. Two moles of an ideal gas expand spontaneously into vacuum. The work done is :-
(1) Zero (2) 2 J
(3) 4 J (4) 8 J
46. One mole of a gas occupying 3 dm^3 expands against a constant external pressure of 1 atm to a volume of 13 lit. The workdone is :-
(1) -10 atm dm^3 (2) -20 atm dm^3
(3) -39 atm dm^3 (4) -48 atm dm^3

ENTROPY/SECOND LAW OF THERMODYNAMICS

47. For which reaction from the following, ΔS will be maximum ?
(1) $\text{Ca(s)} + \frac{1}{2} \text{O}_2\text{(g)} \rightarrow \text{CaO(s)}$
(2) $\text{CaCO}_3\text{(s)} \rightarrow \text{CaO(s)} + \text{CO}_2\text{(g)}$
(3) $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$
(4) $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO(g)}$
48. An adiabatic reversible process is one in which :-
(1) Temperature of the system does not change
(2) The system is not closed to heat transfer
(3) There is no entropy change
(4) None of these
49. Entropy means
(1) Disorderliness (2) Randomness
(3) Orderliness (4) both 1 & 2
50. ΔS for the reaction;
 $\text{MgCO}_3\text{(s)} \rightarrow \text{MgO(s)} + \text{CO}_2\text{(g)}$ will be :
(1) 0 (2) -ve (3) +ve (4) ∞
51. Change in entropy is negative for
(1) Bromine (l) $\rightarrow$ Bromine (g)
(2) $\text{C(s)} + \text{H}_2\text{O(g)} \rightarrow \text{CO(g)} + \text{H}_2\text{(g)}$
(3) $\text{N}_2\text{(g, 10 atm)} \rightarrow \text{N}_2\text{(g, 1 atm)}$
(4) $\text{Fe(at 400 K)} \rightarrow \text{Fe(at 300 K)}$
52. In which reaction ΔS is positive :-
(1) $\text{H}_2\text{O (l)} \rightarrow \text{H}_2\text{O (s)}$
(2) $3\text{O}_2\text{(g)} \rightarrow 2\text{O}_3\text{(g)}$
(3) $\text{H}_2\text{O (l)} \rightarrow \text{H}_2\text{O (g)}$
(4) $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightarrow 2\text{NH}_3\text{(g)}$

53. When the egg is hard boiled, there is-
 (1) Increase in disorder
 (2) Decrease in disorder
 (3) No change in disorder
 (4) ΔG is negative
54. If S° for H_2 , Cl_2 and HCl are 0.13, 0.22 and 0.19 $kJ K^{-1} mol^{-1}$ respectively. The total change in standard entropy for the reaction $H_2 + Cl_2 \longrightarrow 2HCl$ is :
 (1) 30 $JK^{-1} mol^{-1}$ (2) 40 $JK^{-1} mol^{-1}$
 (3) 60 $JK^{-1} mol^{-1}$ (4) 20 $JK^{-1} mol^{-1}$
55. Which has the least entropy :
 (1) Graphite (2) Diamond
 (3) $N_2(g)$ (4) $N_2O(g)$
56. When two gases are mixed the entropy :-
 (1) Remains constant (2) Decreases
 (3) Increases (4) Becomes zero
57. The enthalpy of vaporization for water is 186.5 $kJ mol^{-1}$, the entropy of its vaporization will be-
 (1) 0.5 $kJ K^{-1} mol^{-1}$ (2) 1.0 $kJ K^{-1} mole^{-1}$
 (3) 1.5 $kJ K^{-1} mole^{-1}$ (4) 2.0 $kJ K^{-1} mole^{-1}$
58. The enthalpy of vaporisation of per mole of ethanol (b.p. = 79.5°C and $\Delta S = 109.8 JK^{-1} mol^{-1}$) is :-
 (1) 27.35 $kJ mol^{-1}$ (2) 32.19 $kJ mol^{-1}$
 (3) 38.70 $kJ mol^{-1}$ (4) 42.37 $kJ mol^{-1}$
59. If 900J/g of heat is exchanged at boiling point of water, then what is increase in entropy?
 (1) 43.4 $JK^{-1}mole^{-1}$
 (2) 87.2 $JK^{-1}mole^{-1}$
 (3) 900 $JK^{-1}mole^{-1}$
 (4) Zero
60. 5 mole of an ideal gas expand reversibly from a volume of 8 dm^3 to 80 dm^3 at a temperature of 27°C. The change in entropy is :-
 (1) 41.57 JK^{-1} (2) - 95.73 JK^{-1}
 (3) 95.73 JK^{-1} (4) - 41.57 JK^{-1}
61. In a spontaneous irreversible process the total entropy of the system and surroundings
 (1) Remains constant (2) Increases
 (3) Decreases (4) Zero
62. The total entropy change for a system & its surroundings increases if the process is :
 (1) Reversible (2) Irreversible
 (3) Exothermic (4) Endothermic

63. Calculate the entropy of $Br_2(g)$ in the reaction $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$, $\Delta S^\circ = 20.1 JK^{-1}$ given, entropy of H_2 and HBr is 130.6 and 198.5 $J mol^{-1} K^{-1}$:-
 (1) 246.3 JK^{-1} (2) 123.15 JK^{-1}
 (3) 24.63 JK^{-1} (4) 20 KJK^{-1}
64. Ammonium chloride when dissolved in water leads to cooling sensation. The dissolution of NH_4Cl at constant temperature is accompanied by :-
 (1) Increase in entropy
 (2) Decrease in entropy
 (3) No change in entropy
 (4) No change in enthalpy
65. In which of the following case entropy decreases-
 (1) Solid changing to liquid
 (2) Expansion of a gas
 (3) Crystals dissolve
 (4) Polymerisation
66. Which of the following state function is not zero at standard state :-
 (1) Enthalpy (2) Entropy
 (3) Free energy (4) Work
67. Entropy of an adiabatic reversible process is:-
 (1) Positive (2) Zero
 (3) Negative (4) Constant

GIBBS FREE ENERGY

68. A gas is allowed to expand under reversible adiabatic conditions then what is zero for such a process:-
 (1) $\Delta G = 0$ (2) $\Delta T = 0$
 (3) $\Delta S = 0$ (4) None of these
69. For a reaction at 25°C enthalpy change (ΔH) and entropy change (ΔS) are $-11.7 \times 10^3 Jmol^{-1}$ and $-105 J mol^{-1} K^{-1}$ respectively. The reaction is :
 (1) Spontaneous
 (2) Non spontaneous
 (3) At equilibrium
 (4) Can't say anything
70. The spontaneous nature of a reaction is impossible if :
 (1) ΔH is +ve, ΔS is also +ve
 (2) ΔH is -ve; ΔS is also -ve
 (3) ΔH is -ve ; ΔS is +ve
 (4) ΔH is +ve; ΔS is -ve
71. If $\Delta H > 0$ and $\Delta S > 0$, the reaction proceeds spontaneously when :-
 (1) $\Delta H > 0$ (2) $\Delta H < T \Delta S$
 (3) $\Delta H = T \Delta S$ (4) None

- 72.** The temperature at which the reaction $\text{Ag}_2\text{O(s)} \longrightarrow 2\text{Ag(s)} + \frac{1}{2}\text{O}_2\text{(g)}$ is at equilibrium is;
Given $\Delta H = 30.5 \text{ kJ mol}^{-1}$
and $\Delta S = 0.066 \text{ kJ K}^{-1} \text{ mol}^{-1}$:
(1) 462.12 K (2) 362.12 K
(3) 262.12 K (4) 562.12 K
- 73.** The enthalpy change for a given reaction at 298 K is $-x \text{ cal mol}^{-1}$. If the reaction occurs spontaneously at 298 K, the entropy change at that temperature
(1) Can be negative but numerically larger than $x/298 \text{ Cal K}^{-1} \text{ mol}^{-1}$
(2) Can be negative but numerically smaller than $x/298 \text{ Cal K}^{-1} \text{ mol}^{-1}$
(3) Cannot be negative
(4) Cannot be positive
- 74.** Which of the following is true for the reaction $\text{H}_2\text{O(l)} \rightleftharpoons \text{H}_2\text{O(g)}$ at 100°C and 1 atmosphere
(1) $\Delta S = 0$ (2) $\Delta H = 0$
(3) $\Delta H = \Delta E$ (4) $\Delta H = T\Delta S$
- 75.** For the reaction $\text{Ag}_2\text{O(s)} \longrightarrow 2\text{Ag(s)} + \frac{1}{2} \text{O}_2\text{(g)}$ the value of $\Delta H = 30.56 \text{ kJ mol}^{-1}$ and $\Delta S = 66 \text{ JK}^{-1} \text{ mol}^{-1}$. The temperature at which the free energy change for the reaction will be zero is :-
(1) 373 K (2) 413 K
(3) 463 K (4) 493 K
- 76.** For hypothetical reversible reaction $\frac{1}{2}\text{A}_2\text{(g)} + \frac{3}{2}\text{B}_2\text{(g)} \longrightarrow \text{AB}_3\text{(g)}$; $\Delta H = -20 \text{ kJ}$ if standard entropies of A_2 , B_2 and AB_3 are 60, 40 and $50 \text{ JK}^{-1} \text{ mole}^{-1}$ respectively. The above reaction will be in equilibrium at :-
(1) 400 K (2) 500 K
(3) 250 K (4) 200 K
- 77.** For the precipitation of AgCl by Ag^+ ions and HCl
(1) $\Delta H = 0$ (2) $\Delta G = 0$
(3) $\Delta G = -ve$ (4) $\Delta H = \Delta G$
- 78.** What is the sign of ΔG for the process of ice melting at 283 K ?
(1) $\Delta G > 0$ (2) $\Delta G = 0$
(3) $\Delta G < 0$ (4) None of these
- 79.** What is the free energy change ΔG , when 1.0 mole of water at 100°C and 1 atm pressure is converted into steam at 100°C and 1 atm pressure :-
(1) 540 Cal (2) -9800 Cal
(3) 9800 Cal (4) 0 Cal
- 80.** A reaction $\text{A} + \text{B} \longrightarrow \text{C} + \text{D} + q$ is found to have a positive entropy change, the reaction will be -
(1) Possible at high temperature
(2) Possible only at low temperature
(3) Not possible at any temperature
(4) Possible at any temperature
- 81.** Equilibrium constant of a reaction is related to :
(1) Standard free energy change ΔG°
(2) Free energy change ΔG
(3) Entropy change
(4) None
- 82.** The Vant Hoff equation is :
(1) $\Delta G^\circ = RT \log_e K_p$ (2) $-\Delta G^\circ = RT \log_e K_p$
(3) $\Delta G^\circ = RT^2 \ln K_p$ (4) None
- 83.** If $\Delta G^\circ > 0$ for a reaction then :
(1) $K_p > 1$
(2) $K_p < 1$
(3) The products predominate in the equilibrium mixture
(4) None
- 84.** If the equilibrium constant for a reaction is 10, then the value of ΔG° will be
($R = 8 \text{ JK}^{-1} \text{ mol}^{-1}$, $T = 300 \text{ K}$)
(1) $+ 5.527 \text{ kJ mol}^{-1}$
(2) $- 5.527 \text{ kJ mol}^{-1}$
(3) $+55.27 \text{ kJ mol}^{-1}$
(4) $- 55.27 \text{ kJ mol}^{-1}$
- 85.** The process of evaporation of a liquid is accompanied by :
(1) Increase in enthalpy
(2) Decrease in free energy
(3) Increase in entropy
(4) All
- 86.** For the process, $\text{CO}_2\text{(s)} \longrightarrow \text{CO}_2\text{(g)}$:
(1) Both ΔH and ΔS are +ve
(2) ΔH is negative and ΔS is +ve
(3) ΔH is +ve and ΔS is -ve
(4) Both ΔH and ΔS are -ve
- 87.** Which of the following provide exceptions to third law of thermodynamics
(1) CO (2) ice
(3) CO_2 (4) All the above
- 88.** The Gibbs free energy change of a reaction at 27°C is -26 kCal . and its entropy change is -60 Cals K . ΔH for the reaction is :-
(1) $- 44 \text{ kCals}$. (2) $- 18 \text{ kCals}$.
(3) 34 kals . (4) $- 24 \text{ kCals}$.

89. Which of the following reaction is expected never to be spontaneous :-

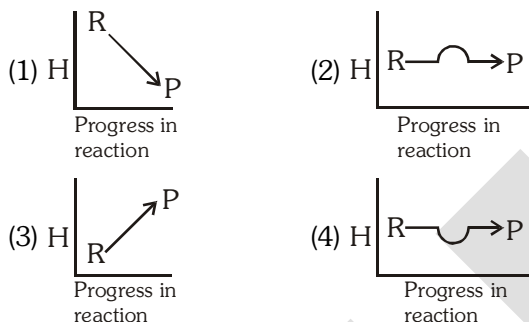
- (1) $2\text{O}_3(\text{g}) \rightarrow 3\text{O}_2(\text{g})$ $\Delta H = -Ve$, $\Delta S = +Ve$
- (2) $\text{Mg}(\text{s}) + \text{H}_2(\text{g}) \rightarrow \text{MgH}_2$ $\Delta H = -Ve$, $\Delta S = -Ve$
- (3) $\text{Br}_2(\text{l}) \rightarrow \text{Br}_2(\text{g})$ $\Delta H = +Ve$, $\Delta S = +Ve$
- (4) $2\text{Ag}(\text{s}) + 3\text{N}_2(\text{g}) \rightarrow 2\text{AgN}_3$ $\Delta H = +Ve$, $\Delta S = -Ve$

THERMOCHEMICAL REACTION

90. The formation of water from $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$ is an exothermic process because :

- (1) The chemical energy of $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$ is more than that of water
- (2) The chemical energy of $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$ is less than that of water
- (3) The temperature of $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$ is higher than that of water
- (4) The temperature of $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$ is lower than that of water

91. Which plot represents for an exothermic reaction:



92. Which one of the following is not applicable for a thermochemical equation :

- (1) It tells about physical state of reactants and products
- (2) It tells whether the reaction is spontaneous
- (3) It tells whether the reaction is exothermic or endothermic
- (4) It tells about the allotropic form (if any) of the reactants

93. The correct thermochemical equation is :

- (1) $\text{C} + \text{O}_2 \longrightarrow \text{CO}_2$; $\Delta H = -94 \text{ kCal}$
- (2) $\text{C} + \text{O}_2 \longrightarrow \text{CO}_2$; $\Delta H = +94.0 \text{ kCal}$
- (3) $\text{C}(\text{s}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$; $\Delta H = -94 \text{ kCal}$
- (4) $\text{C}(\text{s}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$; $\Delta H = +94 \text{ kCal}$

94. The enthalpy changes of formation of the gaseous oxide of nitrogen (N_2O and NO) are positive because of :

- (1) The high bond energy of the nitrogen molecule
- (2) The high electron affinity of oxygen atoms
- (3) The high electron affinity of nitrogen atoms
- (4) The tendency of oxygen to form O^{2-}

95. ΔH for transition of carbon from diamond form to graphite form is -453.5 Cal . This suggests that :

- (1) Graphite is chemically different from diamond
- (2) Graphite is as stable as diamond
- (3) Graphite is more stable than diamond
- (4) Diamond is more stable than graphite

96. Which of the following values of heat of formation indicates that the product is least stable

- (1) -94 kCal
- (2) -231.6 kCal
- (3) $+21.4 \text{ kCal}$
- (4) $+64.8 \text{ kCal}$

97. Heat of formation, ΔH_f° of an explosive compound like NCl_3 is -

- (1) Positive
- (2) Negative
- (3) Zero
- (4) Positive or negative

98. According to the following reaction



- (1) CO is an endothermic compound
- (2) CO is an exothermic compound
- (3) The reaction is endothermic
- (4) None of the above

99. Which of the following represents an exothermic reaction:-

- (1) $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g})$, $\Delta H = 180.5 \text{ kJ}$
- (2) $\text{H}_2\text{O}(\text{g}) + \text{C}(\text{s}) \rightarrow \text{CO}(\text{g}) + \text{H}_2(\text{g})$, $\Delta E = 131.2 \text{ kJ}$
- (3) $2\text{HgO}(\text{s}) + 180.4 \text{ kJ} \rightarrow 2\text{Hg}(\text{l}) + \text{O}_2(\text{g})$
- (4) $2\text{Zn}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{ZnO}(\text{s})$, $\Delta E = -693.8 \text{ kJ}$

100. The heat change during the reaction 24g C and 128g S following the change $\text{C} + \text{S}_2 \rightarrow \text{CS}_2$; $\Delta H = 22 \text{ kCal}$

- (1) 22 kCal
- (2) 11 kCal
- (3) 44 kCal
- (4) 32 kCal

101. Consider the reaction $3\text{O}_2 \rightarrow 2\text{O}_3$; $\Delta H = +Ve$, from the reaction, we can say that :-

- (1) Ozone is more stable than oxygen
- (2) Ozone is less stable than oxygen and ozone decomposes forming oxygen readily
- (3) Oxygen is less stable than ozone and oxygen decomposes forming ozone readily
- (4) None of the above

102. From the reaction $\text{P}(\text{White}) \rightarrow \text{P}(\text{Red})$; $\Delta H = -18.4 \text{ kJ}$, it follows that :-

- (1) Red P is readily formed from white P
- (2) White P is readily formed from red P
- (3) White P can not be converted to red P
- (4) White P can be converted into red P and red P is more stable

FACTORS AFFECTING HEAT OF REACTION

103. In Kirchoff's equation which factor affects the heat of reaction :

- (1) Pressure (2) Temperature
(3) Volume (4) Atomicity

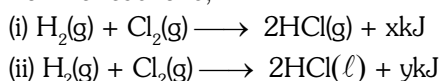
104. For the reaction; $\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) = \text{H}_2\text{O}(\ell)$, $\Delta C_p = 7.63 \text{ Cal deg}$; $\Delta H_{25^\circ\text{C}} = 68.3 \text{ kCal}$, what will be the value (in kCal) of ΔH at 100°C :

- (1) $7.63 \times (373 - 298) - 68.3$
(2) $7.63 \times 10^{-3} (373 - 298) - 68.3$
(3) $7.63 \times 10^{-3} (373 - 298) + 68.3$
(4) $7.63 \times (373 - 298) + 68.3$

105. The enthalpy of a reaction at 273 K is -3.57 kJ . what will be the enthalpy of reaction at 373 K if $\Delta C_p = \text{zero}$:-

- (1) -3.57 (2) Zero
(3) $-3.57 \times \frac{373}{273}$ (4) -375

106. For the reactions,



Which one of the following statement is correct :

- (1) $x > y$ (2) $x < y$
(3) $x = y$ (4) More data required

HEAT OF FORMATION

107. Since the enthalpy of the elements in their standard states is taken to be zero. The heat of formation (ΔH_f) of compounds :

- (1) Is always negative
(2) Is always positive
(3) Is zero
(4) May be positive or negative

108. Reaction $\text{H}_2(\text{s}) + \text{I}_2(\text{g}) \longrightarrow 2\text{HI}$; $\Delta H = 12.40 \text{ kCal}$. According to this, heat of formation of HI will be -

- (1) 12.40 kCal (2) -12.40 kCal
(3) -6.20 kCal (4) 6.20 kCal

109. At 300K the standard enthalpies of formation of $\text{C}_6\text{H}_5\text{COOH}(\text{s})$, $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\ell)$ are -408 , -393 and -286 kJ mol^{-1} respectively. Calculate the heat of combustion of benzoic acid at constant volume :

- (1) $+3201 \text{ kJ}$
(2) $+3199.75 \text{ kJ}$
(3) -3201 kJ
(4) -3199.75 kJ

110. Enthalpy of a compound is equal to its :- (When it is formed from constituent particles)

- (1) Heat of combustion
(2) Heat of formation
(3) Heat of reaction
(4) Heat of solution

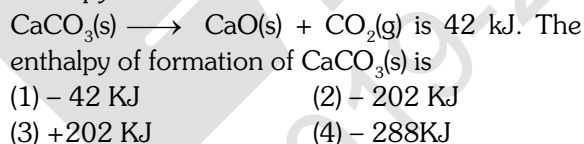
111. Which of the following equations represents standard heat of formation of CH_4 ?

- (1) $\text{C}_{(\text{diamond})} + 2\text{H}_{2(\text{g})} \longrightarrow \text{CH}_{4(\text{g})}$
(2) $\text{C}_{(\text{graphite})} + 2\text{H}_{2(\text{g})} \longrightarrow \text{CH}_{4(\text{g})}$
(3) $\text{C}_{(\text{diamond})} + 4\text{H}_{(\text{g})} \longrightarrow \text{CH}_{4(\text{g})}$
(4) $\text{C}_{(\text{graphite})} + 4\text{H}_{(\text{g})} \longrightarrow \text{CH}_{4(\text{g})}$

112. The enthalpy of formation of ammonia is $-46.0 \text{ kJ mol}^{-1}$. The enthalpy change for the reaction $2\text{NH}_3(\text{g}) \rightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ is :

- (1) 46.0 kJ mol^{-1} (2) 92.0 kJ mol^{-1}
(3) $-23.0 \text{ kJ mol}^{-1}$ (4) $-92.0 \text{ kJ mol}^{-1}$

113. Given enthalpy of formation of $\text{CO}_2(\text{g})$ and $\text{CaO}(\text{s})$ are -94.0 kJ and -152 kJ respectively and the enthalpy of the reaction :



114. Given that standard heat enthalpy of CH_4 , C_2H_4 and C_3H_8 are -17.9 , 12.5 , $-24.8 \text{ kCal mol}^{-1}$. The ΔH for $\text{CH}_4 + \text{C}_2\text{H}_4 \rightarrow \text{C}_3\text{H}_8$ is :

- (1) -55.2 kCal (2) -30.2 kCal
(3) 55.2 kCal (4) -19.4 kCal

115. The standard molar heat of formation of ethane, CO_2 and water(ℓ) are respectively -21.1 , -94.1 and -68.3 kCal . The standard molar heat of combustion of ethane will be

- (1) -372 kCal (2) -162 kCal
(3) -240 kCal (4) -183.5 kCal

116. Two atoms of hydrogen combine to form a molecule of hydrogen gas, the energy of the H_2 molecule is :

- (1) Greater than that of separate atoms
(2) Equal to that of separate atoms
(3) Lower than that of separate atoms
(4) Some times lower and some times higher

117. The ΔH_f° for $\text{CO}_2(\text{g})$, $\text{CO}(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ are -393.5 , -110.5 and $-241.8 \text{ kJ mol}^{-1}$ respectively the standard enthalpy change (in kJ) for the reaction $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g})$ is -

(1) 524.1 (2) 41.2 (3) -262.5 (4) -41.2

118. The enthalpies of combustion of carbon and carbon monoxide are -393.5 kJ and -283 kJ , respectively the enthalpy of formation of carbon monoxide is :
 (1) -676.5 kJ (2) -110.5 kJ
 (3) 110.5 kJ (4) 676.5 kJ

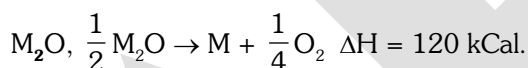
119. The standard heat of formation of $\text{CS}_2(\ell)$ will be; given that the standard heat of combustion of carbon (s), sulphur(s) and $\text{CS}_2(\ell)$ are -393.3 , -293.72 and $-1108.76 \text{ kJ mol}^{-1}$ respectively is
 (1) $-128.02 \text{ kJ mole}^{-1}$
 (2) $+12.802 \text{ kJ mol}^{-1}$
 (3) $+128.02 \text{ kJ mol}^{-1}$
 (4) $-12.802 \text{ kJ mol}^{-1}$

120. The heat of combustion of $\text{CH}_4(\text{g})$, $\text{C}(\text{s})$ and $\text{H}_2(\text{g})$ at 25°C are -212.4 K Cal , -94.0 K Cal and -68.4 K Cal respectively, the heat of formation of CH_4 will be -
 (1) $+54.4 \text{ K Cal}$ (2) -18.4 K Cal
 (3) -375.2 K Cal (4) $+212.8 \text{ K Cal}$

121. Standard enthalpy of formation is zero for .
 (1) $\text{C}_{\text{diamond}}$ (2) $\text{Br}_{(\text{g})}$
 (3) $\text{C}_{\text{graphite}}$ (4) $\text{O}_{3(\text{g})}$

122. The standard heats of formation of $\text{NO}_2(\text{g})$ and $\text{N}_2\text{O}_4(\text{g})$ are 8.0 and $2.0 \text{ kCal mol}^{-1}$ respectively the heat of dimerization of NO_2 in kCal is
 (1) 10.0 (2) -6.0 (3) -12.0 (4) -14.0

123. M is a metal that forms an oxide



When a sample of metal M reacts with one mole of oxygen what will be the ΔH in that case

- (1) 240 kCal. (2) -240 kCal.
 (3) 480 kCal. (4) -480 kCal.

HEAT OF COMBUSTION

124. According to equation,
 $\text{C}_6\text{H}_6(\ell) + 15/2 \text{O}_2(\text{g}) \longrightarrow 6\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\ell);$
 $\Delta H = -3264.4 \text{ kJ mol}^{-1}$ the energy evolved when 7.8 g benzene is burnt in air will be -
 (1) 163.22 kJ (2) 32.64 kJ
 (3) 3.264 kJ (4) 326.4 kJ

125. Heat of combustion of CH_4 , C_2H_6 , C_2H_4 and C_2H_2 gases are -212.8 , -373.0 , -337.0 and -310.5 kCal respectively at the same temperature. The best fuel among these gases is :
 (1) CH_4 (2) C_2H_6 (3) C_2H_4 (4) C_2H_2

126. Given standard enthalpy of formation of CO (-110 kJ mol^{-1}) and CO_2 (-394 kJ mol^{-1}). The heat of combustion when one mole of graphite burns is
 (1) -110 kJ (2) -284 kJ
 (3) -394 kJ (4) -504 kJ

127. The enthalpy of formation for $\text{C}_2\text{H}_4(\text{g})$, $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\ell)$ at 25°C and 1 atm. pressure are 52 , -394 and $-286 \text{ kJ mole}^{-1}$ respectively. The enthalpy of combustion of C_2H_4 will be:-
 (1) $+1412 \text{ kJ mole}^{-1}$ (2) $-1412 \text{ kJ mole}^{-1}$
 (3) $+142.2 \text{ kJ mole}^{-1}$ (4) $-141.2 \text{ kJ mole}^{-1}$

128. The heat of combustion of carbon and carbon monoxide are -394 and -285 kJ mol^{-1} respectively. The heat of formation of CO in kJ mol^{-1} is:-
 (1) $+109$ (2) -109
 (3) $+218$ (4) -218

129. If heat of combustion of ethylene is 1411 kJ when a certain amount of ethylene was burnt 6226 kJ heat was evolved. Then the volume of O_2 (at NTP) that entered into the reaction is :-
 (1) 296.5 ml (2) 296.5 litre
 (3) $6226 \times 22.4 \text{ litre}$ (4) 22.4 litre

130. The heat evolved during the combustion of 112 litre of water gas at STP (mixture of equal volume of H_2 and CO) is : Given
 $\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) = \text{H}_2\text{O}(\text{g}); \Delta H = -241.8 \text{ kJ}$
 $\text{CO}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) = \text{CO}_2(\text{g}); \Delta H = -283 \text{ kJ}$
 (1) 241.8 kJ (2) 283 kJ
 (3) 1312 kJ (4) 1586 kJ

131. A person requires 2870 kCal of energy to lead normal daily life. If heat of combustion of cane sugar is -1349 kCal , then his daily consumption of sugar is :
 (1) 728 g (2) 0.728 g
 (3) 342 g (4) 0.342 g

132. On complete combustion of 2 gm methane 26575 Cal heat is generated. The heat of formation of methane will be (given heat of formation of CO_2 and H_2O are -97000 and -68000 Cal respectively) :
 (1) $+20400 \text{ Cal}$ (2) $+20600 \text{ Cal}$
 (3) -20400 Cal (4) -2000 Cal

133. X gm of ethanal was subjected to combustion in a bomb calorimeter and the heat produced is Y Joules. Then -

(1) $\Delta E_{(\text{combustion})} = -XJ$

(2) $\Delta E_{(\text{combustion})} = -YJ$

(3) $\Delta E_{(\text{combustion})} = -\frac{44Y}{X} \text{ J mol}^{-1}$

(4) $\Delta H_{(\text{combustion})} = -\frac{44Y}{X} \text{ J mol}^{-1}$

134. The following are the heats of reactions -

(i) ΔH_f° of $\text{H}_2\text{O}_{(l)} = -68.3 \text{ kCal mol}^{-1}$

(ii) $\Delta H_{\text{comb}}^\circ$ of $\text{C}_2\text{H}_2 = -337.2 \text{ kCal mol}^{-1}$

(iii) $\Delta H_{\text{comb}}^\circ$ of $\text{C}_2\text{H}_4 = -363.7 \text{ kCal mol}^{-1}$

Then heat change for the reaction $\text{C}_2\text{H}_2 + \text{H}_2 \rightarrow \text{C}_2\text{H}_4$ is -

(1) -716.1 kCal (2) $+ 337.2 \text{ kCal}$

(3) -41.8 kCal (4) -579.5 kCal

135. The heat of combustion of a substance is :-

(1) Always positive

(2) Always negative

(3) Numerically equal to the heat of formation

(4) 1 and 3 both

136. The value of ΔH for the combustion of C(s) is -94.4 kCal . The heat of formation of $\text{CO}_2(\text{g})$ is :-

(1) -49.5 kCal

(2) -94.4 kCal

(3) -188.0 kCal

(4) More data required

137. In the combustion of 0.4 g. of CH_4 , 0.25 kCal. of heat is liberated. The heat of combustion of CH_4 is

(1) -20 kCal

(2) -10 kCal

(3) -2.5 kCal

(4) -5 kCal

138. If $\text{C}_6\text{H}_{12}\text{O}_6(\text{s}) + 9\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$; $\Delta H = -680 \text{ kCal}$ The weight of $\text{CO}_2(\text{g})$ produced when 170 kCal of heat is evolved in the combustion of glucose is:-

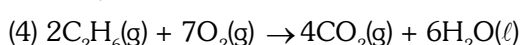
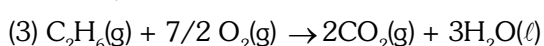
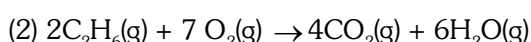
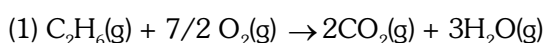
(1) 265 g

(2) 66 g

(3) 11 g

(4) 64 g

139. Which of the following equations corresponds to the enthalpy of combustion at 298 K :-



140. Heat of formation of CO_2 is -94.0 kCal . What would be the quantity of heat liberated, when 3 g of graphite is burnt in excess of oxygen:-

(1) 23.5 kCal

(2) 2.35 kCal

(3) 94.0 kCal

(4) 31.3 kCal

HEAT OF NEUTRALIZATION

141. The amount of heat liberated when one mole of NH_4OH reacts with one mole of HCl is

(1) 13.7 kCal

(2) More than 13.7 kCal

(3) Less than 13.7 kCal

(4) Cannot be predicted

142. If $\text{H}^+ + \text{OH}^- = \text{H}_2\text{O} + 13.7 \text{ kCal}$, then heat of complete neutralisation of one gram mole of H_2SO_4 with strong base will be :

(1) 13.7 Kcal

(2) 27.4 Kcal

(3) 6.85 Kcal

(4) 3.425 KCal

143. Heat of neutralisation of a strong dibasic acid in dilute solution by NaOH is nearly :

(1) $-27.4 \text{ kCal eq}^{-1}$

(2) $-13.7 \text{ kCal eq}^{-1}$

(3) $13.7 \text{ kCal eq}^{-1}$

(4) $-13.7 \text{ kCal mol}^{-1}$

144. The temperature of a 5 ml of strong acid increases by 5°C when 5 ml of a strong base is added to it. If 10 ml of each are mixed temperature should increase by :

(1) 5°C

(2) 10°C

(3) 15°C

(4) Cannot be known

145. The heat of neutralization of HCl by NaOH is $-55.9 \text{ kCal mol}^{-1}$. If the heat of neutralization of HCN by NaOH is $-12.1 \text{ kCal mol}^{-1}$. The energy of dissociation of HCN is

(1) -43.8 kJ

(2) 43.8 kJ

(3) 68 kJ

(4) -68 kJ

146. If water is formed from H^+ ions and OH^- the heat of formation of water is :

(1) -13.7 kCal

(2) 13.7 kCal

(3) -63.4 kCal

(4) More data required

147. The change in the enthalpy of $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ is called :

(1) Heat of neutralisation

(2) Heat of reaction

(3) Heat of hydration

(4) Heat of solution

148. Heat of neutralisation of oxalic acid is $-106.7 \text{ kJ mol}^{-1}$ using NaOH hence ΔH of :



(1) 5.88 kJ mol^{-1}

(2) $-5.88 \text{ kJ mol}^{-1}$

(3) $-13.7 \text{ kCal mol}^{-1}$

(4) 7.5 kJ mol^{-1}

HEAT OF HYDROGENATION

- 149.** The heat of combustion of C_2H_4 , C_2H_6 and H_2 are -1409.5 kJ, -1558.3 kJ and -285.6 kJ. The heat of hydrogenation of ethene is -
 (1) -136.8 kJ (2) -13.68 kJ
 (3) 273.6 kJ (4) 1.368 kJ
- 150.** The enthalpy of combustion of cyclohexane, cyclohexene and H_2 are respectively -3920 , -3800 and -241 kJ mol $^{-1}$. The heat of hydrogenation of cyclohexene is:-
 (1) -121 kJ mol $^{-1}$ (2) 121 kJ mol $^{-1}$
 (3) -242 kJ mol $^{-1}$ (4) 242 kJ mol $^{-1}$

BOND ENERGY/RESONANCE ENERGY

- 151.** Bond energy of a molecule :
 (1) Is always negative
 (2) Is always positive
 (3) Either positive or negative
 (4) Depends upon the physical state of the system
- 152.** Among the following for which reaction heat of reaction represents bond energy of HCl
 (1) $HCl(g) \longrightarrow H^+(g) + Cl^-(g)$
 (2) $HCl(g) \longrightarrow \frac{1}{2} H_2(g) + \frac{1}{2} Cl_2(g)$
 (3) $2HCl(g) \longrightarrow H_2(g) + Cl_2(g)$
 (4) $HCl(g) \longrightarrow H(g) + Cl(g)$
- 153.** The bond energies of F_2 , Cl_2 , Br_2 and I_2 are 155.4 , 243.6 , 193.2 and 151.2 kJ mol $^{-1}$ respectively. The strongest bond is :
 (1) F - F (2) Cl - Cl
 (3) Br - Br (4) I - I
- 154.** Energy required to dissociate 4g of gaseous hydrogen into free gaseous atoms is 208 kCal at 25°C. The bond energy of H—H bond will be :
 (1) 1.04 kCal (2) 10.4 kCal
 (3) 104 kCal (4) 1040 kCal
- 155.** Heat evolved in the reaction $H_2 + Cl_2 \longrightarrow 2HCl$ is 182 kJ. Bond energies of H—H and Cl—Cl are 430 and 242 kJ mol $^{-1}$ respectively. The H—Cl bond energy is :
 (1) 245 kJ mol $^{-1}$
 (2) 427 kJ mol $^{-1}$
 (3) 336 kJ mol $^{-1}$
 (4) 154 kJ mol $^{-1}$

- 156.** The enthalpy change for the reaction $H_2(g) + C_2H_4(g) \rightarrow C_2H_6(g)$ is..... . The bond energies are,
 $H-H = 103$, $C-H = 99$, $C-C = 80$ &
 $C=C = 145$ kCal mol $^{-1}$
 (1) -10 kCal mol $^{-1}$ (2) $+10$ kCal mol $^{-1}$
 (3) -30 kCal mol $^{-1}$ (4) $+30$ kCal mol $^{-1}$
- 157.** Bond dissociation enthalpies of $H_2(g)$ and $N_2(g)$ are 436.0 kJ mol $^{-1}$ and 941.8 kJ mol $^{-1}$ respectively and enthalpy of formation of $NH_3(g)$ is -46 kJ mol $^{-1}$. What is enthalpy of atomization of $NH_3(g)$?
 (1) 390.3 kJ mol $^{-1}$ (2) 1170.9 kJ mol $^{-1}$
 (3) 590 kJ mol $^{-1}$ (4) 720 kJ mol $^{-1}$
- 158.** From the reactions :
 $C(s) + 2H_2(g) \rightarrow CH_4(g)$ $\Delta H = -X$ Kcal
 $C(g) + 4H(g) \rightarrow CH_4(g)$, $\Delta H = -X_1$ Kcal
 $CH_4(g) \rightarrow CH_3(g) + H(g)$ $\Delta H = +Y$ (Kcal)
 Bond energy of C—H bond is -
 (1) $\frac{X}{4}$ kCal mol $^{-1}$
 (2) Y kCal mol $^{-1}$
 (3) $\frac{X_1}{4}$ kCal mol $^{-1}$
 (4) X_1 kCal mol $^{-1}$
- 159.** The enthalpy changes at 298 K in successive breaking of O—H bonds of water are
 $H_2O \longrightarrow H(g) + OH(g)$; $\Delta H = 498$ kJ mol $^{-1}$
 $OH(g) \longrightarrow H(g) + O(g)$; $\Delta H = 428$ kJ mol $^{-1}$
 the bond enthalpy of O—H bond is
 (1) 498 kJ mol $^{-1}$ (2) 428 kJ mol $^{-1}$
 (3) 70 kJ mol $^{-1}$ (4) 463 kJ mol $^{-1}$
- 160.** If ΔH_f° of $ICl_{(g)}$, $Cl_{(g)}$, and $I_{(g)}$ is 17.57, 121.34 and 106.96 J mol $^{-1}$ respectively. Then bond dissociation energy of ICl bond is -
 (1) 35.15 J mol $^{-1}$ (2) 106.69 J mol $^{-1}$
 (3) 210.73 J mol $^{-1}$ (4) 420.9 J mol $^{-1}$
- 161.** Heat of dissociation of benzene to elements is 5535 kJ mol $^{-1}$. The bond enthalpies of C—C, C=C and C—H are 347.3, 615.0 and 416.2 kJ respectively. Magnitude resonance energy of benzene is
 (1) 1.51 kJ (2) 15.1 kJ
 (3) 151 kJ (4) 1511 kJ

SOME OTHER HEAT OF REACTIONS

- 162.** The enthalpy change for the reaction $2\text{C}(\text{graphite}) + 3\text{H}_2(\text{g}) \longrightarrow \text{C}_2\text{H}_6(\text{g})$ is called
 (1) Enthalpy of formation
 (2) Enthalpy of combustion
 (3) Enthalpy of hydrogenation
 (4) Enthalpy of vaporisation
- 163.** $\text{Cl}_2(\text{g}) \longrightarrow 2\text{Cl}(\text{g})$, In this process value of ΔH will be -
 (1) Positive
 (2) Negative
 (3) Zero
 (4) Nothing can be predicted
- 164.** The magnitude of heat of solution on addition of solvent to solution
 (1) Decreases
 (2) Increases
 (3) Remains constant
 (4) Increases or decreases
- 165.** If $\text{H}_2(\text{g}) = 2\text{H}(\text{g})$; $\Delta H = 104 \text{ kCal}$, then heat of atomisation of hydrogen is :
 (1) 52 kCal (2) 104 kCal
 (3) 208 kCal (4) None of these
- 166.** $\text{S}_{(\text{rhombic})} + \text{O}_{2(\text{g})} \longrightarrow \text{SO}_{2(\text{g})}$; $\Delta H = -297.5 \text{ kJ}$
 $\text{S}_{(\text{monoclinic})} + \text{O}_{2(\text{g})} \longrightarrow \text{SO}_{2(\text{g})}$; $\Delta H = -300 \text{ kJ}$
 The data can predict that -
 (1) Rhombic sulphur is yellow in colour
 (2) Monoclinic sulphur has metallic lusture.
 (3) Monoclinic sulphur is more stable
 (4) ΔH transition of S_R to S_M is endothermic
- 167.** The heat of combustion of yellow phosphorous and red phosphorous are -9.91 kJ and -8.78 kJ respectively. The heat of transition of yellow phosphorous to red phosphorous is
 (1) -18.69 kJ (2) $+1.13 \text{ kJ}$
 (3) $+18.69 \text{ kJ}$ (4) -1.13 kJ
- 168.** For the change $\text{C}(\text{diamond}) \longrightarrow \text{C}(\text{graphite})$; $\Delta H = -1.89 \text{ KJ}$, if 6 g of diamond and 6g of graphite are separately burnt to yield CO_2 the heat liberated in first case is :
 (1) Less than in the second case by 1.89 kJ
 (2) Less than in the second case by 11.34 kJ
 (3) Less than in the second case by 14.34 kJ
 (4) More than in the second case by 0.945 kJ
- 169.** $2\text{CO}_{(\text{g})} + \text{O}_{2(\text{g})} \longrightarrow 2\text{CO}_{2(\text{g})} + X \text{ kJ}$
 In the above equation X kJ refers to :
 (1) Heat of formation of CO_2
 (2) Heat of vapourisation
 (3) Heat of reaction
 (4) Heat of sublimation
- 170.** Which of the following reactions represents ΔH (hydration) :-
 (1) $\text{CuSO}_4(\text{s}) + (\text{aq}) \rightarrow \text{CuSO}_4(\text{aq})$; $\Delta H = -x \text{ kJ}$
 (2) $\text{BaCl}_2(\text{s}) + 2\text{H}_2\text{O}(\ell) \rightarrow \text{BaCl}_2 \cdot 2\text{H}_2\text{O}(\text{s})$; $\Delta H = -x' \text{ kJ}$
 (3) $\text{CuSO}_4(\text{s}) + 5\text{H}_2\text{O}(\ell) + (\text{aq}) \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{aq})$; $\Delta H = -y' \text{ kJ}$
 (4) None of the above
- 171.** ΔH for the reaction, $\text{I}_{(\text{g})} + \text{I}_{(\text{g})} \rightarrow \text{I}_{2(\text{g})}$ will be:-
 (1) Zero (2) -ve (3) +ve (4) ∞
- 172.** Given that :
 $\text{A}(\text{s}) \rightarrow \text{A}(\ell)$; $\Delta H = x$, $\text{A}(\ell) \rightarrow \text{A}(\text{g}) - y$
 The heat of sublimation of A will be:-
 (1) $x + y$ (2) $x - y$
 (3) x or y (4) $-(x + y)$

HESS LAW

- 173.** The enthalpy change of a reaction does not depend on
 (1) State of reactants and products
 (2) Nature of reactants and products
 (3) Different intermediate reactions
 (4) Initial and final enthalpy change of reaction
- 174.** From the thermochemical reactions,
 $\text{C}(\text{graphite}) + \frac{1}{2}\text{O}_2 \longrightarrow \text{CO}$; $\Delta H = -110.5 \text{ kJ}$
 $\text{CO} + \frac{1}{2}\text{O}_2 \longrightarrow \text{CO}_2$; $\Delta H = -283.2 \text{ kJ}$
 the heat of reaction of $\text{C}(\text{graphite}) + \text{O}_2 \longrightarrow \text{CO}_2$ is :
 (1) 393.7 kJ
 (2) -393.7 kJ
 (3) -172.7 kJ
 (4) $+172.7 \text{ kJ}$
- 175.** If $\text{H}_2 + \frac{1}{2}\text{O}_2 \longrightarrow \text{H}_2\text{O}$; $\Delta H = -68.39 \text{ kCal}$
 $\text{K} + \text{H}_2\text{O} + \text{water} \longrightarrow \text{KOH}(\text{aq}) + \frac{1}{2}\text{H}_2$; $\Delta H = -48.0 \text{ kCal}$
 $\text{KOH} + \text{water} \longrightarrow \text{KOH}(\text{aq})$ $\Delta H = -14.0 \text{ kCal}$
 the heat of formation of KOH is -
 (1) $-68.39 + 48 - 14.0$
 (2) $-68.39 - 48.0 + 14.0$
 (3) $+68.39 - 48.0 + 14.0$
 (4) $+68.39 + 48.0 - 14.0$

- 176.** Given $\text{C(s)} + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 94.2 \text{ kCal}$
 $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{H}_2\text{O}(\ell) + 68.3 \text{ kCal}$
 $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\ell) + 210.8 \text{ kCal}$
 The heat of formation of methane in Kcal will be
 (1) -45.9 (2) -47.8
 (3) -20.0 (4) -47.3

- 177.** From the following data, the heat of formation of $\text{Ca(OH)}_2(\text{s})$ at 18°C is kCal.
 (i) $\text{CaO(s)} + \text{H}_2\text{O}(\ell) = \text{Ca(OH)}_2(\text{s})$;
 $\Delta H_{18^\circ\text{C}} = -15.26 \text{ kCal} \dots\dots$
 (ii) $\text{H}_2\text{O}(\ell) = \text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$;
 $\Delta H_{18^\circ\text{C}} = 68.37 \text{ kCal} \dots\dots$
 (iii) $\text{Ca(s)} + \frac{1}{2}\text{O}_2(\text{g}) = \text{CaO(s)}$;
 $\Delta H_{18^\circ\text{C}} = -151.80 \text{ kCal} \dots\dots$
 (1) -98.69 (2) -235.43
 (3) 194.91 (4) 98.69

- 178.** If, $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl(g)} ; \Delta H^\circ = -44 \text{ kCal}$
 $2\text{Na(s)} + 2\text{HCl(g)} \longrightarrow 2\text{NaCl(s)} + \text{H}_2(\text{g})$;
 $\Delta H = -152 \text{ kCal}$
 Then, $\text{Na(s)} + 0.5 \text{ Cl}_2(\text{g}) \longrightarrow \text{NaCl(s)} ; \Delta H^\circ = ?$
 (1) 108 kCal (2) 196 kCal
 (3) -98 kCal (4) 54 kCal

- 179.** (i) $\text{S(s)} + 3/2 \text{ O}_2(\text{g}) = \text{SO}_3(\text{g}) + 2x \text{ kCal}$
 (ii) $\text{SO}_2(\text{g}) + 1/2 \text{ O}_2(\text{g}) = \text{SO}_3(\text{g}) + y \text{ kCal}$
 Calculate the heat of formation of SO_2 :
 (1) $(2x + y)$ (2) $-(2x - y)$
 (3) $x + y$ (4) $2x / y$

- 180.** If $\text{S} + \text{O}_2 \longrightarrow \text{SO}_2$; $\Delta H = -298.2$
 $\text{SO}_2 + \frac{1}{2}\text{O}_2 \longrightarrow \text{SO}_3$; $\Delta H = -98.7$
 $\text{SO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4$; $\Delta H = -130.2$
 $\text{H}_2 + \frac{1}{2}\text{O}_2 \longrightarrow \text{H}_2\text{O}$; $\Delta H = -287.3$
 Then the enthalpy of formation of H_2SO_4 at 298 K is -
 (1) -814.4 kJ (2) -650.3 kJ
 (3) -320.5 kJ (4) -433.5 kJ

- 181.** Given that :
 $\text{Zn} + \frac{1}{2}\text{O}_2 \rightarrow \text{ZnO} + 84000 \text{ Cal} \dots\dots\dots 1$
 $\text{Hg} + \frac{1}{2}\text{O}_2 \rightarrow \text{HgO} + 21700 \text{ Cal} \dots\dots\dots 2$
 The heat of reaction (ΔH) for,
 $\text{Zn} + \text{HgO} \rightarrow \text{ZnO} + \text{Hg}$ is :-
 (1) 105700 Cal
 (2) 62300 Cal
 (3) -105700 Cal
 (4) -62300 Cal

- 182.** Given that -
 $2\text{C(s)} + 2\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) \quad \Delta H = -787 \text{ kJ}$
 $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\ell) \quad \Delta H = -286 \text{ kJ}$
 $\text{C}_2\text{H}_2(\text{g}) + \frac{5}{2}\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\ell) \quad \Delta H = -1310 \text{ kJ}$

Heat of formation of acetylene is :-

- (1) + 1802 kJ (2) - 1802 kJ
 (3) - 800 kJ (4) + 237 kJ

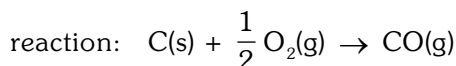
- 183.** Find the heat change in the reaction :
 $\text{NH}_3(\text{g}) + \text{HCl(g)} \rightarrow \text{NH}_4\text{Cl(s)}$
 from the following data
 $\text{NH}_3(\text{g}) + \text{aq} \rightarrow \text{NH}_3(\text{aq}), \quad \Delta H = -8.4 \text{ kCal}$
 $\text{HCl(g)} + \text{aq} \rightarrow \text{HCl(aq)}, \quad \Delta H = -17.3 \text{ kCal}$
 $\text{NH}_3(\text{aq}) + \text{HCl(aq)} \rightarrow \text{NH}_4\text{Cl(aq)}, \Delta H = -12.5 \text{ kCal}$
 $\text{NH}_4\text{Cl(s)} + \text{aq} \rightarrow \text{NH}_4\text{Cl(aq)}, \Delta H = +3.9 \text{ kCal}$
 (1) -42.1 (2) -34.3
 (3) +34.3 (4) +42.1

- 184.** The heat of reaction for



$\text{AO} + \frac{1}{2}\text{O}_2 \rightarrow \text{AO}_2$ is 100 kCal. The heat of reaction for $\text{A} + \text{O}_2 \rightarrow \text{AO}_2$ is:-
 (1) -50 kCal (2) +50 kCal
 (3) 100 kCal (4) 150 kCal

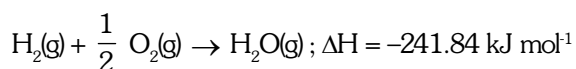
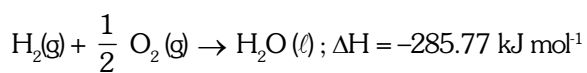
- 185.** $\text{C(s)} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 94.0 \text{ kCal}$
 $\text{CO(g)} + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}), \quad \Delta H = -67.7 \text{ kCal}$
 from the above reactions find how much heat (kCal mole^{-1}) would be produced in the following



- (1) 20.6 (2) 26.3
 (3) 44.2 (4) 161.6

- 186.** Using the following thermochemical data:
 $\text{C(s)} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}), \quad \Delta H = -94.0 \text{ kCal}$
 $\text{H}_2(\text{g}) + 1/2\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\ell), \Delta H = -68.0 \text{ kCal}$
 $\text{CH}_3\text{COOH}(\ell) + 2\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\ell), \Delta H = -210.0 \text{ kCal}$
 The heat of formation of acetic acid is:-
 (1) 116.0 kCal (2) -116.0 kCal
 (3) -114.0 kCal (4) +114.0 kCal

187. The enthalpy of vapourisation of liquid water using the data:



- (1) + 43.93 kJ mol⁻¹ (2) - 43.93 kJ mol⁻¹
 (3) + 527.61 kJ mol⁻¹ (4) - 527.61 kJ mol⁻¹

188. $\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) = \text{H}_2\text{O}(\ell); \Delta H_{298\text{K}} = -68.32 \text{ kCal}$. Heat of vapourisation of water at 1 atm and 25°C is 10.52 kCal. The standard heat of formation (in kCal) of 1 mole of water vapour at 25°C is

- (1) 10.52 (2) -78.84
 (3) +57.80 (4) -57.80

189. The heat of solution of anhydrous CuSO_4 and $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ are - 15.89 and 2.80 kCal mol⁻¹ respectively. What will be the heat of hydration of anhydrous CuSO_4 ?

- (1) -18.69 kCal (2) 18.69 kCal
 (3) -28.96 kCal (4) 28.96 kCal

190. One mole of anhydrous salt AB dissolves in water and liberates 21.0 J mol⁻¹ of heat. The value of $\Delta H_{(\text{hydration})}$ of AB is -29.4 J mol⁻¹. The heat of dissolution of hydrated salt $\text{AB} \cdot 2\text{H}_2\text{O}(\text{s})$ is -

- (1) 50.4 J mol⁻¹
 (2) 8.4 J mol⁻¹
 (3) -50.4 J mol⁻¹
 (5) -8.4 J mol⁻¹

191. Which of the following expressions is true:-

(1) $H_f^0(\text{CO}, \text{g}) = \frac{1}{2} \Delta H_f^0(\text{CO}_2, \text{g})$

(2) $\Delta H_f^0(\text{CO}, \text{g}) = \Delta H_f^0(\text{C, graphite}) + \frac{1}{2} \Delta H_f^0(\text{O}_2, \text{g})$

(3) $\Delta H_f^0(\text{CO}, \text{g}) = \Delta H_f^0(\text{CO}_2, \text{g}) - \frac{1}{2} \Delta H_f^0(\text{O}_2, \text{g})$

(4) $\Delta H_f^0(\text{CO}, \text{g}) = \Delta H_{\text{comb}}^0(\text{C, graphite}) - \Delta H_{\text{comb}}^0(\text{CO}, \text{g})$

EXERCISE-I (Conceptual Questions)**ANSWER KEY**

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	3	4	4	3	3	1	4	1	1	1	4	2	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	4	1	1	4	3	1	3	2	2	4	2	2	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	1	2	1	1	2	3	4	2	2	2	3	1	4	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	2	3	4	3	4	3	1	1	2	3	1	3	1	3
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	2	2	1	1	4	2	4	3	2	4	2	1	2	4	3
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	3	3	4	4	1	2	2	2	4	1	4	1	4	1
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	1	2	3	1	3	4	1	2	4	3	2	4	2	3	1
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	2	4	4	4	2	2	2	4	4	1	3	2	2	3	2
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	3	4	4	4	1	3	2	2	2	3	1	3	3	3	2
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	2	2	2	3	1	3	2	2	1	2	1	1	4	1	1
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	2	4	2	3	2	3	2	3	4	3	3	1	1	3	2
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	4	4	4	3	2	2	1	3	2	2	3	2	3	2	1
Que.	181	182	183	184	185	186	187	188	189	190	191				
Ans.	4	4	1	2	2	3	1	4	1	2	4				

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

- Identify the correct statement for change of Gibbs energy for a system (ΔG_{system}) at constant temperature and pressure.
 - If $\Delta G_{\text{system}} > 0$, the process is spontaneous.
 - If $\Delta G_{\text{system}} = 0$, the system has attained equilibrium.
 - If $\Delta G_{\text{system}} = 0$, the system is still moving in a particular direction.
 - If $\Delta G_{\text{system}} < 0$, the process is not spontaneous.
- Assume each reaction is carried out in an open container. For which reaction will $\Delta H = \Delta E$?
 - $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \rightarrow 2 \text{HBr}(\text{g})$
 - $\text{C}(\text{s}) + 2 \text{H}_2\text{O}(\text{g}) \rightarrow 2 \text{H}_2(\text{g}) + \text{CO}_2(\text{g})$
 - $\text{PCl}_5(\text{g}) \rightarrow \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
 - $2 \text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2 \text{CO}_2(\text{g})$
- The enthalpy and entropy change for reaction $\text{Br}_2(\text{l}) + \text{Cl}_2(\text{g}) \rightarrow 2 \text{BrCl}(\text{g})$ are 30 kJ mol^{-1} and $105 \text{ JK}^{-1} \text{ mol}^{-1}$ respectively. The temperature at which the reaction will be in equilibrium is
 - 285.7 K
 - 273 K
 - 450 K
 - 300 K
- The enthalpy of hydrogenation of cyclohexene is $-119.5 \text{ kJ mol}^{-1}$. If resonance energy of benzene is $-150.4 \text{ kJ mol}^{-1}$, its enthalpy of hydrogenation would be
 - $-508.9 \text{ kJ mol}^{-1}$
 - $-208.1 \text{ kJ mol}^{-1}$
 - $-269.9 \text{ kJ mol}^{-1}$
 - $-358.5 \text{ kJ mol}^{-1}$

AIIMS 2006

- When you make ice cubes, the entropy of water
 - Does not change
 - Increases
 - Decreases
 - May either increase or decrease depending on the process used
- For a phase change $\text{H}_2\text{O}(\text{l}) \xrightarrow{0^\circ\text{C 1bar}} \text{H}_2\text{O}(\text{s})$
 - $\Delta G = 0$
 - $\Delta S = 0$
 - $\Delta H = 0$
 - $\Delta U = 0$
- For a spontaneous process the correct statement is –
 - Entropy of the system always increase
 - Free energy of the system always increases
 - Total entropy change is always negative
 - Total entropy change is always positive

- The enthalpy change (ΔH) for the reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ is -92.38 kJ at 298 K . The internal energy change ΔU at 298 K is
 - -92.38 kJ
 - -87.42 kJ
 - -97.34 kJ
 - -89.9 kJ

AIPMT 2007

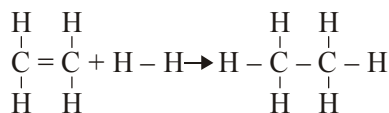
- Consider the following reactions :
 - $\text{H}^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})} = \text{H}_2\text{O}_{(\text{l})}$, $\Delta H = -X_1 \text{ kJ mol}^{-1}$
 - $\text{H}_{2(\text{g})} + \frac{1}{2} \text{O}_{2(\text{g})} = \text{H}_2\text{O}_{(\text{l})}$, $\Delta H = -X_2 \text{ kJ mol}^{-1}$
 - $\text{CO}_{2(\text{g})} + \text{H}_{2(\text{g})} = \text{CO}_{(\text{g})} + \text{H}_2\text{O}_{(\text{l})}$, $\Delta H = -X_3 \text{ kJ mol}^{-1}$
 - $\text{C}_2\text{H}_{2(\text{g})} + \frac{5}{2} \text{O}_{2(\text{g})} = 2\text{CO}_{2(\text{g})} + \text{H}_2\text{O}_{(\text{l})} + X_4 \text{ kJ mol}^{-1}$
 Enthalpy of formation of $\text{H}_2\text{O}_{(\text{l})}$ is :
 - $+X_1 \text{ kJ mol}^{-1}$
 - $-X_2 \text{ kJ mol}^{-1}$
 - $+X_3 \text{ kJ mol}^{-1}$
 - $-X_4 \text{ kJ mol}^{-1}$
- Given that bond energies of H-H and Cl-Cl are 430 kJ mol^{-1} and 240 kJ mol^{-1} respectively and $\Delta_f H$ for HCl is -90 kJ mol^{-1} . Bond enthalpy of HCl is :
 - 245 kJ mol^{-1}
 - 290 kJ mol^{-1}
 - 380 kJ mol^{-1}
 - 425 kJ mol^{-1}

AIPMT 2008

- Bond dissociation enthalpy of H_2 , Cl_2 and HCl are 434 , 242 and 431 kJ mol^{-1} respectively. Enthalpy of formation of HCl is :
 - -93 kJ mol^{-1}
 - 245 kJ mol^{-1}
 - 93 kJ mol^{-1}
 - -245 kJ mol^{-1}
- For the gas phase reaction, $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
Which of the following conditions are correct ?
 - $\Delta H < 0$ and $\Delta S < 0$
 - $\Delta H > 0$ and $\Delta S < 0$
 - $\Delta H = 0$ and $\Delta S < 0$
 - $\Delta H > 0$ and $\Delta S > 0$
- Which of the following are not state functions ?
 - $q + w$
 - q
 - w
 - H-TS
 - (I), (II) and (III)
 - (II) and (III)
 - (I) and (IV)
 - (II), (III) and (IV)

AIPMT 2009

14. From the following bond energies :-
 H - H bond energy : $431.37 \text{ kJ mol}^{-1}$
 C = C bond energy : $606.10 \text{ kJ mol}^{-1}$
 C - C bond energy : $336.49 \text{ kJ mol}^{-1}$
 C - H bond energy : $410.50 \text{ kJ mol}^{-1}$
 Enthalpy for the reaction,



will be :-

- (1) $553.0 \text{ kJ mol}^{-1}$ (2) $1523.6 \text{ kJ mol}^{-1}$
 (3) $-243.6 \text{ kJ mol}^{-1}$ (4) $-120.0 \text{ kJ mol}^{-1}$
15. The values of ΔH and ΔS for the reaction,
 $\text{C}_{(\text{graphite})} + \text{CO}_{2(\text{g})} \rightarrow 2\text{CO}_{(\text{g})}$ are 170 kJ and 170 JK^{-1} , respectively. This reaction will be spontaneous at :-
 (1) 510 K (2) 710 K
 (3) 910 K (4) 1110 K

AIPMT 2010

16. For vaporization of water at 1 atmospheric pressure, the values of ΔH and ΔS are $40.63 \text{ kJ mol}^{-1}$ and $108.8 \text{ JK}^{-1}\text{mol}^{-1}$, respectively. The temperature when Gibbs energy change (ΔG) for this transformation will be zero, is :-
 (1) 393.4 K (2) 373.4 K
 (3) 293.4 K (4) 273.4 K
17. Three moles of an ideal gas expanded spontaneously into vacuum. The work done will be :-
 (1) 3 Joule (2) 9 Joule
 (3) Zero (4) Infinite
18. The following two reactions are known :
 $\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g}) \rightarrow 2\text{Fe}(\text{s}) + 3\text{CO}_2(\text{g})$;
 $\Delta H = -26.8 \text{ kJ}$
 $\text{FeO}(\text{s}) + \text{CO}(\text{g}) \rightarrow \text{Fe}(\text{s}) + \text{CO}_2(\text{g})$;
 $\Delta H = -16.5 \text{ kJ}$
 Correct target equation is
 $\text{Fe}_2\text{O}_3(\text{s}) + \text{CO}(\text{g}) \rightarrow 2\text{FeO}(\text{s}) + \text{CO}_2(\text{g})$, $\Delta H = ?$
 (1) -43.3 kJ (2) -10.3 kJ
 (3) $+6.2 \text{ kJ}$ (4) $+10.3 \text{ kJ}$
19. Standard entropies of X_2 , Y_2 and XY_3 are 60, 40 and $50 \text{ JK}^{-1}\text{mol}^{-1}$ respectively. For the reaction
 $\frac{1}{2}\text{X}_2 + \frac{3}{2}\text{Y}_2 \rightleftharpoons \text{XY}_3$, $\Delta H = -30 \text{ kJ}$ to be at equilibrium, the temperature should be :-
 (1) 500 K (2) 750 K
 (3) 1000 K (4) 1250 K

AIIMS 2010

20. Intensive property is :-
 (1) Moles (2) Volume
 (3) Enthalpy (4) Temperature
21. In which of following there is decrease in entropy :-
 (1) When temperature is raised from 30 K to 150 K .
 (2) When NaHCO_3 changes into $\text{Na}_2\text{CO}_3 (\text{s})$ and $\text{CO}_2(\text{g})$
 (3) $\text{H}_2(\text{g}) \rightarrow 2\text{H}(\text{g})$
 (4) Liquid crystallises into a solid
22. Which of the following is not a state function:-
 (1) Pressure (2) Volume
 (3) Temperature (4) Heat

AIPMT Pre. 2011

23. If the enthalpy change for the transition of liquid water to steam is 30 kJ mol^{-1} at 27°C , the entropy changes for the process would be:
 (1) $10 \text{ J mol}^{-1} \text{ K}^{-1}$
 (2) $1.0 \text{ J mol}^{-1} \text{ K}^{-1}$
 (3) $0.1 \text{ J mol}^{-1} \text{ K}^{-1}$
 (4) $100 \text{ J mol}^{-1} \text{ K}^{-1}$
24. Enthalpy change for the reaction,
 $4\text{H}_{(\text{g})} \rightarrow 2\text{H}_{2(\text{g})}$ is -869.6 kJ
 The dissociation energy of H-H bond is :
 (1) -434.8 kJ (2) 869.6 kJ
 (3) $+434.8 \text{ kJ}$ (4) $+217.4 \text{ kJ}$
25. Which of the following correct option for free expansion of an ideal gas under adiabatic condition ?
 (1) $q = 0$, $\Delta T \neq 0$, $w = 0$
 (2) $q \neq 0$, $\Delta T = 0$, $w = 0$
 (3) $q = 0$, $\Delta T = 0$, $w = 0$
 (4) $q = 0$, $\Delta T < 0$, $w \neq 0$

AIPMT Msins 2011

26. Consider the following processes :-
 $\Delta H(\text{kJ mol}^{-1})$
- | | |
|--|------|
| $\frac{1}{2} \text{A} \rightarrow \text{B}$ | +150 |
| $3\text{B} \rightarrow 2\text{C} + \text{D}$ | -125 |
| $\text{E} + \text{A} \rightarrow 2\text{D}$ | +350 |
- For $\text{B} + \text{D} \rightarrow \text{E} + 2\text{C}$, ΔH will be :
 (1) 325 kJ mol^{-1}
 (2) 525 kJ mol^{-1}
 (3) -175 kJ mol^{-1}
 (4) -325 kJ mol^{-1}

AIIMS 2011

27. The enthalpy of formation of CO(g) , $\text{CO}_2\text{(g)}$, $\text{N}_2\text{O(g)}$ and $\text{N}_2\text{O}_4\text{(g)}$ is -110 , -393 , $+81$ and 10 kJ mol^{-1} respectively. For the reaction $\text{N}_2\text{O}_4\text{(g)} + 3\text{CO(g)} \rightarrow \text{N}_2\text{O(g)} + 3\text{CO}_2\text{(g)}$. ΔH_r is
 (1) -212 (2) $+212$
 (3) $+778$ (4) -778
28. For adiabatic process which is correct-
 (1) $\Delta T = 0$ (2) $w = 0$
 (3) $q = 0$ (4) $\Delta U = 0$
29. Which of the following is not thermodynamic function-
 (1) Internal energy (2) work done
 (3) Enthalpy (4) Entropy
30. Which of the following is intensive property-
 (1) Enthalpy (2) Entropy
 (3) specific heat (4) volume

AIPMT Pre. 2012

31. In which of the following reactions, standard reaction entropy change (ΔS°) is positive and standard Gibb's energy change (ΔG°) decreases sharply with increasing temperature?
 (1) $\text{Mg(s)} + \frac{1}{2} \text{O}_2\text{(g)} \rightarrow \text{MgO(s)}$
 (2) $\frac{1}{2} \text{C graphite} + \frac{1}{2} \text{O}_2\text{(g)} \rightarrow \frac{1}{2} \text{CO}_2\text{(g)}$
 (3) $\text{C graphite} + \frac{1}{2} \text{O}_2\text{(g)} \rightarrow \text{CO(g)}$
 (4) $\text{CO(g)} + \frac{1}{2} \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$
32. Standard enthalpy of vapourisation $\Delta_{\text{vap}}H^\circ$ for water at 100°C is $40.66 \text{ kJ mol}^{-1}$. The internal energy of vaporisation of water at 100°C (in kJ mol^{-1}) is
 (1) $+43.76$ (2) $+40.66$
 (3) $+37.56$ (4) -43.76
33. The enthalpy of fusion of water is $1.435 \text{ kCal mol}^{-1}$. The molar entropy change for the melting of ice at 0°C is:
 (1) $5.260 \text{ Cal mol}^{-1} \text{ K}^{-1}$
 (2) $0.526 \text{ Cal mol}^{-1} \text{ K}^{-1}$
 (3) $10.52 \text{ Cal mol}^{-1} \text{ K}^{-1}$
 (4) $21.04 \text{ Cal mol}^{-1} \text{ K}^{-1}$

AIIMS 2012

34. At equilibrium which is correct :-
 (1) $\Delta G = 0$ (2) $\Delta S = 0$
 (3) $\Delta H = 0$ (4) $\Delta G^\circ = 0$
35. Heat of atomisation of CH_4 is 360 kJ mol^{-1} and C_2H_6 has 620 kJ mol^{-1} . Then bond dissociation energy of C-C bond is :-
 (1) 170 kJ mol^{-1}
 (2) 50 kJ mol^{-1}
 (3) 80 kJ mol^{-1}
 (4) 220 kJ mol^{-1}

AIIMS 2013

36. Which thermodynamic parameter does not depend only on initial and final state ?
 (1) q at constant pressure
 (2) q at constant volume
 (3) w at adiabatic
 (4) w at isothermal

AIPMT 2014

37. For the reaction : $\text{X}_2\text{O}_4\text{(l)} \rightarrow 2\text{XO}_2\text{(g)}$
 $\Delta U = 2.1 \text{ kCal}$, $\Delta S = 20 \text{ cal K}^{-1}$ at 300 K
 Hence ΔG is :-
 (1) 2.7 kCal (2) -2.7 kCal
 (3) 9.3 kCal (4) -9.3 kCal

AIIMS 2014

38. $\Delta S_{\text{Total}} = -40 \text{ kJ mol}^{-1} \times \text{K}^{-1}$
 $\Delta H_{\text{sys}} = 2000 \text{ kJ mol}^{-1}$
 $T = 400 \text{ K}$
 Calculate value of ΔS_{system} ?
 (1) $-35 \text{ kJ mol}^{-1} \times \text{K}$
 (2) $-5 \text{ kJ mol}^{-1} \times \text{K}$
 (3) $-40 \text{ kJ mol}^{-1} \times \text{K}$
 (4) $+5 \text{ kJ mol}^{-1} \times \text{K}$

AIPMT 2015

39. Which of the following statements is **correct** for a reversible process in a state of equilibrium ?
 (1) $\Delta G = 2.30 RT \log K$
 (2) $\Delta G^\circ = -2.30 RT \log K$
 (3) $\Delta G^\circ = 2.30 RT \log K$
 (4) $\Delta G = -2.30 RT \log K$

Re-AIPMT 2015

40. The heat of combustion of carbon to CO_2 is $-393.5 \text{ kJ mol}^{-1}$. The heat released upon formation of 35.2 g of CO_2 from carbon and oxygen gas is:
 (1) -630 kJ (2) -3.15 kJ
 (3) -315 kJ (4) $+315 \text{ kJ}$

AIIMS 2015

41. $A_{2(g)} \rightarrow 2A_{(g)}$ and for this reaction on increasing T value of K_{eq} increases then for this reaction is
 (1) ΔH = positive; ΔS = positive
 (2) ΔH = negative; ΔS = negative
 (3) ΔH = positive; ΔS = negative
 (4) ΔH = negative ; ΔS = positive
42. For endothermic reaction when change in entropy is negative, then reaction is
 (1) not possible at any temperature
 (2) possible at low temperature
 (3) possible at all temperature
 (4) possible at high temperature
43. For the formation of CH_4 of $\Delta U^0 = -x \text{ kJ mol}^{-1}$ then what will be value of $\Delta H^0 \text{ kJ mol}^{-1}$.
 (1) $-x - RT$ (2) $-x + 2RT$
 (3) $-x + RT$ (4) $-2x - 4RT$

NEET-I 2016

44. The correct thermodynamic conditions for the spontaneous reaction at all temperatures is
 (1) $\Delta H > 0$ and $\Delta S > 0$
 (2) $\Delta H > 0$ and $\Delta S < 0$
 (3) $\Delta H < 0$ and $\Delta S > 0$
 (4) $\Delta H < 0$ and $\Delta S < 0$

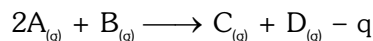
NEET-II 2016

45. For a sample of perfect gas when its pressure is changed isothermally from p_i to p_f , the entropy change is given by
 (1) $\Delta S = nRT \ln\left(\frac{p_f}{p_i}\right)$ (2) $\Delta S = RT \ln\left(\frac{p_i}{p_f}\right)$
 (3) $\Delta S = nR \ln\left(\frac{p_f}{p_i}\right)$ (4) $\Delta S = nR \ln\left(\frac{p_i}{p_f}\right)$

AIIMS 2016

46. For a given reaction at 298K
 $2A_{(g)} + B_{(g)} \longrightarrow 2C_{(g)}$
 If $\Delta U^0 = -10 \text{ kJ mol}^{-1}$ $\Delta S^0 = -45 \text{ J K}^{-1} \text{ mol}^{-1}$
 Then what will be value of ΔG^0 for above reaction?
 (1) $+930 \text{ J mol}^{-1}$ (2) $-25890 \text{ J mol}^{-1}$
 (3) 2000 J mol^{-1} (4) 8500 J mol^{-1}
47. Calculate work done during isothermal reversible process when 5 mol ideal gas is expanded so that its volume is doubled at 400K?
 (1) -11.5 kJ (2) -344 kJ
 (3) 0 (4) -2.8 kJ

48. For a reaction



Where ΔS is -ve; then which of the following is correct?

- (1) reaction is possible only at low temperature
 (2) reaction is possible only at high temperature
 (3) reaction is never possible
 (4) reaction is possible at all temperature
49. If 400 kJ work is done by the system and 150 kJ heat is given to system then what will be effect on internal energy?
 (1) Increases by 250 kJ (2) Decreases by 250 kJ
 (3) Increases by 600 kJ (4) Decreases by 600 kJ
50. For gas 'A' in a calorimeter heat evolved is 250 kJ mol^{-1} . For 0.2 mol of A, temperature rise from 298 K to 300 K. Find out heat capacity of calorimeter:-
 (1) 12.5 kJ K^{-1} (2) 25 kJ K^{-1}
 (3) 50 kJ K^{-1} (4) 100 kJ K^{-1}

NEET(UG) 2017

51. For a given reaction, $\Delta H = 35.5 \text{ kJ mol}^{-1}$ and $\Delta S = 83.6 \text{ J K}^{-1} \text{ mol}^{-1}$. The reaction is spontaneous at : (Assume that ΔH and ΔS do not vary with temperature)
 (1) $T > 425 \text{ K}$ (2) All temperatures
 (3) $T > 298 \text{ K}$ (4) $T < 425 \text{ K}$
52. A gas is allowed to expand in a well insulated container against a constant external pressure of 2.5 atm from an initial volume of 2.50 L to a final volume of 4.50 L. The change in internal energy ΔU of the gas in joules will be:-
 (1) -500 J (2) -505 J
 (3) $+505 \text{ J}$ (4) 1136.25 J

AIIMS 2017

53. For an isolated system :-
 (1) $q = 0$ and $w = 0$
 (2) $q \neq 0$ and $w = 0$
 (3) $q = 0$ and $w \neq 0$
 (4) $q \neq 0$ and $w \neq 0$
54. An ideal gas is expanded against 42 Pascal external pressure by 10 m^3 . If 350 J heat is given to the system then ΔU of system (in J) will be :-
 (1) -50 J (2) -70 J
 (3) $+50 \text{ J}$ (4) $+70 \text{ J}$

55. The heat of combustion of C_4H_{10} is $-2878 \text{ kJ mol}^{-1}$. If the heats of formation of CO_2 and H_2O are $-393.5 \text{ kJ mol}^{-1}$ and $-285.8 \text{ kJ mol}^{-1}$ then the heat of formation of C_4H_{10} is :-
 (1) $-125.0 \text{ kJ mol}^{-1}$ (2) $126.75 \text{ kJ mol}^{-1}$
 (3) $-402.5 \text{ kJ mol}^{-1}$ (4) $402.5 \text{ kJ mol}^{-1}$
56. During adiabatic expansion of an ideal gas in vacuum :-
 (1) $q = 0$ and $\Delta U \neq 0$ (2) $q \neq 0$ and $\Delta U \neq 0$
 (3) $q \neq 0$ and $\Delta U = 0$ (4) $q = 0$ and $\Delta U = 0$

NEET(UG) 2018

57. The bond dissociation energies of X_2 , Y_2 and XY are in the ratio of $1 : 0.5 : 1$. ΔH for the formation of XY is -200 kJ mol^{-1} . The bond dissociation energy of X_2 will be
 (1) 200 kJ mol^{-1} (2) 100 kJ mol^{-1}
 (3) 800 kJ mol^{-1} (4) 400 kJ mol^{-1}

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58. Which of the following pair is an example of extensive property :-
 (1) P and T (2) C_p and E
 (3) E and V (4) T and V
59. An ideal gas expands from 2 L to 10 L adiabatically against a constant external pressure of 10 atm . Then find the change in internal energy:-
 (1) -80 L-atm (2) Zero
 (3) 80 L-atm (4) 40 L-atm
60. The correct set of intensive properties :-
 (1) P , T (2) V , T (3) P , V (4) V , E

61. Which of the following always increases for spontaneous process :-
 (1) ΔH_{system} (2) ΔS_{system}
 (3) $\Delta S_{\text{system}} - \frac{\Delta H_{\text{system}}}{T}$
 (4) $\Delta G_{\text{system}} = \Delta H_{\text{system}} - T\Delta S_{\text{system}}$
62. 1 mole of ideal gas (A) expands from 2 lit to 4 lit through reversible isothermal process. 3 mol of another ideal gas (B) expands from 2 lit to $x \text{ lit}$ and does the same amount of work as done by gas (A) the value of x is :- (Consider that both the gases temperature is same)
 (1) 2 (2) 8 (3) $(8)^{1/3}$ (4) $(4)^{2/3}$
63. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ $\Delta H = -2220 \text{ kJ mol}^{-1}$
 $C_3H_6 + H_2 \rightarrow C_3H_8$ $\Delta H = -124 \text{ kJ mol}^{-1}$
 $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ $\Delta H = -285 \text{ kJ mol}^{-1}$
 Then find heat of combustion of C_3H_6
 (1) $-2059 \text{ kJ mol}^{-1}$ (2) 2059 kJ mol^{-1}
 (3) $-4118 \text{ kJ mol}^{-1}$ (4) $+4118 \text{ kJ mol}^{-1}$
64. A rigid container contain 1 kg N_2 gas at 400 K if final temperature is 800 K then find change in entropy ($J K^{-1}$) [Given : $C_p = 29.099 \text{ joule}$]
 (1) 512.86 (2) 1025.736
 (3) 256.43 (4) 170.9
65. Heat of formation of ethane $-20.2 \text{ kCal mol}^{-1}$ and heat of atomisation of C and H_2 are $179.2 \text{ kCal mol}^{-1}$ and $52.1 \text{ kCal mol}^{-1}$ respectively. If bond energy of $C-H$ bond is $73.3 \text{ kCal mol}^{-1}$. Then calculate approx. B.E. of $C-C$:-
 (1) 96 kCal mol^{-1} (2) $230 \text{ kCal mol}^{-1}$
 (3) 88 kCal mol^{-1} (4) $540 \text{ kCal mol}^{-1}$

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	1	1	2	3	1	4	2	2	4	1	4	2	4	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	3	3	2	4	4	4	4	3	3	3	4	3	2	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	3	1	1	3	4	2	1	2	3	1	1	1	3	4
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	1	3	2	2	1	2	1	2	1	4	3	3	1	1
Que.	61	62	63	64	65										
Ans.	3	4	1	1	1										

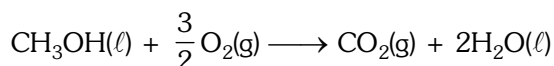
EXERCISE-III (Analytical Questions)

Check Your Understanding

- For the reaction of one mole of zinc dust with one mole of sulphuric acid in a bomb calorimeter, ΔU and w correspond to :-
 (1) $\Delta U < 0$, $w=0$ (2) $\Delta U < 0$, $w < 0$
 (3) $\Delta U > 0$, $w=0$ (4) $\Delta U > 0$, $w > 0$
- Consider the reaction : $N_2 + 3H_2 \rightarrow 2NH_3$ carried out at constant temperature and pressure. If ΔH and ΔU are the enthalpy and internal energy changes for the reaction, which of the following expressions is true ?
 (1) $\Delta H < \Delta U$ (2) $\Delta H > \Delta U$
 (3) $\Delta H = 0$ (4) $\Delta H = \Delta U$
- For a reversible process at $T = 300K$, the volume is increased from $V_i = 1L$ to $V_f = 10L$. Calculate ΔH if the process is isothermal -
 (1) 11.47 kJ (2) 4.98 kJ
 (3) 0 (4) -11.47 kJ
- Assuming that water vapour is an ideal gas, the internal energy change (ΔU) when 1 mol of water is vapourised at 1 bar pressure and $100^\circ C$, (Given : Molar enthalpy of vapourisation of water at 1 bar and $373 K = 41 kJ mol^{-1}$ and $R = 8.3 J mol^{-1} K^{-1}$ will be) :-
 (1) 4.100 kJ mol⁻¹ (2) 3.7904 kJ mol⁻¹
 (3) 37.904 kJ mol⁻¹ (4) 41.00 kJ mol⁻¹
- A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0 mL to 375 mL at a constant temperature of $37.0^\circ C$. As it does so, it absorbs 208 J of heat. The values of q and w for the process will be :-
 ($R = 8.314 J mol^{-1} K^{-1}$) ($\ln 7.5 = 2.01$)
 (1) $q = + 208 J$, $w = - 208 J$
 (2) $q = - 208 J$, $w = - 208 J$
 (3) $q = - 208 J$, $w = + 208 J$
 (4) $q = + 208 J$, $w = + 208 J$
- The entropy change involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 dm³ to a volume of 100 dm³ at $27^\circ C$ is :-
 (1) 32.3 J mol⁻¹ K⁻¹ (2) 42.3 J mol⁻¹ K⁻¹
 (3) 38.3 J mol⁻¹ K⁻¹ (4) 35.8 J mol⁻¹ K⁻¹
- The value of enthalpy change (ΔH) for the reaction $C_2H_5OH(l) + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 3H_2O(l)$ at $27^\circ C$ is $-1366.5 kJ mol^{-1}$. The value of internal energy change for the above reaction at this temperature will be :-
 (1) -1371.5 kJ (2) -1369.0 kJ
 (3) -1364.0 kJ (4) -1361.5 kJ
- For the process $H_2O(l)$ (1 bar, 373K) $\rightarrow$ $H_2O(g)$ (1 bar, 373K), the correct set of thermodynamic parameters is :
 (1) $\Delta G = 0$, $\Delta S = +ve$
 (2) $\Delta G = 0$, $\Delta S = -ve$
 (3) $\Delta G = +ve$, $\Delta S = 0$
 (4) $\Delta G = -ve$, $\Delta S = +ve$
- Which of the following pairs of a chemical reaction is certain to result in a spontaneous reaction?
 (1) endothermic and decreasing disorder
 (2) exothermic and increasing disorder
 (3) endothermic and increasing disorder
 (4) exothermic and decreasing disorder
- The conversion A to B is carried out by the following path :

$$\begin{array}{ccc} C & \rightarrow & D \\ \uparrow & & \downarrow \\ A & & B \end{array}$$
 Given : $\Delta S_{(A \rightarrow C)} = 50 \text{ e.u.}$,
 $\Delta S_{(C \rightarrow D)} = 30 \text{ e.u.}$, $\Delta S_{(B \rightarrow D)} = 20 \text{ e.u.}$
 where e.u. is entropy unit then $\Delta S_{(A \rightarrow B)}$ is
 (1) + 100 e.u.
 (2) + 60 e.u.
 (3) - 100 e.u.
 (4) - 60 e.u.
- In conversion of lime-stone to lime,
 $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
 the values of ΔH° and ΔS° are $+179.1 kJ mol^{-1}$ and $160.2 J K^{-1}$ respectively at 298 K and 1 bar. Assuming that ΔH° and ΔS° do not change with temperature, temperature above which conversion of limestone to lime will be spontaneous is :-
 (1) 1008 K
 (2) 1200 K
 (3) 845 K
 (4) 1118 K

12. In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is

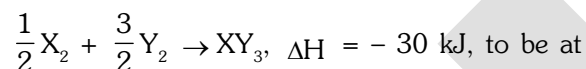


At 298 K standard Gibbs energies of formation for $\text{CH}_3\text{OH}(\ell)$, $\text{H}_2\text{O}(\ell)$ and $\text{CO}_2(\text{g})$ are -166.2 , -237.2 and $-394.4 \text{ kJ mol}^{-1}$ respectively. If standard enthalpy of combustion of methanol is -726 kJ mol^{-1} , efficiency of the fuel cell will be
(1) 90% (2) 97% (3) 80% (4) 87%

13. Identify the correct statement regarding a spontaneous process :-

- (1) For a spontaneous process in an isolated system, the change in entropy is positive
(2) Endothermic processes are never spontaneous
(3) Exothermic processes are always spontaneous
(4) Lowering of energy in the reaction process is the only criterion for spontaneity

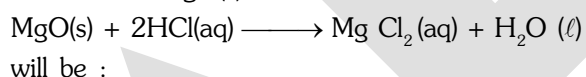
14. Standard entropy of X_2 , Y_2 and XY_3 are 60, 40 and $50 \text{ JK}^{-1} \text{ mol}^{-1}$, respectively. For the reaction,



equilibrium, the temperature will be

- (1) 1250 K (2) 500 K (3) 750 K (4) 1000 K

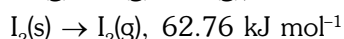
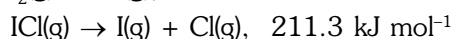
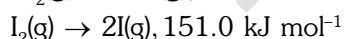
15. The absolute enthalpy of neutralisation of the reaction for $\text{MgO}(\text{s})$:



will be :

- (1) $57.33 \text{ kJ mol}^{-1}$
(2) $-57.33 \text{ kJ mol}^{-1}$
(3) Greater than $-57.33 \text{ kJ mol}^{-1}$
(4) Less than $-57.33 \text{ kJ mol}^{-1}$

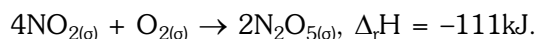
16. The enthalpy changes for the following processes are listed below :



Given that the standard states for iodine and chlorine are $\text{I}_2(\text{s})$ and $\text{Cl}_2(\text{g})$, the standard enthalpy of formation for $\text{ICl}(\text{g})$ is :-

- (1) $-16.8 \text{ kJ mol}^{-1}$ (2) $+16.8 \text{ kJ mol}^{-1}$
(3) $+244.8 \text{ kJ mol}^{-1}$ (4) $-14.6 \text{ kJ mol}^{-1}$

17. Consider the reaction :



If $\text{N}_2\text{O}_{5(\text{s})}$ is formed instead of $\text{N}_2\text{O}_{5(\text{g})}$ in the above reaction, the $\Delta_r H$ value will be :-

(given, ΔH of sublimation for N_2O_5 is 54 kJ mol^{-1})

- (1) -165 kJ (2) $+54 \text{ kJ}$
(3) $+219 \text{ kJ}$ (4) -219 kJ

18. For complete combustion of ethanol, $\text{C}_2\text{H}_5\text{OH}(\ell) + 3\text{O}_{2(\text{g})} \rightarrow 2\text{CO}_{2(\text{g})} + 3\text{H}_2\text{O}(\ell)$, the amount of heat produced as measured in bomb calorimeter, is $1364.47 \text{ kJ mol}^{-1}$ at 25°C . Assuming ideality the Enthalpy of combustion, $\Delta_c H$, for the reaction will be :- ($R = 8.314 \text{ kJ mol}^{-1}$)

- (1) $-1460.50 \text{ kJ mol}^{-1}$
(2) $-1350.50 \text{ kJ mol}^{-1}$
(3) $-1366.95 \text{ kJ mol}^{-1}$
(4) $-1361.95 \text{ kJ mol}^{-1}$

19. If the bond dissociation energies of XY , X_2 and Y_2 (all diatomic molecules) are in the ratio of 1 : 1 : 0.5 and $\Delta_f H$ for the formation of XY is -200 kJ mol^{-1} . The bond dissociation energy of X_2 will be :-

- (1) 200 kJ mol^{-1} (2) 100 kJ mol^{-1}
(3) 800 kJ mol^{-1} (4) 300 kJ mol^{-1}

20. The standard enthalpy of formation ($\Delta_f H^\circ$) at 298K for methane, $\text{CH}_4(\text{g})$, is $-74.8 \text{ kJ mol}^{-1}$. The additional information required to determine the average energy for C-H bond formation would be:-

- (1) Latent heat of vapourization of methane
(2) The first four ionization energies of carbon and electron gain enthalpy of hydrogen
(3) The dissociation energy of hydrogen molecule H_2
(4) The dissociation energy of H_2 and enthalpy of sublimation of carbon

21. The standard enthalpy of formation of NH_3 is $-46.0 \text{ kJ mol}^{-1}$. If the enthalpy of formation of H_2 from its atoms is -436 kJ mol^{-1} and that of N_2 is -712 kJ mol^{-1} , the average bond enthalpy of N-H bond in NH_3 is :-

- (1) $-1102 \text{ kJ mol}^{-1}$
(2) -964 kJ mol^{-1}
(3) $+352 \text{ kJ mol}^{-1}$
(4) $+1056 \text{ kJ mol}^{-1}$

22. The incorrect expression among the following is :-

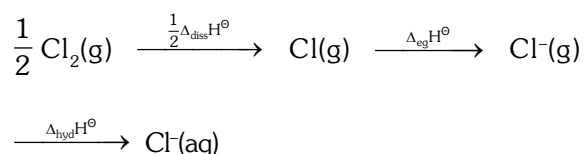
(1) $K = e^{-\Delta G^\circ/RT}$

(2) $\frac{\Delta G_{\text{system}}}{\Delta S_{\text{total}}} = -T$

(3) In isothermal process, $W_{\text{reversible}} = -nRT \ln \frac{V_f}{V_i}$

(4) $\ln K = \frac{\Delta H^\circ - T\Delta S^\circ}{RT}$

23. Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below:



The energy involved in the conversion of $\frac{1}{2} \text{Cl}_2(\text{g})$ to $\text{Cl}^-(\text{aq})$

(using the data $\Delta_{\text{diss}} H_{\text{Cl}_2}^\circ = 240 \text{ kJ mol}^{-1}$,

$\Delta_{\text{eg}} H_{\text{Cl}}^\circ = -349 \text{ kJ mol}^{-1}$, $\Delta_{\text{hyd}} H_{\text{Cl}^-}^\circ = -381 \text{ kJ mol}^{-1}$)

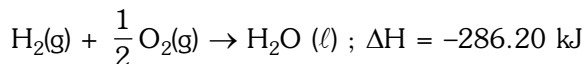
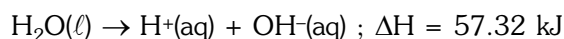
will be:-

(1) -610 kJ mol^{-1} (2) -850 kJ mol^{-1}

(3) $+120 \text{ kJ mol}^{-1}$ (4) $+152 \text{ kJ mol}^{-1}$

24. On the basis of the following thermochemical

data : $[\Delta H_f^\circ (\text{H}_{(\text{aq})}^+) = 0]$



The value of enthalpy of formation of OH^- ion at 25°C is :-

(1) $+228.88 \text{ kJ}$ (2) -343.52 kJ

(3) -22.88 kJ (4) -228.88 kJ

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	3	3	1	3	3	1	2	2	4	2	1	3	3
Que.	16	17	18	19	20	21	22	23	24						
Ans.	2	4	3	3	4	3	4	1	4						

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** There is no reaction known for which ΔG is positive, yet it is spontaneous.
Reason :- For photochemical reactions, ΔG is negative.
 (1) A (2) B (3) C (4) D
- Assertion :-** Absolute value of enthalpy cannot be determined.
Reason :- Enthalpy is defined as $H=E+PV$, and value of internal energy cannot be determined absolutely therefore absolute value of enthalpy can not be determined.
 (1) A (2) B (3) C (4) D
- Assertion :-** When a rubber band is stretched entropy increases.
Reason :- During expansion entropy increases.
 (1) A (2) B (3) C (4) D
- Assertion :-** At constant pressure for the change $H_2O(s) \rightarrow H_2O(g)$ work done is negative.
Reason :- During phase transition work done is always negative.
 (1) A (2) B (3) C (4) D
- Assertion :-** Enthalpy of graphite is lower than that of diamond.
Reason :- Entropy of graphite is greater than that of diamond.
 (1) A (2) B (3) C (4) D
- Assertion :-** The enthalpy of formation of gaseous oxygen molecules at 298K and under a pressure of one atm. is zero.
Reason :- The entropy of formation of gaseous oxygen molecules under the same condition is zero.
 (1) A (2) B (3) C (4) D
- Assertion :-** Many endothermic reactions that are not spontaneous at room temp. becomes spontaneous at high temp.
Reason :- Entropy of the system increases with increase in temperature.
 (1) A (2) B (3) C (4) D
- Assertion :-** Mass, Volume and pressure are extensive properties.
Reason :- Extensive properties depend upon the amount of the substance.
 (1) A (2) B (3) C (4) D
- Assertion :-** For a particular reaction, heat of combustion at constant pressure (q_p) is always greater than that at constant volume (q_v).
Reason :- Combustion reactions are invariably accomplished by increase in no. of moles.
 (1) A (2) B (3) C (4) D
- Assertion :-** At constant temp 0°C and 1atm, the change $H_2O(s) \rightarrow H_2O(l)$ ΔH and ΔE both are zero.
Reason :- During isothermal process H and E both remains constant.
 (1) A (2) B (3) C (4) D
- Assertion :-** The change in internal energy (ΔE) for the vapourization of one mole of water at 1 atm and 373 K is zero.
Reason :- For all isothermal processes $\Delta E=0$
 (1) A (2) B (3) C (4) D
- Assertion :-** Water in liquid state is more stable than ice at room temperature.
Reason :- Water in liquid form has higher entropy than ice.
 (1) A (2) B (3) C (4) D
- Assertion :-** In an isolated system the entropy increases due to spontaneous process.
Reason :- The processes in an isolated system are isothermal.
 (1) A (2) B (3) C (4) D
- Assertion :-** Entropy is always constant for a closed system.
Reason :- Closed system is always reversible
 (1) A (2) B (3) C (4) D

15. **Assertion :-** For an isolated system q is zero.
Reason :- In an isolated system change in U is zero.
 (1) A (2) B (3) C (4) D
16. **Assertion :-** Entropy of system always increases for a spontaneous reaction.
Reason :- Enthalpy of reaction always decreases for spontaneous reaction.
 (1) A (2) B (3) C (4) D
17. **Assertion :-** Catalyst changes Gibbs free energy of system.
Reason :- Catalyst changes preexponential factor of a chemical reaction.
 (1) A (2) B (3) C (4) D
18. **Assertion :-** Entropy increases with increase in temperature of gas.
Reason :- If q_{rev} is same, then entropy of that system is higher for which temperature is higher.
 (1) A (2) B (3) C (4) D
19. **Assertion :-** The value of specific heat at constant pressure is more than that at constant volume.
Reason :- The energy required to raise temperature by a unit at constant pressure is greater because some amount of heat is used in doing work.
 (1) A (2) B (3) C (4) D
20. **Assertion :** Sum of two path functions is always path function.
Reason : Sum of path function depends on path followed.
 (1) A (2) B (3) C (4) D
21. **Assertion :-** Heat is not a state thermodynamic property.
Reason :- Heat given in a process depends on its type.
 (1) A (2) B (3) C (4) D
22. **Assertion :-** For ideal gas in a closed container with adiabatic walls, temperature decreases when work is done by the system.
Reason :- Internal energy is used in work done by the system.
 (1) A (2) B (3) C (4) D
23. **Assertion :-** Adiabatic free expansion of an ideal gas is irreversible.
Reason :- $PV = \text{constant}$ for reversible adiabatic expansion.
 (1) A (2) B (3) C (4) D
24. **Assertion :-** Internal energy at particular temperature does not depend on volume.
Reason :- State function does not depend on volume.
 (1) A (2) B (3) C (4) D
25. **Assertion :-** Graphite is thermodynamically most stable allotrope of carbon.
Reason :- Standard heat of formation of graphite is $+1.9 \text{ kJ/mol}$.
 (1) A (2) B (3) C (4) D
26. **Assertion :** A gas expands from (P_1, V_1, T_1) to (P_2, V_2, T_2) the work done in 2 steps will be more than, if the process is carried out in single step.
Reason : Work done is path dependent quantity.
 (1) A (2) B (3) C (4) D
27. **Assertion :** All spontaneous process in nature are irreversible.
Reason : All reversible process are quasistatic.
 (1) A (2) B (3) C (4) D
28. **Assertion :-** Sum of q and w is different in reversible and irreversible process.
Reason :- Sum of two path function can never state function.
 (1) A (2) B (3) C (4) D
29. **Assertion :** Internal energy of gas remain constant during adiabatic free expansion.
Reason : Work done is zero and heat exchange is zero.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	1	4	3	2	3	2	4	4	4	4	2	1	4	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	
Ans.	4	4	3	1	4	1	1	3	3	3	1	2	4	1	

REDOX REACTION

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E

NEET SYLLABUS

REDOX REACTIONS : Concept of oxidation and reduction, redox reactions oxidation number, balancing redox reactions in terms of loss and gain of electron and change in oxidation numbers.

@neetfocusquizz

OBJECTIVES

After studying this unit, we will be able to :

- *identify redox reactions as a class of reactions in which oxidation and reduction reactions occur simultaneously;*
- *define the terms oxidation, reduction, oxidant (oxidising agent) and reductant (reducing agent);*
- *explain mechanism of redox reactions by electron transfer process;*
- *use the concept of oxidation number to identify oxidant and reductant in a reaction;*
- *classify redox reaction into combination (synthesis), decomposition, displacement and disproportionation reactions;*
- *suggest a comparative order among various reductants and oxidants;*
- *balance chemical equations using*
 - (i) oxidation number*
 - (ii) ion electron method;*

"A person starts to live when he can live outside himself."

Albert Einstein

REDOX REACTION

6.0 Introduction :

Redox reactions shows vital role in non renewable energy sources. In cell reactions where oxidation and reduction both occurs simultaneously will have redox reaction for interconversion of energy.

6.1 Redox Reaction (Oxidation-Reduction) :

Many chemical reactions involve transfer of electrons from one chemical substance to another. These electron-transfer reactions are termed as **oxidation-reduction** or **redox reactions**.

Or

Those reactions which involve oxidation and reduction both simultaneously are known as oxidation reduction or redox reactions.

Or

Those reactions in which increase and decrease in oxidation number of same or different atoms occurs are known as redox reactions.

6.2 Oxidation State :

Oxidation state of an atom in a molecule or ion is the hypothetical or real charge present on an atom due to electronegativity difference.

Or

Oxidation state of an element in a compound represents the number of electrons lost or gained during its change from free state into that compound.

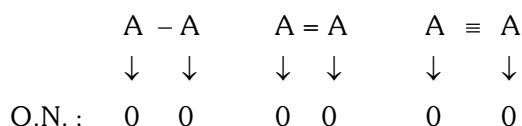
Some important points concerning oxidation number :

- (1) Electronegativity values of no two elements are same –
 $P > H$ $C > H$ $S > C$ $Cl > N$
- (2) Oxidation number of an element may be positive or negative.
- (3) Oxidation number can be zero, whole number or a fractional value.
 Ex. $Ni(CO)_4 \Rightarrow$ O.S of Ni = 0
 $N_3H \Rightarrow$ O.S of N = $-1/3$
 $HCl \Rightarrow$ O.S of Cl = -1
- (4) Oxidation state of same element can be different in same or different compounds.
 Ex. $H_2S \Rightarrow$ O.S of S = -2
 $H_2SO_3 \Rightarrow$ O.S of S = $+4$
 $H_2SO_4 \Rightarrow$ O.S of S = $+6$

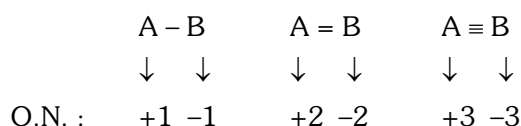
6.3 Some helping rules for calculating oxidation number :

(A) In case of covalent bond :

(i) For homoatomic molecule



(ii) For heteroatomic molecule (EN of B > A)



(iii) The oxidation state of an element in its free state is zero. Example- Oxidation state of Na, Cu, I, Cl, O etc. are zero.

(iv) Oxidation state of atoms present in homoatomic molecules is zero.

Ex. H_2 , O_2 , N_2 , P_4 , S_8 = zero

(v) Oxidation state of an element in any of its allotropic form is zero.

Ex. C_{Diamond} , C_{Graphite} , $S_{\text{Monoclinic}}$, S_{Rhombic} = 0

(vi) Oxidation state of all the components of an alloy are 0.

Ex. $(Na - Hg)$
 $\downarrow \quad \downarrow$
 0 0

(vii) In complex compounds, oxidation state of some neutral molecules (ligands) is zero.

Ex. CO, NO, NH_3 , H_2O .

(viii) Oxidation state of fluorine in all its compounds is -1.

(ix) Oxidation state of IA & II A group elements are +1 and +2 respectively.

(x) Oxidation state of hydrogen in most of its compounds is +1 except in metal hydrides (-1)

Ex. NaH LiH CaH_2 MgH_2
 $\downarrow \downarrow$ $\downarrow \downarrow$ $\downarrow \downarrow$ $\downarrow \downarrow$
 O.S. : +1 -1 +1 -1 +2 -1 +2 -1

(xi) Oxidation state of oxygen in most of its compounds is -2 except in -

(a) Peroxides (O_2^{-2}) → Oxidation state (O) = -1

Ex. H_2O_2 , BaO_2

(b) Super Oxides (O_2^{-1}) → Oxidation state (O) = -1/2

Ex. KO_2
 $\downarrow$
 -1/2

(c) Ozonide (O_3^{-1}) → Oxidation state (O) = -1/3

Ex. KO_3
 $\downarrow$
 -1/3

(d) OF_2 (Oxygen difluoride)

$F - O - F$
 $\downarrow$

Oxidation state (O) = + 2

(e) O_2F_2 (dioxygen difluoride)

$\downarrow$
 Oxidation state (O) = + 1

(xii) Oxidation state of monoatomic ions is equal to the charge present on the ion.

Ex. $Mg^{+2} \rightarrow$ Oxidation state = +2

(xiii) The algebraic sum of oxidation state of all the atoms present in a polyatomic neutral molecule is 0.

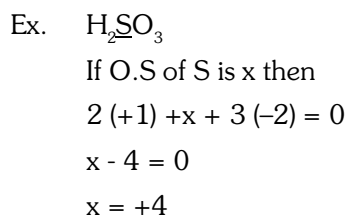
Ex. H_2SO_4

If O.S of S is x then

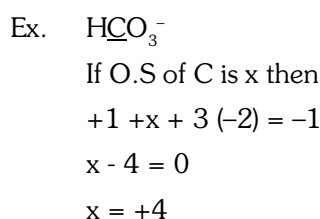
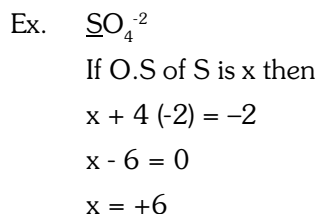
$2(+1) + x + 4(-2) = 0$

$x - 6 = 0$

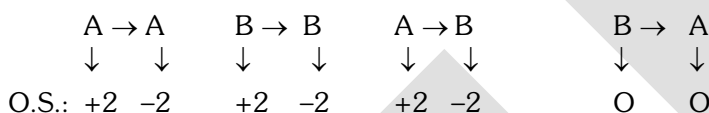
$x = +6$



(xiv) The algebraic sum of oxidation state of all the atoms in a polyatomic ion is equal to the charge present on the ion.



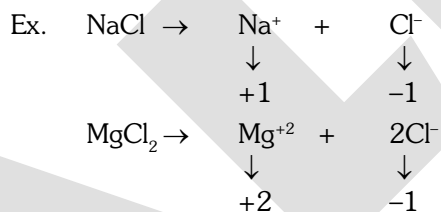
(B) In case of co-ordinate bond (EN of B > A) :



(C) In case of Ionic bond :

Charge on cation = O.S of cation

Charge on anion = O.S of anion



Illustrations

Illustration 1. Oxidation number of cobalt in $[\text{Co}(\text{NH}_3)_6] \text{Cl}_2\text{Br}$ is –
 (1) +6 (2) Zero (3) +3 (4) +2

Solution. Let the oxidation number of Co be x
 Oxidation number of NH_3 is zero
 Oxidation number of Cl is -1
 Oxidation number of Br is -1
 Hence, $x + 6(0) - (1 \times 2) - 1 = 0$
 $\therefore x = +3$

So, the oxidation number of cobalt in the given complex compound is +3.

Illustration 2. The order of increasing oxidation numbers of S in S_8 , $S_2O_8^{-2}$, $S_2O_3^{-2}$, $S_4O_6^{-2}$ is given below –

- (1) $S_8 < S_2O_8^{-2} < S_2O_3^{-2} < S_4O_6^{-2}$ (2) $S_2O_8^{-2} < S_2O_3^{-2} < S_4O_6^{-2} < S_8$
 (3) $S_2O_8^{-2} < S_8 < S_4O_6^{-2} < S_2O_3^{-2}$ (4) $S_8 < S_2O_3^{-2} < S_4O_6^{-2} < S_2O_8^{-2}$

Solution. The oxidation number of S are shown below along with the compounds

S_8	$S_2O_8^{-2}$	$S_2O_3^{-2}$	$S_4O_6^{-2}$
0	+6	+2	+2.5

Hence the order of increasing oxidation state of S is –

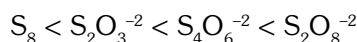


Illustration 3. The oxidation number of Cl in $NOClO_4$ is –

- (1) +11 (2) +9 (3) +7 (4) +5

Solution. The compound may be written as $NO^+ ClO_4^-$.

For ClO_4^- , Let oxidation number of Cl = a

$$a + 4 \times (-2) = -1$$

$$a = +7$$

Hence, the oxidation number of Cl in $NOClO_4$ is +7

Illustration 4. The two possible oxidation states of N atoms in NH_4NO_3 are respectively –

- (1) +3, +5 (2) +3, -5 (3) -3, +5 (4) -3, -5

Solution. There are two N atoms in NH_4NO_3 , but one N atom has negative oxidation states (attached to H) and the other has positive oxidation states (attached to O). Therefore evaluation should be made separately as –

Oxidation states of N is NH_4^+

$$a + 4 \times (+1) = +1$$

$$\therefore a = -3$$

Oxidation states of N in NO_3^-

$$\text{and } a + 3(-2) = -1$$

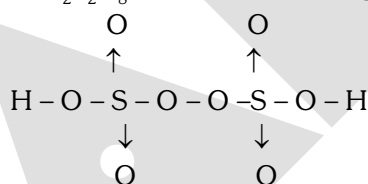
$$\therefore a = +5$$

Here the two oxidation states are -3 and +5 respectively.

Illustration 5. The oxidation states of S in $H_2S_2O_8$ is –

- (1) +8 (2) -8 (3) +6 (4) +4

Solution. In $H_2S_2O_8$, two O atoms form peroxide linkage i.e.



$$2 \times 1 + 2a + 6(-2) + 2(-1) = 0$$

$$\therefore a = +6$$

Thus the oxidation states of S in $H_2S_2O_8$ is +6

Illustration 6. The oxidation number of S in $(CH_3)_2SO$ is –

- (1) 1 (2) 2 (3) 0 (4) 3

Solution. Let the oxidation number of S is 'a'

Oxidation number of $CH_3 = +1$

Oxidation number of O = -2

$$2(+1) + a + (-2) = 0$$

$$a = 0$$

Hence the oxidation no. of S in dimethyl sulphoxide is zero.

BEGINNER'S BOX-1

- In which of the following compounds, the oxidation state of I-atom is highest ?
(1) KI_3 (2) KIO_4 (3) KIO_3 (4) IF_5
- The oxidation number of phosphorus in $\text{Ba}(\text{H}_2\text{PO}_2)_2$ is –
(1) +3 (2) +2 (3) +1 (4) –1
- Oxidation number of Ni in $\text{Ni}(\text{CO})_4$ is –
(1) 0 (2) 4 (3) 8 (4) 2
- Positive oxidation state of an element indicates that it is –
(1) Elementary form (2) Oxidised (3) Reduced (4) Only reductant
- Predict the highest and lowest oxidation state of (a) Ti and (b) Tl in combined state.
(1) a[0, +3] b[0, +2] (2) a[+3, 0] b[+4, 0] (3) a[+4, 0] b[+4, 0] (4) a[+4, +2] b[+3, +1]
- The oxidation state of oxygen atom in potassium superoxide is –
(1) Zero (2) $-1/2$ (3) –1 (4) –2

6.4 APPLICATIONS OF OXIDATION NUMBER :

(A) To compare the strength of acid and base :

Strength of acid $\propto$ Oxidation Number

Strength of base $\propto \frac{1}{\text{Oxidation Number}}$

Example : Order of acidic strength in HClO , HClO_2 , HClO_3 , HClO_4 will be.

Solution : Oxidation Number of chlorine

HClO (Hypo chlorous acid) +1

HClO_2 (Chlorous acid) +3

HClO_3 (Chloric acid) +5

HClO_4 (Perchloric acid) +7

$\therefore$ Strength of acid $\propto$ Oxidation Number

So the order will be -

$\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HClO}$

(B) To determine the oxidising and reducing nature of the substances :

Oxidising agents are the substances which accept electrons in a chemical reaction i.e., electron acceptors are oxidising agent.

Reducing agents are the substances which donate electrons in a chemical reaction i.e., electron donors are reducing agent.

Highest O.S.	+4	+5	+5	+6	+7	+6	+7	+8	+8	+2	+1
Elements	C	N	P	S	Cl	Cr	Mn	Os	Ru	O	H
Lowest O.S.	-4	-3	-3	-2	-1	0	0	0	0	-2	-1

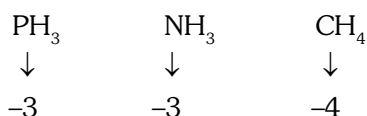
- (a) If effective element in a compound is present in maximum oxidation state then the compound acts as oxidising agent.

Ex. KMnO_4 $\text{K}_2\text{Cr}_2\text{O}_7$ H_2SO_4 SO_3 H_3PO_4 HNO_3 HClO_4

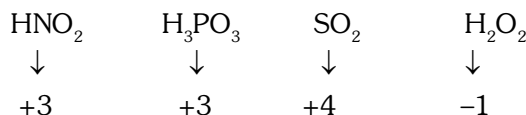
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+7 +6 +6 +6 +5 +5 +7

- (b) If effective element in a compound is present in minimum oxidation state then the compound acts as reducing agent.



- (c) If effective element in a compound is present in intermediate oxidation state then the compound can act as oxidising agent as well as reducing agent.

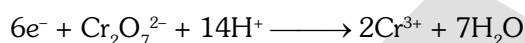


(C) To calculate the equivalent weight of compounds :

The equivalent weight of an oxidising agent or reducing agent is that weight which accepts or loses one mole electrons in a chemical reaction.

(a) Equivalent weight of oxidant = $\frac{\text{Molecular weight}}{\text{No. of electrons gained by one mole}}$

Example : In acidic medium



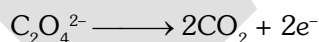
Here atoms which undergoes reduction is Cr. Its O. S. is decreasing from +6 to +3

$$\text{Equivalent weight of } \text{K}_2\text{Cr}_2\text{O}_7 = \frac{\text{Molecular weight of } \text{K}_2\text{Cr}_2\text{O}_7}{3 \times 2} = \frac{M}{6}$$

Note :- [6 in denominator indicates that 6 electrons were gained by $\text{Cr}_2\text{O}_7^{2-}$ as it is clear from the given balanced equation]

(b) Equivalent weight of a reductant = $\frac{\text{Molecular weight}}{\text{No. of electrons lost by one mole}}$

In acidic medium,

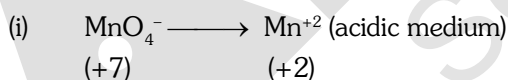


Here, atoms which undergoes oxidation is C. Its oxidation state is increasing from +3 to +4.

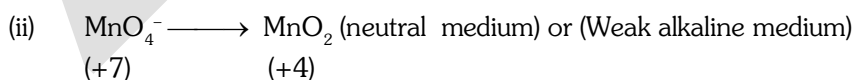
Here, Total electrons lost in $\text{C}_2\text{O}_4^{2-} = 2$ So, equivalent weight of $\text{C}_2\text{O}_4^{2-} = \frac{M}{2}$

- (c) In different conditions a compound may have different equivalent weight because, it depends upon the number of electrons gained or lost by that compound in that reaction.

Example :

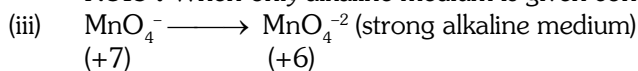


Here 5 electrons are taken by MnO_4^- so its equivalent weight = $\frac{M}{5} = \frac{158}{5} = 31.6$



Here, only 3 electrons are gained by MnO_4^- so its equivalent weight = $\frac{M}{3} = \frac{158}{3} = 52.7$

Note : When only alkaline medium is given consider it as weak alkaline medium.

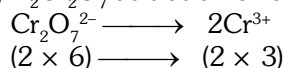


Here, only one electron is gained by MnO_4^- equivalent weight = $\frac{M}{1} = 158$

Note :- KMnO_4 acts as an oxidant in every medium although with different strength which follows the order –

acidic medium > neutral medium > alkaline medium

while, $\text{K}_2\text{Cr}_2\text{O}_7$ acts as an oxidant only in acidic medium as follows



Here, 6 electrons are gained by $\text{K}_2\text{Cr}_2\text{O}_7$ equivalent weight = $\frac{M}{6} = \frac{294}{6} = 49$

(D) To determine the possible molecular formula of compound :

Since the sum of oxidation number of all the atoms present in a compound is zero, so the validity of the formula can be confirmed.

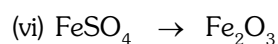
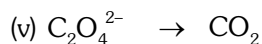
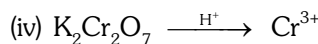
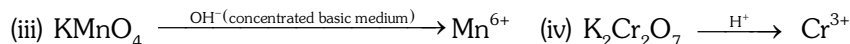
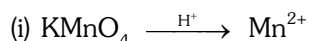
GOLDEN KEY POINTS

SOME OXIDIZING AGENTS/REDUCING AGENTS WITH EQUIVALENT WEIGHT :

Species	Changed to	Reaction	Electrons exchanged or change in O.N.	Eq. wt.
MnO_4^- (O.A.)	Mn^{+2} in acidic medium	$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	5	$E = \frac{M}{5}$
MnO_4^- (O.A.)	MnO_2 in neutral medium or in weak alkaline medium	$\text{MnO}_4^- + 3\text{e}^- + 2\text{H}_2\text{O} \longrightarrow \text{MnO}_2 + 4\text{OH}^-$	3	$E = \frac{M}{3}$
MnO_4^- (O.A.)	MnO_4^{2-} in strong alkaline medium	$\text{MnO}_4^- + \text{e}^- \longrightarrow \text{MnO}_4^{2-}$	1	$E = \frac{M}{1}$
$\text{Cr}_2\text{O}_7^{2-}$ (O.A.)	Cr^{3+} in acidic medium	$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	6	$E = \frac{M}{6}$
MnO_2 (O.A.)	Mn^{2+} in acidic medium	$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^- \longrightarrow \text{Mn}^{2+} + 2\text{H}_2\text{O}$	2	$E = \frac{M}{2}$
Cl_2 (O.A.) in bleaching powder	Cl^-	$\text{Cl}_2 + 2\text{e}^- \longrightarrow 2\text{Cl}^-$	2	$E = \frac{M}{2}$
CuSO_4 (O.A.) in iodometric titration	Cu^+	$\text{Cu}^{2+} + \text{e}^- \longrightarrow \text{Cu}^+$	1	$E = \frac{M}{1}$
$\text{S}_2\text{O}_3^{2-}$ (R.A.)	$\text{S}_4\text{O}_6^{2-}$	$2\text{S}_2\text{O}_3^{2-} \longrightarrow \text{S}_4\text{O}_6^{2-} + 2\text{e}^-$	2 (for two moles)	$E = \frac{2M}{2} = M$
H_2O_2 (O.A.)	H_2O	$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow 2\text{H}_2\text{O}$	2	$E = \frac{M}{2}$
H_2O_2 (R.A.)	O_2	$\text{H}_2\text{O}_2 \longrightarrow \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$ (O.N. of oxygen in H_2O_2 is -1 per atom)	2	$E = \frac{M}{2}$
Fe^{2+} (R.A.)	Fe^{3+}	$\text{Fe}^{2+} \longrightarrow \text{Fe}^{3+} + \text{e}^-$	1	$E = \frac{M}{1}$
I^- (R.A.)	I_2 (in acidic medium)	$2\text{I}^- \longrightarrow \text{I}_2 + 2\text{e}^-$	2 (for two moles)	$E = \frac{M}{1}$
I^- (R.A.)	IO_3^- (in basic medium)	$\text{I}^- + 6\text{OH}^- \longrightarrow \text{IO}_3^- + 3\text{H}_2\text{O} + 6\text{e}^-$	6	$E = \frac{M}{6}$

Illustrations

Illustration 7. Find the n-factor of reactant in the following chemical changes.

**Solution**

(i) In this reaction, KMnO_4 which is an oxidizing agent, itself gets reduced to Mn^{2+} under acidic conditions.

$$n = |1 \times (+7) - 1 \times (+2)| = 5$$

(ii) In this reaction, KMnO_4 gets reduced to Mn^{4+} under neutral or slightly (weakly) basic conditions.

$$n = |1 \times (+7) - 1 \times (+4)| = 3$$

(iii) In this reaction, KMnO_4 gets reduced to Mn^{6+} under basic conditions.

$$n = |1 \times (+7) - 1 \times (+6)| = 1$$

(iv) In this reaction, $\text{K}_2\text{Cr}_2\text{O}_7$ which acts as an oxidizing agent reduced to Cr^{3+} under acidic conditions. (It does not react under basic conditions.)

$$n = |2 \times (+6) - 2 \times (+3)| = 6$$

(v) In this reaction, $\text{C}_2\text{O}_4^{2-}$ (oxalate ion) gets oxidized to CO_2 when it is reacted with an oxidizing agent.

$$n = |2 \times (+3) - 2 \times (+4)| = 2$$

(vi) In this reaction, ferrous ions get oxidized to ferric ions.

$$n = |1 \times (+2) - 1 \times (+3)| = 1$$

(vii) In this reaction, ferric ions are getting reduced to ferrous ions.

$$n = |2 \times (+3) - 2 \times (+2)| = 2$$

Illustration 8. Suppose that there are three atoms A, B, C and their oxidation numbers are 6, -1, -2, respectively. Then the molecular formula of compound will be.

Solution

Since, the charge on a free compound is zero. So

$$+6 = (-1 \times 4) + (-2)$$

$$+6 = -6$$

$$\text{or } +6 = (-1 \times 2) + (-2 \times 2)$$

$$= -2 + (-4) = -6$$

So molecular formula, AB_4C or AB_2C_2 .

BEGINNER'S BOX-2

- Molecular weight of KMnO_4 in acidic medium and neutral medium will be respectively –
 (1) $7 \times$ equivalent weight and $2 \times$ equivalent weight
 (2) $5 \times$ equivalent weight and $3 \times$ equivalent weight
 (3) $4 \times$ equivalent weight and $5 \times$ equivalent weight
 (4) $2 \times$ equivalent weight and $4 \times$ equivalent weight
- In acidic medium, equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ (Molecular weight = M) is –
 (1) $M/3$ (2) $M/4$
 (3) $M/6$ (4) $M/2$

6.5 OXIDATION AND REDUCTION :

There are two concepts of oxidation and reduction.

(A) Classical/old concept :

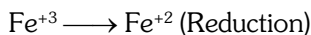
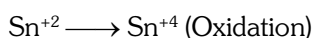
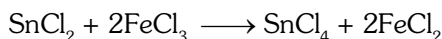
OXIDATION	REDUCTION
(1) Addition of O_2 $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$	Addition of H_2 $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$
(2) Removal of H_2 $\text{H}_2\text{S} + \text{Cl}_2 \rightarrow 2\text{HCl} + \text{S}$ (oxidation of H_2S) $4\text{HI} + \text{O}_2 \rightarrow 2\text{I}_2 + 2\text{H}_2\text{O}$ (oxidation of HI)	Removal of O_2 $\text{CuO} + \text{C} \rightarrow \text{Cu} + \text{CO}$ (reduction of CuO) $\text{H}_2\text{O} + \text{C} \rightarrow \text{CO} + \text{H}_2$ (reduction of H_2O)
(3) Addition of electronegative element $\text{Fe} + \text{S} \rightarrow \text{FeS}$ (oxidation of Fe) $\text{SnCl}_2 + \text{Cl}_2 \rightarrow \text{SnCl}_4$ (oxidation of SnCl_2)	Addition of electropositive element $\text{CuCl}_2 + \text{Cu} \rightarrow \text{Cu}_2\text{Cl}_2$ (reduction of CuCl_2) $\text{HgCl}_2 + \text{Hg} \rightarrow \text{Hg}_2\text{Cl}_2$ (reduction of HgCl_2)
(4) Removal of electropositive element $2\text{NaI} + \text{H}_2\text{O}_2 \rightarrow 2\text{NaOH} + \text{I}_2$ (oxidation of NaI)	Removal of electronegative element $2\text{FeCl}_3 + \text{H}_2 \rightarrow 2\text{FeCl}_2 + 2\text{HCl}$ (reduction of FeCl_3)

(B) Electronic/Modern Concept :

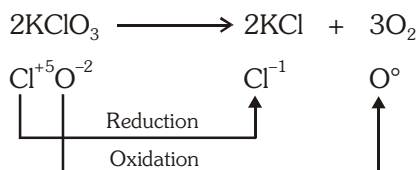
OXIDATION	REDUCTION
(1) De-electronation	Electronation
(2) Oxidation process are those process in which one or more e^- s are lost by an atom, ion or molecule.	Reduction process are those process in which one or more e^- s are gained by an atom, ion or molecule.
(3) Example - (a) $\text{Zn} \rightarrow \text{Zn}^{+2} + 2e^-$ $\text{M} \rightarrow \text{M}^{n+} + ne^-$ (b) $\text{Sn}^{+2} \rightarrow \text{Sn}^{+4} + (4-2)e^-$ $\text{M}^{+n_1} \rightarrow \text{M}^{+n_2} + (n_2-n_1)e^-$ (c) $\text{Cl}^- \rightarrow \text{Cl} + e^-$ $\text{A}^{-n} \rightarrow \text{A} + ne^-$ (d) $\text{MnO}_4^{2-} \rightarrow \text{MnO}_4^- + (2-1)e^-$ $\text{A}^{-n_1} \rightarrow \text{A}^{-n_2} + (n_1-n_2)e^-$	$\text{Cu}^{+2} + 2e^- \rightarrow \text{Cu}$ $\text{M}^{n+} + ne^- \rightarrow \text{M}$ $\text{Fe}^{+3} + (3-2)e^- \rightarrow \text{Fe}^{+2}$ $\text{M}^{+x_1} + (x_1-x_2)e^- \rightarrow \text{M}^{+x_2}$ $\text{O} + 2e^- \rightarrow \text{O}^{2-}$ $\text{A} + xe^- \rightarrow \text{A}^{-x}$ $[\text{Fe}(\text{CN})_4]^{3-} + (4-3)e^- \rightarrow [\text{Fe}(\text{CN})_4]^{-4}$ $\text{A}^{-n_1} + (n_2-n_1)e^- \rightarrow \text{A}^{-n_2}$

6.6 TYPES OF REDOX REACTIONS :

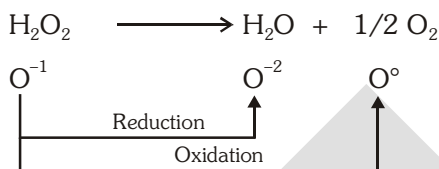
(A) **Intermolecular redox reaction :-** When oxidation and reduction takes place separately in different compounds, then the reaction is called intermolecular redox reaction.



(B) Intramolecular redox reaction :- During the chemical reaction, if oxidation and reduction takes place in single compound then the reaction is called intramolecular redox reaction.



(C) Disproportionation reaction :- When reduction and oxidation takes place in the same element of the same compound then the reaction is called disproportionation reaction.



(D) Comproportionation reaction: Reverse of disproportionation reaction known as comproportionation reaction. **Ex.** $\text{HClO} + \text{Cl}^- \rightarrow \text{Cl}_2 + \text{OH}^-$

BEGINNER'S BOX-3

- Oxidation is defined as –
(1) Gain of electrons
(2) Decrease in positive valency
(3) Loss of electrons
(4) Addition of electropositive element
- Reduction is defined as –
(1) Increase in positive valency
(2) Gain of electrons
(3) Loss of protons
(4) Decrease in negative valency
- In the reaction $\text{MnO}_4^- + \text{SO}_3^{2-} + \text{H}^+ \longrightarrow \text{SO}_4^{2-} + \text{Mn}^{2+} + \text{H}_2\text{O}$
(1) MnO_4^- and H^+ both are reduced
(2) MnO_4^- is reduced and H^+ is oxidised
(3) MnO_4^- is reduced and SO_3^{2-} is oxidised
(4) MnO_4^- is oxidised and SO_3^{2-} is reduced
- The charge on cobalt in $[\text{Co}(\text{CN})_6]^{-3}$ is –
(1) –6
(2) –3
(3) +3
(4) +6
- Which of the following halogen always show only one oxidating state in its compounds?
(1) Cl
(2) F
(3) Br
(4) I

6. Which of the following reactions do not involve oxidation-reduction ?
- (1) $2\text{Rb} + 2\text{H}_2\text{O} \longrightarrow 2\text{RbOH} + \text{H}_2$ (2) $2\text{CuI}_2 \longrightarrow 2\text{CuI} + \text{I}_2$
 (3) $\text{NH}_4\text{Cl} + \text{NaOH} \longrightarrow \text{NaCl} + \text{NH}_3 + \text{H}_2\text{O}$ (4) $3\text{Mg} + \text{N}_2 \longrightarrow \text{Mg}_3\text{N}_2$
7. The fast reaction between water and sodium is the example of –
- (1) Oxidation (2) Reduction (3) Intermolecular redox (4) Intramolecular redox
8. Choose the redox reaction from the following–
- (1) $\text{Cu} + 2\text{H}_2\text{SO}_4 \longrightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$ (2) $\text{BaCl}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{BaSO}_4 + 2\text{HCl}$
 (3) $2\text{NaOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$ (4) $\text{KNO}_3 + \text{H}_2\text{SO}_4 \longrightarrow 2\text{HNO}_3 + \text{K}_2\text{SO}_4$
9. Which of the following is not a redox reaction ?
- (1) $\text{MnO}_4^- \longrightarrow \text{MnO}_2 + \text{O}_2$ (2) $\text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{HCl} + \text{HClO}$
 (3) $2\text{CrO}_4^{2-} + 2\text{H}^+ \longrightarrow \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$ (4) $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Ag} \longrightarrow \text{Mn}^{+2} + 4\text{H}_2\text{O} + 5\text{Ag}^+$
10. In the reaction $6\text{Li} + \text{N}_2 \longrightarrow 2\text{Li}_3\text{N}$
- (1) Li undergoes reduction (2) Li undergoes oxidation
 (3) N undergoes oxidation (4) Li is oxidant
11. $\text{H}_2\text{O}_2 + \text{H}_2\text{O}_2 \longrightarrow 2\text{H}_2\text{O} + \text{O}_2$ is an example of disproportionation because –
- (1) Oxidation number of oxygen only decreases
 (2) Oxidation number of oxygen only increases
 (3) Oxidation number of oxygen decreases as well as increase
 (4) Oxidation number of oxygen neither decreases nor increases

6.7 BALANCING OF REDOX REACTION :

- (A) Oxidation number change method.
 (B) Ion electron method.

(A) Oxidation number change method :

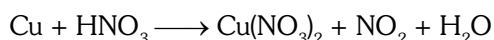
This method was given by Johnson. In a balanced redox reaction, total increase in oxidation number must be equal to total decreases in oxidation number. This equivalence provides the basis for balancing redox reactions.

The general procedure involves the following steps :

- Select the atom in oxidising agent whose oxidation number decreases and indicate the gain of electrons.
- Select the atom in reducing agent whose oxidation number increases and indicate the loss of electrons.
- Now cross multiply i.e. multiply oxidising agent by the number of loss of electrons and reducing agent by number of gain of electrons.
- Balance the number of atoms on both sides whose oxidation numbers change in the reaction.
- In order to balance oxygen atoms, add H_2O molecules to the side deficient in oxygen.
- Then balance the number of H atoms by adding H^+ ions to the side deficient in hydrogen.

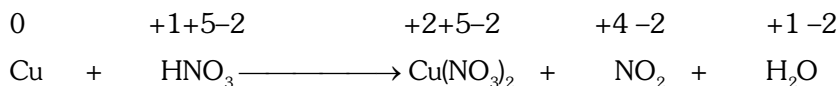
Illustrations

Illustration 9. Balance the following reaction by the oxidation number method –

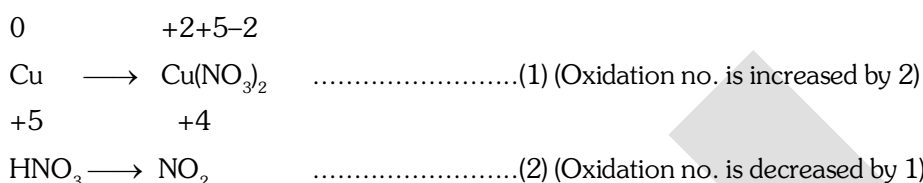


Solution

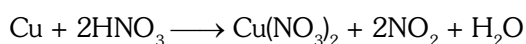
Write the oxidation number of all the atoms.



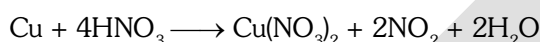
There is change in oxidation number of Cu and N.



To make increase and decrease equal, eq. (2) is multiplied by 2.

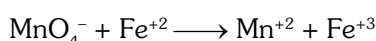


Balancing nitrates ions, hydrogen and oxygen, the following equation is obtained.



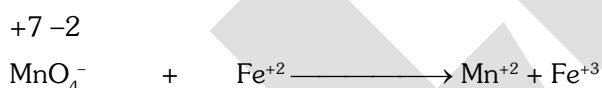
This is the balanced equation.

Illustration 10. Balance the following reaction by the oxidation number method –



Solution

Write the oxidation number of all the atoms.



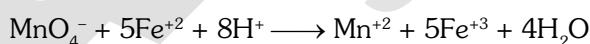
change in oxidation number has occurred in Mn and Fe.



To make increase and decrease equal, eq. (2) is multiplied by 5.



To balance oxygen, $4\text{H}_2\text{O}$ are added to R.H.S. and to balance hydrogen, 8H^+ are added to L.H.S.



This is the balanced equation.

(B) Ion-Electron method :-

This method was given by Jette and La Mev in 1972.

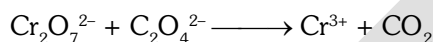
The following steps are followed while balancing redox reaction (equations) by this method.

- (i) Write the equation in ionic form.
- (ii) Split the redox equation into two half reactions, one representing oxidation and the other representing reduction.
- (iii) Balance these half reactions separately and then add by multiplying with suitable coefficients so that the electrons are cancelled. Balancing is done using following substeps.

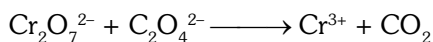
- Balance all other atoms except H and O.
- Then balance oxygen atoms by adding H_2O molecules to the side deficient in oxygen. The number of H_2O molecules added is equal to the deficiency of oxygen atoms.
- Balance hydrogen atoms by adding H^+ ions equal to the deficiency in the side which is deficient in hydrogen atoms.
- Balance the charge by adding electrons to the side which is rich in +ve charge. i.e. deficient in electrons. Number of electrons added is equal to the deficiency.
- Multiply the half equations with suitable coefficients to equalize the number of electrons.
- Add these half equations to get an equation which is balanced with respect to charge and atoms.
- If the medium of reaction is basic, OH^- ions are added to both sides of balanced equation, which is equal to number of H^+ ions in Balanced Equation.

Illustrations

Illustration 11. Balance the following reaction by ion-electron method in acidic medium :



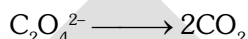
Solution



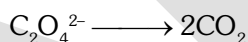
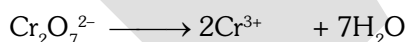
- (a) Write both the half reaction.



- (b) Atoms other than H and O are balanced.



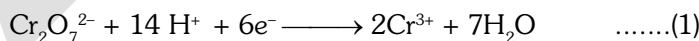
- (c) Balance O-atoms by the addition of H_2O to another side



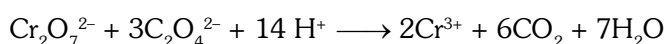
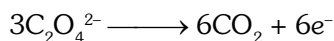
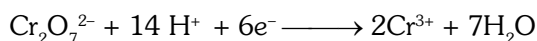
- (d) Balance H-atoms by the addition of H^+ to another side



- (e) Now, balance the charge by the addition of electron (e^-).



- (f) Multiply equations by a constant to get the same number of electrons on both side. In the above case second equation is multiplied by 3 and then added to first equation.



GOLDEN KEY POINTS

• **FOR REDOX REACTIONS :**

$N_1V_1 = N_2V_2$ is always true.

But $(M_1 \times V_1) \times n_1 = (M_2 \times V_2) \times n_2$ (always true where n term represents valency factor).

Illustrations

Illustration 13 Calculate the normality of a solution containing 15.8 g of KMnO_4 in 50 mL acidic solution.

Solution **Normality (N)** = $\frac{W \times 1000}{E \times V_{\text{mL}}}$

where $W = 15.8 \text{ g}$, $V = 50 \text{ mL}$ $E = \frac{\text{molar mass of } \text{KMnO}_4}{\text{Valence factor}} = 158/5 = 31.6$

So, $N = 10$

Illustration 14 Calculate the normality of a solution containing 50 mL of 5 M solution $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium.

Solution Normality (N) = Molarity $\times$ Valency factor = $5 \times 6 = 30 \text{ N}$

Illustration 15 Find the number of moles of KMnO_4 needed to oxidise one mole Cu_2S in acidic medium.

The reaction is $\text{KMnO}_4 + \text{Cu}_2\text{S} \longrightarrow \text{Mn}^{2+} + \text{Cu}^{2+} + \text{SO}_2$

Solution From law of equivalence

equivalents of $\text{Cu}_2\text{S} = \text{equivalents of } \text{KMnO}_4$
moles of $\text{Cu}_2\text{S} \times \text{v.f.} = \text{moles of } \text{KMnO}_4 \times \text{v.f.}$

$$1 \times 8 = n_2 \times 5$$

$$n_2 = \frac{8}{5} = 1.6$$

Illustration 16 The number of moles of oxalate ions oxidized by one mole of MnO_4^- ion in acidic medium.

(A) $\frac{5}{2}$

(B) $\frac{2}{5}$

(C) $\frac{3}{5}$

(D) $\frac{5}{3}$

Solution Equivalents of $\text{C}_2\text{O}_4^{2-} = \text{equivalents of } \text{MnO}_4^-$

$$x (\text{mole}) \times 2 = 1 \times 5 ; x = \frac{5}{2}$$

Illustration 17 What volume of 6 M HCl and 2 M HCl should be mixed to get two litre of 3 M HCl ?

Solution Let, the volume of 6 M HCl required to obtain 2 L of 3M HCl = x L

$\therefore$ Volume of 2 M HCl required = $(2 - x) \text{ L}$

$$\begin{array}{ccccc} M_1V_1 & + & M_2V_2 & = & M_3V_3 \\ 6\text{M HCl} & & 2\text{M HCl} & & 3\text{M HCl} \end{array}$$

$$6 \times (x) + 2 \times (2 - x) = 3 \times 2$$

$$\Rightarrow 6x + 4 - 2x = 6 \Rightarrow 4x = 2$$

$$\therefore x = 0.5 \text{ L}$$

Hence, volume of 6 M HCl required = 0.5 L

Volume of 2M HCl required = 1.5 L

Illustration 18 In a reaction vessel, 1.184 g of NaOH is required to be added for completing the reaction. How many millilitre of 0.15 M NaOH should be added for this requirement ?

Solution Amount of NaOH present in 1000 mL of 0.15 M NaOH = $0.15 \times 40 = 6$ g

$$\therefore 1 \text{ mL of this solution contain NaOH} = \frac{6}{1000} \times 10^{-3} \text{ g}$$

$$\therefore 1.184 \text{ g of NaOH will be present in} = \frac{1}{6 \times 10^{-3}} \times 1.184 = 197.33 \text{ mL}$$

Illustration 19 What weight of Na_2CO_3 of 85% purity would be required to prepare 45.6 mL of 0.235N H_2SO_4 ?

Solution Meq. of Na_2CO_3 = Meq. of H_2SO_4 = 45.6×0.235

$$\therefore \frac{W_{\text{Na}_2\text{CO}_3}}{E_{\text{Na}_2\text{CO}_3}} \times 1000 = 45.6 \times 0.235 \Rightarrow \frac{W_{\text{Na}_2\text{CO}_3}}{106/2} \times 1000 = 45.6 \times 0.235$$

$$\therefore W_{\text{Na}_2\text{CO}_3} = 0.5679 \text{ g}$$

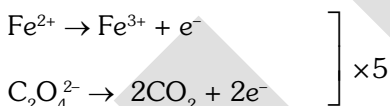
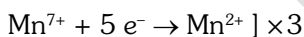
For 85 g of pure Na_2CO_3 , weight of sample = 100 g

$$\therefore \text{For } 0.5679 \text{ g of pure } \text{Na}_2\text{CO}_3, \text{ weight of sample} = \frac{100}{85} \times 0.5679 = 0.6681 \text{ g}$$

Illustration 20 The number of moles of KMnO_4 that will be required to react with 2 mol of ferrous oxalate is

- (A) $\frac{6}{5}$ (B) $\frac{2}{5}$ (C) $\frac{4}{5}$ (D) 1

Solution



3 moles of KMnO_4 = 5 moles of FeC_2O_4

$$\therefore 2 \text{ mol of ferrous oxalate} \equiv \frac{6}{5} \text{ mole of } \text{KMnO}_4$$

Hence, (A) is the correct answer.

Illustration 21 What volume of 6 M HNO_3 is needed to oxidize 8 g of Fe^{2+} to Fe^{3+} , HNO_3 gets converted to NO?

- (A) 8 mL (B) 7.936 mL (C) 32 mL (D) 64 mL

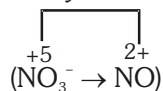
Solution

Meq. of HNO_3 = Meq. of Fe^{2+}

$$\text{or } 6 \times 3 \times V = \frac{8}{56} \times 1000$$

$$V = 7.936 \text{ mL}$$

valency factor = 3



Hence, (B) is the correct answer.

Illustration 22 Which of the following is / are correct?

- (A) g mole weight = molecular weight in g = wt. of 6.02×10^{23} molecules
 (B) mole = N_A molecule = 6.02×10^{23} molecules
 (C) mole = g molecules
 (D) none of the above

Solution **Ans.** (A), (B) and (C)

BEGINNER'S BOX-4

- In the half reaction : $2\text{ClO}_3^- \longrightarrow \text{Cl}_2$
 (1) 5 electrons are gained
 (2) 5 electrons are liberated
 (3) 10 electrons are gained
 (4) 10 electrons are liberated
- The number of electrons required to balance the following equation –
 $\text{NO}_3^- + 4\text{H}^+ + e^- \longrightarrow 2\text{H}_2\text{O} + \text{NO}$ are –
 (1) 5 (2) 4 (3) 3 (4) 2
- Which of the following equations is a balanced one –
 (1) $5\text{BiO}_3^- + 22\text{H}^+ + \text{Mn}^{2+} \longrightarrow 5\text{Bi}^{3+} + 7\text{H}_2\text{O} + \text{MnO}_4^-$
 (2) $5\text{BiO}_3^- + 14\text{H}^+ + 2\text{Mn}^{2+} \longrightarrow 5\text{Bi}^{3+} + 7\text{H}_2\text{O} + 2\text{MnO}_4^-$
 (3) $2\text{BiO}_3^- + 4\text{H}^+ + \text{Mn}^{2+} \longrightarrow 2\text{Bi}^{3+} + 2\text{H}_2\text{O} + \text{MnO}_4^-$
 (4) $6\text{BiO}_3^- + 12\text{H}^+ + 3\text{Mn}^{2+} \longrightarrow 6\text{Bi}^{3+} + 6\text{H}_2\text{O} + 3\text{MnO}_4^-$

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6				
	Ans.	2	3	1	2	4	2				
BEGINNER'S BOX-2	Que.	1	2								
	Ans.	2	3								
BEGINNER'S BOX-3	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	2	3	3	2	3	3	1	3	2
	Que.	11									
	Ans.	3									
BEGINNER'S BOX-4	Que.	1	2	3							
	Ans.	3	3	2							

Build Up Your Understanding

(1) +2 and +3 (2) +1 and +2
(3) +1 and +3 (4) None

- 21.** Compound $\text{YBa}_2\text{Cu}_3\text{O}_7$ is a super conductor. The O.N. of the copper in the compound will be: [O.No. of Y=+3]
(1) +7/3 (2) zero (3) +2 (4) +1
- 22.** The oxidation state of iodine in H_4IO_6^- is :-
(1) +7 (2) -1
(3) +5 (4) +1
- 23.** Amongst the following, identify the species with an atom in + 6 oxidation state:-
(1) MnO_4^- (2) Cr(CN)_6^{3-}
(3) NiF_6^{2-} (4) CrO_2Cl_2
- 24.** The oxidation state of + 1 for phosphorous is found in:-
(1) Phosphorous acid (H_3PO_3)
(2) Orthophosphoric acid (H_3PO_4)
(3) Hypo phosphorous acid (H_3PO_2)
(4) Hypo phosphoric acid ($\text{H}_4(\text{P}_2\text{O}_6)$)
- 25.** In which of the following compounds iron has lowest oxidation state:-
(1) $\text{FeSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
(2) $\text{K}_4[\text{Fe(CN)}_6]$
(3) $[\text{Fe(CO)}_5]$
(4) $\text{Fe}_{0.94}\text{O}$
- 26.** Select the compound in which the oxidation number of oxygen is -1:-
(1) H_2O (2) O_2F_2
(3) Na_2O (4) BaO_2
- 27.** Match List - I (compound) with list - II (Oxidation state of N) and select the correct answer using the codes given below the list:-
List - I List-II
(A) KNO_3 (a) - 1/3
(B) HNO_2 (b) - 3
(C) NH_4Cl (c) 0
(D) NaN_3 (d) + 3
 (e) + 5
- Codes are:-
- | | A | B | C | D |
|-----|---|---|---|---|
| (1) | e | d | b | a |
| (2) | e | b | d | a |
| (3) | d | e | a | c |
| (4) | b | c | d | e |
- 28.** In which of the following pair oxidation number of Fe is same :-
(1) $\text{K}_3[\text{Fe(CN)}_6]$, Fe_2O_3 (2) Fe(CO)_5 , Fe_2O_3
(3) Fe_2O_3 , FeO (4) $\text{Fe}_2(\text{SO}_4)_3$, $\text{K}_4[\text{Fe(CN)}_6]$
- 29.** In the conversion of Br_2 to BrO_3^- the oxidation state of bromine changes from :-
(1) 0 to 5 (2) 1 to 5 (3) 0 to -3 (4) 2 to 5
- 30.** The sum of oxidation states of sulphur in $\text{H}_2\text{S}_2\text{O}_8$ is :-
(1) +2 (2) +6 (3) +7 (4) +12
- 31.** In which of the following compounds of Cr, the oxidation number of Cr is not +6 :-
(1) CrO_3 (2) CrO_2Cl_2
(3) Cr_2O_3 (4) $\text{K}_2\text{Cr}_2\text{O}_7$
- 32.** Oxidation state of cobalt in $[\text{Co(NH}_3)_4(\text{H}_2\text{OCl})\text{SO}_4]$ is
(1) 0 (2) +4 (3) -2 (4) +3
- 33.** Oxidation number of carbon in graphite is :-
(1) Zero (2) +1 (3) +4 (4) +2
- 34.** Oxidation number of 'N' in N_3H (hydrazoic acid) is :-
(1) $-\frac{1}{3}$ (2) -3 (3) +3 (4) $+\frac{2}{3}$
- 35.** Phosphorous has the oxidation state of +3 in :-
(1) Phosphorus acid
(2) Orthophosphoric acid
(3) Meta phosphoric acid
(4) Pyro phosphoric acid
- 36.** The oxidation number of arsenic atom in H_3AsO_4 is :-
(1) -1 (2) -3 (3) +3 (4) +5
- 37.** In substance $\text{Mg(HXO}_3)$, the oxidation number of X is :-
(1) 0 (2) +2 (3) +3 (4) +4
- 38.** Oxidation number of P in KH_2PO_3 is :-
(1) -1 (2) -3 (3) +5 (4) +3
- 39.** The oxidation number of iron in potassium ferricyanide $\text{K}_3[\text{Fe(CN)}_6]$ is :-
(1) Two (2) Six
(3) Three (4) Four

- 40.** The oxidation number of phosphorus in PH_4^+ , PO_2^{3-} , PO_4^{3-} and PO_3^{3-} are respectively :-
 (1) -3, +1, +3, +5 (2) -3, +3, +5, +1
 (3) +3, -3, +5, +1 (4) -3, +1, +5, +3
- 41.** Which of the following compounds are arranged in increasing oxidation number of S :-
 (1) H_2SO_3 , H_2S , H_2SO_4 , $\text{H}_2\text{S}_2\text{O}_3$
 (2) $\text{H}_2\text{S}_2\text{O}_3$, H_2SO_3 , H_2S , H_2SO_4
 (3) H_2S , H_2SO_3 , H_2SO_4 , $\text{H}_2\text{S}_2\text{O}_3$
 (4) H_2S , $\text{H}_2\text{S}_2\text{O}_3$, H_2SO_3 , H_2SO_4
- 42.** Iodine shows the highest oxidation state in the compound :-
 (1) KI (2) KI_3 (3) IF_5 (4) KIO_4
- 43.** The sum of the oxidation states of all the carbon atoms present in the compound $\text{C}_6\text{H}_5\text{CHO}$ is :
 (1) -4 (2) 3 (3) +5 (4) -4/7
- 44.** Oxidation number of sodium in sodium amalgam is
 (1) +1 (2) 0
 (3) -1 (4) +2
- APPLICATIONS OF REDOX REACTIONS**
- 45.** A reducing agent is a substance which can :
 (1) Accept electrons (2) Donate electrons
 (3) Accept protons (4) Donate protons
- 46.** The reaction $\text{H}_2\text{S} + \text{H}_2\text{O}_2 \rightarrow \text{S} + 2\text{H}_2\text{O}$ manifests :
 (1) Oxidising action of H_2O_2
 (2) Reducing nature of H_2O_2
 (3) Acidic nature of H_2O_2
 (4) Alkaline nature of H_2O_2
- 47.** If an element is in its lowest oxidation state, under proper conditions it can act as :
 (1) Reducing agent
 (2) An oxidising agent
 (3) Oxidising as well as reducing agent
 (4) Neither oxidising nor reducing agent
- 48.** In a reaction of H_2O (steam) + C (glowing) $\rightarrow \text{CO} + \text{H}_2$
 (1) H_2O is the reducing agent
 (2) H_2O is the oxidising agent
 (3) carbon is the oxidising agent
 (4) oxidation-reduction does not occur
- 49.** The compound that can work both as an oxidising as well as reducing agent is :
 (1) KMnO_4 (2) H_2O_2
 (3) $\text{Fe}_2(\text{SO}_4)_3$ (4) $\text{K}_2\text{Cr}_2\text{O}_7$
- 50.** Reaction (A) $\text{S}^{2-} + 4 \text{H}_2\text{O}_2 \rightarrow \text{SO}_4^{2-} + 4 \text{H}_2\text{O}$
 (B) $\text{Cl}_2 + \text{H}_2\text{O}_2 \rightarrow 2\text{HCl} + \text{O}_2$
 The true statement regarding the above reactions is :
 (1) H_2O_2 acts as reductant in both the reactions.
 (2) H_2O_2 acts as oxidant in reaction (A) and reductant in reaction (B).
 (3) H_2O_2 acts as an oxidant in both the reactions.
 (4) H_2O_2 acts as reductant in reaction (A) and oxidant in reaction (B)
- 51.** HNO_2 acts as an oxidant with which one of the following reagent:-
 (1) KMnO_4 (2) H_2S (3) $\text{K}_2\text{Cr}_2\text{O}_7$ (4) Br_2
- 52.** In which of the following reaction H_2O_2 acts as reducing agent :-
 (1) $2\text{FeCl}_2 + 2\text{HCl} + \text{H}_2\text{O}_2 \rightarrow 2\text{FeCl}_3 + 2\text{H}_2\text{O}$
 (2) $\text{Cl}_2 + \text{H}_2\text{O}_2 \rightarrow 2\text{HCl} + \text{O}_2$
 (3) $2\text{HI} + \text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{I}_2$
 (4) $\text{H}_2\text{SO}_3 + \text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
- 53.** A sulphur containing species that can not be a reducing agent is :-
 (1) SO_2 (2) SO_3^{2-}
 (3) H_2SO_4 (4) $\text{S}_2\text{O}_3^{2-}$
- 54.** When H_2 reacts with Na, it acts as :-
 (1) Oxidising agent (2) Reducing agent
 (3) Both (4) Cannot be predicted
- 55.** Which one is the oxidising agent in the reaction given below
 $2\text{CrO}_4^{2-} + 2\text{H}^+ \rightarrow \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$
 (1) H^+ (2) $\text{Cr}_2\text{O}_7^{2-}$
 (3) Cr^{++} (4) H_2O
- 56.** In the course of a chemical reaction an oxidant -
 (1) Loses electron
 (2) Gains electron
 (3) Both loses and gain electrons
 (4) Electron change does not occur

57. In the reaction:-



HNO_3 acts as :-

- (1) An oxidising agent (2) An acid
(3) A reducing agent (4) A base

58. A compound contains atoms A, B and C. The oxidation number of A is +2, of B is +5 and of C is -2. The possible formula of the compound is :

- (1) ABC_2 (2) $B_2(AC_3)_2$
(3) $A_3(BC_4)_2$ (4) $A_3(B_4C)_2$

59. Equivalent weight of N_2 in the change $N_2 \rightarrow NH_3$ is

- (1) $\frac{28}{6}$ (2) 28 (3) $\frac{28}{2}$ (4) $\frac{28}{3}$

60. Equivalent weight of NH_3 in the change $N_2 \rightarrow NH_3$ is :

- (1) $\frac{17}{6}$ (2) 17 (3) $\frac{17}{2}$ (4) $\frac{17}{3}$

61. In the reaction, $2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$, the eq. wt. of $Na_2S_2O_3$ is equal to its :

- (1) Mol. wt. (2) Mol. wt./2
(3) 2 x Mol. wt. (4) Mol. wt./6

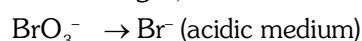
62. In the reaction, $VO + Fe_2O_3 \rightarrow FeO + V_2O_5$, the eq. wt. of V_2O_5 is equal to its :

- (1) Mol. wt. (2) Mol. wt./8
(3) Mol. wt./6 (4) Mol. wt./2

63. The eq. wt. of iodine in, $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$ is :

- (1) Its Mol. wt. (2) Mol. wt./2
(3) Mol. wt./4 (4) None of these

64. Molecular weight of $KBrO_3$ is M. What is its equivalent weight, if the reaction is :

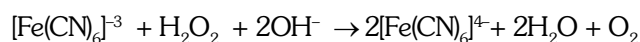


- (1) M (2) M/4
(3) M/6 (4) 6M

65. In the reaction : $A^{n_2} + xe^- \rightarrow A^{n_1}$, here x will be

- (1) $n_1 + n_2$ (2) $n_2 - n_1$
(3) $n_1 - n_2$ (4) $n_1 \cdot n_2$

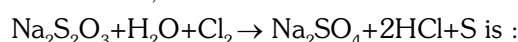
66. What would be the equivalent weight of the reductant in the reaction :



[Given : Fe = 56, C = 12, N = 14, O = 16, H = 1]

- (1) 17 (2) 212
(3) 34 (4) 32

67. The equivalent weight of $Na_2S_2O_3$ as reductant in the reaction,



- (1) (Mol. wt.)/1 (2) (Mol. wt.)/2
(3) (Mol. wt.)/6 (4) (Mol. wt.)/8

68. Equivalent weight of FeC_2O_4 in the change : $FeC_2O_4 \rightarrow Fe^{3+} + CO_2$ is :

- (1) M/3 (2) M/6
(3) M/2 (4) M/1

69. What will be n-factor for $Ba(MnO_4)_2$ in acidic medium? (Where it behaves as oxidant)

- (1) 5 (2) 10 (3) 6 (4) 3

70. The number of mole of oxalate ions oxidised by one mole of MnO_4^- is :

- (1) 1/5 (2) 2/5
(3) 5/2 (4) 5

71. Oxidising product of substance Na_3AsO_3 would be

- (1) $As_2O_3^{3-}$ (2) AsO_3^{3-}
(3) AsO_2^{4-} (4) AsO_4^{3-}

72. In a reaction 4 mole of electrons are transferred to one mole of HNO_3 when it acts as an oxidant. The possible reduction product is :

- (1) (1/2) mole N_2 (2) (1/2) mole N_2O
(3) 1 mole of NO_2 (4) 1 mole NH_3

73. The equivalent weight of $MnSO_4$ is half of its molecular weight when it is converted to :-

- (1) Mn_2O_3 (2) MnO_2
(3) MnO_4^- (4) MnO_4^{2-}

74. In the following change, $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$. If the atomic weight of iron is 56, then its equivalent weight will be :-

- (1) 42 (2) 21 (3) 63 (4) 84

75. $Cr_2O_7^{2-} + I^- + H^+ \rightarrow Cr^{3+} + I_2 + H_2O$

The equivalent weight of the reductant in the above equation is :- (At. wt. of Cr=52, I=127)

- (1) 26 (2) 127 (3) 63.5 (4) 10.4

76. How many moles of KMnO_4 are reduced by 1 mole of ferrous oxalate in acidic medium:-

- (1) $\frac{1}{5}$ (2) $\frac{5}{3}$ (3) $\frac{1}{3}$ (4) $\frac{3}{5}$

77. The number of moles of KMnO_4 reduced by one mole of KI in alkaline medium is :-

- (1) One (2) Two
(3) Five (4) One fifth

REDOX REACTIONS

78. Which one of the following is a redox reaction ?

- (1) $\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr}$
(2) $2\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl}$
(3) $\text{HCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{HNO}_3$
(4) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

79. Which of the following is not a redox change ?

- (1) $2\text{H}_2\text{S} + \text{SO}_2 \rightarrow 2\text{H}_2\text{O} + 3\text{S}$
(2) $2\text{BaO} + \text{O}_2 \rightarrow 2\text{BaO}_2$
(3) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$
(4) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$

80. In the reaction $4\text{Fe} + 3\text{O}_2 \rightarrow 4\text{Fe}^{3+} + 6\text{O}^{2-}$ which of the following statements is incorrect ?

- (1) It is a redox reaction
(2) Metallic iron is a reducing agent
(3) Fe^{3+} is an oxidising agent
(4) Metallic iron is reduced to Fe^{3+}

81. In the reaction, $\text{Cl}_2 + \text{OH}^- \rightarrow \text{Cl}^- + \text{ClO}_4^- + \text{H}_2\text{O}$, chlorine is :

- (1) Oxidised
(2) Reduced
(3) Oxidised as well as reduced
(4) Neither oxidised nor reduced

82. Which is a redox reaction :

- (1) $2\text{CuI}_2 \rightarrow \text{CuI} + \text{I}_2$
(2) $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$
(3) $\text{NH}_4\text{Cl} + \text{NaOH} \rightarrow \text{NH}_3 + \text{NaCl} + \text{H}_2\text{O}$
(4) $\text{Cr}_2(\text{SO}_4)_3 + 6\text{KOH} \rightarrow 2\text{Cr}(\text{OH})_3 + 3\text{K}_2\text{SO}_4$

83. Which of the following example does not represent disproportionation -

- (1) $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$
(2) $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
(3) $4\text{KClO}_3 \rightarrow 3\text{KClO}_4 + \text{KCl}$
(4) $3\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$

84. The decomposition of KClO_3 to KCl and O_2 on heating is an example of :

- (1) Intermolecular redox change
(2) Intramolecular redox change
(3) Disproportionation or auto redox change
(4) Comproportionation

85. Which of the following change represents a disproportionation reaction (s) :

- (1) $\text{Cl}_2 + 2\text{OH}^- \rightarrow \text{ClO}^- + \text{Cl}^- + \text{H}_2\text{O}$
(2) $\text{Cu}_2\text{O} + 2\text{H}^+ \rightarrow \text{Cu} + \text{Cu}^{2+} + \text{H}_2\text{O}$
(3) $2\text{HCuCl}_2 \xrightarrow[\text{Water}]{\text{dilution with}} \text{Cu} + \text{Cu}^{2+} + 4\text{Cl}^- + 2\text{H}^+$
(4) All of the above

86. One mole of iron [55.8 gm], when oxidised to +2 oxidation state gives up :

- (1) $1N_A$ electron (2) $2N_A$ electron
(3) $3N_A$ electron (4) 0.5 mole of electron

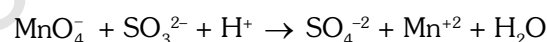
87. How many electrons should X_2H_4 liberate so that in the new compound X shows oxidation number of $-\frac{1}{2}$ (E.N. $\text{X} > \text{H}$)

- (1) 10 (2) 4 (3) 3 (4) 2

88. Which one of the following is not a redox reaction :-

- (1) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
(2) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
(3) $\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + \frac{1}{2}\text{H}_2$
(4) $\text{MnCl}_3 \rightarrow \text{MnCl}_2 + \frac{1}{2}\text{Cl}_2$

89. In the reaction -



- (1) MnO_4^- and H^+ both are reduced
(2) MnO_4^- is reduced and H^+ is oxidised
(3) MnO_4^- is reduced and SO_3^{2-} is oxidised
(4) MnO_4^- is oxidised and SO_3^{2-} is reduced

90. $\text{I}_2 + \text{KI} \rightarrow \text{KI}_3$

In the above reaction:-

- (1) Only oxidation taken place
(2) Only reduction takes place
(3) Both the above
(4) Neither oxidation nor reduction

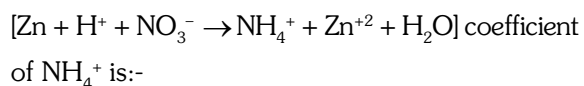
91. Which of the following reaction represents the oxidising behaviour of H_2SO_4 :-
 (1) $2\text{PCl}_5 + \text{H}_2\text{SO}_4 \rightarrow 2\text{POCl}_3 + 2\text{HCl} + \text{SO}_2\text{Cl}_2$
 (2) $2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
 (3) $\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HCl}$
 (4) $2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
92. Select the example of disproportionation reaction
 (1) $\text{BaCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{HCl}$
 (2) $\text{NH}_4\text{NO}_3 \rightarrow \text{N}_2\text{O} + 2\text{H}_2\text{O}$
 (3) $4\text{H}_3\text{PO}_3 \rightarrow \text{PH}_3 + 3\text{H}_3\text{PO}_4$
 (4) $\text{AgCl} + 2\text{NH}_3 \rightarrow \text{Ag}(\text{NH}_3)_2\text{Cl}$
93. Which of the following reaction involves oxidation & reduction :-
 (1) $\text{NaBr} + \text{HCl} \rightarrow \text{NaCl} + \text{HBr}$
 (2) $\text{HBr} + \text{AgNO}_3 \rightarrow \text{AgBr} + \text{HNO}_3$
 (3) $2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
 (4) $\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr}$
94. The reaction $2\text{K}_2\text{MnO}_4 + \text{Cl}_2 \rightarrow 2\text{KMnO}_4 + 2\text{KCl}$ is an example of
 (1) Redox (2) Reduction only
 (3) Neutralization (4) Disproportionation
95. Which of the following reaction involves neither oxidation nor reduction :-
 (1) $\text{CrO}_4^{2-} \rightarrow \text{Cr}_2\text{O}_7^{2-}$ (2) $\text{Cr} \rightarrow \text{CrCl}_3$
 (3) $\text{Na} \rightarrow \text{Na}^+$ (4) $2\text{S}_2\text{O}_3^{2-} \rightarrow \text{S}_4\text{O}_6^{2-}$
96. $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$
 Zn undergoes -
 (1) Reduction
 (2) Oxidation
 (3) Both oxidation and reduction
 (4) Neither oxidation nor reduction

BALANCING OF REDOX REACTIONS

97. Balance the following given half reaction for the unbalanced whole reaction :
 $\text{CrO}_4^{2-} \rightarrow \text{CrO}_2^- + \text{OH}^-$ is :
 (1) $\text{CrO}_4^{2-} + 2\text{H}_2\text{O} + 3\text{e}^- \rightarrow \text{CrO}_2^- + 4\text{OH}^-$
 (2) $2\text{CrO}_4^{2-} + 8\text{H}_2\text{O} \rightarrow \text{CrO}_2^- + 4\text{H}_2\text{O} + 8\text{OH}^-$
 (3) $\text{CrO}_4^{2-} + \text{H}_2\text{O} \rightarrow \text{CrO}_2^- + \text{H}_2\text{O} + \text{OH}^-$
 (4) $3\text{CrO}_4^{2-} + 4\text{H}_2\text{O} + 6\text{e}^- \rightarrow 2\text{CrO}_2^- + 8\text{OH}^-$

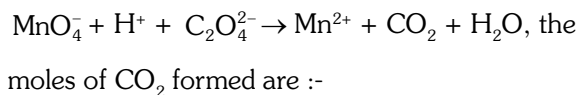
98. Choose the set of coefficients that correctly balances the following equation :
 $x \text{Cr}_2\text{O}_7^{2-} + y\text{H}^+ + z\text{e}^- \rightarrow a \text{Cr}^{3+} + b\text{H}_2\text{O}$
- | | x | y | z | a | b |
|-----|---|----|---|---|---|
| (1) | 2 | 14 | 6 | 2 | 7 |
| (2) | 1 | 14 | 6 | 2 | 7 |
| (3) | 2 | 7 | 6 | 2 | 7 |
| (4) | 2 | 7 | 6 | 1 | 7 |
99. In the reaction: $\text{MnO}_4^- + x\text{H}^+ + n\text{e}^- \rightarrow \text{Mn}^{2+} + y\text{H}_2\text{O}$
 What is the value of n :
 (1) 5 (2) 8
 (3) 6 (4) 3
100. The number of electrons required to balance the following equation -
 $\text{NO}_3^- + 4\text{H}^+ + \text{e}^- \rightarrow 2\text{H}_2\text{O} + \text{NO}$ is
 (1) 5 (2) 4
 (3) 3 (4) 2
101. The molar mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is 249. Its equivalent mass in the reaction (a) and (b) would be
 (a) Reaction $\text{CuSO}_4 + \text{KI} \rightarrow$ product
 (b) Electrolysis of CuSO_4 solution.
 (1) (a) 249 (b) 249 (2) (a) 124.5 (b) 124.5
 (3) (a) 249 (b) 124.5 (4) (a) 124.5 (b) 249
102. $2\text{KMnO}_4 + 5\text{H}_2\text{S} + 6\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 2\text{K}^+ + 5\text{S} + 8\text{H}_2\text{O}$.
 In the above reaction, how many electrons would be involved in the oxidation of 1 mole of reductant?
 (1) Two (2) Five
 (3) Ten (4) One
103. The value of n in : $\text{MnO}_4^- + 8\text{H}^+ + n\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ is
 (1) 5 (2) 4
 (3) 3 (4) 2
104. What is the value of n in the following equation :
 $\text{Cr}(\text{OH})_4^- + \text{OH}^- \rightarrow \text{CrO}_4^{2-} + \text{H}_2\text{O} + n\text{e}^-$
 (1) 3 (2) 6
 (3) 5 (4) 2
105. For the redox reaction
 $\text{Zn} + \text{NO}_3^- \rightarrow \text{Zn}^{2+} + \text{NH}_4^+$ in basic medium,
 coefficients of Zn, NO_3^- and OH^- in the balanced equation respectively are :
 (1) 4, 1, 7 (2) 7, 4, 1
 (3) 4, 1, 10 (4) 1, 4, 10

106. In the balanced equation-



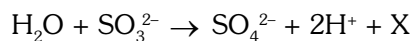
- (1) 4 (2) 3
(3) 2 (4) 1

107. In the balanced equation



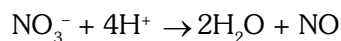
- (1) 2 (2) 4
(3) 5 (4) 10

108. In the following reaction the value of 'X' is



- (1) $4e^-$ (2) $3e^-$
(3) $2e^-$ (4) $1e^-$

109. The number of electrons required to balance the following equation are:



- (1) 2 on right side (2) 3 on left side
(3) 3 on right side (4) 5 on left side

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	3	2	3	3	2	1	2	3	3	4	3	1	2	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	3	2	1	1	1	4	3	3	4	1	1	1	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	4	1	1	1	4	3	4	3	4	4	4	1	2	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	1	2	2	2	2	2	3	1	4	2	1	3	1	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	1	3	2	3	3	1	4	1	2	3	4	2	2	2	2
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	4	2	1	3	4	3	1	1	2	4	2	3	1	3	3
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	4	3	4	1	1	2	1	2	1	3	3	1	1	1	3
Que.	106	107	108	109											
Ans.	4	4	3	2											

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2008

1. Number of moles of MnO_4^- required to oxidize one mole of ferrous oxalate completely in acidic medium will be:-
 (1) 7.5 moles (2) 0.2 moles
 (3) 0.6 moles (4) 0.4 moles

AIPMT 2009

2. Oxidation numbers of P in PO_4^{3-} of S in SO_4^{2-} and that of Cr in $\text{Cr}_2\text{O}_7^{2-}$ are respectively :-
 (1) -3, +6 and +6 (2) +5, +6 and +6
 (3) +3, +6 and +5 (4) +5, +3 and +6

AIIMS 2010

3. Correct order of oxidising strength is :-
 (1) $\text{MnO}_4^- > \text{VO}_2^+ > \text{Cr}_2\text{O}_7^{2-}$
 (2) $\text{Cr}_2\text{O}_7^{2-} > \text{MnO}_4^- > \text{VO}_2^+$
 (3) $\text{Cr}_2\text{O}_7^{2-} > \text{VO}_2^+ > \text{MnO}_4^-$
 (4) $\text{MnO}_4^- > \text{Cr}_2\text{O}_7^{2-} > \text{VO}_2^+$

AIIMS 2011

4. How much amount of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ required for liberation of 2.54 g I_2 when titrated with KI
 (1) 2.5 gm (2) 4.99 gm
 (3) 2.4 gm (4) 1.2 gm

AIPMT Pre. 2012

5. When Cl_2 gas reacts with hot and concentrated sodium hydroxide solution, the oxidation number of chlorine changes from:
 (1) Zero to -1 and zero to +3
 (2) Zero to +1 and zero to -3
 (3) Zero to +1 and zero to -5
 (4) Zero to -1 and zero to +5
6. In which of the following compounds, nitrogen exhibits highest oxidation state?
 (1) N_3H (2) NH_2OH
 (3) N_2H_4 (4) NH_3

AIIMS 2013

7. $\text{K}_2\text{Cr}_2\text{O}_7$ react with hydrazine to form product. The oxidation state of Cr in the product will be :-
 (1) +4 (2) +3
 (3) +5 (4) +2
8. Oxidation state of iron in haemoglobin is :-
 (1) 0 (2) +2
 (3) -2 (4) +3

AIPMT 2014

9. (a) $\text{H}_2\text{O}_2 + \text{O}_3 \rightarrow \text{H}_2\text{O} + 2\text{O}_2$
 (b) $\text{H}_2\text{O}_2 + \text{Ag}_2\text{O} \rightarrow 2\text{Ag} + \text{H}_2\text{O} + \text{O}_2$
 Role of hydrogen peroxide in the above reactions is respectively -
 (1) Oxidizing in (a) and reducing in (b)
 (2) Reducing in (a) and oxidizing in (b)
 (3) Reducing in (a) and (b)
 (4) Oxidizing in (a) and (b)
10. The reaction of aqueous KMnO_4 with H_2O_2 in acidic conditions gives :-
 (1) Mn^{4+} and O_2 (2) Mn^{2+} and O_2
 (3) Mn^{2+} and O_3 (4) Mn^{4+} and MnO_2

AIIMS 2014

11. In weak alkaline medium KMnO_4 converted in to:-
 (1) MnO_2 (2) K_2MnO_4
 (3) $\text{Mn}(\text{OH})_2$ (4) MnO

RE-AIPMT 2015

12. Assuming complete ionization, same moles of which of the following compounds will require the least amount of acidified KMnO_4 for complete oxidation?
 (1) FeC_2O_4 (2) $\text{Fe}(\text{NO}_2)_2$
 (3) FeSO_4 (4) FeSO_3

AIIMS 2015

13. When KMnO_4 reacts with H_2O_2 in acidic medium then which of following is formed
 (1) Mn^{+2} (2) Mn^{+7}
 (3) MnO_2 (4) Mn_2O_3
14. Lowest oxidation state of phosphorous is in
 (1) H_3PO_2 (2) H_3PO_4
 (3) $\text{H}_4\text{P}_2\text{O}_7$ (4) H_3PO_3

NEET-II 2016

15. Hot concentrated sulphuric acid is a moderately strong oxidizing agent. Which of the following reactions does not show oxidizing behaviour ?
 (1) $\text{C} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CO}_2 + 2\text{SO}_2 + 2\text{H}_2\text{O}$
 (2) $\text{CaF}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{HF}$
 (3) $\text{Cu} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$
 (4) $3\text{S} + 2\text{H}_2\text{SO}_4 \rightarrow 3\text{SO}_2 + 2\text{H}_2\text{O}$

NEET(UG) 2018

16. For the redox reaction
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \longrightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$
 the correct coefficients of the reactants for the balanced equation are

	MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+
(1)	16	5	2
(2)	2	5	16
(3)	2	16	5
(4)	5	16	2

AIIMS 2018

17. 2 g FeSO_4 is completely oxidised by acidic 0.05 M KMnO_4 solution then what will be volume of KMnO_4 required

(Fe = 56, S = 32, O = 16)

- (1) 0.10 L (2) 0.05 L
 (3) 0.30 L (4) 0.40 L

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	4	2	4	1	2	2	3	2	1	3	1	1	2
Que.	16	17													
Ans.	2	2													

EXERCISE-III (Analytical Questions)

Check Your Understanding

- In the reaction $\text{CH}_3\text{OH} \rightarrow \text{HCOOH}$, the number of electrons that must be added to the right is :-
(1) 4 (2) 3 (3) 2 (4) 1
- Which statement is wrong :-
(1) Oxidation number of oxygen is +1 in peroxides
(2) Oxidation number of oxygen is +2 in oxygen difluoride
(3) Oxidation number of oxygen is $-\frac{1}{2}$ in superoxides
(4) Oxidation number of oxygen is -2 in most of its compound
- In the reaction $8\text{Al} + 3\text{Fe}_3\text{O}_4 \rightarrow 4\text{Al}_2\text{O}_3 + 9\text{Fe}$, the number of electrons transferred from reductant to oxidant is :-
(1) 8 (2) 4 (3) 16 (4) 24
- In which of the following reaction hydrogen is acting as an oxidising agent :-
(1) With iodine to give hydrogen iodide
(2) With lithium to give lithium hydride
(3) With nitrogen to give ammonia
(4) With sulphur to give hydrogen sulphide
- Oxidation number of Xe in XeF_5^- is :
(1) +1 (2) +2 (3) +3 (4) +4
- Which is the best description of the behaviour of bromine in the reaction given below :-
 $\text{H}_2\text{O} + \text{Br}_2 \rightarrow \text{HOBr} + \text{HBr}$
(1) Both oxidized and reduced
(2) Oxidized only
(3) Reduced only
(4) Proton acceptor only
- The correct order of acidic strength is -
(1) $\text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2 < \text{HClO}$
(2) $\text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4 < \text{HClO}$
(3) $\text{HClO}_4 < \text{HClO} < \text{HClO}_2 < \text{HClO}_3$
(4) $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$

- Consider the following reaction:
$$x\text{MnO}_4^- + y\text{C}_2\text{O}_4^{2-} + z\text{H}^+ \rightarrow x\text{Mn}^{2+} + 2y\text{CO}_2 + \frac{z}{2}\text{H}_2\text{O}$$

The values of x, y and z in the reaction are respectively:-
(1) 5, 2 and 16 (2) 2, 5 and 8
(3) 2, 5 and 16 (4) 5, 2 and 8
- In which of the following reaction H_2O_2 acts as a reducing agent ?
(a) $\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$
(b) $\text{H}_2\text{O}_2 - 2\text{e}^- \rightarrow \text{O}_2 + 2\text{H}^+$
(c) $\text{H}_2\text{O}_2 + 2\text{e}^- \rightarrow 2\text{OH}^-$
(d) $\text{H}_2\text{O}_2 + 2\text{OH}^- - 2\text{e}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O}$
(1) (a), (c) (2) (b), (d) (3) (a), (b) (4) (c), (d)
- What will be the oxidation number of Fe in $\text{K}_3[\text{Fe}(\text{CN})_6]$:-
(1) +4 (2) +3 (3) +2 (4) Zero
- In the following reaction
 $2\text{FeCl}_3 + \text{H}_2\text{S} \rightarrow 2\text{FeCl}_2 + 2\text{HCl} + \text{S}$
(1) FeCl_3 is oxidant
(2) FeCl_3 & H_2S are oxidised
(3) FeCl_3 is oxidised & H_2S is reduced
(4) H_2S is oxidant
- In oxidation reduction reaction
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$ the balance equation coefficient for MnO_4^- , $\text{C}_2\text{O}_4^{2-}$, H^+ are :-
(1) 2, 5, 16 (2) 16, 5, 2
(3) 2, 16, 5 (4) 5, 2, 16
- Which of the following act both as oxidant & reductant:-
(1) H_2S (2) SO_3
(3) H_2O_2 (4) F_2
- Which of the following reaction is spontaneous oxidation-reduction reaction
(1) $\text{Mn}^{+2} + 5\text{Fe}^{+3} + 4\text{H}_2\text{O} \rightarrow \text{MnO}_4^- + 5\text{Fe}^{+2} + 8\text{H}^+$
(2) $\text{MnO}_4^- + 5\text{Fe}^{+3} + 8\text{H}^+ \rightarrow \text{Mn}^{+2} + 5\text{Fe}^{+2} + 4\text{H}_2\text{O}$
(3) $\text{MnO}_4^- + 5\text{Fe}^{+2} + 8\text{H}^+ \rightarrow \text{Mn}^{+2} + 5\text{Fe}^{+3} + 4\text{H}_2\text{O}$
(4) $\text{Mn}^{+2} + 5\text{Fe}^{+2} + 4\text{H}_2\text{O} \rightarrow \text{MnO}_4^- + 5\text{Fe}^{+3} + 8\text{H}^+$
- Oxidation number of chlorine in perchloric acid is:-
(1) +1 (2) +3
(3) +5 (4) +7

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	4	2	4	1	4	3	2	2	1	1	3	3	4

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** O.N. of carbon in $\text{H-C}\equiv\text{N}$ is +4.
Reason :- Carbon always shows an O.N. of +4.
 (1) A (2) B (3) C (4) D
- Assertion :-** In NH_4NO_3 , the oxidation number of the two N-atoms is not equal.
Reason :- One N atom is present in the ammonium ion while the other is present in the nitrate ion.
 (1) A (2) B (3) C (4) D
- Assertion :-** Oxidation state of Hydrogen is +1 in H_2O while -1 in CaH_2 .
Reason :- CaH_2 is a metal hydride and for metal hydrides, hydrogen is assigned the oxidation number of -1.
 (1) A (2) B (3) C (4) D
- Assertion :-** Oxidation number of carbon in CH_2O is zero.
Reason :- CH_2O (formaldehyde) is a covalent compound.
 (1) A (2) B (3) C (4) D
- Assertion :-** Oxidation number of Ni in $[\text{Ni}(\text{CO})_4]$ is zero.
Reason :- Nickel is bonded to neutral ligand, carbonyl.
 (1) A (2) B (3) C (4) D
- Assertion :-** In HClO_4 , Chlorine has the oxidation number of +4.
Reason :- HClO_4 (perchloric) acid has two peroxide linkages.
 (1) A (2) B (3) C (4) D
- Assertion :-** Oxidation number of S in HSO_3^- is +4.
Reason :- Sulphur is in different oxidation state in different compounds.
 (1) A (2) B (3) C (4) D
- Assertion :-** Oxidation number of Carbon in all its compounds is +4.
Reason :- An element has a fixed oxidation state.
 (1) A (2) B (3) C (4) D
- Assertion :-** Oxidation number of Cr in CrO_5 is +6.
Reason :- In CrO_5 , four oxygen atoms are involved in peroxide linkage.
 (1) A (2) B (3) C (4) D
- Assertion :-** Oxidation number of Cr in $[\text{Cr}(\text{CO})_6]$ is zero.
Reason :- Cr is a metal.
 (1) A (2) B (3) C (4) D
- Assertion :-** The oxidation no. of sulphur in $\text{Na}_2\text{S}_4\text{O}_6$ is 2.5
Reason :- Two S-atoms are not directly linked with O-atoms.
 (1) A (2) B (3) C (4) D
- Assertion :-** In the reaction, $\frac{1}{2}\text{O}_2 + \text{F}_2 \rightarrow \text{OF}_2$ Fluorine is oxidant.
Reason :- Fluorine cannot show positive oxidation state.
 (1) A (2) B (3) C (4) D
- Assertion :-** $\text{H}_2\text{S} + \text{Cl}_2 \longrightarrow 2\text{HCl} + \text{S}$
 In the above reaction, Cl has been oxidised to Cl^- while S^{2-} has been reduced to S
Reason :- In a reaction the element whose oxidation number decreases is reduced and the element whose oxidation number increases is oxidised.
 (1) A (2) B (3) C (4) D

14. Assertion :- Nitrous acid (HNO_2) may act as an oxidising agent as well as a reducing agent.

Reason :- The oxidation number of Nitrogen remains same in all the compounds.

(1) A (2) B (3) C (4) D

15. Assertion :- A reducing agent is a substance which accepts electron.

Reason :- A substance which helps in oxidation is known as reducing agent.

(1) A (2) B (3) C (4) D

16. Assertion :- In a redox reaction, the oxidation number of the oxidant decreases while that of reductant increases.

Reason :- Oxidant gains electron(s) while reductant loses electron(s).

(1) A (2) B (3) C (4) D

17. Assertion :- H_2SO_4 can not act as reducing agent.

Reason :- Sulphur can not increase its oxidation number beyond +6.

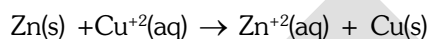
(1) A (2) B (3) C (4) D

18. Assertion :- When Cl_2 react with conc. NaOH form NaCl & NaClO_3

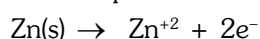
Reason :- Cl_2 is a oxidizing agent.

(1) A (2) B (3) C (4) D

19. Assertion :-



can be split into following half reactions



(Oxidation half reaction)



(Reduction half reaction)

Reason :- Every redox reaction can be split into two reactions, one representing loss of electrons and the other representing gain of electrons.

(1) A (2) B (3) C (4) D

20. Assertion :- MnO_4^- is always reduced to Mn^{+2} .

Reason :- Decrease in oxidation number or gaining of electron means oxidation.

(1) A (2) B (3) C (4) D

21. Assertion :- $\text{KClO}_3 \rightarrow \text{KClO}_4 + \text{KCl}$

This is a disproportionation type reaction.

Reason :- The reaction in which one substance oxidise or reduce is known as disproportionation reaction.

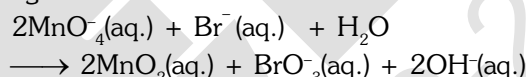
(1) A (2) B (3) C (4) D

22. Assertion :- Equivalent weight of KMnO_4 in acidic medium is $M/5$ (M =molecular weight) while in alkaline medium, it is equal to $M/3$.

Reason :- In acidic medium, 1 mol of MnO_4^- gains 5 mole electrons while in alkaline medium it gains 3 mole electrons.

(1) A (2) B (3) C (4) D

23. Assertion :- Bromide ion is serving as a reducing agent in the reaction.



Reason :- Oxidation number of Br increases from -1 to +5.

(1) A (2) B (3) C (4) D

24. Assertion :- Equivalent weight of NH_3 in the reaction $\text{N}_2 \rightarrow \text{NH}_3$ is $17/3$ while that of N_2 is $28/6$.

Reason :- Equivalent weight

$$= \frac{\text{Molecular weight}}{\text{number of } e^- \text{ lost or gained/mole}}$$

(1) A (2) B (3) C (4) D

25. Assertion :- In acidic medium, equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ is equal to $294/6$.

Reason :- In acidic medium, $\text{Cr}_2\text{O}_7^{2-}$ is reduced in Cr^{+3} .

(1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	1	1	2	1	4	2	4	1	2	1	2	4	3	4
Que.	16	17	18	19	20	21	22	23	24	25					
Ans.	1	1	2	1	4	3	1	1	1	1					

LINE STRUCTURE OF SOME COMPOUNDS				Oxidation state
1.	Hydrogen peroxide	H_2O_2	$\text{H}-\text{O}-\text{O}-\text{H}$	$\text{O} = \dots\dots$
2.	Nitrous acid	HNO_2	$\text{H}-\text{O}-\text{N}=\text{O}$	$\text{N} = \dots\dots$
3.	Nitric acid	HNO_3	$\text{H}-\text{O}-\text{N} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{O} \end{array}$	$\text{N} = \dots\dots$
4.	Hypo chlorous acid	HClO	$\text{H}-\text{O}-\text{Cl}$	$\text{Cl} = \dots\dots$
5.	Chlorous acid	HClO_2	$\text{H}-\text{O}-\text{Cl} \rightarrow \text{O}$	$\text{Cl} = \dots\dots$
6.	Chloric acid	HClO_3	$\text{H}-\text{O}-\text{Cl} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{O} \end{array}$	$\text{Cl} = \dots\dots$
7.	Perchloric acid	HClO_4	$\text{H}-\text{O}-\text{Cl} \begin{array}{l} \nearrow \text{O} \\ \rightarrow \text{O} \\ \searrow \text{O} \end{array}$	$\text{Cl} = \dots\dots$
8.	Hydrazine	N_2H_4	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{N}-\text{N}-\text{H} \end{array}$	$\text{N} = \dots\dots$
9.	Carbonic acid	H_2CO_3	$\text{H}-\text{O}-\text{C} \begin{array}{c} \nearrow \text{O} \\ \searrow \text{O} \end{array}-\text{O}-\text{H}$	$\text{C} = \dots\dots$
10.	Chromium pentoxide	CrO_5	$\begin{array}{c} \text{O} \quad \text{O} \\ \diagdown \quad \diagup \\ \text{Cr} \\ \diagup \quad \diagdown \\ \text{O} \quad \text{O} \end{array}$	$\text{Cr} = \dots\dots$
11.	Nitrosyl chloride/ Tilden's reagent	NOCl	$\text{Cl}-\text{N}=\text{O}$	$\text{N} = \dots\dots$
12.	Chromyl chloride	CrO_2Cl_2	$\begin{array}{c} \text{O} \\ \uparrow \\ \text{Cl}-\text{Cr}-\text{Cl} \\ \downarrow \\ \text{O} \end{array}$	$\text{Cr} = \dots\dots$
13.	Perchloric anhydride	Cl_2O_7	$\begin{array}{c} \text{O} \quad \text{O} \\ \diagdown \quad \diagup \\ \text{O}-\text{Cl}-\text{O}-\text{Cl} \\ \diagup \quad \diagdown \\ \text{O} \quad \text{O} \end{array}$	$\text{Cl} = \dots\dots$
14.	Calcium oxy-chloride/ Bleaching powder	CaOCl_2	$\text{Ca}(\text{O}^*\text{Cl})^{**}\text{Cl}$	$\begin{array}{l} * \text{Cl} = \dots\dots \\ ** \text{Cl} = \dots\dots \end{array}$

OXY ACIDS OF SULPHUR				O.S. of central Sulphur atom
1.	Sulphoxilic acid	H_2SO_2	$H-O-S-O-H$	
2.	Sulphurous acid	H_2SO_3	$\begin{array}{c} O \\ \uparrow \\ H-O-S-O-H \end{array}$	
3.	Sulphuric acid	H_2SO_4	$\begin{array}{c} O \\ \uparrow \\ H-O-S-O-H \\ \downarrow \\ O \end{array}$	
4.	Peroxymonosulphuric acid (Caro's acid)	H_2SO_5	$\begin{array}{c} O \\ \uparrow \\ H-O-S-O-O-H \\ \downarrow \\ O \end{array}$	
5.	Thiosulphurous acid	$H_2S_2O_2$	$\begin{array}{c} S \\ \uparrow \\ H-O-S-O-H \end{array}$	
6.	Thiosulphuric acid	$H_2S_2O_3$	$\begin{array}{c} S \\ \uparrow \\ H-O-S-O-H \\ \downarrow \\ O \end{array}$	
7.	Dithionous acid	$H_2S_2O_4$	$\begin{array}{c} O \quad O \\ \uparrow \quad \uparrow \\ H-O-S-S-O-H \end{array}$	
8.	Pyrosulphurous acid	$H_2S_2O_5$	$\begin{array}{c} O \quad O \\ \uparrow \quad \uparrow \\ H-O-S-S-O-H \\ \downarrow \\ O \end{array}$	
9.	Dithionic acid	$H_2S_2O_6$	$\begin{array}{c} O \quad O \\ \uparrow \quad \uparrow \\ H-O-S-S-O-H \\ \downarrow \quad \downarrow \\ O \quad O \end{array}$	
10.	Pyrosulphuric acid/ Fuming sulphuric acid/ Oleum	$H_2S_2O_7$	$\begin{array}{c} O \quad O \\ \uparrow \quad \uparrow \\ H-O-S-O-S-O-H \\ \downarrow \quad \downarrow \\ O \quad O \end{array}$	
11.	Peroxydisulphuric acid (Marshall's acid)	$H_2S_2O_8$	$\begin{array}{c} O \quad O \\ \uparrow \quad \uparrow \\ H-O-S-O-O-S-O-H \\ \downarrow \quad \downarrow \\ O \quad O \end{array}$	

OXY ACIDS OF PHOSPHOROUS				O.S. of central P atom
1.	Hypophosphorous acid	H_3PO_2	$\begin{array}{c} \text{O} \\ \uparrow \\ \text{H}-\text{P}-\text{O}-\text{H} \\ \\ \text{H} \end{array}$	
2.	Orthophosphorous acid/ Phosphorous acid	H_3PO_3	$\begin{array}{c} \text{O} \\ \uparrow \\ \text{H}-\text{O}-\text{P}-\text{O}-\text{H} \\ \\ \text{H} \end{array}$	
3.	Orthophosphoric acid/ Phosphoric acid	H_3PO_4	$\begin{array}{c} \text{O} \\ \uparrow \\ \text{H}-\text{O}-\text{P}-\text{O}-\text{H} \\ \\ \text{O} \\ \\ \text{H} \end{array}$	
4.	Hypophosphoric acid	$\text{H}_4\text{P}_2\text{O}_6$	$\begin{array}{c} \text{O} \quad \text{O} \\ \uparrow \quad \uparrow \\ \text{H}-\text{O}-\text{P}-\text{P}-\text{O}-\text{H} \\ \quad \\ \text{O} \quad \text{O} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	
5.	Pyrophosphoric acid	$\text{H}_4\text{P}_2\text{O}_7$	$\begin{array}{c} \text{O} \quad \text{O} \\ \uparrow \quad \uparrow \\ \text{H}-\text{O}-\text{P}-\text{O}-\text{P}-\text{O}-\text{H} \\ \quad \\ \text{O} \quad \text{O} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	
6.	Metaphosphoric acid	HPO_3	$\begin{array}{c} \text{O} \\ \uparrow \\ \text{O}=\text{P}-\text{O}-\text{H} \end{array}$	
7.	Peroxydiphosphoric acid	H_3PO_5	$\begin{array}{c} \text{O} \\ \uparrow \\ \text{H}-\text{O}-\text{P}-\text{O}-\text{O}-\text{H} \\ \\ \text{O} \\ \\ \text{H} \end{array}$	
8.	Peroxydiphosphoric acid	$\text{H}_4\text{P}_2\text{O}_8$	$\begin{array}{c} \text{O} \quad \text{O} \\ \uparrow \quad \uparrow \\ \text{H}-\text{O}-\text{P}-\text{O}-\text{O}-\text{P}-\text{O}-\text{H} \\ \quad \\ \text{O} \quad \text{O} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	

IMPORTANT NOTES

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IMPORTANT NOTES

BEHAVIOUR OF GASES

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E

NEET SYLLABUS

Behaviour of gases : Graham's Law of diffusion , Deviation from ideal gas behaviour

@neetfocusquizz

OBJECTIVES

After studying this unit, you will be able to :

- explain the diffusion of different non-reactive gases.
- explain the behaviour of real gases.
- explain describe the conditions required for liquifaction of gases

*The snowflake falls, yet lays not long its feath'ry grasp on Mother Earth
Ere Sun returns it to the vapors Whence it came, or to waters tumbling down the
rocky slope.*

Rod O' Connor

BEHAVIOUR OF GASES

INTRODUCTION :

Gaseous state is that state of matter where repulsive forces are dominating than attractive forces. For study of gases we require gas parameters like p , v , n and temperature. Gases having no interaction between gas molecules are known as ideal gases. They follow kinetic gas assumptions, kinetic gas equation, gas laws and ideal gas equation but certain gases which do not follow above conditions are known as real gases.

(A) SOME IMPORTANT DEFINITIONS :

(a) Diffusion :

- The tendency of inter mixing of non-reacting gases spontaneously without external pressure.

or

- The property of the spontaneous flow of gaseous molecules from high concentration to low concentration without external pressure.

In diffusion, $P_{\text{ext}} = \text{constant}$.

(b) Effusion :

- The property of the flow of gaseous molecules from high concentration to low concentration from a **pin hole** with external pressure.

or

- It is diffusion through **pin hole** which arises due to pressure difference.

In effusion, $P_{\text{ext}} = \text{variable}$

(c) Rate of diffusion (r) :

At constant temperature and pressure, the distance travelled by diffused gas or volume of diffused gas or moles of diffused gas per unit time is called as rate of diffusion.

$$r = \frac{d_{\text{diffused gas}}}{t_{\text{(taken)}}} = \frac{V_{\text{diffused gas}}}{t_{\text{(taken)}}} = \frac{n_{\text{diffused gas}}}{t_{\text{(taken)}}}$$

Graham's law of diffusion (Applicable for non-reacting gases) :

- At constant temperature and pressure, the rate of diffusion is inversely proportional to the square root of the density.

$$r \propto \frac{1}{\sqrt{d}} \quad (\text{at constant } T \text{ and } P)$$

$$r \propto \frac{1}{\sqrt{VD}}$$

$$r \propto \frac{1}{\sqrt{M_w}}$$

$$\frac{r_1}{r_2} = \frac{v_1}{t_1} \times \frac{t_2}{v_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{M_2}{M_1}}$$

- At constant temperature, rate of diffusion is proportional to $\frac{P}{\sqrt{d}}$.

$$\frac{r_1}{r_2} = \frac{v_1}{t_1} \times \frac{t_2}{v_2} = \frac{P_1}{P_2} \sqrt{\frac{d_2}{d_1}} = \frac{P_1}{P_2} \sqrt{\frac{M_2}{M_1}}$$

(B) REAL GASES : Those gases which do not follow ideal gas equation ($PV = nRT$) at all temperature and pressure.

- For real gases two assumptions of the kinetic theory do not hold good. These are

(i) There is no force of attraction between the molecules of a gas.

(ii) Volume of the molecules of a gas is negligibly small in comparison to the space occupied by the gas.

(a) REAL GAS EQUATION OR VAN DER WAAL GAS EQUATION

$$PV = nRT \quad \dots(1) \text{ ideal gas equation}$$

Two major corrections are required for real gases

(i) The intermolecular force of attraction between molecules is not negligible.

$$F_{\text{att}} \neq 0 \text{ so } P_{\text{act.}} = \left(P_{\text{real}} + \frac{n^2 a}{V^2} \right)$$

(ii) The volume of a molecule is not negligible in comparison of total volume of gas.

$$V_1 \neq 0 \text{ so } V_{\text{act}} = (V - nb)$$

$$b = \text{covolume} = \text{excluded volume} = 4 \times \frac{4\pi r^3}{3} \times N_A$$

From equation (i)

$$\left(P + \frac{an^2}{V^2} \right) (V - nb) = nRT$$

- a and b are van der Waal's gas constant which depends on nature of gas but do not depend on the temperature and pressure.

(A) van der Waal's gas constant (a) :

Where **a** is van der Waal gas constant which indicates the **inter molecular** forces of attraction.

$$\text{Pressure} = \frac{an^2}{V^2}$$

$$\text{unit of } a = \text{atm L}^2 \text{ mol}^{-2}$$

$$P_i = P_R + \frac{an^2}{V^2}$$

$$P_i > P_R$$

- $a \uparrow \Rightarrow F_{\text{att}} \uparrow \Rightarrow \text{liquefaction} \uparrow$.
- Order of a is 10^{-1} to 10^{-2} atm L² mol⁻²

(B) van der Waal gas constant (b)

It tells about molecular volume of gas molecules.

$$\text{Unit of } b = \text{L mol}^{-1}$$

$$b = \text{incompressible volume}$$

$$V_i > V_R$$

- If $b \uparrow \Rightarrow \text{effective size of molecule} \uparrow$
 $\Rightarrow \text{Incompressible volume} \uparrow$
 $\Rightarrow \text{compressible volume} \downarrow$
- Order of b is 10^{-2} to 10^{-4} L mol⁻¹

(b) DEVIATION OF REAL GASES FROM IDEAL GAS BEHAVIOUR :

"At very low pressure and high temperature real gases behave as ideal gas." Deviation of real gases from ideal gas behaviour is mathematically denoted by "Z" named as **compressibility factor**.

$$Z = \frac{V_{\text{real}}}{V_{\text{ideal}}} = \frac{V_{\text{obs}}}{nRT} = \frac{P(V_m)_{\text{obs}}}{RT}$$

Case I If $Z = 1$; ideal gas ; it signifies there is no interaction between gas molecules

$$\frac{P(V_m)_{\text{obs}}}{RT} = 1 \Rightarrow P(V_m)_{\text{obs}} = RT$$

Case II If $Z > 1$; positive deviation from ideal gas behaviour; signifies repulsive forces are greater than attractive forces so such gases can't be liquefied easily.

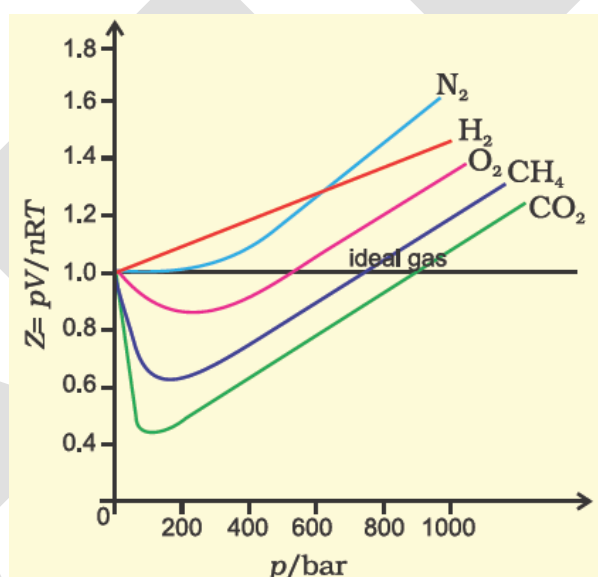
$$\frac{P(V_m)_{\text{obs}}}{RT} > 1 \Rightarrow P(V_m)_{\text{obs}} > RT$$

The gas shows the positive deviation generally at high pressure from ideal gas behaviour.

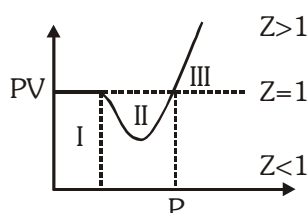
Case III If $Z < 1$; Negative deviation from ideal gases behaviour; it signifies attractive forces are higher than repulsive forces so gas can be liquefied easily.

$$\frac{P(V_m)_{\text{obs}}}{RT} < 1 \Rightarrow P(V_m)_{\text{obs}} < RT$$

The gas show the negative deviation generally at low pressure from ideal gas behaviour.



(c) Effect of pressure on van der Waal gas equation



- **Critical volume (V_c)** : Volume occupied by 1 mol gas at critical temperature & pressure; is called critical volume $V_c = 3b$.

- **Van Der waal gas constants :**

Gas	a (atm L ² mol ⁻²)	b (L mol ⁻¹)
H ₂	0.244	0.0266
N ₂	1.39	0.0391
CH ₄	2.25	0.0428
CO ₂	3.59	0.0427
NH ₃	4.17	0.0371
H ₂ O	5.46	0.0305

- Higher value of "a" is more significant for liquefaction of a gas.
- **Boyle's Temperature (T_b)** : Temperature at which a real gas obeys ideal gas law over an appreciable range of pressure is called Boyle's temperature.

$$T_b = \frac{a}{bR}$$

Illustrations

Illustration 1 The rate of effusion of a gas is proportional to –

- (1) $\frac{P}{\sqrt{d}}$ (2) $\frac{P}{d}$ (3) $\sqrt{\frac{P}{d}}$ (4) $\frac{\sqrt{P}}{d}$

Solution

Ans. (1)

$$\text{Rate of effusion} \propto \frac{1}{\sqrt{d}} \propto P \therefore \text{Rate of effusion} \propto \frac{P}{\sqrt{d}}$$

Illustration 2 If 4g of oxygen diffuse through a very narrow hole, how much hydrogen would have diffused under identical conditions ?

- (1) 16 g (2) 1 g (3) 1/4 g (4) 64 g

Solution

Ans. (2)

$$\frac{r_{O_2}}{r_{H_2}} = \frac{V_{O_2}}{V_{H_2}} ; \frac{n_{O_2}}{n_{H_2}} = \sqrt{\frac{M_{wH_2}}{M_{wO_2}}} \Rightarrow \frac{4/32}{w/2} = \sqrt{\frac{2}{32}} \Rightarrow \frac{1}{4w} = \frac{1}{4} \Rightarrow w = 1$$

Illustration 3 van der Waal's equation of state is obeyed by real gases. For n moles of a real gas, the expression will be –

- (1) $\left(\frac{P}{n} + \frac{na}{V^2}\right)\left(\frac{V}{n-b}\right) = RT$ (2) $\left(P + \frac{a}{V^2}\right)(V-b) = nRT$
 (3) $\left(P + \frac{na}{V^2}\right)(nV-b) = nRT$ (4) $\left(P + \frac{n^2a}{V^2}\right)(V-nb) = nRT$

Solution

Ans. (4)

$$\text{van der Waal's equation is } \left(P + \frac{an^2}{V^2}\right)(V-nb) = nRT$$

Illustration 4 van der Waal's constant 'a' and 'b' are related withrespectively –

- (1) Attractive force and bond energy of molecules
- (2) Volume and repulsive force of molecules
- (3) Shape and repulsive forces of molecules
- (4) Attractive force and volume of the molecules

Solution **Ans. (4)**

'a' is related to attractive forces and 'b' to the volume of the molecules.

BEGINNER'S BOX-1

1. The rate of diffusion of SO_2 and O_2 are in the ratio –
 (1) $1 : \sqrt{2}$ (2) $1 : 32$ (3) $1 : 2$ (4) $1 : 4$
2. In the van der Waal's equation which will have highest value of 'a'.
 (1) CH_4 (2) He (3) NH_3 (4) H_2
3. The unit of the van der Waal's constant 'a' is :
 (1) $\text{atm L}^2 \text{mol}^{-2}$ (2) $\text{atm L}^{-2} \text{mol}^{-2}$ (3) atm L mol^{-1} (4) atm mol L^{-1}

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3							
	Ans.	1	3	1							

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

GRAHAM'S LAW OF DIFFUSION

- Which pair of the gaseous species diffuse through a small jet with the same rate of diffusion at same P and T :
 (1) NO, CO (2) NO, CO₂
 (3) NH₃, PH₃ (4) NO, C₂H₆
- The rate of diffusion of methane at a given temperature is twice that of a gas X. The molecular weight of X is :
 (1) 64 (2) 32 (3) 4.0 (4) 8.0
- The increasing order of effusion among the gases, H₂, O₂, NH₃ and CO₂ is –
 (1) H₂, CO₂, NH₃, O₂ (2) H₂, NH₃, O₂, CO₂
 (3) H₂, O₂, NH₃, CO₂ (4) CO₂, O₂, NH₃, H₂
- Gas A having molecular weight 4 diffuses thrice as fast as the gas B at a given T. The molecular weight of gas B is :
 (1) 36 (2) 12 (3) 18 (4) 24
- Four rubber tubes are respectively filled with H₂, O₂, N₂ and CO₂. The tube which will be reinflated first is :
 (1) H₂ filled tube (2) O₂ filled tube
 (3) N₂ filled tube (4) CO₂ filled tube
- A balloon filled with methane CH₄ is pricked with a sharp point and quickly plunged into a tank of hydrogen at the same pressure. After sometime the balloon will have :
 (1) Enlarged
 (2) Collapsed
 (3) Remained unchanged in size
 (4) Ethylene (C₂H₄) inside it
- Rate of diffusion of hydrogen is :
 (1) Half of He (2) 1.4 times of He
 (3) Double than He (4) four times of He
- A football bladder contains equimolar proportions of H₂ and O₂. The composition by mass of the mixture effusing out of punctured football is in the ratio (H₂ : O₂)
 (1) 1 : 4 (2) 2√2 : 1
 (3) 1 : 2√2 (4) 4 : 1
- If the vapour densities of methane & oxygen are in the ratio 1 : 2, the ratio of rate of diffusion of O₂ & CH₄ is respectively
 (1) 1 : 2 (2) 1 : 1.414
 (3) 2 : 1 (4) 1.414 : 1
- A gas X diffuses three times faster than another gas Y the ratio of their densities i.e., D_x : D_y is
 (1) 1/3 (2) 1/9
 (3) 1/6 (4) 1/12
- The relative rate of diffusion of a gas (Mol wt. = 98) as compared to hydrogen will be :
 (1) 1/7 (2) 1/5
 (3) 1/4 (4) 1
- The relative rate of diffusion of a gas (molecular weight = 128) as compared to oxygen is
 (1) 2 times (2) 1/4 times
 (3) 1/8 times (4) 1/2 times
- Since the atomic weights of carbon, nitrogen and oxygen are 12, 14 and 16 respectively, among the following pairs of gases, the pair that will diffuse at the same rate is :
 (1) Carbon dioxide and nitrous oxide
 (2) Carbon dioxide and nitrogen peroxide
 (3) Carbon dioxide and carbon monoxide
 (4) Carbon dioxide and nitric oxide
- A bottle of dry ammonia and a bottle of dry hydrogen chloride connected through a long tube are opened simultaneously at both ends, the white ammonium chloride ring first formed will be :
 (1) at the centre of the tube
 (2) near the hydrogen chloride bottle
 (3) near the ammonia bottle
 (4) throughout the length of the tube
- 50 ml of a gas A diffuse through a membrane in the same time as for the diffusion of 40 ml of a gas B under identical pressure temperature conditions. If the Molecular weight of A = 64, that of B would be :
 (1) 100 (2) 250
 (3) 200 (4) 80
- If rate of diffusion of A is 5 times that of B, what will be the density ratio of A and B :
 (1) 1/25 (2) 1/5
 (3) 25 (4) 5
- 50 ml of hydrogen diffuses through a small hole from vessel in 20 minutes time. Time taken for 40 ml of oxygen to diffuse out under similar conditions will be :
 (1) 12 min. (2) 64 min
 (3) 8 min (4) 32 min
- The densities of two gases are in the ratio of 1 : 16. The ratio of their rates of diffusion is :
 (1) 16 : 1 (2) 4 : 1
 (3) 1 : 4 (4) 1 : 16
- The rate of diffusion of a gas having molecular weight just double of nitrogen gas is 56 ml per sec the rate of diffusion of nitrogen gas will be :
 (1) 79.19 ml/sec. (2) 112 ml/sec
 (3) 56 ml/sec (4) 90 ml/sec

20. If the four tubes of a car are filled to the same pressure with N_2 , O_2 , H_2 and CO_2 separately then which one will be filled first :
- (1) N_2 (2) O_2
(3) H_2 (4) CO_2
21. Under identical conditions of temperature and pressure the ratio of the rates of effusion of O_2 and CO_2 gases is given by :
- (1) $\frac{\text{rate of effusion of oxygen}}{\text{rate of effusion of } CO_2} = 0.87$
(2) $\frac{\text{rate of effusion of oxygen}}{\text{rate of effusion of } CO_2} = 1.17$
(3) $\frac{\text{rate of effusion of oxygen}}{\text{rate of effusion of } CO_2} = 8.7$
(4) $\frac{\text{rate of effusion of oxygen}}{\text{rate of effusion of } CO_2} = 0.117$
- DEVIATION FROM IDEAL GAS BEHAVIOUR**
22. When does a real gas show behaviour same as ideal gas:
- (1) At low temperature and low pressure
(2) At high temperature and high pressure
(3) At low temperature and high pressure
(4) At high temperature and low pressure
23. In van der Waal's equation of state of the gas law, the constant 'b' is a measure of :
- (1) intermolecular repulsions
(2) intermolecular attraction
(3) volume occupied by the molecules
(4) intermolecular collisions per unit volume
24. The term that accounts for intermolecular force in van der Waals' equation for non ideal gas is :
- (1) RT (2) $V - b$
(3) $\left(P + \frac{a}{V^2}\right)$ (4) $[RT]^{-1}$
25. Pressure of real gas is less than the pressure of ideal gas because :
- (1) No. of collisions increases
(2) Definite shape of molecule
(3) K.E. of molecule increases
(4) Inter molecular forces
26. Which gas can be easily liquefied ? given 'a' for $NH_3 = 4.17$, $CO_2 = 3.59$, $SO_2 = 6.71$, $Cl_2 = 6.49$
- (1) NH_3 (2) Cl_2
(3) SO_2 (4) CO_2
27. At relatively high pressure, van der waals' equation reduces to :
- (1) $PV = RT$ (2) $PV = RT + \frac{a}{V}$
(3) $PV = RT + Pb$ (4) $PV = RT - \frac{a}{V^2}$
28. A real gas most closely approaches the behaviour of an ideal gas at :
- (1) 15 atm and 200 K (2) 1 atm and 273 K
(3) 0.5 atm and 500 K (4) 15 atm and 500 K
29. The compressibility factor of an ideal gas is :
- (1) 0 (2) 1
(3) 2 (4) 4
30. The compressibility of a gas is less than unity at STP therefore :
- (1) $V_m > 22.4 \text{ L}$
(2) $V_m < 22.4 \text{ L}$
(3) $V_m = 22.4 \text{ L}$
(4) $V_m = 44.8 \text{ L}$
31. The values of van der Waal's constant 'a' for the gases O_2 , N_2 , NH_3 and CH_4 are 1.360, 1.390, 4.170 and 2.253 $L^2 \text{ atm mol}^{-2}$ respectively. The gas which can most easily be liquefied is :
- (1) O_2 (2) N_2
(3) NH_3 (4) CH_4

EXERCISE-I (Conceptual Questions)**ANSWER KEY**

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	1	4	1	1	1	2	1	2	2	1	4	1	2	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	2	1	3	2	4	3	3	4	3	3	3	2	2
Que.	31														
Ans.	3														

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT Pre.-2011

1. Two gases A and B having the same volume diffuse through a porous partition in 20 and 10 seconds respectively. The molecular mass of A is 49u. Molecular mass of B will be :-
 (1) 50.00 u (2) 12.25 u
 (3) 6.50 u (4) 25.00 u

AIPMT Pre. - 2012

2. 50 mL of each gas A and of gas B takes 150 and 200 seconds respectively for effusing through a pin hole under the similar condition. If molecular mass of gas B is 36, the molecular mass of gas A will be:
 (1) 20.25 (2) 64 (3) 96 (4) 128

AIPMT Main - 2012

3. A certain gas takes three times as long to effuse out as helium. Its molecular mass will be :
 (1) 64 u (2) 9 u (3) 27 u (4) 36 u

NEET-UG 2013

4. Maximum deviation from ideal gas is expected from :
 (1) $\text{NH}_3(\text{g})$ (2) $\text{H}_2(\text{g})$ (3) $\text{N}_2(\text{g})$ (4) $\text{CH}_4(\text{g})$

AIIMS-2014

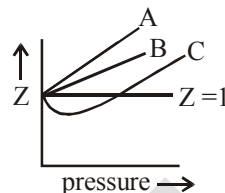
5. The temperature at which intermolecular attraction force balance the intermolecular repulsion force is known as :-
 (1) Boyle temperature
 (2) Critical temperature
 (3) Boiling temperature
 (4) Equilibrium temperature

AIIMS-2016

6. Which is correct for gases showing attraction behaviour?
 (1) $\left(\frac{dV}{dP}\right)_T < 0$ (2) $\left(\frac{d}{dP}(PV)\right)_T > 0$
 (3) $\left(\frac{d}{dP}PV\right)_T < 0$ (4) All of these

AIIMS-2017

7. In following compressibility factor v/s pressure graph which is true :-



- (1) At high pressure, gas B behaves as non ideal
 (2) At moderate pressure, gas B behaves as ideal gas.
 (3) At moderate pressure, gas A behaves as ideal gas.
 (4) At moderate pressure, gas C behaves as ideal gas.

NEET(UG) 2018

8. Given van der Waal's constant for NH_3 , H_2 , O_2 and CO_2 are 4.17, 0.244, 1.36 and 3.59, $\text{L}^2 \text{ atm mol}^{-1}$ respectively which one of the following gases is most easily liquefied?
 (1) NH_3 (2) H_2 (3) O_2 (4) CO_2
9. The correction factor 'a' to the ideal gas equation corresponds to
 (1) density of the gas molecules
 (2) volume of the gas molecules
 (3) electric field present between the gas molecules
 (4) forces of attraction between the gas molecules

AIIMS-2018

10. 1 mole of real gas ($a = 1.4$) occupies a volume of 0.1 L at 300K then the pressure of gas will be :-
 (1) 2460 atm (2) 2320 atm
 (3) 106 atm (4) 212 atm
11. A closed rigid container at 300 K contains N_2 gas at 10 atm pressure and this container can bear maximum 15 atm pressure; at what temperature the container will burst ?
 (1) 350 K (2) 400 K
 (3) 450 K (4) 425 K

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	
Ans.	2	1	4	1	1	3	1	1	4	3	3	

EXERCISE-III (Analytical Questions)

Check Your Understanding

1. When r , P and M represent rate of diffusion, pressure and molecular mass, respectively, then the ratio of the rates of diffusion (r_A/r_B) of two gases A and B, is given as :-
- (1) $(P_A/P_B)^{1/2} (M_A/M_B)$
 (2) $(P_A/P_B) (M_B/M_A)^{1/2}$
 (3) $(P_A/P_B)^{1/2} (M_B/M_A)$
 (4) $(P_A/P_B) (M_A/M_B)^{1/2}$
2. The compressibility factor for a real gas at high pressure is :-
- (1) $1 - \frac{pb}{RT}$ (2) $1 + \frac{RT}{pb}$
 (3) 1 (4) $1 + \frac{pb}{RT}$
3. If Z is a compressibility factor, van der Waal's equation **at low** pressure can be written as :
- (1) $Z = 1 - \frac{Pb}{RT}$ (2) $Z = 1 + \frac{Pb}{RT}$
 (3) $Z = 1 + \frac{RT}{Pb}$ (4) $Z = 1 - \frac{a}{VRT}$

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	
Ans.	2	4	4	

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

1. **Assertion :-** The pressure of real gas is less than the pressure of ideal gas.

Reason :- The intermolecular forces of attraction in real gases are greater than those of ideal gas.

- (1) A (2) B (3) C (4) D

2. **Assertion :-** CO₂ gas will have higher rate of liquification then H₂ gas.

Reason :- Because CO₂ will have higher value of Z so higher inter molecular force of attraction.

- (1) A (2) B (3) C (4) D

3. **Assertion :-** For Vanderwaal real gas, in low pressure region equation of compressibility factor

$$\text{is } Z = 1 - \frac{a}{VRT}$$

Reason :- In low pressure region 'b' is negligible.

- (1) A (2) B (3) C (4) D

4. **Assertion :-** Gases having higher critical temperature can be liquified easily.

Reason :- because attractive forces are higher for gases having higher critical temperature

- (1) A (2) B (3) C (4) D

5. **Assertion :-** At critical condition value of compressibility factor is 3/8.

Reason :- At critical condition attraction forces dominates.

- (1) A (2) B (3) C (4) D

6. **Assertion :-** Critical temperature of CO₂ is 304K. If it is compressed at 320 K. Then it will not liquify.

Reason :- Volume decreases on increasing pressure at a given temperature.

- (1) A (2) B (3) C (4) D

7. **Assertion :** Value of PV increases linearly for real gas on increasing pressure at constant temprature.

Reason : PV for real gas does not depend on pressure when temprature is constant.

- (1) A (2) B (3) C (4) D

8. **Assertion :** Pressure of real gas is less then ideal gas.

Reason : Because attractive forces in real gases are more than ideal gas.

- (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8
Ans.	1	3	1	1	2	2	4	1



PERIODIC TABLE

E

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NEET SYLLABUS

PERIODIC TABLE : Modern periodic law and long form of periodic table, periodic trends in properties of elements- atomic radii, ionic radii, ionization enthalpy, electron gain enthalpy, electronegativity, valence.

@neetfocusquizz

OBJECTIVES

After studying this unit, you will be able to :

- Understand the Periodic Law.
- Understand the significance of atomic number and electronic configuration as the basis for periodic classification;
- Name the elements with $Z > 100$ according to IUPAC nomenclature;
- Classify elements into s, p, d, f blocks and learn their main characteristics;
- Recognise the periodic trends in physical and chemical properties of elements;
- use scientific vocabulary appropriately to communicate ideas related to certain important properties of atoms e.g., atomic/ ionic radii, ionization enthalpy, electron gain enthalpy, electronegativity, valence of elements.

The Periodic Table is arguably the most important concept in chemistry, both in principle and in practice. An awareness of the Periodic Table is essential to anyone who wishes to disentangle the world and see how it is built up from the fundamental building blocks of the chemistry, the chemical elements.

Glenn T. Seaborg

Extended or Long Form of the Periodic Table

		p-Block Elements																0 (18)	
		s-Block Elements				d-Block Elements													
		IA (1)		IIA (2)		IIIB (3)		IVB (4)		VB (5)		VIB (6)		VIIB (7)		VIII (8) (9) (10)		IIB (12)	
Period	↓	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	↓	H 1.0079 Hydrogen																	He 4.0026 Helium
2		Li 6.940 Lithium		Be 9.0122 Beryllium															Ne 20.180 Neon
3		Na 22.990 Sodium		Mg 24.305 Magnesium															Ar 39.948 Argon
4		K 39.098 Potassium		Ca 40.078 Calcium															Kr 83.80 Krypton
5		Rb 85.468 Rubidium		Sr 87.62 Strontium															Xe 131.29 Xenon
6		Cs 132.91 Cesium		Ba 137.33 Barium															Rn 222 Radon
7		Fr 223 Francium		Ra 226 Radium															Uuo 118 Ununoctium
		*Lanthanide Series		**Actinide Series															
6																			
7																			

IUPAC designations of groups of elements are given in brackets

PERIODIC TABLE

1.0 INTRODUCTION :

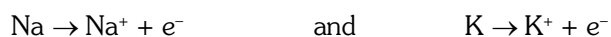
The arrangement of all the known elements according to their properties in such a way that the elements with similar properties are grouped together in a tabular form is called periodic table.

DEVELOPMENT OF PERIODIC TABLE

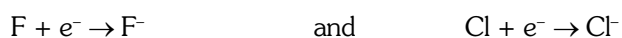
(A) LAVOISIER CLASSIFICATION :

- (i) Lavoisier classified the elements simply in metals and non metals.

Metals are the one which have the tendency of losing the electrons.



Non-metals are the one which have the tendency of gaining the electrons.



- (ii) **Drawback or Limitation :**

- (a) As the number of elements increases, this classification became insufficient for the study of elements.
 (b) There are few elements which have the properties of both metals as well as non-metals and they are called metalloids. Lavoisier could not decide where to place the metalloids.

(B) PROUT'S HYPOTHESIS :

He simply assumed that all the elements are made up of hydrogen, so we can say that

Atomic weight of element = $n \times$ (Atomic weight of one hydrogen atom)

Atomic weight of H = 1

where n = number of hydrogen atom = 1, 2, 3,....

Drawback or Limitation :

- (i) Every element can not be formed by Hydrogen.
 (ii) Atomic weight of all elements were not found as the whole numbers.

Ex. Chlorine (atomic weight 35.5) and Strontium (atomic weight 87.6)

(C) DOBEREINER TRIAD RULE [1817] :

- (i) He made groups of three elements having similar chemical properties called TRIAD.
 (ii) In Dobereiner triad, atomic weight of middle element is nearly equal to the average atomic weight of first and third element.

Ex.	Cl	Br	I	Ca	Sr	Ba	Li	Na	K
	35.5	80.0	127	40	87.6	137	7	23	39

$$\left[x = \frac{35.5 + 127}{2} = 81.2 \right]$$

$$\left[x = \frac{40 + 137}{2} = 88.5 \right]$$

$$\left[x = \frac{7 + 39}{2} = 23 \right]$$

Where x = average atomic weight

- (iii) Other examples – (K, Rb, Cs), (P, As, Sb), (S, Se, Te)

Drawback or Limitation : All the known elements could not be arranged as triads. It is not applicable for d and f-block elements.

(D) NEWLAND OCTAVE RULE [1865]

- He arranged the elements in the increasing order of their atomic mass and observed that the properties of every 8th element was similar to the 1st element. (like in the case of musical vowels notation)
- At that time inert gases were not known.

Sa	Re	Ga	Ma	Pa	Dha	Ni	Sa
1	2	3	4	5	6	7	8

H

Li	Be	B	C	N	O	F
Na	Mg	Al	Si	P	S	Cl
K	Ca					

- The properties of Li are similar to 8th element i.e. Na and Be are similar to Mg and so on.

Drawback or Limitation :

- This rule is valid only upto Ca because after Ca due to presence of d-block element there is a difference of 18 elements instead of 8 elements.
- After the discovery of Inert gas and including them in the periodic table, it has become the 8th element from Alkali metal so this law had to be dropped out.

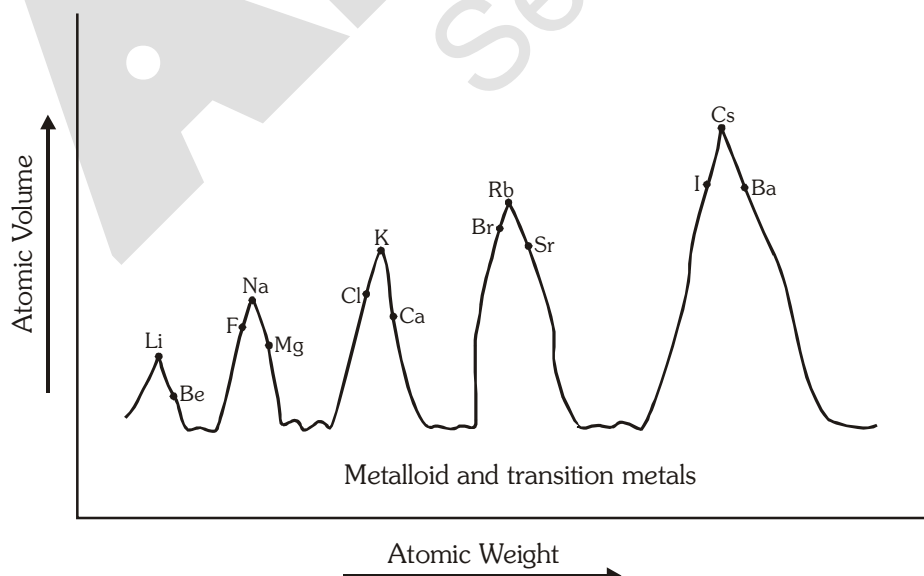
(E) LOTHAR MEYER'S CURVE [1869] :

- He plotted a curve between atomic weight and atomic volume of different elements.
- The following observation can be made from the curve –
 - Most electropositive elements i.e. alkali metals (Li, Na, K, Rb, Cs) occupy the peak positions on the curve.
 - Less electropositive i.e. alkaline earth metal (Be, Mg, Ca, Sr, Ba) occupy the descending position on the curve.
 - Metalloids (Si, Ge, As, Sb, Te, Po, At) and transition metals occupy bottom part of the curve.
 - Most electronegative i.e. halogens (F, Cl, Br, I) occupy the ascending position on the curve.

Note : Elements having similar properties occupy similar position on the curve.

Conclusion : On the basis of this curve Lothar Meyer proposed that the physical properties of the elements are periodic function of their atomic weight and this has become the base of Mendeleev's periodic table.

Periodic function : When the elements are arranged in the increasing order of their atomic weight, elements having similar properties gets repeated after a regular interval.



(F) MENDELEEV'S PERIODIC TABLE [1869] :

(i) **Mendeleev's periodic law :** The physical and chemical properties of elements are the periodic function of their atomic weight.

(ii) **Characteristics of Mendeleev's periodic table :**

- It is based on atomic weight
- 63 elements were known, noble gases were not discovered.
- He was the first scientist to classify the elements in a systematic manner i.e. in horizontal rows and in vertical columns.
- Horizontal rows are called periods and there were 7 periods in Mendeleev's Periodic table.
- Vertical columns are called groups and there were 8 group in Mendeleev's Periodic table.
- Each group upto VII is divided into A & B subgroups. 'A' sub group element are called normal or representative elements and 'B' sub group elements are called transition elements.
- The VIII group consisted of 9 elements in three rows (Transitional metals group).
- The elements belonging to same group exhibit similar properties.

(iii) **Merits or advantages of Mendeleev's periodic table :**

- Study of elements :** First time all known elements were classified in groups according to their similar properties. So study of the properties of elements become easier .
- Prediction of new elements :** It gave encouragement to the discovery of new elements as some gaps were left in it.

Sc (Scandium) Ga (Gallium) Ge (Germanium) Tc (Technetium)

These were the elements for whom position and properties were well defined by Mendeleev even before their discoveries and he left the blank spaces for them in his table.

Ex. Blank space at atomic weight 72 in silicon group was called Eka silicon (means properties like silicon) and element discovered later was named Germanium .

Similarly other elements discovered after mendeleev's periodic table were.

Eka Aluminium – Galium(Ga)	Eka Boron – Scandium (Sc)
Eka Silicon – Germanium (Ge)	Eka Manganese – Technetium (Tc)

- Correction of doubtful atomic weights :** Correction were done in atomic weight of some elements.

$$\text{Atomic weight} = \text{Valency} \times \text{Equivalent weight.}$$

Initially, it was found that equivalent weight of Be is 4.5 and it is trivalent ($V = 3$), so the weight of Be was 13.5 and there is no space in Mendeleev's table for this element. So, after correction, it was found that Be is actually bivalent ($V = 2$). So, the weight of Be became $2 \times 4.5 = 9$ and there was a space between Li and B for this element in Mendeleev's table.

Corrections were done in atomic weight of elements are – **U, Be, In, Au, Pt.**

(iv) **Demerits of Mendeleev's periodic table :**

- Position of hydrogen :** Hydrogen resembles both, the alkali metals (IA) and the halogens (VIIA) in properties so Mendeleev could not decide where to place it.
- Position of isotopes :** As atomic wt. of isotopes differs, they should have placed in different position in Mendeleev's periodic table. But there were no such places for isotopes in Mendeleev's table.

- (c) **Anomalous pairs of elements :** There were some pair of elements which did not follow the increasing order of atomic weights.

Ex. Ar and Co were placed before K and Ni respectively in the periodic table, but having higher atomic weights.

$$\begin{pmatrix} \text{Ar} & \text{K} \\ 39.9 & 39.1 \end{pmatrix}$$

$$\begin{pmatrix} \text{Te} & \text{I} \\ 127.5 & 127 \end{pmatrix}$$

$$\begin{pmatrix} \text{Co} & \text{Ni} \\ 58.9 & 58.6 \end{pmatrix}$$

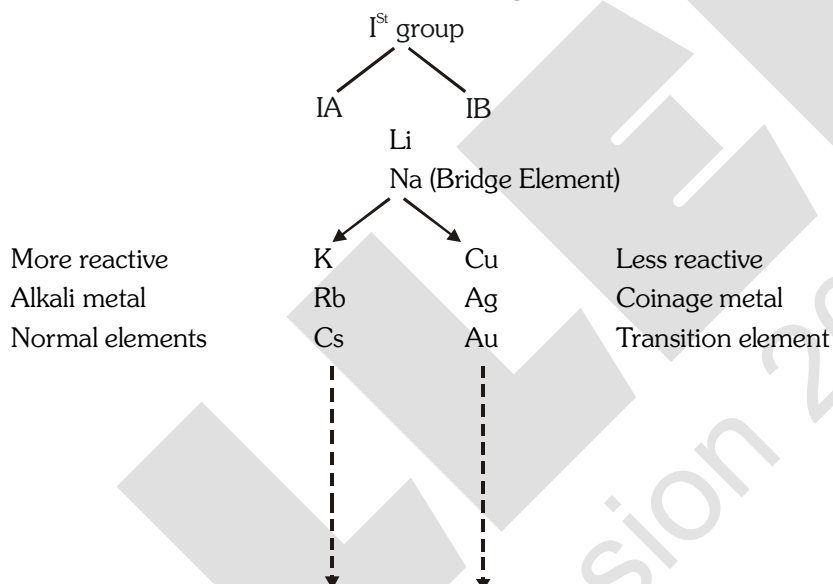
$$\begin{pmatrix} \text{Th} & \text{Pa} \\ 232 & 231 \end{pmatrix}$$

- (d) **Like elements were placed in different groups :**

There were some elements like Platinum (Pt) and Gold (Au) which have similar properties but were placed in different groups in Mendeleev's table.

Pt Au
VIII IB

- (e) **Unlike elements were placed in same group :**



Cu, Ag and Au placed in 1st group along with Na, K etc. While they differ in their properties (Only similar in having ns^1 electronic configuration)

- (f) It was not clear that '**lanthanides** and **Actinides**' are related with IIIA group or IIIB group.
- (g) **Cause of periodicity :** Why physical & chemical properties repeated in a group.

BEGINNER'S BOX-1

- Mendeleev's periodic law is based on -
(1) Atomic number (2) Atomic weight (3) Number of neutrons (4) None of the above
- The first attempt to classify elements systematically was made by -
(1) Mendeleev (2) Newland (3) Lothar Meyer (4) Dobereiner
- Atomic weight of an element X is 39, and that of element Z is 132. atomic weight of their intermediate element Y, as per dobereiner triad, will be
(1) 88.5 (2) 93.0 (3) 171 (4) 85.5
- Which of the following is not a dobereiner triad
(1) Li, Na, K (2) Mg, Ca, Sr (3) Cl, Br, I (4) S, Se, Te

5. The law of triads is applicable to
 (1) C, N, O (2) H, O, N (3) Na, K, Rb (4) Cl, Br, I
6. The law of triads is not applicable on
 (1) Cl, Br, I (2) Na, K, Rb (3) S, Se, Te (4) Ca, Sr, Ba
7. Which of the following set of elements obeys Newland's octave rule –
 (1) Na, K, Rb (2) F, Cl, Br (3) Be, Mg, Ca (4) B, Al, Ga
8. For which of the pair Newland octave rule is not applicable –
 (1) Li, Na (2) C, Si (3) Mg, Ca (4) Cl, Br
9. Which of the following element was present in Mendeleev's periodic table?
 (1) Sc (2) Tc (3) Ge (4) None of these
10. Is Fe, Co, Ni are dobereiner triad ?

1.1 MODERN PERIODIC TABLE (MODIFIED MENDELEEV PERIODIC TABLE) :

- (i) It was proposed by Moseley.
- (ii) Modern periodic table is based on atomic number.
- (iii) Moseley did an experiment in which he bombarded high speed electron on different metal surfaces and obtained X-rays.

He found out that $\sqrt{\nu} \propto Z$ where ν = frequency of X-rays, Z = atomic number.

From this experiment, Moseley concluded that the physical and chemical properties of the elements are periodic function of their atomic number. It means that when the elements are arranged in the increasing order of their atomic number elements having similar properties gets repeated after a regular interval. This is also known as 'Modern Periodic Law'.

- (iv) **Modern periodic law :** The physical & chemical properties of elements are the periodic function of their atomic number.
- (v) **Characteristics of modern periodic table :**
 - (a) 9 vertical columns called groups.
 - (b) I to VIII group + 0 group of inert gases.
 - (c) Inert gases were introduced in periodic table by Ramsay.
 - (d) 7 horizontal rows called periods.

LONG FORM / PRESENT FORM OF MODERN PERIODIC TABLE :

(It is also called as 'Bohr, Bury, Rang & Werner Periodic Table')

- (i) It is based on the Bohr-Bury electronic configuration concept and atomic number.
- (ii) This model is proposed by Rang & Werner
- (iii) 7 periods and 18 groups
- (iv) According to I. U. P. A. C. 18 vertical columns are named as 1st to 18th group.

- (v) The co-relation between the groups in long form of periodic table and in modern form of periodic table are given below –

IA	IIA	IIIB	IVB	VB	VIB	VII B	VIII	IB	IIB	IIIA	IVA	VA	VIA	VIIA	0
1	2	3	4	5	6	7	8 9 10	11	12	13	14	15	16	17	18

- (vi) Elements belonging to same group have same number of electrons in the outermost shell so their properties are similar.

Description of periods

Period	n	Period Sub shell	No. of elements	Element	Name of Period
1.	1	1s	2	${}_1\text{H} - {}_2\text{He}$	Shortest
2.	2	2s, 2p	8	${}_3\text{Li} - {}_{10}\text{Ne}$	Short
3.	3	3s, 3p	8	${}_{11}\text{Na} - {}_{18}\text{Ar}$	Short
4.	4	4s, 3d, 4p	18	${}_{19}\text{K} - {}_{36}\text{Kr}$	Long
5.	5	5s, 4d, 5p	18	${}_{37}\text{Rb} - {}_{54}\text{Xe}$	Long
6.	6	6s, 4f, 5d, 6p	32	${}_{55}\text{Cs} - {}_{86}\text{Rn}$	Longest
7.	7	7s, 5f, 6d, 7p	32	${}_{87}\text{Fr} - {}_{118}\text{Uuo}$	Complete

CONCLUSION

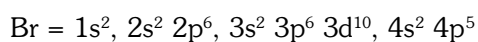
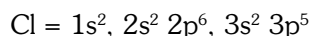
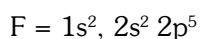
- Period number = outermost shell
- Number of element in a period = Number of electrons in a period subshell

DESCRIPTION OF GROUPS :

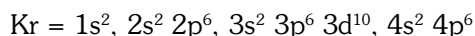
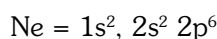
1st/IA/Alkali metals $\text{H} = 1s^1$ $\text{Li} = 1s^2, 2s^1$ $\text{Na} = 1s^2, 2s^2 2p^6, 3s^1$ $\text{K} = 1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^1$ General electronic configuration = ns^1 Number of valence shell $e^- = 1$	2nd/IIA/Alkaline earth metals $\text{Be} = 1s^2, 2s^2$ $\text{Mg} = 1s^2, 2s^2 2p^6, 3s^2$ $\text{Ca} = 1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2$ General electronic configuration = ns^2 (n = Number of shell) Number of valence shell $e^- = 2$
13th/IIIA/Boron Family $\text{B} = 1s^2, 2s^2 2p^1$ $\text{Al} = 1s^2, 2s^2 2p^6, 3s^2 3p^1$ $\text{Ga} = 1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^1$ General electronic configuration = $ns^2 np^1$ Number of valence shell $e^- = 3$	14th/IVA/Carbon Family $\text{C} = 1s^2, 2s^2 2p^2$ $\text{Si} = 1s^2, 2s^2 2p^6, 3s^2 3p^2$ $\text{Ge} = 1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^2$ General electronic configuration = $ns^2 np^2$ Number of valence $e^- = 4$
15th/VA/Nitrogen family/Pnicogen (Used in fertilizer as urea) $\text{N} = 1s^2, 2s^2 2p^3$ $\text{P} = 1s^2, 2s^2 2p^6, 3s^2 3p^3$ $\text{As} = 1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^3$ General electronic configuration = $ns^2 np^3$ Number of valence shell $e^- = 5$	16th/VIA/Oxygen family/Chalcogen (Ore forming) $\text{O} = 1s^2, 2s^2 2p^4$ $\text{S} = 1s^2, 2s^2 2p^6, 3s^2 3p^4$ $\text{Se} = 1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^4$ General electronic configuration = $ns^2 np^4$ Number of valence shell $e^- = 6$

17th/VIIA/Fluorine family/Halogens

(Salt forming)

General electronic configuration = $ns^2 np^5$ Number of valence shell $e^- = 7$ **18th/Zero group/Inert gases / Noble gases**

(Less reactive)

General electronic configuration = $ns^2 np^6$ (except He)Number of valence shell $e^- = 8$ **(vii) NOMENCLATURE OF ELEMENTS :**

(a) IUPAC gave names to elements above atomic number 100 as follows –

0	1	2	3	4	5	6	7	8	9
nil	un	bi	tri	quad	pent	hex	sept	oct	enn

(b) In all the elements suffix is – ium.

Ex.

Atomic No.	IUPAC Name	Symbol	Elemental Name	Symbol
101	Un nil unium	Unu	Mendelevium	Md
102	Un nil bium	Unb	Nobelium	No
103	Un nil trium	Unt	Lawrencium	Lr
104	Un nil quadium	Unq	Rutherfordium	Rf
105	Un nil pentium	Unp	Dubnium	Db
106	Un nil hexium	Unh	Seaborgium	Sg
107	Un nil septium	Uns	Bohrium	Bh
108	Un nil octium	Uno	Hassium	Hs
109	Un nil ennium	Une	Meitnerium	Mt
110	Un un nilium	Uun	Darmstadtium	Ds

(viii) Identification of group, period and block :**(A) When atomic number is given :**

Step I : $71 \geq Z \geq 58 \Rightarrow$ Lanthanoids (6th Period) } f-block
 $103 \geq Z \geq 90 \Rightarrow$ Actinoids (7th Period)

Group number = IIIB (largest group of periodic table)**Step II :** $Z = 104$ to 118 (Period number = 7)**Group number** = last two digits in atomic number of element**Example :** $Z = 104$

Group no. = 4

Step III : **Group number** = 18 + given atomic number – atomic number of next noble gas

If the value of this formula is negative then use 32 instead of 18 in formula.

(B) When electronic configuration is given

Period number (n) = number of outermost shell/Highest shell number.

Block identification :

- If np electron present then p - block ($ns^2 np^{1-6}$)
group number = 12 + np electrons
- If np electron absent then s/f/d block
If $(n-2)f^0 (n-1)d^0 ns^{1-2}$ = s block
group number = ns electrons
If $(n-2)f^{1-14} (n-1)d^{0-1} ns^2$ = f block
group number = IIIB
- If any other configuration or $(n-1)d^{1-10} ns^{0-2}$ (d-block)
group number = (n-1)d electron + ns electron

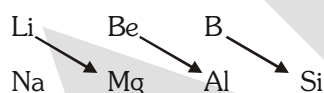
Bohr's Classification

Inert gases	Normal or representative elements	Transition element	Inner transition element
outermost shell complete	outermost shell incomplete	n & n-1 shells incomplete either in atomic or ionic form	n, (n-1), (n-2) shells incomplete
6 element	s & p block element except inert gas 38 element	all d block element except = IIB (Zn, Cd, Hg & Uub) 36 element	f-block elements 28 elements

SOME IMPORTANT POINTS :

- (a) 2nd period elements (Li, Be, B) Shows **diagonal relationship** with 3rd period elements (Mg, Al, Si). Because of same ionic potential value they shows similarity in properties.

$$(\text{Ionic potential } (\phi) = \frac{\text{Charge on cation}}{\text{Radius of cation}})$$

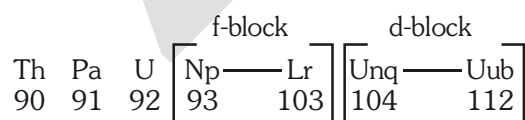


- (b) 3rd period elements (Na, Mg, Al, Si, P, S, Cl) except inert gases are called **typical elements** because they represent the properties of other element of their respective group.

(c) TRANSURANIC ELEMENTS :

Elements having atomic number more than 92 are known as transuranic element.

All transuranic elements are radioactive & artificial.



First man made element is Tc

First man made lanthanoid is Pm

All actinoids are radioactive but all lanthanoids are not artificial / man made (except Pm)

- (d) The group containing most electro positive elements – GROUP IA.
(e) The group containing most electro negative elements – GROUP VIIA
(f) The group containing maximum number of gaseous elements – GROUP ZERO (18th)

- (g) The group in which elements have generally ZERO valency – GROUP ZERO(18th)
- (h) **In the periodic table**
 Number of Gaseous elements – 11 (H, N, O, F, Cl + Noble gases)
 Number of Liquid elements – 6 (Cs, Fr, Ga, Hg, Br, Uub)
 Number of Liquid elements at room temp. – 2
 Bromine is the only non-metal which exists in liquid form.
 Number of Solid elements – 95 (if discovered elements are 112)
- (i) 0/18 group have all the elements in gaseous form.
- (j) 2nd period contains maximum number of gaseous elements. They are 4 (**N, O, F, Ne**)
- (k) IIIB/3rd group is called longest group having 32 elements including 14 Lanthanides and 14 Actinides
 Sc
 Y
 La.....Lanthanides (14)
 Ac.....Actinides (14)

BEGINNER'S BOX-2

- Which of the following is best general electronic configuration of normal element
 (1) $ns^{1-2} np^{0-6}$ (2) $ns^{1-2} np^{1-5}$ (3) $ns^{1-2} np^{0-5}$ (4) $ns^{1-2} np^{1-6}$
- Which of the following set of atomic numbers represents representative element
 (1) 5, 13, 30, 53 (2) 11, 33, 58, 84 (3) 5, 17, 31, 54 (4) 9, 31, 53, 83
- Which of the following electronic configuration does not belongs to same block as others :-
 (1) $[Xe] 4f^{14} 5d^{10} 6s^2$ (2) $[Kr] 4d^{10} 5s^2$ (3) $[Kr] 5s^2$ (4) $[Ar] 3d^6 4s^2$
- The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$. What is the atomic number of next element of the same group which is recently discovered :-
 (1) 20 (2) 119 (3) 111 (4) None
- Which of the following electronic configurations in the outermost shell is characteristic of alkali metals
 (1) $(n-1) s^2 p^6 ns^2 p^1$ (2) $(n-1) s^2 p^6 d^{10} ns^1$ (3) $(n-1) s^2 p^6 ns^1$ (4) $ns^2 np^6 (n-1)d^{10}$
- Which of the following elements belong to alkali metals ?
 (1) $1s^2, 2s^2 2p^2$ (2) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6, 5s^1$
 (3) $1s^2, 2s^2 2p^5$ (4) None of these
- Elements whose atoms have three outermost shells incomplete are called –
 (1) s-block (2) p-block (3) d-block (4) f-block
- Which of the following statement is wrong :-
 (1) All the actinides are synthetic (man made) elements
 (2) In the Lanthanides last electron enters in 4f orbitals
 (3) Np_{93} onwards are transuranic elements
 (4) Lanthanum is d-block element
- Which of the following statement is wrong :-
 (1) Total no. of liquid elements in the periodic table.....Six
 (2) First metal element in the periodic table is....Li
 (3) All type of elements are present in 6th period
 (4) Iodine is a gaseous element.
- An element which is recently discovered is placed in 7th period and 10th group. IUPAC name of the element will be :-
 (1) Unnilseptium (2) Ununnilium (3) Ununbium (4) None

1.2 PERIODICITY

- (A) In a period, the ultimate orbit remain same, but the number of e^- gradually increases.
In a group, the number of e^- in the ultimate orbit remains same, but the values of n increases.

(B) Causes of periodicity :

- (a) The cause of periodicity in properties is due to the same outermost shell electronic configuration coming at regular intervals.
(b) In the periodic table, elements with similar properties occur at intervals of 2, 8, 8, 18, 18 and 32. These numbers are called as magic numbers.

SCREENING EFFECT (σ) AND EFFECTIVE NUCLEAR CHARGE (Z_{eff}) :

- (i) Valence shell e^- suffer force of attraction due to nucleus and force of repulsion due to inner shell electrons.
(ii) The decrease in force of attraction on valence e^- due to inner shell e^- is called screening effect or shielding effect. (i.e. total repulsive force is called shielding effect.)
(iii) Due to screening effect valence shell e^- experiences less force of attraction exerted by nucleus. i.e. total attraction force experienced by valence electrons represented by a number is Z_{eff} .
(iv) There is a reduction in nuclear charge due to screening effect. Reduced nuclear charge is called effective nuclear charge.
(v) If nuclear charge = Z , effective nuclear charge = Z_{eff} , σ (Sigma) = Screening constant or shielding constant.

$$\text{So, } Z_{\text{eff}} = (Z - \sigma)$$

● Slater's rule to know screening constant (σ)

- (a) For single electron species $\sigma = 0$
(b) Screening effect (S.E.) for two e^- species 0.30

Ex. In He ($1s^2$)

Screening effect of one $1s e^-$. where $\sigma = 0.30$

$$\therefore Z_{\text{eff}} = Z - \sigma = 2 - 0.30 = 1.7$$

- (c) Screening effect of each ns and np (Outermost orbit) electrons is 0.35
(d) Screening effect of each $(n - 1)$ penultimate orbit s, p, d electrons is 0.85
(e) Screening effect of each $(n - 2)$ and below all the e^- present in s, p, d, f is 1.0

From top to bottom in a group Z_{eff} remain constant

Group	Element	Li	Na	K	Rb	Cs	Fr
	Z_{eff}	1.30	2.20	2.20	2.20	2.20	2.20
Period	Element	Be	B	C	N	O	F
	Z_{eff}	1.95	2.6	3.25	3.90	4.55	5.20

For same shell shielding effect has the order as $s > p > d > f$ (due to penetration effect)

Z_{eff} for different ions of an element

$$Z_{\text{eff}} \propto \frac{\text{positive charge}}{\text{negative charge}} \quad \begin{array}{l} \text{(i) } Z_{\text{eff}} \text{ for different ions of an element} \\ \text{(ii) } Z_{\text{eff}} \text{ for isoelectronic species.} \end{array}$$

(i) Z_{eff} for different ions of an element

Ex. $N^+ > N > N^- = Z_{\text{eff}}$

(ii) Z_{eff} of isoelectronic species

Ex. $H^- < Li^+ < Be^{+2} < B^{+3}$ ($2e^-$ species)

$N^{3-} < O^{2-} < F^- < Na^+ < Mg^{+2}$ ($10e^-$ species)

1.3 ATOMIC RADIUS

The average distance of valence shell e^- from nucleus is called atomic radius. It is very difficult to measure the atomic radius because –

- The isolation of single atom is very difficult.
 - There is no well defined boundary for the atom. (The probability of finding the e^- is 0 only at infinity).
- So, the more accurate definition of atomic radius is –
- Half the inter-nuclear distance(d) between two atoms in a homoatomic molecule is known as atomic radius.
 - This inter-nuclear distance is also known as bond length. Inter-nuclear distance depends upon the type of bond by which two atoms combine.

Based on the chemical bonds, atomic radius is divided into four categories –

(A) Covalent radius (B) Ionic radius (C) Metallic radius (D) van der Waal's radius

(A) Covalent Radius

(SBCR –Single Bonded Covalent Radius)

- Covalent bonds are formed by overlapping of atomic orbitals.
- Internuclear distance is minimum in this case.
- Covalent radius is the half of the internuclear distance between two singly bonded homo atoms.

Ex. If internuclear distance of $A-A(A_2)$ molecule is (d_{A-A}) and covalent radius is r_A then

$$d_{A-A} = r_A + r_A \quad \text{or} \quad 2r_A$$

$$r_A = \frac{d_{A-A}}{2}$$

Ex. In Cl_2 molecule, internuclear distance is $1.98 \text{ \AA}$ so $r_d = \frac{1.98}{2} = 0.99 \text{ \AA}$

(B) Ionic Radius**(i) Cationic Radius**

- When an neutral atom loses e^- it converts into cation (+ve charged ion)
- Cationic radius is always smaller than atomic radius **because** after losing e^- number of e^- reduces, but number of protons remains same, due to this Z_{eff} increases, hence electrons are pulled towards nucleus and atomic radius decreases, moreover after losing all the electrons from the outer most shell, penultimate shell becomes ultimate shell which is nearer to nucleus so size decreases.

$$(c) \text{ Size of cation } \propto \frac{1}{\text{Magnitude of the charge or } Z_{\text{eff}}}$$

Ex. (i) $Fe > Fe^{+2} > Fe^{+3}$

(ii) $Pb^{+2} > Pb^{+4}$

(iii) $Mn > Mn^{+2} > Mn^{+3} > Mn^{+4} > Mn^{+5} > Mn^{+6} > Mn^{+7}$

(ii) Anionic Radius

- (a) When a neutral atom gains e^- it converts into anion [Negative charge ion]
 (b) Anionic radius is always greater than atomic radius **because** in an anion e^- are more than protons and inter electronic repulsion increases, which also increases screening effect. So effective nuclear charge reduces, so distance between e^- and nucleus increases and size of anion also increases.

Ex. Flourine ($Z=9$)

	F	F^-
Proton	9	9
e^-	9	10

so $\frac{Z}{e} = \frac{9}{9} = 1$ $\frac{9}{10} = 0.9$ As Z_{eff} of F^- is less than F so size of $F^- > F$

(c) Size of isoelectronic species :

- Those species having same number of e^- but different nuclear charge forms isoelectronic series.
- For isoelectronic species the atomic radius increases with decrease in effective nuclear charge

Species	K^+	Ca^{+2}	S^{-2}	Cl^-
Z	19	20	16	17
e	18	18	18	18
$\frac{Z}{e}$	$\frac{19}{18}$	$\frac{20}{18}$	$\frac{16}{18}$	$\frac{17}{18}$

Order of radius : ($S^{-2} > Cl^- > K^+ > Ca^{+2}$), ($N^{3-} > O^{2-} > F^- > Na^+ > Mg^{+2} > Al^{+3}$)

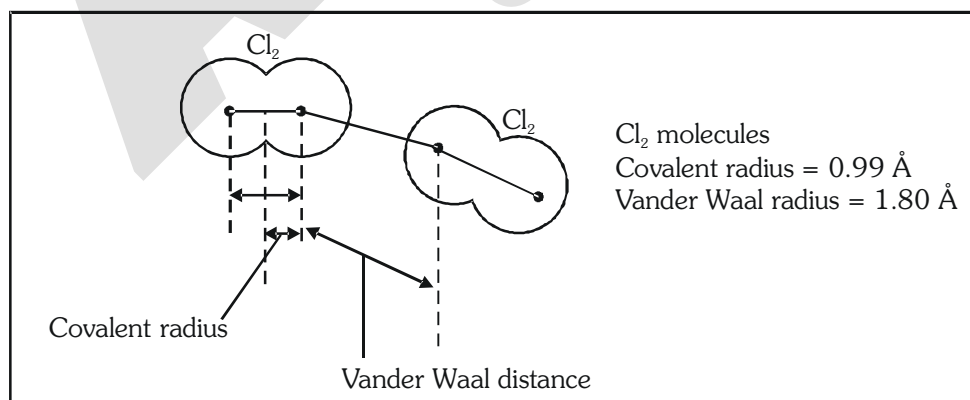


(C) Metallic/Crystal Radius

- (a) Half of the inter nuclear distance between two adjacent metallic atoms in crystalline lattice structure.
 (b) there is no overlapping of atomic orbitals, So **Metallic radius** > **Covalent radius**
 (c) Metallic radius $\propto \frac{1}{\text{Metallic bond strength}}$

(D) Vander Waal's Radius

- (a) Those atoms (like noble gases) which are not bonded with each other, experiences a weak attractive force to come nearer.
 (b) Half of the distance between the nuclei of adjacently placed atoms in solid state of a noble gas is Vander Waal's radius.
 (c) Inert gas have only Vander Waal radius.
 (d) In molecules of nonmetals solid both covalent and van der Waal's radius exists.



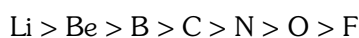
Vander Waal's radius $\cong 2 \times$ covalent radius

Vander Waal's radius > Metallic radius > Covalent radius

● **Factors affecting atomic size are :**

(a) **In a period**

$$\text{Atomic radius} \propto \frac{1}{Z_{\text{eff}}} \propto \frac{\text{negative charge}}{\text{positive charge}}$$



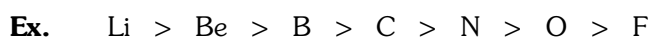
(b) **In a group**

$$\text{Atomic radius} \propto \text{number of shells}$$

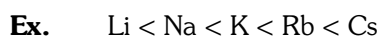


● **Periodic variation of atomic size :**

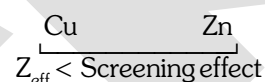
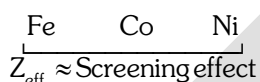
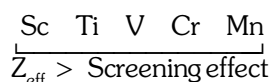
(i) **Across a period :** It decreases from left to right in a period as effective nuclear charge (Z_{eff}) increases



(ii) **In a group :** It increases from top to bottom in a group as number of shell increases

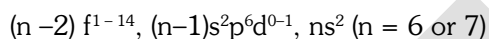


Exceptions : Transition elements



● **Lanthanide Contraction :**

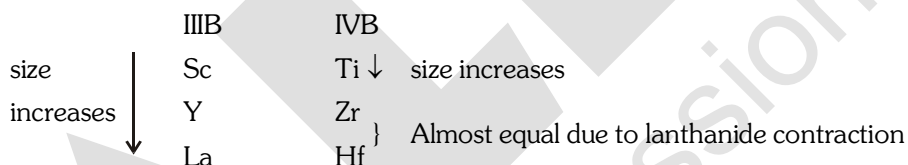
(a) Outermost electronic configuration of inner transition elements is



(b) e^- enters in $(n-2)f$ orbitals

(c) Because of complicated structure of f-orbital and due to poor shielding f electrons, the outermost shell electrons get attracted towards nucleus.

(d) In 1st, 2nd and 3rd transition series, Radii— $3d < 4d \approx 5d$ (**except IIIrd B**)

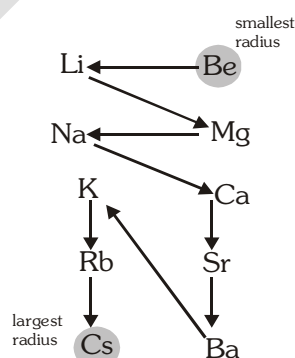


● **Transition contraction :**

Note : While atomic size should increase down the group.

At. size of Ga $\approx$ At. size of Al, due to transition contraction. (Due to poor shielding of d electrons)

● **s-block size variation**



BEGINNER'S BOX-3

- From the given set of species, point out the species from each set having least atomic radius:-
 (1) O^{2-} , F^- , Na^+ (2) Ni, Cu, Zn (3) Li, Be, Mg (4) He, Li^+ , H^-
 Correct answer is :-
 (1) O^{2-} , Cu, Li, H^- (2) Na^+ , Ni, Be, Li^+ (3) F^- , Zn, Mg, He (4) Na^+ , Cu, Be, He
- Which has the lowest anion to cation size ratio-
 (1) LiF (2) NaF (3) CsI (4) CsF
- Arrange the elements in increasing order of atomic radius Na, Rb, K, Mg :-
 (1) Na, K, Mg, Rb (2) K, Na, Mg, Rb (3) Mg, Na, K, Rb (4) Rb, K, Mg, Na
- Which of the following pairs of elements have almost similar atomic radii :-
 (1) Zr, Hf (2) Mo, W (3) Co, Ni (4) All
- If the ionic radii of $K^{\oplus}$ and $F^{\ominus}$ are nearly the same (i.e. 1.34 Å) then the atomic radii of K and F respectively are :-
 (1) 1.34 Å, 1.34 Å (2) 0.72 Å, 1.96 Å (3) 1.96 Å, 0.72 Å (4) 1.96 Å, 1.34 Å
- For the element X, student mansi measured its radius as 102 nm, student Rohit as 203nm. and Ankur as 100 nm. using same apparatus. Their teacher explained that measurements were correct by saying that recorded values by three students were :-
 (1) Crystal, van der Waal and Covalent radii
 (2) Covalent, crystal and van der Waal radii
 (3) van der Waal, ionic and covalent radii
 (4) None is correct.
- Screening effect is not observed in :-
 (1) He^+ (2) Li^{+2} (3) H (4) All of these
- Arrange in orders of atomic and ionic radii :
 (a) Ni, Cu, Zn (b) H^+ , H, H^- (c) Ti, Zr, Hf (d) Be, Li, Na
 (e) Cr, V, Ti, Sc (f) I^+ , I, I^- (g) Sc, Y, La, Ac (h) Cl, Na, Rb
 (i) Cu, Ag, Au (j) B, Be, Al, Mg (k) F, O, Cl, S
- Which statement is false:-
 (1) Screening effect increases down the group (2) Z_{eff} increases down the group
 (3) Z_{eff} . increases in a period (4) All
- The screening effect of d- electrons is :-
 (1) Equal to the p - electrons (2) Much more than p - electrons
 (3) Same as f - electrons (4) Less than p - electrons

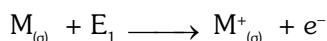
1.4 IONISATION POTENTIAL OR IONISATION ENERGY OR IONISATION ENTHALPY (IP / IE)

- (i) Minimum energy required to remove most loosely bonded outer most shell e^- in ground state from an isolated gaseous atom is known as ionization energy.

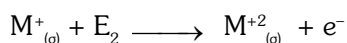
(Isolated $\rightarrow$ Without any bonding with other atom)

(ii) Successive Ionization Energy

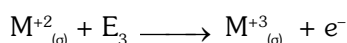
- (a) For an atom $M_{(g)}$ successive ionization energies are as follows -



$E_1 = \text{I}^{\text{st}}$ Ionization energy



$E_2 = \text{II}^{\text{nd}}$ Ionization energy



$E_3 = \text{III}^{\text{rd}}$ ionization energy

$$E_1 < E_2 < E_3 \dots \dots \dots \text{ (Always for an element)}$$

- (b) Electron can not be removed from solid state of an atom, it has to be convert into gaseous form, Energy required for conversion from solid state to gaseous state is called Sublimation energy.
- (c) For any neutral atom ionization energy is always an endothermic process ($\Delta H = +ve$)
- (d) It is measured in eV/atom (electron volt/atom) or Kcal/mole or KJ/mole

FACTORS AFFECTING IONISATION ENERGY**In a period****(i) Effective nuclear charge (Z_{eff})**

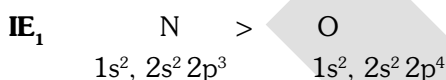
$\text{Ionisation Energy} \propto Z_{\text{eff}} \propto \frac{\text{positive charge}}{\text{negative charge}}$

Ion with high positive oxidation state will have high ionisation energy.

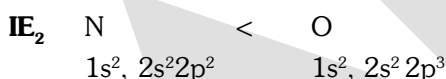
Ex. $\text{Fe}^{+3} > \text{Fe}^{+2} > \text{Fe}$

(ii) Stability of half filled and fully filled orbitals :

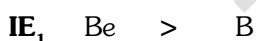
Half filled p^3, d^5, f^7 or fully filled p^6, d^{10}, f^{14} are more stable than others so it requires more energy.



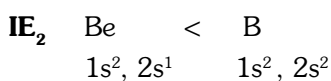
After losing one e^- , O attains electronic configuration of N, so II^{nd} ionisation energy of O is more than N.

**(iii) Penetration power of sub shells**

- (a) Order of attraction of subshells towards nucleus (Penetration power) is $s > p > d > f$
- (b) 's' subshell is more closer to nucleus so more energy will be required to remove e^- from s-subshell as comparison to p, d & f subshells.



After losing one e^- , B attains electronic configuration of Be, so II^{nd} ionisation energy of B is more than Be.

**In a group**

$$\text{Atomic size : IE} \propto \frac{1}{\text{atomic size}}$$

COMPARISON OF IONISATION ENERGY

(i) **In a period** : Z_{eff} increases so removal of electron becomes difficult and ionisation energy increases.

Order of IE of 2nd period elements $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{O} < \text{N} < \text{F} < \text{Ne}$

(ii) **In a group** : Size increase so ionisation energy decrease.

$\text{Li} \quad \text{Na} \quad \text{K} \quad \text{Rb} \quad \text{Cs}$

Size increases, Ionisation Energy decreases

Exception :

- Ionisation Energy $\text{Ga} > \text{Al}$ (due to Transition contraction)
- Ionisation Energy of $5d > 4d$ (due to lanthanide contraction)

Ex. $\text{Hf} > \text{Zr}$

Application of ionisation energy :

(A) Metallic and non metallic character :

Generally for metals Ionisation Energy is low.

For Non-metals Ionisation Energy is high.

$$\text{Metallic character} \propto \frac{1}{\text{IE}}$$

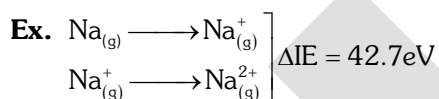
(B) Reactivity of metals :

$$\text{Reactivity of metals} \propto \frac{1}{\text{IE}}$$

Reactivity of metals increases down the group as ionisation energy decreases.

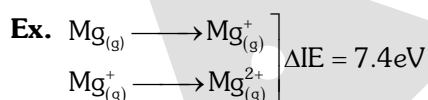
(C) Stability of oxidation states of an element :

(a) If the difference between two successive ionisation energy of an element $\geq 16\text{eV}$, then its lower oxidation state is stable.

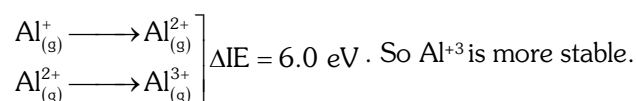
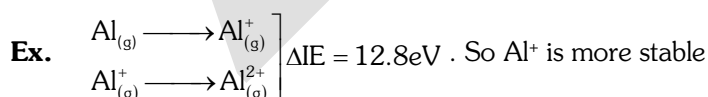


Difference between ionisation energy $> 16\text{eV}$. So Na^{+} is more stable.

(b) If the difference between two successive ionisation energy of an element $\leq 11\text{eV}$, then its higher oxidation state is stable.



Difference of ionisation energy $< 11\text{eV}$. So Mg^{+2} is more stable.



Overall order of stability is $\boxed{\text{Al}^{+3} > \text{Al}^{+} > \text{Al}^{+2}}$

(D) To determine the number of valence electron of an element :

Number of valence electrons = number of lower values of IP before 1st highest jump.

BEGINNER'S BOX-4

- IP₁ and IP₂ of Mg are 178 and 348 K. cal mol⁻¹. The enthalpy required for the reaction $\text{Mg} \rightarrow \text{Mg}^{2+} + 2e^-$ is :-
 (1) + 170 K.cal (2) + 526 K.cal (3) - 170 K.cal (4) - 526 K.cal
- The IP₁, IP₂, IP₃, IP₄ and IP₅ of an element are 7.1, 14.3, 34.5, 46.8, 162.2 eV respectively. The element is likely to be:-
 (1) Na (2) Si (3) F (4) Ca
- Which of the following element has 2nd IP < 1st IP
 (1) Mg (2) Ne (3) C (4) None
- In which of the following the energy change corresponds to first ionisation potential only :-
 (1) $X_{(g)} \rightarrow X^+_{(g)} + e$ (2) $X_{2(g)} \rightarrow X^+_{(g)} + e$
 (3) $X_{(s)} \rightarrow X^+_{(g)} + e$ (4) $X_{(aq)} \rightarrow X^+_{(aq)} + e$
- In the given process which oxidation state is more stable.

$$M_{(g)} \longrightarrow M_{(g)}^+ \quad IE_1 = 7.9 \text{ eV}$$

$$M_{(g)}^+ \longrightarrow M_{(g)}^{+2} \quad IE_2 = 15.5 \text{ eV}$$
 (1) M⁺ (2) M⁺² (3) Both (4) None
- The electronic configuration of some neutral atoms are given below :-
 (A) 1s² 2s¹ (B) 1s² 2s² 2p³ (C) 1s² 2s² 2p⁵ (D) 1s² 2s² 2p⁶ 3s¹
 In which of these electronic configuration would you expect to have highest :-
 (i) IE₁ (ii) IE₂
 (1) C, A (2) B, A (3) C, B (4) B, D
- The correct order of decreasing second ionization energy of Li, Be, Ne, C, B
 (1) Ne>B>Li>C>Be (2) Li>Ne>C>B>Be
 (3) Ne>C>B>Be>Li (4) Li>Ne>B>C>Be
- In which of the following element has highest value of ionisation energy-
 (1) Ti (2) Zr (3) Hf (4) None of these
- What is the correct order of ionisation energy :
 (1) K < Cu < Cu⁺ < K⁺ (2) K < Cu⁺ < Cu < K⁺
 (3) Cu⁺ < K < Cu < K⁺ (4) K⁺ < Cu⁺ < Cu < K

10. Match the column.

Column-I

Valence electronic configuration

- (a) ns¹
 (b) ns²
 (c) ns² np¹
 (d) ns² np²

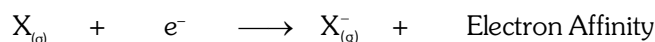
Column-II

Successive ionisation energies

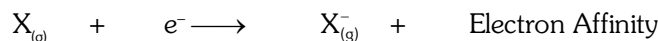
- (p) 19, 27, 36, 48, 270
 (q) 16, 28, 34, 260
 (r) 18, 26, 230, 250
 (s) 14, 200, 220, 240

1.5 ELECTRON AFFINITY/ELECTRON GAIN ENTHALPY (EA/ ΔH_{eg})

- (1) The amount of energy released when an electron is added to the valence shell of an isolated gaseous atom known as Electron affinity.

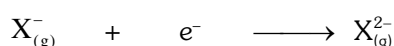


- (2) Generally first electron addition of an isolated gaseous atom is an exothermic process (except stable electronic configuration)

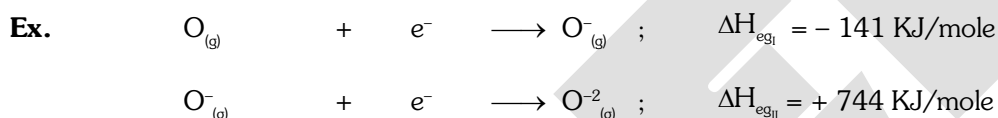


$$\Delta H_{egI} (\text{first electron gain enthalpy}) = -ve$$

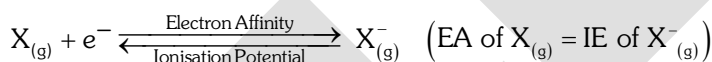
- (3) Second electron addition of an isolated gaseous atom is always an endothermic process due to inter electronic repulsion.



$$\Delta H_{egII} (\text{second electron gain enthalpy}) = \text{positive}$$



- (4) Formation of poly negative anion like O^{2-} , N^{3-} , C^{4-} etc. is always an endothermic process.
 (5) Electron affinity of neutral atom is equal to ionisation energy of its anion.



- (6) IE of neutral atom is equal to electron affinity of its cation



- (7) **Factors affecting electron affinity :**

(A) **Atomic size :** Electron Affinity $\propto \frac{1}{\text{Atomic size}}$

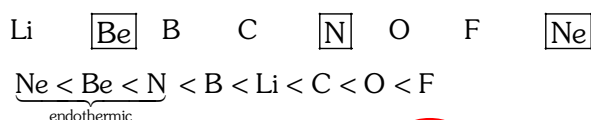
(B) **Effective nuclear charge (Z_{eff}) :** Electron Affinity $\propto Z_{eff} \propto \frac{\text{positive charge}}{\text{negative charge}}$

(C) **Stability of completely filled or half filled orbitals :** Electron affinity of elements having full-filled or half filled configuration is very less or zero so for these elements electron gain enthalpy ($\Delta H_{(eg)}$) will be positive.



- (8) **Variation of electron affinity :**

(i) **In 2nd period -**



(ii) **In Group :**

Electron affinity of 3rd period element is greater than electron affinity of 2nd period elements of the respective group.



Due to small size of fluorine, **electron density around the nucleus increases**. The incoming electron suffers more repulsion. In case of chlorine electron density decreases due to large size, decreasing order of electron affinity

Cl > F > Br > I	S > O > P > N	Si > C > P > N
-----------------	---------------	----------------

Note : N & P have low electron affinity due to stable half filled configuration.

BEGINNER'S BOX-5

- The correct order of electron affinity is :-
 (1) Be < B < C < N (2) Be < N < B < C (3) N < Be < C < B (4) N < C < B < Be
- In the formation of a chloride ion, from an isolated gaseous chlorine atom, 3.8 eV energy is released, which would be equal to :-
 (1) Electron affinity of Cl⁻ (2) Ionisation potential of Cl
 (3) Electronegativity of Cl (4) Ionisation potential of Cl⁻
- O_(g) + 2e⁻ → O²⁻_(g) ΔH_{eg} = 603 KJ/mole. The positive value of ΔH_{eg} is due to :-
 (1) Energy is released to add on 1 e⁻ to O⁻¹ (2) Energy is required to add on 1 e⁻ to O⁻¹
 (3) Energy is needed to add on 1e⁻ to O (4) None of the above is correct
- The electron affinity values for the halogens shows the following trend :-
 (1) F < Cl > Br > I (2) F < Cl < Br < I (3) F > Cl > Br > I (4) F < Cl > Br < I
- The process requiring the absorption of energy is.
 (1) F → F⁻ (2) Cl → Cl⁻ (3) O → O²⁻ (4) H → H⁻
- Second electron affinity of an element is :-
 (1) Always exothermic (2) Endothermic for few elements
 (3) Exothermic for few elements (4) Always endothermic
- Process, Na_(g)⁺ $\xrightarrow{\text{I}}$ Na_(g) $\xrightarrow{\text{II}}$ Na_(s)
 (1) In (I) energy released, (II) energy absorbed (2) In both (I) and (II) energy is absorbed
 (3) In both (I) and (II) energy is released (4) In (I) energy absorbed, (II) energy released
- Which of the following configuration will have least electron affinity.
 (1) ns²np⁵ (2) ns²np² (3) ns²np³ (4) ns²np⁴
- Which of the following will have the most negative electron gain enthalpy and which the least negative ?
 (1) F, Cl (2) Cl, F (3) Cl, S (4) Cl, P
- Which arrangement represents the correct order of electron gain enthalpy (with negative sign) of the given atomic species ?
 (1) S < O < Cl < F (2) O < S < F < Cl (3) Cl < F < S < O (4) F < Cl < O < S

1.6 ELECTRONEGATIVITY (EN)

- (i) The tendency of a covalently bonded atom to attract shared pair of electrons towards itself is called electronegativity.
- (ii) A polar covalent bond of A – B may be broken as
 $A \text{ --- } B \longrightarrow A^{\delta-} \text{ --- } B^{\delta+}$ (Electronegativity A > Electronegativity B)
 depending on their tendency to attract bonded electron.

(iii) **Difference between electronegativity and Electron Affinity :**

Electronegativity	Electron Affinity
• Tendency of an atom in a molecule to attract the bonded electrons • It is not an energetic term • It regularly increases in a period because not depend on stable electronic configuration • It has no unit	• Energy released when an electron is added to neutral isolated gaseous atom • It is an energetic term • It does not increases regularly in a period because depend on stable electronic configuration • It is measured in eV/atom or KJ mol⁻¹ or K cal mole⁻¹

- (iv) EN was explained by Pauling for the first time
 Electronegativity of some other elements are as follows –

Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	H 2.1
Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	F 4.0
K 0.8						Cl 3.0
Rb 0.8						Br 2.8
Cs 0.7						I 2.5
Fr 0.7						

In Pauling's scale, elements having almost same electronegativity are-

N = Cl = 3.0
 C = S = I = 2.5
 P = H = 2.1
 Be = Al = 1.5
 K = Rb = 0.8
 Cs = Fr = 0.7

Note : Small atoms are normally having more electronegativity than larger atoms.

(v) **FACTORS AFFECTING ELECTRONEGATIVITY :**

(A) **Atomic size**

$$\text{Electronegativity} \propto \frac{1}{\text{Atomic size}}$$

Ex. F > Cl > Br > I

(B) **Effective nuclear charge (Z_{eff})**

$$\text{Electronegativity} \propto Z_{\text{eff}} \propto \frac{\text{positive charge}}{\text{negative charge}}$$

Ex. $Mn^{+2} < Mn^{+4} < Mn^{+7}$
 $O^{-2} < O^{-1} < O < O^{+1} < O^{+2}$
 $Fe < Fe^{+2} < Fe^{+3}$

----->
 $Z_{\text{eff}} \uparrow \text{EN} \uparrow$

(C) **% s - character**

$$\text{Electronegativity} \propto \%s\text{-Character}$$

(vi) **PERIODIC TABLE & ELECTRONEGATIVITY :**

- (A) Electronegativity decreases down the group.
 (B) In period on moving from left to right electronegativity increases.
 (C) Electronegativity of Cs and Fr are equal, it is because from ${}_{55}\text{Cs}$ to ${}_{87}\text{Fr}$ only one shell increases but nuclear charge (No. of proton) increases by +32, so effect of nuclear charge balanced the effect of increase in number of shell.

Electronegativity of F > Cl but Electron affinity of Cl > F

- (D) In IIIA group, value of electronegativity is irregular when going down the group, because of transition contraction

Electronegativity of Ga > Electronegativity of Al

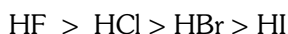
(vii) APPLICATION OF ELECTRONEGATIVITY :**(A) Metallic and non metallic nature :**

Generally metals have low electronegativity and non metals have high electronegativity, so we can say metallic character increases down the group but decreases along a period.

$$\text{Non Metallic Nature} \propto \text{EN}$$

(B) Bond energy : By increasing difference in electronegativity of bonded atoms, bond length decreases and hence bond energy increases

$$\text{Bond energy} \propto \text{Electronegativity difference}$$

**(C) Schoemaker and Stevenson law**

If in a diatomic molecule electronegativities of A – B have more difference. Then actual bond length will be reduced. As per schoemaker & Stevenson– The reduction in bond length depends on the difference in electronegativities of atoms by following manner -

$$d_{A-B} = r_A + r_B - 0.09 (X_A - X_B)$$

Here X_A is E.N. of A & X_B is E.N. of B

Ex. If bond length of $F_2 = 1.44 \text{ \AA}$, Bond length of $H_2 = 0.74 \text{ \AA}$. Find out the bond length of H – F ?
(EN of F is 4.0, EN of H is 2.1)

Solution.

$$d_{H-F} = r_F + r_H - 0.09 (X_F - X_H)$$

$$\therefore r_F = 1.44 / 2 = 0.72 \text{ \AA}, r_H = 0.74 / 2 = 0.37 \text{ \AA}$$

$$\therefore d_{H-F} = 0.72 + 0.37 - 0.09 (4.0 - 2.1)$$

$$= 1.09 - (0.09 \times 1.9) = 1.09 - 0.171 = 0.919 \text{ \AA}$$

(D) Acidic & Basic Strength :**(i) Nature of hydrides :**

$$\text{Stability of molecule} \propto \text{Bond energy}$$

Order of stability of hydrohalides :	$\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
Order of acid strength :	$\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$
In VA group :	$\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{BiH}_3$

Thermal stability decreases

Acidic character increases

(ii) Nature of hydroxides :**(a) As per Gallis,**

- In AOH if electronegativity of A is more than 1.7 (Non metal) then it is acidic in nature.
- If electronegativity of 'A' is less than 1.7 (metal) then AOH will be basic in nature
- If $X_A - X_O \geq X_O - X_H$ (X_A = EN of A) then AO bond will be more polar and will break up as
 $\text{A} - \text{OH} \rightarrow \text{A}^+ + \text{OH}^-$ It shows basic nature

Ex. In NaOH

$$X_O - X_{Na} (2.6) > X_O - X_H (1.4)$$

So hydroxide is basic

- If $X_A - X_O \leq X_O - X_H$
 $\text{A} - \text{O} - \text{H} \rightarrow \text{H}^+ + \text{AO}^-$ It shows Acidic nature

then OH bond will be more polar and will break up as

In ClOH

$$X_O - X_{Cl} (0.5) < X_O - X_H (1.4)$$

So hydroxide is acidic

(viii) ELECTRONEGATIVITY SCALE :

Mulliken scale : According to Mulliken electronegativity is average value of ionisation potential and electron affinity of an element,

$$X_m = \frac{\text{Ionisation Potential} + \text{Electron Affinity}}{2}$$

where X_p is electronegativity on the basis of Pauling scale.

- If ionisation potential and electron affinity are given in eV, then electronegativity by Mulliken on Pauling scale will be

$$X_p = \frac{\text{Ionisation Potential} + \text{Electron Affinity}}{5.6}$$

$$X_p = \frac{X_m}{2.8}$$

- If ionisation potential and electron affinity are given in K.cal/mole then

$$X_p = \frac{\text{Ionisation Potential} + \text{Electron Affinity}}{2 \times 62.5}$$

BEGINNER'S BOX-6

- Which of the following is affected by stable configuration of an atom :-
 (a) Electronegativity (b) Ionisation potential (c) Electron affinity
 Correct answer is :-
 (1) Only electronegativity
 (2) Only ionisation potential
 (3) Electron affinity and ionisation potential
 (4) All of the above
- Which of the following elements have the different value of electronegativity :-
 (1) H (2) S (3) Te (4) P
- Which is the correct order of electronegativity –
 (1) Cl > S > P > Si (2) Si > Al > Mg > Na
 (3) F > Cl > Br > I (4) All
- Electronegativity scale of Pauling is based upon :-
 (1) Bond length (2) Bond energy (3) Atomic radius (4) Covalent radius
- Correct order of electronegativity of N, P, C and Si is :-
 (1) N < P < C < Si (2) N > C > Si > P
 (3) N = P > C = Si (4) N > C > P > Si

6. Outermost electronic configuration of the most electronegative element is :-
 (1) ns^2np^3 (2) ns^2np^6 (3) ns^2 (4) ns^2np^5
7. Electronegativity of the following elements increases in the order.
 (1) O, N, S, P (2) P, S, N, O (3) P, N, S, O (4) S, P, N, O
8. Give the correct order of electronegativity of central atom in following compounds –
 (a) $CH_3 - CH_3$, (b) $CH_2 = CH_2$ (c) $CH \equiv CH$
 The correct order is –
 (1) $a > b > c$ (2) $c > a > b$ (3) $c > b > a$ (4) $b > c > a$

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	2	1	4	2	4	2	3	4	4	No
BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	3	4	3	3	3	2	4	1	4	2
BEGINNER'S BOX-3	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	2	4	3	4	3	1	4		2	4
BEGINNER'S BOX-4	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	2	2	4	1	2	1	4	3	1	
BEGINNER'S BOX-5	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	2	4	2	1	3	4	3	3	4	2
BEGINNER'S BOX-6	Que.	1	2	3	4	5	6	7	8		
	Ans.	3	2	4	2	4	4	2	3		

Build Up Your Understanding

10. In a period the elements are arranged in :-

10. In a period the elements are arranged in :-
 - (1) Decreasing order of nuclear charge
 - (2) Decreasing order of No. of electrons
 - (3) Increasing order of nuclear charge
 - (4) In order of same nuclear charge
11. Which of the following statement is wrong :-
 - (1) 2nd period contain 8 elements
 - (2) 3rd period contains 18 elements
 - (3) 1st period contains two non metals
 - (4) In p-block, metal, nonmetal and metalloids are present
12. Which of the following element was absent in the Mendeleev's periodic table :-
 - (1) Tc
 - (2) Si
 - (3) B
 - (4) F
13. IUPAC name of the element placed just after actinide series :-
 - (1) Unniltrium
 - (2) Unnilpentium
 - (3) Unnilquadium
 - (4) Ununbium
14. Which statement is wrong for the long form of periodic table :-
 - (1) Number of periods are 7 and groups 18
 - (2) No. of valence shell electrons in a period are same
 - (3) IIIrd B group contains 32 elements
 - (4) Lanthanides and actinides are placed in same group
15. The elements which are cited as an example to prove the validity of mendeleev's periodic law are
 - (1) H, He
 - (2) Ga, Sc
 - (3) Co, Ni
 - (4) Zr, Hf
16. Which pair of successive elements follows increasing order of atomic weight in mendeleev's periodic table
 - (1) Argon and potassium
 - (2) Lithium and Beryllium
 - (3) Cobalt and nickel
 - (4) Tellurium and iodine
17. Which of the following statement is false :-
 - (1) Elements of ns^2np^6 electronic configuration lies in 1st to 6th period
 - (2) Typical elements lies in 3rd period
 - (3) The seventh period will accommodate thirty two elements
 - (4) Boron and silicon are diagonally related

18. Among the Lanthanides the one obtained by synthetic method is :-
 (1) Lu (2) Pm (3) Pr (4) Ce

PERIOD, GROUP AND BLOCK

19. Which of the following set of elements belongs to same period :-
 (1) Zn, Cd, Hg (2) Fr, Ra, U
 (3) K, Ca, Ag (4) None
20. The element with atomic number $Z = 115$ will be placed in :-
 (1) 7th period, IA group (2) 8th period, IVA group
 (3) 7th period, VA group (4) 6th period, VB group
21. Elements upto atomic no. 112 have been discovered till now. What will be the electronic configuration of the element possessing atomic no. 108 :-
 (1) $[Rn]5f^{14} 6d^6 7s^2$ (2) $6f^{14} 7d^8 7s^2$
 (3) $[Rn] 5f^{14} 6d^8 7s^0$ (4) $[Xe] 4f^{14} 5d^8 6s^2$
22. In 6th period of the modern periodic table, electronic energy levels are in the order :-
 (1) 6s, 4f, 5d, 6p (2) 6s, 6p, 4f, 5d
 (3) 4f, 5d, 6s, 6p (4) None
23. Out of first 100 elements no. of elements having electrons in 3d orbital (in their complete electronic configuration) are :-
 (1) 80 (2) 100 (3) 40 (4) 60
24. The IUPAC name of the element which is placed after Db_{105} in the periodic table, will be :-
 (1) Un nil pentium (2) Un un nilium
 (3) Un nil hexium (4) Un nil quadium
25. The element with the electronic configuration $ns^2(n-1)s^2p^6d^0(n-2)s^2p^6d^{10}f^7$ lies in the :-
 (1) s - block (2) p - block
 (3) d - block (4) f - block
26. The element with atomic number $Z=118$ will be :-
 (1) Noble gas
 (2) Transition metal
 (3) Alkali metal
 (4) Alkaline earth metal
27. The atom having the valence shell electronic configuration $4s^2 4p^2$ would be in:-
 (1) Group II A and period 3
 (2) Group II B and period 4
 (3) Group IV A and period 4
 (4) Group IV A and period 3

28. The electronic configuration of d-block elements is exhibited by :-
 (1) $ns^{1-2}(n-1)d^{1-10}$ (2) $ns^2 (n - 1) d^{10}$
 (3) $(n - 1)d^{10}s^2$ (4) $ns^2 np^5$
29. The electronic configuration of the element with atomic number 109 if discovered will be:-
 (1) $(n-1)d^7 ns^2$ (2) $(n-1)d^9 ns^2$
 (3) $nd^7 ns^2$ (4) $(n-1)d^5 ns^2 np^2$
30. The element having electronic configuration $4f^{14} 5d^0 6s^2$ belongs to :-
 (1) d-block, 12th group
 (2) f-block, III B group
 (3) f-block, 14th group
 (4) s-block, 2nd group
31. Element with the electronic configuration given below, belong to which group in the periodic table $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6 4d^{10}, 5s^2 5p^3$
 (1) 3rd (2) 5th
 (3) 15th (4) 17th
32. $4d^3 5s^2$ configuration belongs to which group :-
 (1) IIA (2) IIB (3) V B (4) III B
33. Which of the following electronic configuration belongs to inert gas elements :-
 (1) $ns^2 (n - 1)d^{10}$ (2) $ns^2 (n - 1)s^2 p^6$
 (3) $ns^2 np^6$ (4) None
34. From atomic number 58 to 71, elements are placed in :-
 (1) 5th period and III A group
 (2) 6th period and III B group
 (3) separate period and group
 (4) 7th period and IV B group
35. True statement is :-
 (1) All the transuranic elements are synthetic elements
 (2) Elements of third group are called bridge elements
 (3) Element of $1s^2$ configuration is placed in IIA group
 (4) Electronic configuration of elements of a group is same
36. Elements having $ns^2 np^6$ valence shell electronic configuration lies in :-
 (1) '0' gp. and 1st-7th period
 (2) 18th gp. and 2nd-6th period
 (3) 18th gp and 1st-6th period
 (4) All are correct

37. Which of the following match is correct :-
 (1) Last natural element – Uub
 (2) General electronic configuration of IA group – ns^2
 (3) Inert gas elements lies in 2nd – 6th period
 (4) Typical elements – 3rd period elements
38. The electronic configuration of elements X and Z are $1s^2 2s^2 2p^6 3s^2 3p^5$ and $1s^2 2s^2 2p^5$ respectively. What is the position of element X with respect to position of Z in the periodic table -
 (1) Just below Z (2) Just above Z
 (3) Left to the Z (4) right to the Z
39. Which of the following sequence contains atomic number of only representative elements
 (1) 55, 12, 18, 53 (2) 13, 33, 54, 83
 (3) 3, 33, 53, 87 (4) 22, 33, 55, 66
40. Uranium (At No. - 92) is the last natural element in the periodic table. The last element of the periodic table which is recently discovered is Uub. What will be the total number of transuranic elements in the periodic table :-
 (1) 21 (2) 20 (3) 11 (4) 12
41. Which two elements are in same period as well as same group of modern periodic table :-
 (1) Z = 23, Z = 31 (2) Z = 65, Z = 66
 (3) Z = 52, Z = 87 (4) Z = 58, Z = 46
42. Which of the following statement is not correct for given electronic configuration
 $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6 4d^{10} 4f^{14}, 5s^2 5p^6 5d^{10}, 6s^2$
 (1) It belongs to IIB group and 6th period
 (2) It is liquid at room temperature
 (3) It is a transition element
 (4) It is not used in high temperature thermometer
43. General electronic configuration of outermost and penultimate shell is $(n-1)s^2 (n-1)p^6 (n-1)d^x ns^2$. If $n = 4$ and $x = 5$, then number of protons in the nucleus will be :-
 (1) > 25 (2) < 24 (3) 25 (4) 30
44. An ion M^{+3} has electronic configuration [Ar] $3d^{10} 4s^2$ element M belongs to :-
 (1) s-block (2) p-block
 (3) d-block (4) f-block
45. What is the atomic number of element having maximum no. of unpaired e^- in 4p subshell :-
 (1) 33 (2) 17
 (3) 53 (4) 15

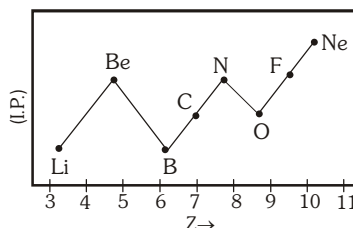
Zeff, SCREENING CONSTANT & ATOMIC RADIUS

46. The formula for effective nuclear charge is (if σ is screening constant)
 (1) $Z - \sigma$ (2) $Z + \sigma$ (3) $Z \sigma^{-1}$ (4) $Z \sigma$
47. According to Slater rule, Effective nuclear charge in group generally :-
 (1) Increases down the group
 (2) Decreases down the group
 (3) Remains constant
 (4) First increases then decreases
48. In sodium atom the screening is due to :-
 (1) $3s^2, 3p^6$ (2) $2s^1$
 (3) $1s^2, 2s^2, 2p^6$ (4) $1s^2, 2s^2$
49. If the difference in atomic size of :
 $Na - Li = x$; $Rb - K = y$; $Fr - Cs = z$
 Then correct order will be:-
 (1) $x = y = z$ (2) $x > y > z$
 (3) $x < y < z$ (4) $x < y < z$
50. The correct order of size would be:-
 (1) $Ni < Pd \approx Pt$ (2) $Pd < Pt < Ni$
 (3) $Pt > Ni > Pd$ (4) $Pd > Pt > Ni$
51. Which of the following order of radii is correct
 (1) $Li < Be < Mg$ (2) $H^+ < Li^+ < H^-$
 (3) $O < F < Ne$ (4) $Na^+ > F^- > O^{2-}$
52. K^+ , Ar, Ca^{2+} and S^{2-} contains -
 (1) Same electronic configuration and atomic volume
 (2) Different electronic configuration but same IP.
 (3) Same electronic configuration but different atomic volume
 (4) None
53. Which of the following is not isoelectronic series :-
 (1) Cl^- , P^{3-} , Ar (2) N^{3-} , Ne, Mg^{+2}
 (3) B^{+3} , He, Li⁺ (4) N^{3-} , S^{2-} , Cl^-
54. Which group of atoms have nearly same atomic radius:-
 (1) Na, K, Rb, Cs (2) Li, Be, B, C
 (3) Fe, Co, Ni (4) F, Cl, Br, I
55. Atomic radii of Fluorine and Neon in Angstrom units are given by :-
 (1) 0.72, 1.60 (2) 1.60, 1.60
 (3) 0.72, 0.72 (4) None of these

56. Which of the following has largest radius :-
 (1) $1s^2 2s^2 2p^6 3s^2$
 (2) $1s^2 2s^2 2p^6 3s^2 3p^1$
 (3) $1s^2 2s^2 2p^6 3s^2 3p^3$
 (4) $1s^2 2s^2 2p^6 3s^2 3p^5$
57. Which of the following order of atomic/ionic radius is not correct :-
 (1) $I > I > I^+$ (2) $Mg^{+2} > Na^+ > F^-$
 (3) $P^{+5} < P^{+3}$ (4) $Li > Be > B$
58. In the lithium atom screening effect of valence shell electron is caused by-
 (1) Electrons of K and L shell
 (2) Electrons of K shell
 (3) Two electrons of 1^{st} and one of 2^{nd} shell
 (4) None
59. Correct order of ionic radii is
 (1) $Ti^{4+} < Mn^{7+}$ (2) $^{37}Cl^- < ^{35}Cl^-$
 (3) $K^+ > Cl^-$ (4) $P^{3+} > P^{5+}$
60. The radius of potassium atom is 0.203 nm. The radius of the potassium ion in nanometer will be :-
 (1) 0.133 (2) 0.231
 (3) 0.234 (4) 0.251
61. S^{-2} is not isoelectronic with :-
 (1) Ar (2) Cl^-
 (3) HS^- (4) Ti^{+3}
62. The best reason to account for the general tendency of atomic diameters to decrease as the atomic numbers increase within a period of the periodic table is the fact that
 (1) Outer electrons repel inner electrons
 (2) Closer packing among the nuclear particles is achieved
 (3) The number of neutrons increases
 (4) The increasing nuclear charge exerts a greater attractive force on the electrons
63. In an anion :-
 (1) Number of proton decreases
 (2) Protons are more than electrons
 (3) Effective nuclear charge is more
 (4) Radius is larger than neutral atom
64. Maximum size of first member of a period is due to
 (1) Maximum number of shells
 (2) Maximum screening effect
 (3) Minimum Z_{eff}
 (4) All
65. Which of the following ion has largest size :-
 (1) F^- (2) Al^{+3} (3) Cs^+ (4) O^{-2}
66. In which of the following pair radii of second species is smaller than that of first species :-
 (1) Li, Na (2) Na^+ , F^-
 (3) N^{-3} , Al^{+3} (4) Mn^{+7} , Mn^{+4}
67. Spot the incorrect order of atomic radii :-
 (1) $r_{Cu} > r_{Zn}$ (2) $r_{Cl} > F$ (3) $r_P > S$ (4) $r_{Sc} > Ti$
68. Which of the following orders of atomic radii are correct :-
 (a) $Li < Be < Na$ (b) $Ni < Cu < Zn$
 (c) $Ti > V > Cr$ (d) $Ti > Zr \approx Hf$
 Correct answer is :-
 (1) All (2) a, b (3) b, c (4) b, d
69. Which electronic configuration of an atom is smallest in size :-
 (1) $3s^2$ (2) $3s^2 3p^3$
 (3) $3s^1$ (4) $3s^2 3p_x^2 3p_y^2 3p_z^1$
70. Decreasing order of size of ions is :-
 (1) $Br^- > S^{-2} > Cl^- > N^{-3}$
 (2) $N^{3-} > S^{-2} > Cl^- > Br^-$
 (3) $Br^- > Cl^- > S^{-2} > N^{-3}$
 (4) $N^{-3} > Cl^- > S^{-2} > Br^-$
71. Which of the following statement is wrong
 (1) According to Slater, Z_{eff} in group remains constant
 (2) In a period atomic size decreases
 (3) Screening effect in a period remains constant
 (4) In a period atomic radius of inert gas element is maximum
72. The covalent and vander Waal's radii of hydrogen respectively are :-
 (1) 0.37 Å, 0.8 Å
 (2) 0.37 Å, 0.37 Å
 (3) 0.8 Å, 0.8 Å
 (4) 0.8 Å, 0.37 Å
73. Which of the following sequence is correct for decreasing order of ionic radius :-
 (1) Se^{-2} , I, Br^- , O^{-2} , F^-
 (2) I, Se^{-2} , O^{-2} , Br^- , F^-
 (3) Se^{-2} , I, Br^- , F^- , O^{-2}
 (4) I, Se^{-2} , Br^- , O^{-2} , F^-
74. Element having maximum number of low shielding electrons :-
 (1) [Xe] $4f^{14}$, $5d^{10}$, $6s^2$, $6p^2$
 (2) [Rn] $5f^{14}$, $6d^1$, $7s^2$
 (3) [Ar] $3d^{10}$, $4s^2$
 (4) [Ne] $3s^2$, $3p^1$

94. The correct order of decreasing first ionisation energy is :-
 (1) $\text{Si} > \text{Al} > \text{Mg} > \text{Na}$ (2) $\text{Si} > \text{Mg} > \text{Al} > \text{Na}$
 (3) $\text{Al} > \text{Si} > \text{Mg} > \text{Na}$ (4) $\text{Mg} > \text{Li} > \text{Al} > \text{Si}$
95. Out of Na^+ , Mg^{+2} , O^{-2} and N^{-3} , the pair of species showing minimum and maximum IP would be.
 (1) Na^+ , Mg^{+2} (2) Mg^{+2} , N^{-3}
 (3) N^{-3} , Mg^{+2} (4) O^{-2} , N^{-3}
96. The element having highest I.P. is the from the two series C, N, O and Si, P, S :-
 (1) P (2) N (3) S (4) O
97. Lowest IP will be shown by the element having the configuration :-
 (1) $[\text{He}] 2s^2$ (2) $1s^2$
 (3) $[\text{He}] 2s^2 2p^2$ (4) $[\text{He}] 2s^2 2p^5$
98. The strongest reducing agent among the following is:-
 (1) Na (2) Mg (3) Al (4) K
99. Which ionisation potential (IP) in the following equations involves the greatest amount of energy:-
 (1) $\text{K}^+ \rightarrow \text{K}^{+2} + e^-$ (2) $\text{Li}^+ \rightarrow \text{Li}^{+2} + e^-$
 (3) $\text{Fe} \rightarrow \text{Fe}^+ + e^-$ (4) $\text{Ca}^+ \rightarrow \text{Ca}^{+2} + e^-$
100. Values of first four ionisation potential of an elements are 68, 370, 400, 485. It belongs to which of the following electronic configuration:-
 (1) $1s^2 2s^1$ (2) $1s^2 2s^2 2p^1$
 (3) $1s^2 2s^2 2p^6 3s^1$ (4) (1) and (3) both
101. (a) $\text{M}_{(g)}^- \rightarrow \text{M}_{(g)}$ (b) $\text{M}_{(g)} \rightarrow \text{M}_{(g)}^+$
 (c) $\text{M}_{(g)}^+ \rightarrow \text{M}_{(g)}^{+2}$ (d) $\text{M}_{(g)}^{+2} \rightarrow \text{M}_{(g)}^{+3}$
 Minimum and maximum I.P. would be of :-
 (1) a, d (2) b, c (3) c, d (4) d, a
102. Which of the following electronic configuration belongs to least and most metallic character respectively:-
 (a) $1s^2 2s^1$ (b) $5s^2 5p^5$
 (c) $3s^2 3p^6 4s^1$ (d) $1s^2 2s^2 2p^5$
 (1) a, b (2) d, c (3) b, a (4) c, d
103. Triad - I [N^{3-} , O^{-2} , Na^+]
 Triad - II [N^+ , C^+ , O^+]
 Choose the species of lowest IP from triad-I and highest IP from triad-II respectively
 (1) N^{3-} , O^+ (2) Na^+ , C^+
 (3) N^{3-} , N^+ (4) O^- , C^+
104. The correct values of ionisation energies (in kJ mol^{-1}) of Be, Ne, He and N respectively are
 (1) 786, 1012, 999, 1256
 (2) 1012, 786, 999, 1256
 (3) 786, 1012, 1256, 999
 (4) 786, 999, 1012, 1256

105. Following graph shows variation of I.P. with atomic number in second period (Li – Ne). Value of I.P. of Na (11) will be :-



- (1) Above Ne
 (2) Below Ne but above O
 (3) Below Li
 (4) Between N and O
106. Which one of the following has highest ionisation potential :-
 (1) Li^+ (2) Mg^+ (3) He (4) Ne
107. In which of the following pairs, the ionisation energy of the first species is less than that of the second
 (1) N, P (2) Be^+ , Be
 (3) N, N^- (4) Ne, Ne^+
108. Consider the following ionisation reactions
 $\text{A(g)} \rightarrow \text{A}^+(\text{g}) + e^-$ IE in (KJ/mol) is A_1
 $\text{A}^+(\text{g}) \rightarrow \text{A}^{+2}(\text{g}) + e^-$ IE in (KJ/mol) is A_2
 $\text{A}^{+2}(\text{g}) \rightarrow \text{A}^{+3}(\text{g}) + e^-$ IE in (KJ/mol) is A_3
 then correct order of IE is :-
 (1) $A_1 > A_2 > A_3$ (2) $A_1 = A_2 = A_3$
 (3) $A_1 < A_2 < A_3$ (4) $A_3 = A_2 < A_1$
109. IE_1 , IE_2 and IE_3 of an element are 10 eV, 15 eV, 45 eV respectively, the most stable oxidation state of the element will be :-
 (1) + 1 (2) + 2 (3) + 3 (4) + 4
110. Select the correct order of I.E. :-
 (1) $\text{Cl}^- > \text{Cl} > \text{Cl}^+$ (2) $\text{Cl}^+ > \text{Cl} > \text{Cl}^-$
 (3) $\text{Cl} > \text{Cl}^+ > \text{Cl}^-$ (4) $\text{Cl}^- > \text{Cl}^+ > \text{Cl}$

ELECTRON AFFINITY

111. In the process $\text{Cl}_{(g)} + e^- \xrightarrow{\Delta H} \text{Cl}^-(\text{g})$, ΔH is
 (1) Positive (2) Negative
 (3) Zero (4) None
112. Process in which maximum energy is released:-
 (1) $\text{O} \rightarrow \text{O}^{-2}$ (2) $\text{Mg}^+ \rightarrow \text{Mg}^{+2}$
 (3) $\text{Cl} \rightarrow \text{Cl}^-$ (4) $\text{F} \rightarrow \text{F}^-$
113. Which of the following is energy releasing process
 (1) $\text{X}^- \rightarrow \text{X}(\text{g}) + e^-$
 (2) $\text{O}^-(\text{g}) + e^- \rightarrow \text{O}^{2-}$
 (3) $\text{O}(\text{g}) \rightarrow \text{O}^+(\text{g}) + e^-$
 (4) $\text{O}(\text{g}) + e^- \rightarrow \text{O}^-(\text{g})$

114. In which of the following process energy is liberated:-

- (1) $\text{Cl} \rightarrow \text{Cl}^+ + e^-$ (2) $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$
 (3) $\text{Cl} + e^- \rightarrow \text{Cl}^-$ (4) $\text{O}^- + e^- \rightarrow \text{O}^{2-}$

115. Element of which atomic number has highest electron affinity:-

- (1) 35 (2) 17 (3) 9 (4) 53

116. The electron affinity

- (1) Of carbon is greater than oxygen
 (2) Of fluorine is less than iodine
 (3) Of fluorine is less than chlorine
 (4) Of sulphur is less than oxygen

117. Which of the following element will form most stable bivalent anion.

- (1) Fluorine (2) Oxygen
 (3) Chlorine (4) Nitrogen

118. Energy absorbed in second electron addition in an atom is called.

- (1) 1stIP (2) 2ndEA
 (3) 1stEA (4) 2ndIP

119. The amount of energy released for the process $\text{X}_{(g)} + e^- \rightarrow \text{X}_{(g)}^-$ is minimum and maximum respectively for :-

- (a) F (b) Cl (c) N (d) B
 Correct answer is :-
 (1) c & a (2) d & b
 (3) a & b (4) c & b

120. Which of the following electronic configuration is expected to have highest electron affinity:-

- (1) $2s^2 2p^0$ (2) $2s^2 2p^2$
 (3) $2s^2 2p^3$ (4) $2s^2 2p^1$

121. Consider the following conversions

- (i) $\text{O(g)} + e^- \longrightarrow \text{O}^-(\text{g})$; ΔH_1
 (ii) $\text{F(g)} + e^- \longrightarrow \text{F}^-(\text{g})$; ΔH_2
 (iii) $\text{Cl(g)} + e^- \longrightarrow \text{Cl}^-(\text{g})$; ΔH_3
 (iv) $\text{Na(g)} \longrightarrow \text{Na}^+(\text{g})$; ΔH_4

incorrect statement is :-

- (1) ΔH_1 and ΔH_2 is less negative than ΔH_3
 (2) ΔH_2 is more negative than ΔH_1
 (3) ΔH_2 , ΔH_3 are negative while ΔH_1 is positive
 (4) ΔH_1 , ΔH_2 and ΔH_3 are negative while ΔH_4 is positive

122. In which of the following process, least energy is required :-

- (1) $\text{F}_{(g)}^- \longrightarrow \text{F}_{(g)} + e^-$
 (2) $\text{P}_{(g)}^- \longrightarrow \text{P}_{(g)} + e^-$
 (3) $\text{S}_{(g)}^- \longrightarrow \text{S}_{(g)} + e^-$
 (4) $\text{Cl}_{(g)}^- \longrightarrow \text{Cl}_{(g)} + e^-$

ELECTRONEGATIVITY

123. The correct set of decreasing order of electronegativity is :-

- (1) Li, H, Na (2) Na, H, Li
 (3) H, Li, Na (4) Li, Na, H

124. Polarity of a bond can be explained by :-

- (1) Electron affinity (2) Ionisation potential
 (3) Electronegativity (4) All of the above

125. Electronegativity values for elements are useful in predicting :-

- (1) Bond energy of a molecule
 (2) Polarity of a bond
 (3) Nature of an oxide
 (4) All

126. Mulliken scale of electronegativity uses the concept of :-

- (1) E. A. and EN of pauling
 (2) E. A. and atomic size
 (3) E.A. and I.P.
 (4) E.A. and bond energy

127. The pair with minimum difference in electronegativity is :-

- (1) F, Cl (2) C, H (3) P, H (4) Na, Cs

128. Least electronegative element is :-

- (1) I (2) Br (3) C (4) Cs

129. In which of the following pairs of elements the electronegativity of first element is less than that of second element :-

- (1) Zr, Hf
 (2) K, Rb
 (3) Cl, S
 (4) None of the above

130. The nomenclature of ICl is iodine chloride because

- (1) Size of I < Size of Cl
 (2) Atomic number of I > Atomic number of Cl
 (3) E.N. of I < E.N. of Cl
 (4) E. A. of I < E. A. of Cl

131. Among the following least and most polar bonds are respectively :-

- (a) C - I (b) N - O (c) C - F (d) P - F
 (1) d and c (2) a and d (3) b and d (4) b and c

132. If the ionisation potential is IP, electron affinity is EA and electronegativity is X then which of the following relation is correct :-

- (1) $2X - EA - IP = 0$
 (2) $2EA - X - IP = 0$
 (3) $2IP - X - EA = 0$
 (4) All of the above

133. The properties which are not common to both groups 1 and 17 elements in the periodic table are :-

- (1) Electropositive character increases down the groups
- (2) Reactivity decreases from top to bottom in these groups
- (3) Atomic radii increases as the atomic number increases
- (4) Electronegativity decreases on moving down the group

134. Electronegativity of an element can be measured using :-

- (1) Pauling's scale
- (2) Mulliken's scale
- (3) Both
- (4) None

135. As we proceed across the period in periodic table, we find there is a decrease in :-

- (1) Ionisation energy
- (2) Electron affinity
- (3) Electronegativity
- (4) Atomic radii

136. Which compound strongly absorb CO_2 ?

- (1) BeO
- (2) K_2O
- (3) H_3PO_4
- (4) P_4O_6

137. The electronegativities of the following elements: H, O, F, S and Cl increase in the order :-

- (1) $\text{H} < \text{O} < \text{F} < \text{S} < \text{Cl}$
- (2) $\text{Cl} < \text{H} < \text{O} < \text{F} < \text{S}$
- (3) $\text{H} < \text{S} < \text{O} < \text{Cl} < \text{F}$
- (4) $\text{H} < \text{S} < \text{Cl} < \text{O} < \text{F}$

138. Which of the following is different from other three oxides :-

- (1) MgO
- (2) SnO
- (3) PbO
- (4) ZnO

EXERCISE-I

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	3	1	1	2	2	4	2	4	3	2	1	3	2	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	2	2	3	1	1	1	3	4	1	3	1	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	3	3	2	1	2	4	1	3	2	2	3	3	2	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	3	3	2	1	2	3	4	3	1	1	2	2	4	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	4	4	4	3	3	3	1	3	4	1	3	1	4	2	1
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	4	3	4	1	3	4	3	2	4	2	2	4	3	2	4
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	1	3	3	2	3	2	1	4	2	3	1	2	1	3	3
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	1	4	3	2	2	2	3	4	3	2	3	2	2	4	2
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	3	2	3	3	4	3	3	4	1	3	2	1	2	3	4
Que.	136	137	138												
Ans.	2	4	1												

EXERCISE-II (Previous Year Questions)**AIPMT/NEET & AIIMS (2006-2018)****AIPMT 2006**

1. Which of the following is the most basic oxide?
 (1) SeO_2 (2) Al_2O_3
 (3) Sb_2O_3 (4) Bi_2O_3

AIPMT 2007

2. Identify the correct order of the size of the following
 (1) $\text{Ca}^{2+} < \text{K}^+ < \text{Ar} < \text{Cl}^- < \text{S}^{2-}$
 (2) $\text{Ar} < \text{Ca}^{2+} < \text{K}^+ < \text{Cl}^- < \text{S}^{2-}$
 (3) $\text{Ca}^{2+} < \text{Ar} < \text{K}^+ < \text{Cl}^- < \text{S}^{2-}$
 (4) $\text{Ca}^{2+} < \text{K}^+ < \text{Ar} < \text{S}^{2-} < \text{Cl}^-$

AIPMT 2008

3. The correct order of decreasing second ionisation enthalpy of Ti(22), V(23), Cr(24) and Mn(25) is :
 (1) $\text{Mn} > \text{Cr} > \text{Ti} > \text{V}$
 (2) $\text{Ti} > \text{V} > \text{Cr} > \text{Mn}$
 (3) $\text{Cr} > \text{Mn} > \text{V} > \text{Ti}$
 (4) $\text{V} > \text{Mn} > \text{Cr} > \text{Ti}$

AIPMT 2009

4. Which of the following oxides is not expected to react with sodium hydroxide ?
 (1) BeO (2) B_2O_3
 (3) CaO (4) SiO_2
5. Amongst the elements with following electronic configurations, which one of them may have the highest ionization energy ?
 (1) $[\text{Ne}]3s^23p^1$ (2) $[\text{Ne}]3s^23p^3$
 (3) $[\text{Ne}]3s^23p^2$ (4) $[\text{Ar}]3d^{10}4s^24p^3$

AIPMT 2010

6. Among the elements Ca, Mg, P and Cl, the order of increasing atomic radii is :-
 (1) $\text{Cl} < \text{P} < \text{Mg} < \text{Ca}$
 (2) $\text{P} < \text{Cl} < \text{Ca} < \text{Mg}$
 (3) $\text{Ca} < \text{Mg} < \text{P} < \text{Cl}$
 (4) $\text{Mg} < \text{Ca} < \text{Cl} < \text{P}$
7. The correct order of the decreasing ionic radii among the following isoelectronic species is :-
 (1) $\text{K}^+ > \text{Ca}^{2+} > \text{Cl}^- > \text{S}^{2-}$
 (2) $\text{Ca}^{2+} > \text{K}^+ > \text{S}^{2-} > \text{Cl}^-$
 (3) $\text{Cl}^- > \text{S}^{2-} > \text{Ca}^{2+} > \text{K}^+$
 (4) $\text{S}^{2-} > \text{Cl}^- > \text{K}^+ > \text{Ca}^{2+}$

8. Which of the following represents the correct order of increasing electron gain enthalpy with negative sign for the elements O, S, F and Cl ?

- (1) $\text{S} < \text{O} < \text{Cl} < \text{F}$ (2) $\text{Cl} < \text{F} < \text{O} < \text{S}$
 (3) $\text{O} < \text{S} < \text{F} < \text{Cl}$ (4) $\text{F} < \text{S} < \text{O} < \text{Cl}$

AIIMS 2010

9. Which is correct order of IP_1 :-
 (1) $\text{Na} > \text{Al}$ (2) $\text{Mg} > \text{Al}$
 (3) $\text{Ga} > \text{Ca}$ (4) $\text{Mg} > \text{Be}$

AIPMT Mains-2011

10. What is the value of electron gain enthalpy of Na^+ if IE_1 of $\text{Na} = 5.1 \text{ eV}$:-
 (1) $+0.2 \text{ eV}$ (2) -5.1 eV
 (3) -10.2 eV (4) $+2.55 \text{ eV}$

AIPMT Pre.-2012

11. Identify the **wrong** statement in the following:
 (1) Atomic radius of the elements increases as one moves down the first group of the periodic table
 (2) Atomic radius of the elements decreases as one moves from left to right in the 2nd period of the periodic table
 (3) Amongst isoelectronic species, smaller the positive charge on the cation, smaller is the ionic radius
 (4) Amongst isoelectronic species, greater the negative charge on the anion, larger is the ionic radius

AIIMS 2013

12. The 1st Ionisation enthalpy of Na, Mg and Si are 496, 737, 776 kJ mol^{-1} respectively then what will be the 1st ionisation enthalpy of Al in kJ mol^{-1} :-
 (1) $> 766 \text{ kJ mol}^{-1}$
 (2) > 496 and $< 737 \text{ kJ mol}^{-1}$
 (3) > 737 and $< 766 \text{ kJ mol}^{-1}$
 (4) $> 496 \text{ kJ mol}^{-1}$

AIPMT 2014

13. Which of the following orders of ionic radii is correctly represented ?
 (1) $\text{H}^- > \text{H}^+ > \text{H}$
 (2) $\text{Na}^+ < \text{F}^- < \text{O}^{2-}$
 (3) $\text{F}^- > \text{O}^{2-} > \text{Na}^+$
 (4) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{N}^{3-}$

14. Be^{2+} is isoelectronic with which of the following ions?

- (1) H^+ (2) Li^+
(3) Na^+ (4) Mg^{2+}

15. Acidity of diprotic acids in aqueous solutions increases in the order :-

- (1) $\text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$
(2) $\text{H}_2\text{Se} < \text{H}_2\text{S} < \text{H}_2\text{Te}$
(3) $\text{H}_2\text{Te} < \text{H}_2\text{S} < \text{H}_2\text{Se}$
(4) $\text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{S}$

16. Reason of lanthanoid contraction is :-

- (1) Negligible screening effect of 'f' orbitals
(2) Increasing nuclear charge
(3) Decreasing nuclear charge
(4) Decreasing screening effect

AIIMS 2014

17. Correct order of atomic radius is :-

- (1) $\text{V} > \text{Ti}$ (2) $\text{Cl} > \text{S}$
(3) $\text{Rb} > \text{Cs}$ (4) $\text{Ne} > \text{Be}$

18. Incorrect order of acidic strength is :-

- (1) $\text{H}_2\text{S} > \text{H}_2\text{Se}$ (2) $\text{HI} > \text{HBr}$
(3) $\text{HBr} > \text{HCl}$ (4) $\text{H}_2\text{Te} > \text{H}_2\text{S}$

AIPMT 2015

19. The species Ar , K^+ and Ca^{2+} contain the same number of electrons. In which order do their radii increase ?

- (1) $\text{Ca}^{2+} < \text{Ar} < \text{K}^+$ (2) $\text{Ca}^{2+} < \text{K}^+ < \text{Ar}$
(3) $\text{K}^+ < \text{Ar} < \text{Ca}^{2+}$ (4) $\text{Ar} < \text{K}^+ < \text{Ca}^{2+}$

20. The number of d-electrons in Fe^{2+} ($Z = 26$) is not equal to the number of electrons in which one of the following?

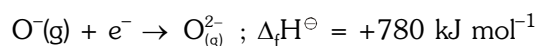
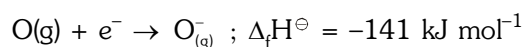
- (1) p-electrons in Cl ($Z = 17$)
(2) d-electrons in Fe ($Z = 26$)
(3) p-electrons in Ne ($Z = 10$)
(4) s-electrons in Mg ($Z = 12$)

21. Because of lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii ? (Numbers in the brackets are atomic numbers).

- (1) Zr (40) and Nb (41)
(2) Zr (40) and Hf (72)
(3) Zr (40) and Ta (73)
(4) Ti (22) and Zr (40)

Re-AIPMT 2015

22. The formation of the oxide ion, $\text{O}^{2-}(\text{g})$, from oxygen atom requires first an exothermic and then an endothermic step as shown below :



Thus process of formation of O^{2-} in gas phase is unfavourable even though O^{2-} is isoelectronic with neon. It is due to the fact that,

- (1) Oxygen is more electronegative
(2) Addition of electron in oxygen results in larger size of the ion
(3) Electron repulsion outweighs the stability gained by achieving noble gas configuration
(4) O^- ion has comparatively smaller size than oxygen atom

23. Which is the correct order of increasing energy of the listed orbitals in the atom of titanium ?

(At. no. $Z = 22$)

- (1) $3s \ 3p \ 3d \ 4s$ (2) $3s \ 3p \ 4s \ 3d$
(3) $3s \ 4s \ 3p \ 3d$ (4) $4s \ 3s \ 3p \ 3d$

AIIMS 2015

24. Smallest ionic radius is :-

- (1) La^{3+} (2) U^{3+}
(3) Yb^{3+} (4) Cf^{3+}

25. Electronic configuration of Al^{3+} with excluding bonded electron in aluminate ion

- (1) $[\text{Ne}]$ (2) $[\text{Ar}]$
(3) $[\text{Ne}]3s^2$ (4) $[\text{Ar}]4s^2$

NEET-I 2016

26. In which of the following options the order of arrangement does not agree with the variation of property indicated against it ?

- (1) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{F}^-$ (increasing ionic size)
(2) $\text{B} < \text{C} < \text{N} < \text{O}$ (increasing first ionisation enthalpy)
(3) $\text{I} < \text{Br} < \text{Cl} < \text{F}$ (increasing electron gain enthalpy)
(4) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ (increasing metallic radius)

AIIMS 2016

27. The biggest gap in electronegativity is :-

- (1) $\text{B} \rightarrow \text{Al}$
(2) $\text{Al} \rightarrow \text{Ga}$
(3) $\text{Ga} \rightarrow \text{In}$
(4) $\text{In} \rightarrow \text{Tl}$

NEET(UG) 2017

28. The element $Z = 114$ has been discovered recently. It will belong to which of the following family/group and electronic configuration ?
- (1) Carbon family, $[\text{Rn}] 5f^{14} 6d^{10} 7s^2 7p^2$
 - (2) Oxygen family, $[\text{Rn}] 5f^{14} 6d^{10} 7s^2 7p^4$
 - (3) Nitrogen family, $[\text{Rn}] 5f^{14} 6d^{10} 7s^2 7p^6$
 - (4) Halogen family, $[\text{Rn}] 5f^{14} 6d^{10} 7s^2 7p^5$

AIIMS 2017

29. If the I.P. of Na, Mg & Si are 496, 737 & 786 kJ/mole respectively then I.P. of Al is
- (1) 760 kJ/mole
 - (2) 756 kJ/mole
 - (3) 577 kJ/mole
 - (4) 986 kJ/mole
30. Which of the following not reacts with NaOH
- (1) As_2O_3
 - (2) Bi_2O_3
 - (3) Sb_2O_3
 - (4) SeO_2

NEET(UG) 2018

31. Which of the following oxides is most acidic in nature?
- (1) MgO
 - (2) BeO
 - (3) BaO
 - (4) CaO
32. The correct order of atomic radii in group 13 elements is
- (1) $\text{B} < \text{Al} < \text{In} < \text{Ga} < \text{Tl}$
 - (2) $\text{B} < \text{Al} < \text{Ga} < \text{In} < \text{Tl}$
 - (3) $\text{B} < \text{Ga} < \text{Al} < \text{Tl} < \text{In}$
 - (4) $\text{B} < \text{Ga} < \text{Al} < \text{In} < \text{Tl}$

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33. In which of the following elements d-orbitals do not have any electrons in their outer electronic configuration :-
- (1) Am
 - (2) Lr
 - (3) Th
 - (4) Lu

EXERCISE-II

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	1	3	3	2	1	4	3	2	2	3	2	2	2	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	4	1	2	1	2	3	2	3	1	2,3	1	1	3	2
Que.	31	32	33												
Ans.	2	4	1												

EXERCISE-III (Analytical Questions)

Check Your Understanding

1. Which of the following is correct match :-

	Atomic number	Group number	Period number
(A)	46	10	6
(B)	58	3	6
(C)	56	2	6
(D)	42	6	5

- (1) Only B, C, D (2) Only A, B, C
(3) Only B, C (4) Only A, C, D

2. ${}_{92}\text{U}$ (III B) changes to ${}_{90}\text{Th}$ by emission of α -particle. Daughter element will be in -
(1) IB (2) IIA (3) IIIB (4) VB

3. Atomic number of Ag is 47. In the same group the atomic number of elements placed above and below Ag will be :-
(1) 37, 67 (2) 29, 79
(3) 39, 69 (4) 29, 65

4. In the general electronic configuration - $(n-2)f^{1-14} (n-1)d^{0-1} ns^2$, if value of $n = 7$ the configuration will be of -
(1) Lanthanides (2) Actinides
(3) Transition elements (4) None

5.

		39		
	A		D	
E		B	C	

On the basis of given part of periodic table, incorrect statement is :-

- (1) A is an alkaline earth metal
(2) Atomic number of B is 103 which belongs to III B group
(3) Atomic number, group no. and period number of D are 72 IVB and 6th respectively
(4) C is a transuranic element.
6. If total 110 elements are present in periodic table then how many of them contain e^- in f subshell :-
(1) 28 (2) 57 (3) 58 (4) 53

7. Which of the following statements is wrong :-
(1) van der Waal's radius of iodine is more than its covalent radius
(2) All isoelectronic ions belong to same period of the periodic table
(3) IE_1 of N is higher than that of O while IE_2 of O is higher than that of N
(4) The electron affinity of N is less than that of P

8. The inter nuclear distance in H_2 and Cl_2 molecules are 74 and 198 Å. respectively. The bond length of HCl may be
(EN of H = 2.1 Cl = 3.0)
(1) 136 Å (2) 272 Å
(3) 135.919 Å (4) 271.919 Å

9. These are 3 elements A, B and C. Their atomic number are Z_1, Z_2, Z_3 respectively. If $Z_3 - Z_1 = 2$ and $\frac{Z_1 + Z_3}{2} = Z_2$ and the electronic configuration

of element C is $[\text{Ar}] 3d^2 4s^2$ then correct order of atomic radius is :-

- (1) $A^{+2} < B^{+3} < C^{+4}$ (2) $A^{+2} = B^{+3} = C^{+4}$
(3) $A^{+2} > B^{+3} > C^{+4}$ (4) $B^{+3} < A^{+2} = C^{+4}$

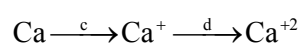
10. $\text{M(g)} \rightarrow \text{M}^+(\text{g}) + e^-, \Delta H = 100 \text{ eV}$
 $\text{M(g)} \rightarrow \text{M}^{2+}(\text{g}) + 2e^-, \Delta H = 250 \text{ eV}$

Which is incorrect statements :-

- (1) I_1 of M(g) is 100 eV (2) I_2 of M(g) is 150 eV
(3) I_2 of M(g) is 250 eV (4) none

11. Successive ionisation energies of an element A are 100 eV, 150 eV, 181 eV, 2000 eV, 2200 eV correct statement regarding A is :-
(1) Element 'A' may be metal
(2) Formula of oxide of A may be A_2O_3
(3) Oxide of element A may be amphoteric
(4) All are correct

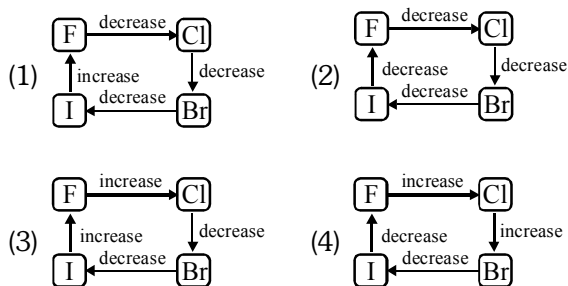
12. $\text{K} \xrightarrow{a} \text{K}^+ \xrightarrow{b} \text{K}^{+2},$



If a, b, c, d are ionisation energies, the which of the following order is not correct -

- (1) $c > a$ (2) $b > a$
(3) $d > c$ (4) $b < d$

13. Which of the following diagrams is correctly related to electron affinity of halogens :-



14. Elements of which group form anions most readily:-

- (1) Oxygen family (2) Nitrogen group
(3) Halogens (4) Alkali metals

15. Which is the weakest base among NaOH, $\text{Ca}(\text{OH})_2$, KOH and $\text{Zn}(\text{OH})_2$:-

- (1) NaOH (2) KOH
(3) $\text{Ca}(\text{OH})_2$ (4) $\text{Zn}(\text{OH})_2$

16. If Electron affinity of an element M is x kJ/mol than ionisation potential of this element :-

- (1) More than x (2) less than x
(3) equal to x (4) more than 2x

17. Identify the incorrect are :-

- (1) Shielding constant (σ) : $\text{Li} < \text{Na} < \text{K} < \text{Rb}$
(2) Z_{eff} : $\text{Li} > \text{Na} > \text{K} > \text{Rb}$
(3) Ionic radius : $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+}$
(4) Atomic size : $\text{Li} < \text{Na} < \text{K} < \text{Rb}$

18. If electronegativity values of element X and Y are 3.8 and 1.8 respectively, then percentage of ionic character in compound XY is :

- (1) 50 (2) 46
(3) 64 (4) 36

19. Arrange Cl, F, F^- , Cl^- in increasing order of ionisation potential ?

- (1) $\text{F}^- < \text{Cl}^- < \text{Cl} < \text{F}$
(2) $\text{Cl}^- < \text{F}^- < \text{Cl} < \text{F}$
(3) $\text{Cl}^- < \text{F}^- < \text{F} < \text{Cl}$
(4) $\text{F}^- < \text{Cl}^- < \text{F} < \text{Cl}$

20. The order of ionisation potential between He^+ ion and H-atom (both species are in gaseous state) is:-

- (1) I.P. (He^+) = I.P. (H)
(2) I.P. (He^+) < I.P. (H)
(3) I.P. (He^+) > I.P. (H)
(4) Cannot be compared

21. Electronic configuration are :-

- A - $1s^2 2s^2 2p^1$
B - $1s^2 2s^2 2p^6 3s^1 3p^2$
C - $1s^2 2s^1 2p^1$
D - $1s^2 2s^2 2p^5 3s^1$

then which among these will belong to the same group in the periodic table ?

- (1) A & B (2) A, B, C
(3) A, B, D (4) A, B, C, D

22. The IE_1 & IE_2 of three elements A, B & C are given as (IE in KJ/mol).

	A	B	C
IE_1	400	550	1150
IE_2	2650	1070	2090

Identify the element which represent a non-metal:-

- (1) A (2) B
(3) Both A & B (4) C

23. The maximum 2nd I. E. is of :-

- (1) Mn (2) Sc (3) Cr (4) Ti

24. In which of the following arrangements the order is NOT according to the property indicated against it?

- (1) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{F}^-$ - increasing ionic size
(2) $\text{B} < \text{C} < \text{N} < \text{O}$ - increasing first ionization energy
(3) $\text{I} < \text{Br} < \text{F} < \text{Cl}$ - increasing electron gain enthalpy (with negative sign)
(4) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ - increasing metallic radius

25. Which one of the following orders represents the correct sequence of the increasing basic nature of the given oxides ?

- (1) $\text{Na}_2\text{O} < \text{K}_2\text{O} < \text{MgO} < \text{Al}_2\text{O}_3$
(2) $\text{K}_2\text{O} < \text{Na}_2\text{O} < \text{Al}_2\text{O}_3 < \text{MgO}$
(3) $\text{Al}_2\text{O}_3 < \text{MgO} < \text{Na}_2\text{O} < \text{K}_2\text{O}$
(4) $\text{MgO} < \text{K}_2\text{O} < \text{Al}_2\text{O}_3 < \text{Na}_2\text{O}$

26. The outer electron configuration of Gd (Atomic No. : 64) is :-

- (1) $4f^4 5d^4 6s^2$ (2) $4f^7 5d^1 6s^2$
(3) $4f^3 5d^5 6s^2$ (4) $4f^8 5d^0 6s^2$

27. The correct order of electron gain enthalpy with negative sign of F, Cl, Br and I, having atomic number 9, 17, 35 and 53 respectively, is :-

- (1) $\text{I} > \text{Br} > \text{Cl} > \text{F}$
(2) $\text{F} > \text{Cl} > \text{Br} > \text{I}$
(3) $\text{Cl} > \text{F} > \text{Br} > \text{I}$
(4) $\text{Br} > \text{Cl} > \text{I} > \text{F}$

- 28.** The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be :-
 (1) - 2.55 eV (2) - 5.1 eV
 (3) - 10.2 eV (4) + 2.55 eV
- 29.** For electron affinity of halogens which of the following is correct :-
 (1) $\text{Br} > \text{F}$ (2) $\text{F} > \text{Cl}$
 (3) $\text{Br} > \text{Cl}$ (4) $\text{F} > \text{I}$
- 30.** The pair of amphoteric hydroxide is
 (1) $\text{Al}(\text{OH})_3$, LiOH
 (2) $\text{Be}(\text{OH})_2$, $\text{Mg}(\text{OH})_2$
 (3) $\text{B}(\text{OH})_3$, $\text{Be}(\text{OH})_2$
 (4) $\text{Be}(\text{OH})_2$, $\text{Zn}(\text{OH})_2$
- 31.** Electronegativity is the measurement of capacity of an atom by which :
 (1) Electrons get repelled
 (2) Electrons get attracted
 (3) Gain of electron
 (4) Loose of proton
- 32.** The ions O^{2-} , F^- , Na^+ , Mg^{2+} and Al^{3+} are isoelectronic. Their ionic radii show
 (1) an increase from O^{2-} to F^- and then decrease from Na^+ to Al^{3+}
 (2) a decrease from O^{2-} to F^- and then increase from Na^+ to Al^{3+}
 (3) a significant increase from O^{2-} to Al^{3+}
 (4) a significant decrease from O^{2-} to Al^{3+}
- 33.** Ionic radii are :-
 (1) Directly proportional to square of effective nuclear charge
 (2) Inversely proportional to effective nuclear charge
 (3) Inversely proportional to square of effective nuclear charge
 (4) Directly proportional to effective nuclear charge
- 34.** Which one of the following arrangements represents the correct order of electron gain enthalpy (with negative sign) of the given atomic species :-
 (1) $\text{F} < \text{Cl} < \text{O} < \text{S}$ (2) $\text{S} < \text{O} < \text{Cl} < \text{F}$
 (3) $\text{O} < \text{S} < \text{F} < \text{Cl}$ (4) $\text{Cl} < \text{F} < \text{S} < \text{O}$
- 35.** Four successive members of the first row transition elements are listed below with their atomic numbers. Which one of them is expected to have the highest third ionization enthalpy :-
 (1) Vanadium ($Z = 23$)
 (2) Manganese ($Z = 25$)
 (3) Chromium ($Z = 24$)
 (4) Iron ($Z = 26$)
- 36.** Which of the following statement is correct ?
 (1) Element having high I.P. have high EA also
 (2) 2nd I.P. of noble gas is less than 1st I.P.
 (3) I.P. of Be is more than 'B'
 (4) 2nd I.P. of 'N' is more than 2nd I.P. of 'O'
- 37.** The incorrect statement is :-
 (1) 2nd I.P. = $\text{Se} > \text{As}$
 (2) 1st I.P. = $\text{C}^{+2} > \text{N}^{+2}$
 (3) 3rd I.P. = $\text{F} > \text{O}$
 (4) Halogens have highest I.P. in respective period
- 38.** Which of the following E.A. order is not correct ?
 (1) $\text{N} < \text{O} < \text{S}$ (2) $\text{Cl} > \text{O} > \text{N} > \text{C}$
 (3) $\text{O} < \text{S} < \text{F} < \text{Cl}$ (4) $\text{B} < \text{C} < \text{Si} < \text{S}$
- 39.** Which one of the following order of given properties is correct ?
 (1) Atomic radius - $\text{Li} < \text{Be} < \text{B}$
 (2) Ionisation potential - $\text{Li} < \text{Be} < \text{B}$
 (3) Electron affinity - $\text{Li} < \text{Be} < \text{B}$
 (4) Electronegativity - $\text{Li} < \text{Be} < \text{B}$
- 40.** The incorrect statement among the following is :-
 (1) Helium has the highest first I.P. in the periodic table
 (2) The process $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$ is endothermic
 (3) The electron affinities of S and P are less than that O and N respectively
 (4) In any period, the first ionisation potential of noble gas is the highest

EXERCISE-III

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	3	2	2	2	4	2	3	3	3	4	4	3	3	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	2	1	3	1	4	3	2	3	2	3	2	4	4
Que.	31	32	33	34	35	36	37	38	39	40					
Ans.	2	4	2	3	2	3	4	2	4	3					

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion** : Two successive ionisation energies of Argon are 56.8 eV and 36.8 eV respectively.
Reason : Z_{eff} of Ar ($3s^23p^6$) is greater than Ar^+ ($3s^23p^5$).
 (1) A (2) B (3) C (4) D
- Assertion** : Electron affinity of fluorine is greater than chlorine.
Reason : Ionisation potential of fluorine is less than chlorine.
 (1) A (2) B (3) C (4) D
- Assertion** : IP of F > EA of F
Reason : F is a highly electronegative element
 (1) A (2) B (3) C (4) D
- Assertion** : Properties of Beryllium is similar to that of Aluminium
Reason : Both the elements belongs to same group
 (1) A (2) B (3) C (4) D
- Assertion** : I.P. of first element in a period is minimum.
Reason : Effective nuclear charge of first element in a period is minimum
 (1) A (2) B (3) C (4) D
- Assertion** : Size of anion is larger than its parent atom.
Reason : Z_{eff} of anion is greater than that of their parent atom.
 (1) A (2) B (3) C (4) D
- Assertion** : Stable electronic configuration does not affects electronegativity.
Reason : EN is the tendency of an atom to attract shared electrons, not to gain electrons.
 (1) A (2) B (3) C (4) D
- Assertion** : Chlorine is most electronegative element.
Reason : Chlorine has tendency to loose electrons.
 (1) A (2) B (3) C (4) D
- Assertion** : Atomic radius increases, descending down the group.
Reason : On going down the group EN increase.
 (1) A (2) B (3) C (4) D
- Assertion** : Atomic radius of inert gases are largest in the period
Reason : Effective nuclear charge of inert gases are minimum
 (1) A (2) B (3) C (4) D
- Assertion** : Second IP of oxygen is greater than that of fluorine
Reason : Oxygen aquires stable half filled electronic configuration after loosing one electron
 (1) A (2) B (3) C (4) D
- Assertion** : Electronegativity of nitrogen is greater than carbon.
Reason : Nitrogen has stable half filled electronic configuration.
 (1) A (2) B (3) C (4) D
- Assertion** : Atomic size of Boron is larger than Beryllium
Reason : Number of shell in Boron is greater than Beryllium
 (1) A (2) B (3) C (4) D
- Assertion** : Alkali metals have least 1st I.P. in the respective period
Reason : Alkali metals have only one electron in the valence shell.
 (1) A (2) B (3) C (4) D
- Assertion** : Atomic size of Na is larger than Mg.
Reason : No. of shell in Mg is more than Na.
 (1) A (2) B (3) C (4) D
- Assertion** : Na^+ and Cl^- have similar ionic radius.
Reason : Z_{eff} in Na^+ and Cl^- are same.
 (1) A (2) B (3) C (4) D

17. **Assertion** : Ionisation potential of Li^+ is greater than He.

Reason : Z_{eff} of Li^+ is greater than He.

(1) A (2) B (3) C (4) D

18. **Assertion** : Size of Ca^{+2} is larger than K^+ .

Reason : Number of electrons in Ca^{+2} is more than K^+ .

(1) A (2) B (3) C (4) D

19. **Assertion** : 2nd IP of alkali metals is maximum in the period.

Reason : Alkali metals have the smallest atomic size in the period.

(1) A (2) B (3) C (4) D

20. **Assertion** : Atomic size along a period decreases.

Reason : Z_{eff} in a period decreases.

(1) A (2) B (3) C (4) D

21. **Assertion** : The 1st IP of Be is greater than that of B.

Reason : 2p orbital is lower in energy than 2s.

(1) A (2) B (3) C (4) D

22. **Assertion** : First ionization energy of nitrogen is lower than oxygen. [AIIMS-2005]

Reason : Across the period effective nuclear charge decreases.

(1) A (2) B (3) C (4) D

23. **Assertion** :- H_2Se is less acidic than H_2S .

Reason :- S is less electronegative than Se.

(1) A (2) B (3) C (4) D

24. **Assertion** : Atomic sizes of Cs and Fr are almost similar.

Reason : Cs and Fr belong to same group.

(1) A (2) B (3) C (4) D

25. **Assertion** :- Zn is not a transition element whereas Sc is

Reason :- Outershell configuration of Zn is $3d^{10}, 4s^2, 4p^0, 4d^0$.

(1) A (2) B (3) C (4) D

EXERCISE-IV

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	4	2	3	1	3	1	4	3	3	1	2	4	2	3
Que.	16	17	18	19	20	21	22	23	24	25					
Ans.	4	1	4	3	3	3	4	4	2	2					

CHEMICAL BONDING

E

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NEET SYLLABUS

Valence electrons, ionic bond, covalent bond, bond parameters, Lewis structure, polar character of covalent bond, valence bond theory, resonance, geometry of molecules, VSEPR theory, concept of hybridization involving s, p and d orbitals and shapes of some simple molecules, molecular orbital theory of homonuclear diatomic molecules (qualitative idea only). Hydrogen bond.

@neetfocusquizz

OBJECTIVES

After studying this unit, you will be able to :

- understand Kössel-Lewis approach to chemical bonding;
- explain the octet rule and its limitations, draw Lewis structures of simple molecules;
- explain the formation of different types of bonds;
- describe the VSEPR theory and predict the geometry of simple molecules;
- explain the valence bond approach for the formation of covalent bonds;
- predict the directional properties of covalent bonds;
- explain the different types of hybridisation involving s, p and d orbitals and draw shapes of simple covalent molecules;
- describe the molecular orbital theory of homonuclear diatomic molecules;
- explain the concept of hydrogen bond.

The meeting of two personalities is like the contact of two chemical substances; if there is any reaction, both are transformed.

Carl Jung

CHEMICAL BONDING

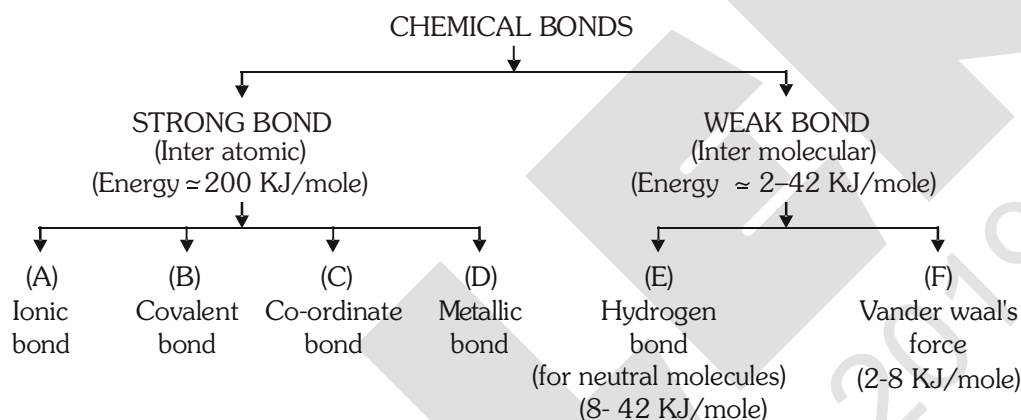
2.0 INTRODUCTION

- It is well known fact that except inert gases, no other element exists as independent atom under ordinary conditions.
- Most of the elements exist as molecules which are cluster of atoms. How do atoms combine to form molecules and why do atoms form bonds? Such doubts will be discussed in this chapter.
- A molecule will only be formed if it is more stable and has a lower energy, than the individual atoms.

Chemical Bond

- A force that acts between two or more atoms to hold them together as a stable molecule.
- It is union of two or more atoms involving redistribution of e^- among them.
- This process accompanied by decrease in energy.
- Decrease in potential energy (P.E.) $\propto$ Strength of the bond.
- Therefore molecules are more stable than atoms.

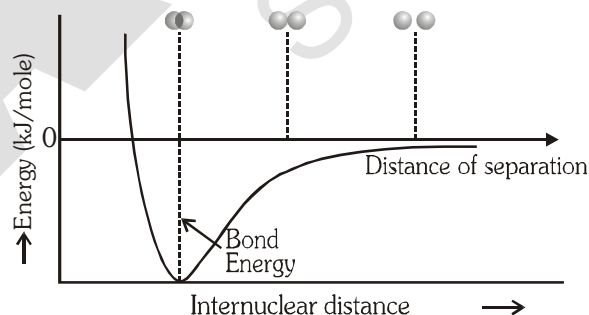
Classification of Bonds



Cause of Chemical Combination

(A) Tendency to acquire minimum energy

- When two atoms approaches to each other. Nucleus of one atom attracts the electron of another atom.
- Two nuclei and electrons of both the atoms repells each other.
- If net result is attraction, the total energy of the system (molecule) decreases and a chemical bond forms.
- So Attraction $\propto 1/\text{energy} \propto$ Stability.
- Bond formation is an exothermic process

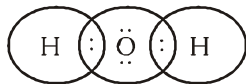


(B) Tendency to acquire noble gas configuration :

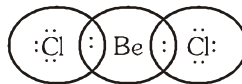
- Atom combines to acquire noble gas configuration.
- Only outermost electrons i.e. ns , np and $(n-1)d$ shells electrons participate in bond formation.
- Inert gas elements do not participate in bond formation, as they have stable electronic configuration hence will have minimum energy. (Stable configuration $1s^2$ or ns^2np^6)

2.1 KOSSEL - LEWIS APPROACH TO CHEMICAL BONDING

- Every atom has a tendency to complete its octet in outermost shell
- H has the tendency to complete its duplet.
- To acquire inert gas configuration atoms lose or gain electron or share electron.
- The tendency of atoms to achieve eight electrons in their outer most shell is known as Lewis octet rule.



Obeys octet rule



Doesn't obey octet rule

Exception of Octet Rule

(a) Incomplete octet molecules :- or (electron deficient molecules)

Compound in which octet is not complete in outer most orbit of central atom.

Example - Halides of IIIA groups, BF_3 , AlCl_3 , BCl_3 , hydride of III A/ 13^{th} group etc.

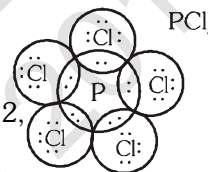


Other examples – BeCl_2 ($4e^-$), HgCl_2 ($4e^-$), $\text{Ga}(\text{CH}_3)_3$ ($6e^-$)

(b) Expansion of octet or (electron efficient molecules)

Compound in which central atom has more than $8e^-$ in outermost orbits.

Example - In PCl_5 , SF_6 and IF_7 the central atom P, S and I contain 10, 12, and 14 electrons respectively.



(c) Pseudo inert gas configuration :-

Cations which contain 18 electrons in outermost orbit

Ex. Ga^{+3} , Cu^+ , Ag^+ , Zn^{+2} , Cd^{+2} , Sn^{+4} , Pb^{+4} etc.

Electronic configuration of Ga - $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^1$

Electronic configuration of Ga^{+3} - $1s^2, 2s^2, 2p^6, \underbrace{3s^2 3p^6 3d^{10}}_{18e^-}$

(d) Cations having electron between 9 to 17 in their outer most shell

Ex. Mn^{+2} , Fe^{+2} , Fe^{+3} , Ti^{+2} etc.

Electronic configuration of Fe - $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^6, 4s^2$

Electronic configuration of Fe^{+3} - $1s^2, 2s^2 2p^6, \overbrace{3s^2 3p^6 3d^5}^{\text{(less than } 18e^-)}$

(e) Odd electron molecules :-

Central atom have an unpaired electron or odd no ($7e^-$, $11e^-$ etc) of electrons in their outer most shell.

e.g. NO, NO₂, ClO₂ etc.

VALENCY

It is defined as the combining capacity of the elements. The word valency is derived from an Italian word "Valentia" which means combining capacity.

Old concept : Given by : **Frankland**

Valency with respect to Hydrogen : Valency of H = 1

It is defined as the number of hydrogen attached with a particular element.

	IA	IIA	IIIA	IVA	VA	VIA	VIIA
	NaH	MgH ₂	AlH ₃	SiH ₄	PH ₃	H ₂ S	HCl
Valency	1	2	3	4	3	2	1

Note : Valency w.r.t. H across the period increases upto 4 and then again decreases to 1.

Valency with respect to oxygen : Valency of 'O' = 2

It is defined as twice the number of oxygen atoms attached with a particular atom.

	IA	IIA	IIIA	IVA	VA	VIA	VIIA
	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₂ O ₅	SO ₃	Cl ₂ O ₇
Valency	1	2	3	4	5	6	7

Note : Valency with respect to oxygen increases from 1 to 7 across the period. Valency w.r.t. 'O' is equal to the group number.

New concept : This concept is based on the electronic configuration. According to this concept valency for IA to IVA group elements is equal to number of valence shell e⁻ and from VA to zero group, it is -

$$[8 - (\text{number of valence } e^-)]$$

$$\text{Valency} = \text{No. of valence } e^-$$

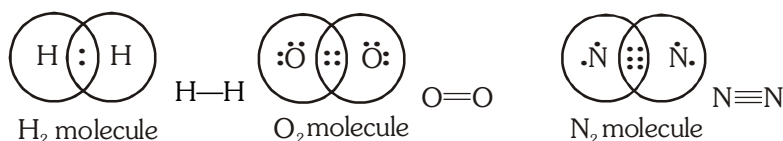
$$\text{Valency} = (8 - \text{no. of valence } e^-)$$

	IA IIA IIIA IVA				VA VIA VIIA 0			
	ns ¹	ns ²	ns ² np ¹	ns ² np ²	ns ² np ³	ns ² np ⁴	ns ² np ⁵	ns ² np ⁶
Valence shell e ⁻	1	2	3	4	5	6	7	8
Valency	1	2	3	4	3	2	1	0
					(8 - 5) = 3			(8 - 8) = 0

Note : All the elements of a group have same valencies because they have same number of valence shell electrons.

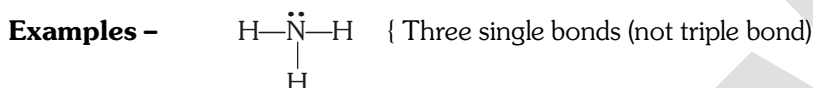
2.2 COVALENT BOND

- A covalent bond is formed by the mutual sharing of electrons between two atoms of electro negative elements to complete their octet. (Except H which completes its duplet)



- The shared pair of electrons should have opposite spins, and are localised between two atoms concerned.
- Sharing of electrons may occurs in three ways –

No. of electrons shared between two atoms	Electron pair	Bond.
2	1	Single bond (—)
4	2	Double bond (==)
6	3	Triple bond (≡)



$\text{N} \equiv \text{N}$ Triple bond. (not three single bonds) $\text{O}=\text{O}$ Double bond (Not two single bonds) $\text{H}-\text{O}-\text{H}$ (Two single bonds.)

Orbital Concept of Covalent Bond

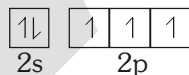
- One orbital can accommodate maximum 2 electrons with opposite spins like $\uparrow\downarrow$
- Half filled orbital or unpaired electron orbital share one electron from another atom, to complete its orbital.
- Tendency to complete orbital or to pair the electron is an essential condition of covalent bond. Completion of octet is not the essential condition of covalent bond.
- Covalency** : It is the number of covalent bonds which an atom makes in a molecule.
- If the outermost orbit has empty orbitals then covalent bonds are formed in excited state.

2.3 VARIABLE VALENCY IN COVALENT BONDS

- Variable valencies are shown by those elements which have empty orbitals in outermost shell.
- Lone pair electrons get excited in the subshell of the same shell to form the maximum number of unpaired electrons. Maximum covalency is shown in excited state.
- The energy required for excitation of electrons is called promotion energy.
- Promotion rule – Excitation of electrons in the same orbit.

Example –

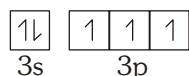
(a) Nitrogen → Ground state



Covalency 3 (NCl_3)

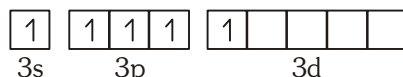
For Nitrogen → Excited states are not possible due to absence of vacant orbital that's why (NCl_5) does not exist

(b) Phosphorus → Ground state



Covalency 3 (PCl_3)

Phosphorus → Excited state



Covalency – 5 (PCl_5)

Note :

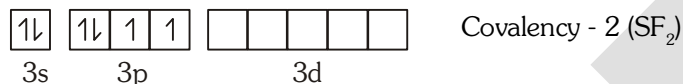
NCl_3 — exists

NCl_5 — doesn't exist (due to absence of d-orbitals in Nitrogen.) While PCl_3 and PCl_5 both exist because 3d orbitals are present in phosphorus.

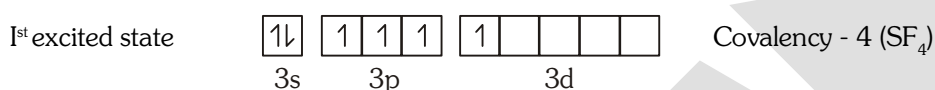
OF_2 — exists, but OF_4 and OF_6 doesn't exist due to absence of d-orbitals While SF_4 and SF_6 exist due to presence of d-orbital which are present in its valence shell.

- It can explain existence of molecules.

(c) Sulphur → Ground state.

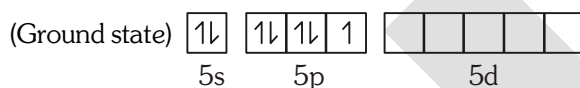


Sulphur → Excited state



So variable covalency of S is 2, 4, & 6.

(d) Iodine has three lone pair of electrons



So it shows three excited states – Maximum number of unpaired electrons = 7

Variable Valencies are 1, 3, 5 and 7

2.4 CHARACTERISTIC OF COVALENT COMPOUND

(a) Physical state :- Covalent compounds are found in all the three states - Gas, Solid & Liquid.

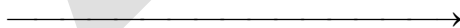
Separate molecules — In gaseous state

Associate molecules — In liquid & solid state

(Due to strong vander waal's force and hydrogen bonding among the molecules.)

As the molecular weight increases physical state changes -

eg. F_2 and Cl_2 Br_2 I_2 , At_2
 gas liquid solid



Top to bottom in a group. Vander waal's force increases between the molecules.

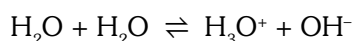
(b) Covalent solid :- Those solids in which atoms are linked together by covalent bonds, forms infinite three dimensional giant structure.

eg. Diamond, Graphite, AlN, SiC, SiO_2 etc.

Molecular solid :- Discrete (separate) molecules are formed by covalent bonds and then the molecules associates due to intermolecular force of attraction. (van der Waal force)

eg. Solid I_2 , dry ice (Solid CO_2) etc.

- (c) **Conductivity** :- Mostly covalent compounds are bad conductor of electricity. But few polar covalent compounds due to self ionisation can conduct electricity. eg. H_2O , liq. NH_3 etc.



Free ions are formed which can conduct electricity.

Exceptions :- Graphite, HCl in water.

- (d) **Solubility** :- Non polar compounds are soluble in non polar solvents. Non polar compounds forms vander waal bond with non polar solvent molecules.

Non polar solvents are C_6H_6 , CCl_4 etc.

- (e) **Isomerism** :- Covalent bond is rigid and directional, so it shows isomerism.

eg. Organic compounds.

- (f) **Reaction** :- Reaction between covalent compounds are slow. Because it involves breaking of old bonds and formation of new bonds.

BEGINNER'S BOX-1

- Which condition favours the bond formation:
 - (1) Maximum attraction and maximum potential energy
 - (2) Minimum attraction and minimum potential energy
 - (3) Minimum potential energy and maximum attraction
 - (4) None of the above
- Which one of the following element will never obey octet rule:
 - (1) Na
 - (2) F
 - (3) S
 - (4) H
- Which is not an exception to octet rule ?
 - (1) BF_3
 - (2) SiCl_4
 - (3) BeI_2
 - (4) ClO_2
- An oxide of chlorine which is an odd electron molecule is :
 - (1) ClO_2
 - (2) Cl_2O_6
 - (3) Cl_2O_7
 - (4) Cl_2O

2.5 VALENCE BOND THEORY (VBT) :

- (A) It was presented by Heitler & London to explain how a covalent bond is formed.

It was extended by Pauling & Slater.

- (B) The main points of theory are –

- To form a covalent bond overlapping occurs between half filled valence shell orbitals of the two atoms.
- Resulting bond acquires a pair of electrons with opposite spins to get stability.
- Orbitals come closer to each other from the direction in which there is maximum overlapping
- So covalent bond has directional character.
- Strength of covalent bond $\propto$ extent of overlapping.
- Extent of overlapping depends on two factors.
 - Nature of orbitals – p, d and f are directional orbitals $\rightarrow$ more overlapping
s-orbital $\rightarrow$ non directional – less overlapping

- (ii) Nature of overlapping – Co-axial overlapping - extent of overlapping more.
Collateral overlapping - extent of overlapping less

Order of strength of Co - axial overlapping – $p - p > s - p > s - s$



- (g) As the value of n increases, bond strength decreases.

$$1 - 1 > 1 - 2 > 2 - 2 > 2 - 3 > 3 - 3$$

$$1s - 2p > 2s - 2p > 3s - 3p$$

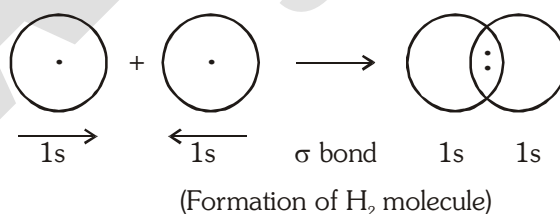
- (h) If n is same $2p - 2p > 2s - 2p > 2s - 2s$
- (i) Electron which is already paired in valency shell can enter into bond formation, if they can be unpaired first and shifted to vacant orbitals of slightly higher energy of the same energy shell.
- (j) This point can explain the trivalency of boron, tetravalency of carbon, pentavalency of phosphorus etc.
- (k) Two types of bonds are formed on account of overlapping.
(A) Sigma (σ) bond (B) Pi (π) bond

Sigma (σ) Bond

- (a) Bond formed between two atoms by the overlapping of half filled orbitals along their axis (end to end overlap) is called sigma bond.
- (b) σ bond is directional.
- (c) σ bond do not take part in resonance.
- (d) Free rotation is possible about a single σ bond.
- (e) Maximum overlapping is possible between electron clouds and hence it is strong bond.
- (f) There can be only one σ bond between two atoms.

Sigma bonds are formed by four types of overlapping

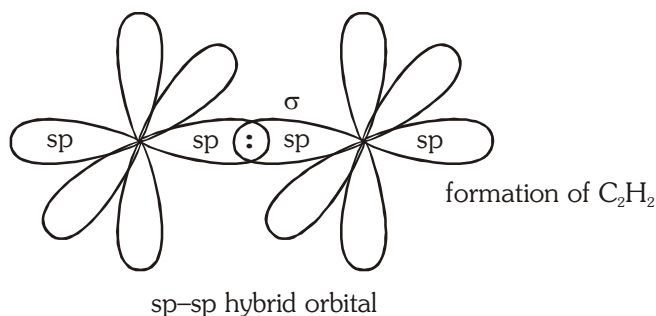
- (i) s - s overlapping – Two half filled s-orbitals overlap along the internuclear axis. Ex. H_2 molecule.



- (ii) s - p overlapping (Formation of HF) – When half fill s-orbital of one atom overlap with half filled p-orbital of other atom.

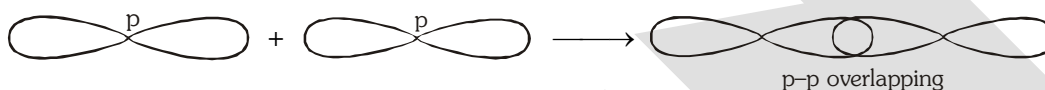


- (iii) Bond between two hybrid orbitals – $sp^3 - sp^3$, $sp^2 - sp^2$, $sp^3 - sp^2$, $sp^3 - sp$ etc.



Note : overlapping of hybrid orbitals form σ bond.

- (iv) p - p overlapping – (Coaxial) – It involves the coaxial overlapping between half filled p-orbitals of two different or same atoms.

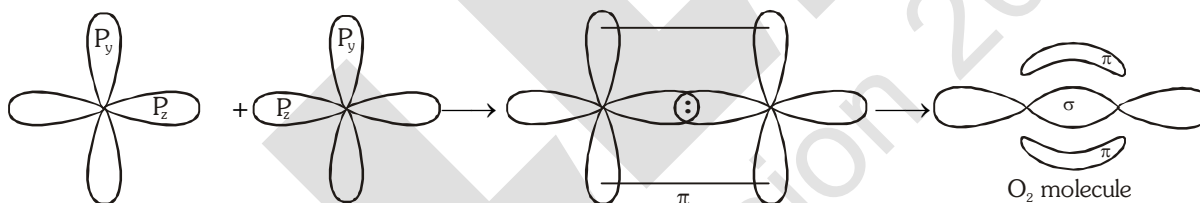


Ex. Formation of Cl_2 , F_2 , Br_2

Pi(π)-Bond

- The bond formed by sidewise (lateral) overlapping are known as π bonds.
- Lateral overlapping is only partial, so formed are weaker and hence more reactive than σ bonds (Repulsion between nucleus is more as orbitals have to come much close to each other for π bonds formation)

Example - Formation of O_2 molecule -



Note : Only two porbitals of oxygen atom have unpaired e^- in each orbital for bonding.

Electron configuration of oxygen is – $1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1$

- (c) Free rotation about a π bond is not possible.
- (d) π bond is weaker than σ bond (Bond energy difference is 63.5 KJ or 15 K cal/mole)
- (e) π bonds are less directional, so do not determine the shape of a molecule.
- (f) π bond takes part in resonance.
- (g) π bond formed by pure or unhybrid orbitals.

Comparison between σ and π bond

σ bond	π bond
1. Formed by axial overlapping	1. Formed by side by side overlapping
2. Involves s-s, s-p, p-p (axial) & hybrid orbitals	2. Involve p-p, p-d & d-d orbital
3. Extent of overlapping is more so stronger	3. Extent of overlapping is less so weaker
4. Free rotation around σ bond is possible	4. Free rotation around π bond is not possible
5. Hybridized or unhybridized orbital forms σ bond	5. Hybridized orbital never forms π bond
6. Independent existence of σ -bond.	6. No independent existence.

BEGINNER'S BOX-2

- According to the valence bond theory, when a covalent bond is formed between two reacting atoms, the potential energy of the system becomes—
 (1) negative (2) positive (3) minimum (4) maximum
- The strongest covalent bond is formed by the overlap of— (If considering for same shell)
 (1) s and p orbitals (2) s and s orbitals
 (3) p and d orbitals (4) p and p collateral orbitals
- In a triple bond there is sharing of :—
 (1) 3—electrons (2) 4—electrons (3) Several electrons (4) 6—electrons
- Which of the following configuration shows second excitation state of Iodine:—
 (1)

1↓	1↓	1	1	1				
----	----	---	---	---	--	--	--	--

 (2)

1↓	1	1	1	1	1	1		
----	---	---	---	---	---	---	--	--

 (3)

1↓	1	1	1	1	1			
----	---	---	---	---	---	--	--	--

 (4)

1	1	1	1	1	1	1		
---	---	---	---	---	---	---	--	--
- Variable covalency is exhibited by:—
 (1) P and S (2) N and O (3) N and P (4) F and Cl
- Which of the following bonds will have directional character
 (1) Ionic bond (2) Metallic bond
 (3) Covalent bond (4) Both covalent & metallic
- Number of σ and π bonds present in $\text{CH}_3 - \text{CH} = \text{CH} - \text{C} \equiv \text{CH}$ are -
 (1) 10 σ , 3 π (2) 10 σ , 2 π (3) 9 σ , 2 π (4) 8 σ , 3 π
- Which of the following statements regarding covalent bond is not true ?
 (1) The electrons are shared between atoms
 (2) The bond is non-directional
 (3) The strength of the bond depends upon the extent of overlapping
 (4) The bond formed may or may not be polar
- Predict the nature of bond

Orbitals	Internuclear axis	Bond
s + s	any axis	
s + p _x	x-axis	
s + p _y	y-axis	
s + p _z	z-axis	
p _x + p _x	x-axis	
p _y + p _y	y-axis	
p _z + p _z	z-axis	
p _x + p _z	y or z-axis	
p _y + p _y	x or z axis	
p _z + p _z	x or y axis	

2.6 HYBRIDISATION

Consider an example of Be compound :-

If it is formed without hybridisation then $\text{Cl} \overset{\text{p}}{\text{---}} \text{Be} \overset{\text{p}}{\text{---}} \text{Cl}$

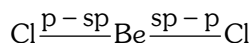
both the Be-Cl bonds should have different parameters and p-p bond strength > s-p bond strength. But practically bond strength and distance of both the Be-Cl bonds are same.

This problem may overcome if hybridisation of s and p-orbital occurs.

Hybridisation

- (1) It is introduced by Pauling, to explain equivalent nature of covalent bonds in a molecule.
- (2) **Definition** : Mixing of different shape and approximate equal energy atomic orbitals, and redistribution of energy to form new orbitals, of same shape & same energy. These new orbitals are called hybrid orbitals and the phenomenon is called hybridisation.

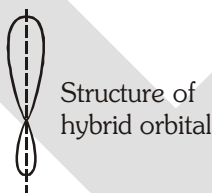
Now after considering s-p hybridisation in BeCl_2



(Bond strength of both the bonds will be equal)

Characteristic of Hybridisation

- (1) Hybridisation is a mixing of orbitals and not electrons. Therefore in hybridisation full filled, half filled and empty orbitals may take part.



- (2) Number of the hybrid orbitals formed is always be equivalent to number of atomic orbital which have taken part in the process of hybridisation.
- (3) Each hybrid orbital having two lobes, one is larger and other is smaller. Bond will be formed from large lobe.
- (4) The number of hybrid orbitals on central atom of a molecule or ion = number of σ bonds + lone pair of electron.
 - (i) The 1st bond between two atoms will be sigma.
 - (ii) The other bond between same two atoms will be pi bond.
 - (iii) The electron pair of an atom which do not take part in bond formation called as lone pair of electron.
- (5) One element can represent many hybridisation state depending on experimental conditions for example, C showing sp, sp^2 and sp^3 hybridisation in its compounds.
- (6) Hybrid orbitals are differentiated as sp, sp^2 , sp^3 etc.
- (7) The repulsion between lp - lp > lp - bp > bp - bp
- (8) The directional properties in hybrid orbital is more than atomic orbitals. Therefore hybrid orbitals form stronger sigma bond. The directional property of different hybrid orbitals will be in following order.

$$\text{sp} < \text{sp}^2 < \text{sp}^3 < \text{sp}^3\text{d} < \text{sp}^3\text{d}^2 < \text{sp}^3\text{d}^3$$

DETERMINATION OF HYBRIDISATION STATE

Method (I) : Number of hybrid orbital = number of σ bond + number of lone pair [surrounding the central atom]

Method (II) : To predict hybridisation following formulae may be used :

$$\text{No. of hybrid orbital} = \frac{1}{2} [Ve^- + SA \pm C]$$

[Ve^- = Total number of valence e^- in the central atom, SA = total number of monovalent atoms; C = charge]

eg. NH_4^+ $\frac{1}{2} [5 + 4 - 1] = 4$ sp^3 hybridisation.

SF_4 $\frac{1}{2} [6 + 4] = 5$ sp^3d hybridisation.

SO_4^{2-} $\frac{1}{2} [6 + 2] = 4$ sp^3 hybridisation.

('O' is divalent so add only charge on anion)

NO_3^- $\frac{1}{2} [5 + 1] = 3$ sp^2 hybridisation.

If such type of e^- pairs are –

two	–	sp	hybridisation
three	–	sp^2	hybridisation
four	–	sp^3	hybridisation
five	–	sp^3d	hybridisation
six	–	sp^3d^2	hybridisation
seven	–	sp^3d^3	hybridisation

S.No.	Formula	Total pair of e^-		Hybridisation	Geometry	Ex.
		bp	lp			
1.	AB_2	2	0	sp	Linear	$BeCl_2, CO_2$
2.	AB_3	3	0	sp^2	Trigonal Planar	BCl_3, BF_3
3.	AB_4	4	0	sp^3	Tetrahedral	CH_4, CCl_4
4.	AB_5	5	0	sp^3d	Trigonal bipyramidal	PCl_5
5.	AB_6	6	0	sp^3d^2	Octahedral	SF_6
6.	AB_7	7	0	sp^3d^3	(Square bipyramidal) Pentagonal bipyramidal	IF_7

Position of lone pair & multiple bond

(i) $sp/sp^2/sp^3$ = Any where

(ii) sp^3d = equatorial

(iii) sp^3d^2 = axial (defined first)

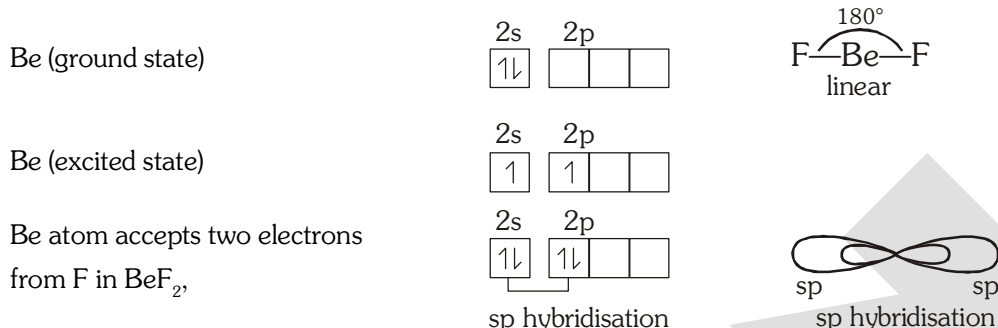
(iv) sp^3d^3 $\begin{cases} \text{Lone pair} = 1 \text{ then equatorial} \\ \text{Lone pair} = 2 \text{ then axial} \end{cases}$

(v) sp^3d hybridisation Axial bond length > equatorial bond length
 sp^3d^3 hybridisation Axial bond length < equatorial bond length] terminal atom same

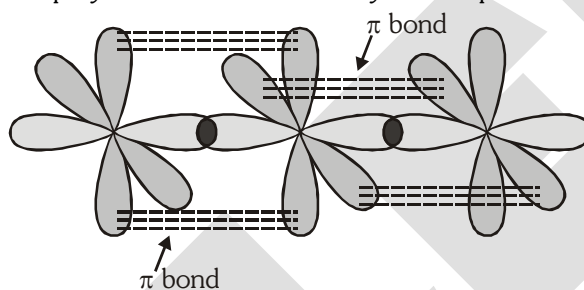
Types of Hybridisation

(A) sp hybridisation :

- In this hybridisation one s- & one p- orbital of an atom are mixed to give two new hybrid orbitals which are equivalent in shape & energy known as sp hybrid orbitals.
- These two sp hybrid orbitals are arranged in a straight line & at bond angle 180° .
- s-character 50%

**CO₂ Molecule (O = C = O) :**

In CO₂ molecule, C has two sp hybrid orbitals & two unhybridised p orbitals.

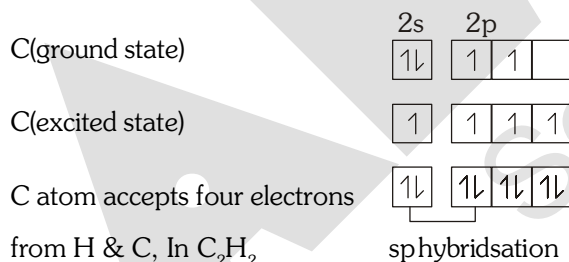
Molecular orbital picture of CO₂

Thus, CO₂ molecule is linear in shape & having 180° bond angle.

The bond length between C-O bond is reduced due to the presence of π bond.

CH $\equiv$ CH [H-C_A $\equiv$ C_B-H]

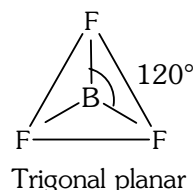
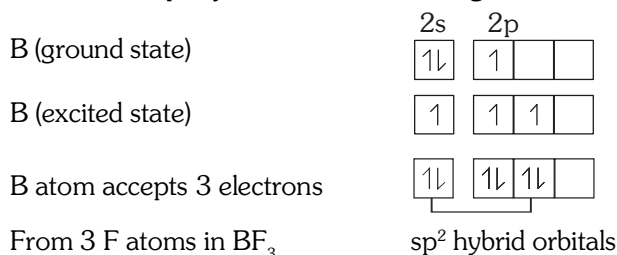
In CH $\equiv$ CH molecules, each C atom contains two sp hybrid orbitals & two unhybridised p orbitals



- sp hybrid orbital of each C overlaps to give sigma bond between C - C.
- The remaining one sp hybrid orbital of each C atom overlaps with s orbital of H, forming sigma bond between C - H.
- The two unhybridised p orbitals of each C atom (p_y and p_x) overlap laterally to form two $\pi(\pi)$ bonds.
- Therefore in H-C_A $\equiv$ C_B-H
 - sigma bond between C_A - C_B is formed sp - sp overlapping
 - sigma bond between C_A - H is formed sp - s overlapping
 - sigma bond between C_B - H is formed sp - s overlapping
 - π bond between C_A-C_B is formed : $p_y - p_y$, $p_x - p_x$ overlapping
- Each C atom forms two sigma bonds but in C₂H₂, total sigma bonds are 3
- Each C atom forms two π bonds. Total π bonds in C₂H₂ are two
- Total number of bonds in acetylene are : $3\sigma + 2\pi$ bond = 5 bonds.

(B) sp^2 Hybridisation :

- (a) In this hybridisation one s & two p orbitals are mixed to give three new sp^2 hybrid orbitals which are in same shape & equivalent energies.
 (b) These three sp^2 hybrid orbitals are at angle of 120° & giving trigonal planar shape.

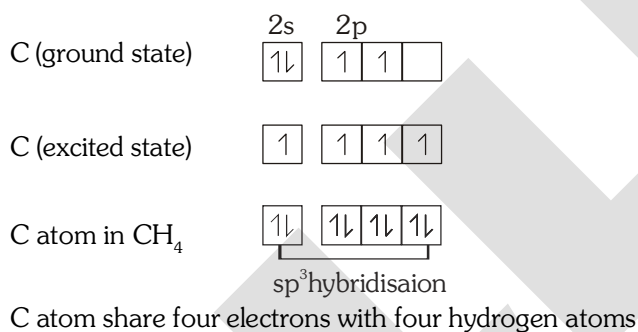


- (c) s - character 33.3 %

- SnX_2 having two σ bonds & one l.p. electron therefore hybridisation is sp^2
- The bond angle in SnX_2 will be less than 120° (due to presence of one l.p.e⁻).
- The shape of SnX_2 molecule is bent.

(C) sp^3 Hybridisation :

- (I) In this hybridisation one s orbital and three p orbitals of an atom of a molecule or ion, are mixed to give four new hybrid orbitals called as sp^3 hybrid orbitals.
 (II) The angle between hybrid orbitals will be $109^\circ 28'$ (6)

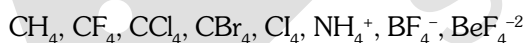


- (III) The shape obtained from these hybrid orbitals would be tetrahedron.

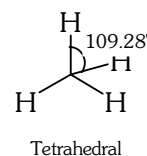
Three following examples represent this condition.

(a) Four sigma bonds with zero lone pair electron :

- (I) The following examples represent this condition.



- (II) In above compounds, bond angle is $109^\circ 28'$ & tetrahedron shape.



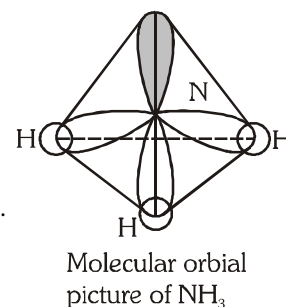
(b) Three sigma bonds & one lone pair of electron :

- (I) This condition is shown by following compounds & ions.



- (II) sp^3 hybridisation, pyramidal shape & bond angle will be less than $109^\circ 28'$.

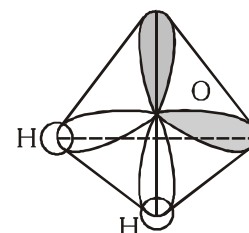
Due to the presence of one lone pair electron on nitrogen it repels bond pair electron more therefore bond angle reduced from $109^\circ 28'$ to 107° . The repulsion between lp - bp > bp - bp.



(c) Two sigma bonds & two lone pair of electrons :

- (I) This condition is shown by following compounds and ions.
 H_2O , OCl_2 , OBr_2 , OF_2 , OI_2 etc.
- (II) In all above examples, the central atom showing sp^3 hybridisation, angular shape and bond angle will be either less than $109^\circ 28'$ or more than $109^\circ 28'$.

In H_2O the hybridisation on O atom is sp^3 , but due to presence of two lone pair electrons they repel each other

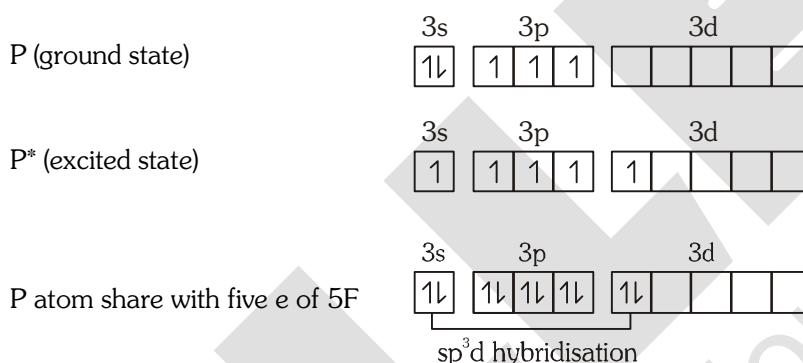


Molecular orbital picture of H_2O

(D) sp^3d Hybridisation :

- (I) In this hybridisation one s orbital, three p orbitals and one d orbital are mixed to give five new hybrid orbitals which are equivalent in shape and energy called as sp^3d hybrid orbitals.
- (II) Out of these five hybrid orbitals, three hybrid orbitals are at 120° angle and two hybrid orbitals are perpendicular to the plane of three hybrid orbitals that is trigonal planar, the shape of molecule becomes trigonal bipyramidal.

For example, PF_5 showing sp^3d hybridisation



- (III) In this hybridisation d_{z^2} orbital is hybridised with s and p orbitals.

In this way five sp^3d hybrid orbitals form five sigma bond with five F atoms and give a molecule of PF_5 , shape of this molecule is trigonal bipyramidal.

Two axial P-Cl bonds are longer than three equatorial P-Cl bonds due to repulsion between 3 equatorial bp of e^- and 2 axial b.p. of e^-

In above hybridisation, there are four conditions.

(a) Five sigma bonds and zero lone pair electron :

The following examples represent this conditions.

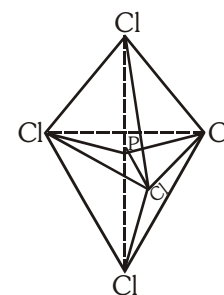
PCl_5 , PBr_5 , AsF_5 , AsCl_5 , SbCl_5 , SbF_5 etc.

The shape of all the above molecules is trigonal bipyramidal.

(b) Four sigma bonds and one lone pair of electron :

The following examples represent this condition.

SF_4 , SeF_4 , TeF_4 , PoF_4 , PF_4^- , SbF_4^- , SCl_4 , SeCl_4 , TeCl_4 etc.

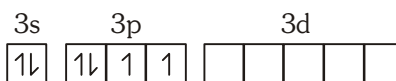


Structure of PCl_5

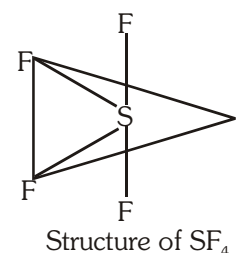
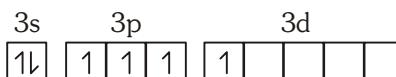
The shape of all above examples will be irregular tetrahedron or See-saw

Example SF_4

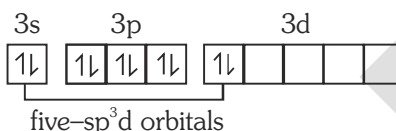
S ground state



S excited state



S atom share with four e^- from 4F atoms

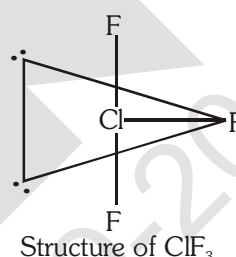


(c) Three sigma bonds & two lone pair of electrons :

The following examples represent this condition.

ClF_3 , BrF_3 , IF_3 , BrCl_3 , ICl_3 etc.

The shape of all above compounds is 'T' shape.



(d) Two sigma bonds & three lone pair of electrons :

The following examples represent this condition.

ICl_2^- , IBr_2^- , ClF_2^- , IF_2^- , BrF_2^- , XeF_2 , I_3^- , Br_3^-

The geometry of above examples will be linear shape.

(E) sp^3d^2 Hybridisation :

(I) In this hybridisation, one s-orbital, three p-orbitals & two d-orbitals (dz^2 , dx^2-y^2) are mixed to give six new hybrid orbitals known as sp^3d^2 hybrid orbitals.

(II) The geometry of molecule obtained from above six hybrid orbitals will be symmetrical octahedral or square bipyramidal.

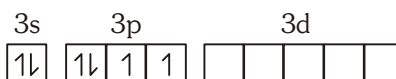
(III) The angle between all hybrid orbitals will be 90° .

Example : SF_6 , AlF_6^{3-} , PF_6^- , ICl_5 , XeF_4 , XeOF_4 , ICl_4^- ,

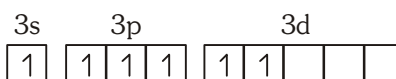
(IV) Two 'd' orbital participates in the hybridisation are $d_{x^2-y^2}$ and d_{z^2} .

SF_6

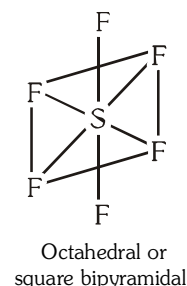
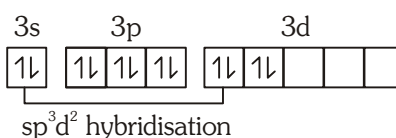
S (ground state)



S (excited state)

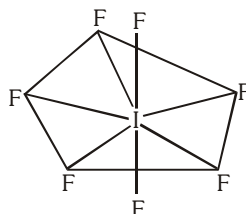


S (after hybridisation)



(F) sp^3d^3 Hybridisation :

- (I) In this hybridisation, one s-orbital, three p-orbitals & three d-orbitals are mixed to give seven new hybrid orbitals known as sp^3d^3 hybrid orbitals.
- (II) These seven sp^3d^3 orbitals are configured in pentagonal bipyramidal shape.
- (III) Five bond angles are of 72° and 10 bond angles of 90° .
- (IV) The following examples showing sp^3d^3 hybridisation – IF_7 & XeF_6 .

**(Pentagonal biipyramidal)****EXAMPLES ON sp^3d HYBRIDISATION**

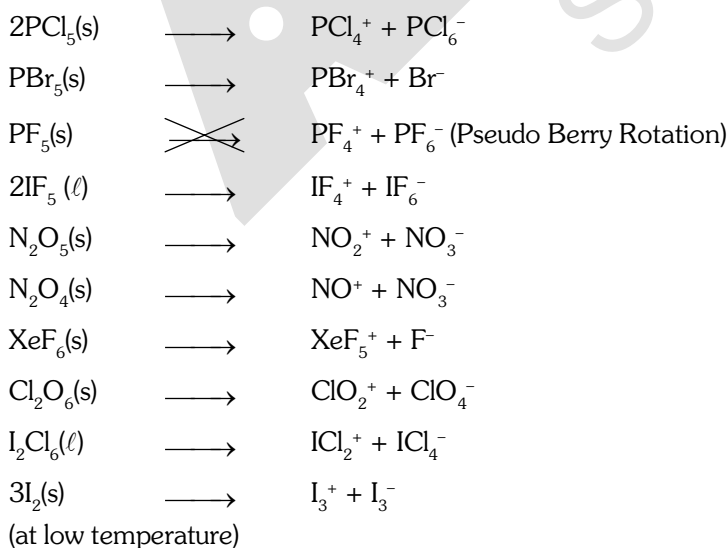
Example	σ bond	I.p.e.	Hybridisation	Bond angle	Shape
PCl_5	5	-	sp^3d	$120^\circ, 180^\circ$ & 90°	Trigonal bipyramidal
SF_4	4	1	sp^3d	$< 180^\circ, < 90^\circ, < 120^\circ$	Irregular tetrahedron
ClF_3	3	2	sp^3d	87.6°	T-shape
IF_3	3	2	sp^3d	87.6°	T-shape
ICl_3	3	2	sp^3d	87.6°	T-shape
Br_3^-	2	3	sp^3d	180°	Linear
ICl_2^-	2	3	sp^3d	180°	Linear
XeF_2	2	3	sp^3d	180°	Linear
PCl_4^+	4	-	sp^3		Tetrahedron
NH_4^+	4	-	sp^3		Tetrahedron
NF_3	3	1	sp^3		Pyramidal
H_3O^+	3	1	sp^3		Pyramidal
SO_3^{2-}	3	1	sp^3		Pyramidal
XeO_3	3	1	sp^3		Pyramidal
H_2O	2	2	sp^3		Angular (V)
NH_2^-	2	2	sp^3		Angular (V)
OF_2	2	2	sp^3		Angular (V)
Cl_2O	2	2	sp^3		Angular (V)
Diamond	4	-	sp^3		Tetrahedron
SiO_2	4	-	sp^3		Tetrahedron
SiC	4	-	sp^3		Tetrahedron
NO_3^-	3	-	sp^2	120°	Trigonal planar
SO_2	2	1	sp^2	$< 120^\circ$	Angular (V)
HCO_3^-	3	-	sp^2	120°	Trigonal planar
$SnCl_2$	2	1	sp^2	$< 120^\circ$	Angular (V)
NO_2^+	2	-	sp	180°	Linear
N_3^-	2	-	sp	180°	Linear

BEGINNER'S BOX-3

- Which of the following is incorrect about hybridization?
 - (1) The concept of hybridization is not applied to isolated atoms.
 - (2) Hybridization is the mixing of at least two non-equivalent atomic orbitals.
 - (3) The number of hybrid orbitals generated is more than the number of pure atomic orbitals that participate in the hybridization process.
 - (4) Hybridization requires an input of energy.
- The hybridization state of the central atom in HgCl_2 is—
 - (1) sp
 - (2) sp^2
 - (3) sp^3
 - (4) dsp^2
- The hybridization state of the central atom in AlI_3 is—
 - (1) dsp^2
 - (2) sp^3
 - (3) sp^2
 - (4) sp
- In C_3O_2 , the hybridization state of C is—
 - (1) sp^2
 - (2) sp
 - (3) sp^3
 - (4) dsp^2
- By hybridization, we mean the hybridization of—
 - (1) electrons
 - (2) atomic orbitals
 - (3) atoms
 - (4) protons
- The d- orbitals involved in sp^3d hybridisation is:—
 - (1) $d_{x^2-y^2}$
 - (2) d_{z^2}
 - (3) d_{xy}
 - (4) d_{xz}
- A sp^3 hybrid orbital contains:—
 - (1) $\frac{3}{4}$ s- character
 - (2) $\frac{1}{4}$ p - character
 - (3) $\frac{3}{4}$ p - character
 - (4) $\frac{1}{2}$ s - character
- In the protonation of NH_3 molecule, following statement is true:—
 - (1) A covalent bond is formed
 - (2) Hydrogen bond is formed
 - (3) Hybridisation state of N is changed
 - (4) Shape of NH_3 molecule is changed
- The shape of sulphate ion is :—
 - (1) Hexagonal
 - (2) Square planar
 - (3) Trigonal bipyramidal
 - (4) Tetrahedral
- In which following compound, central atom has four bond pair and one lone pair:—
 - (1) NH_4^+
 - (2) ICl_4^-
 - (3) SF_4
 - (4) XeF_4
- In which molecule s - p overlapping occurs ?
 - (1) CH_4
 - (2) NH_3
 - (3) H_2O
 - (4) None of these

Hybridisation in solid state :

- Compounds which change their structure in solid state/liquid state.



Hybridisation in radicals :

Radical	hybridisation
CH ₃	
CF ₃	
ClO ₃	
NO ₂	

Existence and Nonexistence :

Identify existing / non existing compounds / ions

- (a) PCl₆⁻ (b) NH₅ (c) PH₅ (d) (CCl₆)²⁻ (e) (SiF₆)²⁻ (f) (SiCl₆)²⁻ (g) ClBr₇
 (h) SH₆ (i) XeH₄ (j) XeOF₄ (k) FCl₃ (l) ClF₃ (m) BH₄⁻ (n) (PI₆)⁻

2.7 VALENCE SHELL ELECTRON PAIR REPULSION THEORY (VSEPR)

- (a) If the central atom possess only bonded pairs of electrons along with identical atoms then shape of the compound is symmetrical and according to Sidgwick & Powl.

eg.	BF ₃	—	120°	—	triangular
	CH ₄	—	109° 28'	—	tetrahedral
	CO ₂	—	180°	—	linear

- (b) If the central atom possess bonded pair of electrons as well as lone pair of electron, then shape of the molecule will be unsymmetrical ie. the original bond angle will disturbed due to repulsion between lone pair of electrons.

Similarly on having different type of side atoms, molecule becomes unsymmetrical due to unequal force of repulsion between e⁻. Order of repulsion is - [ℓp - ℓp > ℓp - bp > bp - bp]

$$\left[\text{Bond angle} \propto \frac{1}{\text{Number of lone pair of electrons}} \right]$$

- (c) By increasing one lone pair of electron, bond angle is decreased approx by 2.5°.

eg.:-	CH ₄	NH ₃	H ₂ O	→	sp ³
	109°	107°	105°		hybridisation

2.8 BOND PARAMETERS**Bond Length**

The average distance between the nucleus of two atoms is known as bond length, normally it is represented in Å. eg. A — B

It depends mainly on electronegativities of constituent atoms.

Case - I. Electronegativity difference is zero then -

$$\text{Bond length} = r_A + r_B \quad \text{or} \quad d_{A-B} = r_A + r_B$$

where

r_A = covalent radius of A

r_B = covalent radius of B

x_A = electronegativity of A

x_B = electronegativity of B

If $r_A = r_B$ then Bond length = $2r_A$ or $2r_B$

Example : - Cl - Cl

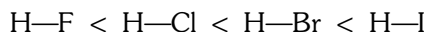
Case - II Electronegativity difference is not equal to zero then -

Bond length is given by Shomaker & Stevenson formula is - Bond length = $r_A + r_B - 0.09 (x_A - x_B) \text{Å}$

$(x_A - x_B)$ = Difference in electronegativities

Factors Affecting Bond Length

(a) **Δ electronegativity** :- Bond length $\propto \frac{1}{\Delta EN}$ (While B.E. $\propto \Delta EN$)



(b) **Bond order or number of bonds** :- Bond length $\propto \frac{1}{\text{Number of bonds or bond order}}$

Bond energy $\propto$ Number of bond

ex.



Bond length

1.54 Å

1.34 Å

1.20 Å

← increasing

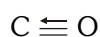
Bond energy

80

140

180-200 K.Cal.

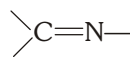
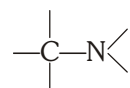
increasing →



1.43 Å

1.20 Å

1.13 Å



1.47 Å

1.28 Å

1.15 Å

(c) **Resonance** :- Due to resonance bond length affected

Ex.1. Benzene

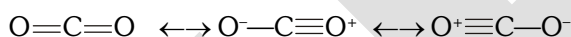
C—C bond length 1.54 Å

C=C bond length 1.34 Å



But bond length is between single and double bond is = 1.39 Å

Ex.2. Bond length of C—O in CO₂ is 1.15 Å Resonance occurs in CO₂ is as follows -



Bond length = 1.15 Å (Between double & triple bond)

(d) **Hybridization** :- Bond length $\propto \frac{1}{s \text{ character}}$

Example :-

s-character increases ↓

Compound	Hybridisation	Bond length
Ethane	sp ³ —sp ³	1.54 Å
$\text{>C}-\text{C}=\text{}$	sp ³ —sp ²	1.51 Å
$\text{>C}-\text{C}\equiv\text{C}$	sp ³ —sp	1.47 Å
C=C—C=C	sp ² —sp ²	1.46 Å
C=C—C≡C	sp ² —sp	1.42 Å
C≡C—C≡C	sp—sp	1.37 Å

Bond Angle

The minimum angle between any two adjacent bonds is known as bond angle.

It is represented in degree (°), min (') and second (")

Factors affecting the bond angle -

Step I : Hybridisation or % 's' character : Bond angle $\propto$ % s character



180° 120° 109.28'

Step II : Lone pair

When hybridisation is same, lone pair are different.

$$\text{Bond angle} \propto \frac{1}{\text{No. of lone pair}}$$

Example :-

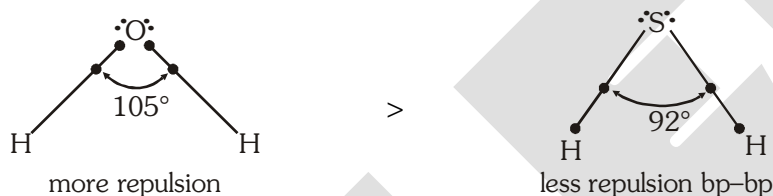
	CH_4	$\ddot{\text{N}}\text{H}_3$	$\text{H}_2\ddot{\text{O}}:$
Hybridisation	sp^3	sp^3	sp^3
Bond angle	109	107	105
	No l.p.	one l.p.	two l.p.

- In the different molecules if central atom have same number of lone pair of electron then bond angle will depend on electronegativities of A & B.

Step III : Central Atom

$$\text{Bond angle} \propto \text{Electronegativity of central atom}$$

In AB_x type of molecules if side atoms are same and EN of central atom increases the bond angle increases.



- Electronegativity of 'O' > Electronegativity of 'S'
- Bond angle of $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3$

Example :-

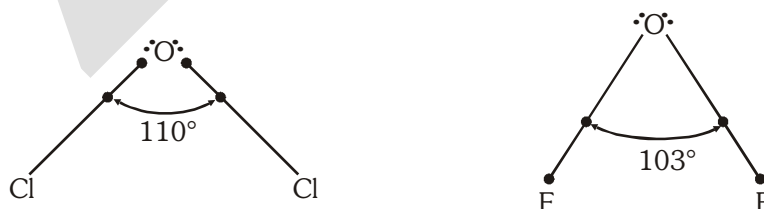
	$\ddot{\text{N}}\text{H}_3$	$\ddot{\text{P}}\text{H}_3$	$\ddot{\text{As}}\text{H}_3$
Bond angle	107°	93°	91°

-
- Electronegativity decreasing.
 - Bond angle will decrease

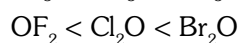
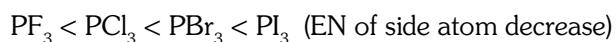
Step IV : Side atom

$$\text{Bond angle} \propto \frac{1}{\text{electronegativity of bonded atom}} \propto \text{size of side atom}$$

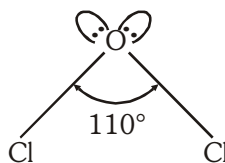
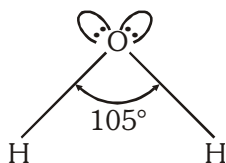
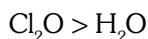
In AB_x type molecules, if central atoms are same and the EN of side atoms increases then bond angle decreases.



Electronegativity of Fluorine is greater than chlorine



Bond angle depends on size of side atom, On increasing size of side atom bond angle increases.



When hybridisation is same, lone pair are same, Central atom is same, bonded atoms are different.

sp^3	OF_2	103 - 105°	Electronegativity of bonded atom is decreasing
sp^3	Cl_2O	109 - 111°	
sp^3	Br_2O	116 - 118°	

Bond Energy (BE)

Bond energy may be defined as -

- Bond formation energy :- Energy released when any bond is formed is known as bond formation energy or bond energy.
- Bond dissociation energy :- Energy required to dissociate any bond is known as Bond dissociation energy.

Calculation of released energy is more difficult than the dissociation energy therefore dissociation energy of bond is calculated and is assumed as bond energy or bond formation energy.

Case-I In diatomic molecule :

Bond energy = bond dissociation energy

Example:- $\text{N}_2 > \text{O}_2 > \text{H}_2 > \text{F}_2$

Case-II For polyatomic molecule :-

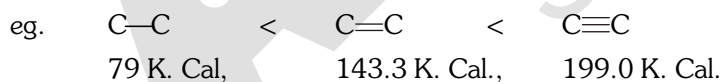
Bond energy $\approx$ Bond dissociation energy (D)

Factors Affecting The Bond Energy

- Electronegativity
- Bond order
- Atomic size
- Bond polarity
- Resonance
- Hybridisation
- Lone pair electron

(a) Δ Electronegativity :- Bond energy $\propto \Delta \text{EN}$ eg. $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$

(b) Bond order :- Bond energy $\propto$ Bond order.



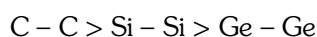
(c) Atomic size :- Bond energy $\propto \frac{1}{\text{Atomic size}}$ eg. $\text{C}\equiv\text{C} < \text{C}\equiv\text{N} < \text{N}\equiv\text{N}$

Exception :- In case of halogen group, order of bond energy is -

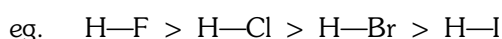


Because of higher electron density and small size of F atoms, repulsion between electrons of two F atoms, weakens the bond energy.

Other example $\text{S}-\text{S} > \text{O}-\text{O}$



(d) Bond Polarity :- Bond energy $\propto$ Bond polarity



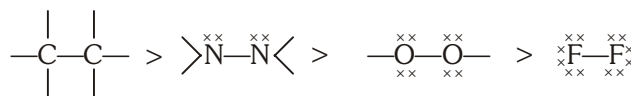
(e) **Resonance** :- Bond energy increases due to resonance.

eg. In benzene bond energy of C—C increases due to π electrons of C = C.

(f) **Hybridisation** :- Bond energy $\propto$ s-character in hybrid orbitals.

eg. $sp-sp > sp^2-sp^2 > sp^3-sp^3$
 s.character - 50 % 33.3 % 25 %

(g) **Lone pair of electrons** :- Bond energy $\propto \frac{1}{\text{lone pair of electrons}}$



Size of F and O atoms are small so their bond energy should be high (small atomic radius) but it is actually less, due to lone pair of electrons present on F and O atoms, which repels each other in F—F and —O—O—type of bonds.

Important Note (Summary) :

(i) Bond strength $\propto$ overlapping (if orbitals are given)

(ii) Bond strength $\propto \frac{1}{\text{size of orbitals}}$

e.g. $1s-2p > 1s-3p > 1s-4p$

(iii) If orbitals are of same shell

Bond strength $\propto$ extent of overlapping $\propto$ directional properties

$2p-2p > 2s-2p > 2s-2s > 2p-2p$ (Side ways)

Axial overlap

(iv) π -bond strength

$2p\pi-2p\pi > 2p\pi-3d\pi > 2p\pi-3p\pi > 3p\pi-3p\pi$

(v) O = O exist but S = S does not exist at room temperature.

(vi) N $\equiv$ N exist but P $\equiv$ P does not exist at room temperature.

(vii) O=C=O exist but O=Si=O does not exist.

BEGINNER'S BOX-4

- Which of the following molecules has the longest nitrogen-nitrogen bond?
 (1) N_2H_4 (2) N_2
 (3) N_2F_2 (4) All have equal bond lengths
- Which of the following molecules has the highest value of carbon-carbon bond energy?
 (1) C_2H_4 (2) C_3H_8 (3) C_2H_2 (4) C_2H_6
- Which of the following has the shortest bond length?
 (1) Br_2 (2) F_2 (3) Cl_2 (4) I_2
- In ethene, the carbon-carbon bond distance is—
 (1) 154 pm (2) 120 pm (3) 134 pm (4) 142 pm
- Carbon-halogen bond is strongest in the following
 (1) CH_3Cl (2) CH_3Br (3) CH_3F (4) CH_3I
- The correct order of bond length is
 (1) $C-C < C=C < C \equiv C$ (2) $C \equiv C < C=C < C-C$
 (3) $C=C < C \equiv C < C-C$ (4) $C=C < C-C < C \equiv C$
- The F—F bond is weak because :
 (1) The repulsion between the nonbonding pairs of electrons of two fluorine atoms is large
 (2) The ionization energy of the fluorine atom is very low
 (3) The length of the F—F bond much larger than the bond lengths in other halogen molecules
 (4) The F—F bond distance is small and hence the internuclear repulsion between the two F atoms is very low
- The correct order of decreasing bond energy is:-
 (1) $O-O > S-S > Se-Se$ (2) $C-C > Si-Si > Ge-Ge$
 (3) $F-F > O-O > N-N$ (4) $F-F > Cl-Cl > Br-Br$

9. The bond length does not affected by:-
 (1) Electron affinity (2) Bond order (3) Hybridisation (4) Resonance
10. In allene structure three carbon atoms are joined by :
 (1) Three sigma bonds and three pi bonds (2) Two sigma bonds and one pi bond
 (3) Two sigma bonds and two pi bonds (4) Three pi bonds only

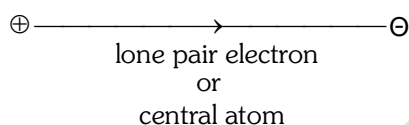
2.9 DIPOLE MOMENT

POLARITY OF BOND (IONIC NATURE IN COVALENT BOND)

- (a) Polarity of any polar covalent bond or molecule is measured in terms of dipole moment.
 (b) For measurement of extent of polarity, Pauling introduced the concept of dipole moment (μ).
 The product of positive or negative charge (q) and the distance (d) between two poles is called dipole moment.

Here - $\mu = q \times d$ (magnitude of charge $\times$ distance),

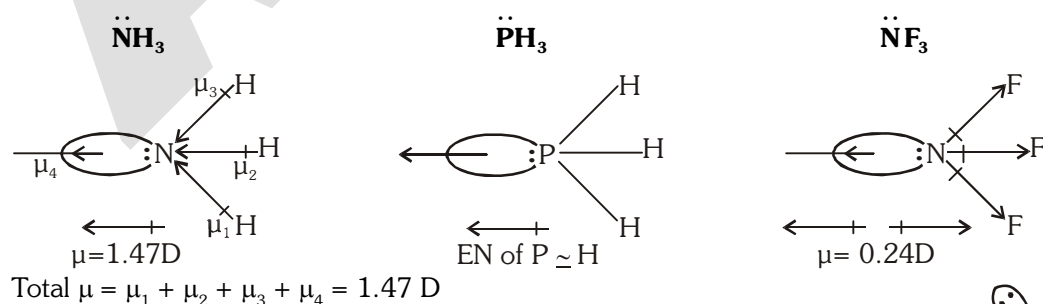
- (c) Dipole moment is a vector quantity i.e. it has both magnitude as well as direction.
 (d) Direction of dipole moment is represented by an arrow pointing from electro +ve to electro -ve element and from central atom to lone pair of electrons.



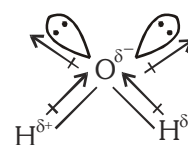
- (e) Unit of dipole moment is Debye
 $1 \text{ Debye} = 1 \times 10^{-18} \text{ e.s.u. cm} = 3.33 \times 10^{-30} \text{ coulomb metre}$
- (f) In the diatomic molecule μ depends upon difference of EN i.e. $\mu \propto \Delta \text{EN}$
 order of μ , $\text{H-F} > \text{H-Cl} > \text{H-Br} > \text{H-I}$
 $\mu = 0$ for H-H , F-F , Cl-Cl , Br-Br , O-O
- (g) For polyatomic molecules μ depends on the vector sum of dipole moments of all the covalent bonds.
 (h) For PCl_5 and SF_6 , etc. $\mu = 0$ due to their symmetrical geometry (According to charge).
 (i) Benzene, naphthalene, diphenyl have $\mu = 0$ due to planar structure.
 (j) If the vector sum is zero, than compound is non-polar compound or symmetrical compound (and it is not essential that individual μ of every bond should be zero).

Example - BX_3 , CCl_4 , SiCl_4 , CH_4 , CO_2 , CS_2 , PCl_5 , SiH_4 etc.

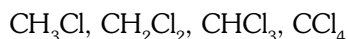
In these examples the bond B-F , C-Cl , C-H , C-O , P-Cl etc. are polar even though compounds are non-polar.



- (k) Dipole moment of H_2O is 1.85 D which is resultant μ of two O-H bonds.
 μ of $\text{H}_2\text{O} > \mu$ of H_2S because electronegativity of oxygen is higher than sulphur.
- (l) Angular structure of molecule have greater dipole moment.



Ex. Write the order of the dipole moment of following compounds ?



Sol. Right order is $\longrightarrow$ CCl_4 < CHCl_3 < CH_2Cl_2 < CH_3Cl
 $\mu = 0$ 1.02 1.55 1.93

Applications of Dipole Moment

(a) To determine polarity and geometry of molecule :-

If $\mu = 0$ compound is non polar and symmetrical

eg. $\text{CO}_2, \text{BF}_3, \text{CCl}_4, \text{CH}_4, \text{BeF}_2$ etc.

If $\mu \neq 0$ compound will be polar and unsymmetrical.

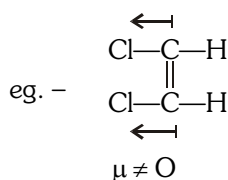
$\text{H}_2\text{O}, \text{SO}_2, \text{NH}_3, \text{Cl}_2\text{O}, \text{CH}_3\text{Cl}, \text{CHCl}_3$ etc.

(b) To calculate % ionic character :-

$$\% \text{ Ionic character} = \frac{\text{Experimental value of } \mu}{\text{Theoretical Value of } \mu} \times 100$$

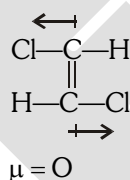
(c) To distinguish cis form or trans form :-

(I) Dipole moment of cis isomers is normally higher than trans isomers.



cis-form

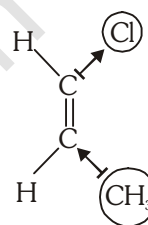
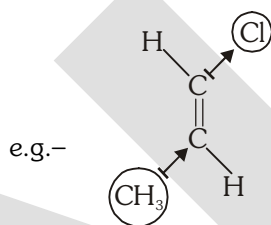
Polar ($\mu \neq 0$)



Trans-form

Non polar ($\mu = 0$)

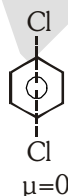
(II) If two groups have opposite inductive effect then trans-isomer will have greater dipole moment.



(d) To locate position of substituents in aromatic compounds.

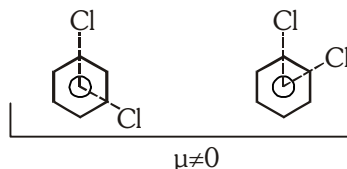
$$\mu \propto \frac{1}{\text{Bond angle}}$$

(I) If same substituents are present in the symmetrical position μ of benzene ring compounds will be zero.



Angle 180°

p-dichloro benzene



Angle 120°

m-dichloro benzene

Angle 60°

o-dichloro benzene

(II) As angle between vector decreases value of μ increases.

Illustrations

Illustration 1. Calculate the μ of HCl ? If bond distance is $1.34 \text{ \AA}$, charge = 4.8×10^{-10} esu and calculate % ionic character if experimental value of $\mu = 1.08 \text{ D}$?

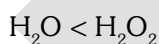
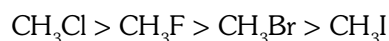
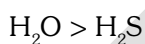
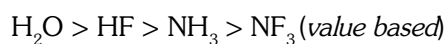
Solution $\mu = q \times d = 4.8 \times 10^{-10} \times 1.34 \times 10^{-8}$

$\mu = 6.4 \times 10^{-18} \text{ esu cm.}$

% Ionic character = $\frac{1.08 \times 10^{-18}}{6.4 \times 10^{-18}} \times 100 = 16.79 \%$

GOLDEN KEY POINTS

● **Some important orders of dipole moments :**



BEGINNER'S BOX-5

- Which of the following contains polar and nonpolar bonds?
 (1) H_2O_2 (2) CH_4 (3) HCN (4) NH_4Cl
- Carbon tetrachloride has no net dipole moment because of—
 (1) Similar electron affinity of C and Cl (2) its regular tetrahedral geometry
 (3) its planar geometry (4) similar sizes of C and Cl atoms
- Which of the following molecules is nonpolar?
 (i) PbCl_4 (ii) BF_3 (iii) SnCl_2 (iv) CS_2
 (1) (i), (ii), (iii) (2) (i), (ii), (iii), (iv) (3) (i), (ii), (iv) (4) (ii), (iii), (iv)
- Which of the following has the highest dipole moment ?
 (1) *o*-Dichlorobenzene (2) *m*-Dichlorobenzene (3) *p*-Dichlorobenzene (4) All have equal values
- Both CO_2 and H_2O contain polar covalent bonds but CO_2 is nonpolar while H_2O is polar because—
 (1) H atom is smaller than C atom
 (2) CO_2 is a linear molecule while H_2O is an angular molecule
 (3) O – H bond is more polar than C – H bond
 (4) CO_2 contains multiple bonds while H_2O has only single bonds

2.10 MOLECULAR ORBITAL THEORY (MOT)

MOT put forward by Hund & Mulliken, which can be applied to explain the properties, which the old VBT (Valence bond theory) was unable to explain. eg. Paramagnetic nature of O_2 molecule, as per VBT ($:O = O:$) it should be diamagnetic.

Definition : The atomic orbital lose their identity during molecule formation (overlapping) and form new orbitals termed as molecular orbitals.

Characteristic of Molecular Orbitals

- Molecular orbital formed by overlapping of atomic orbital of same energy.
- Number of molecular orbital formed = number of atomic orbitals involved in overlapping.
- Half of the molecular orbital have lower energy are called Bonding molecular orbital.
- Half are of higher energy - termed as Antibonding molecular orbital.
- Electronic configuration in various molecular orbital are governed by same three rules.
 - (1) Aufbau's rule
 - (2) Hund's rule
 - (3) Pauli's exclusion principle.
- Wave function for bonding molecular orbital is $\psi_b = \psi_A + \psi_B$
- Wave function for antibonding molecular orbital is $\psi_a = \psi_A - \psi_B$
 ψ_A = wave function of atom A
 ψ_B = wave function of atom B

Comparison of Bonding molecular orbital & Antibonding molecular orbital:

Bonding molecular orbital(BMO)	Antibonding molecular orbital (ABMO)
1. Bonding MO is the result of the linear combination of AO when their wave function are added $\psi_b = \psi_A + \psi_B$	1. ABMO is result of linear combination of AO when their wave function are subtracted $\psi_a = \psi_A - \psi_B$
2. Generally it does not have nodal plane.	2. It always have a nodal plane between two nuclei of bonded atom.
3. Electron density increases between two nuclei resulting attraction between two atoms.	3. Electron density decreases in between two nuclei, leads to repulsion between two atoms.
4. Energy of BMO is less, hence stable.	4. Energy of ABMO is high.
5. Electron placed in a BMO stabilises a molecule.	5. Electron placed in the ABMO destabilises the molecule.

Notation of molecular orbitals

As atomic orbitals are known by letters s, p, d and f depending on their shapes. Similarly for molecular orbital

For bonding molecular orbital - σ, π etc.

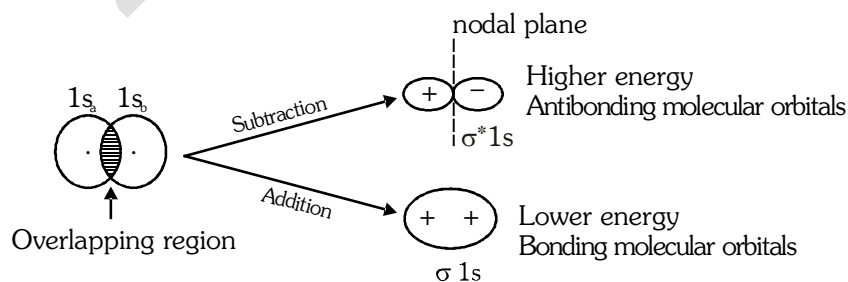
For antibonding molecular orbital- σ^*, π^* etc.

are used for different shapes of electron cloud.

Shapes of Molecular Orbitals (L.C.A.O. Method)

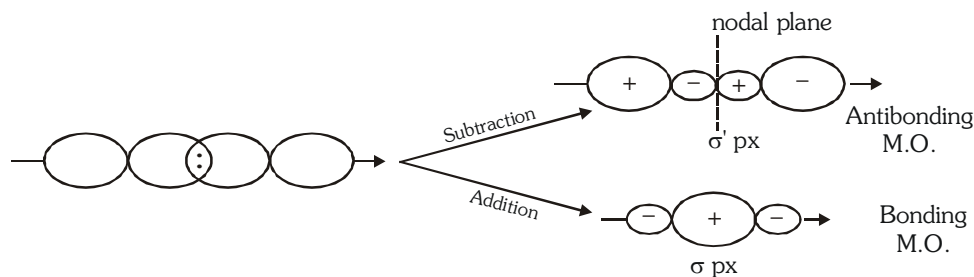
(A) (σ molecular orbital) :- It is formed by two ways -

- Combination of s-orbitals -



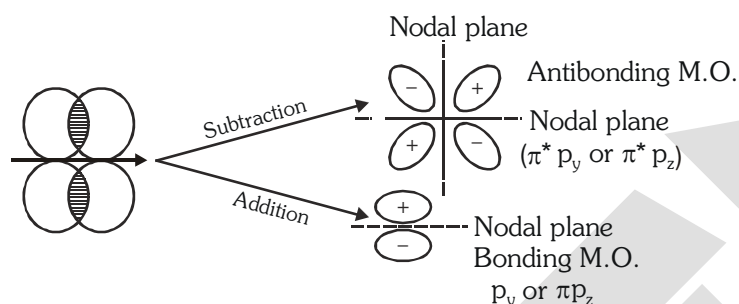
- $\sigma^* 1s$ have one nodal plane

(b) End on overlapping of p-orbitals - (Linearly) :-



• $\sigma^* p_x$ have one nodal plane

(B) π (pi) molecular orbitals :-



Positive sign, represent maximum probability finding of electrons.

$\pi^* p_y$ or $\pi^* p_z$ have two nodal plane

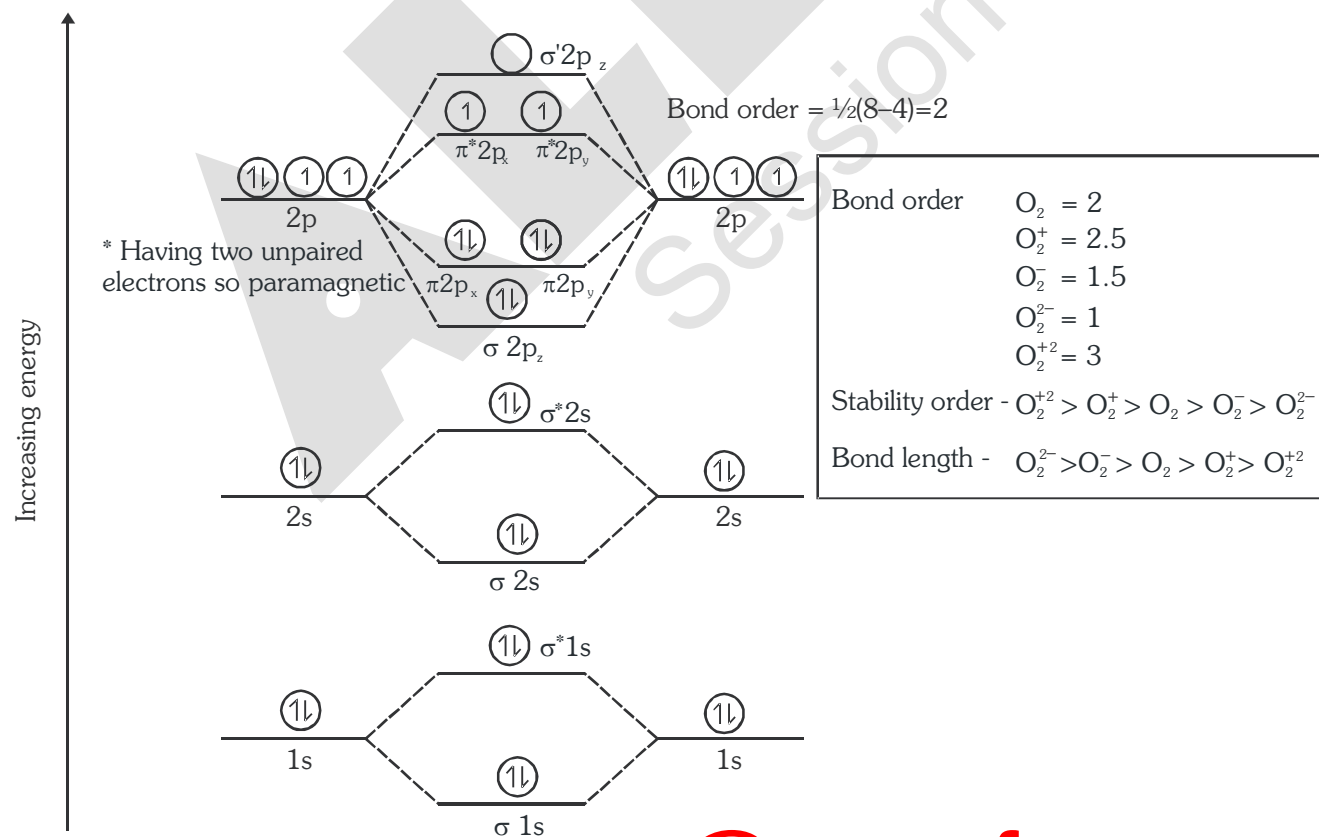
ENERGY LEVEL DIAGRAM OF MOLECULAR ORBITAL

(A) Energy level diagram for O_2, F_2, Ne_2 (Beyond N_2)

On the basis of Aufbau's rule - increasing order of energies of various molecular orbitals is -

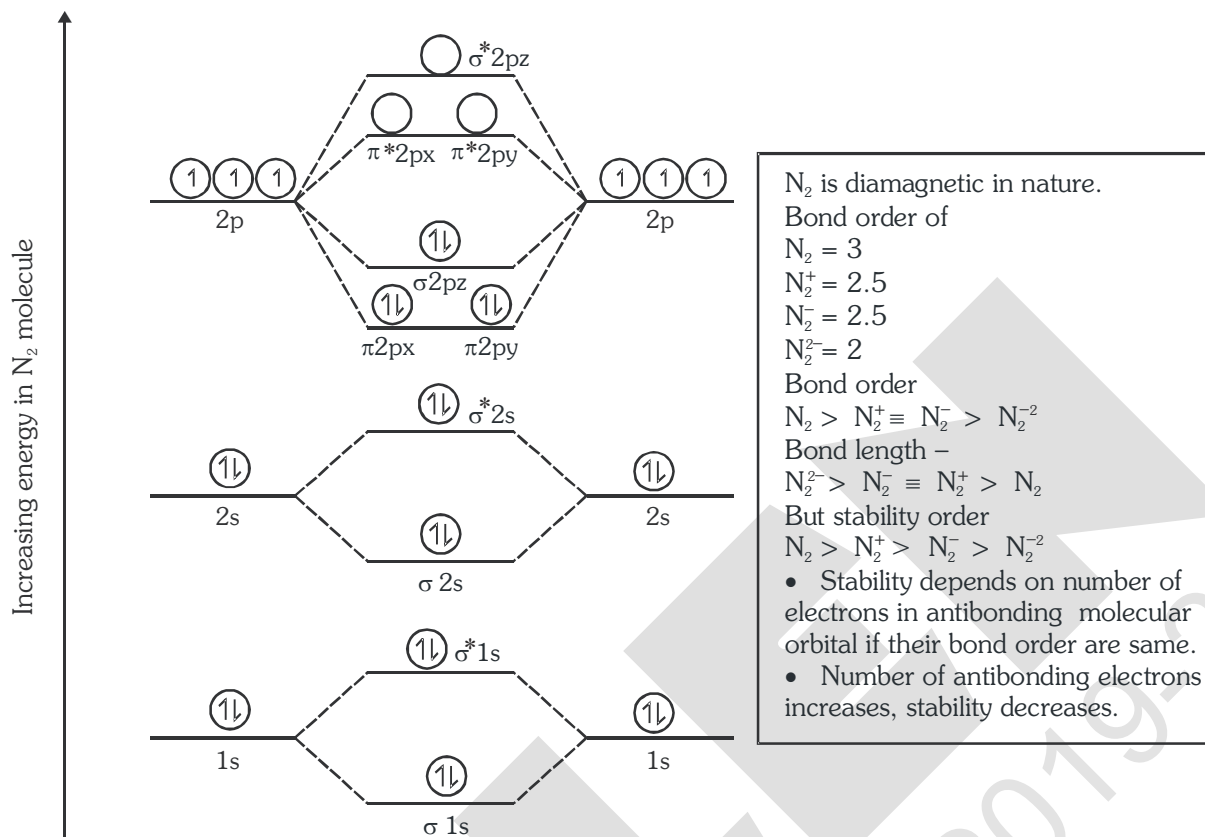
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \pi 2p_x \equiv \pi 2p_y < \pi^* 2p_x \equiv \pi^* 2p_y < \sigma^* 2p_z$

For O_2 molecule



(B) Energy level diagram for B₂, C₂ and N₂ molecules (upto N₂)

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x \equiv \pi 2p_y < \sigma 2p_z < \pi^* 2p_x \equiv \pi^* 2p_y < \sigma^* 2p_z$$

For N₂ molecule**Electronic configuration of molecules and their related properties :-**

For writing electronic configuration of diatomic molecules following two rules to be followed-

- Count the number of electrons present in two atoms and then fill in the appropriate energy level diagram according to Aufbau rule.
- The pairing in $\pi 2p_x$ and $\pi 2p_y$ or $\pi^* 2p_x$ and $\pi^* 2p_y$ will take place only when each molecular orbital of identical energy has one electron.

GOLDEN KEY POINTS

- Bond order** :- $\frac{1}{2}[N_b - N_a]$ N_b = Total number of bonding electron
 N_a = Total number of antibonding electron

Application of bond order :

- Bond length** :- (distance between two nuclei) Bond length $\propto \frac{1}{\text{Bond order}}$
 If $\left. \begin{matrix} N_b > N_a \\ N_b < N_a \\ N_b = N_a \end{matrix} \right\}$ $\begin{matrix} \text{Molecule exists} \\ \text{Molecule do not exists} \end{matrix}$
- Stability of molecule** :- stability $\propto$ Bond order of molecule
- Dissociation energy** :- Bond dissociation energy $\propto$ Bond order.
- Magnetic property** :-
 - When electron in MO are paired — diamagnetic
 - When electron in MO are unpaired — paramagnetic

(a) Hydrogen molecule

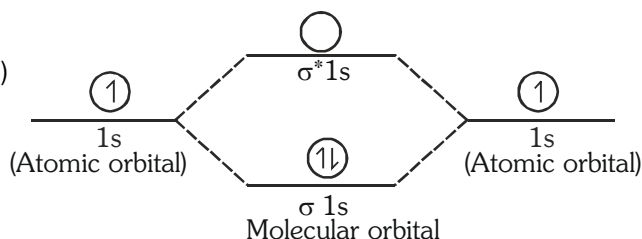
Having two H atoms with one electron each (1s')

M.O. configuration of $H_2 = (\sigma 1s)^2 (\sigma^* 1s)^0$

Bond order = $\frac{1}{2} [N_b - N_a]$
= $\frac{1}{2} [2 - 0] = 1$ ie. single bond.

Having paired electron so diamagnetic.

Stability $\rightarrow$ quite stable (having single bond)



(b) H_2^+ ion —

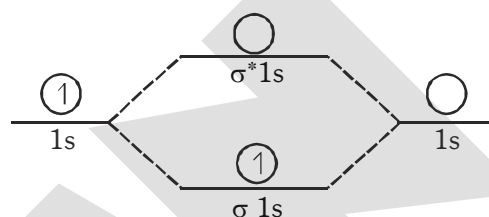
Configuration of $H_2^+ = (\sigma 1s)^1 (\sigma^* 1s)^0$

One electron in bonding molecular orbital.

So paramagnetic

Bond order = $\frac{1}{2} [1 - 0] = \frac{1}{2}$

Less stable. (Incomparision to H_2)



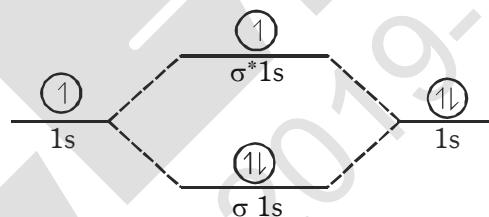
(c) H_2^- anion -

M.O. configuration - $(\sigma 1s)^2 (\sigma^* 1s)^1$

Paramagnetic

Bond order = $\frac{1}{2} [2 - 1] = \frac{1}{2}$

Stability is less than H_2^+ because H_2^- contain an ABMO electron



Stability order	H_2	H_2^+	H_2^-
Bond order	1.0	0.5	0.5

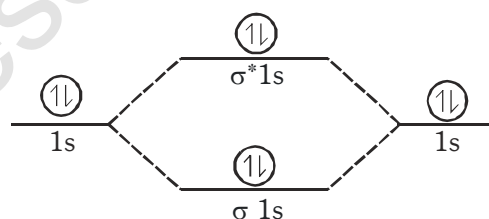
(d) Helium molecule (He_2)

M.O. configuration - $(\sigma 1s)^2 (\sigma^* 1s)^2$

Diamagnetic

Bond order = $\frac{1}{2} [2 - 2] = 0$ (zero)

- Bond order zero indicates no linkage between He atoms. Hence He_2 molecule does not exists.



Comparison between VBT and MOT

1. According to VBT electron moves around only one nucleus	1. According to MOT electron moves under influence of two or more nuclei
2. Orbitals are monocentric	2. Orbitals are polycentric
3. According to VBT O_2 is diamagnetic	3. According to MOT O_2 is paramagnetic

BEGINNER'S BOX-6

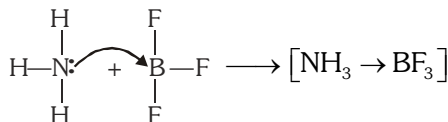
- Which of the following is incorrect regarding the MO theory ?
 - (1) The number of molecular orbitals formed is always equal to the number of atomic orbitals combined.
 - (2) The more stable the bonding molecular orbital, the less stable the corresponding antibonding molecular orbital.
 - (3) In a stable molecule, the number of electrons in bonding molecular orbitals is always equal to that in antibonding molecular orbitals.
 - (4) Like an atomic orbital, each molecular orbital can accommodate up to two electrons with opposite spins in accordance with the Pauli exclusion principle.
- If the z-axis is the molecular axis, then π MOs are formed by the overlap of—
 - (i) p_z and p_z
 - (ii) p_y and p_y
 - (iii) s_z and p_z
 - (iv) p_x and p_x
 - (1) (ii), (iv)
 - (2) (ii), (iii)
 - (3) (i), (ii)
 - (4) (i), (iii)
- If the z-axis is taken as the internuclear axis, then which of the following combinations of atomic orbitals is a nonbonding combination ?
 - (1) s and p_y
 - (2) p_x and d_{yz}
 - (3) p_x and p_y
 - (4) all of these
- Which of the following is the correct order of stability ?
 - (1) $H_2 > H_2^+ > He_2 > He_2^+$
 - (2) $H_2 > He_2^+ > H_2^+ > He_2$
 - (3) $H_2 > H_2^+ > He_2^+ > He_2$
 - (4) $H_2 > He_2 > He_2^+ > H_2^+$
- Bond order in C_2^+ is:-
 - (1) $\frac{1}{2}$
 - (2) $\frac{2}{3}$
 - (3) $\frac{3}{2}$
 - (4) 1
- In which of the following set, the value of bond order will be 2.5:-
 - (1) O_2^+ , NO, NO^{+2} , CN
 - (2) CN, NO^{+2} , CN^- , F_2
 - (3) O_2^+ , NO^{+2} , O_2^{+2} , CN^-
 - (4) O_2^{-2} , O_2^- , O_2^+ , O_2
- Of the following species which has the highest bond order and shortest bond length :
NO, NO^+ , NO^{2+} , NO^-
 - (1) NO only
 - (2) Bond order of NO is highest and bond length of NO^{2+} is shortest
 - (3) NO^+ only
 - (4) NO^{2+} only
- The diamagnetic molecule is
 - (1) Super oxide ion
 - (2) Oxygen molecule
 - (3) Carbon molecule
 - (4) Unipositive ion of nitrogen molecule
- On the basis of molecular orbital theory which molecules does not exist
 - (1) H_2
 - (2) He_2
 - (3) HeH
 - (4) Li_2
- Maximum bond energy will be shown by the species
 - (1) O_2^+
 - (2) O_2
 - (3) O_2^-
 - (4) O_2^{-2}

2.11 CO-ORDINATE BOND

- (1) It is a covalent bond in which the shared e^- pair come from one atom is called coordinate bond.
- (2) Necessary conditions for the formation of coordinate bond are -
 - (a) Octet of donor atom should be complete and should have atleast one lone pair of electron.
 - (b) Acceptor atom should have deficiency of at least one pair of electron.

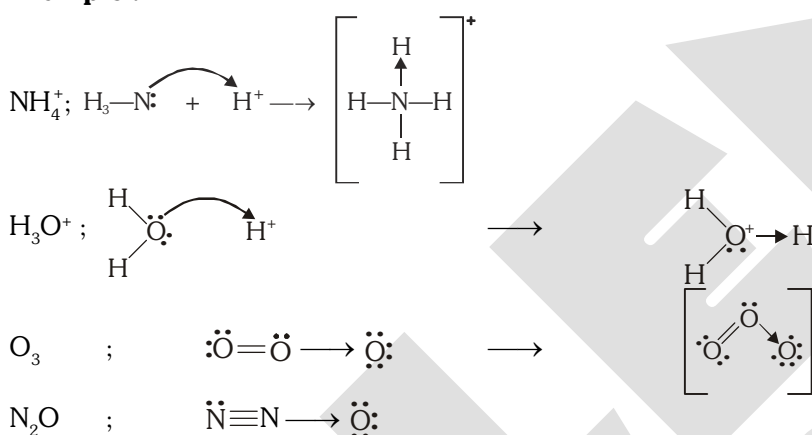


- (3) Atom which provide electron pair for sharing is called donor.
- (4) Other atom which accepts electron pair is called acceptor. That is why it is called donor-acceptor or dative bond.



BF_3 is electron deficient compound.

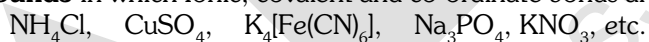
Example :



Metal co-ordinate compounds -



Compounds in which Ionic, covalent and co-ordinate bonds are present, are as follows -



2.12 FORMAL CHARGE

The difference between the valence electrons in an isolated atom and the number of valence electrons assigned to that atom in a given Lewis structure is called that atom's formal charge. The formal charge, abbreviated FC, on an atom in a Lewis structure is given by the following relationship:

Formal charge on a atom in a Lewis structure

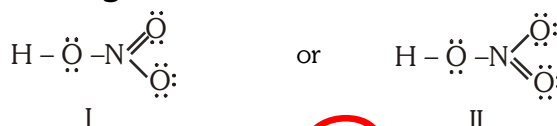
$$= (\text{total number of valence electrons in the isolated atom}) - (\text{Total number of nonbonding electrons}) - \frac{1}{2} (\text{total number of bonding electrons}) \dots (i)$$

$$\text{or} \quad \text{FC} = (\text{Valence electrons}) - (\text{Nonbonding electrons}) - \frac{1}{2} (\text{bonding electrons})$$

$$= (\text{Valence electrons or group number}) - [(\text{Number of unshared } e^-s) + (\text{Number of bonds})]$$

To find the formal charge, we count the bonding electrons as though they are equally shared between the two bonded atoms.

Q. Calculate the formal charges on the various atoms of nitric acid (HNO_3) molecule which has been described by the following Lewis structures:



The H atom : It has one valence electron, zero lone pair (or zero nonbonding electrons), and forms one bond (two bonding electrons).

Substituting in Eq. (i), we write $FC = (1) - (0) - \frac{1}{2}(2) = 0$.

The O atom bonded to H: It has six valence electrons, two lone pairs (or four nonbonding electrons), and from two bonds (or four bonding electrons).

Hence, we write $FC = (6) - (4) - \frac{1}{2}(4) = 0$.

The N atom: It has five valence electrons, zero lone pair (or zero nonbonding electrons), and forms four bonds (or has eight bonding electrons).

Thus, we write $FC = (5) - (0) - \frac{1}{2}(8) = +1$

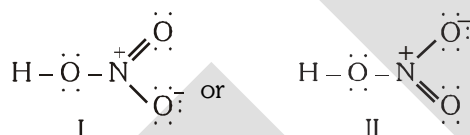
The end O atom in N = O: It has six valence electrons in the free state *but* in the Lewis structure (I or II), it has two lone pairs (or four nonbonding electrons) and forms two bonds (or has four bonding electrons).

Thus, we can write $FC = (6) - (4) - \frac{1}{2}(4) = 0$

The end O atom in N - O : It has six valence electrons in the free state but in the Lewis structure (I or II), it has three lone pairs or six nonbonding electrons and forms one bond (or has two bonding electrons).

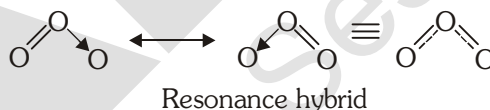
thus, we write $FC = (6) - (6) - \frac{1}{2}(2) = -1$

We can now write the Lewis structures (I and II) for nitric acid molecule including the formal charges as



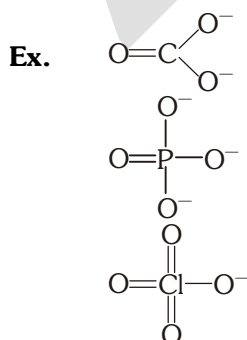
2.13 RESONANCE

- (a) It has been found that the observed properties of certain compounds cannot be satisfactorily explained by writing a single Lewis structure. The molecule is then supposed to have many structures, each of which can explain most of the properties of the molecule but none can explain all the properties of the molecules. The actual structure is in between of all these contributing structures and is called resonance hybrid and the different individual structures are called resonating structures or canonical forms. This phenomenon is called resonance.
- (b) Let us discuss resonance in ozone, according to its resonance structure it should have one single bond (O—O = 1.48Å) but experiments show that both the bonds are same which can be proved by its resonance hybrid as shown below.

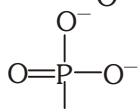


Note : To calculate bond order in the polyatomic molecule or ion use following formula :

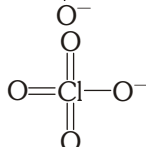
$$\text{Bond order} = \frac{\text{Total number of bonds in a molecule}}{\text{Resonating Structures}}$$



$$\text{Bond order} = \frac{4}{3} = 1.33$$



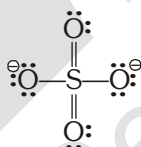
$$\text{Bond order} = \frac{5}{4} = 1.25$$



$$\text{Bond order} = \frac{7}{4} = 1.75$$

BEGINNER'S BOX-7

- Which of the following ions has resonating structures ?
 (1) SO_4^{2-} (2) PO_4^{3-} (3) SO_3^{2-} (4) All of these
- How many resonating structures can be drawn for NO_2 ?
 (1) Six (2) Four (3) Five (4) Two
- Which of the following is true for nitrate anion
 (1) Formal charge on N is zero (2) Bond order of NO bond is $\frac{4}{3}$
 (3) Average formal charge on oxygen is $-\frac{1}{3}$ (4) There are 2 π -bonds in the ion
- Which of the following contains Co-ordinate and covalent bonds:-
 (a) N_2H_5^+ (b) H_3O^+ (c) HCl (d) H_2O
 Correct answer is :-
 (1) a & d (2) a & b (3) c & d (4) Only a
- The correct statement for the reaction -
 $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$:-
 (1) Hybridisation state is changed (2) Bond angle increases
 (3) NH_3 act as a Lewis acid (4) Regular geometry is changed
- The number of coordinate bonds presents in SO_3 molecule are
 (1) 1 (2) 2 (3) 3 (4) 4
- One of the resonating structure of SO_4^{2-} is



Which set of formal charge on oxygen and bond order is correct

- (1) 0.5 and 1.5 (2) 1.5 and 3 (3) 2 and 3 (4) 1.5 and 1.5
- Resonance is not shown by -
 (1) C_6H_6 (2) CO_2 (3) CO_3^{2-} (4) SiO_2
- Bond length of C - O is minimum in -
 (1) CO (2) CO_2 (3) CO_3^{2-} (4) HCOO^-

Subjective Type Questions

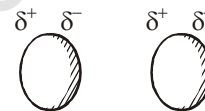
- Discuss resonance and formal charge in N_3^- and N_2O ?
- Give the average formal charge and average bond order of XO bond in the following oxy compounds?
 (a) SO_4^{2-} (b) SO_3^{2-} (c) NO_2^- (d) ClO_2^- (e) ClO_3^-
 (f) ClO_4^- (g) HCO_3^- (h) CO_3^{2-} (i) HSO_3^- (j) PO_4^{3-}

2.14 METALLIC BOND

- (1) The constituent particles of metallic solids are metal atoms which are held together by metallic bond.
- (2) In order to explain the nature of metallic bond Lorentz proposed a simple theory known as electron gas model or electron sea model.
- (3) A metal atom is supposed to consist of two parts, valence electrons and the remaining part (the nucleus and the inner shells) called kernel.
- (4) The kernels of metal atoms occupy the lattice sites while the space between the kernel is occupied by valence electrons.
- (5) Due to small ionisation energy the valence electrons or metal atoms are not held by the nucleus firmly. Therefore, the electrons leave the field of influence of one kernel and come under the influence of another kernel. Thus the electrons are not localised but are mobile.
- (6) The simultaneous attraction between the kernels and the mobile electrons which hold the kernel together is known as metallic bond.

WEAKER FORCES**2.15 Vander Waal's Forces**

- (a) These are weak; non directional, non valence force of attraction among neutral species.
- (b) These are electrical in nature, due to induced polarity caused by temporary displacement of electrons towards one end of the inert atoms, becoming a temporary dipole.
- (c) This temporary dipole in one molecule can induce opposite dipoles in surrounding molecule due to displacement of electrons, one end becomes -ve and other +ve. These partially charged ends, induce surrounding molecules accordingly.
- (d) Strength of vander waal force depends on ease of distortion of electron cloud.



van der Waal's force $\propto$ size of atom or molecule $\propto$ atomic wt. or molecular weight.

- (e) Therefore m.p. and b.p. of noble gas elements (inert gas atom) and halogens increases down the group.

Types of van der Waal's Forces

- (a) **Keesom force or dipole dipole force** - one dipole molecule orient the other dipole molecule, to bring opposite ends close to each other for attraction. This is called orientation effect. Example - HCl, H₂O, NH₃ etc.
- (b) **Debye force or dipole induce dipole force** :- Forces which results from the interaction of a permanent dipole and induced dipole are called Debye force.
eg. When non-polar substance like benzene come in contact with polar molecules like NH₃, induced dipole moment in benzene appears (Induction effect).
- (c) **London dispersion force or Instantaneous dipole instantaneous induce dipole attraction** :-
Due to vibration or moment to atom electron cloud and nuclei shifts temporarily towards opposite ends, leads to attraction between them, eg. In atoms of noble gases and halogens.

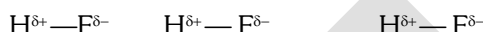
2.16 HYDROGEN BONDING

Definition

- (a) It is an electrostatic attractive force between covalently bonded hydrogen atom of one molecule and an electronegative atom.



- (b) It is not formed in ionic compounds.
 (c) H-bond forms in polar covalent compounds, (not in non-polar).
 (d) It is very weak bond but stronger than vander waal's force.
 (e) It is also known as dipole-dipole attraction.



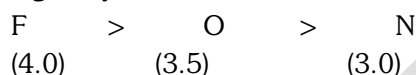
Main condition for H-bonding

- (a) H— should be covalently bonded with high electro -ve element like F, O, N.
 (b) Atomic size of electro -ve element should be small.

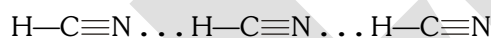
Order of atomic size is -



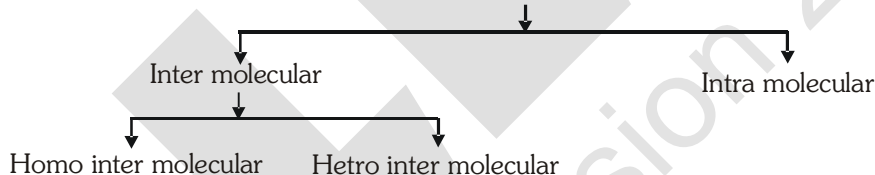
Order of electronegativity is -



- (c) Strength of H-bond $\propto$ Electronegativity of Z (element) $\propto \frac{1}{\text{atomic size of Z}}$
 (d) Hydrogen bonding occurs in HCN, due to $(-C \equiv N)$ triple bond (sp hybridisation), electronegativities of carbon and nitrogen increases.



TYPES OF HYDROGEN BONDING

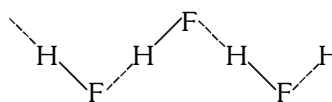
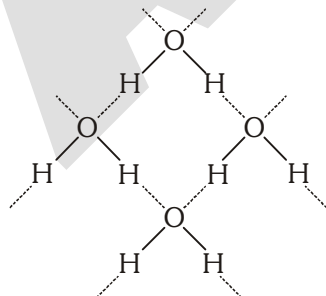


Intermolecular H-Bond

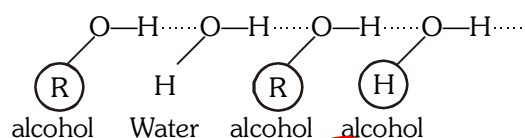
H-bond formation between two or more molecules of either the same or different compounds known as **Inter molecular H-bonding**.

These are of two types :-

- (i) **Homo intermolecular** :- H-bond between molecules of same compound.



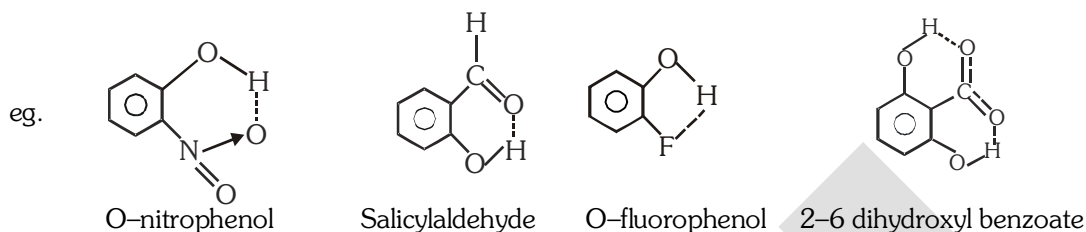
- (ii) **Hetro intermolecular** :- H-bond between molecules of different compounds
Eg. between alcohol and water



Intramolecular H-bond

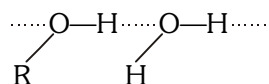
It takes place within the molecule.

- (a) H-bonded with electronegative element of a functional group, form H-bond with another electronegative element present on nearest position on the same molecule.
- (b) This type of H-bond is mostly occurred in organic compounds (Aromatic)
- (c) It results in ring formation (Chelation).

**Effect of H-bond on Physical Properties****(A) Solubility****(a) Inter molecular H-bonding**

- (I) Few organic compounds (Non-polar) are soluble in water (Polar solvent) due to H-bonding.

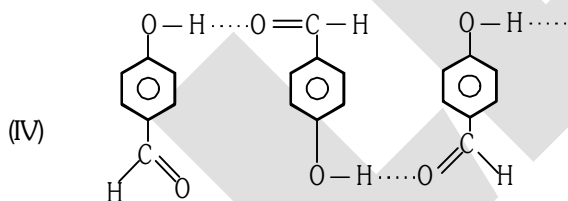
Ex. - Alcohol in water



Other examples - Glucose, Fructose etc. dissolve in water.

- (II) Ketone, ether, alkane etc. are insoluble (no H-bond). Dimethyl ether is soluble in water while diethyl ether is partially soluble, due to bulky ethyl groups H-bonding interrupts.

- (III) Solubility order- $\text{CH}_3\text{OCH}_3 < \text{CH}_3\text{OH}$

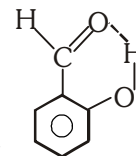


p-hydroxy benzaldehyde.

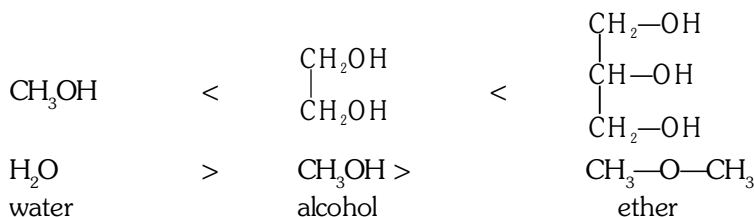
It can form H-bond with water molecule so it can dissolve.

(b) Intra molecular H-bonding

- (I) It decreases solubility as it form chelate by H-bonding, so H- is not free for other molecule.
- (II) It can not form H-bond with water molecule so it can not dissolve.

**(B) Viscosity**

H-bond associates molecules together so viscosity increases.

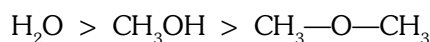


(C) **Surface Tension**

Surface tension of a liquid $\propto$ extent of H-bonding.

(D) **Melting point and boiling point**

(I) Due to intermolecular H-bond M.P. & B.P. of compounds increases.



(II) Boiling points of VA, VIA, VIIA hydrides are as shown below :



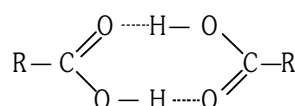
(III) But sudden increase in boiling point of NH_3 , H_2O and HF is due to H-bonding.

(IV) $\text{H}_2\text{O} > \text{HF} > \text{NH}_3$ (BP comparison due to combined effect of strength and number of H bond)

Intramolecular H-bonding gives rise to ring formation, so the force of attraction among these molecules are vander waal force. So M.P. and B.P. are low.

(E) **Molecular weight**

Molecular wt. of CH_3COOH is double of its molecular formula, due to dimer formation occur by H-bonding.



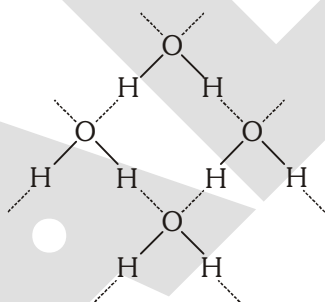
(F) **Physical states**

H_2O is liquid while H_2S is gas.

Water and Ice :- Both have H-bonding even then density of ice is less than water.

Volume of ice is more because of open cage like crystal structure, form by association of water molecules with the help of H-bond.

H_2O becomes solid due to four hydrogen bond among water molecule are formed in tetrahedral manner.

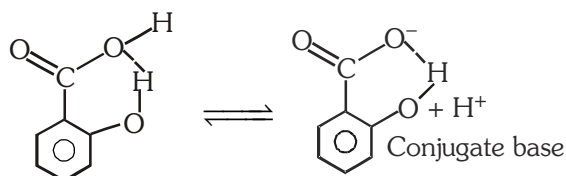


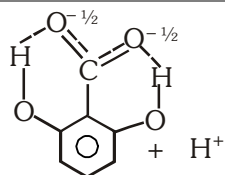
GOLDEN KEY POINTS

Effect of intramolecular H-bonding

Strength of acid

(I) The formation of intramolecular H-bonding in the conjugate base of an acid gives extra stability to conjugate base and hence acid strength increases eg. Salicylic acid is stronger than benzoic acid
2, 6 - dihydroxy benzoic acid > salicylic acid > benzoic acid.





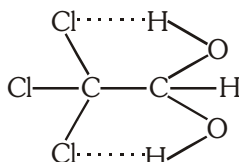
2, 6-dihydroxy benzoate ion.

(II) C_2H_5SH is more acidic than C_2H_5OH . In C_2H_5OH , H-bond forms so H^+ is not free.

(III) HF is weaker acid than HI , due to H-bond in $H-F$, H^+ is not free

● **Stability of chloral hydrate**

If two or more OH group on the same atom are present it will be unstable, but chloral hydrate is stable (due to H-bonding).



Chloral hydrate

BEGINNER'S BOX-8

- Two ice cubes are pressed over each other until they unite to form one block. The force mainly responsible for holding them together is—
 - van der Waals force
 - dipole-dipole interaction
 - H bonding
 - covalent bonding
- The vapor pressure of *o*-nitrophenol at any given temperature is predicted to be—
 - higher than that of *p*-nitrophenol
 - lower than that of *p*-nitrophenol
 - same as that of *p*-nitrophenol
 - higher or lower depending upon the size of the vessel
- The hydrogen bond is strongest in:—
 - $O-H \cdots S$
 - $S-H \cdots O$
 - $F-H \cdots F$
 - $O-H \cdots O$
- H_2O boils at higher temperature than H_2S , because it is capable of forming:—
 - Ionic bonds
 - Covalent bonds
 - Hydrogen bonds
 - Metallic bonds
- Maximum number of H-bonding is shown by
 - H_2O
 - H_2Se
 - H_2S
 - HF
- Which is the weakest among the following types of bonds ?
 - Debye force
 - Metallic bond
 - Dipole-dipole bond
 - Hydrogen bond
- The boiling point of *p*-nitrophenol is higher than that of *o*-nitrophenol because :
 - NO_2 group at *p*-position behaves in a different way from that at *o*-position
 - intramolecular hydrogen bonding exists in *p*-nitrophenol
 - there is intermolecular hydrogen bonding in *p*-nitrophenol
 - p*-nitrophenol has a higher molecular weight than *o*-nitrophenol
- In which molecule the Vander Waals force (dispersion force) is likely to be the most important in determining the m.pt. and b.pt. :
 - Br_2
 - CO
 - H_2S
 - HCl
- Covalent-molecules are usually held in a crystal structure by
 - Dipole-dipole attraction
 - Electrostatic attraction
 - Hydrogen bond
 - Van-der waal's attraction
- In solid argon the atoms are held together
 - by ionic bonds
 - by hydrogen bonds
 - By vander-waals forces
 - By hydrophobic bonds

2.17 IONIC OR ELECTROVALENT BOND

The chemical bond formed between two or more atoms as a result of the complete transfer of one or more electrons from one atom to another is called Ionic or electrovalent bond.

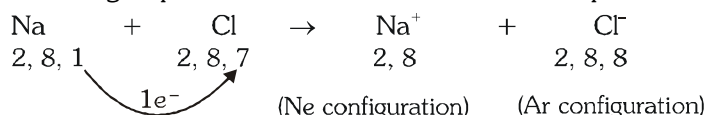
Electro +ve atom loses electron (group IA to IIIA)

Electro -ve atom gains electron (group VA to VIIA)

Electrostatic force of attraction between cation and anion is called ionic bond or electrovalent bond.

Nature of ionic bond $\propto$ Electronegativity difference.

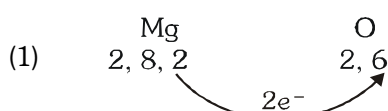
e.g. IA and VIIA group elements form maximum ionic compound.



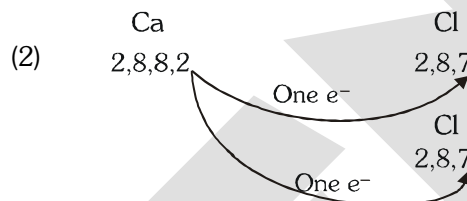
More the distance between two elements in periodic table more will be ionic character of bond.

Total number of electron lose or gained is called electrovalency.

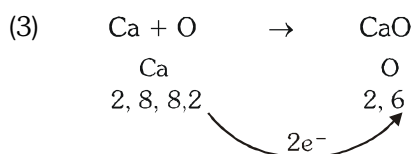
Example –



electrovalency of Mg = 2
electrovalency of O = 2



electrovalency of Ca = 2
electrovalency of Cl = 1



electrovalency of Ca = 2
electrovalency of O = 2

The force of attraction is equal in all direction so ionic bond is non-directional. A definite three dimensional structure is formed called crystal lattice. Energy released during the formation of one mole crystal lattice is called lattice energy.

Ionic compound do not have molecular formula. It has only empirical formula.

e.g. NaCl is empirical formula of sodium chloride

Conditions for Forming Ionic Bonds

Formation of Ionic bond depends upon these three factors –

(a) Ionisation energy :

Amount of energy required to remove an electron from the outermost orbit of an isolated gaseous atom to form the +ve ion or cation. (energy absorbed)

Lesser Ionisation energy $\rightarrow$ Greater tendency to form cation.



(b) Electron affinity :

Amount of energy released when an electron is added to an isolated gaseous atom to form -ve ion (anion) energy released.

Higher electron affinity $\rightarrow$ Greater tendency to form anion



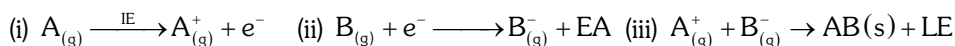
- (c) **Lattice energy** - (Energy released) Amount of energy released when one mole of crystal lattice is formed

Higher lattice energy $\rightarrow$ Greater will be the stability or strength of ionic compound.
or Amount of energy required to break the crystal.

- (d) **Overall lowering of energy :**

Energy must be released during bond formation.

Energy changes are involved in the following steps –



This concludes that for lower value of IE and higher value of EA there is more ease of formation of the ionic compound which is summarised as Born Haber Cycle.

GOLDEN KEY POINTS

● Factors affecting lattice energy

- (1) Magnitude of charge $\rightarrow U \propto z^+ z^-$ (Ionic charge)

Lattice energy $\propto$ Magnitude of charge



- $\rightarrow$
- Lattice energy increases
 - Charge of cation increases

- (2) Size of Cation :- Lattice energy $\propto \frac{1}{r^+ + r^-}$



- $\rightarrow$
- Size of cation increasing
 - Size of anion is constant
 - Lattice energy decreases.

● Representation of formula of compounds :

- (1) Write the symbols of the ions side by side in such a way that positive ion is at the left and negative ion is at the right as A^+B^-
- (2) Write their electrovalencies in figure at the top of each symbol as $A^x B^y$

- (3) Now apply criss cross rule as



Examples : Calcium chloride

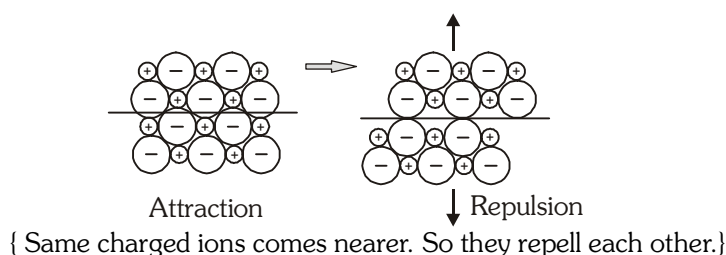


Properties of Ionic Compound

- (a) **Physical state** –

Ionic compounds are hard, crystalline and brittle due to strong electrostatic force of attraction.

Brittleness $\rightarrow$



(b) Isomorphism –

- (1) Two compounds are said to be isomorphous if they have similar no. of electrons i.e. similar configuration of their cation and anion.
- (2) They have similar crystal structure.

Example –

	Na ⁺	F ⁻	Mg ⁺²	O ⁻²
Valency	+1,	-1	+2,	-2
electronic configuration	2, 8	2, 8	2, 8	2, 8
similarly	Ca ⁺²	2Cl ⁻¹	2K ⁺¹	S ⁻²
	2, 8, 8	2, 8, 8 } 2, 8, 8 }	2, 8, 8 } 2, 8, 8 }	2, 8, 8

(c) Boiling point and melting point –

Ionic compounds have high boiling point and melting point due to strong electrostatic force of attraction among oppositely charged ions.

(d) Conductivity –

It depends on ionic mobility. In solid state - No free ions - Bad conductor of electricity.

In fused state or aqueous solution Due to free ions - Good conductor of electricity.

Conductivity order Solid state < fused state < Aqueous solution

(e) Solubility –

Highly soluble in Polar solvents like water.

Less soluble in non polar solvents like benzene.

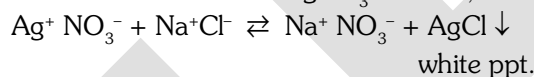
Ex. NaCl form a true solution in water but is colloid in benzene

(f) Ionic reaction –

(a) Ionic compounds shows ionic reaction and covalent compounds shows - molecular reaction.

(b) Ionic reactions are faster than molecular reaction because of free ions.

e.g. When NaCl is added in AgNO₃ solution, white ppt of AgCl is formed at once.



(g) Ionic bond non-directional and does not show stereo isomerism

2.18 Polarisation (Fajan's Rule) (Covalent Nature in Ionic Bond)

When a cation approaches an anion closely the positive charge of a cation attract the electron cloud of the anion towards itself, due to the electrostatic force of attraction between them.

At the same time cation also repel the positively charge nucleus of anion.

Due to this combined effect, cloud of anion is bulged or elongated towards the cation. This is called distortion, deformation or Polarisation of the anion by the cation and anion is called Polarised.

Polarisation Power

The ability of cation to polarised a nearby anion is called Polarisation power of cation.



Polarizability

Ability of an anion to get polarised by the cation.

Polarisation of anion causes some sharing of electron between the ions so ionic bond acquires certain covalent character.

[Polarisation $\propto$ Covalent character]

Magnitude of polarisation depends upon a no. of factors, suggested by Fajan and are known as Fajan's rule.

Fajan's Rule (Factors Affecting Polarisation)**(a) Size of cation** : Polarisation of the anion increases as the size of cation decreases.

$$\text{Polarisation} \propto \frac{1}{\text{size of cation}}$$

eg.

In a group



- Size of cation increases
 - Covalent character decreases
 - Ionic character increases

Greatest polarising power of Be²⁺, shows its maximum covalent characterIn a period — Na⁺, Mg²⁺, Al³⁺, Si⁴⁺

- Cation size decreases
 - Covalent character increases

(b) Size of anion :- If the size of the anion increases for a given cation, the covalent character increases.
Polarisation $\propto$ size of anion.

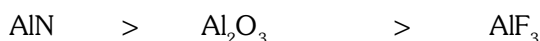
- size of anion increases
 - Covalent character increases
 - Ionic character decreases

(c) Charge on cation and anion :-Polarisation $\propto$ charge on cation or anion**(i) Polarisation (covalent character) $\propto$ Charge on cation**

- Charge on cation increases
 - Covalent character increases
 - Ionic character decreases (M.P. decreases)

Ex. Write the increasing order of M.P. & B.P. of following compounds.**Ans.** (1) Sn⁺⁴ < Sn⁺² (2) Fe⁺³ < Fe⁺² (3) Pb⁺⁴ < Pb⁺²(Charge on cation $\propto$ polarisation power $\propto$ covalent character $\propto \frac{1}{\text{M.P.}}$)**(ii) Polarisation $\propto$ Charge of anion**

- Charge increases
 - Covalent character increases



Charge on anion $\propto$ polarisation $\propto$ covalent nature $\propto \frac{1}{\text{M.P.}}$

(d) **Electronic configuration of cation : -**

Order of Polarisation power : $8e^- < (18+2)e^- < 18e^-$

$\text{CuCl} \longrightarrow \text{Cu}^+ \quad 2, 8, 18 \quad (\text{Covalent})$

$\text{NaCl} \longrightarrow \text{Na}^+ \quad 2, 8 \quad (\text{Ionic})$

Cu^+ and Na^+ both the cation (Pseudo inert & inert respectively) have same charge and size but polarising power of Cu^+ is more than Na^+ because -

$$\begin{array}{ccc} z_{\text{eff}} \text{ of } ns^2p^6 \text{ (inert)} & < & z_{\text{eff}} \text{ of } ns^2p^6d^{10} \text{ (pseudo inert)} \\ \text{Na}^+ & < & \text{Cu}^+ \\ (\text{Ionic}) & & (\text{Covalent}) \end{array}$$

So CuCl has more covalent character than NaCl .

Order of covalent character

- $\text{LiF} < \text{LiCl} < \text{LiBr} < \text{LiI}$
- $\text{SF}_2 < \text{SF}_4 < \text{SF}_6$
- $\text{CaCl}_2 < \text{FeCl}_2 < \text{FeCl}_3$
- $\text{Hg}_2\text{Cl}_2 < \text{HgCl}_2$
- $\text{SrCl}_2 < \text{SnCl}_2 < \text{CdCl}_2$
- $\text{ZnCl}_2 < \text{CdCl}_2 < \text{HgCl}_2$

Note :

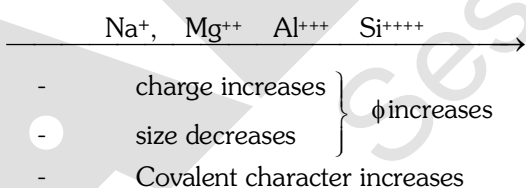
Polarisation power of a cation is usually called ionic potential or charge density.

$$\text{Ionic potential } \phi \text{ (phi)} = \frac{\text{Charge on cation}}{\text{Size of cation}}$$

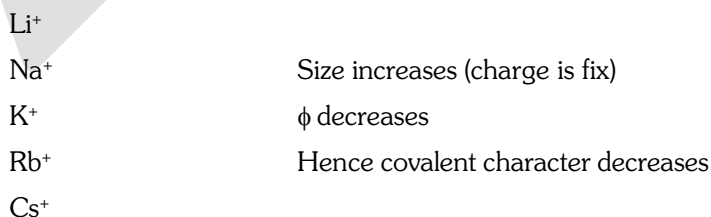
To determine covalent and ionic character of molecule

$$\phi \propto \frac{\text{Covalent Character}}{\text{Ionic character}}$$

From left (larger size) to right (smaller size) in a period ϕ increases so covalent character increases.



From top to bottom in a group ϕ decreases so covalent, character decreases.



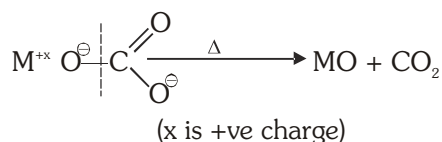
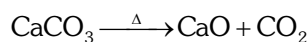
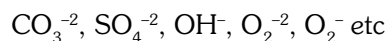
BEGINNER'S BOX-9

1. The electrovalency of the element is equal to the—
(1) number of electrons lost
(2) number of electrons gained
(3) number of electrons transferred
(4) number of electrons lost or gained by the atom of the element during the formation of ions of ionic compound
2. Which of the following polar solvents has the highest dielectric constant?
(1) H_2O (2) D_2O (3) CH_3OH (4) $\text{C}_2\text{H}_5\text{OH}$
3. Which of the following cations possesses neither noble gas nor pseudo noble gas configurations?
(i) Bi^{3+} (ii) Pb^{2+} (iii) Sn^{2+} (iv) Tl^+
(1) (ii), (iii) (2) (i), (iv) (3) (i), (ii), (iii) (4) (i), (ii), (iii), (iv)
4. Ionic bond formation involves :
(1) Elimination of protons (2) Sharing of electrons
(3) Overlapping of orbitals (4) Formation of octets
5. The hydration of ionic compounds involves —
(1) Evolution of heat (2) Weakening of attractive forces
(3) Dissociation into ions (4) All
6. The hydration energy of Mg^{+2} is greater than the hydration energy of
(1) Al^{+3} (2) Mg^{+3} (3) Na^+ (4) Be^{+2}
7. Among the following which compounds will show the highest lattice energy ?
(1) KF (2) NaF (3) CsF (4) RbF
8. The lattice energy of the lithium is in the following order :
(1) $\text{LiF} > \text{LiCl} > \text{LiBr} > \text{LiI}$ (2) $\text{LiCl} > \text{LiF} > \text{LiBr} > \text{LiI}$
(3) $\text{LiBr} > \text{LiCl} > \text{LiF} > \text{LiI}$ (4) $\text{LiI} > \text{LiBr} > \text{LiCl} > \text{LiF}$
9. Among LiCl , BeCl_2 , BCl_3 and CCl_4 , the covalent bond character follows the order :
(1) $\text{LiCl} < \text{BeCl}_2 > \text{BCl}_3 > \text{CCl}_4$ (2) $\text{LiCl} > \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$
(3) $\text{LiCl} < \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$ (4) $\text{LiCl} > \text{BeCl}_2 > \text{BCl}_3 > \text{CCl}_4$

2.19 THERMAL DECOMPOSITION

Thermal stability of metal carbonates

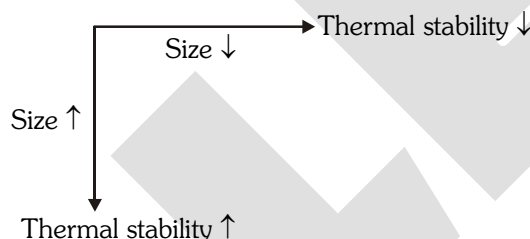
Thermal stability of compound having poly atomic anion



Polarising Power ($\uparrow$) Thermal Stability ($\downarrow$)

$$\text{Thermal stability} \propto \frac{1}{\text{Pol. power}} \propto \frac{\text{size of cation}}{\text{charge of cation}}$$

Compounds having poly atomic anions :



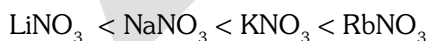
For fluoride, hydride & normal oxide

$$\left[\text{Thermal stability} \propto \frac{1}{\text{size}} \right] \text{ for a group}$$

$$[\text{Thermal stability} \propto \Delta \text{EN}] \text{ for a period}$$

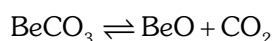
Thermal Stability order

For Example



Note :

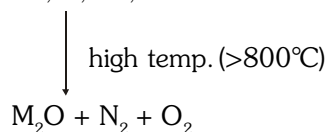
- (i) LiHCO_3 and IIA group bicarbonate does not exist in solid state.
- (ii) Carbonate, Sulphates & hydroxide of Na, K, Rb & Cs do not decompose at high temperature only melt.
- (iii) BeCO_3 is kept in CO_2 atmosphere due to less thermal stability.



Heating Effect

- (a) Metal carbonate $\xrightarrow{\Delta}$ metal oxide + CO_2
- (b) Metal hydroxide $\xrightarrow{\Delta}$ metal oxide + H_2O
- (c) Metal bicarbonate $\xrightleftharpoons{\Delta}$ metal carbonate + CO_2 + H_2O
- (d) Ammonium salts having CO_3^{-2} , PO_4^{-3} , SO_4^{-2} , X-anion (non oxidising or weak oxidising) gives NH_3 gas on decomposition.
 Ammonium salt having $\text{Cr}_2\text{O}_7^{-2}$, ClO_3^{-1} , NO_2^{-1} , NO_3^{-1} (strong oxidising anion) gives N_2 or N_2O gas on decomposition.
- (e) Metal nitrate $\xrightarrow{\Delta}$ metal oxide + NO_2 + O_2

Except : Na, K, Rb, Cs nitrate $\xrightarrow{\text{low temp.}}$ MNO_2 + $\frac{1}{2} \text{O}_2$

**Note :**

- (i) Some less stable metal oxide like Ag_2O & HgO further decompose into metal & oxygen.
- (ii) Metal salts having high percentage of oxygen like KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ & KClO_3 give O_2 gas on decomposition.

Complete following reactions :

$\text{PbCO}_3 \xrightarrow{\Delta}$	$\text{FeCO}_3 \xrightarrow{\Delta}$
$\text{ZnCO}_3 \xrightarrow{\Delta}$	$\text{MgCO}_3 \cdot \text{CaCO}_3 \xrightarrow{\Delta}$
$\text{CuCO}_3 \cdot \text{Cu(OH)}_2 \xrightarrow{\Delta}$	$\text{NaCO}_3 \cdot 10\text{H}_2\text{O} \longrightarrow \text{X} \longrightarrow \xrightarrow{\Delta}$
$2\text{NaHCO}_3 \xrightarrow{\Delta}$	$\text{NH}_4\text{NO}_2 \xrightarrow{\Delta}$
$\text{NH}_4\text{NO}_3 \xrightarrow{\Delta}$	$(\text{NH}_4)_2\text{SO}_4 \xrightarrow{\Delta}$
$(\text{NH}_4)_2\text{CO}_3 \xrightarrow{\Delta}$	$\text{Ba(N}_3)_2 \text{ or } \text{NaN}_3 \xrightarrow{\Delta}$
$\text{Pb(NO}_3)_2 \xrightarrow{\Delta}$	$\text{Zn(NO}_3)_2 \xrightarrow{\Delta}$
$\text{Ca(NO}_3)_2 \xrightarrow{\Delta}$	$\text{LiNO}_3 \xrightarrow{\Delta}$
$\text{NaNO}_3 \xrightarrow{400^\circ\text{C}}$	$\text{PbCl}_4 \xrightarrow{\Delta}$
$\text{FeCl}_3 \xrightarrow{\Delta}$	$\text{AuCl}_3 \xrightarrow{\Delta}$
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \xrightarrow{<200^\circ\text{C}} \text{A} \xrightarrow{>200^\circ\text{C}} \text{B} \xrightarrow{800^\circ\text{C}}$	$\text{BaCl}_2 \cdot 2\text{H}_2\text{O} \xrightarrow{\Delta}$
$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O} \xrightarrow{<200^\circ\text{C}} \text{A} \xrightarrow{>200^\circ\text{C}} \text{B} \xrightarrow{800^\circ\text{C}}$	$\text{AlCl}_3 \cdot 6\text{H}_2\text{O} \xrightarrow{\Delta}$
$\text{FeSO}_4 \cdot 7\text{H}_2\text{O} \xrightarrow{<200^\circ\text{C}} \text{A} \xrightarrow{>200^\circ\text{C}} \text{B} \xrightarrow{750^\circ\text{C}}$	$\text{CaCl}_2 \cdot 6\text{H}_2\text{O} \xrightarrow{\Delta}$
$\text{Fe}_2(\text{SO}_4)_3 \xrightarrow{\Delta}$	$\text{FeCl}_3 \cdot 6\text{H}_2\text{O} \xrightarrow{\Delta}$
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{<200^\circ\text{C}} \text{A} \xrightarrow{>200^\circ\text{C}} \text{B} \xrightarrow{\Delta}$	$\text{MgCl}_2 \cdot \text{H}_2\text{O} \xrightarrow{\Delta}$
$2\text{KMnO}_4 \xrightarrow{\Delta}$	$\text{K}_2\text{Cr}_2\text{O}_7 \xrightarrow{\Delta}$
$2\text{KClO}_3 \xrightarrow{\Delta}$	$(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{\Delta}$

2.20 SOLUBILITY OF IONIC COMPOUNDS

(a) Solubility of ionic compounds :

Soluble in polar solvents like water which have high dielectric constant

Factors affecting solubility of ionic compounds :

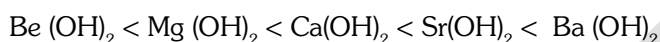
- (i) Dielectric constant of the solvent increases the solubility of compound increases



- (ii) If heat of hydration of ions exceeds the lattice energy (L.E.) of ionic compounds, the ionic compounds will be soluble in water.
- Lattice energy as well as hydration energy depend on the size of ions. Both lattice energy and hydration energy decrease with increase in ionic size.

Two general rule, regarding the solubility is that -

(a) If the anion and the cation are of comparable size, the cationic radius will influence the lattice energy. Since lattice energy decreases much more than the hydration energy with increasing ionic size, solubility will increase as we go down the group.



Solubility increases

(b) If the anion is large compared to the cation, (i.e. compound contain ions with widely different radii) the lattice energy will remain almost constant i.e. change is very small within a particular group. Since the hydration energies decrease down a group, solubility will decrease.



Solubility decreases

Note :

- If common ion is small like Na^+ , Li^+ , O^{2-} , F^- , OH^- , IIA cation then lattice energy dominates.
- If common ion is large like Cs^+ , Rb^+ , Br^- , I^- , polyatomic anion like CO_3^{2-} , SO_4^{2-} then hydration energy dominates.

Some important solubility orders:

Examples

- $\text{BeSO}_4 > \text{MgSO}_4 > \text{CaSO}_4 > \text{SrSO}_4 > \text{BaSO}_4$ (SO_4^{2-} larger)
- $\text{CsF} > \text{CsCl} > \text{CsBr} > \text{CsI}$ [Cs^+ (larger)]
- $\text{BeS} > \text{MgS} > \text{CaS} > \text{SrS} > \text{BaS}$ (S^{2-} larger)

*Important facts about solubility

- All metal chlorides are soluble except Ag, Pb, Hg
- All metal sulphides are insoluble except : IA and ammonium
- Highly insoluble sulphides are of Hg^{2+} , Pb^{2+} , Cu^{2+} , Cd^{2+} , Bi^{3+}
- All hydroxides are insoluble except IA, lower IIA (Ca^{+2} , Sr^{+2} , Ba^{+2}) and ammonium
- Most insoluble hydroxides are of Al^{3+} , Fe^{3+} , Cr^{3+}
- All metal nitrate, Acetate, perchlorate are soluble in water
- Generally metal sulphates are soluble in water except Ag, Hg, Pb, Ca, Sr, Ba
- All alkali metal salts are soluble in water except . LiF , Li_2CO_3 , Li_3PO_4 , $\text{Li}_2\text{C}_2\text{O}_4$

2.21 MELTING POINT

Melting point of ionic compounds

The two factors which mainly influence the melting point of ionic compound are

- (i) Lattice enthalpy (ii) Polarisation

M.P. $\propto$ LE when $\left\{ \begin{array}{l} \text{Cation} = \text{Na}^+, \text{K}^+, \text{Rb}^+, \text{Cs}^+ \\ \text{or} \\ \text{Anion} = \text{F}^-, \text{O}^{2-}, \text{N}^{3-}, \text{C}^{4-}, \text{H}^- \end{array} \right.$

Melting point of ionic compound > covalent compound

Except Giant molecules

↓
Diamond, Carborundum (SiC)
Norbide (B₄C), Silica (SiO₂), Borazone (BN)_x

Order of Hardness : Diamond > Norbide > SiC > Al₂O₃

Some important melting point orders :

Examples



Melting Point and Boiling Point of non-metallic molecules

- (i) $\text{CH}_3\text{—O—CH}_3 < \text{C}_2\text{H}_5\text{OH}$ (boiling point)
- (ii) $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4 < \text{PbH}_4$ (boiling point)
- (iii) $\text{I-Cl} > \text{Br}_2$ (boiling point)
- (iv) $\text{H}_2 < \text{O}_2 < \text{H}_2\text{O}$ (boiling point)
- (v) $\text{CH}_3\text{-OH} < \text{H}_2\text{O}$ (surface tension)
- (vi) $\text{HCl} < \text{HNO}_3 < \text{H}_2\text{SO}_4 < \text{H}_3\text{PO}_4$ (melting point)
- (vii) $\text{H}_2\text{O} < \text{D}_2\text{O}$ (boiling point)
- (viii) O-nitro phenol < p-nitro phenol (boiling point)
- (ix) $\text{H}_2\text{O} \gg \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$ (melting point)
- (x) $\text{NH}_3 \gg \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$ (melting point)
- (xi) $\text{HCl} < \text{HBr} < \text{HI} < \text{HF}$ (boiling point)
- (xii) $\text{HCl} < \text{HBr} < \text{HF} < \text{HI}$ (melting point)

Melting point & boiling point of elements

- (i) $\text{Li} > \text{Na} > \text{K} > \text{Rb} > \text{Cs}$ (melting point)
- (ii) $\text{Be} > \text{Ca} > \text{Sr} > \text{Ba} > \text{Mg}$ (melting point)
- (iii) **3d series**
 Melting point $\propto$ number of unpaired electrons
 $\therefore$ In 3d series (MP) maximum = Cr (chromium)
 In 4d series (MP) maximum = Mo
 In 5d series (MP) maximum = W
- (iv) $3d < 4d < 5d$ (melting point)
- (v) $\text{Zn} > \text{Cd} > \text{Hg}$ (Melting point)
 $\text{He} < \text{Ne} < \text{Ar} < \text{Kr} < \text{Xe}$ (boiling point)
 $\text{F}_2 < \text{Cl}_2 < \text{Br}_2 < \text{I}_2$ (boiling point)
 $\text{N}_2 < \text{P}_4 < \text{As}_4$ (boiling point)
 $\text{O}_2 < \text{S}_8 < \text{Se}_8 < \text{Te}_8$ (boiling point)
 $\text{H}_2 > \text{He}$ (melting point)

Note : Maximum melting point in Boron family $\rightarrow$ Boron

Minimum melting point in Boron family $\rightarrow$ Gallium

Maximum melting point in all elements $\rightarrow$ Carbon

Minimum melting point in all elements $\rightarrow$ Helium

Maximum melting point in Metals $\rightarrow$ Tungsten

Minimum melting point in Metals $\rightarrow$ Mercury

COLOUR OF COMPOUNDS

More the covalent character, more will be the colour intensity.

Colour intensity $\propto \phi$ (Covalent character)

Example :

AgF (Colourless)	AgCl (White)	AgBr (Light yellow)	AgI (Dark yellow)
------------------------------	--------------------------	---------------------------------	-------------------------------

 $\rightarrow$

BEGINNER'S BOX-10

Arrange in order of thermal stability :

1. $\text{BeCO}_3, \text{MgCO}_3, \text{CaCO}_3, \text{SrCO}_3, \text{BaCO}_3$
 $\text{KO}_2, \text{RbO}_2, \text{CsO}_2$
 $\text{NH}_3, \text{PH}_3, \text{AsH}_3, \text{SbH}_3, \text{BiH}_3$
 $\text{H}_2\text{O}, \text{H}_2\text{S}, \text{H}_2\text{Se}, \text{H}_2\text{Te}$
 $\text{BeO}, \text{MgO}, \text{CaO}, \text{SrO}, \text{BaO}$
 $\text{NH}_3, \text{H}_2\text{O}, \text{HF}$
 $\text{NaF}, \text{MgF}_2, \text{AlF}_3$
 $\text{Na}_2\text{SO}_4, \text{MgSO}_4, \text{Al}_2(\text{SO}_4)_3$
 $\text{CCl}_4, \text{SiCl}_4, \text{GeCl}_4, \text{SnCl}_4, \text{PbCl}_4$

2. Give solubility order for :-

- | | |
|-------------------------------------|------------------------------|
| (i) alkali metal hydroxide | (ii) Sodium halides |
| (iii) alkali metal per chlorates | (iv) alkali metal oxides |
| (v) alkaline earth metal carbonates | (vi) alkali metal carbonates |
| (vii) alkali metal bicarbonates | (viii) silver halides |
| (ix) lead (II) halides | (x) mercury (II) halides |

3. Give the order of melting points

- | | | |
|--|---|---|
| • Na_2O , MgO , Al_2O_3 | • CaCl_2 , FeCl_2 , FeCl_3 | • NaF , MgO , ScN , TiC |
| • SnCl_2 , SnCl_4 | • KCl , CuCl , CuCl_2 | |

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4							
	Ans.	3	4	2	1							
BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7	8			
	Ans.	3	1	4	3	1	3	1	2			
BEGINNER'S BOX-3	Que.	1	2	3	4	5	6	7	8	9	10	11
	Ans.	3	1	3	2	2	2	3	4	4	3	4
BEGINNER'S BOX-4	Que.	1	2	3	4	5	6	7	8	9	10	
	Ans.	1	3	2	3	3	2	1	2	1	3	
BEGINNER'S BOX-5	Que.	1	2	3	4	5						
	Ans.	1	2	3	1	2						
BEGINNER'S BOX-6	Que.	1	2	3	4	5	6	7	8	9	10	
	Ans.	3	1	4	3	3	1	3	3	2	1	
BEGINNER'S BOX-7	Que.	1	2	3	4	5	6	7	8	9		
	Ans.	4	4	2	2	2	2	1	4	1		
BEGINNER'S BOX-8	Que.	1	2	3	4	5	6	7	8	9	10	
	Ans.	3	1	3	3	1	1	3	1	4	3	
BEGINNER'S BOX-9	Que.	1	2	3	4	5	6	7	8	9		
	Ans.	4	1	4	4	4	3	2	1	3		

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

OCTET RULE :

- Among the following element group number is not equals to valence electrons -
 (1) Na (2) N
 (3) Ne (4) None
- Which of the following does not act as lewis acid ?
 (1) BF_3 (2) SnCl_4
 (3) CCl_4 (4) SF_4

COVALENT BOND

- The strength of bonds by 2s - 2s, 2p - 2p and 2p - 2s overlapping has the order :-
 (1) $s - s > p - p > s - p$
 (2) $s - s > p - s > p - p$
 (3) $p - p > s - p > s - s$
 (4) $p - p > s - s > p - s$
- In which of the excitation state of chlorine ClF_3 is formed:-
 (1) In ground state
 (2) In third excitation state
 (3) In first excitation state
 (4) In second excitation state
- A sigma bond is formed by the overlapping of :-
 (1) s-s orbital alone
 (2) s and p orbitals alone
 (3) s-s, s-p or p-p orbitals along internuclear axis
 (4) p-p orbital along the sides
- Which overlapping is involved in HCl molecule :-
 (1) s-s overlap (2) p-p overlap
 (3) s-d overlap (4) s-p overlap
- Which is not characteristic of π -bond:-
 (1) π - bond is formed when a sigma bond already formed
 (2) π - bond are formed from hybrid orbitals
 (3) π - bond may be formed by the overlapping of p-orbitals
 (4) π -bond results from lateral overlap of atomic orbitals
- π bond is formed :-
 (1) By overlapping of hybridised orbitals
 (2) Overlapping of s - s orbitals
 (3) Head on overlapping of p -p orbitals
 (4) By p - p collateral overlapping
- Which of the following overlapping is strongest ?
 (1) 1s - 1s (2) 2p - 2p
 (3) 2s - 2p (4) 1s - 2p
- p-p overlapping will be observed in the molecules of:
 (1) Hydrogen (2) Hydrogen bromide
 (3) Hydrogen chloride (4) Chlorine

- Which compound of xenon is not possible
 (1) XeF_2 (2) XeF_4 (3) XeF_5 (4) XeF_6
- Similarity of fluorine and oxygen may not be attributed to-
 (1) Their atomic and ionic radii are closely similar
 (2) The atom of both elements attain an octet of electrons in their valence shell
 (3) Both of them are highly electronegative elements
 (4) Both form stable $p\pi - p\pi$ multiple bonds with themselves
- Higher is the bond order, greater is -
 (1) Bond dissociation energy
 (2) Covalent character
 (3) Bond length
 (4) Paramagnetism

HYBRIDISATION

- In the protonation of H_2O , change occurs in
 (1) Hybridisation state of oxygen
 (2) Shape of molecule
 (3) Hybridisation and shape both
 (4) None
- In the compound
 $\overset{1}{\text{CH}_2}=\overset{2}{\text{CH}}-\overset{3}{\text{CH}_2}-\overset{4}{\text{CH}_2}-\overset{5}{\text{C}}\equiv\overset{6}{\text{CH}}$, the $\text{C}^2 - \text{C}^3$ bond is formed by the overlapping of :-
 (1) $sp - sp^2$ (2) $sp^3 - sp^3$
 (3) $sp - sp^3$ (4) $sp^2 - sp^3$
- Among the following species identify the isostructural pairs:-
 NF_3 , NO_3^- , BF_3 , H_3O^+ , HN_3
 (1) $[\text{NF}_3, \text{NO}_3^-]$ and $[\text{BF}_3, \text{H}_3\text{O}^+]$
 (2) $[\text{NF}_3, \text{HN}_3]$ and $[\text{NO}_3^-, \text{BF}_3]$
 (3) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{NO}_3^-, \text{BF}_3]$
 (4) $[\text{NF}_3, \text{H}_3\text{O}^+]$ and $[\text{HN}_3, \text{BF}_3]$
- Which of the set of species have same hybridisation state but different shapes:-
 (1) NO_2^+ , NO_2 , NO_2^-
 (2) ClO_4^- , SF_4 , XeF_4
 (3) NH_4^+ , H_3O^+ , OF_2
 (4) SO_4^{2-} , PO_4^{3-} , ClO_4^-

- 18.** Which of the following elements can not exhibit sp^3d hybridisation state:-
 (a) C (b) P (c) Cl (d) B
 Correct answer is:-
 (1) a, c (2) a, d (3) b, c (4) b, d
- 19.** Which of the following species are expected to be planar:-
 (a) NH_3 (b) NH_3^{2+} (c) CH_3^+ (d) PCl_3
 The correct answer is:-
 (1) b and c (2) c and d
 (3) b and d (4) a and d
- 20.** In which following set of compound/ion has linear shape
 (1) CH_4 , NH_4^+ , BH_4^- (2) CO_3^{2-} , NO_3^- , BF_3
 (3) NO_2^+ , CO_2 , XeF_2 (4) $BeCl_2$, BCl_3 , CH_4
- 21.** Which order of decreasing bond angle is correct:-
 (1) $CCl_4 > BF_3 > NO_2^+$
 (2) $NH_3 > NCl_3 > NBr_3$
 (3) $Br_2O > Cl_2O > OF_2$
 (4) $PCl_3 > PBr_3 > PI_3$
- 22.** Which of the following set is not correct:-
 (1) SO_3 , O_3 , NH_4^+ all have coordinate bonds
 (2) H_2O , NO_2 , ClO_2^- , all are 'V' shape molecules
 (3) I_3^- , ICl_2^- , NO_2^+ ; all are linear molecules
 (4) SF_4 , SiF_4 , XeF_4 are tetrahedral in shape
- 23.** Shape of a molecule having 4 bond pairs and two lone pairs of electrons, will be :-
 (1) Square planar (2) Tetrahedral
 (3) Linear (4) Octahedral
- 24.** The shape of IF_4^+ will be :-
 (1) Square planar
 (2) Tetrahedral
 (3) Pentagonal bipyramidal
 (4) Distorted tetrahedral
- 25.** Which of the following has pyramidal shape ?
 (1) BF_3 (2) H_3O^+
 (3) NO_3^- (4) CO_3^{2-}
- 26.** A σ bonded molecule MX_3 is T-shaped. The number of non-bonding pairs of electrons is
 (1) 0
 (2) 2
 (3) 1
 (4) Can be predicted if atomic number of M is known.
- 27.** Amongst CO_3^{2-} , AsO_3^{3-} , XeO_3 , ClO_3^- , BO_3^{3-} and SO_3^{2-} the non-planar species are :-
 (1) XeO_3 , ClO_3^- , SO_3^{2-} , AsO_3^{3-}
 (2) AsO_3^{3-} , XeO_3 , CO_3^{2-}
 (3) BO_3^{3-} , CO_3^{2-} , SO_3^{2-}
 (4) AsO_3^{3-} , BO_3^{3-} , CO_3^{2-}
- 28.** The type of hybrid orbitals used by chlorine atom in ClO^- , ClO_2^- , ClO_3^- and ClO_4^- is/are :-
 (1) sp , sp^2 , sp^3 and sp^3d (2) sp and sp^3
 (3) Only sp^3 (4) only sp
- 29.** On the basis of hybridization of one s & one p orbitals they are arranged at :-
 (1) Two orbitals mutually at 90° angle
 (2) two orbitals mutually at 180° angle
 (3) Two orbitals mutually at 120° angle
 (4) Two orbitals mutually at 150° angle
- 30.** Which of the following having a square planar structure is
 (1) NH_4^+ (2) BF_4^- (3) XeF_4 (4) CCl_4
- 31.** When p-character of hybridised orbital (formed by s and p orbitals) increases. Then the bond angle
 (1) Decreases (2) Increases
 (3) Becomes twice (4) Remains unaltered
- 32.** Which orbitals overlap to form bond in OF_2
 (1) $sp^3 - 2p$ (2) $sp^2 - 2p$
 (3) $sp - 2p$ (4) $p - p$
- 33.** In compounds X, all the bond angles are exactly $109^\circ 28'$. X is :
 (1) Chloromethane
 (2) Carbon tetrachloride
 (3) Iodoform
 (4) Chloroform
- 34.** Among the following orbitals/bonds, the angle is minimum between :
 (1) sp^3 bonds
 (2) p_x and p_y orbitals
 (3) $H-O-H$ bond in water
 (4) sp bonds
- 35.** The structure and hybridization of $Si(CH_3)_4$ is :
 (1) bent, sp
 (2) trigonal, sp^2
 (3) octahedral, sp^3d
 (4) tetrahedral, sp^3
- 36.** The AsF_5 molecule is trigonal bipyramidal. The hybrid orbitals used by the As atoms for bonding are:
 (1) $d_{x^2-y^2}$, d_{z^2} , s , p_x , p_y (2) d_{xy} , s , p_x , p_y , p_z
 (3) s , p_x , p_y , p_z , d_{z^2} (4) $d_{x^2-y^2}$, s , p_x , p_y

37. When the hybridization state of carbon atom changes from sp^3 , sp^2 and sp , the angle between the hybridized orbitals.
 (1) decrease considerably
 (2) increase progressively
 (3) decrease gradually
 (4) all of these
38. The hybridization states of the central atoms of the ions I_3^- , ICl_4^- and ICl_2^- are respectively :
 (1) sp^2 , dsp^2 , sp^3
 (2) sp^3d , sp^3d^2 and sp^3d
 (3) sp^3d , sp^3d , dsp^2
 (4) sp , sp , dsp^2
39. Molecular shapes of SF_4 , CF_4 and XeF_4 are :-
 (1) The same, with 2, 0 and 1 lone pairs of electrons respectively
 (2) The same, with 1, 1 and 1 lone pairs of electrons respectively
 (3) Different, with 0, 1 and 2 lone pairs of electrons respectively
 (4) Different, with 1, 0 and 2 lone pairs of electrons respectively
40. Which of the following two are isostructural :-
 (1) XeF_2 , IF_2^- (2) NH_3 , BF_3
 (3) CO_3^{2-} , SO_3^{2-} (4) PCl_5 , ICl_5
41. Select the correct matching :

List I		List II	
A : XeF_4	1. Pyramidal		
B : XeF_6	2. T-shape		
C : XeO_3	3. Distorted octahedral		
D : $XeOF_2$	4. Square planar		

A	B	C	D
(1) 4	3	1	2
(2) 1	2	3	4
(3) 2	1	3	4
(4) 4	1	3	2
42. Which one of the following pair is a correct with respect to molecular formula of xenon compound and hybridization state of xenon in it :
 (1) XeF_4 , sp^3 (2) XeF_2 , sp
 (3) XeF_2 , sp^3d (4) XeF_4 , sp^2
43. The xenon compound(s) that are iso-structural with IBr_2^- and BrO_3^- respectively are :
 (1) Linear XeF_2 and pyramidal XeO_3
 (2) Bent XeF_2 and pyramidal XeO_3
 (3) Bent XeF_2 and planar XeO_3
 (4) Linear XeF_2 and tetrahedral XeO_3

DIPOLE MOMENT

44. Which statement is correct:-
 (1) All the compounds having polar bonds, have dipole moment
 (2) SO_2 is non-polar
 (3) H_2O molecule is non polar, having polar bonds
 (4) PH_3 is polar molecule having non polar bonds
45. BeF_2 has zero dipole moment where as H_2O has a dipole moment because :-
 (1) Water is linear
 (2) H_2O is bent
 (3) F is more electronegative than O
 (4) Hydrogen bonding is present in H_2O
46. Which of the following species are symmetrical :-
 (a) XeF_4 (b) XeF_6
 (c) SO_2 (d) NH_3^{+2}
 Correct answer is:-
 (1) a and b (2) b and c
 (3) c and d (4) a and d
47. Which of the following molecule have zero dipole moment:-
 (1) BF_3 (2) CH_2Cl_2
 (3) NF_3 (4) SO_2
48. The dipole moment of NH_3 is:-
 (1) Less than dipole moment of NCl_3
 (2) Higher than dipole moment of NCl_3
 (3) Equal to the dipole moment of NCl_3
 (4) None of these
49. Which of the following order of polarities of molecules is correct:-
 (1) $HF > NH_3 > PH_3$
 (2) $CH_4 > NH_3 > H_2O$
 (3) $CH_3Cl < CH_2Cl_2 < CHCl_3$
 (4) $BF_3 > BeF_2 > F_2$
50. Which set of molecules is polar :-
 (1) XeF_4 , IF_7 , SO_3 (2) PCl_5 , C_6H_6 , SF_6
 (3) $SnCl_2$, SO_2 , NO_2 (4) CO_2 , CS_2 , C_2H_6
51. Which of the following has symmetrical structure :
 (1) PCl_3 (2) CH_2Cl_2
 (3) $CHCl_3$ (4) CCl_4
52. Species having zero dipole moment :-
 (1) XeF_4
 (2) 1,2,4 trichloro benzene
 (3) SF_4
 (4) CH_2Cl_2

53. What conclusion can be drawn from the fact that BF_3 has no dipole moment but PF_3 does
 (1) BF_3 is not symmetrical but PF_3 is
 (2) BF_3 molecule must be linear
 (3) Atomic radius of P is larger than that of B
 (4) BF_3 molecule must be planar triangular
54. PCl_5 is non polar because :-
 (1) P – Cl bond is non-polar
 (2) Its dipole moment is zero
 (3) P – Cl bond is polar
 (4) P & Cl have equal electronegativity
55. Dipole moment of CO_2 is zero which implies that :
 (1) Carbon and oxygen have equal electronegativities
 (2) Carbon has no polar bond
 (3) CO_2 is a linear molecule
 (4) Carbon has bond moments of zero value
56. The correct order of dipole moment is :
 (1) $\text{CH}_4 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$
 (2) $\text{NF}_3 < \text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$
 (3) $\text{NH}_3 < \text{NF}_3 < \text{CH}_4 < \text{H}_2\text{O}$
 (4) $\text{H}_2\text{O} < \text{NH}_3 < \text{NF}_3 < \text{CH}_4$
57. Which of the following molecule does not possess a permanent dipole moment :
 (1) H_2S (2) SO_2 (3) SO_3^{2-} (4) CS_2
58. Which of the following has the highest value of dipole moment :
 (1) HCl (2) HF (3) HI (4) HBr

MOLECULAR ORBITAL THEORY

59. The ion that is isoelectronic with CO and having same bond order is :-
 (1) CN^- (2) O_2^+ (3) O_2^- (4) N_2^+
60. Which of the following is paramagnetic:-
 (1) O_2^- (2) CN^- (3) CO (4) NO^+
61. In the following which of the two are paramagnetic
 (a) N_2 (b) CO (c) B_2 (d) NO_2
 Correct answer is :-
 (1) a and c (2) b and c
 (3) c and d (4) b and d
62. The bond order of CO molecule on the basis of molecular orbital theory is
 (1) Zero (2) 2 (3) 3 (4) 1
63. The energy of σ_{2s} orbital is greater, than σ_{1s}^* orbital because
 (1) σ_{2s} orbital is bigger than σ_{1s}^* orbital
 (2) σ_{2s} orbital is a bonding orbital where as σ_{1s}^* is an antibonding orbital
 (3) σ_{2s} orbital has a greater value of n than σ_{1s}^* orbital
 (4) None
64. The no. of antibonding electron pair in O_2^- is
 (1) 4 (2) 3
 (3) 8 (4) 10
65. Which of the following species will have the minimum bond energy
 (1) N_2 (2) N_2^-
 (3) N_2^+ (4) N_2^{2-}
66. Which of the following ion has not bond order of 2.5 ?
 (1) O_2^- (2) O_2^+
 (3) N_2^+ (4) N_2^-
67. In a homonuclear molecule which of the following set of orbitals are degenerate ?
 (1) σ_{2s} and σ_{1s} (2) π_{2p_x} and π_{2p_y}
 (3) π_{2p_x} and σ_{2p_z} (4) σ_{2p_z} and $\pi_{2p_x}^*$

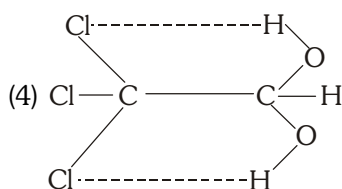
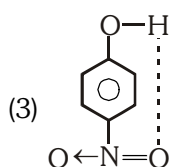
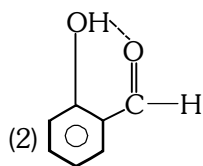
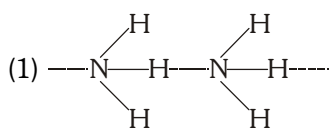
COORDINATE BOND

68. In Co-ordinate bond, the acceptor atoms must essentially contain in its valency shell an orbital:-
 (1) With paired electron
 (2) With single electron
 (3) With no electron
 (4) With three electron
69. The bonds present in N_2O_5 are :-
 (1) Only ionic
 (2) Covalent & coordinate
 (3) Only covalent
 (4) Covalent & ionic
70. Dative bond is present in
 (1) SO_3 (2) NH_3
 (3) K_2CO_3 (4) BF_3
71. Which of the following has no coordinate bond ?
 (1) PH_3 (2) $\text{P}_2\text{H}_6^{+2}$
 (3) P_2H_5^+ (4) PH_4^+
72. The compound containing co-ordinate bond is :
 (1) H_2SO_4 (2) O_3
 (3) SO_3 (4) All of these

WEAK FORCES

73. In dry ice the bond present between two molecules is
 (1) Ionic bond (2) Covalent bond
 (3) Hydrogen bond (4) Vander Waal
74. Intermolecular hydrogen bonds are not present in:-
 (1) $\text{CH}_3\text{CH}_2\text{OH}$ (2) CH_3COOH
 (3) $\text{C}_2\text{H}_5\text{NH}_2$ (4) CH_3OCH_3

75. In which of the following molecule, the shown hydrogen bond is not possible:-



76. Correct order of volatility is:-

- (1) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- (2) $\text{HCl} > \text{HBr} > \text{HI} > \text{HF}$
- (3) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
- (4) $\text{HBr} < \text{HCl} < \text{HI} < \text{HF}$

77. The correct order of volatility is:-

- (1) $\text{NH}_3 < \text{H}_2\text{O}$
- (2) p-nitro phenol < o-nitro phenol
- (3) $\text{CH}_3\text{OH} > \text{CH}_3-\text{O}-\text{CH}_3$
- (4) $\text{HF} > \text{HCl}$

78. The incorrect order of decreasing boiling points is

- (1) $\text{NH}_3 > \text{AsH}_3 > \text{PH}_3$
- (2) $\text{H}_2\text{O} > \text{H}_2\text{Se} > \text{H}_2\text{S}$
- (3) $\text{Br}_2 > \text{Cl}_2 > \text{F}_2$
- (4) $\text{CH}_4 > \text{GeH}_4 > \text{SiH}_4$

79. Acetic acid exists as dimer in benzene due to:-

- (1) Condensation reaction
- (2) Hydrogen bonding
- (3) Presence of carboxyl group
- (4) None of the above

80. Maximum no. of hydrogen bonds formed by a water molecule in ice is

- (1) 4
- (2) 3
- (3) 2
- (4) 1

81. Strongest hydrogen bond is shown by :

- (1) Water
- (2) Ammonia
- (3) Hydrogen fluoride
- (4) Hydrogen sulphide

82. Density of ice is less than that of water because of

- (1) presence hydrogen bonding
- (2) crystal modification of ice
- (3) open porous structure of ice due to hydrogen bonding
- (4) different physical states of these

83. NH_3 has abnormally high boiling point because it has :

- (1) Alkaline nature
- (2) Distorted shape
- (3) sp^3 - Hybridisation
- (4) Hydrogen bonding

84. Which of the following is soluble in water ?

- (1) CS_2
- (2) $\text{C}_2\text{H}_5\text{OH}$
- (3) CCl_4
- (4) CHCl_3

85. KF combines with HF to form KHF_2 . The compound contains the species :

- (1) K^+ , F^- and H^+
- (2) K^+ , F^- and HF
- (3) K^+ and $[\text{HF}_2]^-$
- (4) $[\text{KHF}]^+$ and F_2

86. Which of the following compounds show intramolecular hydrogen bonding :

- (A) o-nitrophenol
- (B) p-nitrophenol
- (C) phenol
- (D) salicylaldehyde
- (1) A & B
- (2) A & C
- (3) A & D
- (4) B & C

87. The pair of molecules forming strongest hydrogen bonds are :

- (1) SiH_4 and SiF_4
- (2) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$ and CHCl_3
- (3) $\text{H}-\text{C}(=\text{O})-\text{OH}$ and $\text{CH}_3-\text{C}(=\text{O})-\text{OH}$
- (4) H_2O and H_2

IONIC BOND

88. Which one is the correct statement with reference to solubility of MgSO_4 in water:

- (1) Hydration energy of MgSO_4 is higher in comparison to its lattice energy
- (2) Ionic potential of Mg^{2+} is very low
- (3) SO_4^{2-} ion mainly contributes towards hydration energy
- (4) Size of Mg^{2+} and SO_4^{2-} are similar

- 89.** Conditions for ionic bond formation is/are :
 (a) Small cation, large anion
 (b) Low IP of cation, high electron affinity of anion
 (c) Large cation, small anion and less charge
 (d) Less lattice energy
 Correct answer is:
 (1) a, d (2) b, c and d
 (3) b and c (4) a, b
- 90.** Capacity of solvent to neutralise charge on ionic compound is called:-
 (1) Solvation energy (2) Dielectric constant
 (3) Dipole moment (4) Solubility
- 91.** The force responsible for dissolution of ionic compound in water is –
 (1) Dipole – dipole forces
 (2) Ion – dipole force
 (3) Ion – ion force
 (4) Hydrogen bond
- 92.** Born Haber cycle is mainly used to determine
 (1) Lattice energy (2) Electron affinity
 (3) Ionisation energy (4) Electronegativity
- 93.** An ionic compound $A^+ B^-$ is most likely to be formed when –
 (1) Ionization energy of A is low
 (2) Electron affinity of B is high
 (3) Electron affinity of B is low
 (4) Both (1) and (2)
- 94.** The pair of elements which on combination are most likely to form an ionic compound is
 (1) Na and Ca (2) K and O
 (3) O and Cl (4) Al and I
- 95.** Choose the compound of each of the following pairs that has the more solubility in water :-
 (a) $BeSO_4$, $BaSO_4$ (b) $NaCl$, $MgCl_2$
 (c) $AgCl$, AgI
 (1) $BeSO_4$, $MgCl_2$, AgI (2) $BeSO_4$, $NaCl$, $AgCl$
 (3) $BaSO_4$, $NaCl$, $AgCl$ (4) $BaSO_4$, $MgCl_2$, AgI
- 96.** Highest melting point would be of
 (1) $AlCl_3$ (2) $LiCl$
 (3) $NaCl$ (4) $BeCl_2$
- 97.** Which of the following substance will have highest b.p.t. ?
 (1) He (2) CsF
 (3) NH_3 (4) $CHCl_3$
- 98.** As compared to covalent compounds electrovalent compounds generally possess
 (1) High m.p. and high b.p.
 (2) Low m.p. and low b.p.
 (3) Low m.p. and high b.p.
 (4) high m.p. and low b.p.
- 99.** The electronic configuration of metal M is $1s^2 2s^2 2p^6 3s^1$. The formula of its oxide will be :
 (1) MO (2) M_2O
 (3) M_2O_3 (4) MO_2
- 100.** Which of the following pairs will form the most stable ionic bond ?
 (1) Na and Cl (2) Mg and F
 (3) Li and F (4) Na and F
- 101.** For two ionic solids, CaO and KI . Identify the wrong statement among the following :
 (1) Lattice energy of CaO is much larger than that of KI
 (2) KI is soluble in benzene
 (3) CaO has higher melting point
 (4) KI has lower melting point
- 102.** Which of the following does not show electrical conduction ?
 (1) diamond
 (2) graphite
 (3) sodium chloride (fused)
 (4) potassium
- 103.** The most covalent halide is:-
 (1) AlF_3 (2) $AlCl_3$
 (3) $AlBr_3$ (4) AlI_3
- 104.** Ionic potential (ϕ) of electropositive element will be highest in which of the following compound:-
 (1) $CsCl$ (2) $MgCl_2$
 (3) AlF_3 (4) SF_6
- 105.** $LiCl$ is soluble in organic solvent while $NaCl$ is not because :-
 (1) Lattice energy of $NaCl$ is less than that of $LiCl$
 (2) Ionisation potential of Li is more than that of Na
 (3) Li^+ has more hydration energy than Na^+ ion
 (4) $LiCl$ is more covalent compound than that $NaCl$
- 106.** The most stable carbonate is
 (1) Li_2CO_3 (2) $BeCO_3$
 (3) $CaCO_3$ (4) $BaCO_3$
- 107.** Correct order of covalent character of alkaline earth metal chloride in
 (1) $BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2$
 (2) $BeCl_2 < CaCl_2 < SrCl_2 < MgCl_2$
 (3) $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2$
 (4) $SrCl_2 > BeCl_2 > CaCl_2 > MgCl_2$
- 108.** Which of the compound is least soluble in water
 (1) AgF (2) $AgCl$
 (3) $AgBr$ (4) AgI

109. CCl_4 is more covalent than LiCl because :

- (1) There is more polarization of Cl in CCl_4
- (2) There is more polarization of Cl in LiCl
- (3) CCl_4 has more weight
- (4) None of above

110. The M.P. of SnCl_4 is less than that of SnCl_2 , the suitable reason for the observed fact is

- (1) There is more charge on Sn^{+4}
- (2) The size of Sn^{+4} is small
- (3) Ionic potential of Sn^{+4} is high
- (4) The shape of SnCl_4 is tetrahedral

111. The correct order of decreasing polarisable ions is:

- (1) Cl^- , Br^- , I^- , F^-
- (2) F^- , I^- , Br^- , Cl^-
- (3) F^- , Cl^- , Br^- , I^-
- (4) I^- , Br^- , Cl^- , F^-

112. Ionic conductances of hydrated M^+ ions are in the order –

- (1) $\text{Li}^+(\text{aq}) > \text{Na}^+(\text{aq}) > \text{K}^+(\text{aq}) > \text{Rb}^+(\text{aq}) > \text{Cs}^+(\text{aq})$
- (2) $\text{Li}^+(\text{aq}) > \text{Na}^+(\text{aq}) < \text{K}^+(\text{aq}) < \text{Rb}^+(\text{aq}) < \text{Cs}^+(\text{aq})$
- (3) $\text{Li}^+(\text{aq}) > \text{Na}^+(\text{aq}) > \text{K}^+(\text{aq}) > \text{Rb}^+(\text{aq}) < \text{Cs}^+(\text{aq})$
- (4) $\text{Li}^+(\text{aq}) < \text{Na}^+(\text{aq}) < \text{K}^+(\text{aq}) < \text{Rb}^+(\text{aq}) < \text{Cs}^+(\text{aq})$

113. Which of the following halides has the highest melting point –

- (1) NaCl
- (2) KCl
- (3) NaBr
- (4) NaF

114. Which of the following does not give an oxide on heating –

- (1) MgCO_3
- (2) Li_2CO_3
- (3) ZnCO_3
- (4) K_2CO_3

115. Which decomposes on heating –

- (1) NaOH
- (2) KOH
- (3) LiOH
- (4) RbOH

116. Which of the following forms metal oxide on heating

- (1) Na_2CO_3
- (2) Li_2CO_3
- (3) K_2SO_4
- (4) NaHCO_3

117. Increasing order of stability of –

- I. K_2CO_3 II. MgCO_3 III. Na_2CO_3
- (1) $\text{I} < \text{II} < \text{III}$
 - (2) $\text{II} < \text{III} < \text{I}$
 - (3) $\text{II} < \text{I} < \text{III}$
 - (4) $\text{I} < \text{III} < \text{II}$

118. Which of the following carbonate will not decompose on heating :-

- (1) BaCO_3
- (2) ZnCO_3
- (3) Na_2CO_3
- (4) Li_2CO_3

119. The decomposition temperature is maximum for:-

- (1) BeCO_3
- (2) CaCO_3
- (3) K_2CO_3
- (4) Li_2CO_3

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	3	3	3	3	4	2	4	1	4	3	4	1	2	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	2	1	3	3	4	1	4	2	2	1	3	2	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	1	2	2	4	3	2	2	4	1	1	3	1	4	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	1	2	1	3	4	1	4	2	3	1	4	2	1	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	3	3	3	2	4	1	2	3	2	1	1	4	4	4	3
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	2	4	2	1	3	3	4	2	3	3	3	1	3	2
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	2	1	4	2	2	3	2	1	2	2	2	1	4	4	4
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	
Ans.	4	3	4	1	3	4	4	4	4	3	2	2	3	3	

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT-2006

- The number of unpaired electrons in a paramagnetic diatomic molecule of an element with atomic number 16 is
(1) 1 (2) 2 (3) 3 (4) 4
- Which of the following species has a linear shape:
(1) O_3 (2) NO_2^-
(3) SO_2 (4) NO_2^+
- Which of the following is not isostructural with $SiCl_4$?
(1) NH_4^+ (2) SCl_4
(3) SO_4^{2-} (4) PO_4^{3-}
- The electronegativity difference between N and F is greater than that between N and H yet the dipole moment of NH_3 (1.5 D) is larger than that of NF_3 (0.2 D). This is because
(1) in NH_3 the atomic dipole and bond dipole are in the opposite directions whereas in NF_3 these are in the same direction
(2) in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in the same direction
(3) in NH_3 the atomic dipole and bond dipole are in the same direction whereas in NF_3 these are in opposite directions
(4) in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in opposite directions
- In which of the following molecules are all the bonds **not** equal :-
(1) NF_3 (2) ClF_3
(3) BF_3 (4) AlF_3

AIPMT-2007

- The correct order of increasing thermal stability of K_2CO_3 , $MgCO_3$, $CaCO_3$ and $BeCO_3$ is
(1) $BeCO_3 < MgCO_3 < CaCO_3 < K_2CO_3$
(2) $MgCO_3 < BeCO_3 < CaCO_3 < K_2CO_3$
(3) $K_2CO_3 < MgCO_3 < CaCO_3 < BeCO_3$
(4) $BeCO_3 < MgCO_3 < K_2CO_3 < CaCO_3$
- In which of the following pairs the two species are iso-structural
(1) SO_3^{2-} and NO_3^-
(2) BF_3 and NF_3
(3) BrO_3^- and XeO_3
(4) SF_4 and XeF_4

- The correct order of C-O bond length among CO , CO_3^{2-} , CO_2 is
(1) $CO < CO_3^{2-} < CO_2$ (2) $CO_3^{2-} < CO_2 < CO$
(3) $CO < CO_2 < CO_3^{2-}$ (4) $CO_2 < CO < CO_3^{2-}$
- In which of the following hydration energy is higher than lattice energy
(1) $MgSO_4$ (2) $CaSO_4$
(3) $BaSO_4$ (4) $SrSO_4$
- Which one of the following orders correctly represents the increasing acid strengths of the given acids :
(1) $HOCIO_3 < HOCIO_2 < HOCIO < HOCl$
(2) $HOCl < HOCIO < HOCIO_2 < HOCIO_3$
(3) $HOCIO < HOCl < HOCIO_2 < HOCIO_3$
(4) $HOCIO_2 < HOCIO_3 < HOCIO < HOCl$

AIPMT-2008

- Four diatomic species are listed below in different sequences. Which of these presents the correct order of their increasing bond order :
(1) $C_2^{2-} < He_2^+ < NO < O_2^-$
(2) $He_2^+ < O_2^- < NO < C_2^{2-}$
(3) $O_2^- < NO < C_2^{2-} < He_2^+$
(4) $NO < C_2^{2-} < O_2^- < He_2^+$
- The angular shape of ozone molecule (O_3) consists of
(1) 1 sigma and 1 pi bonds
(2) 2 sigma and 1 pi bonds
(3) 1 sigma and 2 pi bonds
(4) 2 sigma and 2 pi bonds
- The correct order of increasing bond angles in the following triatomic species is:-
(1) $NO_2^+ < NO_2 < NO_2^-$ (2) $NO_2^+ < NO_2^- < NO_2$
(3) $NO_2^- < NO_2^+ < NO_2$ (4) $NO_2^- < NO_2 < NO_2^+$

AIPMT-2009

- In which of the following molecules/ions BF_3 , NO_2^- , NH_2^- and H_2O , the central atom is sp^2 hybridized :
(1) BF_3 and NO_2^- (2) NO_2^- and NH_2^-
(3) NH_2^- and H_2O (4) NO_2^- and H_2O

15. According to MO theory which of the following lists ranks the nitrogen species in terms of increasing bond order :
- (1) $N_2^- < N_2^{2-} < N_2$ (2) $N_2^- < N_2 < N_2^{2-}$
 (3) $N_2^{2-} < N_2^- < N_2$ (4) $N_2 < N_2^{2-} < N_2^-$
16. In the case of alkali metals, the covalent character decreases in the order :
- (1) $MI > MBr > MCl > MF$
 (2) $MCl > MI > MBr > MF$
 (3) $MF > MCl > MBr > MI$
 (4) $MF > MCl > MI > MBr$
17. What is the dominant intermolecular force or bond that must be overcome in converting liquid CH_3OH to a gas :
- (1) London or dispersion force
 (2) Hydrogen bonding
 (3) Dipole-dipole interaction
 (4) Covalent bonds

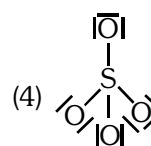
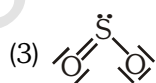
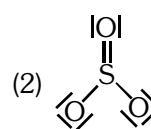
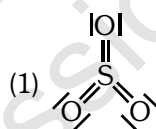
AIPMT-2010

18. Some of the properties of the two species, NO_3^- and H_3O^+ are described below. Which one of them is correct :-
- (1) Isostructural with same hybridization for the central atom.
 (2) Isostructural with different hybridization for the central atom.
 (3) Similar in hybridization for the central atom with different structures.
 (4) Dissimilar in hybridization for the central atom with different structures.
19. In which of the following molecules the central atom does not have sp^3 hybridization :-
- (1) SF_4 (2) BF_4^-
 (3) NH_4^+ (4) CH_4
20. Which one of the following species does not exist under normal conditions :
- (1) Li_2 (2) Be_2^+
 (3) Be_2 (4) B_2
21. In which of the following pairs of molecules/ions, the central atoms have sp^2 hybridization :
- (1) BF_3 and NH_2^- (2) NO_2^- and NH_3
 (3) BF_3 and NO_2 (4) NH_2^- and H_2O
22. Which of the following alkaline earth metal sulphates has hydration enthalpy higher than the lattice enthalpy :
- (1) $SrSO_4$ (2) $CaSO_4$
 (3) $BeSO_4$ (4) $BaSO_4$
23. In which one of the following species the central atom has the type of hybridisation which is not the same as that present in the other three :
- (1) PCl_5 (2) SF_4
 (3) I_3^- (4) $SbCl_5^{2-}$

24. Property of the alkaline earth metals that increases with their atomic number :-
- (1) Electronegativity
 (2) Solubility of their hydroxides in water
 (3) Solubility of their sulphates in water
 (4) Ionization energy

AIPMT Pre-2011

25. Considering the state of hybridization of carbon atoms, find out the molecule among the following which is linear :
- (1) $CH_3-CH=CH-CH_3$
 (2) $CH_3-C \equiv C-CH_3$
 (3) $CH_2=CH-CH_2-C \equiv CH$
 (4) $CH_3-CH_2-CH_2-CH_3$
26. Which of the following has the minimum bond length :
- (1) O_2^+ (2) O_2^- (3) O_2^{2-} (4) O_2
27. Which of the two ions from the list given below that have the geometry that is explained by the same hybridization of orbitals, NO_2^- , NO_3^- , NH_2^- , NH_4^+ , SCN^- :
- (1) NO_2^- and NO_3^- (2) NH_4^+ and NO_3^-
 (3) SCN^- and NH_2^- (4) NO_2^- and NH_2^-
28. Which of the following compounds has the lowest melting point :
- (1) $CaCl_2$ (2) $CaBr_2$
 (3) CaI_2 (4) CaF_2
29. Which of the following structures is the most preferred and hence of lowest energy for SO_3 :



AIIMS-2011

30. In solid ice, oxygen atom is surrounded:
- (1) Tetrahedrally by 4 hydrogen atoms
 (2) Octahedrally by 2 oxygen and 4 hydrogens atoms
 (3) Tetrahedrally 2 hydrogen and 2 oxygens atoms
 (4) Octahedrally by 6 hydrogens atoms
31. Decreasing order of bond angle is :
- (1) $BeCl_2 > NO_2 > SO_2$
 (2) $BeCl_2 > SO_2 > NO_2$
 (3) $SO_2 > BeCl_2 > NO_2$
 (4) $SO_2 > NO_2 > BeCl_2$

AIPMT Pre-2012

32. Which one of the following pairs is isostructural (i.e. having the same shape and hybridization) :
 (1) NF_3 and BF_3 (2) $[\text{BF}_4^-]$ and $[\text{NH}_4^+]$
 (3) $[\text{BCl}_3]$ and $[\text{BrCl}_3]$ (4) $[\text{NH}_3]$ and $[\text{NO}_3^-]$
33. Which of the following species contains three bond pairs and one lone pair around the central atom:
 (1) NH_2^- (2) PCl_3 (3) H_2O (4) BF_3
34. The pair of species with the same bond order is:
 (1) NO , CO (2) N_2 , O_2
 (3) O_2^{2-} , B_2 (4) O_2^+ , NO^+
35. Bond order of 1.5 is shown by:
 (1) O_2^{2-} (2) O_2 (3) O_2^+ (4) O_2^-

AIPMT Mains-2012

36. During change of O_2 to O_2^- ion, the electron adds on which one of the following orbitals :
 (1) σ^* orbital (2) σ orbital
 (3) π^* orbital (4) π orbital
37. Four diatomic species are listed below. Identify the correct order in which the bond order is increasing in them :
 (1) $\text{C}_2^{2-} < \text{He}_2^+ < \text{O}_2^- < \text{NO}$
 (2) $\text{He}_2^+ < \text{O}_2^- < \text{NO} < \text{C}_2^{2-}$
 (3) $\text{NO} < \text{O}_2^- < \text{C}_2^{2-} < \text{He}_2^+$
 (4) $\text{O}_2^- < \text{NO} < \text{C}_2^{2-} < \text{He}_2^+$

NEET-UG-2013

38. Which one of the following molecules contains no π bond :
 (1) NO_2 (2) CO_2 (3) H_2O (4) SO_2
39. XeF_2 is isostructural with :
 (1) BaCl_2 (2) TeF_2 (3) $[\text{ICl}_2^-]$ (4) SbCl_3
40. Dipole induced dipole interactions are present in which of the following pairs :
 (1) SiF_4 and He atoms (2) H_2O and alcohol
 (3) Cl_2 and CCl_4 (4) HCl and He atoms
41. Which of the following is a polar molecule :
 (1) XeF_4 (2) BF_3 (3) SF_4 (4) SiF_4
42. Which of the following is paramagnetic :
 (1) NO^+ (2) CO (3) O_2^- (4) CN^-
43. Identify the correct order of solubility in aqueous medium :
 (1) $\text{Na}_2\text{S} > \text{ZnS} > \text{CuS}$ (2) $\text{CuS} > \text{ZnS} > \text{Na}_2\text{S}$
 (3) $\text{ZnS} > \text{Na}_2\text{S} > \text{CuS}$ (4) $\text{Na}_2\text{S} < \text{CuS} > \text{ZnS}$

AIIMS-2013

44. Total no. of Antibonding electrons present in O_2 will be :-
 (1) 6 (2) 8 (3) 4 (4) 2

AIPMT-2014

45. Which of the following molecules has the maximum dipole moment :
 (1) CO_2 (2) CH_4 (3) NH_3 (4) NF_3
46. Which one of the following species has plane triangular shape :
 (1) N_3^- (2) NO_3^- (3) NO_2^- (4) CO_2

AIIMS-2014

47. Heat of sublimation of Na = x J
 Ionization energy of Na = y J
 Bond dissociation energy of Cl_2 = a J
 e^- gain enthalpy of Cl = -b J
 Enthalpy of formation of NaCl = -c J
 Then what will be lattice energy of $\text{NaCl}_{(s)}$
 (1) $b + \left(x + y + c - \frac{a}{2}\right)$
 (2) $b - \left(x + y + c - \frac{a}{2}\right)$
 (3) $b - \left(\frac{a}{2} + x + y + c\right)$
 (4) $c - \frac{a}{2}x + y + c$

AIPMT-2015

48. The correct bond order in the following species is:-
 (1) $\text{O}_2^{2+} < \text{O}_2^- < \text{O}_2^+$ (2) $\text{O}_2^+ < \text{O}_2^- < \text{O}_2^{2+}$
 (3) $\text{O}_2^- < \text{O}_2^+ < \text{O}_2^{2+}$ (4) $\text{O}_2^{2+} < \text{O}_2^+ < \text{O}_2^-$
49. Which of the following pairs of ions are isoelectronic and isostructural :
 (1) $[\text{ClO}_3^-]$, $[\text{CO}_3^{2-}]$ (2) $[\text{SO}_3^{2-}]$, $[\text{NO}_3^-]$
 (3) $[\text{ClO}_3^-]$, $[\text{SO}_3^{2-}]$ (4) $[\text{CO}_3^{2-}]$, $[\text{SO}_3^{2-}]$
50. Which of the following options represents the correct bond order :
 (1) $\text{O}_2^- < \text{O}_2 < \text{O}_2^+$ (2) $\text{O}_2^- > \text{O}_2 < \text{O}_2^+$
 (3) $\text{O}_2^- < \text{O}_2 > \text{O}_2^+$ (4) $\text{O}_2^- > \text{O}_2 > \text{O}_2^+$

51. Solubility of the alkaline earth's metal sulphates in water decreases in the sequence :-
 (1) $\text{Ca} > \text{Sr} > \text{Ba} > \text{Mg}$ (2) $\text{Sr} > \text{Ca} > \text{Mg} > \text{Ba}$
 (3) $\text{Ba} > \text{Mg} > \text{Sr} > \text{Ca}$ (4) $\text{Mg} > \text{Ca} > \text{Sr} > \text{Ba}$
52. Maximum bond angle at nitrogen is present in which of the following :
 (1) NO_2^- (2) NO_2^+
 (3) NO_3^- (4) NO_2

RE-AIPMT-2015

53. On heating which of the following releases CO_2 most easily :
 (1) MgCO_3 (2) CaCO_3
 (3) K_2CO_3 (4) Na_2CO_3
54. Decreasing order of stability of O_2 , O_2^- , O_2^+ and O_2^{2-} is :-
 (1) $\text{O}_2 > \text{O}_2^+ > \text{O}_2^{2-} > \text{O}_2^-$
 (2) $\text{O}_2^- > \text{O}_2^{2-} > \text{O}_2^+ > \text{O}_2$
 (3) $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$
 (4) $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2 > \text{O}_2^+$
55. In which of the following pairs, both the species are not isostructural :
 (1) NH_3 , PH_3
 (2) XeF_4 , XeO_4
 (3) SiCl_4 , PCl_4
 (4) Diamond, silicon carbide
56. The variation of the boiling points of the hydrogen halides is in the order $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$. What explains the higher boiling point of hydrogen fluoride ?
 (1) The bond energy of HF molecules is greater than in other hydrogen halides
 (2) The effect of nuclear shielding is much reduced in fluorine which polarises the HF molecule
 (3) The electronegativity of fluorine is much higher than for other elements in the group.
 (4) There is strong hydrogen bonding between HF molecules

AIIMS-2015

57. SF_4 & XeF_2 shape respectively are
 (1) Linear and distorted tetrahedral
 (2) See-saw and linear
 (3) T-shape and trigonal bipyramidal
 (4) Tetrahedral and linear
58. In ClF_3 lone pair are present at approx
 (1) 180° (2) 120°
 (3) 90° (4) 60°

59. In solid phase of XeF_6 , it consists of
 (1) $\text{XeF}_5^+ + \text{F}^-$ (2) $\text{XeF}_4 + \text{F}_2$
 (3) $\text{XeF}_5^+ + \text{XeF}_2^-$ (4) $\text{XeF}_2 + 2\text{F}_2$
60. Correct order of bond length
 (1) $\text{CO} \approx \text{CO}_3^{2-} > \text{CO}_2$
 (2) $\text{CO}_2 \approx \text{CO}_3^{2-} > \text{CO}$
 (3) $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$
 (4) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$
61. Correct order of bond energy
 (1) $\text{I}-\text{Cl} < \text{I}_2$
 (2) $\text{Br}-\text{F} < \text{F}-\text{F}$
 (3) $\text{Br}-\text{F} < \text{I}-\text{I}$
 (4) $\text{F}-\text{F} < \text{Cl}-\text{Cl}$

NEET-I 2016

62. Consider the molecules CH_4 , NH_3 and H_2O . Which of the given statements is false ?
 (1) The H-C-H bond angle in CH_4 , the H-N-H bond angle in NH_3 , and the H-O-H bond angle in H_2O are all greater than 90°
 (2) The H-O-H bond angle in H_2O is larger than the H-C-H bond angle in CH_4 .
 (3) The H-O-H bond angle in H_2O is smaller than the H-N-H bond angle in NH_3 .
 (4) The H-C-H bond angle in CH_4 is larger than the H-N-H bond angle in NH_3 .
63. Which one of the following orders is correct for the bond dissociation enthalpy of halogen molecules?
 (1) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$
 (2) $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$
 (3) $\text{Br}_2 > \text{I}_2 > \text{F}_2 > \text{Cl}_2$
 (4) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
64. Predict the correct order among the following :-
 (1) lone pair - lone pair > lone pair - bond pair > bond pair - bond pair
 (2) lone pair - lone pair > bond pair - bond pair > lone pair - bond pair
 (3) bond pair - bond pair > lone pair - bond pair > lone pair - lone pair
 (4) lone pair - bond pair > bond pair - bond pair > lone pair - lone pair

65. Match the compounds given in column I with the hybridisation and shape given in column II and mark the **correct** option.

Column-I		Column-II	
(a)	XeF_6	(i)	Distorted octahedral
(b)	XeO_3	(ii)	Square planar
(c)	XeOF_4	(iii)	pyramidal
(d)	XeF_4	(iv)	Square pyramidal

Code :-

(a)	(b)	(c)	(d)
(1) (i)	(iii)	(iv)	(ii)
(2) (i)	(ii)	(iv)	(iii)
(3) (iv)	(iii)	(i)	(ii)
(4) (iv)	(i)	(ii)	(iii)

NEET-II 2016

66. The correct geometry and hybridization for XeF_4 are:
- (1) Planar triangle, sp^3d^3
 - (2) square planar, sp^3d^2
 - (3) octahedral, sp^3d^2
 - (4) trigonal bipyramidal, sp^3d
67. Among the following which one is a wrong statement?
- (1) SeF_4 and CH_4 have same shape
 - (2) I_3^+ has bent geometry
 - (3) PH_5 and BiCl_5 do not exist
 - (4) $\pi\text{-}\pi$ bonds are present in SO_2
68. The hybridizations of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are
- (1) sp , sp^2 and sp^3
 - (2) sp^2 , sp and sp^3
 - (3) sp , sp^3 and sp^2
 - (4) sp^2 , sp^3 and sp
69. Which of the following fluoro-compounds is most likely to behave as a Lewis base ?
- (1) CF_4
 - (2) SiF_4
 - (3) BF_3
 - (4) PF_3
70. Which of the following pairs of ions is isoelectronic and isostructural ?
- (1) SO_3^{2-} , NO_3^-
 - (2) ClO_3^- , SO_3^{2-}
 - (3) CO_3^{2-} , NO_3^-
 - (4) ClO_3^- , CO_3^{2-}

AIIMS 2016

71. Minimum lone pair on central atom will be present in :-
- (1) XeO_3
 - (2) XeF_4
 - (3) ClF_3
 - (4) HOCl
72. Which of the following has paramagnetic character in gaseous state :-
- (1) S^{-2}
 - (2) S_2
 - (3) S_8
 - (4) S_6
73. In which of the following molecule dipole moment is more than zero but less than one ?
- (1) NH_3
 - (2) NF_3
 - (3) BeF_2
 - (4) BF_3
74. Select the incorrect statement :-
- (1) hybrid orbitals are maximum repelled to each other
 - (2) only half filled atomic orbital are participate in hybridisation.
 - (3) Bond energy of hybrid orbitals are more than unhybrid atomic orbital.
 - (4) Without hybridisation, geometry of molecule is uncertain.
75. Which of the following has tetrahedral structure :-
- (1) XeF_4
 - (2) H_3PO_4
 - (3) SF_4
 - (4) ClF_3
76. Which of the following molecule has more than one lone pair on central atom :-
- (1) SO_2
 - (2) XeF_2
 - (3) PCl_5
 - (4) IF_7
77. In ice crystal nearest and second nearest atoms to water molecule respectively are :-
- (1) 4 & 4
 - (2) 4 & 6
 - (3) 6 & 6
 - (4) 6 & 4
78. Which of the following statement is true regarding H-bond :-
- (1) it is a directional bond
 - (2) bond strength is similar to O-H bond
 - (3) it is attraction between hydrogen and electronegative atom like as F, O, N
 - (4) it does not require minimum one lone pair of electron at electronegative atom

NEET(UG) 2017

79. Ionic mobility of which of the following alkali metal ions is lowest when aqueous solution of their salts are put under an electric field ?
 (1) K (2) Rb (3) Li (4) Na
80. Match the interhalogen compounds of column-I with the geometry in column II and assign the correct code.

Column-I		Column-II	
(a)	XX'	(i)	T-shape
(b)	XX' ₃	(ii)	Pentagonal bipyramidal
(c)	XX' ₅	(iii)	Linear
(d)	XX' ₇	(iv)	Square-pyramidal
		(v)	Tetrahedral

Code :

- | | | | |
|-----------|-------|-------|------|
| (a) | (b) | (c) | (d) |
| (1) (iii) | (i) | (iv) | (ii) |
| (2) (v) | (iv) | (iii) | (ii) |
| (3) (iv) | (iii) | (ii) | (i) |
| (4) (iii) | (iv) | (i) | (ii) |
81. Which of the following pairs of compounds is isoelectronic and isostructural ?
 (1) TeI₂, XeF₂ (2) IBr₂⁻, XeF₂
 (3) IF₃, XeF₂ (4) BeCl₂, XeF₂
82. The species, having bond angles of 120° is :-
 (1) ClF₃ (2) NCl₃ (3) BCl₃ (4) PH₃
83. Which of the following pairs of species have the same bond order ?
 (1) O₂, NO⁺ (2) CN⁻, CO
 (3) N₂, O₂⁻ (4) CO, NO

AIIMS 2017

84. Which of the following has longest bond length
 (1) C = O (2) N = O
 (3) C = C (4) C ≡ N
85. Which of the following has hydrogen bonded water
 (1) BaCl₂·2H₂O (2) CrCl₃·5H₂O
 (3) CuSO₄·5H₂O (4) CaCl₂·6H₂O

86. Which of the following is diamagnetic ?
 (1) NO (2) Cl₂⁺
 (3) S₂²⁻ (4) ClO
87. Number of lone pair in SO₂ on S & O are respectively
 (1) 0, 6 (2) 2, 4
 (3) 2, 5 (4) 1, 5

NEET(UG) 2018

88. Among CaH₂, BeH₂, BaH₂, the order of ionic character is
 (1) BeH₂ < CaH₂ < BaH₂
 (2) CaH₂ < BeH₂ < BaH₂
 (3) BeH₂ < BaH₂ < CaH₂
 (4) BaH₂ < BeH₂ < CaH₂
89. Magnesium reacts with an element (X) to form an ionic compound. If the ground state electronic configuration of (X) is 1s² 2s² 2p³, the simplest formula for this compound is
 (1) Mg₂X₃ (2) MgX₂ (3) Mg₂X (4) Mg₃X₂
90. Consider the following species:
 CN⁺, CN⁻, NO and CN
 Which one of these will have the highest bond order?
 (1) NO (2) CN⁻
 (3) CN⁺ (4) CN
91. Which one of the following elements is unable to form MF₆³⁻ ion ?
 (1) Ga (2) Al
 (3) B (4) In
92. In the structure of ClF₃, the number of lone pairs of electrons on central atom 'Cl' is
 (1) one (2) two
 (3) four (4) three

AIIMS 2018

93. Which statement is correct regarding N₂O₅ :-
 (1) It is anhydride of nitrous acid
 (2) It is brown color gas
 (3) It is paramagnetic
 (4) It exist as NO₂⁺ & NO₃⁻ in solid state
94. In s-p overlapping wrong statement is :-
 (1) Nodal plane formed in ABMO
 (2) Some orbital may have higher energy than combining orbitals
 (3) Combining orbitals should have same symmetry
 (4) Nodal plane is found in BMO

95. Which pair not shows same shape :-
 (1) I_3^{-1} , $BeCl_2$ (2) BF_3 , ClO_3^{-1}
 (3) SO_2 , H_2O (4) XeF_4 , BrF_4^{-}
96. In which at least one σ -bond of np (σ_{np}) orbital is present :-
 (1) B_2 (2) O_2
 (3) C_2 (4) Li_2
97. Which is correct for dipole moment :-
 (1) $NH_3 < NF_3$ (2) $SO_2 < SO_3$
 (3) $NF_3 > BF_3$ (4) $SO_2 > H_2O$
98. Which of the following is a molecular solid :-
 (1) solid SO_2 (2) SiC
 (3) C. (graphite) (4) ZnS
99. The geometry and No. of lone pair on ClF_2^{-} and ClF_4^{-} are respectively :-
 (1) Square planar, 2 & Linear, 3
 (2) Square Planar, 3 & Linear, 2
 (3) Linear, 3 & Square planar, 2
 (4) Tetrahedral, 2 & linear, 3
100. Wrong statement for N_3^{-} is :-
 (1) Isoelectronic with CO_2
 (2) Structure is linear
 (3) Both N-N bond lengths are unequal in hydrogen azide
 (4) Oxidation state for Nitrogen is same as in NH_2OH
101. Bond order of which of following pair is different:-
 (1) F_2^{-} & O_2^{-2} (2) N_2^{-} & O_2^{+}
 (3) C_2 & B_2^{-2} (4) N_2 & O_2^{2+}
102. Correct order for Bond angle is
 (1) $SO_2 > NH_3$ (2) $NH_3 < H_2O$
 (3) $SO_2 < H_2O$ (4) $H_2S > NH_3$
103. Which of the following two species have same shape but different hybridisation
 (1) XeF_2 and ICl_2^{-} (2) NO_2^{+} and SO_2
 (3) $BeCl_2$ and NO_2^{+} (4) SO_2 and NH_2^{-}
104. Which of the following has maximum %s character in N-H bond ?
 (1) N_2H_2 (2) NH_4^{+} (3) NH_3 (4) N_2H_4

EXERCISE-II (Previous Year Questions)

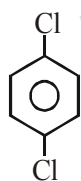
ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	4	2	3	2	1	3	3	1	2	2	2	4	1	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	4	1	3	3	3	4	2	2	1	1	3	1	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	2	2	3	4	3	2	3	3	4	3	3	1	1	3
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	3	3	3	1	4	2	1	3	2	4	2	2	1	3
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	4	2	2	1	1	2	1	1	4	2,3	1	2	2	2	2
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	2	2	3	3	1	2	3	2	3	3	3	4	1	4	2
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	
Ans.	3	2	4	4	2	2	3	1	3	4	1	1	4	1	

EXERCISE-III (Analytical Questions)

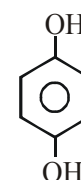
Check Your Understanding

- The thermal stability of alkaline earth metal carbonates MgCO_3 , CaCO_3 , BaCO_3 and SrCO_3 decreases as :
 (1) $\text{CaCO}_3 > \text{SrCO}_3 > \text{MgCO}_3 > \text{BaCO}_3$
 (2) $\text{BaCO}_3 > \text{SrCO}_3 > \text{MgCO}_3 > \text{CaCO}_3$
 (3) $\text{BaCO}_3 > \text{SrCO}_3 > \text{CaCO}_3 > \text{MgCO}_3$
 (4) $\text{MgCO}_3 > \text{CaCO}_3 > \text{SrCO}_3 > \text{BaCO}_3$
- Which of the following are isoelectronic and isostructural NO_3^- , CO_3^{2-} , ClO_3^- , SO_3
 (1) NO_3^- , CO_3^{2-} (2) SO_3 , NO_3^-
 (3) ClO_3^- , CO_3^{2-} (4) CO_3^{2-} , SO_3
- According to molecular orbital theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+ ?
 (1) paramagnetic and bond order $< \text{O}_2$
 (2) paramagnetic and bond order $> \text{O}_2$
 (3) diamagnetic and bond order $< \text{O}_2$
 (4) diamagnetic and bond order $> \text{O}_2$
- The bond order in NO is 2.5 while that in NO^+ is 3. Which of the following statements is true for these two species ?
 (1) Bond length in NO^+ is equal to that in NO
 (2) Bond length in NO is greater than in NO^+
 (3) Bond length in NO^+ is greater than in NO
 (4) Bond length is unpredictable
- Number of σ and π bond in SO_4^{2-} are :-
 (1) 4, 2 (2) 4, 3
 (3) 4, 4 (4) 2, 3
- The paramagnetic property of the oxygen molecule is due to the presence of unpaired electrons present in :-
 (1) $(\sigma 2p_x)^1$ and $(\pi^* 2p_x)^1$
 (2) $(\sigma 2p_x)^1$ and $(\pi 2p_y)^1$
 (3) $(\pi^* 2p_y)^1$ and $(\pi^* 2p_x)^1$
 (4) $(\pi^* 2p_y)^1$ and $(\pi 2p_y)^1$
- The boiling point of methanol is greater than that of methylthiol because :-
 (1) There is intramolecular hydrogen bonding in methanol and intermolecular hydrogen bonding in methylthiol
 (2) There is intermolecular hydrogen bonding in methanol and no hydrogen bonding in methylthiol
 (3) There is no hydrogen bonding in methanol and intermolecular hydrogen bonding in methylthiol
 (4) There is intramolecular hydrogen bonding in methanol and no hydrogen bonding in methylthiol
- In silicon dioxide –
 (1) each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms
 (2) each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms
 (3) silicon atom is bonded to two oxygen atoms
 (4) there are double bond between silicon and oxygen atoms
- The number and type of bond between two carbon atom in calcium carbide are :
 (1) One sigma, one pi
 (2) One sigma, two pi
 (3) Two sigma, one pi
 (4) Two sigma, two pi
- What is the hybridization of oxygen atom in an alcohol molecule?
 (1) sp^3 (2) sp (3) sp^2 (4) sp^3d
- The decreasing values of bond angles from NH_3 (106°) to SbH_3 (91°) down group-15 of the periodic table is due to
 (1) decreasing lp – bp repulsion
 (2) increasing electronegativity
 (3) increasing bp – bp repulsion
 (4) increasing p-orbital character in sp^3
- In which of the following ionisation processes, the bond order has increased and the magnetic behaviour has changed
 (1) $\text{NO} \rightarrow \text{NO}^+$ (2) $\text{O}_2 \rightarrow \text{O}_2^+$
 (3) $\text{N}_2 \rightarrow \text{N}_2^+$ (4) $\text{C}_2 \rightarrow \text{C}_2^+$
- Which of the following hydrogen bonds is the strongest
 (1) $\text{F}-\text{H} \cdots \text{F}$ (2) $\text{O}-\text{H} \cdots \text{O}$
 (3) $\text{O}-\text{H} \cdots \text{F}$ (4) $\text{O}-\text{H} \cdots \text{N}$
- Which of the following species exhibits the diamagnetic behaviour
 (1) O_2^+ (2) O_2 (3) NO (4) O_2^{2-}
- The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+ , Ca^{+2} , Mg^{+2} , Be^{+2}
 (1) $\text{Be}^{+2} < \text{K}^+ < \text{Ca}^{+2} < \text{Mg}^{+2}$
 (2) $\text{K}^+ < \text{Ca}^{+2} < \text{Mg}^{+2} < \text{Be}^{+2}$
 (3) $\text{Ca}^{+2} < \text{Mg}^{+2} < \text{Be}^{+2} < \text{K}^+$
 (4) $\text{Mg}^{+2} < \text{Be}^{+2} < \text{K}^+ < \text{Ca}^{+2}$

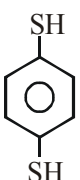
- 16.** Which one of the following constitutes a group of the isoelectronic species ?
 (1) C_2^{2-} , O_2^- , CO, NO
 (2) NO^+ , C_2^{2-} , CN^- , N_2
 (3) CN^- , N_2 , O_2^{2-} , C_2^{2-}
 (4) N_2 , O_2^- , NO^+ , CO
- 17.** Which of the following pairs of species have the same bond order ?
 (1) CN^- and NO^+ (2) CN^- and CN^+
 (3) O_2^- and CN^- (4) NO^+ and CN^+
- 18.** Among the following, the molecule with highest dipole moment is :-
 (1) CH_3Cl (2) CH_2Cl_2
 (3) $CHCl_3$ (4) CCl_4
- 19.** For a diatomic molecule the dipole moment is 1.2D and bond length is 1.0 Å then charge fraction on each atom is :-
 (1) 12% (2) 18%
 (3) 25% (4) 29%
- 20.** Which one of the following molecules is expected to exhibit diamagnetic behaviour ?
 (1) C_2 (2) N_2^-
 (3) O_2 (4) S_2
- 21.** For which of the following molecule significant $\mu \neq 0$
- 

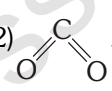
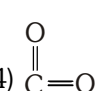
(a)



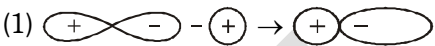
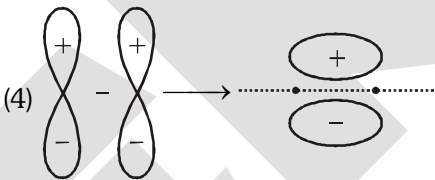
(b)
- 

(c)



(d)
- (1) Only (c) (2) (c) and (d)
 (3) Only (a) (4) (a) and (b)
- 22.** Which one of the following order is correct for the bond energies of halogen molecules :
 (1) $I_2 > Cl_2 > Br_2$ (2) $Br_2 > Cl_2 > I_2$
 (3) $I_2 > Br_2 > Cl_2$ (4) $Cl_2 > Br_2 > I_2$
- 23.** The boiling point of ICl is nearly 40°C higher than that of Br_2 although the two substances have the same relative molecular mass. This is because :-
 (1) ICl is ionic compound
 (2) I-Cl bond is stronger than Br - Br bond
 (3) ICl is polar covalent molecule while Br_2 is non polar
 (4) IP of Iodine is less than that of Br.
- 24.** Nitrogen does not form NF_5 because
 (1) Nitrogen is member of V group
 (2) It contains no empty d-orbital
 (3) The bond energy of $N \equiv N$ is very high
 (4) Inert pair effect exists in the molecule
- 25.** CO_2 is a gas, while SiO_2 is a solid but both are-
 (1) Covalent containing π -bond
 (2) Molecules having $p\pi - d\pi$ bonding
 (3) Acidic
 (4) Discrete molecules
- 26.** Which trihalide ion is unknown due to absence of vacant d-orbital?
 (1) F_3^- (2) Br_3^- (3) I_3^- (4) Cl_3^-
- 27.** Dipole moment is shown by
 (1) 1, 4 - dichlorobenzene
 (2) Cis 1, 2-dichloro ethene
 (3) Trans-1, 2- dichloro ethene
 (4) benzene
- 28.** Experiment shows that H_2O has a dipole moment where as CO_2 has not. Point out the structures which best illustrate these facts -
 (1) $O=C=O$, $H-O-H$
 (2) , $H-O-H$
 (3) $O=C=O$, 
 (4) , 
- 29.** In which of the species, bond order increases on removing one electron:-
 (a) NO (b) CN^- (c) O_2 (d) CO
 Correct answer is -
 (1) b and d (2) a and c
 (3) b, d and c (4) b and c
- 30.** Increasing order of bond length in NO, NO^+ and NO^- is :-
 (1) $NO > NO^- > NO^+$ (2) $NO^+ < NO < NO^-$
 (3) $NO < NO^+ < NO^-$ (4) $NO < NO^+ = NO^-$

31. N_2 and O_2 are converted into monoanions, N_2^- and O_2^- respectively. Which of the following statements is wrong ?
 (1) In N_2^- , N-N bond weakens
 (2) In O_2^- , O-O bond order increases
 (3) In O_2^- , O-O bond order decreases
 (4) N_2^- becomes paramagnetic
32. N_2 and O_2 are converted into monocations, N_2^+ and O_2^+ respectively. Which of the following is wrong ?
 (1) In N_2^+ , N-N bond weakens
 (2) In O_2^+ , the O-O bond order increases
 (3) In O_2^+ , paramagnetism decreases
 (4) N_2^+ becomes diamagnetic
33. The pair of compounds which can form a co-ordinate bond is :-
 (1) $(C_2H_5)_3B$ and $(CH_3)_3N$
 (2) HCl and HBr
 (3) BF_3 and NH_3
 (4) (1) & (3) both
34. In the neutralization process of NH_3 and $AlCl_3$ the compound formed will have the bonding
 (1) Ionic (2) Covalent
 (3) Coordinate (4) Hydrogen
35. Glycerol is more viscous than glycol the reason is :-
 (1) Higher molecular wt.
 (2) More covalent
 (3) More extent of hydrogen bonding
 (4) Complex structure
36. Incorrect order of viscosity :-
 (1) $H_2SO_4 > HNO_3$
 (2) $H_2O > CH_3OH$
 (3) o-nitro phenol > p-nitro phenol
 (4) Glycol > ether
37. Which of the following can form H-Bond with water
 (1) HCN (2) NH_4F
 (3) $NaHCO_3$ (4) All
38. Which of the following does not form a hydrogen bond with water
 (1) $(CH_3)_2CO$ (2) CH_3CN
 (3) CH_3OH (4) C_2H_6
39. A metal M readily forms its sulphate MSO_4 , which is water soluble. It forms an insoluble hydroxide $M(OH)_2$ which is soluble in $NaOH$ solution, then M is :-
 (1) Mg (2) Ca (3) Be (4) Ba

40. Which is most ionic :
 (1) P_2O_5 (2) MnO
 (3) CrO_3 (4) Mn_2O_7
41. Lithium chloride is highly soluble in -
 (1) C_6H_6 (2) H_2O
 (3) D_2O (4) All
42. Which of the following linear combinations of atomic orbitals is incorrectly depicted ?
 (1) 
 (2) 
 (3) 
 (4) 
43. Which of the following molecule contains net π -bond only
 (1) B_2 (2) C_2
 (3) C_2^{-2} (4) Both 1 & 2
44. Pick out the incorrect statement :-
 (1) sp^3d hybridisation involves $d_{x^2-y^2}$ orbital
 (2) Hybridised orbital form σ -bond when overlaps with other orbitals.
 (3) SF_2 molecule is more polar than CS_2
 (4) o-nitrophenol is more volatile than p-nitrophenol.
45. The group of substances in which at melting point covalent bond becomes weak :-
 (1) $NaCl$, KCl , $CaCl_2$
 (2) $I_{2(s)}$, $CH_{4(s)}$, dry ice
 (3) B_4C , diamond, SiC
 (4) All of the above
46. The incorrect statement is :-
 (1) π^* p have two nodal planes
 (2) Bond order of HeH^+ is 0.5
 (3) In NCO^- , C is sp hybridised
 (4) O_3 is polar while O_2 is non polar

47. Solid CH_4 is
 (1) Molecular solid
 (2) Ionic solid
 (3) Covalent solid
 (4) None of these
48. Which compound has the weakest bond
 (1) Diamond
 (2) Neon(solid)
 (3) KCl
 (4) water (ice)
49. Which of the following order is incorrect ?
 (1) $\text{Na}_2\text{O} < \text{K}_2\text{O} < \text{Rb}_2\text{O}$ (basic nature)
 (2) $\text{CH}_4 > \text{SiH}_4 > \text{GeH}_4 > \text{SnH}_4$ (Stability of hydride)
 (3) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3$ (basic nature)
 (4) $\text{N}_2\text{O}_5 > \text{P}_2\text{O}_5 > \text{As}_2\text{O}_5$ (acidic nature)
50. Which order are correct ?
 (a) Thermal stability : $\text{BeSO}_4 < \text{MgSO}_4 < \text{CaSO}_4 < \text{SrSO}_4 < \text{BaSO}_4$
 (b) Basic Nature : $\text{ZnO} > \text{BeO} > \text{MgO} > \text{CaO}$
 (c) Solubility in water : $\text{LiOH} > \text{NaOH} > \text{KOH} > \text{RbOH} > \text{CsOH}$
 (d) Melting point : $\text{NaCl} > \text{KCl} > \text{RbCl} > \text{CsCl} > \text{LiCl}$
 (1) a, b, d (2) b, c
 (3) a, d (4) All correct
51. Which of the following order is not correct ?
 (1) $\text{SF}_2 > \text{SF}_4 > \text{SF}_6$ (ionic character)
 (2) $\text{AlF}_3 < \text{Al}_2\text{O}_3 < \text{AlN}$ (covalent character)
 (3) $\text{CaCl}_2 < \text{SnCl}_2 < \text{CdCl}_2$ (covalent character)
 (4) $\text{ZnCl}_2 < \text{CdCl}_2 < \text{HgCl}_2$ (ionic character)
52. Match the column :-
 (A) $\text{NO}_3^- + \text{HCl}$ (p) Kessome attraction
 (B) $\text{Xe} + \text{H}_2\text{O}$ (q) Debye attraction
 (C) $\text{CH}_3-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{CH}_3 + \text{CH}_3-\text{C}\equiv\text{N}$ (r) London force
 (D) $\text{CO}_2 + \text{CS}_2$ (s) Ion-dipole attraction
 (1) A-s, B-q, C-p, D-r (2) A-q, B-s, C-p, D-r
 (3) A-s, B-p, C-q, D-r (4) A-s, B-q, C-r, D-p
53. Correct order is :-
 (1) $\text{MgO} < \text{NiO} < \text{K}_2\text{O} < \text{Cs}_2\text{O}$ (basic strength)
 (2) $\text{Cs}^+ < \text{K}^+ < \text{Mg}^{2+} < \text{Be}^{2+}$ (ionic mobility in aqueous solution)
 (3) $\text{Li}_2\text{CO}_3 < \text{Na}_2\text{CO}_3 < \text{K}_2\text{CO}_3$ (solubility order)
 (4) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3$ (thermal stability)
54. Which of the following statement is not correct ?
 (1) NO has one unpaired electron in ABMO.
 (2) N-H bond length is higher in N_2H_4 as compared to N_2H_2
 (3) Both PF_5 and IF_5 have identical shape
 (4) Both SO_2 and BrF_3 are polar and planar
55. In which of the following process, hybridisation of central atom remains same ?
 (1) Solidification of PCl_5
 (2) Dimerisation of NO_2
 (3) Trimerisation of SO_3
 (4) BF_3 forms adduct with lewis base

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	1	2	2	1	3	2	1	2	1	4	1	1	4	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	1	3	1	2	4	3	2	3	1	2	3	2	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	4	4	3	3	3	4	4	3	2	1	4	4	1	3
Que.	46	47	48	49	50	51	52	53	54	55					
Ans.	2	1	2	3	3	4	1	3	3	2					

EXERCISE-IV (Assertion & Reason)

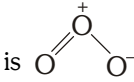
Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

1. **Assertion** : Ionic bonds are formed between metal and nonmetals
Reason : In ionic bonds electrons are shared
 (1) A (2) B (3) C (4) D
2. **Assertion** : D_2O is better solvent for ionic compounds, than H_2O .
Reason : dielectric constant of D_2O is higher than H_2O .
 (1) A (2) B (3) C (4) D
3. **Assertion** : Hydrated ionic radius of Li^+ is maximum in IA group elements.
Reason : Lithium is a metal
 (1) A (2) B (3) C (4) D
4. **Assertion** : $LiCl$ exhibits covalent character.
Reason : Lithium is lightest metal.
 (1) A (2) B (3) C (4) D
5. **Assertion** : K_2CO_3 do not gives CO_2 gas on heating.
Reason : Value of $\sqrt{\phi}$ for K^+ is high.
 (1) A (2) B (3) C (4) D
6. **Assertion** : CO_2 molecule is non-polar while SO_2 is polar.
Reason : Carbon atom is smaller than sulphur.
 (1) A (2) B (3) C (4) D
7. **Assertion** : CH_3OH is soluble in water
Reason : CH_3OH is ionic in nature
 (1) A (2) B (3) C (4) D
8. **Assertion** : Boiling point of H_2O is greater than C_2H_5OH .
Reason : Molecular wt. of H_2O is higher than C_2H_5OH .
 (1) A (2) B (3) C (4) D
9. **Assertion** : NO^+ is more stable than NO^- .
Reason : NO^+ do not have electron in antibonding orbitals.
 (1) A (2) B (3) C (4) D
10. **Assertion** : In MgO electrovalency of Mg is 2.
Reason : Mg shares two electrons with oxygen.
 (1) A (2) B (3) C (4) D
11. **Assertion** : Ionic reactions are faster than molecular reactions.
Reason : Ionic bonds are weaker than covalent bonds.
 (1) A (2) B (3) C (4) D
12. **Assertion** : LiI is more soluble in water than LiF .
Reason : LiI has more ionic character.
 (1) A (2) B (3) C (4) D
13. **Assertion** : Bond energy of $H-H$ bond is greater than $Cl-Cl$ bond.
Reason : H_2 is more covalent than Cl_2 .
 (1) A (2) B (3) C (4) D
14. **Assertion** : Ionic compounds tend to be non-volatile.
Reason : Inter ionic forces in these compounds are strong.
 (1) A (2) B (3) C (4) D
15. **Assertion** : NCl_3 has pyramidal shape
Reason : In NCl_3 central atom is sp^3 hybridised.
 (1) A (2) B (3) C (4) D
16. **Assertion** : BF_3 is planar while NF_3 is non-planar molecule.
Reason : $B-F$ bond is more polar than $N-F$ bond.
 (1) A (2) B (3) C (4) D
17. **Assertion** : Density of ice is greater than water
Reason : In ice H_2O molecules are closely packed.
 (1) A (2) B (3) C (4) D
18. **Assertion** : Bond order of O_2 and BN is same.
Reason : O_2 and BN are isoelectronic
 (1) A (2) B (3) C (4) D

- 19. Assertion** : p-nitrophenol is more viscous than o-nitrophenol.
Reason : In p-nitrophenol, intermolecular H-bonding occurs.
 (1) A (2) B (3) C (4) D
- 20. Assertion** : H_2O_2 is not used as solvent for ionic compounds
Reason : Dielectric constant of H_2O_2 is low
 (1) A (2) B (3) C (4) D
- 21. Assertion** : Ionic compounds exhibits electrical conductivity in solution state.
Reason : In solution state electrons of ionic compounds are free.
 (1) A (2) B (3) C (4) D
- 22. Assertion** : p-nitrophenol is more volatile than o-nitrophenol.
Reason : Molecular wt. of p-nitrophenol is higher than o-nitrophenol.
 (1) A (2) B (3) C (4) D
- 23. Assertion** : N_2^+ is more stable than N_2^-
Reason : N_2^+ has less electrons in antibonding orbitals.
 (1) A (2) B (3) C (4) D
- 24. Assertion** : SO_4^{2-} is square planar in shape
Reason : SO_4^{2-} has sp^3d hybridisation
 (1) A (2) B (3) C (4) D
- 25. Assertion** : NF_3 molecule is more polar than NH_3 molecule.
Reason : NF_3 is pyramidal while NH_3 is trigonal planar.
 (1) A (2) B (3) C (4) D
- 26. Assertion** : OF_4 does not exists.
Reason : Empty d-orbitals are absent in valence shell of oxygen.
 (1) A (2) B (3) C (4) D
- 27. Assertion** : BeCl_2 shows covalent character.
Reason : More polarization of Cl^- by Be^{+2} .
 (1) A (2) B (3) C (4) D
- 28. Assertion** : MgO and NaF are isomorphous
Reason : Crystal structure of MgO and NaF is identical.
 (1) A (2) B (3) C (4) D
- 29. Assertion** : NF_3 molecule is polar.
Reason : N-F bonds are polar.
 (1) A (2) B (3) C (4) D
- 30. Assertion** : NaCl is soluble in non polar solvents.
Reason : NaCl is a non-polar covalent compound.
 (1) A (2) B (3) C (4) D
- 31. Assertion** : Sigma bonds are stronger than π bonds.
Reason : Sigma bonds are covalent bonds.
 (1) A (2) B (3) C (4) D
- 32. Assertion** : CO_2 is non polar while SO_2 is polar molecule.
Reason : S-O bonds are polar while C-O non polar.
 (1) A (2) B (3) C (4) D
- 33. Assertion** : Both H_2O and SnCl_2 are bent molecules.
Reason : Both H_2O and SnCl_2 are sp^2 hybridised.
 (1) A (2) B (3) C (4) D
- 34. Assertion** : NO is paramagnetic in nature.
Reason : Bond order of NO is 2.5.
 (1) A (2) B (3) C (4) D
- 35. Assertion** : Nitrogen is unreactive at room temperature but becomes reactive at elevated temperature (on heating or in the presence of catalysts).
Reason : In nitrogen molecule, there is extensive delocalization of electrons.
 (1) A (2) B (3) C (4) D
- 36. Assertion** : NO_3^- is planar.
Reason : N in NO_3^- is sp^2 and no lone pair at central atom.
 (1) A (2) B (3) C (4) D
- 37. Assertion** : The electronic structure of O_3
- is 
- Reason** :  Structure is not allowed because octet around 'O' can not be expanded.
 (1) A (2) B (3) C (4) D
- 38. Assertion** : LiCl is predominantly a covalent compound.
Reason : Electronegativity difference between 'Li' and 'Cl' is too small.
 (1) A (2) B (3) C (4) D

39. Assertion :- Ozone is a powerful oxidizing agent in comparison to O_2 .

Reason :- Ozone is diamagnetic but O_2 is paramagnetic.

(1) A (2) B (3) C (4) D

40. Assertion :- B_2 molecule is diamagnetic.

Reason :- The highest occupied molecular orbital is of σ type.

(1) A (2) B (3) C (4) D

41. Assertion :- BH_4^- is known while BH_6^{3-} is not.

Reason :- B has very small atomic size.

(1) A (2) B (3) C (4) D

42. Assertion :- Some molecules are polar.

Reason :- The centre of negative charge and positive charge do not coincide each other in some molecule.

(1) A (2) B (3) C (4) D

43. Assertion :- $R_3P = O$ exists but $R_3N = O$ does not exist.

Reason :- P is more electronegative than N.

(1) A (2) B (3) C (4) D

44. Assertion :- ClF_3 has T-shape structure.

Reason :- It has two lone pair arrange at 180° (Angle).

(1) A (2) B (3) C (4) D

45. Assertion :- Bond Dissociation energy is $F_2 > Cl_2$.

Reason :- Cl_2 have more electronic repulsion than F_2 .

(1) A (2) B (3) C (4) D

46. Assertion :- H_2O molecule can form four hydrogen bonds.

Reason :- Two lone pairs and two hydrogens are directly attached with oxygen atoms.

(1) A (2) B (3) C (4) D

47. Assertion :- Bond angle in  is more than expected.

Reason :- It is due to lp-lp repulsion.

(1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	4	2	2	3	2	3	3	3	3	3	3	3	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	3	1	3	3	4	1	4	4	1	1	1	2	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	3	3	2	3	1	1	3	2	4	2	1	3	3	4
Que.	46	47													
Ans.	1	3													

IMPORTANT NOTES

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s-BLOCK ELEMENTS

E

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2.	Physical properties of I(A) & II(A)	115
3.	Chemical properties of I(A) & II(A)	117
4.	Compound of s-Block elements	120
5.	Exercise-I (Conceptual Question)	128
6.	Exercise-II (Previous Years Questions)	132
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8.	Exercise-IV (Assertion & Reason)	134

NEET SYLLABUS

s-BLOCK ELEMENTS (Alkali and Alkaline Earth Metals) : Group I and group II elements:

General introduction, electronic configuration, occurrence, anomalous properties of the first element of each group, diagonal relationship, trends in the variation of properties (such as ionization enthalpy, atomic and ionic radii), trends in chemical reactivity with oxygen, water, hydrogen and halogens; uses. Preparation and Properties of Some important Compounds: Sodium carbonate, sodium chloride, sodium hydroxide and sodium hydrogencarbonate, biological importance of sodium and potassium. Industrial use of lime and limestone, biological importance of Mg and Ca.

@neetfocusquizz

OBJECTIVES

After studying this unit, you will be able to :

- describe the general characteristics of the alkali metals and their compounds;
- explain the general characteristics of the alkaline earth metals and their compounds;
- describe the manufacture, properties and uses of industrially important sodium and calcium compounds including Portland cement
- appreciate the biological significance of sodium, potassium, magnesium and calcium.

If human condition were the periodic table, may be love would be hydrogen at number 1. Death would be helium at number 2.

David Mitchell

s-BLOCK ELEMENTS

3.0 INTRODUCTION

The s-block elements of the Periodic Table are those in which the last electron enters in the outermost s-orbital. As the s-orbital can accommodate only two electrons, two groups (1 & 2) belong to the s-block of the Periodic Table. Group 1 of the Periodic table consists of the elements: lithium, sodium, potassium, rubidium, cesium and francium. They are collectively known as the alkali metals. They are so called because they form hydroxides on reaction with water which are strongly alkaline in nature. The elements of Group 2 include Beryllium, Magnesium, Calcium, Strontium, Barium and Radium. These elements with the exception of **beryllium** are commonly known as the alkaline earth metals. They are so called because their oxides and hydroxides are alkaline in nature and these metal oxides are found in the earth's crust.

3.1 PHYSICAL PROPERTIES OF S-BLOCK ELEMENTS

The atomic, physical and chemical properties of alkali metals are discussed below.

Atomic and Physical Properties of the Alkaline Metals

physical properties of s-Block Elements

ALKALI METALS	ALKALINE EARTH METALS
Physical state	
- One electron in outermost shell & General formula ns^1. - Francium is radioactive element. - All are silvery white - Light soft, malleable and ductile metals with metallic lustre.	- Two electrons in outer most shell & General formula ns^2. - Radium is radioactive element. - All are silvery white. - These metals are harder than alkali metals.
Atomic size	
- Largest in their respective period (except noble gas element) - Size increases from Li to Cs due to addition of an extra shell. $Li < Na < K < Rb < Cs$ IA Li ↓ Na ↓ K ↓ Rb ↓ Cs IIA Be ↓ Mg ↓ Ca ↓ Sr ↓ Ba	- Smaller than IA group elements, since extra charge on nucleus attracts the electron cloud. - Size increases gradually from Be to Ba $Be < Mg < Ca < Sr < Ba$ In s-block elements Be has smallest size, while Cs has largest size.
Melting point and Boiling point	
- Weak interatomic bonds are due to their large atomic radii and presence of only one valence electron hence melting point and boiling point are low. - Decreasing order of melting point and boiling point is $Li > Na > K > Rb > Cs$	- Metallic bond is stronger than IA group due to smaller atomic size and two electrons in valence shell hence melting point and boiling point are higher. - Decreasing order of melting point $Be > Ca > Sr > Ba > Mg$ Boiling point $Be > Ba > Ca > Sr > Mg$
Melting point & Boiling point $\propto$ Strength of metallic bond $\propto$ Number of valence shell e^-	

Ionisation energy (I.E.)

- First ionisation energy (I.E.) is very less because of larger atomic size and only one electron in outer most shell.
- Decreasing order of ionisation energy-
 $\text{Li} > \text{Na} > \text{K} > \text{Rb} > \text{Cs}$
- Second ionisation energy of alkali metals is very high because by losing one electron they achieve inert gas configuration.
- First ionisation energy is higher than IA group because of smaller atomic size and completely filled s-orbital (stable electronic configuration)
- Decreasing order of ionisation energy-
 $\text{Be} > \text{Mg} > \text{Ca} > \text{Sr} > \text{Ba}$
- Second ionisation energy is lesser than IA group.

Oxidation state

- The alkali metals shows only + 1 oxidation state.
(difference between IE_1 and $\text{IE}_2 > 16\text{eV}$)
- Alkaline earth metal shows +2. Oxidation state
(difference between IE_1 and $\text{IE}_2 < 11\text{eV}$)

Electro positive character or metallic character

- **Electropositivity $\propto 1/\text{Ionisation energy}$**
Due to their larger size electron can easily be removed to form M^+ ion. Electro positive property increases from Li to Cs.
- Their atomic size is smaller than IA group so these are lesser electro positive than IA group. Electropositivity increases from Be to Ba

Density ($D = M/V$)

- In a group atomic volume also increase along with atomic weight but atomic weight increases more than atomic volume, so density increases from Li to Cs
Increasing order of density $\text{Li} < \text{K} < \text{Na} < \text{Rb} < \text{Cs}$
 - Density increases from Be to Ba
Increasing order of density $\text{Ca} < \text{Mg} < \text{Be} < \text{Sr} < \text{Ba}$
- Exception :** Density of K is less than Na. Why ?
- Ans.** This is due to presence of vacant d-orbital in the inner shells of K (volume increases, density decreases)

Conductivity

- Due to the presence of loosely held valence electrons which are free to move in a metal structure, these elements are good conductor of heat and electricity.
- These are also good conductor of heat and electricity due to presence of two free electrons.

Conductivity of IA < Conductivity of IIA**Flame test**

- Alkali metals and their salts gives characteristic colour to bunsen flame. The flame energy causes an excitation of the outer most electron which on dropping back to ground state emits absorbed energy as a visible light
Li-Crimson red Na-Golden yellow K-Violet Rb-Red violet Cs-Blue
- Due to small size of Be & Mg outer most electrons are tightly bounded. So not excited to higher level, hence they do not give flame test.
- Other elements gives characteristic colour to flame
Ca-Brick red Sr-Crimson red Ba-apple green

Photo electric effect

- Atomic size of K, Rb and Cs is quite large, so their ionisation energy is very low
- Due to very low ionisation energy their valence shell electrons gets excited even by absorbing visible light. That's why Cs is used in photoelectric cells.
- These elements do not show this property as their atomic size is small hence ionisation energy is higher than IA group.

Standard oxidation potential

- All the alkali metals have high +ve values of standard oxidation potential (tendency of releasing electrons in water or self ionic solutions) So they are good reducing agent, having upper most positions in the electro chemical series.
- Li has highest standard oxidation potential (+3.05 eV) due to its high hydration energy. Such that it converts into. Li^+ ion by losing one electron.
- Order of standard oxidation potential is -
 $\text{Li} > \text{Cs} \approx \text{Rb} \approx \text{K} > \text{Na}$
- They have lower values of standard oxidation potential due to their high IE.
- Increasing order of standard oxidation potential is -
 $\text{Be} < \text{Mg} < \text{Ca} < \text{Sr} < \text{Ba}$

Complex formation tendency

- Only those elements can form complex compounds which have
 - Small cation size
 - High charge density
 - Vacant orbitals to accept electrons.
- Only Li^+ can form complex compound, due to its small size rest alkali metals have very less tendency to form complex compounds.
- Less tendency to form complex compound, but due to small size of cations Be and Mg forms complex compounds like
 $\text{Be}-(\text{BeF}_4)^{-2}$; $\text{Be}_4\text{O}(\text{CH}_3\text{COO})_6$; Mg - Chlorophyll

Chemical properties of s-block elements

Reactivity

- These elements are very reactive, so do not found in free state in nature.
 - Less reactive than alkali metals.
- Reactivity $\propto 1/\text{Ionisation potential}$**
- order of reactivity - $\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs}$
- Order of reactivity :-
 $\text{Be} < \text{Mg} < \text{Ca} < \text{Sr} < \text{Ba}$

Reaction with air

- Alkali metals gets tarnish in air due to the formation of oxide at their surface hence they are kept in kerosene or paraffin wax.
 - Except Be, these metals easily tarnished in air.
 - These elements reacts with moist air to form carbonates
 - Beryllium in powdered form, burns brilliantly on ignition in air.
 - In moist air, except Be all the elements converts into carbonates.
 - In dry air all elements of II-A give nitride and oxide both.
- $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$
- $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$
(moist)
- $2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$
(in air)
- In dry air only Li gives nitride and oxide both while other elements gives only oxides.

Reaction with oxygen

Oxide ion [O²⁻] :

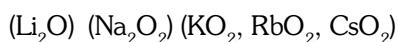
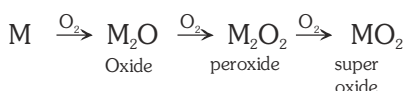
- Li forms mainly Li₂O (Lithium oxide).

Peroxide [O₂⁻²] :

- Na reacts with O₂ to form mainly peroxide (Na₂O₂).

Super oxide [O₂⁻] :

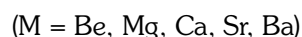
- K, Rb and Cs forms MO₂ type oxides (super oxides) in excess of oxygen. Super oxides are paramagnetic and coloured.



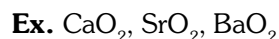
- Stability order of different oxide of a metal is due to Lattice Energy



- Alkaline earth metals reacts with O₂ to form 'MO' type oxides



- In IIA only Ca, Sr, Ba form peroxide.



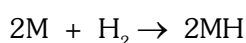
- BeO shows amphoteric property.



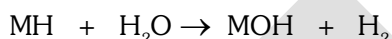
- Basic properties increases from BeO to BaO.

Reaction with hydrogen

- Alkali metals combine with H₂ forming ionic hydrides



- Hydrides of alkali metals are attacked by water to give back hydrogen



Thermal stability decrease, Basic property increases

- Except Be all the alkaline metals forms MH₂ type hydrides, (MgH₂, CaH₂, SrH₂, BaH₂) on heating directly with H₂

- BeH₂ is prepared by action of BeCl₂ with LiAlH₄

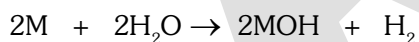


reducing agent

- BeH₂ and MgH₂ are covalent, other are ionic.

Reaction with water

- Alkali metals react vigorously with water forming hydroxides with the liberation of H₂.



- Reactivity with water increases from Li to Cs.

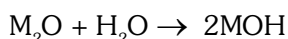
Li → least reactive towards water

Na → reacts vigorously

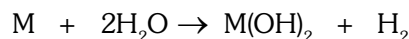
K → reacts producing a flame

Rb, Cs → reacts explosively.

- Monoxides gives strongly alkaline solution with water



- These metals reacts slowly with water gives H₂ and metals hydroxides.



- Be does not reacts with water

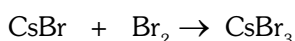
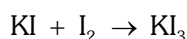
- Mg reacts only with hot water

- Ca, Sr, Ba reacts with cold water but not as energetically as alkali metals.

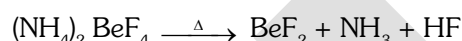
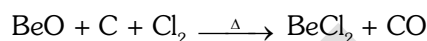
- from Be(OH)₂ to Ba(OH)₂ basic nature increases.

Halides

- Alkali metals reacts directly with halogen to form MX (M – alkali metal, X – Halide ion)
- Ionic nature of MX increases from LiCl to CsCl
- LiCl is covalent in nature (due to polarisation of Cl⁻ ion by small Li⁺ ion). hence its tendency of hydrolysis is more.
- K, Rb and Cs halides reacts with more halogens to gives polyhalides.

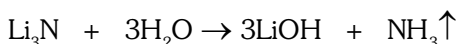


- Alkaline metals reacts with X (Halogen) to form MX₂.
- Order of Ionic nature
BeCl₂ < MgCl₂ < CaCl₂ < SrCl₂ < BaCl₂
- Hydrolysis tendency of these halides decreases from BeCl₂ to BaCl₂ due to decrease in covalent nature.
- BeCl₂ and MgCl₂ are covalent in nature.



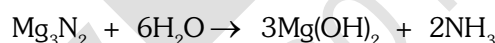
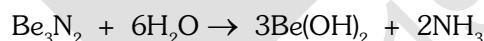
Nitrides

- Only Li reacts directly with N₂ to form nitride which gives NH₃ on reacting with water.



- All elements of II-A burns in N₂ to give M₃N₂ (nitrides)

For example :



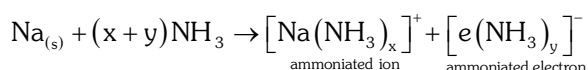
Formation of amalgam

- Alkali metals gives amalgam with Hg.
- These metals reacts with other metals to give mixed metals (alloys)

- Shows same properties.

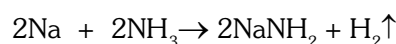
Solubility in liquid ammonia

- All the alkali metals dissolves in NH₃ (liq.) and produces blue coloured solution.
- The blue colour and reducing nature of solution is due to presence of ammoniated electron.
- Solution is a good conductor due to presence of both ammoniated ion and ammoniated electron.



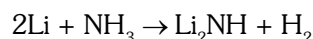
- This dilute solution is paramagnetic in nature.

- Only Ca, Sr and Ba gives blue solution of ammoniated electron.
- Be and Mg are small in size and have high ionisation energy so do not dissolves in liquid NH₃.
- Dark blue colour of solution becomes fade if it is allowed to stand for a long time, it is because of metal amide formation.



(Sodamide)

Exception $\Rightarrow$ Li forms imide



- On increasing the concentration of metal in solution, it converts into bronze colour due to cluster formation of metal ions.

BEGINNER'S BOX-1

- All alkali metal superoxides contain the $[O_2^-]$ ion. They are—
 (1) paramagnetic (2) colored compounds (3) oxidizing agents (4) all of these
- Which of the following is soluble in organic solvents like ethanol?
 (1) LiCl (2) NaCl (3) KCl (4) RbCl
- Which of the following is the correct order of hydrated radii?
 (1) $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$ (2) $Rb^+ < Na^+ < Li^+ < Cs^+ < K^+$
 (3) $Cs^+ < Rb^+ < K^+ < Na^+ < Li^+$ (4) $Li^+ < K^+ < Na^+ < Rb^+ < Cs^+$
- Which of the following has the maximum solubility in water?
 (1) $CsHCO_3$ (2) $NaHCO_3$ (3) $KHCO_3$ (4) $RbHCO_3$
- As compared to potassium, sodium has
 (1) Lower electronegativity (2) Higher ionization potential
 (3) Larger atomic radius (4) Lower melting point
- Which of the following statement is correct regarding alkali metals
 (1) Cation is less stable than the parent atom (2) Cation is smaller in size than the parent atom
 (3) Size of cation and parent atom is the same (4) Cation is greater in size than the parent atom
- Which of the following is not true?
 (1) Group 2 elements are electropositive & strong reducing agents but not as strong as group 1 elements.
 (2) the reducing power of groups 2 elements increases down the group.
 (3) Be has the most negative standard reduction potential.
 (4) The magnesium cation is more easily reduced than the cations of the heavier members of the group.
- Which of the following halides are ionic in nature ?
 (1) BaX_2 (2) CaX_2 (3) SrX_2 (4) All of these
- Which of the following compounds is highly soluble in water ?
 (1) CaF_2 (2) MgF_2 (3) BeF_2 (4) BaF_2
- Which of the following carbonates is the most thermally stable ?
 (1) $BeCO_3$ (2) $MgCO_3$ (3) $CaCO_3$ (4) $BaCO_3$

COMPOUNDS OF s-BLOCK ELEMENTS

3.2 SODIUM CHLORIDE NaCl

Occurrence : Sea water is the main source and also found in salt lakes.

Preparation :

- Sea water $NaCl(2.7 - 2.9\%) \xrightarrow[\text{by solar heat}]{\text{Evaporation}}$ crude NaCl
- It contains impurities – Na_2SO_4 , $MgCl_2$, $CaCl_2$ etc.
- Insoluble impurities removed by filtration.
- Filtrate $\xrightarrow{\text{HCl gas passed}}$ Pure NaCl precipitation (Common ion effect)
- $MgCl_2$ and $CaCl_2$ are more soluble in water so left in solution.

Properties :

- Table salt is slightly hygroscopic due to the presence of magnesium and calcium chlorides in small amounts.
- Reaction with AgNO_3

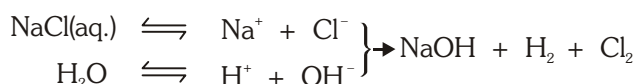
$$\text{NaCl} + \text{AgNO}_3 \rightarrow \text{NaNO}_3 + \text{AgCl}(\text{white ppt.})$$

- Uses**
- As a preservative for pickles, meat and fish.
 - For making freezing mixture with Ice.

3.3 SODIUM HYDROXIDE [Caustic Soda(NaOH)]

Manufacture : By electrolysis of NaCl .

- (a) **Nelson Cell or Diaphragm Cell :** The following reactions takes place –



At cathode (Perforated steel) : $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$ At anode (Carbon) : $2\text{Cl}^-(\text{aq.}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$

- (b) **Castner – Kellner Cell :** (Hg – Cathode Process)



On electrolysis –

At Cathode (Hg)

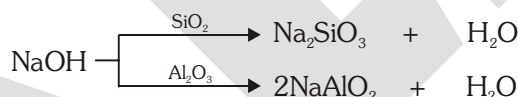


At anode (Graphite)



Properties :

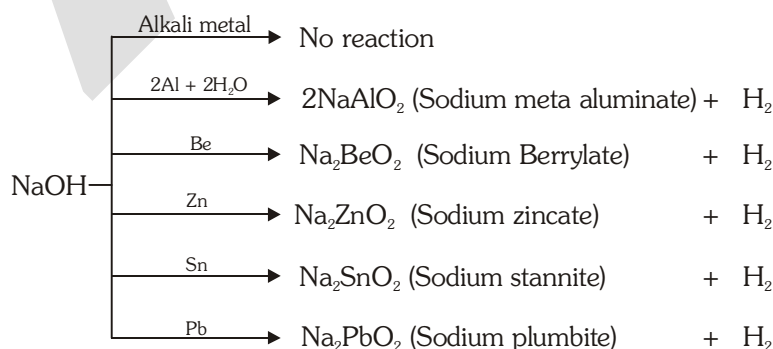
- It is deliquescent white crystalline solid.
- It absorbs CO_2 from air forming Na_2CO_3 .
- NaOH is **strong base**



- iv. **Reaction with non metals :** No reaction with H_2 , N_2 and C



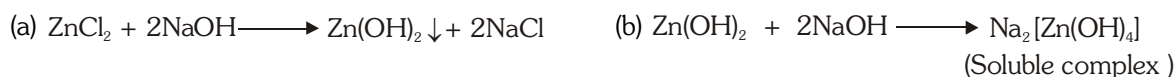
- v. **Reaction with Metal :**



- vi. The hydroxides of aluminium, zinc, lead and tin, however, dissolve in excess of sodium hydroxide giving clear solution which can also be obtained when these metals are acted upon by the concentrated solution of sodium hydroxide.



vii. **Reaction with ZnCl_2 or ZnSO_4**



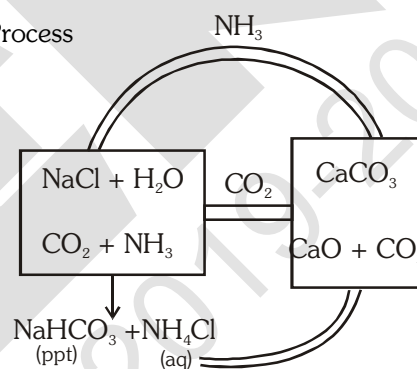
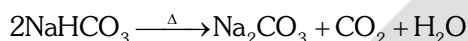
Uses (i) In the manufacture of soap, rayon, dyes, paper and drugs.

(ii) In petroleum refining.

3.4 SODIUM BICARBONATE OR BAKING SODA (NaHCO_3) and SODIUM CARBONATE OR WASHING SODA [$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$]

Preparation : Solvay Process (Commercial Scale)/Ammonia Soda Process

- (i) $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$
- (ii) $\text{NH}_3 + \text{H}_2\text{O} + \text{CO}_2 \longrightarrow \text{NH}_4\text{HCO}_3$
- (iii) $\text{NaCl} + \text{NH}_4\text{HCO}_3 \longrightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3 \text{ (ppt)}$
- (iv) Sodium carbonate is prepared by heating of NaHCO_3



Note : Potassium bicarbonate (KHCO_3) cannot be prepared by Solvay process as it is more soluble in water as compared to NaHCO_3 .

Properties of NaHCO_3 :

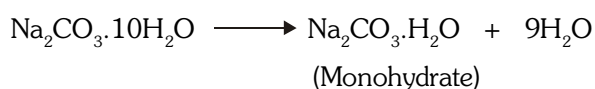
- (i) **Hydrolysis** $\text{NaHCO}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NaOH} + \text{H}_2\text{CO}_3$
- (ii) **Effect of heat** (temp. $> 100^\circ\text{C}$) $2\text{NaHCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2 \uparrow$
- (iii) **Reaction with acids** – gives CO_2 $\text{NaHCO}_3 + \text{HCl} \longrightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$
- (iv) **Reaction with bases** $\text{NaHCO}_3 + \text{NaOH} \longrightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$

Note : Reaction (iii) and (iv) explain **amphoteric** behaviour of NaHCO_3 .

Properties of Na_2CO_3

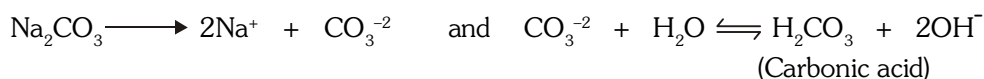
(i) Efflorescence :

$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ when exposed to air it gives out nine out of ten H_2O molecules.



This process is called efflorescence. Hence washing soda loses weight on exposure to air.

(ii) **Hydrolysis :** Aqueous solution of Na_2CO_3 is alkaline in nature due to anionic hydrolysis.



Uses of NaHCO_3

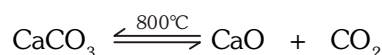
- In the preparation of baking powder.
- In the preparation of effervescent drinks.
- In the fire extinguishers.
- As antacid medicine (removing acidity)

Uses of Na_2CO_3

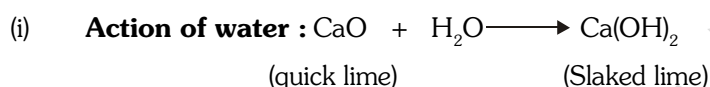
- For making fusion mixture ($\text{Na}_2\text{CO}_3 + \text{K}_2\text{CO}_3$)
- In the manufacture of glass, caustic soda, soap powders etc.
- In laundries and softening of water.

3.5 CALCIUM OXIDE [Quick lime (CaO)]

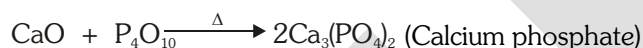
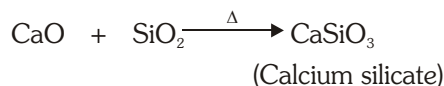
Preparation : By heating limestone at 800°C .



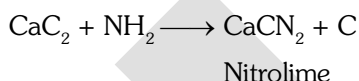
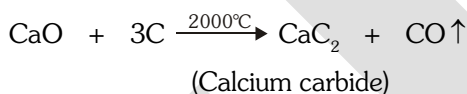
Properties :



(ii) **Basic Nature :**



(iii) **Reaction with carbon :**



Uses of CaO :

- In the manufacture of bleaching powder, cement, glass, calcium carbide etc.
- In the purification of sugar
- As a drying agent for NH_3 and $\text{C}_2\text{H}_5\text{OH}$
- As basic lining in furnaces
- For making Soda lime [$\text{NaOH} + \text{CaO}$]

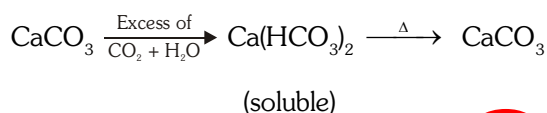
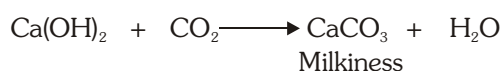
3.6 CALCIUM HYDROXIDE [Slaked lime Ca(OH)_2]

Preparation : By the action of water on quick lime

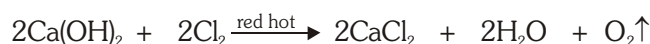
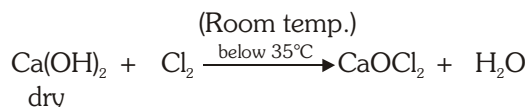


Properties of Ca(OH)_2

(i) **Action of CO_2 :** Lime water turns milky on passing CO_2 gas.



(ii) **Action of Chlorine :**

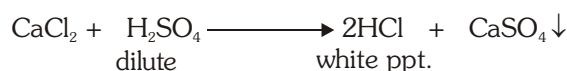


Uses of Ca(OH)_2

- (i) For softening of hard water.
- (ii) For purification of sugar and Coal gas.
- (iii) In the manufacture of bleaching powder, Caustic soda and soda lime
- (iv) In preparation of mortar, plaster and white washing.

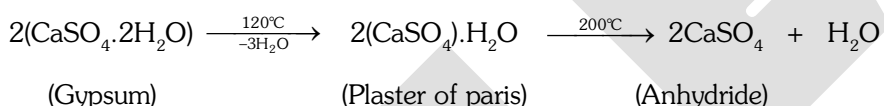
3.7 CALCIUM SULPHATE [Gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$]

Preparation : $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ is naturally occurring calcium sulphate. It can be obtained by the action of dil. H_2SO_4 on a soluble calcium salt below 60°C .



Properties of Gypsum

(i) **Action of heat :**



- (ii) It forms an important fertilizer $(\text{NH}_4)_2\text{SO}_4$

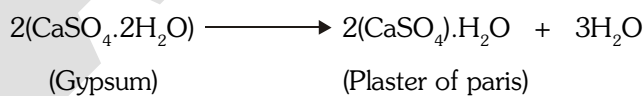


Uses of Gypsum

- (i) In the preparation of plaster of paris
- (ii) Anhydrous CaSO_4 used as drying agent.
- (iii) Anhydrite (CaSO_4) is used for manufacture of sulphuric acid, ammonium sulphate.

3.8 PLASTER OF PARIS [POP] $2(\text{CaSO}_4) \cdot \text{H}_2\text{O}$ or $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$

Preparation : It is obtained when gypsum is heated at 120°C



Properties of POP

- (i) It is a white powder.
- (ii) It has the property of setting to a hard mass when a paste with water is allowed to stand aside for sometime.
- (iii) When it is heated at 200°C , anhydrous CaSO_4 is formed.

Uses of POP

- (i) In surgery for setting broken bones
- (ii) In making casts for toys, statues etc.
- (iii) In making blackboard chalks.

3.9 SOME IMPORTANT POINTS TO REMEMBER

ANOMALOUS BEHAVIOR OF LITHIUM

- On account of its small size it exerts the greatest polarising effect out of all the alkali metals and ions, consequently covalent character is developed.
- Li has the highest ionisation energy and electronegativity as compared to other alkali metals.
- It is not affected by air easily and does not lose its lustre even on melting.
- It is harder and lighter than other alkali metal.
- It reacts slowly with water to liberate hydrogen.
- When burnt in air or oxygen, it forms only monoxide, Li_2O . However, the rest of the alkali metals give peroxide or superoxides.
- Li_2O is less basic oxides than of other alkali metals.
- Lithium hydroxide decomposes when red heated to form Li_2O . Hydroxides of other alkali metals do not decompose.

$$2\text{LiOH} \xrightarrow{\Delta} \text{Li}_2\text{O} + \text{H}_2\text{O}$$
- Li_2CO_3 is less stable, as it decomposes on heating. $\text{Li}_2\text{CO}_3 \xrightarrow{\Delta} \text{Li}_2\text{O} + \text{CO}_2$
- Li_2SO_4 is the only alkali metal sulphate, which does not form double salts Ex. Alum.
- Li when heated in NH_3 forms imide Li_2NH while other alkali metals form amides. MNH_2
- Lithium shows resemblance with magnesium, an element of group IIA.

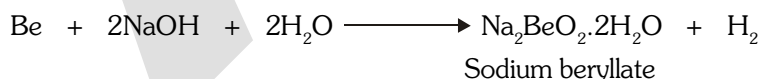
This resemblance is termed as diagonal relationship.

Similarities between lithium and Magnesium

- Both lithium and magnesium are harder and lighter than the other elements in their respective groups.
- Lithium and magnesium reacts slowly with cold water. Their oxides and hydroxides are much less soluble and their hydroxides decomposes on heating. They both form nitride by direct combination with nitrogen, Li_3N and Mg_3N_2 .
- The oxides, Li_2O and MgO do not combine with excess oxygen to give a peroxide or a superoxide.
- The carbonates of lithium and magnesium decomposes easily on heating to form the oxide and CO_2 . Solid bicarbonates are not formed by lithium and magnesium.
- Both LiCl and MgCl_2 are soluble in ethanol.
- Both LiCl and MgCl_2 are deliquescent and crystallise from their aqueous solution as hydrates, $\text{LiCl} \cdot 2\text{H}_2\text{O}$ and $\text{MgCl}_2 \cdot 8\text{H}_2\text{O}$.

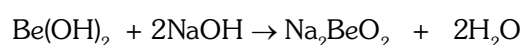
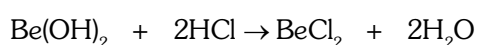
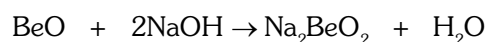
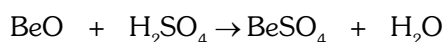
ANOMALOUS BEHAVIOR OF BERYLLIUM

- It is the hardest of all alkaline earth metal as maximum metallic bonding is there due to its smallest size.
- The melting and boiling points of the beryllium are the highest.
- It is least reactive due to highest ionisation potential.
- Due to high charge density its polarising effect is highest and it forms covalent bond.
- It dissolves in alkalis with the evolution of hydrogen



other alkaline earth metals do not react with alkalis.

- Oxides and hydroxides of beryllium are amphoteric in nature.



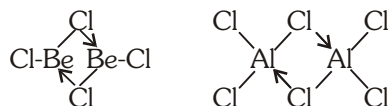
The hydroxide is Insoluble in water and is covalent in nature.

- Like Al_4C_3 , its carbide (Be_2C) on hydrolysis evolves methane.
- Due to its small size it has strong tendency to form complex.
- It shows diagonal relationship with Al.

DIAGONAL SIMILARITY BETWEEN BERYLLIUM AND ALUMINIUM :

In many of its properties, beryllium resembles with aluminium. Thus –

- The two elements have same electronegativity and their charge/ radius ratios.
- Both metals are fairly resistant to the action of acids due to the formation of a protective film of oxide on their surface. Both metals are acted upon by strong alkalis to form soluble complexes, beryllates $[\text{Be}(\text{OH})_4]^{2-}$ and aluminates, $[\text{Al}(\text{OH})_4]^-$.
- The chlorides of both beryllium and aluminium have bridged chloride structures in vapour phase.

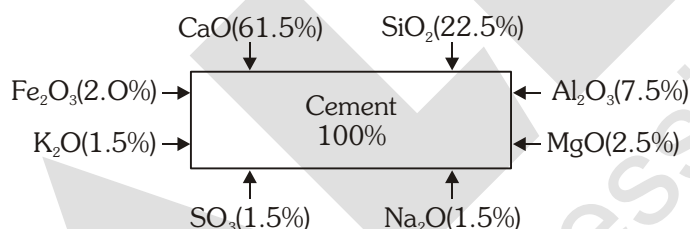


- Salts of these metals form hydrated ions, Ex. $[\text{Be}(\text{OH})_4]^{2-}$ and $[\text{Al}(\text{OH})_6]^{3+}$ in aqueous solutions. Due to similar charge/ radius ratios of beryllium and aluminium ions they have a strong tendency to form complexes. For example beryllium forms tetrahedral complexes such as BeF_4^{2-} and $[\text{Be}(\text{C}_2\text{O}_4)_2]^{2-}$ and aluminium forms octahedral complexes like AlF_6^{3-} and $[\text{Al}(\text{C}_2\text{O}_4)_3]^{3-}$.

3.10 PORTLAND CEMENT

It is a light grey, heavy fine powder, It is a homogenous mixture of silicates and aluminates of calcium, which form more than 90% of the cement are –

- | | | |
|--|---|---|
| (i) Tricalcium silicate | - | $3\text{CaO} \cdot \text{SiO}_2$ |
| (ii) Dicalcium silicate (slowest setting component) | - | $2\text{CaO} \cdot \text{SiO}_2$ |
| (iii) Tricalcium aluminate (fastest setting component) | - | $3\text{CaO} \cdot \text{Al}_2\text{O}_3$ |
| (iv) Tetracalcium aluminoferrite | - | $4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3$ |

Important Points :-**(i) Composition of Cement**

For high quality

- $\Rightarrow \text{SiO}_2 \text{ to } \text{Al}_2\text{O}_3 \text{ Ratio} = 2.5-4$
 $\Rightarrow \text{CaO to Rest all oxide} = 2$

(ii) Raw Materials

- Lime Stone – It provides CaO
- Clay – It provides Al_2O_3 and silica (SiO_2)
- Gypsum – $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

(iii) Setting of cement : When water is mixed to cement and the mixture is left it becomes very hard. This property of cement is called setting.

(iv) Mortar : It is a mixture of cement, sand and water to give a proper consistency.

(v) Concrete : A mixture of cement, Sand gravel and water is known as concrete.

(vi) Reinforced concrete cement (RCC) : When concrete is filled in beams made of iron bars, it is called RCC. Iron imparts extra strength to the structure.

3.11 BIOLOGICAL ROLE OF SODIUM, POTASSIUM, MAGNESIUM AND CALCIUM (FROM BIOLOGY PORTION)

Normally % of abundance in human body – $\text{K} > \text{Na} > \text{Fe} > \text{Cu}$

BEGINNER'S BOX-2

- Potassium carbonate cannot be made by the Solvay process because
 - (1) potassium hydrogen carbonate is unstable
 - (2) potassium hydrogen carbonate is rather too soluble in water to be precipitated
 - (3) potassium carbonate is insoluble in water
 - (4) potassium carbonate is soluble in water
- Plaster of Paris, a white powder, is—
 - (1) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
 - (2) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
 - (3) CaSO_4
 - (4) $\text{CaSO}_4 \cdot \text{H}_2\text{O}$
- Which of the following is incorrect?
 - (1) Cement containing no iron is white.
 - (2) Cement containing excess amount of lime cracks during setting.
 - (3) Setting of cement is an endothermic process.
 - (4) Setting of cement is an example of hydration.
- On passing excess of CO_2 in lime water, its milky appearance disappears because -
 - (1) Soluble $\text{Ca}(\text{OH})_2$ is formed
 - (2) Soluble $\text{Ca}(\text{HCO}_3)_2$ is formed
 - (3) Reaction becomes reversible
 - (4) Calcium compound evaporated
- In the preparation of sodium carbonate which of the following is used -
 - (1) Slaked lime
 - (2) Lime stone
 - (3) Lime
 - (4) quick lime
- When chlorine is passed slow over dry slaked lime $\text{Ca}(\text{OH})_2$ at room temperature, the main product is
 - (1) CaCl_2
 - (2) CaOCl_2
 - (3) $\text{Ca}(\text{ClO}_2)_2$
 - (4) $\text{Ca}(\text{OCl})_2$
- Identify the correct statement -
 - (1) Gypsum contains a lower percentage of Ca than plaster of paris
 - (2) Gypsum is obtained by heating plaster of paris
 - (3) Plaster of paris can be obtained by hydration of gypsum
 - (4) Plaster of paris is obtained by partial oxidation of gypsum
- Quick lime is prepared on a commercial scale by heating ____ in a rotary kiln at 1070 – 1270 K.
 - (1) CaSO_4
 - (2) $\text{Ca}(\text{NO}_3)_2$
 - (3) CaCO_3
 - (4) $\text{Ca}(\text{OH})_2$
- Ca^{2+} ions play an important role in
 - (i) neuromuscular function
 - (ii) interneuronal transmission
 - (iii) cell membrane integrity
 - (iv) blood coagulation
 - (1) (i), (ii)
 - (2) (ii), (iii)
 - (3) (ii), (iii), (iv)
 - (4) (i), (ii), (iii), (iv)

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	4	1	3	1	2	2	3	4	3	4

BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7	8	9	
	Ans.	2	2	3	2	2	2	1	3	4	

EXERCISE-I (Conceptual Questions)**Build Up Your Understanding****PROPERTIES AND COMPOUNDS**

- Correct order of density is –
(1) $\text{Li} > \text{Na}$ (2) $\text{K} > \text{Na}$
(3) $\text{Mg} > \text{Ca}$ (4) $\text{Cs} < \text{Rb}$
- Which is having highest m.p. –
(1) Be (2) Mg (3) Ca (4) Sr
- Weak reductant in alkali metal is –
(1) Li (2) Na (3) K (4) Cs
- The metal used in photoelectric cell is –
(1) Na (2) Cs (3) Mg (4) Ca
- Lithium chloride is highly soluble in –
(1) C_6H_6 (2) H_2O (3) D_2O (4) All
- Which metal will not form superoxide –
(1) Li (2) Be (3) Na (4) All
- More stable hydride is –
(1) $\text{Cs} - \text{H}$ (2) $\text{Rb} - \text{H}$
(3) $\text{K} - \text{H}$ (4) $\text{Li} - \text{H}$
- In which compound hydrogen is electronegative –
(1) CaH_2 (2) CH_4 (3) HCl (4) All
- Which of the following metal will give apple green colour on Bunsen flame –
(1) Ba (2) Sr
(3) Ca (4) K
- The density of –
(1) $\text{Na} > \text{K}$ (2) $\text{Na} = \text{K}$
(3) $\text{K} > \text{Na}$ (4) $\text{Li} > \text{K}$
- Alkali metals salts are –
(1) Diamagnetic and coloured
(2) Diamagnetic and colourless
(3) Paramagnetic and coloured
(4) Paramagnetic and colourless
- Ionic conductances of hydrated M^+ ions are in the order –
(1) $\text{Li}^+(\text{aq}) > \text{Na}^+(\text{aq}) > \text{K}^+(\text{aq}) > \text{Rb}^+(\text{aq}) > \text{Cs}^+(\text{aq})$
(2) $\text{Li}^+(\text{aq}) > \text{Na}^+(\text{aq}) < \text{K}^+(\text{aq}) < \text{Rb}^+(\text{aq}) < \text{Cs}^+(\text{aq})$
(3) $\text{Li}^+(\text{aq}) > \text{Na}^+(\text{aq}) > \text{K}^+(\text{aq}) > \text{Rb}^+(\text{aq}) < \text{Cs}^+(\text{aq})$
(4) $\text{Li}^+(\text{aq}) < \text{Na}^+(\text{aq}) < \text{K}^+(\text{aq}) < \text{Rb}^+(\text{aq}) < \text{Cs}^+(\text{aq})$
- Which of the following halides has the highest melting point –
(1) NaCl (2) KCl
(3) NaBr (4) NaF

- Which of the following does not give an oxide on heating –
(1) MgCO_3 (2) Li_2CO_3
(3) ZnCO_3 (4) K_2CO_3
- When strongly heated in steam, Mg burns brilliantly producing –
(1) $\text{Mg}(\text{OH})_2$ (2) MgO and H_2
(3) MgO and O_2 (4) MgO and O_3
- When magnesium ribbon is heated to redness in an atmosphere of nitrogen and subsequently cooled with water, the gas evolved is –
(1) N_2 (2) NH_3 (3) O_2 (4) CO_2
- Molten potassium chloride conducts electricity due to the presence of –
(1) Free electron
(2) Free ions
(3) Free molecules
(4) Atom of potassium & chloride
- Which of the following element have maximum tendency to form complex compound –
(1) Be (2) Ba (3) Ca (4) Mg
- On heating sodium metal in the current of dry ammonia leads to the formation of which gas –
(1) NaNH_2 (2) NaN_3 (3) NH_3 (4) H_2
- Sodium reacts with water more vigorously than lithium because it –
(1) Has higher atomic weight
(2) Is more electronegative
(3) Is more electropositive
(4) Is a metal
- Which of the following alkali metals has the biggest tendency of the half reaction $\text{M}_{(\text{g})} \longrightarrow \text{M}_{(\text{aq})}^+ + \text{e}^-$
(1) Sodium (2) Lithium
(3) Potassium (4) Cesium
- The strongest reducing agent is –
(1) Be (2) Mg (3) Sr (4) Ba
- Both Be and Al become passive on reaction with conc. nitric acid due to –
(1) The non reactive nature of the metal
(2) The non reactive nature of the acid
(3) The formation of an inert oxide layer on the surface of the metals
(4) None of these

24. Sodium loses its lustre on exposure to moist air due to formation of –
 (1) Na_2O , NaOH and Na_2CO_3
 (2) Na_2O and NaOH
 (3) Na_2O and Na_2CO_3
 (4) NaOH and Na_2CO_3
25. Potassium carbonate when heated to high temperature.
 (1) Gives CO_2 (2) Gives O_2
 (3) Gives CO (4) Gives no gas at all
26. On Flame test K give ----- colour –
 (1) Golden yellow (2) Crimson red
 (3) Violet (4) Apple green
27. An element having electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ will form –
 (1) Acidic oxide (2) Basic oxide
 (3) Amphoteric oxide (4) Neutral oxide
28. Which decomposes on heating –
 (1) NaOH (2) KOH (3) LiOH (4) RbOH
29. Which metal does not form ionic hydride –
 (1) Na (2) Rb (3) Ca (4) Be
30. The element of IA group which combines directly with nitrogen is –
 (1) Li (2) Na (3) K (4) Cs
31. Which of the following releases 0.2 moles of hydrogen on hydrolysis –
 (1) 0.1 mole of LiH (2) 0.2 mole of LiH
 (3) 0.3 mole of LiH (4) 0.4 mole of LiH
32. Which of the following is paramagnetic
 (1) K_2O (2) K_2O_2 (3) KO_2 (4) Na_2O
33. A compound which upon hydrolysis releases ammonia is –
 (1) Li_3N (2) LiNO_3
 (3) NaNO_3 (4) None of these
34. The metal ion which does not give any flame colouration is –
 (1) Li^+ (2) Be^{+2} (3) Na^+ (4) K^+
35. Which of the following exists as hydrated salt –
 (1) NaCl (2) LiCl
 (3) RbCl (4) KCl
36. Strong reductant in IIA and IA group is –
 (1) Ba, Li (2) Li, Be
 (3) Cs, Ba (4) Ba, Cs
37. Which statement will be true for solution, when Ba is dissolved in ammonia:-
 (1) Solution becomes blue
 (2) Solution becomes good conductor
 (3) Solution remains colourless
 (4) Both (1) and (2) are correct
38. In K, Rb and Cs, the decreasing order of reducing power in gaseous state is:-
 (1) $\text{K} > \text{Cs} > \text{Rb}$ (2) $\text{Cs} > \text{Rb} > \text{K}$
 (3) $\text{K} < \text{Cs} < \text{Rb}$ (4) $\text{Rb} > \text{Cs} > \text{K}$
39. The correct order of density of following elements is:- (Be, Mg, Ca, Sr)
 (1) $\text{Be} > \text{Mg} > \text{Ca} > \text{Sr}$ (2) $\text{Ca} > \text{Mg} > \text{Be} > \text{Sr}$
 (3) $\text{Ca} < \text{Mg} < \text{Be} < \text{Sr}$ (4) $\text{Mg} < \text{Ca} < \text{Sr} < \text{Be}$
40. Identify the correct statement elemental sodium:-
 (1) Is a strong oxidising agent
 (2) Can be extracted by electrolysis of aqueous solution
 (3) Its density is lower than K
 (4) Is easily oxidised
41. On addition of metal, colour of liquid NH_3 solutions converts into bronze, the reason is :-
 (1) Ammoniated electrons
 (2) Metal amide formation
 (3) Liberation of NH_3 gas
 (4) Cluster formation of metal ions
42. On allowing ammonia solution of s-block metals to stand for a long time, blue colour becomes fade. The reason is:-
 (1) Formation of NH_3 gas
 (2) Formation of metal amide
 (3) Cluster formation of metal ions
 (4) Formation of metal nitrate
43. Which of the following s-block element reacts with NaOH to give water soluble complex :-
 (1) Al (2) Ca (3) Be (4) Li
44. Which is having least mpt. :-
 (1) Ba (2) Ca (3) Mg (4) Be
45. When Na and Li placed in dry air we get :-
 (1) NaOH , Na_2O , Li_2O
 (2) Na_2CO_3 , Na_2O_2 , Li_2O
 (3) Na_2O , Li_3N , NH_3
 (4) Na_2O , Li_2O , Li_3N
46. Which of the following oxide having O_2^{-2} (peroxide) anion :-
 (1) Na_2O (2) BaO_2 (3) RbO_2 (4) KO_2

47. Generally which of the following properties of IA group metals increases as the atomic number rises:
 (a) Metallic character (b) Ionic radius
 (c) Melting point (d) Density
 (e) Ionisation potential
 Correct answer is :-
 (1) a, b, c (2) a, b, d (3) c, d, e (4) All
48. Which of the following s-block metals do not impart any colour to the flame
 (1) Li, Be (2) Cs, Fr (3) Be, Mg (4) Ba, Ra
49. Which can not be used to generate H_2 :-
 (1) $Al + NaOH$ (2) $Zn + NaOH$
 (3) $Mg + NaOH$ (4) $LiH + H_2O$
50. Only those elements of s-block can produce superoxides which have :-
 (1) High ionisation energy
 (2) High electronegativity
 (3) High charge density
 (4) Low ionisation potential
51. Which does not exist in solid state :-
 (1) $LiHCO_3$ (2) $CaCO_3$
 (3) $NaHCO_3$ (4) Na_2CO_3
52. Alkali metals dissolve in liquid NH_3 then which of the following observations is not true:
 (1) It becomes paramagnetic
 (2) Solution turns into blue due to solvated electrons
 (3) It becomes diamagnetic
 (4) Solution becomes conducting
53. Alkali metals give colour in bunsen flame due to -
 (1) Low electronegativity
 (2) One e^- in outer most orbit
 (3) Smaller atomic radii
 (4) Low ionisation energy
54. Which of the following ions forms a hydroxide that is highly soluble in water ?
 (1) K^+ (2) Zn^{2+} (3) Ni^{2+} (4) Al^{3+}
55. The slaked lime is prepared by adding water to-
 (1) Quick lime (2) Nitrolim
 (3) Lime stone (4) Plaster of paris
56. The plaster of paris is hardened by
 (1) Liberating CO_2 (2) Giving out water
 (3) Combining with water (4) Changing into $CaCO_3$
57. Which of the following alkali metal carbonate is the least stable and decomposes readily
 (1) Li_2CO_3 (2) Na_2CO_3
 (3) K_2CO_3 (4) Cs_2CO_3
58. In the reaction $M + O_2 \longrightarrow MO_2$ (super oxide) the metal is
 (1) Li (2) Na (3) K (4) Ba
59. Li does not resemble other alkali metals in following properties
 (1) Li_2CO_3 decomposes into oxides while other alkali carbonates are thermally stable
 (2) $LiCl$ is predominantly covalent
 (3) Li_3N is stable
 (4) All
60. Be and Al resemble in
 (1) Both become passive on reaction with HNO_3 due to formation of oxide layer
 (2) Their chlorides are lewis acids
 (3) Hydroxides are soluble in alkali as well as in acid
 (4) All
61. Consider the following points
 (a) Cs is the strongest reducing agent in IA group element
 (b) Be does not form peroxide in II A group elements
 (c) The density of potassium is less than sodium
 (d) In alkali metals Li, Na, K and Rb, lithium has the minimum value of M.P.
 Point out that the statement -
 (1) (a) & (b) are correct
 (2) (a), (b) & (c) are correct
 (3) (b) & (c) are correct
 (4) (b), (c) & (d) are correct
62. Mg^{+2} does not form either peroxide or superoxide, because
 (1) Mg^{+2} ion is relatively bigger
 (2) Mg^{+2} ion is relatively smaller
 (3) Mg^{+2} ion is stable
 (4) Mg^{+2} ion is unstable
63. The stability order of oxide, peroxide and superoxide of alkalimetal is
 (1) Normal oxide > super oxide > per oxide
 (2) Normal oxide > per oxide > super oxide
 (3) super oxide > per oxide > normal oxide
 (4) per oxide > normal oxide > super oxide
64. Which of the following is true about Alkali metals
 (1) All form solid bicarbonates
 (2) All form ionic salt like hydride MH
 (3) All form superoxide like KO_2
 (4) All form nitrides

65. Which of the following statement is not correct
 (1) LiOH is amphoteric in nature
 (2) LiCl is soluble in pyridine
 (3) Li_3N is stable while Na_3N doesn't exist even at room temperature
 (4) BeO is amphoteric in nature
66. In between the metals A and B both form oxide but only B forms nitride, when both burn in air so A and B are
 (1) Cs, K (2) Mg, Ca
 (3) Li, Na (4) K, Mg
67. Which of the following statement is not correct
 (1) BeF_2 forms complex ion with NaF in which Be goes with cation
 (2) BeCO_3 is kept in the atmosphere of CO_2 since it is least thermally stable
 (3) Be dissolves in alkali forming $[\text{Be}(\text{OH})_4]^{2-}$
 (4) BeF_2 forms complex ion with NaF in which Be goes with anion
68. CO_2 gas along with solid (Y) is obtained when sodium salt (X) is heated. (X) is again obtained when excess CO_2 gas is passed into aqueous solution of (Y). X and Y are :
 (1) Na_2CO_3 , Na_2O
 (2) Na_2CO_3 , NaOH
 (3) NaHCO_3 , Na_2CO_3
 (4) Na_2CO_3 , NaHCO_3
69. A compound which can be used in space vehicles both to absorb CO_2 and liberate O_2 is :
 (1) NaOH (2) Na_2O
 (3) Na_2O_2 (4) $\text{CaO} + \text{NaOH}$

70. There is loss in weight when mixture of Li_2CO_3 and $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ is heated strongly. This loss is due to :
 (1) Li_2CO_3 (2) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
 (3) both (4) none
- Note : Q.71 to 74 are based on following reaction (s) :**
- $$\text{A} \xrightarrow{\Delta} \text{B (oxide)} + \text{CO}_2$$
- $$\text{B} + \text{H}_2\text{O} \longrightarrow \text{C}$$
- $$\text{C} + \text{CO}_2 \longrightarrow \text{A (milky)}$$
- $$\text{B} + \text{NH}_4\text{Cl} \xrightarrow{\Delta} \text{D (gas)}$$
- $$\text{D} + \text{H}_2\text{O} + \text{CO}_2 \longrightarrow \text{E}$$
- $$\text{E} + \text{NaCl} \longrightarrow \text{F}$$
- $$\text{F} \xrightarrow{\Delta} \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$$
71. Name of the process is :
 (1) solvay (2) ammonia-soda
 (3) both correct (4) none is correct
72. A is :
 (1) $\text{Ca}(\text{HCO}_3)_2$ (2) CaCO_3
 (3) CaO (4) Na_2CO_3
73. B and C are :
 (1) CaO, $\text{Ca}(\text{OH})_2$ (2) $\text{Ca}(\text{OH})_2$, CaCO_3
 (3) CaCO_3 , $\text{Ca}(\text{OH})_2$ (4) $\text{Ca}(\text{OH})_2$, CaO
74. D, E and F are :
 (1) NH_3 , NH_4Cl , NH_4HCO_3
 (2) NH_3 , NH_4HCO_3 , NaHCO_3
 (3) NH_4HCO_3 , Na_2CO_3 , NaHCO_3
 (4) None
75. A wire of an alkaline earth metal X, burnt in air and dipped in water, a gas 'Y' is evolved X and Y are respectively :-
 (1) Na, NO_2 (2) Be, NO_2
 (3) Mg, CO_2 (4) Mg, NH_3

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	1	2	2	1	4	4	1	1	1	2	4	4	4	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	2	1	4	3	2	4	3	1	4	3	2	3	4	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	3	1	2	2	1	4	2	3	4	4	2	3	3	4
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	2	3	3	4	1	3	4	1	1	3	1	3	4	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	3	2	2	2	1	4	1	3	3	3	3	2	1	2	4

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIIMS-2006

1. The pair whose both species are used in antacid medicinal preparations is –
- (1) NaHCO_3 and Mg(OH)_2
 - (2) Na_2CO_3 and $\text{Ca(HCO}_3)_2$
 - (3) $\text{Ca(HCO}_3)_2$ and Mg(OH)_2
 - (4) Ca(OH)_2 and NaHCO_3

AIPMT-2010

2. Compound A on heating gives a colourless gas and a residue that is dissolved in water to obtain B. Excess of CO_2 is bubbled through aqueous solution of B, C is formed which is recovered in the solid form. Solid C on gentle heating gives back A. The compound is :-
- (1) Na_2CO_3
 - (2) K_2CO_3
 - (3) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
 - (4) CaCO_3

AIPMT Mains-2011

3. Which of the following statements is incorrect :-
- (1) NaHCO_3 on heating gives Na_2CO_3
 - (2) Pure sodium metal dissolves in liquid ammonia to give blue solution
 - (3) NaOH reacts with glass to give sodium silicate
 - (4) Aluminium reacts with excess NaOH to give Al(OH)_3

AIIMS-2011

4. Which of the following is not hygroscopic–
- (1) NaCl
 - (2) MgCl_2
 - (3) CaCl_2
 - (4) LiCl

AIIMS-2015

5. Which metal easily give H_2 gas when reacts with NaOH
- (1) Mg
 - (2) Zn
 - (3) Cu
 - (4) Ag

NEET-II 2016

6. The suspension of slaked lime in water is known as
- (1) milk of lime
 - (2) aqueous solution of slaked lime
 - (3) limewater
 - (4) quicklime
7. In context with beryllium, which one of the following statements is **incorrect** ?
- (1) Its salts rarely hydrolyze.
 - (2) Its hydride is electron-deficient and polymeric.
 - (3) It is rendered passive by nitric acid.
 - (4) it forms Be_2C .

AIIMS 2017

8. The % abundance of elements in the human body
- (1) $\text{K} > \text{Fe} > \text{Cu}$
 - (2) $\text{Cu} > \text{Fe} > \text{K}$
 - (3) $\text{Fe} > \text{Cu} > \text{K}$
 - (4) $\text{K} > \text{Cu} > \text{Fe}$

AIIMS 2018

9. Which of the following compound will remain in dissolved state in $\text{NH}_3\text{-NH}_4\text{Cl(aq)}$ solution :-
- (1) BaCO_3
 - (2) SrCO_3
 - (3) CaCO_3
 - (4) MgCO_3
10. Which is incorrect for Solvay process :-
- (1) NH_3 is reused
 - (2) CaCl_2 is one of final product
 - (3) Na_2CO_3 is partially changed into NaHCO_3
 - (4) CaCO_3 is consumed

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	
Ans.	1	4	4	1	2	1	1	1	4	3	

EXERCISE-III (Analytical Questions)

Check Your Understanding

- On dissolving moderate amount of sodium metal in liquid NH_3 at low temperature, which one of the following does **not** occur
 (1) Blue coloured solution is obtained.
 (2) Na^+ ions are formed in the solution.
 (3) Liquid NH_3 becomes good conductor of electricity.
 (4) Liquid ammonia remains diamagnetic.
- Which of the following pair can't exist in solution ?
 (1) NaHCO_3 and NaOH
 (2) Na_2CO_3 and NaOH
 (3) Na_2CO_3 and NaCl
 (4) NaHCO_3 and NaCl
- NaOH is manufactured by electrolysis of brine solution. The products of the reaction are
 (1) Cl_2 and H_2 (2) Cl_2 and Na-Hg
 (3) Cl_2 and Na (4) Cl_2 and O_2
- Sodium carbonate is manufactured by Solvay process, the products that are recycled are
 (1) CO_2 and NH_3
 (2) CO_2 and NH_4Cl
 (3) NaCl and CaO
 (4) CaCl_2 and CaO
- At anode in the electrolysis of fused NaCl :
 (1) Na^+ is oxidised (2) Cl^- is oxidised
 (3) Cl^- is reduced (4) Na^+ is reduced
- Which alkali metal on flame test gives red violet colour
 (1) Li (2) Cs
 (3) Na (4) Rb
- In presence of iron, alkali metal react with liquid ammonia and form
 (1) Metal mixture + H_2
 (2) Iron metal mixture + H_2
 (3) Metal mixture
 (4) Metal amide + H_2

- Photoelectric effect is maximum in
 (1) Cs (2) Na (3) K (4) Li
- When a standard solution of NaOH is left in air for a few hours,
 (1) a precipitate will form
 (2) strength will decrease
 (3) strength will increase
 (4) the concentration of Na^+ ions remain same
 (5) all are wrong
- Which is used in purification of air in the space craft.
 (1) Slaked lime (2) Quick lime
 (3) Potassium superoxide (4) CaCl_2
- The correct order of ionic character of oxides of alkali earth metal :-
 (1) $\text{MgO} > \text{CaO} > \text{SrO} > \text{BaO}$
 (2) $\text{BaO} > \text{SrO} > \text{CaO} > \text{MgO}$
 (3) $\text{CaO} > \text{SrO} > \text{BaO} > \text{MgO}$
 (4) $\text{SrO} > \text{BaO} > \text{MgO} > \text{CaO}$
- Potassium superoxide is used in oxygen cylinders of space craft as it -
 (1) Absorbs O_2
 (2) Eliminate moisture
 (3) Absorbs CO_2 and increases O_2 content
 (4) Forms ozone
- Compounds of alkaline earth metals are generally less soluble in water than that of alkali metals because of :-
 (1) High hydration energy
 (2) More covalent character
 (3) More ionic character
 (4) Less lattice energy

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13
Ans.	4	1	1	1	2	4	4	1	2	3	2	3	2

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion** : In the solution of K in liquid NH_3 , blue colour appears.
Reason : K reacts with NH_3 to form KNH_2
 (1) A (2) B (3) C (4) D
- Assertion** : Na_2O_2 is coloured and paramagnetic
Reason : Na_2O_2 is superoxide
 (1) A (2) B (3) C (4) D
- Assertion** : KHCO_3 can not be obtained by solvay process.
Reason : KHCO_3 is less soluble than NaHCO_3 .
 (1) A (2) B (3) C (4) D
- Assertion** : Mg can burn in the atmosphere of N_2 .
Reason : Mg reacts with N_2 to form nitride.
 (1) A (2) B (3) C (4) D
- Assertion** : Li_2SO_4 do not form double salt like alum.
Reason : Atomic size of Li is too small.
 (1) A (2) B (3) C (4) D
- Assertion** : NaCl when exposed in air it becomes wet.
Reason : NaCl contains hygroscopic impurities like CaCl_2 , MgCl_2 etc.
 (1) A (2) B (3) C (4) D
- Assertion** : Lithium is most reducing element.
Reason : IP of lithium is minimum in the periodic table.
 (1) A (2) B (3) C (4) D
- Assertion** : When cement is mixed with water and left as such, it becomes hard mass.
Reason : Setting of cement is exothermic process.
 (1) A (2) B (3) C (4) D
- Assertion** : Beryllium is most reducing s-block element
Reason : Hydration energy of Be is greater than its I.P.
 (1) A (2) B (3) C (4) D
- Assertion** : Halides of Be dissolve in organic solvents
Reason : Atomic size of Be is smallest in the s-block elements.
 (1) A (2) B (3) C (4) D
- Assertion** : Be exhibit photoelectric effect.
Reason : Be has least IP in the s-block
 (1) A (2) B (3) C (4) D
- Assertion** : Chlorides of Li, Be and Mg are covalent in nature
Reason : Li, Be and Mg have large cationic size in the s-block elements
 (1) A (2) B (3) C (4) D
- Assertion** :- Alkaline earth metal and alkali metal form superoxide.
Reason :- Both have tendency to form single bond.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	
Ans.	2	4	3	1	1	1	3	2	4	2	4	3	4	

HYDROGEN AND ITS COMPOUND

E

S.No.	CONTENTS	Page
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NEET SYLLABUS

HYDROGEN : Occurrence, isotopes, preparation, properties and uses of hydrogen; hydrides (ionic, covalent and interstitial) ; physical and chemical properties of water, heavy water; hydrogen peroxide-preparation, reactions, uses and structure.

@neetfocusquizz

OBJECTIVES

After studying this unit, you will be able to :

- *identify the modes of occurrence and preparation of dihydrogen on a small and commercial scale; describe isotopes of hydrogen;. explain how different elements combine with hydrogen to form ionic, molecular and nonstoichiometric compounds;*
- *describe how an understanding of its properties can lead to the production of useful substances, and new technologies;*
- *understand the structure of water and use the knowledge for explaining physical and chemical properties;*
- *explain how environmental water quality depends on a variety of dissolved substances; difference between 'hard' and 'soft' water and learn about water softening;*
- *acquire the knowledge about heavy water and its importance;. understand the structure of hydrogen peroxide, learn its preparatory methods and properties leading to the manufacture of useful chemicals and cleaning of environment;*
- *understand and use certain terms e.g., electron-deficient, electronprecise, electron-rich, hydrogen economy, hydrogenation etc.*

The two most common elements in the universe are hydrogen and stupidity.

Harlan Ellison

HYDROGEN AND ITS COMPOUNDS

4.0 INTRODUCTION :

Hydrogen is the lightest element and also the lightest gas in the periodic table.

Order of Abundance of H :

Hydrogen is the most abundant element in the universe (70% of the total mass of the universe.)

Order of abundance of H : Universe > Sun > Earth

The planets Jupiter & Saturn consist mainly of H_2 . Similarly about half the mass of the sun & some other stars is made up of hydrogen.

In Sun's atmosphere & in universe, it is found in atomic form. While in earth it is generally found in molecular form. At Sun, the stratosphere is made up of H (atomic hydrogen) & they undergoes fusion & converted into He nuclei & this reaction is exothermic.

It is the ninth element on earth in order of abundance.

Earth does not possess enough gravitational force to retain live hydrogen molecule i.e. why it is not found in earth atmosphere in atomic form.

Hydrogen is the most reactive element in atomic form but it is less reactive in molecular form because of very high bond dissociation energy due to 1s-1s overlapping.

4.1 ISOTOPIC EFFECT :

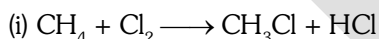
The effect which can change the physical & chemical properties of isotopes is called isotopic effect.

It is because of difference in mass.

In isotopic effect maximum changes occurs in physical properties like melting point, boiling point, bond energy, while minimum changes occurs in chemical properties like state of chemical reaction etc.

Imp. Isotopic effect is found only in hydrogen isotopes. Because there is a large difference in mass.

Ex. Which of the following reaction is fast & why ?

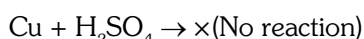
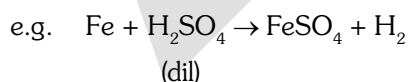


Ans. (i) because C-H bond energy is less in comparison to C-D bond energy.

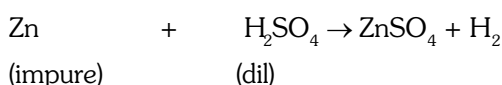
4.2 METHOD OF PREPARATION :

(a) From acids :

The metal which are placed above H_2 in electrochemical series react with dil acids to liberate H_2 .



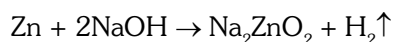
Lab preparation : When impure Zn reacts with dil H_2SO_4 it forms H_2



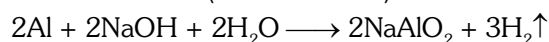
Ex. Why we use impure Zn.

Ans. Because the rate of reaction with pure Zn is very slow.

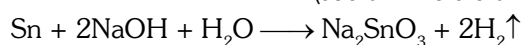
(b) **By alkalis :** Only (Be, Zn, Al, Sn, Pb, Si) (Amphoteric metal) react with boiling NaOH or KOH to evolve H_2 .



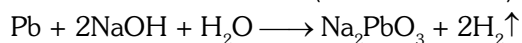
(sodium zincate)



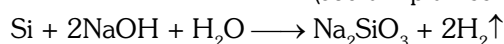
(sodium meta aluminate)



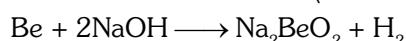
(sodium stannate)



(sodium plumbate)



(sodium silicate)



(sodium beryllate)

(c) **With water :**

(i) With cold water (7°C - 25°C) : Li, K, Ba, Ca, Na, Sr

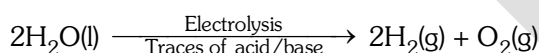
(ii) With hot water (25°C - 90°C) : Mg, Al, Cr, Mn, Zn

(iii) With steam (greater than 100°C) : Fe, Cd, Ni, Sn, Pb

(d) **Commercial or industrial method to prepare H_2 :**

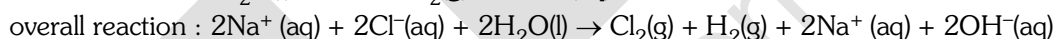
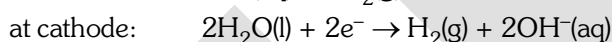
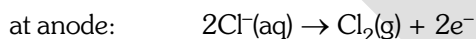
The commonly used processes are outlined below:

(i) Electrolysis of acidified water using platinum electrodes gives hydrogen.

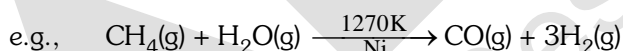
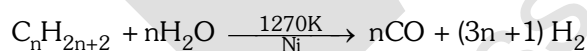


(ii) High purity ($>99.95\%$) dihydrogen is obtained by electrolysis of warm aqueous barium hydroxide solution between nickel electrodes.

(iii) It is obtained as a by product in the manufacture of sodium hydroxide and chlorine by the electrolysis of brine solution. During electrolysis, the reactions that take place are:



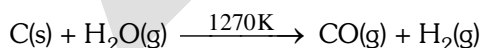
(iv) **From hydrocarbons :** Reaction of steam on hydrocarbons or coke at high temperatures in the presence of catalyst yields hydrogen.



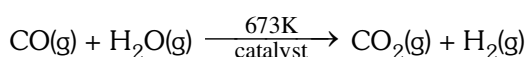
The mixture of CO and H_2 is called *water gas*. As this mixture of CO and H_2 is used for the synthesis of methanol and a number of hydrocarbons, it is also called *synthesis gas* or '**syngas**'. Nowadays 'syngas' is produced from sewage, saw-dust, scrap wood, newspapers etc.

The process of producing 'syngas' from coal is called '*coal gasification*'.

(v) **Bosch process :**

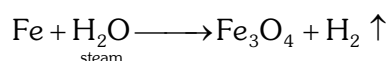


The production of dihydrogen can be increased by reacting carbon monoxide of syngas mixtures with steam in the presence of iron chromate as catalyst.



This is called *water-gas shift reaction*.

(vi) **Lane's process :**



4.3 PHYSICAL PROPERTIES OF HYDROGEN :

- Hydrogen is the lightest, colorless, odourless and tasteless gas. It is sparingly soluble in water. It is inflammable and less reactive gas.
- Its m.p. (-259.2°C) and b.p. (-252°C) are very low indicating less intermolecular attraction. Due to low m.p. liquid hydrogen is used as a **cryogenic fluid** (to produce low temperature).
- H—H bond energy [$104 \text{ Kcal mol}^{-1}$] and 436 KJ/m
- H – H bond length [74 pm] so H_2 is less reactive and require high temp for reaction.

4.4 USES OF HYDROGEN :

- Hydrogenation of vegetable oil to form solid fats i.e. vanaspati ghee.
- In liquid form as a rocket fuel. (Liquid H_2 + Liquid O_2)
- In a air ship of balloons as a mixture of Hydrogen & Helium [$15\% \text{H}_2$ + $85\% \text{He}$]
- Formation of different compounds.

Like $\rightarrow \text{NH}_3$, (Haber process) alkane, alcohol and other hydrocarbons

BEGINNER'S BOX-1

- Hydrogen does not combine with
(1) Antimony (2) Sodium (3) Bismuth (4) Helium
- Which of the following produces hydrolith with dihydrogen
(1) Mg (2) Al (3) Cu (4) Ca
- Hydrogen combines with other elements by
(1) Losing an electron (2) Gaining an electron
(3) Sharing an electron (4) Losing, gaining or sharing electron
- Hydrogen acts as a reducing agent and thus resembles
(1) Halogen (2) Noble gas (3) Radioactive elements (4) Alkali metals
- Hydrogen readily combines with non-metals and thus it shows its
(1) Electronegativity character (2) Electropositive character
(3) Both (1) and (2) (4) None of these
- The oxidation states shown by hydrogen are
(1) -1 only (2) Zero only (3) $+1, -1, 0$ (4) $+1$ only
- Hydrogen readily combines with metals and thus shows its
(1) Electropositive character (2) Electronegative character
(3) Both (1) and (2) (4) None of these
- In which of the compounds does hydrogen have an oxidation state of -1
(1) CH_4 (2) NH_3 (3) HCl (4) CaH_2
- Match List I (Fuels) with List II (composition) and select the correct answer using the codes given below the lists

List I (Fuels)

List II (Composition)

- | | |
|-----------------|---|
| A. Water gas | i. A mixture of CO and N_2 |
| B. Producer gas | ii. Methane |
| C. Coal gas | iii. A mixture of CO and H_2 |
| D. Natural gas | iv. A mixture of CO, H_2 , CH_4 and CO_2 |

- | | A | B | C | D |
|-----|-----|-----|----|----|
| (1) | iii | i | iv | ii |
| (2) | iii | i | ii | iv |
| (3) | i | iii | iv | ii |
| (4) | iii | ii | iv | i |

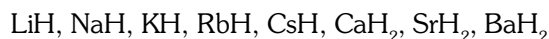
4.5 HYDRIDES

The binary compounds of hydrogen with different elements are called hydrides.

These are of three types :

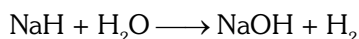
(a) Ionic/Salt like/Saline hydrides :

- ◆ Compounds of hydrogen with s-block elements except beryllium & magnesium are called ionic hydrides.

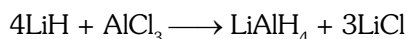


$\text{BeH}_2, \text{MgH}_2$ are covalent polymeric hydride.

- ◆ Structure of these hydrides are similar to rock salt, so they are also called salt like/saline hydrides.
- ◆ Down the group, atomic size $\uparrow$ Lattice energy $\downarrow$ stability $\downarrow$ Melting point $\downarrow$ Boiling point $\downarrow$
- ◆ On electrolysis of these hydrides, hydrogen is liberated at anode.
- ◆ On reaction with water these hydrides will form hydrogen



- ◆ These hydrides form complex hydrides which are very good reducing agents.

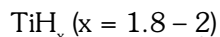
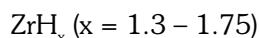


$\text{LiAlH}_4 \longrightarrow$ Lithium aluminium hydride

$\text{NaBH}_4 \longrightarrow$ Sodium borohydride

(b) Metallic / Interstitial hydrides :

- ◆ They are the compounds of d & f-block elements. In these hydrides hydrogen occupies interstitial sites present in metallic lattice, so they are called interstitial hydrides.
- ◆ Properties of these hydrides are similar to parent metals, so they are also known as metallic hydrides.
- ◆ These hydrides are non stoichiometric in nature (i.e. having variable composition)



- ◆ Metals of group 7,8,9 don't form any hydrides so this particular part of periodic table is known as hydride gap.
- ◆ In group-6 only one hydride CrH is formed.

(c) Covalent/Molecular hydrides

- ◆ They are the compounds of hydrogen with p-block elements $\text{CH}_4, \text{NH}_3, \text{H}_2\text{O}, \text{HF}$, etc.
- ◆ These hydrides exist as molecules, so they are also known as molecular hydrides. These hydrides are non-conductor of electricity.

These hydrides are again divides into 3 categories.

(a) Electron deficient hydrides :

They are the hydrides of group 13 elements.

e.g. $\text{BH}_3, \text{AlH}_3, \text{GaH}_3$ – In these hydrides central element does not have complete octet. i.e. why they are called electron deficient compounds.

(b) Electron precise hydrides :

They are the hydrides of group 14 element.

e.g. $\text{CH}_4, \text{SiH}_4, \text{GeH}_4$ – In these type of hydrides central element has $8e^-$ in its outer most shell.

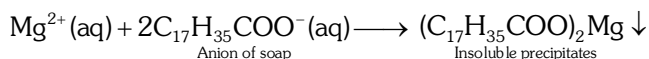
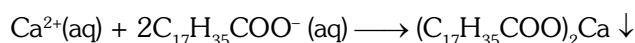
(c) Electron rich hydrides :

These are the hydrides of group 15, 16 and 17

e.g. $\ddot{\text{N}}\text{H}_3, \text{H}_2\ddot{\text{O}}, \text{H}\ddot{\text{F}}:$ – In these hydrides lone pair are present on central element which can be given to others. So they are called electron rich hydrides.

4.6 HARD AND SOFT WATER

Water which produces lather with soap is **soft water** while water which does not produces lather with soap is **hard water**. The hardness of water is caused by presence of bicarbonates, chlorides and sulphates of calcium and magnesium.

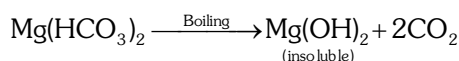
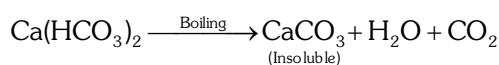


Hardness of water are of two types :

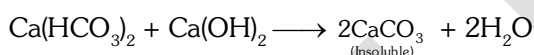
- (a) Temporary hardness (b) Permanent hardness

- (a) **Temporary hardness :** This is due to the presence of bicarbonates of calcium and magnesium.

Temporary hardness in water is easily removed by boiling, as the bicarbonates decompose readily and the insoluble compounds are precipitated.



Temporary hardness can also be removed by **Clark's process** which involves the addition of slaked lime $[\text{Ca}(\text{OH})_2]$.

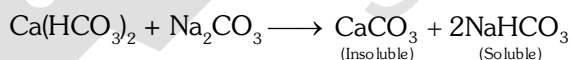
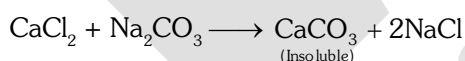


It is essential to add only the calculated amount of $\text{Ca}(\text{OH})_2$ because excess will cause artificial hardness.

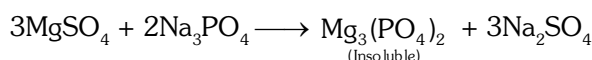
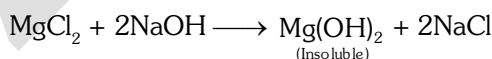
- (b) **Permanent hardness :** This is due to the presence of sulphates or chlorides of both of calcium and magnesium. This type of hardness cannot be removed by boiling or by the addition of slaked lime.

The various water softeners are :

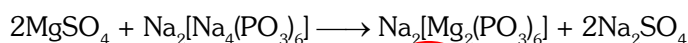
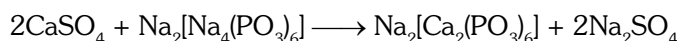
- (i) **Washing soda :** It removes both the temporary and permanent hardness by converting soluble calcium and magnesium compounds into insoluble compounds.



In place of sodium carbonate, caustic soda or sodium phosphate can also be used.



- (ii) **Calgon** : The complex salt of metaphosphoric acid, sodium hexametaphosphate (NaPO_3)₆, is known as **calgon**. It is represented as $\text{Na}_2[\text{Na}_4(\text{PO}_3)_6]$. Calcium and magnesium salts present in hard water react with calgon to give complex salts.

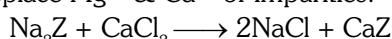


- (iii) **Permutit process :** Permutit is hydrated sodium aluminosilicate $[\text{Na}_2\text{Al}_2\text{Si}_2\text{O}_8 \cdot x\text{H}_2\text{O}]$ or $[\text{Na}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot x\text{H}_2\text{O}]$

Permutit is also known as sodium zeolite (Na_2Z) means Zeolite is $\text{Al}_2\text{Si}_2\text{O}_8 \cdot x\text{H}_2\text{O}$.

In this process when hard water is poured into chamber, it may contain organic impurities like plant. These impurities can be removed by gravel.

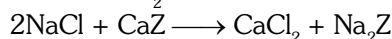
On moving upwards hard water will react with Na_2Z during this reaction Na^+ ions of Na_2Z will replace Mg^{+2} & Ca^{+2} of impurities.



NaCl is dissolved in water & water becomes soft.

This soft water is fit for washing purpose but not fit for drinking purpose.

After some time when Na_2Z is completely converted into CaZ . Process is stopped and for regeneration of Na_2Z . We use NaCl solution.



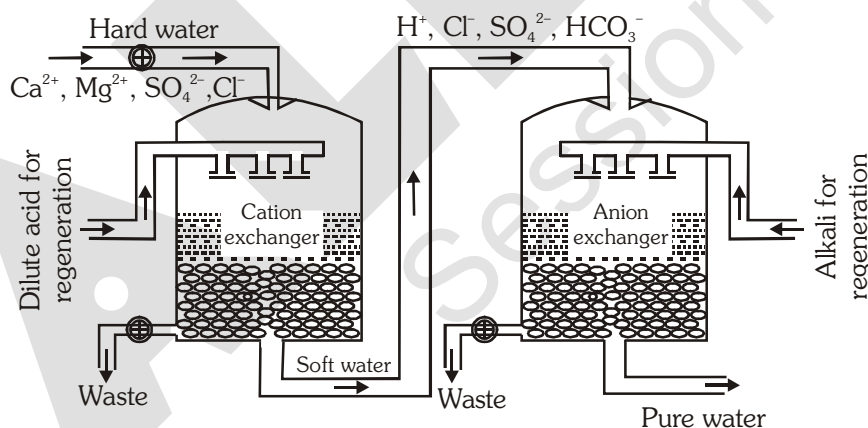
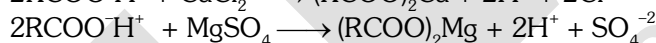
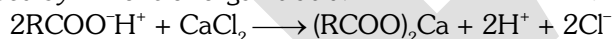
- (iv) **Ion exchange resin :** This process removes both temporary and permanent hardness. Also, by this process we can remove both cation & anion of hardness.

This process contain two chambers.

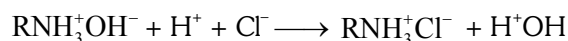
(a) Cation exchange resin : This resin contains granular insoluble organic acid having giant molecules with RCOO^-H^+ group.

(b) Anion exchange resin : This resin contain giant organic molecules with basic groups derived from amines with $\text{RNH}_3^+\text{OH}^-$

Process : When hard water is poured into first chamber the cation of hardness (Mg^{+2} , Ca^{+2}) removed by H^+ ions of organic acid.



This water becomes soften but not used for drinking purpose because this water contain the impurity of acid. To remove anion of hardness this acidic water then passed through another bed containing anion exchanger. This exchanger removes anion like Cl^- , SO_4^{2-} & HCO_3^- .

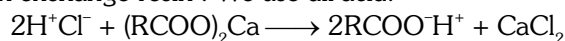


This water is free from impurities & can be used for drinking purpose.

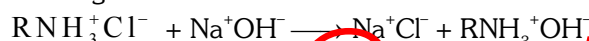
After sometime when both resin gets exhausted process is stopped.

Regeneration of resin :

- (i) Cation exchange resin : We use dil acid.



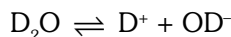
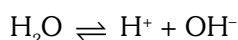
- (ii) Anion exchange resin : We use dil NaOH solution



4.7 HEAVY WATER (D₂O)

Method of preparation :

Repeated electrolysis of H₂O : On electrolysis of water (impure) H₂O dissociate into H⁺ & OH⁻ while a fractional part of D₂O will dissociate into D⁺ & OD⁻



D⁺ & OD⁻ due to more mass have less mobility i.e. why H⁺ & OH⁻ will move towards cathode & anode respectively while D⁺ & OD⁻ will be in solution.

This process is repeated six times.

Properties of Heavy water :

Physical properties :

- (a) Heavy water is a colourless, odourless and tasteless mobile liquid.
- (b) Nearly all the physical constants are higher than the corresponding values of ordinary water.
(dielectric constant order H₂O > D₂O).

Chemical properties :

- (a) Heavy water is chemically similar to ordinary water. However, D₂O reacts more slowly than H₂O in chemical reactions.
- (b) All chemical properties are same, but reactions are slow.

Uses : As a neutron moderator : Fission in uranium-235 is brought by slow speed neutrons. The substances which are used for slowing down the speed of neutrons are called moderators. Heavy water is used for this purpose in nuclear reactors.

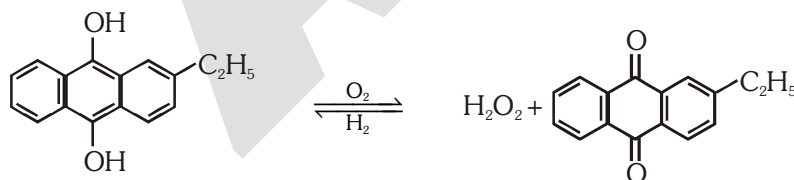
4.8 H₂O₂ (HYDROGEN PEROXIDE)

Method of preparation

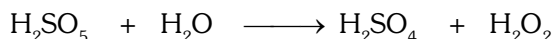
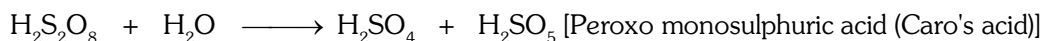
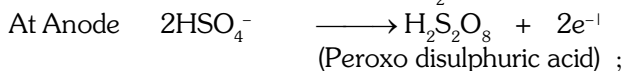
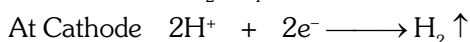
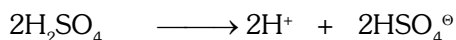
- (i) Acidifying barium peroxide and removing excess water by evaporation under reduced pressure gives hydrogen peroxide.



- (ii) Industrial Method : Auto oxidation of 2 ethyl anthraquinol (cyclic process) :



- (iii) Electrolytic Process : (Used 50%) H₂SO₄ in electrolytic cell using Pt as anode and graphite as cathode.

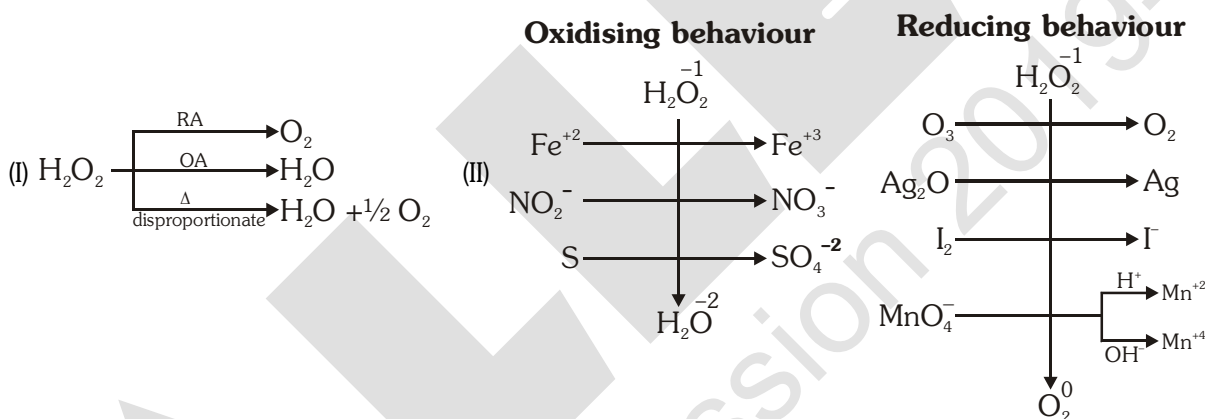
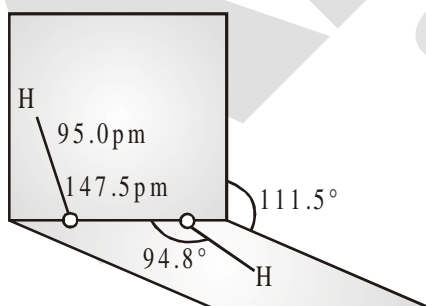


Physical properties :

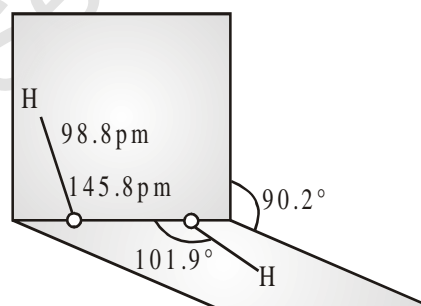
- (i) Pure H_2O_2 is colourless, odourless liquid and impure with bluish layer.
- (ii) It has more Hydrogen bonding than H_2O . So, order of boiling point is $\text{H}_2\text{O}_2 > \text{D}_2\text{O} > \text{H}_2\text{O}$

$$\text{Order of melting point} = \left[\begin{array}{cc} \text{H}_2\text{O}_2 < \text{H}_2\text{O} \\ -0.35^\circ\text{C} & 0^\circ\text{C} \end{array} \right]$$

- (iii) It is soluble in H_2O , alcohol and ether.
- (iv) It has bitter taste and harmful for skin.
- (v) It is a dibasic weak acid.
- (vi) It has oxidising as well as reducing property.
- (vii) H_2O_2 easily decompose in presence of light and temperature, so H_2O_2 always kept in dark bottles and kept at cool places & small amount of inhibitor is added like R-OH, glycerol & Acetone.
- (viii) 30% solution of H_2O_2 is called **Perhydrol**.
- (ix) H_2O_2 decomposes at its boiling point hence its distillation is carried out under reduced pressure.

Chemical properties :**(A) Oxidising and Reducing behaviour of H_2O_2** **Structure of hydrogen peroxide**

(a) Gas phase

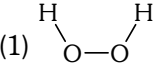
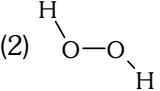
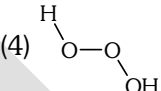


(a) Solid phase

Uses

- | | |
|--|--|
| (i) Bleaching agent | (ii) Hair dying |
| (iii) $\text{H}_2\text{O}_2 + \text{N}_2\text{H}_4$ as Rocket propellant | (iv) H_2O_2 as oxidant and reductant |
| (v) Antiseptic | |

BEGINNER'S BOX-2

- Among the following, identify the compound which cannot act as both oxidising and reducing agents
(1) H_2O_2 (2) H_2 (3) SO_2 (4) HCl
- In which of the following reaction hydrogen peroxide is a reducing agent
(1) $2\text{FeCl}_2 + 2\text{HCl} + \text{H}_2\text{O}_2 \longrightarrow 2\text{FeCl}_3 + 2\text{H}_2\text{O}$ (2) $\text{Cl}_2 + \text{H}_2\text{O}_2 \longrightarrow 2\text{HCl} + \text{O}_2$
(3) $2\text{HI} + \text{H}_2\text{O}_2 \longrightarrow 2\text{H}_2\text{O} + \text{I}_2$ (4) $\text{H}_2\text{SO}_3 + \text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
- The structure of H_2O_2 is
(1)  (2)  (3) $\text{H}-\text{O}-\text{O}-\text{H}$ (4) 
- The oxide that gives hydrogen peroxide on the treatment with a dilute acid is
(1) MnO_2 (2) PbO_2 (3) Na_2O_2 (4) TiO_2
- Temporary hardness may be removed from water by adding
(1) CaCO_3 (2) $\text{Ca}(\text{OH})_2$ (3) CaSO_4 (4) HCl
- Temporary hardness of water can be removed by
(1) Addition of potassium permanganate (2) Boiling
(3) Filtration (4) Addition of chlorine
- When zeolite (Hydrated sodium aluminium silicate) is treated with hard water the sodium ions are exchanged with
(1) OH^- ions (2) SO_4^{2-} ions (3) Ca^{2+} ions (4) H^+ ions
- The velocity of neutrons in nuclear reactor is slowed down by
(1) Heavy water (2) Ordinary water (3) Zinc rod (4) Fused caustic soda
- When temporary hard water containing $\text{Mg}(\text{HCO}_3)_2$ is boiled the ppt. formed is of
(1) MgCO_3 (2) MgO (3) $\text{Mg}(\text{OH})_2$ (4) None of these
- Which of the following can effectively remove all types of hardness of water
(1) Soap (2) Washing soda (3) Slaked lime (4) None of these

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7	8	9	
	Ans.	4	4	4	4	2	3	2	4	1	

BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	7	8	9	10
	Ans.	4	2	2	3	2	2	3	1	3	2

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

BASED ON HYDROGEN

- Out of the following metals which will give H_2 on reaction with NaOH :
 I : Zn, II : Mg,
 III : Al, IV : Be
 (1) I, II, III, IV (2) I, III, IV
 (3) II, IV (4) I, III
- The gas used in the hydrogenation of oils, in presence of nickel as a catalyst is :
 (1) methane (2) ethane
 (3) ozone (4) hydrogen
- Hydrogen has the tendency to loose one e^- and form H^+ , In this respect it resembles with :
 (1) Alkali metal (2) Carbon
 (3) Alkaline earth metal (4) Halogens
- H_2 gas can not be prepared by :-
 (1) $Be + NaOH$ (2) $Na + NaOH$
 (3) $Mg + NaOH$ (4) By (2 & 3) method
- Deuterium, an isotope of hydrogen is :-
 (1) Radioactive (2) Non radioactive
 (3) Heaviest (4) Lightest
- Hydride gap in periodic table is from :-
 (1) Group 7 to group 9
 (2) Group 5 to group 7
 (3) Group 4 to group 6
 (4) Group 7 to group 10
- Which of the following reaction is called water gas shift reaction ?
 (1) $C(s) + H_2O(g) \longrightarrow CO(g) + H_2(g)$
 (2) $3Fe(s) + 4H_2O(\text{steam}) \longrightarrow Fe_3O_4 + 4H_2(g)$
 (3) $CH_4(g) + H_2O(g) \xrightarrow[1270\text{ K}]{Ni} CO(g) + 3H_2(g)$
 (4) $CO(g) + H_2O(g) \xrightarrow[FeCrO_4]{773\text{ K}} CO_2(g) + 3H_2(g)$

BASED ON WATER

- Which is true statement about D_2O and H_2O :-
 (1) D_2O has lower dielectric constant than H_2O
 (2) NaCl is more soluble in D_2O than in H_2O
 (3) both are correct
 (4) none is correct
- The reactions of heavy water are slow
 The reason is :-
 (1) Heavy water is associated
 (2) Heavy water is dissociated
 (3) High bond energy of D-O bond
 (4) Heavy water is of lower mass

- Hard water when passed through ion exchange resin containing RCOOH group, becomes free from :-
 (1) Cl^- (2) SO_4^{2-}
 (3) H_3O^+ (4) Ca^{+2}
- Permutit is a technical name given to :-
 (1) Aluminates of Ca and Na
 (2) Hydrated silicates of Al and Na
 (3) Silicates of Ca and Na
 (4) Silicates of Ca and Mg
- The formula of sodium zeolite which is used in permutit process for softening water is :-
 (1) $Na_2O \cdot Al_2O_3 \cdot Si_2O_4 \cdot xH_2O$
 (2) $Na_2 \cdot Al_2 \cdot Si_2O_4 \cdot xH_2O$
 (3) $Na_2O \cdot AlO_3 \cdot SiO_4 \cdot xH_2O$
 (4) $K_2Al_2SiO_8 \cdot xH_2O$
- The compound sodium hexameta phosphate $Na_2[Na_4(PO_3)_6]$ is called calgon because :-
 (1) It was developed by the scientist
 (2) It was developed first in California
 (3) It refers to calcium gone
 (4) It is based on the name of the company which developed it.
- Permanent hardness in water due to presence of :-
 (1) Ca^{+2}, Mg^{+2} (2) $CaCl_2, MgCl_2$
 (3) $CaCO_3, MgCO_3$ (4) All
- Temporary unstable hardness of water due to presence of :-
 (1) $CaCl_2, MgSO_4$
 (2) Ca^{+2}, Mg^{+2}
 (3) $K^+, CaCO_3$
 (4) $Ca(HCO_3)_2, Mg(HCO_3)_2$
- In which of the following method of the removal of hardness, Ca^{+2} and Mg^{+2} are not separated from sample of hard water ?
 (1) By boiling of temporary hard water
 (2) Addition of sodium carbonate
 (3) Using sodium hexa meta phosphate
 (4) Synthetic resins and zeolite method

BASED ON HYDROGEN PEROXIDE

17. H_2O_2 is used but not as :-
 (1) oxidant, reductant (2) bleaching agent
 (3) antiseptic (4) catalyst
18. Which of the following is a true structure of H_2O_2 :-
- (1) $\text{H}-\overset{180^\circ}{\text{O}}-\text{O}-\text{H}$ (2) $\text{H}-\overset{94^\circ}{\text{O}}-\text{O}-\text{H}$
- (3) $\begin{array}{c} \text{H} \\ \diagup \\ \text{O} \rightarrow \text{O} \\ \diagdown \\ \text{H} \end{array}$ (4) $\begin{array}{c} \text{H} \\ \diagup \\ \text{O}=\text{O} \\ \diagdown \\ \text{H} \end{array}$
19. In the reaction $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$, oxidation state of oxygen changes as :-
 (1) Only -1 to -2
 (2) Only -1 to zero
 (3) Both of the above
 (4) -1 to -3

20. The dipole moment of H_2O_2 is 2.1D. This indicates that the structure of H_2O_2 is :-
 (1) Linear (2) Non-linear
 (3) Symmetrical (4) None
21. Bleaching action of H_2O_2 is due to its :-
 (1) Oxidising nature
 (2) Reducing nature
 (3) Acidic nature
 (4) Thermal instability
22. Correct order of boiling point is :-
 (1) $\text{H}_2 > \text{H}_2\text{O}_2 > \text{D}_2\text{O} > \text{H}_2\text{O} > \text{D}_2$
 (2) $\text{H}_2\text{O}_2 > \text{H}_2 > \text{D}_2\text{O} > \text{H}_2\text{O} > \text{D}_2$
 (3) $\text{H}_2\text{O}_2 > \text{D}_2\text{O} > \text{H}_2\text{O} > \text{D}_2 > \text{H}_2$
 (4) $\text{H}_2\text{O}_2 > \text{D}_2\text{O} > \text{H}_2\text{O} > \text{H}_2 > \text{D}_2$
23. $\text{H}_2\text{O} < \text{H}_2\text{O}_2$ order is incorrect for :-
 (1) Boiling point (2) Acidic nature
 (3) Dipole moment (4) Strength of H-bond

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	4	1	4	2	1	4	1	3	4	2	1	3	2	4
Que.	16	17	18	19	20	21	22	23							
Ans.	3	4	2	3	2	1	3	4							

EXERCISE-II (Previous Year Questions)**AIPMT/NEET & AIIMS (2006-2018)****AIPMT 2010**

1. Some statements about heavy water are given below:
- Heavy water is used as a moderator in nuclear reactor.
 - Heavy water is more associated than ordinary water
 - Heavy water is more effective solvent than ordinary water
- Which of the above statements are correct ?
- (a), (b) and (c)
 - (b) and (c)
 - (a) and (c)
 - (a) and (b)

NEET-I 2016

2. Which of the following statements about hydrogen is **incorrect** ?
- hydrogen has three isotopes of which tritium is the most common.
 - Hydrogen never acts as cation in ionic salts
 - Hydronium ion, H_3O^+ exists freely in solution
 - Dihydrogen does not act as a reducing agent

AIIMS 2017

3. Which of the following is not correct about D_2O .
- 100 ml of its weights is more than 100 gm
 - acts as a moderator
 - boiling point of D_2O is 376 K
 - It is used in exchange reactions for the study of reaction mechanism

AIIMS 2018

4. When $\text{Ba}(\text{OH})_2$ is mixed with H_2O_2 and ClO_2 , then the product formed is
- $\text{Ba}(\text{ClO}_3)_2$
 - $\text{Ba}(\text{ClO}_2)_2$
 - $\text{Ba}(\text{OH})_2$
 - HClO_3

EXERCISE-II (Previous Year Questions)**ANSWER KEY**

Que.	1	2	3	4
Ans.	4	1,4	3	1

EXERCISE-III (Analytical Questions)

- Calgon is an industrial name given to :-
(1) normal sodium phosphate
(2) sodium meta-aluminate
(3) sodium hexametaphosphate
(4) hydrated sodium aluminium silicate
- Hydrogen may be prepared by heating a solution of caustic soda with :-
(1) Mg (2) Zn (3) Fe (4) Ag
- Hydrogen peroxide has a :-
(1) linear structure
(2) closed chain structure
(3) closed book type structure
(4) half open book type structure
- Hydrogen peroxide is a :-
(1) liquid (2) gas
(3) solid (4) semi-solid
- One of the following is an incorrect statement, point it out.
(1) Permanent hardness can be removed by boiling water
(2) Hardness of water effects soap consumption
(3) Temporary hardness is due to bicarbonates of Ca and Mg
(4) Permanent hardness is due to the soluble SO_4^{2-} , Cl^- of Ca and Mg
- Water is said to be permanently hard when it contains:-
(1) Chlorides and sulphates of Mg and Ca
(2) Bicarbonates of Na and K
(3) Carbonates of Na and K
(4) Phosphates of Na and K
- Which is the lightest gas :-
(1) Nitrogen (2) Helium
(3) Oxygen (4) Hydrogen
- The adsorption of hydrogen by platinum is known as :-
(1) Hydrogenation (2) Reduction
(3) Dehydrogenation (4) Occlusion
- False statement for H atom :-
(1) It resembles halogens in some properties
(2) It resembles alkali metals in some property
(3) It can not be placed in 1st group of periodic table
(4) It can be placed in 17th group of periodic table

Check Your Understanding

- Boiling point of water is high due to presence of :-
(1) H-bonding
(2) Bent structure
(3) It high dielectric constant
(4) None
- D_2O is used more in :-
(1) Chemical industry
(2) Nuclear moderator
(3) Pharmaceutical preparation
(4) Insecticide preparations
- Which one of the following removes temporary hardness of water :-
(1) Slaked lime (2) Plaster of paris
(3) CaCO_3 (4) Hydrolith
- Heavy water is used in nuclear reactors as :-
(1) Source of α - particles
(2) Slowing down the speed of high energy neutrons
(3) Transporting heat of the reactor
(4) Heating purposes
- Nascent hydrogen consists of :-
(1) Hydrogen atoms with excess of energy
(2) Hydrogen molecules with excess energy
(3) Hydrogen ions in excited state
(4) Solvated protons
- Match list I with list II and select the correct answer using the codes given below the lists :-

List I	List II
1. Heavy water	A. Bicarbonates of Mg and Ca in water
2. Temporary hard water	B. No foreign ions in water
3. Soft water	C. D_2O
4. Permanent hard water	D. Sulphates and chlorides of Mg and Ca in water

 (1) 1-C, 2-D, 3-B, 4-A (2) 1-B, 2-A, 3-C, 4-D
 (3) 1-B, 2-D, 3-C, 4-A (4) 1-C, 2-A, 3-B, 4-D
- Shape of O_2F_2 is similar to that of :
(1) C_2F_2 (2) H_2O_2 (3) H_2F_2 (4) C_2H_2

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	4	1	1	1	4	4	3	1	2	1	2	1	4
Que.	16														
Ans.	2														

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :** Decomposition of H_2O_2 is a disproportionation reaction.
Reason : H_2O_2 molecule simultaneously undergoes both oxidation and reduction.
 (1) A (2) B (3) C (4) D
- Assertion :** The colour of old lead paintings can be restored by washing with dilute solution of H_2O_2 .
Reason : Black lead sulphide is oxidised by H_2O_2 to white lead sulphate
 (1) A (2) B (3) C (4) D
- Assertion :** The O—O bond length in H_2O_2 is shorter than that in O_2F_2 .
Reason : H_2O_2 is ionic compound
 (1) A (2) B (3) C (4) D
- Assertion :** Hard Water is not fit for washing clothes.
Reason : It contains Ca^{+2} & Mg^{+2} ion which can forms ppt with soap.
 (1) A (2) B (3) C (4) D
- Assertion :** Rate of reaction $CD_4 + Cl_2 \rightarrow$ Slow ; $CH_4 + Cl_2 \rightarrow$ Fast
Reason : Due to isotopic effect.
 (1) A (2) B (3) C (4) D
- Assertion :-** Saline hydrides are non volatile, non conducting & crystalline solids.
Reason : Saline hydrides are compounds of hydrogen with most of the p-block elements.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6
Ans.	1	1	4	1	1	3



CLASSIFICATION AND NOMENCLATURE



S. No.	CONTENTS	Page
1.	Format for IUPAC name	2
2.	Exercise-I (Conceptual Questions)	6
3.	Exercise-II (Previous Years Questions)	11
4.	Exercise-III (Analytical Questions)	14
5.	Exercise-IV (Assertion & Reason)	17

NEET SYLLABUS

General introduction, classification and IUPAC nomenclature of organic compounds.

@neetfocusquizz

OBJECTIVES

After studying this unit, we will be able to :

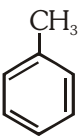
- *understand reasons for tetravalence of carbon and shapes of organic molecules;*
- *write structures of organic molecules in various ways;*
- *classify the organic compounds;*
- *name the compounds according to IUPAC system of nomenclature and also derive their structures from the given names;*

"Don't take rest after your first victory because if you fail in second, more lips are waiting to say that your first victory was just luck."

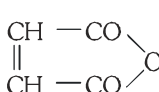
A.P.J. Abdul Kalam

CLASSIFICATION & NOMENCLATURE

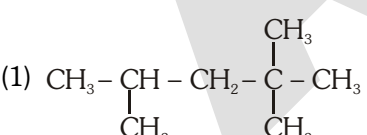
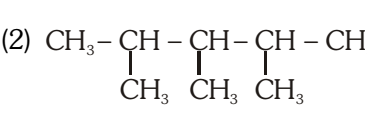
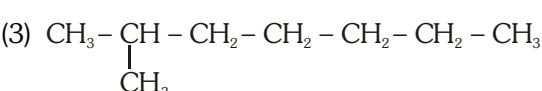
BEGINNER'S BOX-1

- How many 1° , 2° & 3° H atoms are present in  [Toluene] respectively :-
 (1) 3, 0, 5 (2) 3, 5, 0 (3) 4, 3, 0 (4) 0, 5, 3
- What is hybridisation of each carbon atom in following compound
 $\text{HC} \equiv \text{C} - \text{CH} = \text{CH} - \text{CH}_3$
 (1) $sp, sp^2, sp^2, sp^2, sp^3$ (2) sp, sp, sp^2, sp^2, sp^3
 (3) sp, sp, sp^2, sp^3, sp^3 (4) $sp, sp^2, sp^2, sp^3, sp^3$
- Which one is not correct for a homologous series -
 (1) All members have a general formula
 (2) All members have same chemical properties
 (3) All members have same physical properties
 (4) All members have same functional group

BEGINNER'S BOX-2

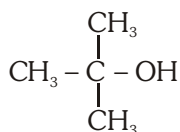
- How many carbon atom are present in third homologue of methyl ether.
 (1) 1 (2) 2 (3) 3 (4) 4
- Which of the following is not a hetero cyclic compound
 (1) Thiophene (2) Furan (3) Benzene (4) Pyridine
- In structure , how many hetero atoms are present ?
 (1) 1 (2) 2 (3) 3 (4) 4

BEGINNER'S BOX-3

- Which of the followings is incorrect name :-
 (1) Isopropyl (2) Ter. butyl (3) Neo butyl (4) Neo pentyl
- Which of the followings is secondary radical :-
 (1) $\text{CH}_2=\text{CH}-$ (2) $(\text{CH}_3)_3\text{C}-$ (3) C_6H_5- (4) $\text{CH}_3-(\text{CH}_2)_2-\text{CH}_2-$
- Which of the followings is isooctane :-
 (1) 
 (2) 
 (3) 
 (4) None

BEGINNER'S BOX-4

1. Common name of given compound is :-



- (1) Neobutyl alcohol (2) Isobutyl alcohol
 (3) Tertiary butyl alcohol (4) Secondary butyl alcohol
2. Which of the following is Crotonic acid ?
 (1) $\text{CH}_2 = \text{CH} - \text{COOH}$ (2) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CHO}$
 (3) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{COOH}$ (4) $\text{CH}_3 - \text{CH} = \text{CH} - \text{COOH}$
3. What is derived name of Neopentyl alcohol :-
 (1) Isopropyl carbinol (2) n-Butyl carbinol
 (3) Tertiary butyl carbinol (4) Ethyl methyl carbinol

Format for IUPAC name :

s - prefix	+	p - prefix	+	word root	+	p - suffix	+	s - suffix
Substituents with locants		cyclo		Alk word according to carbon in parent C chain		-ane -ene -yne		According to main functional group given in priority table

(a) **Locant** : Locants are separated by (,) comma.

- Locants and alphabets are separated by hyphen (-). [2, 3 - dimethyl pentane]
- di, tri, iso, neo and cyclo are neither separated by comma nor by hyphen

(b) **Prefix** :- According to substituents .

Prefix (es) are written in alphabetical order before root word.

Prefix ← $\begin{cases} 1^\circ \text{ or p - prefix} \\ 2^\circ \text{ or sec. - prefix} \end{cases}$

Cyclo is 1° prefix and used for cyclic compound.

2° prefix is used for substituents and written before 1° prefix.

For acyclic compounds : 2° prefix + Root word + 1° suffix + 2° suffix.

Substituents	Prefix
— R	Alkyl group
— X (F, Cl, Br, I)	Halo
— O — N=O	Nitrite
— CH_2OH	Hydroxy methyl
— NHC_2H_5	Ethyl amino

Substituents	Prefix
— OR	Alkoxy
— $\text{N} \begin{smallmatrix} \text{O} \\ \parallel \\ \text{O} \end{smallmatrix}$	Nitro
— N = O	Nitroso
— CH_2Cl	Chloro methyl

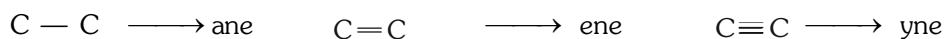
(c) **Word root** : According to number of carbons in parent C-chain.

Number of carbons	Root word
1	Meth
2	Eth
3	Prop
4	But
5	Pent

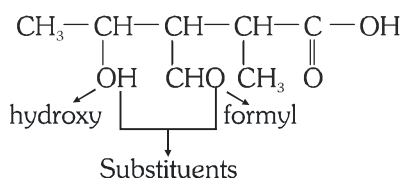
Number of carbons	Root word
6	Hex
7	Hept
8	Oct
9	Non
10	Dec

Number of carbons	Root word
11	Undec
12	dodec
13	tridec

(d) **Primary suffix :-** According to saturation and unsaturation.



(e) **Secondary Suffix :-** According to senior most of F. G.



3-Formyl-4-hydroxy-2-methyl pentanoic acid

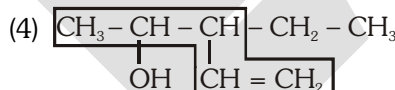
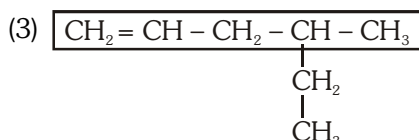
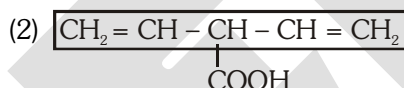
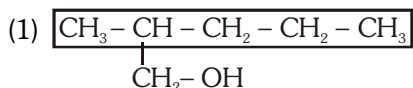
S. NO.	Functional group	Prefix	Suffix
1.	— (C) OOH (carboxylic acid) — COOH	× carboxy	oic acid carboxylic acid
2.	— SO ₃ H (sulphonic acid)	sulpho	sulphonic acid
3.	$\begin{array}{c} \text{O} \\ \\ -(\text{C}) > \text{O} \\ \\ -(\text{C}) > \text{O} \\ \\ \text{O} \end{array}$ (anhydride)	×	oic anhydride
4.	— (C) OOR (ester) — COOR	× alkoxy carbonyl or carbalkoxy	alkyl ----- oate alkyl ----- carboxylate
5.	— (C) OX (acid halide) — COX	× halo formyl	oyl halide carbonyl halide
6.	— (C) ONH ₂ (amide) — CONH ₂	× carbamoyl	amide carboxamide
7.	— (C) N (cyanide) — CN	× cyano	Nitrile carbonitrile
8.	— N ≡ C (isocyanide)	isocyano/carbyl amino	isonitrile/carbyl amine
9.	— (C) HO (aldehyde) — CHO	oxo formyl	al carbaldehyde
10.	— (C) — (Ketone) $\begin{array}{c} \text{O} \\ \\ -(\text{C}) - \end{array}$	keto/oxo	one
11.	— OH (alcohol)	hydroxy	ol
12.	— SH (thio alcohol)	mercapto	thiol
13.	— NH ₂ (amine)	amino	amine

Note : (C) atom written in brackets means that it has been included in the parent chain.

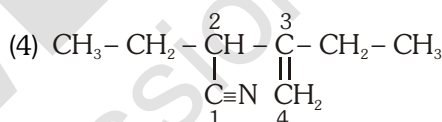
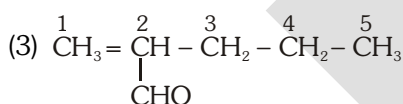
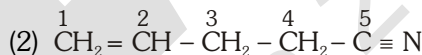
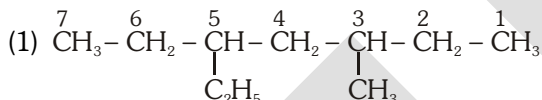
SUBSTITUENTS	PREFIX	SUBSTITUENTS	PREFIX
— R	alkyl	— X	halo
— NH ₂	amino	$\text{— N} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{O} \end{array}$	nitro
— O—N=O	nitrito	— N = O	nitroso
— OCH ₂ CH ₃	ethoxy	— CH ₂ —OH	hydroxy methyl
— CH ₂ —Cl	chloro methyl	— NH—CH ₃	methyl amino
— S—	thio		
$\text{CH}_3\text{—}\overset{\text{O}}{\parallel}\text{C—O—}$	acetoxy/ethanoyloxy	$\text{CH}_3\text{CH}_2\text{—}\overset{\text{O}}{\parallel}\text{C—O—}$	propanoyloxy
$\text{C}_6\text{H}_5\text{—}\overset{\text{O}}{\parallel}\text{C—O—}$	benzoyloxy	—OR	Alkoxy
		—OC ₆ H ₅	Phenoxy

BEGINNER'S BOX-5

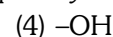
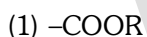
1. Which of the following selected chain is correct :-



2. Which of the following has correct numbering according IUPAC :-

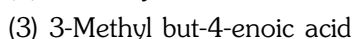
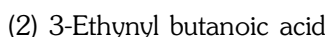
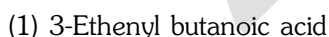
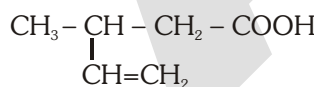


3. Which of the following functional group has highest priority according to priority table :-

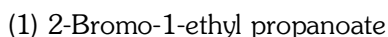
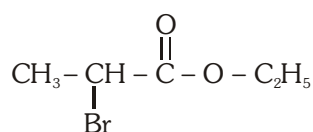


BEGINNER'S BOX-6

1. Correct IUPAC name of compound is :-



2. Correct IUPAC name of compound is :-



3. IUPAC name of $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{O}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{CH}_3$ is :-

- (1) Acetic anhydride (2) Methanoic anhydride
(3) Ethanoic methanoic anhydride (4) Ethanoic anhydride

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3
	Ans.	2	2	3
BEGINNER'S BOX-2	Que.	1	2	3
	Ans.	4	3	1
BEGINNER'S BOX-3	Que.	1	2	3
	Ans.	3	3	1
BEGINNER'S BOX-4	Que.	1	2	3
	Ans.	3	4	3
BEGINNER'S BOX-5	Que.	1	2	3
	Ans.	4	4	1
BEGINNER'S BOX-6	Que.	1	2	3
	Ans.	4	3	4

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

CLASSIFICATION

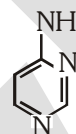
- The hybrid state of C-atoms which are attached to a single bond with each other in the following structure are : $\text{CH}_2=\text{CH}-\text{C}\equiv\text{CH}$
 (1) sp^2 , sp (2) sp^3 , sp
 (3) sp^2 , sp^2 (4) sp^2 , sp^3
- The third member of the homologous series of aliphatic aldehydes has the structure :-
 (1) $\text{CH}_3\text{CH}_2\text{CHO}$ (2) $\text{CH}_3(\text{CH}_2)_2\text{CHO}$
 (3) $\text{CH}_3\text{COCH}_2\text{CH}_3$ (4) CH_3COCH_3
- Molecular formula $\text{C}_4\text{H}_8\text{O}_2$ represents :-
 (1) An acid only
 (2) An ester only
 (3) An alcohol only
 (4) An acid and an ester also
- The higher homologue of dimethylamine ($\text{CH}_3-\text{NH}-\text{CH}_3$) has the structure :-
 (1) $\text{CH}_3-\text{N}-\text{CH}_3$
 CH_3
 (2) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{NH}_2$
 (3) $\text{CH}_3-\text{NH}-\text{CH}_2-\text{CH}_3$
 (4) $\text{CH}_3-\text{CH}-\text{CH}_3$
 NH_2
- The third member of the family of alkenynes has the molecular formula :-
 (1) C_6H_6 (2) C_5H_6
 (3) C_6H_8 (4) C_4H_4
- The number of olefinic bonds in the given compound is/are :-
 $\text{CH}_2=\text{CH}-\text{C}(=\text{O})-\text{CH}=\text{CH}-\text{C}\equiv\text{N}$
 (1) 2 (2) 3
 (3) 1 (4) 4
- The number of acetylinic bonds in the given compound is/are :
 $\text{HC}\equiv\text{C}-\text{C}(=\text{O})-\text{CH}=\text{CH}-\text{C}\equiv\text{N}$
 (1) 2 (2) 3
 (3) 1 (4) 4

- The number of C-atoms in second member of an ester is/are :
 (1) 2 (2) 3
 (3) 4 (4) 5

- Which of the following is an example of symmetrical or simple ether :

- $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$
- $\text{CH}_3-\text{O}-\text{CH}_2-\text{CH}_3$
- $\text{CH}_3-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_2-\text{CH}_3$
- $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{O}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$

- The number of hetero atoms present in the following compound is/are :



- (1) 2 (2) 3
(3) 1 (4) 4

- The minimum number of carbon atoms in an alkane having four primary carbon atoms are :-

- (1) 4 (2) 8
(3) 5 (4) 6

- Which of the following compound has sp -hybridised carbon atom :-

- (1) CH_3COOH (2) CH_3COCH_3
(3) $\text{CH}_3\text{CH}_2\text{CN}$ (4) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$

- In compound $\text{HC}\equiv\text{C}-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_3$, the C_2-C_3 bond is the type of :-

- (1) $\text{sp} - \text{sp}^2$ (2) $\text{sp}^3 - \text{sp}^3$
(3) $\text{sp} - \text{sp}^3$ (4) $\text{sp}^2 - \text{sp}^2$

- Which of the following represents the given mode of hybridization $\text{sp}^2-\text{sp}^2-\text{sp}-\text{sp}$ from left to right :-

- (1) $\text{H}_2\text{C}=\text{CH}-\text{C}\equiv\text{CH}$ (2) $\text{HC}\equiv\text{C}-\text{C}\equiv\text{CH}$
(3) $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$ (4) $\text{H}_2\text{C}=\text{C}=\text{C}=\text{CH}_2$

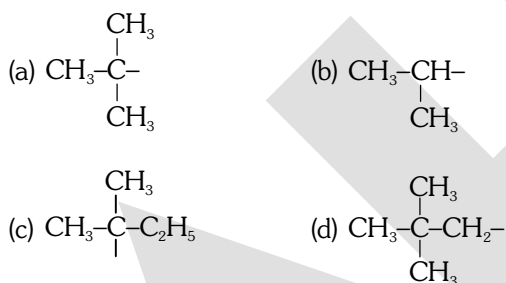
- Which of the following pair of compounds are homologues :-

- (1) 1-Propanol & 2-Propanol
(2) Ethanol & Propanal
(3) Acetone & Acetaldehyde
(4) Acetic acid & Butyric acid

16. Which of the following homologous series has incorrect general formula :-
 (1) Alkyne C_nH_{2n-2}
 (2) Alkanol $C_nH_{2n+2}O$
 (3) Alkanal $C_nH_{2n+1}O$
 (4) Carboxylic acid $C_nH_{2n}O_2$
17. The Cl - C - Cl bond angle in 1,1,2,2 - tetrachloro ethene and tetrachloro methane respectively are:-
 (1) 120° and 109.5° (2) 90° and 109.5°
 (3) 109.5° and 90° (4) 109.5° and 120°
18. Minimum number of carbon atoms present in an ester are :-
 (1) 2 (2) 1 (3) 4 (4) 3
19. Which of the following has general formula C_nH_{2n}
 (1) Only Alkyne
 (2) Only Alkane
 (3) Aromatic hydrocarbon
 (4) Alkene & cyclic Alkane
20. Which compound has alkyne group
 (1) C_7H_{14} (2) $C_{10}H_{22}$
 (3) C_9H_{16} (4) $C_{16}H_{32}$

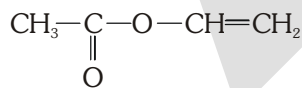
COMMON AND DERIVED NAME

21. Which of the following are tertiary radicals :-



- (1) a and b (2) b and c
 (3) a and c (4) b and d

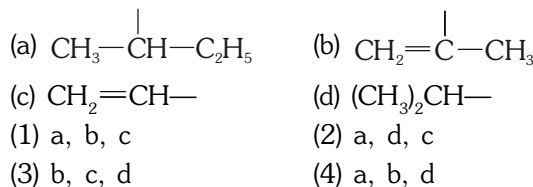
22. Common name of the given compound is :-



- (1) vinyl acetate (2) acryl acetate
 (3) methyl acrylate (4) Vinyl ethanoate

23. A primary amine has amino group ($-\text{NH}_2$) attached to:-
 (1) A primary carbon atom only
 (2) A secondary carbon atom only
 (3) A tertiary carbon atom only
 (4) A primary, secondary or tertiary carbon atom

24. Which of the following are secondary radicals :-



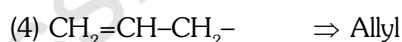
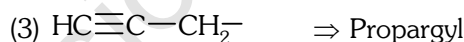
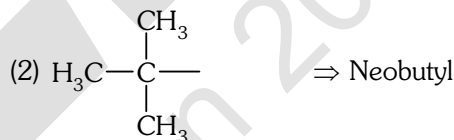
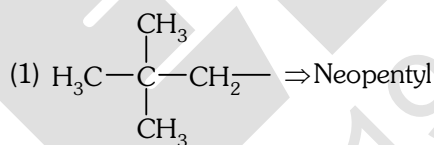
25. Examine the following structures :-



Which of the following statement is correct :-

- (1) A is tertiary alcohol while B is tertiary amine
 (2) A is primary alcohol while B is primary amine
 (3) A is tertiary alcohol while B is primary amine
 (4) A is primary alcohol while B is tertiary amine

26. Which of the following is not a correct match

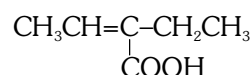


IUPAC NAME

27. The IUPAC name for isobutyl chloride is :-

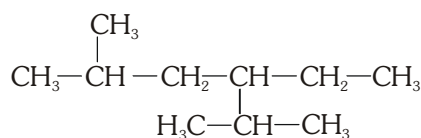
- (1) 2-Methyl-2-chloro butane
 (2) 2-Chloro-2-methyl butane
 (3) 1-Chloro-2-methyl propane
 (4) 2-Methyl-3-chloro propane

28. The IUPAC name of given compound is :-



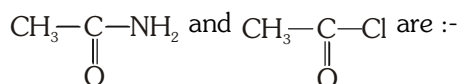
- (1) 3-Carboxy-2-pentene
 (2) 2-Ethylidene butanoic acid
 (3) 2-Ethyl-2-butenic acid
 (4) 3-Ethyl-2-buten-4-oic acid

29. The IUPAC name for the given structure is :-



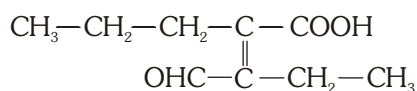
- (1) 3-Isopropyl-4-methylhexane
- (2) 4-Isopropyl-3-methylhexane
- (3) 3-Ethyl-2,5-dimethylhexane
- (4) 2-Ethyl-3-isopropylpentane

30. The IUPAC name for



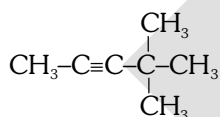
- (1) 1-Amino-1-oxo ethane, 1-chloro ethanal
- (2) 1-Amino ethanal, acetyl chloride
- (3) 1-Oxoethanamine, ethanoyl chloride
- (4) Ethanamide, Ethanoyl chloride

31. The number of carbon atoms in the principle chain of the given compound are :-



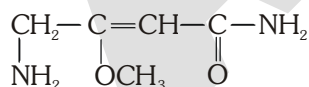
- (1) 7
- (2) 5
- (3) 4
- (4) 6

32. The IUPAC name of given compound is :-



- (1) Methyl tertiarybutyl acetylene
- (2) t-Butyl propyne
- (3) 4,4-Dimethyl-2-pentyne
- (4) 1,3,3,3-Tetramethyl ethyne

33. The IUPAC name of the compound is :-

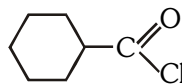


- (1) 4-Amino-2-methoxy-1-amino-2-butene
- (2) 4-Amino-3-methoxy-2-butenamide
- (3) 2-Methoxy-1,4-diamino-2-butenal
- (4) 1-Amino-2-methoxy-3-amino propene

34. The IUPAC name of $\text{CH}_3-\text{CH}_2-\text{NH}-\text{CH}_3$ is :-

- (1) Methyl ethyl amine
- (2) 1-methyl amino ethane
- (3) N-methyl ethan amine
- (4) N-ethyl methan amine

35. The IUPAC name for the compound is :-

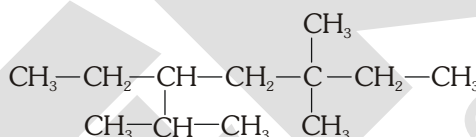


- (1) Cyclohexanoyl chloride
- (2) Cyclohexane carbonyl chloride
- (3) 1-Chloro cyclohexanal
- (4) Chloro cyclohexyl methanal

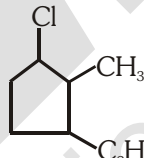
36. The IUPAC name of $\text{HC}\equiv\text{C}-\text{C}(\text{CH}_3)=\text{CH}-\text{CH}_3$ is

- (1) 3-Methyl-2-penten-4-yne
- (2) 3-Methyl-3-penten-1-yne
- (3) 3-Methyl-4-pentyn-1-ene
- (4) 3-Methyl pentenyne

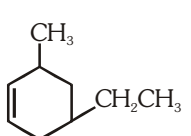
37. The IUPAC name of the structure is :-



- (1) 3-Isopropyl-5,5-dimethyl heptane
- (2) 5-Ethyl-3,3,6-trimethyl heptane
- (3) 3,3-Dimethyl-5-isopropyl heptane
- (4) 3-Ethyl-2,5,5-trimethyl heptane

38.  has the IUPAC name :-

- (1) 3-Chloro-1-ethyl-2-methyl cyclopentane
- (2) 1-Chloro-3-ethyl-2-methyl cyclopentane
- (3) 4-Chloro-1-ethyl-5-methyl cyclopentane
- (4) All are correct

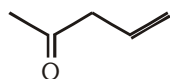
39. The IUPAC name of  is :-

- (1) 1-Methyl-5-ethyl cyclohex-2-ene
- (2) 5-Ethyl-3-methyl cyclohex-1-ene
- (3) 4-Ethyl-6-methyl cyclohex-1-ene
- (4) 1-Ethyl-5-methyl cyclohex-3-ene

40. $\text{H}-\text{C}(=\text{O})-\text{CN}$ in IUPAC called :-

- (1) Cyano methanal
- (2) 2-Oxo ethane nitrile
- (3) Cyano ethanal
- (4) Formonitrile

41. The IUPAC name for the compound is :-



- (1) 2-Acetyl prop-1-ene (2) Pent-1-en-4-one
(3) Pent-4-en-2-one (4) Formyl propene

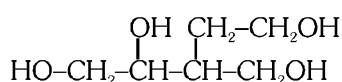
42. Which is incorrect IUPAC name :-

- (1) 3-Pentyne
(2) 3-Methyl-2-butanone
(3) 2-Ethyl-3-methyl-1-butene
(4) 3-Ethyl-2-methyl pentane

43. The IUPAC name of $\text{H}_2\text{N}-\text{C}_6\text{H}_4-\text{OCH}_3$ is :-

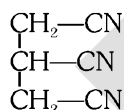
- (1) 1-Methoxy-4-amino benzene
(2) Aminophenyl methyl ether
(3) 4-Methoxy aniline
(4) None of the above

44. The IUPAC name of the given compound is :-



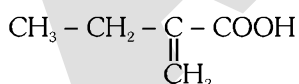
- (1) 3-Hydroxymethyl pentane-1,4,5-triol
(2) 3-Hydroxyethyl butane-1,2,4-triol
(3) 4-Hydroxyethyl-1,2,4-trihydroxy butane
(4) 3-Hydroxymethyl pentane-1,2,5-triol

45. The IUPAC name of the given compound is :-



- (1) 1,2,3-Tricarbonitrile propane
(2) Propane-1,1,1-tricarbylamine
(3) Propane-1,2,3-tricarbonitrile
(4) 3-Cyano pentane dicyanide

46. Number of carbon atoms in the principle carbon chain in the given compound are :-

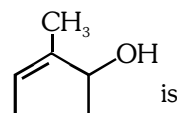


- (1) 4 (2) 3
(3) 2 (4) 5

47. Wrong IUPAC name is :-

- (1) $\text{CH}_3\text{CH}_2\text{CONH}_2$ Propanamide
(2) $\text{CH}_3\text{CH}_2\text{COOCH}_3$ Methyl propanoate
(3) $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}=\text{CH}-\text{CH}_3$ 2-Methyl pent-3-ene
(4) $\text{CH}_3\text{CH}_2-\text{O}-\text{CH}(\text{CH}_2\text{CH}_3)-\text{CH}_3$ 2-Ethoxy butane

48. The IUPAC name of the compound

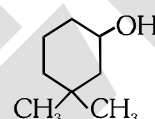


- (1) 2-Methyl cyclopent-1-en-2-ol
(2) 3-Methyl cyclopent-2-en-1-ol
(3) 2-Methyl cyclopent-2-en-1-ol
(4) 3-Methyl cyclopent-1-en-2-ol

49. The IUPAC name of $\text{CH}_3\text{COOCH}_2\text{COOH}$ is :-

- (1) Acetic anhydride
(2) Formyl ethanoate
(3) Butane-2,4-dione
(4) Ethanoic methanoic anhydride

50. The IUPAC name of given compound is :-



- (1) 3,3-Dimethyl-1-hydroxy cyclohexane
(2) 1,1-Dimethyl-3-hydroxy cyclohexane
(3) 3,3-Dimethyl-1-cyclohexanol
(4) 1,1-Dimethyl-3-cyclohexanol

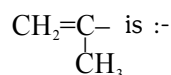
51. IUPAC name of $(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$ is :-

- (1) 2,2-Dimethyl butane (2) 2,3-Dimethyl butane
(3) 2,4-Dimethyl butane (4) 1-Methyl pentane

52. IUPAC name of $\text{CH}_2=\text{CH}-\text{CH}_2-\text{Cl}$ is :-

- (1) Allyl chloride
(2) 1-Chloro-3-propene
(3) 3-Chloro-1-propene
(4) Vinyl chloride

53. The IUPAC name of the following group

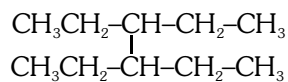


- (1) Isopropenyl (2) 1-Methylethenyl
(3) 2-Methylethynyl (4) None of the above

54. $\text{CH}_3-\text{CH}=\text{CH}-\text{C}\equiv\text{CH}$ has IUPAC name :-

- (1) Pent-2-en-4-yne
(2) Pent-4-yn-2-ene
(3) Pent-1-yn-3-ene
(4) Pent-3-en-1-yne

55. The IUPAC name of the following compound



- (1) 3, 4 - Dimethyl octane
 (2) 3-sec pentyl pentane
 (3) 3, 4 - Diethyl hexane
 (4) 3, 4 - Dimethyl hexane

56. Correct IUPAC name is :-

- (1) 3-Methyl-2-ethylpentane
 (2) 2-Ethyl-3-methylpentane
 (3) 3-Ethyl-2-methylpentane
 (4) 2-Ethyl-2-methylpentane

EXERCISE-I

ANSWER KEY

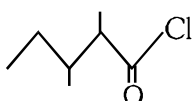
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	4	3	3	1	3	2	4	1	3	3	3	1	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	1	1	4	3	3	1	4	4	3	2	3	3	3	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	3	2	3	2	2	4	2	2	2	3	1	3	4	3
Que.	46	47	48	49	50	51	52	53	54	55	56				
Ans.	2	3	3	4	3	2	3	2	4	3	3				

EXERCISE-II (Previous Year Questions)

AIPMT/NEET & AIIMS (2006-2018)

AIPMT-2006

1. The general molecular formula, which represents the homologous series of alkanols is
 (1) $C_nH_{2n}O_2$ (2) $C_nH_{2n}O$
 (3) $C_nH_{2n+1}O$ (4) $C_nH_{2n+2}O$

2. The IUPAC name of  is :-

- (1) 3,4-Dimethylpentanoyl chloride
 (2) 1-Chloro-1-oxo-2,3-dimethylpentane
 (3) 2-Ethyl-3-methylbutanoyl chloride
 (4) 2,3-Dimethylpentanoyl chloride

AIPMT-2008

3. In the hydrocarbon

$$\begin{array}{ccccccc} CH_3 & - & CH & = & CH & - & CH_2 & - & C \equiv & CH \\ 6 & & 5 & & 4 & & 3 & & 2 & 1 \end{array}$$

 The state of hybridization of carbons 1, 3 and 5 are in the following sequence :-
 (1) sp , sp^2 , sp^3 (2) sp^3 , sp^2 , sp
 (3) sp^2 , sp , sp^3 (4) sp , sp^3 , sp^2

AIPMT-2009

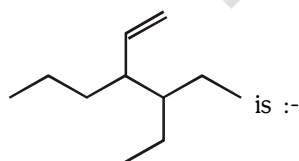
4. The IUPAC name of the compound having the formula $CH \equiv C - CH = CH_2$ is :-
 (1) 1-buten-3-yne (2) 3-buten-1-yne
 (3) 1-butyne-3-ene (4) but-1-yn-3-ene

AIPMT Mains-2010

5. The IUPAC name of the compound $CH_3CH=CHC \equiv CH$ is :-
 (1) Pent-3-en-1-yne (2) Pent-2-en-4-yne
 (3) Pent-1-yn-3-ene (4) Pent-4-yn-2-ene

AIPMT Pre.-2011

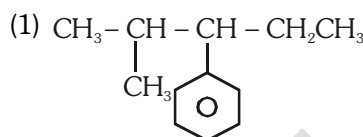
6. The correct IUPAC name of the compound



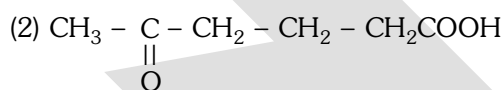
- (1) 4-Ethyl-3-propyl hex-1-ene
 (2) 3-Ethyl-4-ethenyl heptane
 (3) 3-Ethyl-4-propyl hex-5-ene
 (4) 3-(1-ethyl propyl) hex-1-ene

AIPMT Pre.-2012

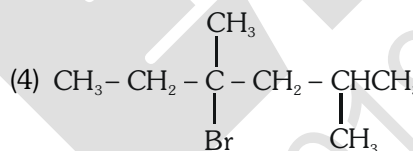
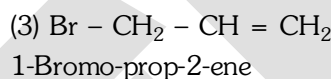
7. Which nomenclature is not according to IUPAC system?



2-Methyl-3-phenylpentane



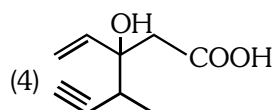
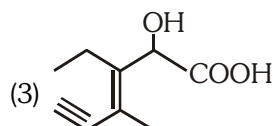
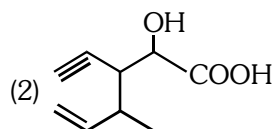
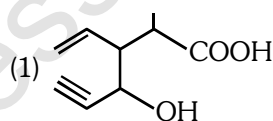
5-Oxohexanoic acid



4-Bromo,2,4-di-methylhexane

NEET UG-2013

8. Structure of the compound whose IUPAC name is 3-Ethyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid is :-

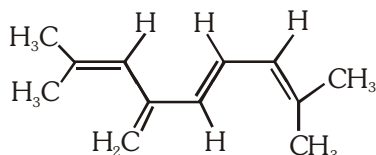


AIPMT-2014

9. Which of the following organic compounds has same hybridization as its combustion product CO_2 ?
- (1) Ethane (2) Ethyne
(3) Ethene (4) Ethanol

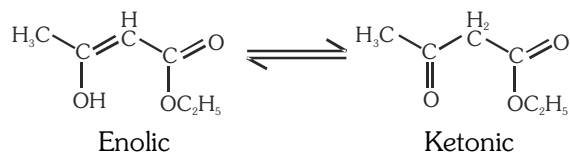
AIPMT-2015

10. The total number of π -bond electrons in the following structure is :-



- (1) 8 (2) 12 (3) 16 (4) 4

11. The enolic form of ethyl acetoacetate as below has:-



- (1) 16 sigma bonds and 1 pi - bond
(2) 9 sigma bonds and 2 pi - bonds
(3) 9 sigma bonds and 1 pi - bond
(4) 18 sigma bonds and 2 pi - bonds

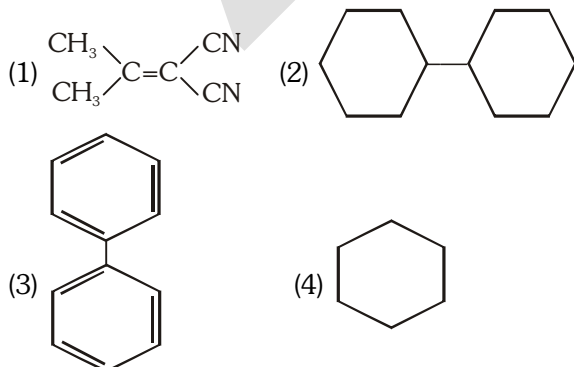
NEET-I 2016

12. The pair of electron in the given carbanion, $\text{CH}_3\text{C} \equiv \text{C}^\ominus$, is present in which of the following orbitals ?

- (1) 2p (2) sp^3
(3) sp^2 (4) sp

NEET-II 2016

13. In which of the following molecules, all atoms are coplanar ?

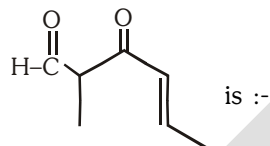


AIIMS 2016

14. In which of the following multiple bond is absent:-
- (1) CH_3CN (2) HCOOH
(3) CH_2N_2 (4) C_3H_6

NEET(UG) 2017

15. The IUPAC name of the compound



- (1) 5-formylhex-2-en-3-one
(2) 5-methyl-4-oxohex-2-en-5-al
(3) 3-keto-2-methylhex-5-enal
(4) 3-keto-2-methylhex-4-enal

AIIMS 2017

16. IUPAC name for given compound is -

- (1) 2-Amino-4-ethyl-5-methyl hexan-1-ol
(2) 2-Amino-4-isopropyl-hexan-1-ol
(3) 5-Amino-3-ethyl-2-methyl hexan-6-ol
(4) 4-Ethyl-2-amino-5-methyl hexan-1-ol

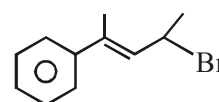
NEET(UG) 2018

17. Which of the following molecules represents the order of hybridisation sp^2 , sp^2 , sp, sp from left to right atoms ?

- (1) $\text{HC} \equiv \text{C} - \text{C} \equiv \text{CH}$
(2) $\text{CH}_2 = \text{CH} - \text{C} \equiv \text{CH}$
(3) $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$
(4) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$

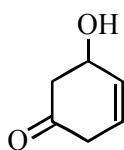
AIIMS 2018

18. IUPAC name of the compound



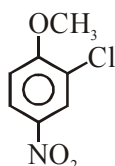
- (1) 2-Phenyl-4-bromopentane
(2) 4-Bromo-2-phenylpent-2-ene
(3) 2-Bromo-4-phenylpentene
(4) 4-Phenyl-2-bromopentane

19. IUPAC name of the given compound is



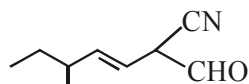
- (1) 3-Hydroxycyclohex-4-en-1-one
- (2) 5-Hydroxycyclohex-3-enone
- (3) 5-Ketocyclohex-2-enol
- (4) 3-Ketocyclohex-5-enol

20. IUPAC name of the given compound



- (1) 2-chloro-4-nitroanisole
- (2) 1-chloro-3-nitro-6-methoxybenzene
- (3) 3-chloro-1-nitro-4-methoxybenzene
- (4) 1-chloro-2-methoxy-5-nitrobenzene

21. Correct IUPAC name of the given compound



- (1) 2-cyano-5-methylhept-3-en-1-al
- (2) 2-Formyl-5-methylhept-3-ene-1-nitrile
- (3) 5-Ethyl-2-cyanohept-3-enal
- (4) 2-Cyano-5-ethylhex-3-enal

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	4	4	1	1	1	3	3	2	1	4	4	3	4	4
Que.	16	17	18	19	20	21									
Ans.	1	2	2	2	1	2									

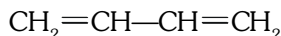
EXERCISE-III (Analytical Questions)

Check Your Understanding

1. The number of π bonds and σ bonds in the structure  respectively are :-

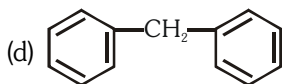
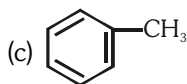
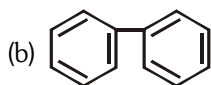
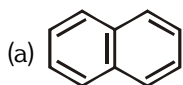
(1) 8, 5 (2) 8, 6
(3) 3, 8 (4) 5, 13

2. The geometry of the compound is :-



(1) Tetrahedral (2) Linear
(3) Planar (4) Pyramidal

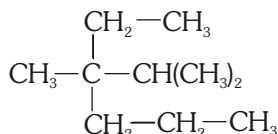
3. Which of the following have only 2° H-atom :



Correct code is :

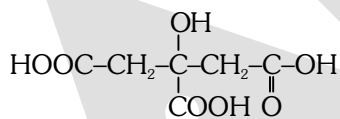
(1) a and b (2) a, b and d
(3) a, c and d (4) a, b, c and d

4. The IUPAC name of the given structure is :-



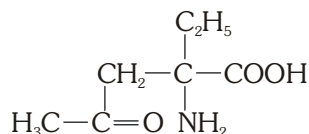
(1) 3-Methyl-3-isopropyl hexane
(2) 3-Isopropyl-3-methyl hexane
(3) 3-Ethyl-2,3-dimethyl hexane
(4) 2,3-Dimethyl-3-ethyl hexane

5. The IUPAC name of the given compound is :



(1) 3-Carboxy-3-hydroxy butane dioic acid
(2) 2-Hydroxy propane-1,2,3-tricarboxylic acid
(3) 3-Hydroxy butane dioic acid
(4) 2-Bis(carboxymethyl)-2-hydroxy ethanoic acid

6. The IUPAC name of given compound is :-



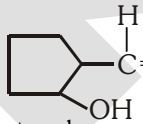
(1) 2-Oxo-4-amino-4-ethyl pentanoic acid
(2) 2-Amino-4-oxo-2-ethyl pentanoic acid
(3) 3-Amino-3-carboxy-5-hexanone
(4) 2-Amino-2-ethyl-4-oxo pentanoic acid

7. The IUPAC name from the incorrect name 4-Amino-3-hydroxy-2-butene is :-

(1) 1-Amino-2-hydroxy-2-butene
(2) 4-Amino-2-buten-3-ol
(3) 1-Amino-2-buten-2-ol
(4) 1-Amino-2-butenol

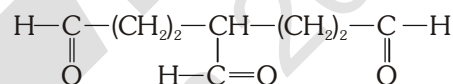
8. The correct IUPAC name of 2-chloro-3-butanol is :-

(1) 3-Chloro-2-hydroxy butane
(2) 3-Chloro-2-butanol
(3) 3-Hydroxy-2-chloro butane
(4) 2-Chloro-3-hydroxy butane

9. The IUPAC name for  is :-

(1) 2-Hydroxy cyclopentanal
(2) 2-Formyl-1-hydroxy cyclopentane
(3) 2-Hydroxy cyclopentane carbaldehyde
(4) Cyclopentane-2-ol-1-al

10. The IUPAC name of the compound is :-



(1) 1,4,7-Triheptanal
(2) 1,3,5-Triformyl pentane
(3) 3-Formyl heptane dial
(4) 1,3,5-Pentane tricarbaldehyde

11. $\text{CH}_3-\text{O}-\text{C}(=\text{O})-\text{CH}_2-\text{COOH}$

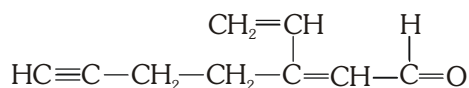
The IUPAC name of the above compound is :-

(1) 2-Acetoxy ethanoic acid
(2) 2-Methoxycarbonyl ethanoic acid
(3) 3-Methoxyformyl ethanoic acid
(4) 2-Methoxyformyl acetic acid

12. The IUPAC name of $\text{Cl}-\text{C}(=\text{O})-\text{OC}_2\text{H}_5$ is :-

(1) Ethoxy formyl chloride
(2) Ethoxy methanoyl chloride
(3) Ethyl chloro methanoate
(4) Ethoxy carbonyl chloride

13. The IUPAC name of the compound is :-

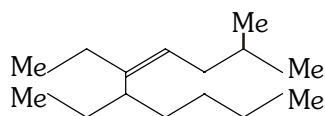


- (1) 3-(1-butyryl)-2,4-pentadienal
- (2) 5-Ethynyl-3-vinyl-2-pentenal
- (3) 3-Vinyl-2-hepten-6-ynal
- (4) 5-Acetyl-3-ethenyl-2-pentenal

14. The IUPAC name of $\text{CH}_3\text{CH}_2\text{NHCHO}$ is :

- (1) N-formyl ethanamine
- (2) Ethyl amino methanal
- (3) N-ethyl methanamide
- (4) N-ethyl methanol

15. The IUPAC name of the structure is :-



- (1) 2,4,5-Triethyl-3-nonene
- (2) 5,6-Diethyl-2-methyl-4-decene
- (3) 2,4,5-Triethyl-3-octene
- (4) 3-Ethyl-5-methyl-3-heptene

16. has the IUPAC name :

- (1) Ethyl-2-keto cyclopentane carboxylate
- (2) 2-Cyclopentanone-1-carbethoxy
- (3) 2-Ethylcarbonate cyclopentanone
- (4) 1-Keto-2-carbethoxy cyclopentanone

17. has the IUPAC name :-

- (1) 3,4-Dimethyl-1-penten-3-ol
- (2) Isopropyl-3-methyl vinyl carbinol
- (3) 2,3-Dimethyl-4-penten-3-ol
- (4) None of the above

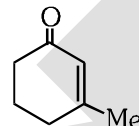
18. Which of the following has wrong IUPAC name

- (1) 5-Ethyl-1-methylcyclohexene
- (2) 5-Bromo-6-chloro-1-cyclohexen-3-yne
- (3) 1,2-Dimethylcyclopropane
- (4) 2-Methylcyclopent-4-en-1-ol

19. The IUPAC name of is :-

- (1) 3,4-Dihydroxy benzoic acid
- (2) 2,3-Dihydroxy benzoic acid
- (3) 4-Carboxy-2-hydroxy phenol
- (4) 4-Carboxy benzene-1,2-diol

20. The IUPAC name of the given compound is :-

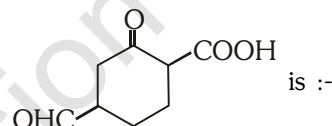


- (1) 3-Methyl-2-cyclohexenone
- (2) 2-Methyl-3-cyclohexenone
- (3) 1-Oxo-3-methyl cyclohexene
- (4) 2-Oxo-6-methyl cyclohexene

21. The IUPAC name of is :-

- (1) 2,3-Dimethyl hexane
- (2) 2-Ethyl-4-methyl pentane
- (3) 3-Ethyl -2-methyl pentane
- (4) 2,4-Dimethyl hexane

22. IUPAC name of



- (1) 6-Oxo-4-formyl cyclohexane carboxylic acid
- (2) 4-Formyl-2-oxo cyclohexane carboxylic acid
- (3) 4-Formyl-2-oxo cyclohexanoic acid
- (4) 1-Carboxy-4-formyl-2-oxo cyclohexane

23. Which is correct IUPAC name :-

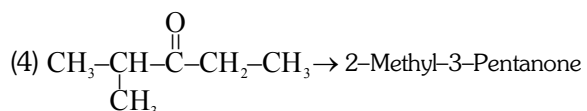
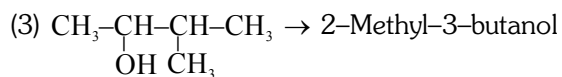
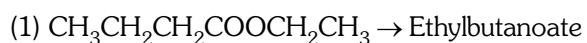
- (1) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{Br}$ 1-Bromopropanone

- (2) 1,3,5-Cyclohexanetrione

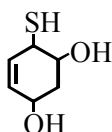
- (3) 2-Ketocyclopentanecarbaldehyde

- (4) All are correct

24. Which of the following compounds has wrong IUPAC name ?

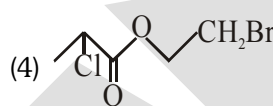
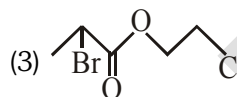
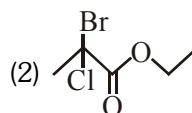
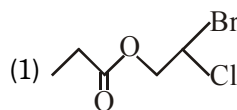


25. Following compound is named as :-



- (1) 6-Mercaptocyclohex-4-ene-1,3-diol
 (2) 1-Mercaptocyclohex-2-ene-4,6-diol
 (3) 1-Mercaptocyclohex-5-ene-2,4-diol
 (4) 4-Mercaptocyclohex-2-ene-1,5-diol

26. Structure of the compound Ethyl 2-bromo-2-chloropropanoate is :-



EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	3	2	3	2	4	3	2	3	4	2	3	3	3	2
Que.	16	17	18	19	20	21	22	23	24	25	26				
Ans.	1	1	4	1	1	4	2	4	3	1	2				

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

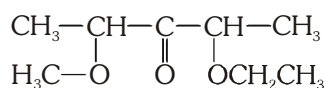
- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

1. **Assertion:-**  is called cyclohexanenitrile.

Reason :- It contains six carbon atoms in the ring and CN as substituent.

- (1) A (2) B (3) C (4) D

2. **Assertion :-** The IUPAC name of



is 2-ethoxy-4-methoxy pentan-3-one.

Reason :- Larger substituents are assigned lower locants.

- (1) A (2) B (3) C (4) D

3. **Assertion :-** The IUPAC name for $\text{HC}\equiv\text{C}-\text{CH}_2-\text{CH}=\text{CH}_2$ is pent-4-en-1-yne.

Reason :- If there is a choice, a triple bond is always given a lower locant than a double bond.

- (1) A (2) B (3) C (4) D

4. **Assertion :-** Benzene is a carbocyclic compound.

Reason :- It has three π bonds in the cycle.

- (1) A (2) B (3) C (4) D

5. **Assertion :-** Number of bond angles of 120° and $109^\circ 28'$ in butenyne are 6 and 2 respectively.

Reason :- It's molecular formula is C_4H_6 .

- (1) A (2) B (3) C (4) D

6. **Assertion :-** Neopentane forms only one mono substituted compound.

Reason :- It has only one type of carbon atoms.

- (1) A (2) B (3) C (4) D

7. **Assertion :-** $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ is a planar compound.

Reason :- It has 9σ bonds and 2π bonds.

- (1) A (2) B (3) C (4) D

8. **Assertion :-** Acetic acid is an unsaturated compound.

Reason :- It has two double bonds.

- (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8
Ans.	4	3	4	2	4	3	2	4

IMPORTANT NOTES

18

ISOMERISM

E

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NEET SYLLABUS

Isomerism (Conformational, Geometrical and Optical Isomerism) :

OBJECTIVES

After studying this unit, we will be able to :

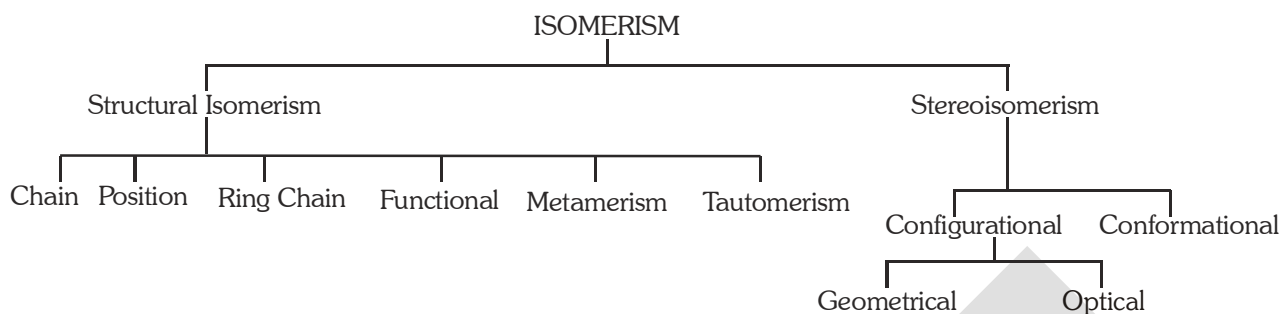
- recognise and write structures of isomers of alkanes, alkenes, alkynes and aromatic hydrocarbons;
- *understand the concept of molecular and structural formula;*
- draw and differentiate between various conformations of ethane;
- *understand the concept of restricted rotation and optical rotation;*

*"Dream is not that which you see while sleeping it is something that;
does not let you sleep."*

A.P.J. Abdul Kalam, Wings of Fire : An Autobiography

ISOMERISM

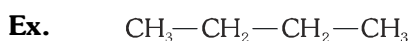
Two or more than two compounds having the same molecular formula but different physical/chemical or both properties are called isomers and the phenomenon is called isomerism.



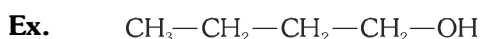
2.1 Structural Isomerism

2.1.1 Chain Isomerism (C.I.)

The compounds which have same molecular formula, same functional group but different arrangement of carbon chain (Parental or side chain) show chain isomerism.



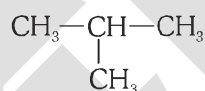
Butane (4C)



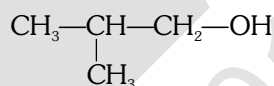
1-Butanol (4C)



Methylcyclobutane



2-Methyl propane (3C)



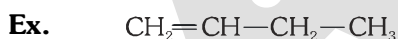
2-Methyl-1-propanol (3C)



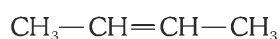
Cyclopentane

2.1.2 Position Isomerism (P.I.)

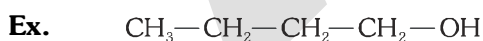
The compounds which have same molecular formula, same functional group, same parent carbon chain but different position of functional group or multiple bond or substituents, show position isomerism.



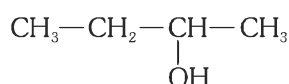
But-1-ene



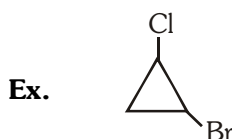
But-2-ene



1-Butanol



2-Butanol



1-Bromo-2-chlorocyclopropane



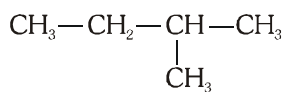
1-Bromo-1-chlorocyclopropane

Example of CI and PI :

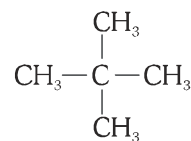
(i) C_5H_{12} has three structural isomers :



Pentane



2-Methyl butane

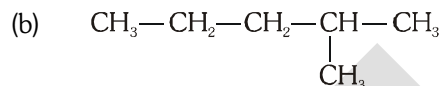


2,2-Dimethylpropane

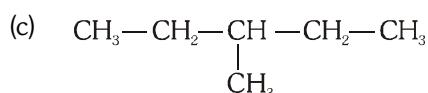
(ii) C_6H_{14} has 5 structural isomers



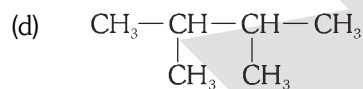
Hexane



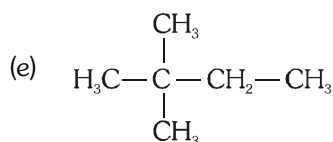
2-Methyl pentane



3-Methyl pentane



2,3-Dimethyl butane

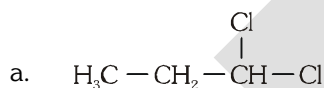


2,2-Dimethyl butane

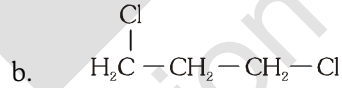
• (a-b), (a-c), (a-d), (a-e) → Chain Isomers
(b-d), (b-e), (c-d), (c-e)

• b-c, d-e → Position Isomers

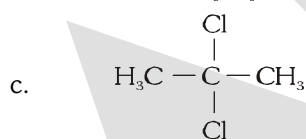
(iii) $C_3H_6Cl_2$ has 4 isomers : Position of chlorine atom is different in all the structure, so these are position Isomers.



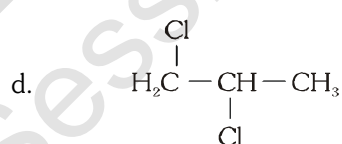
1,1-Dichloropropane



1,3-Dichloropropane



2,2-Dichloropropane

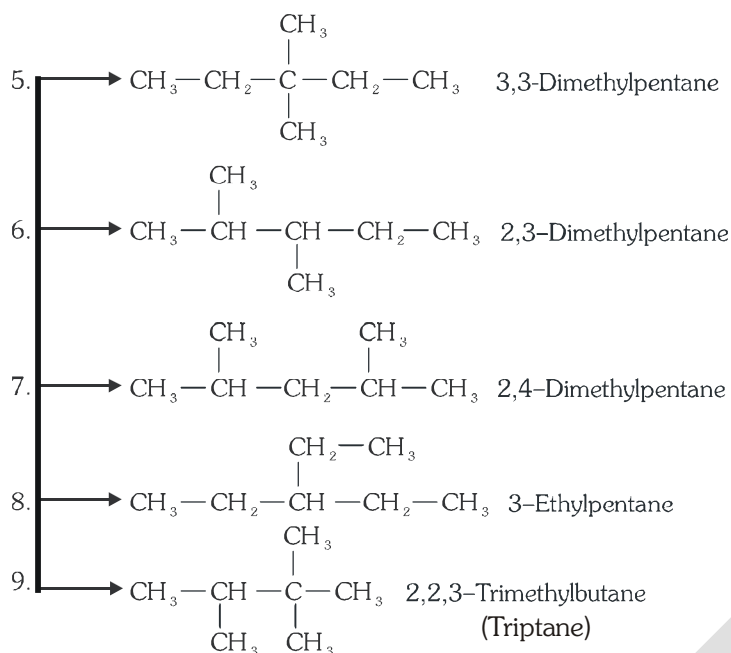


1,2-Dichloropropane

Ex. How many structural isomers of C_7H_{16} are possible ?

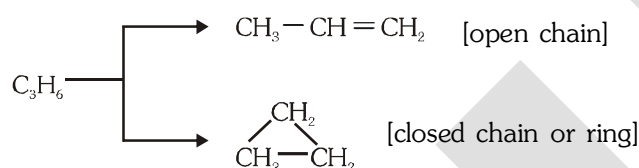
Ans. (9)

1. $\rightarrow CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_3$ Heptane
2. $\rightarrow \begin{array}{c} CH_3-CH_2-CH_2-CH_2-CH-CH_3 \\ | \\ CH_3 \end{array}$ 2-Methylhexane
3. $\rightarrow \begin{array}{c} CH_3-CH_2-CH_2-CH-CH_2-CH_3 \\ | \\ CH_3 \end{array}$ 3-Methylhexane
4. $\rightarrow \begin{array}{c} CH_3-CH_2-CH_2-C-CH_3 \\ | \\ CH_3 \end{array}$ 2,2-Dimethyl pentane



2.1.3 Ring chain isomerism (RCI)

Same molecular formula but different mode of linking (open chain & closed chain) of carbon atoms.



GOLDEN KEY POINTS

- Alkenes with cycloalkane and alkynes and alkadienes with cycloalkenes show Ring-chain Isomerism.
- Ring-chain Isomers are also Functional Isomers but priority must be given to Ring-chain Isomers.

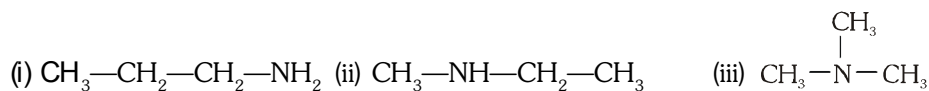
2.1.4 Functional Isomerism

Same molecular formula but different functional groups.

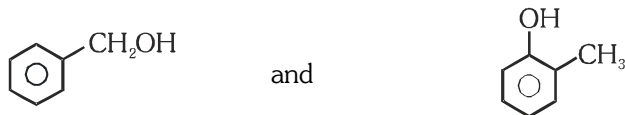
Following compounds show Functional isomerism, as they have same molecular formula and different functional group.

- | | | | | |
|-----------------------------|-------------------|--|-----|---|
| (i) Alcohol and Ether | $\longrightarrow$ | $\text{CH}_3-\text{CH}_2-\text{OH}$ | and | $\text{CH}_3-\text{O}-\text{CH}_3$ |
| (ii) Aldehydes and Ketones | $\longrightarrow$ | $\text{CH}_3-\text{CH}_2-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{H}$ | and | $\text{CH}_3-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{CH}_3$ |
| (iii) Acids and Ester | $\longrightarrow$ | $\text{CH}_3-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{OH}$ | and | $\text{H}-\underset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{O}-\text{CH}_3$ |
| (iv) Cyanide and Isocyanide | $\longrightarrow$ | $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CN}$ | and | $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{NC}$ |
| (v) Nitro and Nitrite | $\longrightarrow$ | $\text{CH}_3-\text{CH}_2-\text{N} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{O} \end{array}$ | and | $\text{CH}_3-\text{CH}_2-\text{O}-\text{N}=\text{O}$ |

(vi) 1°, 2°, 3° Amines



(vii) Alcoholic and Phenolic compounds :

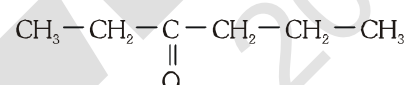
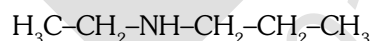
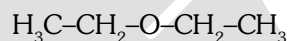
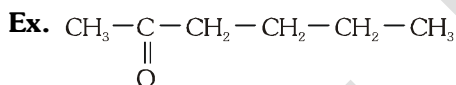
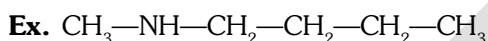
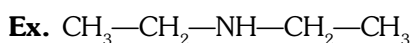
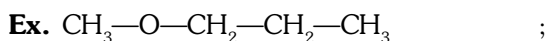
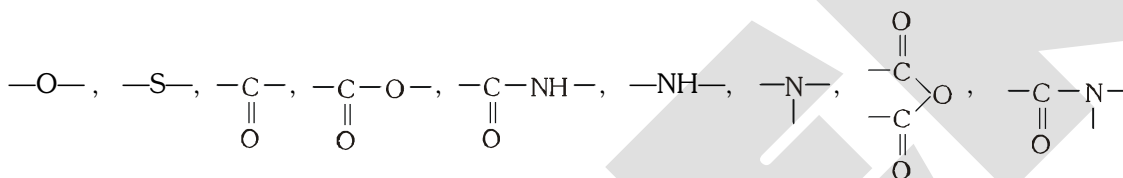


(viii) Alkyl halides do not show Functional isomerism.

(ix) $\text{CH}_3\text{—C}\equiv\text{CH}$ propyne and $\text{CH}_2=\text{CH}=\text{CH}_2$ allene**2.1.5 Metamerism**

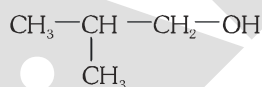
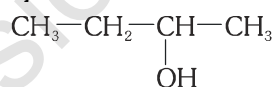
Same molecular formula, same polyvalent Functional group but different alkyl groups attached to polyvalent Functional group.

Polyvalent Functional group [Which have more than one valency] are :

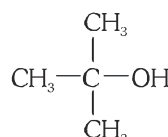
**Examples Based on Structural Isomerism :-**

Ex. 1 $\text{C}_4\text{H}_{10}\text{O}$ Structural isomers $\Rightarrow$ 7 [4 alcohol and 3 ethers] total 7 structural isomers are possible.

Alcohol : $\text{CH}_3\text{—CH}_2\text{—CH}_2\text{—CH}_2\text{—OH}$

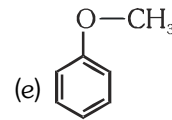
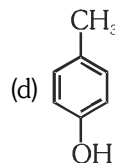
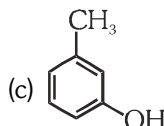
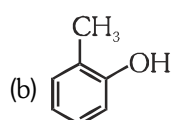
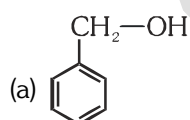


and



Ethers : $\text{CH}_3\text{—O—CH}_2\text{—CH}_2\text{—CH}_3$, $\text{C}_2\text{H}_5\text{—O—C}_2\text{H}_5$, $\text{CH}_3\text{—O—}\overset{\text{CH}_3}{\underset{|}{\text{CH}}}\text{—CH}_3$

Ex. 2 Aromatic isomers of $\text{C}_7\text{H}_8\text{O}$.



- a, b – Functional isomers
- c, d – Position isomers
- a, e – Functional isomers
- b, c – Position isomers
- a, d – Functional isomers
- a, c – Functional isomers

Note : Alcoholic and phenolic groups are Functional isomers.

BEGINNER'S BOX-1

- Structures $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}_2$ and $\text{CH}_3-\underset{\text{CH}_3}{\text{C}}=\text{CH}_2$ are :-
 (1) Chain isomers (2) Position isomers
 (3) Both chain & position isomers (4) Not isomers
- How many minimum carbons required for Chain isomerism and Position isomerism in alkanes ?
 (1) 4, 5 (2) 3, 5 (3) 4, 6 (4) 4, 4
- How many minimum carbons required for Chain isomerism and Position isomerism in alkenes ?
 (1) 4, 5 (2) 3, 4 (3) 4, 4 (4) 5, 5
- How many minimum carbons required for Chain isomerism and Position isomerism in alkynes ?
 (1) 5, 4 (2) 5, 3 (3) 4, 6 (4) 4, 4
- $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$ and $\text{CH}_2-\underset{\text{CH}_2-\text{CH}_2}{\text{CH}_2}$ are :-
 (1) Ring-chain Isomers (2) Chain Isomers (3) Functional Isomers (4) Position Isomers
- $\text{CH}_3-\text{CH}_2-\text{CH}_2-\overset{\text{O}}{\parallel}\text{C}-\text{OH}$ and $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\overset{\text{O}}{\parallel}\text{C}-\text{H}$ are
 (1) Position isomers (2) Functional isomers (3) Identical (4) Chain isomers
- $\text{CH}_3-\text{S}-\text{CH}_2-\text{CH}_3$ and $\text{CH}_3-\text{CH}_2-\text{S}-\text{CH}_3$ are -
 (1) Ring-chain Isomers (2) Chain Isomers (3) Functional Isomers (4) Identical
- $\text{CH}_3-\overset{\text{O}}{\parallel}\text{C}-\text{O}-\overset{\text{O}}{\parallel}\text{C}-\text{C}_6\text{H}_5$ and $\text{C}_6\text{H}_5-\overset{\text{O}}{\parallel}\text{C}-\text{O}-\overset{\text{O}}{\parallel}\text{C}-\text{CH}_3$ are -
 (1) Metamers (2) Chain Isomers (3) Identical (4) Position Isomers

2.2 Stereo Isomerism

Two or more than two compounds having same molecular formula, same structural formula but different arrangements of atoms or groups in space.

(A) Configurational Isomerism : Stereo isomers which have following characteristics.

- Stereo isomer which cannot interconvert at room temperature due to restricted rotation known as Geometrical isomerism.
- Stereo isomers which have different behaviour towards plane polarised light are known as optical isomers.

(B) Conformational Isomerism

2.2.1 Geometrical isomerism (G. I) :

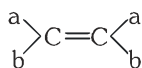
- Alkenes ($>\text{C}=\text{C}<$), oximes ($>\text{C}=\text{N}-\text{OH}$) and azo compounds [$-\text{N}=\text{N}-$] etc., show G. I. due to restricted rotation about double bond and (ii) cycloalkanes show G. I. due to restricted rotation about single bond in ring.

G. I. IN ALKENES :

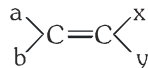
Reason : Restricted rotation about double bond :

Condition for Geometrical isomerism :

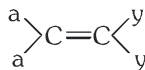
Only those alkenes show G. I. in which "Each sp^2 carbon individually have different atoms or groups"



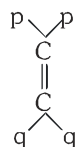
Geometrical isomerism
possible



Geometrical isomerism
possible



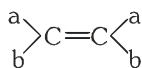
Geometrical isomerism
not possible



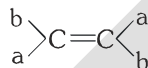
Geometrical isomerism
not possible

Nomenclature Systems of Geometrical isomers**(a) Cis-Trans System**

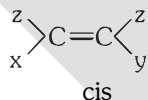
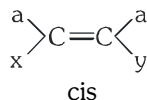
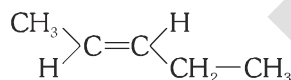
If same groups are at same side then cis and if same groups are at different side then trans.



[Same groups, same side]
cis



[Same groups different side]
trans

Ex.**Ex.**

trans-2-pentene

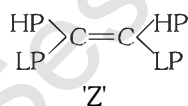
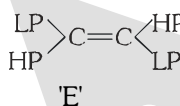


It does not show Geometrical isomers
So no cis-trans

(b) E - Z System :

E (Entgegen) : When high priority groups are at opposite side.

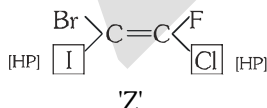
Z (Zusammen) : When high priority groups are at same side.



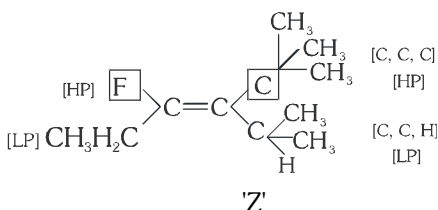
HP - High priority and LP - Low priority

Priority Rules [CIP Sequence Rule] :

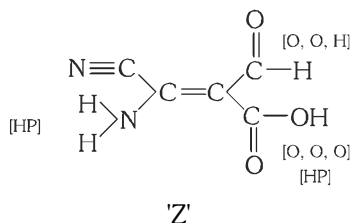
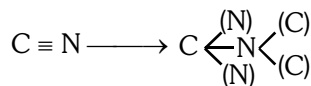
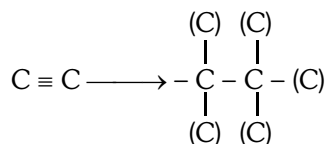
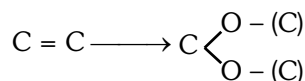
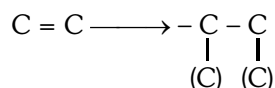
Rule I : Priority is proportional to atomic number of atom which is directly attached to sp^2 carbon.



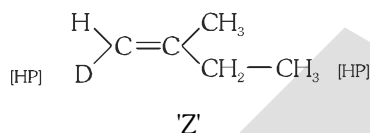
Rule II : If rule-I is failed then consider the atomic number of next atom and so on.



Rule III :- If multiple bond is present then consider them as :-

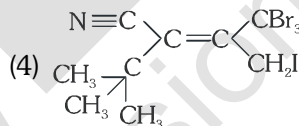
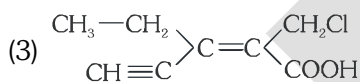
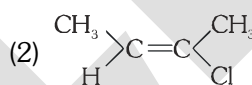
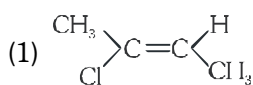


Rule IV : If isotopes are present then consider atomic weight.



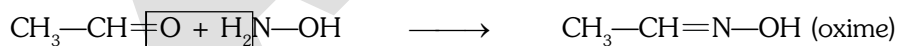
BEGINNER'S BOX-2

1. Assign correct E, Z configuration to following molecules :-

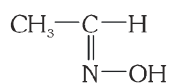


GEOMETRICAL ISOMERS IN OXIMES [$>C = N-OH$] :

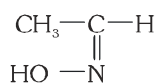
- Oximes show G. I. due to restricted rotation about double bond.
- Only those oximes show Geometrical isomerism in which sp^2 carbon have two different groups.



Ex. Acetaldoximes has two Geometrical isomers -



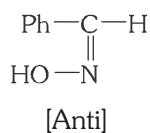
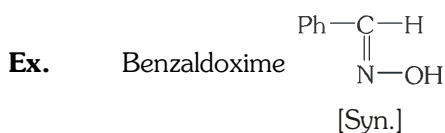
syn

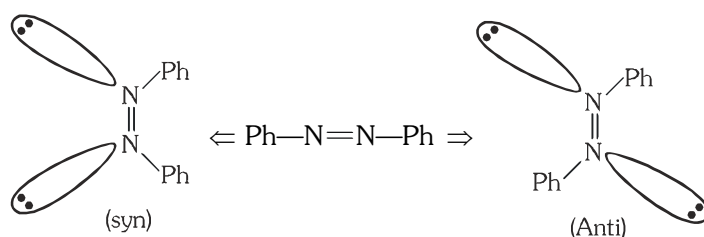


anti

When H and OH are on the same side.

When H and OH are on the opposite side



GEOMETRICAL ISOMERS IN AZO COMPOUNDS : ($-\ddot{\text{N}}=\ddot{\text{N}}-$)

GEOMETRICAL ISOMERS IN CYCLOALKANES : Cycloalkanes show Geometrical isomerism due to restricted rotation about single bond. Only those cyclo alkanes show Geometrical isomers in which atleast two different carbons have two different groups.



NUMBER OF GEOMETRICAL ISOMERS IN POLYENES :



- (a) If $\text{R}_1 \neq \text{R}_2$ then number of Geometrical isomers = 2^n
 [n = number of double bonds which show G.I.]



As $n = 3$, number of Geometrical isomers = $2^3 = 8$

- (b) If $\text{R}_1 = \text{R}_2$ then number of Geometrical isomers = $2^{n-1} + 2^{p-1}$

where $p = \frac{n}{2}$ (when n is even) and $p = \frac{n+1}{2}$ (n is odd)

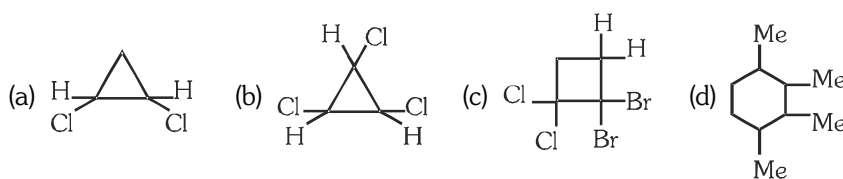


Number of Geometrical isomers = $2^2 + 2^1$
 $= 4 + 2 = 6$

BEGINNER'S BOX-3

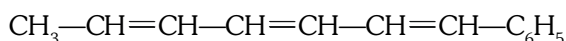
- Which of the following shows Geometrical isomerism –
 - $\text{CH}_3-\text{CH}_2-\text{CH}=\text{N}-\text{OH}$
 - $\text{H}_2\text{C}=\text{N}-\text{OH}$
 - $\text{CH}_3-\text{C}(\text{N}=\text{OH})-\text{CH}_3$
 - $\text{CH}_3-\text{C}(\text{N}=\text{OH})-\text{CH}_2\text{CH}_3$
 - (1) a, c
 - (2) c, d
 - (3) a, d
 - (4) b, c
- Which of the following show Geometrical isomerism –
 - (1) 1,1-Diphenyl-1-butene
 - (2) 1,1-Diphenyl-2-butene
 - (3) 2,3-Dimethyl-2-butene
 - (4) 3-Phenyl-1-butene

3. Which of the following show Geometrical isomerism –



- (1) a, b, d (2) a, c, d (3) a, b, c (4) b, c, d

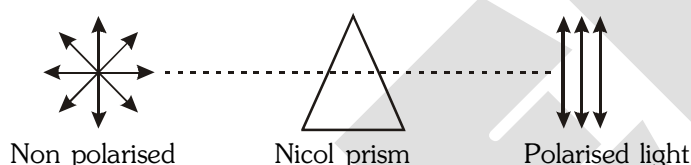
4. Calculate total number of geometrical isomers in the following compound



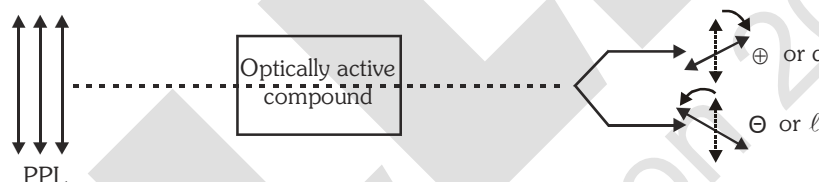
- (1) 2 (2) 4 (3) 6 (4) 8

2.2.2 OPTICAL ISOMERISM :

Optical isomers : Two or more than two compound have same molecular formula, same structural formula but different behaviour towards PPL (plane polarised light).



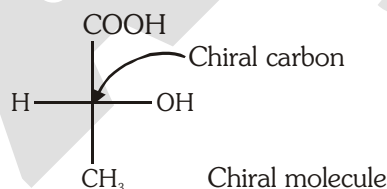
Optical activity : Tendency to rotate plane of PPL in a particular direction. If a compound rotates plane of PPL in clockwise direction then it will be dextrorotatory or d or (+) and if a compound rotates plane of PPL in anticlockwise direction then it will be levorotatory or ℓ or (-).



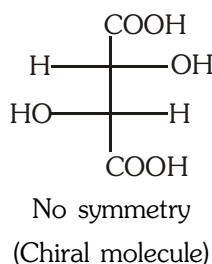
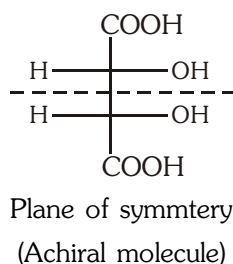
Chiral carbon : Carbon which is bonded to four different atoms or groups.



Chiral molecule : (i) A molecule having only one chiral centre then it will be always chiral. Except flipping



(ii) A molecule having atleast two chiral carbons and no symmetry then it will be chiral molecule.



Essential and sufficient condition for optical activity is :

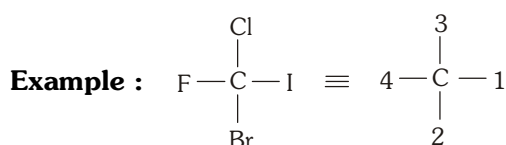
"Molecule should be **chiral**"
or
"Molecule should be **unsymmetrical**"
or
"**Nonsuperimposable** mirror images"

Absolute Configuration (R, S configuration) :

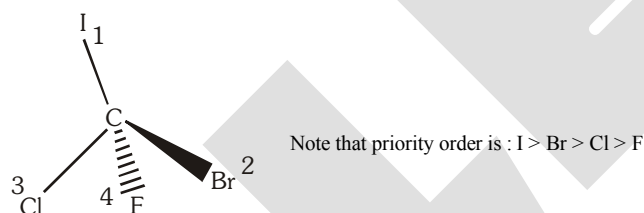
The actual three dimensional arrangement of groups in a molecule containing asymmetric carbon is termed as **absolute configuration**.

(R) (S) nomenclature is assigned as follows :

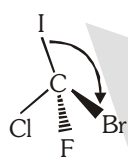
- Each group attached to stereocentre is assigned a priority on the basis of CIP sequence rule.



- After assigning priorities, the least priority group is written at remotest valency (going away).

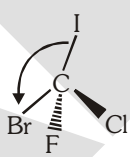


- Now the order from top priority to second priority and then to the third priority is determined. If it gives a clockwise direction then it is termed R configuration and if the anticlockwise direction is obtained then it is assigned S configuration. For example.



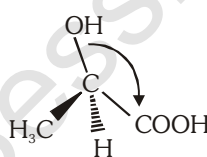
R-configuration

(clockwise from top priority)



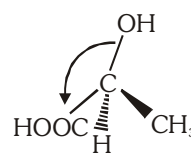
S-configuration

(Anti clockwise from top priority)



R-configuration

(clockwise from top priority)



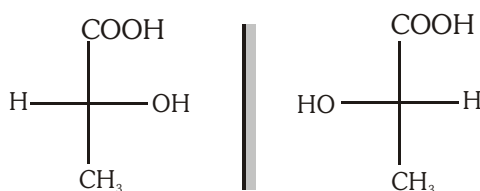
S-configuration

(Anti clockwise from top priority)

Enantiomers :

Stereoisomers which are mirror images to each other and not superimpose to each other.

Nonsuperimposable mirror images are called as enantiomers.

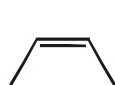


object (I) mirror image (II)

I and II are nonsuperimposable mirror images, so I and II are enantiomers.

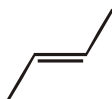
- (i) Enantiomers have chiral structures.
- (ii) Enantiomers are optically active substances. They rotate the plane polarized light in opposite directions but to the equal extent.

Diastereomers : Stereoisomers which are not mirror images of each others.



cis-2 butene

(I)

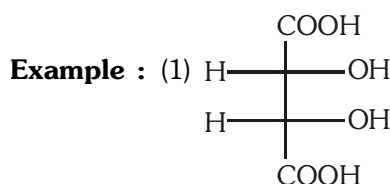


trans-2-butene

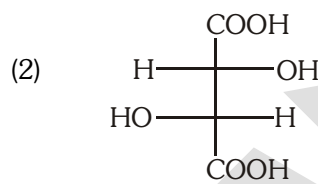
(II)

I and II are not mirror images, so I and II are diastereomers.

EXAMPLES OF OPTICAL ISOMERISM



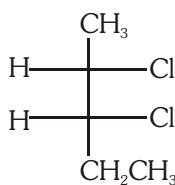
Example :



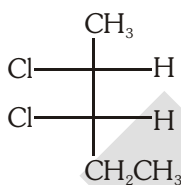
(2)

(1) and (2) are diastereomers.

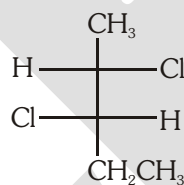
Example :



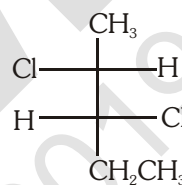
(I)



(II)



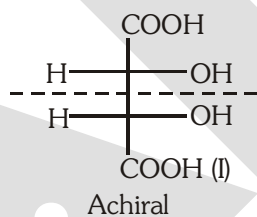
(III)



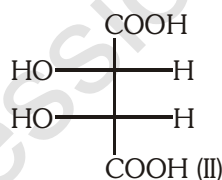
(IV)

I, II = Enantiomers,
I, III = Diastereomers,
II, III = Diastereomers.

III, IV = Enantiomers
II, IV = Diastereomers
I, IV = Diastereomers



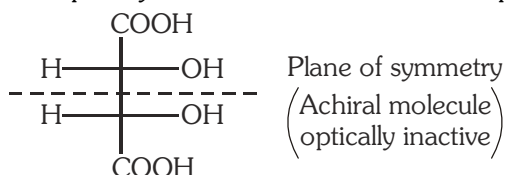
Achiral


$$\text{COOH (II)}$$

Example :

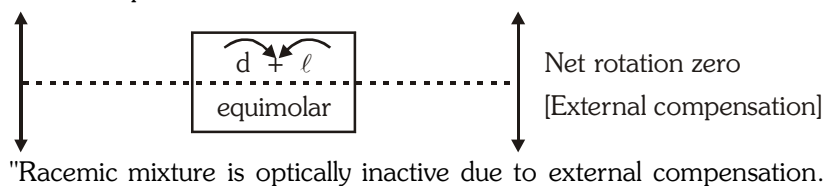
I and II are identical

Meso compounds : Compounds having atleast two chiral carbons and plane of symmetry/cos/symmetry is called meso compounds. It is optically inactive due to internal compensation.



Plane of symmetry
(Achiral molecule)
(optically inactive)

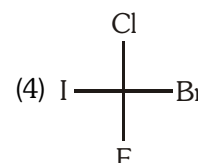
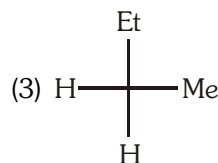
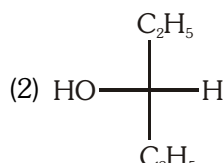
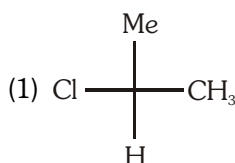
Racemic mixture : Equimolar mixture of d and *l* enantiomers is called as racemic mixture. (*d**l* or +).



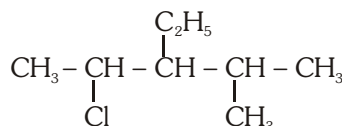
"Racemic mixture is optically inactive due to external compensation.

BEGINNER'S BOX-4

1. Which of the following molecule is chiral :-
 (1) Isobutane (2) Neopentane (3) Sec. butylchloride (4) All
2. Which of the following molecule has chiral carbon :-



3. How many chiral carbon atoms are present in following molecule :-



(1) 1

(2) 2

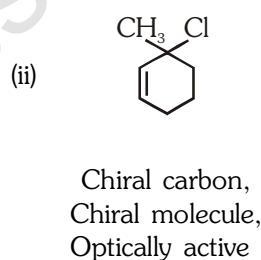
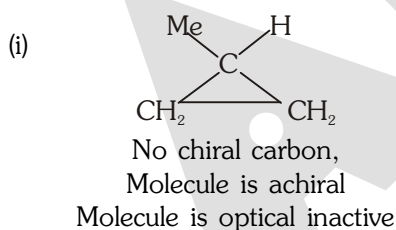
(3) 3

(4) 4

Number of stereoisomers :

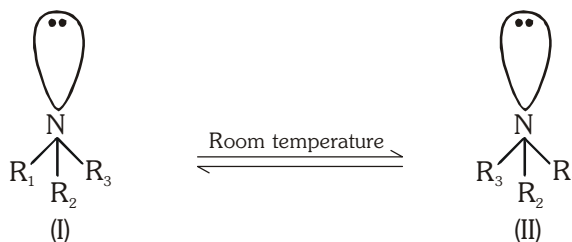
S.N.	Nature of compounds	No. of optically active isomers (a)	No. of meso compounds (m)	Total no. of Stereoisomer (a + m)
1	Compounds having dissimilar ends	2^n (n = No. of chiral centre)	0	$(2^n + 0)$
2	Compounds having similar ends with even chiral centres	2^{n-1}	$2^{\frac{n}{2}-1}$	$(2^{n-1} + 2^{\frac{n}{2}-1})$
3	Compounds having similar ends with odd chiral centres	$2^{n-1} - 2^{\frac{n-1}{2}}$	$2^{\frac{n-1}{2}}$	2^{n-1}

Optical Isomerism in cyclic compounds :



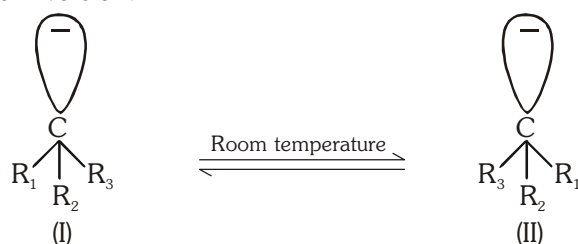
Special Point :

1. Chiral nitrogen containing/tertiary/open chain amine does not show optical isomerism
 Reason :- Rapid umbrella inversion.



Energy required for this interconversion is available at room temperature. So I and II are interconvertible.

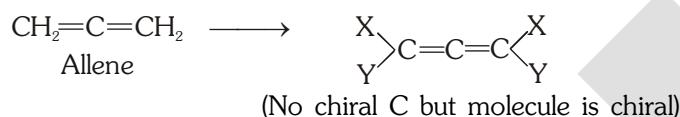
2. Chiral C containing carbanion does not show optical isomerism.
Reason :- Rapid umbrella inversion.



Energy required for this interconversion is available at room temperature. So I and II are interconvertible.

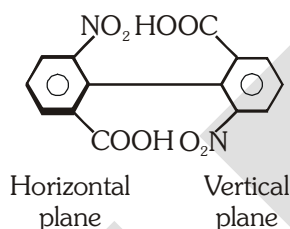
Note :- Only 2nd period elements show flipping.

3. Substituted Allenes do not have chiral carbons but molecule is chiral, so show optical isomerism.



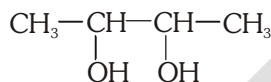
Only those substituted allenes will be optically active in which "each sp^2 C have different atoms or group".

4. Ortho substituted biphenyl compounds do not have any chiral carbon but due to chiral molecule, they are optically active.



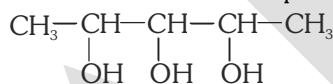
BEGINNER'S BOX-5

- 1.** Calculate total number of optical isomers in following compound :-



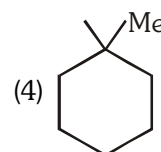
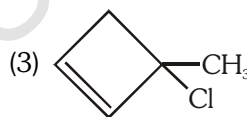
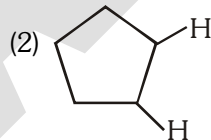
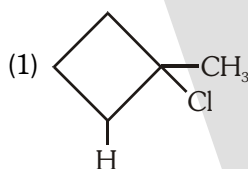
- (1) 4 (2) 3 (3) 2 (4) 1

- 2.** Calculate total number of optical isomers in following compound :-



- (1) 2 (2) 4 (3) 3 (4) 8

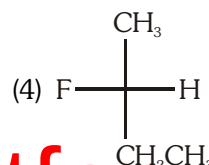
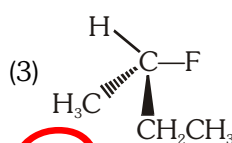
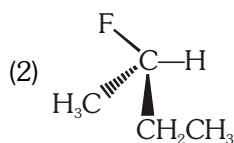
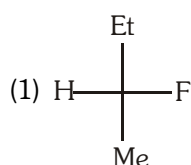
- 3.** Which of the following is optically active :-



- 4.** Which of the following is not optically active –

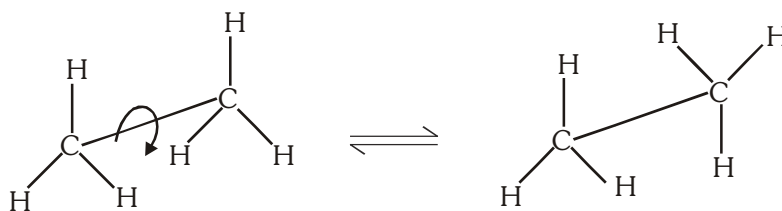
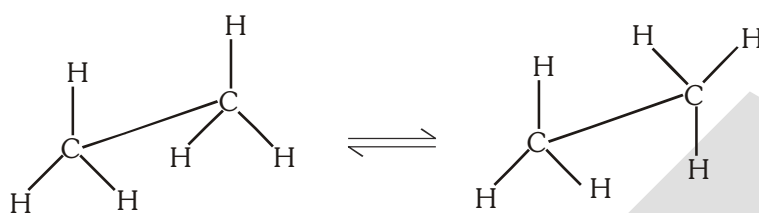
- (1) $\text{CH}_3\text{—CH=CH—CH}_3$ (2) $\text{CH}_2=\text{C}=\text{CH}_2$
- (3) $\begin{array}{c} \text{Cl} \\ \diagup \\ \text{C}=\text{C}=\text{C} \\ \diagdown \\ \text{Br} \end{array}$ (4) $\begin{array}{c} \text{Me} \\ \diagup \\ \text{C}=\text{C}=\text{C} \\ \diagdown \\ \text{Et} \end{array}$

5. Which of the following has 'S' configuration :-

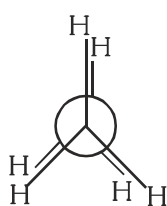


(B) Conformational Isomerism

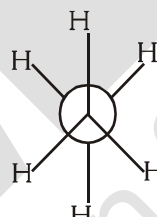
The different arrangement of atoms in space that result from the free rotation around single bond, are called conformations. The phenomenon is called conformational isomerism

**Conformations of ethane [CH₃—CH₃] :**

(Saw horse projection)



(III)

 $\xrightleftharpoons{60^\circ}$ 

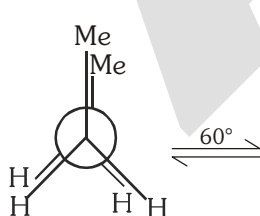
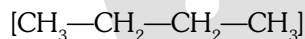
(IV)

(Newman projection)

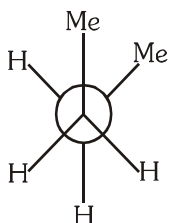
Stability order : Staggered > Eclipsed

Dihedral Angle : Dihedral angle in eclipsed form of ethane is 0° .

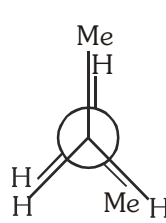
Dihedral angle in staggered form of ethane is 60° .

Conformation of Butane

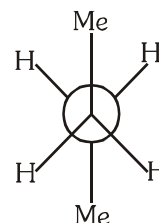
(I)

 $\xrightleftharpoons{60^\circ}$ 

(II)

 $\xrightleftharpoons{60^\circ}$ 

(III)

 $\xrightleftharpoons{60^\circ}$ 

(IV)

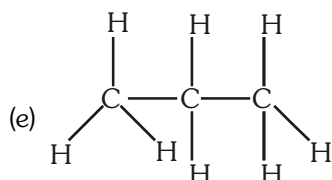
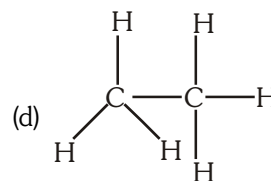
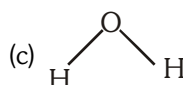
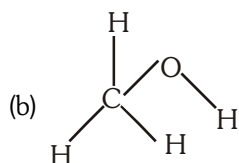
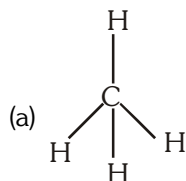
I (Fully eclipsed form) : In this form distance between 2 methyl groups is minimum so maximum repulsion or minimum stable.

IV (Anti or staggered) : In this form distance between 2 methyl groups is maximum so minimum repulsion or maximum stable.

Stability order : IV > II > III > I

BEGINNER'S BOX-6

- 1.** Which of the following shows conformational isomerism.



- (1) b, d, e (2) a, b, c, e (3) b, c, d (4) c, d, e
- 2.** Most stable conformation of butane is :-
(1) Partial eclipsed (2) Full eclipsed (3) Staggered (4) Gauche
- 3.** Which of the following is not a pair of isomers :
(1) Propyne and Cyclopropene (2) Propyne and Propadiene
(3) Propene and Cyclopropene (4) 1-Propanol and Methoxy ethane
- 4.** What is dihedral angle in staggered form of ethane :-
(1) 30° (2) 45° (3) 75° (4) 60°

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7	8
	Ans.	1	3	3	1	1	2	4	3

BEGINNER'S BOX-2	Que.	1	2	3	4
	Ans.	Z	E	E	E

BEGINNER'S BOX-3	Que.	1	2	3	4
	Ans.	3	2	1	4

BEGINNER'S BOX-4	Que.	1	2	3	
	Ans.	3	4	2	

BEGINNER'S BOX-5	Que.	1	2	3	4	5
	Ans.	2	2	3	2	3

BEGINNER'S BOX-6	Que.	1	2	3	4
	Ans.	1	3	3	4

EXERCISE-I (Conceptual Questions)

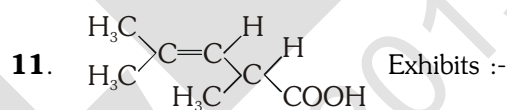
Build Up Your Understanding

STRUCTURAL ISOMERISM

- $\text{CH}_3\text{CHOHCH}_2\text{CHO}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ constitute a pair of :-
 (1) Position isomers
 (2) Metamers
 (3) Optical isomers
 (4) Functional isomers
- The minimum number of carbon atoms present in an organic compound to show chain isomerism is
 (1) 2 (2) 3 (3) 5 (4) 4
- The minimum number of carbon atoms present in an organic compound to be able to show position isomerism is :-
 (1) 3 (2) 4 (3) 2 (4) 5
- Which of the following compound is isomeric with propanoic acid :-
 (1) $\text{CH}_3-\text{C}(=\text{O})-\text{OC}_2\text{H}_5$
 (2) $\text{CH}_2(\text{OH})-\text{CH}_2-\text{C}(=\text{O})-\text{H}$
 (3) $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3$
 (4) $\text{CH}_3\text{O}-\text{CH}_2-\text{CH}_2\text{OH}$
- $\text{CH}_3-\text{NH}-\text{C}_2\text{H}_5$ and $(\text{CH}_3)_3\text{N}$ show which type of isomerism :-
 (1) Position (2) Functional
 (3) Chain (4) None
- $\text{CH}_3-\underset{\text{Cl}}{\text{CH}}-\text{CH}_2-\underset{\text{H}}{\text{C}}=\text{O}$ and $\text{CH}_3-\text{CH}_2-\text{CH}_2-\underset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{Cl}$ are constitute a pair of :-
 (1) Position isomers
 (2) Metamers
 (3) Optical isomers
 (4) Functional group isomers
- The minimum number of carbon atoms in ketone to show position isomerism :-
 (1) 3 (2) 4 (3) 5 (4) 6

- Which are metamers :-
 (1) $\text{CH}_3-\text{O}-\text{CH}_2\text{CH}_2\text{CH}_3$, $\text{CH}_3-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_3$
 (2) $\text{C}_2\text{H}_5-\text{O}-\text{C}_2\text{H}_5$, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 (3) $\text{CH}_3-\text{O}-\text{C}_2\text{H}_5$, $\text{CH}_3-\text{CH}_2-\text{O}-\text{CH}_3$
 (4) $\text{CH}_3-\underset{\text{O}}{\underset{||}{\text{C}}}-\text{CH}_3$, $\text{CH}_3-\text{CH}_2-\underset{\text{O}}{\underset{||}{\text{C}}}-\text{H}$
- Which similarity is necessary for isomerism-
 (1) Molecular formula
 (2) Structure formula
 (3) Physical formula
 (4) Chemical formula
- Number of structural isomers of C_6H_{14} is -
 (1) 3 (2) 4
 (3) 5 (4) 6

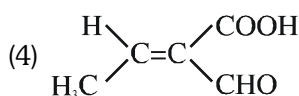
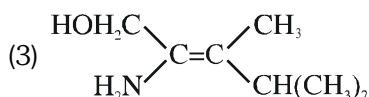
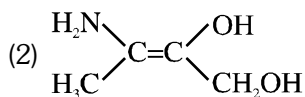
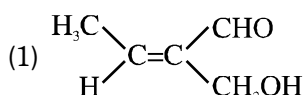
GEOMETRICAL AND OPTICAL ISOMERISM



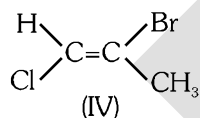
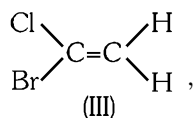
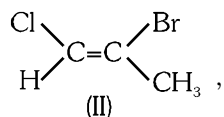
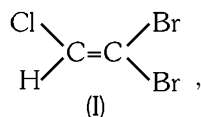
- (1) Tautomerism
 - (2) Optical isomerism
 - (3) Geometrical isomerism
 - (4) Geometrical and optical isomerism
12. The isomerism shown by Benzaldoxime
 $\left[\text{C}_6\text{H}_5-\text{CH}=\text{N}-\text{OH} \right]$ is :-
 (1) Optical (2) Geometrical
 (3) Metamerism (4) All of these

13. Which of the following has Z-configuration :
- (1) $\begin{array}{c} \text{H}_3\text{C} \\ \diagdown \\ \text{C} \\ \diagup \\ \text{H} \end{array} = \begin{array}{c} \text{C}_2\text{H}_5 \\ \diagdown \\ \text{C} \\ \diagup \\ \text{H} \end{array}$
 - (2) $\begin{array}{c} \text{Br} \\ \diagdown \\ \text{C} \\ \diagup \\ \text{HOCH}_2 \end{array} = \begin{array}{c} \text{CH}(\text{CH}_3)_2 \\ \diagdown \\ \text{C} \\ \diagup \\ \text{CH}_2-\text{CH}_3 \end{array}$
 - (3) $\begin{array}{c} \text{Cl} \\ \diagdown \\ \text{C} \\ \diagup \\ \text{Br} \end{array} = \begin{array}{c} \text{H} \\ \diagdown \\ \text{C} \\ \diagup \\ \text{D} \end{array}$
 - (4) All the above

14. Which of the following has E-configuration :-



15. Which is a pair of geometrical isomers :-



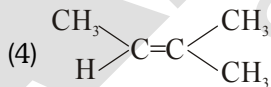
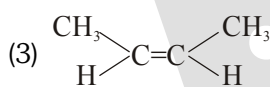
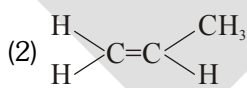
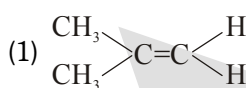
(1) I and II

(2) I and III

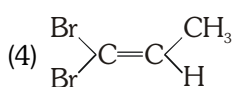
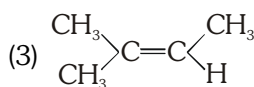
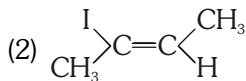
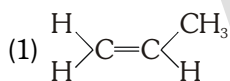
(3) II and IV

(4) III and IV

16. Which can show 'Geometrical isomerism' :-



17. Geometrical isomerism is shown by :



18. The simplest alkanol exhibiting optical activity is

(1) n-butyl alcohol

(2) Isobutyl alcohol

(3) s-butyl alcohol

(4) t-butyl alcohol

19. Meso-tartaric acid $\begin{bmatrix} \text{COOH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{COOH} \end{bmatrix}$ is optically

inactive due to the presence of :-

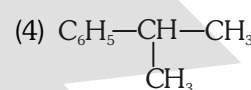
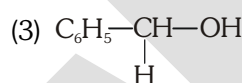
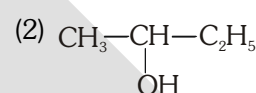
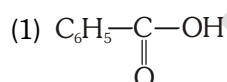
(1) Molecular symmetry

(2) Molecular asymmetry

(3) External compensation

(4) Two asymmetric carbon atoms

20. Which is optically active molecule :-



21. The number of stereo isomers of glucose (a six carbon sugar) is :-

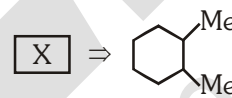
(1) 8

(2) 12

(3) 16

(4) 24

22. Number of chiral carbon atoms in the compound X, Y and Z respectively would be :



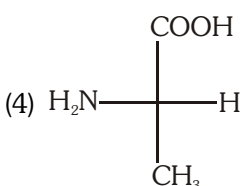
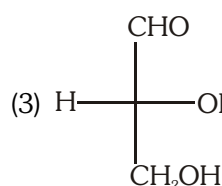
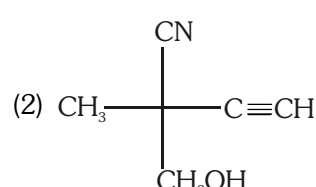
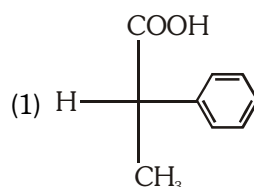
(1) 2, 1, 1

(2) 1, 1, 1

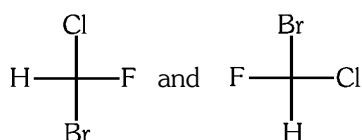
(3) 2, 0, 2

(4) 2, 0, 1

23. Identify R configuration :

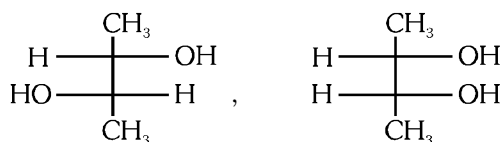


24. The following two compounds are



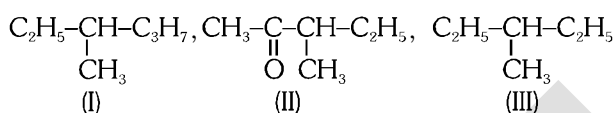
- (1) Enantiomers (2) Diastereomers
(3) Identical (4) Epimers

25. If optical rotation produced by the compound [A] is $+65^\circ$, then produced by the compound [B] is-



- [A] [B]
(1) $+65^\circ$ (2) -65°
(3) Zero (4) Unpredictable

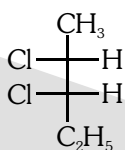
26. Among the following structure I to III



It is true that :-

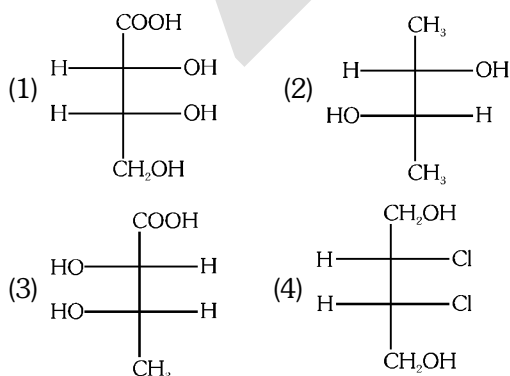
- (1) All three are chiral compounds
(2) Only I and II are chiral compounds
(3) Only II is chiral compound
(4) Only I and III are chiral compounds

27. The absolute configuration of the compound is :

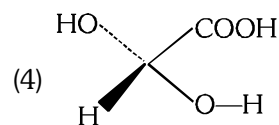
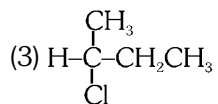
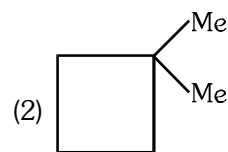
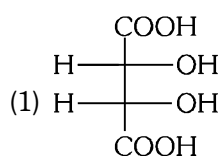


- (1) 2S, 3R (2) 2S, 3S
(3) 2R, 3S (4) 2R, 3R

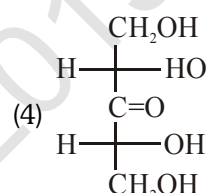
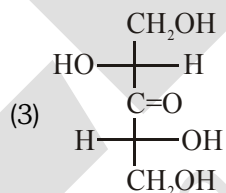
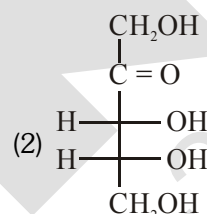
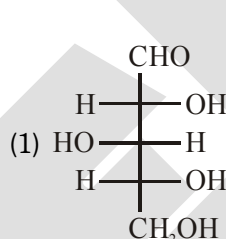
28. Which one of the following is a meso-compound.



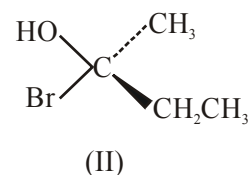
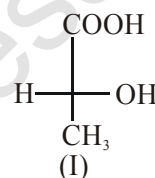
29. Which of the following is optically active :-



30. Amongst the following, which one could be the structure of an optically inactive monosaccharide having the molecular weight 150 :-

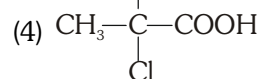
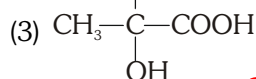
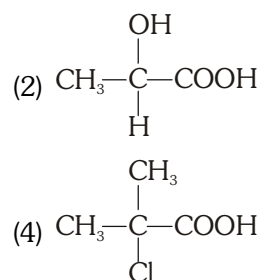
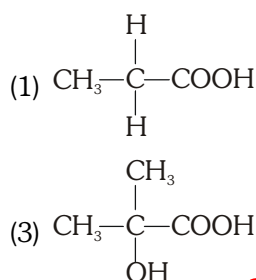


31. The correct configuration assigned for compounds (I) and (II) respectively are :-



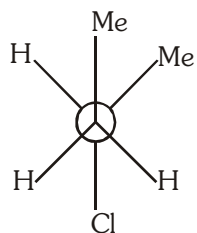
- (1) R, R (2) S, S
(3) R, S (4) S, R

32. Which compound is optical active -

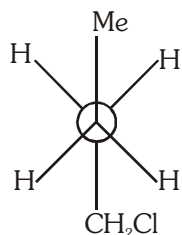


CONFORMATIONAL ISOMERISM

33. The pair of structures represents :-



- (1) Enantiomers
(3) Conformers



- (2) Position isomers
(4) None

34. Rotational angle required to get maximum stable conformer from minimum stable conformer in n-butane is :

- (1) 360° (2) 180°
(3) 120° (4) 240°

35. Which conformation of butane will have the minimum energy :-

- (1) Gauche
(2) Anti/staggered
(3) Eclipsed
(4) None

EXERCISE-I

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	4	3	2	2	4	3	1	1	3	2	2	4	4	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	3	1	2	3	4	3	1	3	2	3	4	3	4
Que.	31	32	33	34	35										
Ans.	1	2	2	2	2										

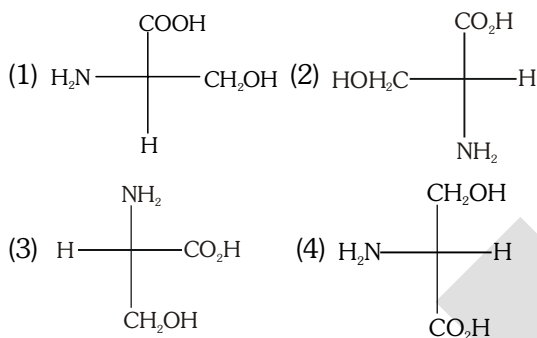
EXERCISE-II (Previous Year Questions)**AIPMT/NEET & AIIMS (2006-2018)****AIPMT-2006**

1. Which of the following is not chiral :-
 (1) 2-Butanol
 (2) 2,3-Dibromo pentane
 (3) 3-Bromo pentane
 (4) 2-Hydroxy propanoic acid

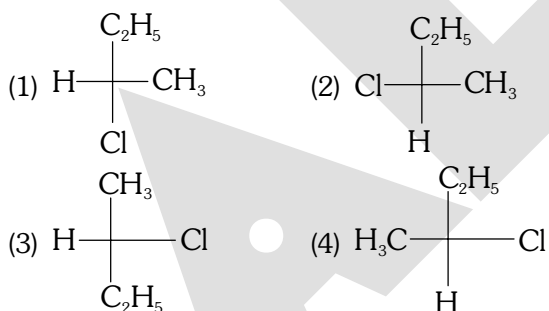
AIIMS-2006

2. Among the following which one can have a meso form -
 (1) $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{Cl})\text{C}_2\text{H}_5$
 (2) $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$
 (3) $\text{C}_2\text{H}_5\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$
 (4) $\text{HOCH}_2\text{CH}(\text{Cl})\text{CH}_3$

3. Among the following L-serine is -

**AIPMT-2007**

4. $\text{CH}_3-\text{CHCl}-\text{CH}_2-\text{CH}_3$ has a chiral centre which one of the following represents its R configuration

**AIPMT-2008**

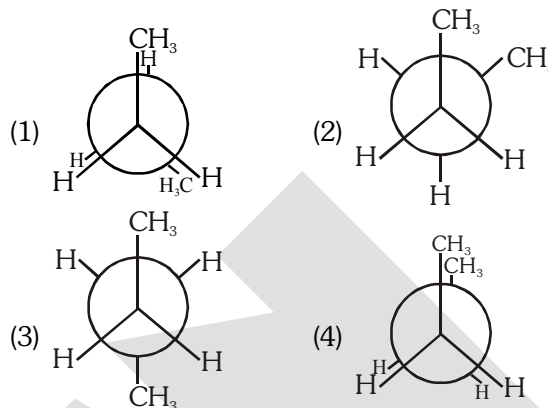
5. How many stereoisomer does this molecule have
 $\text{CH}_3\text{CH}=\text{CHCH}_2\text{CHBrCH}_3$
 (1) 8 (2) 2
 (3) 4 (4) 6

AIPMT-2009

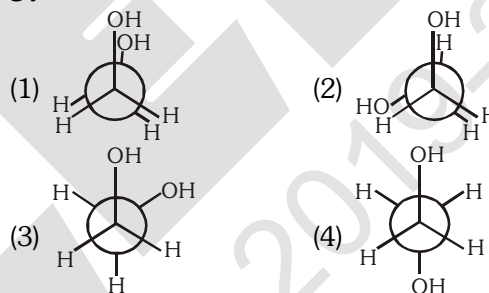
6. Which of the following compounds will exhibit cis-trans (geometrical) isomerism ?
 (1) 1-Butanol (2) 2-Butene
 (3) 2-Butanol (4) 2-Butyne

AIPMT-2010

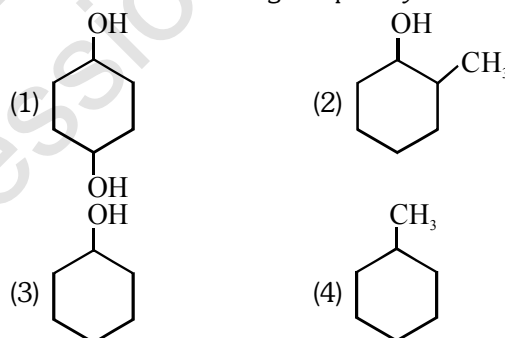
7. In the following the most stable conformation of n-butane is :-



8. Which of the following conformers for ethylene glycol is most stable :-

**AIIMS-2010**

9. Which of the following is optically active :-

**AIIMS-2011**

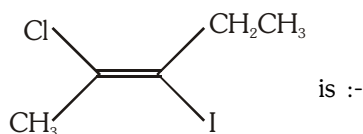
10. Find the number of stereo isomers of 1,2-dihydroxy cyclopentane:
 (1) 1 (2) 2
 (3) 3 (4) 4

AIPMT Pre.-2012

11. Which of the following acids does not exhibit optical isomerism?
 (1) Lactic acid (2) Tartaric acid
 (3) Maleic acid (4) α -amino acids

AIPMT Mains-2011

12. The IUPAC name of the following compound



- (1) cis-2-chloro-3-iodo-2-pentene
(2) trans-2-chloro-3-iodo-2-pentene
(3) cis-3-iodo-4-chloro-3-pentene
(4) trans-3-iodo-4-chloro-3-pentene

Re-AIPMT-2015

13. Two possible stereo-structures of $\text{CH}_3\text{CHOH.COOH}$, which are optically active, are called :-

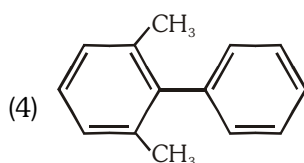
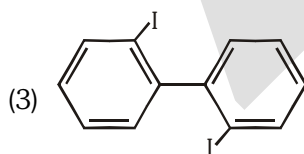
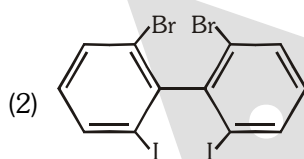
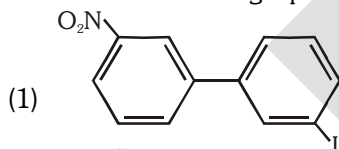
- (1) Enantiomers
(2) Mesomers
(3) Diastereomers
(4) Atropisomers

14. The number of structural isomers possible from the molecular formula $\text{C}_3\text{H}_9\text{N}$ is :

- (1) 2 (2) 3
(3) 4 (4) 5

NEET-I 2016

15. Which of the following biphenyls is optically active?

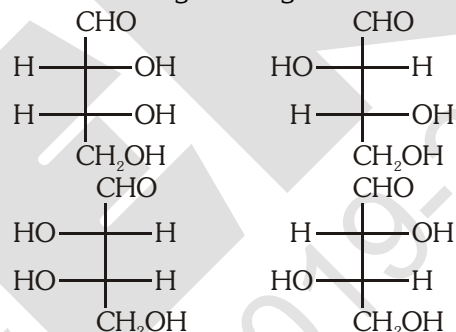


16. The **correct** statement regarding the comparison of staggered and eclipsed conformation of ethane, is :-

- (1) The staggered conformation of ethane is less stable than eclipsed conformation, because staggered conformation has torsional strain
(2) The eclipsed conformation of ethane is more stable than staggered conformation, because eclipsed conformation has no torsional strain
(3) The eclipsed conformation of ethane is more stable than staggered conformation even through the eclipsed conformation has torsional strain
(4) The staggered conformation of ethane is more stable than eclipsed conformation, because staggered conformation has no torsional strain.

NEET-II 2016

17. The **correct** corresponding order names of four aldoses with configuration given below

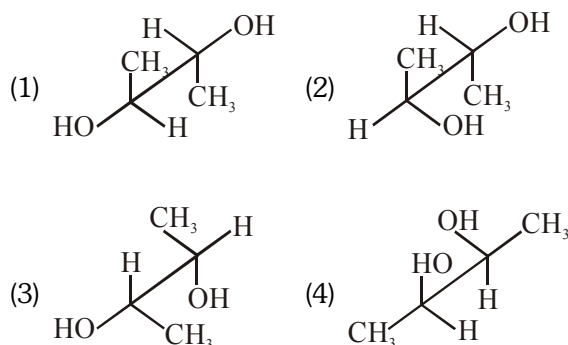
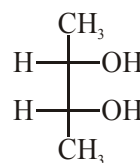


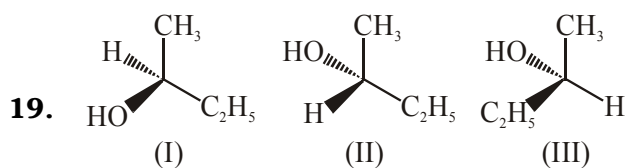
respectively, is :-

- (1) L-erythrose, L-threose, D-erythrose, D-threose
(2) D-erythrose, D-threose, L-erythrose, L-threose
(3) L-erythrose, L-threose, L-erythrose, D-threose
(4) D-threose, D-erythrose, L-threose, L-erythrose

AIIMS 2016

18. Give sawhorse representation for following compound





Find out relation in above structures

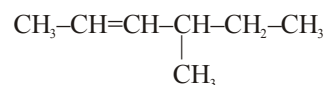
- (1) I and II are identical
 (2) II and III are identical
 (3) I and III are enantiomer
 (4) II and III are enantiomer
20. Which of the following statement is incorrect?
- (1) It is possible to separate eclipsed and staggered form of ethane at room temperature.
 (2) Eclipsed form has torsional strain.
 (3) Ethane has infinite number of conformer.
 (4) Energy difference between staggered form and eclipsed form of ethane is 12.5 kJ/mole.

NEET(UG) 2017

21. With respect to the conformers of ethane, which of the following statements is **true** ?
- (1) Bond angle changes but bond length remains same
 (2) Both bond angle and bond length change
 (3) Both bond angles and bond length remains same
 (4) Bond angle remains same but bond length changes

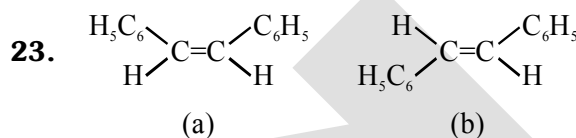
AIIMS 2017

22. How many total number of stereo isomers are possible in following compound :-



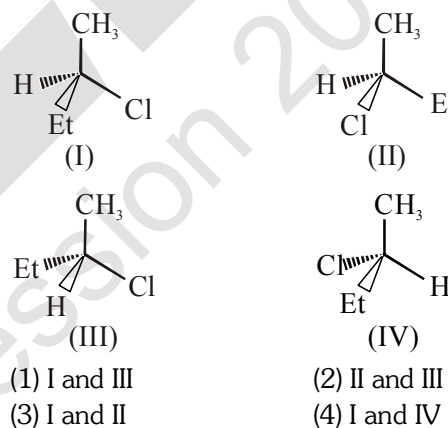
- (1) 4 (2) 2 (3) 6 (4) 3

AIIMS 2018



Compare the melting point (MP) and dipole moment (μ) of the above compounds :-

- (1) MP and dipolemoment of b > a
 (2) MP of b > a but dipolemoment of b < a
 (3) MP and dipolemoment both of b < a
 (4) MP of a > b but dipolemoment of b > a
24. Which of the following isomers are not pair of enantiomers:-



EXERCISE-II (Previous Year Questions)

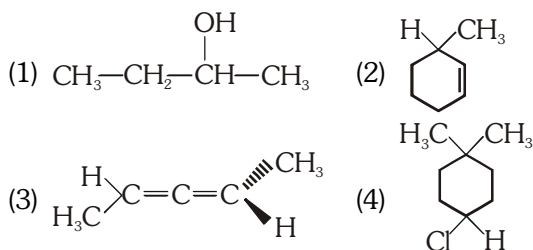
ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	3	2	3	2	3	3	2	3	3	2	1	3	2
Que.	16	17	18	19	20	21	22	23	24						
Ans.	4	2	1	4	1	3	1	2	2						

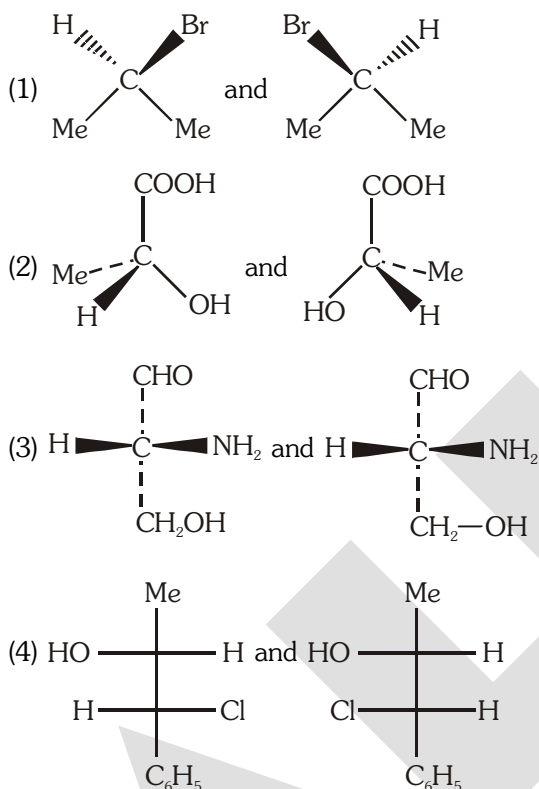
EXERCISE-III (Analytical Questions)

Check Your Understanding

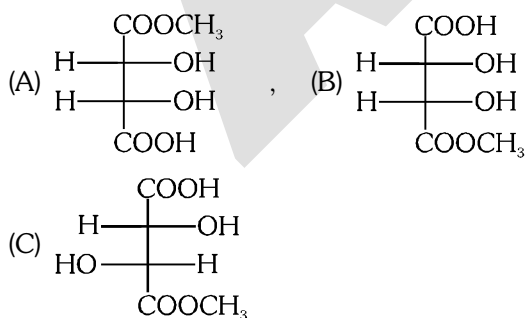
1. Which of the following compound is optically inactive:



2. Which of the following pairs are diastereomers :



3. The correct statement about the compound (A), (B) and (C) is :-

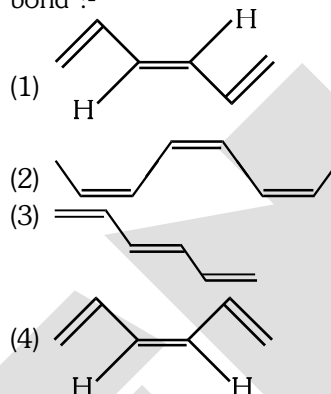


- (1) (A) and (B) are identical
(2) (A) and (B) are diastereomers
(3) (A) and (C) are enantiomers
(4) (A) and (B) are enantiomers

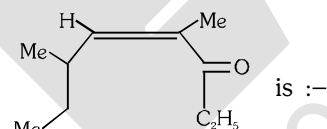
4. Which of the following does not contain any asymmetric carbon but can show enantiomerism:-

- (1) Lactic acid (2) 1,3-pentadiene
(3) Tartaric acid (4) 2,3-pentadiene

5. Which of the following represents the structure having cis arrangement around each double bond :-

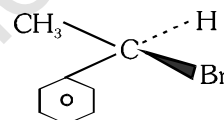


6. The IUPAC name of compound



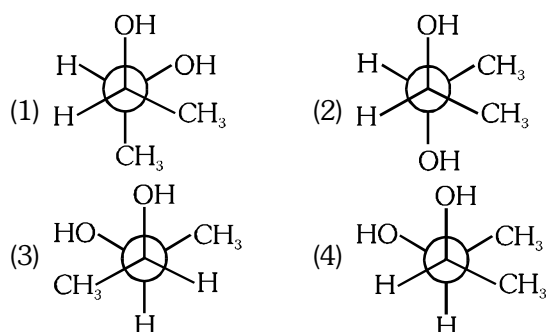
- (1) (Z)-4,6,7-Trimethyl hept-4-en-3-one
(2) (E)-4,6-Dimethyl oct-4-en-3-one
(3) (Z)-4,6-Dimethyl oct-4-en-3-one
(4) (E)-4,6-Dimethyl oct-4-en-6-one

7. The complete IUPAC name of the compound :-

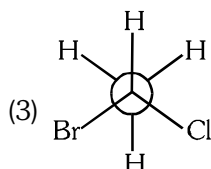
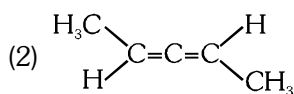
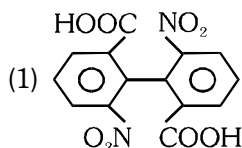


- (1) (R)-1-Bromo-1-phenyl ethane
(2) (S)-1-Bromo-1-phenyl ethane
(3) (E)-1-Bromo-1-phenyl ethane
(4) (Z)-1-Bromo-1-phenyl ethane

8. Which one of the following is the most stable conformation of 2, 3-butanediol :-



9. Which of the following shows optical isomerism:-



(4) All the above

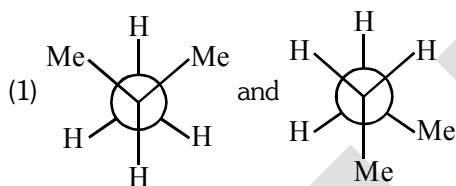
10. How many isomers of $C_5H_{11}OH$ will be primary alcohols (exclude stereoisomers) :-

(1) 2 (2) 3 (3) 4 (4) 6

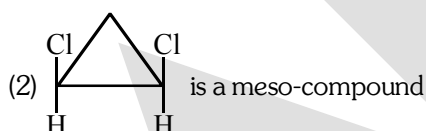
11. The minimum number of carbon atoms in ketone to show metamerism :-

(1) 3 (2) 4 (3) 5 (4) 6

12. Which is incorrect statement :-



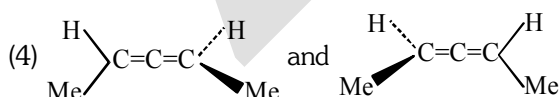
are conformations



is a meso-compound

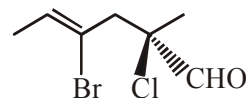


are Geometrical isomers



are Enantiomers

13. The IUPAC name of the given compound is :-



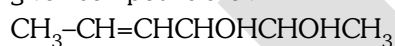
(1) (2R, 4Z)-4-bromo-2-chloro-2-methyl-4-hexenal

(2) (2R, 4E)-4-bromo-2-chloro-2-methyl-4-hexenal

(3) (2Z, 5R)-3-bromo-5-chloro-5-formyl-2-hexene

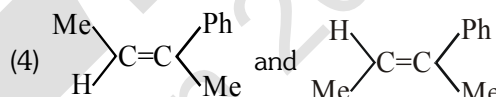
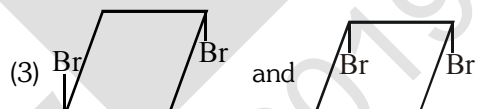
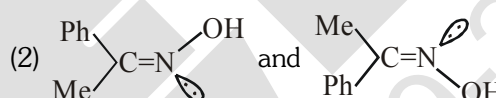
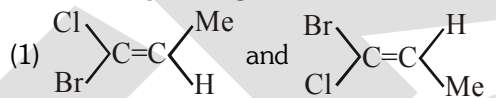
(4) (2S, 4E)-4-bromo-2-chloro-2-methyl-4-hexenal

14. The total number of configurational isomers of the given compound are :-



(1) 2 (2) 4 (3) 6 (4) 8

15. Which is a pair of geometrical isomers :-



16.

(1) Enantiomers

(2) Position isomers

(3) Geometrical isomers

(4) Homomers

17. Which of the following is not a metamer of $C_4H_{10}O$

(1) Diethyl ether

(2) Methyl n-propyl ether

(3) 2-Methoxy propane

(4) Isobutyl alcohol

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	4	4	4	2	3	1	3	4	3	3	1	1	4	4
Que.	16	17													
Ans.	4	4													

EXERCISE-IV (Assertion & Reason)

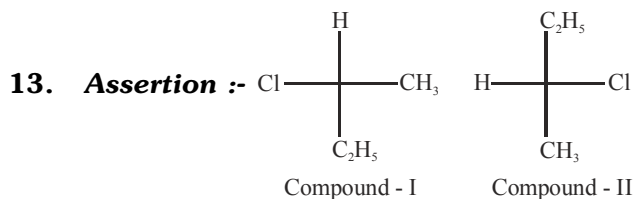
Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** The melting point of fumaric acid is higher than that of maleic acid.
Reason :- The molecules of fumaric acid are more symmetric than those of maleic acid and hence it gets closely arranged in the crystal lattice.
 (1) A (2) B (3) C (4) D
- Assertion :-** trans-1-chloro propene has higher dipole moment than cis-1-chloro propene.
Reason :- The resultant of two vectors in trans-1-chloropropene is more than in cis-1-chloro propene.
 (1) A (2) B (3) C (4) D
- Assertion :-** Trihydroxyglutaric acid (COOH—CHOH—CHOH—CHOH—COOH) exists in four stereoisomeric forms, two of which are optically active while the other two are meso-forms.
Reason :- It contains two asymmetric and one pseudo-asymmetric carbon atom.
 (1) A (2) B (3) C (4) D
- Assertion :-** All meso form of compound are optical isomer.
Reason :- Meso form is optically inactive.
 (1) A (2) B (3) C (4) D
- Assertion :-** Benzaldehyde forms two oximes on reacting with NH_2OH .
Reason :- The two oximes arise due to geometrical isomerism around $\text{C}=\text{N}$ bond.
 (1) A (2) B (3) C (4) D
- Assertion :-** The boiling point of cis 1, 2-dichloro ethene is higher than that of corresponding trans-isomer.
Reason :- Cis- 1,2-dichloro ethene has higher dipole moment as compared to that of the trans-isomer.
 (1) A (2) B (3) C (4) D
- Assertion :-** Meso tartaric acid is optically inactive.
Reason :- Its optical inactivity is due to external compensation.
 (1) A (2) B (3) C (4) D
- Assertion :-** 1,2-propadiene exhibits optical isomerism.
Reason :- Its mirror image is non-superimposable.
 (1) A (2) B (3) C (4) D
- Assertion :-** Ethanol cannot show position isomerism.
Reason :- Ethanol cannot show isomerism.
 (1) A (2) B (3) C (4) D
- Assertion :-** $\text{CH}_3\text{—CH}_2\text{—Br}$ and $\text{CH}_3\text{—CH}_2\text{—I}$ are Functional isomers.
Reason :- The above compounds are homologues.
 (1) A (2) B (3) C (4) D
- Assertion :-** Neopentyl bromide and 2-bromo-2-methyl butane are enantiomers.
Reason :- Both have same molecular formula as well as structure formula.
 (1) A (2) B (3) C (4) D
- Assertion :-** Organic compounds which do not contain chiral carbon atoms can be optically active.
Reason :- Presence of chiral carbon is an essential condition for optical activity.
 (1) A (2) B (3) C (4) D



Compound-I and II are enantiomers.

Reason :- Non superimposable mirror images are enantiomers.

(1) A (2) B (3) C (4) D

14. **Assertion :-** cis-3-bromopropenoic acid is more stable than its trans form.

Reason :- Dipole moment of cis form is greater than trans form

(1) A (2) B (3) C (4) D

15. **Assertion :-** All hydrogen atoms in $\text{CH}_2=\text{C}=\text{CH}_2$ lie in one plane.

Reason :- All the carbon atoms in it are sp^2 hybridised.

(1) A (2) B (3) C (4) D

16. **Assertion :-** 4-Methylhex-2-ene shows stereoisomerism.

Reason :- It can show Geometrical isomerism.

(1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	1	1	2	1	1	3	4	3	4	4	3	1	4	4
Que.	16														
Ans.	2														

REACTION MECHANISM - I

E

S. No.	CONTENTS	Page
1.	Concepts to Understand reaction mechanism	45
2.	Type of bond cleavage	45
3.	Type of attacking reagent	45
4.	Reaction Intermediate	46
5.	Electronic effect	47
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7.	Resonance effect	51
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NEET SYLLABUS

Electronic displacements in a covalent bond : inductive effect, electromeric effect, resonance and hyper conjugation. Homolytic and heterolytic fission of a covalent bond : free radicals, carbocations, carbanions; electrophiles and nucleophiles.

OBJECTIVES

After studying this unit, we will be able to :

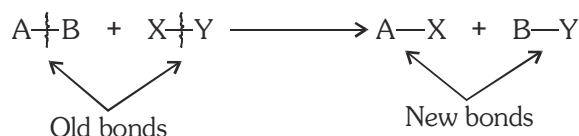
- understand the concept of organic reaction mechanism;
- explain the influence of electronic displacements on structure and reactivity of organic compounds;

*"When learning is purposeful, creativity blossoms. When creativity blossoms;
thinking emanates. When thinking emanates, knowledge is fully lit.
When knowledge is lit, economy flourishes."*

A.P.J. Abdul Kalam

REACTION MECHANISM : PART-I

- ❑ **Reaction :** Breaking of old bond and formation of new bond is known as chemical reaction



A sequential account of each step, describing details of electron movement, energetics during bond cleavage and bond formation, and the rates of transformation of reactants into products (kinetics) is referred to as reaction mechanism. Reactants are of two types substrate and reagent.

Substrate is that reactant which supplies carbon to the new bond and the other reactant is called reagent. If both reactants supply carbon to the new bond then choice is arbitrary and in that case the molecule on which attention is focused is called substrate.

3.0 CONCEPTS TO UNDERSTAND REACTION MECHANISM :

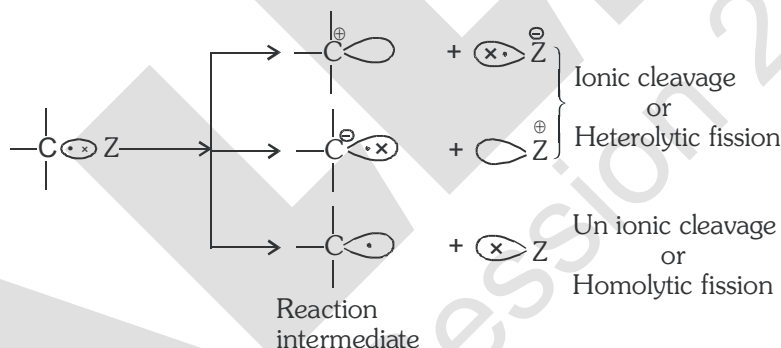
- (1) Bond cleavage (2) Attacking reagent (3) Reaction intermediate (4) Electronic effect

3.1 TYPE OF BOND CLEAVAGE :

(a) Heterolytical cleavage/fission : Cleavage in which unequal distribution of electrons takes place during the bond cleavage is known as heterolytical cleavage. Due to unequal distribution of electrons, ions are formed. That's why it is also known as ionic cleavage or heterolytical cleavage.

(b) Homolytical cleavage/fission : Cleavage in which equal distribution of e^- s takes place during the chemical reaction, is known as homolytical cleavage.

Due to equal distribution of electrons, without charge unpaired electron containing species are formed, which are known as free radicals and cleavage is known as unionic cleavage/homolytical fission.



3.2 TYPES OF ATTACKING REAGENTS

These are of two types :

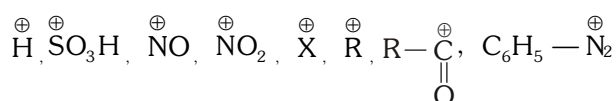
(a) Electrophilic reagent or electrophiles:

Electrophilic (electro + philic)
 (electron + loving)

The reagent which attacks on the negative part of the molecule or having attraction for electrons are called electrophiles.

Electrophiles may be positively charged or neutral.

(i) Positively charged electrophiles :



(ii) Neutral electrophiles :- central atom e^- deficient

(a) All Lewis acids as :

$\text{BF}_3, \text{AlCl}_3, \text{SO}_3, \text{ZnCl}_2, \text{BeCl}_2, \text{FeCl}_3, \text{SnCl}_2, \text{CO}_2, \text{SnCl}_4.$

(b) Free radicals, carbenes and nitrenes act as electrophiles.

(b) Nucleophilic reagent or nucleophiles

Which attacks on the positive site of the substrate or loves nucleus or having attraction towards nucleus.

Nucleophilic

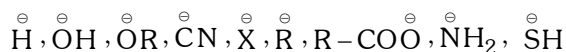
(Nucleo + philes)



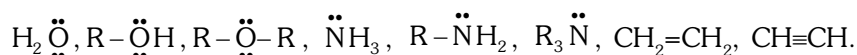
(Nucleus + loving)

Nucleophiles may be negatively charged ions or possess a lone pair of electron or πe^- .

(i) Negatively charged nucleophiles.



(ii) All Lewis base which contains lone pairs or πe^-



(iii) $\text{R}^* - \text{Mg} - \text{X}$, LiAlH_4^* , NaBH_4^*

The star (*) indicates the atom which donates electrons to the substrate.

Ambident nucleophile :- Nucleophiles which have two sites of electron rich centre or in which two or more atoms bear a lone pair of electrons.

Examples :- $\text{K}^+\text{O}^--\text{N}=\text{O}$, $\text{NH}_2-\ddot{\text{O}}\text{H}$, $\text{Na}^+\text{C}^-\equiv\text{N}$

3.3 REACTION INTERMEDIATE**□ Carbocation :**

Cation in which positive charge is present on carbon atom is called carbocation.

- ◆ Due to electron deficiency it acts as an electrophile and always attack on electron richer site.
- ◆ It is incomplete octet species because it has six electron in outer most shell.
- ◆ All electrons are paired.

□ Carbanions : Anion in which negative charge is present on carbon atom is called carbanion.

- ◆ It has eight electron in outermost shell so it is complete octet species.
- ◆ It is an electron richer species because it has extra electron.
- ◆ Due to presence of non bonding electrons it acts as a nucleophile.

□ Free Radical :

- ◆ Electrically neutral species in which unpaired electron is present on carbon atom is known as carbon free radical.
- ◆ It has seven electron or odd electron in outermost shell of unpaired electron containing carbon.
- ◆ It is electron deficient species due to incomplete octet.

□ Carbenes ($\text{CH}_2:$) :

Carbenes are neutral carbon species in which the carbon atom is bonded to two monovalent atoms or groups and carries two nonbonded electrons.

- It behaves as an electrophile.
- It is neutral.
- $6 e^-$ in outermost shell.
- $4 e^-$ are bonded and two are nonbonded e^- .

□ Nitrenes ($-\ddot{\text{N}}:$)

Nitrenes are neutral nitrogen species in which the nitrogen is bonded to one monovalent atom or group and carries four non-bonded electrons.

- It is monovalent radical.
- It is neutral.
- $6 e^-$ in outermost shell.
- Two are bonded and four are nonbonded electrons.

BEGINNER'S BOX-1

1. Which of the following is ambident nucleophile :-
- (1) NH_2OH (2) $\text{NCO}^\ominus$ (3) $\text{NO}_2^\ominus$ (4) All of these
2. $\text{CH}_3\text{CH}_2\text{-Cl}$ undergoes homolytic fission to produce :-
- (1) $\text{CH}_3\dot{\text{C}}\text{H}_2$ and $\dot{\text{C}}\text{H}_2\text{Cl}$ (2) $\text{CH}_3\dot{\text{C}}\text{H}_2$ and $\text{Cl}^\ominus$
- (3) $\text{CH}_3\overset{\oplus}{\text{C}}\text{H}_2$ and $\text{Cl}^\ominus$ (4) $\text{CH}_3\overset{\ominus}{\text{C}}\text{H}_2$ and $\text{Cl}^\oplus$
3. Which of the following intermediate has complete octet :-
- (1) Carbocation (2) Carbanion (3) Free radical (4) Carbene

3.4 ELECTRONIC EFFECTS :

There are four effects which affect the chemical reaction due to transfer of electron

- (1) Inductive effect (2) Mesomeric effect
(3) Hyperconjugation (4) Electromeric effect

3.4.1 INDUCTIVE EFFECT (I-EFFECT) :

- ◆ Polarity induced in non polar σ bond due presence of adjacent polar bond is known as inductive effect.

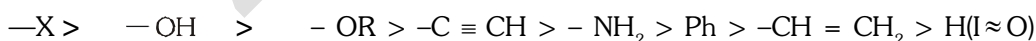
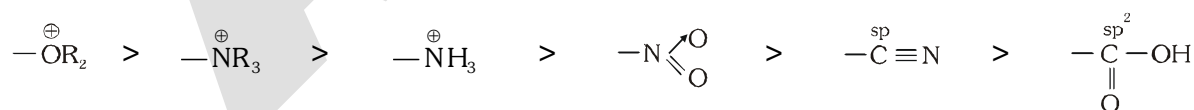
GOLDEN KEY POINTS

- In I-effect there is partial displacement of $e^\ominus$.
- After 3 or 4 C-atom I-effect is considered to be zero.
- Inductive effect decreases on increasing distance.

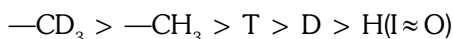
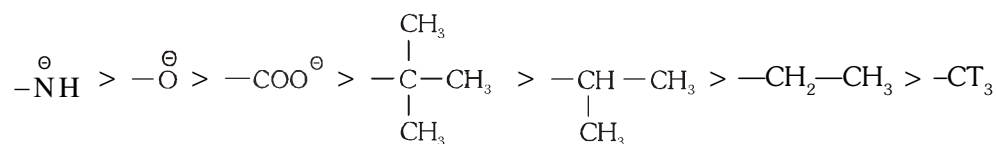
So **Magnitude of I effect** $\propto \frac{1}{\text{distance}}$

- I-effect of hydrogen is considered as zero.

-I groups :



+I groups :

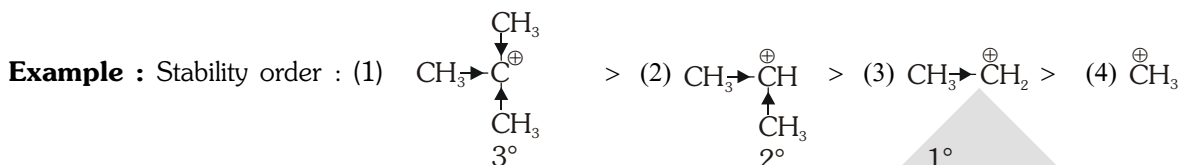


APPLICATION OF I-EFFECT

(1) Stability of carbocation :

$$\text{Energy} \propto \text{charge} \propto \frac{1}{\text{stability}}$$

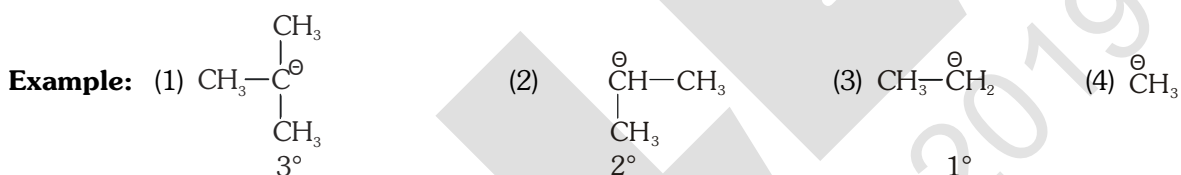
$$\text{Stability of carbocation} \propto \frac{+I \text{ effect}}{-I \text{ effect}}$$



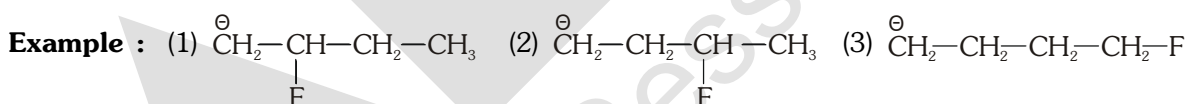
Reason : More no. of +I group.
more stable carbocation.
so stability order $1 > 2 > 3 > 4$.

(2) Stability of carbanion :

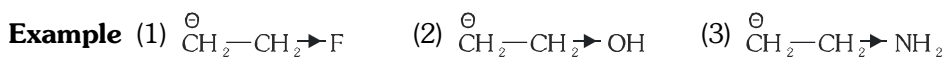
$$\text{Stability of Carbanion} \propto \frac{-I \text{ effect}}{+I \text{ effect}}$$



More No. of +I group.
Less stable carbanion.
So stability order $4 > 3 > 2 > 1$



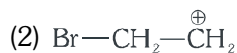
Minimum distance of -F.
Maximum -I of -F.
Minimum negative charge.
Maximum stable.
So stability order $1 > 2 > 3$



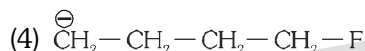
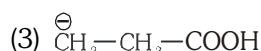
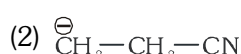
Maximum -I of F.
Negative charge will be minimum.
Maximum stable.
So stability order $1 > 2 > 3$

BEGINNER'S BOX-2

1. Most stable carbocation is :

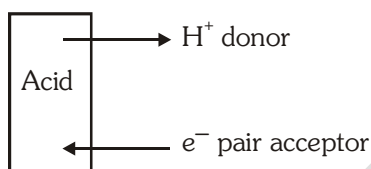


2. Most stable carbanion is :



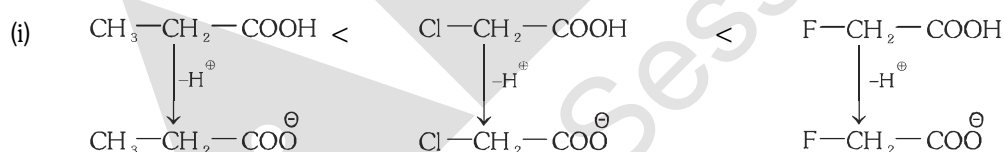
(3) Acidic and basic strength :

◆ Acidic strength :



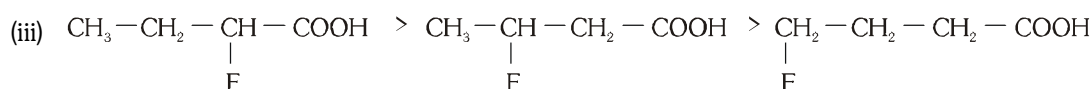
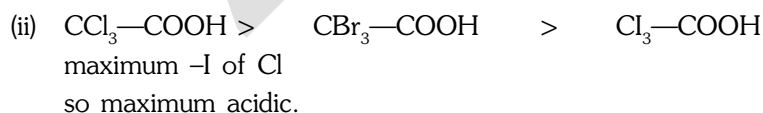
Acidic strength $\propto$ Stability of conjugate base (anion) $\propto \frac{-I \text{ effect}}{+I \text{ effect}}$

Example :

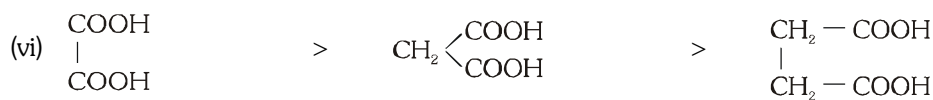
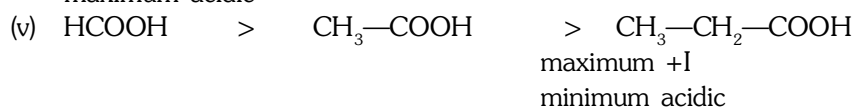
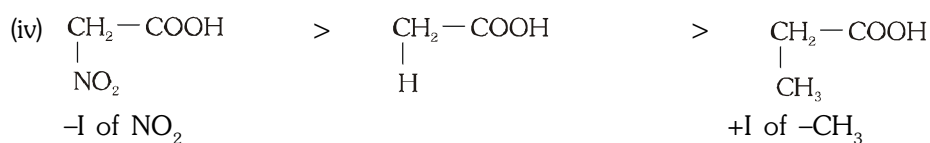


+I of $-\text{CH}_3$
 anion is less stable

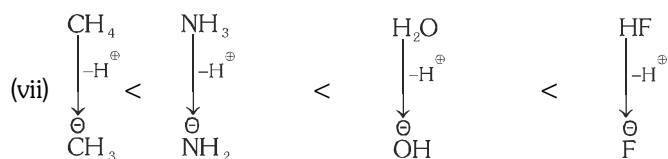
Maximum -I of -F
 Maximum stable anion
 Corresponding acid is maximum acidic.



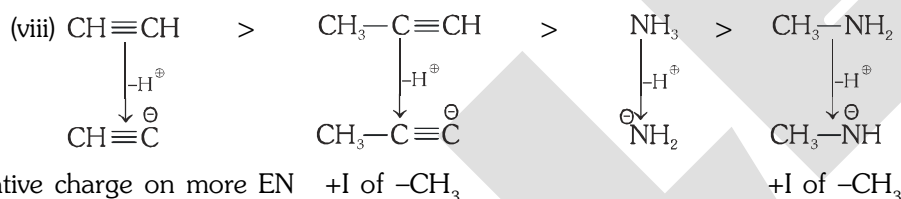
minimum distance of F from $-\text{COOH}$
 maximum -I of F.
 So maximum acidic.



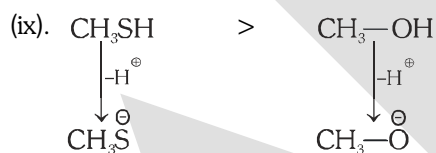
minimum distance of -COOH from other
 maximum -I of -COOH on other
 maximum acidic



Negative charge on maximum E.N.
Maximum stable anion
So corresponding acid is most acidic

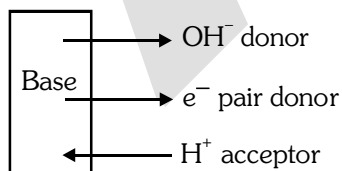


negative charge on more EN atom and no +I
anion is maximum stable
so corresponding acid is most acidic

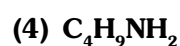
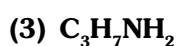
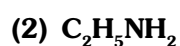
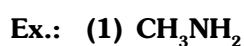


negative charge on big size atom
more stable anion
so corresponding acid is more acidic

◆ **Basic strength :**



$$\text{Basic strength} \propto +I \text{ effect} \propto \frac{1}{-I \text{ effect}}$$



Ans. $4 > 3 > 2 > 1$

Maximum +I

so maximum basic

BEGINNER'S BOX-3

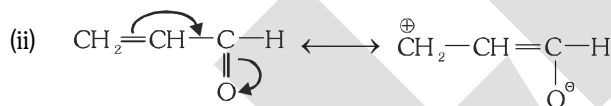
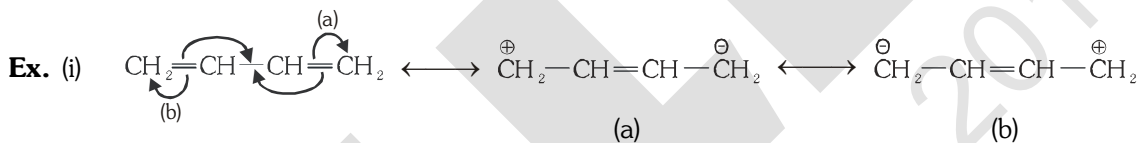
- 1.** Which of the following is most acidic :
- (1) $\text{Cl}_3\text{C}-\text{COOH}$ (2) $\text{Cl}_2\text{CH}-\text{COOH}$ (3) $\text{ClCH}_2-\text{COOH}$ (4) CH_3COOH
- 2.** Which of the following is most acidic :
- (1) $\begin{array}{c} \text{CH}_2 - \text{COOH} \\ | \\ \text{F} \end{array}$ (2) $\begin{array}{c} \text{CH}_2 - \text{COOH} \\ | \\ \text{Cl} \end{array}$
- (3) $\begin{array}{c} \text{CH}_2 - \text{COOH} \\ | \\ \text{Br} \end{array}$ (4) $\begin{array}{c} \text{CH}_2 - \text{COOH} \\ | \\ \text{I} \end{array}$
- 3.** Which is most basic among the following :
- (1) $\text{CH}_3\text{CH}_2-\text{NH}_2$ (2) $\text{Cl}-\text{CH}_2\text{CH}_2-\text{NH}_2$
- (3) $\text{O}_2\text{N}-\text{CH}_2\text{CH}_2-\text{NH}_2$ (4) $^-\text{O}-\text{CH}_2\text{CH}_2-\text{NH}_2$

3.4.2 RESONANCE OR MESOMERIC EFFECT

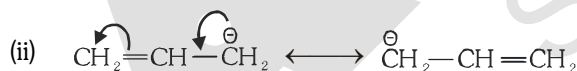
Delocalization of πe^- is called as resonance or complete transfer of πe^- from one shell to another shell is called as **Resonance**.

Conditions For Resonance

- (1)** If there are two π bonds in conjugation then e^- of one π bond are transferred towards another π bond.



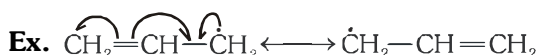
- (2)** If there is lone pair or a negative charge and π bond are in conjugation then lone pair of e^- or negative charge are transferred towards π bond.



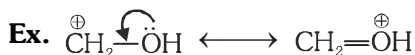
- (3)** If there is positive charge (vacant orbital) and π bond are in conjugation then e^- of π bond are transferred towards positive charge.



- (4)** If there is free e^- and π bond are in conjugation.



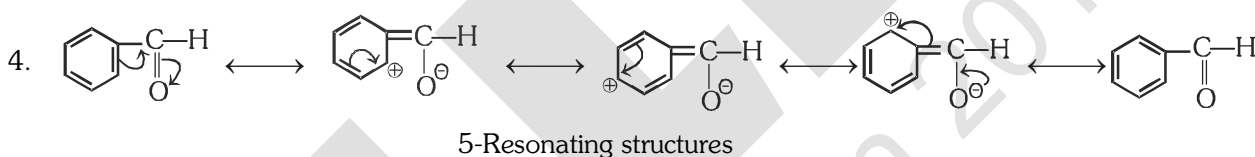
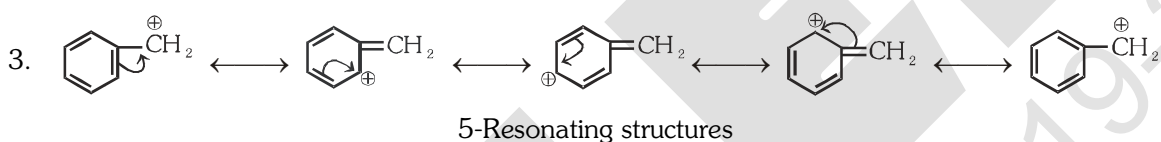
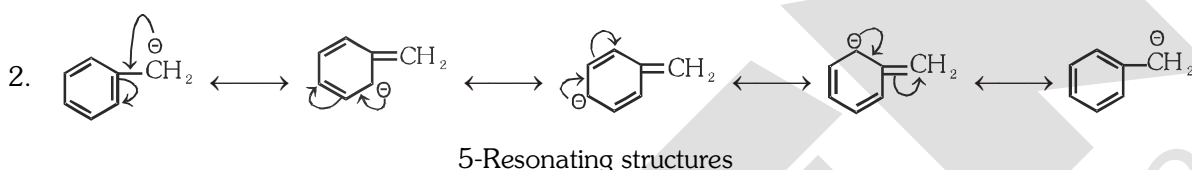
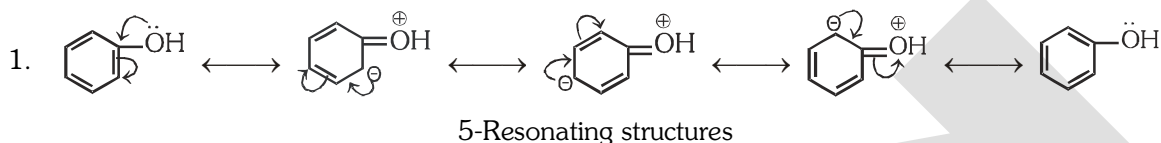
- (5)** If there is lone pair or negative charge and positive charge (vacant orbital) are in conjugation then e^- of lone pair or negative charge are transferred towards positive charge.



GOLDEN KEY POINTS

- In resonance only e^- are delocalised not atoms.
- The number of e^- or number of unpaired or paired e^- in all resonating structures should be same.
- It is permanent effect.
- All the resonating or canonical structures must follow the Lewis structures.

Draw resonating structures :

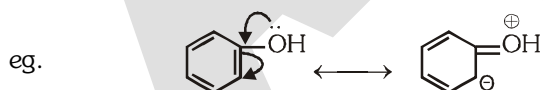


M-effect : Delocalisation of electron in conjugated system (due to presence of EWG or EDG) is known as 'M' effect.

(1) +M effect :- Group that donates the electron pair to conjugated system is known as +M effect exerting groups and the phenomena is known as +M effect.

+M group : Lone pair containing group like

e.g. $-\ddot{N}H_2$, $-\ddot{O}H$, $-\ddot{O}R$, $-\ddot{N}R_2$, $-\ddot{S}H$, $-\ddot{N}HR$, $-\ddot{X}$, $-\ddot{N}HCOCH_3$

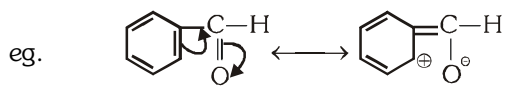


-OH group lone pair donor

So + M of -OH group

(2) -M effect :- Group, that withdraws electron pair from the conjugated system, is known as -M effect exerting groups and the phenomena is known as -M effect.

-M group : $-\text{CHO}$, $-\text{COOH}$, $-\text{COOR}$, $-\text{COR}$, $-\text{NO}_2$, $-\text{CN}$, $-\text{COX}$, $-\text{CONH}_2$, $-\text{SO}_3\text{H}$



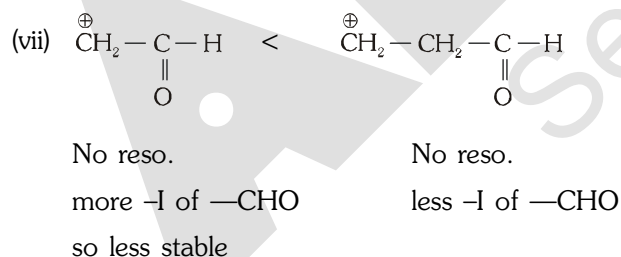
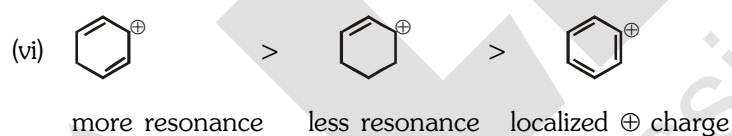
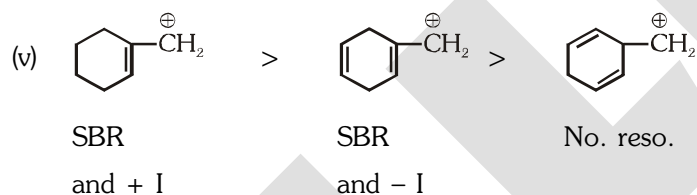
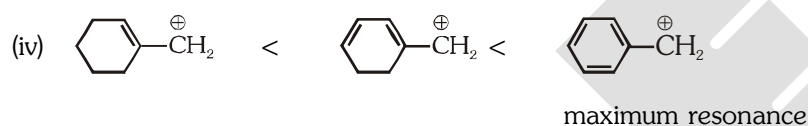
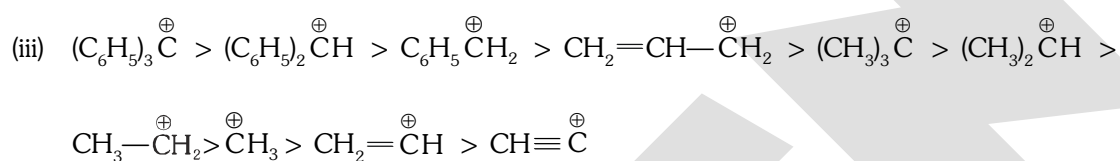
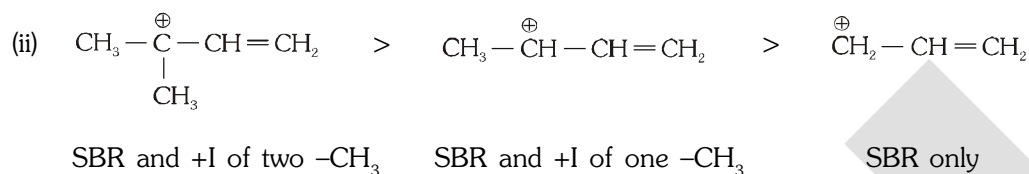
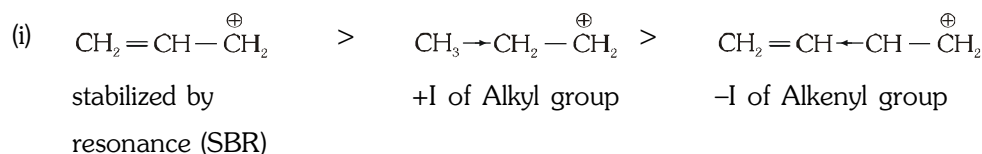
-CHO group withdrawing e^- .

So -CHO is -M group

APPLICATIONS OF RESONANCE EFFECT :

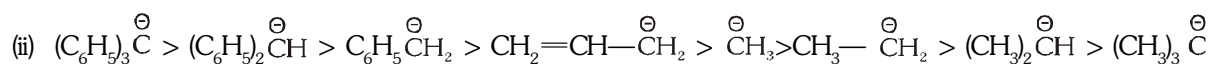
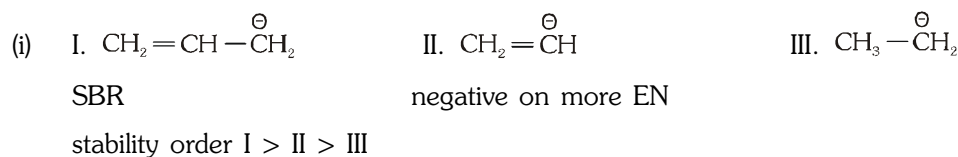
(1) Stability of carbocation.

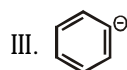
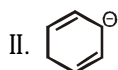
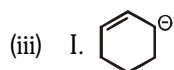
Ex. Give stability order for :-



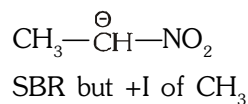
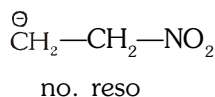
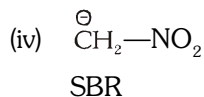
(2) Stability of carbanion :

Ex. Give stability order of :

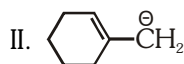
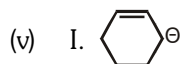




resonance stable more resonance stable localized $\ominus$ charge
 stability order II > I > III



stability order I > III > II

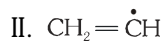
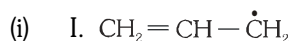


SBR and +I

SBR

SBR and more +I

stability order II > I > III

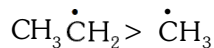
(3) Stability of free radicals.**Ex.** Give stability order for :

less resonance

no resonance

more resonance

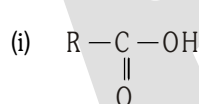
stability order III > I > II

**(4) Stability of resonating structures (R.S) Rules :**

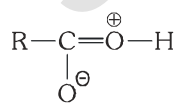
(i) Complete octet R.S. is more stable than incomplete octet.

(ii) Nonpolar R.S. is more stable than polar resonating structures.

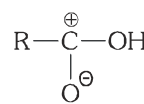
(iii) For charged R.S. negative charge on more EN and positive charge on less EN is more stable.

Ex. Arrange the following for stability order.

>



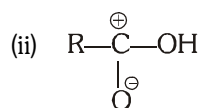
>



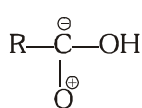
complete octet, nonpolar

complete octet, polar

incomplete octet



>

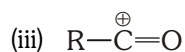
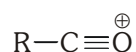


less EN atom

more EN atom with

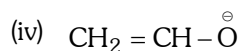
with incomplete octet

incomplete octet

 $\angle$ 

incomplete octet

complete octet

 $\geq$ 

(negative charge on more EN)

(negative charge on less EN)

(5) Aromaticity (Huckel's rule) : Cyclic, planar and completely conjugated system with $(4n+2)\pi$ electrons (where $n = 0, 1, 2, 3, \dots$) is known as aromatic compounds, these compound gains extra stability which is known as aromaticity.

Note : $[4n + 2]$ π electrons. (Odd number of π electron pairs) means

If $n = 0$ 2π electrons or 1 pair

n = 1 6 π electrons or 3 pairs

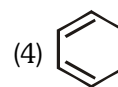
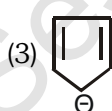
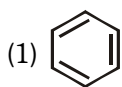
$$n = 2 \qquad 10\pi \text{ electrons} \qquad \text{or } 5 \text{ pairs}$$
$$n = 3 \qquad 14\pi \text{ electrons} \qquad \text{or } 7 \text{ pairs}$$

S.No	Compound	Cyclic	Plannar	Cyclic Resonance	Huckel Rule (4n+2) πe^-	Aromatic Yes/No
1.		✓	✓	✓	$2\pi e^-$	Yes
2.		✓	✓	✓	$4\pi e^-$	No
3.		✓	✓	✓	$4\pi e^-$	No
4.		✓	✓	✓	$6\pi e^-$	Yes
5.		✓	✓	✓	$6\pi e^-$	Yes
6.		✓	✗	✗	$4\pi e^-$	No
7.		✓	✓	✓	$6\pi e^-$	Yes
8.		✓	✓	✓	$6\pi e^-$	Yes

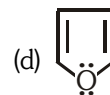
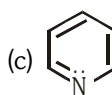
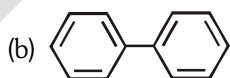
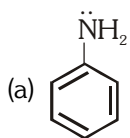
9.		✓	✓	✓	$6\pi e^-$	Yes
10.		✓	✓	✓	$6\pi e^-$	Yes
11.		✓	✓	✓	$6\pi e^-$	Yes
12.		✓	✓	✓	$10\pi e^-$	Yes
13.		✓	✓	✓	$14\pi e^-$	Yes
14.		✓	✓	✓	$6\pi e^-$	Yes
15.		✓	✗	✗	$6\pi e^-$	No
16.		✓	✓	✓	$6\pi e^- + 6\pi e^-$	Yes

BEGINNER'S BOX-4

1. Which of the following is nonaromatic



2. Which of the following compounds are aromatic in nature



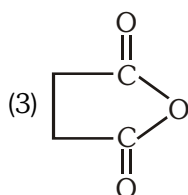
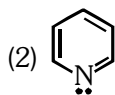
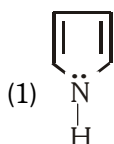
(1) Only a

(2) a and b

(3) a,b,d

(4) all of these

3. Which of the following is not aromatic compound



(4) All of these

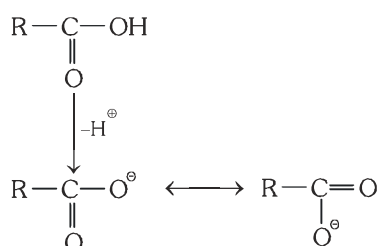
(6) Acidic and Basic strength :

(a) Acidic strength : $\left(\text{Acidic strength} \propto \text{stability of conjugate base (anion)} \propto \frac{-M, -I}{+M + I} \right)$

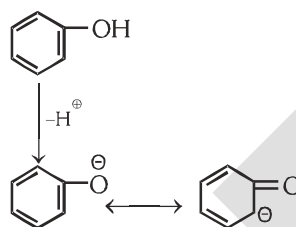
Illustrations

Illustration 1. Carboxylic acids are more acidic than phenols, why ?

Solution.



2, equal R.S. more stable anion
so corresponding acid is more acidic.
Here, carboxylate ion is more stable
than phenoxide ion.

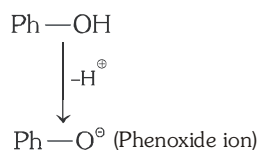


5, unequal R.S. less stable anion since $-ve$ charge is being shared by oxygen is less electronegative carbon.

Note : The molecule having equivalent R.S. has more stability than the molecule having non equivalent R.S.

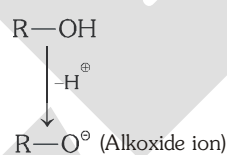
Illustration 2. Phenol is more acidic than alcohols why ?

Solution.



anion stable by resonance

So, corresponding acid is more acidic.



no resonance.

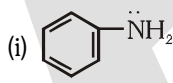
less stable anion

(b) Basic strength order :- Tendency to donate the electron pair by an atom or group is known as its basic strength. Compounds in which electron pair is delocalised will be less basic while, Those in which electron pair is localised will be more basic.

$$\left(\text{Basic strength} \propto \text{H}^{\oplus} \text{ accepting tendency} \propto \text{l.p. donating tendency} \propto \frac{+M, +I}{-M, -I} \right)$$

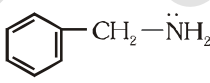
Ex.

Give basic strength order :



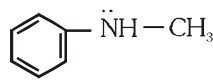
I

l.P. is stabilized by
resonance
basic order —



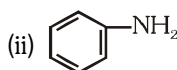
II

no reso. of l.p.
so maximum basic
 $II > III > I$



III

l.p. is stabilized by resonance
and +I of CH_3



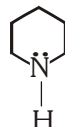
I

SBR
basic order —



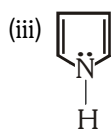
II

localized l.p.
on more EN
 $III > II > I$



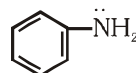
III

localized l.p.
on less EN

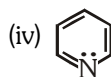


more resonance of l.p.
so less basic

<



less resonance of l.p.
so more basic



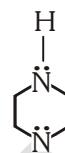
I
l.p. on
more EN
so minimum basic
So, basic order



II
more -I
of oxygen



III
+I

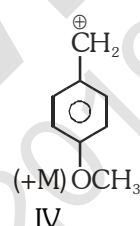
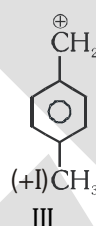
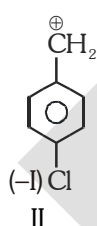
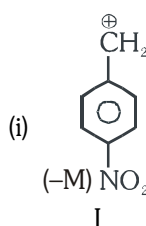


IV
less -I of
nitrogen

III > IV > II > I

Illustrations

Illustration 3. Give stability order of :

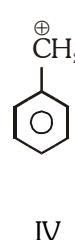
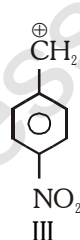
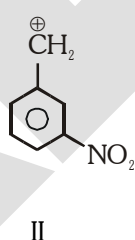
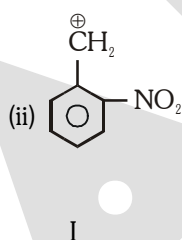


charge ↑
So stability order

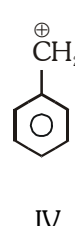
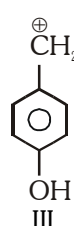
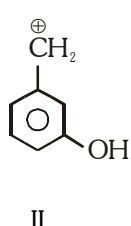
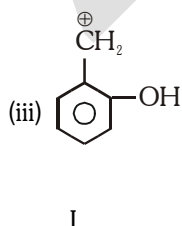
charge ↑
IV > III > II > I

charge ↓

charge ↓



$-M_o = -M_p$ and $-M_m = 0$
but $-I_o > -I_m > -I_p$, $-M$ and $-I$ increases positive charge.
stability order is IV > II > III > I



$-OH$ group shows $+M$ effect
 $+M_o = +M_p$ and $+M_m = 0$ but $-I_o > -I_m > -I_p$ and $+M \gg -I$
So $+M$ stabilize the carbocation by decreasing positive charge
Stability order III > I > IV > II

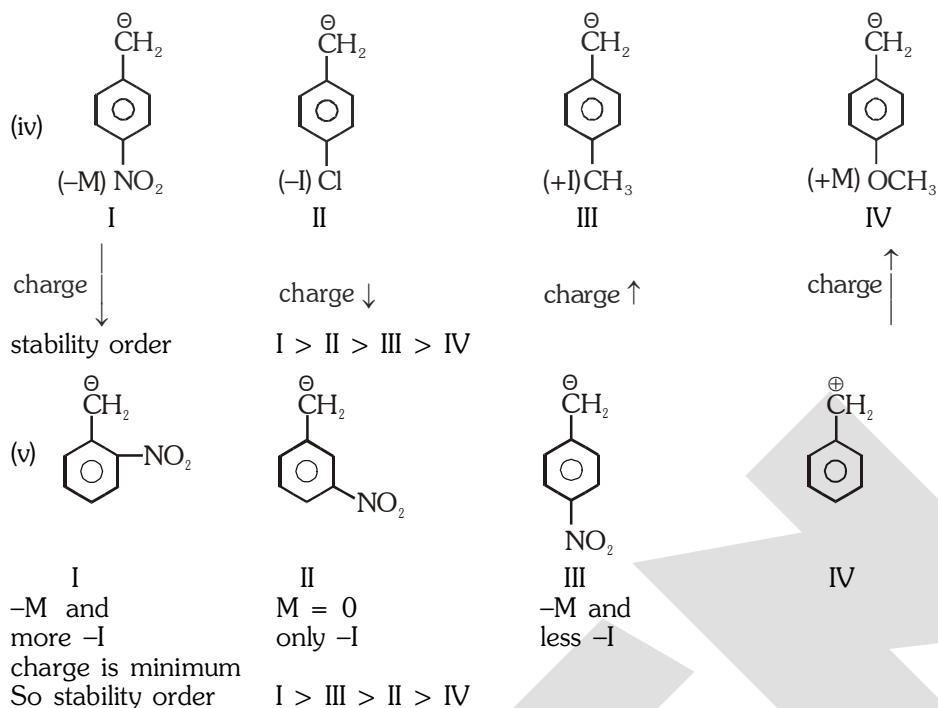
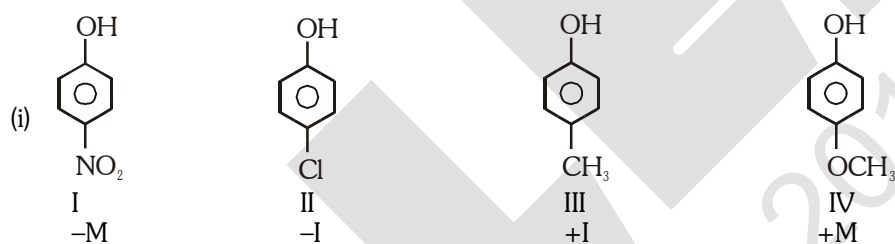
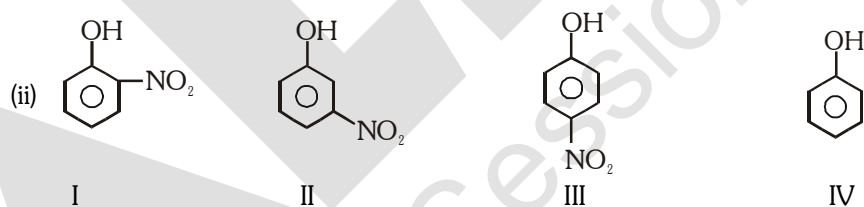


Illustration 4. Give acidic strength order for :



So acidic order is $I > II > III > IV$



–M and
more –I

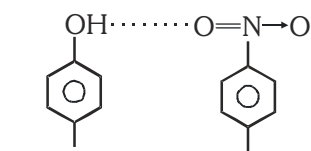
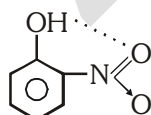
M = 0
only -I

–M and
less –I

more acidic only less acidic

Acidic order should be $I > III > II > IV$ but correct order is $III > I > II > IV$

Reason : Due to intramolecular H-bonding in ortho nitrophenol, it is less acidic than para nitrophenol.


$$\text{O} \leftarrow \text{N}=\text{O} \cdots \text{H}-\ddot{\text{O}}$$

intermolecular H-bonding

more association

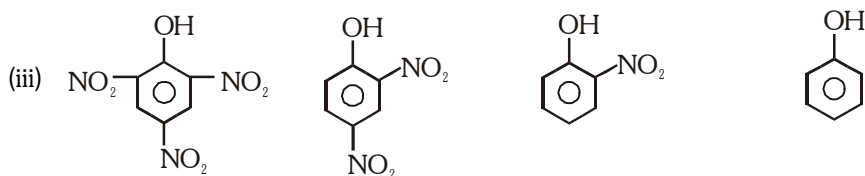
more B.P.

less volatile

more acidic

more soluble in water.

- intramolecular H-bonding
- less association
- less B.P.
- more volatile
- less acidic
- less soluble in water.



I

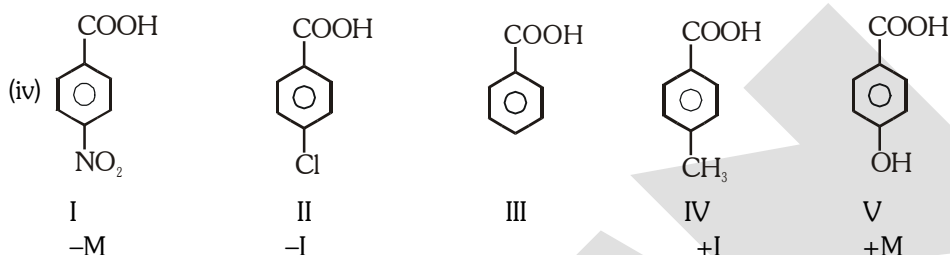
II

III

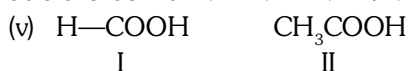
IV

maximum -M and
maximum -I effect
so maximum acidic

Acidic order $I > II > III > IV$



acidic order is $I > II > III > IV > V$



Acidic order is $I > III > II$



ortho effect

only -I

-I and -M

Acidic order is $\Rightarrow o > p > m > \text{benzoic acid}$



ortho effect

only -I

-I and +M

Acidic order is $\Rightarrow o > m > \text{benzoic acid} > p$

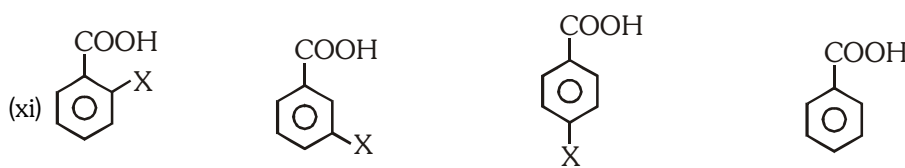


ortho effect

only +I

+I and +H

Acidic order is $\Rightarrow o > \text{benzoic acid} > m > p$



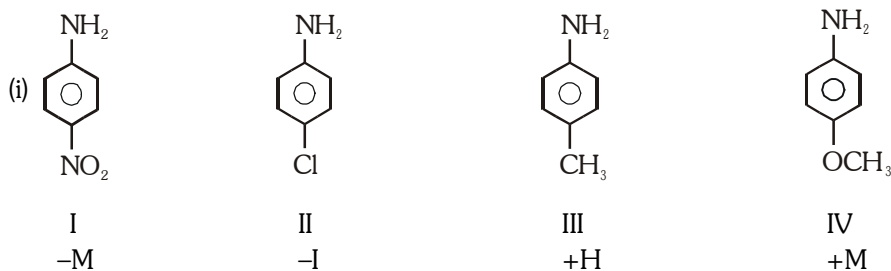
ortho effect

-I (more)

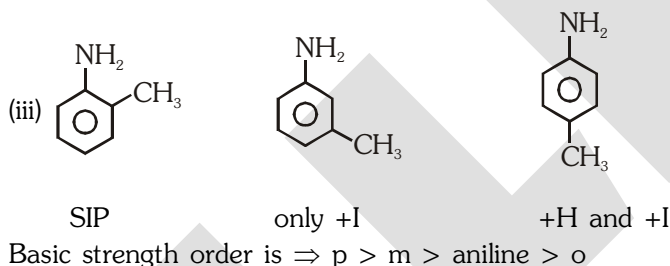
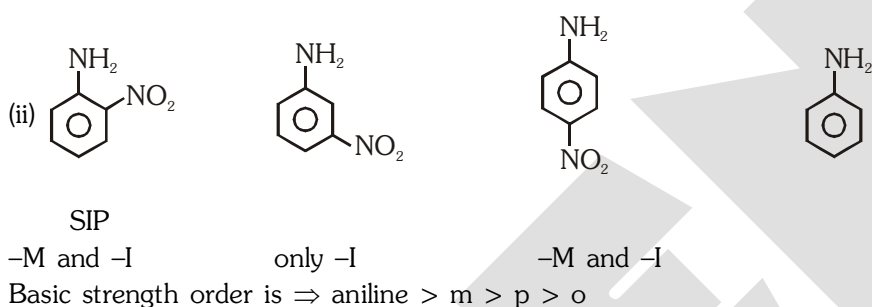
-I (less)

Acidic order is $\Rightarrow o > m > p > \text{benzoic acid}$

Illustration 5. Give basic strength order for :



Basic strength order IV > III > II > I

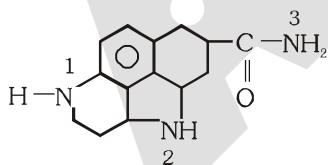


BEGINNER'S BOX-5

1. Most acidic among the following is :



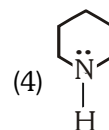
2. Which nitrogen is most basic :-



- | | | | |
|-------|-------|-------|---------------------------|
| (1) 1 | (2) 2 | (3) 3 | (4) All are equally basic |
|-------|-------|-------|---------------------------|

3. Which is strongest base ?

- | | | |
|-------------|-------------|--------------|
| (1) pyrrole | (2) Aniline | (3) Pyridine |
|-------------|-------------|--------------|



4. Which is weakest base :

- | | |
|------------------------|---------------------------|
| (1) $C_6H_5-CH_2-NH_2$ | (2) $C_6H_5-CH_2-NH-CH_3$ |
| (3) $O_2N-CH_2-NH_2$ | (4) $CH_3-NH-CHO$ |

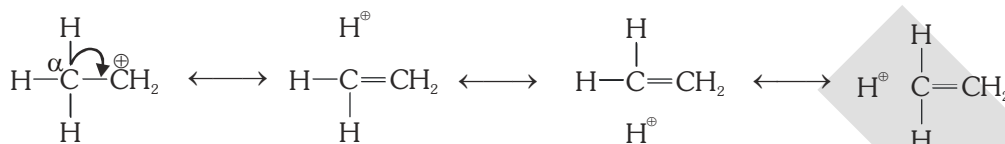
3.4.3 HYPERCONJUGATION EFFECT (H-EFFECT)

Complete transfer of e^- of C-H σ bond towards π bond or positive charge or free electron is called as H-effect (permanent effect). It is also called as No bond resonance (given by Nathen and Baker).

❑ **CONDITIONS OF H-EFFECT :**

1. If there is C-H σ bond and positive charge are in conjugation

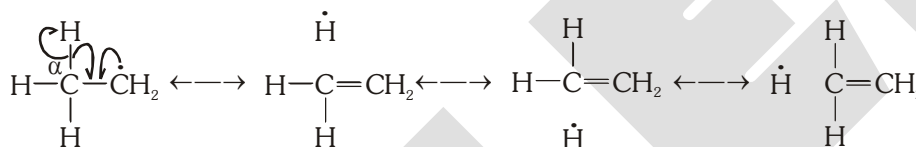
Carbon which is attached to positively charged carbon is called as α -C and H which is attached to α -C is called as α -H. So if number of α -H are more, then there will be more number of hyperconjugating structures, so more stable will be the carbocation.



all are called as hyperconjugating structures or canonical structures.

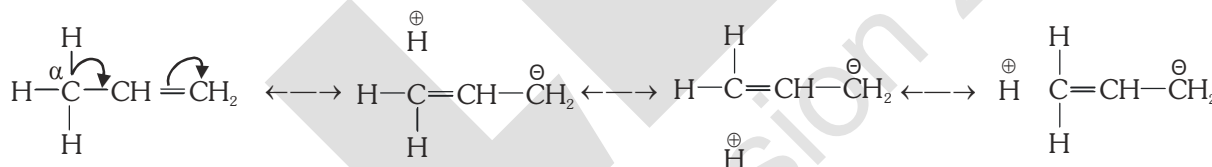
2. If there is C-H σ bond and free e^- are in conjugation then there will be H-effect.

Carbon, which is attached to C having unpaired e^- , is called as α -C and H which are attached to α -C are called as α -H.

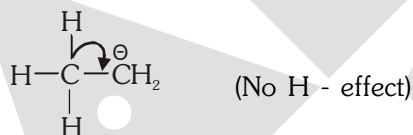


3. If there is C-H σ bond and π bond are in conjugation then there will be H-effect.

sp^3 carbon which is attached to double bonded C is called as α -C and H attached to α -C is called as α -H.



Note : If there is C-H σ bond and negative charge in conjugation then there will be no H-effect.



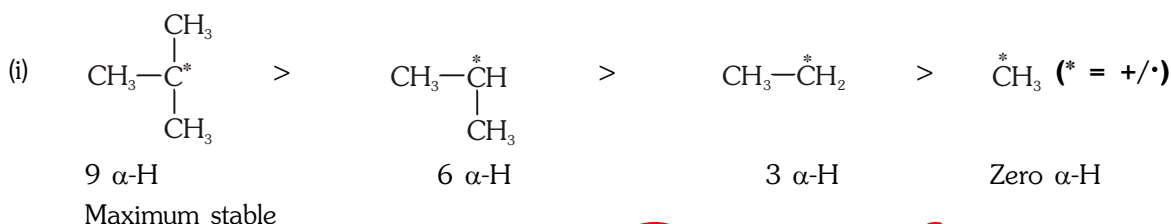
no shifting of C-H σ bond, because anion is having complete octet. ($8e^-$)

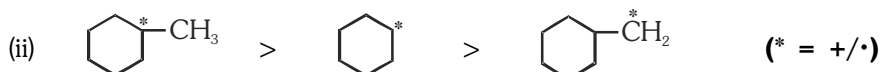
❑ **APPLICATION OF H-EFFECT**

❑ **Stability of carbocation / Free Radical /Alkene**

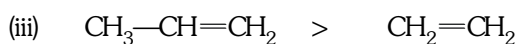
◆ Stability $\propto$ No. of canonical structures $\propto$ No. of α H.

Example : Give stability order for :-



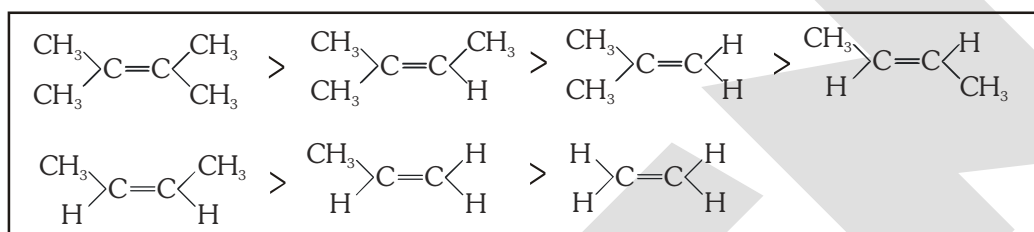
7 α -H4 α -H1 α -H

Maximum stable

3 α -HZero α -H

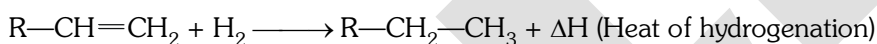
more stable

(iv) Stability order of alkenes will be



GOLDEN KEY POINTS

☐ **Heat of hydrogenation :**



Heat evolved when any unsaturated hydrocarbon is hydrogenated is called heat of hydrogenation (ΔH)

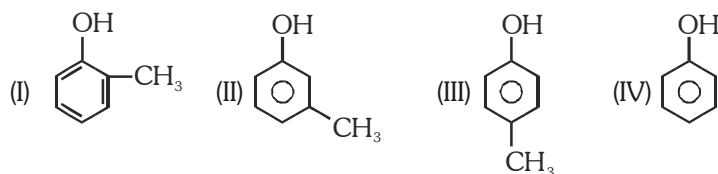
(If alkene is more reactive towards hydrogen then it will evolve more ΔH)

So, $\text{Heat of hydrogenation} \propto \frac{1}{\text{stability of alkene}} \propto \frac{1}{\text{number of } \alpha\text{-H}}$

BEGINNER'S BOX-6

- 1.** Which of the following alkene is most stable ($R = CH_3$).
- (1) $R_2C=CR_2$ (2) $R-CH=CR_2$ (3) $R-CH=CH-R$ (4) $R-CH=CH_2$
- 2.** Which of the following has minimum heat of hydrogenation.
- (1) ethene (2) Propene (3) cis-2-butene (4) trans-2-butene
- 3.** Which of the following is most stable.
- (1) Conjugated alkadiene ($CH_2=CH-CH=CH_2$)
- (2) Isolated alkadiene ($CH_2=CH-CH_2-CH=CH_2$)
- (3) Cumulated alkadiene ($CH_2=C=CH_2$)
- (4) All are equal.

4. What is the order of acidic strength of given molecules



(1) $I > II > III > IV$

(2) $IV > III > II > I$

(3) $IV > II > III > I$

(4) $I > II > IV > III$

3.4.4 ELECTROMERIC EFFECT : (E Effect)

Complete transfer of a shared pair of π -electrons from one atom to another atom in presence of attacking reagent, is known as 'E' effect.

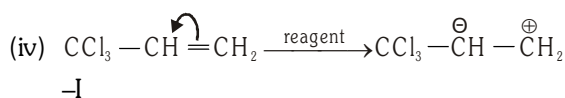
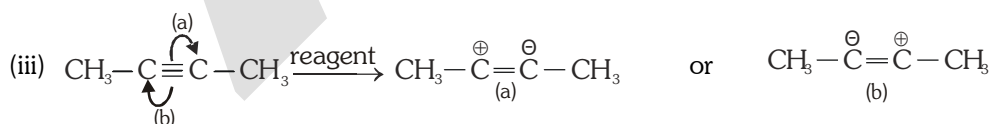
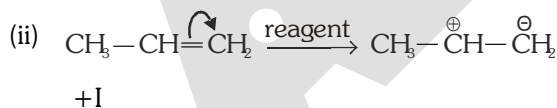
- (i) **Positive Electromeric Effect (+ E effect)** : In this effect the π -electrons of the multiple bond are transferred to that atom to which the reagent gets attached. For example :



- (ii) **Negative Electromeric Effect (-E effect)** : In this effect the π -electrons of the multiple bond are transferred to that atom, to which the attacking reagent does not get attached. For example.

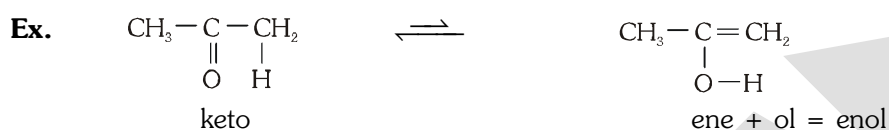
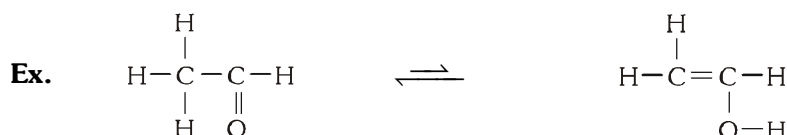
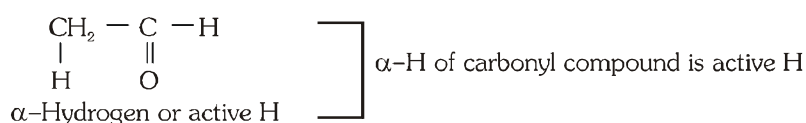


Shifting of π electrons :



3.5 TAUTOMERISM OR DESMOTROPISM

- Tautomers have same molecular formula but different structural formula due to migration of active hydrogen from one polyvalent atom to another polyvalent atom. This phenomena is known as tautomerism.
- Desmotropism means bond turning. [Desmos = Bond ; Tropos = Turn]

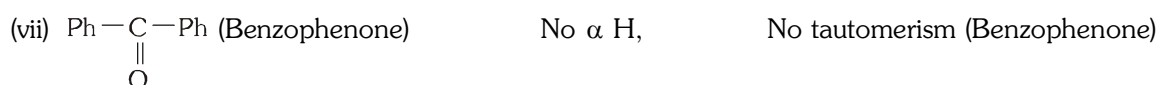
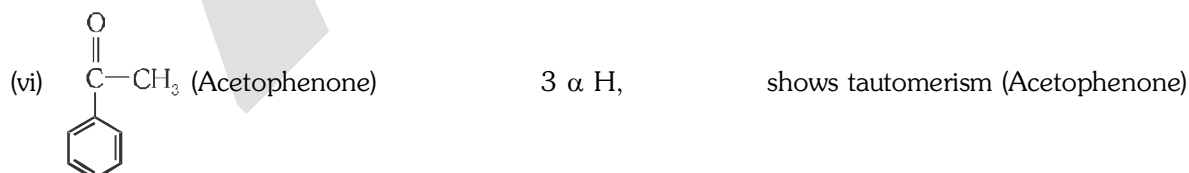


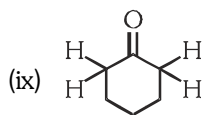
Note : (1) Tautomers exist in dynamic equilibrium.

(2) By shifting of H-atom, π bond also changes its position.

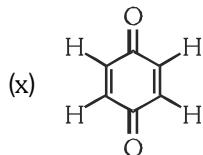
(I) Condition for Tautomerism :

(a) For carbonyl compounds :- Carbonyl compounds having at least one active-H (α -H) show tautomerism.



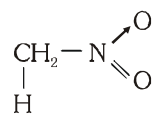


4 α H, shows tautomerism

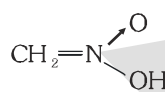


α -H, attached sp^2 carbon does not initiate in tautomerism

(b) For nitro compounds : Nitro compounds having at least one active-H (α - H) show tautomerism



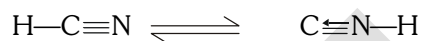
Nitro form



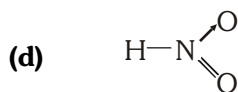
Acinitro form

(acidic form so soluble in base)

(c) $H-C\equiv N$ and $H-N\equiv C$ are tautomers [also Functional isomers] while $R-C\equiv N$ and $R-N\equiv C$ are only Functional isomers.



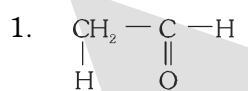
Active H



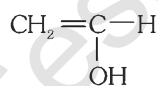
and $H-O-N=O$ are tautomers.

Note : Nitro compounds with at least one α -H are soluble in NaOH.

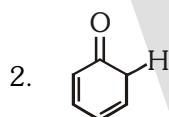
(II) Enol Content :



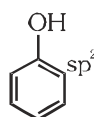
"keto" ($\approx 99\%$)



"enol" ($\approx 1\%$)



"keto" ($\approx 1\%$)



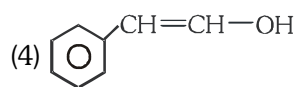
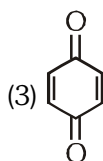
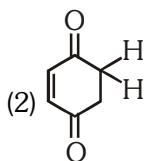
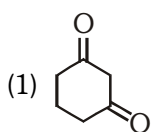
"enol" (stable by resonance and aromatic nature) ($\approx 99\%$)

GOLDEN KEY POINTS

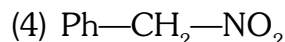
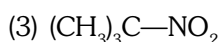
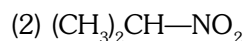
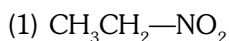
- More active the H, more will be its participation in tautomerism.
- Stability of enol form depends on (i) Resonance and (ii) H - Bond (iii) Aromaticity.

BEGINNER'S BOX-7

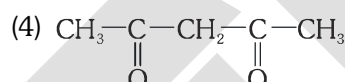
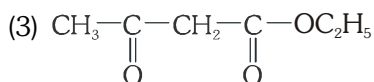
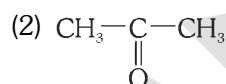
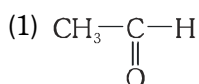
- 1.** Which of the following does not show keto enol tautomerism.



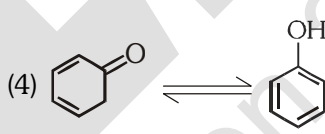
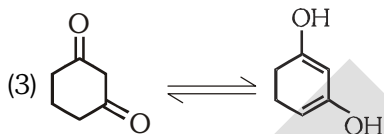
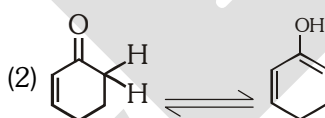
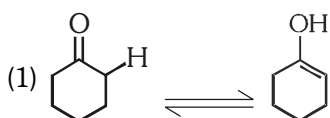
- 2.** Which of the following is not soluble in NaOH.



- 3.** Which of the following has highest enol content.



4. In which of the following reaction most stable enol is present.



ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	
	Ans.	4	1	2	
BEGINNER'S BOX-2	Que.	1	2		
	Ans.	1	1		
BEGINNER'S BOX-3	Que.	1	2	3	
	Ans.	1	1	4	
BEGINNER'S BOX-4	Que.	1	2	3	
	Ans.	4	4	3	
BEGINNER'S BOX-5	Que.	1	2	3	4
	Ans.	3	2	4	4
BEGINNER'S BOX-6	Que.	1	2	3	4
	Ans.	1	4	1	3
BEGINNER'S BOX-7	Que.	1	2	3	4
	Ans.	3	3	4	4

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

ATTACKING REAGENT

- Which of the following species is an electrophile
 (1) RNH_2 (2) SO_3
 (3) $\text{NO}_3^\ominus$ (4) ROH
- Which of the following acts as a nucleophile?
 (1) $\text{NO}_2^\oplus$ (2) $:\text{CCl}_2$
 (3) $\text{NH}_2^\ominus$ (4) $\cdot\text{CH}_3$

REACTION INTERMEDIATES

- Which of the following contains only three pair of electrons :
 (1) Carbanion (2) Carbocation
 (3) Carbon free radical (4) None
- Carbanion is a :-
 (1) Base (2) Nucleophile
 (3) Both the above (4) None

I-EFFECT

- $\text{CH}_3^\ominus$ is less stable than :-
 (1) $\text{CH}_3-\text{CH}_2^\ominus$ (2) $\text{CH}_3-\text{CH}^\ominus-\text{CH}_3$
 (3) $\text{CH}_2^\ominus-\text{NO}_2$ (4) $\text{CH}_3-\text{CH}^\ominus-\text{C}_2\text{H}_5$
- Decreasing order of -I effect of the triad $[-\text{NO}_2, -\text{NH}_3^\oplus, -\text{CN}]$ is :-
 (1) $-\text{NH}_3^\oplus > -\text{NO}_2 > -\text{CN}$
 (2) $-\text{NH}_3^\oplus > -\text{CN} > \text{NO}_2$
 (3) $-\text{CN} > -\text{NO}_2 > -\text{NH}_3^\oplus$
 (4) $-\text{NO}_2 > -\text{CN} > -\text{NH}_3^\oplus$
- Most stable carbanion is :-
 (1) $\text{HC}\equiv\text{C}^\ominus$ (2) $\text{H}_2\text{C}=\text{C}^\ominus$
 (3) $\text{CH}_3-\text{C}(\text{CH}_3)_2-\text{CH}_2^\ominus$ (4) $\text{CH}_3-\text{C}^\ominus(\text{CH}_3)=\text{CH}$

- The correct order of stability of given carbanions will be :-
 $\text{CH}_3-\text{CH}_2^\ominus$ (I) $\text{CH}_2=\text{CH}^\ominus$ (II) $\text{HC}\equiv\text{C}^\ominus$ (III)
 (1) $\text{I} > \text{II} > \text{III}$ (2) $\text{III} > \text{II} > \text{I}$
 (3) $\text{I} > \text{III} > \text{II}$ (4) $\text{II} > \text{I} > \text{III}$

- Which is most basic among the following :-

- CH_3NH_2
- $\text{CH}_3\text{CH}_2\text{NH}_2$
- NH_3
- $(\text{CH}_3)_2\text{CHNH}_2$

- Which of the following has maximum pK_a :-

- CH_3FCOOH
- CH_2ClCOOH
- CH_3COOH
- HCOOH

- Which of the following is most acidic

- Methoxy acetic acid
- Acetic acid
- Chloro acetic acid
- Trifluoroacetic acid

- Which of the following show + I-effect :-

- OH (2) $-\text{OCH}_3$ (3) $-\text{CH}_3$ (4) $-\text{Cl}$

- Among the following the most highly ionised in water is:

- $\text{CH}_3\text{CH}_2\text{CHClCOOH}$
- $\text{CH}_3\text{CH}_2\text{CCl}_2\text{COOH}$
- $\text{CH}_3\text{CHClCH}_2\text{COOH}$
- $\text{CH}_2\text{ClCH}_2\text{CH}_2\text{COOH}$

- The strongest acid amongst the following compounds is ?

- $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{CO}_2\text{H}$
- $\text{ClCH}_2\text{CH}_2\text{CH}_2\text{COOH}$
- CH_3COOH
- HCOOH

- Which of the following acids is stronger than acetic acid :-

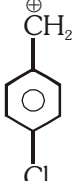
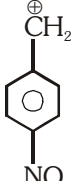
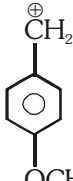
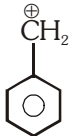
- Propanoic acid (2) HCOOH
 (3) Butyric acid (4) $(\text{CH}_3)_2\text{CHCOOH}$

16. Which of the following acids have the lowest pK_a value :-

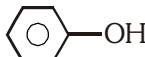
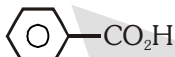
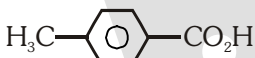
- (1) $\text{CH}_3-\overset{\text{Cl}}{\underset{|}{\text{CH}}}-\text{COOH}$
 (2) $\text{Cl}-\text{CH}_2-\text{CH}_2-\text{COOH}$
 (3) CCl_3COOH
 (4) CHCl_2COOH

R- OR M-EFFECT

17. Most stable carbocation is :-

- (1)  (2) 
 (3)  (4) 

18. Most acidic compound is :-

- (1)  (2) 
 (3)  (4) 

19. Which resonating structure of vinyl chloride is least stable :-

- (1) $\text{CH}_2=\text{CH}-\ddot{\text{Cl}}:$
 (2) $\overset{\ominus}{\text{C}}\text{H}_2-\text{CH}=\overset{\oplus}{\text{C}}\text{H}-\text{Cl}$
 (3) $\overset{\ominus}{\text{C}}\text{H}_2-\overset{\oplus}{\text{C}}\text{H}-\text{Cl}$
 (4) All have equal stability

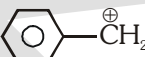
20. The stabilization due to resonance is maximum in

- (1)  (2) 
 (3)  (4) 

21. In which of the following compounds carbon-chlorine bond distance is minimum :-

- (1) CH_3-Cl
 (2) $\text{C}_6\text{H}_5-\text{CH}_2-\text{Cl}$
 (3) $\text{CH}_2=\text{CH}-\text{Cl}$
 (4) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{Cl}$

22. Consider the following carbocations

- (a) $\text{CH}_3\text{O}-\text{C}_6\text{H}_4-\overset{\oplus}{\text{C}}\text{H}_2$ (b) 
 (c) $\text{CH}_3-\text{C}_6\text{H}_4-\overset{\oplus}{\text{C}}\text{H}_2$ (d) $\text{CH}_3-\overset{\oplus}{\text{C}}\text{H}_2$

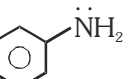
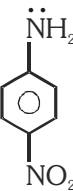
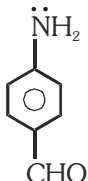
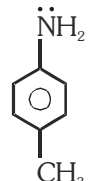
The relative stabilities of these carbocations are such that :-

- (1) $d < b < c < a$
 (2) $b < d < c < a$
 (3) $d < b < a < c$
 (4) $b < d < a < c$

23. Among the following, the strongest base is :-

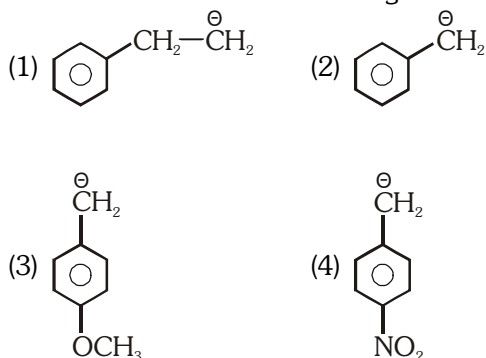
- (1) $\text{C}_6\text{H}_5\text{NH}_2$
 (2) $p\text{-NO}_2-\text{C}_6\text{H}_4\text{NH}_2$
 (3) $m\text{-NO}_2-\text{C}_6\text{H}_4\text{NH}_2$
 (4) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$

24. Arrange in decreasing order of basic strength :

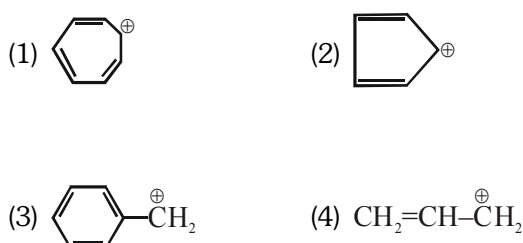
- (I)  (II) 
 (III)  (IV) 

- (1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{II} > \text{III} > \text{I} > \text{IV}$
 (3) $\text{IV} > \text{I} > \text{III} > \text{II}$ (4) $\text{IV} > \text{I} > \text{II} > \text{III}$

25. The most stable carbanion among the following is



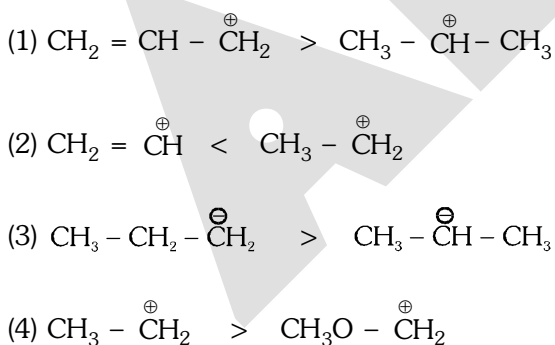
26. Which of the following is most stable carbocation:-



27. The oxygen atom in phenol -

- (1) exhibits only inductive effect
- (2) exhibits only resonance effect
- (3) has more dominating resonance effect than inductive effect
- (4) has more dominating inductive effect than resonance effect

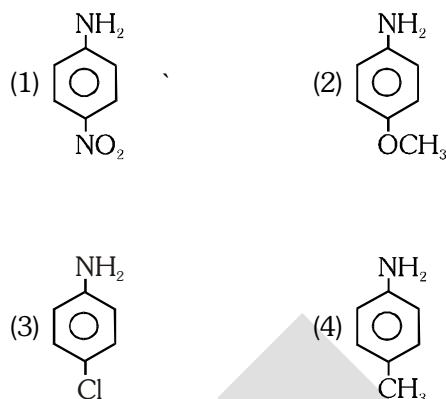
28. Which is incorrect stability order :-



29. Mesomeric effect is due to :-

- (1) Delocalization of σ e⁻s
- (2) Delocalization of π e⁻s
- (3) Migration of H - atom
- (4) Migration of proton

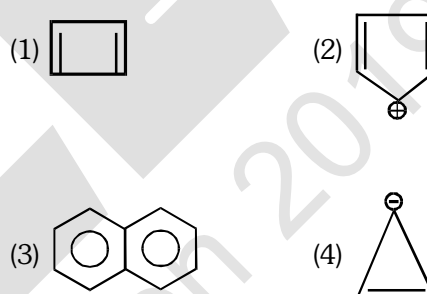
30. Which of the following is least basic :-



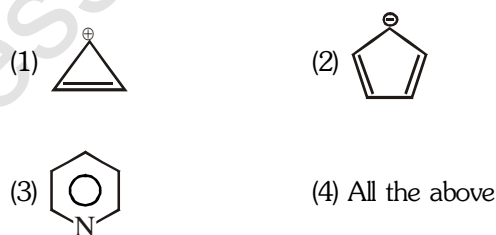
31. Among the following the pKa is minimum for :-

- (1) $\text{C}_6\text{H}_5\text{OH}$
- (2) HCOOH
- (3) $\text{C}_2\text{H}_5\text{OH}$
- (4) $\text{CH}_3\text{C}\equiv\text{CH}$

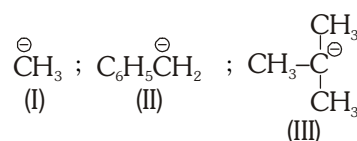
32. Among the following the aromatic compound is -



33. Which is aromatic compound among the following

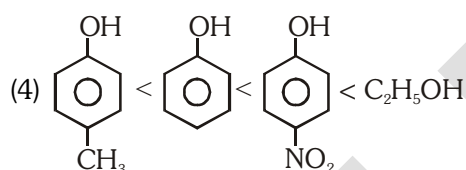
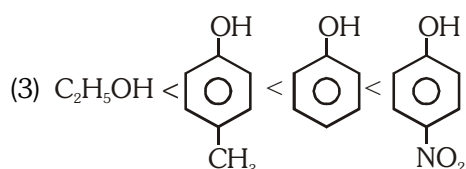
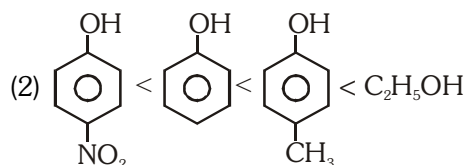
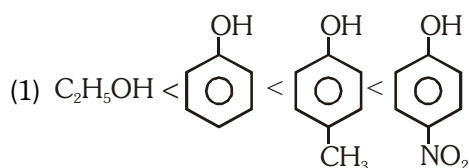


34. Select the correct option for stability of following carbanions :

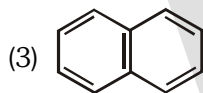
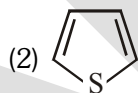
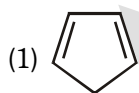


- (1) I > II > III
- (2) II > I > III
- (3) III > II > I
- (4) II > III > I

35. Correct increasing order of acidity of the following phenols is :-



36. The non aromatic compound among the following is :-

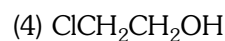
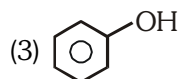
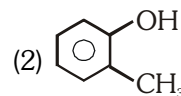
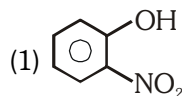


37. The correct order of acidic strength of the following compounds is :-

A. Phenol B. p-Cresol
C. m-Nitrophenol D. p- Nitrophenol

- (1) $C > B > A > D$
(2) $D > C > A > B$
(3) $B > D > A > C$
(4) $A > B > D > C$

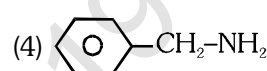
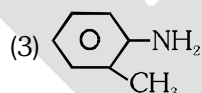
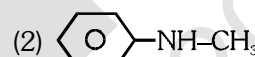
38. Which one of the following compounds is most acidic:-



39. Which of the following is most acidic :-

- (1) phenol
(2) benzyl alcohol
(3) m-chloro phenol
(4) cyclohexanol

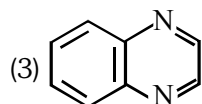
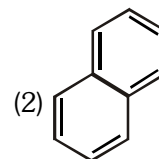
40. Which of the following is the strongest base :-



41. The dipole moment of vinyl chloride is lower than that of methyl chloride. This is due to :-

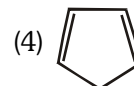
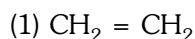
- (1) Resonance effect
(2) Inductive effect
(3) Electromeric effect
(4) Hyperconjugation

42. Which is Aromatic compound :-



(4) 2 and 3 both

43. Which of the following is the most acidic compound :-



44. Among the following the strongest acid is :-

- (1) CH_3COOH
- (2) $\text{C}_6\text{H}_5\text{COOH}$
- (3) $m\text{-CH}_3\text{OC}_6\text{H}_4\text{COOH}$
- (4) $p\text{-CH}_3\text{OC}_6\text{H}_4\text{COOH}$

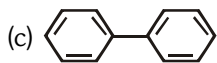
45. The least reactive chlorine is present in -

- (1) Methyl chloride
- (2) Allyl chloride
- (3) Ethyl chloride
- (4) Vinyl chloride

46. Which one of the following resonating structures of 1-methoxy-1,3-butadiene is least stable :-

- (1) $\text{CH}_2=\text{CH}=\text{CH}=\text{O}^+-\text{CH}_3$
- (2) $\text{CH}_2=\text{CH}-\text{CH}^+-\text{CH}=\text{O}-\text{CH}_3$
- (3) $\text{CH}_2^+-\text{CH}=\text{CH}=\text{CH}-\text{O}-\text{CH}_3$
- (4) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}-\text{O}-\text{CH}_3$

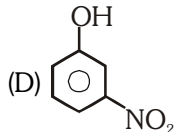
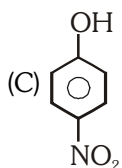
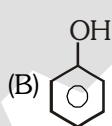
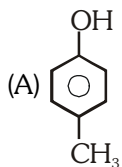
47. Four structures are given in options (a) to (d). Examine them and select the aromatic structures.



- (1) a and d
- (3) a and b

- (2) b and c
- (4) a and c

48. Order of acidic strength of the following compound will be :



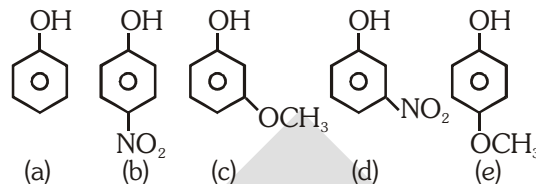
- (1) $\text{C} > \text{D} > \text{B} > \text{A}$
- (3) $\text{A} > \text{B} > \text{C} > \text{D}$

- (2) $\text{D} > \text{C} > \text{B} > \text{A}$
- (4) $\text{B} > \text{A} > \text{C} > \text{D}$

49. Phenol is less acidic than

- (1) Ethanol
- (2) o-Nitrophenol
- (3) o-Methylphenol
- (4) o-Methoxyphenol

50. Mark the correct order of decreasing acid strength of the following compounds.



- (1) $e > d > b > a > c$
- (2) $b > d > a > c > e$
- (3) $e > d > c > b > a$
- (4) $b > d > c > a > e$

HYPERCONJUGATION

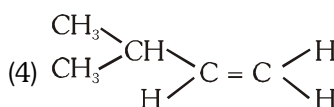
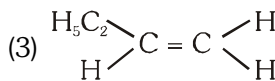
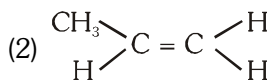
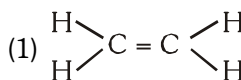
51. Which of the following compounds exhibits hyperconjugation :

- (1) Phenol
- (2) Ethyne
- (3) Ethanol
- (4) Propene

52. Which of the following is least stable :-

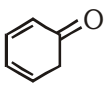
- (1) $\text{CH}_3-\text{CH}^+-\text{CH}_3$
- (2) $\text{CH}_3-\text{CH}_2-\text{CH}_2^+$
- (3) $\text{CH}_3-\text{C}^+(\text{CH}_3)-\text{CH}_3$
- (4) $\text{CH}_3-\text{C}^+(\text{CH}_3)-\text{CH}-\text{C}_6\text{H}_5$

53. Which of the following is most stable alkene :-

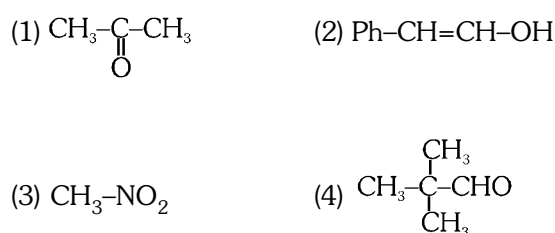


54. The correct order of increasing stability of the given alkenes is
- (1) 1-pentene > cis-2-pentene > trans-2-pentene > 2-methyl-2-butene
 - (2) 1-pentene > trans-2-pentene > cis-2-pentene > 2-methyl-2-butene
 - (3) 1-pentene < cis-2-pentene < trans-2-pentene < 2-methyl-2-butene
 - (4) 1-pentene < trans-2-pentene < cis-2-pentene < 2-methyl-2-butene

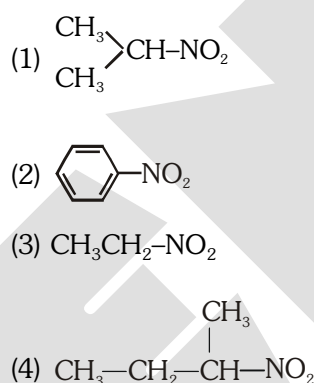
TAUTOMERISM

55. Tautomerism is due to :-
- (1) Delocalization of sigma electrons
 - (2) Delocalization of pi electrons
 - (3) Migration of active-H-atom
 - (4) None is correct
56. Which of the following will lead to maximum enolisation :-
- (1) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$
 - (2) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{C}(=\text{O})-\text{H}$
 - (3) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}(\text{Br})-\text{C}(=\text{O})-\text{CH}_3$
 - (4) 
57. Urea $\left[\text{H}_2\text{N}-\text{C}(=\text{O})-\text{NH}_2 \right]$ molecule exhibits (isomerism):-
- (1) Chain
 - (2) Position
 - (3) Geometrical
 - (4) Tautomerism

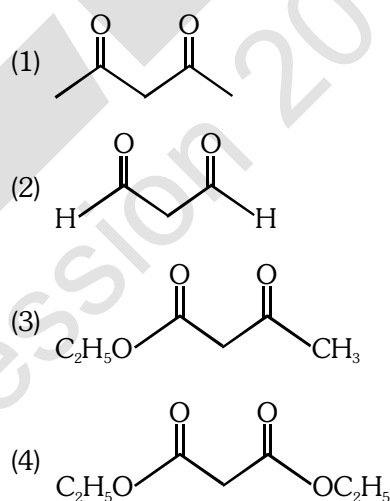
58. Tautomerism is not observed in :-



59. Which of the following is not soluble in NaOH solution :



60. Which of the following has highest enol content.



EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	3	2	3	3	1	1	2	4	3	4	3	2	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	4	3	4	3	1	4	3	4	1	3	4	2	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	3	4	2	3	1	2	1	3	4	1	4	4	3	4
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	4	1	2	4	4	2	2	3	3	4	4	4	2	2

EXERCISE-II (Previous Year Questions)**AIPMT/NEET & AIIMS (2006-2018)****AIPMT 2006**

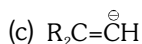
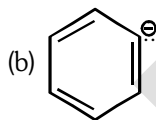
1. Which of the following is more basic than aniline
 (1) Diphenyl amine (2) Triphenyl amine
 (3) p-nitro aniline (4) Benzyl amine

AIPMT 2007

2. Which of the following presents the correct order of the acidity in the given compounds :
 (1) $\text{FCH}_2\text{COOH} > \text{CH}_3\text{COOH} > \text{BrCH}_2\text{COOH} > \text{ClCH}_2\text{COOH}$
 (2) $\text{BrCH}_2\text{COOH} > \text{ClCH}_2\text{COOH} > \text{FCH}_2\text{COOH} > \text{CH}_3\text{COOH}$
 (3) $\text{FCH}_2\text{COOH} > \text{ClCH}_2\text{COOH} > \text{BrCH}_2\text{COOH} > \text{CH}_3\text{COOH}$
 (4) $\text{CH}_3\text{COOH} > \text{BrCH}_2\text{COOH} > \text{ClCH}_2\text{COOH} > \text{FCH}_2\text{COOH}$

AIPMT 2008

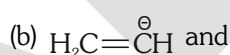
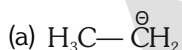
3. The stability of carbanions in the following:-



is in the order of:-

- (1) (d) > (b) > (c) > (a) (2) (a) > (c) > (b) > (d)
 (3) (a) > (b) > (c) > (d) (4) (b) > (c) > (d) > (a)

4. Basic strength of:-

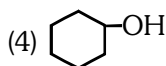
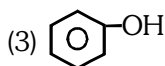
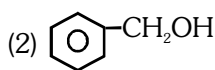
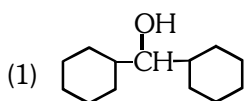


is in the order of:-

- (1) (a) > (c) > (b) (2) (a) > (b) > (c)
 (3) (b) > (a) > (c) (4) (c) > (b) > (a)

AIPMT 2010

5. Which one of the following compounds has the most acidic nature ?



6. Given are cyclohexanol (I), acetic acid (II), 2, 4, 6-trinitrophenol (III) and phenol (IV). In these the order of decreasing acidic character will be :-

- (1) III > IV > II > I
 (2) III > II > IV > I
 (3) II > III > I > IV
 (4) II > III > IV > I

AIPMT Mains-2010

7. Among the following four compounds :-

- (a) phenol (b) methyl phenol
 (c) metanitrophenol (d) paranitrophenol,

The acidity order is :

- (1) c > d > a > b (2) c > d > c > b
 (3) b > a > c > d (4) d > c > a > b

8. Which of the following species is not electrophilic in nature :-

- (1) BH_3 (2) H_3O^+
 (3) NO_2^+ (4) Cl^+

AIIMS 2010

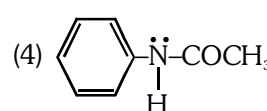
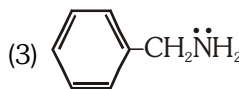
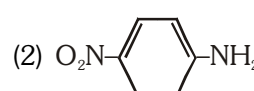
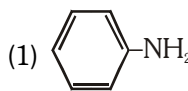
9. The correct order of pKa is

- (I) CH_3COOH
 (II) $\text{CH}_3\text{CH}_2\text{COOH}$
 (III) $\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$
 (IV) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{COOH}$

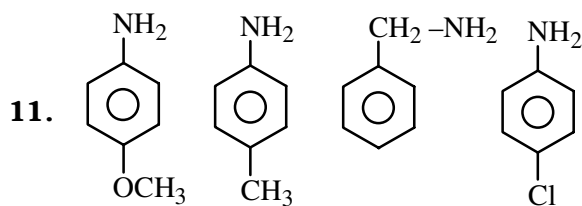
- (1) IV > III > II > I
 (2) III > IV > II > I
 (3) IV > II > III > I
 (4) II > I > III > IV

AIPMT Mains-2011

10. Which of the following compounds is most basic ?

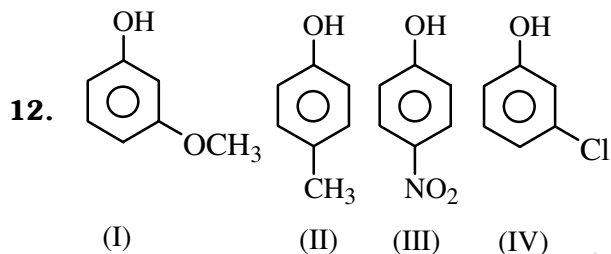


AIIMS 2011



(I) (II) (III) (IV)
The correct decreasing order of pK_b is:-

- (1) I > II > III > IV
(2) III > IV > II > I
(3) II > III > IV > I
(4) IV > II > I > III



(I) (II) (III) (IV)
The correct decreasing order of pK_a is:-

- (1) II > I > IV > III (2) IV > II > III > I
(3) III > II > IV > I (4) IV > I > II > III

AIPMT Pre.-2012

13. The correct order of decreasing acid strength of trichloroacetic acid (A), trifluoroacetic acid (B), acetic acid (C) and formic acid (D) is:

- (1) A > B > C > D (2) A > C > B > D
(3) B > A > D > C (4) B > D > C > A

AIIMS 2012

14. pK_a increases in benzoic acid, substituent "x" bonded at para position then "x" is :-

- (1) $-\text{COOH}$ (2) $-\text{NO}_2$
(3) $-\text{CN}$ (4) $-\text{OCH}_3$

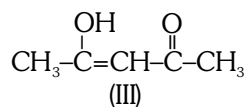
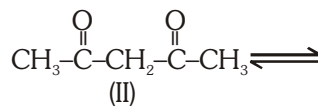
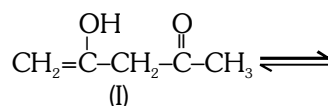
NEET-UG 2013

15. The radical,  is aromatic because it

has :-

- (1) 6p-orbitals and 7 unpaired electrons
(2) 6p-orbitals and 6 unpaired electrons
(3) 7p-orbitals and 6 unpaired electrons
(4) 7p-orbitals and 7 unpaired electrons

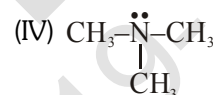
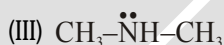
16. The order of stability of the following tautomeric compounds is :-



- (1) II > III > I (2) I > II > III
(3) III > II > I (4) II > I > III

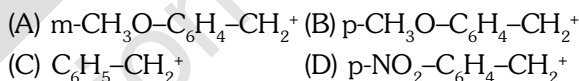
AIIMS 2014

17. Correct decreasing order of pK_b value of following compounds:-



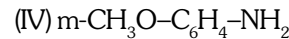
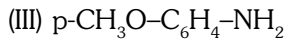
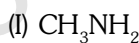
- (1) I > IV > II > III (2) I > II > IV > III
(3) IV > I > II > III (4) I > II > III > IV

18. Decreasing order of stability of following carbocations is :-



- (1) A > B > C > D (2) B > C > D > A
(3) C > B > A > D (4) B > C > A > D

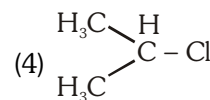
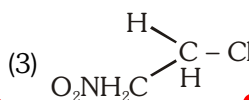
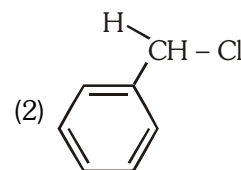
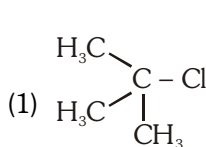
19. Decreasing order of basic strength is :-



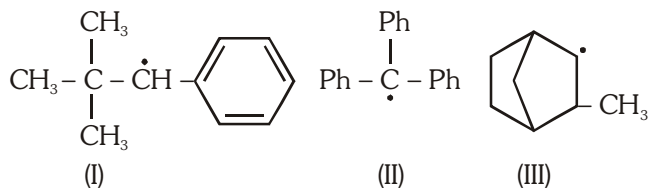
- (1) I > II > III > IV (2) II > IV > I > III
(3) II > IV > III > I (4) IV > I > II > III

AIPMT 2015

20. In which of the following compounds, C-Cl bond ionisation shall give most stable carbonium ion?



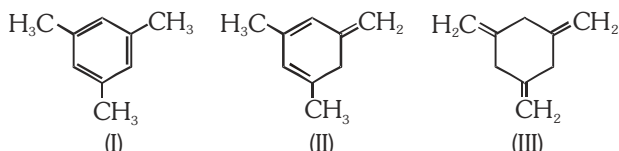
21. Consider the following compounds



Hyperconjugation occurs in :-

- (1) II only (2) III only (3) I and III (4) I only

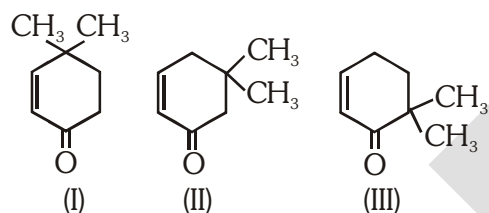
22. Given :-



The enthalpy of the hydrogenation of these compounds will be in the order as :-

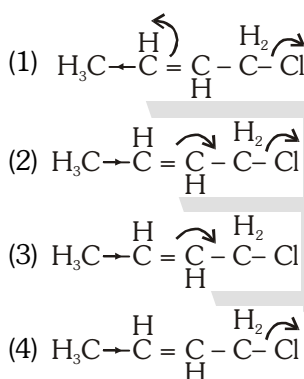
- (1) III > II > I (2) II > III > I
(3) II > I > III (4) I > II > III

23. Which of the given compounds can exhibit tautomerism?



- (1) I and III (2) II and III
(3) I, II and III (4) I and II

24. Which of the following is the most correct electron displacement for a nucleophilic reaction to take place?

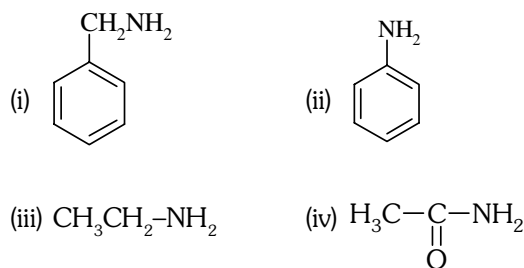


Re-AIPMT 2015

25. Which of the following statements is not correct for a nucleophile ?

- (1) Nucleophiles attack low e^- density sites
(2) Nucleophiles are not electron seeking
(3) Nucleophile is a Lewis acid
(4) Ammonia is a nucleophile

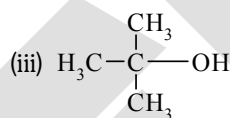
26. Correct order of K_b is



- (1) iv > iii > ii > i (2) iii > i > ii > iv
(3) i > ii > iii > iv (4) ii > iii > iv > i

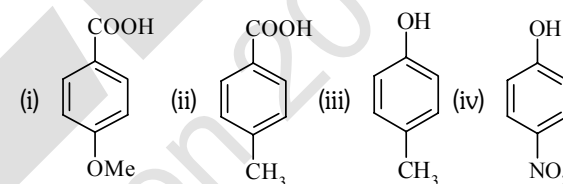
AIIMS 2015

27. Correct order of pK_a is



- (1) i > ii > iii (2) ii > i > iii
(3) iii > ii > i (4) ii > iii > i

28. Correct order of acidic strength is



- (1) i > ii > iv > iii (2) iii > iv > ii > i
(3) ii > i > iv > iii (4) iv > ii > iii > i

NEET-I 2016

29. The correct statement regarding a carbonyl compound with a hydrogen atom on its alpha carbon, is :-

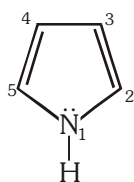
- (1) a carbonyl compound with a hydrogen atom on its alpha-carbon never equilibrates with its corresponding enol.
(2) a carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as aldehyde-ketone equilibration.
(3) a carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as carbonylation.
(4) a carbonyl compound with a hydrogen atom on its alpha-carbon rapidly equilibrates with its corresponding enol and this process is known as keto-enol tautomerism.

30. The **correct** statement regarding the basicity of arylamines is :-

- (1) Arylamines are generally less basic than alkylamines because the nitrogen lone-pair electrons are delocalized by interaction with the aromatic ring π electron system.
- (2) Arylamines are generally more basic than alkylamines because the nitrogen lone-pair electrons are not delocalized by interaction with the aromatic ring π electron system.
- (3) Arylamines are generally more basic than alkylamines because of aryl group.
- (4) Arylamines are generally more basic than alkylamines, because the nitrogen atom in arylamines is sp -hybridized.

NEET-II 2016

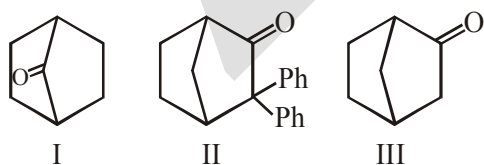
31. In pyrrole



The electron density is maximum on :-

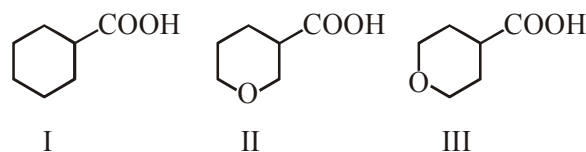
- (1) 2 and 4
- (2) 2 and 5
- (3) 2 and 3
- (4) 3 and 4

32. Which among the given molecules can exhibit tautomerism ?



- (1) Both I and II
- (2) Both II and III
- (3) III only
- (4) Both I and III

33. The **correct** order of strengths of the carboxylic acids



is

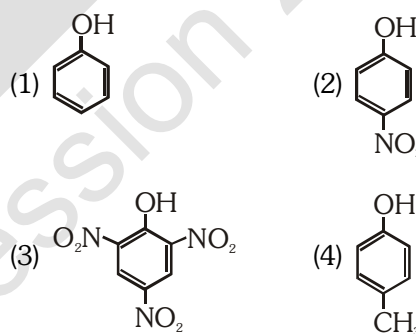
- (1) III > II > I
- (2) II > I > III
- (3) I > II > III
- (4) II > III > I

NEET(UG) 2017

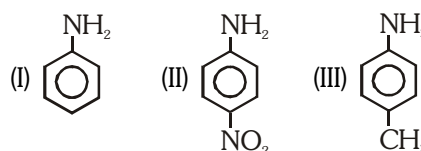
34. Which one is the correct order of acidity ?

- (1) $\text{CH} \equiv \text{CH} > \text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH}_2 = \text{CH}_2 > \text{CH}_3 - \text{CH}_3$
- (2) $\text{CH} \equiv \text{CH} > \text{CH}_2 = \text{CH}_2 > \text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH}_3 - \text{CH}_3$
- (3) $\text{CH}_3 - \text{CH}_3 > \text{CH}_2 = \text{CH}_2 > \text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH} \equiv \text{CH}$
- (4) $\text{CH}_2 = \text{CH}_2 > \text{CH}_3 - \text{CH} = \text{CH}_2 > \text{CH}_3 - \text{C} \equiv \text{CH} > \text{CH} > \text{CH} \equiv \text{CH}$

35. Which one is the most acidic compound ?



36. The **correct** increasing order of basic strength for the following compounds is :



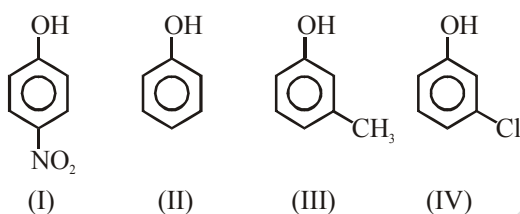
- (1) III < I < II
- (2) III < II < I
- (3) II < I < III
- (4) II < III < I

37. The **correct** statement regarding electrophile is :-

- (1) Electrophile is a negatively charged species and can form a bond by accepting a pair of electrons from another electrophile
- (2) Electrophiles are generally neutral species and can form a bond by accepting a pair of electrons from a nucleophile
- (3) Electrophile can be either neutral or positively charged species and can form a bond by accepting a pair of electrons from a nucleophile
- (4) Electrophile is a negatively charged species and can form a bond by accepting a pair of electrons from a nucleophile

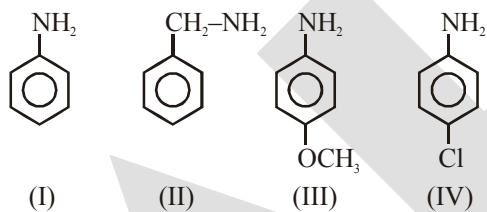
AIIMS 2017

38. Arrange followings in correct decreasing order of pK_a :-



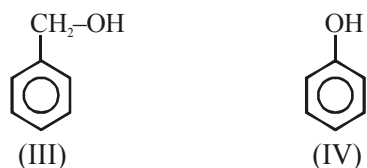
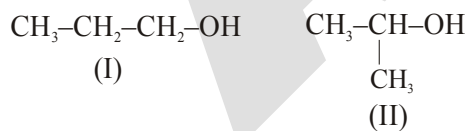
- (1) I > II > IV > III (2) III > IV > II > I
- (3) III > II > IV > I (4) III > I > II > IV

39. Arrange followings in correct decreasing order of pK_b :-



- (1) II > III > IV > I (2) II > III > I > IV
- (3) III > II > IV > I (4) IV > I > III > II

40. Arrange following in correct increasing order of pK_a



- (1) IV < III < I < II (2) IV < III < II < I
- (3) I < II < III < IV (4) II < III < IV < I

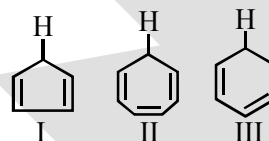
NEET(UG) 2018

41. Which of the following is correct with respect to -I effect of the substituents ? (R = alkyl)

- (1) $-\text{NH}_2 < -\text{OR} < -\text{F}$
- (2) $-\text{NR}_2 < -\text{OR} < -\text{F}$
- (3) $-\text{NH}_2 > -\text{OR} > -\text{F}$
- (4) $-\text{NR}_2 > -\text{OR} > -\text{F}$

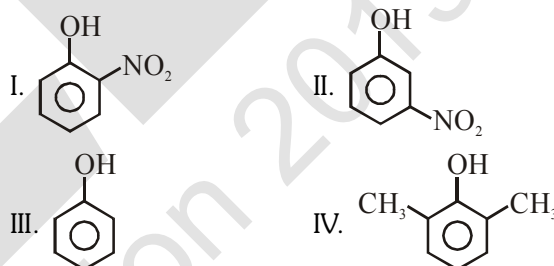
AIIMS 2018

42. Arrange the following compounds in their decreasing order of acidic strength



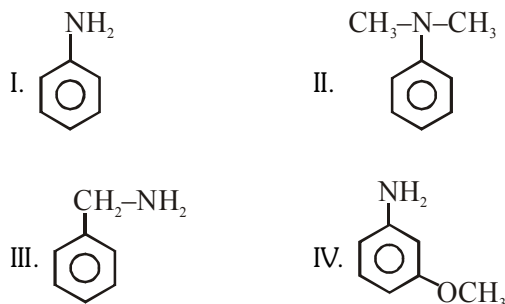
- (1) I > II > III
- (2) I > III > II
- (3) II > III > I
- (4) II > I > III

43. Arrange the following decreasing order of their acidic strength



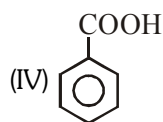
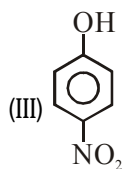
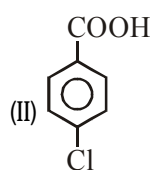
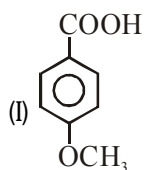
- (1) I > II > III > IV
- (2) I > II > IV > III
- (3) II > I > III > IV
- (4) IV > I > II > III

44. Arrange the following in decreasing order of basic strength



- (1) I > II > III > IV
- (2) III > II > IV > I
- (3) III > II > I > IV
- (4) IV > II > III > I

45. Arrange the following acids in decreasing order of their acidic strength :-



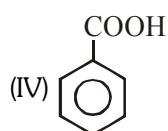
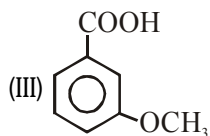
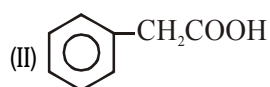
(1) III > II > IV > I

(2) II > IV > I > III

(3) I > III > II > IV

(4) IV > II > III > I

46. Compare the acidic strength of the given acids



(1) III > IV > II > I

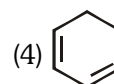
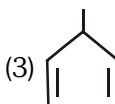
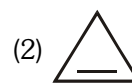
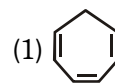
(2) IV > III > II > I

(3) III > IV > I > II

(4) II > IV > I > III

47. Which of the following molecule on reaction with

$\text{CH}_3\text{CH}_2\text{O}^-\text{Na}^+$ gives aromatic species :-



EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	3	3	2	3	2	4	2	4	3	4	1	3	4	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	1	4	1	2	2	1	3	2	3	2	3	3	4	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	3	4	1	3	3	3	3	4	1	1	2	1	3	2
Que.	46	47													
Ans.	1	3													

EXERCISE-III (Analytical Questions)

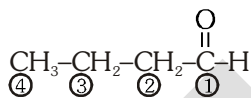
Check Your Understanding

1. Consider the following statements :-

- (a) $\text{CH}_3\text{O}^+\text{CH}_2$ is more stable than CH_3CH_2^+
- (b) Me_2CH^+ is more stable than $\text{CH}_3\text{CH}_2\text{CH}_2^+$
- (c) $\text{CH}_2=\text{CH}-\text{CH}_2^+$ is more stable than $\text{CH}_3\text{CH}_2\text{CH}_2^+$
- (d) $\text{CH}_2=\text{CH}^+$ is more stable than CH_3CH_2^+

of these statements :-

- (1) a and b are correct
 (2) c and d are correct
 (3) a, b and c are correct
 (4) b, c and d are correct
2. Which of the following carbon has most acidic hydrogen.

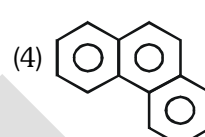
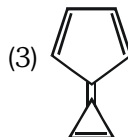
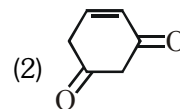
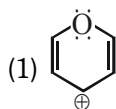


- (1) C_1-H (2) C_2-H
 (3) C_3-H (4) C_4-H

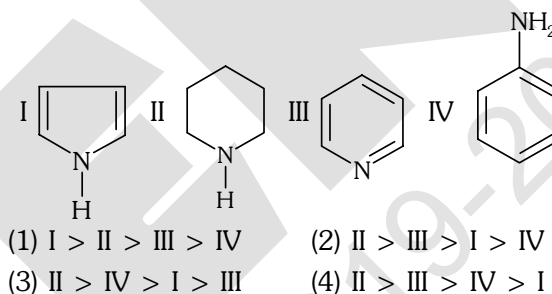
3. Which of the following has the most acidic proton:

- (1) $\text{CH}_3-\overset{\text{O}}{\text{C}}-\text{CH}_3$
- (2) $\text{CH}_3-\overset{\text{O}}{\text{C}}-\text{H}$
- (3) $\text{CH}_3-\overset{\text{O}}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\text{C}}-\text{CH}_3$
- (4) $\text{CH}_3-\overset{\text{O}}{\text{C}}-\overset{\text{O}}{\text{C}}-\text{CH}_3$

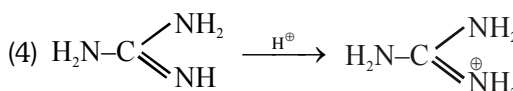
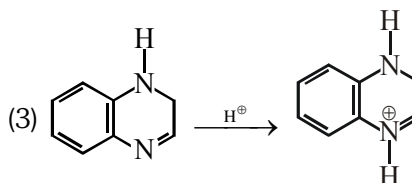
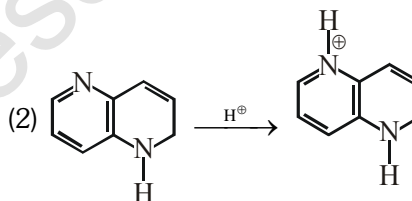
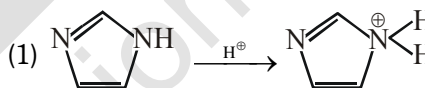
4. Which is not aromatic compound :-



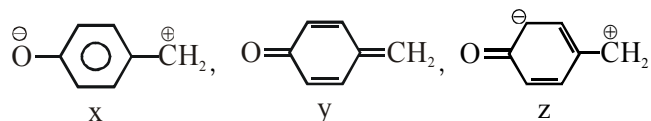
5. The correct order of decreasing basic strength is



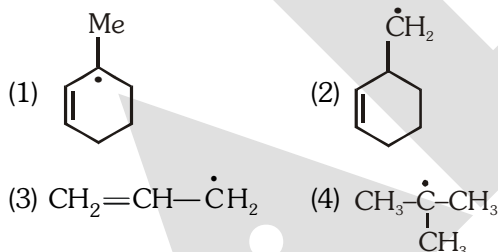
6. Which of the following acid-base reaction is not feasible :-



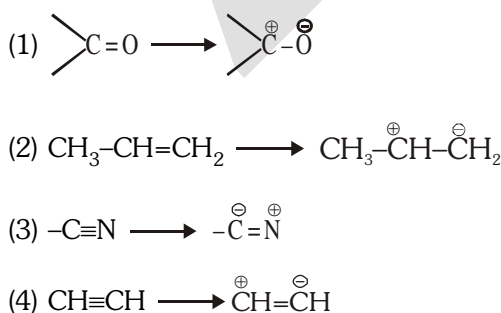
7. Select the correct option for stability of following resonance structures :-



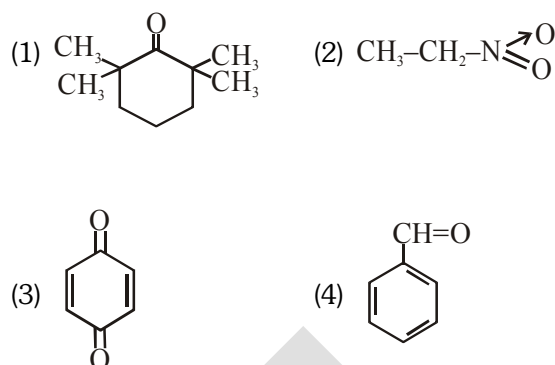
- (1) $x > y > z$ (2) $z > y > x$
 (3) $y > x > z$ (4) $y > z > x$
8. (a) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$
 (b) $\text{CH}_3-\text{CH}=\text{CH}_2$
 (c) CH_3-CH_3
 C—C single bond distance would be :-
 (1) $a < b < c$
 (2) $a > b < c$
 (3) $a < b > c$
 (4) $a < b > c$
9. Hydrogen attached to sp^3 carbon in cyclopentadiene can be easily removed as what :-
 (1) Hydride ion
 (2) Hydrogen molecule
 (3) Proton
 (4) Hydrogen atom
10. Which one is most stable free radical :-



11. Which is wrong electromeric effect :-



12. Tautomerism is exhibited by :-



13. Identify the compound that exhibits tautomerism :-

- (1) 2-Pentanone (2) Phenol
 (3) 2-Butene (4) Lactic acid

14. Arrange the following in correct order of acidic strength :-

- (I) CH_3-NO_2 (II) $\text{NO}_2-\text{CH}_2-\text{NO}_2$
 (III) $\text{CH}_3-\text{CH}_2-\text{NO}_2$ (IV) $\text{NO}_2-\text{CH}(\text{NO}_2)-\text{NO}_2$

- (1) $\text{IV} > \text{II} > \text{I} > \text{III}$
 (2) $\text{IV} > \text{II} > \text{III} > \text{I}$
 (3) $\text{III} > \text{I} > \text{II} > \text{IV}$
 (4) $\text{III} > \text{I} > \text{IV} > \text{II}$

15. The correct order of stability of following carbon free radical is :-

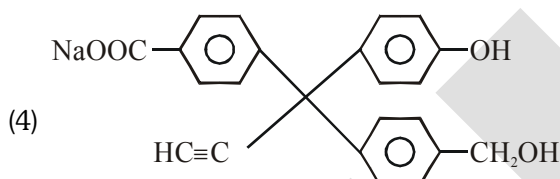
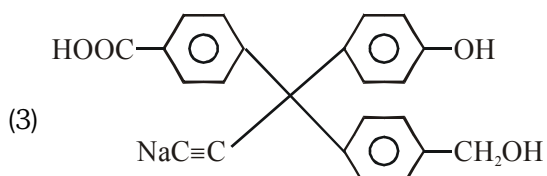
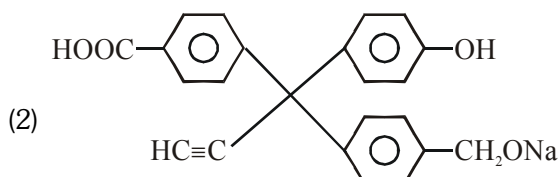
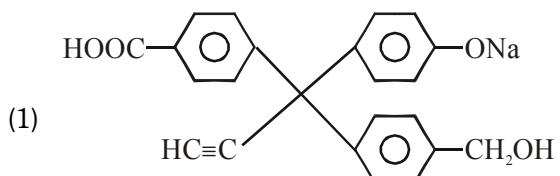
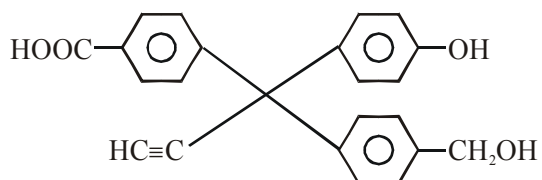


- (1) $1 > 2 > 3 > 4$
 (2) $4 > 3 > 2 > 1$
 (3) $2 > 3 > 1 > 4$
 (4) $2 > 1 > 3 > 4$

16. Which of the following is the strongest acid :-

- (1) Carbolic acid (2) Carbonic acid
 (3) Picric acid (4) Acetic acid

17.



18. The correct order of acidic strength is :-

- (1) $\text{CH}_3\text{COOH} < \text{HCN} < \text{H}_2\text{O} < \text{C}_2\text{H}_5\text{OH}$
- (2) $\text{CH}_3\text{COOH} > \text{HCN} > \text{H}_2\text{O} > \text{C}_2\text{H}_5\text{OH}$
- (3) $\text{HCN} > \text{CH}_3\text{COOH} > \text{H}_2\text{O} > \text{C}_2\text{H}_5\text{OH}$
- (4) $\text{CH}_3\text{COOH} > \text{HCN} > \text{C}_2\text{H}_5\text{OH} > \text{H}_2\text{O}$

19. Which of the following orders of acid strength is correct :-

- (1) $\text{RCOOH} > \text{ROH} > \text{HOH} > \text{HC}\equiv\text{CH}$
- (2) $\text{RCOOH} > \text{HOH} > \text{ROH} > \text{HC}\equiv\text{CH}$
- (3) $\text{RCOOH} > \text{HOH} > \text{HC}\equiv\text{CH} > \text{ROH}$
- (4) $\text{RCOOH} > \text{HC}\equiv\text{CH} > \text{HOH} > \text{ROH}$

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	4	2	4	1	3	1	3	1	3	2	1,2	1	4
Que.	16	17	18	19											
Ans.	3	4	2	2											

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** Tertiary carbonium ions are generally formed more easily than primary carbonium ions.
Reason :- Hyperconjugative as well as inductive effect of additional alkyl groups stabilize tertiary carbonium ions.
 (1) A (2) B (3) C (4) D
- Assertion :-** Carbanions have ammonia like pyramidal shape.
Reason :- The carbon atom carrying negative charge has an octet of electrons.
 (1) A (2) B (3) C (4) D
- Assertion :-** Pyrrole is a weaker base than its hydrogenated product pyrrolidine.
Reason :- Lone pair of electrons on nitrogen is delocalized in pyrrole.
 (1) A (2) B (3) C (4) D
- Assertion :-** A mixture of o-nitrophenol and p-nitrophenol can be separated by steam distillation.
Reason :- o-nitrophenol is steam volatile but p-nitrophenol is not.
 (1) A (2) B (3) C (4) D
- Assertion :-** The carbocation $\text{CF}_3 - \text{CH}_2^+$ is less stable than CF_3^+ .
Reason :- In case of $\text{CF}_3 - \text{CH}_2^+$, the strongly electron withdrawing $-\text{CF}_3$ group intensifies the +ve charge but in case of CF_3^+ , the lone pairs of electrons on each of the three F-atoms overlap with the empty p-orbital of the carbocation carbon atom, thereby dispersing the +ve charge.
 (1) A (2) B (3) C (4) D
- Assertion :-** Acidity of the following compounds decreases in the order $\text{HC}\equiv\text{CH} > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{CH}_3$
Reason :- Acidity of the compounds increase as the electronegativity of the carbon increases to which H is attached.
 (1) A (2) B (3) C (4) D
- Assertion :-** The acidity of alcohols follows the order : $1^\circ > 2^\circ > 3^\circ$.
Reason :- The + I effect of the additional alkyl groups favours the cleavage of O-H bond.
 (1) A (2) B (3) C (4) D
- Assertion :-** Phenol is less acidic than p-nitro phenol.
Reason :- Phenolate ion is more stable than p-nitro phenolate ion.
 (1) A (2) B (3) C (4) D
- Assertion :-** Pyrrole is more basic than pyridine.
Reason :- In pyrrole, nitrogen is sp^3 -hybridized.
 (1) A (2) B (3) C (4) D
- Assertion :-** Aniline is a weaker base than benzyl amine.
Reason :- In aniline, mesomeric interaction occurs between benzene ring and amino group.
 (1) A (2) B (3) C (4) D
- Assertion :** Cyclopentadienyl anion is much more stable than allyl anion.
Reason : Cyclopentadienyl anion is aromatic in character.
 (1) A (2) B (3) C (4) D
- Assertion :** Acetic acid is stronger acid than ethyl alcohol.
Reason : In acetate ion both C-O bond lengths are equal.
 (1) A (2) B (3) C (4) D

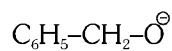
13. **Assertion :** Tropylium ion is more stable than benzyl cation.

Reason : Tropylium ion is anti-aromatic in nature.

(1) A (2) B (3) C (4) D

14. **Assertion :-** Phenol is more acidic than benzyl alcohol.

Reason :- $\text{C}_6\text{H}_5-\text{O}^\ominus$ is more stable than



(1) A (2) B (3) C (4) D

15. **Assertion :-** Picric acid is stronger acid than phenol.

Reason :- Picric acid has $-\text{COOH}$ group, while phenol has $-\text{OH}$ group.

(1) A (2) B (3) C (4) D

16. **Assertion :-** Pyridine is less basic than trimethyl amine.

Reason :- Nitrogen atom in pyridine is sp^2 hybridized, while in trimethylamine it is sp^3 .

(1) A (2) B (3) C (4) D

17. **Assertion :-** Enol form of cyclohexane-1,3,5-trione is more stable than its keto form.

Reason :- It contains α -hydrogen atoms.

(1) A (2) B (3) C (4) D

18. **Assertion :-** $\text{C}_2\text{H}_5\text{NO}_2$ shows functional isomerism as well as tautomerism.

Reason :- Nitroethane shows tautomerism due to presence of α -hydrogens and functional isomerism with ethyl nitrite.

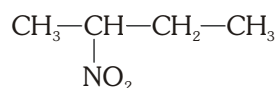
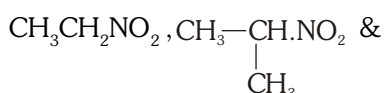
(1) A (2) B (3) C (4) D

19. **Assertion :** Tautomers have different functional group.

Reason : Tautomers have same molecular formula.

(1) A (2) B (3) C (4) D

20. **Assertion :-** The following compound are given below –



all three compounds are soluble in NaOH.

Reason :- All above compound have lower boiling than CH_3NO_2 .

(1) A (2) B (3) C (4) D

21. **Assertion :-** Phenol is stronger acid than alcohols.

Reason :- Phenol is stabilized by resonance whereas alcohols are not.

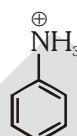
(1) A (2) B (3) C (4) D

22. **Assertion :-** Formic acid is a stronger acid than benzoic acid.

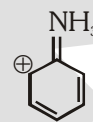
Reason :- pK_a of formic acid is higher than that of benzoic acid.

(1) A (2) B (3) C (4) D

23. **Assertion :-** Here two structures are given



I



II

I and II are not acceptable canonical structure.

Reason :- Carbonium ions are less stable than ammonium ions.

(1) A (2) B (3) C (4) D

24. **Assertion :-** Benzene hexachloride is an aromatic compound.

Reason :- It contain 3π bonds in the ring.

(1) A (2) B (3) C (4) D

25. **Assertion :-** $\text{Cl}-\underset{\text{C}_2\text{H}_5}{\overset{\text{H}}{\text{C}}}-\text{CH}_3$ and $\text{H}-\underset{\text{CH}_3}{\overset{\text{C}_2\text{H}_5}{\text{C}}}-\text{Cl}$ are enantiomers.

Reason :- Non-superimposable mirror images are known as enantiomers.

(1) A (2) B (3) C (4) D

26. **Assertion :-** Dimethyl amine has more basic strength than trimethyl amine.

Reason :- It is due to solvation effect.

(1) A (2) B (3) C (4) D

27. **Assertion :-** In $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}=\text{CH}_2$, the C-C-C bond angles are different.

Reason :- The C has steric effect of $-\text{CH}_3$ group.

(1) A (2) B (3) C (4) D

28. Assertion :- Benzyl amine is more basic than ethyl amine.

Reason :- Phenyl group has +I effect.

(1) A (2) B (3) C (4) D

29. Assertion :- o-nitrophenol is more acidic than m-nitrophenol.

Reason :- At o-position, -I effect of $-\text{NO}_2$ is more than at m-position.

(1) A (2) B (3) C (4) D

30. Assertion :- N, N-diethylethane is more basic than N, N-dimethyl methane.

Reason :- +I of ethyl is more than methyl group.

(1) A (2) B (3) C (4) D

31. Assertion :- Dimethyl amine has more basic strength than trimethyl amine.

Reason :- It is due to solvation effect.

(1) A (2) B (3) C (4) D

32. Assertion :- In $\begin{array}{c} \text{CH}_3 \\ | \\ \text{C}=\text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$, the C-C-C bond

angles are different.

Reason :- The C has steric effect of $-\text{CH}_3$ group.

(1) A (2) B (3) C (4) D

33. Assertion :- Benzyl amine is more basic than ethyl amine.

Reason :- Phenyl group has +I effect.

(1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	2	1	1	1	1	3	3	4	1	1	2	3	1	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	1	2	3	2	3	2	4	1	1	1	4	2	1
Que.	31	32	33												
Ans.	1	1	4												

IMPORTANT NOTES

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REACTION MECHANISM - II

E

S.No.	CONTENTS	Page
1.	Type of reactions	87
2.	Addition reactions	87
3.	Substitution reactions	98
4.	Elimination reactions	113
5.	Exercise-I (Conceptual Questions)	118
6.	Exercise-II (Previous Years Questions)	124
7.	Exercise-III (Analytical Questions)	138
8.	Exercise-IV (Assertion & Reason)	142

NEET SYLLABUS

Types of organic reaction (free radical reaction of Alkane, addition of hydrogen, halogen, water, hydrogen halides (Markovnikov's addition and peroxide effect) in alkene and alkyne, mechanism of electrophilic substitution - Nitration sulphonation, halogenation, Friedel Craft's alkylation and acylation; directive influence of functional group in mono-substituted benzene, mechanism of substitution reactions, mechanism of nucleophilic addition)

@neetfocusquizz

OBJECTIVES

After studying this unit, we will be able to :

- *recognise the types of organic reactions;*
- *predict the formation of the addition products of unsymmetrical alkenes and alkynes on the basis of electronic mechanism;*
- *understand mechanism of electrophilic substitution reactions of benzene;*

*"Thinking should become your capital asset, no matter whatever ups and downs;
you come across in your life."*

A.P.J. Abdul Kalam

REACTION MECHANISM : PART-II

4.0 TYPE OF REACTIONS :

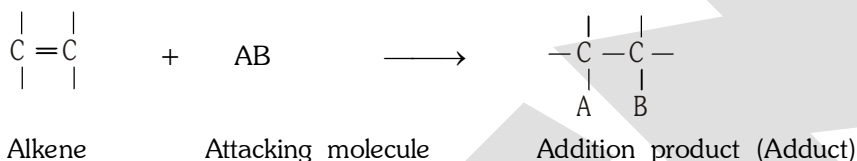
Reactions are of mainly four types :

1. Addition reactions.
2. Substitution reactions.
3. Elimination reactions.
4. Isomerisation reactions

4.1 ADDITION REACTIONS : It is also of three types :

- (A) Electrophilic addition reactions
- (B) Free radical addition reactions
- (C) Nucleophilic addition reactions

(A) Electrophilic addition reaction :- Because of the presence of $>C=C<$ bond in molecules, alkenes generally take part in the **addition reactions**.



From mechanism point of view, the addition in alkenes is generally **electrophilic in nature** which means that attacking reagent which carries the initial attack is an electrophile (E^+). This is quite expected also as there is high electron density in the double bond. The mechanism proceeds in two steps.

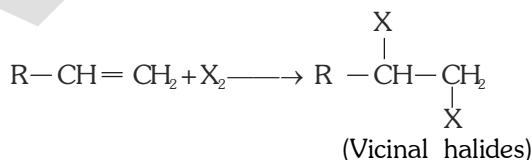
Step I : The π -electron cloud of the double bond causes the polarisation of the attacking molecule ($E-Nu$) which cleaves to release the electrophile (E^+) for the attack. The double bond simultaneously undergoes electromeric effect and the attack by the electrophile is accomplished in slow step (also called rate determining step) to form a **carbocation** intermediate.



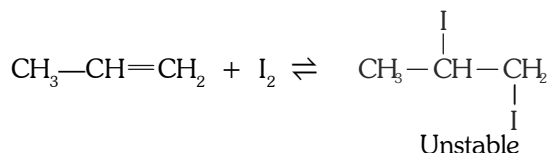
Step II : The nucleophile ($:Nu^-$) released in the slow step combines with the carbocation to give the desired addition product in the fast step.

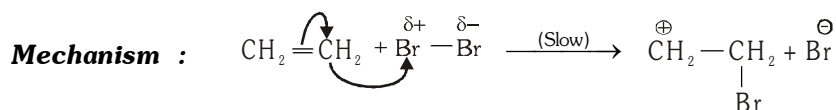
Reactivity for Electrophilic addition reaction $\propto$ stability of carbocation formed in RDS

(1) Addition of Halogen : It is a electrophilic addition reaction.

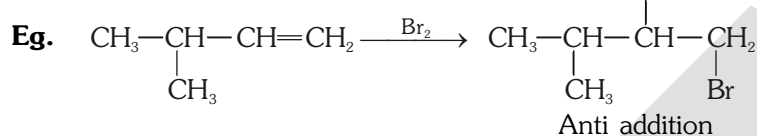
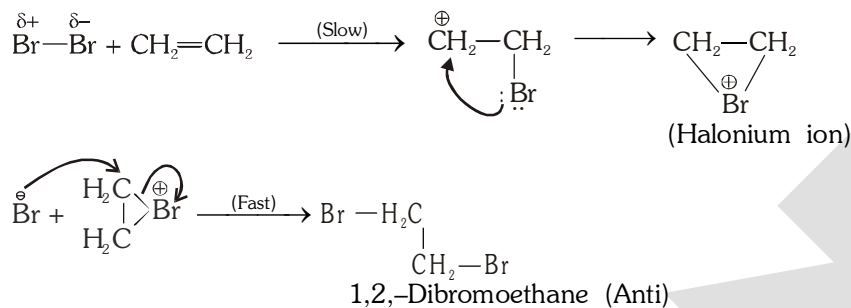


- (a) The addition of Br_2 on alkenes provides a useful test for unsaturation in molecule. The brown colour of the bromine being rapidly discharged. Thus decolorization of 5% Br_2 in CCl_4 by a compound suggest unsaturation in it. Colourless dibromo compound is formed.
- (b) I_2 reacts with alkenes to form Vicinal di-iodides which are unstable and I_2 gets eliminated to give original alkene.



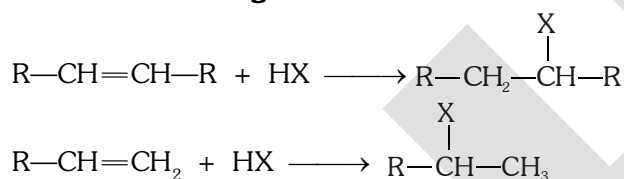


It is interesting to note that product which is mainly formed as a result of addition is **trans** in nature whereas the cis isomer is obtained in relatively smaller proportions. Since carbocation intermediate is planar (sp^2 hybridised), both cis and trans addition products must be formed almost in equal proportions. The trans product can be justified in case a cyclic halonium ion is formed by the initial electrophile attack.



No carbocation rearrangement and anti addition product.

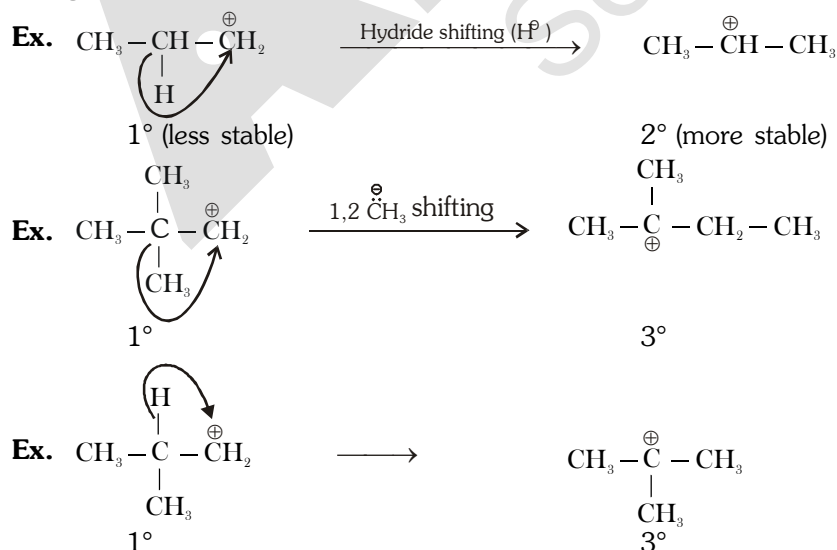
(2) Addition of halogen acid :

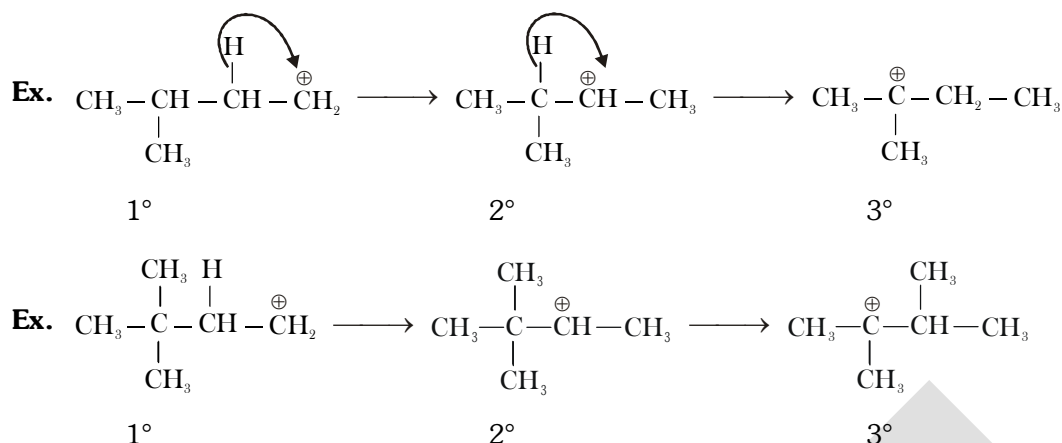


GOLDEN KEY POINTS

- The order of reactivity of hydrogen halide is : **HI > HBr > HCl**
- Addition on alkene proceeds via the formation of more stable carbonium ion.
- Addition of HX on unsymmetrical alkenes ($\text{R}-\text{CH}=\text{CH}_2$) takes place according to Markovnikov's rule. Carbocation rearrangement is observed in the reaction.

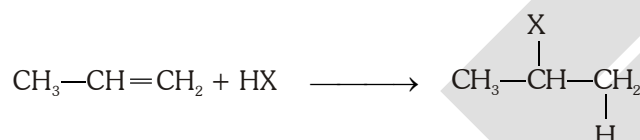
Rearrangement in carbocation



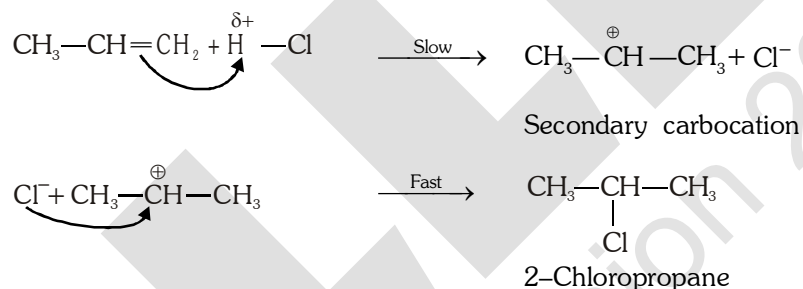


Markovnikov's Rule States :

(a) First Rule : When molecule of HX add up on unsymmetrical unsaturated hydrocarbon, the electrophile (H^+) goes to the unsaturated carbon atom bearing more number of hydrogen atoms.



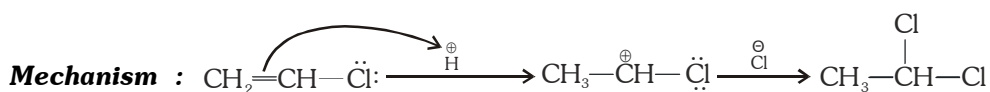
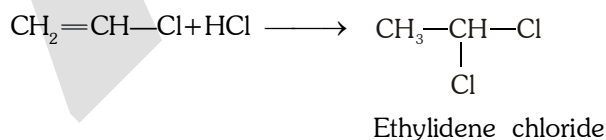
Mechanism : It is electrophilic addition and is illustrated by the action of HCl to propene.



Primary carbocation ($\text{CH}_3\text{—CH}_2\text{—}\overset{\oplus}{\text{CH}}_2$) is also formed but only in very small proportion since it is less stable than the secondary carbocation. Markovnikov's rule is based on stability of carbocation.

Note ; The electrophilic addition of HX to unsymmetrical alkenes always occurs through the formation of a more stable carbocation intermediate.

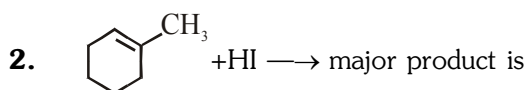
(b) Second Rule : In the addition of HX to vinyl halide and analogous compounds, the halogen attaches itself to the carbon atom, on which the halogen atom is already present.



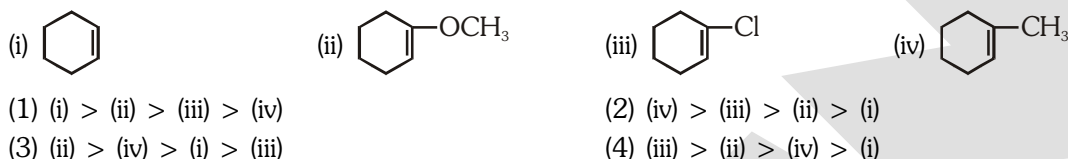
All polar reagents of the general structure $\overset{\oplus}{Y} \overset{\ominus}{Z}$ (such as $\overset{\oplus}{H}-\overset{\ominus}{X}$, $\overset{\oplus}{H}-\overset{\ominus}{OH}$, $\overset{\oplus}{H}-\overset{\ominus}{SO_3H}$, $\overset{\oplus}{X}-\overset{\ominus}{OH}$) add on unsymmetrical unsaturated compound in accordance with Markovnikov's rules. Such additions are called normal Markovnikov's rule, where as additions in the opposite manner are referred to as abnormal or **anti Markovnikov's additions**.

BEGINNER'S BOX-1

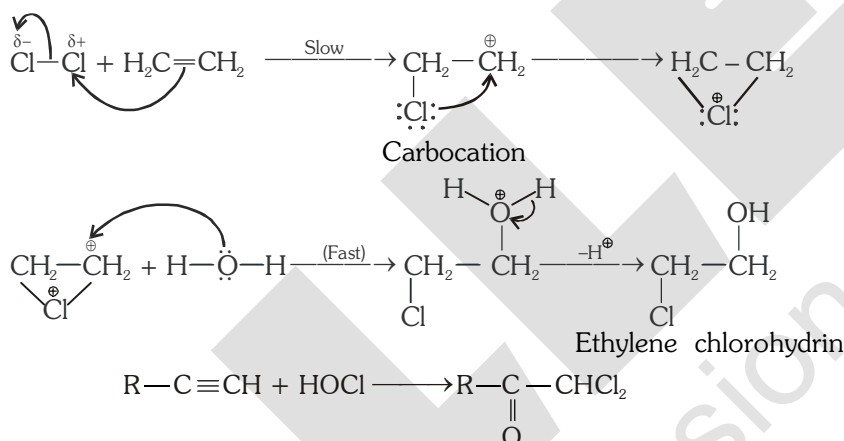
1. The intermediate in the Electrophilic addition-reaction is :-
 (1) Carbocation (2) Carbanion (3) Free radical (4) Carbene



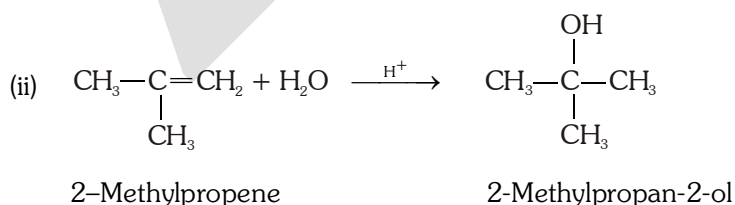
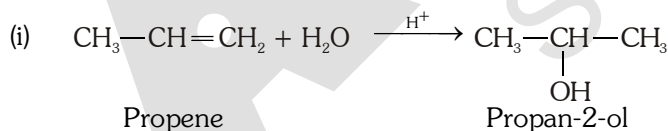
3. Give reactivity order towards EAR.



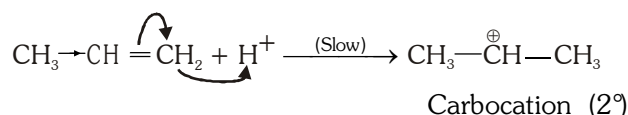
(3) Addition of Hypohalous acid (or X_2/H_2O , or HOX) : It is a electrophilic addition and follows Markovnikov's rule, and anti addition.

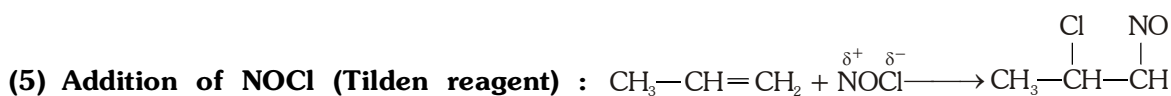
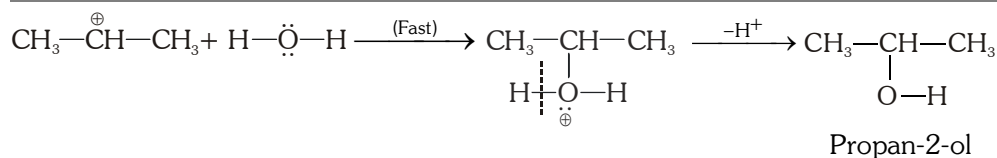


(4) Addition of water (Hydration of alkenes) : Propene and higher alkenes react with water in the presence of acid to form alcohol. This reaction is known as acidic **hydration reaction**. Intermediate in this reaction is carbocation, so rearrangement may take place.



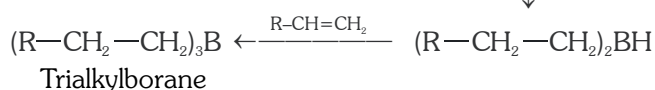
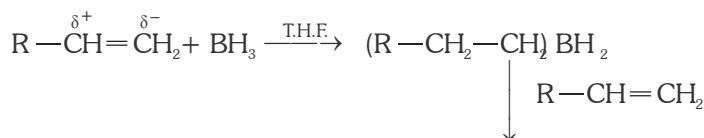
Mechanism :



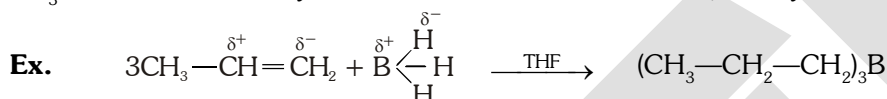


Propylene nitrosochloride

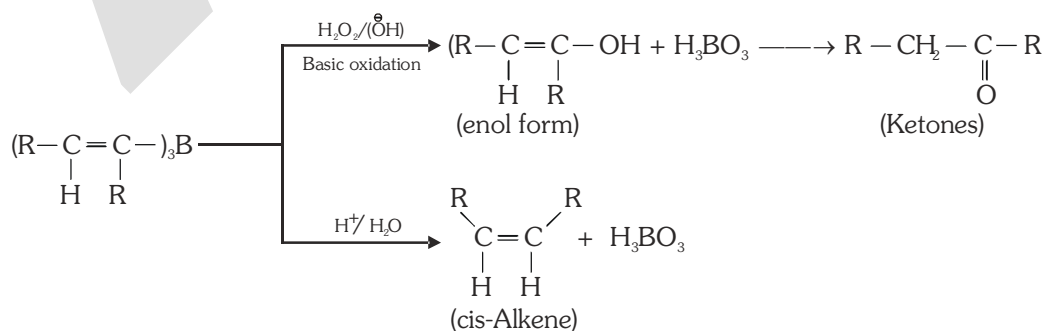
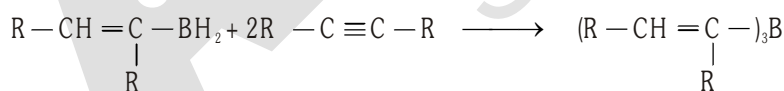
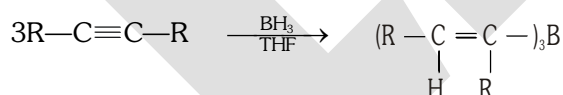
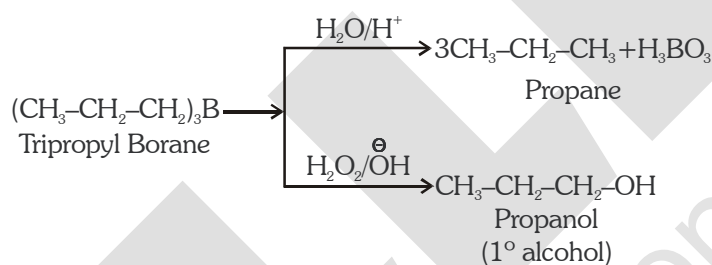
(6) Hydroboration Oxidation : Borane readily reacts with alkenes giving trialkyl boranes. The reaction is called hydroboration.



BH_3 does not exist freely as monomer so a solvent THF (tetra hydro furane) is used to stabilised it.

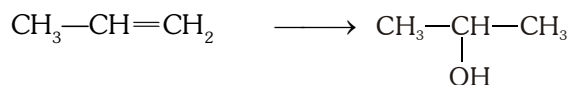


BHR₂ also can be taken.



Note : The overall process appears to be addition of water according to anti Markovnikov's rule and involves syn. addition.

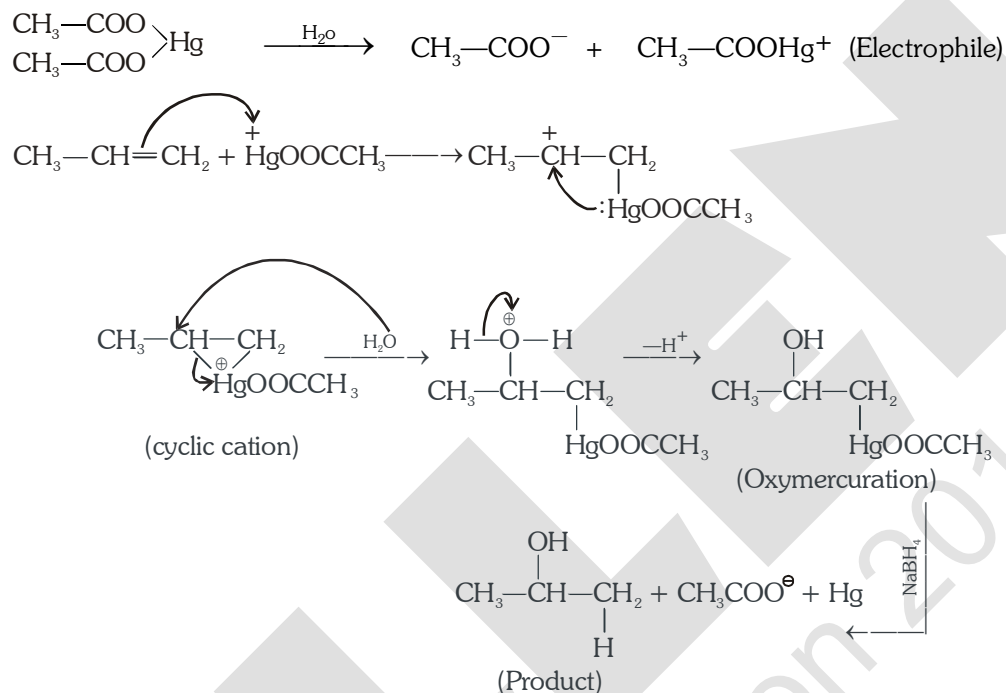
- (7) **Oxymercuration – demercuration** : Mercuric acetate in water is treated with an alkene. The addition product on reduction with sodium Boro hydride in aqueous NaOH Solution gives alcohol. It follows the Markovnikov's rule.



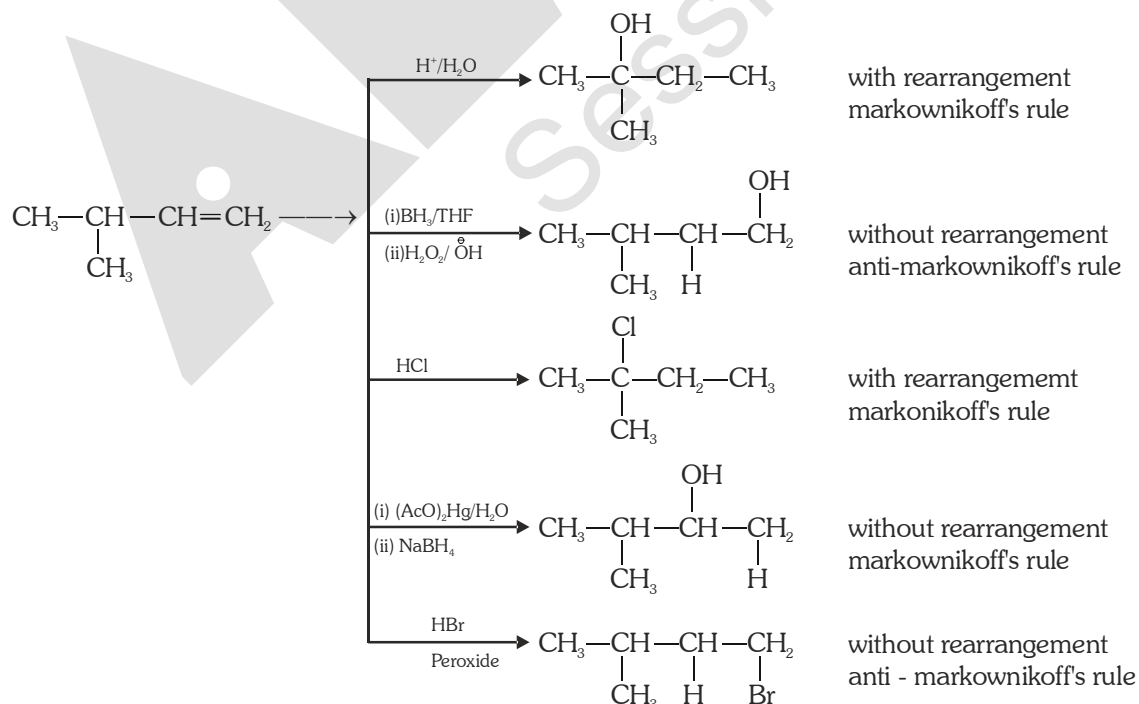
(i) $(\text{AcO})_2\text{Hg}/\text{H}_2\text{O}$ (Mercuric acetate) or $(\text{CH}_3\text{COO})_2\text{Hg}/\text{H}_2\text{O}$

(ii) NaBH_4

Mechanism :

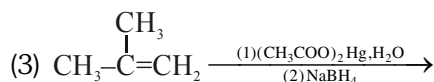
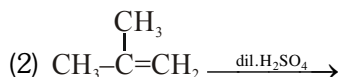
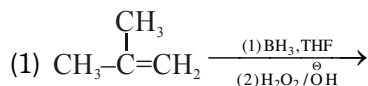


Note : Intermediate is cyclic cation so there is no rearrangement.



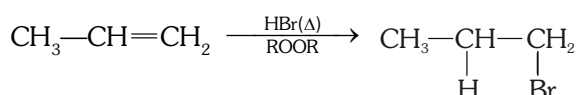
BEGINNER'S BOX-2

1. What is the product formed when acetylene reacts with hypochlorous acid ?
(1) CH_3COCl (2) ClCH_2CHO (3) Cl_2CHCHO (4) ClCH_2COOH
2. Primary alcohol can be formed as major product by



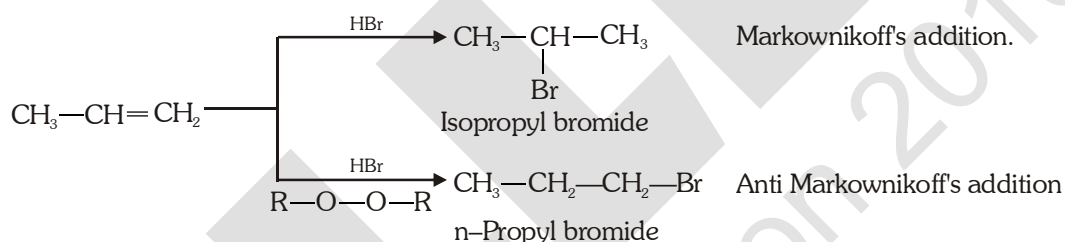
(4) 2 & 3 both

(B) Free radical addition reactions :- Addition of HBr on alkene or alkyne in presence of peroxide.



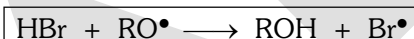
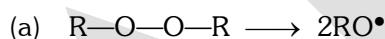
Anti Markovnikov's rule or peroxide effect or Kharasch rule

- (i) In the presence of peroxides the addition of HBr on unsaturated unsymmetrical compound takes place contrary to Markovnikov's rule. This is called peroxide effect and is due to the difference in the mechanism of the addition.
- (ii) In the normal Markovnikov's addition the mechanism is ionic.
- (iii) In the presence of peroxide the addition of HBr takes place via free radicals.
- (iv) Peroxide effect is not observed in case of H-F, HCl and HI. Reactions follows electrophilic addition mechanism.

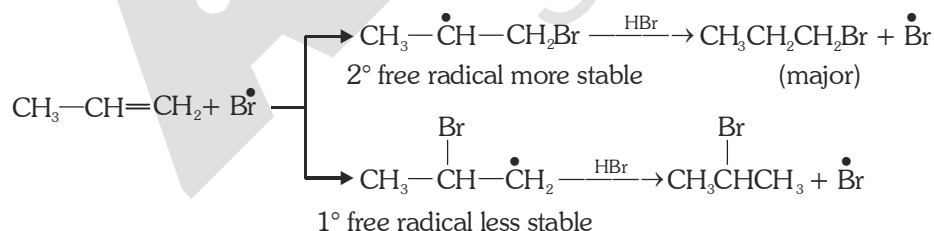


Mechanism :

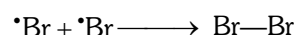
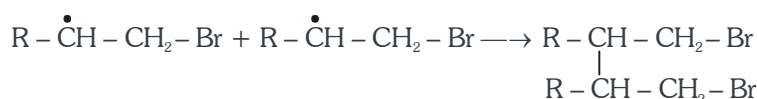
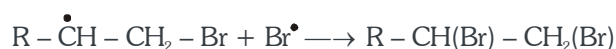
(i) Chain initiation -

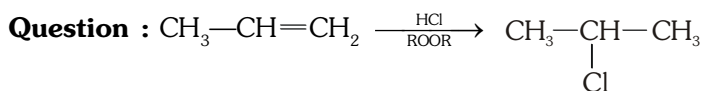


(ii) Chain propagation



(iii) Chain termination :



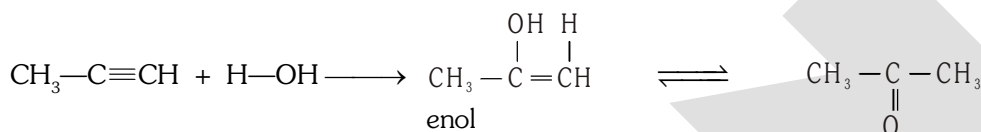


Ans. no effect simple EAR

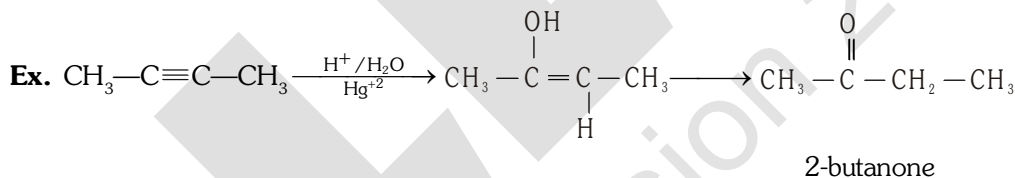
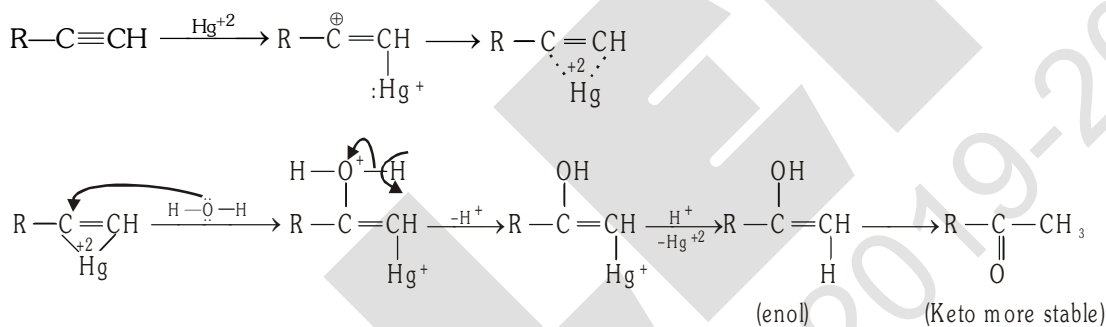
(C) Nucleophilic Addition Reaction :-

(C₁) NAR in Alkyne : In these reactions some heavy metal cation like Hg^{+2} , Pb^{+2} , Ba^{+2} are used. These cation attracts the π e^- of alkynes and decrease the e^- density and hence a nucleophile can attack an alkynes.

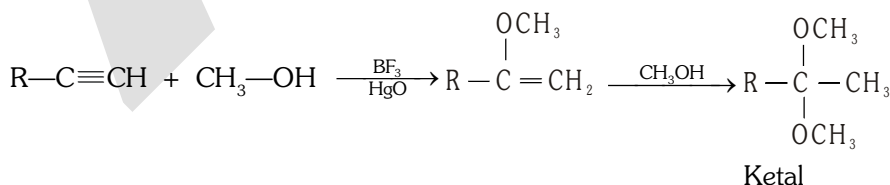
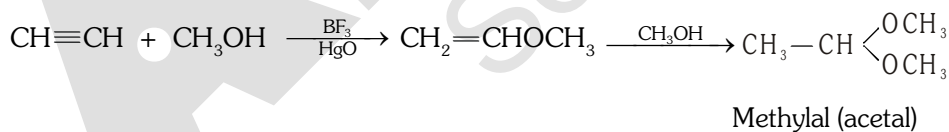
(1) Addition of dil. H_2SO_4 (Hydration) : The addition of water takes place in the presence of Hg^{+2} and H_2SO_4 [1% HgSO_4 + 40% H_2SO_4].



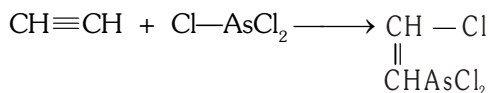
Mechanism :



(2) Addition of alcohols : In presence of BF_3 and HgO alkynes react with alcohols and form acetal and ketal

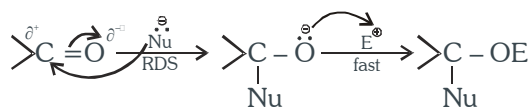


(3) Addition of AsCl_3 : In presence of AlCl_3 or HgCl_2 acetylene combines with AsCl_3 to yield Lewisite gas. It is four times poisonous than mustard gas.



2-Chlorovinyl dichloro arsine (Lewisite gas)

(C₂) NAR in Aldehyde & Ketone : Due to strong electronegativity of oxygen, the mobile π electrons pulled strongly towards oxygen, leaving the carbon atom deficient of electrons. Carbon is thus readily attacked by Nu. The negatively charged oxygen is attacked by electron deficient (electrophile) E⁺.



Reactivity of carbonyl group	$\propto$	Magnitude of δ^+ ve charge
	$\propto$	-I effect
	$\propto$	$\frac{1}{+I \text{ effect}}$

Illustrations

Illustration 1. Arrange the following for reactivity in decreasing order

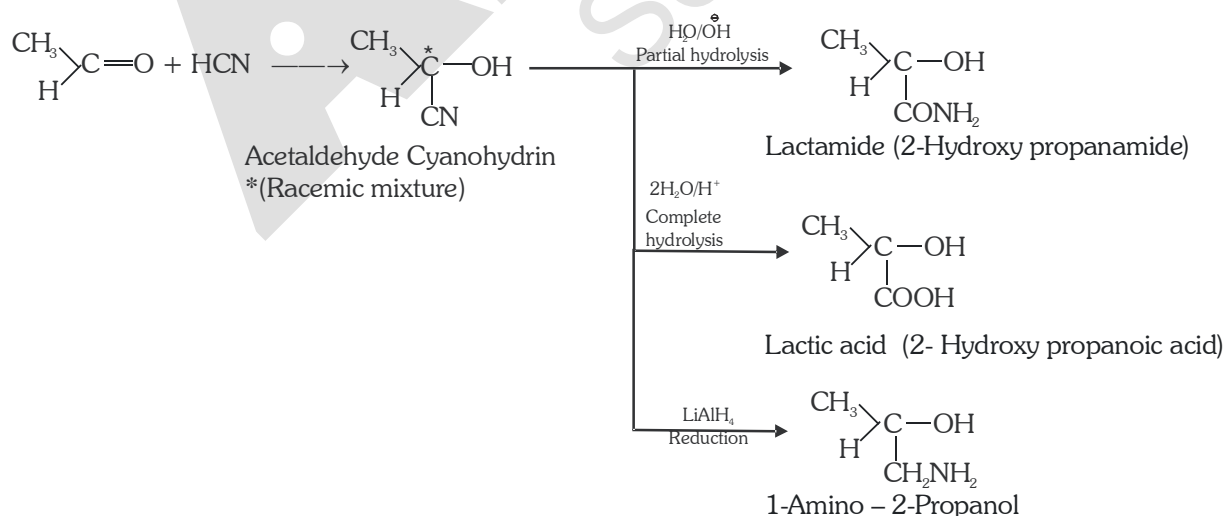
- (a) I. $\begin{array}{c} H \\ >C=O \\ H \end{array}$ II. $\begin{array}{c} CH_3 \\ >C=O \\ H \end{array}$ III. $\begin{array}{c} CH_3 \\ >C=O \\ CH_3 \end{array}$
- (b) I. $ClCH_2CHO$ II. NO_2CH_2CHO III. CH_3CHO IV. CH_3CH_2CHO
- (c) I. CH_3CHO II. $ClCH_2CHO$ III. $HCCl_2CHO$ IV. CCl_3CHO
- (d) I. $\begin{array}{c} CH_3 \\ >C=O \\ CH_3 \end{array}$ II. $\begin{array}{c} CH_3CH_2 \\ >C=O \\ CH_3 \end{array}$ III. $\begin{array}{c} (CH_3)_2CH \\ >C=O \\ CH_3 \end{array}$ IV. $\begin{array}{c} CCl_3 \\ >C=O \\ CH_3 \end{array}$

Solution.

- (a) I > II > III (b) II > I > III > IV
- (c) IV > III > II > I (d) IV > I > II > III

CH_3- is +I group, decreases the intensity of δ^+ ve charge on C-atom of $>C=O$ group.
Cl - is -I group increases the intensity of δ^+ ve charge on C-atom of $>C=O$ group.

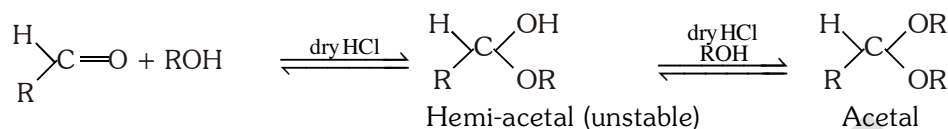
(1) Addition of HCN :



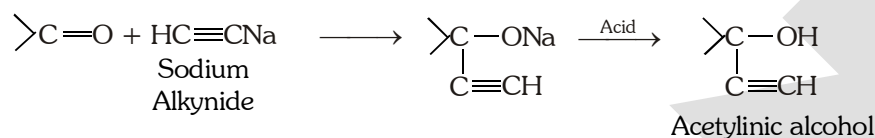
(2) Addition of NaHSO_3 : This reaction is utilized for the separation of carbonyl compounds from non-carbonyl compounds.



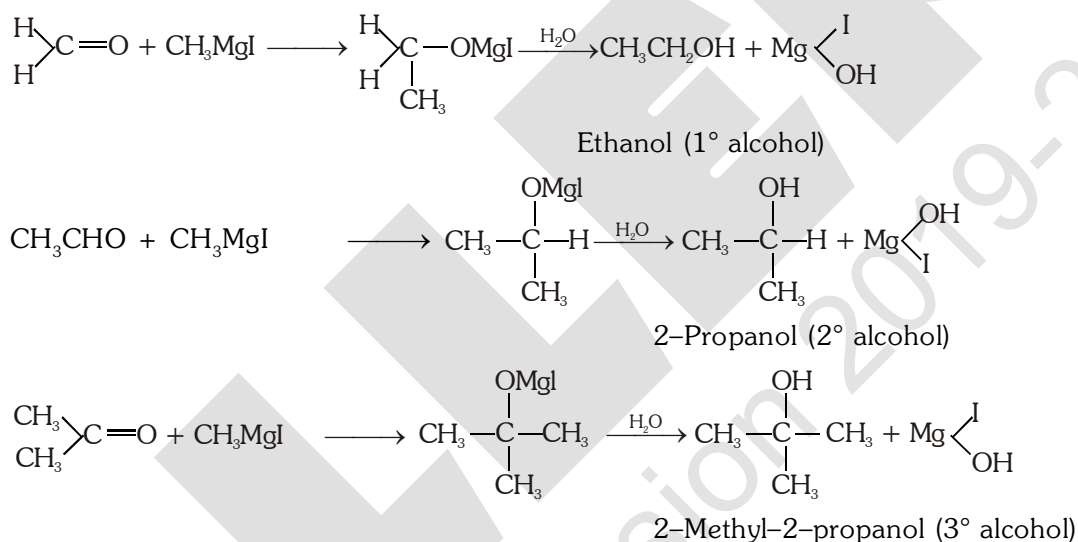
(3) With Alcohol :



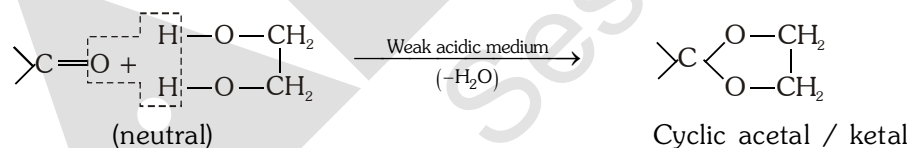
(4) Reaction with sodium alkynide :



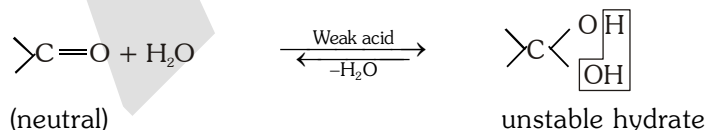
(5) Reaction with Grignard reagent :



(6) Reaction with glycol :

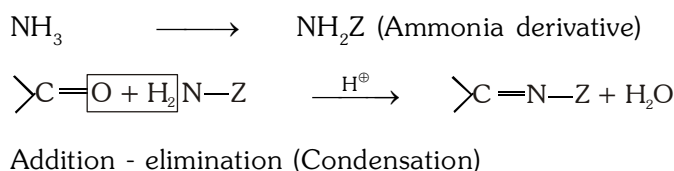


(7) Reaction with H_2O : It is a reversible reaction.

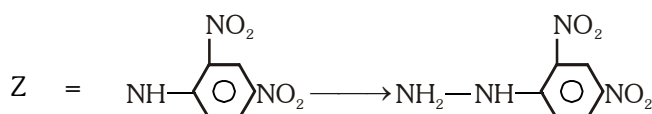
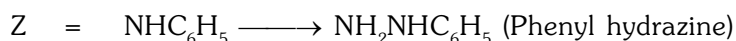
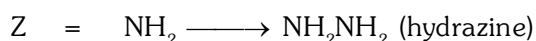


Note : Chloral ($\text{Cl}_3\text{C-CH=O}$) forms stable hydrate [$\text{CCl}_3-\text{CH(OH)}_2$] (chloral hydrate)

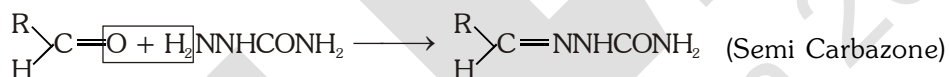
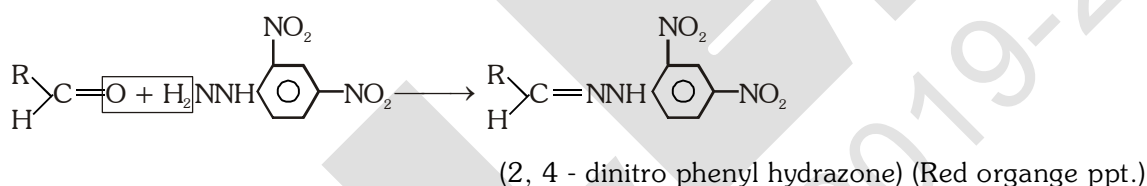
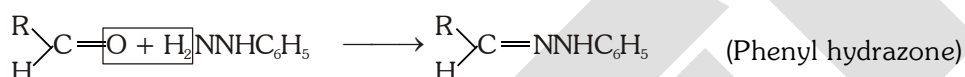
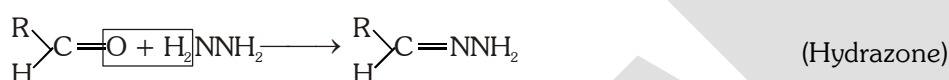
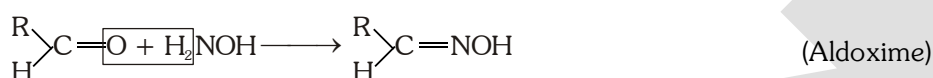
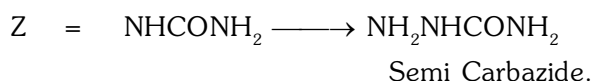
(8) Reaction with ammonia derivatives : These are condensation or addition elimination reaction. These proceeds well in weakly acidic medium.



Ammonia derivatives (NH_2Z) :



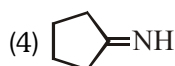
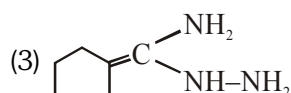
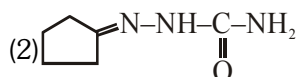
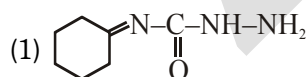
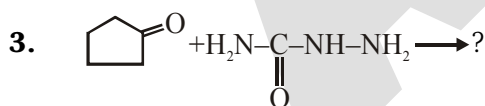
2, 4-Dinitro phenyl hydrazine (2,4-DNP) Brady's reagent.



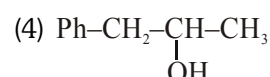
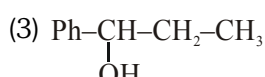
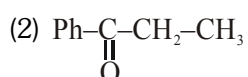
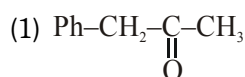
BEGINNER'S BOX-3

1. Which compound form more stable hydrate with H_2O ?
 (1) CCl_3CHO (2) CH_3CHO (3) CH_3COCH_3 (4) $\text{CH}_3\text{COC}_2\text{H}_5$

- 2.** The formation of cyanohydrin from a ketone is an example of :-
- (1) Electrophilic addition (2) Nucleophilic addition
(3) Nucleophilic substitution (4) Electrophilic substitution



4. $\text{Ph}-\text{C}\equiv\text{C}-\text{CH}_3 \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4} \text{A}$, A is



4.2 SUBSTITUTION REACTIONS : Reactions in which one atom or a group of substrate is replaced by other atom or group are called as substitution reactions.

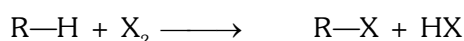
On the basis of reaction conditions and attacking species, substitution reaction is also of three types :

- (A) Free radical substitution reactions
- (B) Electrophilic substitution reactions
- (C) Nucleophilic substitution reactions

(A) Free radical substitution reactions : Substitution reaction in alkanes show free radical mechanism.

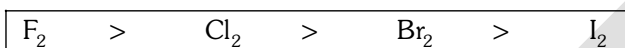
They give following substitution reaction.

(a) Halogenation : Replacement of H-atom by halogen atom

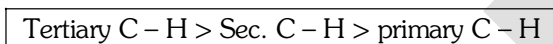


Halogenation is made on exposure to (halogen + alkane) mixture to UV or at elevated temp.

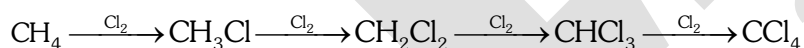
The reactivity order for halogens shows the order.



Reactivity order of hydrogen atom in alkane is

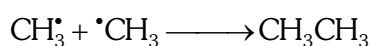
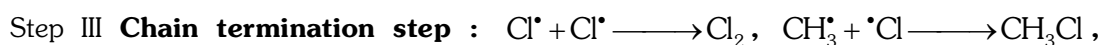
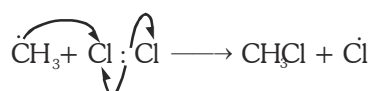
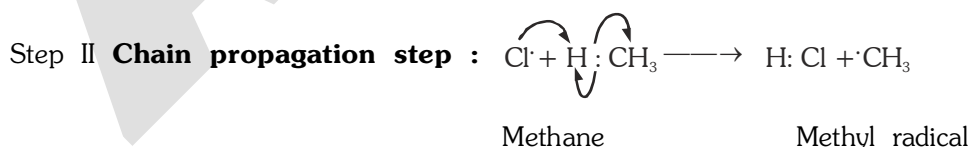
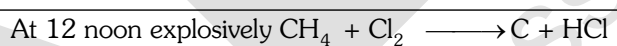


- (i) **Fluorination :** Reacts explosively even in dark. Fluorination can be achieved without violence when alkane is treated with F_2 diluted with an inert gas like N_2 .
- (ii) **Chlorination :**



The monochloro derivative of alkane is obtained as major product by taking alkane in large excess.

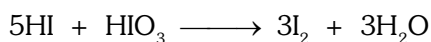
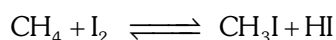
When chlorine is in excess then perchloro derivative is obtained as major product.



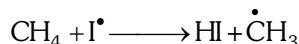
- (iii) **Bromination :** Br_2 reacts with alkanes in a similar manner but less vigorously.

- (iv) **Iodination** : Iodine reacts with alkanes reversibly. HI formed as the by product is a powerful reducing agent and is capable of reducing the CH_3I to CH_4 .

Iodination may be carried out in the presence of an oxidising agent such as HIO_3 , HNO_3 , HgO etc. which decompose HI,

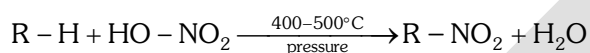


Iodination is very slow because energy of activation of the reaction is very large

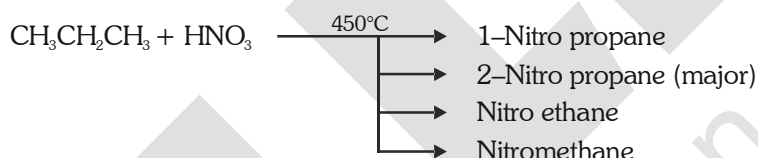


- (b) **Nitration** : (Vapour phase nitration) This involves the substitution of a hydrogen atom of alkane with $-\text{NO}_2$ group.

At ordinary temperature, alkanes do not react with HNO_3 . But reacts with vapours of Conc. HNO_3 at 450°C and in pressure.

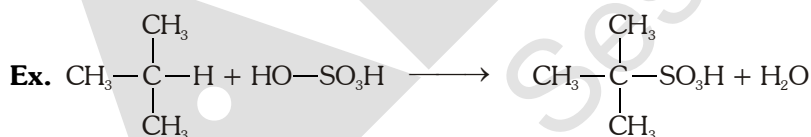


Since the reaction is carried at high temperature and in pressure, so the $\text{C}-\text{C}$ bonds of alkanes also break during the reaction and a mixture of nitroalkanes is formed.



- (c) **Sulphonation** : Replacement of H atom of alkane by $-\text{SO}_3\text{H}$ is known sulphonation.

Alkane react with fuming H_2SO_4 or oleum ($\text{H}_2\text{S}_2\text{O}_7$).

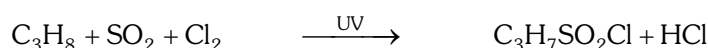


2-Methyl propane

The reactivity order for sulphonation is tert. H > Sec. H > prim. H

Note : The reaction is observed in higher alkanes and the alkanes having 3°H .

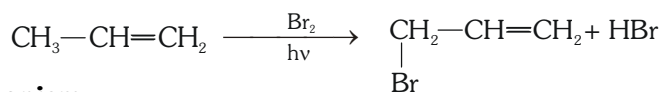
- (d) **Chlorosulphonation (Reed reaction)** : Reaction with a mixture of SO_2 and Cl_2 at ordinary temp. in the presence of UV light is called chlorosulphonation.



Propane sulphonyl Chloride

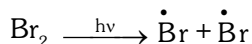
GOLDEN KEY POINTS

Allylic or benzylic substitution by Br_2 (low concentration)/ $h\nu$ or NBS/ $h\nu$

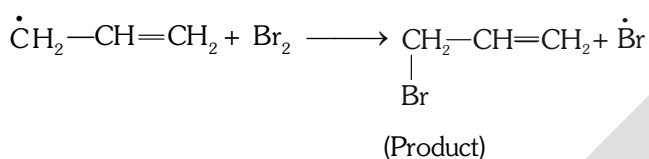
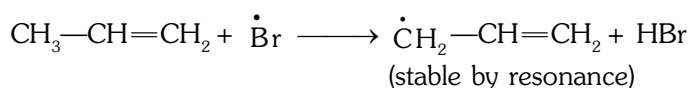


Mechanism :

I Chain initiation step :

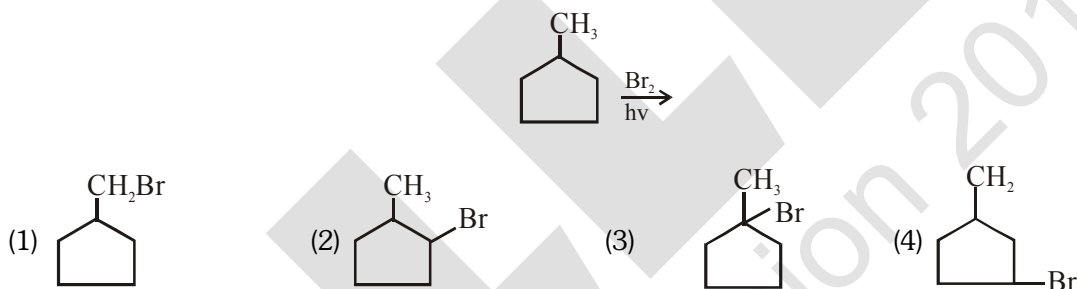


II Chain propagation step :



BEGINNER'S BOX-4

1. In the following reaction, the major product is :-



2. The bond dissociation energy at the C-H bond for the compound :-

(I) CH_3H (II) $\text{CH}_3-\text{CH}_2-\text{H}$ (III) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{H}$ (IV) $\text{C}_6\text{H}_5-\text{H}$

(1) $\text{I} > \text{II} > \text{III} > \text{IV}$

(2) $\text{IV} > \text{III} > \text{II} > \text{I}$

(3) $\text{IV} > \text{I} > \text{II} > \text{III}$

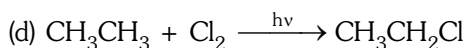
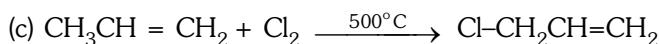
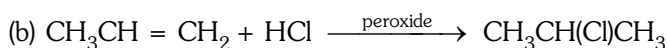
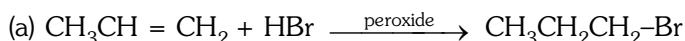
(4) $\text{II} > \text{I} > \text{IV} > \text{III}$

3. Arrange the following in correct order of reactivity towards $\text{Cl}_2/h\nu$ -

(A) CH_4 (B) CH_3CH_3 (C) $\text{CH}_3\text{CH}_2\text{CH}_3$ (D)

(1) $\text{A} > \text{B} > \text{C} > \text{D}$ (2) $\text{D} > \text{C} > \text{B} > \text{A}$ (3) $\text{B} > \text{C} > \text{A} > \text{D}$ (4) $\text{C} > \text{B} > \text{D} > \text{A}$

4. Which of the following are free radical reactions:-



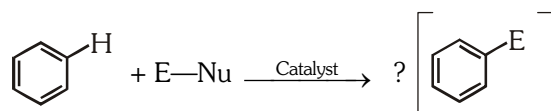
(1) Only d

(2) a, c

(3) a, b, d

(4) a, c, d

(B) Electrophilic substitution reaction [ESR] : Characteristic reaction of arenes is ESR

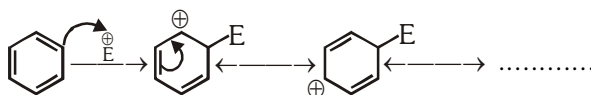


Mechanism :

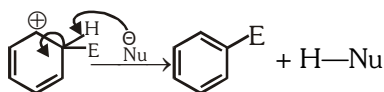
Formation of $\mathbb{E}$



Attack of $\oplus$ E

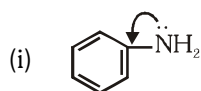


Abstraction of $\mathbb{H}^\oplus$

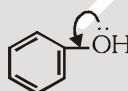


Illustrations

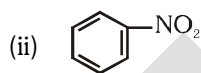
Illustration 2. Give reactivity order for electrophilic substitution reaction.



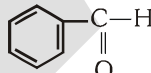
less EN of nitrogen
so more +M so more e^- density
so more reactive



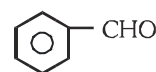
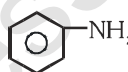
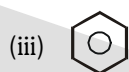
less +M of -OH
due to more EN



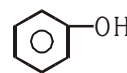
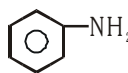
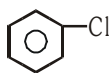
More -M of $-\text{NO}_2$ & more -I
so e^- density decrease
(more) so less reactive



less -M of -CHO & less -I
so e^- density decreases (less)
so more reactive.

 $+H \uparrow$
$$+M \uparrow$$
$$\begin{array}{c} -M \\ \downarrow \end{array}$$

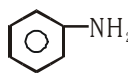
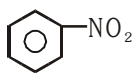
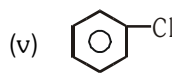
ESR order III > II > I > IV

 $+H \uparrow$
$$\begin{array}{c} -I > +M \\ \downarrow \end{array}$$

more +M
due to less
EN

less +M ↑

ESR order III > IV > I > II

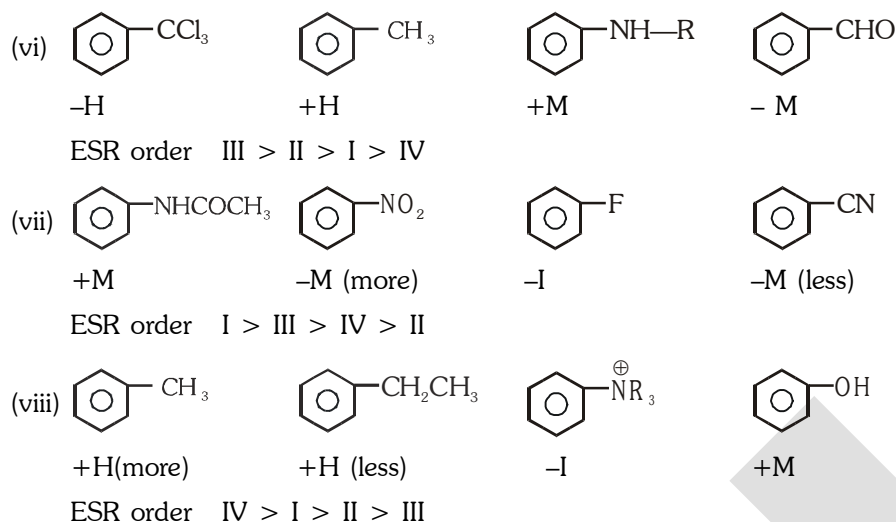


- I

– M

+ M

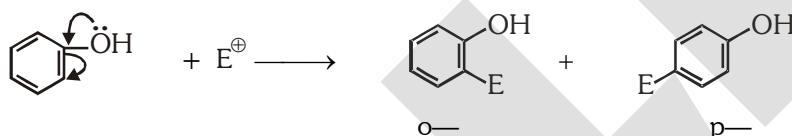
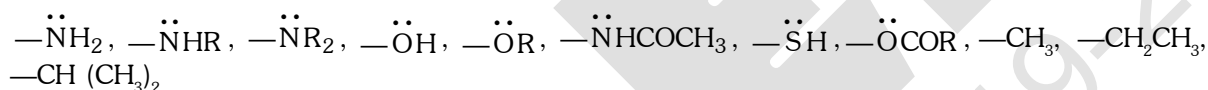
ESR order III > IV > I > II

**Note:**

(1) ortho/para directing group or activating group : Group which direct electrophile on ortho and para position is called as o/p directing group.

These group increases electron density or increases reactivity of benzene ring so are called activating group.

These groups are :

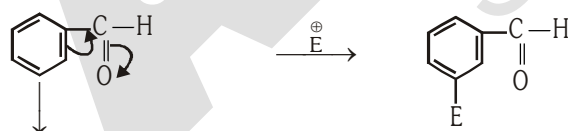


due to +M/+H of these groups electron density at ortho and para position is increased so electrophile easily attack on ortho/para position.

(2) Meta directing or deactivating group : Due to -M/-H of groups electron density at ortho and para position is less but more at meta position so electrophile attack on meta position. So, groups which direct electrophile on meta position are called as meta directing groups.

These groups decrease electron density in benzene ring and decrease reactivity of benzene ring so are called as deactivating group.

These groups are :

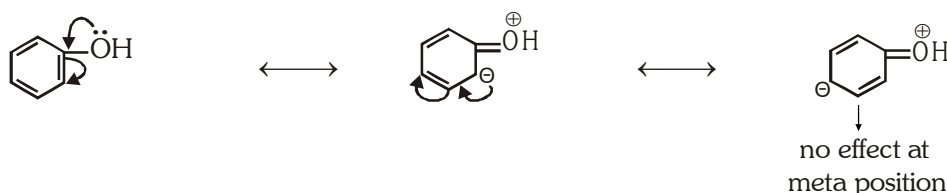


More e^- density at meta position

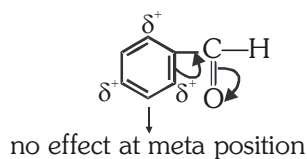
(3) Halogens are o/p directing group due to +M effect but are deactivating group due to -I > +M.

(4) M and H effect does not depend on distance while I-effect depends on distance In given example

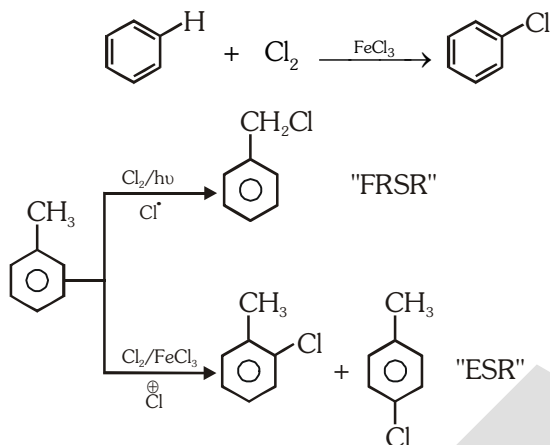
(5) M-effect at meta position is considered zero.



Similarly :

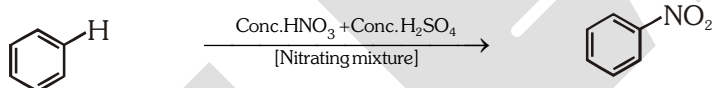


(i) Halogenation :

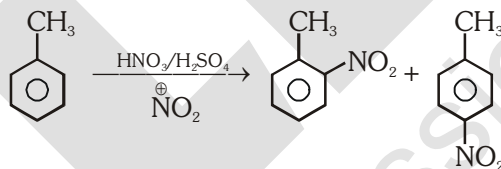
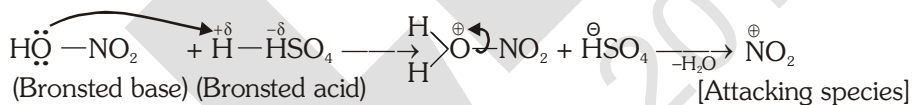


Note: CH₃ group in toluene is o/p directing and activating group.

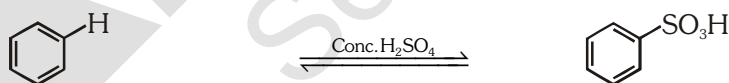
(ii) Nitration :



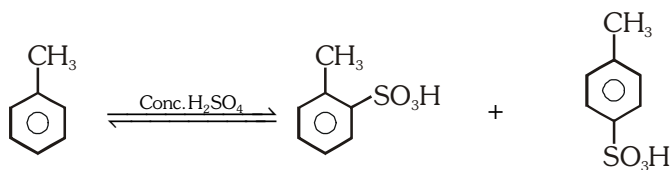
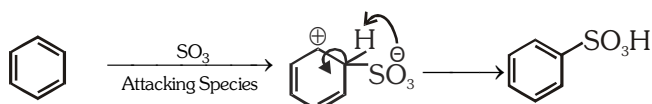
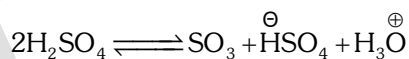
Formation of $\oplus \mathbb{E}$



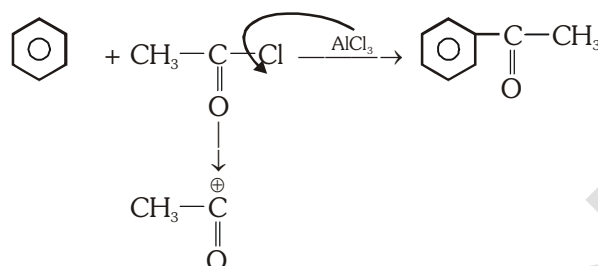
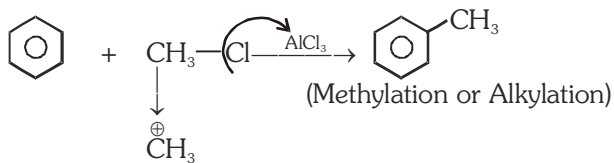
(iii) Sulphonation :



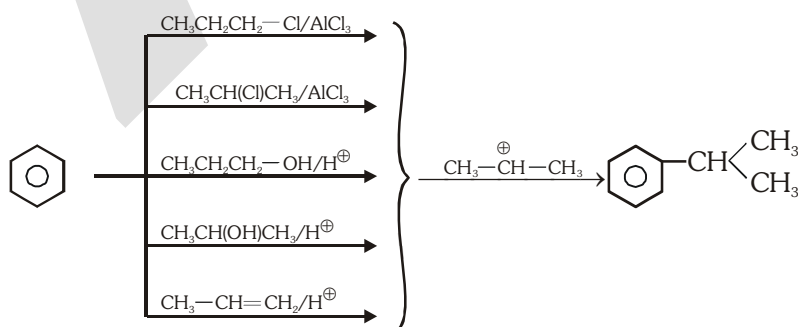
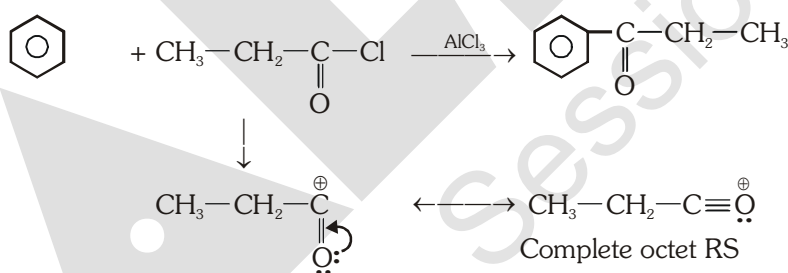
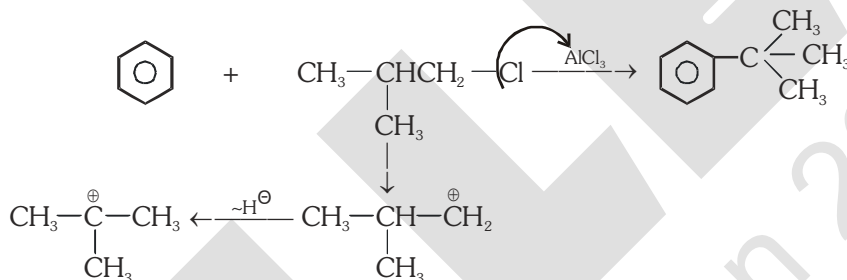
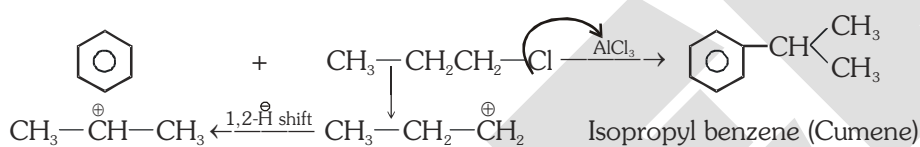
Mechanism :

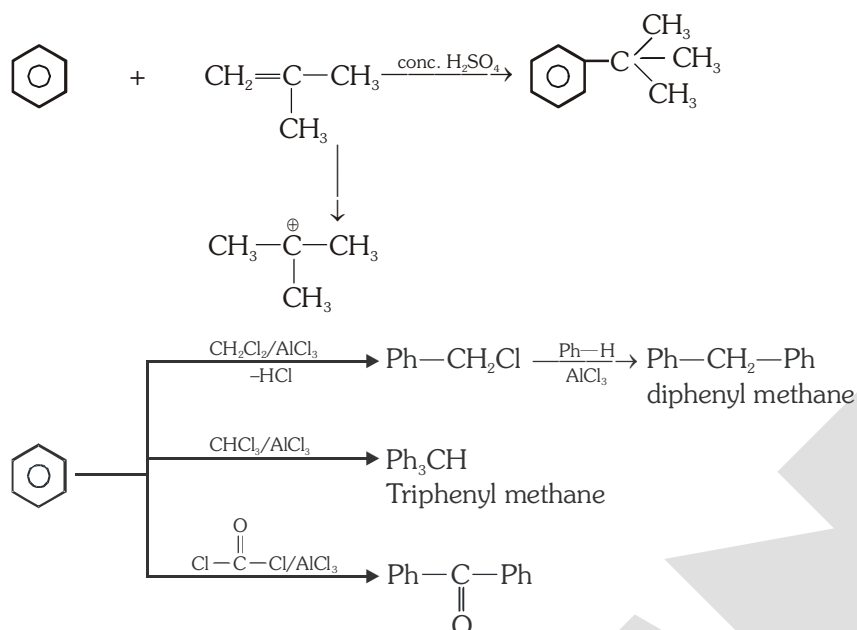


(iv) **Friedel crafts reaction [FCR]** : Alkylation or acylation of arenes in presence of lewis acid $[\text{FeCl}_3, \text{AlCl}_3 \text{ or } \text{ZnCl}_2 \dots]$ is called as FCR.

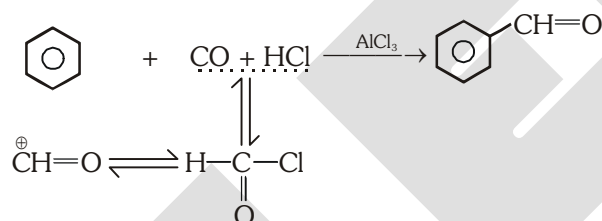


Intermediate carbocation is formed in FCR so rearrangement is possible.

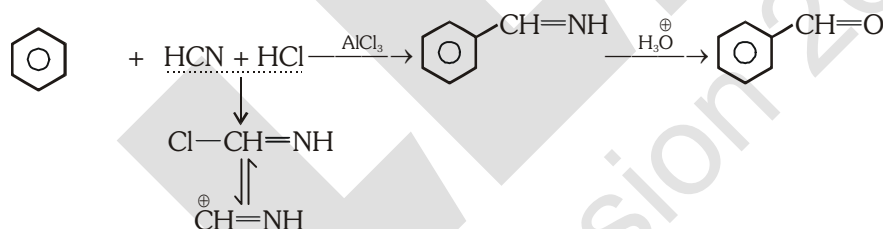




(vi) Gatterman's Koch Reaction :



(vii) **Gatterman's Aldehyde synthesis :**



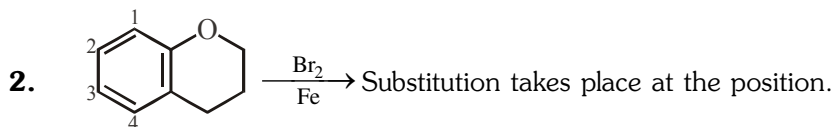
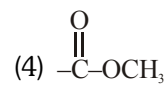
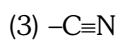
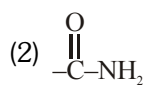
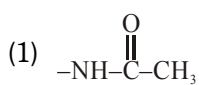
GOLDEN KEY POINTS

The important electrophiles used in the aromatic substitution are the following -

Electrophilic	Source	Name of substitution reaction
Cl^+ (Chloronium)	$\text{Cl}_2 + \text{AlCl}_3$ or FeCl_3	Chlorination
Br^+ (Bromonium)	$\text{Br}_2 + \text{AlBr}_3$ or FeBr_3	Bromination
NO_2^+ (Nitronium)	(conc. $\text{HNO}_3 + \text{H}_2\text{SO}_4$)	Nitration
SO_3 (Sulphurtrioxide)	conc. H_2SO_4 , fuming sulphuric acid	Sulphonation
R^+ (Alkyl carbonium)	$\text{RX} + \text{AlX}_3$ (X= Cl or Br) $\text{ROH} + \text{H}^+$	Friedel crafts (Alkylation)
$\text{R}-\overset{\oplus}{\text{C}}=\text{O}$ (Acyl carbonium)	$\text{RCOCl} + \text{AlCl}_3$	Friedel crafts (Acylation)

BEGINNER'S BOX-5

1. Which of the following group is ortho para director :-



(1) 1

(2) 2

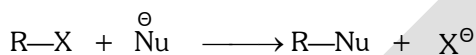
(3) 3

(4) Both (1) and (3)

(C) **Nucleophilic substitution reaction (S_N)** : Due to electronegativity difference the $\text{—}\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{—X}$ bond is polarised bond. $\left(\overset{\delta+}{\text{C}} \text{—} \overset{\delta-}{\text{X}} \right)$

Thus the C-atom of the $\overset{\delta+}{\text{C}}\text{—}\overset{\delta-}{\text{X}}$ bond becomes centre to attack by a nucleophile ($\text{Nu}^\ominus$).

$\text{X}^\ominus$ ion from R—X molecule is substituted by a $\text{Nu}^\ominus$. i.e. S_N reaction are the most common reactions in R—X .



Two mechanisms are observed in S_N reaction :

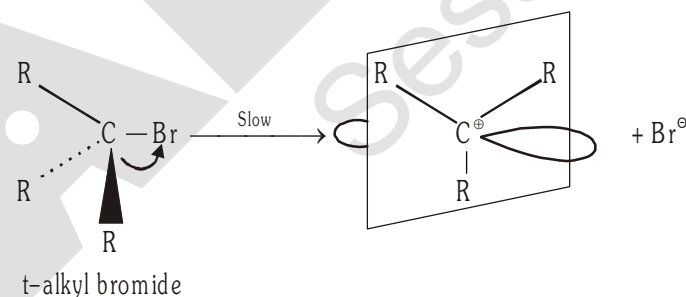
(a) S_{N1} mechanism

(b) S_{N2} mechanism

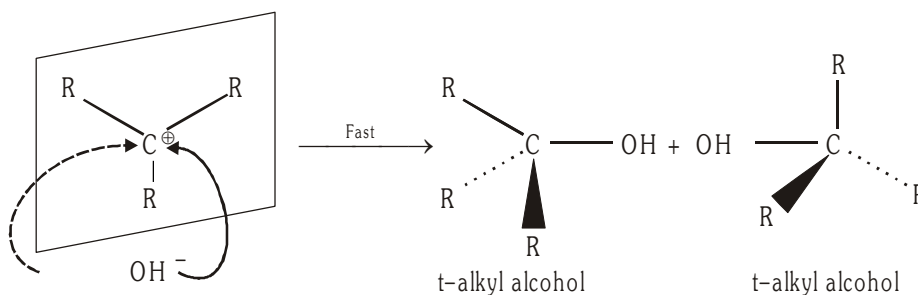
Mechanism of S_{N1} and S_{N2} :

S_{N1} Mechanism : S_{N1} stands for uni molecular nucleophilic substitution. The mechanism involves two steps. Consider the hydrolysis of tert. butyl bromide with aqueous NaOH.

Step 1 : The alkyl halide ionises to give a planar carbonium ion. The carbonium ion is planar because the central positively charged carbon is sp^2 hybridized.



Step-2 : The nucleophile can attack the planar carbonium ion from either side to give the product.



- (i) Ionisation is the rate determining step because it is the slow step. In other words, the rate at which alcohol is formed depends upon the concentration of tertiary alkyl halide alone.

$$\therefore \text{Rate} = K[\text{R}_2\text{C}=\text{Br}]$$

It is obvious that the reaction follows first order kinetics, therefore reaction is called S_N1 .

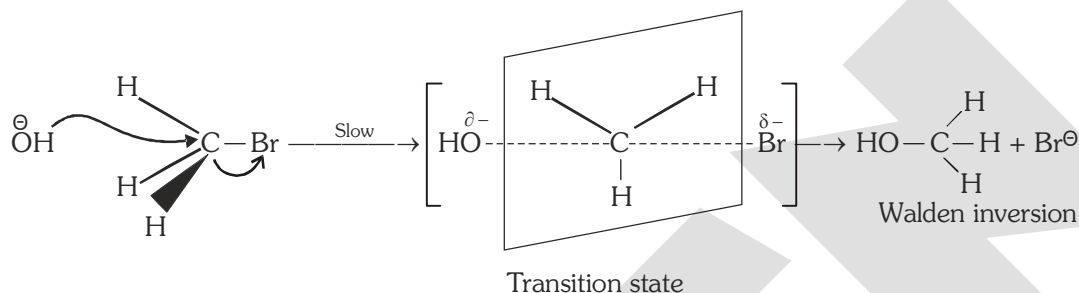
- (ii) The reactivity order for S_N1 reaction $\propto$ stability of carbocations formed by halides.

- ∴ reactivity order of halides (S_N1) varies as follows :

Benzyl halide > Allyl 3° halide > Allyl 2° halide > Allyl 1° halide > 3° halide > 2° halide > 1° halide > methyl halide.

- (iii) Remember that in case alkyl halide is optically active, S_N1 reactions lead to racemisation.

S_N2 mechanism : S_N2 stands for bimolecular nucleophilic substitution. In this type of nucleophilic substitution reaction, bond making and bond breaking process occur simultaneously.



- (i) Reactivity of alkyl halides in S_N2 substitution is governed by steric factors. The bulkier the group, the less reactive it will be.
- (ii) Reactivity order of alkyl halide varies as follows :

***allyl halide* > CH_3X > 1°halide > 2°halide > 3°halide**

- (iii) The order of reactivity among 1° alkyl halides is : $\text{CH}_3\text{X} > \text{C}_2\text{H}_5\text{X} > \text{C}_3\text{H}_7\text{X}$ etc.

Remember that in case alkyl halide is optically active, S_N1 reactions lead to Walden inversion.

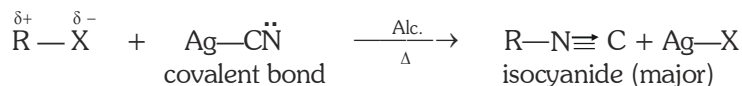
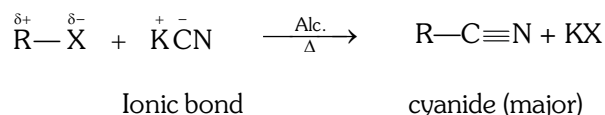
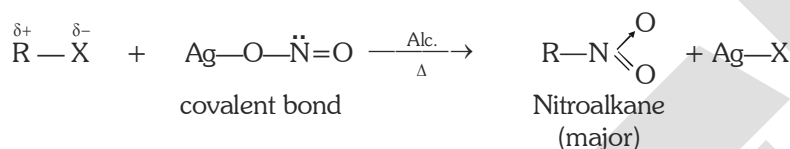
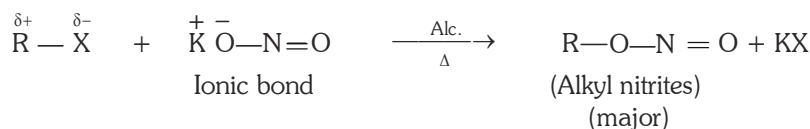
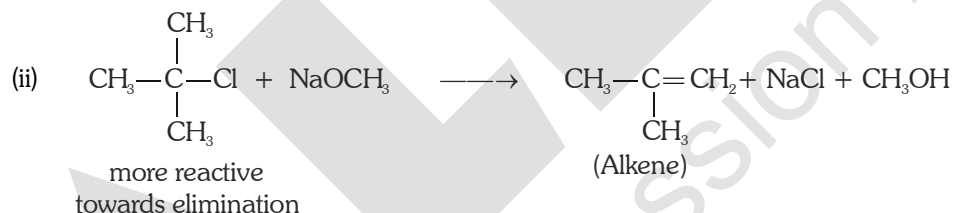
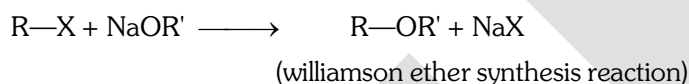
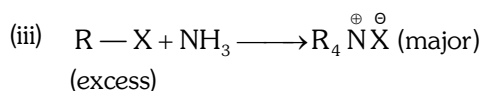
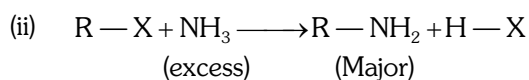
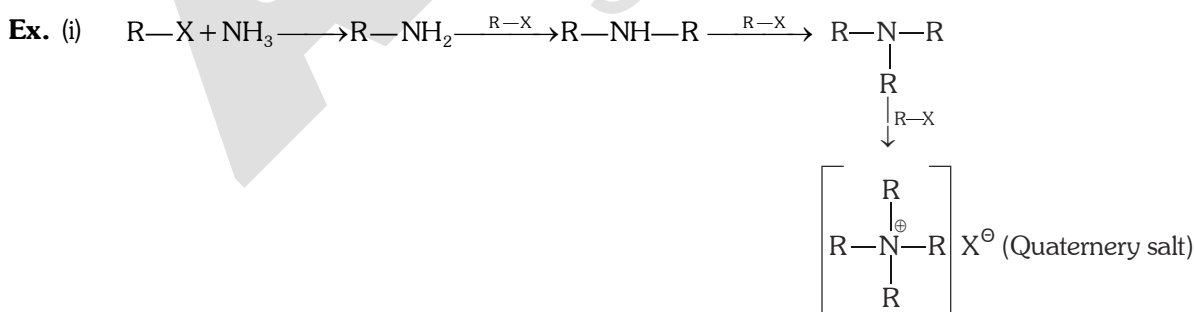
- (iv) For a given alkyl group the order of reactivity is - (for S_N1 and S_N2 both) : **$RI > RBr > RCl > RF$**

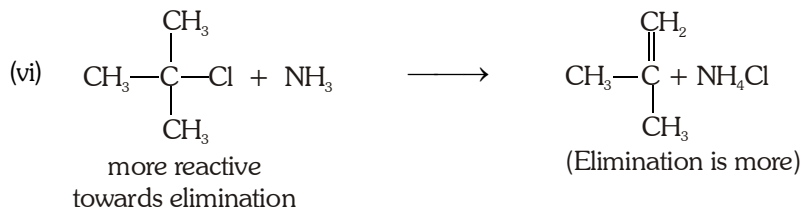
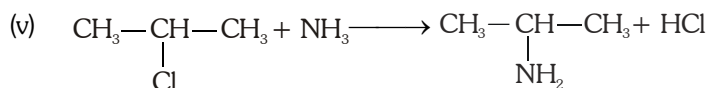
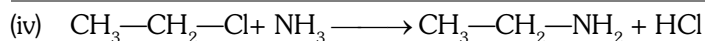
- (v) In addition to substitution reaction alkyl halide also undergo elimination reactions to form alkene with the removal of a molecule of hydrogen halide (dehydrohalogenation). In dehydrohalogenation, hydrogen and halogen atoms are eliminated from two adjacent carbon atoms, the reaction also known as β -elimination it may proceed by E^1 or E^2 mechanism (analogous to S_N1 and S_N2 mechanism).

The order of elimination reaction is : **3° halides** > **2° halides** > **1° halides**

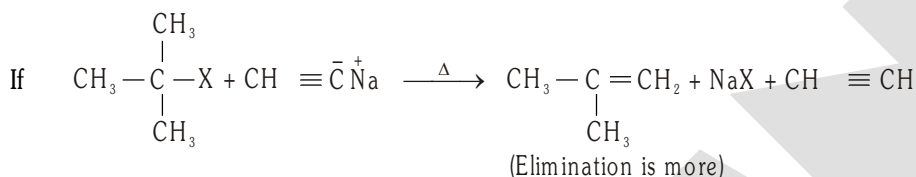
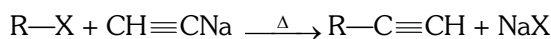
(1) nucleophilic substitution reaction (S_N) in alkyl halide :

	Replacement of	Product
(a)	HOH (Warm)	$\text{R-OH (alcohol) + HX}$
(b)	$\text{KOH aq. } \Delta$	$\text{R-OH (alcohol) + KX}$
(c)	moist AgO_2	$\text{R-OH (alcohol) + AgX}$
(d)	$\text{KSH alc. } \Delta$	R-SH + KX Alkane thiol (Mercaptane)
(e)	NaSR'	RSR' + NaX Thioether
(f)	$\text{R' COOAg } \Delta$	R' COOR + AgX (Alkyl alkanoate) ester

(g) Reaction with KCN and AgCN :**(h) Reaction with KNO₂ and AgNO₂ :****(i) Reaction with NaOR' (Sodium alkoxide) :****(j) Reaction with NH₃ :**

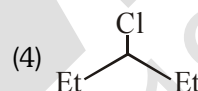
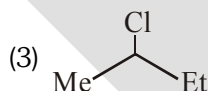
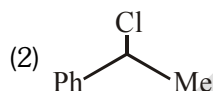
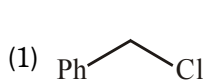


(k) Reaction with $\text{CH}\equiv\text{CNa}$:

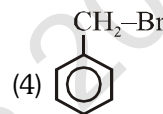
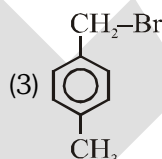
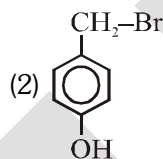
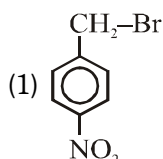


BEGINNER'S BOX-6

1. Which is most reactive for $\text{S}_{\text{N}}1$ reaction :-

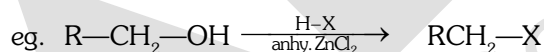


2. Which is most reactive for $\text{S}_{\text{N}}1$?

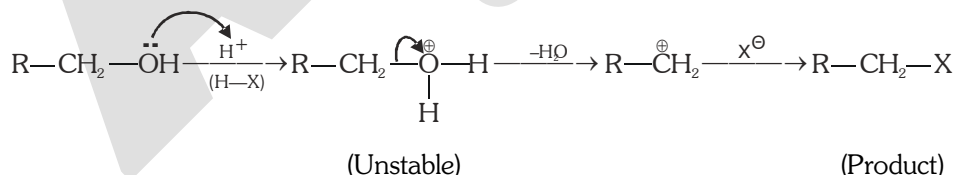


NSR reaction of alcohol :-

(a) Reaction with HX :

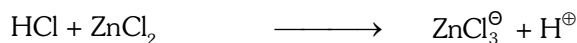


Mechanism :



In this reaction, intermediate carbocation is formed so rearrangement can take place.

anhy. ZnCl_2 act as dehydrating agent and absorbs H_2O from the reaction so good yield of halide is obtained. Also it generates H^+ from HCl .

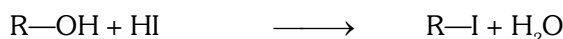


Reactivity order for alcohol :

Reactivity $\propto$ stability of intermediate carbocation, so reactivity order : **Tert. alc.** > **Sec. alc.** > **Pri. alc.**

Reactivity order of $H-X$ is : **$HI > HBr > HCl$**

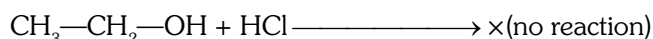
HI is maximum reactive so it reacts readily with 1° , 2° and 3° alcohols.



HCl and also 1° alcohol are less reactive so $ZnCl_2$ or some amount of H_2SO_4 is needed to increase the reactivity.



At normal condition :

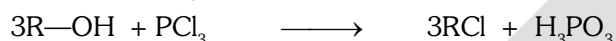
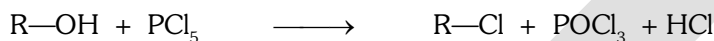


Note : $[HCl_{(conc.)} + ZnCl_{2 (anhydrous)}]$ is called as **lucas reagent**, alcohol gives turbidity with lucas reagent.

Reactivity towards lucas reagent (difference in 1° , 2° and 3° alcohol).

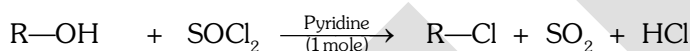
	1° alcohol	2° alcohol	3° alcohol
Time to give turbidity	in 30 min. on heating	in 5 min.	in 2-3 second

(b) Reaction with phosphorus halides :



PBr_3 and PI_3 are less stable, thus for bromides and iodides, $(P + Br_2)$ Or $(P + I_2)$ mixture is used.

(c) Reaction with thionyl chloride - (Darzen's procedure) :



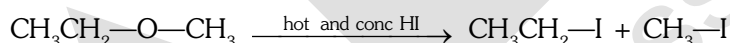
One mole One mole

NSR reaction in ether :-

Reaction with HX : Reactivity of HX **$HI > HBr > HCl$**

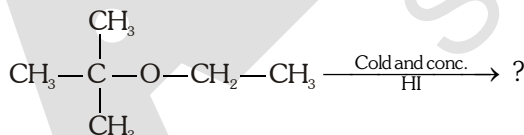
(A) Reaction with cold conc. HX : Ethers forms oxonium salt with cold and conc. HCl (less reactive)
Cold conc. HI and HBr (more reactive) break C-O bond.

(B) Reaction with conc. HI or conc. HBr :



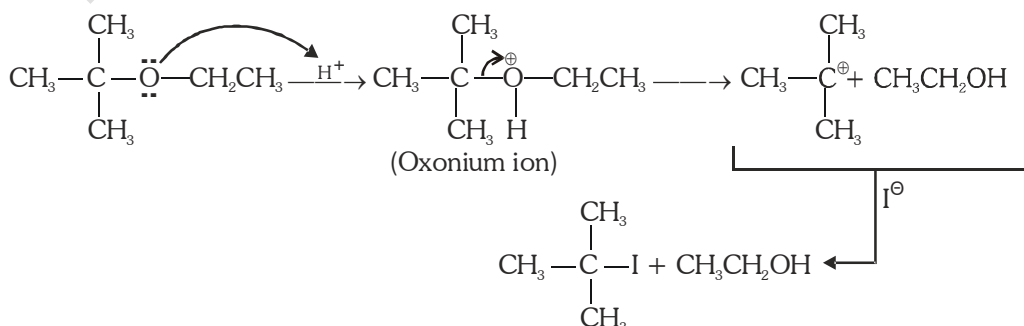
Illustrations

Illustration 3.



Solution.

Mechanism



If oxonium ion gives more stable carbocation [$\text{Ph}\overset{\oplus}{\text{C}}\text{H}_2$, $\text{CH}_2=\text{CH}-\overset{\oplus}{\text{C}}\text{H}_2$, $(\text{CH}_3)_3\overset{\oplus}{\text{C}}$] then $\text{S}_\text{N}1$ reaction occurs.

If oxonium ion gives less stable carbocation [$\text{Ph}^\oplus$, $\text{CH}_2=\overset{\oplus}{\text{C}}\text{H}$, $\text{CH}_3\overset{\oplus}{\text{C}}\text{H}_2$, $(\text{CH}_3)_2\overset{\oplus}{\text{C}}\text{H}$] then $\text{S}_\text{N}2$ reaction occurs, and $\text{X}^\ominus$ attacks at less hindered carbon

Illustration 4. $\text{CH}_3\text{CH}_2-\text{O}-\text{CH}_2\text{Ph} \xrightarrow[\text{HI}]{\text{Cold conc.}} \text{CH}_3\text{CH}_2-\text{OH} + \text{PhCH}_2-\text{I}$, write mechanism of given reaction.

Solution. **Mechanism :**

$$\text{CH}_3\text{CH}_2-\ddot{\text{O}}-\text{CH}_2\text{Ph} \xrightarrow{\text{H}^\oplus} \text{CH}_3\text{CH}_2-\overset{\oplus}{\text{O}}(\text{H})-\text{CH}_2\text{Ph}$$

$$\text{CH}_3\text{CH}_2\text{OH} + \text{Ph}\overset{\oplus}{\text{C}}\text{H}_2 \xrightarrow{\text{I}^\ominus} \text{PhCH}_2\text{I} + \text{CH}_3\text{CH}_2-\text{OH} \quad (\text{S}_\text{N}1)$$

Illustration 5. $\text{CH}_3\text{CH}_2-\text{O}-\text{CH}_3 \xrightarrow[\text{conc. and cold HI}]{\text{conc. and cold HI}} ?$

Solution.

$$\text{CH}_3\text{CH}_2-\ddot{\text{O}}-\text{CH}_3 + \text{H}^\oplus \rightarrow \text{CH}_3\text{CH}_2-\overset{\oplus}{\text{O}}(\text{H})-\text{CH}_3 \xrightarrow{\text{I}^\ominus} \text{CH}_3\text{I} + \text{CH}_3\text{CH}_2\text{OH} \quad (\text{S}_\text{N}2)$$

Oxonium ion gives less stable carbocation
 $\text{S}_\text{N}2$ reaction $\text{I}^\ominus$ attacks at less hinderd carbon.

Illustration 6. $\text{CH}_3-\text{CH}_2-\text{O}-\text{Ph} \xrightarrow[\text{HBr}]{\text{cold and conc.}} ?$

Solution. **Mechanism :**

$$\text{CH}_3-\text{CH}_2-\ddot{\text{O}}-\text{Ph} \xrightarrow{\text{H}^\oplus} \text{CH}_3-\text{CH}_2-\overset{\oplus}{\text{O}}(\text{H})-\text{Ph} \xrightarrow{\text{Br}^\ominus} \text{CH}_3\text{CH}_2\text{Br} + \text{PhOH}$$

Note : If excess of HI/Δ is used then two moles of alkyl hallides are formed.

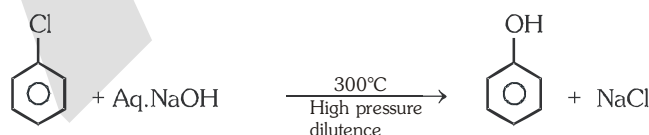


Illustration 7. $\text{C}_2\text{H}_5-\text{O}-\text{C}_2\text{H}_5 \xrightarrow[\text{hot and conc. HBr}]{\text{hot and conc. HBr}} ? + ?$

Solution. $\text{C}_2\text{H}_5-\text{Br} + \text{C}_2\text{H}_5-\text{Br}$

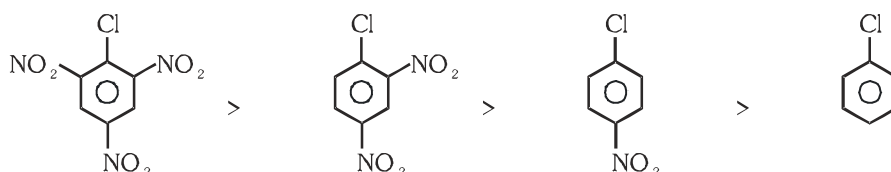
Aromatic nucleophilic substitution :

NSR reaction in halobenzene :-



Presence of deactivating group at ortho and para position makes the nucleophilic substitution easier.

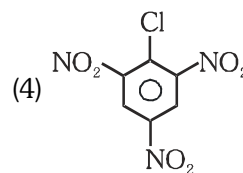
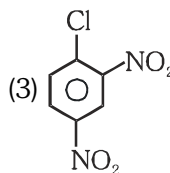
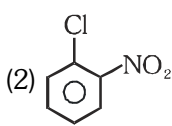
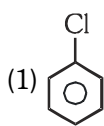
Reactivity Order : (Towards nucleophilic substatitution)



Illustrations

Illustration

Which of the following undergoes Hydrolysis most easily :



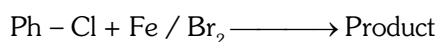
Solution.

If there is more e^- withdrawing groups then there will be more nucleophilic substitution reaction.

Ans. (4)

Illustration

The product in the following reaction is :



(1) o-bromo-chloro benzene

(2) p-bromo-chloro benzene

(3) (1) and (2) both

(4) 2, 4, 6-tribromo chloro benzene

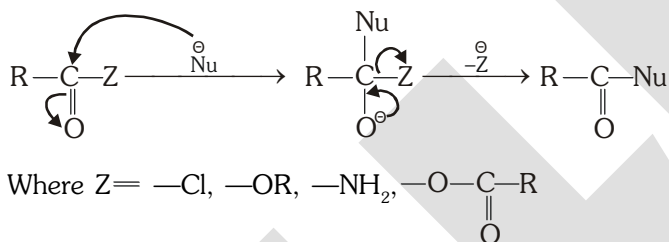
Solution.

Since $-\text{Cl}$ group is deactivating and o/p directing group so o- and p- products are formed.

Ans. (3)

NSR in Acid derivatives :

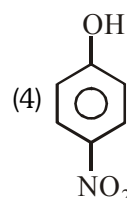
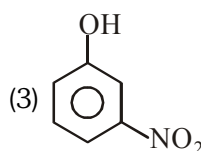
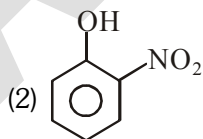
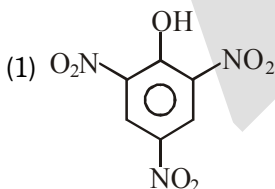
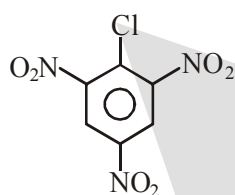
Example : Hydrolysis of acid derivatives :



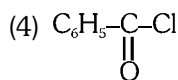
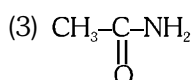
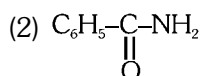
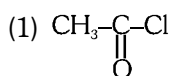
here Z is a good leaving group

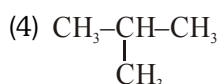
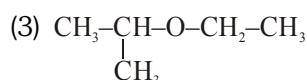
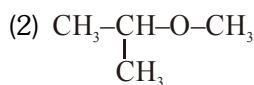
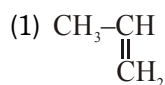
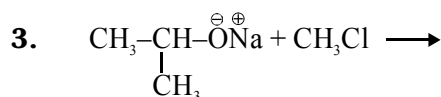
BEGINNER'S BOX-7

1.



2. Which one of the following is least reactive with water





4.3 ELIMINATION REACTIONS :

Elimination

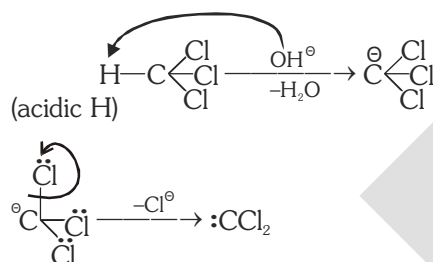
- α -elimination
or 1, 1-elimination

β-elimination
or 1, 2-elimination

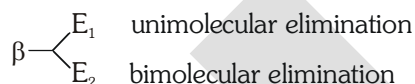
α -Elimination (1, 1-Elimination) : Removal of H and X from one C-atom



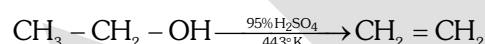
Mechanism :



α , β Elimination (β -elimination) : Removal of H and X from adjacent C-atoms

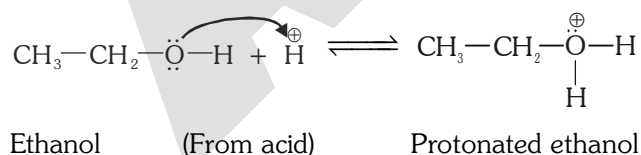


(a) Unimolecular elimination (E_1) :-

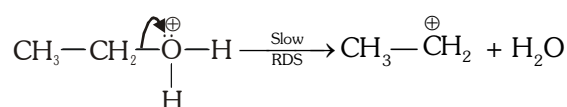


Mechanism of Reaction: The acidic dehydration of alcohol proceeds through the formation of a carbocation intermediate and is explained as follows :

Step I : Alcohol being a Lewis base accepts a proton (H^+) from the acid in a reversible step as follows:



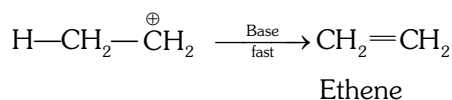
Step II : Due to presence of positive charge on electronegative oxygen, its electron accepting tendency increases. As a result C – O bond becomes weak and cleaves as follows :



Ethyl carbocation

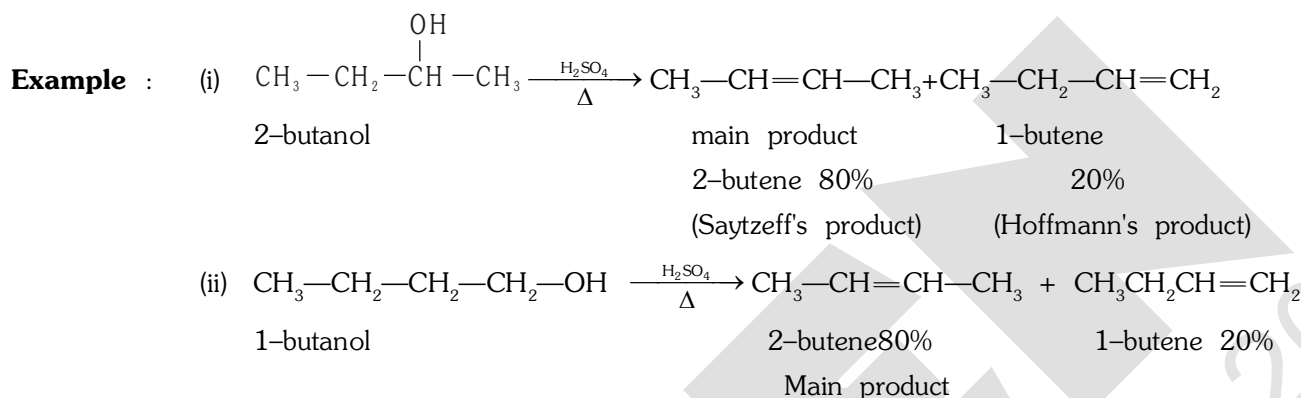
This is a slow and is regarded as **rate determining step** in E₁ reaction.

Step III : Base removes $H\alpha$ (proton) from carbocation and changes it into ethene in a fast step as follows:

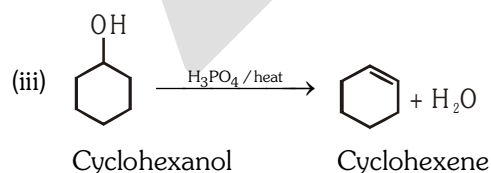
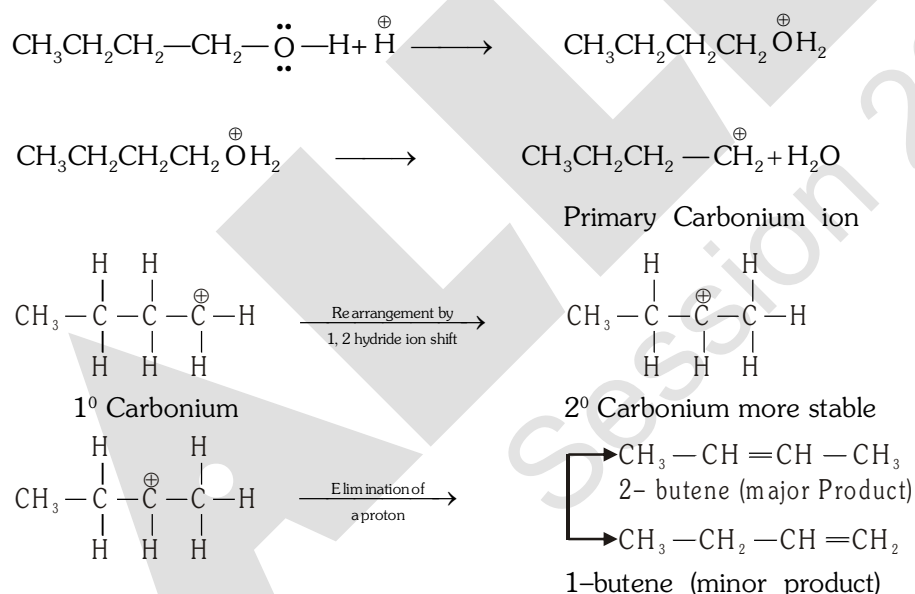


Saytzeff rule : When two possible alkenes are obtained by the elimination reaction then that alkene containing maximum number of alkyl group on double bonded C-atoms is called Saytzeff's product and formed as major product.

Note : The alkene having less number of alkyl groups on double bonded C-atoms is called Hofmann's product.



Mechanism : Acid catalyzed dehydration of alkanols proceeds via the formation of more stable carbonium ion.



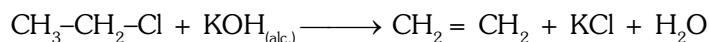
Reactivity order of acidic dehydration of alcohols is : $3^\circ > 2^\circ > 1^\circ R-OH$

- ◆ Rate of reaction $\propto$ [substrate]
- ◆ Molecularity of reaction = 1 (So reaction is called as E_1)
- ◆ In reaction intermediate carbocation is formed, so carbocation rearrangement is possible.

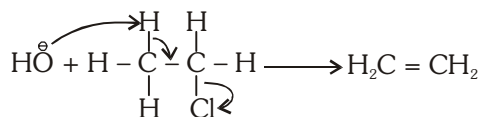
(b) **Bimolecular elimination (E_2) :**

Example :

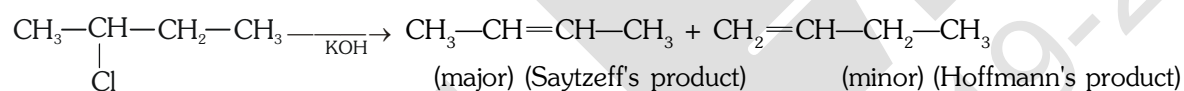
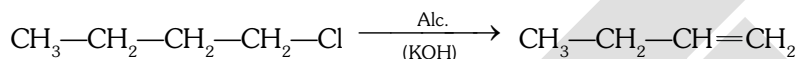
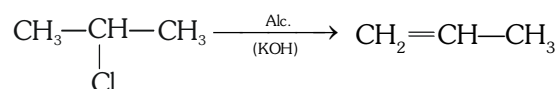
(i) Dehydrohalogenation of halides by alcoholic KOH/ NaNH_2 :



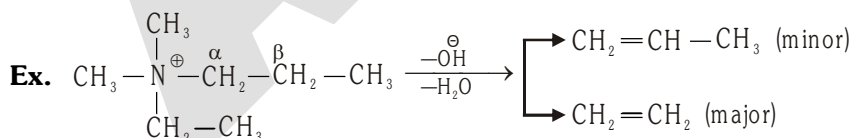
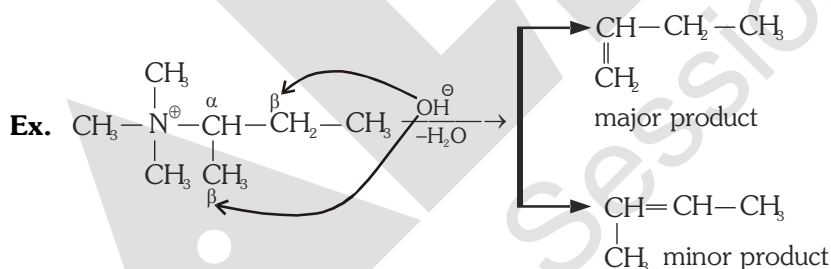
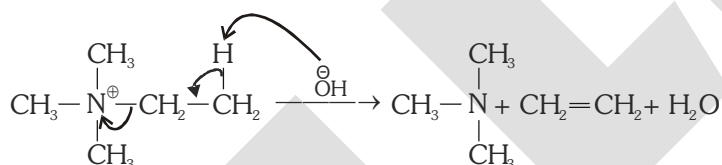
Mechanism :



- ◆ Rate of reaction $\propto$ [substrate] [base]
- ◆ Order of reaction = 2 (So reaction is E₂)
- ◆ In E₂ reaction intermediate (carbocation) is not formed. So there will be no carbocation rearrangement.



(ii) Pyrolysis of tetra alkyl ammonium ion :



Note : Hoffmann's product is formed as major product.

Competition between substitution and elimination reactions

Reactivity order of alkyl halides : E_r Reaction : $1^\circ < 2^\circ < 3^\circ$

E_o- Reaction : 1° < 2° < 3°

$$S_N1\text{-Reaction: } 1^\circ < 2^\circ < 3^\circ$$

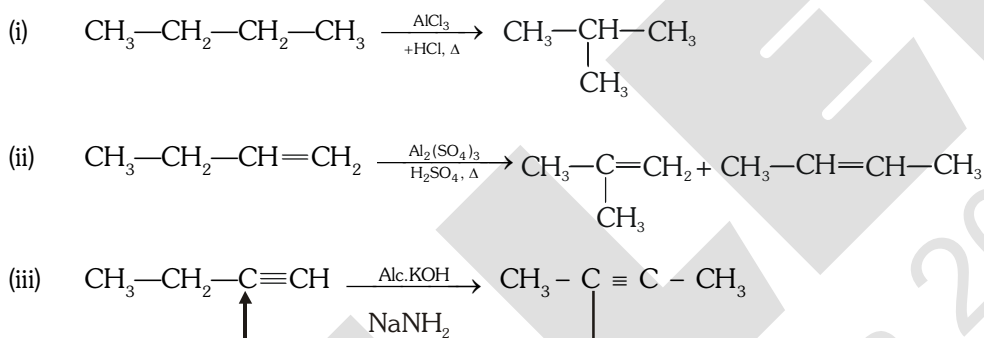
S_N2-Reaction : 1° > 2° > 3°

GOLDEN KEY POINTS

- SN^2/E^2 is favoured by high conc. of good nucleophile or strong base. (CH_3O^- , HO^-)
Rate of Reaction $\propto$ (Substrate) (Reagent)
- SN^1/E^1 is favoured by low conc. of poor nucleophile or weak base (CH_3OH , H_2O)
- If an alkyl halide, undergoes SN^2/SN^1 then SN^2 reaction will be favoured by high conc. of good nucleophile (negatively charged) in presence of polar aprotic solvent whereas SN^1 – reaction is favoured by low conc. of poor nucleophile (neutral) in presence of polar protic solvent.
Polar protic solvent : H_2O , CH_3OH , HCOOH
Polar aprotic solvent : DMSO , CH_3CN , $\text{C}_2\text{H}_5\text{O}-\text{C}_2\text{H}_5$, DMF

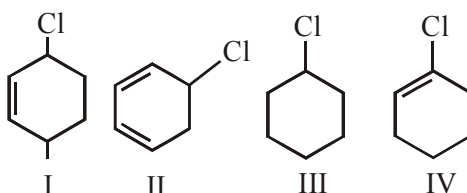
4.4 ISOMERIZATION REACTIONS : These reaction involves the interconversion of one isomer into the another isomer.

For example :



BEGINNER'S BOX-8

- Acidic dehydration of alcohol involves :-
 - E_1 elimination
 - Carbocation rearrangement if possible
 - Saytzeff's product is formed as major product
 - All
- Arrange the following in order of their reactivity toward dehydrohalogenation :-



- (1) $\text{II} > \text{I} > \text{III} > \text{IV}$
- (2) $\text{III} > \text{II} > \text{I} > \text{IV}$
- (3) $\text{IV} > \text{III} > \text{I} > \text{II}$
- (4) $\text{I} > \text{II} > \text{III} > \text{IV}$

S.N.	Class of compounds	Types of reactions
(i)	Alkane	Free radical substitution
(ii)	Alkene, alkyne	Electrophilic addition
(iii)	Alkyl halide	Nucleophilic substitution
(iv)	Aldehyde, ketone	Nucleophilic addition
(v)	Acid and their derivatives	Nucleophilic substitution
(vi)	Aromatic compounds	Electrophilic substitution

ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3
	Ans.	1	1	3

BEGINNER'S BOX-2	Que.	1	2
	Ans.	3	1

BEGINNER'S BOX-3	Que.	1	2	3	4
	Ans.	1	2	2	2

BEGINNER'S BOX-4	Que.	1	2	3	4
	Ans.	3	3	2	4

BEGINNER'S BOX-5	Que.	1	2	
	Ans.	1	4	

BEGINNER'S BOX-6	Que.	1	2	
	Ans.	2	2	

BEGINNER'S BOX-7	Que.	1	2	3	
	Ans.	1	2	2	

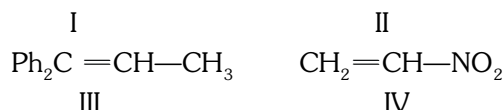
BEGINNER'S BOX-8	Que.	1	2	
	Ans.	4	1	

EXERCISE-I (Conceptual Questions)

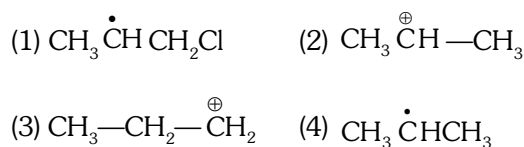
Build Up Your Understanding

ELECTROPHILIC ADDITION REACTION

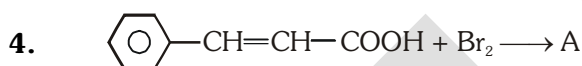
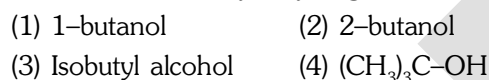
1. Correct reactivity order for EAR of following compounds is



- (1) IV > I > II > III (2) III > II > I > IV
(3) II > III > I > IV (4) II > III > IV > I
2. The intermediate during the addition of HCl to propene in the presence of peroxide is

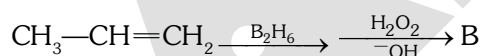
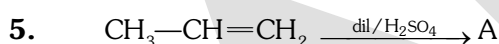


3. Hydroboration of isobutylene followed by oxidative alkaline hydrolysis gives



the number of chiral carbons in 'A' are

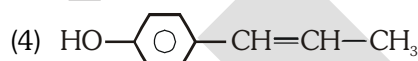
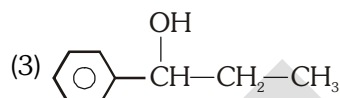
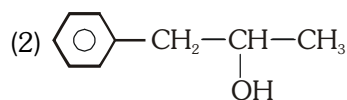
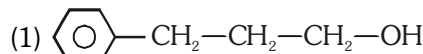
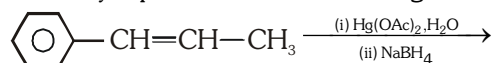
- (1) 1 (2) 2
(3) 3 (4) 4



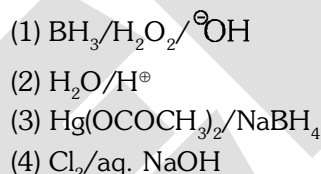
Wrong statement about the product is

- (1) A and B have the same functional group
(2) A and B are position isomers.
(3) A and B show chain isomerism
(4) Mixed ether is the isomer of both A and B
6. Which of the following alkene is most reactive for hydration
- (1) ethene
(2) propene
(3) 1-butene
(4) 2-methyl propene

7. The major product of the following reaction is



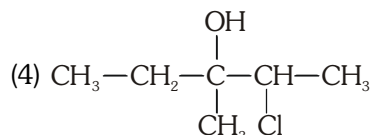
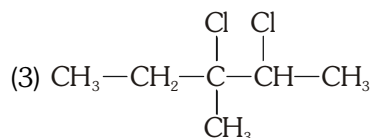
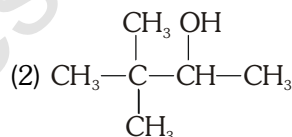
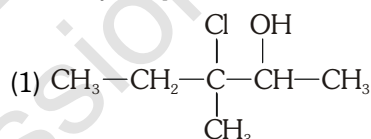
8. $\text{Cyclohexene}-\text{CH}_3 \xrightarrow{\text{A}} \text{Cyclohexane}-\text{CH}_2-\text{CH}_2-\text{OH}$; Reagent 'A' is



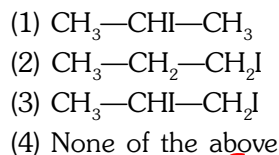
9. Which of the following alkenes on hydration gives a tertiary alcohol



10. The predominant product formed when 3-methyl-2-pentene reacts with HOCl is



11. Propene on addition with HI, gives



12. What is the main product of this reaction?



- (1) $\text{CH}_3 - \overset{\text{Cl}}{\underset{|}{\text{C}}} = \text{CH}_2$
 (2) $\text{CH}_3 - \underset{\text{Cl}}{\underset{|}{\text{CH}}} - \underset{\text{Cl}}{\underset{|}{\text{CH}_2}}$
 (3) $\text{CH}_3 - \text{CH}_2 - \text{CH} \begin{matrix} \text{Cl} \\ \diagup \\ \text{Cl} \end{matrix}$
 (4) $\text{CH}_3 - \overset{\text{Cl}}{\underset{\text{Cl}}{\underset{|}{\text{C}}}} - \text{CH}_3$

13. Propan-1-ol can be prepared from propene by

- (1) $\text{H}_2\text{O}/\text{H}_2\text{SO}_4$
 (2) $\text{Hg}(\text{OAc})_2/\text{H}_2\text{O}_2$ followed by NaBH_4 .
 (3) B_2H_6 followed by H_2O_2 .
 (4) $\text{CH}_3\text{CO}_2\text{H}/\text{H}_2\text{SO}_4$

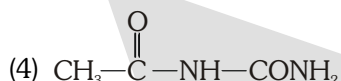
14. 3-Phenyl propene on reaction with HBr gives (as a major product)

- (1) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$
 (2) $\text{C}_6\text{H}_5\text{CH}(\text{Br})\text{CH}_2\text{CH}_3$
 (3) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
 (4) $\text{C}_6\text{H}_5\text{CH}(\text{Br})\text{CH}=\text{CH}_2$

NUCLEOPHILIC ADDITION REACTION

15. Acetaldehyde reacts with semicarbazide, product will be :

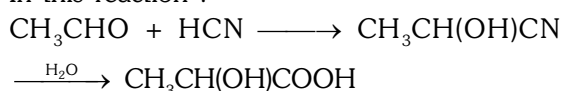
- (1) $\text{CH}_3\text{CH}=\text{NNH}-\text{CO}-\text{NH}_2$
 (2) $\text{CH}_3\text{CH}=\text{NCONHNH}_2$
 (3) $\text{CH}_3\text{CH}=\text{NHNH}_2$



16. Cyanohydrin of the following compound on hydrolysis gives compound that can show optical isomerism :

- (1) HCHO (2) CH_3CHO
 (3) CH_3COCH_3 (4) All the above

17. In this reaction :



an asymmetric centre is generated. The acid obtained would be :-

- (1) D-isomer
 (2) L-isomer
 (3) 50% D + 50% L-isomer
 (4) 20% D + 80% L-isomer

18. When two mole of alcohol reacts with 1 mole of ketone, it gives :-

- (1) Hemiacetal (2) Acetal
 (3) Hemiketal (4) Ketal

19. When acetone reacts with Grignard reagent followed by hydrolysis, it gives :

- (1) 1°-alcohol (2) 2°-alcohol
 (3) 3°-alcohol (4) Methyl alcohol

FREE RADICAL ADDITION REACTION

20. Reaction of HBr with propene in the presence of peroxide gives

- (1) 3-bromo propane (2) Allyl bromide
 (3) n-propyl bromide (4) Isopropyl bromide

21. Isobutene $\xrightarrow[\text{H}_2\text{O}_2]{\text{HBr}}$ "product". The product is

- (1) Isobutyl bromide (2) Tert. butyl bromide
 (3) Tert. butyl alcohol (4) isobutyl alcohol

FREE RADICAL SUBSTITUTION REACTION

22. Alkane reacts with which of the following halogens in dark

- (1) F_2 (2) Cl_2 (3) I_2 (4) Br_2

23. The nitrating agent for the nitration of alkanes is:

- (1) Conc. HNO_3
 (2) Mixture of conc. HNO_3 and conc. H_2SO_4
 (3) Acetyl nitrate
 (4) HNO_3 vapours at high temperature

24. The chain propagating step is fastest in the reaction of an alkane with

- (1) Fluorine free radical (2) Chlorine free radical
 (3) Iodine free radical (4) Bromine free radical

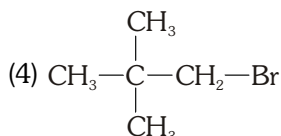
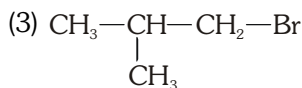
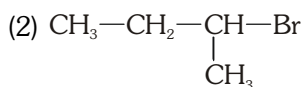
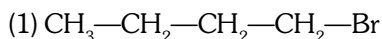
25. In the nitration of propane, the product obtained in maximum yield is

- (1) 1-nitropropane (2) 2-nitropropane
 (3) Nitroethane (4) Nitromethane

26. Only two isomeric monochloro derivatives are possible for (exclude stereo isomers)

- (1) n-butane
 (2) 2, 4-dimethyl pentane
 (3) benzene
 (4) 2-methyl butane

27. What is the chief product obtained when n-butane is treated with bromine in the presence of light at 130°C ?

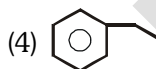
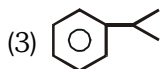
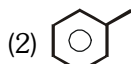
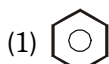


ELECTROPHILIC SUBSTITUTION REACTION

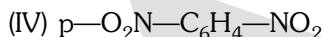
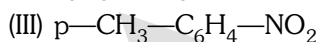
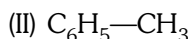
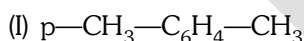
28. The strongest deactivating effect on aromatic ring is

- (1) $-\text{CH}_2\text{Cl}$ (2) $-\text{OCH}_3$
(3) $-\text{CH}_3$ (4) $-\text{CCl}_3$

29. Which of the following is maximum reactive towards E.S.R. :-



30. Correct order of reactivity of following compound with an electrophile :-



- (1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{II} > \text{I} > \text{IV} > \text{III}$
(3) $\text{III} > \text{II} > \text{I} > \text{IV}$ (4) $\text{IV} > \text{III} > \text{II} > \text{I}$

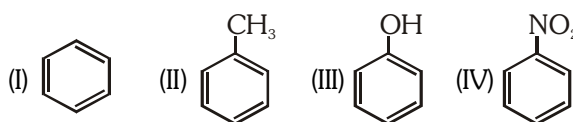
31. Toluene is more reactive than benzene towards electrophilic reagents due to :-

- (1) Inductive effect only
(2) Hyperconjugative effect only
(3) Both inductive as well as hyperconjugative effects
(4) Strong mesomeric effect

32. Nitration of benzene is

- (1) nucleophilic substitution
(2) nucleophilic addition
(3) electrophilic substitution
(4) electrophilic addition

33. Consider the following compounds :



Correct order of their reactivity in electrophilic substitution reactions would be :-

- (1) $\text{I} > \text{II} > \text{III} > \text{IV}$ (2) $\text{IV} > \text{III} > \text{II} > \text{I}$
(3) $\text{III} > \text{II} > \text{I} > \text{IV}$ (4) $\text{III} > \text{IV} > \text{I} > \text{II}$

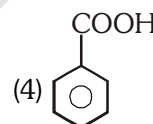
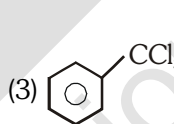
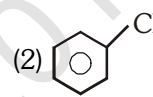
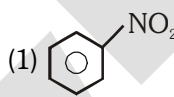
34. The active species in the nitration of benzene is

- (1) NO_2^+ (2) HNO_3
(3) NO_3^- (4) NO_2^-

35. The function of anhydrous AlCl_3 in the Friedel craft's reaction

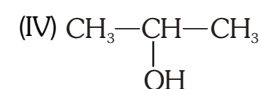
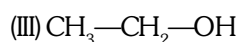
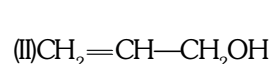
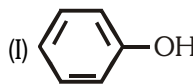
- (1) To absorb water
(2) To absorb HCl
(3) To produce electrophile
(4) To produce Nucleophile

36. In which of the following compound the electrophile attack on o- and p- positions :



NUCLEOPHILIC SUBSTITUTION REACTION

37. The correct reactivity order of alcohols towards H-X will be



- (1) $\text{II} > \text{I} > \text{III} > \text{IV}$ (2) $\text{IV} > \text{III} > \text{II} > \text{I}$
(3) $\text{II} > \text{IV} > \text{I} > \text{III}$ (4) $\text{II} > \text{IV} > \text{III} > \text{I}$

38. Which of the following product will be obtained when neopentyl alcohol is treated with conc. HCl in presence of ZnCl_2 .

- (1) t-butyl chloride (2) isobutylene
(3) t-pentyl chloride (4) Neo pentyl chloride

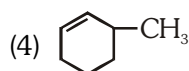
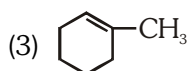
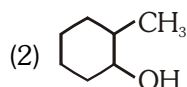
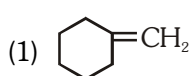
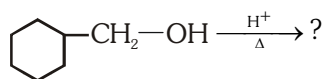
39. In S_N1 the first step involves the formation of
 (1) free radical (2) carbanion
 (3) carbocation (4) final product
40. To form alkane isonitrile, alkyl halide is reacted with:
 (1) KCN (2) AgCN
 (3) NaCN (4) NH_4CN
41. The most reactive towards S_N1 is
 (1) $PhCH_2Cl$ (2) $PhCl$
 (3) $PhCHCl(CH_3)$ (4) $p-NO_2C_6H_4CH_2Cl$
42. Alkyl fluorides are synthesised by
 (1) Finkelstein reaction (2) Swart reaction
 (3) Kolbe reaction (4) Wurtz reaction
43. The products of reaction of alcoholic silver nitrite with ethyl bromide are
 (1) Ethane (2) Ethene
 (3) Ethyl alcohol (4) Nitro ethane
44. For the reaction,
 $C_2H_5OH + HX \xrightarrow{ZnX_2} C_2H_5X$, the order of reactivity is
 (1) $HI > HCl > HBr$
 (2) $HI > HBr > HCl$
 (3) $HCl > HBr > HI$
 (4) $HBr > HI > HCl$
45. The reaction, $CH_3Br + OH^- \longrightarrow CH_3OH + Br^-$ obeys the mechanism
 (1) S_N1 (2) S_N2
 (3) E^1 (4) E^2
46. Butanenitrile may be prepared by heating
 (1) Propyl alcohol with KCN
 (2) Butyl alcohol with KCN
 (3) Butyl chloride with KCN
 (4) Propyl chloride with KCN
47. The given reaction is an example of
 $C_2H_5Br + KCN(aq.) \longrightarrow C_2H_5CN + KBr$
 (1) Elimination
 (2) Nucleophilic substitution
 (3) Electrophilic substitution
 (4) Redox change
48. Most readily hydrolysed halide is
 (1) C_6H_5Cl (2) $(C_6H_5)_2CHCl$
 (3) $C_6H_5CH_2Cl$ (4) $(C_6H_5)_3CCl$
49. An alkyl halide may be converted in to an alcohol by
 (1) Addition (2) Substitution
 (3) Dehydrohalogenation (4) Elimination
50. Compound is most reactive towards NaOH in
 (1) CH_3Cl (2) $CH_2=CHCl$
 (3) C_6H_5Cl (4) $C_6H_5CH_2Cl$
51. Most stable carbocation formed from $(CH_3)_3C-Br$, $(C_6H_5)_3CBr$, $(C_6H_5)_2CHBr$ and $C_6H_5CH_2Br$ would be
 (1) $C_6H_5^+CH_2$ (2) $(CH_3)_3C^+$
 (3) $(C_6H_5)_3C^+$ (4) $(C_6H_5)_2CH^+$
52. S_N1 reaction on optically active substrates mainly gives
 (1) Retention in configuration
 (2) Inversion in configuration
 (3) Racemise product
 (4) No product
53. The hydrolysis of alkyl halides by aqueous NaOH is best termed as
 (1) electrophilic substitution reaction
 (2) electrophilic addition reaction
 (3) nucleophilic addition reaction
 (4) nucleophilic substitution reaction
54. Reaction of sodium ethoxide and ethyl iodide will give :-
 (1) Ether (2) Ethyl alcohol
 (3) Acetaldehyde (4) Acetic acid
55. The least reactive chlorine is present in
 (1) Methyl chloride (2) Allyl chloride
 (3) Ethyl chloride (4) Vinyl chloride
56. Arrange the following compounds in the increasing order of their S_N2 reactivity?
- $\begin{array}{c} CH_3 \\ | \\ CH_3-C-X \\ | \\ CH_3 \end{array}$ (a)
 $\begin{array}{c} CH_3 \\ | \\ CH_3-CH-X \\ | \\ CH_3 \end{array}$ (b)
 CH_3-CH_2-X (c)
 CH_3-X (d)
- (1) (a) < (b) < (c) < (d)
 (2) (a) < (c) < (d) < (b)
 (3) (d) < (c) < (b) < (a)
 (4) (b) < (d) < (c) < (a)

57. Which alcohol produces turbidity with Lucas reagent most slowly
 (1) 2-Butanol (2) t-Butyl alcohol
 (3) Isobutyl alcohol (4) Diphenylcarbinol
58. The preparation of ethers from alcohols by using sulphuric acid is called :-
 (1) Williamson's ether Synthesis
 (2) Williamson's continuous etherification process
 (3) Ziesel's method
 (4) Zerewitinoff method
59. The reaction of ethyl iodide with sodium ethoxide is
 (1) An electrophilic substitution reaction
 (2) A nucleophilic addition reaction
 (3) A nucleophilic substitution reaction
 (4) A free radical substitution reaction
60. The Williamson synthesis involves :-
 (1) A nucleophilic addition
 (2) An electrophilic substitution
 (3) SN^2 displacement
 (4) SN^1 displacement
61. In the Williamson synthesis of ethers given by the general equation -
 $R-X + R'ONa \longrightarrow R-O-R'$ the yield from $R-X$ follows the sequence :-
 (1) $CH_3X > 1^\circ > 2^\circ > 3^\circ$
 (2) $CH_3X < 1^\circ < 2^\circ < 3^\circ$
 (3) $CH_3X < 1^\circ < 2^\circ > 3^\circ$
 (4) $CH_3X > 1^\circ < 2^\circ < 3^\circ$
62. With conc. HBr ethyl phenyl ether yields :-
 (1) Phenol and ethyl bromide.
 (2) Bromobenzene and ethanol
 (3) Phenol and ethane
 (4) Bromobenzene and ethane
63. An unknown alcohol is treated with the "Lucas reagent" to determine whether the alcohol is primary, secondary or tertiary. Which alcohol reacts fastest and by what mechanism :-
 (1) secondary alcohol by $SN1$
 (2) tertiary alcohol by $SN1$
 (3) secondary alcohol by $SN2$
 (4) tertiary alcohol by $SN2$

ELIMINATION REACTION

64. The reactivity of alkyl halides in E^2 elimination reactions follows the order
 (1) $R-I < R-Br < R-Cl < R-F$
 (2) $R-F < R-Cl < R-Br < R-I$
 (3) $R-I > R-Cl > R-Br < R-F$
 (4) $R-I < R-Br < R-F < R-Cl$
65. The unimolecular elimination involves formation of
 (1) A free radical (2) A carbanion
 (3) A carbocation (4) A biradical
66. Which of the following alkyl bromides will eliminate HBr fastest
 (1) Ethyl bromide (2) Propyl bromide
 (3) Isopropyl bromide (4) t-Butyl bromide
67. 1-phenyl-2-chloropropane on treating with alc. KOH gives mainly :
 (1) 1-phenylpropene
 (2) 2-phenylpropene
 (3) 1-phenylpropan-2-ol
 (4) 1-phenylpropan-1-ol
68. Which of the following alkyl halides gives a mixture of alkenes on dehydrohalogenation
 (1) n-Propyl halide
 (2) Isopropyl halide
 (3) s-Butyl bromide
 (4) t-Butyl bromide
69. Arrange the following alkanols A, B and C in order of their reactivity towards acid catalysed dehydration:-
 (A) $CH_3-\underset{\substack{| \\ CH_3}}{CH}-CH_2-CH_2-OH$
 (B) $CH_3-\overset{\substack{OH \\ |}}{C}-CH_2-CH_3$
 (C) $CH_3-\overset{\substack{CH_3 \\ |}}{CH}-\overset{\substack{OH \\ |}}{CH}-CH_3$
 (1) $A > B > C$ (2) $B > A > C$
 (3) $B > C > A$ (4) $C > B > A$

70. The major product in the following reaction is



71. The major product obtained from the heating of 3,3-dimethyl-1-butanol with H_2SO_4 is

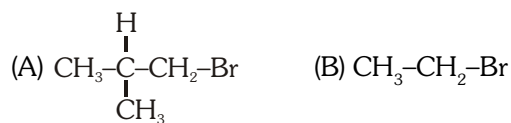
(1) 3,3-dimethyl-1-butene

(2) 2,3-dimethyl-2-butene

(3) 2,3-dimethyl-1-butene

(4) cis and trans isomers of product (2)

72. Arrange the following alkyl halides in decreasing order of the rate of elimination reaction with alcoholic KOH.



(1) $\text{A} > \text{B} > \text{C}$

(2) $\text{C} > \text{B} > \text{A}$

(3) $\text{B} > \text{C} > \text{A}$

(4) $\text{A} > \text{C} > \text{B}$

EXERCISE-I (Conceptual Questions)

ANSWER KEY

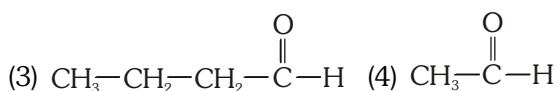
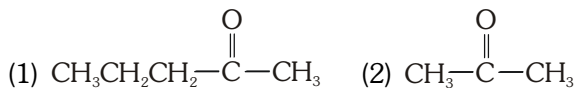
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Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	3	4	3	3	1	1	4	1	2	1	2	4	2	1
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	3	3	1	3	2	4	3	3	2	3	2	4	2	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	2	4	2	4	3	3	4	1	4	1	3	2	3	3
Que.	61	62	63	64	65	66	67	68	69	70	71	72			
Ans.	1	1	2	2	3	4	1	3	3	3	2	4			

EXERCISE-II (Previous Year Questions)

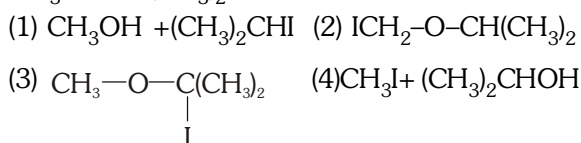
AIPMT/NEET & AIIMS (2006-2018)

AIPMT 2006

1. Nucleophilic addition reaction will be most favoured in :



2. The major organic products of reaction are $\text{CH}_3\text{O}-\text{CH}(\text{CH}_3)_2 + \text{HI} \longrightarrow$

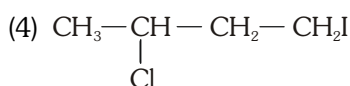
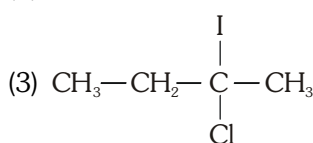
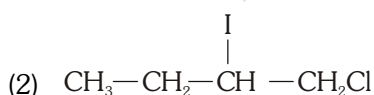
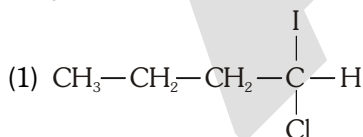


AIIMS 2006

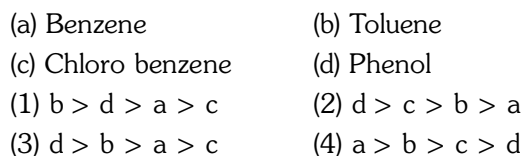
3. The correct increasing order of the reactivity of halides for $\text{S}_{\text{N}}1$ reaction is -
- (1) $\text{CH}_3-\text{CH}_2-\text{X} < (\text{CH}_3)_2\text{CH}-\text{X} < \text{CH}_2=\text{CH}-\text{CH}_2-\text{X} < \text{PhCH}_2-\text{X}$
 (2) $(\text{CH}_3)_2\text{CH}-\text{X} < \text{CH}_3-\text{CH}_2-\text{X} < \text{CH}_2=\text{CH}-\text{CH}_2\text{X} < \text{PhCH}_2-\text{X}$
 (3) $\text{PhCH}_2-\text{X} < (\text{CH}_3)_2\text{CH}-\text{X} < \text{CH}_3-\text{CH}_2-\text{X} < \text{CH}_2=\text{CH}-\text{CH}_2-\text{X}$
 (4) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{X} < \text{Ph}-\text{CH}_2-\text{X} < (\text{CH}_3)_2\text{CH}-\text{X} < \text{CH}_3-\text{CH}_2-\text{X}$

AIPMT 2007

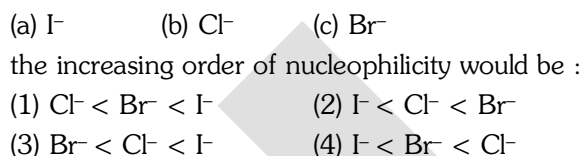
4. Predict the product 'C' obtained in the following reaction of 1-butyne :



5. The order of decreasing reactivity towards electrophilic reagent for the following :

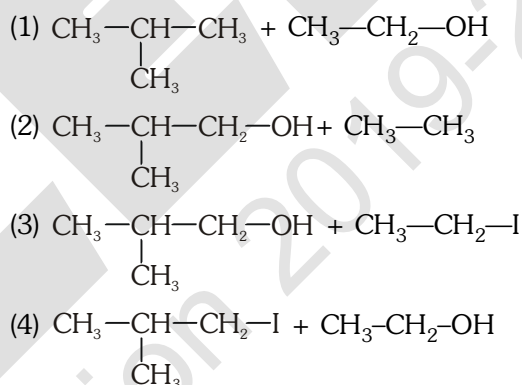


6. For the following :



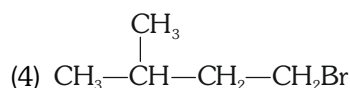
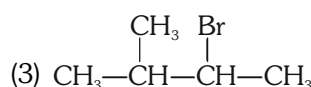
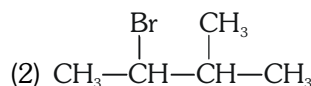
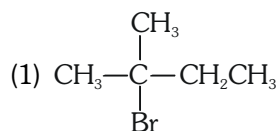
7. In the reaction : $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_3 + \text{HI} \longrightarrow \dots\dots\dots$

which of following compounds will be formed :

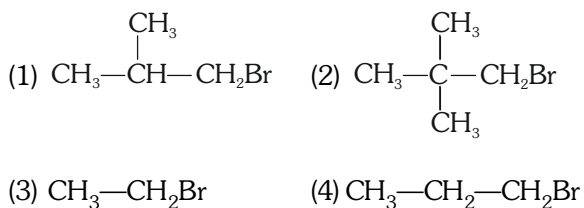


AIPMT 2008

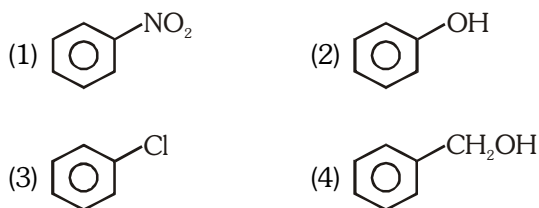
8. $\text{H}_3\text{C}-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}=\text{CH}_2 + \text{HBr} \longrightarrow \text{A}$
 A (Predominantly) is :



9. In a S_N2 substitution reaction of the type $R-Br + Cl^- \xrightarrow{DMF} R-Cl + Br^-$, which one of the following has the highest relative rate ?



10. Which one of the following is most reactive towards electrophilic attack ?



11. The relative reactivities of acyl compounds towards nucleophilic substitution are in the order of :-
- (1) Acid anhydride > Amide > Ester > Acyl chloride
 (2) Acyl chloride > Ester > Acid anhydride > Amide
 (3) Acyl chloride > Acid anhydride > Ester > Amide
 (4) Ester > Acyl chloride > Amide > Acid anhydride

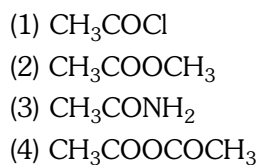
AIPMT 2009

12. Benzene reacts with CH_3Cl in the presence of anhydrous $AlCl_3$ to form :-
- (1) Xylene (2) Toluene
 (3) Chlorobenzene (4) Benzylchloride
13. Nitrobenzene can be prepared from benzene by using a mixture of conc. HNO_3 and conc. H_2SO_4 . In the mixture, nitric acid acts as a/an :-
- (1) Catalyst (2) Reducing agent
 (3) Acid (4) Base
14. Which of the following reactions is an example of nucleophilic substitution reaction ?
- (1) $RX + Mg \longrightarrow RMgX$
 (2) $RX + KOH \longrightarrow ROH + KX$
 (3) $2RX + 2Na \longrightarrow R-R + 2NaX$
 (4) $RX + H_2 \longrightarrow RH + HX$

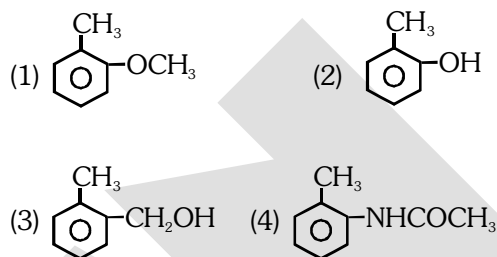
AIPMT 2010

15. Which one is most reactive towards S_N1 reaction?
- (1) $C_6H_5CH_2Br$ (2) $C_6H_5CH(C_6H_5)Br$
 (3) $C_6H_5CH(CH_3)Br$ (4) $C_6H_5C(CH_3)(C_6H_5)Br$

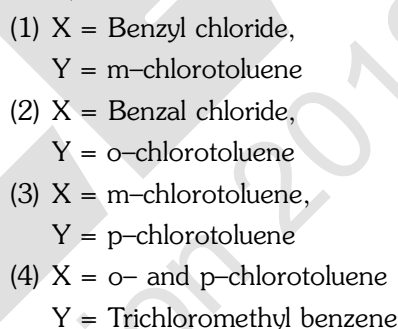
16. Among the given compounds, the most susceptible to nucleophilic attack at the carbonyl group is :-



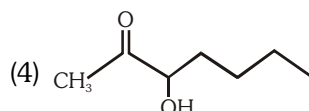
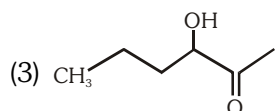
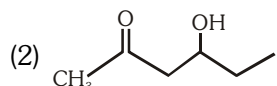
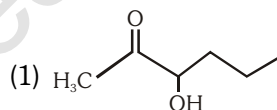
17. Which one is most reactive towards electrophilic reagent ?



18. The reaction of toluene with Cl_2 in presence of $FeCl_3$ gives 'X' and reaction in presence of light gives 'Y'. Thus, 'X' and 'Y' are :-



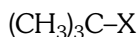
19. Which one of the following compounds will be most readily dehydrated :-



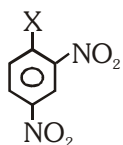
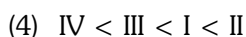
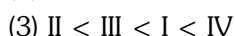
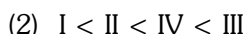
20. The correct order of increasing reactivity of C-X bond towards nucleophile in the following compounds is :-



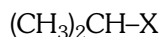
(I)



(III)



(II)



(IV)

AIIMS 2010

21. 2-Butene $\xrightarrow{\text{HBr}}$

Major products of above reaction will be :-

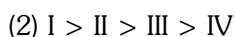
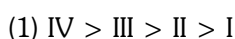
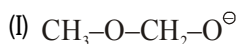
(1) 1

(2) 2

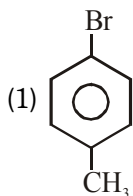
(3) 4

(4) 3

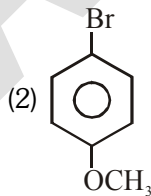
22. Arrange the following in correct order of nucleophilicity :-



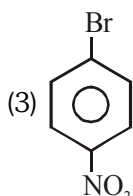
23. Which of the following is maximum reactive towards SN¹ reaction :-



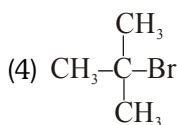
(1)



(2)



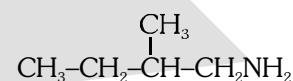
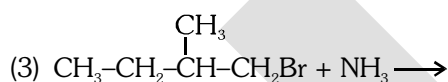
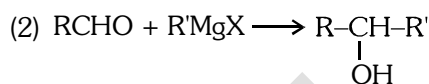
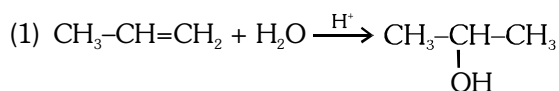
(3)



(4)

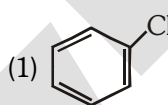
AIPMT Pre-2011

24. Which one is a nucleophilic substitution reaction among the following ?

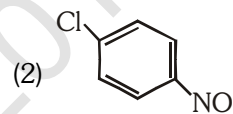


AIPMT Mains-2011

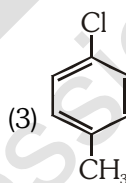
25. Which of the following compounds undergoes nucleophilic substitution reaction most easily ?



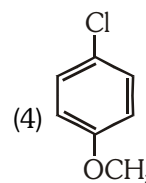
(1)



(2)

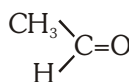


(3)

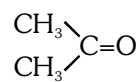


(4)

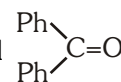
26. The order of reactivity of phenyl magnesium bromide (PhMgBr) with the following compounds :-



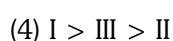
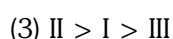
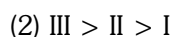
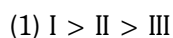
(I)



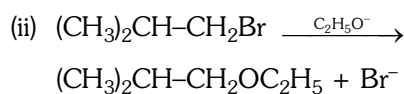
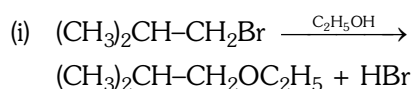
(II)



(III)



27. Consider the reaction :

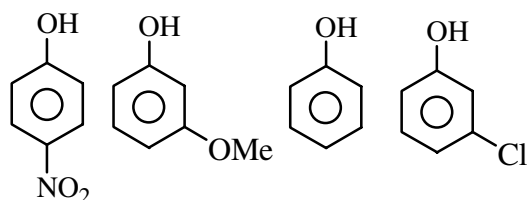


The mechanisms of reaction (i) and (ii) are respectively :-

- (1) S_{N^2} and S_{N^1} (2) S_{N^1} and S_{N^2}
(3) S_{N^1} and S_{N^1} (4) S_{N^2} and S_{N^2}

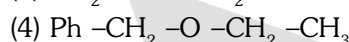
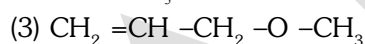
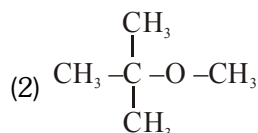
AIIMS 2011

28. The correct increasing order of reactivity for following molecules towards electrophilic aromatic substitution.



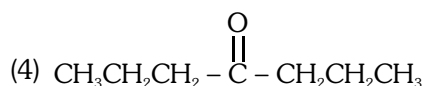
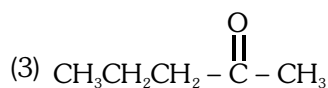
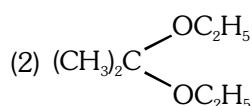
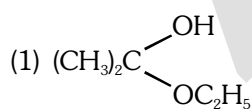
- | | | | |
|-------------------------|------|-------------------------|------|
| (I) | (II) | (III) | (IV) |
| (1) $I < IV < II < III$ | | (2) $I < IV < III < II$ | |
| (3) $I < III < II < IV$ | | (4) $I < III < IV < II$ | |

29. S_N2 reaction readily occurs in:—

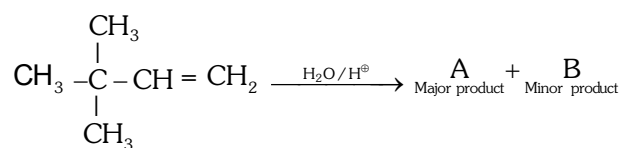


AIPMT Pre.-2012

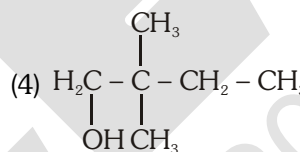
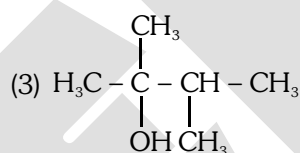
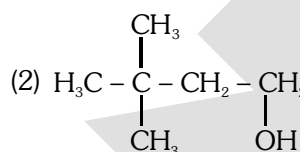
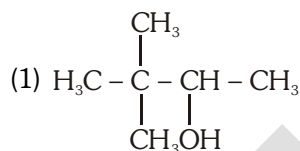
30. Acetone is treated with excess of ethanol in the presence of hydrochloric acid. The product obtained is:



31. In the following reaction:



The major product is :-



32. Among the following compounds the one that is most reactive towards electrophilic nitration is:

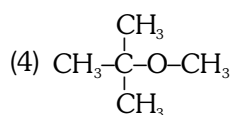
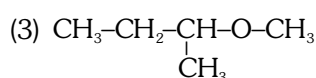
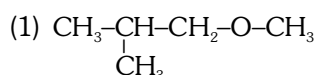
- (1) Toluene (2) Benzene
(3) Benzoic Acid (4) Nitrobenzene

NEET-UG 2013

33. Which of the following compounds will not undergo Friedel-Craft's reaction easily :-

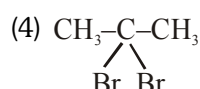
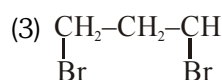
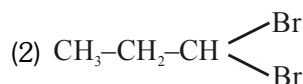
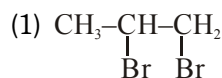
- (1) Toluene (2) Cumene
(3) Xylene (4) Nitrobenzene

34. Among the following ethers, which one will produce methyl alcohol on treatment with hot concentrated HI ?

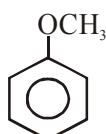


AIIMS 2014

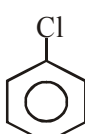
42. $\text{CH}_3\text{C}\equiv\text{CH} + 2\text{HBr} \longrightarrow \text{Product ?}$



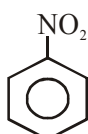
43. Increasing order of reactivity towards E.S.R. for following compounds is ?



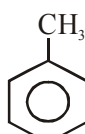
I



II



III

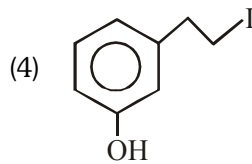
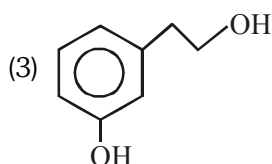
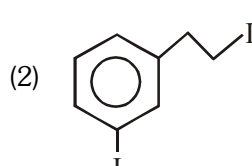
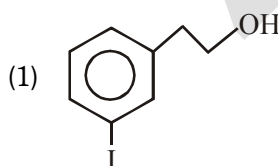
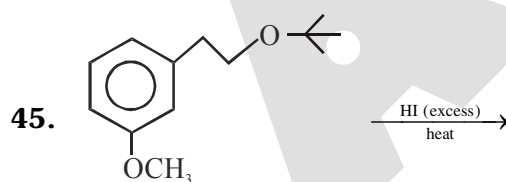


IV

- (1) $\text{III} < \text{II} < \text{IV} < \text{I}$ (2) $\text{II} < \text{III} < \text{IV} < \text{I}$
(3) $\text{III} < \text{II} < \text{I} < \text{IV}$ (4) $\text{III} < \text{I} < \text{II} < \text{IV}$

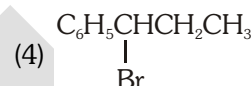
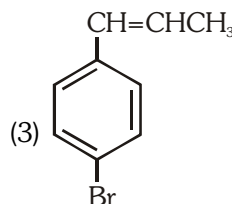
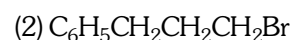
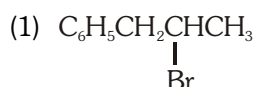
44. Correct order of nucleophilicity :-

- (1) $\text{CH}_3\text{O}^- > \text{CH}_3^- > \text{NH}_2^- > \text{CH}_3\text{COO}^-$
(2) $\text{CH}_3^- > \text{NH}_2^- > \text{CH}_3\text{O}^- > \text{CH}_3\text{COO}^-$
(3) $\text{NH}_2^- > \text{CH}_3^- > \text{CH}_3\text{O}^- > \text{CH}_3\text{COO}^-$
(4) $\text{CH}_3^- > \text{CH}_3\text{O}^- > \text{NH}_2^- > \text{CH}_3\text{COO}^-$



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46. The reaction of $\text{C}_6\text{H}_5\text{CH}=\text{CHCH}_3$ with HBr produces:-



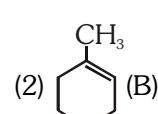
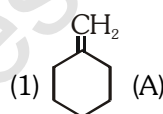
47. Treatment of cyclopentanone with

methyl lithium gives which of the following species?

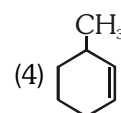
- (1) Cyclopentanonyl cation
(2) Cyclopentanonyl radical
(3) Cyclopentanonyl biradical
(4) Cyclopentanonyl anion

Re-AIPMT 2015

48. In the reaction with HCl , an alkene reacts in accordance with the Markovnikov's rule, to give a product 1-chloro-1-methylcyclohexane. The possible alkene is :-



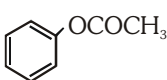
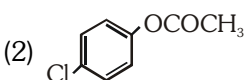
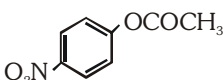
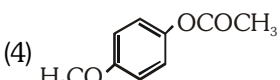
(3) (A) and (B)



49. Reaction of carbonyl compound with one of the following reagents involves nucleophilic addition followed by elimination of water. The reagent is :

- (1) hydrocyanic acid
(2) sodium hydrogen sulphite
(3) a Grignard reagent
(4) hydrazine in presence of feebly acidic solution

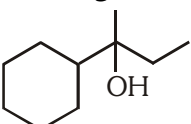
50. Which one of the following esters gets hydrolysed *most easily* under alkaline conditions ?

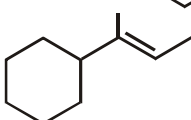
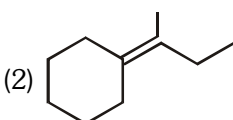
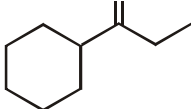
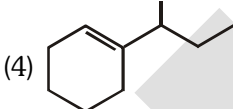
- (1)  (2) 
 (3)  (4) 

51. In an S_N1 reaction on chiral centres, there is :

- (1) 100% retention
 (2) 100% inversion
 (3) 100% racemization
 (4) inversion more than retention leading to partial racemization

52. Which of the following is not the product of

dehydration of  ?

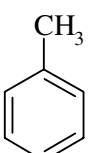
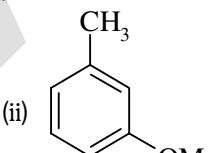
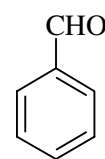
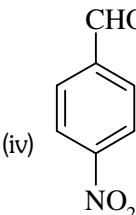
- (1)  (2) 
 (3)  (4) 

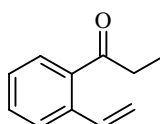
53. Which of the following reaction(s) can be used for the preparation of alkyl halides ?

- (I) $\text{CH}_3\text{CH}_2\text{OH} + \text{HCl} \xrightarrow{\text{anh. ZnCl}_2}$
 (II) $\text{CH}_3\text{CH}_2\text{OH} + \text{HCl} \longrightarrow$
 (III) $(\text{CH}_3)_3\text{COH} + \text{HCl} \longrightarrow$
 (IV) $(\text{CH}_3)_2\text{CHOH} + \text{HCl} \xrightarrow{\text{anh. ZnCl}_2}$
 (1) (IV) only (2) (III) and (IV) only
 (3) (I), (III) and (IV) only (4) (I) and (II) only

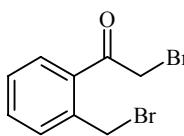
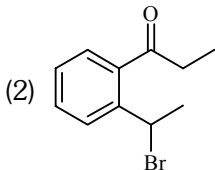
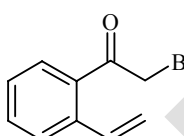
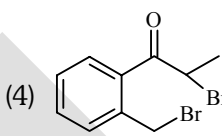
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54. Correct order of reactivity towards ESR is

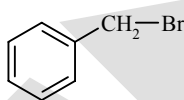
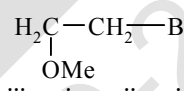
- (i)  (ii) 
 (iii)  (iv) 
 (1) i > iii > ii > iv (2) ii > i > iii > iv
 (3) i > iv > iii > ii (4) iv > ii > iii > i

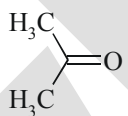
55.  $\xrightarrow{\text{HBr}}$ product.

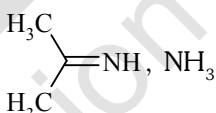
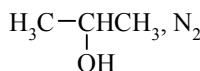
Product of given reaction is

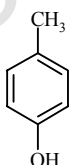
- (1)  (2) 
 (3)  (4) 

56. Correct order of reactivity towards S_N1 is

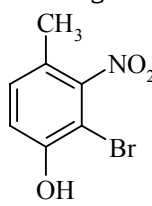
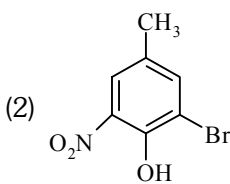
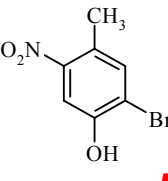
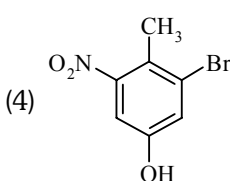
- (i)  (ii) $\text{CH}_3\text{CH}_2\text{Br}$
 (iii) $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{CH}_2-\text{Br}$ (iv) 
 (1) i > ii > iv > iii (2) iii > iv > ii > i
 (3) iv > ii > i > iii (4) ii > iv > iii > i

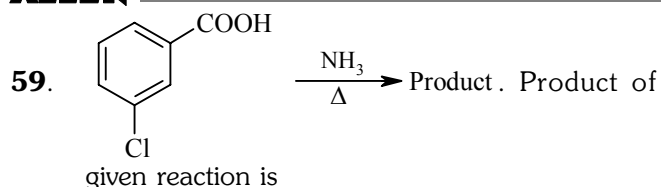
57.  $\xrightarrow[\text{KOH/CH}_2(\text{OH})_2]{\text{NH}_2\text{NH}_2/\Delta}$ product is

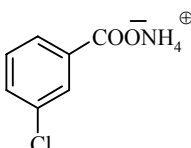
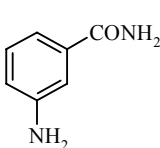
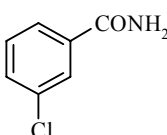
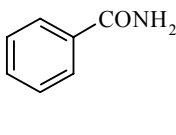
- (1)  (2) $\text{CH}_3\text{CH}_2\text{CH}_3, \text{N}_2$
 (3) $\text{CH}_3\text{CH}_2\text{CH}_3, \text{NH}_3$ (4) 

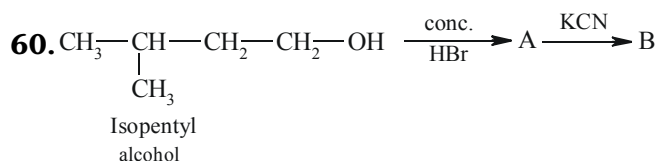
58.  $\xrightarrow{\text{Br}_2} \xrightarrow[\text{H}_2\text{SO}_4]{\text{HNO}_3}$ product.

Product of given reaction is

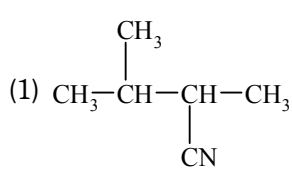
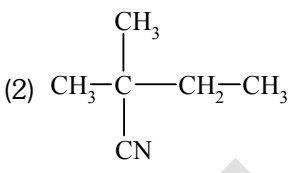
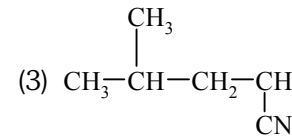
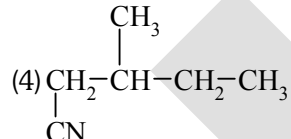
- (1)  (2) 
 (3)  (4) 



- (1)  (2) 
(3)  (4) 



What is B ?

- (1)  (2) 
(3)  (4) 

61. Correct order of reactivity towards elimination reaction

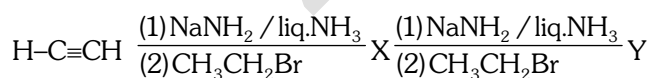
- (i) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2-\text{Br}$
(ii) $\text{CH}_3\text{CH}_2\text{CH}_2-\text{Br}$
(iii) $\text{C}_6\text{H}_5\text{CH}_2\underset{\text{Br}}{\text{CH}}-\text{CH}_3$

- (iv) $\text{H}_3\text{C}-\underset{\text{Br}}{\text{CH}}-\text{CH}_3$

- (1) i>ii>iii>iv (2) ii>i>iii>iv
(3) iii>iv>i>ii (4) iii>i>iv>ii

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62. In the reaction



X and Y are :

- (1) X = 1-Butyne ; Y = 3-Hexyne
(2) X = 2-Butyne ; Y = 3-Hexyne
(3) X = 2-Butyne ; Y = 2-Hexyne
(4) X = 1-Butyne ; Y = 2-Hexyne

63. Which of the following reagents would distinguish cis-cyclopentane-1,2-diol from its trans-isomer?

- (1) Acetone
(2) Ozone
(3) MnO_2
(4) Aluminium isopropoxide

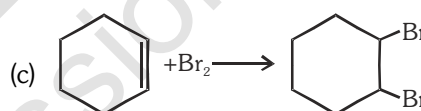
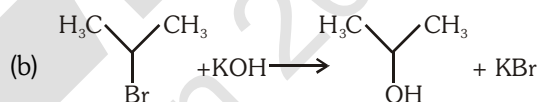
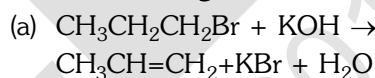
64. The product formed by the reaction of an aldehyde with a primary amine is :-

- (1) Schiff base
(2) Ketone
(3) Carboxylic acid
(4) Aromatic acid

65. Consider the nitration of benzene using mixed conc. H_2SO_4 and HNO_3 . If a large amount of KHSO_4 is added to the mixture, the rate of nitration will be:-

- (1) faster
(2) slower
(3) unchanged
(4) doubled

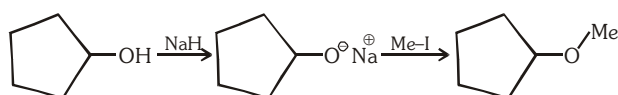
66. For the following reactions :-



Which of the following statement is **correct** ?

- (1) (a) and (b) are elimination reaction and (c) is addition reaction
(2) (a) is elimination, (b) is substitution and (c) is addition reaction
(3) (a) is elimination, (b) and (c) are substitution reactions
(4) (a) is substitution, (b) and (c) are addition reaction

67. The reaction



Can be classified as :-

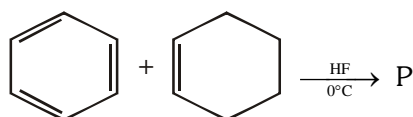
- (1) Williamson ether synthesis reaction
(2) Alcohol formation reaction
(3) Dehydration reaction
(4) Williamson alcohol synthesis reaction

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68. Which of the following compounds shall not produced propene by reaction with HBr followed by elimination or direct only by elimination reaction ?

- (1) $\text{H}_2\text{C}=\text{C}=\text{O}$ (2) $\text{H}_3\text{C}-\overset{\text{H}_2}{\underset{\text{H}_2}{\text{C}}}-\text{CH}_2\text{Br}$
 (3) $\text{H}_2\text{C}-\underset{\text{H}_2}{\underset{\text{H}_2}{\text{C}}}-\text{CH}_2$ (4) $\text{H}_3\text{C}-\overset{\text{H}_2}{\underset{\text{H}_2}{\text{C}}}-\text{CH}_2\text{OH}$

69. In the given reaction



the product P is :-

- (1) (2)
 (3) (4)

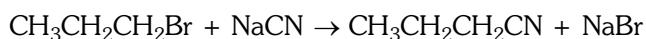
70. The compound that will react most readily with gaseous bromine has the formula

- (1) C_4H_{10} (2) C_2H_4
 (3) C_3H_6 (4) C_2H_2

71. Which of the following can be used as the halide component for Friedel-Crafts reaction ?

- (1) Chloroethene (2) Isopropyl chloride
 (3) Chlorobenzene (4) Bromobenzene

72. Consider the reaction

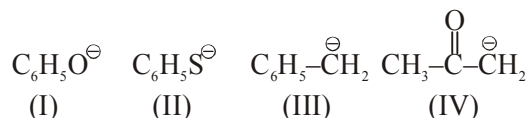


This reaction will be the fastest in

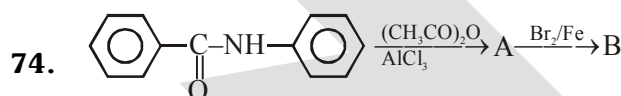
- (1) N,N'-dimethylformamide (DMF)
 (2) water
 (3) ethanol
 (4) methanol

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73. Arrange followings in their nucleophilicity order.



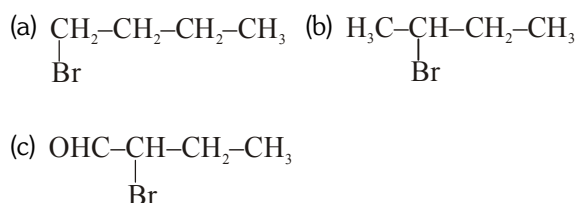
- (1) $\text{II} > \text{III} > \text{IV} > \text{I}$
 (2) $\text{I} > \text{IV} > \text{II} > \text{III}$
 (3) $\text{II} > \text{IV} > \text{III} > \text{I}$
 (4) $\text{III} > \text{IV} > \text{I} > \text{II}$



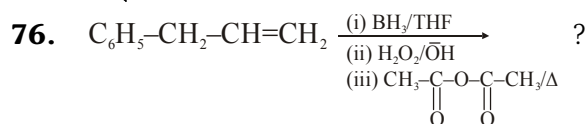
B is :-

- (1)
 (2)
 (3)
 (4)

75. Arrange the following compounds in order of reactivity towards $\text{S}_{\text{N}}1$ reaction.

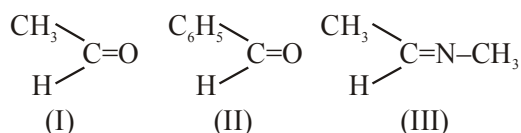


- (1) $\text{a} > \text{b} > \text{c}$
 (2) $\text{b} > \text{a} > \text{c}$
 (3) $\text{c} > \text{b} > \text{a}$
 (4) $\text{a} > \text{c} > \text{b}$

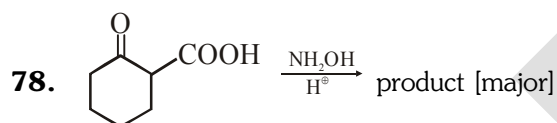


- (1) $\text{C}_6\text{H}_5-\text{CH}_2-\underset{\begin{array}{c} \diagdown \\ \text{O} \\ \diagup \end{array}}{\text{CH}-\text{CH}_2}$
- (2) $\text{C}_6\text{H}_5-\text{CH}_2-\underset{\begin{array}{c} | \\ \text{O}-\overset{\parallel}{\text{C}}-\text{CH}_3 \end{array}}{\text{CH}-\text{CH}_3}$
- (3) $\text{C}_6\text{H}_5-\underset{\begin{array}{c} | \\ \text{C}-\text{CH}_3 \\ || \\ \text{O} \end{array}}{\text{CH}}-\text{CH}_2-\text{CH}_3$
- (4) $\text{C}_6\text{H}_5-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{O}-\underset{\begin{array}{c} || \\ \text{O} \end{array}}{\text{C}}-\text{CH}_3$

77. Arrange following compounds in reactivity order towards nucleophilic addition reaction

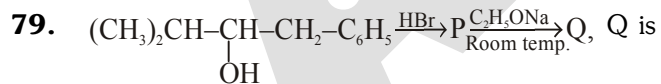


- (1) $I > II > III$ (2) $III > II > I$
(3) $II > III > I$ (4) $I > III > II$



Product will be –

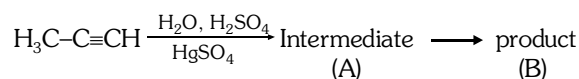
- (1) 
- (2) 
- (3) 
- (4) 



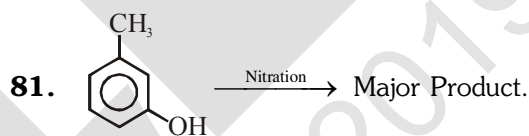
- (1) $(\text{CH}_3)_2\text{CH}-\text{CH}=\text{CH}-\text{C}_6\text{H}_5$
- (2) $(\text{CH}_3)_2\text{CH}-\underset{\text{OC}_2\text{H}_5}{\text{CH}}-\text{CH}_2-\text{C}_6\text{H}_5$
- (3) $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\underset{\text{OC}_2\text{H}_5}{\text{CH}}-\text{C}_6\text{H}_5$
- (4) $\text{CH}_2=\underset{\text{CH}_3}{\text{C}}-\text{CH}_2-\text{CH}_2-\text{C}_6\text{H}_5$

NEET(UG) 2017

80. Predict the correct intermediate and product in the following reaction :

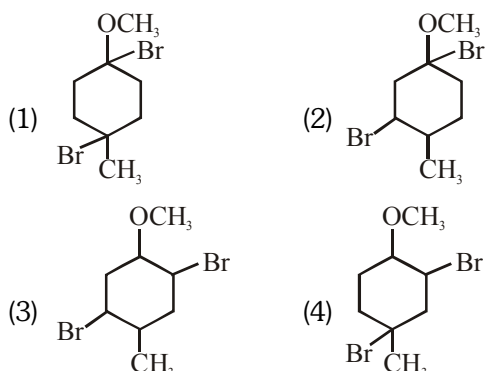
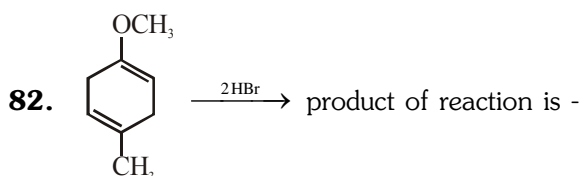


- (1) A : $\text{H}_3\text{C}-\underset{\text{OH}}{\text{C}}=\text{CH}_2$ B : $\text{H}_3\text{C}-\underset{\text{SO}_4}{\text{C}}=\text{CH}_2$
- (2) A : $\text{H}_3\text{C}-\underset{\text{O}}{\text{C}}-\text{CH}_3$ B : $\text{H}_3\text{C}-\text{C}\equiv\text{CH}$
- (3) A : $\text{H}_3\text{C}-\underset{\text{OH}}{\text{C}}=\text{CH}_2$ B : $\text{H}_3\text{C}-\underset{\text{O}}{\text{C}}-\text{CH}_3$
- (4) A : $\text{H}_3\text{C}-\underset{\text{SO}_4}{\text{C}}=\text{CH}_2$ B : $\text{H}_3\text{C}-\underset{\text{O}}{\text{C}}-\text{CH}_3$

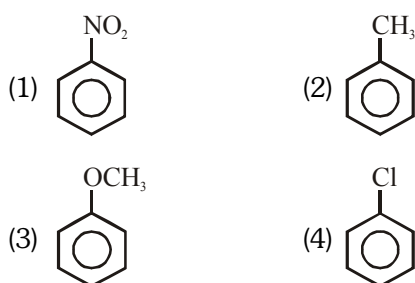
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What is major product ?

- (1) 
- (2) 
- (3) 
- (4) 



83. Which of the following shows fastest electrophilic substitution reaction :-

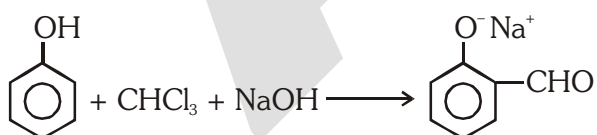


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84. Nitration of aniline in strong acidic medium also gives m-nitroaniline because

- (1) In spite of substituents nitro group always goes to only m-position.
- (2) In electrophilic substitution reactions amino group is meta directive.
- (3) In absence of substituents nitro group always goes to m-position
- (4) In acidic (strong) medium aniline is present as anilinium ion.

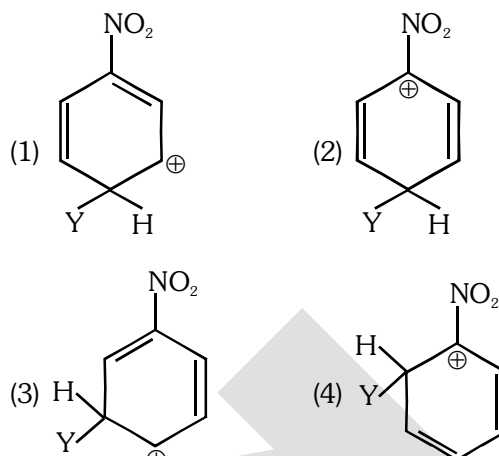
85. In the reaction



the electrophile involved is

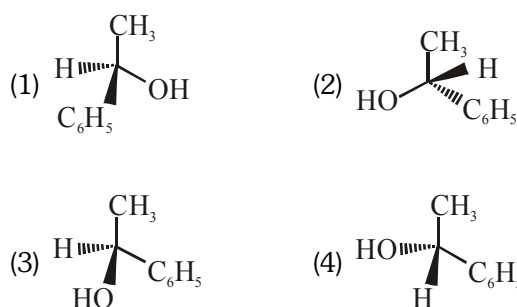
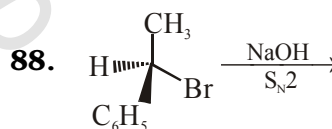
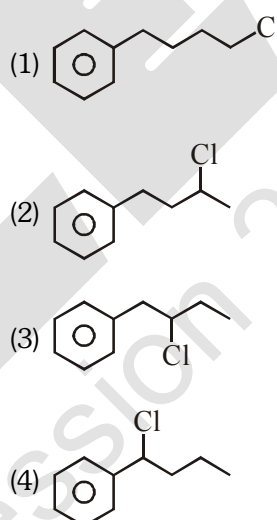
- (1) dichloromethyl cation (CHCl_2^+)
- (2) formyl cation (CHO^+)
- (3) dichloromethyl anion (CHCl_2^-)
- (4) dichlorocarbene ($:\text{CCl}_2$)

86. Which of the following carbocations is expected to be most stable ?

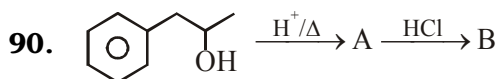
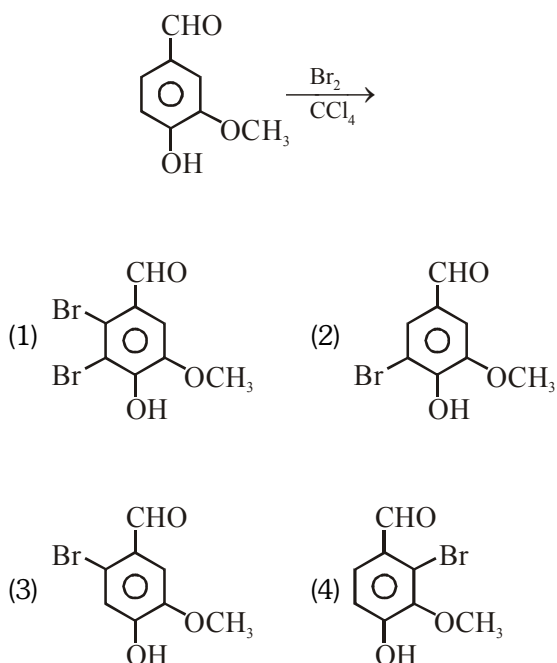


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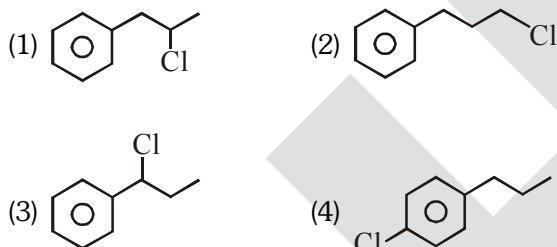
87.  product



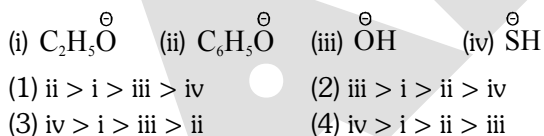
89. The product of the given reaction



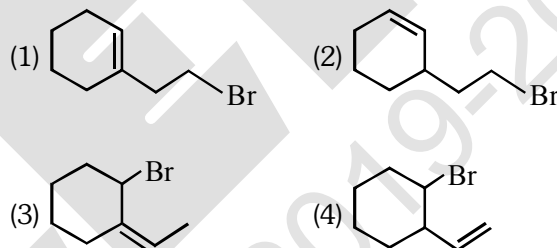
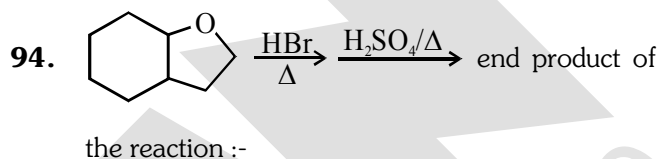
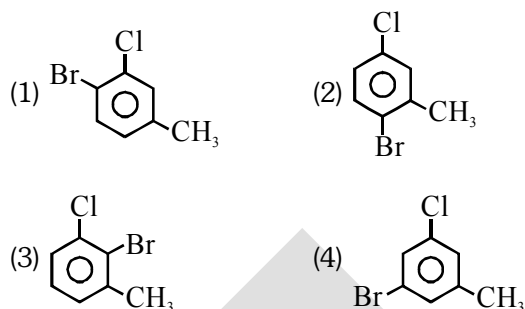
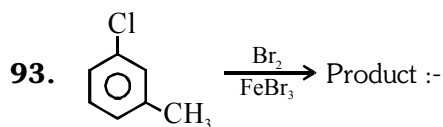
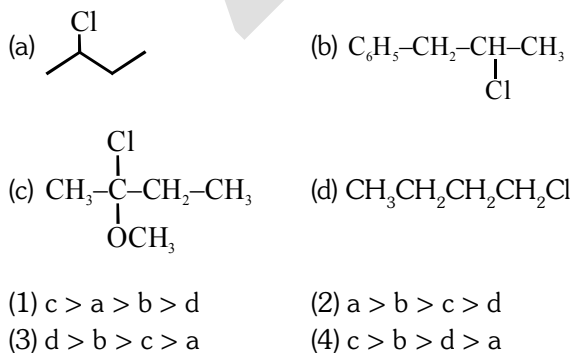
The product 'B' is :



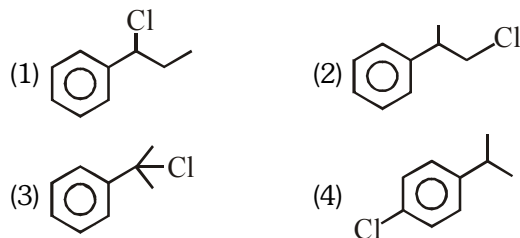
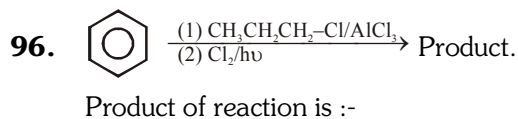
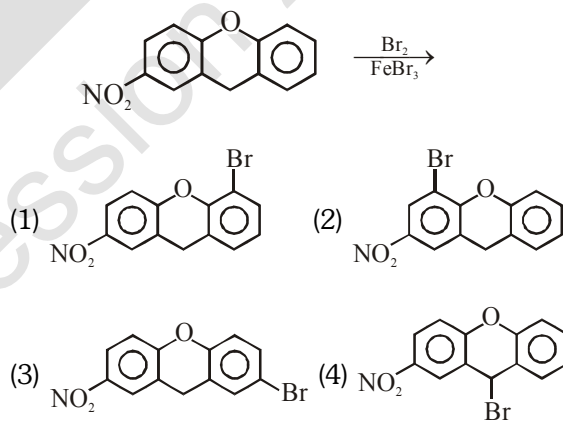
91. The nucleophilicity order of the given nucleophiles :-

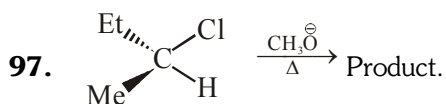


92. Arrange the following in decreasing order of reactivity towards $\text{S}_{\text{N}}1$ reaction :-

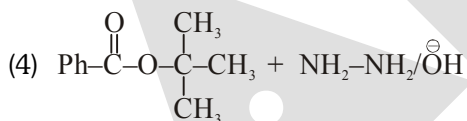
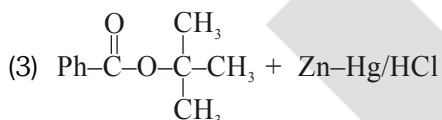
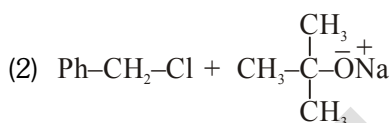
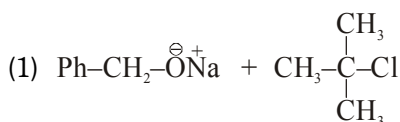
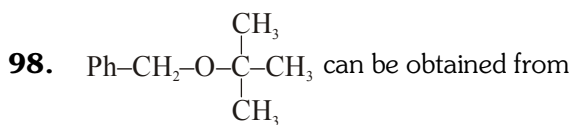
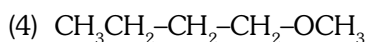
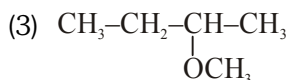
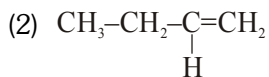
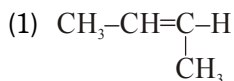


95. Major product of the given reaction :-

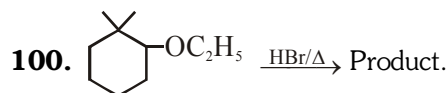
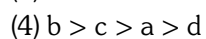
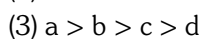
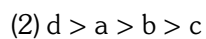
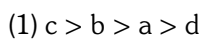
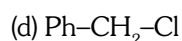
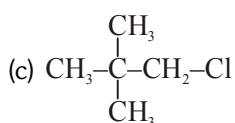
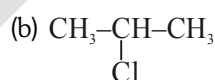




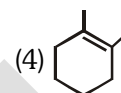
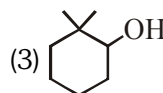
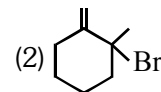
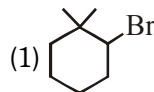
Product of reaction is :-



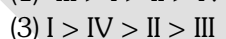
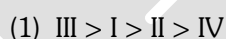
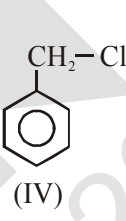
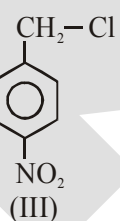
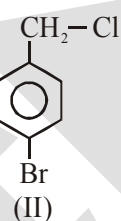
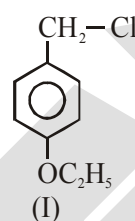
99. Arrange the following in decreasing order of reactivity for $\text{S}_\text{N}2$



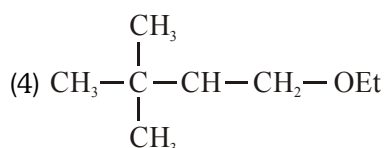
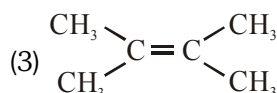
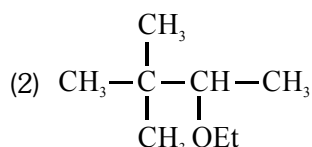
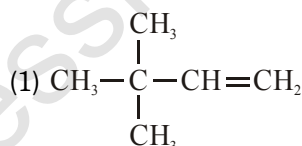
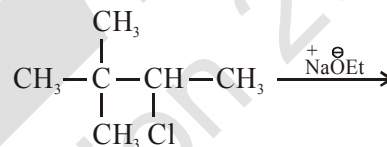
Product of reaction is :-

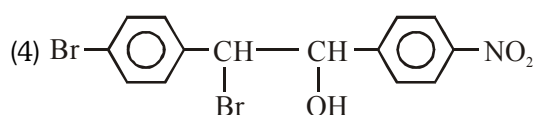
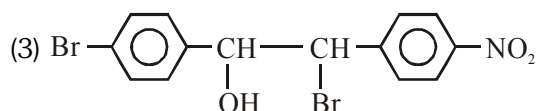
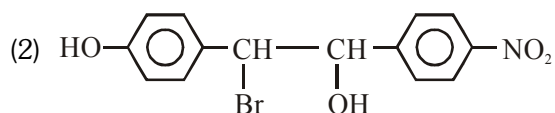
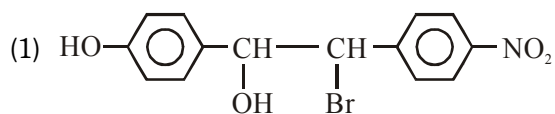
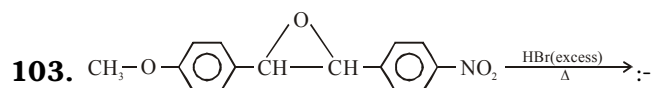


101. Compare the rate of $\text{S}_\text{N}1$ reaction :-



102. Which of the following is major product in the given reaction:-





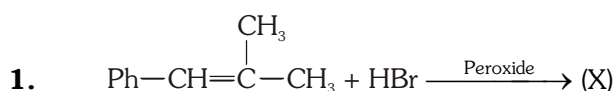
EXERCISE-II (Previous Year Questions)

ANSWER KEY

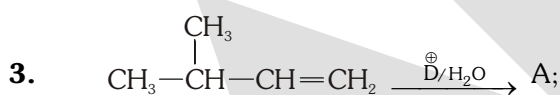
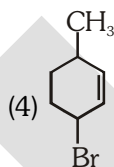
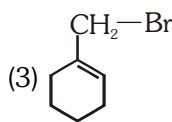
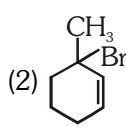
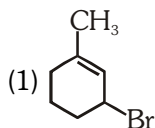
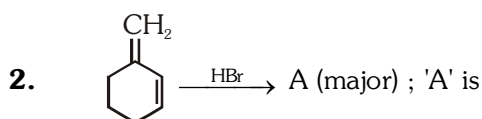
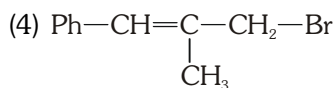
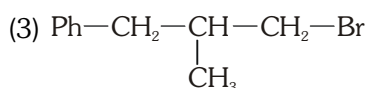
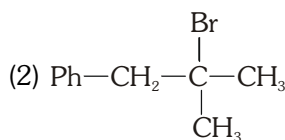
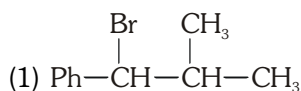
Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	4	1	3	3	1	3	1	3	2	3	2	4	2	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	4	2	2	2	1	4	3	2	1	4	2	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	3	1	4	4	1	2	3	4	3	1	4	4	1	2	4
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	4	3	4	3	4	4	3	2	2	1	2	2	3	3
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	4	1	1	1	2	2	1	1	1	3	2	1	1	1	2
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	4	1	2	1	3	1	1	3	4	4	3	2	3	2	3
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103		
Ans.	3	1	2	1	3	3	1	2	2	3	3	1	2		

EXERCISE-III (Analytical Questions)

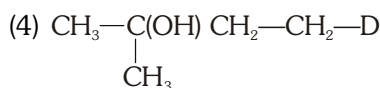
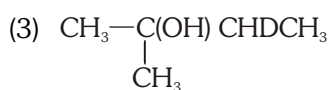
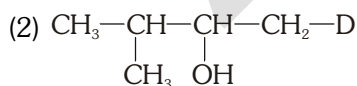
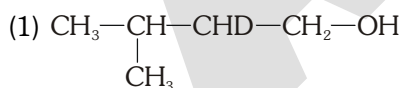
Check Your Understanding



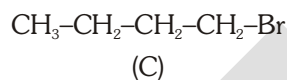
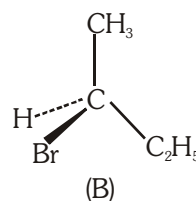
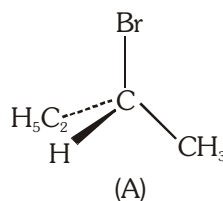
'X' is



A (major product) is



4. The addition of HBr to 1-Butene gives a mixture of products A, B and C.

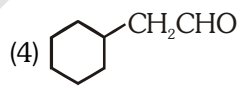
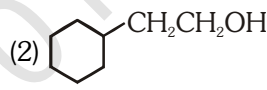
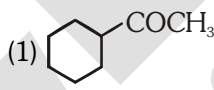


The mixture consists of :-

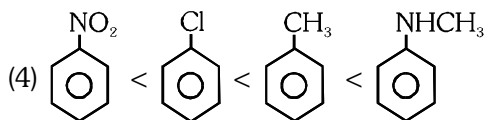
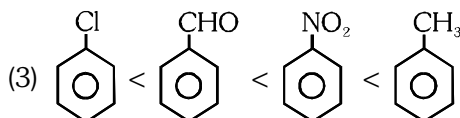
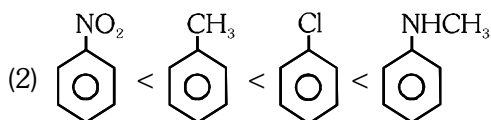
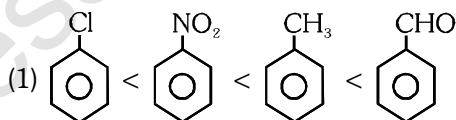
- (1) A and B as major and C as minor products
- (2) B as major, A and C as minor products
- (3) B as minor, A and C as major products
- (4) A and B as minor and C as major products.

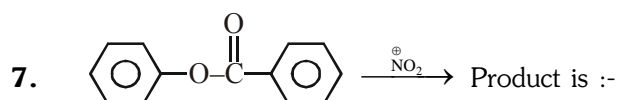
5. Hydration of $\text{Cyclohexyl}-\text{C}\equiv\text{CH}$ in presence of

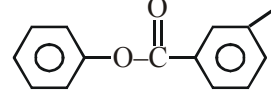
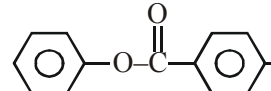
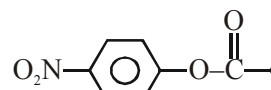
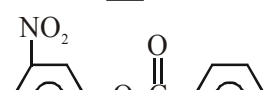
$\text{H}_2\text{SO}_4 / \text{HgSO}_4$ gives



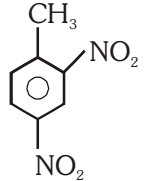
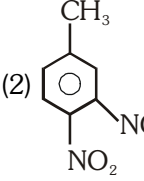
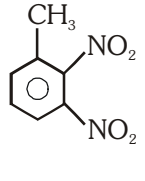
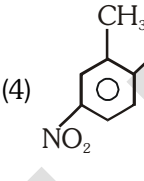
6. Identify correct order of reactivity for electrophilic substitution reaction of the following compounds :-





- (1) 
 (2) 
 (3) 
 (4) 

8. p- nitro toluene on further nitration gives

- (1) 
 (2) 
 (3) 
 (4) 

9. Which of the following reaction gives t-butyl benzene:

- (1) $C_6H_6 + t\text{-butyl chloride} \xrightarrow{AlCl_3}$
 (2) $C_6H_6 + (CH_3)_2C=CH_2 \xrightarrow{HF}$
 (3) $C_6H_6 + t\text{-butyl alcohol} \xrightarrow{H_2SO_4}$
 (4) All of them

10. Halogenation of alkanes gives a mixture of monohalo products. The ease of substitution follows the order

- (1) Tertiary H > primary H > secondary H
 (2) Primary H < secondary H < tertiary H
 (3) Primary H > secondary H > tertiary H
 (4) Secondary H > tertiary H < primary H

11. The number of different substitution products possible when ethane is allowed to react with bromine in sunlight are

- (1) 9
 (2) 6
 (3) 8
 (4) 5

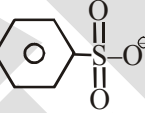
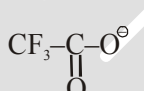
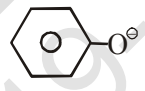
12. Chlorination of toluene in the presence of light and heat followed by treatment with aqueous NaOH gives

- (1) o-cresol
 (2) p-cresol
 (3) 2, 4-dihydroxy toluene
 (4) Benzyl alcohol

13. Which gives maximum yield of C_2H_5Cl

- (1) $C_2H_6 + Cl_2 \xrightarrow[\text{excess}]{h\nu, \text{light}} C_2H_5Cl + HCl$
 (2) $C_2H_6 + Cl_2 \xrightarrow[\text{excess}]{h\nu, \text{light}} C_2H_5Cl + HCl$
 (3) $C_2H_6 + Cl_2 \xrightarrow[\text{light}]{h\nu} C_2H_5Cl$
 (4) $C_2H_6 + Cl_2 \xrightarrow{\text{Dark}} C_2H_5Cl + HCl$

14. Which of the following has maximum nucleophilicity :-

- (1) 
 (2) CH_3-O^-
 (3) 
 (4) 

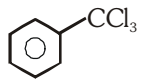
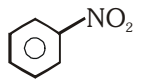
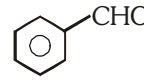
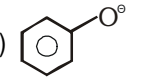
15. Consider thiol anion (RS^-) and alkoxy anion (RO^-). Which of the following statement is correct ?

- (1) RS^- is less basic and less nucleophilic than RO^-
 (2) RS^- is less basic but more nucleophilic than RO^-
 (3) RS^- is more basic and more nucleophilic than RO^-
 (4) RS^- is more basic but less nucleophilic than RO^-

16. Alkaline hydrolysis of 2,2-dichloropropane gives

- (1) Acetone
 (2) 2,2-propane-diol
 (3) Propane
 (4) Propan-2-ol

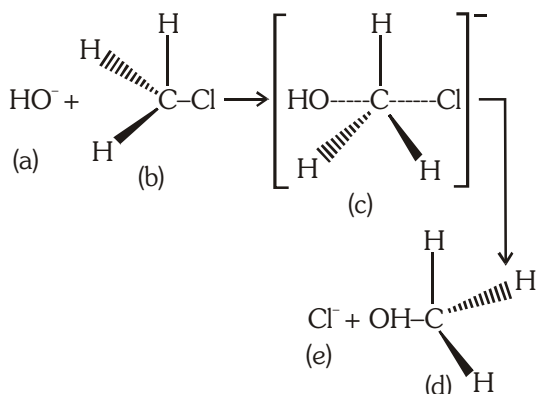
17. Electrophile NO_2^+ attacks the following :

- (1) 
 (II) 
 (III) 
 (IV) 

in which cases NO_2^+ will attack at meta position

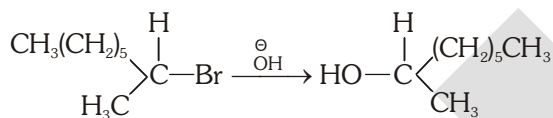
- (1) II and IV
 (2) I, II and III
 (3) III and IV
 (4) I only

18. Which of the following statement is correct for S_N2 mechanism



- (1) In (c) carbon atom is sp^3 hybridized
- (2) In (c) carbon atom is sp^2 hybridised
- (3) (a) and (e) are electrophiles
- (4) (c) is more stable than (d)

19. Following reaction is



- (1) E_1
- (2) S_N1
- (3) E_2
- (4) S_N2



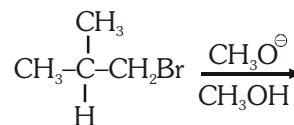
In the above reaction the attack of a nucleophile would be from which side

- (1) On the front side of Cl of C_1 carbon
- (2) On C_2 carbon
- (3) On the rear side of Cl of C_1 carbon
- (4) On C_3 carbon

21. Which of the following ether can not be prepared by Williamson's method :-

- (1) Di-tert-butyl ether
- (2) Ethyl-tert-butyl ether
- (3) Anisole
- (4) 1 & 2 Both

22. The major product formed in the following reaction is :-



- (1) $CH_3-\overset{CH_3}{\underset{H}{\underset{|}{C}}}-CH_2OCH_3$
- (2) $CH_3-\overset{CH_3}{\underset{OCH_3}{\underset{|}{CH}}}-CH_2CH_3$
- (3) $CH_3-\overset{CH_3}{\underset{CH_2}{\underset{|}{C}}}=CH_2$
- (4) $CH_3-\overset{CH_3}{\underset{OCH_3}{\underset{|}{C}}}-CH_3$

23. Major product of the reaction :-



would be :-

- (1) $(CH_3)_2C-OC_2H_5$
- (2) $(CH_3)_3C-C_2H_5$
- (3) $(CH_3)_2C=CH_2$
- (4) $CH_3-CH=CH-C_2H_5$

24. Among the following the most reactive towards alcoholic KOH is

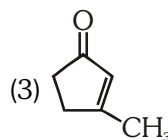
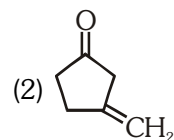
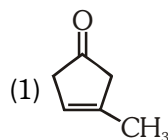
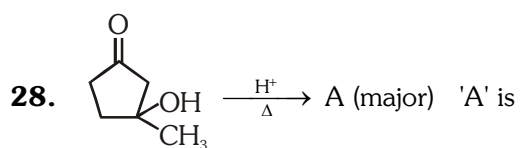
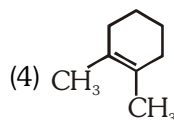
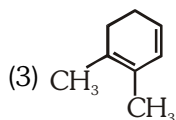
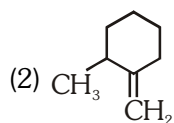
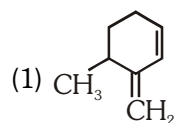
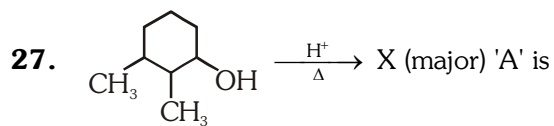
- (1) $CH_2=CHBr$
- (2) $CH_3COCH_2CH_2Br$
- (3) CH_3CH_2Br
- (4) $CH_3CH_2CH_2Br$

25. The major product obtained on treatment of $CH_3CH_2CH(F)CH_3$ with CH_3O^-/CH_3OH is

- (1) $CH_3CH_2CH(OCH_3)CH_3$
- (2) $CH_3CH=CHCH_3$
- (3) $CH_3CH_2CH=CH_2$
- (4) $CH_3CH_2CH_2CH_2OCH_3$

26. Dehydrohalogenation of neopentyl chloride with alc. KOH mainly gives

- (1) 2-methyl-1-butene
- (2) 2-methyl-2-butene
- (3) 2,2-dimethyl-1-propanol
- (4) 2-pentene



(4) All

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	1	4	1	1	4	3	1	4	2	1	4	2	2	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28		
Ans.	1	2	2	4	3	1	3	3	2	3	2	4	3		

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

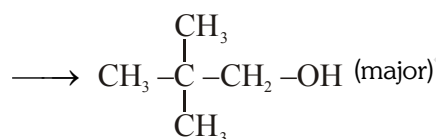
- Assertion :-** trans 2-butene gives racemic mixture when treated with Cl_2 .
Reason :- Addition of Cl_2 is anti.
 (1) A (2) B (3) C (4) D
- Assertion :-** Addition of HBr in presence of peroxide on alkene is free radical addition. This reaction is not shown by HCl.
Reason :- Due to less bond energy of HCl than HBr, homolysis of HCl can not occurs.
 (1) A (2) B (3) C (4) D
- Assertion :-** Addition of CBrCl_3 in presence of peroxides takes place faster to 2-ethyl-1-hexene than to 1-octene.
Reason :- Intermediate free radical in case of 2-ethyl-1-hexene is more stable.
 (1) A (2) B (3) C (4) D
- Assertion :-** Ketone gives 2,4-DNP test easily than aldehyde.
Reason :- Ketone is more reactive than aldehyde.
 (1) A (2) B (3) C (4) D
- Assertion :-** p-nitrophenol gives more electrophilic substituted compound than m-methoxy phenol
Reason :- Methoxy group shows only negative I-effect
 (1) A (2) B (3) C (4) D
- Assertion :** Rates of nitration of benzene and hexadeuterobenzene are different.
Reason : C-H bond is stronger than C-D bond.
 (1) A (2) B (3) C (4) D
- Assertion :-** Major product obtained from FCR of benzene with n-butyl chloride is tert. Butyl benzene.
Reason :- Intermediate 1° carbocation rearrangements in more stable 3° carbocation.
 (1) A (2) B (3) C (4) D
- Assertion :-** When CH_3Cl and AlCl_3 are used in F.C.R. the electrophile is $\text{Cl}^\oplus$.
Reason :- The first attack on benzene is of $\text{Cl}^\oplus$.
 (1) A (2) B (3) C (4) D
- Assertion :-** The correct reactivity order towards sulphonation is –
 meta xylene > Toluene > chlorobenzene > Nitrobenzene
Reason :- m-xylene has the highest electron density & hence most reactive.
 (1) A (2) B (3) C (4) D
- Assertion :-** Iodine monochloride (ICl) reacts with benzene in presence of a lewis acid such as anhyd. AlCl_3 to give iodobenzene and not chlorobenzene.
Reason :- Iodine is less electronegative than chlorine in the reaction.
 (1) A (2) B (3) C (4) D
- Assertion :-** Vinyl halides can not be used in place of alkyl halides in friedel crafts alkylation.
Reason :- Vinyl halides does not give electrophile easily.
 (1) A (2) B (3) C (4) D
- Assertion :-** If the hydrogen atoms of benzene are replaced by deuterium atoms then the reactivity of benzene ring is decreased for ESR.
Reason :- The bond strength of C-D is more than C-H and reaction depends on the bond strength of C-D and C-H.
 (1) A (2) B (3) C (4) D
- Assertion :-** Benzyl chloride is more reactive than p-chloro toluene towards aqueous NaOH.
Reason :- The C-Cl bond in benzyl chloride is more polar than C-Cl bond in p-chloro toluene.
 (1) A (2) B (3) C (4) D

14. **Assertion :-** $\text{CH}_3\text{CH}_2\text{OCH}_2\text{Cl}$ reacts faster when treated with water than $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_2\text{CH}_2\text{Cl}$.
Reason :- Carbonium ion formed by the ionisation of $\text{CH}_3\text{CH}_2\text{OCH}_2\text{Cl}$ is stabilized by resonance.
 (1) A (2) B (3) C (4) D

15. **Assertion :-** p-Chloro aniline and aniliniumchloride can be distinguished by AgNO_3 .
Reason :- p-Chloro aniline is less basic than aniline.
 (1) A (2) B (3) C (4) D

16. **Assertion :-** Phenoxide ion is a poor nucleophile than a methoxide ion.
Reason :- In methoxide ion oxygen is attached to sp^3 carbon.
 (1) A (2) B (3) C (4) D

17. **Assertion :-** $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{CH}_2-\text{Br} + \text{NaOH}$



Reason :- It follows with formation of more stable carbanion
 (1) A (2) B (3) C (4) D

18. **Assertion :-** Aryl sulphonic acid gives phenol on reacting with NaOH at high temperature followed by :
Reason :- This acidification reaction is electrophilic substitution reaction.
 (1) A (2) B (3) C (4) D

19. **Assertion :-** Aryl halides are more reactive than alkyl halide towards NSR .
Reason :- Intermediate carbocation is more stable obtained from Aryl halide.
 (1) A (2) B (3) C (4) D

20. **Assertion :-** In SN^1 reaction if the alkyl halide is optically active then the product is racemic mixture.
Reason :- All tertiary halides gives racemic mixture.
 (1) A (2) B (3) C (4) D

21. **Assertion :-** Chlorobenzene does not react with NaOH where as ethyl chloride reacts.
Reason :- The partial double bond between C and Cl causes less reactivity towards nucleophilic substitution.
 (1) A (2) B (3) C (4) D

22. **Assertion :-** Benzyl bromide when kept in acetone water it produces benzyl alcohol.
Reason :- The reaction follows SN^2 mechanism
 (1) A (2) B (3) C (4) D

23. **Assertion :-** Anisole on reaction with HI gives phenol and CH_3I .
Reason :- Phenyl-oxygen bond is stronger than methyl-oxygen bond in anisole and hence is not cleaved by HI .
 (1) A (2) B (3) C (4) D

24. **Assertion :-** In Lucas test, 3° alcohols react immediately.
Reason :- An equimolar mixture of anhydrous ZnCl_2 and conc. HCl is called Lucas reagent.
 (1) A (2) B (3) C (4) D

25. **Assertion :-** t-butyl methyl ether is prepared by reaction of t-butyl bromide with methoxide ion.
Reason :- Reaction follow SN^2 -mechanism.
 (1) A (2) B (3) C (4) D

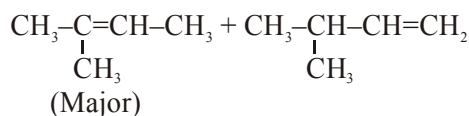
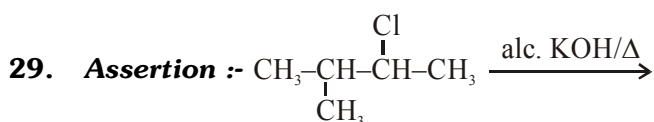
26. **Assertion :-** Treatment of 1,3-dibromopropane with zinc dust produces cyclopropane.
Reason :- The reaction of alkyl halide with Zn dust is nucleophilic substitution.
 (1) A (2) B (3) C (4) D

27. **Assertion :-** 2-Bromobutane on reaction with sodium ethoxide in ethanol gives 2-butene as a major product.
Reason :- 2-Butene is more stable than 1-butene.
 (1) A (2) B (3) C (4) D

28. **Assertion :-** Aldehyde are more reactive than Ketone.
Reason :- Nucleophilic addition reaction in carbonyl compound

$$\propto \frac{1}{\text{Steric hindrance}} \text{ and } \propto \text{positive charge at carbon of carbonyl group.}$$

(1) A (2) B (3) C (4) D



Reason :- In E_2 reaction product formation always takes place by saytzeff rule.

(1) A (2) B (3) C (4) D

30. **Assertion :-** The product of reaction of 2-Methyl cyclo hexanol with HBr at room temperature is 1-Bromo-1-methyl cyclo hexane.

Reason :- Generally dehydration occur and product is 1-methyl cyclo hexene.

(1) A (2) B (3) C (4) D

31. **Assertion :-** Compound $\text{CH}_3-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{OH}$ can

not give ether with conc. H_2SO_4 .

Reason :- It favours the formation of alkene over ether.

(1) A (2) B (3) C (4) D

32. **Assertion :-** In friedel craft's reaction of 1-chloropropane gives isopropyl benzene as product.

Reason :- The primary carbocation rearranges to form secondary carbocation.

(1) A (2) B (3) C (4) D

33. **Assertion :-** Propene reacts with diborane to form tripropyl borane.

Reason :- It follows anti markowni koff's rule.

(1) A (2) B (3) C (4) D

34. **Assertion :-** 2-Bromopentane react with alcoholic KOH to form 2-pentene.

Reason :- Alcoholic KOH follows E_1 , mechanism.

(1) A (2) B (3) C (4) D

35. **Assertion :-** p-Nitrobenzyl ethyl ether on heating with conc. HBr and gives p-Nitrobenzyl bromide.

Reason :- p-Nitrobenzyl carbocation is highly stable.

(1) A (2) B (3) C (4) D

36. **Assertion :-** Phthalic anhydride react with phenol in presence of NaOH gives pink colour.

Reason :- It forms phenolphthalein.

(1) A (2) B (3) C (4) D

37. **Assertion :-** Benzophenone on reaction with NaHSO_3 gives stable addition product.

Reason :- Equilibrium of this reaction lies on the right side.

(1) A (2) B (3) C (4) D

38. **Assertion :-** $\text{C}_6\text{H}_5\text{MgBr}$ on reaction with CO_2 gives benzoic acid

Reason :- CO_2 is a good electrophile

(1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	3	1	4	4	4	4	4	1	1	1	4	1	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	2	4	3	4	3	1	3	1	2	4	3	1	1	3	3
Que.	31	32	33	34	35	36	37	38							
Ans.	1	1	3	3	3	1	4	3							

HYDROCARBONS

E

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NEET SYLLABUS

Alkanes - Physical properties, chemical reactions, combustion and pyrolysis. Alkenes - Structure of double bond, physical properties, methods of preparation, chemical reactions, ozonolysis, oxidation. Alkynes - Structure of triple bond, physical properties, methods of preparation, chemical reactions, acidic character of alkynes, benzene, resonance, aromaticity, carcinogenicity and toxicity.

@neetfocusquizz

OBJECTIVES

After studying this unit, we will be able to :

- learn about various methods of preparation of hydrocarbons;
- distinguish between alkanes, alkenes, alkynes and aromatic hydrocarbons on the basis of physical and chemical properties;
- appreciate the role of hydrocarbons as sources of energy and for other industrial applications;
- comprehend the structure of benzene, explain aromaticity;
- predict the directive influence of substituents in monosubstituted benzene ring;
- learn about carcinogenicity and toxicity.

"Once your mind stretches to a new level it never goes back to its original dimension"

A.P.J. Abdul Kalam

HYDROCARBONS

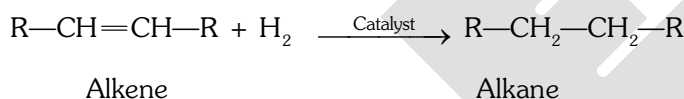
5.0 ALKANES

5.1.1 Introduction of Alkanes

- Branched and unbranched aliphatic saturated open chain hydrocarbons are called member of alkanes.
- CH_4 is also known as Marsh gas (fire damp).
- Calore gas : Mixture of n-butane and isobutane.
- LPG (Liquefied petroleum gas) : liquid propane, isobutane.
- Natural gas : 80% methane + 10% ethane + 10% propane + small amounts of H_2 , N_2 , CO_2 etc.
- Water gas : $\text{CO} + \text{H}_2$ (1:1)
- Synthesis gas : $\text{CO} + 3\text{H}_2$ (1:3)

5.1.2 General Methods of Preparation

- (1) From alkenes and alkynes (Sabatier and Senderens reaction) or (By hydrogenation of alkenes and alkynes) :** Alkenes and alkynes on catalytic hydrogenation give alkanes.



Catalyst :

- Pd/Pt at ordinary temperature and pressure
- Ni, 200–300° C (sabatier)
- Raney Nickel at room temp.
- Methane can not be prepared by this method

- (2) From alkyl Halides (By reduction) :**



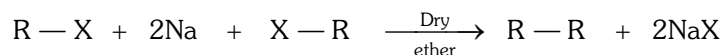
Catalyst :

- | | | |
|---------------------------------|--|---|
| (i) $\text{Zn} + \text{HCl}$ | (ii) $\text{Zn} + \text{CH}_3\text{COOH}$ | (iii) $\text{Zn}-\text{Cu}$ couple in $\text{C}_2\text{H}_5\text{OH}$ |
| (iv) $\text{Red P} + \text{HI}$ | (v) $\text{Al} - \text{Hg} + \text{ethanol}$ | |

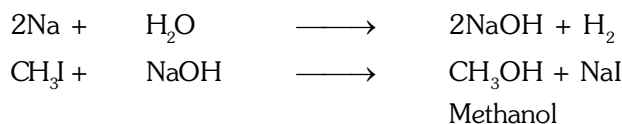
GOLDEN KEY POINTS

- Alkyl halides can also be reduced to alkane by H_2/Pd or LiAlH_4 or H_2/Ni .
- Halogen atom of alkyl halide is replaced by hydrogen atom to obtain an alkane.

(3) From alkyl halide (By Wurtz reaction): A solution of alkyl halide in ether on heating with sodium gives alkane.

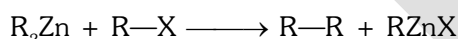
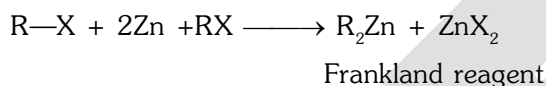


- (a) Two moles of alkyl halide are treated with Na in presence of dry ether. If ether is wet then we obtain alcohol.

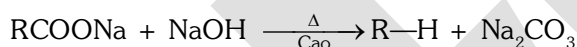


- (b) Methane can not be prepared by this method. The alkane produced is higher and symmetrical i.e. it contains double the number of carbon atoms present in the alkyl halide taken.
- (c) Two different alkyl halides, on wurtz reaction give all possible alkanes.
- (d) The separation of mixture into individual members is not easy because their B.P. are near to each other and thus wurtz reaction is not suitable for the synthesis of alkanes containing odd number of carbon atom.

(4) From Frankland Reagent:



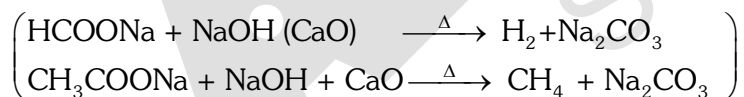
(5) From Carboxylic Acid (By decarboxylation) : Sodium salt of saturated monocarboxylic acid on dry distillation with sodalime give alkane.



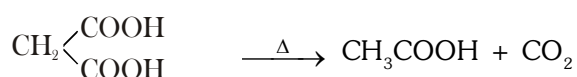
Note :- Sodalime $\Rightarrow$ NaOH + CaO

GOLDEN KEY POINTS

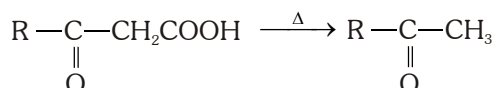
- The process of elimination of Carbon-di-oxide from Carboxylic acid called decarboxylation.
- The alkane formed by decarboxylation contains one carbon atom less than the original acid.
- Decarboxylation of sodium formate gives H_2



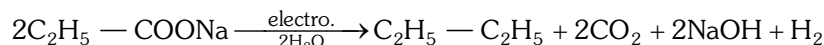
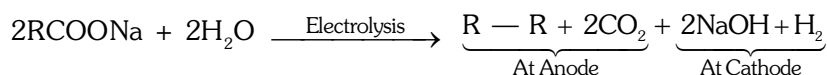
- If in a compound two carboxylic groups are present and they are attached to same carbon atom then decarboxylation of one of the carboxylic groups takes place simply on heating.



- $CH_3-CH_2-CH_3$ can be prepared by Butanoic acid and 2-Methyl propanoic acid.
- β -Keto acids are decarboxylated readily simply on heating (soda lime is not required)

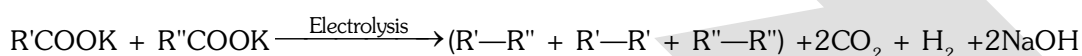


(6) From carboxylic acid (By Kolbe's electrolysis process) : Alkanes are formed on electrolysis of concentrated aqueous solution of sodium or potassium salt of saturated monocarboxylic acids.



GOLDEN KEY POINTS

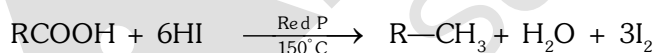
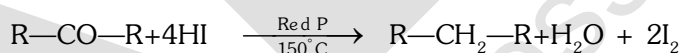
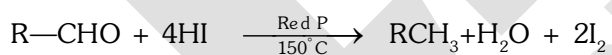
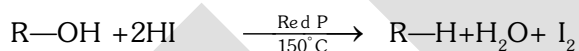
- Electrolysis of an acid salt gives symmetrical alkane, However in case of a mixture of Carboxylic acid salts, all probable alkanes are formed.



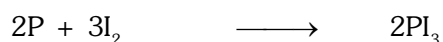
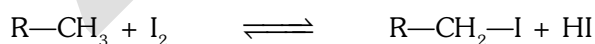
- At anode alkane and CO_2 gas is formed while at cathode NaOH and H_2 gas is formed.
- The concentration of NaOH in solution is increased with time so pH of solution is also increased.

(7) From alkanol, alkanals, Alkanone and alkanic acid (By reduction) :

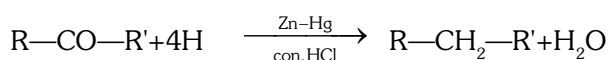
The reduction of either of the above compound in presence of red P and HI gives corresponding alkane.

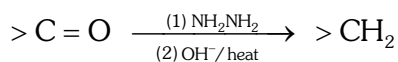


In the above reaction I_2 is formed which may react with alkane to form alkyl halide. So red P is added in the reaction to remove I_2 formed in the reaction.

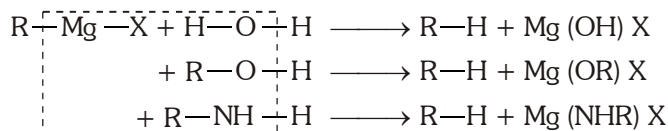


(8) From alkanones (By Clemmensen's method) : Carbonyl compound may also be reduced with Zinc amalgam and concentrated HCl (Zn-Hg/HCl), this reaction is called Clemmensen reduction.



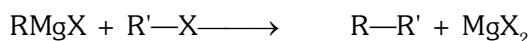
(9) From alkanals and alkanones (By Wolf Kishner reaction) :**(10) From G.R. :**

(a) Formation of alkanes with same number of C atoms : Grignard reagent reacts with the compounds having active hydrogen to form alkane.

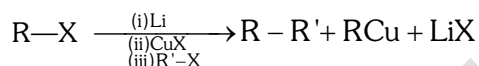


This reaction is used to determine the number of active H-atoms in the compound this is known as Zerewitnoff's method.

(b) G.R. react with alkyl halide to give higher alkanes :



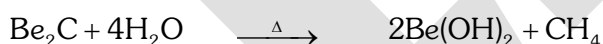
(11) Corey-House Synthesis : This method is suitable for the preparation of unsymmetrical alkanes i.e. those of type $R-R'$



Note: In Corey-house reaction symmetrical and unsymmetrical alkane both can be formed.

(12) From metal carbide (By hydrolysis) :

Only CH_4 can be obtained by the hydrolysis of Be or Al carbides

**BEGINNER'S BOX-1**

- If two moles of Isopropyl chloride reacts with Na in presence of dry ether. Which alkane is obtained.
 - Hexane
 - 2, 3-Dimethyl butane
 - Isopentane
 - Neopentane
- If isopropyl chloride and ethyl chloride both react with Na in presence of dry ether which alkanes are obtained.
 - n-Butane
 - 2-Methyl butane
 - 2, 3-Dimethyl butane
 - All of them
- Which of the following compound can not be obtained from single alkyl halide by wurtz reaction.
 - ethane
 - butane
 - isobutane
 - hexane
- How many acids can be taken to obtain isobutane by decarboxylation ?
 - 4
 - 3
 - 2
 - 5

5. Give reactivity order for decarboxylation ?
 (I) $\text{CH}_3\text{—CH}_2\text{—COOH}$ (II) $\text{CH}_2=\text{CH—COOH}$ (III) $\text{CH}\equiv\text{C—COOH}$
 (1) $\text{I} > \text{II} > \text{III}$ (2) $\text{III} > \text{II} > \text{I}$ (3) $\text{III} > \text{I} > \text{II}$ (4) None is correct
6. Which of the following does not give alkane with R—Mg—X .
 (1) Ph—OH (2) C_6H_6 (3) CH_3COOH (4) HCl
7. Which of the following reaction can not be used to obtained propane in good yield.
 (1) Wurtz reaction (2) Corey-house reaction
 (3) Decarboxylation of acid salt (4) All of them

5.1.3 Physical & Chemical Properties of alkane

Physical properties

- (i) **Solubility** : Alkanes being non polar and thus insoluble in water but soluble in nonpolar solvents

Ex. C_6H_6 , CCl_4 , ether etc.

- (ii) **Boiling point** - $\propto$ molecular weight (for n-alkanes)

$\therefore$ $\text{Vander Waals force of attraction} \propto \text{molecular weight} \propto \text{surface area of molecule.}$

i.e. boiling point Pentane < hexane < heptane

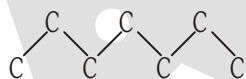
Also boiling point $\propto \frac{1}{\text{number of side chain}}$

because the shape approaches to spherical which results in decrease in van der Waal's forces (as surface area decreases)

Thus boiling point n-Pentane > Isopentane > neopentane

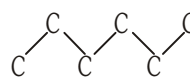
- (iii) **Melting Point** : M.P. of alkanes do not show regular trend. Alkanes with even number of carbon atoms have higher M.P. than their adjacents alkanes of odd number of carbon atoms.

The abnormal trend in M.P. is due to the fact that alkanes with odd carbon atoms have their end carbon atom on the same side of the molecule and in even carbon atom alkane the end Carbon atom on opposite side. Thus alkanes with even carbon atoms are packed closely in crystal lattice to permit greater intermolecular attractions.



Odd number of carbon

<

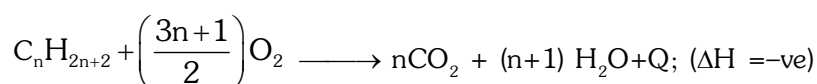


Even number of carbon

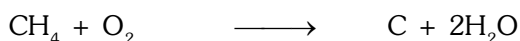
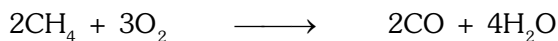
Chemical Properties

(1) Oxidation :

- (a) **Complete oxidation or combustion** : Alkanes burn readily with non-luminous flame in presence of air or oxygen to give CO_2 and water with evolution of heat. Therefore, alkanes are used as fuels.



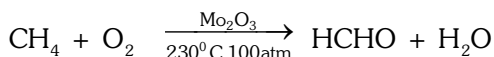
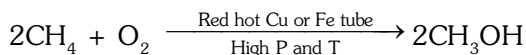
- (b) **Incomplete oxidation :** In limited supply of air, alkane gives carbon black and CO.



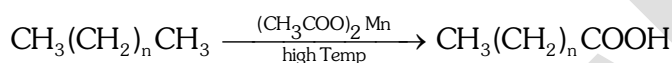
C-black (used in printing ink)

- (c) **Catalytic oxidation :**

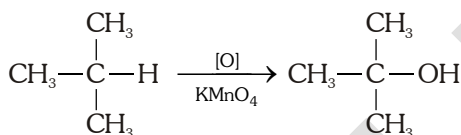
- (i) Alkanes are easily converted to alcohols and aldehydes under controlled catalytic oxidation.



- (ii) Alkanes on oxidation in presence of manganese acetate give fatty acids.

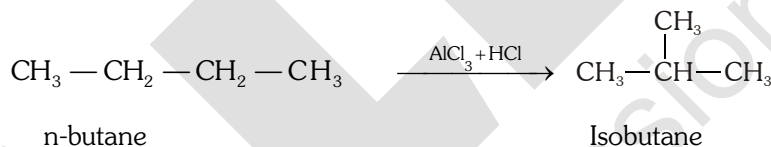


- (iii) Tertiary alkanes are oxidized to give tertiary alcohols by KMnO_4 .



- (2) **Substitution Reactions :** Substitution reaction in alkanes shows free radical mechanism.

- (3) **Isomerization:** Unbranched chain alkanes on heating with $\text{AlCl}_3 + \text{HCl} / 200^\circ\text{C}$ are converted into branched chain alkanes

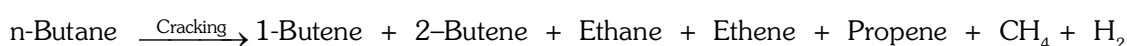
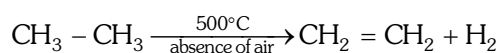
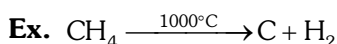


Branched chain alkanes converted to more branched alkane.



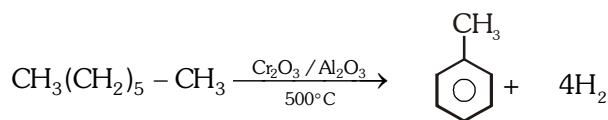
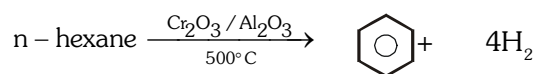
Isomerisation of alkanes is of great importance in petroleum industry to increase the octane number of petrol (gasoline).

- (4) **Pyrolysis or Cracking or thermal decomposition :** When alkanes are heated to $500-700^\circ\text{C}$ they are decomposed into lower hydrocarbon. This decomposition is called pyrolysis.



(5) **Aromatization:**

Unbranched higher alkanes (from 6 to 10 carbon atoms) when heated in presence of oxides of Cr, Mo, V on Al_2O_3 support at 500°C aromatic hydrocarbons are formed.



n-heptane

Toluene

5.1.4 PETROLEUM

Flash Point : Flash point of an oil is that minimum temperature at which the oil gives so much vapour that it starts burning in the presence of air when it comes in contact with a spark. Flash point depends on local temperature of countries.

Flash point of kerosene in **India** $\Rightarrow 44^\circ\text{C}$, in **France** $\Rightarrow 35^\circ\text{C}$, in **England** $\Rightarrow 23^\circ\text{C}$

Knocking : Preignition of the fuel-air mixture in the cylinder ahead of the flame causes knocking. knocking reduces efficiency of the engine and also damages the cylinder and piston of the engine.

Octane Number (Quality of Gasoline) : Octane no. is a scale which is used to determine the quality of a fuel in an internal engine. Two pure hydrocarbons are selected as standard.

- (i) n - heptane – straight chain hydrocarbon knocks very badly have octane no. zero
- (ii) Iso-octane – branched hydrocarbon, good antiknocking properties, have octane no. 100.

Octane no. of a fuel : The percentage of iso-octane by volume in a mixture of iso-octane and n-heptane which has the same antiknocking properties as the fuel under examination.

For example – Let knocking of a fuel is same as a mixture of 70% iso-octane and 30% n-heptane, then its octane no. is 70.

Order of quality of Gasoline or petrol : (i) Length of chain $\uparrow$, the octane no. $\downarrow$
(ii) branches $\uparrow$ octane no. $\uparrow$

Straight chain alkane < branched chain alkane < olefins < cycloalkane < aromatic compound

Octane number can be increased by : (i) Cracking (ii) Isomerization (iii) Aromatization

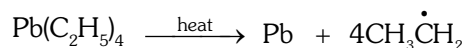
Antiknocking agents – (Gasoline additives) : Quality of a fuel is increased by :

- (i) By using fuel of higher octane no.
- (ii) By addition of certain compounds to the gasoline which reduce knocking. These are called antiknocking agents.

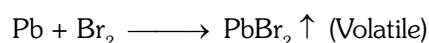
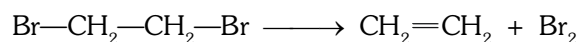
Ex. Tetraethyl lead (TEL) – $\text{Pb}(\text{C}_2\text{H}_5)_4$

Tetra methyl lead – $\text{Pb}(\text{CH}_3)_4$

In cylinder of engine $\text{Pb}(\text{C}_2\text{H}_5)_4$ decomposes in ethyl radicals which combine with the radicals produced due to irregular combustion and this prevents knocking.



but combustion of TEL with petrol gives Litharge (PbO) which deposited in cylinder walls and piston has jammed. So Ethylene dibromide is added with petrol.



BEGINNER'S BOX-2

1. Which of the following reactions of alkanes involve free radical intermediates

- | | |
|------------------|----------------------|
| (1) Halogenation | (2) Pyrolysis |
| (3) Nitration | (4) All of the above |

2. $(\text{CH}_3)_3\text{CMgCl}$ on reaction with D_2O produces

- | | |
|--------------------------------|--------------------------------|
| (1) $(\text{CH}_3)_3\text{CD}$ | (2) $(\text{CH}_3)_3\text{OD}$ |
| (3) $(\text{CD}_3)_3\text{CD}$ | (4) $(\text{CD}_3)_3\text{OD}$ |

3. The compound with the highest boiling point is—

- | | |
|--------------------------|---------------|
| (1) n-hexane | (2) n-pentane |
| (3) 2,2-dimethyl propane | (4) propane |

4. Photochemical chlorination of alkane is initiated by a process of

- | | |
|---------------|------------------|
| (1) pyrolysis | (2) substitution |
| (3) Homolysis | (4) Peroxidation |

5. Isomerization in alkane may be brought about by using :

- | |
|--|
| (1) Al_2O_3 |
| (2) Fe_2O_3 |
| (3) AlCl_3 and HCl |
| (4) Concentrated H_2SO_4 |

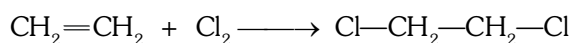
6. Bromination of an alkane as compared to chlorination proceeds

- | |
|---|
| (1) At a slower rate |
| (2) At a faster rate |
| (3) With equal rates |
| (4) With equal or different rate depends upon the temperature |

5.2 ALKENES (OLEFINS)

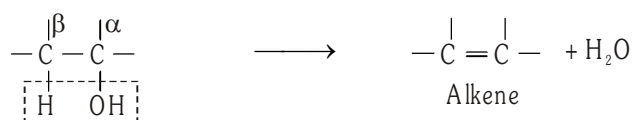
5.2.1 Introduction of Alkenes

Alkene are also called olefins (oil forming) since the first member ethylene (C_2H_4) was found to form an only liquid on reaction with chlorine.



5.2.2 General Methods of Preparation

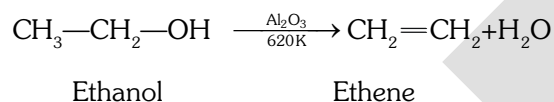
(1) **From Alcohols** : Alkenes can be prepared from monohydric alcohols or alkanols by the loss of H_2O and the reaction is known as **dehydration reaction**.



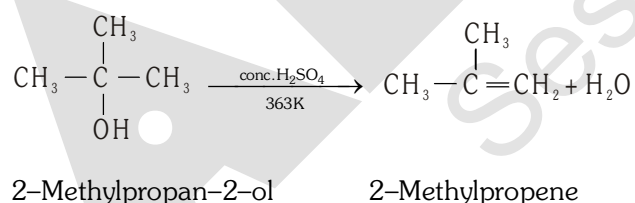
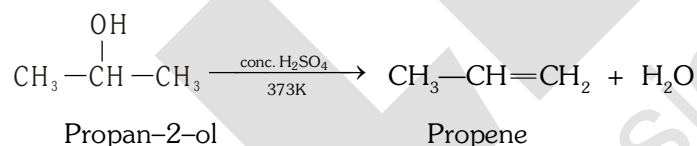
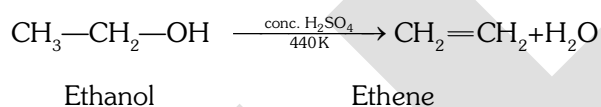
Alcohol

The dehydration can be carried with Al_2O_3 or with **mineral acid** upon heating.

(a) **Dehydration with Al_2O_3** : Ethene is prepared by heating ethanol with Al_2O_3 at 620 K.

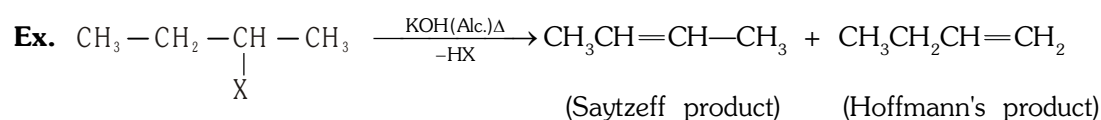
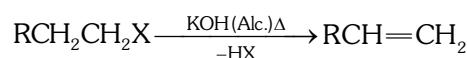


(b) **Dehydration with mineral acid** : Alcohols upon heating with conc. H_2SO_4 form alkenes and the reaction is called **acidic dehydration**.



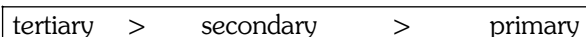
From the above reactions, it is clear that the order of acidic dehydration in different alcohols is Tertiary > Secondary > Primary

(2) **From Alkyl halide (By dehydrohalogenation)**: Removal of HX from a substrate by alcoholic KOH or $NaNH_2$

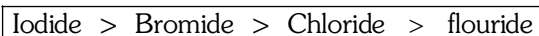


The ease of dehydrohalogenation show the order

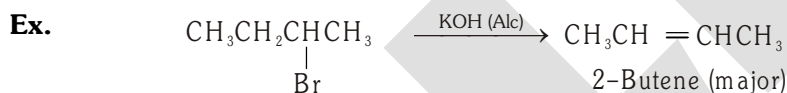
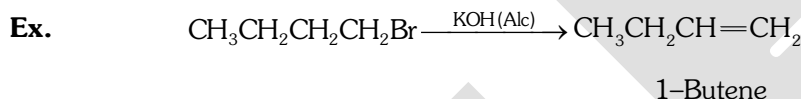
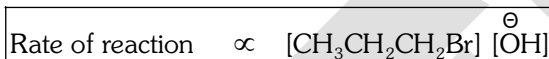
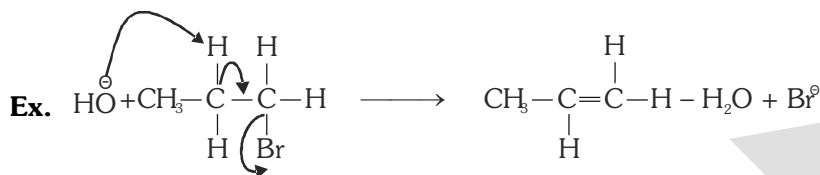
For alkyl group



For halogen in halide

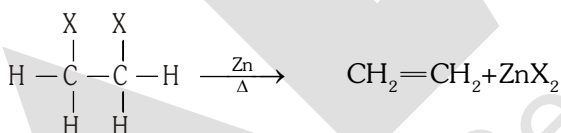


It is single step and synchronous process. Removal of proton, the formation of multiple bond between C α and C β and the release of the leaving group X take place simultaneously. (E₂ mechanism)

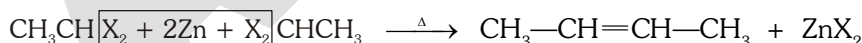


(3) **From Alkyl dihalide (By dehalogenation of Vicinal or Gem dihalide)** : Removal of X₂ from a substrate by Zn dust or Zn—Cu in alcoholic Solution..

(a) From **Vicinal dihalide** : same number carbon alkene is obtained.



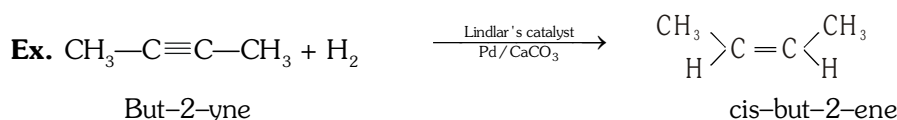
(b) From **gem dihalide** : Higher alkene obtained



(4) **By the controlled hydrogenation of alkynes** :

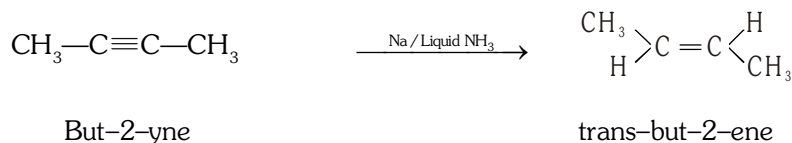
Alkynes can be converted into alkenes as a result of **controlled reduction** in two ways:

(a) **By the use of Lindlar's catalyst** : Lindlar's catalyst is a mixture of palladium catalyst deposited over barium sulphate or calcium carbonate. The catalytic mixture is slightly poisoned by quinoline or sulphur and allows the reduction or hydrogenation of alkyne with hydrogen only upto the alkene stage. The major product is cis-Alkene.

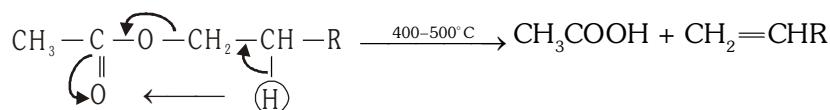


In place of Lindlar's catalyst Nickel-boride (Ni-B also called P-2 catalyst) can also be used.

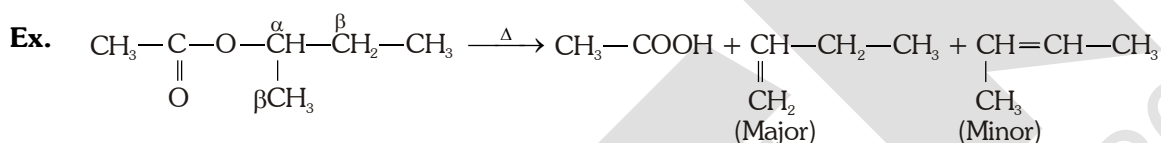
- (b) **By the action of sodium in liquid ammonia :** This is known as **Birch reduction** and the major product is a **trans alkene** i.e., the two hydrogen atoms get attached on the opposite side of the double bond. For example,



(5) By Pyrolysis of ester :



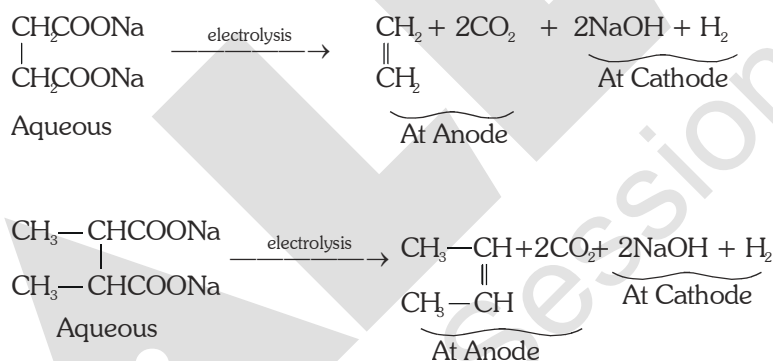
Hoffmann's Rule : Less substituted or less stable alkene is major product.



In the reaction to form an alkene a β -hydrogen from alkyl ester is attracted by oxygen atom of keto group.

(6) By Pyrolysis of tetra alkyl ammonium hydroxide :

- (7) By Kolbe's method:** Electrolysis of potassium or sodium salt of saturated dicarboxylic acid gives alkene.



5.2.3 PHYSICAL & CHEMICAL PROPERTIES OF ALKENES

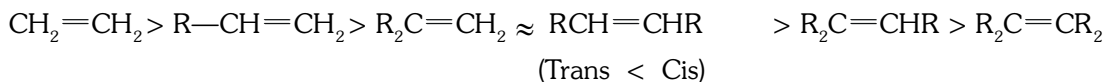
Physical Properties

- (1) All are colourless and have no characteristic odour. Ethene has pleasant smell.
- (2) Lower members (C_2 to C_4) gaseous medium (C_5 to C_{17}) liquid and higher members are solid.
- (3) The B.P., M.P. and specific gravity show a regular increase with increase in molecular weight
- (4) The increase in branching in carbon chain decreases the B.P. among isomeric alkenes.
- (5) The B.P. and M.P. of alkenes are slightly higher than the corresponding alkanes because the intermolecular forces of attraction are stronger due to the presence of easily polarizable π bond.
- (6) Insoluble in water because they can not form H-bond with water molecule, they dissolve freely in organic solvent like benzene, CHCl_3 , CCl_4 etc.

Chemical Properties : Alkenes are more reactive than alkane this is because -

- The π electrons of double bond are located much far from the carbon nuclei and are thus less firmly bound to them.
- π bond is weaker than σ bond and more easily broken.

The reactivity order for alkenes -



The reactivity order of alkenes has been delt in terms of heat of hydrogenation of alkene, more is the heat of hydrogenation ($\Delta H = -ve$), more is the reactivity, the reactivity of alkene is however also related to

- Steric hinderance
- Hyperconjugation

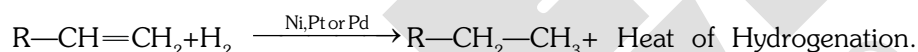
Alkenes give the following type of reactions :

- Addition reaction.
- Oxidation reaction.
- Substitution reaction.
- Polymerization Reaction.
- Isomerisation

(A) Addition Reaction :

[A₁] Free Radical Addition

(1) Addition of H₂ :



- Reaction is exothermic, Heat released in reaction is called heat of hydrogenation.

$$(b) \quad \text{Stability of alkene} \propto \frac{1}{\text{heat of hydrogenation}} \propto \frac{1}{\text{reactivity of alkene with H}_2}$$

- The process is used to obtain vegetable (saturated fats) ghee from hydrogenation of oil.

[A₂] Electrophilic Addition Reactions:

(B) Oxidation Reaction: Alkenes are easily oxidised by oxidising agents. Oxidising agents attack on double bond and product formed during oxidation depends on oxidising agents.

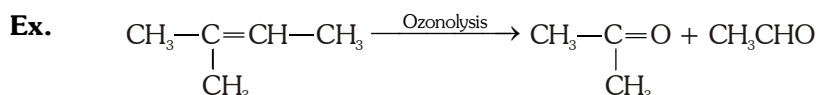
(1) Combustion:



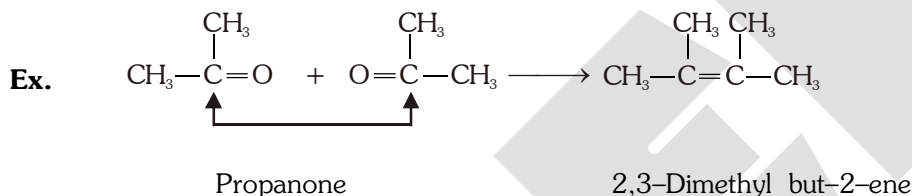
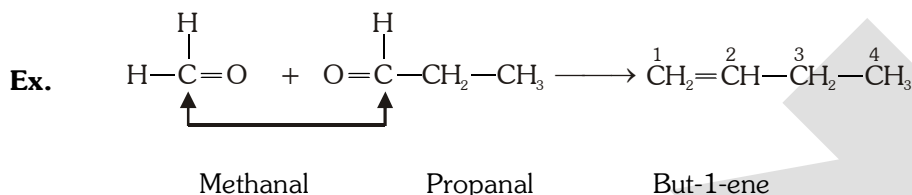
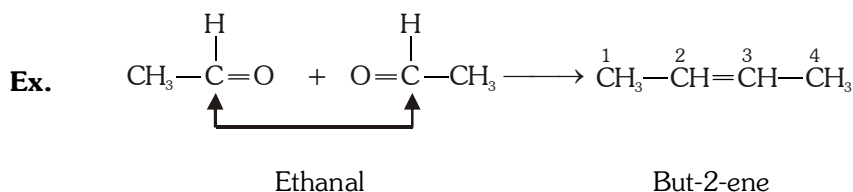
One mole of alkene requires $\frac{3n}{2}$ moles of O₂ for complete combustion.

(2) Ozonolysis : (A test for unsaturation in molecule)

- The addition of ozone on the double bonds and subsequent a reductive hydrolysis of the ozonide formed is termed as ozonolysis.
- Ozonides are explosive compound.
- On warming with Zn and H₂O, ozonides cleave at the site of the double bond, the products are carbonyl compound (aldehyde or ketone) depending on the nature of the alkene.



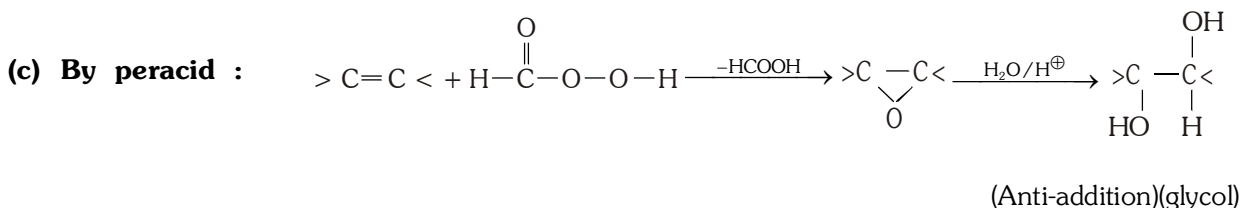
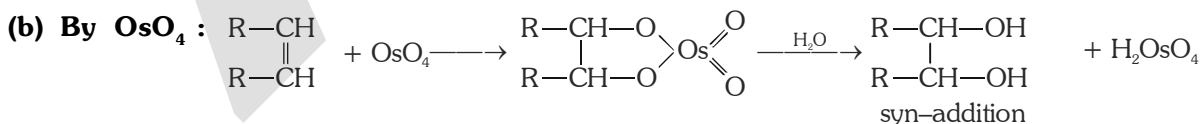
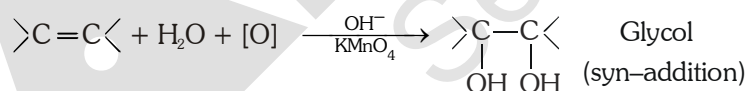
- (iv) Ozonolysis of alkenes helps in locating the position of double bond in an alkene. It can be achieved by joining together the carbon atoms of the two carbonyl compounds formed as the products of ozonolysis with double bond.

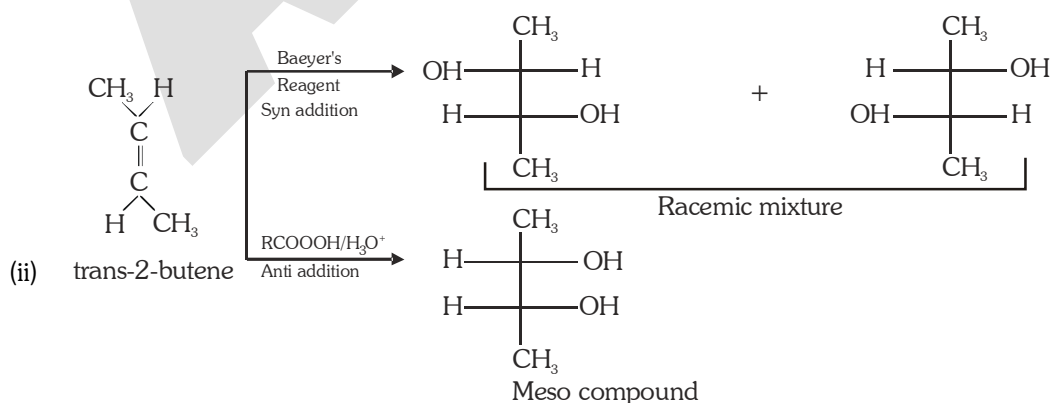
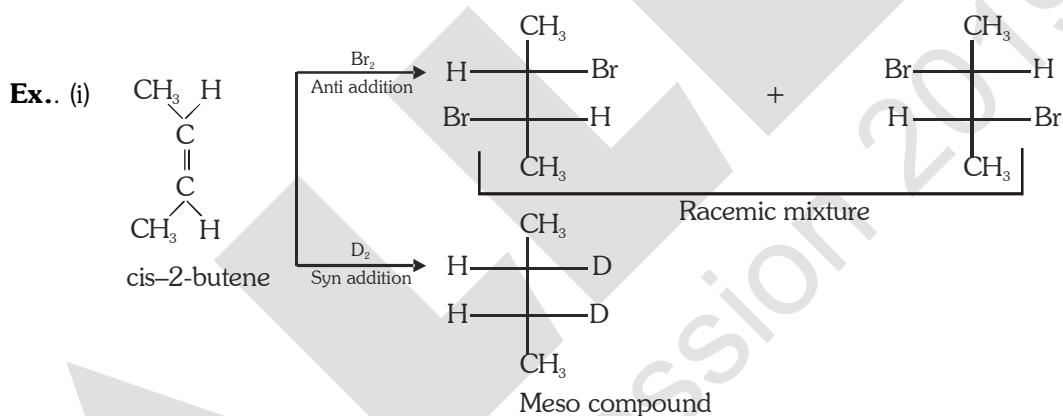
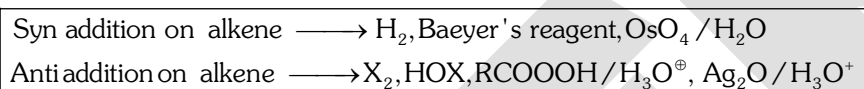
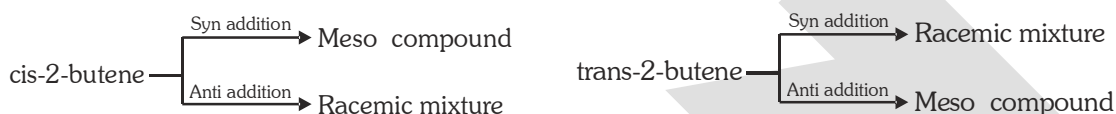
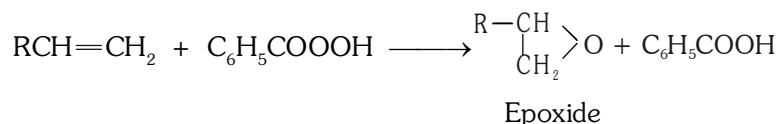
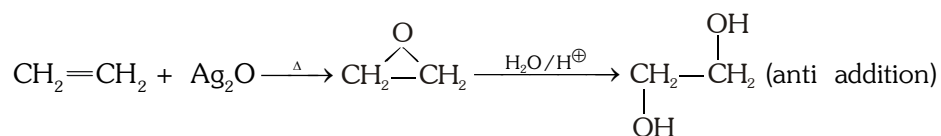


It may be noted that reaction with bromine water or Baeyer's reagent detects the presence of double bond (or unsaturation) in an alkene while ozonolysis helps in locating the position of the double bond.

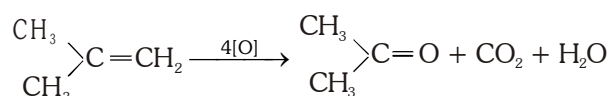
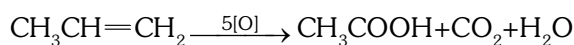
(3) Hydroxylation : Oxidation of carbon-carbon double bond to $\begin{array}{c} | \qquad | \\ -\text{C}-\text{C}- \\ | \qquad | \\ \text{OH} \quad \text{OH} \end{array}$ is known as hydroxylation.

(a) Oxidation by Baeyer's reagent (A test for unsaturation) : Alkenes on passing through dilute alkaline 1% cold KMnO_4 (i.e., Baeyer's reagent) decolourise the pink colour of KMnO_4 and gives brown ppt of MnO_2 . The reaction involves syn addition.



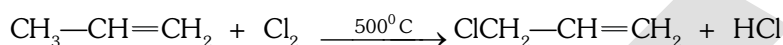
(d) By $\text{Ag}_2\text{O}/\Delta$:(a) Alkenes react with oxygen in the presence of Ag catalyst at $250^\circ\text{--}400^\circ\text{C}$ to form epoxide.

- (4) **Oxidation by strong oxidising agent (Oxidative cleavage):** The alkenes themselves are readily oxidised to acid or ketone by means of acid permanganate. If HCOOH is formed, it further oxidized to CO₂ and H₂O. Keep it in mind that no further oxidation of ketones will takes place.



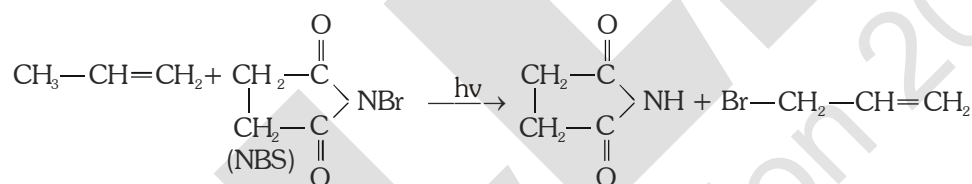
(C) Substitution Reaction (Allylic Substitution):

When alkenes are treated with low concentration of Cl₂ or Br₂ at high temperature or with NBS/hν one of their allylic hydrogen is replaced by halogen atom. Allylic position is the carbon adjacent to one of the unsaturated carbon atoms. It is free radical substitution.

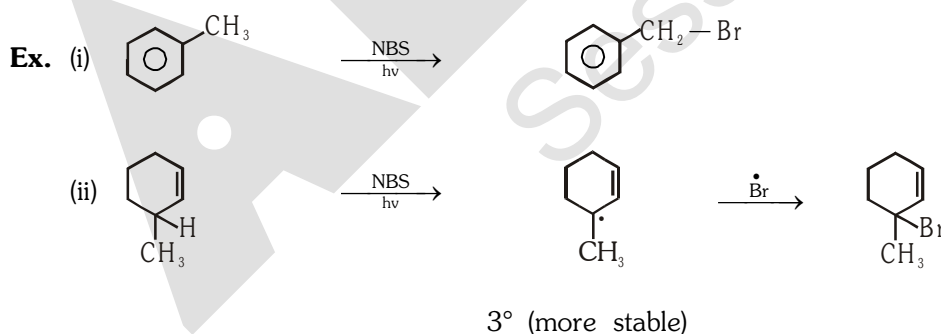


Allyl chloride
(3-Chloro-1-propene)

N-Bromosuccinimide (NBS) is an important reagent used for allylic bromination and benzylic substitution.



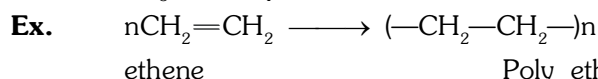
Substitution reaction is not given by ethene.



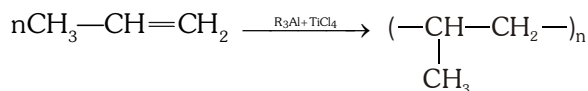
(D) Polymerization:

- (i) Two or more than two molecules of same compound unit with each other to form a long chain molecule with same empirical formula. This long chain molecule having repeating structural units called polymer, and the starting simple molecule as monomer and process is called addition polymerization.
- (ii) Molecular weight of polymer is simple multiple of monomer.
- (iii) Polymerization can be carried out by free radical or ionic mechanism.

- (iv) The presence of oxygen initiates free radical mechanism.
- (v) Addition polymerization can also be carried out by ionic mechanism by using Ziegler - Natta Catalysts ($R_3Al + TiCl_4$)



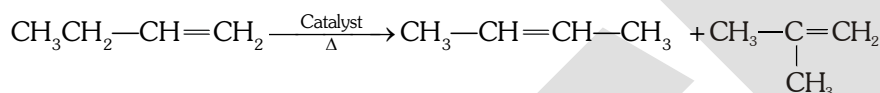
used in the manufacture of insulating Coating, sheeting and moulded products.



Polypropene or Koylene (Plastic)

(E) Isomerisation :

Alkene on heating to 500° to 700 °C or on heating in presence of catalyst $[\text{AlCl}_3 \text{ or } \text{Al}_2(\text{SO}_4)_3]$ undergo isomerisation.



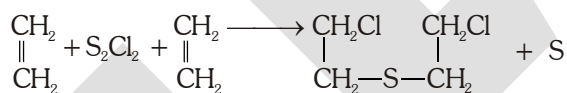
1-Butene

2-Butene

Isobutylene

Uses:

- (1) In plastic formation.
- (2) In oxy ethylene welding
- (3) As food preservatives and ripening fruits.
- (4) As general anaesthetic (C_2H_4 with 10% O_2)
- (5) In preparation of mustard gas



2,2' or (β , β') dichloro diethyl-sulphide
(mustard gas)

BEGINNER'S BOX-3

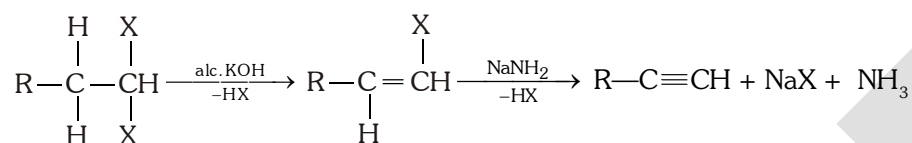
- The treatment of $(\text{CH}_3)_2\text{C}=\text{CHCH}_3$ with boiling KMnO_4 produces
 - $\text{CH}_3\text{COCH}_3 + \text{CH}_3\text{COOH}$
 - $\text{CH}_3\text{COCH}_3 + \text{CH}_3\text{CHO}$
 - $\text{CH}_3\text{CHO} + \text{CO}_2$
 - CH_3COCH_3 only
- An alkene on treating with hot acidified KMnO_4 gives 4-oxopentanoic acid. The alkene is
 - Pentene
 - 2-Pentene
 - 1-Methyl cyclobutene
 - 1, 2-Dimethyl cyclopropene
- The addition of Br_2 to cis-2-butene produces
 - (+) -2, 3-dibromobutane only
 - (-) -2, 3-dibromobutane only
 - racemic-2, 3-dibromobutane
 - meso -2, 3-dibromobutane
- The addition of HCl to 3, 3, 3-trichloropropene gives
 - $\text{Cl}_3\text{CCH}_2\text{CH}_2\text{Cl}$
 - $\text{Cl}_3\text{CCH}(\text{Cl})\text{CH}_3$
 - $\text{Cl}_2\text{CHCH}(\text{Cl})\text{CH}_2\text{Cl}$
 - $\text{Cl}_2\text{CHCH}_2\text{CHCl}_2$
- When ethene reacts with bromine in aqueous sodium chloride solution. The product(s) obtained is (are)
 - Ethylene dibromide only
 - Ethylene dibromide and 1-bromo-2-chloro ethane
 - 1-bromo-2-chloroethane only
 - Ethylene dichloride only

5.3.1 Introduction of Alkynes

Alkynes are unsaturated hydrocarbons and characterised by the presence of a triple bond between the two carbon atoms ($C \equiv C$). The carbon-carbon triple bond is also called acetylenic bond. It consists of a strong σ and two weak π bonds. Alkynes are isomers of alkadienes and cycloalkenes.

5.3.2 General Methods of Preparation

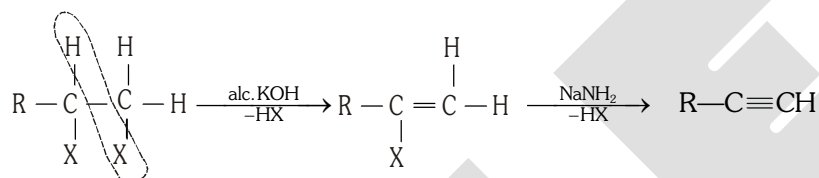
(1) From Gem dihalides (by dehydrohalogenation) : Dehydrohalogenation agents are : $NaNH_2$ (Sodamide) or Alc. KOH or ROH + RNa.



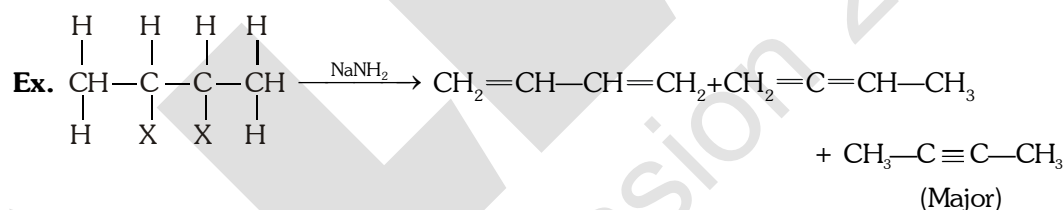
(Stable by resonance)

(Vinyl halide)

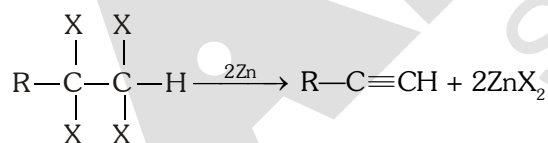
(2) From Vicinal dihalides (by dehydrohalogenation) :



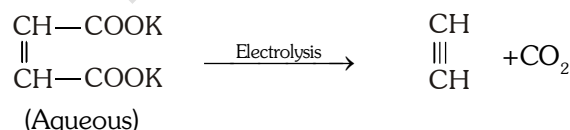
(a) Elimination of Vic. dihalides gives also alkadiene (1, 2 and 1, 3 alkadienes) but the major product is alkyne.



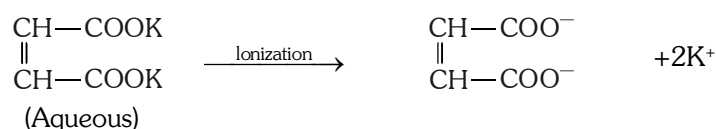
(3) Dehalogenation of tetrahalo alkane : By heating 1, 1, 2, 2 - tetra halo alkane with Zn dust.



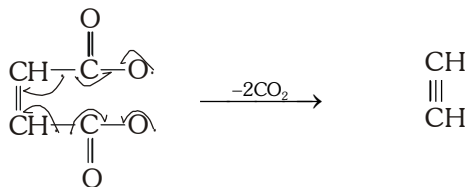
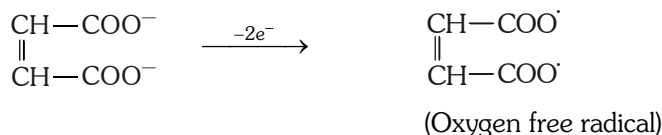
(4) From Kolbe's electrolysis : By the electrolysis of aqueous Solution. of sodium or potassium fumarate or maleate, acetylene is formed at anode.



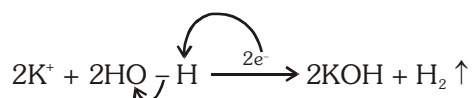
Mechanism :



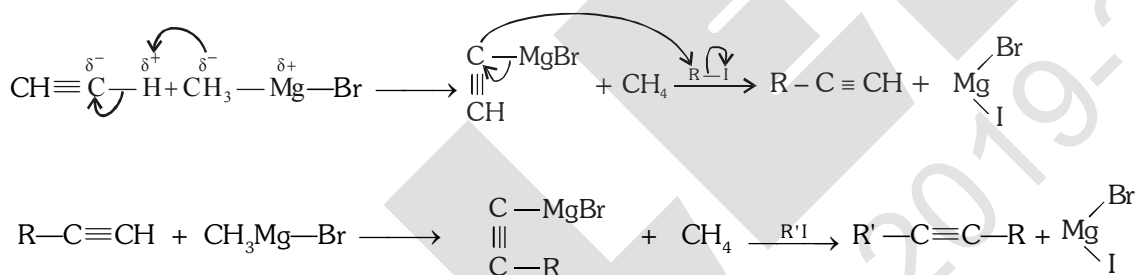
at anode (Alkyne and CO_2 gas is formed)



at cathode (KOH and H_2 gas is formed)



(5) **Preparation of higher alkynes by Grignard reagent** : By this method lower alkyne is converted in to higher alkyne

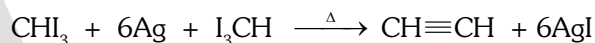


(6) **Preparation of Ethyne or Acetylene:**

(a) **From Metal carbide [Laboratory method]** : Acetylene is prepared in the laboratory by the action of water on calcium carbide.



(b) **From haloform [CHI_3 , CHCl_3]** : Pure acetylene is obtained when iodoform or chloroform is heated with Silver powder



5.3.3 Physical & Chemical Properties of Alkynes

Physical Properties

- (i) First three members [C_2 to C_4] are gases, from C_5 to C_{12} are liquid and after that they are solid.
- (ii) Alkynes are slightly soluble in H_2O but soluble in CCl_4 , benzene, acetone and alcohol.
- (iii) B.P., M.P. and densities of alkynes are comparatively more than alkenes and alkanes due to more polarisation.

$\text{B.P.} \propto \text{mol.wt.} \propto \frac{1}{\text{number of side chains}}$

Chemical Properties

The chemical properties of alkynes are due to two factors

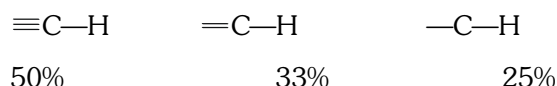
- (a) **Presence of π electrons** : Due to presence of loosely bonded π electrons, alkynes like alkenes, undergo easily electrophilic addition reaction.

Carbon-carbon triple bond is less reactive than the carbon-carbon double bond towards electrophilic addition reactions.

In addition to electrophilic additions, alkynes also undergo nucleophilic addition with nucleophiles

- (b) **Presence of acidic hydrogen atom** : The hydrogen atom attached to the triple bonded carbon can be removed by a strong base and hence acetylene and 1-alkynes are considered as weak acids.

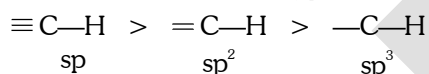
Explanation : The amounts of s-character in various types of C—H bonds is as-



Since s electrons are closer to the nucleus than the p electrons, the electrons present in a bond having more s-character will be closer to nucleus. Due to high s-character of the C—H bond in alkyne (s=50%) the electrons constituting this bond are more strongly held by the carbon nucleus, with the result the H present on $\equiv\text{C}-\text{H}$ can be easily removed as proton.

GOLDEN KEY POINTS

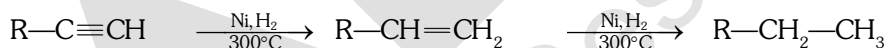
- The acidic nature of the three types of $-\text{C}-\text{H}$ bonds as



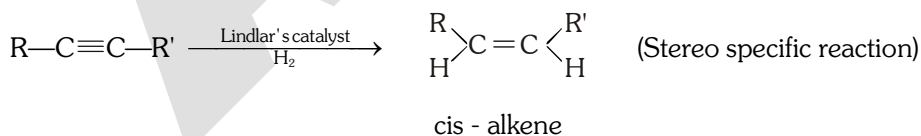
- Relative acidic order $\text{H}_2\text{O} > \text{ROH} > \text{HC} \equiv \text{CH} > \text{HNH}_2 > \text{CH}_2 = \text{CH}_2 > \text{CH}_3 - \text{CH}_3$

(1) Addition reaction

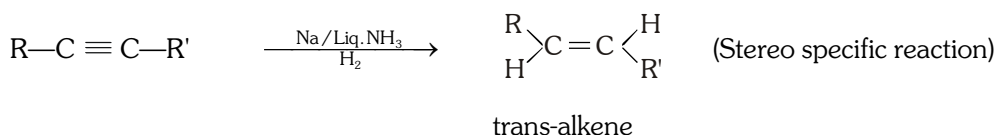
- (a) **Addition of hydrogen** : Alkynes reacts with hydrogen in presence of a catalyst. In presence of Pt., Pd or Ni alkynes give alkanes with H_2



In presence of Lindlar's catalyst [Pd / CaCO_3 + quinoline or Nickel boride] alkynes give cis - alkene

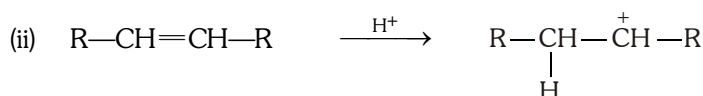
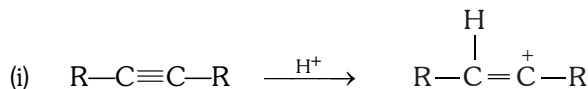


In presence of Na/ NH_3 alkynes give trans-alkene. **(Birch Reduction)**



- (b) Electrophilic addition :** Addition reactions where the addition is initiated by electrophile (positive group). The characteristic reaction of alkynes is electrophilic addition but the reactivity of alkynes towards electrophilic addition is less than alkenes because in $C\equiv C$, the π electrons are tightly held by carbon nuclei and so they are less easily available for reaction with electrophiles.

Another reasons is : The intermediates when an electrophile attack on alkene and alkynes are :

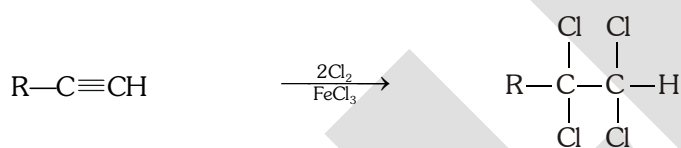


Stability of intermediates :

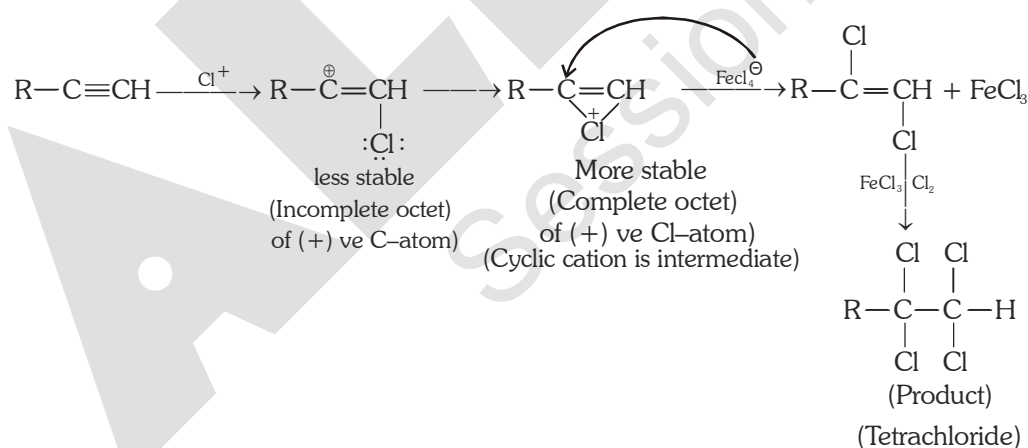
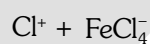


So we can say that alkenes are more reactive towards electrophilic addition reaction.

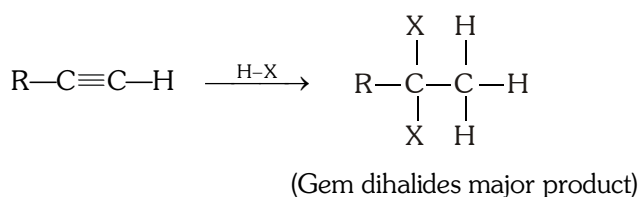
- (i) Addition of Halogens :** Alkynes react with Cl_2 or Br_2 in dark in presence of metal halide and form di and tetra halo derivatives.



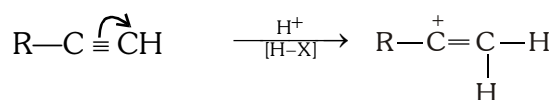
Mechanism :



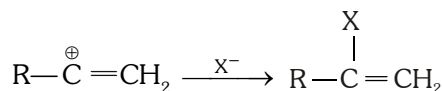
- (ii) Addition of halogen acids (H - X) :** Addition according to Markovnikov's Rule.



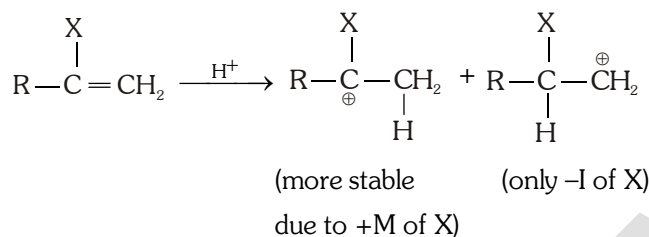
Mechanism :



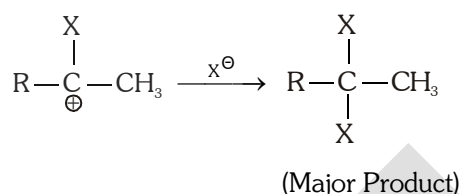
(Intermediate is carbocation)



Further



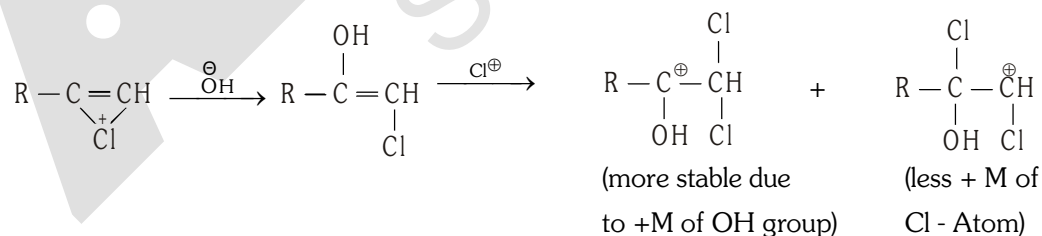
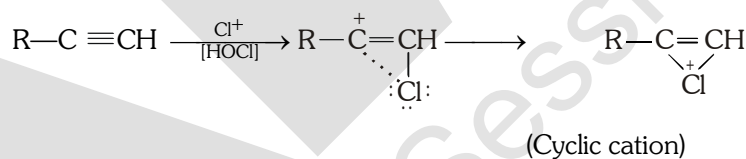
so



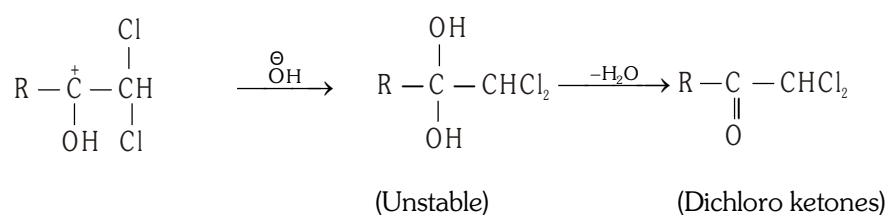
- (iii) Addition of HOX :** Alkynes react with hypohalous acids according to Markovnikov's rule and form gem diol, which are unstable, lose a molecule of water and form dihalo aldehyde or dihalo ketones.

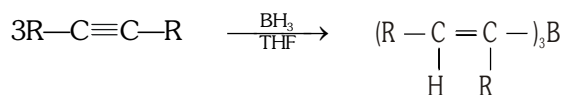


Mechanism :

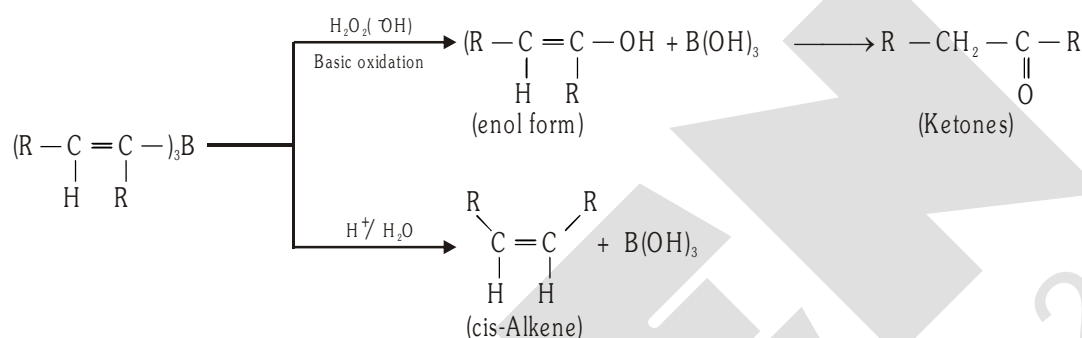
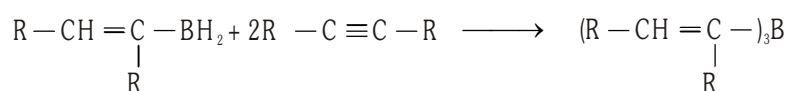
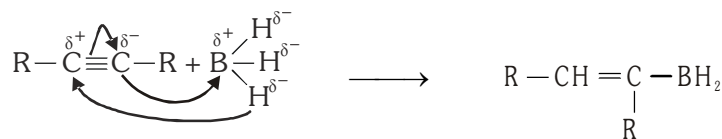


so

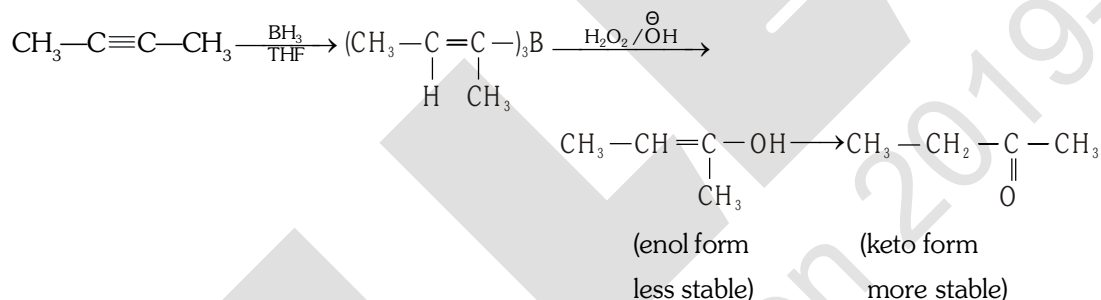




Since BH_3 is not available as monomer so a solvent THF is used for the stability of BH_3 .

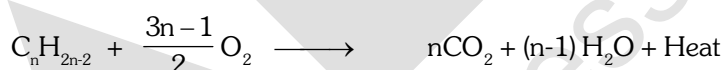


Ex.



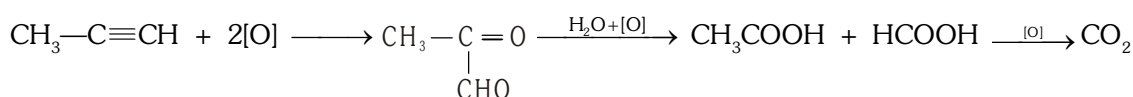
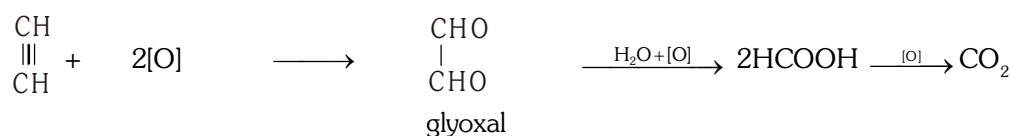
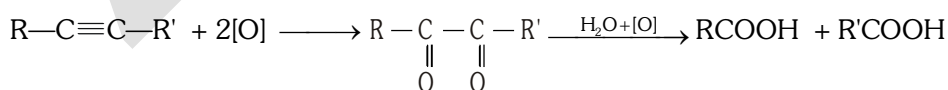
(2) Oxidation Reactions

(a) Combustion :

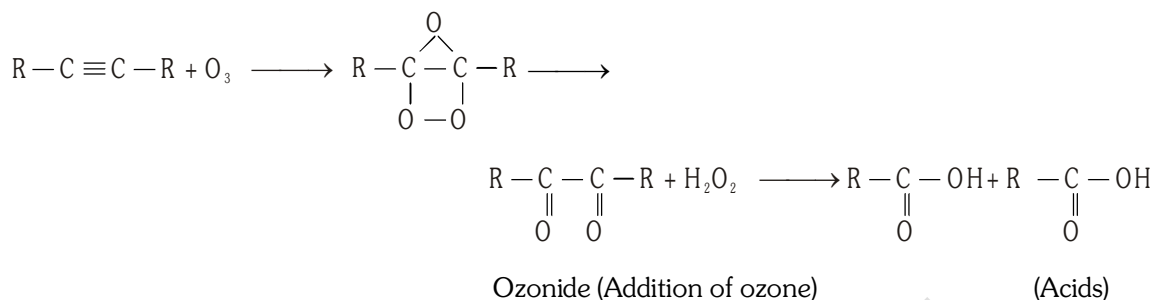


The combustion of acetylene is used for welding and cutting of metals in which oxy-acetylene flame having high temp (3000°C) is produced.

(b) **Oxidation with acidic $KMnO_4$:** In presence of acidic $KMnO_4$, alkynes are oxidised to monocarboxylic acids.

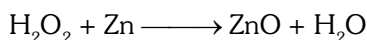


(c) **Oxidation with ozone (O₃)** : In the ozonolysis both sp-C-atoms are converted into $\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—C—} \\ \parallel \quad \parallel \\ \text{O} \quad \text{O} \end{array}$ group.

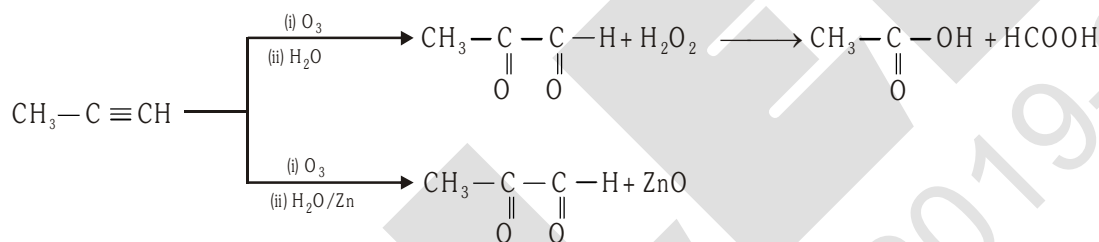


In this reaction H₂O₂ is oxidant which oxidise $\begin{array}{c} \text{R—C—C—R} \\ \parallel \quad \parallel \\ \text{O} \quad \text{O} \end{array}$ into acids.

But if we use some amount of Zn as reductant with H₂O then it reduce H₂O₂ so oxidation does not take place



Ex.

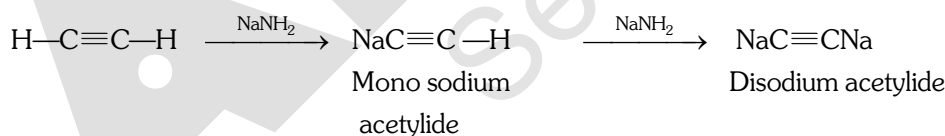


(3) Substitution Reaction : (Formation of metallic derivatives)

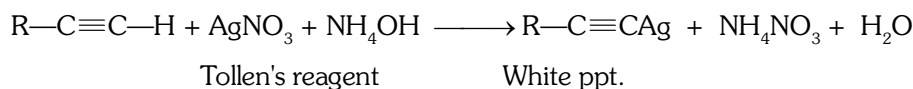
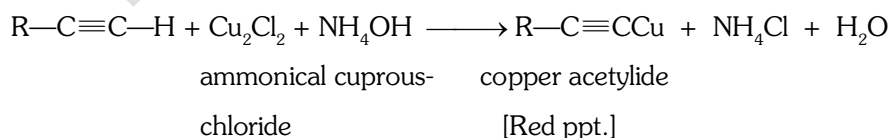
Only 1-alkynes give substitution reaction and show acidic characters $\begin{array}{c} \delta^- \\ \text{C} \\ \delta^+ \end{array} \text{—H}$

Acetylene is dibasic acid where as propyne is monobasic means acetylene can give two H⁺ where as propyne can give one H⁺.

(a) **Formation of sodium acetylides** : Acetylene and 1-alkynes react with sodamide to form acetylides

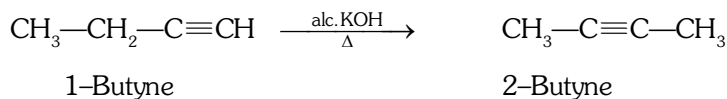


(b) **Formation of copper and silver acetylides** : Copper and silver acetylides are obtained by passing 1-alkynes in the ammonical Solution. of cuprous chloride and silver nitrate (Tollen's reagent) respectively.

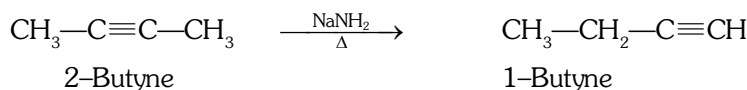


These reactions are used for detecting the presence of acetylenic hydrogen. These are test to distinguish alkenes and alkynes or 1-alkynes and 2-alkynes.

(4) **Isomerisation:** When alkyne-1 is heated with alc. KOH alkyne-2 is obtained.



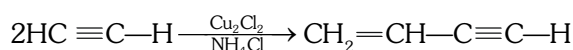
When alkyne -2 is heated with NaNH_2 alkyne -1 is obtained



(5) **Polymerisation :**

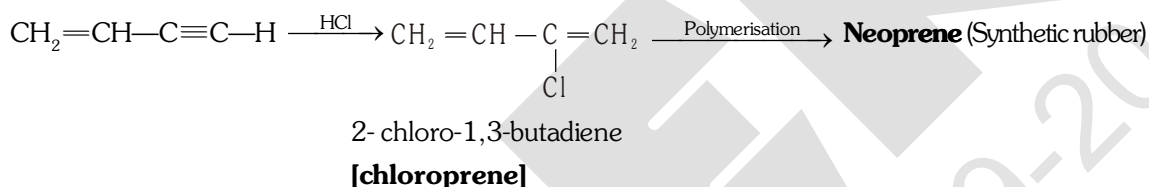
(a) **Linear polymerisation :**

Dimerisation : When two molecules of acetylene passed through a Solution. of Cu_2Cl_2 and NH_4Cl a vinyl acetylene is obtained.

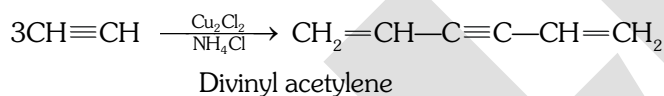


mono vinyl acetylene (butenyne)

When vinyl acetylene react with HCl then chloroprene is obtained.



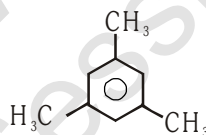
Trimerisation : 3 molecules of acetylene.



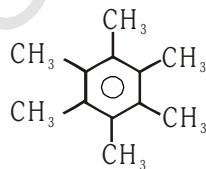
(b) **Cyclic polymerisation :** When alkyne is passed through red hot metallic tube, cyclic polymerisation takes place with the formation of aromatic compound



Benzene



Mesitylene



Hexa methyl benzene

Uses of Acetylene

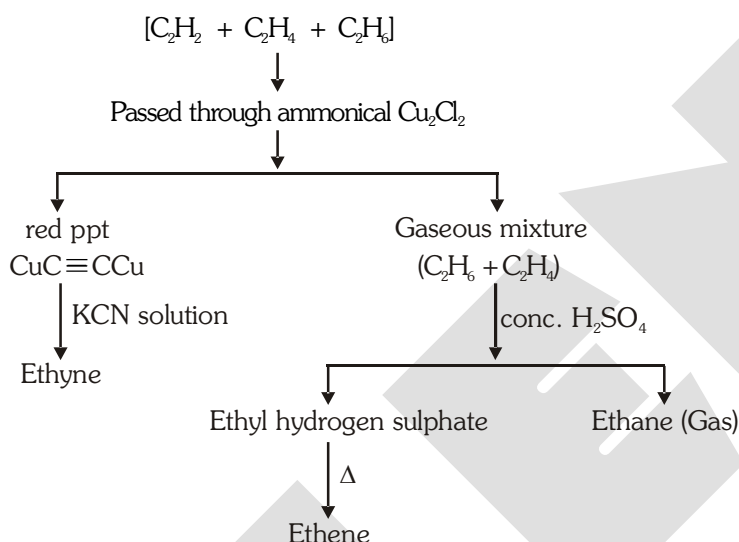
- (i) Oxyacetylene flame used in welding and cutting
- (ii) Acetylene is used as an illuminant
- (iii) Acetylene is used for artificial ripening of fruits
- (iv) Used for manufacture of acetaldehyde, acetic acid, ethyl alcohol, westron, westrosol, PVC, PVA, Chloroprene, butadiene, Lewisite etc.
- (v) It is used as a general anaesthetic.

Test for alkynes :

- (i) Decolourization of Br_2 in CCl_4 Solution..
- (ii) Decolourisation of 1% alkaline KMnO_4 Solution..
- (iii) 1- alkynes give white ppt. with ammonical AgNO_3 and red ppt with ammonical cuprous chloride Solution..

Note : (i) and (ii) tests are used for determination of unsaturation (i.e, presence of double or triple bond in any compound). (iii) Test is used for distinguish between alkenes and 1-alkynes or 1-alkyne and 2-alkyne.

Separation of ethane, ethene and ethyne :



BEGINNER'S BOX-4

1. Propyne and HCl in presence of peroxide gives :
 (1) 1,1-dichloro propane (2) 1,2-dichloro propane (3) 1,3-dichloro propane (4) 2,2-dichloro propane
2. $\text{CH}_3-\text{C}\equiv\text{CH} \xrightarrow[\text{(ii) H}_2\text{O}_2 / -\text{OH}]{\text{(i) BD}_3} \text{A}$, A is
 (1) $\text{CH}_3-\underset{\text{D}}{\text{CH}}-\text{CH}_2-\text{OH}$ (2) $\text{CH}_3-\underset{\text{O}}{\text{C}}-\text{CH}_2-\text{D}$ (3) $\text{CH}_3-\underset{\text{D}}{\text{CH}}-\text{CHO}$ (4) $\text{CH}_3-\underset{\text{D}}{\text{CH}}-\underset{\text{O}}{\text{C}}-\text{D}$
3. Which of the following decolourise 1% alkaline KMnO_4
 (1) Propene (2) Propyne (3) 2-butyne (4) All of them
4. 1-Alkyne and 2-Alkyne can not be distinguished by :
 (1) Ammonical AgNO_3 (2) Baeyer's reagent (3) Br_2 in CCl_4 (4) 2 and 3 both
5. $\text{CH}\equiv\text{CH} \xrightarrow{\text{CH}_3\text{MgBr}} \text{A (gas)} + \text{B}$, $\text{B} \xrightarrow{\text{CH}_3-\text{I}} \text{C}$. C is :
 (1) CH_4 (2) CH_3-CH_3 (3) $\text{CH}\equiv\text{C}-\text{CH}_3$ (4) $\text{CH}_3-\text{C}\equiv\text{C}-\text{I}$

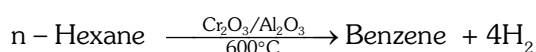
5.4 AROMATIC HYDROCARBONS [ARENES]

5.4.1 Introduction of Arenes

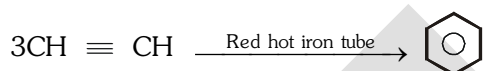
- Arenes are cyclic, planar and follow Huckel's rule.
- There should be cyclic resonance in Arenes.
- General formula of Arenes are $C_n H_{2n-6y}$ where n = Number of carbons
 y = Number of rings
- Characteristic reaction of arenes is Electrophilic substitution reaction (ESR).
- Arenes are cyclic unsaturated compounds but do not give test of unsaturation with Br_2/CCl_4 or alk. $KMnO_4$.
- Main source of Arenes is coaltar.
- They have higher percentage of carbon so burn with smoky flame.

5.4.2 General Methods of Preparation

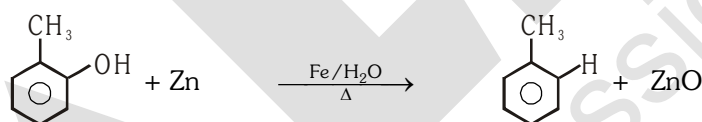
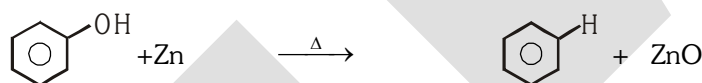
(i) From alkanes (By cyclisation or Aromatisation) Hydroforming or catalytic reforming



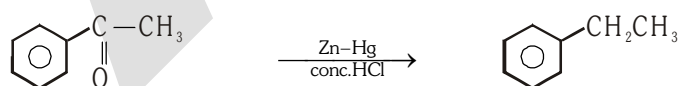
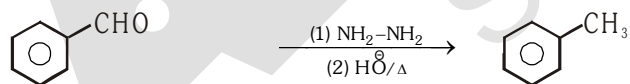
(ii) From alkyne (By cyclic polymerisation) When acetylene is passed through red hot metallic tube cyclic polymerisation takes place and benzene is formed.



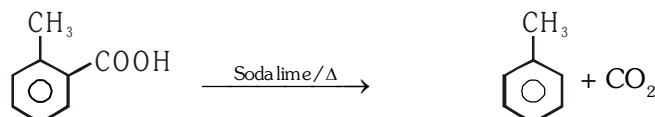
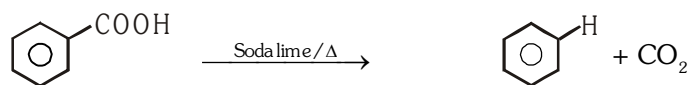
(iii) From Phenolic compounds (By deoxygenation) :



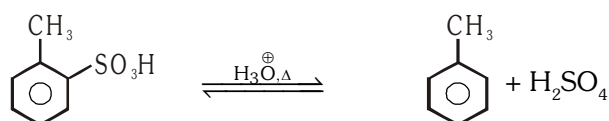
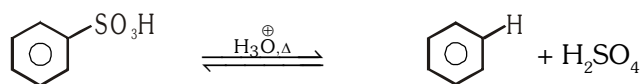
(iv) From carbonyl compounds (By reduction) :



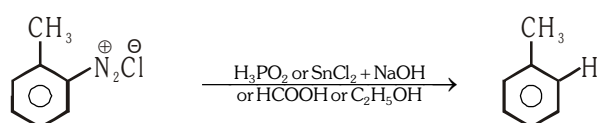
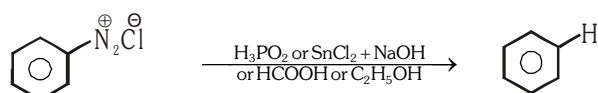
(v) From carboxylic acids (By decarboxylation) :



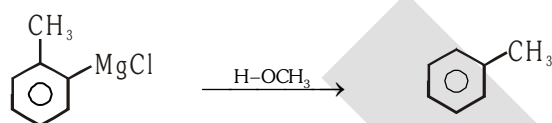
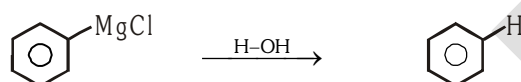
(vi) From sulphonic acids (By hydrolysis) :



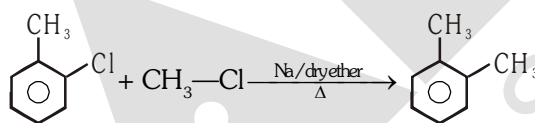
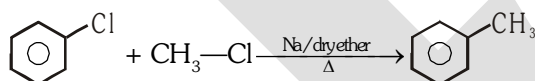
(vii) By diazonium salts :



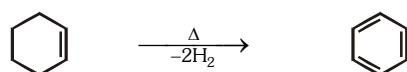
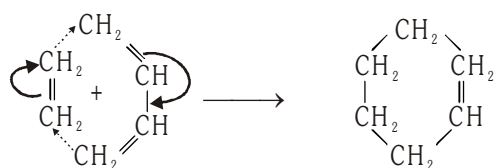
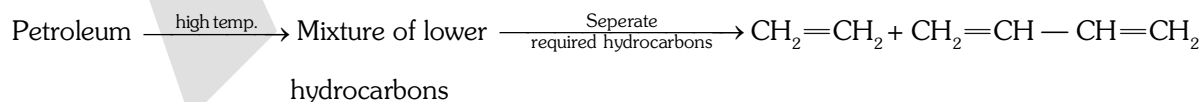
(viii) By Grignard reagent :



(ix) By Wurtz fittig reaction :



(x) From petroleum :



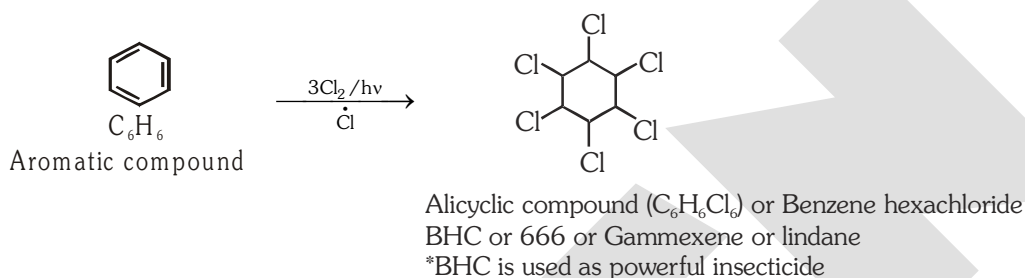
5.4.3 Physical & Chemical Properties of Arenes

Physical properties :

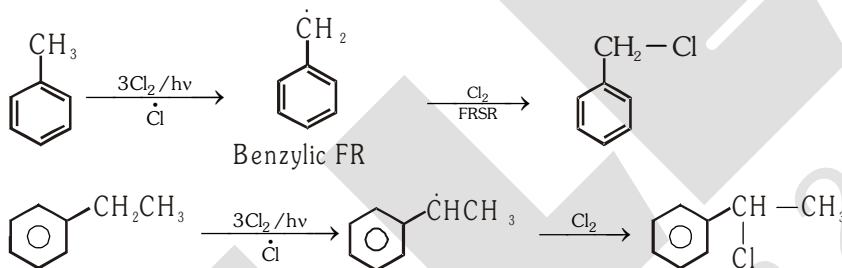
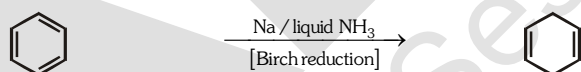
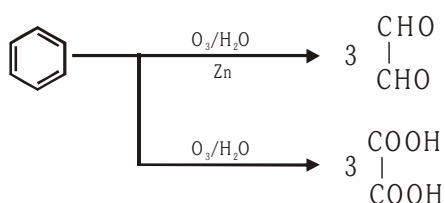
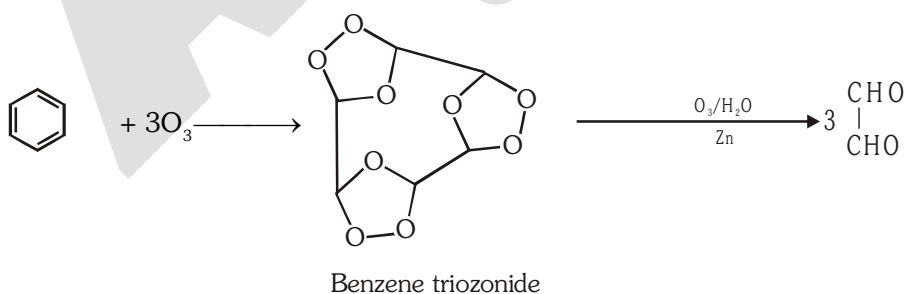
- (i) Benzene is colourless liquid [B.P. is 80°C]
- (ii) Benzene is insoluble in H_2O and density less than H_2O
- (iii) Benzene is used as a solvent and it is soluble in organic solvents.
- (iv) It is highly inflammable and burns with smoky flame.

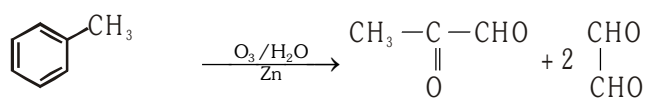
Chemical properties : Benzene show following types of reaction –

- (A) Addition reactions (B) Electrophilic substitution reactions (C) Oxidation reactions

(A) Addition reaction :**(i) Addition of X_2** 

Formation of BHC from C_6H_6 is an example of free radical addition reaction.

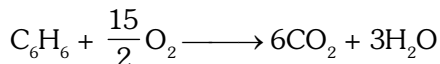
**(ii) Addition of H_2 :****(a) Complete hydrogenation :****(b) Partial hydrogenation :****(iii) Addition of O_3 :**



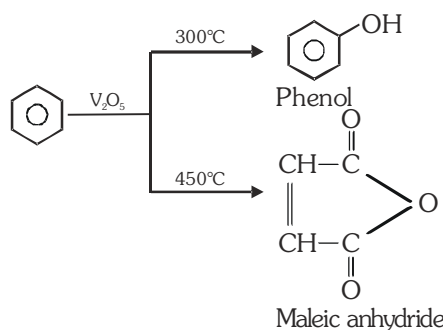
Addition of 3 mole Cl_2 or 3 mole H_2 or 3 mole O_3 on benzene show presence of 3 double bonds in benzene. Benzene does not give addition reaction with Br_2/CCl_4 or alkaline KMnO_4

(C) Oxidation reactions :

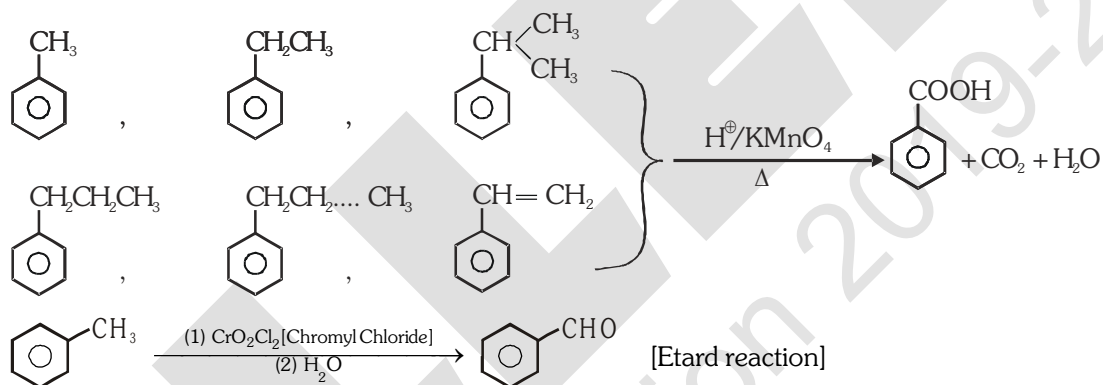
(i) Combustion :



(ii) Catalytic Oxidation :



(iii) Side chain Oxidation : Atleast one benzylic -H containing alkyl benzene gives benzoic acid in presence of strong oxidising agent.



Note : t-butyl benzene does not give benzoic acid in presence of H^+/KMnO_4 , since t-butyl benzene does not contain benzylic-H



ANSWER KEY

BEGINNER'S BOX-1	Que.	1	2	3	4	5	6	7
	Ans.	2	4	3	3	2	2	1

BEGINNER'S BOX-2	Que.	1	2	3	4	5	6	
	Ans.	4	1	1	3	3	1	

BEGINNER'S BOX-3	Que.	1	2	3	4	5	
	Ans.	1	3	3	1	2	

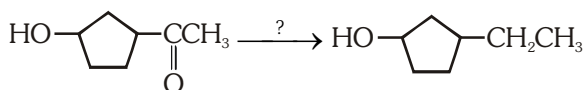
BEGINNER'S BOX-4	Que.	1	2	3	4	5	
	Ans.	4	3	4	4	3	

EXERCISE-I (Conceptual Questions)

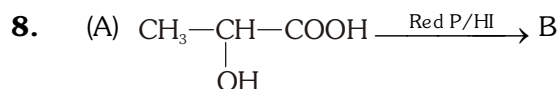
Build Up Your Understanding

ALKANES

- The order of reactivity of alkyl halides in Wurtz reaction is
 - (1) $R-I > R-Br > R-Cl$
 - (2) $R-I < R-Br < R-Cl$
 - (3) $R-Br > R-I < R-Cl$
 - (4) $R-I > R-Cl > R-Br$
- Kolbe's electrolysis of a mixture of pot. Propanoate and pot. 3-Methylbutanoate gives
 - (1) Butane and isobutane
 - (2) Butane and 2,5-dimethylhexane
 - (3) Butane, 2,5-dimethylhexane and isohexane
 - (4) Butane and isohexane
- The Corey-House alkane synthesis is carried out by treating an alkyl halide with
 - (1) Lithium metal
 - (2) Copper metal
 - (3) Lithium metal followed by reaction with cuprous iodide and then treating the product with an alkyl halide
 - (4) Cuprous iodide followed by reaction with alkyl halide
- Which of the following acids on decarboxylation gives isobutane
 - (1) 2,2-Dimethyl butanoic acid
 - (2) 2,2-dimethyl propanoic acid
 - (3) 3-Methyl pentanoic acid
 - (4) 2-Methyl butanoic acid
- Which of the following compound is not suitable to obtain from wurtz reaction ?
 - (1) ethane
 - (2) butane
 - (3) isobutane
 - (4) hexane
- When ethyl chloride and n-propyl chloride undergoes wurtz reaction which is not obtained
 - (1) n-butane
 - (2) n-pentane
 - (3) n-hexane
 - (4) isobutane
- Which of the following reagent can be used for following conversion

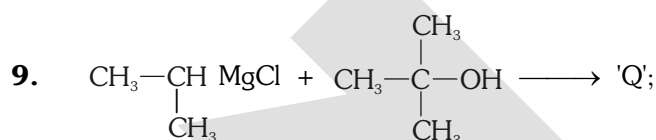


- (1) $\text{Zn} - \text{Hg}/\text{HCl}$
- (2) $\text{Red P} + \text{HI}$
- (3) $\text{NH}_2 - \text{NH}_2/\text{OH}^-$
- (4) All of them



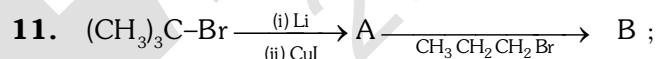
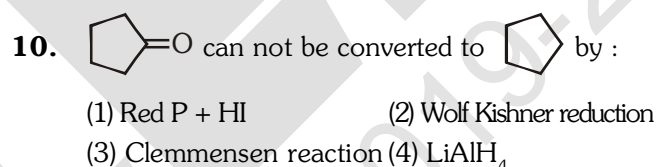
'B' loses its optical activity because of

- (1) Chirality of the molecule destroyed
- (2) Symmetry of molecule is destroyed
- (3) Spatial arrangement is changed
- (4) Racemic mixture is formed



What is 'Q' ?

- (1) isobutane
- (2) isopropane
- (3) tert. butyl chloride
- (4) propane



'B' is

- (1) $\text{CH}_3(\text{CH}_2)_2\text{C}(\text{CH}_3)_3$
- (2) $(\text{CH}_3)_3\text{C}-\text{C}(\text{CH}_3)_3$
- (3) $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$
- (4) $(\text{CH}_3)_2\text{CH}_2\text{C}(\text{CH}_3)_3$

12. Which of the following reactions does not involve a C—C bond formation

- (1) Hydrolysis of a Grignard reagent
- (2) Combination of two alkyl free radicals
- (3) Corey-House synthesis of alkanes
- (4) $\text{RNa} + \text{R}-\text{Br} \longrightarrow \text{R}-\text{R} + \text{NaBr}$

13. The highest boiling point is expected for

- (1) Isooctane
- (2) 2,2,3,3-tetramethylbutane
- (3) n-octane
- (4) n-butane

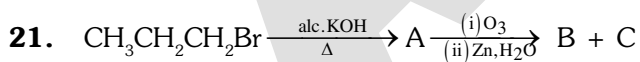
14. Pyrolysis of alkanes is a

- (1) Nucleophilic addition reaction
- (2) Free radical substitution reaction
- (3) Electrophilic addition reaction
- (4) Free radical elimination reaction

15. The antiknocking compound is
 (1) TEL (2) Diethylzinc
 (3) Dimethylcadmium (4) Tetramethyl tin
16. Which of the following reactions of methane is incomplete combustion :-
 (1) $2\text{CH}_4 + \text{O}_2 \xrightarrow{\text{Cu}/523\text{K}/100\text{atm.}} 2\text{CH}_3\text{OH}$
 (2) $\text{CH}_4 + \text{O}_2 \xrightarrow{\text{Mo}_2\text{O}_3} \text{HCHO} + \text{H}_2\text{O}$
 (3) $\text{CH}_4 + \text{O}_2 \longrightarrow \text{C(s)} + 2\text{H}_2\text{O(l)}$
 (4) $\text{CH}_4 + 2\text{O}_2 \longrightarrow \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(l)}$

ALKENE

17. Ozonolysis of 3-Methyl-1-butene gives a mixture of
 (1) Propanal and ethanal
 (2) Propanone and ethanal
 (3) 2-Methylpropanal and methanal
 (4) Butanone and methanal
18. Which alkene gives acetone only on ozonolysis
 (1) Isobutylene
 (2) 2,3-Dimethyl-1-butene
 (3) 2,3-Dimethyl-2-butene
 (4) 3,3-Dimethyl-1-butene
19. Oxidation of isobutylene with acid potassium permanganate gives
 (1) Acetone + CO_2
 (2) Acetic acid
 (3) Acetic acid + CO_2
 (4) Acetic acid + acetone
20. Which of the following reactions is used for locating the position of double bond in an alkene
 (1) Hydroboration (2) Hydroxylation
 (3) Chlorohydroxylation (4) Ozonolysis



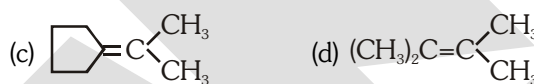
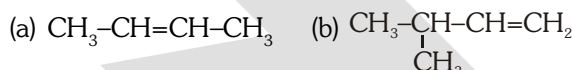
In the above reaction A, B and C are given by the set

- (1) Propylene, acetone, formaldehyde
 (2) Propene, ethanal, methanal
 (3) Propyne, acetaldehyde, formaldehyde
 (4) Propylene, propionaldehyde, formaldehyde
22. Which one of the following has the smallest heat of hydrogenation per mole
 (1) 1-butene (2) Trans-2-butene
 (3) Cis-2-butene (4) Propene

23. An alkene "A" contains three C-C, eight C-H σ -bonds and one C-C π -bond. "A" on ozonolysis gives two moles of an aldehyde of molar mass 44. Which of the following is the IUPAC name of A.
 (1) But-1-ene (2) But-2-ene
 (3) Pent-2-ene (4) But-2-yne

24. The molecules having dipole moment are :-
 (1) 2,2-dimethylpropane
 (2) trans pent-2-ene
 (3) cis-hex-3-ene
 (4) 2 and 3

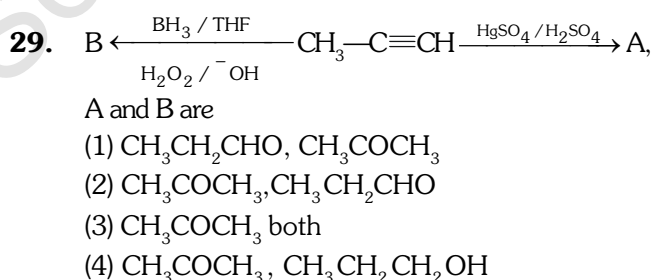
25. Which of the following alkenes on ozonolysis give a mixture of ketones only?



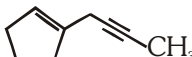
- (1) a and b (2) b and c
 (3) b and d (4) c and d

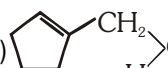
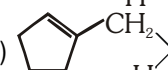
ALKYNES

26. Kolbe's electrolysis of sodium maleate at anode gives.
 (1) Only acetylene (2) Only CO_2
 (3) Acetylene + CO_2 (4) CO_2 + H_2
27. The pH of solution in Kolbe's electrolysis
 (1) increase with time (2) decrease with time
 (3) remains constant (4) nothing can be said
28. Which of the following compound will not give a precipitate with Tollen's reagent
 (1) ethyne (2) 1-butyne
 (3) 3-methyl-1-butyne (4) 1-pentene



30. $\text{MeCH}_2\text{C}\equiv\text{CH} \xrightarrow{\text{NH}_3 / \text{NaNH}_2} \text{A} \xrightarrow{\text{Et Br}} \text{B}$,
 A and B are
 (1) $\text{MeCH}_2\text{C}\equiv\text{CNa}$, $\text{MeCH}_2\text{C}\equiv\text{C-Et}$
 (2) $\text{MeCH}_2\text{CH=CH}_2$, $\text{MeCH}_2\text{-CH=CH-Et}$
 (3) $\text{MeCH}_2\text{CH=CHNH}_2$, $\text{MeCH}_2\text{CH=CH-NHBr}$
 (4) $\text{MeCH}_2\text{C}\equiv\text{C-NH}_2$, $\text{MeC}\equiv\text{C-NH-Br}$

31.  $\xrightarrow[\text{Liq. NH}_3]{\text{Na}}$ product will be :

- (1) 
 (2) 
 (3) 
 (4) 

32. To distinguish between propene and propyne, the reagent would be -

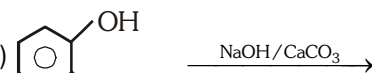
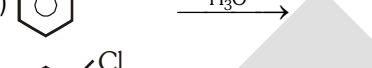
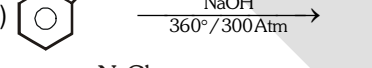
- (1) Bromine (2) Alkaline KMnO_4
 (3) Ammonical silver nitrate (4) Ozone

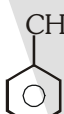
33. When treated with ammonical cuprous chloride, which one among the following forms red precipitate ?

- (1) C_2H_6 (2) C_2H_4 (3) C_2H_2 (4) C_6H_6

AROMATIC HYDROCARBONS

34. In which of the following reactions benzene is obtained

- (1) 
 (2) 
 (3) 
 (4) 

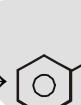
35. A $\xleftarrow{\text{Br}_2}$  $\xrightarrow{\text{KMnO}_4}$ B ; A and B

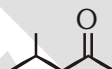
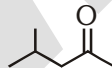
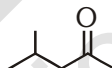
respectively are

- (1) o-bromo styrene, benzoic acid
 (2) p-bromostyrene, benzaldehyde
 (3) m-bromostyrene, benzaldehyde
 (4) Styrene dibromide, benzoic acid.

36. The ozonolysis product of 1, 2-dimethyl benzene is/are :-

- (1) Only $\text{CH}_3-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{CH}_3$
 (2) $\text{CH}_3-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{H} + \text{CH}_3-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{CH}_3$
 (3) $\text{H}-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{H} + \text{CH}_3-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{CH}_3$
 (4) $\text{CH}_3-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{CH}_3 + \text{H}-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{H} + \text{CH}_3-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{H}$

37. Benzene $\xrightarrow{\text{A}}$  $\xrightarrow{\text{B}}$  ; A and B respectively are

- (1) $\text{Zn} - \text{Hg} + \text{HCl}$, 
 (2)  Cl , LiAlH_4
 (3)  Cl , NaBH_4
 (4)  Cl , $\text{Zn} - \text{Hg} + \text{HCl}$

38. Which of the following reaction does not gives benzoic acid

- (1) $\text{C}_6\text{H}_5-\text{CH}_3 \xrightarrow[\text{KMnO}_4]{\text{Acidic}}$
 (2) $\text{C}_6\text{H}_5-\text{CH}_3 \xrightarrow[\text{H}_2\text{O}]{\text{CrO}_2\text{Cl}_2} \xrightarrow{[\text{O}]}$
 (3) $\text{C}_6\text{H}_5\text{C}\equiv\text{CH} \xrightarrow[\text{KMnO}_4]{\text{Acidic}}$
 (4) $\text{C}_6\text{H}_5-\text{C}(\text{CH}_3)_2-\text{CH}_3 \xrightarrow[\text{KMnO}_4]{\text{Acidic}}$

EXERCISE-I (Conceptual Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	1	3	3	2	3	4	3	1	4	4	1	1	3	4	1
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	3	1	4	2	2	2	4	4	3	1	4	2	1
Que.	31	32	33	34	35	36	37	38							
Ans.	2	3	3	2	4	4	4	4							

EXERCISE-II (Previous Year Questions)

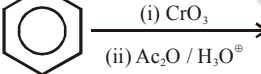
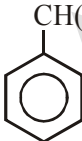
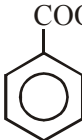
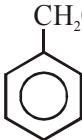
AIPMT 2007

1. Reduction of aldehydes and ketones into hydrocarbons using zinc amalgam and conc. HCl is called
 - (1) Cope reduction
 - (2) Dow reduction
 - (3) Wolf-kishner reduction
 - (4) Clemmensen reduction
2. Which of the compounds with molecular formula C_5H_{10} yields acetone on ozonolysis
 - (1) 3-Methyl-1-butene
 - (2) Cyclopentene
 - (3) 2-Methyl-1-butene
 - (4) 2-Methyl-2-butene

AIPMT 2010

- 3.** Liquid hydrocarbons can be converted to a mixture of gaseous hydrocarbons by :-
- (1) Hydrolysis
 - (2) Oxidation
 - (3) Cracking
 - (4) Distillation under reduced pressure

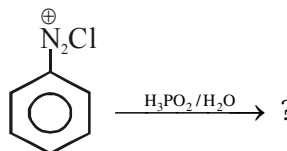
AIIMS 2013

4.  Product A is :-
- (1) 
- (2) 
- (3) 
- (4) 

AIPMT/NEET & AIIMS (2006-2018)

AIIMS 2014

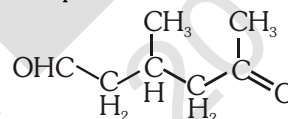
5. The product of the following reaction is :-



- (1) 
- (2) 
- (3) 
- (4) 

AIPMT 2015

- 6.** A single compound of the structure :-



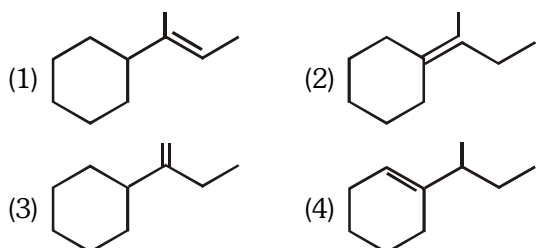
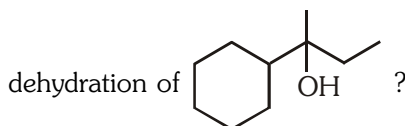
is obtainable from ozonolysis of which of the following cyclic compounds ?

- (1) 
- (2) 
- (3) 
- (4) 

Re-AIPMT 2015

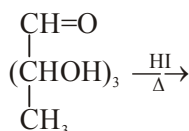
7. 2,3-Dimethyl-2-butene can be prepared by heating which of the following compounds with a strong acid ?
- (1) $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_2-\text{CH}_3$
 - (2) $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}=\text{CH}_2$
 - (3) $(\text{CH}_3)_2\text{CH}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}=\text{CH}_2$
 - (4) $(\text{CH}_3)_3\text{C}-\text{CH}=\text{CH}_2$

8. The oxidation of benzene by V_2O_5 in the presence of air produces :
- (1) benzoic acid (2) benzaldehyde
(3) benzoic anhydride (4) maleic anhydride
9. Which of the following is not the product of



AIIMS 2016

10. Identify product of the reaction



- (1) n-Pentane (2) 2-Pentane
(3) 1,3 Pentadiene (4) 3-Pentene

NEET(UG) 2018

11. Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is
- (1) $\text{CH}\equiv\text{CH}$ (2) $\text{CH}_2=\text{CH}_2$
(3) CH_3-CH_3 (4) CH_4
12. The compound C_7H_8 undergoes the following reactions :

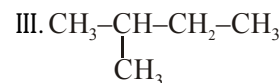
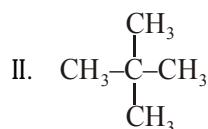


The product 'C' is

- (1) m-bromotoluene
(2) o-bromotoluene
(3) 3-bromo-2,4,6-trichlorotoluene
(4) p-bromotoluene

AIIMS 2018

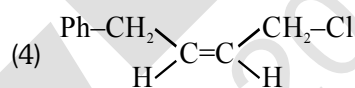
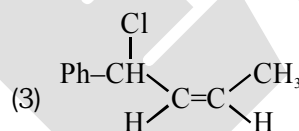
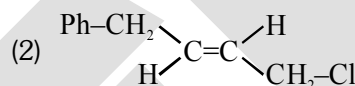
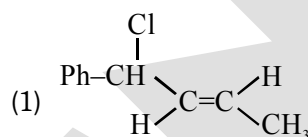
13. Arrange the following alkanes in decreasing order of their boiling point
- I. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$



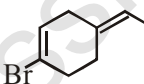
- (1) I > II > III (2) II > I > III
(3) I > III > II (4) III > I > II

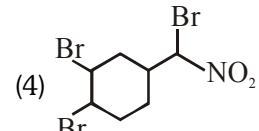
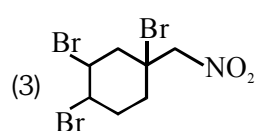
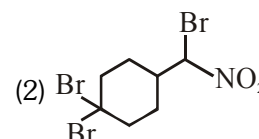
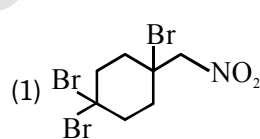
14. $\text{Ph}-\text{CH}_2-\text{C}\equiv\text{C}-\text{CH}_3 \xrightarrow[\text{Liq NH}_3]{\text{Na}} \text{A} \xrightarrow{\text{Cl}_2/h\nu} \text{B}$

The final product 'B' is :-



15. Benzene is :-
- (1) Weakly attracted in the magnetic field
(2) Strongly attracted in the magnetic field
(3) Strongly repelled in magnetic field
(4) Weakly repelled in magnetic field

16.  $\xrightarrow[\Delta]{2\text{HBr}}$ Product.
- Product of reaction is :-



EXERCISE-II (Previous Year Questions)

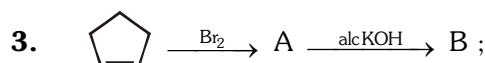
ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	4	4	3	4	2	4	4	4	4	1	4	1	3	1	4
Que.	16														
Ans.	1														

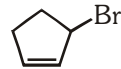
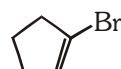
EXERCISE-III (Analytical Questions)

Check Your Understanding

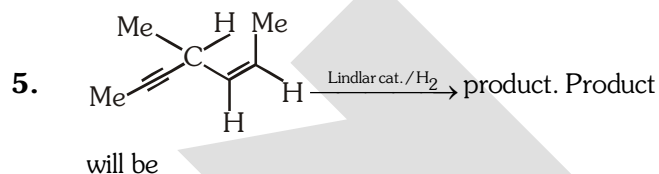
- The volume of methane evolved by treatment of 16.6 g of methyl magnesium iodide with water at S.T.P. is
 (1) 224 mL (2) 2.24 L
 (3) 0.224 L (4) 22.4 L
- 10 litre of an alkane X require 35 litre of O_2 for complete combustion. X forms only one monochloro derivative Y. The action of alcoholic KOH on Y yields
 (1) $CH_3CH_2CH_2OH$
 (2) $CH_2=CH_2$
 (3) $CH_3CH=CH_2$
 (4) $CH_3-\underset{\text{OH}}{\text{CH}}-CH_3$



What is the structure of B

- (1)  (2) 
 (3)  (4) 

- Which of the following is maximum reactive towards NBS
 (1) $PhCH_3$
 (2) $Ph-CH_2-CH_3$
 (3) $PhCH_2-CH=CH_2$
 (4) $Ph-\underset{\text{CH}_3}{\text{CH}}-CH=CH_2$



- (1) An optically active compound
 (2) An optically inactive compound
 (3) A racemic mixture
 (4) A diastereomeric mixture
- Which of the following reactions show that there are three C=C bond in benzene
 (a) $C_6H_6 + Br_2 \xrightarrow{FeBr_3}$
 (b) $C_6H_6 + HNO_3 \xrightarrow{H_2SO_4}$
 (c) $C_6H_6 + 3O_3 \longrightarrow$
 (d) $C_6H_6 + 3H_2 \xrightarrow{Ni}$
 (1) a, c (2) b, d (3) b, c, d (4) c, d

EXERCISE-III (Analytical Questions)

ANSWER KEY

Que.	1	2	3	4	5	6
Ans.	2	2	4	4	2	4

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (C) If Assertion is True but the Reason is False.
 (D) If both Assertion & Reason are false.

- Assertion :-** The reactivity order for decarboxylation is

$$\text{CH}_3-\text{CH}_2-\text{COOH} > \text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{COOH}$$

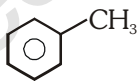
Reason :- Reactivity depends on stability of intermediate carbocation.
 (1) A (2) B (3) C (4) D
- Assertion :-** Baeyer's reagent can be used to distinguish cyclopentane from propene.
Reason :- Baeyer's reagent is decolourised by propene but not by cyclopentane
 (1) A (2) B (3) C (4) D
- Assertion :-** Increasing order of heat of hydrogenation is

$$\text{CH}_2=\text{CH}-\text{CH}_3 < \text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3 < \text{CH}_3-\underset{\text{CH}_3}{\text{C}}=\underset{\text{CH}_3}{\text{C}}-\text{CH}_3$$

Reason :- Highly substituted alkenes have less stability.
 (1) A (2) B (3) C (4) D
- Assertion :-** In the following reaction

$$\text{CH}_2=\text{CH}_2 + \text{NaNH}_2 \rightarrow \text{CH}_2=\overset{-}{\text{CH}}\overset{+}{\text{Na}} + \text{NH}_3$$

reaction is always in forward direction.
Reason :- Ethene is stronger acid than ammonia.
 (1) A (2) B (3) C (4) D
- Assertion :-** Ethyne and its derivatives give white precipitate with ammonical silver nitrate solution.
Reason :- Alkynes give white ppt with ammonical silver nitrate solution due to nucleophilic substitution reaction.
 (1) A (2) B (3) C (4) D
- Assertion :-** Acetylene gives Red ppt when treated with ammonical Cu_2Cl_2 .
Reason :- Red ppt. is obtained due to formation of Cu.
 (1) A (2) B (3) C (4) D
- Assertion :-** 1-alkyne and 2-alkyne can be distinguished by Baeyer's reagent.
Reason :- 1-alkyne decolourise Baeyer's reagent while 2-alkyne does not decolourise.
 (1) A (2) B (3) C (4) D
- Assertion :-** Benzene does not decolourise Br_2 -water.
Reason :- Benzene is stabilized by aromaticity and no addition of Br_2 .
 (1) A (2) B (3) C (4) D
- Assertion :-** Benzene reacts with Cl_2 in presence of light to form BHC.
Reason :- Reaction undergoes electrophilic substitution.
 (1) A (2) B (3) C (4) D
- Assertion :-**



+ $\text{Cl}_2 \xrightarrow{\text{Catalyst}}$ product.
 In product benzene ring is remain unchanged.
Reason :- Hexachloro benzene is an aromatic compound.
 (1) A (2) B (3) C (4) D
- Assertion :-** Acetylene gives glyoxal on reductive ozonolysis while benzene does not give glyoxal.
Reason :- Benzene does not show reductive ozonolysis reaction.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11
Ans.	3	1	4	4	4	3	4	1	3	2	4

ENVIRONMENTAL CHEMISTRY, PURIFICATION METHODS, QUALITATIVE AND QUANTITATIVE ANALYSIS

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NEET SYLLABUS

Environmental pollution : Air, water and soil pollution, chemical reactions in atmosphere, smog, major atmospheric pollutants; acid rain, ozone and its reactions, effects of depletion of ozone layer, greenhouse effect and global warming pollution due to industrial wastes; green chemistry as an alternative tool for reducing pollution, strategy for control of environmental pollution.

@neetfocusquizz

OBJECTIVES

After studying this unit, we will be able to :

- understand the meaning of environmental chemistry;
- define atmospheric pollution, list reasons for global warming. green house effect and acid rain;
- identify causes for ozone layer depletion and its effects;
- give reasons for water pollution and know about international standards for drinking water;
- describe causes of soil pollution;
- suggest and adopt strategies for control of environmental pollution;
- appreciate the importance of green chemistry in day to day life.

"Failure will never overtake me if my definition to succeed is strong enough"

A.P.J. Abdul Kalam

ENVIRONMENTAL CHEMISTRY, PURIFICATION METHODS, QUALITATIVE & QUANTITATIVE ANALYSIS

ENVIRONMENTAL CHEMISTRY

The branch of science which deals with the chemical phenomena occurring in the environment is called as environmental chemistry.

INTRODUCTION

The environment means surroundings. It has 4 following component.

Atmosphere

Function of the atmosphere :

- It **contain all the gases** which are essential for the life on the earth.
- It is a **carrier of water vapour** which are needed for all life.
- Ozone (O_3) is present in stratosphere which is responsible for absorption of 99.5% of harmful U.V. radiations.
- It maintain heat balance of the earth by absorbing infrared radiation, coming from the sun and re-emit from the earth.

ENVIRONMENTAL POLLUTION & ENVIRONMENTAL POLLUTANT

The **addition of any undesirable material** to air, water and soil by a natural source (or due to human activity) which affects the **quality of environment** is called as environmental pollution.

The undesirable material which is added to the environmental is called as pollutant.

Causes of pollution :

- Fast growth.
- Excessive trialisation.
- Rapid urbanisation.
- Use of pesticides in agriculture.

Types of Pollutants

Bio Degradable and Non Bio Degradable Pollutants

(A) Bio degradable pollutants : The materials (such as cow dungs) which are easily decomposed by the micro-organism. These are not harmful, but in the excess in environment, they do not undergo degradation completely and thus become pollutant. Eg. Discarded vegetables

(B) Non bio degradable pollutants :- The material (such as Hg, Al, DDT) which do not undergo degradation (or degrade very slowly) but their presence even in very small amount in the environment is very harmful. They may react with other compounds present in the environment and produce more toxic compound. Eg. Plastic materials, DDT, heavy metal, many chemicals, nuclear wastes etc.

TYPES OF POLLUTION

Depending on the Part of the Environment Polluted

(A) Air pollution **(B)** Water pollution **(C)** Soil pollution (or land pollution)

(A) Air pollution : It is defined as the undesirable materials into the atmosphere either due to natural phenomena (or due to human activity on the earth). Which affect the quality of the air and also affect the life on the Earth.

Major source of air pollution :

(i) Natural sources :

Ex. due to CO , H_2S , SO_2 , Forest fire

(ii) Man made air pollution or source due to human activity :**Natural sources :**

Ex. Burning of fossil fuels which produce some poisonous gases as CO, CO₂, CH₄, oxide of nitrogen .

Combustion of gasoline in the automobiles : The automobiles emit the CO, oxide of nitrogen (NO, NO₂)

Deforestation : Due to this % of CO₂ is increased and % of O₂ is decreased.

Fast industrialisation : The smoke of carbon and CO, CO₂, SO₂, H₂S, NO, NO₂ are coming out from the industries. The industries are responsible for 20% of total air pollution.

Agriculture activities : The pesticides are added in the soil. They give a foul smell and affect the health of animals and human being.

Wars : The nuclear weapons are used in war which emit the radiation.

Air pollutants :

(A) CO as pollutant : It is mainly realised by automobile exhaust due to incomplete combustion of carbon. It binds to haemoglobin to form carboxyhaemoglobin, which is about **300 times** more stable than the oxygen-haemoglobin complex. In blood, if the concentration of carboxyhaemoglobin reaches 3-4 percent then oxygen carrying capacity is greatly reduced.

Sink of CO : A large amount of CO are added in the atm. But the level of CO does not rise too much as CO is converted in CO₂ by the micro organism which is present in the soil.

The micro organism (bacteria) act as sink for CO.

Harmful effect of CO :

(i) The CO is poisonous because it combine with haemoglobin of R.B.C. about 300 times easily than O₂, to form carboxy haemoglobin.

(ii) Oxygen deficient results into headache, weak eyesight, nervouness and cardiovascular disorder.

(iii) In pregnant women who have the habit of smoking the increased CO level in blood may induce premature birth, spontaneous abortions and deformed babies.

(B) Hydro carbon : Hydrocarbons are carcinogenic, i.e., they cause cancer. They harm plants by causing ageing, breakdown of tissues and shedding of leaves, flowers and twigs.

(a) Natural sources :

(i) Due to decomposition of organic matter in soil.

**(b) Man made sources :**

(i) Due to burning of stationary fuel.

(ii) Evaporation of organic solvent.

(C) Sulphur compounds : Produced by burning of fossil fuel. The most common species, sulphur dioxide, is a gas that is poisonous to both animals and plants.

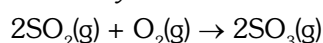
Harmful effect of Sulphur compounds :

(i) Even a low concentration of sulphur dioxide causes respiratory diseases e.g., asthma, bronchitis, emphysema in human beings.

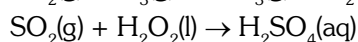
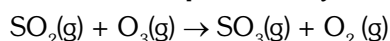
(ii) Causes irritation to the eyes, resulting in tears and redness.

(iii) High concentration of SO₂ leads to stiffness of flower buds which eventually fall off from plants.

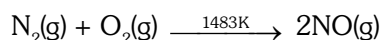
Sulphur dioxide Uncatalysed oxidation of sulphur dioxide is slow. However, the presence of particulate matter in polluted air catalyses the oxidation of sulphur dioxide to sulphur trioxide.



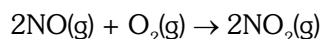
The reaction can also be promoted by ozone and hydrogen peroxide.



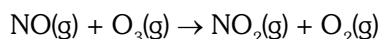
(D) Nitrogen compound : At high altitudes when lightning strikes, O_2 and N_2 combine to form oxides of nitrogen. NO_2 is oxidised to nitrate ion, NO_3^- which is washed into soil, where it serves as a fertilizer. In an automobile engine, (at high temperature) when fossil fuel is burnt, dinitrogen and dioxygen combine to yield significant quantities of nitric oxide (NO) and nitrogen dioxide (NO_2) as given below:



NO reacts instantly with oxygen to give NO_2



Rate of production of NO_2 is faster when nitric oxide reacts with ozone in the stratosphere.



Harmful effects of nitrogen compounds

- (i) The irritant red haze in the traffic and congested places is due to oxides of nitrogen.
- (ii) Higher concentrations of NO_2 damage the leaves of plants and retard the rate of photosynthesis.
- (iii) Nitrogen dioxide is a lung irritant that can lead to an acute respiratory disease in children.
- (iv) It is toxic to living tissues also. It is also harmful to various textile fibres and metals.

(E) Carbon dioxide : Normally it forms about 0.03 per cent by volume of the atmosphere. If the amount of carbon dioxide crosses the delicate proportion of 0.03 percent, the natural greenhouse balance may get disturbed. Carbon dioxide is the major contributor to global warming.

BEGINNER'S BOX-1

1. Domestic waste will lead to -

(1) Biodegradable pollution	(2) Nondegradable pollution
(3) Thermal pollution of soil	(4) Air pollution
2. The major source of BOD in the river Ganga is

(1) Leaf litter	(2) Fishes	(3) Human waste	(4) Aquatic plants
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3. If a lake is contaminated with DDT, its highest concentration would be found in

(1) Primary consumer	(2) Secondary consumer	(3) Tertiary consumer	(4) None of these
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4. The most harmful air pollutant produced by automobiles is

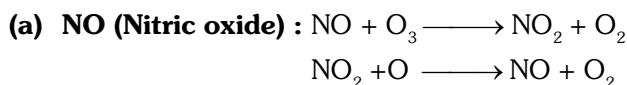
(1) HNO_2	(2) NO	(3) SO_2	(4) CO
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5. Sewage water can be purified by

(1) Aquatic plant	(2) Micro organism	(3) Penicillin	(4) Fishes
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STRATOSPHERIC POLLUTION

Depletion of Ozone Layer

Due to human activity 2 compounds **NO** and **CFC** are responsible for depletion of O_3 layer.

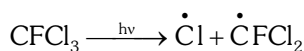
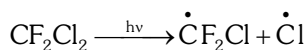


The NO react with O_3 so decrease the amount of O_3 and forms NO_2 which react with oxygen atoms available in the stratosphere and producing back NO.

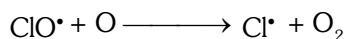
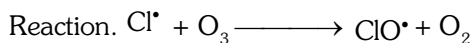
Thus no NO is consumed but O_3 gets depleted.

(b) Chlorofluoro carbons (CFC) or freons :

The freons decomposes in the presence of U.V. radiation coming from the Sun.



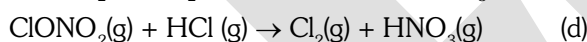
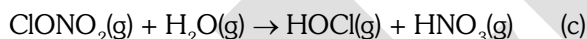
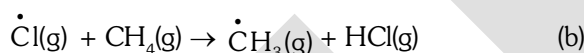
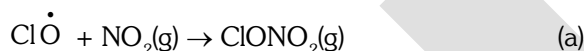
The reactive chlorine atoms then destroy the ozone layer through the following sequence of reaction. Which are repeated because chlorine atom are regenerated in the second reaction :



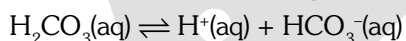
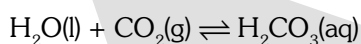
The one molecule CFC can destroy more than one thousand O_3 molecules in the stratosphere.

(c) Effect of depletion of O_3 layer : Due to depletion of O_3 layer, U.V. radiation fall on the Earth.

- The U.V. radiation, damage the cornea and lens of the eyes.
- The U.V. radiation affect the plant proteins so reduce the chlorophyll.
- The U.V. radiation, up set the heat balance of the Earth.
- **Ozone hole :** It was found that a unique set of conditions was responsible for the ozone hole. In summer season, nitrogen dioxide and methane react with chlorine monoxide (reaction a) and chlorine atoms (reaction b) forming chlorine sinks, preventing much ozone depletion, whereas in winter, special type of clouds called polar stratospheric clouds are formed over Antarctica. These polar stratospheric clouds provide surface on which chlorine nitrate formed (reaction a) gets hydrolysed to form hypochlorous acid (reaction c). It also reacts with hydrogen chloride produced as per (reaction b) to give molecular chlorine.

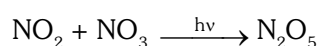
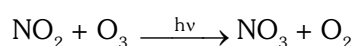
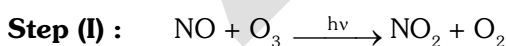


Acid Rain : The rain containig H_2SO_4 , HNO_3 (and small amount of HCl) which are formed from the **oxide of S** and **N_2** present in the air is called as acid rain. Normally rain water has a pH of 5.6 due to the presence of H^+ ions formed by the reaction of rain water with carbon dioxide present in the atmosphere.



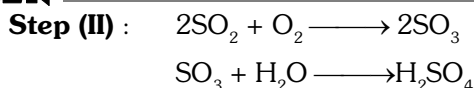
When the pH of the rain water drops below 5.6, it is called acid rain.

Formation of acid rain : The oxide of nitrogen undergo oxidation reaction. The reaction with the water vapour present in the atm to form HNO_3 .



HNO_3 come down with rain to Earth.

The SO_3 react with water vapour and form H_2SO_4 .



Ammonium salts are also formed and can be seen as an atmospheric haze (aerosol of fine particles). Aerosol particles of oxides or ammonium salts in rain drops result in wet-deposition. SO_2 is also absorbed directly on both solid and liquid ground surfaces and is thus deposited as dry-deposition.

Harmful effect of acid-rain

- It causes respiratory ailments in human beings and animals.
- It **corrodes water pipe**. So heavy metal (like Fe, Pb, Cu) are mixed with water which have toxic effect.
- The acid rain **increase the acidity of the lake**. Which is harmful to aquatic ecosystem.
- It is harmful for **agriculture, trees, plants** as it dissolves and washes away nutrients needed for their growth and also damages buildings.

Taj Mahal and Acid Rain : Taj Mahal is reported to be affected by SO_2 and other air pollutants released by oil refinery of Mathura. Acid rain reacts with marble, CaCO_3 of Taj Mahal ($\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$) causing damage to this wonderful monument that has attracted people from around the world.

Green House Effect : The warming of earth due to remission of sun's energy absorbed by the earth. The remission of earth's energy is absorbed by CO_2 molecules and H_2O vapour present near the earth's surface and then its radiation back to the earth, is called as green house effect. So the temp. of the earth is increased is called as global warming.

Important Green House Gases : Methane, water vapour, nitrous oxide, CFCs, ozone and CO_2 .

Particulates in Atmospheric Pollution

Smoke is composed of particulate matter. Particulates are the tiny solid or liquid particles suspended in air. These particles are usually individually invisible to the naked eye. Collectively, however, small particles often form a haze that restricts visibility.

Particulates in the atmosphere may be **viable** or **non-viable** :-

(i) Viable particulates - These are the minute living organisms that are dispersed in atmosphere.

Eg. bacteria, fungi, moulds, algae etc.

(ii) Non-viable - These are formed either by the breakdown of larger materials or by the condensation of minute particles and droplets. There are four types of non-viable particulates in the atmosphere : **mists, smoke, fumes and dust**.

- Mists** are produced by particles of spray liquids and the condensation of vapours in air. Examples are portions of herbicides and insecticides that miss their targets and travel through the air to form mists.
- Smoke** denotes very small soot particles produced by burning and combustion of organic matter. Oil smoke, tobacco smoke and carbon smoke are typical examples of this type of particulate emission.
- Fumes** are condenses vapours : fumes of metals are the well-known particulates of this type. Examples of this category also include metallurgical fumes and alkali fumes.
- Dust** consists of the particles produced during crushing, grinding and attribution of solid materials. Non-viable dust particulates in the atmosphere consist of ground limestone, sand tailings from floatation, pulverised coal, cement, fly ash and silica dust.

The effect of particulate pollutants are largely dependent on the particle size. Airborne particles such as dust, fumes, mist etc., are dangerous for human health. Particulate pollutants bigger than **5 microns are likely to lodge in the nasal passage**, whereas particles of about **10 micron enter into lungs easily**. Lead used to be a major air pollutant emitted by vehicles. Leaded petrol used to be the primary source of air-borne lead emission in Indian cities. Lead interferes with the development and maturation of red blood cells.

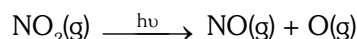
Smogs

Smogs which describe the "smoke-fog" like condition, are the best-known examples of air pollution that occurs in many cities throughout the world.

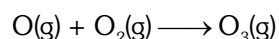
There are two types of smogs :

- (i) **Classical smogs** - Which occur in cool humid climate and are the result of buildup of **sulphur oxides** and **particulate matter** from fuel combustion. Chemically it is a reducing mixture so also called **reducing smog**.
- (ii) **Photochemical smogs** - Which occur in warm, dry and sunny climate and result from the action of sunlight on the nitrogen oxides and hydrocarbons produced by automobiles and factories. Photochemical smog is an **oxidising smog** having a high concentration of oxidising agents whereas classical smog is chemically reducing smog with high concentrations of SO_2 . Mostly in those cities which have very large populations and high vehicular density.

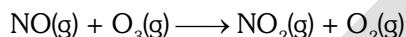
Formation of photochemical smog : The chemistry of formation of photochemical smog centres around nitric oxide (NO). At the high temperatures, in the petrol and diesel engines of cars and trucks, N_2 and O_2 react to form a small quantity of NO, which is emitted into troposphere with the exhaust gases. This NO oxidised in air to NO_2 which in turn absorbs energy from sunlight and breaks up into nitric oxide and free oxygen atom. (Photochemical decomposition)



Oxygen atoms are very reactive and can combine with O_2 to form ozone;



The O_3 formed in the above reaction reacts rapidly with the $\text{NO}(\text{g})$ formed in reaction to regenerate NO_2 . NO_2 is brown gas at high levels can contribute to haze.



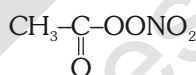
Both NO and O_3 are strong oxidising agents and can react with the unburnt hydrocarbons in the polluted air to produce chemicals such as formaldehyde, acrolein and peroxyacetyl nitrate (PAN) that cause the eyes to water and burn and are harmful to the respiratory system. The brownish haze of photochemical smog is largely attributed to the brown colour of NO_2 .



Acrolein and peroxyacetyl nitrate (PAN) are particularly noxious.



Acrolein



Peroxyacetyl nitrate (PAN)

Effects of Photochemical Smog

The common components of photochemical smog are ozone, nitric oxide, acrolein, formaldehyde and peroxyacetyl nitrate (PAN). Photochemical smog causes serious health problems. Both ozone and PAN act as powerful eye irritants. Ozone and nitric oxide irritate the nose and throat and their high concentration causes headache, chest pain, dryness of the throat, cough and difficulty in breathing. Photochemical smog leads to cracking of rubber and extensive damage to plant life. It also causes corrosion of metals, stones, building materials, rubber and painted surfaces.

Control of Photochemical Smog

If we control the primary precursors of photochemical smog, such as NO_2 and hydrocarbons, the secondary precursors such as ozone and PAN, the photochemical smog will automatically be reduced. Usually catalytic converters are used in the automobiles, which prevent the release of nitrogen oxide and hydrocarbons to the atmosphere. Certain plants **e.g., Pinus, Juniparus, Quercus, Pyrus** and **Vitis** can metabolise nitrogen oxide and therefore, their plantation could help in this matter.

(B) WATER POLLUTION

The quality of drinking water is very important for human welfare. The pollution of water by sewage has been linked to the spread of diseases such as cholera and typhoid fever. Potassium permanganate or bleaching powder are used in water tanks.

Point Source : Easily identified source or place of pollution is called as point source. e.g., municipal and industrial discharge pipes

Non-point Sources : Those where a source of pollution cannot be easily identified, e.g., agricultural run off (from farm, animals and crop-lands), acid rain, storm-water drainage (from streets, parking lots and lawns) etc.

Table lists the major water pollutants and their sources.

Pollutant	Source
Microorganisms	Domestic sewage
Organic wastes	Domestic sewage, animal waste, decaying animals and plants and discharge from food processing factories
Plant nutrients	Chemical fertilizers
Toxic heavy metals	Industries and chemical factories
Sediments	Erosion of soil by agriculture and strip mining
Pesticides	Chemicals used for killing insects, fungi and weeds
Radioactive substances	Mining of uranium containing minerals
Heat	Cooling water used by industrial plants (which is discharged as hot water)

In addition, industrial wastes also contaminate water.

- (i) **Heavy Metals :** Metals such as Cd, Pb and Hg may be present in industrial or mining waste. These metals can prove poisonous to humans – Cadmium and mercury can damage kidneys, liver brain and central nervous system. All of these metals are cumulative poisons the body does not excrete them and their concentration builds up.
- (ii) **Detergents and Fertilizers :** These may contain phosphates as additives. The addition of phosphates to water, in the form of the phosphate anion PO_4^{3-} , encourages the formation of algae, which reduces the dissolved oxygen concentration of water. The process, known as eutrophication, impedes the development of higher life forms, such as fish.
- (iii) **Acid-polluted water (pH < 3) :** This is deadly to most forms of aquatic life. Water downstream from a mine may be contaminated by acid mine drainage, the result of microbial oxidation of discarded waste material at the mine site. Acid mine water principally contains sulphuric acid produced by the oxidation of iron pyrites (FeS_2). Industrial wastes and acid rain may also contribute to the acidity of natural waters.
- (iv) **Polychlorinated biphenyls (PCBs) :** These chemicals are relatively recent additions to the list of contaminants of water. Having high stabilities, PCBs find many applications, for example they are used as fluids in transformer capacitors. PCBs are resistant to oxidation and their release into the environment causes skin disorders in humans. They are reported to be carcinogenic.

Chemical Pollutant

Acids (like sulphuric acid) from mine drainage and salts from many different sources including raw salt used to melt snow and ice in the colder climates (sodium and calcium chloride) are water soluble chemical pollutants. The organic chemicals are another group of substances that are found in polluted water. Petroleum products pollute many sources of water e.g., major oil spills in oceans.

The quality of water is of vital concern for mankind since it is directly linked with human welfare. There are some international standards for drinking water, which must always be obeyed if water is to be used for drinking purposes. These are :

Fluoride : Soluble fluoride is often added to drinking water to bring it up to a concentration of 1 ppm or 1 mg dm^{-3} . The F^- ions make the enamel on teeth much harder by converting hydroxyapatite, $[\text{3}(\text{Ca}_3(\text{PO}_4)_2 \cdot \text{Ca}(\text{OH})_2)]$, the enamel on the surface of the teeth, into much harder fluorapatite, $[\text{3}(\text{Ca}_3(\text{PO}_4)_2 \cdot \text{CaF}_2)]$. However, F^- ion concentration above 2 ppm causes brown mottling of teeth. At the same time, excess fluoride (over 10 ppm) causes harmful effect to bones and teeth.

Lead : The upper limit for the concentration of lead ions in drinking water is 50 ppb. If water is relatively acidic and lead pipes are used for water transport, then the water is liable to get contaminated with lead.

pH : The pH of drinking water should be between 5.5 and 9.5. A decrease in the pH of the water increases the solubility of metal ions.

Other Metals : The maximum recommended levels of common metals in drinking water are as follows.

Metal	Max. concentration (ppm or mg dm ⁻³)
Zn	5
Fe	0.2
Mn	0.05
Cu	3
Cd	0.005
Al	0.2

Sulphate : Sulphate is harmless at moderate levels, but excessive sulphate (> 500 ppm) is thought to have a laxative effect.

Nitrate : Excess nitrate in drinking water can lead to methemoglobinemia (blue-baby syndrome). A maximum limit of 50 ppm for the nitrate ion in drinking water has been set.

Causes of Water Pollution

(i) Pathogens: The most serious water pollutants are the disease causing agents called pathogens. Pathogens include bacteria and other organisms that enter water from domestic sewage and animal excreta. Human excreta contain bacteria such as *Escherichia coli* and *Streptococcus faecalis* which cause gastrointestinal diseases.

(ii) Organic wastes: The other major water pollutant is organic matter such as leaves, grass, trash etc. They pollute water as a consequence of run off. Excessive phytoplankton growth within water is also a cause of water pollution. These wastes are biodegradable.

Importance of Dissolved Oxygen in Water

The large population of bacteria decomposes organic matter present in water. They consume oxygen dissolved in water. The amount of oxygen that water can hold in the solution is limited. In cold water, dissolved oxygen (DO) can reach a concentration up to 10 ppm (parts per million), whereas oxygen in air is about 200,000 ppm. That is why even a moderate amount of organic matter when decomposes in water can deplete the water of its dissolved oxygen. The concentration of dissolved oxygen in water is very important for aquatic life. If the concentration of dissolved oxygen of water is below 6 ppm, the growth of fish gets inhibited. Oxygen reaches water either through atmosphere or from the process of photosynthesis carried out by many aquatic green plants during day light. However, during night, photosynthesis stops but the plants continue to respire, resulting in reduction of dissolved oxygen. The dissolved oxygen is also used by microorganisms to oxidise organic matter.

Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD)

If too much of organic matter is added to water, all the available oxygen is used up. This causes oxygen dependent aquatic life to die. Thus, anaerobic bacteria (which do not require oxygen) begin to break down the organic waste and produce chemicals that have a foul smell and are harmful to human health. Aerobic (oxygen requiring) bacteria degrade these organic wastes and keep the water depleted in dissolved oxygen. Thus, the amount of oxygen required by bacteria to break down the organic matter present in a certain volume of a sample of water, is called **Biochemical Oxygen Demand (BOD)**. The amount of BOD in the water is a measure of the amount of organic material in the water, in terms of how much oxygen will be required to break it down biologically. Clean water, in terms of how much oxygen will be required to break it down biologically. Clean water would have BOD value of less than 5 ppm whereas highly polluted water could have a BOD value of 17 ppm or more.

Chemical Oxygen Demand (COD) :

COD is the oxygen requirement by chemical $K_2Cr_2O_7$ for oxidation of total organic matter (biodegradable + non biodegradable) in water.

Note : COD value is always higher than BOD value.

BEGINNER'S BOX-2

- Major pollutant in Jet plane emission is -
 (1) SO_2 (2) CFC (3) CO (4) CCl_4
- What is B.O.D.
 (1) The amount of O_2 utilised by organisms in water
 (2) The amount of O_2 utilized by micro organisms for decomposition
 (3) The total amount of O_2 present in water
 (4) All of the above
- What is the intensity of sound in normal conversation
 (1) 10 – 20 decibal (2) 30 – 60 decibal (3) 70 – 90 decibal (4) 120 – 150 decibal
- Which of the following is absent in polluted water
 (1) Hydrilla (2) Water hyacinth (3) Larva of stone fly (4) Blue green algae
- Maximum green house gas released by which country
 (1) India (2) France (3) U.S.A (4) Britain

(C) LAND OR SOIL POLLUTION

Most of the land pollution is caused by pesticides and other chemicals which are added to the soil grow better crops. Often, a pesticide poisons many more organisms than those intended. Some of these poison pass through food chains and eventually reach harmful proportions. Solid wastes are another cause of land pollution.

Pesticides

Pesticides are substances that are used to kill or block the reproductive processes of unwanted organisms. Pesticides are basically synthetic toxic chemicals with ecological repercussions. **Eg. DDT, Malathion**

The repeated use of the same or similar pesticides give rise to pests that are resistant to that group of pesticides thus making the pesticides ineffective. Therefore, as insect resistance of DDT increased, other organic toxins such as **Aldrin** and **Dieldrin** were introduced in the market by pesticide industry. Most of the organic toxins are water insoluble and nonbiodegradable. These high persistent toxins are, therefore, transferred from lower trophic level to higher trophic level through food chain. Over the time, the concentration of toxins in higher animals reach a level which causes serious metabolic and physiological disorders.

In respon chlorinated org; to high persistence of ; toxins, a new series of less persistent or more bio-degradable products called **organo-phosphates** and **carbamates** have been introduced in the market. But these chemicals are severe nerve toxins and hence more harmful to humans. As a result, there are reports of some pesticides related deaths of agricultural field workers. Insects have become resistant to these insecticides also. The insecticide industry is engaged in developing new groups of insecticides.

These days, the pesticide industry has shifted its attention to **herbicides** such as **sodium chlorate ($NaClO_3$)**, **sodium arsinite (Na_3AsO_3)** and many others. During the first half of the last century, the shift from mechanical to chemical weed control had provided the industry with flourishing economic market. But one must remember that these are also not environment friendly.

Most herbicides are toxic to mammals but are not as persistent as organo-chlorides. These chemicals decompose in a few months. Like organo-chlorides, these too become concentrated in the food web. Some herbicides cause birth defects. Studies show that cornfields sprayed with herbicides are more prone to insect attack and plant disease than fields that are weeded manually.

INDUSTRIAL WASTE

Industrial solid wastes : (i) Biodegradable - Biodegradable wastes are generated by cotton mills, food processing units, paper mills and textile factories.

(ii) Non-biodegradable - These wastes are generated by thermal power plants which produce fly ash; integrated iron and steel plants which produce blast furnace slag and steel melting slag.

Industries manufacturing aluminium, zinc and copper produce mud and tailings. Fertilizer industries produce gypsum. Hazardous wastes such as inflammables, composite explosives or highly reactive substances are produced by industries dealing in metals, chemicals, drugs, pharmaceuticals, dyes, pesticides, rubber goods etc.

STRATEGY FOR CONTROL OF ENVIRONMENTAL POLLUTION

The management of waste : The production and improper disposal of waste are causes for a great deal of environmental pollution. In addition to the household waste which included sewage and municipal garbage, many toxic industrial wastes from manufacturing processes require treatment and/or safe disposal.

Recycling : When materials are recycled, there are several benefits, apart from saving on the cost of raw materials, waste disposal costs are reduced. Examples of recycling by industry are :

1. the collection and recycling of glass (in bottle banks);
2. the use of scrap metal in the manufacture of steel;
3. the recovery of energy from burning combustible waste.

Thus recycling converts waste into wealth.

Green Fuel : Fuel obtained from plastic waste has high octane rating. It contains no lead and is known as "green fuel".

Biogas : A pilot plant has been set up, where after removing ferrous metals, plastic, glass, paper etc. from garbage, it is mixed with water. It is then cultured with bacterial species for producing methane, commonly known as biogas. The remaining product is used as manure and biogas is used to produce electricity.

Digestion : Anaerobic digestion occurs when microorganisms degrade wastes in the absence of oxygen. It may be used to treat sewage sludge, but the process can also be used to degrade a variety of toxic organic wastes. Carbon dioxide and methane, which may be used as a fuel, are the products. The overall process is the conversion of the organic material into carbon dioxide and methane i.e.

Dumping : Ocean dumping of sewage sludge has been widely practiced in the seas around the world. However, the practice of application to sludge the land is increasing. The sludge contains nitrogen and phosphorus which make it useful as a fertilizer. Urban areas produce sludge with high toxic metal content, so the amount of such sludge dumped in this way must be carefully controlled.

Incineration : Incineration converts organic materials to CO_2 and H_2O . It may serve to destroy household waste, chemical waste and biological waste (e.g. from hospitals). A high temperature is required usually in excess of 1000°C , and a plentiful supply of oxygen. Exhaust gases must be filtered. The process greatly reduced the volume of waste-an inorganic ash is left behind, which is disposed of as landfill.

GREEN CHEMISTRY (A NEW ROUTE TO PROTECTION OF ENVIRONMENT)

Green chemistry is a way of thinking and is about utilising the existing knowledge and principles of chemistry and other sciences to reduce the adverse impact on environment. Green chemistry is a production process that would bring about minimum pollution or deterioration to the environment. The byproducts generated during a process, if not used gainfully, add to the environmental pollution. Such processes are not only environmentally unfriendly but also cost-ineffective. The waste generation and its disposal both are economically unsound. Utilisation of existing knowledge base for reducing the chemical hazards along with the developmental activities is the foundation of green chemistry.

Green chemistry, in a nutshell, is a cost effective approach which involves reduction in material, energy consumption and waste generation.

It is well known that organic solvents such as benzene, toluene, carbon tetrachloride etc., are highly toxic. It may be worthwhile to carry out synthetic reactions in aqueous medium since water has high specific heat and low volatility. Water is cost effective, noninflammable and devoid of any carcinogenic effects.

Green Chemistry in day-to-day Life

(i) Dry Cleaning of Clothes

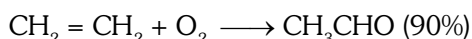
Tetra chloroethene ($\text{Cl}_2\text{C}=\text{CCl}_2$) was earlier used as solvent for dry cleaning. The compound contaminates the ground water and is also a suspected carcinogen. The process using this compound is now being replaced by a process, where liquefied carbondioxide, with a suitable detergent is used. Replacement of halogenated solvent by liquid CO_2 will result in less harm to ground water. These days hydrogen peroxide (H_2O_2) is used for the purpose of bleaching clothes in the process of laundry, which gives better results and makes use of lesser amount of water.

(ii) Bleaching of Paper

Chlorine gas was used earlier for bleaching paper. These days, hydrogen peroxide (H_2O_2) with suitable catalyst, which promotes the bleaching action of hydrogen peroxide, is used.

(iii) Synthesis of Chemicals

Ethanal (CH_3CHO) is now commercially prepared by one step oxidation of ethene in the presence of ionic catalyst in aqueous medium with a yield of 90%.



Some important points :

- Stratosphere - In this ozone is present and it extends from 10 - 50 km from sea level.
- The lowest region of atmosphere in which human beings along with other organisms live is called troposphere. which extends upto 10 km from sea level
- Bhopal gas tragedy - A major gas tragedy took place on the night of december 2, 1984, when a dense cloud of deadly methyl isocyanate (MIC) gas leaked from a storage tank of the Union Carbide Ltd. plant in Bhopal.

PURIFICATION METHODS

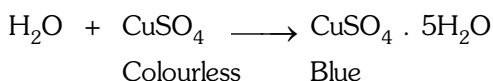
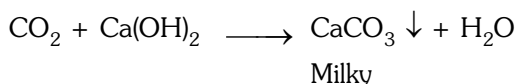
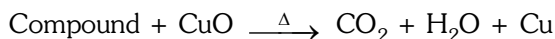
Distillation Techniques :

Type	Conditions	Examples
(A) Simple distillation	(i) When liquid sample has non volatile impurities (ii) When boiling point difference is 80°K or more.	(i) Mixture of chloroform (BP = 334K) and Aniline (BP = 457K) (ii) Mixture of Ether (BP = 308K) & Toluene (BP = 384K) (iii) Hexane(342K) and Toulene(384K)
(B) Fractional distillation	When BP difference is 10°	(i) Crude oil in petroleum industry (ii) Acetone (329) and Methyl alcohol(338K)
(C) Distillation under reduced pressure (Vacuum distillation)	When liquid boils at higher temperature and it may decompose before BP is attained.	(i) Concentration of sugar juice (ii) Recovery of glycerol from spent lye. (iii) Glycerol
(D) Steam distillation	When the substance is immiscible with water and steam volatile. $P = P_1 + P_2$ Vapour pressure Vapour pressure Vapour pressure of Organic liquid of water	(i) Aniline is separated from water (ii) Turpentine oil (iii) Nitro Benzene (iv) Bromo Benzene (v) Naphthalene (vi) o-Nitrophenol

QUALITATIVE AND QUANTITATIVE ANALYSIS OF ORGANIC COMPOUND

Qualitative analysis : Detection of element present in compound :-

(i) Detection of C and H : Compound heated with cupric oxide and produced vapour is passes through lime water or anhydrous CuSO_4 . If lime water turn milky, it shows presence of CO_2 or carbon. If anhydrous CuSO_4 turn blue than it shows presence of H_2O or hydrogen.

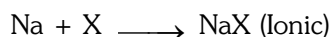
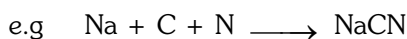


(ii) Detection of Nitrogen, Sulphur and halogen :- [Lassaigne's Test]

Preparation of lassaigne's solution or sodium extract.

Compound is heated with sodium in combustion tube and poured in cold water and heated for some time it gives lassaigne's solution or sodium extract.

In lassaigne solution elements present in compound get converted from covalent to ionic.

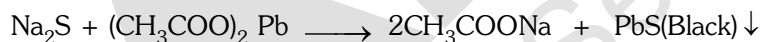


(i) Test of Nitrogen : Sodium extract is boiled with FeSO_4 which gives sodium hexacyano ferrate (II). Now few amount of FeCl_3 is added which gives prussian blue coloured ferro ferri cyanide that shows presence of Nitrogen.



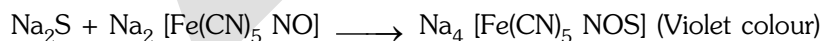
(ii) Test of sulphur : Sulphur is present in form of Na_2S

(a) Sodium Extract is acidified with acetic acid and lead acetate is added to it which gives black ppt of PbS and shows the presence of Sulphur

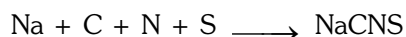


(b) Nitroprusside test

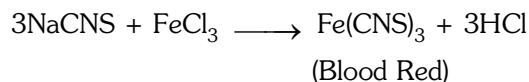
Sodium extract + Sodium Nitroprusside $\longrightarrow$ Violet colour.



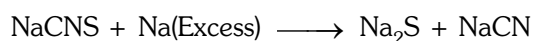
Note : If Nitrogen and sulphur both are present in compound then they form sodium thiocyanate in sodium extract.



during test of Nitrogen in place of prussian blue, we get blood red colour of Fe(CNS)_3



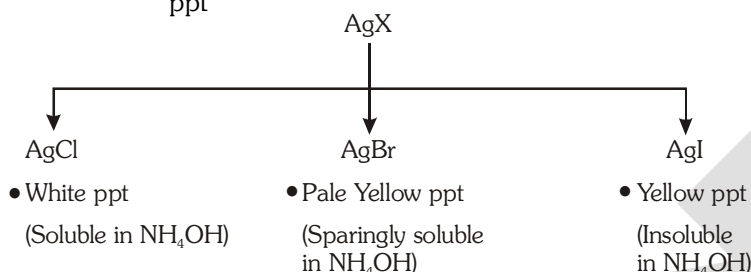
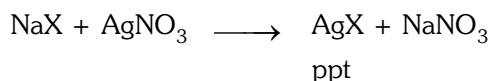
Therefore sodium extract is prepared in presence of excess of sodium which decomposes sodium thiocyanate.



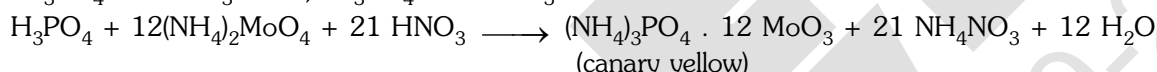
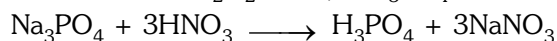
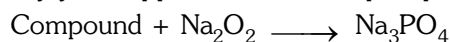
- (iii) **Test of halogen** : Before the test of halogen sodium extract is boiled with conc. HNO_3 to decompose the Na_2S and NaCN in form of H_2S and HCN



Now sodium extract is treated with silver nitrate which gives precipitate of AgX .



- (iv) **Test of phosphorous** : Compound is heated with an oxidising agent [Na_2O_2] so the phosphorus in compound is converted in to phosphate which on further reaction with ammonium molybdate in presence of HNO_3 gives canary yellow ppt of ammonium phospho molybdate and shows the presence of phosphorus.



Quantitative analysis : Calculation of percentage of element in compound.

- (i) **Estimation of carbon and hydrogen** : [Leibigs method]

A known mass of organic compound is burnt in presence of excess of CuO so carbon and hydrogen oxidise to CO_2 and H_2O respectively.

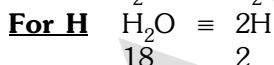
Now above vapour is passed through weighed u-tube having anhyd. CaCl_2 followed by KOH solution. anhyd. CaCl_2 absorbs H_2O while KOH absorbs CO_2 .

Now we can calculate the amount of H_2O and CO_2 produced in combustion.

Calculation of % of C and H :-

If mass of organic compound is W gm.

mass of H_2O or CO_2 produced is m gm



$\therefore$ 18 gm of H_2O contains 2 gm of hydrogen

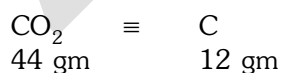
$\therefore$ 1 gm of H_2O contains $\frac{2}{18}$ gm of hydrogen

$\therefore$ m gm of H_2O contains $\frac{2}{18} \times m$ gm of hydrogen

Percentage of hydrogen in W gm organic substance =

$$\% \text{ of H} = \frac{2}{18} \times \frac{m}{W} \times 100$$

For C



$\therefore$ 44 gm CO_2 contain 12 gm carbon

1 gm CO_2 contain $\frac{12}{44}$ gm carbon

m gm CO_2 contain $\frac{12}{44} \times m$ gm carbon

$$\% \text{ of C in W gm organic substance} = \boxed{\% \text{ of C} = \frac{12}{44} \times \frac{m}{W} \times 100}$$

Estimation of Nitrogen :

These are two method for the estimation of nitrogen

- (i) Duma's method [in form of N_2]
- (ii) Kjeldhal's method [in form of NH_3]

(1) Duma's method :

Organic compound is heated with CuO to give free Nitrogen, CO_2 and H_2O .



Now the above mixture is collected over Aq. KOH in nitrometer tube which absorb all the gases except N_2 .
Now volume of nitrogen is calculated at STP.

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \quad V_1 = \text{Volume of } N_2 \text{ gas.}$$

P_1 = Pressure of N_2 - Aqueous tension

T_1 = Room temperature

$$\boxed{V_2 = \frac{P_1 V_1 \times 273}{T_1 \times 760}}$$

Calculation of % of Nitrogen

$\therefore$ 22400 ml of N_2 at STP weight = 28 gm

$\therefore$ V ml of N_2 at STP weight = $\frac{28}{22400} \times V$ gm

$$\% \text{ of Nitrogen in W gm organic substance} = \boxed{\% \text{ of N} = \frac{28}{22400} \times \frac{V}{W} \times 100}$$

(2) Kjeldhal's method : Compound containing nitrogen is heated with H_2SO_4 so compound gets converted in to ammonium sulphate.

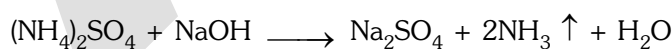
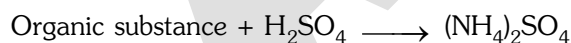
Now above mixture is heated with excess of NaOH which liberates Ammonia gas.

Ammonia gas is absorbed in excess of standard solution of H_2SO_4 .

The amount of ammonia produced is determined by estimating the amount of H_2SO_4 consumed.

Estimation of H_2SO_4 is done by titrating the H_2SO_4 left after absorption of ammonia with standard alkali solution.

The difference between the initial amount of acid taken and that left after absorption of ammonia is the amount of H_2SO_4 used.



If Normality of H_2SO_4 taken = N_1

Volume of H_2SO_4 taken = V_1

If Normality of NaOH used = N_2

Volume of NaOH used = V_2

$\therefore$ Milli equivalent of H_2SO_4 used for neutralisation of produced NH_3 = $(N_1 V_1 - N_2 V_2)$

Which is equal to the milli equivalent of NH_3

$\therefore$ 1000 ml of 1N NH_3 contain 14 gm Nitrogen.

$\therefore (N_1V_1 - N_2V_2)$ ml of 1N NH_3 contain $\frac{14}{1000} \times (N_1V_1 - N_2V_2)$ gm Nitrogen.

If mass of organic substance is W g.

$$\text{Then \% of N} = \frac{14}{1000} \times \frac{(N_1V_1 - N_2V_2)}{W} \times 100$$

$$\boxed{\% \text{ of N} = \frac{1.4}{W} \times (N_1V_1 - N_2V_2)} \quad \text{or} \quad \boxed{\% \text{ of N} = \frac{1.4}{W} \times NV}$$

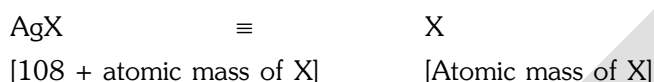
N = Normality of H_2SO_4 V = Volume of H_2SO_4 used in neutralisation.

(3) Estimation of halogen : (by carius method)

A known mass of organic compound is heated with fuming HNO_3 in presence of AgNO_3 . The halogen gets ppt in the form of AgX which can be filtered, washed and dried and weighed.

If mass of organic substance = W gm

mass of ppt = m gm



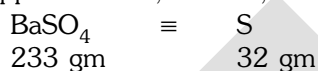
mass of halogen in m gm AgX is = $\frac{\text{Atomic mass of X}}{\text{Molar mass of AgX}} \times m$ gm

$$\boxed{\% \text{ of halogen} = \frac{\text{Atomic mass of X}}{\text{Molar mass of AgX}} \times \frac{m}{W} \times 100}$$

(4) Estimation of sulphur : (by Carius method)

Sulphur is estimated in the form of BaSO_4 .

Now ppt is filtered, washed, dried and weighed.



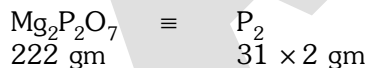
mass of sulphur in m gm ppt of BaSO_4 = $\frac{32}{233} \times m$ gm

% of sulphur in W gm organic sub.

$$\boxed{\% \text{ of S} = \frac{32}{233} \times \frac{m}{W} \times 100}$$

(5) Estimation of phosphorus : (by Carius method)

Phosphorus is estimated in the form of magnesium pyrophosphate ($\text{Mg}_2\text{P}_2\text{O}_7$)



mass of sulphur in m gm $\text{Mg}_2\text{P}_2\text{O}_7$ is

$$\frac{62}{222} \times m \text{ gm}$$

% of P in W gm organic sub.

$$\boxed{\% \text{ of P} = \frac{62}{222} \times \frac{m}{W} \times 100}$$

SOME SOLVED EXAMPLES

Q.1 0.3960 gm of an organic compound on combustion gives 0.792 gm CO₂ and 0.324 gm of H₂O. Calculate the % of C and H

Ans. % of C = $\frac{12}{44} \times \frac{\text{Wt. of CO}_2}{\text{Wt. of compound}} \times 100$

$$= \frac{12}{44} \times \frac{0.792}{0.396} \times 100 = 54.55\%$$

% of H = $\frac{2}{18} \times \frac{\text{Wt. of H}_2\text{O}}{\text{Wt. of compound}} \times 100$

$$= \frac{2}{18} \times \frac{0.324}{0.396} \times 100 = 9.09\%$$

Q.2 0.25 gm of an organic compound at NTP gives 31 ml of N₂ gas by Duma's method. Find out of % of N

Ans. % of N = $\frac{28}{22400} \times \frac{V}{W} \times 100$

$$= \frac{28}{22400} \times \frac{31}{0.25} \times 100 = 15.5\%$$

Q.3 31.7 ml of moist N₂ was obtained from 0.2033 gm of an organic compound in Duma's method at 14°C and 758 mm pressure. If aq. tension at 14°C = 14 mm then calculate % of N

Ans. P₁ = Pressure of N₂ – Aq. tension = 758 – 14 = 744 mm

P₂ = 760 mm T₂ = 273

V₁ = 31.7

T₁ = 273 + 14

T₂ = 287

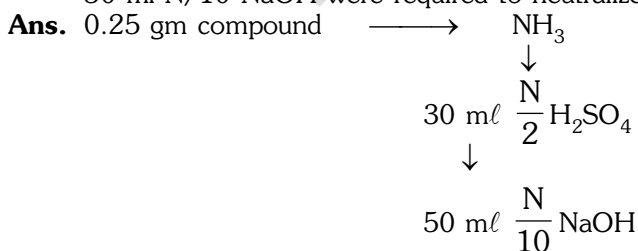
$$V_2 = \frac{P_1 V_1}{T_1} \times \frac{T_2}{P_2} = \frac{744 \times 31.7 \times 273}{287 \times 760} = 29.51$$

$$\% \text{ N} = \frac{28}{22400} \times \frac{V}{W} \times 100 = \frac{28}{22400} \times \frac{29.51}{0.2033} \times 100 = 18.15\%$$

Q.4 30 ml 0.25 N H₂SO₄ are used in neutralizing NH₃ obtained from 0.75 gm of an organic compound in Kjeldhal's method find out % of N in the compound

Ans. N% = $\frac{1.4}{W} \times NV = \frac{14}{0.75} \times 30 \times 0.25 = 14\%$

Q.5 In Kjeldhal's method, NH₃ evolved from 0.25 gm of an organic compound was passed into 30 ml N/2 H₂SO₄. 50 ml N/10 NaOH were required to neutralize the unreact acid, calculate % of N.



$$50 \text{ ml} \times \frac{N}{10} = 5 \text{ ml N NaOH} \equiv 5 \text{ ml N H}_2\text{SO}_4$$

Remaining acid = 5 ml N H₂SO₄

Consumed acid

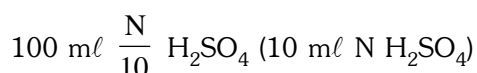
$$= \left(30 \text{ ml} \times \frac{N}{2} - 5 \text{ ml N} \right) = 10 \text{ ml N H}_2\text{SO}_4$$

$$\% \text{ N} = \frac{1.4}{W} \times NV = \frac{1.4}{0.25} \times 10 = 56 \%$$

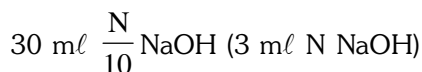
Q.6 0.35 gm of an organic compound was analysed by Kjeldhal's method. Ammonia evolved was absorbed in

100 ml $\frac{N}{10}$ H₂SO₄. Unused acid required 30 ml $\frac{N}{10}$ NaOH for neutralisation calculate % of N

Ans. W=0.35 gm $\rightarrow$ NH₃
 $\downarrow$



$\downarrow$



Remaining acid 3 ml N H₂SO₄

Consumed acid (10 ml N - 3 ml N) = 7 ml N H₂SO₄

$$\% \text{ of N} = \frac{1.4}{W} \times NV = \frac{1.4}{0.35} \times 1 \times 7 = 28 \%$$

Q.7 In Duma's method estimation of nitrogen, 0.3g of an organic compound gave 50 ml of nitrogen collected at 300 K temperature and 715 mm pressure. Calculate the percentage composition of nitrogen in the compound. (Aqueous tension at 300K = 15 mm)

Ans. Volume of nitrogen collected at 300K and 715 mm pressure is 50 ml

Actual pressure = 715 - 15 = 700 mm

$$\text{Volume of nitrogen at STP} = \frac{273 \times 700 \times 50}{300 \times 760} = 41.9 \text{ ml}$$

22,400 ml of N₂ at STP weighs = 28 g

$$41.9 \text{ ml of nitrogen weighs} = \frac{28 \times 41.9}{22400} \text{ g}$$

$$\text{Percentage of nitrogen} = \frac{28 \times 41.9 \times 100}{22400 \times 0.3} = 17.46\%$$

Q.8 During estimation of nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 0.5 g of the compound in Kjeldahl's estimation of nitrogen, neutralized 10 ml of 1 M H₂SO₄. Find out the percentage of nitrogen in the compound.

Ans. 1 M of 10 ml H₂SO₄ = 1M of 20 ml NH₃ 1000 ml of 1M ammonia contains 14 g nitrogen.

20 ml of 1 M ammonia contains $\frac{14 \times 20}{1000}$ g nitrogen.

$$\text{Percentage of nitrogen} = \frac{14 \times 20 \times 100}{1000 \times 0.5} = 56.0\%$$

Q.9 In carius method of estimation of halogen, 0.15g of an organic compound gave 0.12 g of AgBr. Find out the percentage of bromine in the compound.

Ans. Molar mass of AgBr = 108 + 80 = 188 g mol⁻¹
188g AgBr contains 80 g bromine.

0.12 g AgBr contains $\frac{80 \times 0.12}{188}$ g bromine.

Percentage of bromine = $\frac{80 \times 0.12 \times 100}{188 \times 0.15} = 34.04\%$

Q.10 In sulphur estimation, 0.157g of an organic compoud gave 0.4813 g of barium sulphate. What is the percentage of sulphur in the compound ?

Ans. Molecular mass of BaSO₄ = 137 + 32 + 64 = 233g
233g BaSO₄ contains 32 g sulphur

0.4813g BaSO₄ contains $\frac{32 \times 0.4813}{233}$ g sulphur

Percentage of sulphur = $\frac{32 \times 0.4813 \times 100}{233 \times 0.157} = 42.10\%$

ANSWER KEY

BEGINNER'S BOX-1

Que.	1	2	3	4	5
Ans.	1	3	3	4	2

BEGINNER'S BOX-2

Que.	1	2	3	4	5
Ans.	2	2	2	3	3

EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

1. Which is not a renewable source
(1) Forest (2) Coal
(3) Water (4) Forest organism
2. Noosphere is synonyms of
(1) Environment (2) Atmosphere
(3) Hydrosphere (4) Stratosphere
3. When biosphere turns into human dominated environment it is called
(1) Noosphere (2) Troposphere
(3) Mesosphere (4) Man sphere
4. The living organisms on or around the earth constitute
(1) Biome
(2) Biosphere
(3) Community
(4) Biocoenosis
5. What is the correct sequence of atmospheric layers starting from earth
(1) Stratosphere troposphere, mesosphere, thermosphere
(2) Troposphere, stratosphere, mesosphere, thermosphere
(3) Mesosphere, troposphere, stratosphere, thermosphere
(4) Thermosphere, mesosphere, stratosphere, troposphere
6. On earth all living organisms constitute
(1) Community (2) Biome
(3) Association (4) Biosphere
7. Which of the following is the non conventional source of energy
(1) Coal
(2) Petroleum
(3) Electricity from nuclear power plants
(4) Solar radiations
8. Petroleum resource is
(1) Renewable
(2) Non renewable
(3) Synthetic & biodegradable
(4) Infinite & unconventional
9. The main aim of plant conservation is -
(1) To conserve the necessary ecological activities and life supporting systems
(2) To conserve species diversity and range of genetic material
(3) Both the above
(4) None of the above
10. Which will not cause any atmospheric pollution
(1) Hydrogen (2) Sulphur dioxide
(3) Carbon dioxide (4) Carbon monoxide
11. Which of the following is the main factor of water pollution
(1) Smoke (2) Industrial waste
(3) Detergent (4) Ammonia
12. Main air pollutant among the following is
(1) CO (2) CO₂
(3) N₂ (4) Sulphur
13. Which is more important for water pollution
(1) Sound (2) SO₂
(3) Salts of arsenic (4) Sewage
14. Pollution can be controlled by
(1) Sewage treatment
(2) Checking atomic blasts
(3) Manufacturing electrically operated vehicles
(4) All the above
15. In cities like Bombay and Calcutta the major air pollutants are
(1) Ozone
(2) Carbon monoxide and oxides of Sulphur
(3) Hydrocarbons and not air
(4) Algal spores and marsh gas
16. Which is the greatest air pollutant these days
(1) Factories (2) Motor vehicles
(3) Domestic appliances (4) animals
17. Removal of the soil by the action of wind and water is known as
(1) Erosion (2) Fossilization
(3) Leaching (4) Calcification
18. Acid rain occurs due to atmospheric pollution of
(1) SO₂ (2) NH₃
(3) CO₂ (4) N₂O

- 19.** Photochemical smog was first observed in -
 (1) London (2) Los Angeles
 (3) Paris (4) Tokyo
- 20.** Ozone layer of upper atmosphere is being destroyed by
 (1) Sulphur dioxide (2) Carbon dioxide
 (3) Chlorofluorocarbon (4) Smog
- 21.** Most hazardous metal pollutant of automobile exhaust is
 (1) Hg (2) Cd (3) Pb (4) Cu
- 22.** SO_2 pollution is indicated by
 (1) Grasses (2) Mosses (3) Lichens (4) Fossils
- 23.** B.O.D. is connected with
 (1) Organic matter (2) Microbes
 (3) Both (4) None
- 24.** Acid rain is due to increase in atmospheric concentration of
 (1) Ozone and dust (2) CO_2 and CO
 (3) SO_3 and CO (4) SO_2 and NO_2
- 25.** Soil erosion is greater when
 (1) No rain occurs
 (2) Winds do not blow
 (3) The rainfall is evenly distributed
 (4) The Rainfall is received in heavy down pour
- 26.** Soil erosion can be prevented by
 (1) Over grazing
 (2) Removal of vegetation
 (3) Afforestation (Plantation)
 (4) Increasing bird population
- 27.** Soil conservation is the process where
 (1) Soil is aerated
 (2) Soil erosion is allowed
 (3) Soil is protected against loss
 (4) Sterile soil is converted into fertile soil
- 28.** Ether and benzene can be separated by :-
 (1) Filtration (2) Distillation
 (3) Crystallization (4) Sublimation

EXERCISE-I (Conceptual Questions)**ANSWER KEY**

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	1	1	2	2	4	4	2	1	1	2	1	3	4	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28		
Ans.	2	1	1	2	3	3	3	3	4	4	3	3	2		

EXERCISE-II (Previous Year Questions)

AIPMT 2008

1. An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gave C, 38.71% and H, 9.67%. The empirical formula of the compound would be :-
(1) CHO (2) CH₄O (3) CH₃O (4) CH₂O

AIPMT Pre.-2011

2. Which one of the following statement is not true ?
(1) pH of drinking water should be between 5.5–9.5
(2) Concentration of DO below 6 ppm is good for the growth of fish
(3) Clean water would have BOD value of less than 5 ppm
(4) Oxides of sulphur, nitrogen and carbon are the most widespread air pollutant
3. In Dumas' method of estimation of nitrogen 0.35g of an organic compound gave 55 mL of nitrogen collected at 300 K temperature and 715 mm pressure. The percentage composition of nitrogen in the compound would be :
(Aqueous tension at 300 K = 15 mm)
(1) 15.45 (2) 16.45 (3) 17.45 (4) 14.45
4. The Lassaigne's extract is boiled with conc HNO₃ while testing for the halogens. By doing so it :-
(1) decomposes Na₂S and NaCN, if formed
(2) helps in the precipitation of AgCl
(3) increases the solubility product of AgCl
(4) increases the concentration of NO₃⁻ ions

AIPMT Pre.-2012

5. Which one of the following statements regarding photochemical smog is not correct?
(1) Photochemical smog is formed through photochemical reaction involving solar energy
(2) Photochemical smog does not cause irritation in eyes and throat.
(3) Carbon monoxide does not play any role in photochemical smog formation
(4) Photochemical smog is an oxidising agent in character

AIIMS 2013

6. Which of the following is not a green house gas:-
(1) Hydrogen (2) Carbon dioxide
(3) Methane (4) Nitrous oxide or N₂O

AIPMT 2014

7. Which one of the following is **not** a common component of Photochemical Smog?
(1) Ozone (2) Acrolein
(3) Peroxyacetyl nitrate (4) Chlorofluorocarbons

AIPMT/NEET & AIIMS (2006-2018)

8. In the Kjeldahl's method for estimation of nitrogen present in a soil sample, ammonia evolved from 0.75 g of sample neutralized 10 mL of 1 M H₂SO₄. The percentage of nitrogen in the soil is :
(1) 37.33 (2) 45.33
(3) 35.33 (4) 43.33

AIIMS 2014

9. Concentration of Peroxy Acetyl Nitrate (PAN) is maximum in :-
(1) Fog (2) Smoke
(3) Smog (4) Photochemical smog

AIPMT 2015

10. In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is :-
(1) 18.20 (2) 16.76
(3) 15.76 (4) 17.36

AIIMS 2015

11. If BOD of a water body has a 20 ppm then what can you conclude?
(1) polluted (2) light polluted
(3) clean water (4) highly polluted

NEET(UG) 2017

12. The most suitable method of separation of 1 : 1 mixture of ortho and para-nitrophenols is :
(1) Chromatography (2) Crystallisation
(3) Steam distillation (4) Sublimation
13. Which of the following is a sink for CO ?
(1) Micro organism present in the soil
(2) Oceans
(3) Plants
(4) Haemoglobin

AIIMS 2017

14. Which of the following is Herbicide
(1) Sodium chlorate
(2) Organo carbonates
(3) Potassium dihydrogen phosphate
(4) Ammonium nitrate

NEET(UG) 2018

15. Which oxide of nitrogen is **not** a common pollutant introduced into the atmosphere both due to natural and human activity ?
(1) N₂O₅ (2) NO₂
(3) N₂O (4) NO

EXERCISE-II (Previous Year Questions)

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	3	2	2	1	2	1	4	1	4	2	4	3	1	1	1

EXERCISE-III (Analytical Questions)**Check Your Understanding**

- The term biosphere is used for the zone of the earth where life exists
 - On the lithosphere
 - In the hydrosphere
 - In the lithosphere and hydrosphere
 - In the lithosphere, hydrosphere and atmosphere
- The population of India is 15% of the world but its annual energy consumption is only
 - 0.2%
 - 2.0%
 - 10%
 - 25%
- Environmental laning organisation is
 - CSIR
 - CEPHERI
 - ICAR
 - NEERI
- Which of the following atmospheric pollutants is not produced by the exhaust of motor vehicle in Delhi
 - SO₂
 - Hydrocarbon gases
 - Fly ash
 - CO
- If water pollution continues at its present rate, it will eventually
 - Stop water cycle
 - Prevent precipitation
 - Make oxygen molecules unavailable to water plants.
 - Make nitrate molecules unavailable to water plants.
- Recent reports of acid rains industrial cities are due to the effect of atmospheric pollution by
 - Excessive release of NO₂ and SO₂ by burning of fossil fuels.
 - Excessive release of CO₂ by burning of fuel like wood and charcoal, cutting of forests and increased animal population.
 - Excessive release of NH₃ by industrial plants and coal gas.
 - Excessive release of CO in atmosphere by incomplete combustion of cock, charcoal and other carbonaceous fuels in pancy of oxygen,
- Pollution is a change in physical, chemical or biological characters of our land and water that may be
 - Desirable and harmful to human
 - Desirable and useful to human
 - Undesirable and harmful to human
 - undesirable and useful to human
- An increase in CO₂ concentration in the atmosphere will result in
 - Adverse effects of natural vegetation
 - Global warming
 - Temperature decrease in global atmosphere
 - Genetic disorders in plants and animals
- Identify the incorrect statement from the following :-
 - Oxides of nitrogen in the atmosphere can cause the depletion of ozone layer
 - Ozone absorbs the intense ultraviolet radiation of the sun
 - Depletion of ozone layer is because of its chemical reactions with chlorofluoro alkanes
 - Ozone absorbs infrared radiation

EXERCISE-III (Analytical Questions)**ANSWER KEY**

Que.	1	2	3	4	5	6	7	8	9
Ans.	4	2	4	3	3	1	3	2	4

EXERCISE-IV (Assertion & Reason)

Target AIIMS

Directions for Assertion & Reason questions

These questions consist of two statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

- (A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (C) If Assertion is True but the Reason is False.
- (D) If both Assertion & Reason are false.

1. **Assertion :-** Classical smog is different from photo chemical smog.
Reason :- Because classical smog is oxidising and photo chemical smog is reducing.
 (1) A (2) B (3) C (4) D

EXERCISE-IV (Assertion & Reason)

ANSWER KEY

Que.	1	
Ans.	3	

IMPORTANT NOTES

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